

Michigan Public Service Commission
DTE Electric Company
2022/2023 Forecast Energy Allocation Schedule

Case No.: U-20836
Exhibit: A-17
Schedule: G1.1
Witness: M. Asghar
Page: 1 of 1

(a)		(b)	
Line No.	Rate Description	COS Class	Forecast Nov. 2022 - Oct. 2023 Energy (MWh)
1	<u>Residential</u>		
2	Residential	D1 & Other	15,992,982.8
3	Time-of-Day	D1.2	200,586.6
4	Space Heating	D2	322,679.6
5	Residential O.P.L.	E1/E2/D9	7,613.2
6	Total Residential		16,523,862.3
7			
8	<u>Secondary</u>		
9	General Service	D3 & Other	8,248,820.7
10	Large General Service	D4	2,172,452.8
11	Secondary Schools	D3.2	321,421.6
12	Commercial O.P.L.	E1/E2/D9	29,812.9
13	Traffic and Signal Lights	E1/E2/D9	65,081.3
14	Total Secondary		10,837,589.3
15			
16	<u>Primary</u>		
17	Primary Supply	D11 & Other	
18	Transmission Customer Owned Substation		1,349,173.4
19	Transmission DTE Owned Substation		3,049,276.5
20	Subtransmission Customer Owned Substation		440,883.8
21	Subtransmission DTE Owned Substation		1,751,555.7
22	Primary Distribution		6,432,738.4
23	Total Primary Supply		13,023,627.7
24			
25	Schools	D6.2	
26	Transmission Customer Owned Substation		0.0
27	Transmission DTE Owned Substation		0.0
28	Subtransmission Customer Owned Substation		4,880.2
29	Subtransmission DTE Owned Substation		31,426.0
30	Primary Distributuon		331,706.9
31	Total Schools		368,013.1
32			
33	Interruptible Primary Supply	D8	
34	Transmission Customer Owned Substation		130,210.5
35	Transmission DTE Owned Substation		0.0
36	Subtransmission Customer Owned Substation		3,854.9
37	Subtransmission DTE Owned Substation		68,223.7
38	Primary Distribution		413,646.3
39	Total Interruptible Primary		615,935.4
40			
41	Combined Alt. Metal Mltg. and Process Heat	R1.1/R1.2	
42	Transmission Customer Owned Substation		0.0
43	Transmission DTE Owned Substation		20,861.4
44	Subtransmission Customer Owned Substation		5,569.3
45	Subtransmission DTE Owned Substation		79,153.9
46	Primary Distribution		391,208.4
47	Secondary Distribution		45,331.0
48	Total Combined Alt. Metal Mltg. And Process Heat		542,124.1
49			
50	Interruptible Rider	R10	
51	Transmission Customer Owned Substation		283,215.6
52	Transmission DTE Owned Substation		288,007.8
53	Subtransmission Customer Owned Substation		13,252.7
54	Subtransmission DTE Owned Substation		47,815.7
55	Primary Distribution		20,685.8
56	Total Interruptible Rider		652,977.7
57			
58	Total Primary (lines 23 + 31 + 39 + 48 + 56)		15,202,677.9
59			
60	Street Lighting	E1/E2/D9	146,765.3
61			
62	TOTAL LOAD (lines 6 + 14 + 58 + 60)		42,710,894.7

Michigan Public Service Commission
DTE Electric Company
Demand and Energy Allocation
Percentages By Rate Class
2022/2023 Forecast Power Supply
Power Plant Energy Production
Allocation Schedule 100 - Total Energy

Case No.: U-20836
 Exhibit: A-17
 Schedule: G1.2
 Witness: M. Asghar
 Page: 1 of 11

Line No.	(a) Rate Description	(b) COS Class	(c) Nov. 2022 - Oct. 2023 Total Energy Schedule 100 Allocation Factors
1	<u>Residential</u>		
2	Residential	D1 & Other	0.37445
3	Time-of-Day	D1.2	0.00470
4	Space Heating	D2	0.00755
5	Residential O.P.L.	E1/E2/D9	0.00018
6	Total Residential		0.38688
7			
8	<u>Secondary</u>		
9	General Service	D3 & Other	0.19313
10	Large General Service	D4	0.05086
11	Schools - Secondary	D3.2	0.00753
12	Commercial O.P.L.	E1/E2/D9	0.00070
13	Traffic and Signal Lights	E1/E2/D9	0.00152
14	Total Secondary		0.25374
15			
16	<u>Primary</u>		
17	Primary Supply	D11 & Other	0.30493
18	Schools - Primary	D6.2	0.00862
19	Interruptible Primary Supply	D8	0.01442
20	Combined Alt. Metal Mltg. and Process Heat	R1.1/R1.2	0.01269
21	Interruptible Rider	R10	0.01529
22	Total Primary		0.35594
23			
24	Street Lighting	E1/E2/D9	0.00344
25			
26	TOTAL		1.00000

Michigan Public Service Commission
DTE Electric Company
Demand and Energy Allocation
Percentages By Rate Class
2022/2023 Forecast Power Supply
Production System Output
Allocation Schedule 200A - 12CP

Case No.: U-20836
 Exhibit: A-17
 Schedule: G1.2
 Witness: M. Asghar
 Page: 2 of 11

(a)		(b)	(c)
Line			Nov. 2022 - Oct. 2023
No.		COS Class	12CP Demand
Rate Description			Schedule 200A
			Allocation Factors
1	Residential		
2	Residential	D1 & Other	0.42992
3	Time-of-Day	D1.2	0.00438
4	Space Heating	D2	0.00760
5	Residential O.P.L.	E1/E2/D9	0.00012
6	Total Residential		0.44202
7			
8	<u>Secondary</u>		
9	General Service	D3 & Other	0.19572
10	Large General Service	D4	0.04669
11	Schools - Secondary	D3.2	0.00669
12	Commercial O.P.L.	E1/E2/D9	0.00049
13	Traffic and Signal Lights	E1/E2/D9	0.00107
14	Total Secondary		0.25066
15			
16	<u>Primary</u>		
17	Primary Supply	D11 & Other	0.24257
18	Schools - Primary	D6.2	0.00815
19	Interruptible Primary Supply	D8	0.01219
20	Combined Alt. Metal Mltg. and Process Heat	R1.1/R1.2	0.01091
21	Interruptible Rider	R10	0.03109
22	Total Primary		0.30491
23			
24	Street Lighting	E1/E2/D9	0.00240
25			
26	TOTAL		1.00000

Michigan Public Service Commission
DTE Electric Company
Demand and Energy Allocation
Percentages By Rate Class
2022/2023 Forecast Power Supply
Production System Output
Allocation Schedule 200B - 4CP

Case No.: U-20836
 Exhibit: A-17
 Schedule: G1.2
 Witness: M. Asghar
 Page: 3 of 11

(a)		(b)	(c)
Line			Nov. 2022 - Oct. 2023
No.		COS Class	4CP Demand
Rate Description			Schedule 200B
			Allocation Factors
1	<u>Residential</u>		
2	Residential	D1 & Other	0.50006
3	Time-of-Day	D1.2	0.00376
4	Space Heating	D2	0.00548
5	Residential O.P.L.	E1/E2/D9	0.00000
6	Total Residential		0.50930
7			
8	<u>Secondary</u>		
9	General Service	D3 & Other	0.20332
10	Large General Service	D4	0.04547
11	Schools - Secondary	D3.2	0.00658
12	Commercial O.P.L.	E1/E2/D9	0.00000
13	Traffic and Signal Lights	E1/E2/D9	0.00087
14	Total Secondary		0.25624
15			
16	<u>Primary</u>		
17	Primary Supply	D11 & Other	0.21584
18	Schools - Primary	D6.2	0.00841
19	Interruptible Primary Supply	D8	0.00601
20	Combined Alt. Metal Mltg. and Process Heat	R1.1/R1.2	0.00420
21	Interruptible Rider	R10	0.00000
22	Total Primary		0.23446
23			
24	Street Lighting	E1/E2/D9	0.00000
25			
26	TOTAL		1.00000

Michigan Public Service Commission
DTE Electric Company
Demand and Energy Allocation
Percentages By Rate Class
2022/2023 Forecast Power Supply
Input to Transmission Substation
Allocation Schedule 201 - 12CP

Case No.: U-20836
Exhibit: A-17
Schedule: G1.2
Witness: M. Asghar
Page: 4 of 11

(a)		(b)	(c)
Line			Nov. 2022 - Oct. 2023
No.		COS Class	12CP Demand
Rate Description			Schedule 201
			Allocation Factors
1	<u>Residential</u>		
2	Residential	Residential Secondary	0.44835
3	Residential O.P.L.	E1/E2/D9	0.00010
4	Total Residential		0.44845
5			
6	<u>Secondary</u>		
7	General Service	Commercial Secondary	0.28045
8	Commercial O.P.L.	E1/E2/D9	0.00038
9	Traffic and Signal Lights	E1/E2/D9	0.00109
10	Total Secondary		0.28192
11			
12	<u>Primary</u>		
13	Primary Distribution	Primary	0.22382
14	Subtransmission	Subtransmission	0.04373
15	Transmission	Transmission	-
16	Total Primary		0.26755
17			
18	Street Lighting	E1/E2/D9	0.00207
19			
20	TOTAL		1.00000

Michigan Public Service Commission
DTE Electric Company
Demand and Energy Allocation
Percentages By Rate Class
2022/2023 Forecast Power Supply
Input to Subtransmission Lines
Allocation Schedule 202

Case No.: U-20836
Exhibit: A-17
Schedule: G1.2
Witness: M. Asghar
Page: 5 of 11

(a)		(b)	(c)
Line			Nov. 2022 - Oct. 2023
No.		COS Class	Demand
Rate Description			Schedule 202
			Allocation Factors
1	<u>Residential</u>		
2	Residential	Residential Secondary	0.50887
3	Residential O.P.L.	E1/E2/D9	0.00031
4	Total Residential		0.50918
5			
6	<u>Secondary</u>		
7	General Service	Commercial Secondary	0.25770
8	Commercial O.P.L.	E1/E2/D9	0.00097
9	Traffic and Signal Lights	E1/E2/D9	0.00066
10	Total Secondary		0.25933
11			
12	<u>Primary</u>		
13	Primary Distribution	Primary	0.18987
14	Subtransmission	Subtransmission	0.03787
15	Transmission	Transmission	-
16	Total Primary		0.22774
17			
18	Street Lighting	E1/E2/D9	0.00375
19			
20	TOTAL		1.00000

Michigan Public Service Commission
DTE Electric Company
Demand and Energy Allocation
Percentages By Rate Class
2022/2023 Forecast Power Supply
Primary Voltage Substation
Allocation Schedule 203A

Case No.: U-20836
 Exhibit: A-17
 Schedule: G1.2
 Witness: M. Asghar
 Page: 6 of 11

Line	(a)	(b)	(c) Nov. 2022 - Oct. 2023 Demand Schedule 203A Allocation Factors
No.	Rate Description	COS Class	
1	<u>Residential</u>		
2	Residential	Residential Secondary	0.52890
3	Residential O.P.L.	E1/E2/D9	0.00032
4	Total Residential		0.52922
5			
6	<u>Secondary</u>		
7	General Service	Commercial Secondary	0.26784
8	Commercial O.P.L.	E1/E2/D9	0.00101
9	Traffic and Signal Lights	E1/E2/D9	0.00068
10	Total Secondary		0.26953
11			
12	<u>Primary</u>		
13	Primary Distribution	Primary	0.19734
14	Subtransmission	Subtransmission	-
15	Transmission	Transmission	-
16	Total Primary		0.19734
17			
18	Street Lighting	E1/E2/D9	0.00390
19			
20	TOTAL		1.00000

Michigan Public Service Commission
DTE Electric Company
Demand and Energy Allocation
Percentages By Rate Class
2022/2023 Forecast Power Supply
Primary Voltage Substation
Allocation Schedule 203B

Case No.: U-20836
 Exhibit: A-17
 Schedule: G1.2
 Witness: M. Asghar
 Page: 7 of 11

(a)		(b)	(c)
Line			Nov. 2022 - Oct. 2023
No.		COS Class	Demand
Rate Description			Schedule 203B
			Allocation Factors
1	<u>Residential</u>		
2	Residential	Residential Secondary	-
3	Residential O.P.L.	E1/E2/D9	-
4	Total Residential		0.00000
5			
6	<u>Secondary</u>		
7	General Service	Commercial Secondary	-
8	Commercial O.P.L.	E1/E2/D9	-
9	Traffic and Signal Lights	E1/E2/D9	-
10	Total Secondary		0.00000
11			
12	<u>Primary</u>		
13	Primary Distribution	Primary	-
14	Subtransmission	Subtransmission	0.46880
15	Transmission	Transmission	0.53120
16	Total Primary		1.00000
17			
18	Street Lighting	E1/E2/D9	-
19			
20	TOTAL		1.00000

Michigan Public Service Commission
DTE Electric Company
Demand and Energy Allocation
Percentages By Rate Class
2022/2023 Forecast Power Supply
Primary Voltage Substation
Allocation Schedule 203C

Case No.: U-20836
Exhibit: A-17
Schedule: G1.2
Witness: M. Asghar
Page: 8 of 11

(a)		(b)	(c)
Line			Nov. 2022 - Oct. 2023
No.		COS Class	Demand
Rate Description			Schedule 203C
			Allocation Factors
1	<u>Residential</u>		
2	Residential	Residential Secondary	0.49452
3	Residential O.P.L.	E1/E2/D9	0.00030
4	Total Residential		0.49483
5			
6	<u>Secondary</u>		
7	General Service	Commercial Secondary	0.25044
8	Commercial O.P.L.	E1/E2/D9	0.00094
9	Traffic and Signal Lights	E1/E2/D9	0.00064
10	Total Secondary		0.25202
11			
12	<u>Primary</u>		
13	Primary Distribution	Primary	0.18452
14	Subtransmission	Subtransmission	0.03047
15	Transmission	Transmission	0.03453
16	Total Primary		0.24951
17			
18	Street Lighting	E1/E2/D9	0.00364
19			
20	TOTAL		1.00000

Michigan Public Service Commission
DTE Electric Company
Demand and Energy Allocation
Percentages By Rate Class
2022/2023 Forecast Power Supply
Primary Lines
Allocation Schedule 204

Case No.: U-20836
Exhibit: A-17
Schedule: G1.2
Witness: M. Asghar
Page: 9 of 11

(a)		(b)	(c)
Line			Nov. 2022 - Oct. 2023
No.		COS Class	Demand
Rate Description			Schedule 204
			Allocation Factors
1	<u>Residential</u>		
2	Residential	Residential Secondary	0.58854
3	Residential O.P.L.	E1/E2/D9	0.00036
4	Total Residential		0.58890
5			
6	<u>Secondary</u>		
7	General Service	Commercial Secondary	0.29805
8	Commercial O.P.L.	E1/E2/D9	0.00112
9	Traffic and Signal Lights	E1/E2/D9	0.00076
10	Total Secondary		0.29993
11			
12	<u>Primary</u>		
13	Primary Distribution	Primary	0.10732
14	Subtransmission	Subtransmission	-
15	Transmission	Transmission	-
16	Total Primary		0.10732
17			
18	Street Lighting	E1/E2/D9	0.00385
19			
20	TOTAL		1.00000

Michigan Public Service Commission
DTE Electric Company
Demand and Energy Allocation
Percentages By Rate Class
2022/2023 Forecast Power Supply
Secondary Line Transformers
Allocation Schedule 205

Case No.: U-20836
 Exhibit: A-17
 Schedule: G1.2
 Witness: M. Asghar
 Page: 10 of 11

(a)		(b)	(c)
Line			Nov. 2022 - Oct. 2023
No.		COS Class	Demand
Rate Description			Schedule 205
			Allocation Factors
1	<u>Residential</u>		
2	Residential	Residential Secondary	0.65895
3	Residential O.P.L.	E1/E2/D9	0.00040
4	Total Residential		0.65936
5			
6	<u>Secondary</u>		
7	General Service	Commercial Secondary	0.33368
8	Commercial O.P.L.	E1/E2/D9	0.00125
9	Traffic and Signal Lights	E1/E2/D9	0.00085
10	Total Secondary		0.33579
11			
12	<u>Primary</u>		
13	Primary Distribution	Primary	-
14	Subtransmission	Subtransmission	-
15	Transmission	Transmission	-
16	Total Primary		0.00000
17			
18	Street Lighting	E1/E2/D9	0.00486
19			
20	TOTAL		1.00000

Michigan Public Service Commission
DTE Electric Company
Demand and Energy Allocation
Percentages By Rate Class
2022/2023 Forecast Power Supply
Secondary Distribution Lines
Allocation Schedule 300

Case No.: U-20836
Exhibit: A-17
Schedule: G1.2
Witness: M. Asghar
Page: 11 of 11

Line No.	(a) Rate Description	(b) COS Class	(c) Nov. 2022 - Oct. 2023 Demand Schedule 300 Allocation Factors
1	<u>Residential</u>		
2	Residential	Residential Secondary	0.72849
3	Residential O.P.L.	E1/E2/D9	0.00020
4	Total Residential		0.72869
5			
6	<u>Secondary</u>		
7	General Service	Commercial Secondary	0.26777
8	Commercial O.P.L.	E1/E2/D9	0.00064
9	Traffic and Signal Lights	E1/E2/D9	0.00043
10	Total Secondary		0.26884
11			
12	<u>Primary</u>		
13	Primary Distribution	Primary	-
14	Subtransmission	Subtransmission	-
15	Transmission	Transmission	-
16	Total Primary		0.00000
17			
18	Street Lighting	E1/E2/D9	0.00247
19			
20	TOTAL		1.00000

**Michigan Public Service Commission
DTE Electric Company
Current and Historical Credit Ratings**

Case No.: U-20836
Exhibit: A-18
Schedule: H1
Witness: T. J. Lepczyk
Page: 1 of 1

**Line
No.**

1	Standard & Poor's						
2		Current	Ratings at Year End				
3		Ratings	2020	2019	2018	2017	2016
4	DTE Electric						
5	Corporate Credit Rating	A-	A-	A-	BBB+	BBB+	BBB+
7	Senior Secured Debt	A	A	A	A	A	A
8	Commercial Paper	A-2	A-2	A-2	A-2	A-2	A-2
9	Outlook	Stable	Stable	Stable	Stable	Stable	Stable
10							
11	DTE Energy						
12	Corporate Credit Rating	BBB+	BBB+	BBB+	BBB+	BBB+	BBB+
13	Senior Unsecured Debt	BBB	BBB	BBB	BBB	BBB	BBB
14	Commercial Paper	A-2	A-2	A-2	A-2	A-2	A-2
15	Outlook	Stable	Stable	Stable	Stable	Stable	Stable
16							
17							
18							
19	Moody's Investors Service						
20		Current	Ratings at Year End				
21		Ratings	2020	2019	2018	2017	2016
22	DTE Electric						
24	Senior Secured Debt	Aa3	Aa3	Aa3	Aa3	Aa3	Aa3
25	Commercial Paper	P-1	P-1	P-1	P-1	P-1	P-1
26	Outlook	Stable	Stable	Stable	Stable	Stable	Stable
27							
28	DTE Energy						
29	Senior Unsecured Debt	Baa2	Baa2	Baa2	Baa1	Baa1	Baa1
30	Commercial Paper	P-2	P-2	P-2	P-2	P-2	P-2
31	Outlook	Stable	Stable	Stable	Stable	Stable	Stable
32							
33							
34							
35	Fitch Ratings						
36		Current	Ratings at Year End				
37		Ratings	2020	2019	2018	2017	2016
38	DTE Electric						
39	Senior Secured Debt	A+	A+	A+	A+	A+	A+
40	Commercial Paper	F2	F2	F2	F2	F2	F2
41	Outlook	Stable	Stable	Negative	Negative	Negative	Negative
42							
43	DTE Energy						
44	Senior Unsecured Debt	BBB	BBB	BBB+	BBB+	BBB+	BBB+
45	Commercial Paper	F2	F2	F2	F2	F2	F2
46	Outlook	Stable	Stable	Negative	Negative	Negative	Negative

Michigan Public Service Commission
DTE Electric Company
Recent Utility Corporate Bond Issuances

Case No.: U-20836
Exhibit: A-18
Schedule: H2
Witness: T. J. Lepczyk
Page: 1 of 7

Line No.	(a) Offer Date	(b) Issuer	Credit Ratings			(f) Maturity	(g) Tenor	(h) Amount (\$MM)	(i) Coupon	(j) Spread
			(c) Secured (?)	(d) Moody's	(e) S&P					
1	10/01/19	NextEra Energy Capital Holdings	N	Baa1	BBB+	11/01/29	10	1000	2.750%	T+115 bp
2	10/01/19	MidAmerican Energy Co	N	Aa2	A+	04/15/29	10	250	3.650%	T+75 bp
3	10/01/19	MidAmerican Energy Co	N	Aa2	A+	04/15/50	31	600	3.150%	T+107 bp
4	10/02/19	AES Gener	N	Ba2	BB	10/07/79	60	450	6.350%	na
5	10/03/19	NextEra Energy Capital Holdings	N	Baa1	BBB+	09/01/22	3	450	1.950%	T+61 bp
6	10/29/19	DTE Energy Co	N	Baa2	BBB	11/01/22	3	500	2.250%	T+65 bp
7	10/29/19	DTE Energy Co	N	Baa2	BBB	03/01/30	10	300	2.950%	T+115 bp
8	11/04/19	Xcel Energy Inc	N	Baa1	BBB+	12/01/29	10	500	2.600%	T+90 bp
9	11/04/19	Xcel Energy Inc	N	Baa1	BBB+	12/01/49	30	500	3.500%	T+125 bp
10	11/04/19	Commonwealth Edison Co	Y	A1	A	11/15/49	30	300	3.200%	T+97 bp
11	11/05/19	Consolidated Edison Co of New York Inc	N	A3	A-	11/15/59	40	600	3.700%	T+140 bp
12	11/06/19	Vistra Operations Co LLC	N	Ba1	BBB-	07/15/24	5	300	3.550%	T+150 bp
13	11/06/19	Vistra Operations Co LLC	N	Ba1	BBB-	01/30/27	7	800	3.700%	T+200 bp
14	11/13/19	Entergy Mississippi Inc	Y	A2	A	06/01/49	30	135	3.850%	T+105 bp
15	11/14/19	Edison International	N	Baa3	BBB-	11/15/22	3	300	3.125%	T+155 bp
16	11/14/19	Edison International	N	Baa3	BBB-	11/15/24	5	500	3.550%	T+195 bp
17	11/18/19	Dominion Energy Gas Holdings LLC	N	Baa1	BBB+	11/15/24	5	600	2.500%	T+88 bp
18	11/18/19	Dominion Energy Gas Holdings LLC	N	Baa1	BBB+	11/15/29	10	600	3.000%	T+123 bp
19	11/18/19	Dominion Energy Gas Holdings LLC	N	Baa1	BBB+	11/15/29	10	300	3.900%	T+160 bp
20	11/18/19	Arizona Public Service Co	N	A2	A-	12/01/49	30	300	3.500%	T+120 bp
21	11/18/19	Ameren Illinois Co	Y	A1	A	03/15/50	30	300	3.250%	T+98 bp
22	11/20/19	Virginia Electric & Power Co	N	A2	A	12/01/49	30	550	3.300%	T+110 bp
23	11/21/19	Duke Energy Florida LLC	N	A3	A-	11/26/21	2	200	3m\$+25.0 bp	na
24	11/21/19	Duke Energy Florida LLC	Y	A3	A-	12/01/29	10	700	2.500%	T+73.0 bp
25	12/02/19	Dominion Energy Inc	N	Ba1	BBB-	Perp	Perp	800	4.650%	na
26	12/03/19	Wisconsin Electric Power	N	A2	A-	12/15/24	5	300	2.050%	T+52 bp
27	12/03/19	AEP Texas Inc	N	Baa1	A-	01/15/50	30	450	3.450%	T+130 bp
28	01/06/20	Duke Energy Carolinas LLC	Y	Aa2	A	02/01/30	10	500	2.450%	T+68 bp
29	01/06/20	Duke Energy Carolinas LLC	Y	Aa2	A	08/15/49	30	400	3.200%	T+88 bp
30	01/06/20	Southern California Edison Co	Y	A3	A-	08/01/29	10	100	2.850%	T+90 bp
31	01/06/20	Southern California Edison Co	Y	A3	A-	02/01/50	30	500	3.650%	T+140 bp
32	01/06/20	Southern California Gas Co	Y	Aa2	A+	02/01/30	10	650	2.550%	T+77 bp
33	01/06/20	The Southern Co	N	Baa3	BBB	01/30/80	60	1000	4.950%	na
34	01/07/20	Public Service Electric & Gas Co	Y	Aa3	A	01/15/30	10	300	2.450%	na
35	01/07/20	Public Service Electric & Gas Co	Y	Aa3	A	01/01/50	30	300	3.150%	T+85 bp
36	01/07/20	Eversource Energy	N	Baa1	BBB+	01/15/50	30	350	3.450%	T+115 bp
37	01/08/20	Georgia Power Co	N	Baa1	A-	07/30/23	4	700	2.100%	T+50 bp
38	01/08/20	Georgia Power Co	N	Baa1	A-	09/15/29	10	300	2.650%	T+95 bp
39	01/08/20	Georgia Power Co	N	Baa1	A-	01/30/50	30	500	3.700%	T+135 bp
40	01/28/20	Nevada Power Co	N	A2	A+	05/01/30	10	425	2.400%	T+78 bp
41	01/28/20	Nevada Power Co	N	A2	A+	08/01/50	31	300	3.125%	T+102 bp
42	02/11/20	DTE Electric Co	Y	Aa3	A	03/01/30	10	600	2.250%	T+68 bp
43	02/11/20	DTE Electric Co	Y	Aa3	A	03/01/50	30	500	2.950%	T+90 bp
44	02/18/20	Commonwealth Edison Co	Y	A1	A	03/01/30	10	350	2.200%	T+68 bp
45	02/18/20	Commonwealth Edison Co	Y	A1	A	03/01/50	30	650	3.000%	T+100 bp
46	02/18/20	FirstEnergy Corp	N	Baa3	BBB-	03/01/25	5	300	2.050%	T+70 bp
47	02/18/20	FirstEnergy Corp	N	Baa3	BBB-	03/01/30	10	600	2.650%	T+110 bp
48	02/18/20	FirstEnergy Corp	N	Baa3	BBB-	03/01/50	30	850	3.400%	T+140 bp
49	03/03/20	Entergy Louisiana LLC	Y	A2	A	03/15/51	31	350	2.900%	T+130 bp
50	03/03/20	American Electric Power Co Inc	N	Baa1	BBB+	03/01/30	10	400	2.300%	T+130 bp
51	03/03/20	American Electric Power Co Inc	N	Baa1	BBB+	03/01/50	30	400	3.250%	T+165 bp
52	03/04/20	Southern California Edison Co	Y	A3	A-	03/05/30	10	400	2.250%	T+125 bp
53	03/04/20	Southern California Edison Co	Y	A3	A-	02/01/50	30	700	3.650%	T+170 bp
54	03/05/20	Entergy Texas Inc	Y	Baa1	A	09/30/49	30	175	3.550%	T+138 bp
55	03/10/20	Duke Energy Indiana LLC	Y	Aa3	A	04/01/50	30	550	2.750%	T+165 bp

Michigan Public Service Commission
DTE Electric Company
Recent Utility Corporate Bond Issuances

Case No.: U-20836
Exhibit: A-18
Schedule: H2
Witness: T. J. Lepczyk
Page: 2 of 7

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Offer Date	Issuer	Credit Ratings			Maturity	Tenor	Amount (\$MM)	Coupon	Spread
			Secured (?)	Moody's	S&P					
56	03/13/20	Ohio Power Co	N	A2	A-	04/01/30	10	350	2.600%	T+170 bp
57	03/13/20	Entergy Arkansas LLC	Y	A2	A	06/01/28	8	100	4.000%	T+175 bp
58	03/17/20	Oncor Electric Delivery Co	N	A2	A+	05/15/30	10	400	2.750%	T+175 bp
59	03/17/20	Oncor Electric Delivery Co	N	A2	A+	05/15/50	30	400	3.700%	T+205 bp
60	03/17/20	Consumers Energy Co	Y	Aa3	A	08/01/51	31	575	3.500%	T+200 bp
61	03/17/20	Dominion Energy Inc	N	Baa2	BBB	03/15/25	5	400	3.300%	T+265 bp
62	03/17/20	Dominion Energy Inc	N	Baa2	BBB	03/15/27	7	350	3.600%	T+275 bp
63	03/17/20	Union Electric Co	Y	A2	A	03/15/30	10	465	2.950%	T+200 bp
64	03/20/20	Berkshire Hathaway Energy Co	N	A3	A-	04/15/25	5	1250	4.050%	T+355 bp
65	03/23/20	NSTAR Electric Co	N	A1	A+	04/01/30	10	400	3.950%	T+325 bp
66	03/24/20	Florida Power & Light Co	Y	Aa2	A+	04/01/25	5	1100	2.850%	T+238 bp
67	03/25/20	Berkshire Hathaway Energy Co	N	A3	A-	07/15/30	10	1100	3.700%	T+285 bp
68	03/25/20	Berkshire Hathaway Energy Co	N	A3	A-	07/15/50	30	900	4.250%	T+285 bp
69	03/26/20	Consolidated Edison Co of New York Inc	N	Baa1	A-	04/01/30	10	600	3.350%	T+255 bp
70	03/26/20	Consolidated Edison Co of New York Inc	N	Baa1	A-	04/01/50	30	1000	3.950%	T+255 bp
71	03/26/20	Northwest Natural Gas Co	Y	A2	AA-	03/15/50	30	150	3.600%	T+225 bp
72	03/27/20	Xcel Energy Inc	N	Baa1	BBB+	06/01/30	10	600	3.400%	T+270 bp
73	03/30/20	Southern California Edison Co	Y	A3	A-	08/01/25	5	600	3.700%	T+275 bp
74	03/30/20	Exelon Corp	N	Baa2	BBB	04/15/30	10	1250	4.050%	T+338 bp
75	03/30/20	Exelon Corp	N	Baa2	BBB	04/15/50	30	750	4.700%	T+338 bp
76	03/30/20	Wisconsin Power & Light Co	N	A2	A	04/01/50	30	350	3.650%	T+235 bp
77	03/30/20	Oklahoma Gas & Electric	N	A3	A-	04/01/30	10	300	3.250%	T+260 bp
78	03/30/20	AEP Transmission Co LLC	N	A2	A-	04/01/50	30	525	3.650%	T+235 bp
79	03/31/20	DTE Electric Co	Y	Aa3	A	03/01/31	11	600	2.625%	T+195 bp
80	03/31/20	Ameren Corp	N	Baa1	BBB	01/15/31	11	800	3.500%	T+285 bp
81	03/31/20	Dominion Energy Inc	N	Baa2	BBB	04/01/30	10	1500	3.375%	T+280 bp
82	03/31/20	San Diego Gas & Electric	Y	A2	A	04/15/50	30	400	3.320%	T+200 bp
83	03/31/20	Idaho Power Co	Y	A1	A-	03/01/48	28	230	4.200%	T+210 bp
84	03/31/20	Edison International	N	Baa3	BBB-	04/03/25	5	400	4.950%	T+463 bp
85	04/01/20	NextEra Energy Capital Holdings Inc	N	Baa1	BBB+	05/01/25	5	1250	2.750%	T+240 bp
86	04/01/20	The Southern Co	N	Baa2	BBB+	04/30/30	10	1000	3.700%	T+310 bp
87	04/01/20	PPL Capital Funding Inc	N	Baa2	BBB+	04/15/30	10	1000	4.125%	T+350 bp
88	04/02/20	Evergy Kansas Central Inc	Y	A2	A	04/15/50	30	500	3.450%	T+220 bp
89	04/06/20	PacifiCorp	Y	A1	A+	09/15/30	10	400	2.700%	T+205 bp
90	04/06/20	PacifiCorp	Y	A1	A+	03/15/51	31	600	3.300%	T+205 bp
91	04/06/20	Tucson Electric Power Co	N	A3	A-	06/15/50	30	350	4.000%	T+275 bp
92	04/07/20	Narragansett Electric	N	A3	A-	04/09/30	10	600	3.395%	T+265 bp
93	04/07/20	Avangrid Inc	N	Baa1	BBB	04/15/25	5	750	3.200%	T+275 bp
94	04/07/20	NiSource Inc	N	Baa2	BBB+	05/01/30	10	1000	3.600%	T+290 bp
95	04/08/20	American Water Capital Corp	N	Baa1	A	05/01/30	10	500	2.800%	T+210 bp
96	04/08/20	American Water Capital Corp	N	Baa1	A	05/01/50	30	500	3.450%	T+210 bp
97	04/08/20	IPALCO Enterprises Inc	Y	Baa3	BBB-	05/01/30	10	475	4.250%	T+350 bp
98	04/13/20	Essential Utilities Inc	N	Baa2	A-	04/15/30	10	500	2.704%	T+195 bp
99	04/13/20	Essential Utilities Inc	N	Baa2	A-	04/15/50	30	600	3.351%	T+195 bp
100	04/29/20	ONE Gas Inc	N	A2	A	05/15/30	10	300	2.000%	T+145 bp
101	04/29/20	Consumers Energy Co	Y	Aa3	A	05/01/60	40	525	2.500%	T+130 bp
102	05/05/20	NextEra Energy Capital Holdings	N	Baa1	BBB+	06/01/30	10	2000	2.250%	T+160 bp
103	05/06/20	Public Service Electric & Gas Co	Y	Aa3	A	05/01/50	30	375	2.700%	T+130 bp
104	05/08/20	Public Service Co of Colorado	Y	A1	A	01/15/31	11	375	1.900%	T+125 bp
105	05/08/20	Public Service Co of Colorado	Y	A1	A	01/15/51	31	375	2.700%	T+140 bp
106	05/11/20	Southwestern Public Service Co	Y	A3	A	05/01/50	30	350	3.150%	T+175 bp
107	05/11/20	ITC Holdings Corp	Y	Baa2	BBB+	05/14/30	10	700	2.950%	T+225 bp
108	05/12/20	Appalachian Power Co	N	Baa1	A-	05/01/50	30	500	3.700%	T+235 bp
109	05/12/20	Exelon Generation Co LLC	N	Baa2	BBB+	06/01/25	5	900	3.250%	T+295 bp
110	05/13/20	Entergy Corp	N	Baa2	BBB	06/15/30	10	600	2.800%	T+225 bp

Michigan Public Service Commission
DTE Electric Company
Recent Utility Corporate Bond Issuances

Case No.: U-20836
Exhibit: A-18
Schedule: H2
Witness: T. J. Lepczyk
Page: 3 of 7

Line No.	(a) Offer Date	(b) Issuer	(c) Secured (?)	Credit Ratings		(f) Maturity	(g) Tenor	(h) Amount (\$MM)	(i) Coupon	(j) Spread
					Moody's	S&P				
111	05/13/20	Entergy Corp	N	Baa2	BBB	06/15/50	30	600	3.750%	T+250 bp
112	05/13/20	Duke Energy Corp	N	Baa1	BBB+	06/01/30	10	500	2.450%	T+185 bp
113	05/14/20	Puget Energy Inc	N	Baa3	BBB-	06/15/30	10	650	4.100%	T+350 bp
114	05/15/20	CMS Energy Corp	N	Baa2	BBB-	06/01/50	30	500	4.750%	na
115	05/15/20	AES Corp/The	N	Baa3	BBB-	07/15/25	5	900	3.300%	T+300 bp
116	05/15/20	AES Corp/The	N	Baa3	BBB-	07/15/30	10	700	3.950%	T+338 bp
117	05/18/20	Duke Energy Ohio Inc	Y	A2	A	06/01/30	10	400	2.125%	T+140 bp
118	05/18/20	Evergy Metro Inc	Y	A2	A+	06/01/30	10	400	2.250%	T+155 bp
119	05/18/20	Piedmont Natural Gas Co Inc	N	A3	A-	06/01/50	30	400	3.350%	T+195 bp
120	05/19/20	Kentucky Utilities Co	Y	A1	A	06/01/50	30	500	3.300%	T+190 bp
121	05/19/20	National Fuel Gas Co	N	Baa3	BBB-	01/15/26	6	500	5.500%	T+525 bp
122	05/19/20	Entergy Mississippi LLC	Y	A2	A	06/01/51	31	170	3.500%	T+215 bp
123	05/19/20	Arizona Public Service Co	N	A2	A-	05/15/50	30	600	3.350%	T+195 bp
124	05/26/20	Interstate Power and Light Co	N	Baa1	A-	06/01/30	10	400	2.300%	T+165 bp
125	06/01/20	Southwest Gas Corp	N	A3	A-	06/15/30	10	450	2.200%	T+155 bp
126	06/01/20	PECO Energy Co	Y	Aa3	A	06/15/50	30	350	2.800%	T+135 bp
127	06/02/20	CenterPoint Energy Houston Electric LLC	Y	A2	A	07/01/50	30	300	2.900%	T+143 bp
128	06/02/20	The East Ohio Gas Company	N	A2	BBB+	06/15/25	5	500	1.750%	T+100 bp
129	06/02/20	The East Ohio Gas Company	N	A2	BBB+	06/15/30	10	500	2.000%	T+135 bp
130	06/02/20	The East Ohio Gas Company	N	A2	BBB+	06/15/50	30	800	3.000%	T+160 bp
131	06/02/20	Baltimore Gas & Electric Co	N	A3	A	06/15/50	30	400	2.900%	T+147 bp
132	06/03/20	FirstEnergy Corp	N	Baa3	BBB-	01/15/26	6	300	1.600%	T+125 bp
133	06/03/20	FirstEnergy Corp	N	Baa3	BBB-	09/01/30	10	450	2.250%	T+150 bp
134	06/08/20	Duke Energy Florida LLC	Y	A1	A	06/15/30	10	500	1.750%	T+90 bp
135	06/08/20	Northern States Power Co/MN	N	Aa3	A	06/01/51	31	700	2.600%	T+105 bp
136	06/10/20	Sempra Energy	N	Ba1	BBB-	Perp	Perp	900	4.875%	4.875%
137	06/10/20	Pinnacle West Capital Corp	N	A3	BBB+	06/15/25	5	500	1.300%	T+95 bp
138	06/12/20	Black Hills Corp	N	Baa2	BBB+	06/15/30	10	400	2.500%	T+185 bp
139	06/16/20	Pacific Gas & Electric Co	N	Baa3	BBB-	06/16/22	2	500	3m\$L+148 bp	3m\$L+148 bp
140	06/16/20	Pacific Gas & Electric Co	N	Baa3	BBB-	06/16/22	2	2500	1.750%	T+155 bp
141	06/16/20	Pacific Gas & Electric Co	N	Baa3	BBB-	08/01/27	7	1000	2.100%	T+155 bp
142	06/16/20	Pacific Gas & Electric Co	N	Baa3	BBB-	02/01/31	11	2000	2.500%	T+175 bp
143	06/16/20	Pacific Gas & Electric Co	N	Baa3	BBB-	08/01/40	20	1000	3.300%	T+180 bp
144	06/16/20	Pacific Gas & Electric Co	N	Baa3	BBB-	08/01/50	30	1925	3.500%	T+200 bp
145	06/17/20	Idaho Power Co	Y	A1	A-	07/15/30	10	80	1.900%	T+120 bp
146	06/23/20	Niagara Mohawk Power Corp	N	A3	A-	06/27/30	10	600	1.960%	T+125 bp
147	06/23/20	Niagara Mohawk Power Corp	N	A3	A-	06/27/50	30	500	3.025%	T+155 bp
148	06/29/20	AEP Texas Inc	N	Baa1	A-	07/01/30	10	600	2.100%	T+150 bp
149	07/28/20	Florida Power & Light Co	N	A1	A	07/28/23	3	1250	3m\$L+38 bp	3m\$L+38.0 bp
150	08/03/20	DTE Energy Co	N	Baa2	BBB	06/01/25	5	800	1.050%	T+85 bp
151	08/04/20	Public Service Electric & Gas Co	N	Aa3	A	08/01/50	30	375	2.050%	T+90 bp
152	08/05/20	Tucson Electric Power Co	N	A3	A-	08/01/30	10	300	1.500%	T+100 bp
153	08/11/20	Eversource Energy	N	Baa1	BBB+	08/15/25	5	300	0.800%	T+55 bp
154	08/11/20	Eversource Energy	N	Baa1	BBB+	08/15/30	10	600	1.650%	T+100 bp
155	08/11/20	Eversource Energy	N	Baa1	BBB+	01/15/50	29	300	3.450%	T+125 bp
156	08/12/20	NiSource Inc	N	Baa2	BBB+	08/15/25	5	1250	0.950%	T+70 bp
157	08/12/20	NiSource Inc	N	Baa2	BBB+	02/15/31	11	750	1.700%	T+103 bp
158	08/12/20	Public Service Electric & Gas Co	N	Baa1	BBB+	08/15/25	5	550	0.800%	T+55 bp
159	08/12/20	Public Service Electric & Gas Co	N	Baa1	BBB+	08/15/30	10	550	1.600%	T+95 bp
160	08/12/20	Public Service Co of New Hampshire	Y	A1	A+	09/01/50	30	150	2.400%	T+110 bp
161	08/17/20	Duke Energy Progress LLC	N	A2	A-	02/18/22	2	700	3mL+18bp	3m\$L+18 bp
162	08/17/20	Duke Energy Progress LLC	Y	Aa3	A	08/15/50	30	600	2.500%	T+110 bp
163	08/17/20	Southern Co Gas Capital Corp	N	Baa1	A-	01/15/31	10	500	1.750%	T+110 bp
164	08/17/20	Oglethorpe Power Corp	N	Baa1	BBB+	08/01/50	30	450	3.750%	T+240 bp
165	08/24/20	Entergy Corp	N	Baa2	BBB	09/15/25	5	800	0.900%	T+70 bp

Michigan Public Service Commission
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Recent Utility Corporate Bond Issuances

Case No.: U-20836
Exhibit: A-18
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Page: 4 of 7

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166	08/24/20	Alabama Power Company	N	A1	A	09/15/30	10	\$600	1.450%	T+85 bp
167	09/08/20	Arizona Public Service Co	N	A2	A-	09/15/50	30	\$400	2.650%	T+125 bp
168	09/08/20	Entergy Arkansas LLC	Y	A2	A	06/15/51	31	\$675	2.650%	T+125 bp
169	09/09/20	Duke Energy Corp	N	Baa1	BBB+	9/15/2025	5	\$650	0.900%	T+63 bp
170	09/09/20	Duke Energy Corp	N	Baa1	BBB+	6/1/2030	10	\$350	2.450%	T+105 bp
171	09/10/20	Dominion Energy Inc	N	Baa2	BBB	9/15/2023	3	\$1,000	3mL+53bp	3m\$L+53.0 bp
172	09/14/20	WEC Energy Group Inc	N	Baa1	BBB+	09/15/23	3	\$700	0.550%	T+40 bp
173	09/14/20	Southern California Gas Co	N	A2	A	09/14/23	3	\$300	3mL+35bp	3m\$L+35 bp
174	09/15/20	The Southern Co	N	Baa3	BBB	01/15/51	30	\$1,250	4.000%	4.000%
175	09/15/20	The Southern Co	N	Baa3	BBB	10/15/60	40	\$750	4.200%	4.200%
176	09/16/20	Liberty Utilities Finance GP1	N	NR	BBB	09/15/30	10	\$600	2.050%	T+140 bp
177	09/17/20	EDP Finance	N	Baa3	BBB-	01/24/28	7	\$850	1.710%	T+125 bp
178	09/17/20	Duquesne Light Holdings, Inc	N	Baa3	BBB-	10/01/30	10	\$350	2.532%	T+185 bp
179	09/22/20	Xcel Energy Inc	N	Baa1	BBB+	10/15/23	3	\$500	0.500%	T+38 bp
180	09/22/20	San Diego Gas & Electric	Y	A2	A	10/01/30	10	\$800	1.700%	T+105 bp
181	09/23/20	CenterPoint Energy Resources Corp	N	A3	BBB+	10/01/30	10	\$500	1.750%	T+108 bp
182	09/23/20	Oncor Electric Delivery Co LLC	N	A2	A+	10/01/25	5	\$450	0.550%	T+40 bp
183	09/23/20	New York State Electric & Gas Corp	N	A3	A-	10/01/30	10	\$200	1.950%	T+130 bp
184	09/28/20	Entergy Texas Inc	N	Baa1	A	03/15/31	10	\$600	1.750%	T+113 bp
185	09/28/20	Southern California Edison Co	Y	A3	A-	02/01/26	5	\$350	1.200%	T+97 bp
186	09/28/20	PPL Electric Utilities Corp	Y	A1	A	09/28/23	3	\$250	3mL+25 bp	3m\$L+25 bp
187	09/29/20	DTE Energy Co	N	Baa2	BBB	11/01/22	2	\$750	0.550%	T+45 bp
188	09/29/20	National Rural Utilities Cooperative Finance	N	A1	A	03/15/31	10	\$400	1.350%	T+73 bp
189	09/22/20	DTE Energy Co	N	Baa3	BBB-	10/15/80	60	\$230	4.375%	4.375%
190	10/01/20	New England Power Co	N	A3	A-	10/06/50	30	\$400	2.807%	T+135 bp
191	10/01/20	Union Electric Co	N	A2	A	03/15/51	30	\$550	2.625%	T+118 bp
192	10/05/20	WEC Energy Group Inc	N	Baa1	BBB+	10/15/27	7	\$500	1.375%	T+85 bp
193	10/05/20	WEC Energy Group Inc	N	Baa1	BBB+	10/15/30	10	\$450	1.080%	T+105 bp
194	10/27/20	Berkshire Hathaway Energy Co	N	A3	A-	05/15/31	10	\$500	1.650%	T+90 bp
195	10/27/20	Berkshire Hathaway Energy Co	N	A3	A-	05/15/51	30	\$1,500	2.850%	T+128 bp
196	11/09/20	Consolidated Edison Co of New York Inc	Senior	Baa1	A-	12/01/60	40 FXD	\$600	3.000%	T+128 bp
197	11/09/20	Ameren Illinois Co	Senior Secured	A1	A	11/15/30	10 FXD	\$375	1.550%	T+65 bp
198	11/09/20	Entergy Louisiana LLC	Senior Secured	A2	A	03/15/51	30 FXD	\$300	2.900%	T+95 bp
199	11/09/20	Entergy Louisiana LLC	Senior Secured	A2	A	12/15/30	10 FXD	\$300	1.600%	T+70 bp
200	11/12/20	Pacific Gas and Electric Co	Senior Secured	Baa3	BBB-	11/15/21	1 FRN	\$1,450	0.000%	T+138 bp
201	11/16/20	Connecticut Light and Power Co/The	Secured	A1	A+	12/01/25	5 FXD	\$400	0.750%	T+35 bp
202	11/16/20	Rochester Gas and Electric Corp	Secured	A1	A	12/01/30	10 FXD	\$105	1.850%	T+95 bp
203	11/16/20	Arizona Public Service Co	Senior	A2	A-	08/15/29	9 FXD	\$105	2.600%	T+75 bp
204	11/17/20	Entergy Louisiana LLC	FMB	A2	A	11/17/23	3NC1 FXC	\$1,100	0.620%	T+40 bp
205	11/17/20	Alliant Energy Finance LLC	Senior	Baa2	BBB+	03/15/26	5 FXD	\$200	1.400%	T+105 bp
206	11/17/20	Alexander Funding Trust	Senior Secured	Baa3	BBB-	11/15/23	3 FXD	\$900	1.841%	T+163 bp
207	11/17/20	NRG Energy	Senior Secured	Baa3	BBB-	12/02/25	5 FXD	\$500	2.000%	T+163 bp
208	11/17/20	NRG Energy	Senior Secured	Baa3	BBB-	12/02/27	7 FXD	\$900	2.450%	T+185 bp
209	11/18/20	American Electric Power	Senior	Baa2	BBB+	11/01/23	3NC1 FXC	\$450	0.750%	T+55 bp
210	11/18/20	American Electric Power	Senior	Baa2	BBB+	11/01/25	5 FXD	\$450	1.000%	T+65 bp
211	11/18/20	American Electric Power	Senior	Baa2	BBB+	11/01/23	3NC1 FRN	\$600	0.000%	T+48 bp
212	11/19/20	AES Corp/The	Senior	Ba1	BBB-	01/15/26	5 FXD	\$800	1.375%	T+100 bp
213	11/19/20	AES Corp/The	Senior	Ba1	BBB-	01/15/31	10 FXD	\$1,000	2.450%	T+160 bp
214	11/19/20	Massachusetts Electric Co	Senior	A3	A-	11/24/30	10 FXD	\$500	1.729%	T+88 bp
215	11/30/20	Consolidated Edison	Senior	Baa2	BBB+	12/01/23	3 FXD	\$650	0.650%	T+47 bp
216	12/01/20	Southern California Edison Co	FMB	A3	A-	12/03/21	1 FRN	\$900	0.000%	T+27 bp
217	12/01/20	Virginia Electric and Power Co	Senior	A2	BBB+	12/15/50	30 FXD	\$900	2.450%	T+85 bp
218	12/02/20	Consumers Energy Co	FMB	Aa3	A	06/01/23	2 FXD	\$300	0.350%	T+20 bp
219	12/07/20	Washington Gas Light Co	Senior	A3	A-	09/15/49	29 FXD	\$100	3.650%	T+110 bp
220	01/05/21	Southern California Edison Co	FMB	A3	A-	06/01/30	9 FXD	\$150	2.250%	T+85 bp

Michigan Public Service Commission
DTE Electric Company
Recent Utility Corporate Bond Issuances

Case No.: U-20836
Exhibit: A-18
Schedule: H2
Witness: T. J. Lepczyk
Page: 5 of 7

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Offer Date	Issuer	Credit Ratings			Maturity	Tenor	Amount (\$MM)	Coupon	Spread
			Secured (?)	Moody's	S&P					
221	01/05/21	Southern California Edison Co	FMB	A3	A-	02/01/51	30 FXD	\$750	2.950%	T+127 bp
222	01/05/21	Ohio Power Co	Senior	A3	A-	01/15/31	10 FXD	\$450	1.625%	T+68 bp
223	01/05/21	Southern Power Co	Senior	Baa1	BBB+	01/15/26	5 FXD	\$400	0.900%	T+57 bp
224	02/01/21	National Rural Utilities Cooperative Finan	Senior	A2	A	02/08/24	3 FXD	\$500	0.350%	T+20 bp
225	02/09/21	National Fuel Gas Co	Senior	Baa3	BBB-	03/01/31	10 FXD	\$500	2.950%	T+180 bp
226	02/17/21	NextEra Energy Capital Holdings Inc.	Senior	Baa1	BBB+	02/22/23	2NC1 FRN	\$1,650	0.000%	T+27 bp
227	02/22/21	Public Service Company of Colorado	FMB	A1	A	06/15/31	10 FXD	\$750	1.875%	T+58 bp
228	02/22/21	Georgia Power Co.	Senior	Baa2	A-	03/15/51	30 FXD	\$750	3.250%	T+107 bp
229	02/22/21	National Rural Utilities Cooperative Finan	Senior	A2	A	06/15/26	5 FXD	\$600	1.000%	T+45 bp
230	02/23/21	Southern Co	Senior	Baa2	BBB+	02/26/24	3 FXD	\$600	0.600%	T+38 bp
231	02/23/21	Southern Co	Senior	Baa2	BBB+	03/15/28	7 FXD	\$400	1.750%	T+80 bp
232	02/23/21	Southwestern Public Service Co	Senior	A3	A	05/01/50	29 FXD	\$250	3.150%	T+95 bp
233	02/24/21	Ameren Corp	Senior	Baa1	BBB	03/15/28	7 FXD	\$450	1.750%	T+75 bp
234	02/26/21	Centerpoint Energy Respurces Corp	Senior	A3	BBB+	03/02/23	2NC1 FXC	\$700	0.700%	T+60 bp
235	02/26/21	Centerpoint Energy Respurces Corp	Senior	A3	BBB+	03/02/23	2NC1 FRN	\$1,000	0.000%	T+50 bp
236	03/01/21	PECO Energy Co	FMB	Aa3	A	03/15/51	30 FXD	\$375	3.050%	T+82 bp
237	03/02/21	Commonwealth Edison Inc.	FMB	A1	A	03/15/51	30 FXD	\$700	3.125%	T+90 bp
238	03/02/21	Entergy	Senior	Baa2	BBB	06/15/28	7 FXD	\$650	1.900%	T+87 bp
239	03/02/21	Entergy	Senior	Baa2	BBB	06/15/31	10 FXD	\$650	2.400%	T+107 bp
240	03/02/21	Public Service Electric & Gas Co.	Senior	Aa3	A	03/15/26	5 FXD	\$450	0.950%	T+32 bp
241	03/02/21	Public Service Electric & Gas Co.	Senior	Aa3	A	03/01/51	30 FXD	\$450	3.000%	T+77 bp
242	03/03/21	Atlantic City Electric	FMB	A2	A	03/15/31	10 FXD	\$350	2.300%	T+83 bp
243	03/08/21	Eversource Energy	Senior	Baa1	BBB+	03/15/31	10 FXD	\$350	2.550%	T+100 bp
244	03/08/21	One Gas	Senior	A3	BBB+	03/11/23	2NC1 FXC	\$1,000	0.850%	T+70 bp
245	03/08/21	One Gas	Senior	A3	BBB+	03/11/23	2NC1 FRN	\$800	0.000%	T+61 bp
246	03/08/21	One Gas	Senior	A3	BBB+	03/11/24	3NC1 FXC	\$700	1.100%	T+80 bp
247	03/08/21	Piedmont Natural Gas Co Inc	Senior	A3	BBB+	03/15/31	10 FXD	\$350	2.500%	T+95 bp
248	03/08/21	Southwestern Electric Power Co	Senior	Baa2	A-	03/15/26	5 FXD	\$500	1.650%	T+80 bp
249	03/08/21	CenterPoint Energy Houston Electric LLC	Senior Secured	A2	A	04/01/31	10 FXD	\$400	2.350%	T+80 bp
250	03/08/21	CenterPoint Energy Houston Electric LLC	Senior Secured	A2	A	04/01/51	30 FXD	\$700	3.350%	T+105 bp
251	03/08/21	Pacific Gas and Electric Co	Senior Secured	Baa3	BBB-	03/10/23	2NC1 FXC	\$1,500	1.367%	T+120 bp
252	03/08/21	Pacific Gas and Electric Co	Senior Secured	Baa3	BBB-	06/01/31	10 FXD	\$450	3.250%	T+167 bp
253	03/08/21	Pacific Gas and Electric Co	Senior Secured	Baa3	BBB-	06/01/41	20 FXD	\$450	4.200%	T+200 bp
254	03/09/21	Appalachian Power Co	Senior	Baa1	A-	04/01/31	10 FXD	\$400	2.700%	T+120 bp
255	03/09/21	Entergy Mississippi LLC	Senior Secured	A2	A	06/01/51	30 FXD	\$200	3.500%	T+115 bp
256	03/15/21	NextEra Energy	Senior	Baa1	BBB+	03/01/23	2 FXD	\$2,000	0.650%	T+50 bp
257	03/15/21	NextEra Energy	Senior	Baa1	BBB+	03/01/23	2 FRN	\$500	0.000%	T+54 bp
258	03/15/21	Tampa Electric Co.	Senior	A3	BBB+	03/15/31	10 FXD	\$400	2.400%	T+83 bp
259	03/15/21	Tampa Electric Co.	Senior	A3	BBB+	03/15/51	30 FXD	\$400	3.450%	T+108 bp
260	03/16/21	WEC Energy Group Inc	Senior	Baa1	BBB+	03/15/24	3 FXD	\$600	0.800%	T+48 bp
261	03/17/21	FirstEnergy Transmission LLC	Senior	Baa2	BB+	09/15/28	7 FXD	\$500	2.866%	T+0 bp
262	03/22/21	DTE Electric Company	FMB	Aa3	A	04/01/28	7 FXD	\$575	1.900%	T+58 bp
263	03/22/21	DTE Electric Company	FMB	Aa3	A	04/01/51	30 FXD	\$425	3.250%	T+90 bp
264	03/23/21	Northern States Power Co/MN	FMB	Aa3	A	04/01/31	10 FXD	\$425	2.250%	T+65 bp
265	03/23/21	Northern States Power Co/MN	FMB	Aa3	A	04/01/52	31 FXD	\$425	3.200%	T+85 bp
266	03/24/21	Entergy Arkansas LLC	FMB	A2	A	06/15/52	31 FXD	\$400	3.350%	T+105 bp
267	03/24/21	Southern California Edison Co	FMB	A3	A-	04/03/23	2 FXD	\$350	0.700%	T+60 bp
268	03/24/21	Southern California Edison Co	FMB	A3	A-	04/03/23	2 FRN	\$400	0.000%	T+64 bp
269	03/24/21	Southern California Edison Co	FMB	A3	A-	04/01/24	3NC2 FXC	\$700	1.100%	T+80 bp
270	03/24/21	Dominion Energy	Senior	Baa2	BBB	04/15/26	5 FXD	\$600	1.450%	T+65 bp
271	03/24/21	Dominion Energy	Senior	Baa2	BBB	04/15/41	20 FXD	\$500	3.300%	T+110 bp
272	03/24/21	Southern California Edison Co	Senior	A3	A-	04/01/24	3NC2 FRN	\$400	0.000%	T+83 bp
273	03/29/21	Duke Energy Carolinas LLC	FMB	Aa3	A	04/15/31	10 FXD	\$550	2.550%	T+85 bp
274	03/29/21	Duke Energy Carolinas LLC	FMB	Aa3	A	04/15/51	30 FXD	\$450	3.450%	T+105 bp
275	04/07/21	Monongahela Power Co	FMB	A3	BBB	05/15/27	6 FXD	\$200	3.550%	T+120 bp

Michigan Public Service Commission
DTE Electric Company
Recent Utility Corporate Bond Issuances

Case No.: U-20836
Exhibit: A-18
Schedule: H2
Witness: T. J. Lepczyk
Page: 6 of 7

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Offer Date	Issuer	Secured (?)	Credit Ratings		Maturity	Tenor	Amount (\$MM)	Coupon	Spread
				Moody's	S&P					
276	04/07/21	Northern Natural Gas Co	Senior	A2	A	10/16/51	31 FXD	\$550	3.400%	T+110 bp
277	04/15/21	Essential Utilities Inc	Senior	Baa2	A-	05/01/31	10 FXD	\$400	2.400%	T+88 bp
278	04/27/21	Indiana Michigan Power Co	Senior	A3	A-	05/01/51	30 FXD	\$450	3.250%	T+100 bp
279	04/29/21	Toledo Edison Co/The	FMB	A2	BBB	05/01/28	7 FXD	\$150	2.650%	T+138 bp
280	05/04/21	AEP Texas Inc	Senior	Baa2	A-	05/15/51	30 FXD	\$450	3.450%	T+120 bp
281	05/05/21	Florida Power & Light Co	Senior	A1	A	05/10/23	2NC1 FRN	\$1,000	0.000%	T+25 bp
282	05/06/21	Tucson Electric Power Co	Senior	A3	A-	05/01/51	30 FXD	\$325	3.250%	T+105 bp
283	05/10/21	American Water Capital Corp	Senior	Baa1	A	06/01/31	10 FXD	\$550	2.300%	T+73 bp
284	05/10/21	American Water Capital Corp	Senior	Baa1	A	06/01/51	30 FXD	\$550	3.250%	T+93 bp
285	05/11/21	CenterPoint Energy Inc (CNP)	Senior	Baa2	BBB	05/13/24	3NC1 FRN	\$700	0.000%	T+65 bp
286	05/11/21	CenterPoint Energy Inc (CNP)	Senior	Baa2	BBB	06/01/26	5 FXD	\$500	1.450%	T+70 bp
287	05/11/21	CenterPoint Energy Inc (CNP)	Senior	Baa2	BBB	06/01/31	10 FXD	\$500	2.650%	T+105 bp
288	05/13/21	Spire Missouri Inc	Senior	A1	A	06/01/51	30 FXD	\$305	3.300%	T+90 bp
289	05/20/21	Mid-Atlantic Interstate Transmission LLC	Senior	A3	BB+	05/15/28	7 FXD	\$150	4.100%	T+125 bp
290	05/24/21	NSTAR Electric Co	Senior	A1	A	06/01/51	30 FXD	\$300	3.100%	T+83 bp
291	05/24/21	OGE Energy Corp	Senior	Baa1	BBB+	05/26/23	2NC1 FXC	\$500	0.703%	T+55 bp
292	05/24/21	Oklahoma Gas and Electric Co	Senior	A3	A-	05/26/23	2NC1 FXC	\$500	0.553%	T+40 bp
293	05/25/21	Emera US Finance LP	Senior	Baa3	BBB-	06/15/24	3 FXD	\$300	0.833%	T+53 bp
294	05/25/21	Emera US Finance LP	Senior	Baa3	BBB-	06/15/31	10 FXD	\$450	2.639%	T+108 bp
295	06/01/21	Pacific Gas and Electric Co	FMB	Baa3	BBB-	06/15/28	7 FXD	\$800	3.000%	T+180 bp
296	06/01/21	Public Service Co of New Hampshire	FMB	A1	A+	06/15/31	10 FXD	\$350	2.200%	T+62 bp
297	06/02/21	NextEra Energy Capital Holdings Inc	Senior	Baa1	BBB+	06/15/28	7 FXD	\$1,500	1.900%	T+65 bp
298	06/03/21	Puget Energy Inc	Secured	Baa3	BBB-	06/15/28	7 FXD	\$500	2.379%	T+108 bp
299	06/03/21	Consolidated Edison Co of New York Inc	Senior	Baa1	A-	06/15/31	10 FXD	\$750	2.400%	T+80 bp
300	06/03/21	Consolidated Edison Co of New York Inc	Senior	Baa1	A-	06/15/61	40 FXD	\$750	3.600%	T+130 bp
301	06/07/21	Alabama Power Co	Senior	A1	A	07/15/51	30 FXD	\$600	3.125%	T+88 bp
302	06/07/21	Baltimore Gas and Electric Co	Senior	A3	A	06/15/31	10 FXD	\$600	2.250%	T+70 bp
303	06/07/21	Duke Energy	Senior	Baa2	BBB	06/10/23	2 FRN	\$500	0.000%	T+25 bp
304	06/07/21	Duke Energy	Senior	Baa2	BBB	06/15/31	10 FXD	\$1,000	2.550%	T+100 bp
305	06/07/21	Duke Energy	Senior	Baa2	BBB	06/15/41	20 FXD	\$750	3.300%	T+115 bp
306	06/07/21	Duke Energy	Senior	Baa2	BBB	06/15/51	30 FXD	\$750	3.500%	T+125 bp
307	06/08/21	Jersey Central Power & Light Co	Senior	A3	BB+	03/01/32	11 FXD	\$500	2.750%	T+125 bp
308	06/08/21	Wisconsin Electric Power Co	Senior	A2	A-	06/15/28	7 FXD	\$300	1.700%	T+50 bp
309	06/09/21	Southern California Edison Co	FMB	A3	A-	06/01/31	10 FXD	\$450	2.500%	T+105 bp
310	06/09/21	Southern California Edison Co	FMB	A3	A-	06/13/22	1 FRN	\$475	0.000%	T+35 bp
311	06/09/21	Southern California Edison Co	FMB	A3	A-	06/01/51	30 FXD	\$450	3.650%	T+148 bp
312	06/09/21	Union Electric Co	FMB	A2	A	03/15/32	11 FXD	\$525	2.150%	T+67 bp
313	06/21/21	Connecticut Light and Power Co/The	FMB	A1	A+	07/01/31	10 FXD	\$425	2.050%	T+60 bp
314	06/21/21	PPL Electric Utilities Corp	FMB	A1	A	06/24/24	3NC1 FRN	\$650	0.000%	T+33 bp
315	06/22/21	Ameren Illinois Co	FMB	A1	A	06/15/23	2 FXD	\$100	0.375%	T+15 bp
316	06/22/21	Ameren Illinois Co	FMB	A1	A	06/15/51	30 FXD	\$350	2.900%	T+80 bp
317	06/22/21	Duquesne Light Holdings Inc	Senior	Baa3	BBB-	01/07/32	11 FXD	\$400	2.775%	T+130 bp
318	06/24/21	Mississippi Power Co	Senior	Baa1	A-	06/28/24	3 FRN	\$200	0.000%	T+30 bp
319	06/24/21	Mississippi Power Co	Senior	Baa1	A-	07/30/51	30 FXD	\$325	3.100%	T+105 bp
320	07/07/21	PacifiCorp	FMB	A1	A+	06/15/52	31 FXD	\$750	2.900%	T+98 bp
321	07/07/21	Enel Finance International NV	Senior	Baa1	BBB+	07/12/26	5 FXD	\$1,250	1.375%	T+70 bp
322	07/07/21	Enel Finance International NV	Senior	Baa1	BBB+	07/12/28	7 FXD	\$1,000	1.875%	T+85 bp
323	07/07/21	Enel Finance International NV	Senior	Baa1	BBB+	07/12/31	10 FXD	\$1,000	2.250%	T+100 bp
324	07/07/21	Enel Finance International NV	Senior	Baa1	BBB+	07/12/41	20 FXD	\$750	2.875%	T+110 bp
325	07/20/21	MidAmerican Energy Co	Secured	A2	A+	08/01/52	31 FXD	\$500	2.700%	T+85 bp
326	08/02/21	Consumers Energy	FMB	A1	A	08/15/52	31 FXD	\$300	2.650%	T+78 bp
327	08/02/21	AEP Transmission Co LLC	Senior	A2	A-	08/15/51	30 FXD	\$450	2.750%	T+90 bp
328	08/05/21	Commonwealth Edison Co.	FMB	A1	A	09/01/51	30 FXD	\$450	2.750%	T+87 bp
329	08/05/21	Southern California Edison Co.	FMB	A3	A-	08/01/23	2 FXD	\$400	0.700%	T+50 bp
330	08/05/21	Southern California Edison Co.	FMB	A3	BBB+	08/01/24	3 FXD	\$450	0.975%	T+60 bp

Michigan Public Service Commission
DTE Electric Company
Recent Utility Corporate Bond Issuances

Case No.: U-20836
Exhibit: A-18
Schedule: H2
Witness: T. J. Lepczyk
Page: 7 of 7

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Offer Date	Issuer	Credit Ratings			Maturity	Tenor	Amount (\$MM)	Coupon	Spread
			Secured (?)	Moody's	S&P					
331	08/09/21	Duke Energy Progress LLC	FMB	Aa3	A	08/15/31	10 FXD	\$650	2.000%	T+70 bp
332	08/09/21	Duke Energy Progress LLC	FMB	Aa3	A	08/15/51	30 FXD	\$450	2.900%	T+93 bp
333	08/09/21	San Diego Gas & Electric Co	FMB	A1	A	08/15/51	30 FXD	\$750	2.950%	T+100 bp
334	08/10/21	Dominion Energy	Senior	Baa2	BBB	08/15/31	10 FXD	\$1,000	2.250%	T+92 bp
335	08/10/21	Eversource Energy	Senior	Baa1	BBB+	08/15/23	2 FRN	\$350	0.000%	T+25 bp
336	08/10/21	Eversource Energy	Senior	Baa1	BBB+	08/15/26	5 FXD	\$300	1.400%	T+60 bp
337	08/10/21	NSTAR Electric Co	Senior	A1	A	08/15/31	10 FXD	\$300	1.950%	T+67 bp
338	08/11/21	Arizona Public Service Co	Senior	A2	A-	12/15/31	10 FXD	\$450	2.200%	T+88 bp
339	08/11/21	Entergy Texas Inc	Senior	Baa1	A	09/01/26	5 FXD	\$130	1.500%	T+75 bp
340	08/11/21	Public Service Co of Oklahoma	Senior	Baa1	A-	08/15/31	10 FXD	\$400	2.200%	T+90 bp
341	08/11/21	Public Service Co of Oklahoma	Senior	Baa1	A-	08/15/51	30 FXD	\$400	3.150%	T+115 bp
342	08/17/21	Public Service Electric and Gas Co	Senior	Aa3	A	08/15/31	10 FXD	\$425	1.900%	T+68 bp
343	08/18/21	Southwest Gas Corp	Senior	Baa1	A-	08/15/51	30 FXD	\$300	3.180%	T+128 bp
344	08/23/21	Black Hills Corp	Senior	Baa2	BBB+	08/23/24	3NC1 FXC	\$600	1.037%	T+60 bp
345	09/07/21	PECO Energy Co	FMB	Aa3	A	09/15/51	30 FXD	\$375	2.850%	T+88 bp
346	09/07/21	Southern Co Gas Capital Corp	Senior	Baa1	A-	09/30/51	30 FXD	\$450	3.150%	T+123 bp
347	09/08/21	Cleco Power	Senior	A3	BBB+	06/15/23	2 FRN	\$325	0.000%	T+50 bp
348	09/08/21	Puget Sound Energy	Senior	A2	A-	09/15/51	30 FXD	\$450	2.893%	T+95 bp
349	09/09/21	Ohio Power	Senior	A3	A-	10/01/51	30 FXD	\$600	2.900%	T+103 bp
350	09/13/21	Wisconsin Power and Light	Senior	A3	A	09/16/31	10 FXD	\$300	1.950%	T+63 bp
351	09/21/21	New York State Electric & Gas Corp	Senior	Baa1	A-	10/01/31	10 FXD	\$350	2.150%	T+85 bp
352	09/28/21	Entergy Louisiana LLC	Secured	A2	A	10/01/24	3NC1 FXC	\$1,000	0.950%	T+45 bp

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)
Line No.	Descripton	2008 Actuals	2009 Actuals	2010 Actuals	2011 Actuals	2012 Actuals	2013 Actuals	2014 Actuals	2015 Actuals	2016 Actuals	2017 Actuals	2018 Actuals	2019 Actuals	2020 Actuals	Total Historical Actual Realized Benefits
1	Customer Service														
2	Cuts and Restores	N/A	N/A	N/A	N/A	N/A	N/A	3,383	5,044	6,902	4,544	4,786	6,202	2,986	33,847
3	Consumer Affairs	-	-	-	-	-	-	N/A	N/A	N/A	N/A	79	79	79	237
4	Billing	-	-	-	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-
5	Meter Reading	217	1,323	912	1,369	2,618	4,657	6,477	7,935	10,897	12,569	13,938	14,757	14,796	92,464
6	Cust Care - Call Center	-	-	-	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-
7	Cust Care - Misc Reads	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	659	99	74	831
8	Total Customer Service	217	1,323	912	1,369	2,618	4,657	9,860	12,979	17,798	17,113	19,461	21,136	17,935	127,379
9															
10	Distribution Operations														
11	OK on Arrivals	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4,560	5,632	3,664	13,857
12	Transformer Overload	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-
13	Redispatch	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-
14	Primary Reads	-	-	-	-	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-
15	Meter Test	-	-	-	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-
16	Turn-Ons/Turn-Offs	N/A	N/A	N/A	N/A	N/A	N/A	3,736	4,361	5,936	1,950	4,974	5,158	4,123	30,238
17	Call Back	-	-	-	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-
18	Power Quality	-	-	-	-	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-
19	Total Distribution Ops	-	-	-	-	-	-	3,736	4,361	5,936	1,950	9,534	10,790	7,787	44,095
20	O&M Expenditures	(2,822)	(2,129)	(1,785)	(1,675)	(1,652)	(1,537)	(852)	(1,042)	(873)	(2,615)	(4,683)	(4,725)	(6,560)	(32,952)
21	Net DO Savings	(2,822)	(2,129)	(1,785)	(1,675)	(1,652)	(1,537)	2,884	3,319	5,064	(665)	4,851	6,065	1,227	11,144
22															-
23	Uncollectibles	N/A	N/A	N/A	N/A	N/A	N/A	534	752	1,018	674	879	714	285	4,856
24															-
25	Theft	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	339	339
26															-
27	Load Research	-	0	4	17	22	31	43	43	65	66	68	69	70	499
28	Total Benefits	(2,604)	(806)	(874)	(306)	965	3,119	13,278	17,051	23,945	17,188	25,258	27,984	19,856	144,055

Michigan Public Service Commission
DTE Electric Company
Advanced Metering Infrastructure (AMI) - Historical Actual Realized Benefits Comparison to Forecasted Benefits
(\$000)

Case No.: U-20836
Exhibit: A-19
Schedule: I1.1
Witness: P. L. Smith
Page: 2 of 2

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)
<u>Line</u> <u>No.</u>	<u>Descripton</u>	<u>2008 Forecast</u>	<u>2009 Forecast</u>	<u>2010 Forecast</u>	<u>2011 Forecast</u>	<u>2012 Forecast</u>	<u>2013 Forecast</u>	<u>2014 Forecast</u>	<u>2015 Forecast</u>	<u>2016 Forecast</u>	<u>2017 Forecast</u>	<u>2018 Forecast</u>	<u>2019 Forecast</u>	<u>2020 Forecast</u>	<u>Total Historical Forecasted Benefits</u>
1	Customer Service														
2	Cuts and Restores	4	10	66	341	664	913	1,485	2,366	3,029	3,322	3,471	3,592	3,718	22,981
3	Consumer Affairs	-	-	-	-	-	-	38	39	39	38	42	41	43	279
4	Billing	-	-	-	58	107	151	251	356	457	452	495	491	506	3,323
5	Meter Reading	3,211	3,335	4,247	6,583	8,707	10,167	12,093	15,425	19,281	21,515	22,907	22,825	23,586	173,882
6	Cust Care - Call Center	-	-	-	82	153	215	357	507	651	644	705	700	720	4,733
7	Cust Care - Misc Reads	0	1	2	27	56	80	127	191	237	259	270	280	290	1,820
8	Total Customer Service	3,216	3,346	4,315	7,091	9,686	11,525	14,351	18,883	23,694	26,229	27,890	27,929	28,863	207,019
9															
10	Distribution Operations														
11	OK on Arrivals	1	1	5	50	103	141	199	283	372	432	457	473	490	3,007
12	Transformer Overload	2	3	9	100	197	249	341	469	566	619	650	673	696	4,574
13	Redispatch	0	1	4	33	64	96	176	288	366	400	417	432	447	2,723
14	Primary Reads	-	-	-	-	331	351	481	620	767	925	1,228	1,411	1,460	7,572
15	Meter Test	-	-	-	115	180	249	386	467	483	428	370	229	79	2,986
16	Turn-Ons/Turn-Offs	2	6	69	372	700	918	1,330	1,933	2,455	2,732	2,864	2,963	3,067	19,412
17	Call Back	-	-	-	52	110	170	294	365	441	521	539	558	578	3,628
18	Power Quality	-	-	-	-	93	96	200	310	321	442	458	474	491	2,884
19	Total Distribution Ops	4	12	86	721	1,777	2,271	3,407	4,735	5,770	6,499	6,983	7,213	7,308	39,206
20	O&M Expenditures	(866)	-	(2,983)	(4,207)	(5,596)	(4,252)	(4,545)	(4,914)	(5,038)	(3,547)	(3,541)	(3,482)	(3,477)	(46,447)
21	Net DO Savings	(861)	12	(2,897)	(3,485)	(3,819)	(1,982)	(1,138)	(179)	733	2,952	3,442	3,732	3,831	340
22															
23	Uncollectibles	2	4	22	109	212	305	646	1,219	1,599	1,730	1,803	1,866	1,931	11,449
24															
25	Theft	2	5	16	127	233	309	1,178	3,007	4,218	4,514	4,684	4,847	5,017	28,157
26															
27	Load Research	0	0	1	10	20	28	44	65	82	92	103	100	103	650
28	Total Benefits	2,359	3,367	1,458	3,852	6,332	10,186	15,082	22,995	30,326	35,517	37,921	38,474	39,746	247,615

Michigan Public Service Commission
DTE Electric Company
Projected AMI Benefits
(\$000)

Case No.: U-20836
Exhibit: A-19
Schedule: I1.2
Witness: P. L.Smith
Page: 1 of 1

		(a)	(b)	(c)	(d)	(e)
Line No.	Category	Description	Projected Calendar Year			Grand Total
			12 mos. ended 12/31/2021	12 mos. ended 12/31/2022	12 mos. ended 12/31/2023	
						(e) = (b) + (c) + (d)
1	Operational Efficiencies	Meter Reading	15,402	15,711	16,025	47,137
2		Customer Care Improvements	158	158	158	474
3		Remote Disconnect & Reconnect Automation	9,241	10,822	10,502	30,565
4		Outage Management Efficiencies	1,704	2,799	2,923	7,426
5	Revenue Management	Uncollectibles	651	651	651	1,954
6	Revenue Protection	Theft Detection	144	144	144	431
7	Load Research	Load Research	72	74	75	221
8	Energy Savings	Conservation Tools (DTE Insight)	274	350	348	972
9		Demand Response	236	2,278	6,796	9,310
10	Total Forecasted AMI Benefits		27,882	32,987	37,621	98,491

Michigan Public Service Commission
DTE Electric Company
Proposed Nuclear Surcharge
Projected Test Period - 12 Months Ending October 31, 2023
(\$000)

Case No.: U-20836
Exhibit: A-20
Schedule: J1
Witness: J. C. Davis
Page: 1 of 1

	(a)	(b)	(c)		
Line No.	Description	Projected Test Period	Reference		
1	Proposed Nuclear Surcharge:				
2	Site Security & Radiation Protection	30,195	From Line 11, col. (h); Per U-14399 Order		
3	Nuclear Decommissioning Funding	2,867	U-20561 Exh A-20, Sch. J1		
4	Low Level Radioactive Waste Disposal Funding	6,000	Per U-10102 Order; U-20561 Exh. A-20, Sch. J1		
5	Proposed Nuclear Surcharge	39,062	Sum Lines 2 thru 4		
6	Current Authorized Nuclear Surcharge	38,728	Per U-20561 Order		
7	Change in Nuclear Surcharge	334	Line 5 less Line 6		
	(d)	(e)	(f)	(g)	(h)
	Description	2020 Historical 1/	2021	2022	Projected 12 mo. ending 10/31/23
8	Site Security & Radiation Protection:				
9	Site Security	16,346	16,853	17,342	17,761
10	Radiation Protection	11,444	11,799	12,141	12,434
11	Total Site Security & Radiation Protection	27,790	28,652	29,483	30,195
12	Annual Inflation Adjustment		3.10%	2.90%	2.90%
13	No. of Months in Period		12	12	10
14	Pro-rated Inflation Rate		3.10%	2.90%	2.42%

(sponsored by Witness T. Uzenski on Exh. A-13, Sch.C5.15)

1/ Eliminated from Nuclear Power Base O&M expense - see Exh A-13, Sch C5.3, Col. (d)



DTE Electric Nuclear Decommissioning Study
for the Fermi 2 Power Plant
in Response to
Michigan Public Service Commission Order U-20162, dated May 2, 2019



SECTION ONE

1 Executive Summary

As requested by Commission Order U-20162, dated May 02, 2019, DTE Electric is filing this Fermi 2 nuclear decommissioning study to assess the adequacy of the existing funding for the end-of-life decommissioning of the Fermi 2 Power Plant. DTE Electric finds that DTE Electric's existing practice of annual review and update of the Fermi 2 nuclear decommissioning cost estimates is a reasonable and prudent approach further supported by the Fermi 2 site-specific decommissioning cost estimate performed in 2019 and 2020. DTE Electric finds a reasonable and prudent basis, at this time, to leave unchanged the existing Nuclear Decommissioning Funding component of the Nuclear Surcharge of approximately \$3 million per year.

DTE Electric finds reasonable and prudent basis for the following assumptions:

- The Fermi 2 Power Plant will permanently cease operations no later than March 20, 2045 and immediately begin planning and preparation for the nuclear decommissioning method "DECON."
- DTE Electric will have concluded the Fermi 2 Power Plant's radiological decommissioning and terminated the operating license no later than 2058.
- DTE Electric will be responsible for the safety, security and management of Fermi 2's spent fuel until 2105.
- The cost to decommission the Fermi 2 Power Plant is approximately \$2.2 billion in 2019 dollars. This cost to decommission includes the costs necessary to complete the Fermi 2 radiological decommissioning (terminate the license), Fermi 2 Independent Spent Fuel Installation (ISFSI) decommissioning and Fermi 2 non-radiological decommissioning (site restoration) as well as post-shutdown spent fuel management (spent fuel management program).
- A projected compound cost of escalation of 3.8% applied to the Fermi 2 nuclear decommissioning cost of approximately \$2.2 billion is a reasonable and prudent forecast.
- The Fermi 2 Nuclear Decommissioning Trust (NDT) has, as of December 31, 2019, approximately 76% of the amounts required to complete the Fermi 2 nuclear decommissioning.
- The DTE Electric Nuclear Decommissioning Master Trust Agreement (MTA) governs the Fermi 2 NDT. The NDT is an external trust held outside DTE Electric's normal control. The NDT governance is consistent with all Commission orders, state and federal laws and regulations.
- DTE Electric provides the Nuclear Regulatory Commission (NRC) reasonable funding assurance exclusively through the NDT (an external sinking fund, as defined by the NRC). The Nuclear Surcharge, as a non-bypassable charge, is a predicate for DTE Electric's continued exclusive reliance on the NDT to provide NRC reasonable funding assurance.
- A projected compound post-tax rate of return of 4.8% applied to the Fermi 2 NDT balances is a reasonable and prudent forecast.
- Continued application of the existing Nuclear Decommissioning Funding component of the Nuclear Surcharge of approximately \$3 million per year is a reasonable and prudent cost to provide reasonable assurance of sufficient NDT funds to complete the Fermi 2 Power Plant nuclear decommissioning no later than the milestone date of December 31, 2044.

The following notable events have occurred since the Commission Order U-20162, date May 02, 2019:



- In Case No. U-20561, as described in the direct testimony of Witness Jeffrey C. Davis, DTE Electric proposed no change to the Nuclear Decommissioning portion of the Nuclear Surcharge and described the projected expenditures forecasted to complete this nuclear decommissioning study. The Commission approved DTE Electric's proposed no change to the Nuclear Decommissioning portion of the Nuclear Surcharge in the Commission's Order U-20561, dated May 8, 2020.
- DTE Electric contracted with Energy Solutions, LLC (EnergySolutions) to perform an inventory analysis of the Fermi 2 Power Plant and provide a Fermi 2 Power Plant site-specific decommissioning cost estimate (SDCE) report. DTE Electric contracted with General Electric-Hitachi (GEH), Excel Services, Barpellam, Inc and Holtec International to provide Fermi 2 plant analysis to support the EnergySolutions scope of work. DTE Electric contracted with Callan LLC to provide industry decommissioning data and analytics. DTE Electric personnel performed or provided work direction for the entire scope of this study.

DTE Electric intends this study to provide readers with a policy-level understanding of its Fermi 2 Power Plant operations, perspective towards the U.S. commercial nuclear industry and regulatory context. The Fermi 2 Power Plant has history dating back to 1968 and a future through 2107; DTE Electric respects the thoughtful policy decisions of those prior and appreciates the opportunity to advocate public policies to ensure our obligations to decommission the Fermi 2 Power Plant are reasonably assured.



SECTION TWO

2 Table of Contents

1 Executive Summary	2
2 Table of Contents	4
3 Lists of Tables and Figures	8
3.1 List of tables	8
3.2 List of figures	8
4 Introduction	10
4.1 Company overview	10
DTE Electric	10
DTE Electric – Nuclear Generation	10
4.2 Fermi 2 Power Plant overview	11
Site location	11
Site characteristics	11
Fermi 2 Power Plant characteristics	12
Fermi 2 ISFSI characteristics	14
Fermi 1	18
Fermi 3	18
4.3 Nuclear industry overview	18
Industry's decade in review (2010 – 2019)	19
Emergence of nuclear decommissioning as an industry	21
Consolidated Interim Storage Facility	23
5 Remaining life of nuclear assets	24
5.1 Fermi 2 Power Plant	24
Nuclear Regulatory Commission (NRC)	24
Fermi 2 NRC licensing basis	24
Fermi 2 Power Plant end of life conclusion	26
5.2 Independent Spent Fuel Storage Installation (ISFSI)	26
2010 NRC Waste Confidence Decision	27
Nuclear Regulatory Commission (NRC)	28
Fermi 2 ISFSI licensing basis	29



Fermi 2 ISFSI end of life conclusion	30
5.3 Other facilities	30
Fermi 1	30
Fermi 3	30
Small modular reactors or other nuclear technologies	30
6 Regulatory dynamics	31
6.1 Federal regulations overview	31
NRC regulations	31
FERC regulations	33
IRS regulations	33
Integrated regulatory framework	33
6.2 State of Michigan regulations overview	36
Settlement Agreement in MPSC Case No U-6150 (Reopened), dated August 26, 1986.....	36
MPSC Order U-8567, dated January 13, 1987.....	37
MPSC Order U-10102, dated January 21, 1994.....	37
MPSC Order U-14399, dated December 22, 2005	37
MPSC Order U-16472, dated October 20, 2011.....	37
MPSC Order U-17767, dated December 11, 2015	37
MPSC Order U-20162, dated May 2, 2019.....	38
MPSC Order U-20561, dated May 8, 2020.....	38
6.3 Standard Contract	38
Yucca Mountain Project	39
Standard Contract	40
Spent nuclear fuel (SNF) fees.....	40
Department of Energy (DOE) litigation	41
Nuclear Waste Fund	41
7 Decommissioning cost estimates	42
7.1 Decommissioning cost estimate overview	42
Purpose	42
Approaches to cost estimation	42
Elements of cost estimates.....	43
7.2 Existing Fermi 2 decommissioning cost estimate.....	45



Inventory	46
Decommissioning schedule.....	47
7.3 2020 Fermi 2 site-specific decommissioning cost estimate.....	48
Inventory	49
Decommissioning schedule.....	50
7.4 Fermi 2 NRC minimum decommissioning fund requirement.....	51
7.5 DTE Electric conclusion	52
8 Nuclear Decommissioning Trust (NDT).....	53
8.1 DTE Electric Master Trust Agreement overview	53
8.2 Fermi 2 Nuclear Decommissioning Trusts	54
NRC radiological decommissioning.....	55
NRC ISFSI decommissioning.....	55
NRC irradiated fuel management.....	55
8.3 DTE Electric Nuclear Surcharge	56
Nuclear Surcharge overview	56
8.4 Nuclear Decommissioning Funding Portion of Nuclear Surcharge.....	57
9 NDT funding adequacy.....	59
9.1 Nuclear Decommissioning Trust funding adequacy overview	59
9.2 Nuclear Decommissioning Cost Estimate	60
9.3 Inflation rates (cost escalation)	61
DTE Electric rate of inflation methodology.....	61
Historical rate of inflation for nuclear decommissioning cost estimates.....	61
DTE Electric forecasted rate of inflation.....	66
9.4 Fermi 2 NDT balances.....	67
9.5 Administrative and incidental expenses.....	70
Taxes	71
9.7 Fund earnings rate.....	72
DTE Electric rate of return methodology.....	73
DTE Electric historical NDT rate of return.....	73
DTE Electric forecasted rate of return	77
9.8 Net rate of return	78
9.9 Adequacy of annual provision	78



Total annual Nuclear Decommissioning Funding provision	78
Minimum annual Nuclear Decommissioning Funding provision allocation to license termination.....	79
Minimum annual Nuclear Decommissioning Funding provision allocation to ISFSI decommissioning	80
Minimum annual Nuclear Decommissioning Funding provision allocation to other decommissioning	81
Recommended annual Nuclear Decommissioning Funding provision and contributions	82
10 Rate making considerations.....	83
11 References	84



SECTION THREE

3 Lists of Tables and Figures

3.1 List of tables

Table 7.1 Fermi 2 SDCE low-level radioactive waste (LLRW) material inventory (tons)	49
Table 7.2 Fermi 2 SDCE LLRW material inventory (volume) relative to the NRC NUREG/CR-6174 Reference BWR	49
Table 9.1 Summary Fermi 2 SDCE costs in thousands of 2019 dollars	60
Table 9.2 Fermi 2 SDCE by cost element in thousands of 2019 dollars	60
Table 9.3 NUREG-1307, Revision 17 (2019) cost escalation factors for Fermi 2 Power Plant	62
Table 9.4 Fermi 2 NDT balances as of December 31, 2019 in thousands of dollars.	68
Table 9.5 Fermi 2 Nuclear Decommissioning Fund (NDF) amounts by purpose as of December 31, 2019 in thousands of dollars	68
Table 9.6 Nuclear Decommissioning Fund (NDF) funding levels by purpose as of December 31, 2019 in thousands of dollars	69
Table 9.7 Fermi 2 NDT 2015 – 2019 tax expenses in thousands of dollars	72
Table 9.8 Fermi 2 NDT 2015 – 2019 administrative and incidental expenses, excluding taxes, in thousands of dollars	72
Table 9.9 Historical Fermi 2 Nuclear Decommissioning Trust (NDT) pre-tax rate of return (%).	74
Table 9.10 Fermi 2 NDT balances (in billions) as of December 31, 2013 – 2019	75
Table 9.11 Recommended annual Nuclear Decommissioning Funding provision and contributions to the Fermi 2 NDT plans ...	82

3.2 List of figures

Figure 4.1 Fermi 2 Power Plant site location, Frenchtown Township, Monroe County, Michigan	11
Figure 4.2 Fermi 2 Power Plant site	12
Figure 4.3 Simplified schematic of generic Boiling Water Reactor (BWR) and a Pressurized Water Reactor (PWR)	13
Figure 4.4 Cutaway of a drywell torus containment structure and example of a GE BWR/4 with a Mark I containment structure	14
Figure 4.5 Fermi 2 Independent Spent Fuel Installation (ISFSI) site	15
Figure 4.6 Holtec HI-STORM 100 dry storage system	16
Figure 4.7 Fermi 2 Vertical Cask Transport (VCT) vehicle	17
Figure 4.8 Fermi 2 Cask Transfer Facility (CTF)	18
Figure 9.1 The Fermi 2 Power Plan NRC minimum decommissioning funding	63
Figure 9.2 Core Consumer Price Index [CPI] (Series ID CPGRLE01USA657N) growth rate from previous period (1986 – 2018)	64
Figure 9.3 Personal Consumption Expenditures: Chain-type Price Index [PCEPI] (1986 – 2018)	65
Figure 9.4 Fermi 2 reactor services costs index (2010 – 2019)	66
Figure 9.5 Fermi 2 NDT assets as invested on December 31, 2019	69
Figure 9.6 Fermi 2 Qualified Fund assets as invested on December 31, 2019	70
Figure 9.7 Fermi 2 Nuclear Decommissioning Trust (NDT) index	75



Figure 9.8 S&P 500 index (SPX) December 31, 1985 through March 31, 2020.....	76
Figure 9.9 U.S. Aggregate Bond Index (LBUSTRUU) December 31, 1985 through March 31, 2020	77



SECTION FOUR

4 Introduction

The decommissioning plan assumptions and strategies presented in this study are to provide reasonable assurance of adequate annual funding provisions and eventual available funds for the decommissioning of the DTE Electric Fermi 2 Power Plant by the year 2045. This study does not constitute a commitment by DTE Electric to perform decommissioning work exactly as presented in this document. As is customary in the commercial nuclear industry, DTE Electric expects to refine and improve the decommissioning plan and cost estimates as the Fermi 2 Power Plant approaches end of life in 2045.

4.1 Company overview

DTE (NYSE: DTE) is a Detroit-based diversified energy company involved in the development and management of energy-related businesses and services throughout the United States. DTE's operating units include DTE Gas – a natural gas utility engaged in the purchase, storage, transmission, distribution and sale of natural gas to approximately 1.2 million customers throughout Michigan and DTE Electric – an electric utility servicing approximately 2.2 million customers in southeastern Michigan. Within DTE Electric's diverse portfolio of generation assets is the Fermi 2 Power Plant.

DTE Electric

DTE Electric (DTEE or Company) is a direct wholly-owned subsidiary of DTE Energy (DTE) and is an electric utility engaged in the generation, purchase, distribution and sale of electricity to approximately 2.2 million customers in southeastern Michigan. Founded in 1903 as the Detroit Edison Company, DTE Electric is the largest electric utility in Michigan and one of the largest in the United States with over 11,000 megawatts of electricity generation capacity.

DTE Electric is subject to the regulatory jurisdiction of various agencies, including, but not limited to the Michigan Public Service Commission (MPSC or Commission) and the Nuclear Regulatory Commission (NRC). The MPSC has regulatory jurisdiction pertaining to rates, recovery of certain costs, including the costs of generating facilities, conditions of service, accounting and operating-related matters. The NRC has regulatory jurisdiction over all phases of the operation, construction, licensing and decommissioning of DTE Electric's nuclear plant operations.

DTE Electric – Nuclear Generation

DTE Electric's Nuclear Generation organization is responsible for the safe operation of the Fermi 2 Power Plant and is the licensee of the Fermi 1 nuclear plant.

The Company's Fermi 2 nuclear power plant represents approximately 30% of Michigan's total nuclear generation capacity. At 1.2 million kilowatts, Fermi 2 is capable of producing enough electricity to serve a city of about one million people. The Fermi 2 Power Plant has been providing safe, carbon-free, reliable, cost-effective power to our communities for more than 30 years and has approximately 25 years remaining until its extended operating license expires.



4.2 Fermi 2 Power Plant overview

The power plant is designated as Fermi 2. The DTE Electric Company (DTE Electric or DTEE) is the sole owner and operator of the Fermi 2 Power Plant.

In this section, we describe the Fermi 2 Power Plant's site location, site characteristics, plant characteristics and independent spent fuel storage installation (ISFSI) characteristics. We then discuss the Fermi 1 reactor and the option for the Fermi 3 unit. We intend each of these discussions to provide insights into the unique characteristics of the Fermi 2 Power Plant and deserving of thoughtful deliberations when considering the funding provisions of Fermi 2 Power Plant's eventual nuclear decommissioning.

Site location

The Fermi 2 Power Plant is located on the western shore of Lake Erie at Lagoon Beach, Frenchtown Township, in Monroe County, Michigan. The Fermi 2 Power Plant is one of three U.S. nuclear plants located on Lake Erie; Davis-Besse and Perry nuclear power plants are located in Ohio. Fermi 2 Power Plant is the only U.S. nuclear plant where Canada is within the 10-mile emergency planning zone.

Figure 4.1 Fermi 2 Power Plant site location, Frenchtown Township, Monroe County, Michigan



The plant is approximately eight miles east-northeast of Monroe, Michigan, approximately 28 miles southwest of downtown Detroit, approximately 33 miles southeast of Ann Arbor and approximately 25 miles northeast of Toledo, Ohio. The Fermi 2 Power Plant is one of the operating U.S. commercial nuclear plants within 50 miles of major population centers; approximately 4.7 million people live within 50 miles of the Fermi 2 Power Plant.

Michigan-wide consideration: Michigan is unaffiliated with a low-level radioactive waste (LLRW) compact. As such, the nuclear power plants located in Michigan do not have government-guaranteed access to a LLRW disposal site.

Site characteristics

The Fermi 2 site is on approximately 1,260 acres of land owned solely by DTEE and is bounded on the north by Swan Creek, on the east by Lake Erie, on the south by Point Aux Peaux Road and on the west by Toll Road.



The Fermi 2 site is home to the Detroit River International Wildlife Refuge Lagoon Beach Unit, which is 685 acres of hardwood swamp and coastal wetlands.

The site's Lake Erie shoreline is protected against erosion by shore piling and concrete tetrapod armor.

The General Service Water (GSW) intake for Fermi 2 is a shoreline structure located adjacent to the Fermi 1 intake channel. This channel is protected by two rock jetties that extend into Lake Erie. The original design of these GSW intake jetties extended approximately 800' into Lake Erie; however, due to erosion the jetties only extend approximately 150' into Lake Erie. The GSW intake jetties are undergoing a multi-year restoration project using a concrete tetrapod armor design that is forecasted to complete in approximately 2024. Once restored the north GWS jetty will extend approximately 574' into Lake Erie and the south GSW jetty will extend approximately 726' into Lake Erie; each jetty will be approximately 82' wide at lake level and 102' wide at lake bottom.

Transportation facilities are readily available to the Fermi 2 site. Interstate Highways I-75 and I-275 are approximately five miles west of the site. More immediate access to the site is available from the Dixie Highway, which runs north and south approximately two miles west of the site. From Dixie Highway, Enrico Fermi Drive (a paved private access road) enters the site on the western boundary and serves as the site's main entrance. The site also has rail access via a rail spur connected to the Canadian National Railroad's main line located approximately four miles west of the site.

Fermi 2 Power Plant characteristics

Fermi 2 uses a General Electric (GE) single-cycle, forced-circulation Boiling Water Reactor (BWR) of the BWR/4 class, with a pressure-suppression Mark I containment. The Fermi 2 design power rating is 3,486 MW-thermal (MWt) with gross electric output of 1,235 MW-electric (MWe) and a net electrical output of 1,170 MWe.

The principal structures of the Fermi 2 Power Plant are the Reactor Building, Turbine Building, Auxiliary Building, Residual Heat Removal (RHR) Complex, Radwaste Building, On-site Storage Facility (OSSF), two natural-draft hyperbolic circulating water cooling towers, Independent Spent Fuel Storage Installation (ISFSI) and office buildings.

Figure 4.2 Fermi 2 Power Plant site

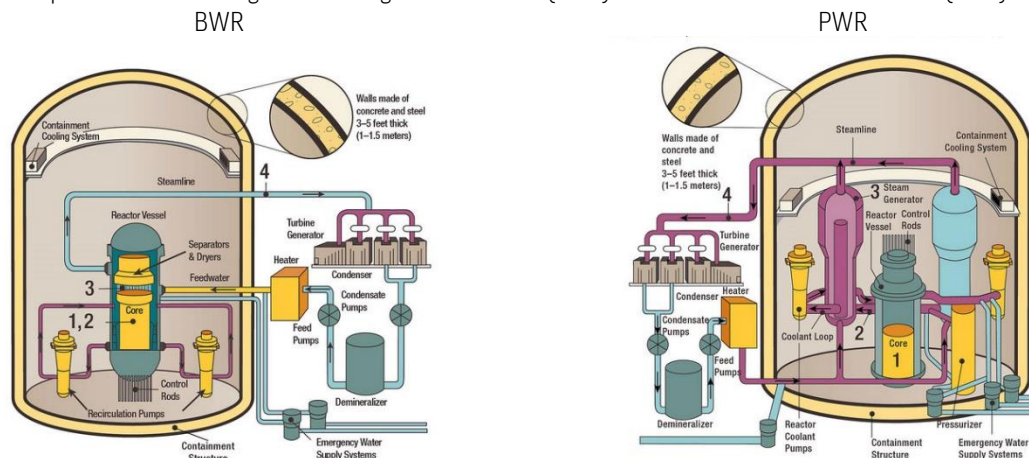


Differentiating characteristics: In United States, there are 96 operating commercial nuclear reactors. Of those 96, 64 are pressurized water reactor (PWR) designs and 32 are BWRs – again, the Fermi 2 Power plant is a BWR design. Of the 32 U.S. BWRs, 22 have the Mark I containment structure; only 14 are of GE BWR/4 class with a Mark I containment similar to the



Fermi 2 Power Plant. Of the 14 GE BWR/4's with Mark I containments, only seven have a design power rating greater than 3,200 MWt (Browns Ferry Units 1,2,3; Peach Bottom Units 2,3, Fermi 2 and Hope Creek). Of these seven BWR/4 Mark-I designs, only Fermi 2 and Hope Creek use natural draft cooling towers.

Figure 4.3 Simplified schematic of generic Boiling Water Reactor (BWR) and a Pressurized Water Reactor (PWR).



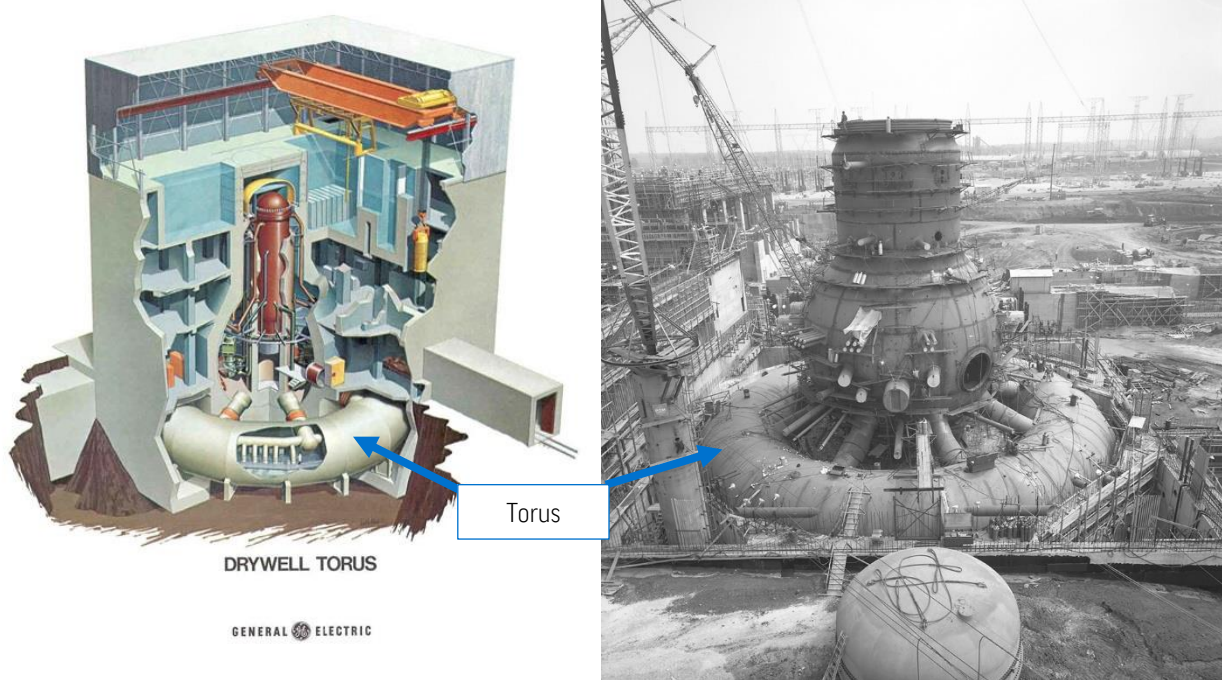
<https://www.nrc.gov/images/bwrsm.jpg>

<https://www.nrc.gov/images/pwrsm.jpg>

As shown in Figure 4.3, a BWR such as the Fermi 2 Power Plant uses a single steam-water loop where the reactor core heats water (figure reference 1,2), which then converts to steam (3) which the steamline directs towards the main turbine (4). In contrast, a PWR uses a two-loop system where the reactor core heats water under pressure (1) which then exchanges heat with a lower pressure water system (2) that converts to steam (3) which the steamline directs towards the main turbine (4). As the BWR steamline drives radioactive steam-water throughout the power plant, BWRs typically have considerably more radioactively contaminated material and equipment than a corresponding PWR.

The primary characteristic of the Mark I containment is the torus containment structure, shown in Figure 4.4; 22 U.S. nuclear plants share the Mark I containment design with the Fermi 2 Power Plant. The torus containment structure is a donut-shaped suppression chamber located beneath the reactor vessel and contains a reservoir of water. The purpose of the torus containment structure is to absorb energy and provide the reactor with emergency cooling water during accident scenarios. The Fermi 2 torus containment structure is a continuous, leaktight steel pressure vessel with a major diameter of approximately 113 feet (ft), the interior diameter of the torus is approximately 30 ft and typically contains approximately one million gallons of water during operations; a special coating material protects the interior of the steel torus shell from corrosion.

Figure 4.4 Cutaway of a drywell torus containment structure and example of a GE BWR/4 with a Mark I containment structure
Cutaway Example BWR/4



Perhaps the most recognizable structure associated with nuclear power plants is the natural-draft cooling tower; however, only about a quarter of the nuclear power plants in the U.S. use natural draft cooling towers. The two Fermi 2 natural-draft cooling towers are reinforced concrete hyperbolic structures approximately 400 ft tall with a diameter of approximately 450' at the exterior base. The water flowing through the cooling towers and associated circulating water system is not contaminated; water vapor emanates from the top of the towers during operations.

Fermi 2 ISFSI characteristics

The Fermi 2 ISFSI consists of a dry fuel storage system, equipment and facilities. The Fermi 2 dry fuel storage system is the Holtec International (Holtec) HI-STORM 100 (an acronym for Holtec International Storage Module) dry storage system. The Fermi 2 ISFSI equipment consists of Vertical Cask Transport (VCT) equipment and material handling equipment. The Fermi 2 ISFSI facilities consist of an ISFSI Equipment Building, ISFSI Storage Pad, ISFSI Fabrication Pad, ISFSI Transfer Pad and ISFSI Cask Transfer Facility (CTF).



Figure 4.5 Fermi 2 Independent Spent Fuel Installation (ISFSI) site



Fermi 2 ISFSI dry fuel storage system: The Fermi 2 ISFSI uses a Holtec HI-STORM 100 dry storage system (nominal 68 fuel assemblies per cask). The HI-STORM 100 unit consists of a sealed metallic canister, referred to as a multi-purpose canister (MPC), contained within a storage overpack.

The HI-STORM 100 system accommodates a spectrum of spent nuclear fuel assemblies in a single overpack design by using different MPCs. The MPC is a cylindrical prismatic structure with honeycomb square cross-section storage cavities (storage locations); the number of storage locations depends on the type of fuel. The external dimensions of all Holtec MPCs are identical to allow the use of a single overpack. Each MPC has different internals (baskets) to accommodate distinct fuel characteristics. Holtec identifies each MPC by the maximum number of fuel assemblies the MPC is capable of receiving; for example, the MPC-24 has a maximum capacity of 24 pressurized water reactor (PWR) fuel assemblies and the MPC-68 has a maximum capacity of 68 BWR fuel assemblies. The Fermi 2 ISFSI uses the Holtec MPC-68 and MPC-68M design.

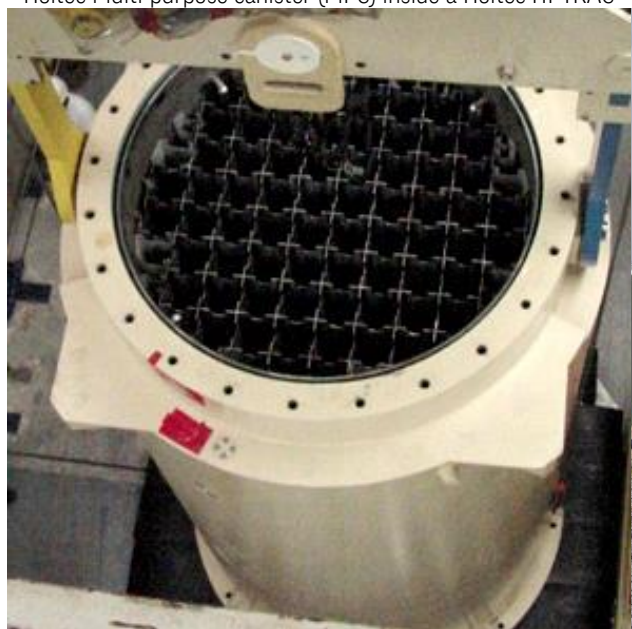
The HI-STORM 100 system is a storage only system and is not suitable for transport.

The HI-STORM 100 system provides spent nuclear fuel (SNF) and radioactive material confinement, radiation shielding, criticality control and passive heat removal independent of any other facility, structures or components. The Fermi 2 HI-STORM 100 system is capable of accepting discharged SNF after approximately seven years of cooling in the spent fuel pool, subject to heat balancing requirements of the Fermi 2 HI-STORM 100 system.

Fermi 2 transports the loaded MPCs from the Fermi 2 Reactor Building to the ISFSI cask transfer pad using the Holtec HI-TRAC (an acronym for Holtec International Transfer Cask) system.



Figure 4.6 Holtec HI-STORM 100 dry storage system
Holtec Multi-purpose canister (MPC) inside a Holtec HI-TRAC



Holtec HI-STORM 100 dry storage cask overpacks



Fermi 2 ISFSI equipment: The Fermi 2 ISFSI uses a Konecranes Vertical Cask Transport (VCT) which is a single failure proof, American Society of Mechanical Engineers Rules for Construction of Overhead and Gantry Cranes (Top Running Bridge, Multiple Girder) ASME NOG-1-2004 Standard compliant wheeled vehicle used for handling spent fuel casks and the transfer of loaded multi-purpose canister(s) (MPC) from HI-TRAC transfer cask(s) into HI-STORM over pack. The VCT consists of a vehicle mainframe, lifting towers, cask-restraining system inhibiting casks from lateral movement, drive and control system, crossbeams and reeved crane system.



Figure 4.7 Fermi 2 Vertical Cask Transport (VCT) vehicle
VCT loading a HI-STORM 100 unit at the CTF



VCT transferring a HI-STORM 100 unit

Fermi 2 ISFSI facilities: The ISFSI Equipment Building provides enclosed parking for the VCT and storage of other ISFSI material handling equipment. The building is an approximately 100' by 55' insulated and fire-protected pre-fabricated steel-panel structure located adjacent to the ISFSI Storage Pad.

The ISFSI Storage Pad provides a level resting surface for dry fuel storage casks. The pad is a 141' by 141' square reinforced concrete structure that is two feet thick and designed to accommodate 64 (expandable to 96) dry fuel storage cask locations. The pad is compliant with 10 CRF Part 72 licensing requirements for the independent storage of spent nuclear fuel (SNF), high-level radioactive waste and reactor-related greater than Class C waste and with ACI 349 "Code Requirements for Nuclear Safety-Related Concrete Structures," 2001. The pad includes a surrounding fence and is dependent on existing Fermi 2 security programs to control physical access to the pad.

The ISFSI Fabrication Pad provides a level surface for completing the manufacture of the dry fuel storage casks.

The ISFSI CTF provides a seismically secure location for fuel transfer. The facility is a subterranean reinforced concrete structure embedded in soils adjacent to the ISFSI haul path.

Figure 4.8 Fermi 2 Cask Transfer Facility (CTF)



Fermi 1

Fermi 2 shares a site with the permanently shut down nuclear reactor facility Fermi 1 Power Plant. DTE is the licensee and holder of NRC Fermi 1 Facility Operating License DPR-9, Docket 50-16.

Fermi 1 was a fast breeder reactor cooled by sodium and operated at essentially atmospheric pressure. The Fermi 1 licensee at the time, Power Reactor Development Company (PRDC), decided to decommission Fermi 1 in November of 1972. Fermi 1 does not have spent fuel or Special Nuclear Material (SNM) stored on-site as the fuel was all shipped to the Savannah River Project facility in 1973 and all SNM was certified to be removed from the Fermi 1 site on November 6, 1975.

Fermi 3

Fermi 2 would share a site with the proposed Fermi 3 Power Plant. The NRC issued DTE Electric a combined (construction and operating) license on May 1, 2015. The Fermi 3 Power Plant is licensed as a GE-Hitachi Economic Simplified Boiling-Water Reactor (ESBWR) with a rated power of 4,500 MWt. No structures have been added or removed to the Fermi 2 site in anticipation of Fermi 3 construction. As of December 31, 2019, DTE Electric had not committed to constructing the Fermi 3 Power Plant.

4.3 Nuclear industry overview

The U.S. commercial nuclear industry ("industry") has undergone significant change in the past decade. Throughout the 2000s, the industry was stable with improving capacity. The 2010s were more volatile to the industry: one new reactor began commercial operations, four reactors began construction – two of which later cancelled, nine reactors permanently shut down, seven reactors announced shutdowns between 2020 and 2025 and seven reactors announced imminent shutdowns but remained operational after state policy changes. As reactors shutdown, companies began to realize different opportunities within the industry and nuclear decommissioning and spent fuel storage emerged as a new competitive market.

In this section, we review the changes in the industry since 2010, we then discuss novel business agreements that have formed between utilities/owners and decommissioning businesses and then we discuss novel business plans for storing spent nuclear fuel. We intend each these discussions to provide summary background and help provide context of future opportunities and risks associated with Fermi 2 Power Plant's eventual nuclear decommissioning.



Industry's decade in review (2010 – 2019)

2010: The U.S. commercial nuclear fleet (“fleet”) of 104 operating nuclear reactors reached a peak generation of approximately 807 million MWh in 2010 [EIA19]. The industry was beginning to focus on extending the licensing life of its reactors to a total of 60 years through NRC license renewal; the Oyster Creek Nuclear Generating Station (OCNGS or “Oyster Creek”) received the NRC’s first license renewal approval in 2009. Also, the industry was uprating the existing fleet with the NRC approving four power uprate applications in 2009 adding approximately 56 megawatts electric (MWe) of capacity, reviewing an additional 15 power uprate applications with potential to add 1,145 MWe of capacity, and expecting an additional 39 power uprate applications through 2014 [SECY1070]. The industry was also progressing towards new nuclear generating assets with the Tennessee Valley Authority (TVA) restarting construction of the Watts Bar Unit 2 nuclear plant in 2007 [SECY1015] and the NRC had 18 combined license (COL) applications (for 28 units) under review. The last nuclear power plants to shut down were Zion Nuclear Power Station (ZNPS or “Zion”), Units 1 & 2 and Millstone Nuclear Power Station, Unit 1 in 1998 [EIA10].

Exelon announced in December 2010 its intention to shut down Oyster Creek in 2019 [EIA11]. Oyster Creek permanently ceased operations on September 17, 2018 [ML18263A163].

2012: By 2012, the industry had received 71 NRC license renewal approvals with another 15 license renewal applications under NRC review and was anticipating an additional 14 license renewal application submissions through 2016. [SECY1284] The Energy Information Agency (EIA) began projecting [EIA12] the restart of construction of the Bellefonte Nuclear, Unit 1 based on TVA’s 2011 Integrated Resource Plan (IRP) [TVA11]. Southern Nuclear Generation and South Carolina Electric & Gas Company (SCE&GC) received the first of the NRC COLs in 2012 to start the construction of the Vogtle Units 3 & 4 and the Virgil C. Summer (“VC Summer”) Units 2 & 3 respectively [EIA13].

Omaha Public Power District (OPPD) in August 2012 awarded Exelon Generation Company, LLC, a subsidiary of Exelon Corporation (“Exelon”), a contract to manage the Fort Calhoun Station [ML122541041]. The agreement was reportedly worth 20-years and \$400 million for Exelon to manage the nuclear plant [OPPD15]; additionally OPPD reportedly awarded a second contract to Exelon worth approximately \$1 million per year to use Exelon’s management model and operating procedures. OPPD retained ownership of the nuclear plant under the arrangement [OPPD18].

Dominion Resources, Inc, a subsidiary of Dominion Energy (“Dominion”), announced in October 2012 its intention to shutdown Kewaunee Power Station in 2013. Kewaunee permanently ceased operations on May 7, 2013 [ML13135A209].

2013: Duke Energy Florida (DEF), a subsidiary of Duke Energy, announced in February 2013 its intention to immediately shutdown Crystal River, Unit 3 (CR-3). CR-3 permanently ceased operations on February 20, 2013 [ML13056A005].

Southern California Edison (SCE) announced in June 2013 its intention to immediately shutdown operations at San Onofre Nuclear Generating Station (SONGS), Units 2 & 3. SONGS permanently ceased operations on June 28, 2013 [ML13183A391].

Entergy Nuclear Vermont Yankee, LLC, a subsidiary of Entergy Corporation (“Entergy”), announced in September 2013 its intention to shutdown Vermont Yankee Nuclear Power Station in 2014. Vermont Yankee permanently ceased operations on December 29, 2014 [ML15085A047].

2015: DTE Electric receives in May 2015 the Fermi 3 COL from the NRC [ML14323B045].



Entergy Nuclear, a subsidiary of Entergy Corporation (“Entergy”), announced in October 2015 its intention to shutdown Pilgrim Nuclear Power Station in 2019. Pilgrim permanently ceased operations on May 31, 2019 [ML19161A033].

Entergy announced in November 2015 the early retirement of the James A. FitzPatrick Nuclear Power Plant in New York. However - in January 2017, Entergy announced FitzPatrick would continue operations in the form of a planned sale and license transfer to Exelon; Exelon had agreed to purchase FitzPatrick from Entergy for approximately \$110 million following New York’s adoption of the Clean Energy Standard [ML17012A280].

2016: Exelon reportedly notified the New York State Public Service Commission in June 2016 of its intent to retire early the R.E. Ginna Nuclear Power Plant and the Nine Mile Point Nuclear Station, Unit 1 absent New York’s adoption of zero emission credits (ZECs). However - Exelon continues to operate both units following New York’s adoption of the Clean Energy Standard [EXN16].

Exelon announced its intention in June 2016 the early retirement of the Clinton Power Station and the two-unit Quad Cities Generating Station, both located in Illinois. However - in December 2016, Exelon later announced both plants would continue operating due to Illinois’ adoption of the Energy Future Jobs Bill [ML16349A311],[ML16349A314].

OPPD announced in June 2016 its intention to shut down the Fort Calhoun Station in 2016. Fort Calhoun permanently ceased operations on October 2016 [ML16319A254].

Pacific Gas & Electric Company, a subsidiary of PG&E Corporation (PG&E), submitted a joint proposal to the California Public Utility Commission (CPUC) in August 2016 to shut down the two-unit Diablo Canyon Power Plant once the unit’s respective operating licenses expire in 2024 and 2025. In January 2018, the CPUC issued an order accepting the proposed 2024/2025 closure of the Diablo Canyon Power Plant [CPUC18].

Entergy announced in December 2016 its intention to shut down the Palisades Nuclear Plant in October 2018. Entergy amended this announcement in September 2017 to update that Palisades would shut down in the spring of 2022 [ML17271A233].

DTE Electric receives in December 2016 the Fermi 2 renewed license from the NRC [ML16270A546].

2017: Entergy announced in January 2017 an agreement with the State of New York to shut down the two-unit Indian Point Nuclear Power Plant no later than April 30, 2020 and April 30, 2021 for Indian Point, Unit 1 and Indian Point, Unit 2 respectively [ML17068A245].

Exelon announced in May 2017 its intention to shut down the Three Mile Island Nuclear Station in 2019. Three Mile Island permanently ceased operations on September 20, 2019 [ML19269E480].

SCE&GC announced in July 2017 its intention to abandon construction of the two VC Summer reactors [ML18198A299].

2018: PSEG Power, LLC, a subsidiary of Public Service Enterprise Group Incorporated (“PSEG”), filed in February 2018 a Form 8-K notifying the U.S. Securities and Exchange Commission (SEC) of the intention cease capital investments in the Hope Creek Generating Station and the two-unit Salem Nuclear Plant, both in New Jersey, absent New Jersey’s adoption of zero emission credits (ZECs). PSEG filed in April 2019 an additional notification of its intention to deactivate both plants absent the awarding



of ZEC certificates to both plants [PSEG19]. However - PSEG continues to operate both plants following New Jersey's approval to issue ZEC certificates to both plants.

First Energy Solutions Corp (FES), a subsidiary of FirstEnergy Corporation, announced in March 2018 the early retirement of the Davis-Besse Nuclear Power Station, Perry Nuclear Power Plant (both located in Ohio) and the two-unit Beaver Valley Power Station located in Pennsylvania. FES filed for relief under Chapter 11 bankruptcy in March 2018. However - in July 2019, FES announced Perry and Davis-Besse nuclear plants would continue operations due to Ohio's adoption of the Ohio Clean Air Program [ML19207A097]. Beaver Valley Power Plant is likely to continue operations as well.

NextEra Energy Resources, LLC (NextEra) announced in July 2018 an agreement with Interstate Power and Light (using the trade name Alliant Energy), a subsidiary of Alliant Energy Corporation, to shorten the term of the Duane Arnold Energy Center (Duane Arnold or DAEC) purchased power agreement. NextEra Energy Duane Arnold, LLC, NextEra, announced in January 2019 its intention to shut down of the Duane Arnold Energy Center (Duane Arnold or DAEC) in the fourth quarter of 2020 [ML19023A196].

2019: The U.S. fleet of 98 operating nuclear power reactors achieved a new peak generation of approximately 809 million MWe in 2019; the U.S. fleet ended 2019 with 96 operating nuclear reactors as two reactors (Pilgrim and Three Mile Island) permanently ceased operation in 2019 [NEI20].

2020: As of January 1, 2020, the U.S. fleet has 96 operating nuclear reactors. Four nuclear plants (6 reactors) are planning to shut down by 2025. Eleven nuclear plants (16 reactors) are operating because of state policy changes in Illinois, New York, New Jersey, Connecticut, Ohio and Pennsylvania. Two reactors (Vogtle Units 3 & 4) are planning to start commercial operation no later than 2025.

Emergence of nuclear decommissioning as an industry

Over the past decade, U.S. commercial nuclear plant licensees have adopted new business models for accomplishing decommissioning. The NRC acknowledges [ML20101H123] four general models:

1. The Licensee Model: The licensee (from when the plant was operating) maintains the license in decommissioning and performs the decommissioning. The licensee could be an electric utility (i.e. regulated cost recovery) or a non-utility company (i.e. a merchant plant). For example, PG&E is using this model to decommission Humboldt Bay Nuclear Power Plant.
2. The Decommissioning Contract Model: The licensee maintains the license in decommissioning and manages a decommissioning contractor. For example, SCE is using this model to decommission SONGS.
3. The Temporary License Transfer Model: The licensee requests a transfer of the license (and its obligations) to a decommissioning company. At the completion of the decommissioning, the license and property transfer back to the original licensee. For example, Exelon is using this model to decommission Zion.
4. The Permanent License Transfer Model: The license requests transfer of the license as part of an asset sale of the nuclear power plant, associated land and spent nuclear fuel to a decommissioning company. For example, Exelon used this model to transfer the obligation to decommission Oyster Creek to a decommissioning company.



Hybrid models have emerged and may continue to develop as opportunities dictate. The general trend of the more recently developed models is to transfer more of the legal obligation to decommission nuclear power plants from public electric utilities or large publicly traded companies to smaller private companies.

The Licensee Model: Humboldt Bay Nuclear Power Plant is owned and operated by Pacific Gas and Electric (PG&E). PG&E has retained the property, plant and spent fuel and ISFSI ownership, operations and obligations through the end of decommissioning. Although contracts awarded for specific activities, PG&E remains the primary responsible party for the completion of the decommissioning activities.

PG&E is an electric utility (i.e. a regulated generator) and Humboldt Bay Nuclear Plant is under the jurisdiction of the California Public Utility Commission (CPUC).

Decommissioning Contract Model: San Onofre Nuclear Generating Station (SONGS) is owned by Southern California Edison (SCE) and San Diego Gas and Electric (SDGE) and is operated by SCE. The SONGS agreement is the original plant owners retain the property, plant and spent fuel and ISFSI rights/ownership, operations and obligations through the end of decommissioning. The original plant owners contract the plant's decommissioning to a single entity who is then contractually obligated to complete the decommissioning activities.

Southern California Edison is an electric utility (i.e. a regulated generator) and SONGS is under the jurisdiction of CPUC.

Temporary License Transfer Model: Zion Nuclear Power Station, Units 1 & 2 (Zion or ZNPS) was owned and operated by Exelon Generation Company (Exelon). The Zion asset sale agreement (ASA) transfers plant ownership, operations and obligations to a third party in exchange for the plant's nuclear decommissioning trust fund; the original owner retains the property and spent fuel and ISFSI ownership, operations and obligations.

The ASA is outlined for the Nuclear Regulatory Commission (NRC) in the Application for License Transfers and Conforming License Amendments, dated January 25, 2008 [ML080310521]. The NRC approved the Zion ASA in Order Approving the Transfer of License and Conforming Amendment, dated May 4, 2009 [ML090930039].

Exelon is a non-utility merchant generator and Zion is not under the jurisdiction of a public service commission.

Permanent License Transfer Model: Oyster Creek Nuclear Generating Station (OCNGS or Oyster Creek) was owned and operated by Exelon. The Oyster Creek asset purchase sale agreement (PSA) transfers the property, plant and spent fuel and ISFSI ownership, operations and obligations to a third party in exchange for the plant nuclear decommissioning trust fund.

The PSA is outlined for the NRC in the Application for Order Approving Direct Transfer of Renewed Facility Operating License and General License and Proposed Conforming License Amendment, dated August 31, 2018 [ML18243A489]. The NRC approved the Oyster Creek ASA in Order Approving the Transfer of License and Conforming Amendment, dated June 20, 2019 [ML19095A458].

Exelon is a non-utility merchant generator and Oyster Creek is not under the jurisdiction of a public service commission.

Hybrid Temporary/Permanent License Transfer Model: Crystal River Unit 3 Nuclear Generating Plant (CR-3) is owned and operated by Duke Energy Florida (Duke Energy). The proposed CR-3 decommissioning services agreement and purchase and



sale agreement (DSA/PSA) transfers spent fuel and ISFSI ownership, operations and obligations as well as the plant operations and obligations to a third party; the original owner retains the property and the plant nuclear decommissioning trust fund.

The DSA/PSA is outlined for the NRC in the Application for Order Consenting to Direct Transfer of Control of Licenses and Approving Conforming License Amendment, dated June 14, 2019 [ML19170A209]. The NRC approved the CR-3 DSA/PSA in Order Approving the Transfer of License and Conforming Amendment, dated April 01, 2020 [ML20069A023].

Duke Energy is an electric utility (i.e. a regulated generator) and CR-3 is under the jurisdiction of the Florida Public Service Commission (FPSC).

Consolidated Interim Storage Facility

The use of a Consolidated Interim Storage Facility (CISF) is a recommendation of the Blue Ribbon Commission on America's Nuclear Future [BRC12]. A CISF is a large-scale, centralized ISFSI capable of accepting the dry-stored spent nuclear fuel (SNF) from multiple commercial nuclear power plants. The Nuclear Waste Policy Act (NWPA) prohibits the Department of Energy (DOE) from accepting title of commercial SNF prior to certain milestones for the establishment of a permanent geological repository. Under the current NWPA, the CISF applicants would be limited, with few exceptions, to contract directly with commercial nuclear power plant owners themselves rather than the DOE to store SNF at their CISF [ML19127A026].

The NRC is reviewing two CISF applications submitted by private commercial entities:

Holtec HI-STORE CIS (Consolidated Interim Storage Facility) - Lea County, New Mexico: Holtec International (Holtec) submitted an application [ML18059A251], dated March 30, 2017, to the NRC to construct and operate an away from reactor (AFR) ISFSI, named HI-STORE CIS, initially authorized to store 500 canisters of SNF for an initial 40-year license duration. Holtec proposes to use its subterranean HI-STORM UMAX canister storage system currently deployed at SONGS, Callaway and Humboldt Bay nuclear power plants at the HI-STORE CIS. This HI-STORM UMAX system is also physically compatible with the all other Holtec canister storage systems such as HI-STORM and HI-STAR; however, the only multi-purpose canisters (MPC) included in the HI-STORM UMAX licensing basis are of the Holtec MPC-37 and MPC-89 design.

Waste Control Specialists LLC Consolidated Interim Storage Facility - Andrews County, Texas: Waste Control Specialists LLC (WCS) submitted an application [ML16132A533], dated April 28, 2016, to the NRC to construct and operate monitored retrievable storage (MRS) installation or ISFSI initially authorized to store 5,000 metric tons of uranium (MTU) in dry-cask storage for an initial 40-year license duration. Interim Storage Partners LLC (ISP), a joint venture between WCS and Orano CIS LLC (a subsidiary of Orano USA), revised the April 28, 2016 application [ML18166A003] on June 8, 2018. ISP proposes to use dry storage technology from Orano TN (a subsidiary of Orano USA) and NAC International (NAC) to target stranded ISFSIs at sites of former reactors such as Yankee Rowe, Connecticut Yankee and Rancho Seco nuclear plants [ML18206A550].



SECTION FIVE

5 Remaining life of nuclear assets

The Fermi 2 Power Plant has been providing DTE Electric customers with safe, reliable, clean and cost-effective electricity since commercial operations began in 1988. In 2019, the Fermi 2 Power Plant set its single-year generation record at 10.3 million MWh, achieved a 98.9% capacity factor and caused avoided emissions of approximately 10 million metric tons of carbon dioxide. The Fermi 2 Power Plant has the capacity to be a key contributor for DTE Electric's emission-free generation fleet for the next two decades.

In this section, we address the remaining life of DTE Electric's nuclear assets located at the Fermi 2 site: Fermi 2 Power Plant, Fermi 2 Independent Spent Fuel Storage Installation (ISFSI), other assets such as the Fermi 1 Power Plant and the Fermi 3 combined operating license (COL). We intend this discussion to DTE Electric's report supporting the "remaining life of nuclear generating units" as listed in Paragraph 11, Bullet d of the Michigan Public Service Commission (MPSC) Settlement Agreement U-6150 (re-opened), dated August 26, 1986. This report is critical to understanding the time horizons DTE Electric and the MPSC have to address any potential nuclear decommissioning funding shortfalls.

5.1 Fermi 2 Power Plant

In this portion of the section, we discuss the anticipated remaining life of the Fermi 2 Power Plant – the nuclear generating unit located at the Fermi 2 site. The discussion reviews Fermi 2 Power Plant's historical and future Nuclear Regulatory Commission (NRC) licensing periods. DTE Electric defines the remaining operating life of the Fermi 2 Power Plant as approximately 25 years as limited by its current NRC extended operating license expiring on March 20, 2045.

Nuclear Regulatory Commission (NRC)

The Nuclear Regulatory Commission (NRC) is the independent federal agency responsible for the safe use of radioactive materials for beneficial civilian purposes while protecting people and the environment. The NRC regulates commercial nuclear power plants and other uses of nuclear materials, such as in nuclear medicine, through licensing, inspection and enforcement of its requirements. Prior to the NRC's creation in 1975 (The Energy Reorganization Act of 1974), the Atomic Energy Commission (AEC) had responsibility for nuclear regulation; historic reference to the AEC is not changed.

The operating life of commercial nuclear power plants is determined by the expiration date of the plant's NRC operating license. The initial term of an NRC operating license is 40 years. Commercial nuclear power plants may apply to the NRC and request a license renewal for an additional 20 years to extend operations from 40 years to 60 years. Recent to 2017, commercial nuclear power plants may apply to the NRC and request a subsequent license renewal for a second additional 20 years to extend operations from 60 years to 80 years. A commercial nuclear power plant's NRC license must be terminated within 60 years of the permanent cessation of operations.

Fermi 2 NRC licensing basis

DTE Electric is the holder of the Fermi 2 NRC operating license NPF-43, Docket Number 50-341. DTE Electric is a direct wholly owned subsidiary of the DTE Energy Company (DTE). Prior to 2013, DTE Electric was the Detroit Edison Company (Detroit Edison or DECo); historic reference to the Detroit Edison Company is not changed.



Construction Permit: In April 1969, the Detroit Edison Company submitted an application to the Atomic Energy Commission (AEC) to construct the Enrico Fermi Atomic Power Plant Unit 2. The AEC granted Exemption Authorizations for construction prior to receipt of the construction permit for (1) reactor building foundations and walls below grade in October 1970 and (2) for steel sections below grade in March 1971. The AEC approved the Fermi Power Plant construction permit in September 1972.

Initial Facility Operating License: The Atomic Energy Act of 1954 and NRC regulations limit commercial power reactor licenses to an initial 40 years but also allow licenses to be renewed. Political, economic, antitrust and other considerations – not a nuclear technology limitation - determined the basis for the initial 40-year license term.

The NRC issued the initial Facility Operating License (NPF-33) for the Fermi 2 Power Plant on March 20, 1985 to allow Fermi 2 to load fuel and perform low-power testing; this license issuance marked the start of the initial 40-year term for the Fermi 2 Power Plant. The NRC superseded NPF-33 with Facility Operating License NPF-43 on July 10, 1985 to allow Fermi 2 to operate at 100% power.

The Fermi 2 Power Plant started commercial operations on January 23, 1988. Michigan Public Service Commission established the requirements for Detroit Edison to certify Fermi 2 Power Plant's commercial operation status in MPSC Order U-7760, dated April 1, 1986. The Commission concurred with Detroit Edison's certification in MPSC Opinion U-7660, dated February 23, 1988.

License renewal: A commercial nuclear power plant licensee may apply to review its license as early as 20 years before the expiration of its current license. There is no limit on how late a licensee may apply for a license renewal; and if the licensee submits the renewal application at least five years before the expiration of its current license, the plant can continue to operate until the NRC completes its review. In the case of an insufficient renewal application submission prior to at least five years before the current license expiration, the plant may have to stop operating if the license expires before the NRC makes a renewal decision.

The amended 10 CFR 54 "Requirements for Renewal of Operating Licenses For Nuclear Power Plants" established the NRC's regulatory process for license renewal where the focus of the NRC is on managing the adverse effects of aging to ensure that important passive, long-lived systems, structures and components will continue to perform their intended function in the period of extended operation. [80 FR 22461].

The revised environmental protection regulations in 10 CFR 51 "Environmental Protection Regulations for Domestic Licensing and Related Regulatory Functions" facilitate the environmental review for license renewal.

The NRC approved the Fermi 2 Power Plant license renewal on December 15, 2016. With the NRC's decision, Fermi 2's licensed life extended 20 years from March 20, 2025 to March 20, 2045. DTE Electric submitted the Fermi 2 Power Plant license renewal to the NRC in April 2014.

Subsequent license renewal (SLR): The NRC defines subsequent license renewal (SLR) to be the period of extended operations from 60 years to 80 years. The NRC first issued SLR evaluation and guidance in 2017 as NUREG-2192 and NUREG-2191.

Turkey Point Nuclear Generating Station (Turkey Point) which is owned and operated by Florida Power & Light (FPL), a wholly owned subsidiary of NextEra Energy, became the first nuclear plant to receive the NRC's approval for SLR in December 2019.



FLP submitted the Turkey Point SLR application in January 2018, which was approximately 15 years prior to the expiration of Turkey Point's extended operating licenses. With the NRC's decision [ML19305C879], Turkey Point's two reactors (Units 3 and 4) licensed-life extended from 2032/2033 to 2052/2053 respectively.

Peach Bottom Nuclear Generating Station (Peach Bottom) which is operated by Exelon Generation Company, LLC, a subsidiary of Exelon Corporation (Exelon) and co-owned between Exelon and PSEG Nuclear, LLC, a subsidiary of PSEG Power, LLC, itself a subsidiary of Public Service Enterprise Group, LLC, received the NRC's approval for SLR in March 2020 [ML20010F285]. Exelon submitted the Peach Bottom SLR application in July 2018, which was approximately 15 years prior to the expiration of Peach Bottom's extended operating licenses. With the SLR, Peach Bottom's two reactors (Units 2 and 3) licensed-life extended from 2033/2034 to 2053/5054 respectively.

Dominion Energy Virginia, a subsidiary of Dominion Energy (Dominion), submitted an SLR application for its two-unit Surry Nuclear Power Plant (Surry) in October 2018, which was approximately 15 years prior to the expiration of Surry's extended operating licenses.

In the context of DTE Electric, a subsequent license renewal would extend the licensed-life of the Fermi 2 Power Plant from 2045 to 2065. DTE Electric has not committed to the subsequent license renewal option for the Fermi 2 Power Plant.

Decommissioning: NRC regulations at 10 CFR 50.82(a)(3) state commercial nuclear power plants are to complete decommissioning within 60 years of permanent cessation of operations. Licensees may petition the NRC to permit completion of decommissioning beyond 60 years when necessary to protect public health and safety; factors the NRC will consider in evaluating alternatives include unavailability of waste disposal capacity and other site-specific factors affecting the licensee's capability to carry out decommissioning, including the presence of other nuclear facilities at the site.

There are three NRC-approved strategies [53 FR 24018] for decommissioning:

- DECON where the decommissioning strategy is to immediately dismantle the nuclear facility
- SAFSTOR where the decommissioning strategy is to defer dismantling where the nuclear facility is maintained and monitored in a condition that allows radioactivity to decay
- ENTOMB where the decommissioning strategy is to permanently encase the radioactivity contaminants on site in a structurally sound material such as concrete.

Fermi 2 Power Plant end of life conclusion

Consistent with the previous discussion, this study assumes Fermi 2 Power Plant will permanently cease operations on March 20, 2045 coinciding with when the Fermi 2 renewed operating license expires. This study assumes DTE Electric will use the DECON strategy to decommission the Fermi 2 Nuclear Power Plant.

5.2 Independent Spent Fuel Storage Installation (ISFSI)

In this section, we discuss the anticipated remaining life of the Fermi 2 Independent Spent Fuel Storage Installation (ISFSI). The discussion reviews the Fermi 2 ISFSI's historical and future Nuclear Regulatory Commission (NRC) licensing periods. DTE Electric concludes the remaining life of the Fermi 2 ISFSI as approximately 85 years, ending March 20, 2105, as limited by the current 2010 NRC Waste Confidence Decision.



2010 NRC Waste Confidence Decision

The NRC Waste Confidence Decisions are periodic assessments to determine the NRC's degree of assurance that radioactive wastes produced by nuclear power plants... "can be safely disposed of, to determine when such disposal or offsite storage will be available, and to determine whether radioactive wastes can be safely stored onsite past the expiration of the existing facility licenses until offsite disposal or storage is available." [44 FR 61372]

The NRC issued the following initial Waste Confidence Findings on August 31, 1984:

- (1) The Commission finds reasonable assurance that safe disposal of HLW [high level waste] and SNF [spent nuclear fuel] in a mined geologic repository is technically feasible;
- (2) The Commission finds reasonable assurance that one or more mined geologic repositories for commercial HLW and SNF will be available by the years 2007-2009 and that sufficient repository capacity will be available within 30 years beyond the expiration of any reactor operating license to dispose of existing commercial HLW and SNF originating in such reactor and generated up to that time;
- (3) The Commission finds reasonable assurance that HLW and SNF will be managed in a safe manner until sufficient repository capacity is available to assure the safe disposal of all HLW and SNF;
- (4) The Commission finds reasonable assurance that, if necessary, spent fuel generated in any reactor can be stored safely and without significant environmental impacts for at least 30 years beyond the expiration of that reactor's operating license at that reactor's spent fuel storage basin, or at either onsite or offsite independent spent fuel storage installations (ISFSIs);
- (5) The Commission finds reasonable assurance that safe independent onsite or offsite spent fuel storage will be made available if such storage capacity is needed [49 FR 34658].

Based on the above findings, the NRC promulgated 10 CFR 51.23(a) to provide a generic determination that for at least 30 years beyond the expiration of reactor operating licenses, no significant environmental impacts would result from the storage of spent fuel in reactor facility storage pools or ISFSIs located at reactor or away-from-reactor sites.

The NRC reviewed the findings in 1989-1990, which resulted in a revision of Findings 2 and 4:

- (2) The Commission finds reasonable assurance that at least one mined geologic repository will be available within the first quarter of the twenty-first century, and sufficient repository capacity will be available within 30 years beyond the licensed life for operation (which may include the term of a revised or renewed license) of any reactor to dispose of the commercial HLW and SNF originating in such reactor and generated up to that time;
- (4) The Commission finds reasonable assurance that, if necessary, spent fuel generated in any reactor can be stored safely and without significant environmental impacts for at least 30 years beyond the licensed life for operation (which may include the term of a revised or renewed license) of that reactor at its spent fuel storage basin, or at either onsite offsite ISFSIs. [55 FR 38474]

The NRC again reviewed its Waste Confidence Findings in 1999 and confirmed the existing findings.



The NRC again reviewed its Waste Confidence Findings in 2010, which again resulted in a revision of Findings 2 and 4:

(2) The Commission finds reasonable assurance that sufficient mined geologic repository capacity will be available to dispose of the commercial high-level radioactive waste and spent nuclear fuel generated by any reactor when necessary.

(4) The Commission finds reasonable assurance that, if necessary, spent fuel generated in any reactor can be stored safely and without significant environmental impacts for at least 60 years beyond the licensed life for operation (which may include the term of a revised or renewed license) of that reactor in a combination of storage in its spent fuel storage basin and either onsite or offsite independent spent fuel storage installations.

Within the 2010 Waste Confidence Decision [75 FR 81037], the NRC stated they had assumed that the Department of Energy would not build the permanent geological radioactive waste repository at Yucca Mountain and that the Waste Confidence Findings reflects the uncertainty regarding the timing of the availability of a geological repository; however, the U.S. Court of Appeals vacated the 2010 Waste Confidence Decision in 2012 on the basis that the NRC did not perform a sufficient analysis of the environmental risks associated with the updated Decision [USCA12].

The NRC issued the revised Waste Confidence Decision in 2014 [79 FR 56238] adopting the findings of the Generic Environmental Impact Statement (GEIS) for Continued Storage of Spent Nuclear Fuel [NUREG-2157]. The updated Waste Confidence Decision states in summary that spent nuclear fuel can be safely managed during short-term (up to 60 years), long-term (100 years incremental to the initial 60 years) and indefinite timeframes. Although the NRC is confident that a geological repository will be ready by the end of the short-term timeframe, the long-term timeframe allows consideration for a delay in opening the geological repository and the GEIS also considers that a geological repository may never be built. The NRC also changed the name from “waste confidence” to “continued storage of spent nuclear fuel.”

Consistent with the waste confidence discussion, this study assumes that the Fermi 2 ISFSI casks (containing spent nuclear fuel and high-level waste generated at the Fermi 2 Power Plant) will remain safely stored on-site at the Fermi 2 ISFSI for an additional 60 years beyond the expiration of the Fermi 2 renewed operating license (i.e. March 20, 2105).

Nuclear Regulatory Commission (NRC)

The NRC regulates storage of spent nuclear fuel at independent spent fuel storage installations (ISFSIs) under two licensing options: (1) site-specific license or (2) general license. Under the site-specific license, the NRC issues an ISFSI-only license, valid up to 40 years, that contains technical requirements and operating conditions for the ISFSI and specifies what the licensee is authorized to store at the ISFSI. Under the general license, the NRC authorizes a nuclear power plant licensee to store spent fuel in NRC-approved ISFSI casks at a site licensed to operate a power reactor under 10 CFR 50; the Fermi 2 ISFSI operates under a general license.

General license for SNF storage in an ISFSI at a reactor site: The attributes of a general license for an ISFSI are:

- General licensing provisions specified in 10 CFR Part 72, Subpart K;
- Limited to 10 CFR Part 50 reactor licensees;
- Effective without filing an application or issuance of licensing document;
- Not transferrable unless the 10 CFR Part 50 reactor license is transferred;



- Authorized for use of only NRC-certified casks (i.e. NRC has issued a Certificate of Compliance (CoC) for the cask);
- Reactor licensee is responsible for assuring generically-approved cask design and site requirements are compatible; and
- Relies on extending reactor licensee experience, qualifications, and programs to ISFSI activities.

The general license for an ISFSI does not have an expiration date per se.

Certificate of Compliance: General licensees, such as DTE Electric, use the Certificate of Compliance (CoC) for in service ISFSI dry casks. The holder of the CoC is the dry cask manufacturer and not the general licensee. The NRC issues new and amended CoC for each specific dry cask model using an application review process. The CoC contains the requirements and operating conditions (fuel specifications, cask leak testing, surveillance, and other requirements) for the cask design and specifies what the licensee may store in the cask system. The initial term for a CoC is 20-years from issuance; CoC renewal terms are in 40-year increments.

Fermi 2 ISFSI licensing basis

DTE Electric uses a general license under the Fermi 2 Power Plant's 10 CFR 50 operating license to store spent nuclear fuel onsite at the Fermi 2 Independent Spent Fuel Storage Installation (ISFSI). DTE Electric has design approval for the Fermi 2 ISFSI to use the Holtec HI-STORM 100 dry cask design per NRC Certificate of Compliance (CoC) Number 1014, Amendment 5 and Amendment 10 [ML082030116],[ML19156A068].

The NRC issued the Holtec HI-STORM 100 CoC Number 1014 to Holtec International on May 5, 2000 following an approximately four-year review process. The initial 20-year term of the HI-STORM 100 CoC was May 31, 2000 through June 1, 2020. In January 2020, Holtec International submitted its application to the NRC to renew the HI-STORM 100 CoC term for 40 additional years (i.e. through June 1, 2060) [ML20049A083]. DTE Electric expects Holtec International, as the CoC holder, to seek 40-year renewals indefinitely in accordance with 10 CFR 72.240.

ISFSI general licensees are limited by the greater of the expiration date of the CoC or the term of the CoC applied to the in service date of the ISFSI cask. For the Fermi 2 HI-STORM 100 system:

1. ISFSI casks placed in service (loaded with spent fuel) prior to June 1, 2020 are licensed for the 20 years following their in-service date.

Once the NRC approves the Holtec HI-STORM 100 CoC renewal application:

2. ISFSI casks placed in service prior to June 1, 2020 will be licensed through at least June 1, 2060.
3. ISFSI casks placed into service after June 1, 2020 will be licensed for 40 years following their in-service date.

Following NRC approval of the Holtec HI-STORM 100 CoC subsequent renewal application:

4. ISFSI casks placed into service before June 1, 2060 will be licensed through at least June 1, 2100.

DTE Electric expects the NRC to approve CoC renewals consistent with the GEIS for Continued Storage of Spent Nuclear Fuel.



Fermi 2 ISFSI end of life conclusion

Consistent with the discussion above, the Fermi 2 ISFSI will cease operations by March 20, 2105, which is 60 years following the shutdown of the Fermi 2 Power Plant. The assumption used in this study is that the Department of Energy will have accepted transfer of all of Fermi 2's spent nuclear fuel and high-level waste for geological deposit by March 20, 2105 and that the decommissioning of the Fermi 2 ISFSI would occur shortly thereafter.

5.3 Other facilities

Fermi 1

Fermi 2 Power Plant shares its site with the permanently shut down nuclear reactor facility Fermi 1 Power Plant. DTE Electric is the licensee and holder of NRC Fermi 1 Facility Operating License DPR-9, Docket 50-16.

The Fermi 1 licensee at the time, Power Reactor Development Company (PRDC), decided to decommission Fermi 1 in November of 1972. Fermi 1 does not have spent nuclear fuel or special nuclear material (SNM) stored on-site as all the fuel was shipped to the Savannah River Project facility in 1973 and all SNM was certified to be removed from the Fermi 1 site on November 6, 1975 [ML102290154].

Fermi 1 has been in a decommissioning state since PDRC declared permanent cessation of operations in November of 1972. Fermi 1 is currently in SAFSTOR status. NRC regulations at 10 CFR 52.82(a)(3) limit the time allowed for DTE Electric to complete the decommissioning activities required for Fermi 1 license termination to no more than 60 years of permanent cessation of operations.

The non-radiological structures such as the Turbine Building and the offices support Fermi 2 operations.

Fermi 1 Nuclear Plant end of life conclusion

Consistent with the discussion above, the Fermi 1 Nuclear Plant reached the end of its life in 1972 – as such, this study assumes DTE Electric will have removed the Fermi 1 structures required for license termination no later than the end of 2032. This study assumes that DTE Electric will remove structures used for the Fermi 2 Power Plant operations after the Fermi 2 Power Plant permanently ceases operations in 2045.

Fermi 3

This study does not consider nor does it preclude the construction or operation of the Fermi 3 Power Plant.

Small modular reactors or other nuclear technologies

This study does not consider nor does it preclude the construction or operation of the small modular reactors (SMRs) or other nuclear technologies at the Fermi 2 site.



SECTION SIX

6 Regulatory dynamics

In this section we discuss the regulatory dynamics of nuclear decommissioning and providing reasonable assurance of funding for nuclear decommissioning. We will begin the discussion with an overview of the applicable federal regulations concerning this decommissioning study. We next discuss regulations or orders promulgated for the state of Michigan through the Michigan Public Service Commission. We then complete the section with a discussion on the Nuclear Policy Waste Act of 1982, as amended and the obligations of the Standard Contract.

6.1 Federal regulations overview

The Nuclear Regulatory Commission (NRC), Federal Energy Regulatory Commission (FERC), and Internal Revenue Service (IRS) regulate aspects of nuclear decommissioning. In this section we review nuclear decommissioning regulatory constructs by federal agency. We will then discuss key aspects of the integrated nuclear decommissioning regulations.

NRC regulations

The Nuclear Regulatory Commission (NRC) regulates nuclear decommissioning trusts at 10 CFR 50.75 and 10 CFR 72.30. The NRC regulates the funding of post-shutdown spent fuel management at 10 CFR 50.54(bb). The NRC regulates nuclear decommissioning activities with the definition at 10 CFR 50.2 and at 10 CFR 50.82. Key updates to the Federal Register include:

1. 53 FR 24018 - - Created 50.2 definition and 50.75. Supplemental info DECON, SAFSTOR
2. 61 FR 39278 - - Made significant amendments to decommissioning rules
3. 63 FR 50465 - - Financial assurance requirements
4. 67 FR 78332 - - Amended rules governing decommissioning trust funds for merchant plants
5. 73 FR 62220 - - NRC discussion on denial to allow pre-shutdown withdrawals
6. 76 FR 35512 - - Mandate to provide for ISFSI funding assurance

The NRC issues regulatory guides to describe to the public the methods that NRC staff considers acceptable for use in implementing specific parts of the NRC's regulations, to explain the techniques that the staff uses in evaluating specific problems or postulated accidents and to provide specific guidance to applicants. The NRC regulatory guides supporting nuclear decommissioning include:

1. Regulatory Guide 1.184, Decommissioning Nuclear Reactors, Rev 1 [ML13144A840]
2. Regulatory Guide 1.185, Standard Format and Content for Post-Shutdown Decommissioning Activities Report, Rev 1 [ML13140A038]
3. Regulatory Guide 1.179, Standard Format and Content of License Termination Plans for Nuclear Power Reactors, Rev 2 [ML19128A067]
4. Regulatory Guide 1.202, Standard Format and Content of Decommissioning Cost Estimates for Nuclear Power Reactors, Rev 0 [ML050230008]



5. Regulatory Guide 1.159 Assuring the Availability of Funds for Decommissioning Nuclear Reactors, Rev 2 – [ML112160012]

The NRC prepares and issues other publications such as Regulatory Issue Summary (RIS) reports, reports prepared by NRC staff (NUREG) or contractors (NUREG/CR), and written issues papers the NRC staff submits to the NRC Commission to inform them on policy, rulemaking, and adjudicatory matters (SECY).

1. Regulatory Issue Summary (RIS) 2001-07, 10 CFR 50.75 Reporting and Recordkeeping for Decommissioning Planning, Rev 1 [ML083440158]
2. SECY-13-0066 Staff Findings on the Table of Minimum Amounts Required to Demonstrate Decommissioning Funding Assurance [ML13127A234]
3. NUREG-1307, Report on Waste Disposal Charges: Changes in Decommissioning Waste Disposal Costs at Low-Level Waste Burial Facilities, Rev. 17 [ML19037A405]
4. NUREG-1577, Standard Review Plan on Power Reactor Licensee Financial Qualifications and Decommissioning Funding Assurance, Rev. 1 [ML17033B536]
5. NUREG-1700, Standard Review Plan for Evaluating Nuclear Power Reactor License Termination Plans, Rev. 2 [ML18116A124]
6. NUREG-1713 Standard Review Plan for Decommissioning Cost Estimates for Nuclear Power Reactors [ML043510113]
7. NUREG-0586 Final Generic Environmental Impact Statement (GEIS) on Decommissioning of Nuclear Facilities [ML18057B048]

The NRC is undertaking proposed rulemaking for Proposed Rule: Regulatory Improvements for Production and Utilization Facilities Transitioning to Decommissioning (RIN 3150-AJ59). The proposed rulemaking consolidates NRC nuclear decommissioning guidance and past practices as well as incorporates a graded approach to the radiological risks associated with nuclear power plants that have permanently ceased operations. Key documents for this proposed rulemaking include:

1. 80 FR 72358 – Advanced notice of proposed rulemaking
2. 82 FR 55954 – Final regulatory analysis
3. SECY-18-0055 Proposed Rule: Regulatory Improvements for Production and Utilization Facilities Transitioning to Decommissioning (RIN 3150-AJ59) [ML18012A021]
4. SECY-18-0055 – Draft Proposed Rule: Regulatory Improvements for Production and Utilization Facilities Transitioning to Decommissioning [ML18012A022]
5. SECY-18-0055 – Regulatory Analysis for the Proposed Rule: Regulatory Improvements for Production and Utilization Facilities Transitioning to Decommissioning [ML18012A024]
6. SECY-18-0055 – Regulatory Guidance of Proposed Rule: Regulatory Improvements for Production and Utilization Facilities Transitioning to Decommissioning [ML18012A228]
7. DG-1346 Proposed RG 1.235 Emergency Planning for Decommissioning Nuclear Reactors [ML17311B018]
8. DG-1347 Revision in RG 1.184 Decommissioning of Nuclear Power Reactors [ML17347A794]
9. DG-1348 Revision to RG 1.159 Assuring the Availability of Funds for Decommissioning Nuclear Reactors [ML17348B485]



10. DG-1349 Revision to RG 1.185 Standard Format and Content for Post-Shutdown Decommissioning Activities Report [ML17353A727]

FERC regulations

The Federal Energy Regulatory Commission (FERC) regulates nuclear decommissioning trusts at 18 CFR 35.32 and 18 CFR 35.33. These FERC regulations only apply to trust amounts collected from FERC-jurisdictional customers. Key updates to the Federal Register include:

1. 60 FR 34109 – FERC Order 580 – Adds Subpart E 35.32 and 35.33
2. 62 FR 33342 – FERC Order 580-1 – Final update to Order 580
3. 70 FR 34340 – FERC Order 658 – Relaxes annual reporting requirements

IRS regulations

The Internal Revenue Service (IRS) regulates nuclear decommissioning trusts at 26 CFR 1.468A. These IRS codes and regulations only apply to “qualified” trusts:

1. 98 Stat. 604 – Deficit Reduction Act of 1984 – instituted the 468A qualified funds [DRA84]
2. 106 Stat. 2776 (see 3024) – Energy Policy Act of 1992 – eliminated Black Lung restrictions [EPA92]
3. 106 Stat. 2776 (see 3034) – Energy Policy Act of 1992 – changed the USC numbering of Black Lung [EPA92]
4. 119 Stat. 594 (see 1007) – Energy Policy Act of 2005 – fund transfers and ruling amounts [EPA05]
5. 75 FR 80697 – Clarified spent fuel management status
6. 81 FR 95929 – Clarified ISFSI construction and self-dealing

Integrated regulatory framework

NRC minimum formula - NRC regulations at 10 CFR 50.75 define the requirements for commercial reactor license holders such as DTE Electric to demonstrate reasonable assurance of funds for decommissioning while a facility is operating. The formula at 10 CFR 50.75(c) establishes the minimum amount required to demonstrate reasonable assurance of funds for decommissioning reactors by type, subject to footnote 1 of 10 CFR 50.75(c):

Amounts are based on activities related to the definition of “Decommission” in § 50.2 of this part and do not include the cost of removal and disposal of spent fuel or of nonradioactive structures and materials beyond that necessary to terminate the license.

where § 50.2 refers to NRC regulations at 10 CFR 50.2 which defines “Decommission” as:

Decommission means to remove a facility or site safely from service and reduce residual radioactivity to a level that permits—

- (1) Release of the property for unrestricted use and termination of the license; or
- (2) Release of the property under restricted conditions and termination of the license.



Two studies performed by Pacific Northwest National Laboratory (PNNL) for the NRC in 1978 and 1980 form the basis for the NRC minimum decommissioning fund formula in 10 CFR 50.75(c). The 10 CFR 50.75(c) formula uses a base of 1986 dollars (10 CFR 50.75(c)(1)), which licensees must recalculate annually to current dollars by using an adjustment factor defined in 10 CFR 50.75(c)(2) to account for escalation changes in labor costs, energy costs and low-level waste burial costs [ML13063A190]. The NRC publishes and makes publicly available biennial revisions to the NRC report NUREG-1307, "Report on Waste Burial Changes" for licensees to use in determining the adjustment factor.

The result of the 10 CFR 50.75(c) calculation updates the 1986 base dollars to a current year minimum amount that the licensee must certify and hold in an approved financial instrument. The total amounts held in the approved financial instruments, which includes expected earnings and rate collections, must be equal to or greater than the amount calculated from application of the 10 CFR 50.75(c) calculation.

The NRC assessed the continued use of the NRC minimum formula as an adequate measurement to establish a common minimum standard in the 2013 NRC report SECY-13-0066 [ML13127A234], "Staff Findings on the Table of Minimum Amounts Required to Demonstrate Decommissioning Funding Assurance." This review was in response to the NRC's Office of the Inspector General (OIG) follow-up audit of the NRC's decommissioning fund program OIG-06-A-07 determination that the NRC delayed implementation of a prior OIG audit report (OIG-99A-16) recommendation to assess the adequacy of the NRC minimum formula. The SECY-0066 also addressed the Government Accountability Office (GAO) report GAP-12-258, "NRC's Oversight of Nuclear Power Reactors' Decommissioning Funds Could Be Further Strengthened" note that the NRC minimum formula is generally less than site-specific cost estimates.

Nuclear Decommissioning Trusts for radiological decommissioning - The NRC regulations at 10 CFR 50.75(e) specify the general requirements for the methods for providing the NRC with reasonable assurance of the availability of funds for nuclear decommissioning. The most common method of meeting this financial assurance requirement is with a trust, typically referred to as a Nuclear Decommissioning Trust (NDT), which places the funding amounts into an account segregated from the licensee assets and outside the licensee's or its affiliates' administrative control. The licensee may deposit amounts into the NDT using the prepayment or external sinking fund method.

IRS and FERC regulations provide additional requirements for NDTs. Most notably, the IRS regulations have a specific tax treatment for qualified NDT amounts which necessitated licensees establish distinct NDTs for qualified amounts and for nonqualified amounts. FERC regulations require licensees differentiate with each trust the amounts deposited from FERC-jurisdiction customers and the amounts deposited from state-jurisdiction customers.

Given this integrated regulatory framework, licensees that serve state- and FERC-jurisdictional customers would minimally establish two separate NDTs (qualified and nonqualified), each with two plans (state-jurisdiction and FERC jurisdiction). The license would then report the total amounts of these four NDT funds:

1. Qualified state
2. Qualified FERC
3. Nonqualified state
4. Nonqualified FERC

to the NRC as the method to meet the financial assurance requirement.



Nuclear Decommissioning Trusts for ISFSI decommissioning - NRC regulations at 10 CFR 72.30(e) specify the general requirements for the methods for providing the NRC with reasonable assurance of the availability of funds for Independent Spent Fuel Storage Installation (ISFSI) decommissioning. A common method of meeting this financial assurance requirement is with the NDT. The licensee may deposit amounts into the NDT using the prepayment or external sinking fund method.

NRC regulations as summarized in Regulatory Issue Summary RIS-2001-07, Revision 1, "10 CFR 50.75 Reporting and Recordkeeping for Decommissioning Planning" do not preclude licensees from using the same NDT to hold amounts reported for radiological decommissioning funding assurance and other decommissioning amounts; however, the licensee must be able to identify and account for the amounts within each trust associated with radiological decommissioning funding assurance and those amounts for other decommissioning. NRC regulations at 10 CFR 72.30(g) establishes licensees must only use amounts reported for ISFSI decommissioning for ISFSI decommissioning and must monitor those accumulated amounts for market changes. As such, licensees may deposit amounts collected for ISFSI decommissioning reasonable assurance into the licensee's NDT, provided the licensee has taken action to segregate the ISFSI amounts from the radiological decommissioning amounts and maintain separate books for each amount.

IRS and FERC regulations apply equally to 10 CFR 72.30(e) amounts and 10 CFR 50.75(e); therefore, licensees using the external sinking fund method and collecting amounts from state and FERC jurisdictional customers would establish an ISFSI decommissioning pool within both the qualified and nonqualified trusts as applicable. The licensee would then report these four NDT funds:

1. Qualified state ISFSI
2. Qualified FERC ISFSI
3. Nonqualified state ISFSI
4. Nonqualified FERC ISFSI

to the NRC as the method to meet the financial assurance requirement.

Nuclear Decommissioning Trusts for other decommissioning - NRC regulations at 10 CFR 50.54(bb) specify the general requirements to provide the NRC with irradiated spent fuel program and funding plans to manage irradiated fuel at the reactor following the permanent cessation of operations. NRC regulations for irradiated fuel management are silent as to methods for providing the NRC with reasonable assurance of availability of funds for post-shutdown irradiated fuel management program; however, the NRC has accepted the use of the NDT, coupled with prepayment or external sinking fund as the funding plan for the post-shutdown irradiated spent fuel management plan.

NRC regulations do not preclude licensees from using the same NDT to hold amounts reported for radiological decommissioning and other decommissioning amounts; however, the licensee must be able to identify and account for the amounts within each trust associated with radiological decommissioning funding assurance and those amounts for other decommissioning. As such, licensees may deposit amounts collected for other decommissioning such as funding the post-shutdown irradiated spent fuel management into the licensee's NDT, provided the licensee has taken action to segregate the irradiated spent fuel management amounts from the radiological decommissioning amounts and maintain separate books for each amount.



IRS and FERC regulations apply equally to other decommissioning amounts and 10 CFR 50.54(bb) decommissioning amounts, as such, there is no requirement or benefit of separating the amounts for other decommissioning and the amounts for 10 CFR 50.54(bb) activities.

6.2 State of Michigan regulations overview

The Michigan Public Service Commission (MPSC or Commission) is the State of Michigan's state regulatory body charged with protecting the public by ensuring safe, reliable, accessible energy and telecommunications services at reasonable rates for Michigan residents. The Commission regulates the rates and services of investor-owned natural gas and electric utilities (such as DTE Gas and DTE Electric), authorizes energy infrastructure, implements state law governing competition and other aspects of the energy and telecommunications industries. The MPSC has regulatory jurisdiction pertaining to rates, recovery of certain costs, including the costs of generating facilities, conditions of service, accounting and operating-related matters.

MPSC orders govern (1) the establishment of the Fermi 2 decommissioning trusts, (2) the mechanism to fund the trusts, (3) the framework to report the adequacy of the trusts and (4) the use of the trust funds. In this section we discuss the MPSC's regulatory history of Fermi 2 nuclear decommissioning funding orders issues by the MPSC.

Settlement Agreement in MPSC Case No U-6150 (Reopened), dated August 26, 1986

The Settlement Agreement in MPSC Case No U-6150 (Reopened), dated August 26, 1986, is the reference MPSC governance for the Fermi 2 decommissioning trusts. The parties to this settlement agreement include The Detroit Edison Company, MPSC Staff, Michigan Attorney General and the Association of Business Advocating Tariff Equity. The Detroit Edison Company agreed to amongst other stipulations:

1. Establish two external Fermi 2 decommission trusts: a qualified trust and a nonqualified trust
2. Assume the retirement date is consistent with the term of the unit's operating license
3. Allocate the cost to decommission the unit to all its Michigan jurisdictional retail customers through a surcharge
4. Beginning in 1990 and at three-year intervals thereafter and no later than March 31, report to the Commission on the adequacy of the existing Fermi 2 annual provision. This report is to include the following elements:
 - a. fund adequacy in light of the amount needed for decommissioning
 - b. inflation rates
 - c. fund earning rates
 - d. remaining life of the nuclear generating units
 - e. requirements of other taxing or regulatory bodies
 - f. administrative and incidental expenses of the fund

The 1986 Settlement Agreement also ordered the Fermi 2 decommissioning trusts must meet the several objectives including:

1. The trustee is to optimize for tax earnings over the life of the trusts, giving consideration to liquidity, risk, diversification and other prudent investment objectives
2. The trustee is to not invest in any securities issued by the Detroit Edison Company, its successors or assignees
3. No part of the trust assets may be used for or diverted to any purpose other than to fund, in whole or in part the costs of nuclear plant decommissioning or to pay administrative or other incidental expenses including taxes



4. Once decommissioning has been completed, excess funds are to be refunded to the ratepayers as determined by the Commission
5. If the trust is insufficient to cover the costs of nuclear plant decommissioning, Detroit Edison may petition the Commission for relief
6. The trustee is to keep accurate and detailed books and records. The trustee's books and records are to be open to inspection and audit at reasonable times by the Commission

MPSC Order U-8567, dated January 13, 1987

The Commission Order U-8567, dated January 13, 1987, adopted Detroit Edison's proposed stipulations to the 1986 Settlement Agreement for the Nuclear Decommissioning Surcharge (NDS) rate schedule and Detroit Edison's model trust agreements. The parties to this stipulation included The Detroit Edison Company, MPSC Staff, Michigan Attorney General and the Association of Business Advocating Tariff Equity. The surcharge was to start within 30 days of the commercial operation of the Fermi 2 Power Plant.

MPSC Order U-10102, dated January 21, 1994

The Commission Order U-10102, dated January 21, 1994 amended the NDS established in MPSC Order U-8567. The Commission ordered Detroit Edison to assume an escalation rate of 6% and an after-tax earnings rate of 7% and to include the cost of removing Fermi 2's non-radioactive structures and materials in the Fermi 2 decommissioning cost estimate.

The Commission Order U-10102 also established the Low-Level Radioactive Waste (LLRW) fund. The Commission acknowledged that LLRW generated during operations is not a recognized decommissioning cost. The Commission approved Detroit Edison an incremental annual provision in the NDS of approximately \$4 million. The Commission also approved Detroit Edison to accumulate the LLRW funds in the nonqualified decommissioning trust. The LLRW funds were for the exclusive use to expedite the disposal of LLRW generated during operations should Detroit Edison have access to an LLRW disposal site.

MPSC Order U-14399, dated December 22, 2005

The Commission Order U-14399, dated December 22, 2005 the Commission ordered site security and radiation protection expenses removed from base rates and transferred to the NDS tariff applied to all customers within Detroit Edison's service territory. This 2005 order built upon the Commission's framework established Order U-11452, dated March 8, 1999, which updated the NDS tariff as it applied to Detroit Edison's retail access customers to include an incremental surcharge for Fermi 2 site security and radiation protection expenses.

MPSC Order U-16472, dated October 20, 2011

The Commission Order U-16472, dated October 20, 2011 adopted Detroit Edison's proposal to assume a Fermi 2 retirement date of 2045, consistent with a renewed operating license. The Commission also agreed with Detroit Edison's recommendation to maintain the existing assumed escalation rate of 6% and after-tax earnings rate of 7%. Resulting from this order the Nuclear Decommissioning Surcharge decreased approximately \$20 million from \$33 million to \$13.5 million.

MPSC Order U-17767, dated December 11, 2015

The Commission Order U-17767, dated December 11, 2015 formally updated the name of the annual funding provision from "Nuclear Decommissioning Surcharge" to "Nuclear Surcharge." The Commission agreed with DTE Electric's position that the



proposed name change more accurately reflected the nature of a surcharge that included components for not only Fermi 2's end-of-life decommissioning but also Fermi 2 site security and radiation protection expenses and the LLRW funding provisions. The Commission also adopted DTE Electric's proposal to reduce the Fermi 2 decommissioning portion of the surcharge from approximately \$13 million to \$2.9 million.

The Fermi 2 Nuclear Decommissioning Funding portion of the Nuclear Surcharge was unchanged and uncontested in DTE Electric's 2016 MPSC Case No U-18014. The Fermi 2 nuclear decommissioning portion of the Nuclear Surcharge was unchanged and uncontested in DTE Electric's 2017 MPSC Case No U-18255.

MPSC Order U-20162, dated May 2, 2019

The Commission Order U-20162, dated May 2, 2019, adopted DTE Electric's proposed Nuclear Surcharge where the Fermi 2 Nuclear Decommissioning Funding portion of the Nuclear Surcharge remained unchanged at \$2.9 million. The Commission also agreed with DTE Electric's positions: (1) that Nuclear Decommissioning Trust funds have specific purposes and (2) reallocation of funds for other purposes is not straightforward. The Commission also ordered DTE Electric to perform an update to the Fermi 2 nuclear decommissioning study.

MPSC Order U-20561, dated May 8, 2020

The Commission Order U-20561, dated May 8, 2020 adopted DTE Electric's proposed Nuclear Surcharge where the Fermi 2 Nuclear Decommissioning Funding portion of the Nuclear Surcharge remained unchanged at \$2.9 million. The Commission also agreed with DTE Electric's position that it was reasonable and prudent to recover expenditures associated with this decommissioning study projected in the Project Evaluation Review Committee (PERC) O&M mechanism.

6.3 Standard Contract

The Nuclear Waste Policy Act of 1982 (NWPAA) [NWPAA], as amended, establishes the U.S. national policy for the financing, siting, licensing, operating and decommissioning of a national geological repository for the disposal of civilian-generated nuclear waste as well as defense-related radioactive materials. Highlights of the NWPAA, as amended, include:

- The Department of Energy (DOE) has the responsibility to site, operate and decommission the geological repository.
- Yucca Mountain is to be the only site eligible for the DOE to consider for the geological repository.
- The DOE was to take title of owners' and generators' spent nuclear fuel (SNF) and high-level waste (HLW) beginning no later than January 31, 1998.
- The DOE has a responsibility to establish a fee applied to electricity generated or sold by commercial nuclear power plant owners to cover all costs associated with the disposal of the nuclear waste.
- Owners and generators of SNF and HLW have responsibility to make payments for the fees.
- Owners and operators of SNF and HLW must execute the Standard Contract with the DOE. The Standard Contract establishes the *quid pro quo* where in exchange for the owners and generators' payments of fees as established by the NWPAA, the DOE accepts title of SNF and HLW as expeditiously as possible following commencement of a geological repository and no later than January 31, 1998.
- The Department of Treasury was to establish a special Nuclear Waste Fund (NWF) for the collection of the fees.



In this section we discuss the aspects of the Standard Contract of particular interest to the Fermi 2 Power Plant decommissioning including the Yucca Mountain Project, the Standard Contract, the spent nuclear fuel fees, DTE Electric's ongoing litigation with the Department of Energy and the Nuclear Waste Fund.

Yucca Mountain Project

The Yucca Mountain Project (YMP) is the DOE's collective effort to site, license and construct a geological repository at Yucca Mountain, Nevada. A brief history of DOE's efforts at the Yucca Mountain site includes:

- 1983 – DOE identified Yucca Mountain as one of nine potentially acceptable sites for the first geological repository
- 1984 – DOE issued Draft Environmental Assessments (EAs) for each of the nine sites
- 1986 – DOE issued the final Yucca Mountain EA along with four other high potential sites.
- 1986 – DOE ranked Yucca Mountain as the number one overall potential site based on its Multi-attribute Utility Analysis
- 1987 – Congress amended the NHPA to redirect DOE to focus only on Yucca Mountain. Congress took this action out of concern about escalating costs and confidence in Yucca Mountain. [YMP06]
- 1998 – DOE issued the *Viability Assessment of a Repository at Yucca Mountain* [ML033580473]
- 2002 – Secretary of Energy recommended Yucca Mountain as the site of the first geological repository; however, as allowed by the NHPA, the State of Nevada objected to the recommendation, which required the U.S. Congress to override Nevada objection through passage of the Yucca Mountain Development Act. [YMP02]
- 2005 – Environmental Protection Agency (EPA) proposed to modify its regulations to limit radiation protection exposures over the period from 10,000 years to 1 million years in the future. [70 FR 49014]
- 2008 – DOE submitted the license application for a high-level waste geological waste repository at Yucca Mountain [ML081560407]. DOE also issued The Report to the President and the Congress by the Secretary of Energy on the Need for a Second Repository [YMP08].
- 2009 – The President and Secretary of Energy announced that a repository at Yucca Mountain was not a workable solution and that DOE would terminate the repository program. The Secretary announced a Blue Ribbon Commission on America's Nuclear Future to evaluate alternatives to Yucca Mountain. DOE submitted to the NRC its request to withdraw the Yucca Mountain license application from review [ML100621397]; the NRC denied the request [ML101800299].
- 2011 – Congress did not appropriate monies for Yucca Mountain Project or the NWF; consequently, DOE closed the Office of Civilian Radioactive Waste Management (and the Yucca Mountain Project).
- 2012 – Blue Ribbon Commission submits its final recommendations for consideration to the President and the Congress. The recommendations did not include restarting the Yucca Mountain Project.
- 2015 – NRC suspended review of the Yucca Mountain license application after congressionally appropriated funds were exhausted.
- 2019 – Congress did not include in its appropriations the President's requests for monies to restart the Yucca Mountain Project [OIG19].
- 2020 – The President did not request monies restart the Yucca Mountain Project in its Fiscal Year 2021 budget proposal – ending a three-year attempt to restart the Project.

The Government has spent over 30 years attempting to site and license a geological repository and yet the Yucca Mountain Project today is in a state of suspension and the Government has no working alternatives. Based on the history of the Yucca



Mountain Project, it would not be reasonable or prudent to assume the Government will have an operating geological repository – let alone two or more – for the indefinite future or any time sooner than necessary.

Standard Contract

The Standard Contract is the agreement between the Department of Energy (DOE) and nuclear power plant operators that establishes the terms for both DOE and nuclear power plant operators to meet the obligations set forth in the NWPA. Under the contract, the DOE is to take title to, transport and dispose of spent nuclear fuel (SNF) and high-level radioactive waste (HLW) delivered to the DOE by the nuclear power plant operators; DOE was to begin taking delivery of SNF and HLW no later than January 31, 1998. In exchange for DOE's obligations, nuclear power plant operators are to pay for the DOE's radioactive waste disposal activities through a fee (also known as a spent nuclear fuel fee) authorized under the NWPA. The NWPA mandates that the Nuclear Regulatory Commission (NRC) cannot issue or renew an operating license unless the nuclear power plant operator has entered into the Standard Contract and the Secretary of Energy has affirmed the nuclear power plant operator is in good faith standing. The Standard Contract is non-negotiable.

The Detroit Edison Company executed its Standard Contract with the DOE in June 1983 [USCFC03].

The DOE is in partial breach of the Standard Contract as the DOE has yet to accept delivery of any SNF or HLW from the Fermi 2 Power Plant. Fermi 2 Power Plant's entire inventory of SNF and HLW remains onsite, either in wet storage (Spent Fuel Pool) or in dry storage (ISFSI).

Spent nuclear fuel (SNF) fees

Article VIII of the Standard Contract (Fees and Terms of Payment) establishes a fee of 1.0 mill per kilowatt-hour (1M/kWh) electricity generated or sold. Article VIII also gives DOE an obligation to review the adequacy of the fees and adjust the 1M/kWh fee, if necessary and subject to a 90-day review of the U.S. Congress, to assure full cost recovery by the U.S. government.

According to the terms of the Standard Contract between DTE Electric and the DOE and the NWPA, the fee of 1.0 mill per kilowatt-hour (1M/kWh) of electricity generated and sold from Fermi 2 was to fund the costs associated with the disposal of spent nuclear fuel. DTE Electric was to pay the fee directly to the U.S. Government (Government). Through June 30, 2018, DTE Electric had paid approximately \$172.1 million to the Government through the fee (reference DTE Electric's June 29, 2019 filing in MPSC Case No. U-15244).

The DOE suspended collection of the spent nuclear fuel (SNF) fee in 2014 following the U.S. Court of Appeals (Court) November 2013 decision in *National Association of Regulatory Utility Commissioners v United States Department of Energy*, 736 F3d 517 [USCA13]. The Court found the DOE has no viable strategy for geological disposal of nuclear waste; therefore, DOE could not charge nuclear plant operators for the cost of that disposal. The Court did not provide nuclear plant operators relief from the obligation to pay for the cost of their waste disposal. The Court also allows the DOE to resume the fees once again after the DOE is able to conduct a sufficient assessment because either: (1) the DOE revives the Yucca Mountain Project or (2) Congress enacts an alternative plan.

DTE Electric reported to the Michigan Public Service Commission (MPSC) in its April 23, 2014 filing in MPSC Case No U-17593 that it would cease collections for the fee in its 2015 PSCR case (U-17680) and reconcile the 2014 fees paid in its 2014 PSCR



reconciliation (U-17319R) case. DTE Electric has suspended the collection of the fee since 2015. As discussed above, the Standard Contract binds to DTE Electric to resume payments of the fee once the DOE conducts its sufficient fee assessment.

Department of Energy (DOE) litigation

DTE Electric is a party in the litigation against the DOE for both past and future costs associated with DOE's failure to accept spent nuclear fuel under the schedule set forth in the Federal Nuclear Waste Policy Act of 1982 and partial breach of the Standard Contract.

In July 2012, DTE Electric entered into a settlement agreement with the federal government for costs associated with DOE's delay in acceptance of spent nuclear fuel from the Fermi 2 Power Plant for permanent storage. The settlement agreement, including extensions, provides for a claims process and payment of delay-related costs experienced by DTE Electric through 2019. DTE Electric's claims through 2019 are being settled and paid on a timely basis. The settlement proceeds reduce the cost of the ISFSI assets and provide reimbursement for related operating expenses. DTE Electric has not made a determination in regards to future legal action beyond the current settlement agreement.

Although the federal government continues to maintain it has a legal obligation to accept spent nuclear fuel from Fermi 2 Power Plant for permanent storage, there is no certainty as to the future legal positions or obligations of the federal government. Issues relating to long-term waste disposal policy and to the disposition of funds contributed by DTE Electric ratepayers to the federal waste fund await future governmental action.

The Company annually files an update on its litigation with the DOE in the MPSC Case No. U-15244 docket.

Nuclear Waste Fund

Nuclear Waste Fund (NWF), a separate fund in Treasury of the United States, for the purpose of radioactive waste disposal activities including licensing, construction and operation of a repository and transportation, readying or packaging of spent nuclear fuel (SNF) or high-level waste (HLW) for disposal in the repository. The Government deposits the fees collected through the Standard Contract into the NWF.

The NWF balance was approximately \$40.9 billion as of September 30, 2019 [OIG19]; however, U.S. Congressional budgeting practices prevent the use of the NWF for its intended purpose [BRC12].



SECTION SEVEN

7 Decommissioning cost estimates

In this section we discuss nuclear decommissioning cost estimates which are an element of nuclear decommissioning funding adequacy reviews. We first present an overview of the decommissioning costs estimates. We then discuss the basis and conclusions of DTE Electric's existing Fermi 2 decommissioning cost estimate, followed by a discussion on the basis and conclusions of DTE Electric's 2020 Fermi 2 site-specific decommissioning cost estimate. We then review the Fermi 2 Nuclear Regulatory Commission minimum decommissioning fund requirements for the Fermi 2 Power Plant. We close out the discussion with DTE Electric's conclusion and adopted Fermi 2 nuclear decommissioning cost estimates for use in the Fermi 2 nuclear decommissioning funding adequacy review performed later in this study.

7.1 Decommissioning cost estimate overview

In this section we present an overview of nuclear decommissioning cost estimates and best practices the industry has developed to reasonably estimate the costs of decommissioning nuclear facilities. This review includes discussion on the purpose of the nuclear decommissioning cost estimate, approaches or methods the industry uses to develop the cost estimates and the elements that constitute a nuclear decommissioning cost estimate. Our intention with this section to provide basic knowledge and concepts to support the reader's understanding of DTE Electric's Fermi 2 nuclear decommissioning cost estimates discussed later in this section; additional discussion is available through notable resources such as the Nuclear Regulatory Commission (NRC) Regulatory Guide (RG) 1.202, Revision 0, Standard Format and Content of Decommissioning Cost Estimates for Nuclear Power Reactors, NRC NUREG-1700, Revision 2, Standard Review Plan for Evaluating Nuclear Power Reactor License Termination Plans and the 2015 The Practice of Cost Estimation for Decommissioning of Nuclear Facilities published by the Organization for Economic Co-operation and Development (OECD) Nuclear Energy Agency (NEA).

Purpose

The purpose of the decommissioning cost is estimate:

1. Establish overall decommissioning costs. The decommissioning plan assumptions and strategies presented in this study and cost estimate are to provide reasonable assurance of adequate annual funding provisions and eventual available funds for the decommissioning of the DTE Electric Fermi 2 Power Plant by the year 2045.
2. Inform the long-term financing process
3. Later, when the Fermi 2 Power Plant nears the end of its operational life, the cost estimate potentially informs part of the basis for the detailed decommissioning planning.

This cost estimate does not constitute a commitment by DTE Electric to perform decommissioning work exactly as presented in this document. Consistent with Nuclear Regulatory Commission (NRC) Regulatory Guide 1.159, Revision 2, and indeed the nuclear industry, DTE Electric expects to refine and improve the decommissioning plan and cost estimates as the Fermi 2 Power Plant approaches end of life in 2045.

Approaches to cost estimation

The nuclear industry generally recognizes five approaches to cost estimating [NEA15]:



Bottom-up technique is where estimators use a work statement and specifications or a set of drawings to extract “take off” material quantities required to be dismantled and removed (inventory) and unit cost factors (costs per unit of productivity – per unit volume or per unit weight). The Fermi 2 site-specific decommissioning estimate (SDCE) commissioned for this study is an example of a bottom-up estimate.

Specific analogy is where estimators use the known cost of an item or project from prior estimates as the basis for the cost of a similar item or project in the new estimate. Estimators make adjustments to these known costs to account for differences in relative complexities of performance, design and operational characteristics. The existing Fermi 2 decommissioning cost estimate uses the 2002 Fermi 2 decommissioning cost estimate report as the basis document; this 2002 report is an example of a specific analogy estimate.

Parametric estimating is where estimators use historical databases on similar systems or subsystems and statistical analysis to find correlations between cost drivers and other system parameters to develop unit cost factors. DTE Electric uses parametric estimating for routine maintenance activities such as breaker replacements. Parametric estimates also form the foundation of unit cost factors used in other estimating techniques.

Cost review and update is where estimators develop estimates by reviewing previous estimates of the same or similar projects for internal logic, completeness of scope, assumptions and estimating methodology. This existing Fermi 2 decommissioning cost estimate (DCE) applies an annual cost review and update to the 2002 Fermi 2 decommissioning cost estimate report.

Expert opinion is where estimators do not have other techniques or data available. Estimators iteratively consult with experts until a converged cost estimate emerges.

As discussed above, DTE Electric uses the above techniques or combination of techniques regularly through the normal course of business and specifically for the Fermi 2 decommissioning cost estimates presented in this study.

Elements of cost estimates

Cost estimates have four basic elements: (1) basis of estimate, (2) structure of estimate, (3) work breakdown structure and schedule and (4) uncertainty analysis.

Basis of estimate

The basis of estimate is the framework or scope of assumptions estimators rely on to develop the decommissioning cost estimates. Information that forms the basis of the estimate include:

- Assumptions and exclusions;
- boundary conditions and limitations – legal and technical. For example, the regulatory framework we discussed in Section 6;
- decommissioning strategy such as the DECOM strategy;
- end point state such as “greenfield” vs “brownfield;”
- facility description and site characterization such as our discussion in Sections 4 and 5;
- spent fuel management such as our discussion in Section 6.



Structure of the estimate

Grouping elements of costs into categories assists in better understanding how these elements impact overall cost. Decommissioning cost estimates often group into activity-dependent, period-dependent, collateral and contingency costs.

Activity-dependent costs are expenditures associated with performing decommissioning activities such as demolition and decontamination.

Period-dependent costs are those expenditures associated with project duration and includes expenditures for site security, program management, energy usage, engineering and licensing.

Collateral costs are those expenditures associated with special items such as mobilization, equipment, insurance and property taxes.

Contingency is a specific provision for unforecastable elements of cost within the defined project scope. The elements in decommissioning cost estimates typically assume ideal conditions such as work is able to progress without delay or interruption. NRC regulations at 10 CFR 72.30(b)(2)(ii) require the use of an adequate contingency factor to demonstrate financial assurance; NRC acceptance criteria in NUREG-1700, Revision 2, Standard Review Plan for Evaluating Nuclear Power Reactor License Termination Plans, includes a review for contingency of approximately 25%.

Work breakdown structure and schedule

The work breakdown structure, chart of accounts and schedule interconnect to describe the scope of the nuclear decommissioning project.

The work breakdown structure (WBS) is a hierarchical decomposition of the scope of the project. Each descending level of the WBS represents an increasingly detailed definition of the project work. A WBS dictionary describes the scope of the work and the major deliverable of the WBS element. The industry-standard WBS structure adopted for this study includes the following five high-level WBS elements:

1. Planning, shutdown, transition and pool-to-pad campaign
2. Radiological decommissioning
3. ISFSI decommissioning
4. Spent fuel management
5. Site restoration,

where each high-level WBS then has sub-WBS elements; for example:

4. Spent fuel management
 - 4.1 Construct expansion ISFSI storage pad
 - 4.2 ISFSI operations and maintenance
 - 4.2.1 Maintenance
 - 4.2.2 Facilities, grounds and snow removal
 - 4.3 Spent nuclear fuel and high-level waste disposition



Similar to a WBS structure, the decommissioning cost estimate describes the chart of accounts used in the estimating process. The industry-standard chart of accounts adopted in this study aligns with the NDT funds and includes the following accounts:

1. Radiological decommissioning
2. ISFSI decommissioning
3. Spent fuel management
4. Site restoration

The scheduled duration of the decommissioning project affects costs most importantly through period-dependent resources or costs. Key schedule milestones align to deliverables from the WBS structure include:

- Permanent removal of the fuel from the reactor core. This milestone marks the permanent cessation of operations and opens the decommissioning project windows.
- All fuel placed into dry storage at the site's ISFSI. This milestone allows the licensee to request license amendments to reduce the site's security footprint and emergency response plan requirements and opens the spent fuel management WBS.
- License termination. This milestone marks the closure of the radiological decommissioning WBS and allows the licensee to reduce staffing to support only ISFSI operations. At this milestone (because DTE Electric's regulatory obligation to radiologically decommission the Fermi 2 Power Plant would be complete), DTE could appropriate excess nuclear decommissioning trust (NDT) funds from radiological decommissioning to spent fuel management program.
- DOE acceptance of spent nuclear fuel and high-level waste. This milestone allows the licensee to demobilize almost all staffing and support and begin decommissioning of the ISFSI and also marks the closure of the spent fuel management program WBS and opening of the ISFSI decommissioning WBS.
- ISFSI termination. This milestone completes the licensee's work to decommission the site and closeout remaining costs. At this milestone (because DTE Electric's regulatory obligation to restore the Fermi 2 Power Plant site and decommission ISFSI would be complete), DTE Electric would return excess NDT funds to the appropriate customers.

Uncertainty analysis

The purpose of this Fermi 2 nuclear decommissioning study is to provide reasonable assurance that the Fermi 2 NDT will have sufficient funds to radiologically decommission and restore to greenfield status the Fermi 2 Power Plant site. Risk analysis addresses those issues beyond the project scope such as change in regulations regarding emergency response requirements, security requirements, worker exposure limits, site release limits, waste transportation or change in waste disposal acceptance criteria that would impact DTE Electric's reasonable assurance. DTE Electric used a qualitative risk analysis tool known as a risk register to outline these potential external risks (negative outcomes) and opportunities (positive outcomes) relative to the Fermi 2 nuclear decommissioning scenario presented in this study.

7.2 Existing Fermi 2 decommissioning cost estimate

In this section we discuss the existing Fermi 2 decommissioning cost estimate, which DTE Electric last reviewed with the Michigan Public Service Commission (MPSC) in the 2018 Report on the Adequacy of the Existing Annual Provision of Nuclear Plant Decommissioning and throughout the 2018 DTE Electric main rate case proceeding MPSC Case No. U-20162. This decommissioning cost estimate has proven a reasonable and prudent approach for almost twenty years.



The basis for the existing Fermi 2 decommissioning cost estimate (DCE) is NRC NUREG/CR-6174 *Revised Analyses of Decommissioning for the Reference Boiling Water Reactor Power Station* [ML14008A186] and NUREG/CR-6270 *Estimating Boiling Water Reactor Decommissioning Costs*, modified for factors specific to Fermi 2 Power Plant. DTE Electric performed the original decommissioning study in approximately 2002 [DTE02], which established many parameters of the Fermi 2 nuclear decommissioning project including the estimated Fermi 2 inventory, decommissioning strategy, staffing levels and schedule of decommissioning; this 2002 study was comprehensive including estimated radiological decommissioning, post-shutdown spent fuel stewardship and non-radiological decommissioning (greenfielding), well documented, and consistent with NRC-accepted cost estimate methods for the Post Shutdown Decommissioning Activity Report. Also as required by regulations, DTE Electric has performed annual reviews of this 2002 study since adoption (including adding parameters and cost estimates for the Fermi 2 independent spent fuel installation (ISFSI) decommissioning) and updated the DCE to current dollars.

The use of the reference BWR and DECON scenario was a reasonable and prudent methodology to estimate DTE Electric's Fermi 2 Power Plant decommissioning costs without expending undo effort and resources. DTE Electric's intent for the Fermi 2 DCE was to provide reasonable assurance of sufficient funding to decommission the Fermi 2 Power Plant, an activity not expected to occur until 2045 and not to develop a definitive decommissioning plan and estimate. DTE Electric, as is customary in the nuclear industry, always intended to update and refine the Fermi 2 decommissioning cost estimate as the Fermi 2 Power Plant approached the end of its operating life.

DTE Electric reported to the MPSC in the January 2018 Report on the Adequacy of the Existing Annual Provision of Nuclear Plant Decommissioning that the annual review of Fermi 2 Power Plant DCE yielded an updated decommissioning cost estimate of approximately \$2.04 billion as of December 31, 2017. DTE Electric's annual review of the Fermi 2 Power Plant DCE yielded an updated decommissioning cost estimate of approximately \$2.2 billion as of December 31, 2019. These estimates are inclusive of the total costs estimated for the Fermi 2 Power Plant radiological decommissioning (also known as license termination), ISFSI decommissioning, post-shutdown spent fuel management (AKA spent fuel management program) and greenfielding (AKA site restoration).

Inventory

The Fermi 2 Power Plant DCE uses the inventory developed for the reference reactor as published in NRC NUREG/CR-6174 *Revised Analyses of Decommissioning for the Reference Boiling Water Reactor Power Station* as its base and then makes adjustments for known differences between the Fermi 2 Power Plant and the reference plant.

Reference plant

The NUREG/CR-6174 *Revised Analyses of Decommissioning for the Reference Boiling Water Reactor Power Station* reference plant was the Washington Public Power Supply System's 1155 MWe BWR with a Mark II containment system: Washington Nuclear Project Number 2 (WNP-2), located on the Hanford. The NRC had previously performed a decommissioning inventory of the WNP-2 plant in support of NRC NUREG/CR-0672 *Technology, Safety and Costs of Decommissioning a Reference Boiling Water Reactor Power Station* [PNNL80].

Fermi 2-specific considerations

The Fermi 2 DCE used a system-wide comparison of the Fermi 2 Power Plant equipment against the equipment at the reference BWR. The comparison concluded that Fermi 2 decommissioning would be approximately 1.3 times more complex than the reference BWR; key aspects driving this conclusion were:



- The Fermi 2 Power Plant is a GE designed BWR/4 with a Mark I containment and the reference plant is a GE designed BWR/5 with a Mark II containment. The Mark I (and associated torus containment structure) has more equipment to support its functions than the Mark II (wet well containment structure)
- Of comparative systems: Fermi 2 Power Plant had 950 valves. The reference plant had 750.
- The Fermi 2 Power Plant had an estimated 11,450 potentially contaminated valves. The reference BWR had an estimated 8,750 potentially contaminated valves.
- Fermi 2 Power Plant systems Torus Water Management, Emergency Equipment Cooling, Standby Feedwater, Post Accident Sampling, Reactor Protection System and radiation/activity monitors did not have functional equivalents at the reference BWR.
- Fermi 2 Power Plant has almost double the emergency cooling system pumps than the reference BWR.

Decommissioning schedule

The Fermi 2 DCE uses the high-level decommissioning schedule developed for the reference reactor DECON scenario as published in NRC NUREG/CR-6174 *Revised Analyses of Decommissioning for the Reference Boiling Water Reactor Power Station* (1996) and makes adjustments for known differences between the Fermi 2 Power Plant and the DECON scenario.

Reference DECON scenario

The reference DECON scenario divided the decommissioning schedule into four periods:

1. Period 1 – Pre-decommissioning Engineering and Planning, 2.5 years prior through shutdown
2. Period 2 – Reactor Deactivation, shutdown through 1.2 years post-shutdown
3. Period 3 – Safe Storage and Spent Fuel Management, 1.2 years – 3.4 years post-shutdown
4. Period 4 – Dismantlement, 3.4 years – 5 years post-shutdown

The reference DECON scenario did not contemplate spent fuel management programs of Independent Spent Fuel Installation (ISFSI) decommissioning beyond license termination. The reference DECON scenario did not contemplate ISFSI decommissioning costs.

Fermi 2-specific considerations

The original 2002 Fermi 2 DCE relied on an informal study performed in 1997 to determine required adjustments to the reference DECON scenario:

1. Period 1 – decommissioning Engineering and Planning, 5 years prior through shutdown
2. Period 2/Transition year – Reactor Deactivation, shutdown through 1-year post-shutdown
3. Period 3 – Safe Storage and Spent Fuel Management, 1 year – 5 years post-shutdown
4. Period 4 – Dismantlement, 5 years – 9 years post-shutdown
5. Period 5 – ISFSI monitoring, 9 years – 20 years post-shutdown

The Fermi 2-specific schedule considerations have been informally updated since 2002 to account for industry and Fermi 2-specific changes including:

- Fermi 2's nuclear fuel will require approximately seven years, not five years, in the spent fuel pool to sufficiently cool down prior to placing the spent fuel into dry cask storage. This change was an outcome of updating forecasted decay



heat rates of the Fermi 2 nuclear fuel and accounting for projected burnup exposures with 24-month operating cycles.

- Fermi 2 decommissioning would rely on a “rip and ship” model, where dismantled contaminated equipment is shipped offsite for waste minimization and disposal, rather than the extensive decontamination efforts envisioned in the reference DECON scenario to preserve schedule. This change was the outcome of DTE Electric’s experience with the Fermi 1 Power Plant decommissioning and industry trends.
- Fermi 2’s ISFSI will require capacity of 128, not 96, dry-storage cask locations. This change was an outcome of the Fermi 2 license renewal, which extended the operating life of the Fermi 2 Power Plant from 40 years to 60 years and the associated increase for the 20 additional years of spent nuclear fuel.
- Fermi 2’s spent nuclear fuel will require at least sixty years, not twenty, of post-shutdown stewardship. This requirement change was an outcome of the 2010 update to the Waste Confidence Decision that extended the acceptable duration of operating an on-site ISFSI from thirty years to sixty years post-shutdown. Further reinforcing this change is the decades-long inaction by the US Government to pursue a permanent geological repository.

7.3 2020 Fermi 2 site-specific decommissioning cost estimate

In this section we discuss the 2020 Fermi 2 site-specific decommissioning cost estimate (SDCE) [DTE20], which DTE Electric commissioned EnergySolutions, LLC to prepare in partial response to the Michigan Public Service Commission (MPSC or Commission) U-20162 finding that an updated Fermi 2 nuclear decommissioning study would be beneficial to the Commission.

Whereas the 2002 Fermi 2 decommissioning study used a specific analogy technique with the reference boiling water reactor (BWR) described in the 1996 NRC study as its basis, the 2020 Fermi 2 SDCE uses a bottom-up technique with the Fermi 2 Power Plant specific inventory as its basis. In concert with this bottom-up technique, DTE Electric re-baselined the constituent elements of the Fermi 2 nuclear decommissioning cost estimates for license termination, independent spent fuel installation (ISFSI) decommissioning, spent fuel management program and site restoration. Similar to the 2002 Fermi 2 decommissioning cost study, this 2020 Fermi 2 SDCE is also consistent with NRC-accepted cost estimate methods for the Post Shutdown Decommissioning Activity Report.

The 2020 Fermi 2 SDCE developed using a bottom-up technique from Fermi 2 specific inventories is a reasonable and prudent effort given the Commission’s U-20162 discussion of DTE Electric’s use of the 2002 Fermi 2 decommissioning study basis. DTE Electric’s intent for the 2020 Fermi 2 SDCE is, again, to provide reasonable assurance of sufficient funding to decommission the Fermi 2 Power Plant, an activity not expected to occur until 2045. DTE Electric, as is customary in the nuclear industry, intends to update and refine the Fermi 2 decommissioning cost estimate as the Fermi 2 Power Plant approaches the end of its operating life.

The 2020 Fermi 2 Power Plant SDCE yields a nuclear decommissioning cost estimate for the Fermi 2 Power Plant of approximately \$2.2 billion as of December 31, 2019. This estimate establishes the total costs estimated to the Fermi 2 Power Plant

- license termination as approximately \$1.56 billion,
- ISFSI decommissioning as approximately \$0.01 billion,
- spent fuel management program as approximately \$0.51 billion and



- site restoration as approximately \$0.10 billion.

Inventory

The Fermi 2 SDCE material inventory uses material take offs, a process which involved reviewing approximately one thousand Fermi 2 Power Plant drawings and technical documents to determine the equipment, dimensions and other characteristics for each structure at the Fermi 2 site to calculate building-specific volumes and weights of each waste stream.

Table 7.1 provides the estimated material inventory summary for the Fermi 2 Power Plant with the total amount of waste estimated at approximately 1.0 million tons. The estimated Class A low-level radioactive waste (LLRW) disposal requirements are approximately 446,488 tons, the estimated Class B/C LLRW disposal requirements are approximately 505 tons and estimated the greater-than-Class-C (GTCC) LLRW disposal requirements are approximately 107 tons. The estimated non-radioactive (Class F) waste disposal requirements are approximately 592,000 tons.

Table 7.1 Fermi 2 SDCE low-level radioactive waste (LLRW) material inventory (tons)

Waste class	Waste stream	In-situ waste (tons)	Transportation and disposal protocol
Class A	Concrete	365,785	Rail transport to licensed LLRW landfill
	Metal & debris ¹	60,603	Rail transport to licensed LLRW landfill
	Soil & non-inventory material ²	20,100	Rail transport to licensed LLRW landfill
Class A total (tons)		446,488	
Class B/C	Metal	505	Truck transport to licensed LLRW landfill
Class B/C total (tons)		505	
Greater-than-C	Metal	107	Stored at ISFSI & transferred to DOE
GTCC total (tons)		107	
Clean (Class F)	Concrete	400,600	Process and reuse onsite
	Metal & debris	191,400	Truck transport to local landfills
Class F total		592,000	
Total estimated waste (tons)		1,039,100	

Note 1: Includes dry active waste (DAW) and Class A segments of the reactor vessel

Note 2: Includes estimated soil and other waste materials (such as tools) derived from other than plant inventory

Table 7.2 provides a comparison of the estimated total LLRW inventory volumes for the Fermi 2 Power Plant at approximately 344,592 cubic yards (cyds) relative to the NRC NUREG/CR-6174 Reference BWR at approximately 19,790 cyds. The Fermi 2 SDCE Class A LLRW volume of approximately 333,369 cyds is consistent with the "rip and ship" model used in the SDCE. The increased Class B/C and GGTC LLRW waste volumes are as expected given the specific size, model and construction characteristics of the Fermi 2 Power Plant relative to the Reference BWR.

Table 7.2 Fermi 2 SDCE LLRW material inventory (volume) relative to the NRC NUREG/CR-6174 Reference BWR

Waste class	Reference BWR (NUREG/CR-6174)	Fermi 2 Power Plant (Fermi 2 SDCE)
Class A	9,776	333,369
Class B/C	9,770	10,202
GTCC	244	1,021
Total waste volume (cyds)	19,790	344,592



The NRC describes the classification of low-level radioactive waste at 10 CFR 61.55 where Class A is the lowest classification of LLRW and GTCC is the highest classification. The GTCC LLRW requires geological disposal whereas Class A, B and C LLRW is suitable for near surface disposal, provided the waste form meets regulations.

Decommissioning schedule

The Fermi 2 SDCE uses the DECON scenario; a continuation of the DTE Electric decommission strategy used in the existing Fermi 2 DCE. As we have discussed in Section 4, industry experience with the SAFSTOR options have demonstrated that the DECON scenario is a more reasonable and prudent strategy.

NRC decommissioning levels

The Fermi 2 SDCE uses decommissioning “levels” to summarize the status of Fermi 2 Power Plant nuclear fuel. Level Zero commences once the Fermi 2 reactor is permanently shut down and extends until the plant enters Level 1. Because the status of the site’s nuclear fuel informs many of the NRC’s emergency response and site security regulations, the NRC proposed [ML18012A024] the following decommissioning levels to provide a more efficient NRC regulatory framework during the decommissioning process by using a graded approach commensurate with the reductions in radiological risk associated with each level:

1. Level 1 commences after the NRC’s docketing of Fermi 2 Power Plant’s certification of permanent cessation of operations and permanent removal of the fuel from the reactor vessel. In this level, Fermi 2 Power Plant spent nuclear fuel (SNF) in the spent fuel pool (SFP) is still susceptible to a zirconium fuel cladding fire within 10 hours under adiabatic heat up conditions. At this level, the Fermi 2 Power Plant would approximately maintain its existing operating emergency response plan and site security plan.
2. Level 2 commences once the fuel in the SFP has decayed and cooled sufficiently that it cannot heat up to clad ignition temperature within 10 hours under adiabatic conditions. DTE Electric forecasts entering Level 2 approximately 18 months following shutdown of the plant. At this level, DTE Electric can and would request to change its emergency response plan to reflect the significantly lower risk to the public.
3. Level 3 commences once DTE Electric has removed the entire SNF inventory from the SFP and into dry cask storage. Consistent with Fermi 2 Power Plant’s current licensing basis, DTE Electric forecasts entering Level 3 approximately 5.5 years after entering Level 2. At this level, DTE Electric can and would request to change its emergency response plan and site security plan to reflect the significantly lower risk to the public.
4. Level 4 commences once the Department of Energy (DOE) has accepted title of the entire Fermi 2 SNF inventory and has removed the inventory from the Fermi 2 Power Plant site. Consistent with the discussion in Section 7, DTE Electric forecasts entering Level 4 approximately 53 years after entering Level 3. At this level, DTE Electric can and would decommission the Independent Spent Fuel Installation (ISFSI) and terminate the emergency response plan and site security plan.

Fermi 2 decommissioning periods

The Fermi 2 SDCE uses five overlapping project duration windows to divide the Fermi 2 decommissioning project:

1. Period 1 (2045 – 2051): Pre-planning, shut down & transition, and pool-to-pad campaign During this duration window the Fermi 2 Power Plant transitions from Level Zero through Level 1 to Level 2. Costs allocate to Period 2 and Period 4 as appropriate.



2. Period 2 (2052 – 2059): License termination During this duration window the Fermi 2 Power Plant is at Level 3. Costs approximately correspond to 10 CFR 50.75(c) requirements.
3. Period 3 (2053 – 2060): Site restoration During this duration window the Fermi 2 Power Plant is at Level 3.
4. Period 4 (2045 – 2105): Dry fuel/CTCC storage & transfer to DOE During this duration window the Fermi 2 Power Plant transitions from Level Zero through Levels 1 and 2 to Level 3. Costs approximately correspond to 10 CFR 50.54(bb) requirements.
5. Period 5 (2106 – 2108): ISFSI decommissioning During this duration window the Fermi 2 Power Plant is at Level 4. Costs approximately correspond to 10 CFR 72.30(a) requirements.

7.4 Fermi 2 NRC minimum decommissioning fund requirement

In this section, we review the most recent DTE Electric Fermi 2 decommissioning cost estimates performed in accordance with Nuclear Regulatory Commission (NRC) regulations. NRC regulations require formal submission of the Fermi 2 radiological decommissioning funding assurance amounts every two years and formal submission of the Fermi 2 Independent Spent Fuel Storage Installation (ISFSI) decommissioning amounts every three years.

Radiological decommissioning 10 CFR 50.75(b) and (c)

The 2019 NRC minimum decommissioning cost estimate, pursuant to NRC regulations at 10 CFR 50.75(b) and (c), was approximately \$1.12 billion [ML19087A224]. The labor, energy and burial factors used in DTE Electric's calculation of the NRC minimum requirement for the Fermi 2 decommissioning financial assurance are: Labor (L) = 2.75, Energy (E) = 2.39 and Waste Burial (B) = 28.314. The source of these factors is in accordance with the NUREG-1307, Revision 17 guidance. The Waste Burial (B) factor is consistent with Michigan's unaffiliated status with multi-state low-level radioactive waste (LLRW) disposal compacts.

The NRC minimum decommissioning cost estimate does not represent actual cost of decommissioning for specific reactors but is a reference level established to ensure the licensee (DTE Electric) demonstrates adequate financial responsibility for the bulk of funds necessary for a safe decommissioning early in the facility life (reference [53 FR 24018] page 24030). The NRC Staff had concluded that the NRC minimum amounts required to demonstrate decommissioning assurance were adequate for the NRC's regulatory purpose [ML13127A234].

ISFSI decommissioning 10 CFR 72.30

The 2020 NRC Independent Spent Fuel Storage Installation (ISFSI) decommissioning cost estimate, pursuant to NRC regulations at 10 CFR 72.30(b), was approximately \$9.6 million [DTE20A]. The estimate uses 2014 ISFSI design characteristics and 2014 forecasted dry cask usage to develop the projected ISFSI inventory and associated disposal costs.



7.5 DTE Electric conclusion

DTE Electric concludes a reasonable and prudent cost estimate for the various aspects of the Fermi 2 decommissioning are:

- \$1.560 billion for radiological decommissioning (license termination)
- \$0.013 billion for Independent Spent Fuel Installation (ISFSI) decommissioning
- \$0.511 billion for post-shutdown spent fuel management (spent fuel storage and transfer)
- \$0.096 billion for non-radiological decommissioning (site restoration)
- \$2.179 billion for total Fermi 2 decommissioning costs

DTE Electric supports this conclusion with the Fermi 2 site-specific decommissioning cost estimate completed in 2020.



SECTION EIGHT

8 Nuclear Decommissioning Trust (NDT)

DTE Electric, as a nuclear plant licensee, must provide financial assurance for radiological decommissioning through meeting the requirements of Nuclear Regulatory Commission (NRC) regulations at 10 CFR 50.75 while the Fermi 2 Power Plant is operating. NRC regulations at 10 CFR 50.82 govern the decommissioning process including the use of decommissioning funds. DTE Electric meets the Fermi 2 NRC financial assurance requirements with a nuclear decommissioning trust (NDT) which holds contributions collected from DTE Electric customers through the Nuclear Surcharge provision. The Fermi 2 NDT consists of a Nuclear Decommissioning Fund and an LLRW Fund.

In this section we discuss the structure of the DTE Electric Nuclear Decommissioning Master Trust [DTE13], its Funds and the allowable uses for such. We then discuss the DTE Electric Nuclear Surcharge, which is an exclusive funding mechanism provided by the Michigan Public Service Commission (MPSC) to collect amounts from DTE Electric customers to hold in the trusts. Our intention with this discussion to provide relevant background information to support the reader's understanding of the framework DTE Electric has developed, with oversight from the Michigan Public Service Commission (MPSC), to protect and ensure that amounts collected for the nuclear decommissioning of the Fermi 2 Power Plant are reasonably assured to be available for their intended purpose.

8.1 DTE Electric Master Trust Agreement overview

DTE Electric has ownership interests in the nuclear facilities at Fermi 1 Power Plant and Fermi 2 Power Plant, located in Monroe County, Michigan. The Michigan Public Service Commission (MPSC) and Federal Energy Regulatory Commission (FERC) each have ordered DTE Electric to include in rates amounts to collect from customers which are to be placed and held in an external trust fund in order to provide adequate funds to decommission the Fermi 2 Power Plant. DTE Electric has an executed Nuclear Decommissioning Master Trust Agreement between DTE Electric (Company) and State Street Bank and Trust Company (Trustee).

The Nuclear Decommissioning Master Trust Agreement consolidated the Company's Nuclear Decommissioning Trust Agreement for the Fermi 1 Nuclear Power Plant, the Non-Qualified Nuclear Decommissioning Trust Agreement for the Fermi 2 Nuclear Power Plant and the Qualified Nuclear Decommissioning Trust Agreement for the Fermi 2 Nuclear Power Plant to provide for the combined investment of separate trusts. The Master Trust consists of three Funds identified as: Nonqualified Fermi Unit 1 Fund, Nonqualified Fermi Unit 2 Fund and Qualified Fermi Unit 2 Fund. The Trustee maintains books and records necessary to reflect each Fund separately from each other Fund and maintains subaccounts (funds or plans) within each fund as the Company directs.

The Funds are for the exclusive purpose of providing funds for the decommissioning of the Company's nuclear units identified in their respective titles. The Master Trust Agreement defines decommissioning as:

to remove Nuclear Units safely from service and to reduce residual radioactivity to a level that permits release of property for unrestricted use and termination of license, or as otherwise directed by the Nuclear Regulatory Commission ("NRC"). Decommission also includes other decommissioning activities including but not limited to activities, such as fuel stewardship and site restoration. With respect to funds or accounts specifically identified for



disposal of radioactive materials, decommission also means the disposal of such radioactive materials, whether or not Fermi 2 has ceased to operate. Decommission also includes other activities and costs, if any, which may be included in the definition of decommission pursuant to Internal Revenue Service regulations and rulings implementing Section 468A of the Code.

None of the assets of the Funds are subject to attachment, garnishment, execution or level in any matter for the benefit of the Company's creditors.

The Nonqualified Fermi Unit 1 Funds serves a standby trust in accordance with Appendix A of 10 CFR 30 because the Company is using a Parent Company Guarantee to provide a portion of financial assurance for Fermi 1 decommissioning. The Company does not classify Fermi 1 as utility plant, so there is no regulatory asset. The Nonqualified Fermi Unit 1 Fund can not be used for the Fermi 2 Power Plant decommissioning; therefore, we do not discuss the Nonqualified Fermi Unit 1 Fund any further.

8.2 Fermi 2 Nuclear Decommissioning Trusts

The Fermi 2 Nuclear Decommissioning Trust consists of the two Mater Decommissioning Trust accounts for Fermi 2 Power Plant:

1. The Qualified Fermi 2 Unit Fund (Qualified Fund)
2. The Nonqualified Fermi 2 Unit Fund (Nonqualified Fund)

where qualified and nonqualified denotes the different tax code applicability. The Qualified Fund is invested as a single account and the Nonqualified Fund is invested as a separate single account. Both the Qualified and Nonqualified Funds consists of distinct plans to segregate amounts within by jurisdiction and purpose. DTE Electric maintains separate books and records for each distinct plan.

The Qualified Fund consists of four plans:

1. Fermi 2 Qualified MPSC fund
2. Fermi 2 Qualified FERC fund
3. ISFSI Decommissioning fund
4. Fermi 2 Qualified Other Decommissioning fund

The Nonqualified Fund consists of four plans:

1. Fermi 2 Nonqualified MPSC fund
2. Fermi 2 Nonqualified FERC fund
3. Fermi 2 Nonqualified Other Decommissioning fund
4. LLRW fund.

The Fermi 2 Qualified MPSC fund, Fermi 2 Qualified FERC fund, ISFSI Decommissioning fund, Fermi 2 Qualified Other Decommissioning fund, Fermi 2 Nonqualified MPSC fund, Fermi 2 Nonqualified FERC fund, and Fermi 2 Nonqualified Other Decommissioning fund are the seven plans that collectively equate to the Nuclear Decommissioning Fund and are for the



exclusive use of the end-of-life decommissioning of the Fermi 2 Power Plant. The LLRW fund is for the use of the disposal of low-level radioactive waste (LLRW) material generated during plant operations.

NRC radiological decommissioning

Radiological decommissioning are those activities required to terminate the Fermi 2 Power Plant's license. DTE Electric has certified to the Nuclear Regulatory Commission (NRC) that the Fermi 2 Qualified MPSC fund, Fermi 2 Qualified FERC fund, Fermi 2 Nonqualified MPSC fund and Fermi 2 Nonqualified FERC fund amounts are available for the Fermi 2 radiological decommissioning, pursuant to NRC regulations at 10 CFR 50.75(b) and (c).

Fermi 2 Qualified MPSC fund

The Fermi 2 Qualified MPSC fund is a qualified trust fund under IRS code 468A and is for the exclusive use of radiological decommissioning of the Fermi 2 Power Plant and subject to NRC minimum assurance regulations at 10 CFR 50.75(b). The amounts in this fund originated from customers within the MPSC jurisdiction.

Fermi 2 Qualified FERC fund

The Fermi 2 Qualified FERC fund is a qualified trust fund under IRS code 468A and is for the exclusive use of radiological decommissioning of the Fermi 2 Power Plant and subject to NRC regulations at 10 CFR 50.75(b). The amounts in this fund originated from customers within the FERC jurisdiction.

Fermi 2 Nonqualified MPSC fund

The Fermi 2 Nonqualified MPSC fund is not a qualified trust fund under IRS code 468A but is for the exclusive use of radiological decommissioning of the Fermi 2 Power Plant and subject to NRC regulations at 10 CFR 50.75(b). The amounts in this fund originated from customers within the MPSC jurisdiction.

Fermi 2 Nonqualified FERC fund

The Fermi 2 Nonqualified FERC fund is not a qualified trust fund under IRS code 468A but is for the exclusive use of radiological decommissioning of the Fermi 2 Power Plant and subject to NRC regulations at 10 CFR 50.75(b). The amounts in this fund originated from customers within the FERC jurisdiction.

NRC ISFSI decommissioning

The ISFSI Decommissioning fund is a qualified trust fund under IRS code 468A and is for the exclusive use of decommissioning the Fermi 2 Power Plant ISFSI and subject to NRC regulations at 10 CFR 72.30. The amounts in this fund originated from customers within the MPSC jurisdiction.

NRC irradiated fuel management

Irradiated fuel management are the activities required for post-shutdown fuel stewardship following the certification of permanent cessation of operations and permanent defueling of the Fermi 2 reactor. DTE Electric intends to use portions of the Fermi 2 Qualified Other fund and Fermi 2 Nonqualified Other fund amounts as available for irradiated fuel management, pursuant to NRC regulations at 10 CFR 50.54(bb).

Fermi 2 Qualified Other Decommissioning fund

The Fermi 2 Qualified Other Decommissioning fund is a qualified trust fund under IRS code 468A and is for the exclusive use of decommissioning the Fermi 2 Power Plant. DTE Electric does not report this fund as available to the NRC – as such, this fund is



for the use of post-shutdown fuel stewardship and non-radiological decommissioning (“greenfielding” or site restoration) of the Fermi 2 Power Plant. The monies in this fund originated from customers within both the MPSC jurisdiction and FERC jurisdiction.

Fermi 2 Nonqualified Other Decommissioning fund

The Fermi 2 Nonqualified Other Decommissioning fund is not a qualified trust under IRS code 468A. The fund is for the exclusive use of decommissioning the Fermi 2 Power Plant. DTE Electric does not report this fund as available to the NRC – as such, this fund is available for the use of post-shutdown fuel stewardship and non-radiological decommissioning (“greenfielding” or site restoration) of the Fermi 2 Power Plant. After the permanent cessation of operations, the NRC will require DTE Electric to report funds available for post-fuel shutdown fuel stewardship per 10 CFR 72. The amounts in this fund originated from customers within both the MPSC jurisdiction and FERC jurisdiction.

8.3 DTE Electric Nuclear Surcharge

The exclusive source of amounts for the Fermi 2 NDT is a surcharge collected from the MPSC jurisdictional customer base. We discuss the surcharge in this section.

Nuclear Surcharge overview

The Nuclear Surcharge is a non-bypassable charge applied to all electricity usage within DTE Electric's jurisdiction. The Michigan Public Service Commission (MPSC or Commission) reviews the DTE Electric proposed annual Nuclear Surcharge provision during DTE Electric's rate case proceedings. The Commission then authorizes the amount DTE Electric is to collect within the electric rate case order. Attachment B of Order U-17767, dated December 11, 2015 provides a summary of the Nuclear Surcharge:

On January 1987 MPSC Order authorized the establishment of an external trust fund to finance the decommissioning of Fermi 2 Power Plant when its operating license expires. The Order approves a decommissioning surcharge on customer bills under which the funds are collected. Pursuant to Commission Order U-10102 dated January 21, 1994, a revised surcharge became effective with service rendered on and after January 22, 1994. In the same order, the Commission authorized the establishment of an external fund to finance the disposal of low-level radioactive waste during the operating life of Fermi 2 Power Plant. Pursuant to an order in Case No. U-14399, costs associated with site security and radiation protection services were removed from base rates and transferred to the Nuclear Surcharge.

The Nuclear Surcharge then has three separate components (two for Fermi 2 operations and one for Fermi 2 decommissioning):

1. Site Security and Radiation Protection Funding – DTE Electric records this portion of the Nuclear Surcharge to offset current year site security and radiation protection O&M costs.
2. Low-level Radioactive Waste Funding (LLRW) – DTE Electric deposits this portion of the Nuclear Surcharge into the LLRW Fund, used to offset the operational expenses resulting from the disposal of LLRW generated during Fermi 2 operations.
3. Nuclear Decommissioning Funding – DTE Electric deposits this portion of the Nuclear Surcharge into the Nuclear Decommissioning Fund.



As the scope of this study is Fermi 2 decommissioning, we use the balance of this section to discuss the Nuclear Decommissioning Funding portion of the Nuclear Surcharge.

8.4 Nuclear Decommissioning Funding Portion of Nuclear Surcharge

The Nuclear Decommissioning Funding (NDF) portion of the Nuclear Surcharge is the exclusive DTE Electric mechanism authorized by the Commission to provide contributions to the Nuclear Decommissioning Fund. The Commission authorizes the annual surcharge amount DTE Electric for the Nuclear Decommissioning Funding.

NRC regulations

The Nuclear Decommissioning Funding portion of the Nuclear Surcharge is a *non-bypassable charge* according to the NRC definition at 10 CFR 50.2:

Non-bypassable charges mean those charges imposed over an established time period by a Government authority that affected persons or entities are required to pay to cover costs associated with the decommissioning of a nuclear power plant. Such charges include, but are not limited to, wire charges, stranded cost charges, transition charges, exit fees, other similar charges, or the securitized proceeds of a revenue stream.

With the non-bypassable charge, the Fermi 2 Nuclear Decommissioning Trust qualifies as an external sinking fund as summarized in revision 2 of the NRC Regulatory Guide 1.159 "Assuring the Availability of Funding for Decommissioning Nuclear Reactors:"

External Sinking Fund—A fund established and maintained by setting funds aside periodically, in an account segregated from licensee assets and outside the administrative control of the licensee and its subsidiaries or affiliates, in which the total amount of funds would be sufficient to pay decommissioning costs at the time permanent termination of operations is expected. An external sinking fund may be in the form of a trust, escrow account, government fund, certificate of deposit, deposit of government securities, or other payment acceptable to the NRC.

The DTE Electric exclusive use of the external sinking fund method to provide the NRC with reasonable assurance of funding availability is consistent with NRC regulations at 10 CFR 50.75(e)(1)(ii):

...The external sinking fund method may be used as the exclusive mechanism relied upon for providing financial assurance for decommissioning in the following circumstances:

(B) By a licensee whose source of revenues for its external sinking fund is a "non-bypassable charge," the total amount of which will provide funds estimated to be needed for decommissioning pursuant to §§ 50.75(c), 50.75(f), or 50.82 of this part.

Because the Nuclear Decommissioning Funding Portion of the Nuclear Surcharge is the exclusive method DTE Electric has identified to the NRC to provide reasonable assurance, DTE Electric must continue to collect at least a nominal annual amount to maintain this method. Suspension or discontinuance of the Nuclear Decommissioning Portion of the Nuclear Surcharge would require NRC notification (10 CFR 50.75(f)(1).



Allocation of the Nuclear Decommissioning Funding provision

The MPSC approves the total annual provision for Nuclear Decommissioning Funding; DTE Electric determines the allocation of the amounts collected from the annual Nuclear Decommissioning Funding provision to each NDT plan.

DTE Electric must factor minimum annual funding commitments as well as each fund's adequacy into the allocation of the amounts collected from the annual Nuclear Decommissioning Funding provision. DTE Electric has the following approximate annual funding commitments with the NRC:

1. \$12 per year allocated to radiological decommissioning [ML19087A224]
2. \$382,000 per year allocated to ISFSI decommissioning [ML17089A789].

DTE Electric last updated the allocation in 2017 [DTE17] following the NRC approval of the Fermi 2 license renewal. The current allocation of the Nuclear Decommissioning Funding provision is approximately:

- 14% or ~\$420,000 per year to ISFSI Decommissioning fund. This amount is to ensure DTE Electric contributes at least the NRC commitment of \$382,000 per year to the ISFSI Decommissioning fund regardless of DTE Electric sales.
- 0% or ~\$12 per year to Fermi 2 Qualified MPSC fund. This amount is to ensure DTE Electric contributes at least \$1/month to satisfy the NRC that DTE Electric is maintaining the NDT's status as an external sinking fund.
- 86% or ~\$2,580,000 per year to Fermi 2 Qualified Other Decommissioning fund. This amount is to ensure DTE Electric contributes the balance of the Nuclear Decommissioning Funding provision to post-fuel stewardship and site restoration as these activities are underfunded.

We address the recommended future allocation of the Nuclear Decommissioning Funding provision in the Funding Adequacy discussion later in this report.



SECTION NINE

9 NDT funding adequacy

In this section, we will discuss the elements used to calculate DTE Electric's 2020 Fermi 2 Nuclear Decommissioning Trust (NDT) funding adequacy. Our intention with this discussion is to support the reader's understanding and walk through the elements of assessing the funding adequacy of the Fermi 2 Nuclear Decommissioning Trust. The goal of this section's discussion is to provide reasonable assurance that DTE Electric's annual Nuclear Decommissioning Funding provision is adequate.

9.1 Nuclear Decommissioning Trust funding adequacy overview

In this section we provide an overview of the DTE Electric Nuclear Decommissioning Trust funding adequacy review where we introduce the mathematical model for determining the required minimum annual Nuclear Decommissioning Funding provision for reasonable assurance and the elements required to use the model.

We use a simple model to calculate the required annual Nuclear Decommissioning Funding provision of the Nuclear Surcharge NS_{NDF}

$$NS_{NDF} = (SDCE_f - NDT_f) \left(\frac{i}{(1+i)^t - 1} \right), \quad 9.1$$

where,

$SDCE_i$ is the Fermi 2 site-specific cost estimate (SDCE) at the time of expected shut down (i.e. December 31, 2044), given no contributions where

$$SDCE_f = SDCE_0(1 + r)^t \text{ where}$$

$SDCE_0$ is the Fermi 2 site-specific cost estimate (SDCE) at time of reference for the review (i.e. December 31, 2019)

r is the projected rate of escalation from starting date t_0 to time of expected shutdown t_i

t is the time t in years growth from starting date t_0 to time of expected shutdown t_i

NDT_i is the balance of Fermi 2 Nuclear Decommissioning Fund at the time of expected shutdown (i.e. December 31, 2044), assuming no contributions where

$$NDT_f = NDT_0(1 + i)^t \text{ where}$$

NDT_0 is the balance of Fermi 2 Nuclear Decommissioning Fund at time of reference for the review (i.e. December 31, 2019)

i is the projected effective rate of growth from starting date t_0 to time of expected shutdown t_i

t is the time t in years growth from starting date t_0 to time of expected shutdown t_i

We'll make two important notes for this model:

1. The rate of growth applied to assets is the effective rate of growth. This is to say, if we assume:
 $NDT_0 = 100$; $i = 7\%$; and $t = 2$ years; then the expected available NDT balance at the end of two years would be \$114.49 after paying for taxes and administrative and incidental expenses.



2. This model uses a single constant rate of inflation and a single constant effective rate of return on assets to calculate the required annual collections to achieve the minimum amounts within the specified time period, which is a reasonable model to provide reasonable assurance without overburdening reviewers with additional inputs; however we must exercise care when updating the elements of the model, particularly, the rates of inflation and escalation, where, because of compounding effects – just a tenth of a percent change in either rate element can result in significant changes in the calculated required annual collections. Fundamentally, for any set of potential rates using the same net difference, the lower the rate of inflation, the less annual collections required; for example, keeping all other elements constant, using 0% rate of inflation and 0% effective rate of growth on assets will result in a lower calculated annual collection than using a 4% rate of inflation and a 4% effective rate of growth on assets even though both calculations use a net zero difference between the rate of inflation and the effective rate of growth on assets.

We use the balance of this discussion to establish the values for each element of the funding adequacy model discussed above:

- Fermi 2 SDCE at December 31, 2019
- Rate of escalation as applied to the SDCE from December 31, 2019 through December 31, 2044
- Fermi 2 Nuclear Decommissioning Fund balance as of December 31, 2019
- Effective rate of growth as applied to the NDT and its funds from December 31, 2019 through December 31, 2044 which must account for applied taxes and incidental and administrative expenses.

9.2 Nuclear Decommissioning Cost Estimate

For this funding adequacy review, we adopt the Fermi 2 Site-Specific Decommissioning Cost Estimate (SDCE) as discussed in Section 7.4. This adoption is a reasonable and prudent approach given the Michigan Public Service Commission (MPSC) requested this updated study precisely to obtain a decommissioning cost estimate using more Fermi 2 site-specific considerations. The total estimated cost to decommission the Fermi 2 Power Plant is approximately \$2.2 billion as of December 31, 2019 and detailed in Tables 9.1 and 9.2.

Table 9.1 Summary Fermi 2 SDCE costs in thousands of 2019 dollars.

Fermi 2 decommissioning activity	Cost estimate
License termination	\$1,559,579.2
ISFSI decommissioning	12,608.5
Spent fuel management program	510,843.3
Site restoration	96,141.3
Total Fermi 2 site-specific decommissioning cost estimate (SDCE)	\$2,179,172.3

Table 9.2 Fermi 2 SDCE by cost element in thousands of 2019 dollars

Fermi 2 decommissioning activity	Labor	Materials	Disposal	Other	Total
License termination	\$487,126.2	\$121,724.1	\$585,460.4	\$365,268.6	\$1,559,579.2
ISFSI decommissioning	6,223.4	2,139.2	455.0	3,790.9	12,608.5
Spent fuel management program	236,337.4	82,600.0	25,048.6	166,857.3	510,843.3
Site restoration	26,864.7	39,723.2	13,755.3	15,798.1	96,141.3
Total Fermi 2 SDCE	\$756,551.7	\$246,186.5	\$624,719.3	\$551,714.9	\$2,179,172.3



We retain the basis of the SDCE as well, most notably, we maintain that Fermi 2 Power Plant will shut down on March 20, 2045; therefore, the amounts in the Fermi 2 NDT balances must be sufficient no later than December 31, 2044, which is consistent with the existing Fermi 2 decommissioning cost estimate.

As we have discussed prior, because of each power plant's unique characteristics, strategies and situations there are pitfalls associated with simple benchmarking of the industry's nuclear decommissioning cost estimates; however, the Commission expressed interest in better understanding how the Fermi 2 decommissioning cost estimate related to those throughout the industry and to that end DTE Electric worked with Callan LLC (Callan) to provide a draft benchmarking analysis [CAL20] of the various elements used by DTE Electric in modeling nuclear decommissioning funding adequacy. The Callan analysis demonstrates that using industry information available to Callan, DTE Electric's Fermi 2 decommissioning cost estimates were within the industry's expected range for BWRs as of 2018, after scaling for known differences in published estimates.

9.3 Inflation rates (cost escalation)

As we have discussed earlier in this section, the inflation rate or cost escalation rate used in our nuclear decommissioning funding adequacy model is an important factor to consider. The requirement of the Settlement Agreement in Michigan Public Service Commission (MPSC) Case No. U-6150 (Re-opened), dated August 26, 1986 states DTE Electric is to include a review of inflation rates in DTE Electric's report on the adequacy of the annual Nuclear Decommissioning Funding provision; we intend this section to satisfy the inflation rates review.

For this section we discuss the projected rate of inflation (or cost escalations) for the Fermi 2 SDCE where we begin with a review of the incumbent rate of inflation approved by the Michigan Public Service Commission (MPSC or Commission) and then transition to a historical perspective of actual rate of inflation experienced by the different elements of nuclear decommissioning activities. We close out the section with a recommendation for a reasonable and prudent going-forward rate of inflation for use in this funding adequacy review.

DTE Electric rate of inflation methodology

DTE Electric's existing nuclear decommissioning funding adequacy model uses the rate of inflation of 6%. The Michigan Public Service Commission last approved the use of this 6% rate of inflation model in Order U-20162, dated May 2, 2019; the MPSC has consistently approved the use of the 6% rate of inflation since Commission Order U-10102, dated January 21, 1994.

The Commission's approval of the 6% rate of inflation has been demonstrated as reasonable (through 2019) when applied to DTE Electric's annual review and update of the 2002 Fermi 2 nuclear decommissioning cost estimate. As we have discussed in prior sections of this study, applying the 6% rate of inflation in DTE Electric's annual review and update of the 2002 Fermi 2 decommissioning cost estimate resulted in a total decommissioning cost estimate within 0.5% of the 2020 Fermi 2 site-specific decommissioning cost estimate as of December 31, 2019.

Historical rate of inflation for nuclear decommissioning cost estimates

Regulatory guidance for nuclear decommissioning cost escalation factors

The NRC issues a bi-annual publication named NUREG-1307, Report on Waste Burial Charges Changes in Decommissioning Waste Disposal Costs at Low-Level Waste Burial Facilities [ML19037A405], which provides the regulatory guidance on cost escalation factors for the NRC minimum nuclear decommissioning cost estimates. In this report, the NRC publishes the results



of updated Pacific Northwest National Laboratory (PNNL) analyses of cost escalation for low-level radioactive waste (LLRW) disposal based on LLRW burial facilities rate changes. The report also specifies the sources to obtain labor and energy escalation rates. Using Revision 17 of this report, we present the relevant Fermi 2 Power Plant decommissioning cost escalation factors in Table 9.3:

Table 9.3 NUREG-1307, Revision 17 (2019) cost escalation factors for Fermi 2 Power Plant

Component	Cost escalation factor	Compound cost escalation (%) (1986 - 2018)	Basket weighting	Basket-weighted cost escalation factor	Basket-weighted compound cost escalation (%) (1986 - 2018)
Labor (L_x)	2.75	3.211	0.65	1.79	
Energy (E_x)	2.39	2.759	0.13	0.31	
Burial (B_x)	28.34	11.016	0.22	6.23	
Total				8.33	6.85

Table 9.3 shows that the basket-weighted compound cost escalation for the NRC minimum cost estimates for a BWR in the Midwest was approximately 6.9% for the period 1986 - 2018. The labor component (L_x) cost escalation factor uses the U.S. Bureau of Labor Statistics (BLS) Total compensation for Private industry workers in Midwest, Index (Series ID CIU2010000000230I) as its reference. The energy component (E_x) cost escalation factor uses a combination of the Producer Price Index (PPI) for industrial electrical power (PPI Commodity Code 543) and the PPI for light fuel oils (PPI Commodity Code 0573). The compound cost escalation for the 1986 - 2018 period is the conversion of the cost escalation factor into an annualized cost escalation rate. The basket-weighting is the fraction of that components costs to the total decommissioning cost estimate for the NRC's reference BWR. The basket-weighted cost escalation factor is the result of multiplying each component's cost escalation factor by its basket weighting; the sum of these products (8.33) is the total cost escalation factor to convert the 1986-dollar decommissioning cost estimate into 2018 dollars. The basket-weighted compound cost escalation of 6.85% is the equivalent compound cost escalation rate for the 32-year period 1986 - 2018 (March 2019 submittal to the NRC).

Figure 9.1 The Fermi 2 Power Plan NRC minimum decommissioning funding.

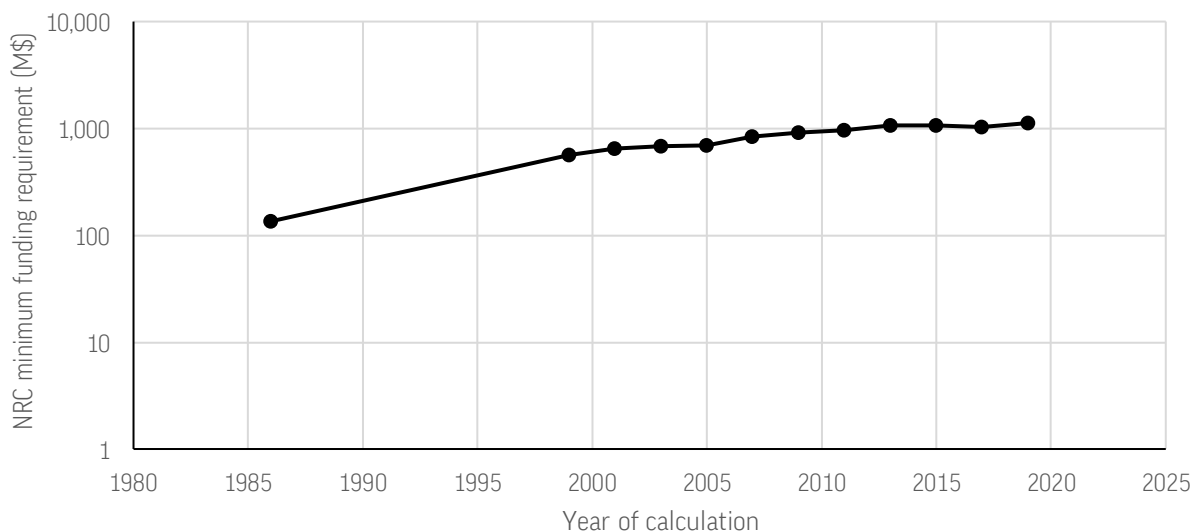


Figure 9.1 shows that the period from 1986 – 1998 experienced the highest rate of cost escalation. Although it is plausible, we will experience a similar period of high inflation, we believe it is reasonable and prudent to focus within the 20-year period between 1998 and 2018. The 20-year compound growth rate of the NRC minimum funding requirement was approximately 3.5% for the years 1998 – 2018.

U.S. inflation trends

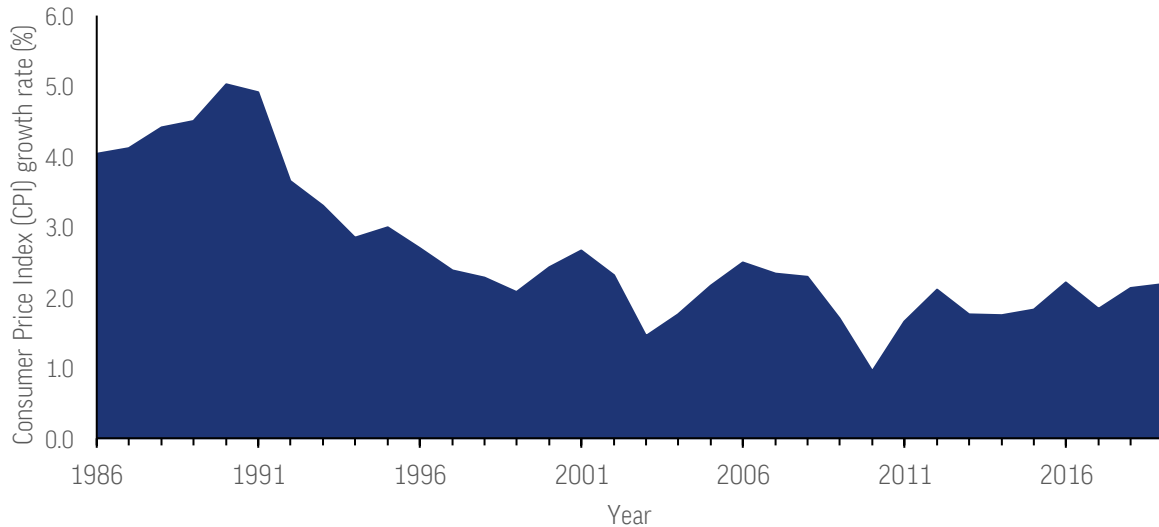
We use two common U.S. inflation measures to better understand the historical context of the growth shown in decommissioning costs over time: Consumer Price Index (CPI) and Personal Consumption Expenditures (PCE).

CPI is a measure of the average change over time in the prices paid by urban consumers for a market basket of consumer goods and services. Indexes are available for the U.S. and various geographic areas. Core CPI excludes cost changes due to food and energy prices. The U.S. Bureau of Labor and Statistics (BLS) maintains the CPI and Core CPI calculations, which are available as seasonally-adjusted or not seasonally-adjusted.

Figure 9.2 shows that the Core CPI annual, not seasonally-adjusted, rates were highest from 1986 through 1992 which follows the general escalation trend as the calculated NRC minimum funding amounts.



Figure 9.2 Core Consumer Price Index [CPI] (Series ID CPGRLE01USA657N) growth rate from previous period (1986 – 2018)

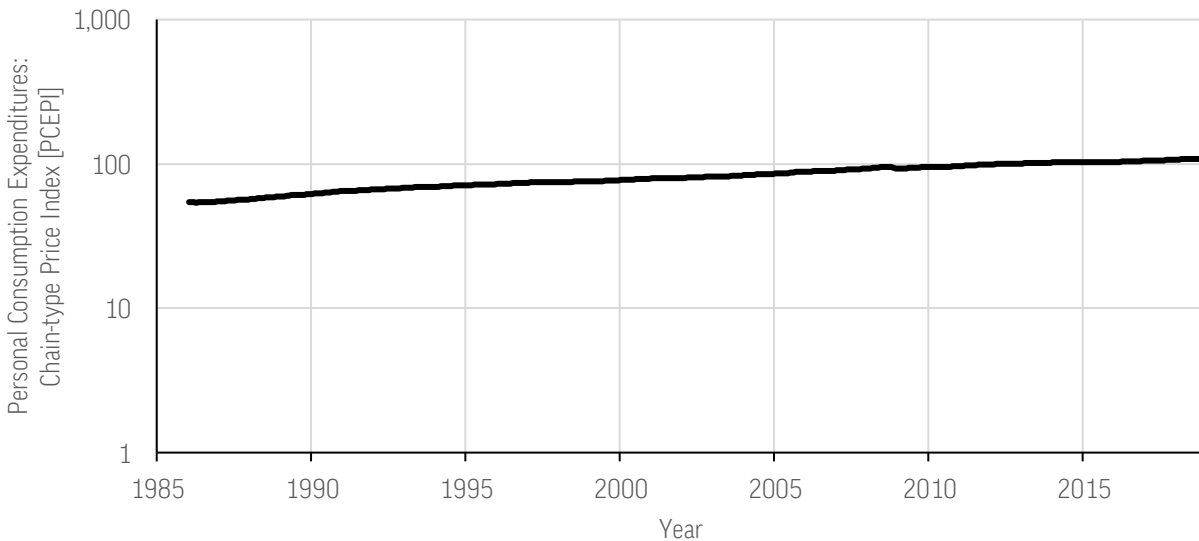


PCE is a measure of the prices that people living in the United States, or those buying on their behalf, pay for goods and services. PCE is similar to CPI in that it is a measure of inflation and follows similar trends; however, PCE tracks the prices of a different basket of goods than CPI and notably PCE uses substitution effect (a model that assumes consumers will alter purchasing habits if one good experiences higher inflation than similar goods) whereas CPI does not. PCE inflation tends to be lower than CPI inflation. Core PCE excludes costs of food and energy. The Bureau of Economic Analysis maintains the PCE and Core PCE calculations which are available as seasonally-adjusted or not seasonally-adjusted. The Federal Reserve states its goal for inflation in terms of PCE.

Figure 9.3 shows that the Personal Consumption Chain-type Price Index [PCEPI] follows the same general escalation trend as the NRC minimum funding amounts – the highest escalation rates occurred between approximately 1986 and 1992. To align with our decommissioning cost escalation data from prior, the PCEPI 32-year compound rate was approximately 2.2% for the years 1986 – 2018 and the PCEPI 16-year compound rate was approximately 1.8% for years 1986 - 2018. To closeout the discussion, we note the PCEPI 10-year compound rate was approximately 1.6% for years 2010 – 2019.



Figure 9.3 Personal Consumption Expenditures: Chain-type Price Index [PCEPI] (1986 - 2018)



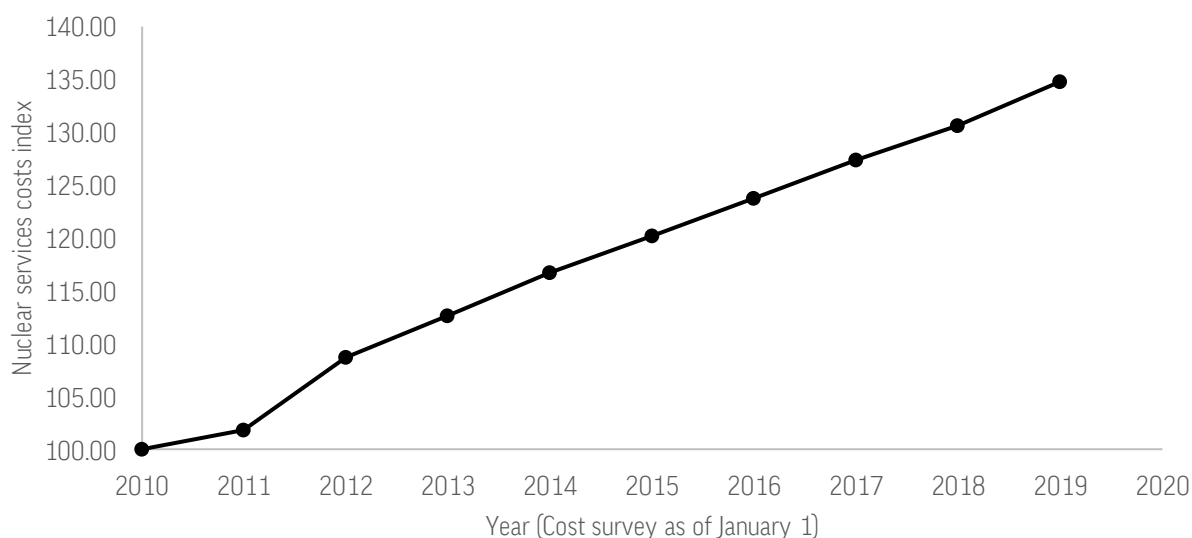
The Board of Governors of the Federal Reserve System confirmed to the U.S. Congress its long-term PCE target is 2% in the February 2020 Monetary Policy Report. The shorter-term Federal Reserve expectation for PCE changes range between 2.0% and 2.2% through 2022.

Fermi-specific cost escalation trends

We now review historical cost escalation more specific to the Fermi 2 Power Plant and its circumstances.

Using a similar methodology as NUREG-1307, Figure 9.4 shows the indexed costs associated with nuclear services for the 10-year period 2010 - 2019. Nuclear services are suppliers that provide specialized boiling water reactor (BWR) services such as project management, technical engineering, fuel handling and BWR vessel maintenance. During decommissioning activities, reactor services would support activities such as project management, engineering, fuel handling and containment vessel segmentation. As reactor services is a specialty service with a limited market, DTE Electric would expect minimal substitution effect and higher cost escalation than the broader labor market, which is demonstrated in Figure 9.4 as the 9-year escalation rate for reactor services was approximately 3.4% for the period 2010 - 2019.

Figure 9.4 Fermi 2 reactor services costs index (2010 – 2019)



The costs associated with nuclear-specific services as well as radioactive waste disposal have a demonstrated history of escalating at a rate greater than the general rate of inflation.

DTE Electric forecasted rate of inflation

Note: This discussion involves “forward looking statements” developed by DTE Electric in February of 2020 as applied to NDT costs.

We use Federal Reserve guidance for forecasting cost escalation. The Federal Reserve published a study in 2010 of inflation forecasting techniques and outcomes and determined that naïve forecasts or expectation forecasts demonstrated the most accuracy over more complicated analysis. In fact, it is the expert opinion of the Governors of the Federal Reserve that determines the Federal Reserve’s forecasted escalation of the PCE. The Federal Reserve uses a long-term target (or projection) of 2% growth in PCE over the long-term [FMOC19] – it is this 2% basis that we use to support the forecasted rate of inflation.

From our historical review of cost escalations, we can reasonably conclude that nuclear decommissioning costs (and other nuclear services) will continue to escalate at a cost greater than the target PCE. Nuclear decommissioning services, while an emerging market, will always be limited in growth potential due to the small number of potential customers; as such it is reasonable to assume that the in future, like current, substitution effect will be minimal. Listed here are the takeaways from the previous historical discussion:

- The 1986 – 2019 compound escalation rate in the NRC minimum nuclear decommissioning funding amount for the Fermi 2 Power Plant was 6.8%. These escalation rates are inclusive of waste burial, labor and energy costs.
- The 1999 – 2019 compound escalation rate for the NRC minimum nuclear decommissioning funding amount for the Fermi 2 Power Plant was approximately 3.5%. These escalation rates are inclusive of waste burial, labor and energy



costs; approximately double (2x) the personal consumption expenditure price index (PCEPI) compound escalation rate of approximately 1.8% for the same period. A reasonable forecasted rate cost escalation for similar decommissioning scope is then approximately 3.9%.

- The 2010 – 2019 compound escalation rate for specialty nuclear services labor was approximately 3.4%; approximately 2.3x the PCEPI compound escalation rate of approximately 1.5% for the same period. A reasonable forecasted rate cost escalation for similar specialty nuclear services scope is the approximately 4.6%.

We also have from Callan's survey of U.S. BWR owners that their nuclear decommissioning cost escalation projections vary from a low of approximately 2.5% to a high of 6% with 3.9% as the median escalation rate. This survey could be considered a close approximation to the industry's expert opinion of projected nuclear decommissioning cost escalation rates.

We remain confident that the past use of the 6% cost escalation rate for DTE Electric's Fermi 2 Power Plant nuclear decommissioning funding adequacy models was reasonable and prudent. As we had discussed prior, the 6% factor applied to the annual review and update of the 2002 Fermi 2 decommissioning cost estimate yielded results similar to the 2020 Fermi 2 site-specific decommissioning cost estimate. DTE Electric has used the 6% inflation model to lower the annual funding provision twice in the since 2011. Additionally, the historical growth in the Fermi 2 Power Plant's NRC minimum decommissioning funding amounts also support the 6% model. Continued use of the 6% inflation could also be considered reasonable and prudent, given the success of the model over a time of relatively stable and low inflation rates. That said, we do find it reasonable to propose an alternative recommendation in this study given we are now only 25 years away from the decommissioning of the Fermi 2 Power Plant and have a newly developed bottom-up site-specific decommissioning cost estimate using 2019 dollars.

Our recommendation is to use a 3.8% escalation factor in this study's nuclear decommissioning funding adequacy model. Modeling at 3.8% would be reasonable and prudent considering the lower inflation rates of the past twenty years and the Federal Reserve's projection for inflation rates to remain stable around 2% for the long term. We have demonstrated that nuclear decommissioning costs (inclusive of disposal, energy and labor costs) as well as services specific to working at a BWR facility escalate at rates 1.9x to 2.3x greater than the PCEPI measure. This recommendation is also supported by a survey of other BWR owners, where the median escalation factor projected was 3.9%.

9.4 Fermi 2 NDT balances

In this section we review balances for each fund or plan within the Fermi 2 Nuclear Decommissioning Trust (NDT). We review the balances to provide the reader with an understanding of where Fermi 2 amounts are deposited which then informs later analysis with respect to projected rate of return on NDT assets and current NDT shortfalls.

The total Fermi 2 NDT balance as of December 31, 2019 was approximately \$1.658 billion (\$1,658,358.4 thousand). We show the balances of the Qualified Fund at approximately \$1.488 billion and the Nonqualified Fund at approximately \$0.171 billion in Table 9.4 along with the balances of each plan within the Funds.



Table 9.4 Fermi 2 NDT balances as of December 31, 2019 in thousands of dollars.

Qualified Fund		Nonqualified Fund	
Plan name	Fund balance (as of 12/31/2019)	Plan name	Fund balance (as of 12/31/2019)
Qualified MPSC	\$1,394,721.7	Nonqualified MPSC	\$144,020.3
Qualified FERC	6,600.0	Nonqualified FERC	6,404.7
ISFSI	4,823.5		
Qualified Other Decommissioning	81,561.3	Nonqualified Other Decommissioning	11,882.0
		LLRW	8,371.8
Total Qualified Fund	\$1,487,706.6	Total Nonqualified Fund	\$170,678.8

We recast the plan balances from Table 9.4 into purpose-based nuclear decommissioning amounts in Table 9.5 where Qualified MPSC and Qualified FERC funds are the qualified amounts set aside for the Fermi 2 Power Plant license termination, Nonqualified MPSC and Nonqualified FERC funds are the nonqualified amounts set aside for license termination; ISFSI fund is the qualified amount set aside for the Fermi 2 ISFSI decommissioning; Qualified Other Decommissioning fund is the qualified amount set aside for spent fuel management and site restoration and Nonqualified Other Decommissioning is the nonqualified amount set aside for spent fuel management and site restoration; we exclude the LLRW fund from Table 9.5 because the LLRW fund is set aside for disposal of LLRW generated during plant operations and is not a decommissioning fund. Also recall our discussion from Section 6, that transfer of amounts from one fund to another fund is inflexible due to the integrated and interlocking regulatory framework.

Table 9.5 Fermi 2 Nuclear Decommissioning Fund (NDF) amounts by purpose as of December 31, 2019 in thousands of dollars

Purpose	Qualified amounts	Nonqualified amounts	Total amounts
License termination	\$1,401,321.8	\$150,425.0	\$1,551,746.8
ISFSI decommissioning	4,823.5	0.0	4,823.5
Spent fuel management and site restoration	81,561.3	11,882.0	93,443.3
Total Nuclear Decommissioning Fund	\$1,487,706.6	\$162,307.0	\$1,650,013.7

Table 9.6 presents the Fermi 2 NDF funding shortfalls as of December 31, 2019, where the Nuclear Decommissioning Fund (NDF) balance of \$1.65 billion is approximately \$0.53 billion short of the \$2.18 billion required to nuclear decommission the Fermi 2 Power Plant per the 2020 Fermi 2 site-specific decommissioning cost estimate; approximately 97% of the projected shortfall is attributable to spent fuel management and site restoration, which is only approximately 15% funded as of December 31, 2019. The ISFSI decommissioning fund is approximately 38% funded; however, the projected shortfall is only approximately \$7.8 million. The license termination fund is essentially fully funded at approximately 99% with a projected shortfall of approximately \$7.8 million. We find Table 9.6 useful as it demonstrates the amounts of funds that must be generated (in 2019 \$s) by the end of the Fermi 2 Power Plant's operating life in 2045 either through the annual Nuclear Decommissioning Funding mechanism, through fund growth or a combination of fund growth and annual collections.



Table 9.6 Nuclear Decommissioning Fund (NDF) funding levels by purpose as of December 31, 2019 in thousands of dollars

Purpose	NDF balance (as of 12/31/2019)	Estimate (as of 12/31/2019)	Shortfall
License termination	\$1,551,746.8	\$1,559,579.2	(\$7,832.4)
ISFSI decommissioning	4,823.5	12,608.5	(7,784.9)
Spent fuel management and site restoration	93,443.3	606,984.6	(513,541.3)
Total Nuclear Decommissioning Fund	\$1,650,013.7	\$2,179,172.3	(\$529,158.7)

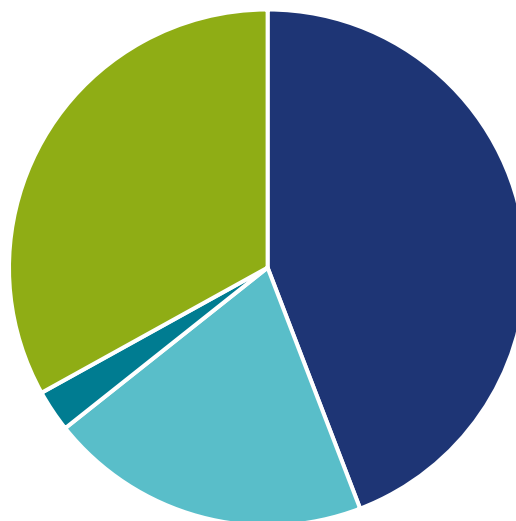
NDT asset allocation

The MPSC Settlement Agreement U-6150 (reopened), dated August 26, 1986, defines the objective of the trusts investments is to "optimize after tax earnings over the life of the trust, giving consideration to liquidity, risk, diversification, and other prudent investment objectives." The DTE Energy Investment Committee is responsible for the asset allocation and investment manager selection for the Fermi 2 NDT. The Investment Committee currently maintains an equity target of 65% within a policy range of 40 to 80% for the NDT.

Figure 9.5 presents the Fermi 2 NDT assets as the amounts DTE Electric had invested as of December 31, 2019. Approximately 44% (~\$732.6 million) of the NDT assets were in domestic equity and approximately 20% (~\$334.0 million) of assets were in international equity. Approximately 3% (~\$43.6 million) of the NDT assets were in alternative investments such as private equity. Approximately 33% (~\$548.2 million) of the NDT assets were in fixed income, such as municipal bonds, and cash.

Figure 9.5 Fermi 2 NDT assets as invested on December 31, 2019

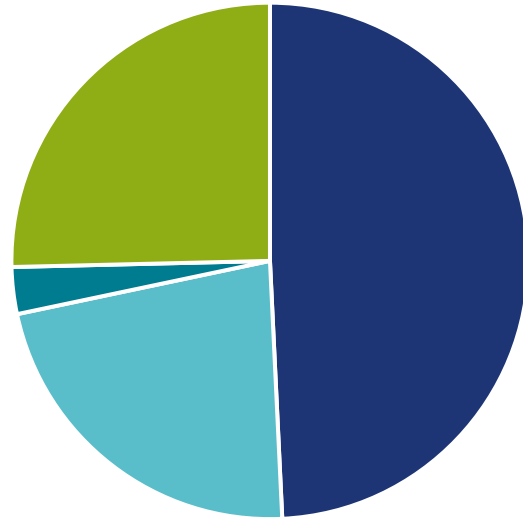
- Domestic equity: \$732.6M (44%)
- International equity: \$334.0M (20%)
- Alternatives: \$43.6M (3%)
- Fixed income + cash: \$548.2M (33%)



The Fermi 2 Qualified Fund and the Fermi 2 Nonqualified Fund are each separate investment accounts within the NDT. The Fermi 2 Qualified Fund investment allocation is predominantly equity assets (both US and non-US equities) with the balance in fixed income and cash assets as shown in Figure 9.6. The Fermi 2 Nonqualified Fund investment allocation is entirely fixed income and cash assets.

Figure 9.6 Fermi 2 Qualified Fund assets as invested on December 31, 2019

- Domestic equity: \$732.6M (49%)
- International equity: \$334.0M (22%)
- Alternatives: \$43.6M (3%)
- Fixed income + cash: \$377.5M (25%)



9.5 Administrative and incidental expenses

In this section we discuss administrative and incidental expenses (including taxes) of the Fermi 2 NDT. The requirement of the Settlement Agreement in Michigan Public Service Commission (MPSC) Case No. U-6150 (Reopened), dated August 26, 1986 states DTE Electric is to include a review of administrative and incidental expenses of the fund in DTE Electric's report on the adequacy of the annual Nuclear Decommissioning Funding provision; we intend this section to satisfy the administrative and incidental expenses review.

Nuclear Regulatory Commission (NRC) regulations at 10 CFR 50.75(h)(2) permits the distributions from the Fermi 2 NDT for ordinary administrative costs (including taxes) and other incidental expenses of the trusts such as legal, accounting, actuarial and trustee expenses in connection with the operation of the trusts. The Settlement Agreement in MPSC Case No. U-6150 (Reopened), Section 13(c.) also permits such distributions:

"No part of the assets of the trust may be used for, or diverted to, any purpose other than to fund, in whole or part, the costs of nuclear plant decommissioning or to pay administrative and other incidental expenses, including taxes, if applicable, of the fund."

The DTE Electric Nuclear Decommissioning Master Trust Agreement has provisions that allow for the payment of administrative and incidental expenses in Article II, Section 2.03:

"The Trustee shall pay, as directed by the Company, the administrative costs and other incidental expenses of a Nonqualified Fund, including all federal, state, and local taxes, if any, imposed directly on the Nonqualified Fund, legal expenses, accounting expenses, actuarial expenses and trustee expenses, from the assets of the Nonqualified Fund and shall pay, as directed by the Company, the administrative costs and other incidental expenses of a Qualified Fund, as defined in the Special Terms, from the assets of the Qualified Fund."



Exhibit “A” Special Terms of the Qualified Nuclear Decommissioning Reserve Fund of the DTE Electric Nuclear Decommissioning Master Trust Agreement states in Section 1. (a):

“Administrative costs and other incidental expenses of the Qualified Fund” shall mean all ordinary and necessary expenses incurred in connection with the operation of a Qualified Fund, as provided in Treas. Reg. Section 1.468A-5(a)(3)(ii)(A) or any corresponding future Treasury Regulation, including without limitation, federal, state and local income tax, legal expenses, accounting expenses, actuarial expenses and trustee expenses.”

Taxes

DTE Electric’s Fermi 2 NDT activity is subject to federal and state income tax assessments. Taxable income is the amount of income used to calculate how much tax DTE Electric or the trust owes to the government in a given year. Taxable income includes earning from appreciated assets that have been sold during the year, income from dividends and interest less deductions for trustee fees, accounting fees, taxes, etc. Income from tax exempt bonds are not subject to federal income tax and excluded from calculation of the taxable income.

The Fermi 2 NDT has two separate accounts or Funds: (1) the Fermi 2 Qualified Fund is a tax-advantaged trust that satisfies the requirements of Internal Revenue Code (IRC) § 468A and (2) the Fermi 2 Nonqualified Fund is not similarly tax-advantaged and is not subject to the requirements of § 468A.

Federal treatment of nuclear decommissioning trust contributions

The Fermi 2 Qualified Fund meets the requirements of IRC § 468A and U.S. Treasury regulations at 26 CFR 1.468A-5. The tax-advantaged attributes of a qualified trust include the ability for DTE Electric to deduct amounts contributed into the Fermi 2 Qualified Fund, which allows DTE Electric to fund the Fermi 2 Power Plant decommissioning liability on a pre-tax basis; additionally, any realized investment gains are subject to a federal income tax rate of 20% rather than DTE Electric’s federal corporate tax rate. Prior to contributing or changing the amounts contributed to the Fermi 2 Qualified Funds, DTE Electric had to first request and receive from the Internal Revenue Service (IRS) a “Schedule of Ruling Amounts.” The IRS Schedule of Ruling Amounts potentially limits the amount of allowable annual contributions to the qualified trust.

The Fermi 2 Nonqualified Fund does not meet the requirements of IRC § 468A and U.S. Treasury regulations at 26 CFR 1.468A-5 and is treated as a grantor trust of DTE Electric for income tax purposes under U.S. Treasury regulations at 26 CFR 1.671-1.678. As a nonqualified fund, contributions paid into the Fermi 2 Nonqualified Fund are not deductible; additionally, any realized investment gains are subject to tax at DTE Electric’s federal corporate tax rate.

Amended schedule of ruling amounts

In March 2017, DTE Electric filed an application to the Internal Revenue Service (IRS) for an Amended Schedule of Ruling Amounts for the Fermi 2 Power Plant retroactive to January 2016. DTE Electric filed for two reasons: (1) as requested by DTE Electric in MPSC Case No. U-17767, the Commission ordered the Nuclear Decommissioning Funding provision of the Nuclear Surcharge reduced by greater than 50% (from ~\$13 million to ~\$3 million) and (2) the NRC approved DTE Electric’s application for a 20-year license extension. On July 10, 2017 the IRS approved DTE Electric’s request [IRS17]. As a result of this IRS Ruling, 100% of DTE Electric’s surcharge collections related to end-of-life decommissioning are eligible for deposit in the qualified trust, which carries tax-advantaged attributes of a current federal tax deduction and a slightly lower federal tax rate.



State of Michigan treatment of nuclear decommissioning trust contributions

DTE Electric contributions to the Fermi 2 Qualified Fund are deductible from the State of Michigan Corporate Income Tax (CIT). DTE contributions to the Fermi 2 Nonqualified Fund are not deductible and would be subject to the Michigan CIT at the statutory rate of 6%. For both the Fermi 2 Qualified Fund and Fermi 2 Nonqualified Fund, any realized investment gains are subject to the Michigan CIT rate of 6%.

Historical taxes

Table 9.7 provides the total tax expenses for the Fermi 2 NDT and its component Funds. The 2019 tax expenses total approximately \$9.9 million with approximately \$9.5 million attributable to the Fermi 2 Qualified Fund and \$0.4 million attributable to the Fermi 2 Nonqualified Fund.

Table 9.7 Fermi 2 NDT 2015 – 2019 tax expenses in thousands of dollars

	2015	2016	2017	2018	2019
Fermi 2 Qualified Fund	11,567	9,115	17,635	18,346	9,470
Fermi 2 Nonqualified Fund	448	0	617	1,522	448
Fermi 2 Nuclear Decommissioning Trust	12,015	9,115	18,252	19,868	9,918

Other administrative and incidental expenses

Table 9.8 provides the other administrative and incidental costs, excluding taxes. The 2019 NDT administrative and incidental expenses totaled approximately \$4.4 million and included expenses for the investment manager, custody expense and professional fees. The 2019 expenses were approximately 0.3% of the Fermi 2 NDT's ending balance.

Table 9.8 Fermi 2 NDT 2015 – 2019 administrative and incidental expenses, excluding taxes, in thousands of dollars

	2015	2016	2017	2018	2019
Fermi 2 Qualified Fund	3,223	3,161	3,819	4,038	4,099
Fermi 2 Nonqualified Fund	345	349	316	385	301
Fermi 2 Nuclear Decommissioning Trust	3,568	3,511	4,136	4,423	4,401

We incorporate the administrative and incidental expenses into the Fermi 2 NDT fund earnings rate discussed in the next section.

9.7 Fund earnings rate

As we have discussed earlier in this section, the Fermi 2 NDT fund earnings rate or rate of return on assets used in our nuclear decommissioning funding adequacy model is another important factor to consider. The requirement of the Settlement Agreement in Michigan Public Service Commission (MPSC) Case No. U-6150 (Re-opened), dated August 26, 1986 states DTE Electric is to include a review of fund earnings rate in DTE Electric's report on the adequacy of the annual Nuclear Decommissioning Funding provision; we intend this section to satisfy the fund earning rate review.

For this section we discuss the projected rate of return for the Fermi 2 Nuclear Decommissioning Trust (NDT) assets where we begin with a review of the incumbent rate of return approved by the Michigan Public Service Commission (MPSC or Commission) and then transition to a historical perspective of actual rate of return experienced by the NDT assets. We close



out the section with a recommendation for a reasonable and prudent going-forward rate of return on NDT assets for use in this funding adequacy review.

Throughout this section we analyze different types of rates of return where:

- Pre-tax rate of return is the return on assets prior to the accounting for taxes the investor must pay.
- Post-tax rate of return is the return on assets after the investor pays taxes. The reader may also think of post-tax rate of return as the effective return after accounting for administrative and incidental expenses and taxes.

DTE Electric rate of return methodology

DTE Electric's existing nuclear decommissioning funding adequacy model uses an effective rate of return of 7%. The Michigan Public Service Commission last approved the use of this 7% rate of return model in Order U-20162, dated May 2, 2019; the MPSC has consistently approved the use of the 7% after-tax rate of return since Commission Order U-10102, dated January 21, 1994.

The use of the 7% fund earnings rate was reasonable and prudent for use in the funding adequacy model because of its use in conjunction with 6% cost escalation to produce a modeled net 1% real rate of return.

DTE Electric historical NDT rate of return

Regulatory guidance for nuclear decommissioning trust fund earnings rate

Nuclear Regulatory Commission (NRC) regulations at 10 CFR 50.75(e)(1)(ii) allows licensees to take credit for projected earnings on external sinking funds of up to two percent annual real rate of return.

DTE Electric-specific nuclear decommissioning trust fund earnings rate

Table 9.9 provides the Fermi 2 NDT's *pre-tax* rates of return dating back to 1989, approximately the trust's inception. DTE Electric does not calculate annual post-tax rates of return. DTE Electric calculated rates of returns for the two separate accounts (Fermi 2 Qualified Fund and Fermi 2 Nonqualified Fund) starting in 2009.



Table 9.9 Historical Fermi 2 Nuclear Decommissioning Trust (NDT) pre-tax rate of return (%).
Individual Fund (Qualified or Nonqualified) pre-tax rates of return are unavailable prior to 2009.

	1989
Nuclear Decommissioning Trust	8.1
Fermi 2 Qualified Fund	
Fermi 2 Nonqualified Fund	

	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999
Nuclear Decommissioning Trust	7.2	10.8	7.2	7.6	1.2	11.5	12.4	21.6	18.4	10.5
Fermi 2 Qualified Fund										
Fermi 2 Nonqualified Fund										

	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
Nuclear Decommissioning Trust	-2.3	-2.1	-7.2	18.0	8.8	3.7	10.5	7.8	-18.8	17.4
Fermi 2 Qualified Fund										18.3
Fermi 2 Nonqualified Fund										12.8

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Nuclear Decommissioning Trust	12.2	1.2	11.2	14.2	4.9	-0.5	7.1	15.6	-5.6	21.0
Fermi 2 Qualified Fund	12.2	-0.7	11.9	17.0	4.2	-1.1	8.1	17.1	-6.5	22.8
Fermi 2 Nonqualified Fund	3.8	11.4	6.8	-2.3	10.1	3.9	1.0	5.2	1.7	6.9

Figure 9.7 illustrates the concept of a Fermi 2 NDT index as a tool to visualize the compound effect of pre-tax rates of return throughout the history of the NDT. The NDT's pre-tax 31-year compound rate of return approximately is 7.2% dating back to NDT's inception; however, the NDT's pre-tax 2000-2019 20-year compound rate of return was only approximately 5.4% resulting from the impacts of the Dot-com Bust of the late 1990s and the Housing Market Bust of the late 2000s..



Figure 9.7 Fermi 2 Nuclear Decommissioning Trust (NDT) index

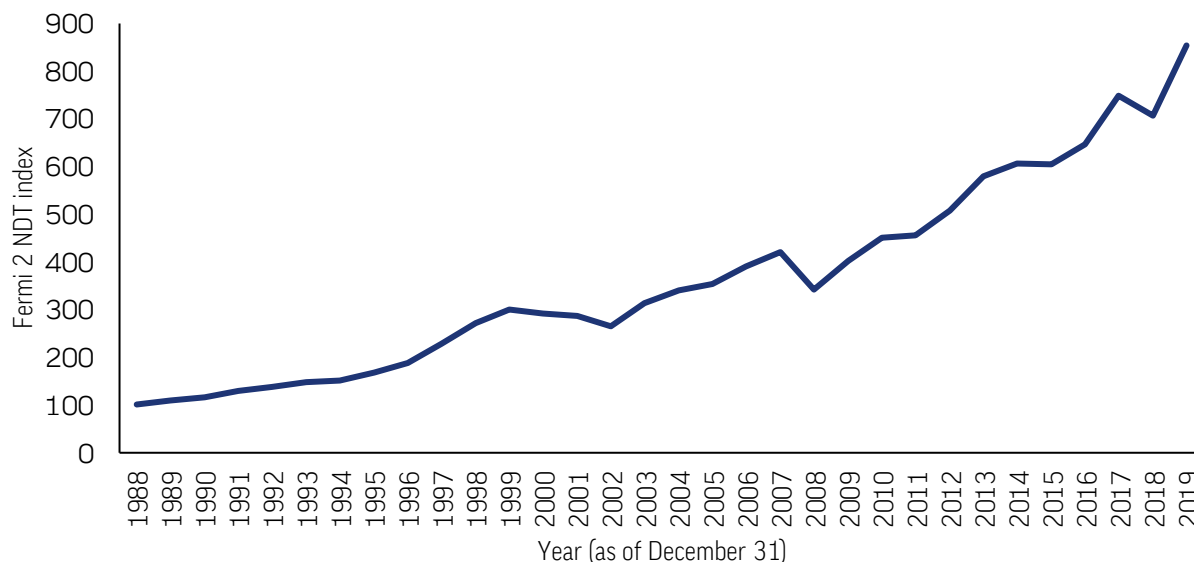


Table 9.10 provides the Fermi 2 NDT fund balances as of December 31, 2013 through 2019 to provide an understanding of the effects of line items such as fund contributions (from the annual Nuclear Decommissioning Funding provision of the Nuclear Surcharge), dividends, administrative and incidental expenses and taxes have had on the NDT balance. Please recall the NDT balances also include the LLRW Fund whose amounts are for the exclusive use of LLRW disposal during Fermi 2 Power Plant operations. DTE Electric used the December 31, 2013 NDT balance of approximately \$1.188 billion in rate case U-17767 to support the current annual Nuclear Decommissioning Funding provision of the Nuclear Surcharge of approximately \$2.9 million. The NDT balance has grown at a compound annual effective rate of approximately 4.9% over this time, factoring contributions.

Table 9.10 Fermi 2 NDT balances (in billions) as of December 31, 2013 – 2019

	2013	2014	2015	2016	2017	2018	2019
Fermi 2 Qualified Fund	1.045	1.081	1.068	1.147	1.323	1.220	1.488
Fermi 2 Nonqualified Fund	0.144	0.156	0.166	0.171	0.166	0.155	0.171
Fermi 2 NDT balance	1.188	1.238	1.234	1.318	1.489	1.375	1.658

U.S. market trends

We provide historical context of general U.S. market trends using common indexes for U.S. equities and bonds. The S&P 500 (SPX) is a stock market index that measures the performance of 500 large companies in U.S. stock exchanges. The U.S. Aggregate Bond Index (LBUSTRUU) is a bond market index that represents investment grade bonds traded in the U.S. and is a proxy for the U.S. bond market performance. The investments of NDT assets serve the purpose of providing reasonable assurance of funding for the nuclear decommissioning of the Fermi 2 Power Plant, which is a different purpose than the two indexes mentioned; we provide these indexes only to show general trends in the U.S. markets and not to provide a benchmark for NDT performance.



Figure 9.8 trends the closing values of S&P 500 Index (SPX) from December 31, 1985 through March 30, 2020; we also note a few key milestone dates in recent investment history. The growth of this index is volatile where the compound annual growth rate from December 31, 1985 through 2019 was approximately 8.4% and approximately 4.0% from December 31, 1999 through 2019 and approximately 11.2% from December 31, 2009 through 2019. We also note the SPX had fallen approximately 20% between December 31, 2019 and March 31, 2020 due to COVID-19 uncertainty.

Figure 9.8 S&P 500 index (SPX) December 31, 1985 through March 31, 2020

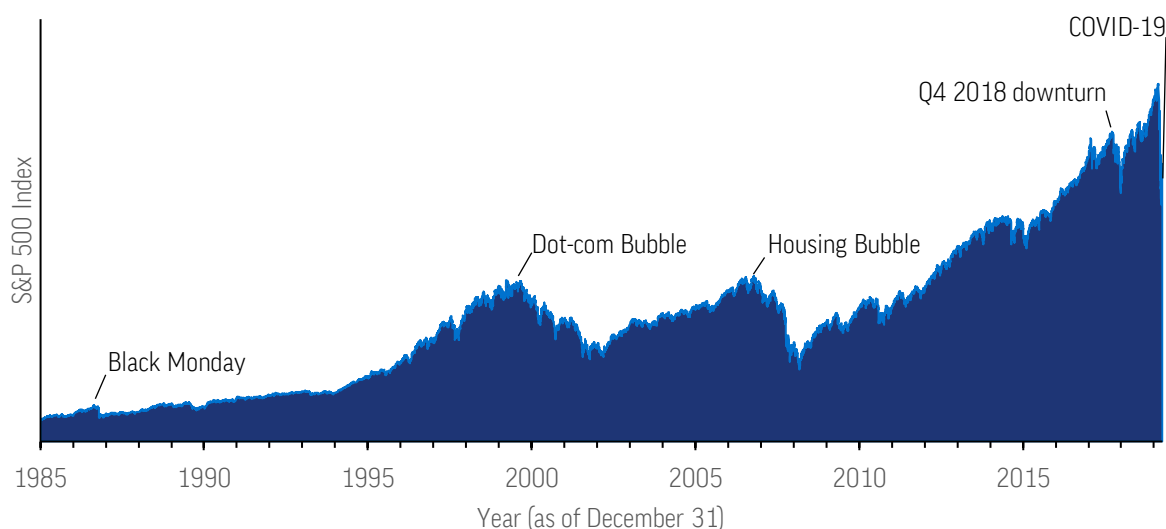
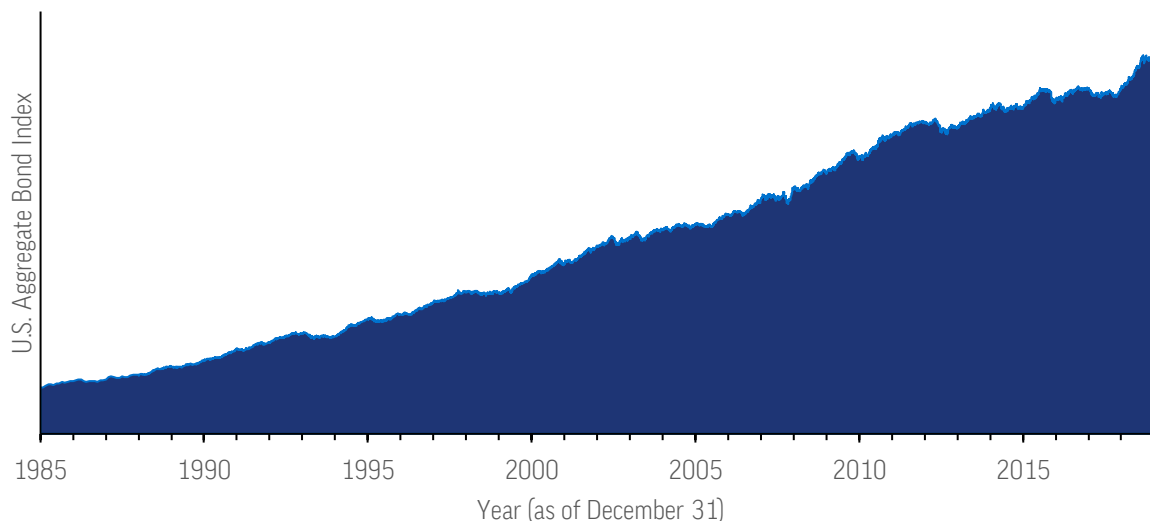


Figure 9.9 trends the closing values of the U.S. Aggregate Bond Index (LBUSTRUU) from December 31, 1985 through March 30, 2020. The growth of this bond index has been more consistent but deaccelerating growth over time than the SPX where the compound annual growth rate from December 31, 1985 through 2019 was approximately 6.4% and approximately 5.0% from December 31, 1999 through 2019 and approximately 3.7% from December 31, 2009 through 2019.



Figure 9.9 U.S. Aggregate Bond Index (LBSTRUU) December 31, 1985 through March 31, 2020



DTE Electric forecasted rate of return

Note: This discussion involves “forward looking statements” developed by DTE Electric in February of 2020.

The continued use of the 7% as the long-term after-tax effective return on NDT investments would not be reasonable or prudent with the understanding the NDT fund earnings rate was only approximately 7.2% on a *pre-tax* basis since approximately trust inception and only approximately 5.4% on a *pre-tax* basis since 1999.

As a lower post-tax effective rate of return on NDT assets would be reasonable and prudent, DTE Trust Investments performed an analysis to recommend a new after-tax rate return to apply to NDT assets. As an outcome of their analysis, DTE Trust Investments recommends a projected after-tax effective rate of return range of 4.8% to 4.9% for the Fermi 2 Qualified Fund and 2.4% to 3.0% for the Fermi 2 Nonqualified Fund as a reasonable and prudent for the purposes of forecasting the NDT balances through December 31, 2044; DTE Trust Investments calculates the total Fermi 2 NDT after-tax effective rate of return as a range of 4.6% to 4.8% through December 31, 2044.

Our recommendation is to use a 4.8% after-tax effective rate of return factor on NDT assets in this study’s nuclear decommissioning funding adequacy model. The 4.8% rate is reasonable and prudent considering the allocation of NDT assets between equities and fixed income investments, historical performance of the NDT assets and the analysis of DTE Trust Investments.



9.8 Net rate of return

The Michigan Public Service Commission (MPSC or Commission) has supported a 1% differential for the rate of inflation (cost escalation) and the fund earns rate (after-tax effective rate of return on assets) dating back to 1994 and MPSC Case No. U-10102, dated January 21, 1994. We believe this 1% differential has a demonstrated record of reasonableness and prudence.

We recommend maintaining the 1% differential between the modeled rate of inflation and after-tax rate of return with reduction in the rate of inflation from the existing 6% to 3.8% and a corresponding reduction to the after-tax effective rate of return from 7% to 4.8% as a reasonable and prudent measure. The 1% net earnings differential is consistent with DTE Electric's most recent funding adequacy reports submitted to the Nuclear Regulatory Commission.

9.9 Adequacy of annual provision

In this section we complete the calculations within the funding adequacy model we presented in Section 9.1 to determine an adequate annual Nuclear Decommissioning Funding provision for the Nuclear Surcharge. We will also use the model to determine the required allocation of the annual provision to each of the Nuclear Decommissioning Fund's individual plans, subject to DTE Electric's regulatory filings.

The Michigan Public Service Commission established the amount of the existing annual provision for the Fermi 2 Nuclear Decommissioning Funding portion of the Nuclear Surcharge at approximately \$2.9 million in MPSC Order U-17767, December 11, 2015. The Commission explicitly reaffirmed in the \$2.9 million annual provision in MPSC Order U-20261, date May 2, 2019.

Total annual Nuclear Decommissioning Funding provision

We complete our simple model to calculate the required annual Nuclear Decommissioning Funding provision of the Nuclear Surcharge NS_{NDF}

$$NS_{NDF} = (SDCE_f - NDT_f) \left(\frac{i}{(1+i)^t - 1} \right) \sim (\$5,536.4 - \$5,327.5) \left(\frac{0.048}{(1+0.048)^{25} - 1} \right) \sim \$4.5, \quad 9.2$$

where,

$SDCE_t$ of \$5,536.4 million is the Fermi 2 site-specific cost estimate (SDCE) at the time of expected shut down (i.e. December 31, 2044), given no contributions where

$$SDCE_f = SDCE_0(1+r)^t \sim \$2,179.2(1+0.038)^{25} \sim \$5,536.4 \text{ where}$$

$SDCE_0$ is the Fermi 2 site-specific cost estimate (SDCE) at time of reference for the review (i.e. December 31, 2019) of approximately \$2,179.2 million

r is the projected rate of escalation of 3.8% from starting date t_0 to time of expected shutdown t_i

t is the time t in years growth of 25 years from starting date t_0 to time of expected shutdown t_i

NDT_t of \$5,327.5 million is the balance of Fermi 2 Nuclear Decommissioning Fund at the time of expected shutdown (i.e. December 31, 2044), assuming no contributions where

$$NDT_f = NDT_0(1+i)^t \sim \$1,650.0(1+0.048)^{25} \sim \$5,327.5 \text{ where}$$



NDT_0 is the balance of Fermi 2 Nuclear Decommissioning Fund at time of reference for the review (i.e. December 31, 2019) of approximately \$1,650.0 million
 i is the projected effective rate of growth of 4.8% from starting date t_0 to time of expected shutdown t_f
 t is the time t in years growth of 25 years from starting date t_0 to time of expected shutdown t_f

Our model projects the required annual Nuclear Decommissioning Funding provision of the Nuclear Surcharge as approximately \$4.5 million. The existing annual provision for the Fermi 2 Nuclear Decommissioning Funding portion of the Nuclear Surcharge at approximately \$2.9 million, which is approximately \$1.6 million per year less than the approximately \$4.5 million calculated here.

We understand from Section 9.4 the two funds partially funded nuclear decommissioning funds are ISFSI decommissioning and spent fuel management and site restoration while the license termination is essentially fully funded. Within this context, we must investigate the reasonableness and prudence of potentially recommending an approximately \$1.6 million increase in the annual Nuclear Decommissioning Funding provision.

Minimum annual Nuclear Decommissioning Funding provision allocation to license termination

The Fermi 2 licensing termination funding is approximately 99% funded with a shortfall of approximately \$7.8 million, meaning – if the Fermi 2 Power Plant were to have permanently ceased operations on December 31, 2019 the Fermi 2 NDT was sufficiently funded to reasonably assure DTE Electric could complete its regulatory obligations to radiologically decommission and terminate the license of the Fermi 2 Power Plant without additional contributions to the NDT. The existing annual Nuclear Decommissioning Funding contribution to license termination (radiological decommissioning) is approximately \$12; however; applying our simple model to only the license termination decommissioning funding, we calculate the required annual contributions to the license termination fund as approximately (\$22.6) million:

$$NS_{Rad} = (SDCE_{Rad_f} - NDT_{Rad_f}) \left(\frac{i}{(1+i)^t - 1} \right) \sim (\$3,962.2 - \$5,010.2) \left(\frac{0.048}{(1+0.048)^{25} - 1} \right) \sim (\$22.6), \quad 9.3$$

where,

$SDCE_{Rad_f}$ of \$3,962.2 million is the Fermi 2 site-specific cost estimate (SDCE) for radiological decommissioning (license termination) at the time of expected shut down (i.e. December 31, 2044), given no contributions where

$$SDCE_{Rad_f} = SDCE_{Rad_0} (1 + r)^t \sim \$1,559.6 (1 + 0.038)^{25} \sim \$3,5962.2 \text{ where}$$

$SDCE_{Rad_0}$ is the Fermi 2 site-specific cost estimate (SDCE) for radiological decommissioning at time of reference for the review (i.e. December 31, 2019) of approximately \$1,559.6 million
 r is the projected rate of escalation of 3.8% from starting date t_0 to time of expected shutdown t_f
 t is the time t in years growth of 25 years from starting date t_0 to time of expected shutdown t_f

NDT_{Rad_f} of \$5,010.2 million is the balance of Fermi 2 radiological decommissioning (license termination) fund at the time of expected shutdown (i.e. December 31, 2044), assuming no contributions where

$$NDT_{Rad_f} = NDT_{Rad_0} (1 + i)^t \sim \$1,551.7 (1 + 0.048)^{25} \sim \$5,010.2 \text{ where}$$

NDT_{Rad_0} is the balance of Fermi 2 radiological decommissioning fund at time of reference for the review (i.e. December 31, 2019) of approximately \$1,551.7 million
 i is the projected effective rate of growth of 4.8% from starting date t_0 to time of expected shutdown t_f



t is the time t in years growth of 25 years from starting date t_0 to time of expected shutdown t_f

With a negative annual funding requirement, we have reasonable assurance of adequate funding for the regulatory required radiological decommissioning and license termination aspects of the Fermi 2 Power Plant nuclear decommissioning expected to start in 2045. Also, with a negative annual funding requirement associated with these funds, we understand that one or more other aspects of the nuclear decommissioning will have the offsetting positive requirement.

We recommend maintaining the existing annual Nuclear Decommissioning Funding contributions of \$12 per year to license termination funds, specifically the Fermi Qualified MPSC fund. While the Fermi 2 Power Plant continues to operate, we believe it is reasonable and prudent to maintain the Fermi 2 NDT's *external sinking fund* status and the Nuclear Surcharge's a *non-bypassable* charge status with the Nuclear Regulatory Commission (NRC). The Nuclear Decommissioning Funding provision of the Nuclear Surcharge must contribute at least \$1 per month to the Fermi 2 NDT to maintain the status which is consistent with DTE Electric's most recent Fermi 2 decommissioning funding assurance filing with the NRC.

Minimum annual Nuclear Decommissioning Funding provision allocation to ISFSI decommissioning

The Fermi 2 ISFSI decommissioning funding is approximately 38% funded with a shortfall of approximately \$7.8 million. The existing annual Nuclear Decommissioning Funding contributions to ISFSI decommissioning are approximately \$0.38 million. Applying our simple model to only the ISFSI decommissioning funding, we calculate the required annual contributions to the ISFSI fund as approximately \$0.354 million:

$$NS_{ISFSI} = (SDCE_{ISFSI_f} - NDT_{ISFSI_f}) \left(\frac{i}{(1+i)^t - 1} \right) \sim (\$32.03 - \$15.57) \left(\frac{0.048}{(1+0.048)^{25} - 1} \right) \sim \$0.35, \quad 9.3$$

where,

$SDCE_{ISFSI_f}$ of \$32.03 million is the Fermi 2 site-specific cost estimate (SDCE) for ISFSI decommissioning at the time of expected shut down (i.e. December 31, 2044), given no contributions where

$$SDCE_{ISFSI_f} = SDCE_{ISFSI_0} (1 + r)^t \sim \$12.61 (1 + 0.038)^{25} \sim \$32.03 \text{ where}$$

$SDCE_{ISFSI_0}$ is the Fermi 2 site-specific cost estimate (SDCE) for ISFSI decommissioning at time of reference for the review (i.e. December 31, 2019) of approximately \$2,179.2 million

r is the projected rate of escalation of 3.8% from starting date t_0 to time of expected shutdown t_f

t is the time t in years growth of 25 years from starting date t_0 to time of expected shutdown t_f

NDT_{ISFSI_f} of \$15.57 million is the balance of Fermi 2 ISFSI fund at the time of expected shutdown (i.e. December 31, 2044), assuming no contributions where

$$NDT_{ISFSI_f} = NDT_{ISFSI_0} (1 + i)^t \sim \$4.82 (1 + 0.048)^{25} \sim \$15.73 \text{ where}$$

NDT_{ISFSI_0} is the balance of Fermi 2 ISFSI fund at time of reference for the review (i.e. December 31, 2019) of approximately \$4.82 million

i is the projected effective rate of growth of 4.8% from starting date t_0 to time of expected shutdown t_f

t is the time t in years growth of 25 years from starting date t_0 to time of expected shutdown t_f

We recommend maintaining the existing Nuclear Decommissioning Funding contributions to ISFSI decommissioning at approximately \$0.38 million. At \$0.38 million per year, there would be reasonable assurance of the minimum contributions of approximately \$0.35 million in a scenario of lower-than-expected electricity sales. DTE Electric is providing the NRC with



reasonable assurance of ISFSI decommissioning funding using the Fermi 2 ISFSI qualified fund as an external sinking fund as described in 10 CFR 72.20(e)(5) and 10 CFR 50.75(e)(1)(i)(A) and (B). We find it essential to continue the annual Nuclear Decommissioning Funding contributions to the ISFSI fund to maintain the existing method of providing financial assurance; otherwise, NRC regulations would require DTE Electric to change the funding assurance method to another option available under 10 CFR 72.30(e) such as prepayment method.

Minimum annual Nuclear Decommissioning Funding provision allocation to other decommissioning

The Fermi 2 other decommissioning funding (spent fuel management and site restoration) is approximately 15% funded with a shortfall of approximately \$513.5 million. The existing annual Nuclear Decommissioning Funding contributions to other decommissioning funds is approximately \$2.5 million.

Applying our simple model to only the other decommissioning funding, we calculate the required annual contributions to the other decommissioning funds as approximately \$26.7 million:

$$NS_{Other} = (SDCE_{Other_f} - NDT_{Other_f}) \left(\frac{i}{(1+i)^t - 1} \right) \sim (\$1,542.1 - \$301.7) \left(\frac{0.048}{(1+0.048)^{25} - 1} \right) \sim \$26.7 \quad 9.3$$

where,

$SDCE_{Other_f}$ of \$1,542.1 million is the Fermi 2 site-specific cost estimate (SDCE) for other decommissioning at the time of expected shut down (i.e. December 31, 2044), given no contributions where

$$SDCE_{Other_f} = SDCE_{Other_0} (1 + r)^t \sim \$0.607(1 + 0.038)^{25} \sim \$1,542.1 \text{ where}$$

$SDCE_{Other_0}$ is the Fermi 2 site-specific cost estimate (SDCE) for other decommissioning at time of reference for the review (i.e. December 31, 2019) of approximately \$607.0 million
 r is the projected rate of escalation of 3.8% from starting date t_0 to time of expected shutdown t_f
 t is the time t in years growth of 25 years from starting date t_0 to time of expected shutdown t_f

NDT_{Other_f} of \$301.7 million is the balance of Fermi 2 Other Decommissioning funds at the time of expected shutdown (i.e. December 31, 2044), assuming no contributions where

$$NDT_{Other_f} = NDT_{Other_0} (1 + i)^t \sim \$93.4(1 + 0.048)^{25} \sim \$301.7 \text{ where}$$

NDT_{Other_0} is the balance of Fermi 2 ISFSI fund at time of reference for the review (i.e. December 31, 2019) of approximately \$93.4 million
 i is the projected effective rate of growth of 4.8% from starting date t_0 to time of expected shutdown t_f
 t is the time t in years growth of 25 years from starting date t_0 to time of expected shutdown t_f

Annual contributions of approximately \$26.7 million to other decommissioning would be reasonable and prudent to provide reasonable assurance of funding for other decommissioning activities such as spent fuel management and site restoration; however, we know from the previous calculations that the radiological decommissioning and license termination funds have an offsetting negative funding requirement which tempers the requirement to increase contributions to other decommissioning funds to the approximately \$26.7 million as we understand that proposed NRC rulemaking [NRC-2015-0070] would change NRC regulations to allow the use of license termination funds to offset spent fuel management costs.

Because we have a reasonable understanding that NRC regulations will allow for the eventual use of license termination funds for the purpose of post-shutdown spent fuel management, we recommend maintaining the existing annual Nuclear



Decommissioning Funding contributions of approximately \$2.5 million to the other decommissioning funds, specifically the Fermi 2 Qualified Other Decommissioning fund.

Recommended annual Nuclear Decommissioning Funding provision and contributions

We recommend maintaining the annual Nuclear Decommissioning Funding provision of the Nuclear Surcharge as approximately \$2.9 million as provided in Table 9.11. As we have discussed throughout this study, our purpose is to provide reasonable assurance of adequate funding for the nuclear decommissioning (including license termination, ISFSI decommissioning, spent fuel management and site restoration) of the Fermi 2 Power Plant following its cessation of operations in 2045. We summarize the recommended contributions from the approximately \$2.9 million annual Nuclear Decommissioning Funding provisions to individual Fermi 2 NDT plans:

Table 9.11 Recommended annual Nuclear Decommissioning Funding provision and contributions to the Fermi 2 NDT plans

Fermi 2 NDT plan	Purpose	Annual provision (\$)
Qualified MPSC	License termination (radiological decommissioning)	12
ISFSI	ISFSI decommissioning	382,000
Qualified Other Decommissioning	Spent fuel management and site restoration	2,484,549
Nuclear Decommissioning Funding		2,866,561



SECTION TEN

10 Rate making considerations

The nuclear decommissioning of the Fermi 2 Power Plant is an obligation that spans generations. True funding assurance only occurs across generations with supportive public policy provided by the Michigan Public Service Commission (Commission) – supportive public policy that includes the Commission’s long-standing policy that the decommissioning of the Fermi 2 Power Plant should be an obligation of all customers and applied through the non-bypassable Nuclear Surcharge, use of external trust to place the assets outside the normal control of contemporary concerns and rate orders, laws and regulations that ensure the proper audit and end use of the assets for their original purpose. The purpose of this study is to provide the Commission with information and context DTE Electric believes to be relevant to continued supportive public policy.

We recommend the Commission:

1. Approve the continued collections of the annual Nuclear Decommissioning Funding provision of the Nuclear Surcharge at approximately \$2.9 million per year as a reasonable and prudent request. This recommendation is no change to the status quo.
2. Approve of DTE Electric’s Fermi 2 site-specific decommissioning cost estimate (SDCEC) of approximately \$1.179 billion in 2019 dollars as a reasonable and prudent SDCE.
3. Approve of DTE Electric’s use of 3.8% as the compound escalation rate for the Fermi 2 nuclear decommissioning costs as a reasonable and prudent forecast. This recommendation will change the status quo of 6%.
4. Approve of DTE Electric’s use of 4.8% as the compound *post-tax* growth rate for the amounts in the Fermi 2 Nuclear Decommissioning Trust (NDT). This recommendation will change the status quo of 7%.
5. Find this study completes DTE Electric’s obligation to present the Commission with an updated Fermi 2 nuclear decommissioning study, as ordered in Case No. U-20162 on May 2, 2019.



SECTION 11

11 References

We have provided in this study a comprehensive list of references. Reference citations are denoted by “[]” and are listed as applicable in the Reference section of this study. Citations Nuclear Regulatory Committee (NRC) Agencywide Documents Access and Management System (ADAMS) official record keeping system start with “ML” and are not relisted in this section of this study. Citations of the Federal Register use the format “## FR #####,” for example: “55 FR 38474” as in, starting page 38474 of Volume 55 of the U.S. Federal Register and are not relisted in section.

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	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
Line No.	Organization	Number of Incumbents	Company's Average Base Salary	Average Market Base Salary	Percent Company More/(Less) Than Market	Company's Average Total Cash	Average Market Total Cash	Percent Company More/(Less) Than Market	Percent Company Base More/(Less) Than Total Cash Market
1	Administration	158	\$ 65,408	\$ 66,213	(1.2%)	\$ 69,401	\$ 69,232	0.2%	(5.5%)
2	Communications	45	\$ 98,661	\$ 96,358	2.4%	\$ 110,482	\$ 105,944	4.3%	(6.9%)
3	Continuous Improvement	69	\$ 133,708	\$ 126,320	5.8%	\$ 150,891	\$ 143,233	5.3%	(6.7%)
4	Corporate & Government Affairs	41	\$ 118,595	\$ 113,715	4.3%	\$ 133,758	\$ 131,251	1.9%	(9.6%)
5	Customer Service	299	\$ 83,997	\$ 85,084	(1.3%)	\$ 92,309	\$ 93,293	(1.1%)	(10.0%)
6	Electrical Distribution	449	\$ 110,474	\$ 109,938	0.5%	\$ 122,254	\$ 124,201	(1.6%)	(11.1%)
7	Engineering	497	\$ 104,950	\$ 113,708	(7.7%)	\$ 115,160	\$ 124,398	(7.4%)	(15.6%)
8	Executive	34	\$ 386,353	\$ 395,000	(2.2%)	\$ 610,775	\$ 590,153	3.5%	(34.5%)
9	Facilities	38	\$ 99,804	\$ 102,058	(2.2%)	\$ 110,919	\$ 113,329	(2.1%)	(11.9%)
10	Finance	371	\$ 110,470	\$ 104,637	5.6%	\$ 123,334	\$ 117,397	5.1%	(5.9%)
11	Fossil Generation	416	\$ 111,996	\$ 112,583	(0.5%)	\$ 124,123	\$ 126,168	(1.6%)	(11.2%)
12	Human Resources	157	\$ 110,617	\$ 110,045	0.5%	\$ 123,832	\$ 121,950	1.5%	(9.3%)
13	Information Technology	511	\$ 117,334	\$ 113,366	3.5%	\$ 129,726	\$ 125,663	3.2%	(6.6%)
14	Legal	37	\$ 163,119	\$ 160,416	1.7%	\$ 186,947	\$ 193,011	(3.1%)	(15.5%)
15	Marketing	241	\$ 109,395	\$ 107,968	1.3%	\$ 121,041	\$ 122,023	(0.8%)	(10.3%)
16	Nuclear Generation	416	\$ 118,085	\$ 118,666	(0.5%)	\$ 129,957	\$ 138,086	(5.9%)	(14.5%)
17	Supply Chain	121	\$ 100,700	\$ 96,272	4.6%	\$ 111,765	\$ 106,570	4.9%	(5.5%)
18									
19									
20	Total	3,900		Weighted Avg.	0.1%		Weighted Avg.	(0.8%)	(11.5%)
21									
22	Incumbents without a job match	26							
23									
24	Total Incumbents	3,926							
25									
26	Proportion of jobs matched	99.3%							

Michigan Public Service Commission
DTE Electric Company
2021 Annual Incentive Plan and Rewarding Employees Plan Metrics:
DTE Electric Company

Case No.: U-20836
Exhibit: A-21
Schedule: K2
Witness: M. S. Cooper
Page: 1 of 1

Line	(a)	(b)	(c)	(d)	(e)	(f)	(g)
No.	Category	Measure	Weight		Threshold	Target	Maximum
			AIP	REP			
1	Financial Performance						
2		DTE Electric Earnings	15.00%	15.00%	\$814	\$840	\$882
3							
4		DTE Electric Cash From Operations	15.00%	15.00%	\$1,775	\$1,972	\$2,169
5							
6		DTE Energy Operating Earnings Per Share	10.00%	10.00%	\$6.17	\$6.47	\$6.77
7							
8	Total Financial Measures		40.00%	40.00%			
9							
10	Customer Satisfaction						
11		Net Promoter Score	12.00%	12.00%	43%	45%	47%
12							
13		MPSC Customer Complaints	8.00%	8.00%	1,967	1,905	1,760
14							
15			20.00%	20.00%			
16	Safety & Engagement						
17		DTE Electric Employee Engagement-Gallup	5.00%		4.18	4.32	4.43
18							
19		DTE Electric OSHA Recordable Incident Rate	5.00%	7.50%	0.84	0.67	0.50
20							
21		DTE Electric OSHA DART Rate	5.00%	7.50%	0.39	0.32	0.20
22							
23			15.00%	15.00%			
24	Operating Excellence						
25		Fossil Power Plant Reliability (ROF)	5.00%	5.00%	7.8%	6.8%	5.8%
26							
27		Blue Sky CAIDI	5.00%	5.00%	108	103	93
28							
29		SAIDI Excluding Major Event Days	10.00%	10.00%	149	137	123
30							
31		Nuclear On-Line Unit Capacity Factor (UCF)	5.00%	5.00%	97.6%	98.5%	98.8%
32							
33			25.00%	25.00%			
34							
35	Total Operating Measures		60.00%	60.00%			
36							
37	Total		100.00%	100.00%			

Michigan Public Service Commission
DTE Electric Company
2021 Annual Incentive Plan and Rewarding Employees Plan Metrics:
Nuclear Generation

Case No.: U-20836
Exhibit: A-21
Schedule: K3
Witness: M. S. Cooper
Page: 1 of 1

Line No.	(a) Category	(b) Measure	(c) Weight		(e) Threshold	(f) Target	(g) Maximum
			AIP	REP			
1	Financial Performance						
2		DTE Electric Operating Earnings	5.00%	5.00%	\$814	\$840	\$882
3							
4		Nuclear Operations & Maintenance Expense	15.00%	15.00%	\$140	\$133	\$126
5							
6	Total Financial Measures		20.00%	20.00%			
7							
8	Safety & Engagement						
9		Nuclear Generation Employee Engagement-Gallup	5.00%		4.18	4.32	4.45
10							
11		Nuclear Total Industrial Safety Accident Events	5.00%	7.50%	1 Incident	N/A	0 Incidents
12							
13		DTE Electric OSHA Recordable Incident Rate	5.00%	7.50%	0.84	0.67	0.50
14							
15			15.00%	15.00%			
16							
17	Operating Excellence						
18		Nuclear On-Line Unit Capabability Factor	25.00%	25.00%	97.6%	98.5%	98.8%
19							
20		Nuclear Operational Focus	15.00%	15.00%	82.8	86.2	91.0
21							
22		Nuclear Annualized Work Management Index	15.00%	15.00%	96.5%	99.4%	100.0%
23							
24		Nuclear Radiation Protection Index	10.00%	10.00%	75.0%	77.5%	80.0%
25							
26			65.00%	65.00%			
27							
28	Total Operating Measures		80.00%	80.00%			
29							
30	Total		100.00%	100.00%			

Michigan Public Service Commission
DTE Electric Company
2021 Annual Incentive Plan and Rewarding Employees Plan Metrics:
DTE Energy Corporate Services LLC

Case No.: U-20836
Exhibit: A-21
Schedule: K4
Witness: M. S. Cooper
Page: 1 of 1

Line No.	(a) Category	(b) Measure	(c) Weight AIP	(d) Weight REP	(e) Threshold	(f) Target	(g) Maximum
1	Financial Performance						
2		DTE Energy Operating Earnings Per Share	20.00%	20.00%	\$6.17	\$6.47	\$6.77
3							
4		DTE Energy Cash From Operations	20.00%	20.00%	\$2,736	\$3,040	\$3,344
5							
6	Total Financial Measures		40.00%	40.00%			
7							
8	Customer Satisfaction						
9		Net Promoter Score	12.00%	12.00%	43%	45%	47%
10							
11		MPSC Customer Complaints	8.00%	8.00%	1,967	1,905	1,760
12							
13			20.00%	20.00%			
14	Safety & Engagement						
15		DTE Energy Employee Engagement-Gallup	5.00%		4.18	4.32	4.43
16							
17		DTE Energy OSHA Recordable Incident Rate	5.00%	7.50%	0.79	0.64	0.46
18							
19		DTE Energy OSHA DART Rate	5.00%	7.50%	0.39	0.32	0.20
20							
21			15.00%	15.00%			
22	Operating Excellence:Electric						
23		Fossil Power Plant Reliability (ROF)	6.25%	6.25%	7.8%	6.8%	5.8%
24							
25		SAIDI Excluding Major Event Days	6.25%	6.25%	149	137	123
26							
27		Nuclear On-Line Unit Capacity Factor (UCF)	12.50%	12.50%	97.6%	98.5%	98.8%
28							
29			25.00%	25.00%			
30							
31	Total Operating Measures		60.00%	60.00%			
32							
33	Total		100.00%	100.00%			

(a)		(b)	(c)	(d)	(e)	(f)	(g)
Line No.	Description	Weight			Threshold	Target	Maximum
		DTE Electric	Nuclear	DTE Energy Corporate Services LLC			
1	Total DTE Energy Shareholder Return vs. Peer Group	80%	20%	80%	25th Percentile	50th Percentile	75th Percentile
2	DTE Electric Return on Equity	20%		20%	Board Approved	Board Approved	Board Approved
3	INPO Index		60%		94.6	96.3	98.1
4	On-Line UCF		20%		97.8	98.5	98.8
5	Total	100%	100%	100%			

Line No.	(a) Description	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		Long Term Incentive Plan			Annual Incentive Plan			Rewarding Employees Plan			Total Incentive Plans		
		Expense	Benefit	Net	Expense	Benefit	Net	Expense	Benefit	Net	Expense	Benefit	Net
1	Financial Performance												
2	Total DTE Energy Shareholder Return vs. Peer Group	17,643		17,643							17,643		17,643
3													
4	DTE Electric Return on Equity	4,344		4,344							4,344		4,344
5													
6	Nuclear Operations & Maintenance Expense				67		67	627		627	694		694
7													
8	DTE Electric Operating Earnings				302		302	2,900		2,900	3,202	0	3,202
9													
10	DTE Electric Cash From Operations				2,382	(8,285)	(5,903)	2,938	(10,219)	(7,281)	5,320	(18,504)	(13,184)
11													
12	DTE Energy Operating Earnings per share				2,568		2,568	4,732		4,732	7,300		7,300
13													
14	DTE Energy Cash From Operations				279		279	2,691		2,691	2,970		2,970
15													
16	Total Financial Measures	21,987	0	21,987	5,598	(8,285)	(2,688)	13,888	(10,219)	3,670	41,473	(18,504)	22,969
17													
18	LTIP Operating Measures												
19	INPO Index	798	(2,023)	(1,225)							798	(2,023)	(1,225)
20													
21	Nuclear On-Line Unit Capacity Factor (UCF)	266	(674)	(408)							266	(674)	(408)
22													
23	Total LTIP Operating Measures	1,064	(2,698)	(1,633)							1,064	(2,698)	(1,633)
24													
25	Customer Satisfaction												
26	Net Promoter Score				1,159	(1,727)	(568)	2,434	(3,626)	(1,192)	3,593	(5,352)	(1,760)
27													
28	MPSC Customer Complaints				773	(73)	699	1,623	(154)	1,468	2,395	(228)	2,167
29													
30	Total Customer Satisfaction Measures				1,931	(1,800)	132	4,056	(3,780)	276	5,988	(5,580)	408
31													
32	Safety & Engagement												
33	Employee Engagement-Gallup				498	(17,629)	(17,132)	528	0	528	1,025	(17,629)	(16,604)
34													
35	OSHA Recordable Incident Rate				483	(551)	(68)	1,257	(1,434)	(177)	1,740	(1,985)	(245)
36													
37	OSHA DART Rate				498	(568)	(70)	1,541	(1,758)	(217)	2,039	(2,326)	(287)
38													
39	Nuclear Total Industrial Safety Accident Events				15	(17)	(2)	284	(324)	(40)	298	(340)	(42)
40													
41	Total Employee Engagement Measures				1,493	(18,765)	(17,272)	3,610	(3,516)	94	5,102	(22,281)	(17,178)
42													
43	Operating Excellence												
44	Fossil Power Plant Reliability (ROF)				590	(555)	34	1,136	(1,069)	66	1,726	(1,625)	101
45													
46	Blue Sky CAIDI				55	(776)	(721)	528	(7,479)	(6,951)	583	(8,255)	(7,672)
47													
48	SAIDI Excluding Major Event Days				645	(9,133)	(8,488)	1,664	(23,568)	(21,905)	2,308	(32,701)	(30,393)
49													
50	Nuclear On-Line Unit Capacity Factor (UCF)				1,198	(3,037)	(1,839)	2,689	(6,816)	(4,127)	3,888	(9,854)	(5,966)
51													
52	Nuclear Operational Focus				44	(111)	(67)	568	(1,439)	(871)	612	(1,550)	(938)
53													
54	Nuclear Annualized Work Management Index				44	(111)	(67)	568	(1,439)	(871)	612	(1,550)	(938)
55													
56	Nuclear Radiation Protection Index				29	(74)	(45)	378	(959)	(581)	408	(1,033)	(626)
57													
58	Total Operating Excellence				2,605	(13,799)	(11,194)	7,530	(42,769)	(35,239)	10,135	(56,567)	(46,433)
59													
60	Total Operating Measures	1,064	(2,698)	(1,633)	6,029	(34,363)	(28,334)	15,196	(50,065)	(34,869)	22,290	(87,126)	(64,836)
61													
62	Total	23,052	(2,698)	20,354	11,627	(42,648)	(31,022)	29,084	(60,284)	(31,199)	63,763	(105,630)	(41,867)

Michigan Public Service Commission
DTE Electric Company
Operating AIP and REP Performance Measure Results
2016 Through 2020

Case No.: U-20836
Exhibit: A-21
Schedule: K7
Witness: M. S. Cooper
Page: 1 of 1

Line No	(a) Description	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
		AIP					Average	REP					Average	Combined Average
		2016	2017	2018	2019	2020		2016	2017	2018	2019	2020		
1	DTE Electric													
2	Less than Threshold	5	2	5	4	3		5	2	5	4	3		
3	Between Threshold and Target	2	1	2	3	0		1	1	2	3	0		
4	Target	0	1	0	0	1		0	1	0	0	1		
5	Between Target and Maximum	3	6	3	4	6		3	4	2	2	5		
6	Maximum	4	4	4	2	4		4	4	4	2	4		
7		14	14	14	13	14		13	12	13	11	13		
8														
9	Sum of Performance for all measures	1,200%	1,675%	1,201%	1,124%	1,544%		1,003%	1,239%	970%	774%	1,273%		
10														
11	Average Performance	85.7%	119.6%	85.8%	86.5%	110.3%	97.6%	77.1%	103.2%	74.6%	70.3%	97.9%	84.6%	
12														
13	Nuclear Generation													
14	Less than Threshold	2	0	1	3	1		2	0	1	3	1		
15	Between Threshold and Target	1	1	1	0	1		1	1	1	0	0		
16	Target	0	1	1	0	1		0	1	1	0	1		
17	Between Target and Maximum	2	2	2	3	1		1	1	1	1	1		
18	Maximum	2	4	2	2	3		2	3	2	2	3		
19		7	8	7	8	7		6	6	6	6	6		
20														
21	Sum of Performance for all measures	640%	1,169%	732%	803%	786%		483%	758%	584%	433%	660%		
22														
23	Average Performance	91.4%	146.1%	104.6%	100.3%	112.3%	111.0%	80.6%	126.4%	97.4%	72.2%	110.0%	97.3%	
24														
25	DTE LLC													
26	Less than Threshold	5	2	5	5	5		5	2	5	5	5		
27	Between Threshold and Target	1	1	2	3	0		1	1	2	3	0		
28	Target	0	1	0	0	1		0	1	0	0	1		
29	Between Target and Maximum	4	5	3	4	6		3	4	2	3	5		
30	Maximum	4	5	4	5	5		4	4	4	4	5		
31		14	14	14	17	17		13	12	13	15	16		
32														
33	Sum of Performance for all measures	1,219%	1,672%	1,254%	1,631%	1,732%		1,003%	1,230%	1,013%	1,207%	1,439%		
34														
35	Average Performance	87.1%	119.5%	89.6%	96.0%	101.9%	98.8%	77.1%	102.5%	77.9%	80.5%	90.0%	85.6%	
36														
37	Total													
38	Less than Threshold	12	4	11	12	9		12	4	11	12	9		
39	Between Threshold and Target	4	3	5	6	1		3	3	5	6	0		
40	Target	0	3	1	0	3		0	3	1	0	3		
41	Between Target and Maximum	9	13	8	11	13		7	9	5	6	11		
42	Maximum	10	13	10	9	12		10	11	10	8	12		
43		35	36	35	38	38		32	30	32	32	35		
44														
45	Sum of Performance for all measures	3,059%	4,516%	3,187%	3,558%	4,063%		2,489%	3,227%	2,567%	2,414%	3,373%		
46														
47	Average Performance	87.4%	125.4%	91.1%	93.6%	106.9%	100.9%	77.8%	107.6%	80.2%	75.4%	96.4%	87.5%	94.2%
48														
49														
50	Measures at Target and Above													
51	Target	0	3	1	0	3		0	3	1	0	3		
52	Between Target and Maximum	9	13	8	11	13		7	9	5	6	11		
53	Maximum	10	13	10	9	12		10	11	10	8	12		
54		19	29	19	20	28		17	23	16	14	26		
55														
56	Total Measures	35	36	35	38	38		32	30	32	32	35		
57														
58	Percentage of Measures at Target and Above	54.3%	80.6%	54.3%	52.6%	73.7%	63.1%	53.1%	76.7%	50.0%	43.8%	74.3%	59.6%	61.3%

Michigan Public Service Commission
DTE Electric Company
Projected Value of Tree Trimming Surge Program
(\$000,000)

Case No.: U-20836
Exhibit: A-22
Schedule: L1
Witness: S. M. Hartwick
Page: 1 of 6

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Line No.	Description		2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
1	Surge Program O&M Costs												
2	Maintenance and Staff		147.3	178.9	183.8	148.0	156.4	116.1	119.3	122.5	126.0	128.2	128.1
3	Herbicide		2.0	2.1	2.1	2.2	2.2	2.3	2.4	2.4	2.5	2.6	2.7
4	Reactive Maintenance Costs		19.0	16.2	15.0	8.9	7.4	7.7	8.2	8.5	8.8	8.8	8.9
5	Total Tree Trimming Program O&M		168.4	197.1	200.9	159.1	166.0	126.1	129.8	133.4	137.3	139.6	139.7
6	Credit to Regulatory Asset	In. 5 - In.17	(70.5)	(58.2)	(67.0)	(52.7)	(43.7)	-	-	-	-	-	-
7	Credit to Tree Trim Incremental Surge		-	(40.0)	(30.0)	-	-						
8	Total Tree Trimming Program Cost		97.9	98.9	103.9	106.4	122.3	126.1	129.8	133.4	137.3	139.6	139.7
9	Tree Trimming Storm O&M		15.0	15.0	13.2	11.2	10.0	10.5	11.1	11.6	12.0	12.0	12.2
10	Other DO Tree-related O&M		11.3	11.3	10.0	8.5	7.5	7.9	8.4	8.8	9.0	9.1	9.2
11	Total Surge Program-Related O&M		124.2	125.3	127.1	126.1	139.9	144.5	149.3	153.8	158.3	160.6	161.1
12	Surge Funding	-In. 6	70.5	58.2	67.0	52.7	43.7	-	-	-	-	-	-
13	Status Quo O&M Costs												
14	Maintenance and Staff		76.8	80.7	85.0	88.1	103.5	112.8	110.6	114.4	118.4	120.5	123.8
15	Herbicide		2.0	2.1	2.1	2.2	2.2	2.3	2.4	2.4	2.5	2.6	2.7
16	Reactive Maintenance Costs		19.0	16.2	16.8	16.1	16.6	17.1	16.9	16.6	16.4	16.5	16.9
17	Total Tree Trimming Program O&M		97.9	98.9	103.9	106.4	122.3	132.2	129.8	133.4	137.3	139.6	143.3
18	Total Tree Trimming Program Cost		97.9	98.9	103.9	106.4	122.3	132.2	129.8	133.4	137.3	139.6	143.3
19	Tree Trimming Storm O&M		15.0	15.0	15.7	16.3	17.1	17.6	17.5	17.4	17.5	17.7	18.1
20	Other DO Tree-related O&M		11.3	11.3	11.8	12.2	12.9	13.3	13.2	13.1	13.2	13.4	13.7
21	Total Status Quo-Related O&M		124.2	125.3	131.4	134.9	152.3	163.1	160.5	164.0	167.9	170.7	175.1
22	Total O&M Savings	In. 21 - In.11	-	-	4.3	8.8	12.5	18.5	11.2	10.2	9.7	10.0	14.0
23	Surge Program Capital Costs												
24	Tree Trim Reactive Capital		2.9	2.9	2.5	2.1	1.9	2.0	2.1	2.2	2.3	2.3	2.3
25	Tree Trim Storm Capital		17.2	17.2	15.1	12.8	11.4	12.0	12.7	13.3	13.7	13.8	13.9
26	Other DO Tree-related Capital		69.5	69.6	61.2	52.0	46.3	48.7	51.4	53.8	55.4	55.7	56.4
27	Total Surge Program-Related Capital		89.5	89.7	78.8	67.0	59.6	62.7	66.2	69.3	71.3	71.7	72.7
28	Status Quo Capital Costs												
29	Tree Trim Reactive Capital		2.9	2.9	3.0	3.1	3.3	3.4	3.3	3.3	3.3	3.4	3.5
30	Tree Trim Storm Capital		17.2	17.2	17.9	18.6	19.6	20.1	20.0	19.9	20.0	20.3	20.8
31	Other DO Tree-related Capital		69.5	69.6	72.5	75.3	79.2	81.5	80.9	80.6	80.9	82.2	84.0
32	Total Status Quo-Related Capital		89.5	89.7	93.4	97.0	102.1	105.0	104.3	103.9	104.2	105.9	108.2
33	Total Capital Savings	In. 32 - In.27	-	-	14.6	30.1	42.5	42.2	38.1	34.6	32.9	34.2	35.5

Michigan Public Service Commission
DTE Electric Company
Projected Value of Tree Trimming Surge Program
(\$000,000)

Case No.: U-20836
Exhibit: A-22
Schedule: L1
Witness: S. M. Hartwick
Page: 2 of 6

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Line No.	Description		<u>2032</u>	<u>2033</u>	<u>2034</u>	<u>2035</u>	<u>2036</u>	<u>2037</u>	<u>2038</u>	<u>2039</u>	<u>2040</u>	<u>2041</u>	<u>2042</u>
1	Surge Program O&M Costs												
2	Maintenance and Staff		131.9	135.3	139.2	141.5	141.2	145.4	149.1	153.9	157.0	156.2	161.7
3	Herbicide		2.7	2.8	2.9	3.0	3.1	3.2	3.3	3.3	3.4	3.5	3.6
4	Reactive Maintenance Costs		9.2	9.5	9.7	10.0	10.3	10.6	10.9	11.2	11.6	11.9	12.2
5	Total Tree Trimming Program O&M		143.8	147.6	151.8	154.6	154.6	159.2	163.3	168.5	172.0	171.6	177.6
6	Credit to Regulatory Asset	In. 5 - In.17	-	-	-	-	-	-	-	-	-	-	-
7	Credit to Tree Trim Incremental Surge												
8	Total Tree Trimming Program Cost		143.8	147.6	151.8	154.6	154.6	159.2	163.3	168.5	172.0	171.6	177.6
9	Tree Trimming Storm O&M		12.5	12.9	13.3	13.6	14.0	14.5	14.9	15.3	15.7	16.2	16.7
10	Other DO Tree-related O&M		9.4	9.7	10.0	10.3	10.6	10.9	11.2	11.5	11.9	12.2	12.6
11	Total Surge Program-Related O&M		165.8	170.2	175.1	178.5	179.3	184.5	189.4	195.3	199.6	200.0	206.8
12	Surge Funding	-In. 6	-	-	-	-	-	-	-	-	-	-	-
13	Status Quo O&M Costs												
14	Maintenance and Staff		131.5	128.1	132.1	134.5	136.6	140.0	142.2	146.9	149.9	152.4	154.4
15	Herbicide		2.7	2.8	2.9	3.0	3.1	3.2	3.3	3.3	3.4	3.5	3.6
16	Reactive Maintenance Costs		16.9	16.7	16.8	17.1	17.5	17.7	18.0	18.3	18.7	19.2	19.6
17	Total Tree Trimming Program O&M		151.1	147.6	151.8	154.6	157.2	160.9	163.4	168.5	172.0	175.1	177.6
18	Total Tree Trimming Program Cost		151.1	147.6	151.8	154.6	157.2	160.9	163.4	168.5	172.0	175.1	177.6
19	Tree Trimming Storm O&M		18.2	18.3	18.5	18.9	19.4	19.7	20.1	20.5	21.1	21.7	22.1
20	Other DO Tree-related O&M		13.7	13.8	14.0	14.3	14.6	14.9	15.1	15.5	15.9	16.3	16.6
21	Total Status Quo-Related O&M		183.0	179.6	184.3	187.8	191.3	195.4	198.6	204.5	209.0	213.2	216.4
22	Total O&M Savings	In. 21 - In.11	17.2	9.4	9.2	9.3	12.0	10.9	9.2	9.2	9.4	13.1	9.5
23	Surge Program Capital Costs												
24	Tree Trim Reactive Capital		2.4	2.5	2.5	2.6	2.7	2.8	2.8	2.9	3.0	3.1	3.2
25	Tree Trim Storm Capital		14.3	14.8	15.2	15.6	16.1	16.5	17.0	17.5	18.0	18.6	19.1
26	Other DO Tree-related Capital		58.0	59.7	61.5	63.2	65.1	67.0	68.9	70.9	73.0	75.1	77.2
27	Total Surge Program-Related Capital		74.8	76.9	79.2	81.5	83.8	86.3	88.8	91.3	94.0	96.7	99.5
28	Status Quo Capital Costs												
29	Tree Trim Reactive Capital		3.5	3.5	3.5	3.6	3.7	3.8	3.8	3.9	4.0	4.1	4.2
30	Tree Trim Storm Capital		20.9	20.9	21.2	21.7	22.2	22.6	23.0	23.5	24.2	24.8	25.3
31	Other DO Tree-related Capital		84.4	84.6	85.8	87.8	90.0	91.3	93.0	95.1	97.8	100.5	102.3
32	Total Status Quo-Related Capital		108.8	109.0	110.6	113.1	115.9	117.6	119.9	122.6	126.0	129.5	131.9
33	Total Capital Savings	In. 32 - In.27	34.0	32.0	31.4	31.6	32.1	31.4	31.1	31.2	32.0	32.8	32.3

Michigan Public Service Commission
DTE Electric Company
Projected Value of Tree Trimming Surge Program
(\$000,000)

Case No.: U-20836
Exhibit: A-22
Schedule: L1
Witness: K. M. Vangilder
Page: 3 of 6

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Line No.	Description		2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
1	Regulatory Asset												
2	Beginning Balance	(1)	118.6	189.1	215.9	251.5	272.9	243.1	169.7	127.7	85.7	43.7	43.7
3	Annual Deferral	pg. 1-2, ln. 6	70.5	58.2	67.0	52.7	43.7	-	-	-	-	-	-
4	Amortization 1/	ln. 20	-	(31.4)	(31.4)	(31.4)	(73.4)	(73.4)	(42.0)	(42.0)	(42.0)	-	-
5	Ending Balance		189.1	215.9	251.5	272.9	243.1	169.7	127.7	85.7	43.7	43.7	43.7
6	Deferred Tax Liability	- ln. 5 x ln. 11	(49.0)	(55.9)	(65.2)	(70.7)	(63.0)	(44.0)	(33.1)	(22.2)	(11.3)	(11.3)	(11.3)
7	Net Rate Base		140.1	160.0	186.4	202.2	180.1	125.8	94.6	63.5	32.4	32.4	32.4
8	Average Balance	(1)	114.0	150.0	173.2	194.3	191.2	153.0	110.2	79.1	47.9	32.4	32.4
9	Return on Permanent Capital	(2)	8.76%	8.76%	8.76%	8.76%	8.76%	8.76%	8.76%	8.76%	8.76%	8.76%	8.76%
10	Return on Deferral	ln. 8 x ln. 9	10.0	13.1	15.2	17.0	16.7	13.4	9.7	6.9	4.2	2.8	2.8
11	Tax Rate	(3)	25.9%										
12													
13	Regulatory Asset Amortization (4)												
14	U-21015 Securitization		-	(31.4)	(31.4)	(31.4)	(31.4)	(31.4)	-	-	-	-	-
15	2021		-	-	-	-	(6.4)	(6.4)	(6.4)	(6.4)	(6.4)	-	-
16	2022		-	-	-	-	(11.6)	(11.6)	(11.6)	(11.6)	(11.6)	-	-
17	2023		-	-	-	-	(13.4)	(13.4)	(13.4)	(13.4)	(13.4)	-	-
18	2024		-	-	-	-	(10.5)	(10.5)	(10.5)	(10.5)	(10.5)	-	-
19	2025		-	-	-	-	-	-	-	-	-	-	-
20	Total Amortization Expense		-	(31.4)	(31.4)	(31.4)	(73.4)	(73.4)	(42.0)	(42.0)	(42.0)	-	-

- (1) Beginning balances provided by Witness Hartwick
- (2) Derived from permanent capital rates on Exh. A-14 Sch. D1 and adjusted to pre-tax using the conversion factors
- (3) Composite tax rate provided by DTE Energy Tax Department
- (4) Assumes 5-year amortization starting in 2025 for the un-securitized portion of 2021 and all subsequent spend.
5- year amortization of the \$156.9 million approved via the U-21015 order dated June 23, 2021 begins in 2022.

Michigan Public Service Commission
DTE Electric Company
Projected Value of Tree Trimming Surge Program
(\$000,000)

Case No.: U-20836
Exhibit: A-22
Schedule: L1
Witness: K. M. Vangilder
Page: 4 of 6

Line	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
No.	Description		2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042
1	Regulatory Asset												
2	Beginning Balance	(1)	43.7	43.7	43.7	43.7	43.7	43.7	43.7	43.7	43.7	43.7	43.7
3	Annual Deferral	pg. 1-2, ln. 6	-	-	-	-	-	-	-	-	-	-	-
4	Amortization 1/	ln. 20	-	-	-	-	-	-	-	-	-	-	-
5	Ending Balance		43.7	43.7	43.7	43.7	43.7	43.7	43.7	43.7	43.7	43.7	43.7
6	Deferred Tax Liability	- ln. 5 x ln. 11	(11.3)	(11.3)	(11.3)	(11.3)	(11.3)	(11.3)	(11.3)	(11.3)	(11.3)	(11.3)	(11.3)
7	Net Rate Base		32.4	32.4	32.4	32.4	32.4	32.4	32.4	32.4	32.4	32.4	32.4
8	Average Balance	(1)	32.4	32.4	32.4	32.4	32.4	32.4	32.4	32.4	32.4	32.4	32.4
9	Return on Permanent Capital	(2)	8.76%	8.76%	8.76%	8.76%	8.76%	8.76%	8.76%	8.76%	8.76%	8.76%	8.76%
10	Return on Deferral	ln. 8 x ln. 9	2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8
11	Tax Rate	(3)											
12													
13	Regulatory Asset Amortization (4)												
14	U-21015 Securitization		-	-	-	-	-	-	-	-	-	-	-
15	2021		-	-	-	-	-	-	-	-	-	-	-
16	2022		-	-	-	-	-	-	-	-	-	-	-
17	2023		-	-	-	-	-	-	-	-	-	-	-
18	2024		-	-	-	-	-	-	-	-	-	-	-
19	2025		-	-	-	-	-	-	-	-	-	-	-
20	Total Amortization Expense		-	-	-	-	-	-	-	-	-	-	-

(1) Beginning balances provided by Witness Hartwick
(2) Derived from permanent capital rates on Exh. A-14 Sch. D1 and adjusted to pre-tax using the conversion factors
(3) Composite tax rate provided by DTE Energy Tax Department
(4) Assumes 5-year amortization starting in 2025 for the un-securitized portion of 2021 and all subsequent spend.
5- year amortization of the \$156.9 million approved via the U-21015 order dated June 23, 2021 begins in 2022.

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(\$000,000)

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Page: 5 of 6

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Line No.	Description		2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
1	Rate Base Savings Calculation												
2	Beginning PP&E	(1)	10.7	10.3	9.8	23.7	52.2	91.5	129.0	160.6	187.3	210.9	234.4
3	Added	pg. 3-4, ln. 9	-	-	14.6	30.1	42.5	42.2	38.1	34.6	32.9	34.2	35.5
4	Depreciation	Depreciation Workpaper	(0.4)	(0.4)	(0.7)	(1.6)	(3.1)	(4.8)	(6.4)	(7.9)	(9.3)	(10.6)	(12.0)
5	Ending PP&E		10.3	9.8	23.7	52.2	91.5	129.0	160.6	187.3	210.9	234.4	258.0
6	Average PP&E	(ln. 2 + ln. 5)/2	10.5	10.1	16.8	37.9	71.8	110.3	144.8	173.9	199.1	222.7	246.2
7	Average DTL	Depreciation Workpaper	(0.1)	(0.2)	(0.2)	(0.4)	(0.9)	(1.6)	(2.5)	(3.6)	(4.8)	(6.1)	(7.4)
8	Average Rate Base		10.4	9.9	16.5	37.5	71.0	108.7	142.3	170.3	194.2	216.5	238.8
9	Revenue Requirement Savings												
10	Average Rate Base	ln. 8	10.4	9.9	16.5	37.5	71.0	108.7	142.3	170.3	194.2	216.5	238.8
11	Return on Permanent Capital	pg. 1-2, ln.	8.8%	8.8%	8.8%	8.8%	8.8%	8.8%	8.8%	8.8%	8.8%	8.8%	8.8%
12	Return on Rate Base	ln. 10 x ln. 11	0.9	0.9	1.4	3.3	6.2	9.5	12.5	14.9	17.0	19.0	20.9
13	Depreciation	ln. 4	0.4	0.4	0.7	1.6	3.1	4.8	6.4	7.9	9.3	10.6	12.0
14	O&M Savings	pg. 1-2, ln. 22	-	-	4.3	8.8	12.5	18.5	11.2	10.2	9.7	10.0	14.0
15	Property Taxes	Property Tax Workpaper	0.3	0.3	0.3	0.3	0.6	1.4	2.6	3.6	4.5	5.3	5.9
16	Cost Savings		1.6	1.6	6.7	14.0	22.4	34.3	32.6	36.6	40.4	44.9	52.9
17	Surge Expense	pg. 1-2, ln. ln. 5 - ln. 17	(71)	(58)	(67)	(53)	(44)	-	-	-	-	-	-
18	Less Credit to Regulatory Asset	pg. 1-2, ln. 6	71	58	67	53	44	-	-	-	-	-	-
19	Revenue Requirement		1.6	1.6	6.7	14.0	22.4	34.3	32.6	36.6	40.4	44.9	52.9
20	Deferral Return	pg. 3-4, ln. 10	(10.0)	(13.1)	(15.2)	(17.0)	(16.7)	(13.4)	(9.7)	(6.9)	(4.2)	(2.8)	(2.8)
21	Deferral Amortization	pg. 3-4, ln. 20	-	(31.4)	(31.4)	(31.4)	(73.4)	(73.4)	(42.0)	(42.0)	(42.0)	-	-
22	Deferral Costs		(10.0)	(44.5)	(46.5)	(48.4)	(90.1)	(86.8)	(51.7)	(48.9)	(46.2)	(2.8)	(2.8)
23	Total Savings	ln. 19 + ln. 22	(8.4)	(42.9)	(39.8)	(34.4)	(67.7)	(52.5)	(19.0)	(12.4)	(5.8)	42.0	50.1
24	<u>Net Present Value of Program Savings</u>	(3)	<u>82.0</u>										
	Discount Rate	(2)	6.8%										

(1) Beginning balances provided by Witness Hartwick
(2) Discount Rate provided by DTE Energy Regulatory Accounting
(3) NPV is calculated 2022-2042

Michigan Public Service Commission
DTE Electric Company
Projected Value of Tree Trimming Surge Program
(\$000,000)

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Page: 6 of 6

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Line No.	Description		2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042
1	Rate Base Savings Calculation												
2	Beginning PP&E	(1)	258.0	278.5	295.8	311.1	325.4	338.9	350.4	360.4	369.2	377.5	385.3
3	Added	pg. 3-4, ln. 9	34.0	32.0	31.4	31.6	32.1	31.4	31.1	31.2	32.0	32.8	32.3
4	Depreciation	Depreciation Workpaper	(13.4)	(14.8)	(16.0)	(17.3)	(18.6)	(19.9)	(21.1)	(22.4)	(23.7)	(25.0)	(25.6)
5	Ending PP&E		278.5	295.8	311.1	325.4	338.9	350.4	360.4	369.2	377.5	385.3	392.0
6	Average PP&E	(ln. 2 + ln. 5)/2	268.2	287.2	303.5	318.3	332.2	344.7	355.4	364.8	373.3	381.4	388.6
7	Average DTL	Depreciation Workpaper	(8.7)	(10.1)	(11.4)	(12.8)	(14.2)	(15.6)	(17.1)	(18.5)	(19.9)	(21.3)	(22.7)
8	Average Rate Base		259.5	277.1	292.0	305.5	318.0	329.0	338.3	346.3	353.4	360.0	365.9
9	Revenue Requirement Savings												
10	Average Rate Base	ln. 8	259.5	277.1	292.0	305.5	318.0	329.0	338.3	346.3	353.4	360.0	365.9
11	Return on Permanent Capital	pg. 1-2, ln.	8.8%	8.8%	8.8%	8.8%	8.8%	8.8%	8.8%	8.8%	8.8%	8.8%	8.8%
12	Return on Rate Base	ln. 10 x ln. 11	22.7	24.3	25.6	26.8	27.9	28.8	29.6	30.3	31.0	31.5	32.1
13	Depreciation	ln. 4	13.4	14.8	16.0	17.3	18.6	19.9	21.1	22.4	23.7	25.0	25.6
14	O&M Savings	pg. 1-2, ln. 22	17.2	9.4	9.2	9.3	12.0	10.9	9.2	9.2	9.4	13.1	9.5
15	Property Taxes	Property Tax Workpaper	6.6	7.3	7.9	8.4	8.9	9.3	9.8	10.2	10.6	11.1	11.5
16	Cost Savings		60.0	55.7	58.7	61.7	67.3	68.9	69.7	72.1	74.7	80.7	78.8
17	Surge Expense	pg. 1-2, ln. ln. 5 - ln.17	-	-	-	-	-	-	-	-	-	-	-
18	Less Credit to Regulatory Asset	pg. 1-2, ln. 6	-	-	-	-	-	-	-	-	-	-	-
19	Revenue Requirement		60.0	55.7	58.7	61.7	67.3	68.9	69.7	72.1	74.7	80.7	78.8
20	Deferral Return	pg. 3-4, ln. 10	(2.8)	(2.8)	(2.8)	(2.8)	(2.8)	(2.8)	(2.8)	(2.8)	(2.8)	(2.8)	(2.8)
21	Deferral Amortization	pg. 3-4, ln. 20	-	-	-	-	-	-	-	-	-	-	-
22	Deferral Costs		(2.8)	(2.8)	(2.8)	(2.8)	(2.8)	(2.8)	(2.8)	(2.8)	(2.8)	(2.8)	(2.8)
23	Total Savings	ln. 19 + ln. 22	57.1	52.9	55.9	58.9	64.5	66.1	66.9	69.3	71.9	77.9	75.9
24	Net Present Value of Program Savings	(3)	<u>82.0</u>										
	Discount Rate	(2)	6.8%										

- (1) Beginning balances provided by Witness Hartwick
(2) Discount Rate provided by DTE Energy Regulatory Accounting
(3) NPV is calculated 2022-2042

DTE Electric Company

2021 Distribution Grid Plan Final Report

September 30, 2021
MPSC Case No. U-20147

The DTE logo is displayed in a bold, blue, sans-serif font.

Table of Contents

Table of Contents.....	i
1 Executive summary	1
1.1 Introduction	1
1.2 Severe Weather of Summer 2021 and the impact on the DGP	9
2 Stakeholder engagement and input	14
2.1 Stakeholder engagement	14
2.1.1 Customer focus groups.....	14
2.1.2 Community leader engagement.....	18
2.1.3 Technical stakeholder engagement	22
2.1.4 Feedback on the Draft DGP Filing	25
2.2 Energy and environmental justice.....	26
2.2.1 Methodology	26
2.2.2 Actions	28
3 Grid modernization process.....	30
3.1 Methodology.....	31
3.2 Define DTEE objectives.....	32
3.3 Scenario planning.....	34
3.3.1 Electrification scenario.....	36
3.3.2 Increasing catastrophic-level storms scenario.....	38
3.3.3 Distributed generation/distributed storage growth scenario.....	40
3.4 Current state assessment and grid needs analysis.....	42
3.4.1 Scenario impact analysis	43
3.4.2 Key categories of investment.....	44
3.5 Short-term investment plan and long-term vision.....	53
4 Distribution planning processes and tools.....	56
4.1 Introduction	56
4.2 Forecasting	57
4.2.1 Integrated forecasting solution	59
4.2.2 Data integrity and management.....	59
4.2.3 Scenario and base model development.....	60
4.2.4 8760 generation system-level forecasts.....	61
4.2.5 8760 distribution-level forecasts	61

4.3	Non-wire alternatives.....	62
4.4	Interconnection process	69
4.5	Hosting capacity analysis	73
4.6	Integrated generation (resource) planning and distribution planning alignment	76
4.6.1	Areas of past successful collaborations	76
4.6.2	Areas of continuing collaboration	78
5	Benefit cost analysis (BCA)	82
5.1	Benefit cost analysis approach	83
5.2	GPM next steps and other considerations	90
5.3	Investment impact on reliability and avoided costs	92
5.4	Customer benefits	97
6	DGP investment summary.....	99
7	Distribution system overview	106
7.1	System overview.....	106
7.2	Reliability	110
7.2.1	Background	110
7.2.2	SAIFI, SAIDI and CAIDI	112
7.2.3	Causes of interruptions	117
7.2.4	MAIFI.....	119
7.2.5	CEMIn and CELIDt	122
7.3	Resiliency.....	124
7.3.1	Incident command system	125
7.3.2	Storm response	126
7.3.3	Response to major (non-storm) events.....	137
7.3.4	Low probability, high impact (LPHI) events	140
7.3.5	Flood protection program.....	141
7.4	Power quality.....	142
7.4.1	Defining power quality and power quality events	143
7.4.2	Addressing distribution system power quality events	145
7.4.3	Addressing subtransmission system power quality events.....	148
7.4.4	Other power quality issues	149
8	Asset health assessment.....	151
8.1	Substation power transformers.....	154
8.2	Network banks (Netbanks)	157

8.3 Circuit breakers	158
8.4 Subtransmission disconnect switches	162
8.5 Circuit switchers	165
8.6 Relays	167
8.7 Switchgear	171
8.8 Poles and pole top hardware	175
8.9 Small wire	178
8.10 Fuse cutouts.....	179
8.11 Three-phase reclosers.....	181
8.12 4.8 kV & 13.2 kV SCADA distribution pole top switches	184
8.13 40kV automatic pole top switches (40kV APTS)	187
8.14 Overhead capacitors	189
8.15 Overhead regulators.....	191
8.16 System cable.....	193
8.17 Underground residential distribution (URD) cable.....	197
8.18 Manholes	200
8.19 Pontiac 8.3kV system vaults.....	202
8.20 Advanced metering infrastructure	205
9 Infrastructure resilience and hardening	211
9.1 Capital replacement programs.....	212
9.1.1 Pole and pole top maintenance and modernization	213
9.1.2 Automatic pole top switch	216
9.1.3 Subtransmission disconnect switch	217
9.1.4 Circuit switcher	218
9.1.5 Breakers.....	219
9.1.6 System cable	220
9.1.7 Underground residential distribution (URD) cable	222
9.1.8 SCADA pole top devices	222
9.1.9 Substation regulators.....	223
9.2 Substation risk	224
9.2.1 Substation outage risk model.....	224
9.2.2 Recent examples of substation events and future risks.....	228
9.2.3 Programs to address substation risk	230
9.3 4.8kV Hardening program	233

9.3.1 Introduction to the 4.8kV Hardening program	233
9.3.2 4.8kV Hardening program prioritization.....	235
9.3.3 4.8kV Hardening program progress and effectiveness.....	236
9.3.4 4.8kV Hardening program plan and timeline	238
9.3.5 Alternatives methods to improve to 4.8kV reliability	239
9.4 Frequent Outage program.....	243
9.5 4.8kV relay improvements	248
9.6 Mobile fleet.....	249
10 Tree Trimming program.....	251
10.1 Introduction	251
10.2 The need for tree trimming	252
10.3 Tree Trim Surge program	256
10.4 Miles completed and spend.....	258
10.5 ETPP reliability results.....	259
10.6 Program Improvements.....	261
10.7 Resources	263
10.8 Conclusion	265
11 Infrastructure Redesign and Modernization	266
11.1 Distribution load relief projects.....	266
11.1.1 Load relief prioritization	268
11.1.2 Consideration of non-wire alternatives	272
11.1.3 Projects to address distribution loading.....	272
11.2 Subtransmission Redesign and Rebuild Program.....	285
11.2.1 Subtransmission prioritization	288
11.2.2 Subtransmission projects and programs	292
11.3 4.8kV Conversion	305
11.3.1 Introduction to the 4.8kV Conversion program	305
11.3.2 4.8kV Conversion prioritization	310
11.3.3 4.8kV Conversion plan and timeline	312
11.3.4 Large current and near-term conversion projects.....	313
11.3.5 4.8kV isolation down areas (ISO down) conversion	322
11.3.6 4.8kV ISO down conversion prioritization.....	323
11.3.7 4.8kV ISO down conversion plan and timeline	323
11.4 CODI Program	325

11.5 8.3kV system conversion and consolidation	333
11.6 Strategic undergrounding	335
11.7 Impact of generation on the distribution system	343
11.7.1 Generation benefits for the distribution system	344
11.7.2 Evaluating the impacts of generation retirement on the grid.....	348
11.7.3 Future of distribution grid with flexible generation resources.....	351
12 Technology & automation	352
12.1 ADMS.....	354
12.2 SOC modernization	362
12.3 Grid telecommunications	365
12.4 Distribution sensing and monitoring.....	372
12.4.1 Smart fault indicators/line sensors	374
12.4.2 Low-cost substation monitoring	376
12.4.3 ION meter data integration.....	377
12.5 Distribution automation.....	381
12.5.1 Circuit automation.....	383
12.5.2 Substation automation program	385
12.6 CVR/VVO	387
12.6.1 First CVR/VVO pilot.....	388
12.6.2 Second CVR/VVO pilot.....	390
12.7 NWA pilots and other technology pilots	392
12.7.1 Hancock substation NWA pilot.....	396
12.7.2 Fisher substation NWA pilot.....	397
12.7.3 Port Austin substation NWA pilot	400
12.7.4 Omega substation NWA pilot.....	402
12.7.5 Veridian	403
12.7.6 O'Shea solar and storage	405
12.7.7 Other technology pilots	406
12.8 DER control.....	408
12.8.1 Large and medium-sized DER monitoring & control.....	409
12.8.2 Upgrade of DR-SOC monitored solar sites to RTAC RTU.....	410
12.8.3 Viper recloser sensing/data	410
12.8.4 Small DER monitoring	410
12.8.5 DERA monitoring integration	411

12.8.6 ICCP connectivity – DERs in a DERA.....	411
12.8.7 ICCP connectivity – load shed in a DERA.....	412
12.8.8 AMI and de-aggregated load data.....	412
12.8.9 DERMS module.....	412
12.8.10 DERMS module modelling.....	413
12.8.11 OSI enterprise customer agreement.....	413
12.9 Technology Strategy Roadmap	413
12.9.1 Goal 1: Improve employee productivity through mobile technology.....	415
12.9.2 Goal 2: Deliver best-in-class work management and integrated scheduling.....	418
12.9.3 Goal 3: Drive top-quartile DO asset management.....	420
12.9.4 Goal 4: Transform capabilities for distribution planning processes.....	421
12.9.4.1 Interconnection process enablement	422
12.9.4.2 Hosting capacity enablement.....	424
12.9.4.3 Network model.....	424
12.9.4.4 Distribution load forecasting.....	427
12.9.4.5 Situational awareness analytics.....	429
12.9.5 Goal 5: Modernize grid management.....	430
12.9.6 Technology strategy roadmap capital investment summary.....	432
13 Base capital.....	434
13.1 Emergent replacement	434
13.1.1 Emergent replacement during storm conditions	435
13.1.2 Emergent replacement during non-storm conditions.....	438
13.1.3 Emergent replacement for substation reactive	440
13.1.4 Measures to reduce emergent capital spend.....	441
13.1.5 Emergent spend forecast.....	442
13.2 Connections, relocations and other	442
13.2.1 Connections and new load.....	443
13.2.2 Relocation.....	443
13.2.3 Electric System equipment.....	444
13.2.4 NRUC and improvement blankets.....	444
13.2.5 General plant, tools & equipment, and miscellaneous.....	445
13.2.6 Other known major projects	445
14 Preventative maintenance	447
15 Performance-based rate making	451

15.1 Background and context.....	451
15.2 Benchmarking	453
15.3 DTEE’s proposed PBR plan	456
16 Enablers.....	466
16.1 Distribution design standards.....	466
16.2 Workforce and resource planning.....	470
16.3 Cybersecurity	476
16.4 Physical security	477
17 Key next steps.....	479
18 Conclusion	484
Appendix I Distribution Plan Requirements	486
Appendix II Strategic Capital Programs excluded from Global Prioritization Model prioritization	496
Appendix III Historical Storm Events	498
Appendix IV Reliability Indices for Service Centers	503
Appendix V Pilot guidelines and information	531
Appendix VI Distribution Design Standards and Distribution Design Orders.....	540
Appendix VII Customer focus groups	545
Appendix VIII Community leader engagement	545
Appendix IX DTEE/ICF Grid Modernization Study 2021-2035.....	545
Table of Exhibits	546
Glossary.....	555

1 Executive summary



1.1 Introduction

The energy landscape is changing across the nation, including right here in Michigan. When the groundwork for our company's electric infrastructure system was laid more than a century ago as one of the first in the country, it was built as a one-way distribution system – providing customers the energy required for a much simpler day-to-day life, while also acting as the engine of progress for industry. Our foundation was formed by the Edison Illuminating Co. of Detroit in the late 1880s and has served our customers well through the turn of the century, the rise of the Detroit automotive industry and the Digital Revolution. While what is now known as DTE Energy continued to build and grow the energy infrastructure throughout the 20th century to meet the needs of its customers and the expansion that occurred in each era, portions of our electric infrastructure that are more than 90 years old are still in operation. This aging equipment will be challenged to meet the growing needs of our 2.2 million electric customers across southeast Michigan for the rest of the 21st century. Equipment that has served customers well beyond its expected life, is now incurring costly maintenance and replacement plans and operational constraints that are hindering reliability for our customers.

Our customers are evolving too, and are more connected than ever, leading to additional grid challenges. Our customers' homes are now "smart" and host dozens of electronic devices that they rely on to give them the information needed to live, work and learn each and every day – thermostat and appliance controls, virtual working and learning platforms, and charging for their electric vehicles. Some of our customers are even generating energy on their own through distributed energy resources (DER) in the form of rooftop solar panels and energy storage. As we

look to the future, we expect our customers' dependence on these devices and the information they provide to grow exponentially, which will challenge the current grid – moving it away from a one-way distribution system, to a system that requires a two-way flow that allows customers to send energy and information back into the grid. This evolution requires even more grid technology and integration to ensure the electric reliability that our customers require and deserve.

While our customers are more dependent on technology than ever, we also know that climate change – and its impacts on the energy grid – is one of the defining issues of our era and top of mind for DTE and our customers. DTE Electric (DTEE) is taking bold steps to reduce our company's impacts on climate change by significantly increasing our investments in renewable energy and cutting carbon emissions in half by 2030¹ to reach our goal of net zero carbon emissions by 2050. Michigan's energy landscape is also undergoing a transformation. While the state has formulated new sustainability goals that encourage and enable the electrification of transportation to reduce environmental impact, our customers and communities are looking to us to continue to reduce carbon emissions and deliver energy through a grid that's more reliable and resilient than in the past, a grid that can also handle the increasingly severe weather we are seeing season after season.

In order to deliver the safe, clean, reliable and affordable electricity our customers deserve, DTEE is reexamining the grid infrastructure that is fundamental to supporting these changes. The same innovative vision and planning that went into developing the electric grid in Detroit and the surrounding communities over a century ago is needed today to ensure our grid will support the evolving needs of our customers and our state.

DTEE has made significant progress in upgrading aging infrastructure, accelerating tree trimming efforts, installing modern equipment to detect, prevent and manage outages and improving safety since the filing and implementation of its Distribution Operations Five-Year (2018-2022) Investment and Maintenance Plan with the MPSC in 2018. The results of these strategic investments speak for themselves, as highlighted below. While we are making marked progress where investments have been made, the data on asset conditions and needs, shows significant

¹ Based on a 2005 baseline

additional investment is required to avoid further deterioration and costly reactive spending in order to mitigate outage risks and to prepare for the grid of the future.

- **Grid hardening and modernization:** In areas where grid hardening work is complete, like our Customer Excellence program (which includes a robust program of infrastructure/equipment updates and tree trimming), our customers are seeing a 50-70% improvement in reliability. We want all our customers in Southeast Michigan to experience this level of service from DTEE, so we've developed an accelerated plan for the next five years to ensure we can provide our customers with the safe, reliable and affordable energy they deserve. The continued investment and infrastructure updates outlined in our updated plan will help us realize a 60% improvement in SAIDI, moving us from third quartile in 2020 to first quartile by 2030.

Modernizing infrastructure will also make the grid more resilient to the increasingly severe weather that is impacting Michigan, helping to significantly reduce reactive costs – a projected \$65 million annually by 2025 – and time spent on outages caused by equipment failures and weather. Our experience shows that reactive equipment replacements due to failures can cost up to five times as much as proactive replacement programs.

- **Tree trimming:** Historically, trees cause two-thirds of the time (outage minutes) our customers spend without power. We began our tree trim “surge” to clear the trees farther away from our wires and improve reliability in 2016, when we implemented our Enhanced Tree Trimming Program (ETTP). The data is showing that those circuits that have been trimmed to our new ETTP specification, on average, have 60% fewer outages, shorter outage durations and fewer wire downs and safety hazards. At the beginning of 2018, 23% of our system was trimmed to the ETTP specification. By the end of 2021, 72% of the system will be trimmed to ETTP, including 89% of the city of Detroit. We will continue this “surge” until all of the system has been trimmed to the ETTP by the end of 2025 to meet the industry benchmark five-year ETTP cycle. The ETTP program and maintenance of the required clearances will improve system resilience and provide increased reliability for our entire customer base.

In addition to the reliability and safety improvements, the Tree Trimming program has been a force for growth within the community. DTEE has implemented two training programs – one

at the Vocational Village and Parnall Correctional Facility in Jackson, and a Tree Trimming Academy in Detroit. These programs provide career opportunities for members of the communities we serve.

- **Future technology:** DTEE's Advanced Distribution Management System (ADMS) is a new technology platform that will provide teams across DTEE with a complete and integrated view of multiple energy management systems, allowing DTEE to diagnose problems quicker, leading to faster outage response for our customers. The implementation of the ADMS, which is the foundational platform for our grid modernization efforts, began in 2018 and is being implemented in phases. To date we have completed implementation of the Generation Management System (GMS); the Energy Management System (EMS) to manage the subtransmission system; and connections to the transmission system and Network Management System (NMS) to improve data quality. Over the next several months, DTEE intends to continue implementation of the ADMS with the rollout of the new Outage Management System (OMS) which will improve outage restoration response, along with the first of the Distribution Management System (DMS) components, eMap, which will provide a schematic of the as-operated network model of the electrical system. The remaining ADMS applications are planned for launch in 2022.

The changes to the grid will also require even more technology investment. The past smart/advanced meter infrastructure (AMI) project is an essential component of a modernized grid, and one that DTEE has already invested in. Other grid enhancements will be needed as part of modernization, including the likely addition of a distributed energy resource management system (DERMS), as well as increased grid sensing and automation technology. Some of the solutions to the grid challenges will include localized new technology, like non-wire alternatives (NWA), and the potential for microgrids. These technologies have the potential to defer traditional utility investments and provide localized solutions to enhance reliability and resiliency.

- **Electric ESOC:** In June 2019, construction began on DTEE's new Electric System Operations Center (ESOC) at the Company's downtown Detroit headquarter complex. The completion of the facility in the fourth quarter of 2021, will allow for co-location of dispatchers and system

operators who utilize the new ADMS, which will lead to improved management of significant system disruptions and quicker response time for our customers. The facility includes electronic displays and improved console designs to meet dispatcher and operator needs, while enhancing situational awareness, and improving safety and reliability of the electric grid for our customers.

- **Infrastructure redesign to support economic growth:** Our customers have increasing demands of the electric grid as they look to new areas to live and grow their businesses and communities, while moving more of their lives and operations toward electrification. Since the filing of DTEE's 2018 five-year distribution grid plan, we have started the planning for and are in the early phases of construction for new substations and the conversion of circuit miles to higher voltage levels. These projects, once complete, will support economic growth, customer demand and improve reliability for the customers and communities we serve in areas like Almont, Ann Arbor, Detroit, Dexter, Northville, Shelby Township and Van Buren. These projects will allow us to add about 700 MW of additional capacity to the distribution grid – enough to power 180,000 residential homes or 110,000 electric vehicles – that will accommodate new load that will flex and grow with customers' expectations for the grid of the future.

Customers, both residential and commercial, are moving toward electrification of their vehicles, and we'll be able to support this demand as well as growth from areas like economic and community development projects. For example, as part of the City of Detroit Infrastructure (CODI) program, DTEE began construction on a new 13.2 kV Corktown substation in 2021, with plans to energize it in the first quarter of 2022. Meeting this schedule will accommodate significant new customer load as part of economic development activities planned in Detroit's Corktown area. In addition, it enables the future conversion and transfer of load from nearby lower voltage Kent and Gibson substations, which will allow for the decommissioning of these aging substations. DTEE's new Distribution Grid Plan builds on the 2018 plan and establishes the foundation for near-term change in DTEE's distribution system for the next five years, as well as a more transformative vision for future improvements to the grid over the next 10-15 years. This transformation will significantly improve reliability and resilience for our customers and increase the ability to

effectively integrate new clean energy technologies such as electric vehicles, rooftop and large-scale solar, and batteries.

We have taken a strategic approach to prioritize and sequence investments, evaluate alternatives, and have included a Performance Based Regulation (PBR) proposal in this plan to ensure progress and transparency in measuring positive impacts for our customers. Our PBR proposal will be included in DTEE's next rate case.

DTEE began its planning efforts by reviewing the MPSC goals, and then developed planning objectives across our company, building on the planning principles that were used to guide our recent integrated resource planning. Five planning objectives were defined, which are described below.



SAFE

Build, operate and maintain the distribution grid and generation fleet in a manner that ensures public and workforce safety, operational risk management and appropriate fail-safe modes and is compliant with state and federal requirements



RELIABLE AND RESILIENT

Build, operate and maintain the power system within acceptable standards to withstand sudden disturbance or unanticipated failure of elements. Ensure the grid and diverse generation resources are integrated, with secure supply resources, and can quickly recover from high impact, low frequency events



AFFORDABLE

Provide efficient and cost-effective service along with diverse and flexible generation resources by optimizing the system and benefiting all customers



CUSTOMER ACCESSIBILITY AND COMMUNITY FOCUS

Provide flexible and accessible technology and grid options, and information that empowers and engages customers. Provide effective and timely communication with customers and stakeholders. Favor plans that support the diversity of Michigan communities, suppliers and workforce



CLEAN

Build, operate and maintain the resource fleet and grid platforms in an environmentally sustainable manner by achieving low carbon aspirations and clean -energy goals. Provide a grid that facilitates a transition to a decarbonized economy

The planning objectives provide guidance and support for the four strategic pillars in our investment strategy.

Pillar	Program / Project Examples
 Infrastructure Resilience and Hardening	• 4.8kV Hardening program • Asset replacement programs stemming from engineering asset condition assessments • Pole and pole top maintenance and modernization (PTMM)
 Infrastructure Redesign	• Conversion of 4.8kV system • Major <u>subtransmission</u> upgrade and redesign projects • System loading projects
 Technology and Automation	• Advanced Distribution Management System (ADMS) • Construction of a new Electric System Operations Center (ESOC) and launch operations in 2021 • Distribution Automation – installation of remote monitoring and grid control devices
 Tree Trimming	• Enhanced Tree Trim Program (ETTP)

These planning objectives and investment pillars align to the long-term strategic shift planned by both the electric utility industry and DTEE. These objectives support our customers' expectations for cleaner and more reliable energy, the state of Michigan's sustainability goals, and the DTE Energy corporate goals for clean energy and decarbonization.

Knowing that our customers' requirements of the grid of the future are much more complex than in the past, DTEE took a different approach to developing the five-year plan and longer-term vision for the distribution grid. Starting with the planning objectives above, the Company worked with an external consultant, ICF, to develop three scenarios. Each scenario is distinct from the others and reflects plausible future outcomes. Although the future will likely blend elements from each scenario, examining the grid impacts of the three distinct scenarios supports an understanding of the constraints on our existing distribution and subtransmission system and provides guidance on required future investments.

Scenario	Description	Implications
 Electrification Scenario	High electrification of transportation, buildings, and industrial processes	Distribution overloads begin to occur at 2% adoption on quite a few substations
 Increasing CAT Storm Scenario	Increased catastrophic (CAT) storm frequency and intensity threat to aging grid reliability	Current system loading and designs flexibility will be further challenged
 DG/DS Scenario	High adoption of distributed generation (DG) solar PV and distributed storage (DS) behind the meter (BTM)	Distribution voltage, thermal and protection issues will increasingly arise at relatively low adoption levels DTEE Needs a DER operational & control strategy for DER, Including MISO coordination

The conclusions of the scenario analysis, grid impacts and modernization needs identified that in order to modernize and absorb the impacts of electrification, storms and DERs, there will need to be an increase in grid investment. While these future grid impacts will present challenges, they also provide DTEE with a great opportunity to rebuild and modernize the distribution system into a more reliable and resilient grid that better serves our customers and other stakeholders.

DTEE used the four pillars and the scenario-based planning to develop a strategic five-year capital investment level of \$7 billion. While this is a significant investment, the results mean DTEE will be able to provide a safer, more resilient and robust system for our customers. The cost of inaction to customers and DTEE, particularly in situations of long-duration outages, is much greater. These costs aren't always quantifiable on energy bills, because they include the experiential burden incurred by our customers every time they are impacted by an outage event. These impacts are important when considering tradeoffs of affordability, as vulnerable customers may be the greatest impacted if we do not take more aggressive steps to invest in the distribution grid to make it more reliable and resilient in the future.

The plan is guided by our objectives of providing safe, clean and reliable electricity to our customers that is affordable and accessible to all. We also fully embrace the Commission's view that distribution planning should relate to a "core objective of 'customer needs' that binds everything together."² Our customers are at the heart of our plans for a modern and more resilient

² MPSC August 20, 2020 order, Docket No. 20147, at p. 40.

grid. This is why DTEE reached out to our customers and other stakeholders to gain feedback through multiple channels to listen to their concerns and gain input on their perceptions of DTE as a company overall, the electric reliability they currently experience and their expectations for the grid of the future.

Almost all stakeholders told DTEE that this transition to a modernized grid must be done, and it must be done equitably. Our customers told us, and we agree, that this investment must be structured in a way that is fair and just for all customers. DTEE will continue to look for efficient ways to work and keep bills affordable for all while we upgrade and strengthen our infrastructure – a long-term investment that will improve the quality of power customers in all communities receive.

The DTEE DGP is not the end of the distribution planning process, rather it is just a step on the journey to a modernized grid. This report lays out a detailed five-year investment plan and a 10- to 15- year vision for the grid. It also identifies the next steps in the process, the continuing work that is required for the longer-term vision to become a reality.

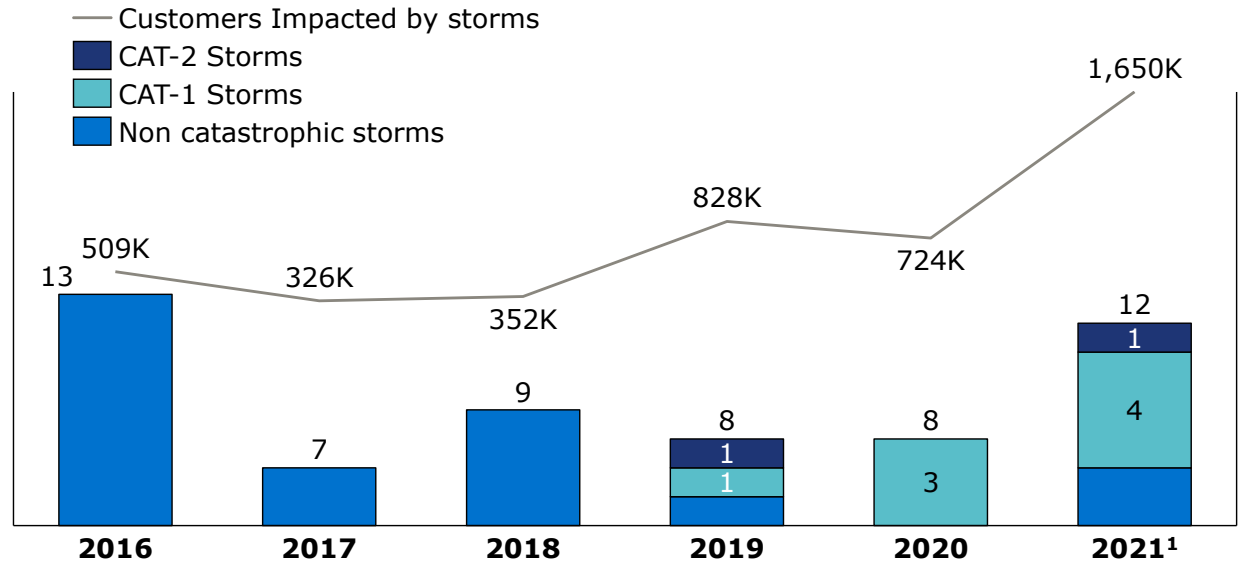
Change can be daunting and modernizing the grid for our customers will present challenges as well as opportunities along the way. We are confident that by listening to our customers and other stakeholders, employing the talent of our employees and partners, and applying our skills in project management and continuous improvement, we will build an efficient modernized grid. We are committed to bringing about the ingenuity that went into developing the electric grid in Detroit and surrounding communities over 100 years ago, in order to build and operate a clean, affordable, reliable, resilient, and accessible grid that will support the evolving needs of our customers and our state well into the future.

1.2 Severe Weather of Summer 2021 and the impact on the DGP

During the summer of 2021, DTEE's customers, communities, and employees distinctly felt the impact of severe weather on the electrical system. From mid-June through mid-September, DTEE experienced twelve storms (roughly defined as weather impacting at least 25,000 customers), including four CAT-1 storms (110,000 customers impacted), and one CAT-2 storm (220,000 or more customers impacted). The storm which started on August 9th impacted almost 500,000 customers and was one of the largest storms in company history. The combination of frequency

and intensity of severe weather, and impact on the system was unprecedented. A summary of the 2021 storms in comparison with recent years is shown below.

June Through September Storms



The weather patterns in Southeast Michigan during this period reflect the challenges faced by the distribution system. During two different storms, the area experienced rainfall in excess of 4 inches over a 48-hour period. There were at least five confirmed tornados, and wind gusts during the August 10th storm approached 75 miles per hour, while winds during three other storms exceeded 60 miles per hour.

For every storm this summer, DTEE followed the storm restoration process described in Section 7.3.2. This process included ramp-up planning (using weather forecasts to prepare for potential outages in advance of the weather system entering the area) and requesting additional resources from other areas to assist with the restoration effort. Over the eleven storms experienced this summer, nearly 5,500 mutual assistance line workers aided in the restoration effort, including over 1,500 who worked during the August CAT-2 storm.

Due to the frequency of storms, a specific challenge this summer was the post-storm recovery phase. The storm restoration efforts include making temporary repairs in order to provide power to customers as quickly as possible. In the weeks following a severe storm, crews go back to these temporary repairs and make permanent repairs or replacements. This summer, the time between

completing restoration for one storm and the beginning of another storm averaged less than five days. This did not give the crews enough time to make the permanent repairs or replacements needed and left some areas of the system vulnerable to repeat outages.

Near-term changes due to 2021 Summer Storms

DTEE acknowledges that the frequency and intensity of storms over the summer caused frustration and hardship for many customers. With 1.65 million customers impacted by storms, many were without power during multiple storms, and many were without power for an extended duration. Operating a safe and reliable grid is a primary goal of distribution planning, which is reflected by our Planning Objectives, which consider Safe and Reliable two of the critical factors by which we prioritize investments. While the draft DGP filed in August included a robust plan to improve storm resilience, DTEE has augmented some of the short-term investments in the DGP to address opportunities identified during the storms, and to accelerate key programs to improve reliability for our customers more quickly. Key initiatives as a result of summer storm activity are:

- **Tree Trimming:** The tree trimming surge is being accelerated and \$70 million of investments are being pulled ahead into the next few years. The impact of this change is that the 'surge' will be completed by the end of 2024, a year ahead of the original schedule. Once the surge is complete, the entire system will be trimmed to the latest specifications. This additional investment is being done without impact to customer rates. More details on the Tree Trimming program are found in Section 9.
- **Frequent Outage:** Due to the frequency of storms and the total number of customers impacted by storm, there are areas of the system that experienced multiple outages. To improve the reliability in these areas, the frequent outage program has increased investment levels by \$10 million in 2021 and \$20 million in 2022.
- **Customer Communication:** A frequent source of frustration from DTEE's customers during the summer storms was the inaccurate information about outage status provided to customers through DTEE channels (Outage Center website and DTE Energy Mobile app). Some customers were told that their power had been restored when it had not. Other customers received inaccurate restoration estimates. In order to improve the quality and accuracy of the information customers received during the restoration process, additional technology projects are being implemented to align data across multiple sources of information and ensure that customers get accurate, error-free information during the restoration process.
- **Strategic Undergrounding:** The draft DGP outlined an approach to strategically underground equipment in areas where 1) it operationally makes sense and 2) the investments are cost-effective. In this final DGP filing, DTEE has made progress in

identifying a potential circuit for an undergrounding pilot, which will help DTEE to understand customer acceptance, drive down costs if the program expands, and ensures that a strategic undergrounding option is cost-effective and affordable for our customers.

- **Outage credits:** DTEE fundamentally understands that outages have an economic impact on our customers. For example, one of the tools DTEE uses to value strategic investments is the Lawrence Berkeley National Laboratory (LBNL) ICE calculator, which estimates the economic costs of outages on customers. This tool has been used in past rate cases and is described in Section 6 on the benefit cost approach to prioritization. Due to the impacts of the storms this summer, DTEE increased the credits paid to customers who experienced frequent or lengthy outages, including proactively paying a \$100 credit to customers who remained without power until Monday August 16th during the CAT-2 storm.³

DGP and planning for storm resilience

DTEE's approach to distribution planning in the DGP incorporated the need to make the grid more reliable and resilient across multiple phases of the planning process. The overall grid modernization process is described in Section 3 and starts with identifying our customers' needs. As part of this initial process DTEE engaged with stakeholders and with customers through focus groups. Throughout these conversations DTEE listened to what was important to customers – a grid that provides reliable electric services even in the face of severe weather. These experiences helped affirm the importance of two of the core DTEE Electric Planning Objectives - Safe and Reliable.

To develop a near-term plan and long-term vision for the distribution system, the Company introduced scenario-based planning as a tool to provide insight into potential future outcomes and to inform the investment strategy. DTEE developed three distinct scenarios to identify trends likely to impact the grid over the next 15 years. Consistent with the 2019 Michigan Statewide Energy Assessment (SEA) and Michigan Hazard Mitigation Plan (MHMP), DTEE developed a scenario which considered increasing frequency of catastrophic-level storms. This scenario is based on the potential for climate change to increase the threat and type of severe weather that DTEE's infrastructure will be subject to, particularly if there are changes in the form of extreme heat, flooding, severe winds, tornados, or other weather patterns. The scenario planning process also

³ The voluntary \$100 credit was not in lieu of, but rather was in addition to any payments made under R460.744, R460.745, or R460.746.

acknowledged that much of the current infrastructure is at or near end of life, including poles, URD, system cable, and substation breakers.

The combination of the customer needs, the current state of the system, and a potential future scenario with more severe weather threats helped inform both the 5-year near-term investment plan and a longer-term trajectory to develop a system that is robust and resilient.

Specifically, DTEE's 5-year plan includes significant investments in projects and programs that will improve reliability to achieve 2nd quartile All-Weather reliability performance by 2025. Investments in tree trimming, 4.8kV hardening, pole top maintenance and modernization (PTMM), and increased penetration of substation and circuit Automation, on top of benefits of increased situational awareness from the ADMS project, will reduce both the frequency and duration of outages. These programs will upgrade large areas of the system over the next 5-years by reducing tree-related outages and replacing aging overhead equipment with stronger poles, fiberglass crossarms, and polymer insulators which all have increased strength in the face of inclement weather.

The longer-term plan also addresses reliability and resiliency by rebuilding the grid more completely through programs such as the 4.8kV conversion program and subtransmission rebuild and redesign program. These programs will continue to fully rebuild circuits to our latest, more reliable and resilient standard. For example, DTEE constructs all new circuits to National Electric Safety Code (NESC) Grade B construction standards for all new overhead construction. While other utilities still build to Grade C construction, Grade B construction standards allow the electrical system to withstand higher ice accumulation and wind speeds.

DTEE's planning process created a robust strategic investment portfolio that transforms the grid into a system that will provide what our customers are asking for – a reliable and resilient grid that can withstand both current weather and a future with more severe and frequent weather.

2 Stakeholder engagement and input



2.1 Stakeholder engagement

At the heart of the electric grid is our customers. Their input is essential to our planning and grid analysis, so DTEE reached out to our customers and other stakeholders to receive their direct feedback about their perceptions of DTE and how they and their constituents feel about the electric reliability they receive. Third-party research companies were engaged to help conduct open access webinars, customer focus group sessions and one-on-one conversations with state and community leaders in order to gain broad input from all stakeholders through proven research methods. This voice of the customer research was conducted from November 2020 through July 2021.

This summer, our customers and other stakeholders have also been providing more informal feedback about the impact of the season's severe weather on the electric grid. We've been listening and conversing with them through multiple channels, including regular federal, state and local stakeholder meetings; meetings with community partners; and interactions with customers at Community Vans in impacted neighborhoods.

2.1.1 Customer focus groups

DTEE invited a cross section of residential and commercial customers to participate in multiple Zoom-based qualitative focus group discussions to provide insights used to inform the Distribution Grid Plan. The customer focus groups participants were randomly selected from customer data samples based on income, geography and recent levels of reliability. These sessions were held in April and May with the intention of gaining feedback on the following:

- Perceptions of DTEE's current grid reliability
- Awareness of positive or negative trends to that reliability
- Anticipated trends in terms of electricity usage at a household- or business-level over the next five years
- Support for grid improvement (and willingness to fund actions that do more than maintain status quo)

Four key themes were consistent throughout the discussions – energy usage, perceptions of DTEE, reactions to grid benefits and equity.

Energy usage

Both residential and commercial customers felt their energy usage has been fairly consistent for the past five years. Lifestyle or domicile changes were noted as drivers of increased usage among residential customers.

However, expectations for future usage varied between residential and commercial customers. Most residential customers anticipated their future energy usage will stay consistent over the next five years. The remainder anticipated their usage would increase due to continued expanded use of personal electronics and potentially an electric vehicle.

Commercial customers, on the other hand, expected an increase in usage due to requiring more electrical-intensive appliances. Businesses that expected their electrical usage to stay consistent over the next five years were often those that have recently taken action to become more energy efficient, with the intention that those savings will offset any potential rise in usage.

Regardless of their personal usage, both residential and commercial customers consistently expected regional electrical use to increase over the next ten years, driven by personal electronics and technology advancements – particularly electric vehicles (EV).

When it comes to electric vehicles (EV), virtually no residential customer who participated in the focus group sessions anticipated adopting an EV within the next ten years. On the contrary, commercial customers recognized that electric vehicles will become the dominant type of vehicle for the future.

“Our cars are over ten years old, but I think the way the world is going, we may not be able to buy a non-electric vehicle...” –Residential Customer

Current perceptions of DTEE

At the time the focus groups were conducted, in spring of 2021, prior to the summer’s storm events, perceptions of DTEE were fairly positive in terms of reliability, innovation and environmental stewardship. As demonstrated across participants in each focus group, most customers’ views of DTEE are driven almost exclusively by personal reliability experiences.

Customers found it almost impossible to talk about DTEE’s performance or infrastructure activities on a regional or company-wide level because they simply don’t pay that much attention to DTEE unless activities/updates specifically impact their home and reliability. Residential customers’ confidence in DTEE’s ability to meet future demand was highly correlated to the company’s ability to provide that customer’s household with reliable power right now. Commercial customers tended to take a more regional view about reliability and were often more cognizant of issues in parts of the grid that may not affect their account directly.

Both residential and commercial customers cited the lack of visible, or specifically identifiable, maintenance or upgrades to the electrical grid as evidence that DTEE does not aggressively maintain or upgrade its distribution grid. They didn’t know what they should be looking for other than the presence of DTEE work trucks or crews in their area doing “something.” To the customer, visible presence equals maintenance.

“We’re in the dark, literally, about what DTE is doing to make improvement. They are, but it should be visible to the customer. We see the trucks and sometimes see the tree trimming, but there’s no bigger picture for us.”
– Residential Customer

Recognizing that they may be missing DTEE’s activities when it comes to maintenance and infrastructure upgrades, many residential customers suggested that DTEE regularly communicate to them with infrastructure updates that represent modernization/improved reliability in their

immediate communities so they better understand what is actually taking place, even if they don't personally see the work happening.

"I don't have any concerns. When we flip the switch, the lights go on, and it's always been like that. I think DTE is generally very responsive. I'm a satisfied customer."

– Commercial Customer

Reactions to grid modernization benefits

Before talking about specific changes to the distribution grid, residential respondents were given an "off-the-shelf" overview of what the electric industry anticipates over the next ten years via a video produced by the Edison Electric Institute. Commercial respondents were given a visual overview (two diagrams highlighting the difference between a traditional and advanced grid), as well as a list of benefits to enhance clarity and further drive the discussion.

Initial reactions to the smart grid varied by customer type. Residential customers were open to the need for maintenance and modernization of today's grid but were less excited about the more "intelligent" elements of a smart grid – largely because the majority did not anticipate widescale adoption of distributed generation resources in the next ten years. Not surprisingly, initial reactions from commercial customers suggest they were more open to the need for modernization but were more immediately concerned about the cost implications and potential bill increases.

Across both customer groups, even among those who were enthusiastic about a grid overhaul, it wasn't until they were made aware of the risks associated with not modernizing the grid that customers ultimately saw enough value in grid modernization that they would be willing to pay more to modernize and ensure continued reliability.

"I think you would do yourself a disservice if you didn't have holistic conversation about both the positives and the negatives [of improving the grid] and you need to find a path in the middle. Yes, you pay a little more, but you improve [the grid]."

– Commercial Customer

Equity

The issue of equity surfaced spontaneously in every residential discussion group, often at different times and focused on different aspects of the grid plan.

One equity concern was the potential inability of lower-income households to pay for an increase in their electric costs. A second concern was whether industrial customers would be paying their fair share of the modernization, or whether they would benefit off the backs of residential customers. A third concern was making sure the new grid was built out in a way that did not benefit more affluent areas first. On the contrary, equity was rarely an issue raised in the business group sessions.

“I could absorb the cost, but I’d like to see [grid improvements] be balanced across all walks of life and make sure that everyone has equal access to power.”

– Residential Customer

The report for the customer focus groups is included in Appendix VII – Customer Focus Groups.

2.1.2 Community leader engagement

DTEE engaged 35 influential community leaders from our electric service territory to gather feedback on their constituents’ energy usage and anticipated needs over the next five to 15 years. These leaders included community activists, community services organizations, faith-based leaders, and elected and non-elected public sector officials. Each leader was asked to speak on behalf of their constituents and discuss their views on the roles DTEE and energy play for their community members. Interviewees were provided a brief overview of DTEE’s Distribution Grid Plan and were asked to provide initial reactions to help inform the final plan.

Findings from leader interviews focused on their constituents’ current electricity usage, perceptions of DTEE, reactions to the grid plan and anticipated benefits to their constituents.

Current electricity usage

Most stakeholders believe their constituent's energy usage is average or above average, almost none responded with below average. Stakeholders were led to this conclusion for several reasons, including constituents living in older homes that are less energy efficient, and constituents having larger families requiring more energy. Community leaders from more affluent areas also cited a growing number of electric vehicles in their neighborhoods. There were mixed responses on how energy usage has shifted over the last year due to the pandemic, but most believed that residential usage has increased, although less so in areas where the digital divide is still prevalent. Usage among businesses, however, is believed to have decreased due to the pandemic shutdown.

Overall, there is consensus that there will be increased demand for energy in the next five to 15 years, with beliefs that the top contributing factors to increases in usage include:

- Heavier reliance on technology, particularly as televisions and other devices continue to remain a significant source of home entertainment and work from home options are expanded.
- Economic growth in various neighborhoods, including increased housing stock, businesses, and energy drainers like indoor agriculture.

In anticipation of greater usage, affordability and sustainability are top of mind for community leaders. There is significant concern around affordability, especially given the current economic situation brought on by COVID-19. There was also sentiment that energy affordability impacts the State of Michigan's ability to attract businesses

Current perceptions of DTEE

DTEE is seen as an increasingly reliable service provider. Recent winters seem to be milder, so outages are not as top of mind. New substations and tree-trimming are helping to decrease outages and durations. Natural gas and nuclear power plants will help retain reliability as more coal plants are closed.

Community leaders flagged that reliability is only a major issue for customers during an outage. Many of these customers experience outages during severe weather, which they see as a factor outside of DTEE's control. Certain neighborhoods continue to have reliability issues due to infrastructure, but many areas have been addressed.

Overall, DTEE is not seen as a leader in innovation, in part due to community leaders' lack of awareness. There was some recognition that work is likely being done, but most stakeholders could not speak to it personally, or if they could, they felt their constituents could not. Most thought that DTEE could do more to communicate what it is doing relative to preparing for the future of energy, particularly in light of rapidly changing technologies.

While innovation was viewed as movement toward clean energy by many community leaders, others identified an opportunity for DTEE to innovate relative to affordability. Innovation is not only making technological advances to produce cleaner energy, but also determining how it can be done in an affordable and equitable manner. Community leaders believe that DTEE has an opportunity to develop innovative business models and policy solutions to continue as a leader among utilities. Concern was also expressed around the accessibility of clean energy technologies to Detroiters.

Reactions to DTEE's grid plan

While all community leaders had a positive reaction to the high-level summary of the grid plan, there was hesitation on taking a firm stance without additional information and hearing from company leadership. Components of the plan, particularly tree trimming, were welcomed by stakeholders at large. Stakeholders were interested in learning more details about the plan, including:

- How plant retirements will be addressed in terms of capacity.
- How the increasing use of electric transportation fits into the plan.
- How battery technologies fit into the plan in terms of renewable energy.
- How the land use required for solar energy be managed.
- How technology be incorporated into customer issue reporting.
- The extent that underground lines rather than overhead lines are being considered.

Flagging the technical nature of the language used was common, as was the lack of specificity around why each action is necessary. Stakeholders hope to see more "straight talk" language from DTEE that clearly outlines the costs and benefits of upgrading the grid and moving to a cleaner energy future. It is important to note that community leaders were provided with only a one-slide summary of key components of the plan during the interviews.

Stakeholders hope to see their community businesses and constituencies play a role in the conversation to green energy. Concerns were expressed about people of color being able to participate in potential job opportunities that would be made available. Other significant questions were around alternative energy sources and adequately addressing capacity needs in certain geographic areas.

Anticipated benefits to constituents

There was general knowledge that tree trimming and preventative maintenance would make service more reliable. While needed to avoid outages, community leaders relayed that customers are often unhappy with tree trimming due to the appearance of trees or reduction of shade on the property. Keeping up with demand while “going green” is a key priority for community leaders, and they understand that updating grid infrastructure will allow for the transition to more renewables without sacrificing reliability.

Community leader concerns

Community leaders identified two key concerns if DTEE does not invest in the grid:

- The inability to advance toward net zero and a cleaner energy future; and
- The inability to provide reliable service in the face of extreme weather, the popularization of electric vehicles, and growing demand for renewable energy sources.

Additionally, it was considered that without a modern electric grid, Michigan is at risk of being passed over for investments and economic growth. Leaders stated that many residents are unaware of the ripple effect an unreliable grid will cause, not only for themselves at home, but for the region in terms of business, tourism, and attracting residents. There was sentiment that our state’s economy will increasingly be linked to the availability of reliable service, and electric vehicle charging stations.

The negative effects of failing to invest in the electric grid were highlighted as the most persuasive arguments for investment. The threat of decreased reliability and decreased opportunity for renewable energy were top motivators for supporting investments. The added benefits of having a modern electric grid were persuasive as well, but not to the same extent as facing the reality of what would happen without an updated grid.

There is serious hesitation around increasing electric rates to fund grid modernization. While there is consensus that updates are needed, the full burden falling on residents is in question. Several respondents indicated that there was no argument to convince residents, particularly low-income families, that a rate increase is warranted. Community leaders warned of the impacts rate increases have on vulnerable populations. There is a call for policy solutions to help DTEE make investments in the grid in an equitable way. A sliding rate scale, a base fee + usage costs, and bill caps were discussed as potential solutions. The need for energy efficiency in older homes, often owned by senior citizens and low-income populations, was also repeatedly flagged. A role for the federal and state governments, as well as DTEE's internal policies, were mentioned when discussing the transition of infrastructure. Finally, it was mentioned that municipalities might be willing to share costs with DTEE in cases where capacity and reliability are significant issues.

Stakeholders are interested in easy-to-understand information and access to data for themselves and their constituents. Many stakeholders have an interest in understanding DTEE's work around innovation and environmental stewardship. Community leaders believe residents will be more favorable toward DTEE and understanding of rate increases if they feel engaged and knowledgeable about what is happening and why. Understanding the bigger picture will be key for leaders and customers alike.

The report for the community leader engagement discussions is included in Appendix VIII – Community leader engagement.

2.1.3 Technical stakeholder engagement

DTEE reached out to stakeholders who had intervened in recent cases and invited them to attend a series of webinars. These webinars introduced the distribution grid plan development process, provided progress on the plan development and solicited input on the plan. The webinars ranged from 90 minutes to two hours, and were held November 19, 2020, and June 18 and July 29, 2021. Invitations to the first webinar were sent by email from the Company using a list of intervenors. The latter two sessions were also advertised in collaboration with the MPSC Staff using the Licensing and Regulatory Affairs (LARA) alert process. Additionally, stakeholders who have been active in recent cases and participants in the first webinar were invited to an individual discussion. Ten of these discussions were held between March and June 2021.

In general, the stakeholders were interested and engaged in DTEE's distribution planning process. Common themes and questions from these discussions, along with guidance on where DTEE addressed the feedback is shown in Exhibit 2.1.3.1 below.

Exhibit 2.1.3.1 Feedback from Individual DGP Stakeholder Discussions

Common Questions	DGP Section where question is addressed
Scenario planning – How do the scenarios interact, what are the underlying assumptions, and what happens when multiple scenarios come to fruition?	Section 3 – The scenarios and scenario planning process are described in detail in this section.
Energy and environmental justice – How is DTEE addressing these issues in the DGP?	Section 2.2 – DTEE’s approach to address energy and environmental justice is described here.
Availability of granular data – Can more granular data on system performance be shared?	As part of the energy and environmental justice approach, DTEE plans to overlay reliability with census-tract level data. More detail on the approach is found in both Section 2.2 and Section 15 – Performance Based Rate Making.
Undergrounding to improve resilience – Has DTEE considered this approach to improve customer reliability and resiliency?	Section 11.6 – Strategic Undergrounding. DTEE is developing pilot circuits and areas for targeted undergrounding
Ability of the grid to incorporate DERs – Linked to the scenario planning effort, how much DER does the grid have capacity for and where are the constraints?	Section 4.5 describes the Hosting Capacity Analysis pilot, with results expected by the end of 2021.
DER control strategy – How is DTEE evaluating centralized vs. localized control of DERs?	Section 12.8 describes the DER control strategy developed to meet system and regulatory needs.
Cybersecurity - Given some recent cyber security incidents, how is DTEE preparing the grid to be more resilient to potential threats?	DTEE has a cross-functional team engaged in upgrading and preparing the grid for any potential threats. Details that can be shared are discussed in Section 16.3
NWA pilots – Will the pilots proposed address concerns beyond capacity, and is there any consideration of using NWAs to facilitate customer connections in overloaded areas?	Section 12.7 contains details on a suite of NWA pilots, which primarily focus on capacity constraints and incorporating new technologies. Rapid deployment of utility scale batteries is being explored in one pilot to relieve loading constraints.

2.1.4 Feedback on the Draft DGP Filing

The Draft DGP was filed and became available on the U-20147 docket on August 2nd, 2021. During the period between the draft and final filings, DTEE received feedback from stakeholders through both filed comments in the docket, written comments sent directly to DTEE, and a series of individual discussion to solicit feedback directly. Through these channels MPSC Staff, 5 Lakes Energy / Doug Jester, Attorney General's Staff, the NRDC, ABATE, Michigan EIBC / Advanced Energy Economy (AEE), and Soulardarity all provided input on the Draft plan.

Overall, the feedback was positive, with some common themes emerging which can be incorporated into future distribution planning efforts. A high level summary of common feedback is listed below.

General Plan Feedback and Observations:

- Common concern of many stakeholders remains reliability and resiliency improvements, reduction of storm related outages
- Several stakeholders express concerns about the impact of increased investment on affordability
- Scenario based planning was useful, though there is still some clarification needed between scenario based planning and detailed forecasting
- The plan included a concrete suite of NWAs to pilot technologies over the next few years
- The stakeholder engagement process from November through the draft plan filing in August was appreciated

Areas of future interest:

- Energy and environmental justice, including involving stakeholders in the further evolution of the plan
- Community resiliency, including how to structure programs to identify and strengthen key facilities – an issue which extends beyond the electric utilities
- Resiliency and interdependency on other infrastructure systems such as telecommunications
- Next steps on benefit cost analysis and future changes to DTEE's Global Prioritization Model

- Interest in mechanisms to share more data, particularly around outages and load forecasts.

2.2 Energy and environmental justice

DTEE's parent company, DTE Energy, is coordinating a corporate-wide, comprehensive Energy and Environmental Justice plan in conjunction with its broader Environmental, Social and Governance (ESG) initiatives. DTE has established a cross-organization Energy and Environmental Justice Committee to explore ways the company may be able to better serve customers living in highly impacted communities as defined by the Michigan Environmental Justice Screen (MIEJScreen) tool (see methodology below), across a variety of company initiatives. This Committee will also establish processes for external stakeholders to provide feedback on DTE's approach to Energy and Environmental Justice and programming. Additionally, DTE has taken an active role on the Michigan Advisory Council on Environmental Justice

One of DTEE's goals is to incorporate more equity considerations in our grid modernization and clean energy efforts. This aspiration aligns with our objectives of providing all customers and communities we serve with energy that is safe, reliable and resilient, affordable, clean, and community focused and accessible.

For the purposes of the Distribution Grid Plan, we will focus on energy justice as it relates to the long-term planning of the grid to meet the evolving needs of all our customers equitably.

2.2.1 Methodology

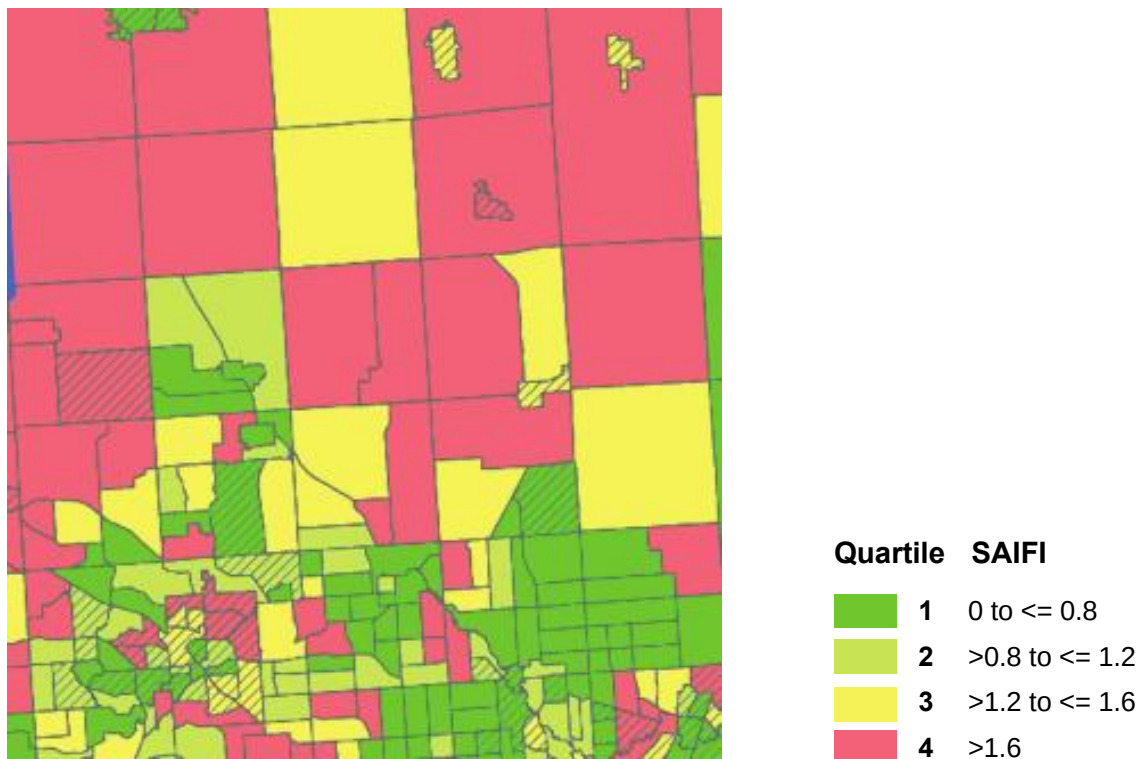
To assess our current state, DTEE intends to use the forthcoming MIEJScreen, which will provide a consistent data set and approach across the state for defining highly impacted communities within our service territory. The screening tool will generate a scale for rating each census-tract level community based on a comprehensive list of indicators that include environmental exposures and effects, sensitive populations, and socioeconomic factors.

DTEE plans to overlay electric reliability data with the MIEJScreen data to identify areas which are considered highly impacted communities and have experienced poor reliability relative to other customers. Customer-focused reliability data will include the three-year average for both number of outages (System Average Interruption Frequency Index – SAIFI) and the duration of the interruptions (System Average Interruption Duration Index – SAIDI) at the census-tract level.

To quantify the current state, the metrics will be categorized into different ranges based on DTEE's average system performance.

The data in Exhibit 2.2.1.1 shows what a hypothetical example of SAIFI by census-tract with highly impacted communities overlaid on a map might look like. This data is illustrative and will be updated with actual data once the MIEJScreen is available. Each census tract is color coded by SAIFI quartile within DTEE's service territory, and the impacted communities are identified with cross-hatching.

Exhibit 2.2.1.1 – Illustrative SAIFI by census tract with highly impacted communities identified



A system-wide analysis and map will be completed in Fall 2021, or when the final tool is released to the public, which will ensure that DTE is using a data set and methodology consistent across the state for any EJ screening initiatives. Once we have identified the overlap of impacted communities with poor reliability, we will review planned investments in those communities, and

make adjustments in the investments, individual programs, or timing to support improved reliability and resiliency more quickly.

2.2.2 Actions

DTEE will be completing this analysis to identify gaps and drive action. The results of the analysis will support the development of an action plan to review our investment programs and processes including investment prioritization, with a focus on reducing the burden of outages on highly impacted communities. Currently there are several initiatives being developed to support this action plan:

- Address areas of poor service in highly impacted communities, as indicated by red or potentially yellow areas with hash marks as illustrated in the image above. We will do this by:
 - o Understanding which circuits are in the highly impacted communities with poor service
 - o Understanding the reasons for outages on those circuits
 - o Scheduling the corresponding work via the hardening, tree trim, pole top maintenance and modernization, or other reliability and resiliency programs, as needed
 - o Understanding if larger strategic projects are needed and incorporating them into the capital plan
 - o Monitoring these areas closely for improvement and follow up
- Assess our global prioritization model and determine how to adapt it to better support highly impacted communities as we invest going forward
- Ensure our trouble and storm restoration strategies are optimized to prioritize highly impacted communities due to the impacts of a longer sustained outage could have on these communities.

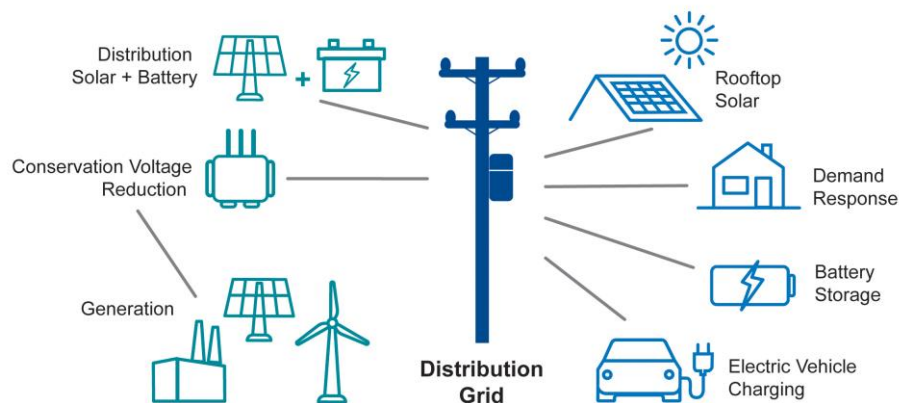
The advancement and emphasis on DTE's Energy and Environmental Justice plan will ensure that the investments in grid modernization support the needs of all of our customers. The goal is to work across the organization to drive greater focus on equity and to build the framework to deliver outcomes that will provide benefits, or reduce burdens, on highly impacted communities. DTEE will be proactive by analyzing and addressing potential opportunities to alleviate the burden

on highly impacted communities with the utilization of the MIEJScreen. DTEE's organizational approach to Energy and Environmental Justice is evolving, but efforts will focus on planning through an equity lens to deploy projects that are reflective of the communities we serve and create desirable outcomes for the future.

3 Grid modernization process



Customer demands and requirements are anticipated to rapidly evolve over the next 15 years as the electric industry undergoes a fundamental shift in the current grid landscape. This change stems from a complex multitude of factors including the growth of DERs within the distribution system, the effects of climate change, and the increasing decarbonization movement away from fossil fuels toward electrification including EVs. While these coming grid impacts pose unique challenges, they also provide DTEE an opportunity to adapt to customers' evolving needs by reshaping the distribution system into a more reliable and resilient grid that provides more value to our customers. Through the grid modernization process, DTEE will work to create a more interconnected grid that is safe, reliable and resilient, clean, accessible and affordable for our customers.



DTEE understands that the future will fundamentally change the nature of the electric grid, to one that is different than the historic paradigm where power is centrally generated, transported via

high voltage transmission lines, then sent through the distribution system to be utilized by the customer. These changes will move the grid toward a more distributed model where power flows both ways through the distribution system. This will require DTEE to address new and complex issues such as two-way power flow, intermittent generation, and DER aggregation and integration. In order to manage these new grid characteristics, greater visibility and data on demand will be needed by both customers and grid operators.

The grid modernization process is designed to facilitate DTEE's transition from the current paradigm to a more interconnected grid. The process calls for a robust, multiphase near-term and long-term investment strategy. Over the next five years, the Company's investments will target safety and reliability improvements along with those investments that are critical to support the longer-term customer desires and resulting grid needs that were identified through this process.

3.1 Methodology

The DTEE grid modernization process was developed using the DOE's DSPx grid modernization framework.⁴ The Company began using the DSPx framework when developing the future investment plan in 2019. Since then, the framework's methodology has helped guide DTEE by providing tools and resources for the planning process. In preparation for the 2021 DGP efforts, the Company used the DSPx framework to help assess current and near-term planned activities and develop a set of recommended strategic investments. The approach that DTEE used in 2020-21 for the development of the Company's current grid modernization strategic process is shown in Exhibit 3.1.1:

Exhibit 3.1.1 Five Step Approach to DTEE's Grid Modernization Planning Process



- 1. Define DTEE objectives** – The team solidified internal distribution grid and integrated resource planning objectives and aligned with MPSC planning objectives.

⁴ *Modern Distribution Grid (DSPx)*. (2019 November) Pacific Northwest National Lab. [Modern-Distribution-Grid Volume 1 v2 0.pdf \(pnnl.gov\)](https://www.pnnl.gov/publications/modern-distribution-grid-volume-1-v2-0.pdf)

2. **Develop plausible scenarios for the future through 2035** – DTEE worked with an external consultant to develop three plausible long-term scenarios that lay out potential grid changes over the next 15 years, as well as an analysis of each scenario’s grid impacts.
3. **Assess current system capabilities** – The team worked with the consultant to assess the current state of the distribution system. Both the current and near-term strategic distribution system investment plans were also reviewed.
4. **Determine grid needs based on scenarios and current state** – The team analyzed the gaps between the grid needs and the current grid capabilities, and from this developed a list of potential “no-regrets”⁵ investments that have positive payoffs irrespective of how the scenarios continue to develop in the future.
5. **Develop near-term plans and long-term vision** – DTEE enhanced its existing five-year distribution investment plan, as well as the longer-term 2026-2035 vision and plan for the grid.

Each of these steps will be discussed further in the subsequent sections.

3.2 Define DTEE objectives

As the first step in the grid modernization process, DTEE created unified planning objectives across both the distribution planning process and the integrated resource planning process. In August 2020, MPSC issued an order in Case No. U-20147 that reinforced what distribution planning objectives are foundational to successfully meeting the needs of our customers and communities. Exhibit 3.2.1 below shows an abbreviated overview of the MPSC’s distribution planning objectives:

⁵ As described in section 3.4.1 Scenario Impact Analysis, “no-regrets investments” are those investments determined to 1) mitigate the grid impacts of one or more scenarios; and 2) meet core safety and reliability requirements

Exhibit 3.2.1 MPSC Distribution Planning Objectives

U-20146 - Distribution Investment and Maintenance Plans		
Distribution Planning Objectives		
Customer Needs	Safety	Reduce risks due to equipment failures or outdated practices, third parties damage or inclement weather. Safety is the Commission's top priority.
	Reliability / Resiliency	The ability to withstand and respond to major weather events and other disruptions and reduce how often and how long customers experience outages. Cybersecurity and physical security also play a key role in ensuring reliability and resiliency.
	Cost-effectiveness /Affordability	Data-driven, value-based approaches to determine when to repair versus replace aging equipment, integrate new technologies in an optimal manner, and provide planning tools and information to encourage efficient siting and operations of customer resources. Ensure long-term affordability for customers through reasonable and prudent investment strategies, considering alternatives and longer-term operational savings.
	Accessibility	Accommodate service to new or expanding customers without causing major network upgrades due to an underlying infrastructure challenge. Planning to assess system conditions under different scenarios to guide siting new projects or accommodating changing load patterns due to customer resources.

MPSC order no. U-20147 confirmed and defined the distribution planning objectives for customer needs under four categories: safety, reliability and resilience, cost-effectiveness/affordability, and accessibility. DTEE reviewed the MPSC objectives, as well as the planning principles that were used for generation planning in the Company's 2019-2020 Integrated Resource Plan (IRP). From this the IRP and distribution planning teams have developed the following DTEE planning objectives:

- Safe
- Reliable and resilient
- Affordable
- Customer accessibility and community focus
- Clean

Exhibit 3.2.2 DTEE Planning Objectives



SAFE

Build, operate, and maintain the distribution grid and generation fleet in a manner that ensures public and workforce safety, operational risk management, and appropriate fail-safe modes and is compliant with State and Federal requirements



RELIABLE AND RESILIENT

Build, operate, and maintain the power system within acceptable standards to withstand sudden disturbance or unanticipated failure of elements. Ensure the grid and diverse generation resources are integrated, with secure supply resources, and can quickly recover from high impact, low frequency events



AFFORDABLE

Provide efficient and cost-effective service along with diverse and flexible generation resources by optimizing the system and benefiting all customers



CUSTOMER ACCESSIBILITY AND COMMUNITY FOCUS

Provide flexible and accessible technology and grid options, and information that empowers and engages customers. Provide effective and timely communication with customers and stakeholders. Favor plans that support the diversity of Michigan communities, suppliers, and workforce



CLEAN

Build, operate, and maintain the resource fleet and grid platforms in an environmentally sustainable manner by achieving low carbon aspirations and clean energy goals. Provide a grid that facilitates a transition to a decarbonized economy

The team ensured the Company's objectives aligned with the MPSC's planning objectives to provide a solid basis for evaluating grid needs in both the five-year plan and longer-term vision. Both DTEE and the MPSC hold customer needs as the fundamental objective for distribution planning. DTEE recognizes that customer needs will be evolving in the coming years to support decarbonization efforts and the integration of DER into the distribution system. In turn, DTEE will look to adapt to these evolving needs by modernizing how the grid is planned and operated.

3.3 Scenario planning

Scenario-based planning is an established methodology for making informed decisions when the future is highly uncertain. To better understand the uncertainties surrounding the fundamental changes to the grid, three future scenarios were developed to assess the potential changes that could impact the electric grid as well as their implications over the next 15 years. This process

enables a closer examination of both the engineering considerations and the identification of overlapping grid needs (i.e. technologies that will be beneficial across multiple scenarios).

The purpose of scenario-based planning is twofold: 1) Assess the robustness of current plans, assuring that current investments are appropriate to serve current needs even as the strategy shifts to incorporate emerging, future needs; and 2) Guide future investment strategies, including identifying areas for further engineering analysis to determine specific actions that may be required.

Grid modernization “plausible” scenario planning is quite different in intent and outputs from probabilistic scenario planning. In this context, plausible indicates that the scenario is directionally correct and seemingly possible; however, the exact forecast is not intended to be precise. The grid modernization scenarios were not input into a model to predict the probability of one or multiple scenarios, rather a plausible future state is developed to understand how the grid is impacted as that scenario progresses along the forecast. Similarly, in the analysis step of a plausible scenario, the grid impacts are determined as a model output, as opposed to a statistical “least-cost solution” output. The team defined the three plausible scenarios to identify the impacts to the grid, and in turn determine the functionalities and technologies that address or mitigate the grid impacts, which is described in detail in Section 3.4.1 Scenario impact analysis. As described in that section, there are technologies that support the grid impacts across multiple scenarios. This overlap helped the team identify no-regrets investments that will be effective no matter how the various scenarios continue to develop in the future. In addition, scenarios may also have offsetting grid impacts. For example, large scale adoption of PV with storage can potentially offset increased substation loading resulting from increased EV charging. Determining the appropriate technology solution to mitigate these impacts is difficult to ascertain given the nature of uncertainty regarding time, scope and magnitude of customer adoption and usage patterns. DTEE recognizes the need to address these and other gaps in future iterations of the grid modernization process. Specifically, as the team monitors the specific signposts over time, DTEE will alter its planning activities to better accommodate the more definitive trajectory.

Furthermore, this process did not intend to determine a definitive 15-year investment plan based on probability or stochastic analysis. Given the uncertainties with customer decision making regarding EV and distributed generation/distributed storage (DG/DS) adoption, as well as an

everchanging and uncertain national regulatory environment, DTEE will develop and continue to refine the long-term investments as we monitor and analyze for the appearance of signposts associated with the scenarios.

Under each scenario DTEE has identified directionally plausible forecasts specific to each scenario (e.g. increased load from electrification), key uncertainties, signposts, and the associated grid impacts that may result. Plausible forecasts represent a possible outlook on the specific drivers for a scenario. The purposes of the plausible forecasts outlined below is not to predict with specificity the numerical value of that scenario; rather, as the directionally plausible forecast plays out, the teams can identify stresses that are seen on the system. Key uncertainties highlight unknowns that affect the scenario's course, such as technology, price or regulatory changes. Signposts are actions that confirm the trajectory of a scenario toward a more certain future. The grid impacts explain how a scenario may impact grid infrastructure given current conditions.

Consistent with DTEE objectives, the scenarios that were chosen are shown below:

Exhibit 3.3.1 DTEE 2035 Grid Modernization Scenarios

Scenario	Description
 Electrification	High electrification of transportation, buildings, and industrial processes
 Increasing CAT Storm	Increased frequency and intensity of catastrophic (CAT) storm threats to electric infrastructure
 DG/DS	High adoption of distributed generation (DG) solar PV and distributed storage (DS) as batteries behind the meter (BTM)

3.3.1 Electrification scenario

The electrification scenario focuses on transitioning vehicles, buildings and industrial processes away from fossil fuels via electrification. Under the electrification scenario, the increased movement toward decarbonization prompts residential and corporate customers to adopt EVs and major cities partially transform their public bus fleet.

Directionally plausible forecast⁶

DTEE's internal forecast from July 2020 includes up to 30% light duty EV growth annually from 2021-2035, reaching 17% market penetration by 2035. A summary of electrification load forecasts for each type of EV chargers is shown below in Exhibit 3.3.1.1:

Exhibit 3.3.1.1 Electrification Scenario Forecasts (2025-2035)

Key Drivers	2025	2030	2035
EV AC L1/L2	383 MW (3.8% peak)	1,425 MW (14.3% peak)	5,286 MW (53.7% peak)
EV DC L1	7.1 MW (0.1% peak)	12 MW (0.1% peak)	19 MW (0.2% peak)
EV DC L2	10 MW (0.1% peak)	50 MW (0.5% peak)	100 MW (1% peak)

Key uncertainties

Key uncertainties under the electrification scenario include EV ownership costs, charging flexibility/availability, community decarbonization initiatives and government regulated targets, including the restriction of fossil fuel use.

Signposts

Signposts for the electrification scenario include EV growth of more than 30%, EV fleets beginning development, public policy extending tax credits and public transportation electrification.

Grid impacts

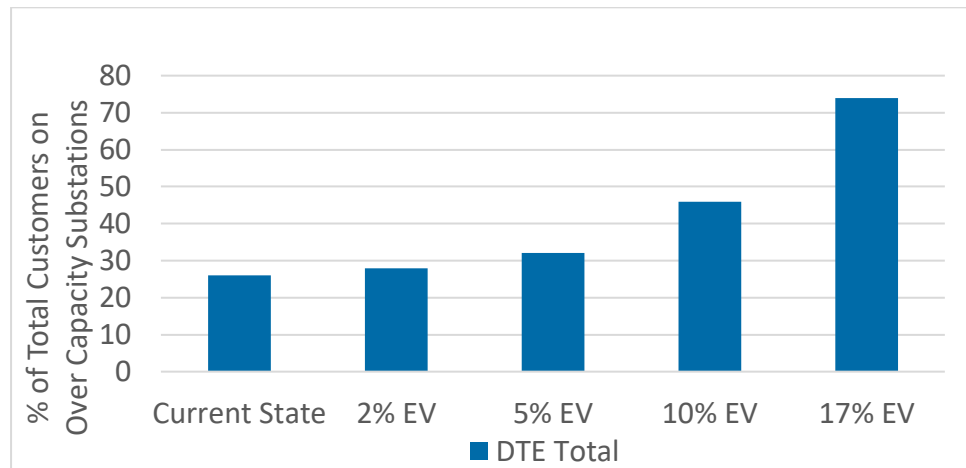
The forecasted electrification will significantly impact the distribution system, as it is expected to add additional load to circuits both on- and off-peak. This will pose additional challenges to the distribution grid which has limited existing capacity⁷ in areas to accommodate the load growth.

⁶ The Directionally Plausible Forecast is not intended to predict the specific impact on peak load, but rather to understand how the increased load stresses the system as it moves along the trajectory of this forecast.

⁷ Capacity refers to distribution assets' ability to deliver electrical power, not the ability to generate the power. It is a measure of how much load DTEE's distribution system can accommodate without negatively impacting operational parameters (e.g. thermal limits). For more information see Section 11 - Infrastructure Redesign and Hardening

Exhibit 3.3.1.2 below shows the percent of total customers on substations that are over their nominal design capacity at varying levels of EV adoption, without increased investment to support the additional load. Substations that are over capacity for extended periods of time pose risks of accelerated equipment aging and failures. Additionally, substations that operate over capacity restrict the flexibility and resiliency of the grid, as load cannot be transferred between locations for planned or unplanned work.

Exhibit 3.3.1.2 Substation Loading⁸



3.3.2 Increasing catastrophic-level storms scenario

The increasing catastrophic (CAT) level storms scenario attempts to capture the risk to DTEE's infrastructure stemming from the increasing intensity and frequency of severe weather. The Company differentiates CAT level storms into two categories: CAT-1 level storms, which affect 5% of the customer base, and CAT-2 level storms, which affect 10% of the customer base.

Directionally plausible forecast⁹

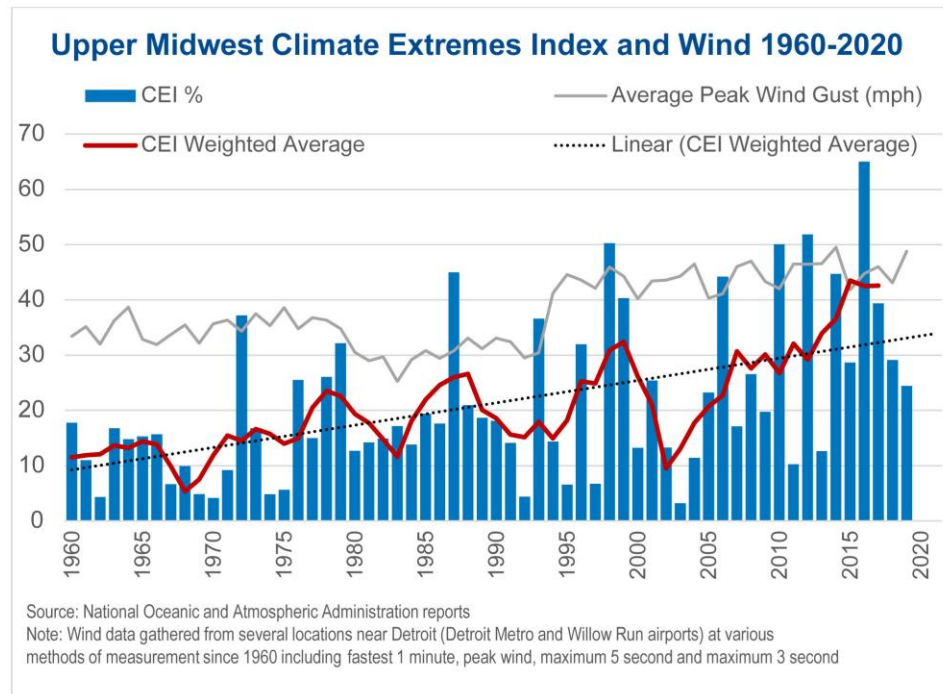
Since 2009, DTEE has experienced 35 CAT-1 level storms (average duration of 4.19 days), as well as 10 CAT-2 level storms (average duration of 5.95 days). One impact of climate change is

⁸ Assuming 12 kW L2 Chargers. Charging set to occur 70% at midnight, 10% at peak, and 20% throughout the rest of the day.

⁹ The Directionally Plausible Forecast is not intended to predict the specific quantity or intensity of future CAT storms, but rather to understand how the increase in CAT storms stresses the system as it moves along the trajectory of this forecast.

the potential for these CAT storm events to occur more frequently and with greater intensity. This is clearly indicated by the MPSC statewide energy assessment, which is included Exhibit 3.3.2.1 below:

Exhibit 3.3.2.1 Upper Midwest Extreme Weather is Increasing in Frequency and Intensity



Specific to Michigan, the National Weather Service has found that severe winds occur more frequently in the southern half of the lower peninsula than anywhere else in the state. In addition, the 2019 Michigan Hazard Mitigation Plan (MHMP) identified severe winds and ice storms as high frequency hazards for the regions of Michigan encompassing the DTEE service area. The MHMP further reported an average of 395 annual severe wind events across Michigan with an upward risk trend. In addition, extreme high temperatures and flooding negatively impact electrical equipment (e.g. cables, breakers and transformer failures), which can compound the effects of summer storms. DTEE is vulnerable to the increasing intensity and frequency of severe weather events due to aging infrastructure, system configuration and high operational utilization of the existing system.

Key uncertainties

Key uncertainties that affect the increasing CAT storm scenario involve future weather patterns, the ability of the grid to sustain weather events, and communities establishing resiliency targets and projects.

Signposts

Signposts that DTEE will look for to confirm the trajectory of the scenario include increasing storm severity and frequency, communities implementing resiliency initiatives and projects, and legislative approval of a bill(s) to support infrastructure projects.

Grid Impacts

The increased likelihood of catastrophic storms will threaten the Company's physical grid infrastructure. In addition to widespread outages that can disrupt service to customers for extended periods of time, storms can damage equipment, resulting in costly emergent work. This will increase the need for certain structural and operational changes (e.g. redundancy, automation) to enhance system reliability and resiliency.

3.3.3 Distributed generation/distributed storage growth scenario

Lastly, the Distributed Generation/Distributed Storage (DG/DS) scenario focuses on adoption of distributed generation and storage technologies that can reduce customer electricity demand and/or export power back to the grid. Both DG and DS are classified as subsets of DER technologies in MPSC Case U-20147, which defines DERs as:

“A source of electric power and its associated facilities that is connected to a distribution system. DER includes both generators and energy storage technologies capable of exporting active power to a distribution system.”

While there is more than one type of DG, solar PV is typical. Under the DG/DS scenario, residential customers and commercial and industrial (C&I) customers rapidly adopt DG/DS. The scenario also assumes some C&I adoption of Combined Heat and Power (CHP).

Directionally plausible forecast¹⁰

In this scenario, driven by government subsidies in the form of tax credits and decreased cost, distributed customer-sited solar PV could potentially grow an average of 23% annually from 2020-2030 and 15% from 2030-2035. Additionally, in the future battery storage is expected to be paired with approximately 60% of new solar PV systems, resulting in potentially 395 MW of added capacity by 2035. In this scenario, DER will be able to participate in wholesale markets (directly or through aggregations via Federal Energy Regulatory Commission (FERC) order 2222). Plausible forecasts are summarized in Exhibit 3.3.3.1 below:

Exhibit 3.3.3.1 DG/DS Scenario Forecast (2025-2035)

Key Drivers	2025	2030	2035
Solar PV – Residential	66 MW (0.6% peak)	210 MW (2.1% peak)	422 MW (4.3% peak)
Solar PV – C&I	52 MW (0.5% peak)	151 MW (1.5% peak)	304 MW (3.1% peak)
Storage – Residential	15 MW (0.1% peak)	94 MW (0.9% peak)	254 MW (2.6% peak)
Storage – C&I	10 MW (0.1% peak)	50 MW (0.5% peak)	141 MW (1.4% peak)
CHP	33 MW (0.3% peak)	53 MW (0.5% peak)	76 MW (0.8% peak)

Key uncertainties

The key uncertainties in the DG/DS scenario are customer adoption rates, federal and state policies (tax credits, MW targets, DG/DS carveouts) and MISO's implementation of FERC Order No. 2222¹¹.

¹⁰ The Directionally Plausible Forecast is not intended to predict the specific impact on peak load, but rather to understand how this load stresses the system as it moves along the trajectory of this forecast.

¹¹ Through FERC Order 2222/2222A, Distributed Energy Resources (DERs) that were previously only able to participate in retail programs are now able to participate in the wholesale market via aggregations

ignposts

The signposts the Company will look for include MISO developing operational coordination with DTEE to implement FERC Order No. 2222, growth of solar + storage installations in the Company's service territory and Michigan increasing the solar PV cap and adopting storage targets within the next 15 years.

Grid impacts

Moderate to high levels of DG/DS on one or more circuits on a substation can cause several issues, including potentially problematic reverse power flow, mainline and lateral voltage and thermal issues, as well as protection malfunctions. At very high adoption rates, customer DG/DS can create issues with power back feeding into the subtransmission system, as other utilities have experienced. As a result, the Company will have to develop grid controls to ensure stability, that may include a need to curtail generation at those sites or invest in imperative system upgrades. Customer DG/DS adoption will also impact loading on secondary systems including service transformers, creating voltage and thermal issues. These system issues arising under different levels of DG/DS are expected to be widespread given the age, design, and condition of the DTEE distribution system. As a result, the ability for customers to interconnect even modest amounts of new distributed generation may be limited in many parts of the DTEE system absent system upgrades and/or significant efforts to optimize charging, as well as DG and storage utilization to relieve grid constraints. See Section 12.8 for further discussion on DER Control.

3.4 Current state assessment and grid needs analysis

DTEE worked with the consultant to conduct a deep dive of the current state and near-term investment plan. This included third-party interviews with several Company SMEs, a review of the 2018 grid plan as well as the initial plans for this filing and a detailed review of the Company's informational technology and operational technology (IT/OT) strategy roadmap. The team then applied the DOE DSPx investment prioritization to DTEE's current state and existing strategic investment plan. This step includes understanding asset health, risks and replacement cycles as described in detail in Section 8 Asset health assessment. This more clearly illustrates the near-term needs under DTEE's four strategic investment pillars, which formed the basis of the current state assessment.

3.4.1 Scenario impact analysis

Each scenario presents technical and operational challenges that are either unique to that scenario or common across scenarios. Results from the analysis of the grid impacts, discussed above, helped to identify grid needs by evaluating these impacts against a comprehensive catalog of planning and operational functionalities adapted from the DSPx framework. This tool consists of approximately 30 unique grid functionalities and dozens more corresponding technologies associated with a modernized electric grid¹². In the scope of DTEE's grid modernization effort, grid functionalities are both the 1) physical grid infrastructure and 2) engineering and operational capabilities needed to mitigate impacts identified from each of the three scenarios. To enable a grid functionality, DTEE has a suite of technologies at its disposal, including hardware and software, planning tools, infrastructure upgrades, etc.

For instance, "Physical Grid" represents a foundational functionality under the DTEE adapted DSPx framework. A durable, modernized physical grid is crucial to withstand the grid impacts of increasing severe weather, as well as to expand system capacity to better enable electrification and DER growth. For "Physical Grid" the enabling technologies include voltage conversion, conductor/equipment capacity increases, and hardening. Another example, the functionality "Reliability Management", involves a number of processes and systems that enable distribution operators to discover, locate, and resolve power outages in an informed and efficient manner. Technologies that enable "Reliability management" include the Outage Management System (OMS) and Fault Location, Isolation and Restoration (FLISR).

Ultimately, no-regrets investments are the output of the scenario impact analysis. However, it is critical to note that not all technologies identified in the framework were categorized as "no regrets" investments. Specific to this analysis, and as defined in Section 3.1, no regrets investments were determined to be those functionalities and technologies from the adapted DSPx framework that met the following requirements: 1) effectively addressed grid impacts in one or more scenarios and 2) met core safety and reliability requirements. To date, DTEE's investment plans have largely been centered around the second requirement, such as investments in infrastructure resilience, hardening and redesign.

¹² [Modern-Distribution-Grid Volume II v2 0.pdf \(pnml.gov\)](#)

In addition to recommending investments corresponding to existing projects and programs, the Company also identified investments that expand on existing efforts, such as a prioritized and accelerated voltage conversion plan, as well as new grid needs, including DER control and monitoring and load forecasting. The Company's planning approach ensures that its chosen investments are sound irrespective of how the future scenarios develop.

3.4.2 Key categories of investment

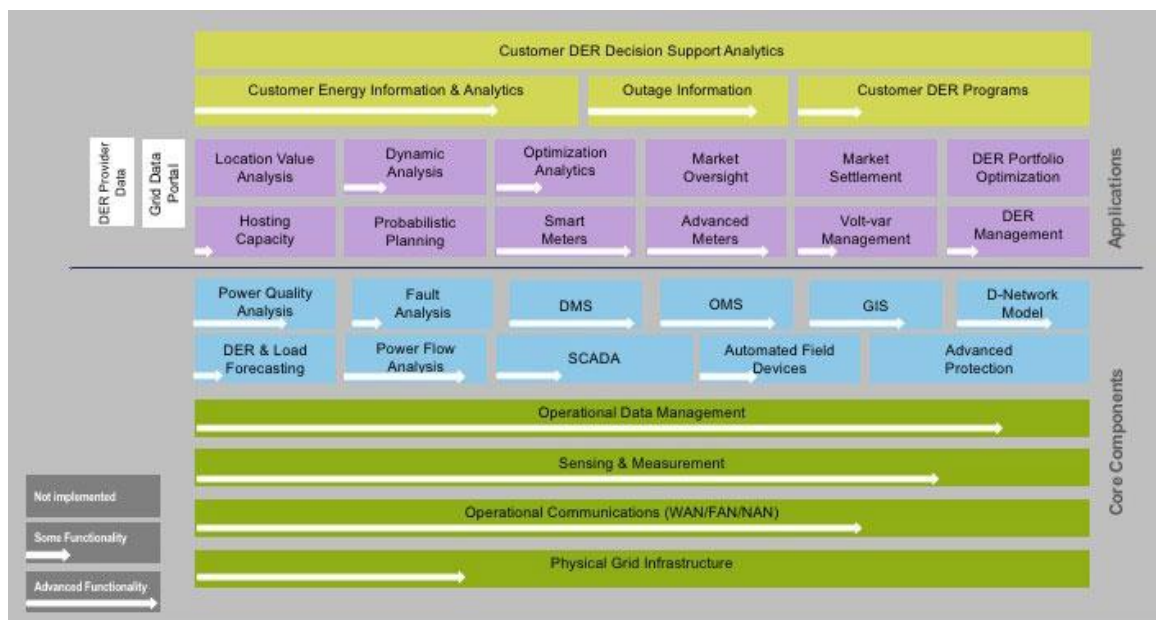
As further described in Section 3.5, DTEE organized more than a dozen cross-functional workgroups to develop no-regrets investment opportunities based on the three scenarios and core grid safety and reliability needs. The no-regrets investment opportunities have been consolidated across five key categories. For each category, a description of the current state and steps needed to progress toward the desired future state are discussed, with recommendations guided by the scenarios. Each section is representative of the largest initiatives taking place and is not meant to comprehensively cover all the no-regrets investment opportunities that were identified, or all the active grid modernization efforts underway at DTEE. A more comprehensive look at the Company's no regrets investments can be found in the sections corresponding to the four pillars of DTEE's strategic investment framework, as further described in section 6 – DGP investment summary. The investment categories are listed below:

- Physical grid infrastructure
- Observability
- Analytics & computing platforms
- Controls
- Communications

As part of grid modernization, DTEE plans to implement each of the no-regrets investments, however, the exact timing and sequence of these investment decisions beyond the next five years is uncertain and will be evaluated against enumerated signposts and changing scenario-driven grid impacts. DTEE's near term five-year plan positions DTEE for the future by supporting the Company's ongoing safety and reliability investments in its underlying infrastructure and addressing known scenario impacts.

The current state of the DTEE grid modernization effort, including the grid improvement projects that are currently in progress, is represented conceptually below in Exhibit 3.4.2.1. This figure is adapted from Volume 3 of the DSPx framework and has been included only to conceptually show the status of DTEE's core grid modernization platform and application layers. The graphic organizes grid modernization investments into two main categories: core components and applications. Per the DSPx framework, the core components shown comprise the essential technologies, as discussed in the scenario impact analysis, that provide a foundation for a modern distribution system¹³. The applications represent the various additional technologies that can layer on top of the foundation and add additional functionality as determined to be necessary by DTEE. The length of the arrow in the box represents the relative status of the Company's efforts in that category now or as projected considering the near term planned investments. For example, "smart meters" has a longer arrow due to DTEE's AMI network being well established.

Exhibit 3.4.2.1 DTEE Grid Modernization Near-Term Status



¹³ *Modern Distribution Grid Decision Guide Volume III*. (2017 June) Pacific Northwest National Lab. [Modern-Distribution-Grid-Volume-III.pdf \(pnnl.gov\)](#)

Physical grid infrastructure

DTEE has identified the need to invest in physical infrastructure (e.g. substations, poles and wires) upgrades as the foundation to meeting all future grid needs. The oldest DTEE distribution infrastructure, the legacy 4.8 kV system, has a design basis that is incompatible with a modern distribution grid platform. The system has its strengths in resiliency, however, due to its ungrounded design, fault locating can be challenging and may complicate restoration efforts. The lower distribution voltage inherent in the 4.8kV system is also becoming increasingly capacity constrained for meeting current electric load demands. As discussed in Section 11.3.2 – Conversion Prioritization, approximately 20% of 4.8 kV substations are currently over their firm rating. However, capacity constraints are not limited to the 4.8kV system. Currently, approximately 20% of 13.2kV substations are also operating above rated capacity. Firm rating of a substation is defined as the limit to carry load that a substation can have during a failure or outage of a single component (N-1). Loading beyond the firm rating does not overload equipment in normal conditions but introduces the risk of loss of power for customers in case of equipment failure for the duration of time when loads are beyond the firm rating. Options and risks are assessed against affordability to manage risk while the substation operates beyond the firm rating. In some cases, the limited increased risk will be managed for some amount of time to maintain affordability for the customers. As noted in the Electrification Scenario above, these constraints negatively affect the operational flexibility of the distribution system due to limited capacity available to support contingency operations and will only be exacerbated by even moderate levels of EV adoption.

An additional driver of the reliability and operational challenges is the age of many of DTEE's grid assets. On average, the Company's 4.8kV and 13.2kV substations are 68- and 43-years old, respectively. While not the only factor, aging equipment can increase the probability and severity of failure, resulting in service interruptions and the need for emergency replacements and repair. The cycle of emergent work — breaking and fixing — can undermine longer-term strategic investment plans, many of which are necessary for further grid modernization. This is because emergent work is typically more costly than the equivalent planned equipment repair or replacement. Further detail on the age and health of our assets can be found in Section 8 – Asset Health Assessment.

Simply put, in addition to impacting reliability today, an aging and constrained grid platform will be unable to reliably accommodate electrification growth and our customers' needs. In order to

continue to support these customer needs, DTEE plans to significantly expand system capacity as discussed in Section 11 - Infrastructure Redesign and Modernization. DTEE's existing 4.8kV conversion program serves as the foundation of the Company's system capacity expansion efforts. However, the pace and magnitude of anticipated load growth are driving the need for the Company to accelerate this conversion work, beginning with substations facing the greatest capacity constraints and highest safety and reliability impacts.

The Company has initiated a study examining the feasibility of introducing a higher distribution voltage (25 kV+) onto the DTEE grid as discussed in Section 17 – Next Steps. In addition to already high system loading on the 4.8kV and 13.2kV systems, the EV adoption discussed in Section 3.3.1 Electrification scenario will translate to a significant increase in the number of substations over their firm ratings. Moving to a higher voltage class could provide DTEE with significantly more power delivered per line mile, lower cost per kW of capacity, improved voltage control, fewer substations and circuits, and an overall more capable system that can better accommodate DER growth and electrification.

Observability

In addition to physical infrastructure upgrades, a modern distribution grid needs operational visibility into physical variables and events for all grid conditions that may need to be addressed. Building out a robust situational awareness platform is essential to reliably plan and operate the grid. Sensing capability enhancement forms one of the core components of DTEE's grid modernization strategy. Over the last decade, DTEE has made significant strides in boosting its sensing coverage throughout the distribution system. The Company's investments in substation monitoring (DSCADA), line sensing and advanced metering infrastructure (AMI) and the new ESOC are proving to be indispensable assets as they are used to support a range of DTEE's internal functions, including grid visibility, forecasting, customer metering and billing, and supporting distribution planning and operations. Further details on the DTEE sensing capabilities can be found in Section 12.4 – Sensing.

Despite the addition of sensing assets in recent years, work remains to both enhance the Company's observational capabilities and translate those observations into situational awareness and actionable intelligence that help customers. The increase of DERs on the Company's system will create a need for enhanced, granular monitoring and sensing capabilities because DERs

interact with the grid in fundamentally different ways than conventional technologies. Consequently, DERs will need to be managed carefully. Everyday grid operations such as power quality management will become more dynamic as customers adopt DG and transportation is electrified, resulting in a wider range of less predictable loading conditions.

As part of longer-term grid planning, DTEE is planning investments to improve observability on the distribution system. Observability is needed to monitor system attributes such as voltage, current and harmonics as well as increase awareness of real-time system conditions, including locating faults. Currently, DTEE has installed several thousand line sensors on the distribution system; however, these older vintage sensors have limited functionality as technology has evolved since installation. While these installed sensors can detect faults and read current, they are incapable of measuring voltage and harmonic data. Next generation line sensors are capable of measuring voltage, current and harmonics and will target monitoring locations with high penetration of DER or where voltage measurements are critical. Once installed, these sensors can also be reconfigured to a higher voltage class to maintain the system through planned conversion work. Ultimately, the exact technology solution and deployment strategy will include considerations for other grid needs, including system protection, and the capabilities of existing and planned sensing technologies.

Analytics & computing platforms

DTEE has made significant investments in its analytics and computing platforms, including mining data quality from AMI meter data, constructing a modernized ESOC and the ongoing implementation of its ADMS platforms. These investments substantially improve the Company's ability to efficiently monitor and operate the grid, respond to emergency conditions and outages and facilitate the integration of DER's into the distribution system. In addition, the Company is making the necessary investments to improve data quality and integrity through monitoring, validation and automated corrections tools. These data quality enhancements will enable the Company to fully realize the capabilities of its investments in the ADMS platform and better leverage AMI data through robust data analytics projects. These investments provide DTEE with the data and analysis for remote and autonomous operations on the system, which improves reliability.

While DTEE has made significant progress in improving its analytics and computing platforms in recent years, the Company recognizes that more work will be required to keep pace with the increasing volumes of data and associated analytics requirements driven by DER growth. As DER growth continues, grid impacts such as voltage regulation issues, reverse power flow and overloads will become more area and time specific. Accurate distribution system models and loading data will become increasingly important as planners work to forecast areas of DER adoption and identify distribution abnormalities well in advance of when they occur. In tandem, uncertainties surrounding consumer EV charging and consumption patterns can further complicate these planning and forecasting efforts. The coming expansion of DER technologies and their associated grid impacts will require expanded data collection and an advanced analytics platform to assist the Company with complex decisions relating to planning and operating the grid.

The Company has identified key investments to meet this challenge, as discussed further in Section 12.9 – Technology Roadmap. This includes tools and support for a more accurate mapping of the distribution system, forecasting, the interconnection process and hosting capacity. Technology investments in the Company’s digital mapping and network model will increase alignment of this digital representation of the grid to physical field conditions. This will be crucial as increasing volumes of data driven from the growth of DER become integrated into the Company’s computing platforms.

In addition to the above improvements, forecasting tools will need to integrate and leverage more granular data sources to support planners with identifying potentially constrained areas well in advance of potential problems. As described in Section [4.2 Forecasting](#), distribution-level forecasts have historically been developed spatially at the substation and circuit level, and for single annual peak demand. While this approach has historically performed well in predicting past load growth in DTEE’s service territory, DER technologies such as EV and PV have complex load profiles that vary by the hour, day, month and year. As such, the Company is currently planning to develop robust “8760” forecasts to have accurate load forecasting for its circuits and substations all 8,760 hours of the year. These forecasts will be able to modularly account for newer technologies such as EVs, PVs, and storage, as well as enable DTEE to adequately identify when and where distribution improvements will be needed. DTEE plans to develop these tools through an integrated forecasting process that will account for both system-wide resource and distribution planning. With distributed generation and storage growth also comes an increase in

interconnection requests and studies. To adapt to the increased volume of interconnection requests, the Company will look to acquire tools to automate portions of the existing process and analysis. For more information please see sections [4.4 Interconnection process](#) and [4.5 Hosting capacity analysis](#).

Controls

With significantly higher penetration of customer assets on the grid, DTEE will need increased coordination between utility and customer assets to operate a distribution grid of the future. Accommodating growing customer desires to adopt technologies such as renewable generation, storage and electric vehicles will require significant improvements in grid controls. This includes controlling switching devices to rapidly reconfigure the grid during outages, as well as coordinating certain DER assets (e.g. DER aggregations, energy storage, utility owned solar facilities) to maintain system integrity and reliability. Integrating technologies for our customers, such as electric vehicles, distributed generation and storage will require active management and control of these assets to maintain reliable grid operations. To this end, the Company has invested in remotely controlled devices on the existing subtransmission and 13.2kV systems, including automatic and nonautomatic pole top switches, line breakers and reclosers. These controlled devices help further enable distribution automation on the 13.2 kV and subtransmission systems. SCADA technology currently enables remote control of about 32% of distribution substations. At a circuit level, “loop schemes” work to facilitate automation by automatically transferring power into adjacent sections of the circuits when an outage is detected and are installed on about 5% of distribution circuits. These loop schemes are a standard for all new or converted circuits. Efforts are also underway to automate specific aspects of the 4.8 kV system where possible. More information on these automation efforts can be found in Section 12.5- Distribution Automation. As a separate but related measure, the Company is also piloting conservation voltage reduction (CVR) / Volt Var Optimization (VVO) to help reduce peak demand and energy consumption. While currently in its early stages, DTEE is expecting its CVR/VVO scheme to allow the SOC to remotely balance line voltage and system reactive power to reduce line losses and to improve system efficiency.

Increased coordination and control of DTEE’s physical grid assets and various interconnected resources will be crucial in a future where electrification, severe weather and DG/DS growth pose

heightened risks to the systems reliability and resiliency. As discussed in Section 12.8 - DER Control Strategy, the increased interconnection of distributed generation introduces voltage variability that can adversely impact power quality on circuits with high DER penetration. In tandem, increasing storms will further threaten grid reliability and increase outages unless effective control schemes are able to both sectionalize the grid and transfer load to adjacent circuits. Increasing control capabilities to adapt to these unique and at times overlapping impacts supports DTEE's safety and reliability priorities and will assist the Company in integrating more DERs into its distribution system.

To properly enable a future where DTEE can adapt to DER grid impacts and mitigate the threats to reliability posed by severe storms the Company has been undertaking certain investments to improve its control capabilities. First, remote and automatic circuit reconfiguration will allow the new ESOC to remotely sectionalize the grid and transfer load onto adjacent circuits when the need arises (e.g. outages). This need will be addressed as part of DTEE's automation strategy in Section 12.5.1, which further outlines DTEE's goal to extend circuit automation to the entire overhead distribution system. Next, the Company recognizes that distributed resource management is needed to monitor and control DERs to ensure thermal and power quality standards are met. To address this, DTEE has developed a DER control strategy centered around managing reliability concerns originating from the increasing number of DERs in the distribution system.

Communications

DTEE has made investments in telecommunications technologies that enable field device observability and control. While much has been done to improve the grid telecommunications, the current architecture must be enhanced to meet the throughput, latency and reliability standards that a modern grid requires.

DTEE has built an existing communication architecture from the subtransmission system down to individual meters. The Company has been implementing a tiered design to serve communication needs now and in the future in a highly resilient manner to provide a high quality of service:

- Tier 1 consists of microwave and fiber backbones for the Tier 1 and Tier 2 systems that traverse between critical facilities such as data centers, power plants, service centers and subtransmission stations.
- Tier 2 consists of wireless point-to-point and point-to-multi-point systems that allow data collection from the Tier 3 system, distribution substations and pole top devices.
- Tier 3 consists of wireless mesh networks that support end point SCADA and AMI meter Communications.

In addition, DTEE has built a new Network Operations Center (NOC) co-located with the ESOC that will monitor all communications required for the Company's operational systems.

As DER growth continues, so too does the number of connected devices and associated volumes of data that must be sent securely and in near real time. While the Company has made headway with its three-tiered implementation, many of the older assets within the current telecommunications architecture are incompatible with modern systems and incapable of fielding the data requirements necessary to support DER growth. The radio-based communications system was not initially designed to support large volumes of data, and many areas of DTEE's service territory lack high bandwidth communications. The Company has also had to adapt to several external regulatory changes from the FCC. The FCC spectrum reallocation has negatively impacted utility communications on wireless point-to-point and point-to-multi-point systems, as well as the planned phase out of 3G cellular networks, which is the basis for the current AMI architecture. DTEE is also susceptible to network blackouts caused by using leased Internet Service Providers and leased Cellular Service Providers that do not build their networks with the same level resilience that a utility would. This has resulted in the strategic shift to build DTEE privately owned and maintained networks to securely control and operate assets requiring reliable telecommunication systems.

DTEE is expanding upon the privately owned tiered telecommunications system design to ensure that appropriate bandwidth and latency are available for the support of DER growth, grid distribution automation, and evolving customer needs. A large part of this strategy involves deploying fiber with enough coverage to support critical sites such as data centers, service centers, power plants, substations and renewables sites. In addition to fiber deployments, wireless point-to-point and point-to-multi-point systems will supplement telecommunications

needs to intelligent edge devices. The tiered telecommunications system design places emphasis on system redundancy and resiliency, better allowing the Company to operate during outages. For more information on DTEE's telecommunications strategy see Section 12.3 – Grid Telecommunications.

3.5 Short-term investment plan and long-term vision

This grid modernization process was the foundation of DTEE's five-year distribution investment plan, as well as a longer-term 2026-2035 vision and plan for the grid. Grid modernization is an organization-wide process, with all relevant business units within the Company actively contributing to develop DTEE's grid modernization process. More than a dozen workgroups of subject matter experts from various business units within the Company identified gaps between the current and future state for specific functional areas of the grid. The grid modernization scenario work informed the workgroups of additional no-regrets investment opportunities that should be considered in the five-year investment plan for that functional area. Each workgroup created the corresponding five-year investment plans to integrate the necessary no-regrets investment recommendations, along with the longer-term 2026-2035 vision. This five-year plan contains the expected system impacts from their investment plan, as well as details on strategic projects and programs. The process by which all the investments from the various workgroups were prioritized is described in Section 5.1 Benefit Cost Analysis Approach.

Exhibit 3.5.1 below contains a selection of the grid needs from the various categories discussed in Section 3.4.2 – Key categories of investment.

Each grid need is covered within a section of the report, and the table below serves as an index to where it can be found, as well as whether DTEE will be addressing it in the five-year plan, the longer-term 2026-2035 vision or both. This table is representative and is not intended to comprehensively cover all the no-regrets investment opportunities that were identified, or all the active grid modernization efforts underway at DTEE. Rather, this table is simply meant to highlight the largest initiatives taking place. Due to evolving customer needs and the uncertainty surrounding how the scenarios will develop, DTEE will continue to evaluate the signposts to assess the integration of the recommended no-regrets investment opportunities into the Company's longer-term vision and plan for the grid. Grid modernization is a continually evolving effort, and as signposts point to a more certain future, DTEE will take the necessary steps to

modify the distribution system into a more reliable and resilient grid that better serves the evolving needs of our customers.

Exhibit 3.5.1 Mapping of Grid Needs to DGP Sections

Timeframe Grid Need	Bucket	Section of DGP Report	2021-2025	2026-2035
Acceleration of the asset upgrade/conversion strategy	Infrastructure	Section 11.3 - Conversion Plan	X	X
Evaluation of a higher voltage class	Infrastructure	Section 17 – Next Steps	X	X
4.8 kV Fault Detection	Observability	Section 9.5 - 4.8 kV Relay Improvements Section 12.5 - Distribution Automation	X	X
Boosting overall coverage of distribution system	Observability	Section 12.4 – Distribution Sensing and Monitoring	X	X
More accurate grid mapping	Analytics & Computing Platforms	Section 12.9 – Technology Strategy Roadmap	X	X
Load and DER forecasting solution	Analytics & Computing Platforms	Section 4.2 Forecasting	X	
Interconnection process tools	Analytics & Computing Platforms	Section 4.4 – Interconnection process	X	
Hosting capacity analysis	Analytics & Computing Platforms	Section 4.5 - Hosting Capacity Analysis	X	

Remote and automatic circuit reconfiguration	Controls	Section 12.5 - Distribution Automation	X	X
Distributed resource management	Controls	Section 12.8 - DER control	X	
Telecommunications strategy	Communications	Section 12.3 - Grid telecommunications	X	X

4 Distribution planning processes and tools



4.1 Introduction

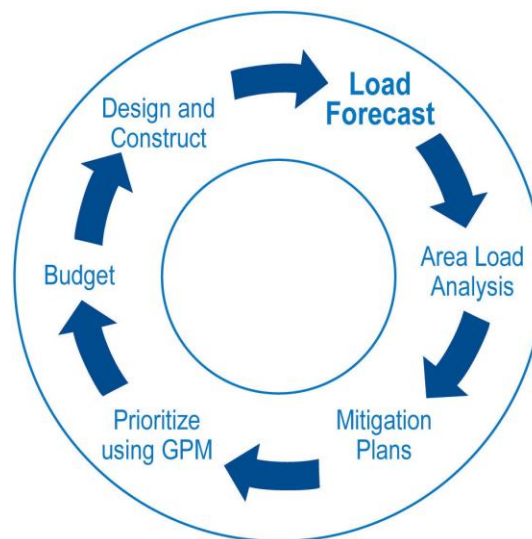
For many years, traditional distribution planning processes have supported utilities in meeting their core objectives to provide safe, reliable, and affordable delivery of electricity. As has been noted in prior sections of the DGP, DTEE is looking at the evolving landscape and preparing for a transformation of the grid. This grid modernization is primarily being driven by changing customer needs, advancements in technology, impacts of severe weather, and electrification led by decarbonization efforts. The confluence of these factors has prompted the Company to examine current planning processes, tools, and practices as we prepare investment plans and the grid moves to a more complex configuration. As described in Section 3.3, Scenario planning, DTEE has started using scenario planning to understand the potential impacts from evolving needs to support our long-term vision and plan. The Company is also focusing on six key distribution planning processes. This section of the report focuses on five of those processes, predominantly through a distribution lens: 1) forecasting, 2) non-wire alternatives (NWA), 3) interconnections, 4) hosting capacity analysis (HCA) and 5) integrated resource planning (IRP) and distribution planning (DP) alignment. The sixth process is stakeholder engagement, which was discussed in Section 2. As noted in Section 3 - Grid modernization process, many of the distribution planning processes are worked collaboratively across different groups within DTEE. In addition, these processes are at varying stages of maturity and will require continued investment in analytic tools as well as process changes to achieve the desired goals and objectives.

4.2 Forecasting

This section of the report describes the process to achieve aligned generation system level and distribution level load forecasts. Currently, the distribution load forecast within DTEE is developed using data, assumptions, and methodologies created within Distribution Operations, with advisory input from other departments in the Company. Distribution level forecasts are typically based on the previous year's actuals, adjusted for projected load growth.

As shown on the diagram below, the distribution level load forecast is an input into the distribution planning cycle and is used to identify grid impacts. If an issue is identified, mitigation plans are then developed and prioritized using the Global Prioritization Model (GPM) as detailed in section 5.

Exhibit 4.2.1 Current State of Annual Distribution Planning Cycle



The distribution level load forecasting process starts with gathering circuit and substation SCADA data. From this point, the forecasting process uses annual single historic peak demand (typically summer demand) from committed loads and applies a growth factor based on demographic and other economic data. While the distribution-level single historic peak demand load forecast approach has been sufficient in identifying and addressing grid impacts in the past, the potential for increased adoption of distributed energy resources (DERs) and electric vehicles (EVs) will require a change in forecasting approach to account for demand and load shifting. To date, the

incorporation of the impacts of DERs and EVs to the load forecasts has occurred only at a generation system-level, not at a circuit/substation-level. Furthermore, all interconnection (section 4.4), energy waste reduction (EWR), or demand response (DR) requests at the distribution level are typically evaluated for grid impacts on a case-by-case and manual basis.

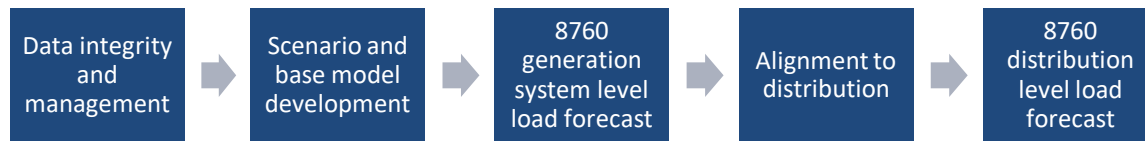
DTEE recognizes that load forecasting at the distribution level will need to advance and become more robust. The Distribution Operations (DO) team is working with the Corporate Energy Forecasting (CEF) team to develop a forecasting solution that accounts for the impact of DERs and EVs into the distribution load forecast. DTEE is working toward a transition to an integrated distribution and generation load forecast and will adopt a more integrated and progressively granular approach to load forecasting. The forecasting process will inform long-term planning at both the generation system and distribution levels. This approach is consistent with the commission's May 27, 2021 Order in Case No. U-20633, Integration of Resource, Distribution, and Transmission Planning, to align planning processes, data, assumptions, and methodologies for a more consistent, integrated and scenario-based approach.

Distribution load forecasts in the past have always been developed spatially at the substation/circuit level, and at a single annual peak demand. The distribution-level load forecast in the near term (five years) will change over time due to the dynamic characteristics of DER and EV load profiles. A load profile is a variation of electrical load versus time typically driven by customer type, temperature, and seasonality. Due to more complex load profiles for EVs and DERs (e.g. EVs typically charging at night and solar photovoltaics (PVs) intermittency due to cloud cover), load shifting will likely become more prevalent as adoption increases. DTEE is therefore pursuing the capability to conduct 8760 hourly load forecasting to help analyze and address any grid impacts from these evolving technologies. A project has been deployed to support integrated resource planning which will utilize hourly load shapes. In the next few years, the distribution planning group will coordinate and leverage the work done in integrated resource planning to adapt those hourly shapes for generation/sales to provide meaningful demand load profiles at the substation/circuit level. There will be further analysis into creating and refining hourly profiles for PVs, EVs, storage and other new or emerging technologies. This new aligned process will be called integrated forecasting solution (IFS).

4.2.1 Integrated forecasting solution

The development of IFS can be described in five broad process steps.

Exhibit 4.2.1.1 Integrated Forecasting Solution (IFS) Steps



To enable and support 8760 load forecasting at the substation/circuit level that incorporates EVs and DERs, DTEE has near term plans to invest in some of these steps.

With the incorporation of the identified investments and the move to a more integrated planning process over the next few years, DTEE plans to phase out the “Alignment to Distribution” process as it currently exists as an iterative corrective process. As we address the identified gaps that need to be bridged, DTEE expects that the distribution level load forecasts will transition to 8760 hourly load forecasting at the substation/circuit level in alignment with the 8760 system-level forecast using consistent data sets, assumptions, and methodologies. Investing in a more aligned load forecasting model will help avoid the risk of an under- or over-investment in capital assets for the future and ensure alignment between the generation and distribution load forecasts.

4.2.2 Data integrity and management

An accurate load forecast is essential to various types of planning activities; therefore, it is paramount to take steps to ensure data sources are current and consistent. The maturation of DTEE’s automated metering infrastructure (AMI) data has enabled the Corporate Energy Forecasting team to advance their modeling capabilities and has resulted in more granular forecasts with steady improvements in accuracy. These modeling approaches, along with the further refinements in accessing and expanding the use of SCADA data at the circuit/substation level, will ensure alignment and consistency across data platforms.

Although each area of planning may require unique forecast outputs, the goal of an integrated forecasting solution is to utilize the same inputs improving the alignment of load forecasts across

business unit planning. There is a coordinated effort across the Company to ensure all data and data requirements are organized and standardized so that each process that makes use of data does so consistently, accurately and in a timely manner.

4.2.3 Scenario and base model development

The expected proliferation of new or emerging technologies has added complexity to the forecasting and utility planning processes. In addition to these technologies, the COVID-19 pandemic has altered the consumption patterns of many customers over the past year, and it is unknown at this time whether these load patterns will continue in the future. Having access to more granular interval data will increase the understanding of customer behavior driven by new trends or emerging technology adoption and provide useful insights to the business.

As adoption of these technologies expands, it will be important to understand where the expected load growth and resource needs will occur geographically within DTEE's service territory. Propensity studies are expected to deliver insights at the substation and circuit level so distribution planners can appropriately target cost-effective investments as well as potentially mitigate or defer grid impacts through innovative demand-side programs or rate structures. DTEE envisions the IFS model to be modular in order to be able to develop assumptions and potential impacts between forecast iterations for the following components of change:

- Weather
- EVs
- EWR
- PVs
- Storage
- CVR/VVO
- Other, as appropriate

These components are forecasted independently, thereby allowing DTEE to layer and aggregate the assumptions into a load forecast at a circuit/substation level. This allows both a clearer understanding of the individual impact from each forecast component on the overall forecast at an hourly level and provides the flexibility to adjust individual components, as needed, in various scenarios and sensitivities for further analysis.

4.2.4 8760 generation system-level forecasts

Prior to the recent ability to access and process customer AMI data, modeling an hourly system forecast relied on complex estimation processes. With the increased availability and accessibility of AMI data, model estimation processes can be eliminated in favor of direct calculations using metered data. The metered data is used to generate a bottom-up hourly system load forecast. Although many areas of forecasting have been improved with AMI data resulting in better planning, the opportunity to align the generation system forecasts with the aggregation of distribution level forecast will be the focus of our work in the near future as described in the next section.

4.2.5 8760 distribution-level forecasts

Leveraging AMI data in distribution level analysis has largely been underutilized, presenting a future opportunity to expand the use of this data through new tools and models. In order to forecast at the substation and circuit-level, DTEE plans to employ the same methodology used to produce the bottom-up hourly generation system load forecast to the distribution load forecast. The load forecasts will use the same set of assumptions including weather data, load shapes by customer class (i.e., residential, commercial and industrial) and end-use (i.e., EVs and PVs) to accurately calibrate unique substation and circuit load characteristics to the variables that impact changes in their load, thus providing linkage to the system-level 8760 forecast developed for Integrated Resource Planning.

DTEE will leverage the existing suite of software packages utilized for generation load planning and introduce a more scalable and sophisticated package. In the next few years, MetrixIDR, Itron's automated hourly load forecasting software program, will be implemented. This program has the capability to reconcile distribution-level with system-level forecasts using a multitude of analytical methods and model templates to generate hourly load shape forecasts across an entire system. Additionally, as noted above, incremental component loads can be layered to adjust the base hourly forecast for each of the Company's substations and circuits.

A summary of some of the process improvements DTEE plans to address in the near-term to enable 8760 forecasting at a substation/circuit level include:

- Load allocation methodology will be technical instead of billing-centric
- Aggregation of net power delivered and received with behind the meter DER will be modeled at the meter level
- Reconciliation of AMI data with applicable SCADA data will improve overall accuracy of the modeling
- Advanced hourly modeling techniques will be implemented to estimate values where SCADA data does not exist
- Acquisition or development of automation and validation tools for new load and load types will be employed
- Acquisition or development of propensity modeling to predict adoption levels of DER and EVs will be developed

The steps or processes that need to be accomplished in the next several years will be foundational to improve our overall forecasting framework and planning processes. As this effort progresses and more possible gaps emerge, DTEE will refine this framework. Using AMI data, along with the iterative refinement of the models with new or updated information established through new sources of data, is expected to improve the accuracy and timeliness of the distribution forecast. As DTEE dives further into these various investments, other issues and challenges will likely surface. DTEE expects to have aligned load forecasts at the distribution and generation system levels in the next five years and will provide an update in the next DGP. The timing of the IFS process improvements and investments is outlined in Section 12 – Technology & automation of the DGP.

4.3 Non-wire alternatives

A non-wire alternative (NWA) is defined in MPSC Case U-20147 as:

An electricity grid investment or project that uses distribution solutions such as distributed energy resources (DER), energy waste reduction (EWR), demand response (DR), and grid software and controls, to defer or replace the need for distribution system upgrades.

As will be described in Section 12.7 – NWA Pilots, DTEE is pursuing a suite of NWA projects over the next five years. These projects have been developed through an initial NWA development process or through external partnerships and grants and have been evaluated by distribution engineers and partners across the organization. Many of the process and tools used to identify these projects are still being enhanced and refined. In order to more fully integrate NWAs into the distribution planning process, DTEE will incorporate learnings from the pilots to advance capabilities in the following areas: technology, processes, and people.

When approaching NWAs, DTEE considered the following framing questions posed by Mr. DeMartini through the MI Power Grid stakeholder process as instructed by the MPSC. High-level responses to these questions are provided below with additional information for each individual pilot presented in Section 12.7.

- **Why are non-wires alternatives being pursued? What are the pressing issues?**

DTEE is developing processes and identifying opportunities for which non-wire alternatives can provide more value to customers than conventional solutions. Through internal evaluation, examination of trends related to NWA, and by recommendation of the August 20, 2020 Order in Case U-20147, DTEE primarily focuses on NWA projects to address situations in which circuits or substation equipment is or may become overloaded to help delay or offset planned traditional upgrades.

- **What are the desired outcomes?**

The goal of the NWAs is to optimize utility distribution expenditures in addressing distribution capacity constraints. Specific system issues to address through the NWA pilots are outlined in Section 12.7. DTEE also seeks to learn about new technologies, process, and people implications through the application of these pilots as outlined further below.

- **What are the range of potential solutions?**

As discussed in the technology section below, DTEE is focusing on demand-side solutions, but is also integrating energy storage and distributed generation controls into the suite of NWA pilots.

In several instances, EWR and DR can solve the distribution system capacity constraints without additional investment in more expensive technologies such as energy storage. However, as the electric generation and distribution industry continues to evolve, costs of technologies such as energy storage may continue to decline and become economic alternatives to traditional solutions. At the same time, the potential scenarios described in Section 3, such as increased DG/DS or electrification, could provide resources on the grid that could potentially be controlled to provide additional grid benefits.

- **Pricing, programs and procurements (3Ps)? What is the role of customers, DER developers, utilities, aggregators and others?**

Customers are a central part of the NWAs, as are DTEE's programs and partners to deliver EWR, DR, and storage solutions as discussed further in the "people" section below. Pricing, programs, and procurement are discussed further for the individual NWA pilots.

- **Are the benefits and costs of NWAs accruing to all customers on an equitable basis?**

DTEE conducted cost-effectiveness screening for the NWA pilots to ensure that the NWA would be less costly than traditional grid solutions, such as a substation upgrade. While individual participating customers benefit from investment in targeted NWA solutions, such as lower energy bills due to EWR, customers overall benefit from the deferred or displaced investment in costly infrastructure. Additional evaluation will further assess cost, benefit, and equity considerations. DTEE will also continue to refine its NWA selection criteria and process to ensure communities identified through the EJ screening tool have adequate opportunities to be included in NWAs.

Technology

Learning about new technologies, including understanding realized costs and benefits once implemented, is a primary goal of a new pilot. Once a technology has been vetted and evaluated, it can be added to the set of tools available for an engineer in the distribution planning process.

In developing NWA pilot projects, the Company has focused its efforts on evaluating technologies with the potential for benefits to customers and current technical viability. The technologies currently being evaluated are the following:

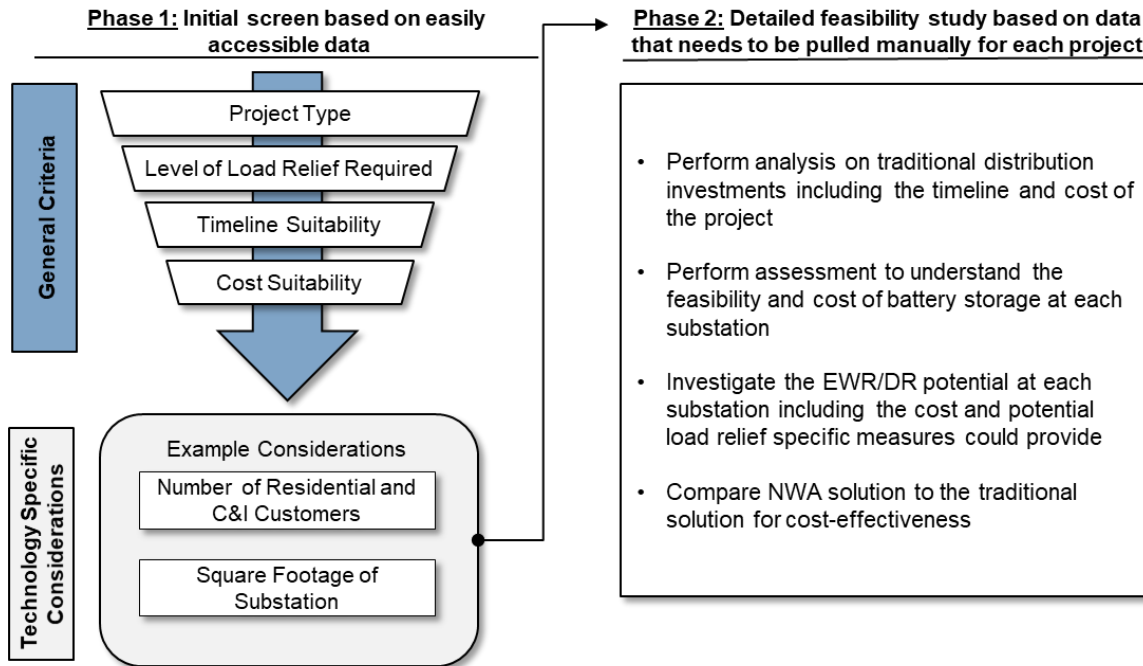
- **Energy waste reduction (EWR)** – geo-targeting of both Residential and Commercial & Industrial customers using the existing portfolio of EWR programs
- **Demand response (DR)** – geo-targeting of residential customers with Interruptible Air Conditioning (IAC) and Smart Thermostat programs
- **Energy storage** – utility-scale or distributed storage to manage peak, including testing of operational procedures and cybersecurity protocols
- **Solar photovoltaics (PV)** – utility-scale and rooftop
- **Electric vehicle (EV) charging** – testing grid impact of fast charging, as well as charge management and cybersecurity
- **Microgrid controls** – create microgrids using planned DERs, DR and grid controls, and implement and test cybersecurity protocols

In addition to learnings from these pilots, DTEE will supplement primary technical knowledge with learnings gained through benchmarking, workgroups and research initiatives.

Process

DTEE has developed a two-phased screening process for selecting potential projects that may be suitable for NWA as shown in Exhibit 4.3.1

Exhibit 4.3.1 NWA Screening Process

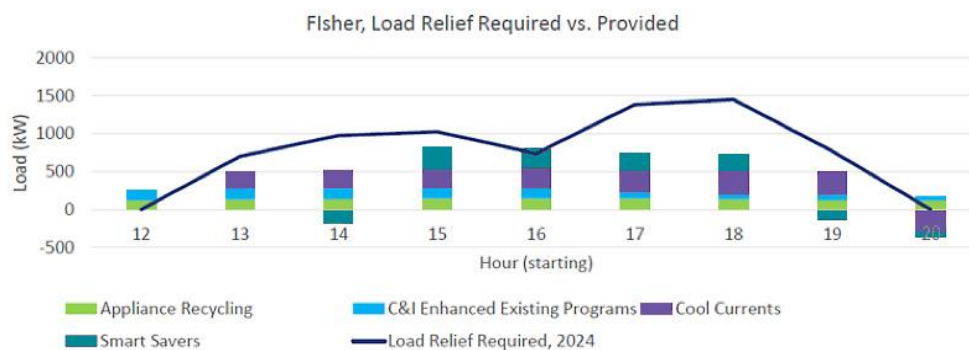


The first phase was developed and has been integrated into the distribution planning process. This ensures that distribution engineers consider NWA options when developing any substation or circuit projects developed for load relief, as described in Section 11.1 – Distribution Load Relief Projects. The screen involves reviewing a general set of criteria as well as technology-specific considerations. For example, as NWAs are currently focused on addressing load relief issues, traditional projects that also address safety, outage event volume, and asset health concerns are not considered good candidates for NWA projects as the technologies in the NWA portfolio do not provide those grid benefits. Other criteria in the first phase screening involves magnitude, duration, and frequency of overload. These screens will be improved in the future based on EPRI's December 2020 report "Screening of Non-Wires Alternatives in Distribution Planning," the Company's pilot experience, and other factors.

The second phase, which is a detailed feasibility study, is much more resource-intensive than the first phase. Detailed load profiles at the circuit and substation are collected for potential projects that pass through the phase one screening. DTEE then uses the DER Estimator Tool to assess the potential for NWA programs to address the overload. This tool considers the load profile, the

count of residential and commercial customers on the circuit as well as the types of C&I customers. Currently, portfolios of EWR and DR programs can be tested against the load profile and the expected participation rates based on historical data and customer demographics to determine the likely outcome for a portfolio of programs. A sample output of the tool is shown in Exhibit 4.3.2. This tool currently only evaluates EWR/DR options for NWA as it was developed as part of pilot projects to evaluate the effectiveness of focusing the EWR/DR programs, which had been applied system-wide, to geo-targeted areas. The tool is being enhanced to incorporate additional technologies, such as energy storage, into the optimization process. Even without these enhancements in place, DTEE was able to leverage partnership opportunities to integrate energy storage and distributed generation as part of this initial round of NWAs.

Exhibit 4.3.2 – Example DER Estimator Tool Output



In the chart above, the “Load Relief Required” line shows the magnitude and duration of the overload. In this case the overload reaches 1.5MW at 6pm (18:00 hours). The portfolio tested in this approach uses geo-targeted Residential EWR (appliance recycling), Commercial & Industrial EWR, Interruptible Air Conditioning (Cool Currents), and Smart Thermostats (Smart Savers) as the portfolio of technologies. In this example, the EWR/DR programs can address about half of the overload.

If the DER Estimating tool develops a feasible portfolio of NWA programs to address the overloading need, the NWA solution is compared to the traditional solution and considered for implementation if it provides customer benefit and defers the traditional investment.

The screening and evaluation process for NWAs will continue to evolve over time as tools are developed and other processes detailed in this section are matured. For example, development

of a full 8760 forecast by circuit and substation will provide a readily available starting point for evaluation of magnitude and duration of potential overloads, and further refine the suitability, size, and duration of one or more NWA solutions to resolve the constraint. In addition, DTEE continues to develop internal tools to aid in the visualization of areas of the system that may allow for NWA opportunities, including those with current or planned deployments of DERs or penetration of EVs.

The solutions identified in this round of NWAs will rely on existing programs and vendor arrangements to deliver the NWA solutions, or will leverage existing partnerships. Programming will be adapted to allow for geo-targeting to reach the targeted customers. Going forward, procurement processes for new NWA solutions will be developed to ensure all solutions to the specific grid issues are tested in the marketplace. This could be conducted through use of RFIs and RFPs depending on the level of project development and available information. As cost and capabilities may rapidly evolve, this will also ensure up-to-date market information is included in the screening and evaluation process for NWAs.

People

While developing new technologies and incorporating them in the new planning process is important, change efforts can fail if the impact of these new technologies and processes on people is not considered. There are at least three groups of people impacted by the evolving effort to incorporate NWAs into the distribution planning process:

- **Employees involved in traditional distribution investments** – The team of engineers, designers, project managers and front-line employees will need to adopt new tools, processes and technologies to implement an NWA solution. For example, a project manager that typically executes a traditional construction project may also be deploying EWR or DR to complete a solution. Front-line employees will need new or additional training to understand new equipment on the grid and how to safely operate and maintain it
- **Internal partnerships** – The current suite of pilots involve collaboration across DTEE organizations including the EWR/DR team and the Renewables team. Solutions will be increasingly reliant on the Integrated Forecasting Solution to leverage the 8760 forecasts by circuit and substation

- **External stakeholders and project partners** – Leveraging technologies such as targeted EWR/DR or distributed solar and storage require outreach to communities, local organizations, energy technology and service providers, and other stakeholders in order to successfully implement programs

The prospect of incorporating new technologies into the suite of tools at the disposal of the distribution planning process is an exciting change. The possibility of meeting grid needs with these new strategies, at lower cost to our customers, is aligned with DTEE's planning objectives of creating a reliable and affordable grid. As new technologies evolve and become more cost effective to meet grid needs, DTEE will continue to evaluate, incorporate, and determine how best to manage the change involved in the design, implementation, and acceptance of these technologies. As part of developing this plan, a suite of NWA projects has been developed using the process and tools described above. More details can be found in Section 12.7 – NWA projects and other technology pilots.

4.4 Interconnection process

As described in Section 3 – Grid modernization process, DERs are expected to be an increasingly common technology deployed on the electric grid. In particular, the DG/DS scenario described in Section 3.1.2 considers directionally plausible forecasts where customer-owned generation, predominantly expected to be rooftop solar PVs and storage in the form of batteries, is expected to increase in adoption throughout DTEE's service territory. The first step for a customer who wants to own and connect those resources to the grid is to file an application for interconnection. Data from interconnection requests provides critical information to DTEE. As the number of applications for DER increases, a critical component in providing accessibility to the grid will be to ensure that the process to connect to the distribution grid is streamlined and efficient. Enhancements to the interconnection process to effectively capture and utilize information like the location, size and type of DER will help distribution planning efforts.

This section of the DGP describes the processes to improve the customer experience with interconnecting to the Company's electrical system. This includes the activities around compliance with new state and federal rules, as well as, the interconnection portal and study processes.

The distribution interconnection process covers a wide array of distributed generation and energy storage projects such as rooftop solar, combined heat and power applications, renewable landfill gas, and specialized applications like regenerative dynamometers. Any project that produces power while connected to the electrical grid needs to be carefully studied to ensure the operability, safety, and reliability of the electrical grid including the safety of the crews and operators that work on the grid. The interconnection process is designed to ensure that the interconnecting technology preserves the safety and reliability of the distribution grid, and consists of major activities like application review, engineering studies, and finally, testing and commissioning after the project is constructed.

The application review ensures that information provided is consistent across all documentation and that all necessary technical information is available for an engineering study. The engineering study reviews the impacts to the electric system of the proposed project and identifies any facility changes that would be needed to safely interconnect the project. The final testing and commissioning phase ensures that necessary operating controls and protections, such as anti-islanding protection, are in place, working as designed, and that the site can meet the requirements of the parallel operating agreement. Not all projects continue through all these phases to completion, and some projects and customers experience challenges navigating the existing process.

Over the past three years, solar installers have assisted approximately 5,000 DTEE customers in installing rooftop solar projects at their homes. DTEE has improved its processes and DER connection knowledge as the volume of applications continue to increase each year. DTEE hosts yearly webinars with developers to share knowledge and answer questions regarding requirements to interconnect to the DTEE grid. Suggestions received from the participants have been implemented. A dedicated interconnection hotline was established to help answer customers' questions. Along with emails being sent regarding approvals and/or corrections that need to be made, DTEE has also been making proactive calls to customers whose projects need revision to move forward through the process. Enhancements to the DTE webpages ([Interconnection Process | DTE Energy](#) & [Rooftop Solar and Private Generation | DTE Energy](#)) as well as the web-based interconnection application tool (PowerClerk) have been enhanced so the customer can see the status of their project throughout the process. In another example of streamlining the processes, DTEE changed its process to require customers sign the Parallel

Operating Agreement (POA) before the anti-island test and the meter re-programming to help ensure the project can be completed timely with all required documentation.

In the next five years, efforts will be undertaken to further improve the integration of interconnection processes with existing new customer service processes. DTEE uses Clean Power Research's PowerClerk software for new customer interconnection applications and to track the process through study and interconnection. A number of investments will be made to further improve the timeliness and accuracy of each customer interaction. Currently, application fees and study fees are collected through a paper check process. This introduces significant time delays in the application process while checks are sent through the mail and manually recorded. An electronic payment processing option will be added to the processes to minimize or eliminate the need for paper checks and mail deliveries, reducing the time needed to validate payment.

To implement this, multiple PowerClerk enhancements will take place with regards to automating updates, upgrading forms to accommodate new processes, and developing more user-focused application forms and processes, to provide additional transparency for customers. Additionally, the existing system has a separate customer login that is not integrated into the Company's customer portal login system. An interface will be developed to allow the customer to use one login for interconnections as well as other Company services. Contractors and developers that install DER projects will be allowed to see all their projects regardless if they are load additions or interconnections. Furthermore, this will perform data validation of key application items such as customer address and account to minimize any potential conflicts of rates, status of account, or service installations.

Internally, process integrations to automatically create, update, and status workorders for payment processing and internal work tracking will be created within the work management system, Maximo, to minimize process delays and manual efforts. Data integration to the larger DTEE's data repository will allow for more comprehensive analytics and analysis of interconnection status and trends as well as the ability to link interconnection activities to other work and projects to identify project management and design synergies.

Finally, the interconnection portal will be further integrated with distribution planning tools to automatically perform some screens and data checks to validate if projects need specific

upgrades that are common, or if known constraints are violated by projects prior to the screening processes. These anomalies can then be flagged for additional analysis. Once implemented, these improvements will lead to a more cohesive interconnection process that will closely align with the new business and new service processes, thereby, allowing for greater synergies to be leveraged, reducing interconnection processing times, cost and level of effort and, subsequently, improving customer satisfaction.

Along with these improvements, there are a number of investments that will be undertaken in the short term, to align with changes in the statewide interconnection process as well as alignment to changes at FERC and MISO relating to orders such as FERC 841, FERC 842, FERC 845, and FERC 2222.

Initially, all work will be focused to streamline the process and meet the new or proposed interconnection rules requirements. The new interconnection rules require that all qualifying projects go through a series of screenings for various grid impacts. A number of these screens can be readily implemented into the existing systems but are currently highly manual tasks. Other screens will require integration such as those with the planning tool to perform initial evaluations of grid constraints. Specifically, automating screens around loading, transformer size compatibility, and voltage constraints will require the planning tool integration. Other screens, such as the validation of a site being inside the service territory, will need to leverage geospatial service provided by the mapping systems.

The new interconnection rules define a batch study process for larger projects. These studies combine multiple DER applications into one larger study and will require planning and interconnection tools to be modified to identify the interrelated impacts of the projects and separate out the drivers and upgrades required. As part of near-term investments, this requires study preparation tools and databases that collect the interconnection information and models and validates that the appropriate information is available for the study to be completed on time and without rework. This includes system information such as the interconnecting DER models, site plans, equipment specifications as well as the system models for study, the scenarios for different loading conditions, the system operating constraints, and device settings for existing and planned DER configurations that will be impacting the study scope. These tools will also provide the scenario archives for the studies and help produce the report outputs that are sent to

customers and developers. The investment levels and timeline for the next five years for these initiatives which enable the Interconnection process are outlined in Section 12 – Technology & automation.

4.5 Hosting capacity analysis

According to the definition in Case No. U-20147, hosting capacity analysis (HCA) is the amount of DER that can be accommodated without adversely impacting operational criteria such as power quality, reliability, and safety under existing grid control and operations without requiring infrastructure upgrades. In addition, in Case No. U-20147, the MPSC has requested that DTEE adopt a phased implementation approach for HCA pilots.

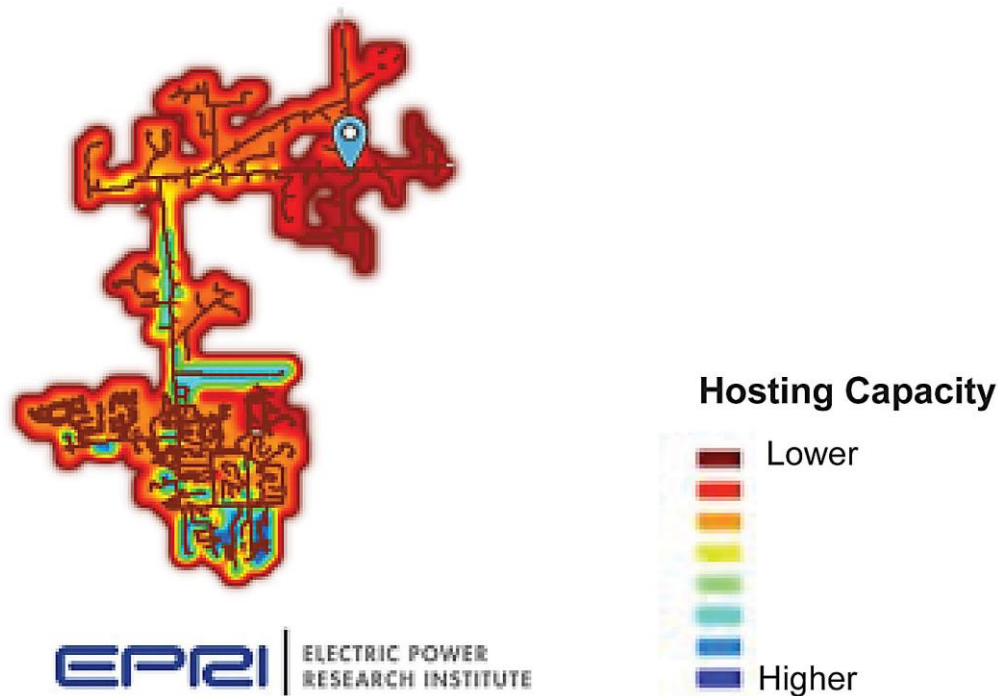
Like the interconnections process, HCA is linked to the DG/DS Scenario described in Section 3.1.2 and would also be impacted by increased electrification as outlined in the Electrification scenario. As the adoption rate of DERs on the system increases, HCA will be a critical tool for distribution planning to identify initial grid constraints. Furthermore, a hosting capacity map can give DER developers the ability to better select sites and possibly reduce interconnection applications for non-viable areas with grid constraints. However, as DTEE has noted in prior discussions and filings, HCA is not a substitute for the developer submitting an application to study interconnecting to the grid. An application to interconnect is still necessary to identify impacts caused by the interconnect and propose solutions to mitigate those impacts.

The approach to hosting capacity at DTEE will be implemented in several phases. The first phase, which is expected to be completed by the end of 2021, will be a high level, go/no-go map of the Company's electrical system. This go/no-go map will indicate the regions of the system where interconnection of DER on the distribution system will be likely to have few if any upgrades to the electrical infrastructure beyond what is necessary to interconnect the DER. Investments in this initial phase will include the development of a tool to maintain and update the hosting capacity and to display this information to customers and developers on the dteenergy.com website.

Further enhancements in this near-term period, which is likely to be completed in 2022 and 2023, will include the integration of more advanced and detailed hosting capacity tools into the processes. This will be initially rolled out to areas of high DER penetration or those areas where known constraints require a more detailed analysis to properly inform prospective applicants of

the limitations in these areas. These more advanced methods include tools such as EPRI DRIVE and CYME integration capacity analysis. Eventually this will link into the investments in the application and screening processes to allow the hosting capacity to visualize the areas where qualifying and standard projects are likely to pass the interconnection screens. This will also include updates to the hosting capacity that are determined from forecasting scenarios, other planning or interconnection studies and will increase the resolution of the hosting capacity and the frequency of which it is updated.

**Exhibit 4.5.1 Illustrative Image of Hosting Capacity on a Distribution Circuit
in the Second Phase**



Initial investments that will be needed in order to support the implementation of FERC 841, FERC 842 and FERC 2222 compliance are still being evaluated. These orders could have an impact on the implementation of the interconnection process and HCA. The following gives a summary of these orders:

- Order 841 under the Federal Power Act (FPA) directs regional grid operators (i.e. Regional Transmission Organizations (RTO)/Independent System Operators (ISO)) to remove barriers for participation of energy storage resources in wholesale markets and allows participation of distribution connected storage resources > 100KW
- Order 842 states newly interconnecting large and small generating facilities, both synchronous and non-synchronous, to install, maintain, and operate equipment capable of providing primary frequency response and voltage ride through as a condition of interconnection
- Order 2222 requires regional grid operations (RTO/ISO) to allow both heterogeneous and homogeneous DER aggregations ($\geq 100\text{kW}$) including demand response and energy efficiency resources, to participate in wholesale power market including energy, capacity, and auxiliary services

These orders combined place requirements on local distribution facilities, such as DTEE, to require frequency response from new interconnection agreements and facilitate distribution facility connected energy storage and other DER's participation in the wholesale markets. While investments needed to support the implementation of these orders are dependent on RTO/ISO compliance plans, it is currently known that there will be process and tool impacts in operational and planning areas. This includes aggregation review and approval processes related to record keeping, operational coordination requirements for distribution market participants, as well as investments in validation and reconciliation of wholesale market resources that are on the distribution systems. These upgrades will also allow for more comprehensive reporting and tracking of the new timelines and process steps that are included in the new interconnection rules and FERC/MISO processes.

To date, DTEE has commissioned two small studies of hosting capacity analysis to determine feasibility and limitations of providing hosting capacity analysis, along with options for displaying information in ways that are accurate and useful for customers and the DER development community. By the end of 2021, DTEE anticipates making a high level go/no-go map available for public use. Some of the remaining key activities in this area include ensuring that data used to generate a map is accurate, current, and can be updated on a regular basis, along with selecting a customer-friendly presentation method for publishing on DTE Energy's website. Investments in

refining the map are expected in 2022 while integration links to the interconnection and screening process to be established in future rule sets are expected to be in 2023 and beyond. The investment levels and timeline for these initiatives are described in Section 12 – Technology & automation.

4.6 Integrated generation (resource) planning and distribution planning alignment

DTEE recognizes the benefits of increase alignment and coordination across the distribution, generation, and transmission planning processes. Increased alignment and collaboration across the planning processes benefits customers, as well as the Company. This alignment is an evolving process that will continue to develop and mature over time.

This section of the Distribution Grid Plan describes the progress achieved in aligning the Integrated Resource Planning (IRP) and Distribution Operations (DO) planning processes within DTEE, as well as the engagement with our transmission partner, International Transmission Company (ITC). The section also lays out the ongoing collaboration and coordination efforts between the IRP, DO, and DTEE Transmission teams, ITC and others.

4.6.1 Areas of past successful collaborations

DTEE identified enhanced alignment between generation planning and distribution planning as an opportunity, and throughout 2020 and 2021, the IRP and DO planning teams have been focusing on putting processes in place to do so. The combined teams collaboratively developed common planning objectives, building on the planning principles that were developed for the 2017 Certificate of Necessity and 2019 IRP. The outcome of this effort resulted in the identification of planning objectives that are meaningful to our customers and stakeholders. These planning objectives are 1) safe, 2) reliable and resilient, 3) affordable, 4) customer accessibility and community focus, and 5) clean. Common planning objectives provide an overall foundational approach to guide and align the planning processes.

Another successful example of collaboration between the IRP and DO teams was the work involved in determining the efficacy of Conservation Voltage Reduction / Volt Var Optimization (CVR/VVO) as a resource in the Company's 2019 IRP. As detailed in the CVR/VVO section (Section 12.6 – CVR/VVO), to support CVR/VVO in the IRP as a generation resource alternative,

a pilot was commissioned to study and understand the viability and value of using CVR/VVO. DTEE is working to verify the early results of the first phase of this pilot and plans to further test more advanced CVR/VVO functionality when the Volt/Var Control (VVC) module of the ADMS is implemented in 2022. In addition, for the 2019 IRP, the IRP team worked with the DO team to identify a system-wide avoided transmission and distribution capacity value that one kW of peak load reduction in any location of the DTEE electric system would provide by deferring or eliminating upgrades to the transmission and distribution system.

Michigan is faced with a unique situation where significant elements of the electric grid are operated and controlled by separate entities that have specific obligations to their respective customers, employees and shareholders. To bridge the processes across the two companies, DTEE is in regular communication with ITC regarding how transmission can support future distribution system needs including customer connections, capacity planning and DERs impacts on transmission systems, and also how DTEE can support ITC transmission planning. Additionally, the two companies met as part of our distribution grid planning process to share our work and observations.

Several recent examples of ongoing collaboration include the transmission projects at MISO relating to the City of Detroit Cable project (MTEP #15981), the Nitro project (MTEP #12443), the Stone Pool – Temple project (MTEP #17998), and the Cato-Corktown project (MTEP #20167). The first two projects in this list are examples of transmission investments providing both transmission and distribution system benefits. The City of Detroit project, specifically, involved DTEE accelerating plans for the Islandview and Promenade conversion projects (Section 11.3 – 4.8kV Conversion). The last two projects are driven almost entirely by distribution system reliability enhancement needs. These projects resulted from significant planning and coordination between DTEE and ITC on how the transmission system can support the distribution grid. This planning took place within the regular planning cadence between the two companies and was reviewed within the MISO MTEP stakeholder process. The MISO stakeholder process provided the framework for stakeholders to publicly ask questions about these investments and propose alternatives.

4.6.2 Areas of continuing collaboration

Distribution Operations has been working on several fronts with different teams within DTEE to address the current challenges it is facing with the transformation of the grid. Some of these initiatives include:

4.6.2.1 Peak load reduction with CVR/VVO

Based on the CVR/VVO collaboration in the 2019 IRP, DTEE continues to evaluate the CVR/VVO pilot to reduce peak demand or energy consumption as well as any additional potential benefits to the distribution grid. These benefits include improving grid efficiency as well as providing sufficient capacity for demand. More about this initiative can be found in Section 12.6 - CVR/VVO of the Distribution Grid Plan.

4.6.2.2 DER load forecasting

As mentioned in Section 4.2, the Corporate Energy Forecasting and DO teams are developing methodologies to enhance modeling capabilities that will improve load forecast granularity, allow for the integration of new and emerging technologies in the forecast, and produce 8760 hourly load forecasts at the generation system and substation/circuit levels. Both generation and distribution planning teams will be able to use the same forecasting data sets and have consistent assumptions. The incorporation of DER and EV adoption into the forecasts through propensity and Time of Use (TOU) studies is one example that will help improve the robustness and consistency of both forecasts thereby increasing alignment between the planning processes.

4.6.2.3 Energy waste reduction and demand response

There are ongoing collaborations between the EWR, DR and DO teams to understand the grid opportunities of NWAs. Processes are being developed and refined across teams to identify, design and deploy potential pilots that will use EWR/DR resources. The successful completion of the Hancock pilot and subsequent rollout of the Fisher and other pilots are crucial for DTEE to understand the efficacy of NWAs as scalable resources to defer or replace traditional grid investments, as appropriate. Those learnings are being incorporated by the teams to develop the selection criteria and suitability of the technologies to address peak reduction. Furthermore, in the future, as results of the learnings from the Hancock and Fisher pilots emerge, evaluating the use

of EWR and DR and other resources to defer or replace a distribution traditional investment will be incorporated into Distribution Operations planning processes. A further elaboration of these processes and pilots are detailed in two separate sections of this plan; Section 4.3 - Non-wire alternatives and Section 12.7 - NWA pilots.

4.6.2.4 FERC 841 and 2222

Orders 841 and 2222 are designed to create new opportunities in the coming years for DERs and DER aggregations by providing these resources a path to participation in the wholesale markets. Compliance with Order 2222 in particular will be a complex undertaking, and the framework for MISO's compliance with the order is currently under development. As a result, DTEE is still early in understanding how to incorporate these changes into our grid and planning processes, and the resulting impacts will need to be incorporated over time as the participation models and the level of interest by potential market participants become clearer.

While there is still uncertainty around the specific impacts arising from these orders, what is clear is that each of the planning processes will need to closely align to effectively manage the grid for DTEE's customers and other stakeholders. The impacts and expected growth of DER will likely require tools, systems and processes to automate aspects of the interconnection and planning processes as DER seek to interact with the grid in new ways and in larger numbers. As discussed in the Grid Modernization Section 3 – Grid Modernization Process, high penetration of DERs and DER aggregations, enabled by FERC orders, will also reinforce the need to better align integrated resource, transmission and distribution planning.

4.6.2.5 ITC and MISO coordination

DTEE will continue to work collaboratively with ITC to ensure regular communication via already well-established protocols. DTEE recognizes that transmission owners and distribution owners must coordinate and plan for system needs on a regular basis. With many anticipated generation retirements, rapid integration of new renewable resources and continued distribution grid modernization efforts, DTEE recognizes there is a need for continued close collaboration with ITC and MISO to ensure optimized deployment of transmission solutions. In addition, the MISO process allows for a broad group of stakeholders to provide inputs and feedback on transmission, and generation investments in a transparent format to facilitate regional and sub-regional

transmission expansion planning. Meetings are held on a regular cadence with these entities so that any system change that could impact the grid and affect these organizations are addressed in a timely manner.

Coordinated planning begins with having common electric system engineering models and integrated planning assumptions between companies. DTEE has aligned its annual distribution system reliability assessment model development process to the MISO Transmission Expansion Plan (MTEP). Annually, the MISO MTEP model development process solicits feedback from stakeholders on changes to the transmission system and distribution system through common models. This includes new generation additions, grid topology changes, load forecast changes, and new interconnections. DTEE both provides and receives information through this process working collaboratively with ITC and MISO to develop two-, five- and ten-year system models. Through this process ITC communicates their planned and approved system upgrades. ITC also presents their future system plans that are represented in the planning models. DTEE shares load forecast information and communicates current and anticipated future distribution system requirements within the MISO process and within regular DTEE and ITC planning meetings. The transmission and distribution reliability planning processes work in parallel to evaluate system needs. As new distribution system needs are confirmed, DTEE works collaboratively with ITC to evaluate if those system needs are best served by the transmission or distribution system. MISO utilizes these inputs to facilitate the annual transmission system expansion planning process where stakeholders participate in an open and transparent forum to arrive at the most efficient solution for customers considering all alternatives.

There are also opportunities to collaborate with ITC on both a technical and policy level on emerging issues such as refinement of ITC's proposed EV fleet proposal, implementation of Order 2222, coordination of T&D infrastructure and associated jurisdiction classifications to meet specific customer requests, and alternative analyses and transmission system impacts in DTEE's next IRP.

Examples of more routine on-going collaborations between ITC and DTEE include mitigating equipment clearance issues, resolving easement relocation requests, decommissioning and coordinating shutdowns on equipment. Coordination meetings are conducted on a regular

cadence and established protocols are updated judiciously when necessary between companies to ensure that the grid is planned and operated in a safe, reliable and accessible manner.

4.6.2.6 Stakeholder input

DTEE increased stakeholder engagement throughout the 2019 IRP process, and that approach, along with the learnings and best practices, has been integrated into the current distribution plan. The stakeholder engagement process and interactions will be iterative, and learnings from the DGP will then be incorporated into the next IRP process. The outreach events were designed to increase transparency, educate the participants, and provide an opportunity for questions and feedback. DTEE appreciates the engagement of and feedback obtained from stakeholders in this process. As was mentioned in Section 2- Stakeholder Engagement, at the heart of the grid analysis and planning is the customer.

DTEE recognizes and acknowledges that the planning processes are iterative and will continue to explore opportunities where generation, distribution, and transmission planning processes may be able to support each other and benefit customers. The implementation and improvement of processes, technology, and tools will significantly evolve over time. Utilities, regulators, and stakeholders should allow an appropriate lead time to invest in and implement the necessary tools, staff training and grid capabilities to enable more integrated planning. The planning teams frequently share information cross-functionally, collaborate on projects and initiatives, and work effectively together. The distribution and generation planning teams will continue to build on this foundation to strengthen communication, collaboration, and alignment around forecasts and scenarios, as well as common planning objectives.

5 Benefit cost analysis (BCA)



A robust BCA framework is a key component of the DSPx platform and is integral to the evaluation and prioritization of projects and programs. Just as the first and most important step in the DSPx process is defining objectives, a BCA framework should provide a link between investments and how they are expected to achieve desired objectives. A key challenge of BCA when applied to grid modernization is that investments can support multiple objectives, and the benefits can be difficult to disaggregate and quantify, also known as joint and interdependent benefits. For example, investments to replace aging infrastructure and increase system capacity, such as voltage conversions, not only improve reliability but also provide flexibility to enable greater customer adoption of EVs and DERs.

DTEE has developed the Global Prioritization Model (GPM) as a comprehensive method for evaluating strategic investments. The GPM aligns with the DSPx's "best-fit, most reasonable-cost" evaluation method, which is most relevant for core grid investments that have joint and interdependent benefits or are driven by standards and policy mandates. The GPM has been used to prioritize DTEE's strategic investments in rate cases and internal annual planning processes since 2018. The GPM has been shared with the MPSC and stakeholders during its development and ongoing use. While aspects of the GPM may need to be modified over time, it currently provides a rigorous and accepted methodology for evaluating investments to meet the grid needs identified in our grid modernization process.

5.1 Benefit cost analysis approach

As mentioned above, a BCA framework should be aligned with the overall objectives for distribution planning. The GPM was developed and aligned to the original key objectives of reducing risks, improving reliability, and managing costs. As discussed in Section 3 – Grid Modernization Process, current and expected changes in customer needs and the needs of the grid resulted in a reevaluation of the original objectives. DTEE’s current Planning Objectives are shown below in Exhibit 5.1.1. The five current objectives of Safe, Reliable and Resilient, Affordable, Customer Accessibility and Community Focus, and Clean are an evolution of the original three objectives, and align with the Commission’s overarching objectives for the electric distribution system of Safety, Reliability and Resilience, Cost Effectiveness and Affordability, and Accessibility.

Exhibit 5.1.1 DTEE Planning Objectives



SAFE

Build, operate, and maintain the distribution grid and generation fleet in a manner that ensures public and workforce safety, operational risk management, and appropriate fail-safe modes and is compliant with State and Federal requirements



RELIABLE AND RESILIENT

Build, operate, and maintain the power system within acceptable standards to withstand sudden disturbance or unanticipated failure of elements. Ensure the grid and diverse generation resources are integrated, with secure supply resources, and can quickly recover from high impact, low frequency events



AFFORDABLE

Provide efficient and cost-effective service along with diverse and flexible generation resources by optimizing the system and benefiting all customers



CUSTOMER ACCESSIBILITY AND COMMUNITY FOCUS

Provide flexible and accessible technology and grid options, and information that empowers and engages customers. Provide effective and timely communication with customers and stakeholders. Favor plans that support the diversity of Michigan communities, suppliers, and workforce



CLEAN

Build, operate, and maintain the resource fleet and grid platforms in an environmentally sustainable manner by achieving low carbon aspirations and clean energy goals. Provide a grid that facilitates a transition to a decarbonized economy

One layer deeper than the Planning Objectives, the seven impact dimensions are used to specifically evaluate and rank strategic investment projects and programs in a quantitative manner. The seven impact dimensions and their alignment to DTEE's objectives is shown in Exhibit 5.1.2. Currently, none of the seven impact dimensions align to the two planning objectives of 'clean' and 'customer accessibility and community focus'. Future steps to develop additional dimensions or enhance the current GPM methodology to better align with these objectives are discussed in section 5.2. Two near-term activities that will help provide data and insights that can be leveraged have already been discussed in the DGP. The enhancements to forecasting, described in Section 4.2 will help provide a future forecast for DER adoption at the distribution level and will help identify areas that could need investment. This may become increasingly important if our Grid modernization scenarios of increased electrification and increased DG/DS materialize. The other activities that will align with the 'customer accessibility and community focus' objective is the development of an Energy and environmental justice screening methodology, which was discussed in Section 2.2. Exhibit 5.1.3 shows a mapping of key programs to the current impact dimensions. As shown in the table, all programs will deliver benefits across multiple objectives.

Exhibit 5.1.2 Program Impact Dimensions

	Impact Dimension	Major Drivers
Safe	Safety	- Reduction in wire down events - Reduction in secondary network cable manhole events - Reduction in major substation events
	Load Relief	- System capability to meet area load growth and system operability needs - Elimination of system overload or over firm
	Regulatory Compliance	- MPSC staff's recommendation (March 30, 2010 report) on utilities' pole inspection program - Docket U-12270 – Service restoration under normal conditions within 8 hours - Docket U-12270 – Service restoration under catastrophic conditions within 60 hours - Docket U-12270 – Service restoration under all conditions within 36 hours - Docket U-12270 – Same circuit repetitive interruption of fewer than five within a 12-month period
	Major Event Risk	Reduction in extensive substation outage events that lead to a large amount of stranded load for more than 24 hours
Reliable & Resilient	Reliability	- Reduction in number of outage events experienced by customers - Reduction in restoration duration for outage events
Affordable	O&M Cost Avoidance	- Trouble event reduction and truck roll reduction - Preventive maintenance spend reduction
	Reactive Capital Avoidance	- Trouble event reduction and truck roll reduction - Reduction in capital replacement during equipment failures

Exhibit 5.1.3 Selected Programs Project and Alignment to Objectives¹⁴

Planning Objective	Safe				Reliable and Resilient		Affordable
Program	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Cost	Reactive Capital
Tree Trimming to the Enhanced Specification	X		X		X	X	X
4.8/8.3 kV Conversion	X	X		X	X	X	X
Substation Risk	X	X		X	X		X
System Loading		X		X			
System Cable Replacement	X			X	X		X
Breaker Replacement	X			X	X	X	X
Line Sensors					X	X	
ADMS	X	X		X	X	X	X
Automation	X			X	X	X	X
Subtransmission Redesign	X	X		X	X	X	X
4.8 kV System Hardening	X				X	X	X
Frequent Outage (CEMI) Program	X		X		X	X	X
URD Cable Replacement					X	X	X
Pole/Pole Top Hardware	X		X		X	X	X
Station Upgrade				X	X	X	X

A limited number of strategic investments are excluded from the prioritization model due to their unique benefits and specific situations addressed. Upgrades to our facilities, such as the

¹⁴ Further enhancements to GPM to measure alignment to the objectives of Clean and Customer Accessibility and Community Focus are discussed in Section 5.2

construction of ESOC and ASOC, are necessary to fully realize the benefits of the ADMS platform and to provide full redundancy in the case of an emergency, but they do not directly provide quantifiable benefits in the dimensions measured by GPM. Other examples include programs to replace disconnect switches (Section 9.1.3 – Subtransmission Disconnect Switch) and the Pontiac vault projects (Section 11.5 – 8.3 kV System Conversion and Consolidation), which are necessary to meet operational safety needs. Over 88% of strategic capital is prioritized using the GPM method. A list of the projects and programs excluded from the model are included in Appendix II. In addition, projects that are excluded from the Global Prioritization Model are subject to close examination of their criticality before being deemed as “must fund” in the capital plan.

The process to prioritize strategic projects and programs in the GPM starts with detailed analyses based on historical data, engineering assessments and field feedback, which are performed annually to quantify how a project or program will deliver benefits within each impact dimension. The quantified benefits are then compared to the programs’ costs to derive their benefit-cost ratios.

Unit measurements used for benefit-cost analysis are different for each impact dimension. Reliability benefits are captured in customer minutes of interruption reduction. O&M and reactive capital benefits are captured in avoided costs. Safety, load relief, regulatory compliance and major event risk benefits are rated in indexed scores.

To determine a program’s overall benefit-cost score, the benefit-cost ratio in each dimension is indexed to a score of 0-100. The overall benefit-cost score is calculated as the summation of the indexed score in each dimension, multiplied by the dimension weight. Exhibit 5.1.4 lists the different weights given to the impact dimensions on a scale of 1 to 10. The weightings are anchored on safety, prioritizing the safety of our employees, customers, and community. The other dimensions have weightings set relative to dimension of safety.

Exhibit 5.1.4 Impact Dimension Weights

Impact Dimension	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Cost	Reactive Capital
Weight	10	4	4	4	3	3	3

The benefit-cost scores for strategic capital programs, ranked from highest to lowest, are illustrated in Exhibits 5.1.5 and 5.1.6. Tree trimming to the enhanced specification under the surge program, excluded from the exhibits as non-capital program, continues to provide the highest customer benefits of any program in the five-year investment portfolio.

Exhibit 5.1.5 Overall Benefit-Cost Scores for Strategic Capital Programs and Projects

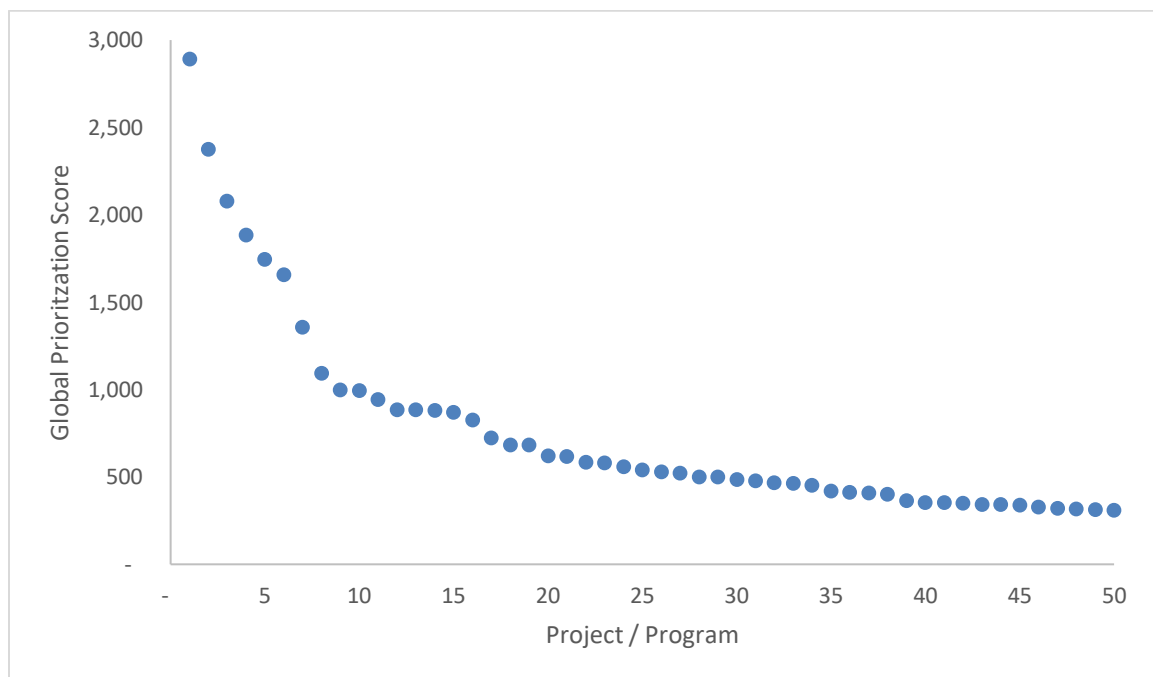


Exhibit 5.1.6 Top 50 Strategic Capital Programs and Projects Based on Benefit-Cost Prioritization Ranking

Rank	Capital Program/ Project
1	ADMS
2	CODI: Charlotte Network Conversion
3	Substation Risk: Port Huron
4	System Loading: Mandy 307 Load Transfer
5	4.8kV CC: ISO KERN
6	4.8 kV Hardening
7	4.8kV CC: ISO CAMDN
8	Poles / PTM
9	Substation Risk: Novi Decommission
10	CODI: Garfield Upgrade
11	4.8kV CC: ISO GLBRT
12	4.8kV CC: ISO BRAZL
13	Subtransmission Redesign & Rebuild: Trunk 4217
14	4.8kV CC: ISO VENOY
15	4.8kV CC: ISO BIDDLE
16	Automation
17	4.8 kV Conversion: Hawthorne Conversion
18	CODI: Amsterdam Upgrade
19	Customer Excellence/ CEMI
20	8.3 kV CC: Pontiac Projects
21	CODI: Howard Conversion
22	Subtransmission Redesign & Rebuild: Trunk 4266
23	Cable Replacement
24	Substation Risk: Apache
25	Subtransmission Redesign & Rebuild: Tie 3416 (Badax Breaker & OH Reconductoring)

Rank	Capital Program / Project
26	System Loading: Marlette/Tacoma/Brown City/Shaw
27	Line Sensors
28	APTS
29	4.8 kV CC: I-94 (Promenade)
30	4.8kV CC: ISO WATFD
31	URD Replacement
32	PQ Meter Fault Locating
33	4.8 kV CC: Belle Isle (Island View)
34	4.8 kV Conv/Cons - Birmingham
35	Relay Replacement: Northeast
36	Subtransmission Redesign & Rebuild: Trunk 7106
37	Subtransmission Redesign & Rebuild: Trunk 7333
38	Subtransmission Redesign & Rebuild: TRK 3508
39	Breaker Replacement
40	Subtransmission Redesign & Rebuild: Trunk 2255
41	4.8 kV Conversion: Belleville Circuit Conversion
42	System Loading: Cody Circuits
43	Relay Replacement: Navarre
44	System Loading: OTSGO/CAPAC/SHAW
45	Subtransmission Redesign & Rebuild: Boyne Station
46	Subtransmission Redesign & Rebuild: Tie 3416 (OH PIGON)
47	Subtransmission Redesign & Rebuild: Thumb Electric Fault Isolation
48	System Loading: Macomb/Golf
49	Subtransmission Redesign & Rebuild: Trunk 3509
50	System Loading: Richmond/Armada

^[1] CODI: City of Detroit (Downtown) Infrastructure

5.2 GPM next steps and other considerations

The benefit-cost scores of programs and projects and their prioritization ranking align DTEE's Planning Objectives of Safe, Reliable and Resilient, and Affordable and provide a solid foundation for DTEE's strategic investment decisions. As DTEE's Planning Objectives have recently evolved, GPM may evolve over time to ensure that both are completely aligned. The process of updating the GPM with additional objectives would involve a robust set of discussions and inputs from stakeholders, however, some initial direction has been considered.

Customer Accessibility and Community Focus

As discussed in Section 2.2 – Energy and Environmental Justice, DTEE is coordinating a comprehensive approach to Energy and Environmental Justice as it relates to the reliability and resiliency of the grid. As a first step, DTEE will overlay recent reliability performance on the grid with the MIEJScreen when it is released in the fall. Adding an additional dimension to GPM to enhance the priority of investments that address reliability in communities identified by the MIEJScreen tool would help align the investment strategy with the Energy and Environmental justice approach.

Clean

Grid investments that enable the transition to a decarbonized economy could be given enhanced prioritization in the GPM. Increased penetration of EVs and DER may require additional capacity, increased sensing and monitoring, reactive power management, and control capabilities. As mentioned above, however, core grid investments such as conversion to a higher voltage, already bring with them the benefits of the ability to integrate more DERs. Many projects in DTEE's plan through 2025 will enable a higher adoption of DERs (such as ADMS, conversions, load relief projects, automation, CVR/VVO, sensing, and many investments in the technology strategy), based on the grid benefits they deliver in the current framework. DTEE will continue to monitor the signposts under our scenarios for EV and DG/DS adoption and consider other dimensions to include in the GPM to meet the objective of Clean.

Additionally, there are other key considerations that impact capital funding decisions:

- Capital spend profiles for new projects are subject to key development milestones, especially in the conceptual and early development stage, including land availability and property purchases, municipal approvals and construction permits, right-of-way and easements and major equipment long-lead items from manufacturing companies. Additionally, while communities in Michigan are developing plans to improve infrastructure, DTEE will work with them to coordinate projects, look for construction efficiencies between DTEE and municipal projects, while minimizing road and other community impacts. While DTEE takes proactive measures, such as advanced planning and project monitoring, to mitigate some of these execution risks, some of these early stage milestones are out of the Company's control and can introduce schedule delays or cost increases. Therefore, DTEE's plan is designed to be flexible to accommodate these unpredictable variations in timing and cost.
- Funding decisions for programs and projects need to consider the implication on resources and workforce planning. Resources required from engineering, design, project management, scheduling and construction need to be evaluated not only by project type (substation, overhead or underground) but also by region and service center. Resource gaps need to be understood and addressed before funding decisions are made. Funding decisions on programs and projects also need to consider the Company's capability to effectively manage the work, as DTEE partners with industry vendors in project execution. DTEE bears the responsibility to oversee the project scope, schedule and cost, and to ensure adherence to DTEE standards.
- As DTEE continues to make investments in distribution infrastructure, the effectiveness of the capital spend is examined frequently. The benefit-cost scores of the programs and projects may change over time as new performance data and field experience become available. The prioritization ranking of the programs and projects may change accordingly.
- Some capital replacement programs are funded annually, despite having lower benefit-cost scores. This is done to avoid an acceleration of asset failures and a large number of assets reaching end-of-life concurrently, thus exceeding available resources to replace them.

- Last but not least, some programs and projects may not receive near-term funding due to their lower benefit-cost scores. This does not mean these programs or projects are not important. Rather, all the programs and projects identified in this report provide system improvements and are good candidates for funding over the next five years. While the strategic capital investment is primarily driven by the Global Prioritization Model, DTEE may adjust the annual plans based on changing circumstances. For example, system limitations including the availability of shutdowns (de-energizing parts of the system to execute work), may be limited in specific geographic areas. Project timing may need to be spread out in order to be able to execute the work.

5.3 Investment impact on reliability and avoided costs

The investments detailed in this Distribution Grid Plan will provide significant customer benefits including improved safety, operability of the distribution system, and reliability and resiliency. The investments will also reduce long-term emergent capital and O&M costs and enhance the distribution system's ability to support increased electrification and adoption of DERs.

Based on the capital investments and investments in tree trimming described in Section 6, DTEE projects reliability improvements will reach second quartile in All Weather SAIDI and first quartile All Weather SAIFI by 2025. Additionally, these investments will prevent further deterioration of reliability that would occur if strategic investments were not made. Exhibit 5.3.1 and Exhibit 5.3.2 illustrate the All-Weather reliability improvements resulting from the investments proposed in this DGP (the Five-Year Investment scenario) and the potential degradation if strategic investments are not made (the Constrained scenario). The Constrained Investment scenario illustrates the reduced reliability improvement if no strategic investments were made on distribution infrastructure beyond the "base capital" program set forth in Section 13.2 – Base Capital. The SAIDI and SAIFI baselines shown in the Exhibits are the three-year average values between 2018 and 2020. DTEE believes that the projections for Constrained Investment scenario could even prove to be optimistic, as some assets are approaching a condition in which there are accelerations in the current rate of equipment failures (e.g. system cable, URD).

Exhibit 5.3.1 System All Weather SAIDI

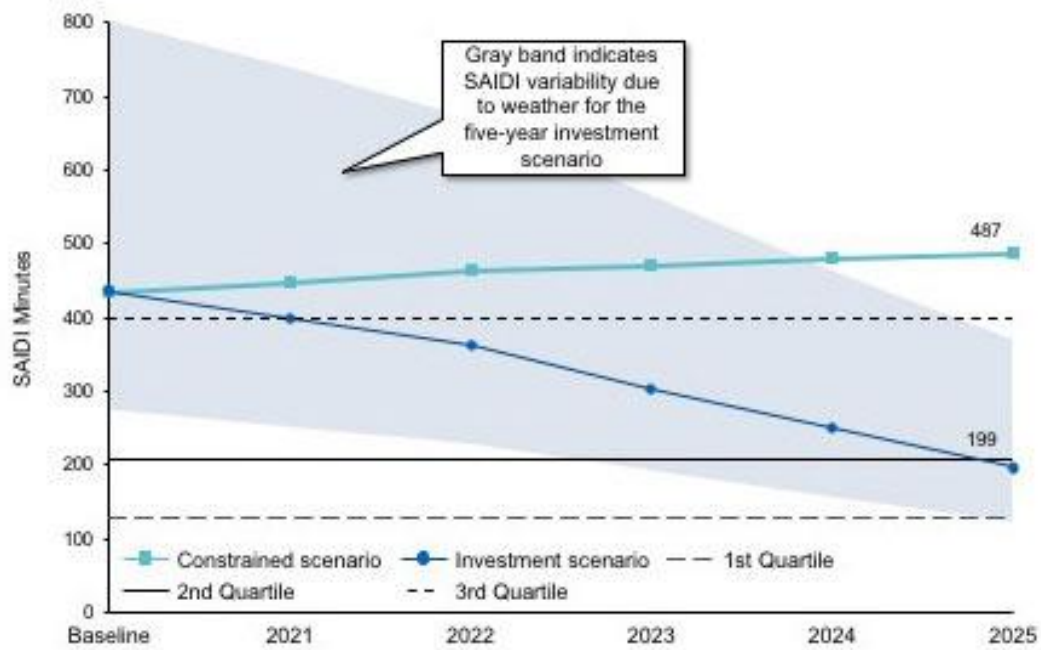
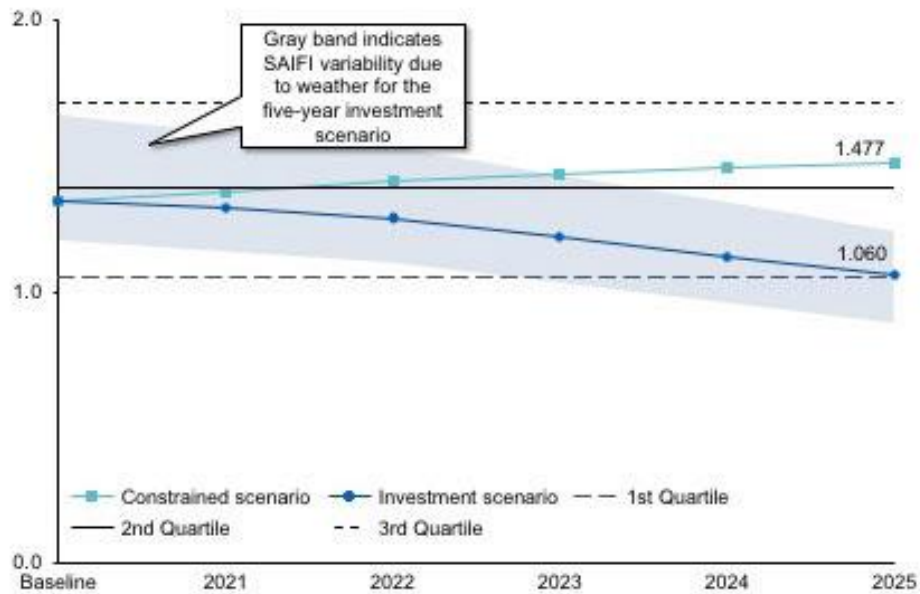


Exhibit 5.3.2 All Weather SAIFI



In addition to the improvements to all weather reliability, DTEE expects to see improvements in reliability excluding major event days (ex-MEDs). A major event day (MED) is a day in which SAIDI exceeds a threshold value based on the calculations set forth by the Institute of Electrical and Electronics Engineers, Inc. (IEEE). Removing MEDs from the reliability calculations better represents the day to day operation of the system and the associated reliability. DTEE expects to reach second quartile SAIDI ex-MEDs by 2025 and first quartile SAIFI ex-MEDs by 2025. Exhibits 5.3.3 and 5.3.4 illustrate the ex-MED reliability improvements.

Exhibit 5.3.3 SAIDI Excluding Major Event Days (ex-MEDs)

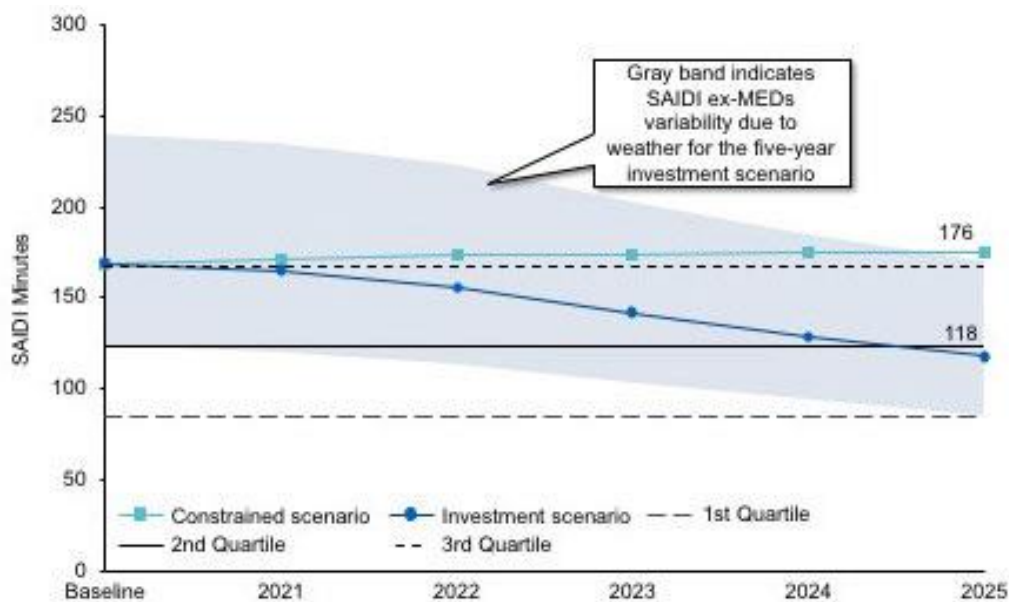
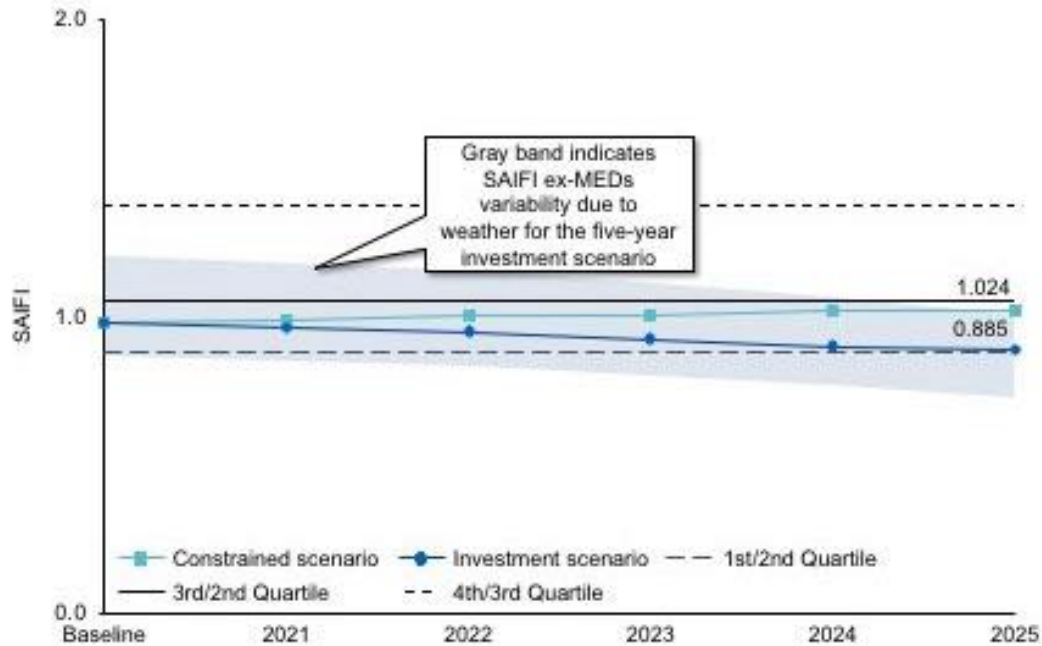
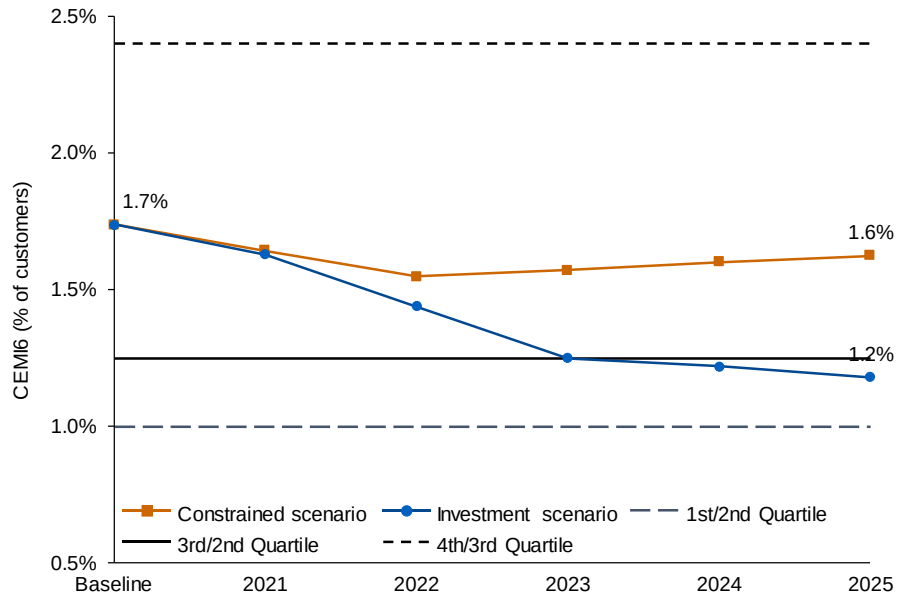


Exhibit 5.3.4 SAIFI Excluding Major Event Days (ex-MEDs)



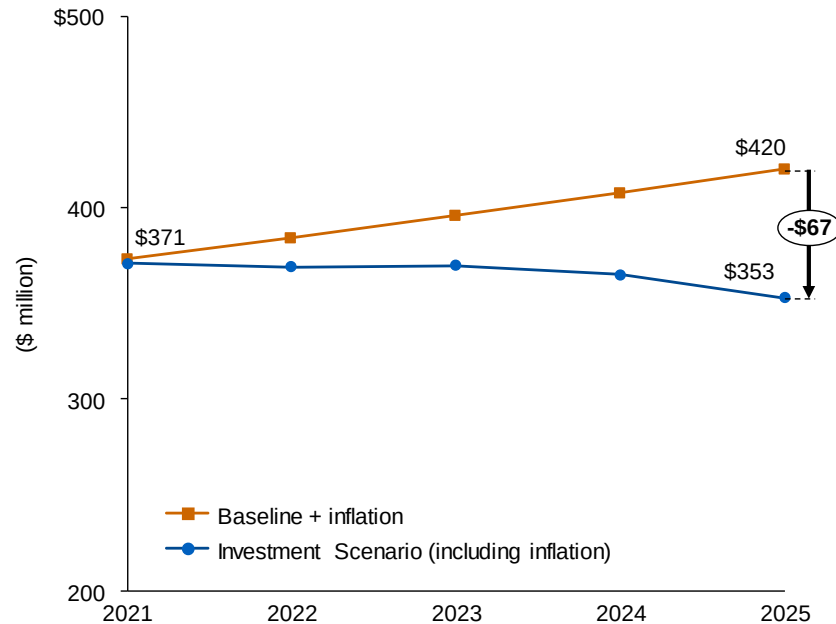
The reliability improvements discussed previously in this section are system level improvements. Additional metrics that help demonstrate the impact of our investments on our customers are CEMI4 and CEMI6, which measure the percentage of customers who have experienced at least four or six outages in a calendar year, respectively. Customers in this category experience some of the worst reliability on the system. It is important that the investments in the DGP not only avoid large outages that can drive SAIDI and SAIFI, but also target the long tail of outages to ensure that all customers experience reliability improvements. The overall portfolio of investment in DTEE's DGP ensures that the percentage of our customers experiencing frequent outages declines over the next five years as well. DTEE expects the percentage of customers experiencing six or more outages (CEMI6+) to be 32% lower in 2025 versus the baseline performance, which will also bring system performance into second quartile (based on 2017 benchmarking data from the Edison Electric Institute). The trajectory of expected CEMI performance is shown in Exhibit 5.3.5 below.

Exhibit 5.3.5 CEMI6



Reliability improvements detailed above are due to a reduction in the number of outages and duration of outages customers will experience. The reduction of outage events, and non-outage events, due to the investments detailed in this DGP will also lead to a direct reduction in the need for emergent expenditures. The reduction in emergent capital spend over the next five-years is expected to not only off-set the cost of inflation, but also reduce the need for emergent capital on an absolute basis. Exhibit 5.3.6 shows our baseline assumption, which is the previous 3-year average plus inflation, and the emergent capital projection based on the savings from investments in this DGP versus our baseline assumption.

Exhibit 5.3.6 Emergent Capital Spend Projection



5.4 Customer benefits

The Investments detailed in the DGP have a direct impact on the electric system in terms of reliability, operability, and avoided costs as shown in the preceding sections. More importantly, the system improvements driven by DTEE’s DGP will directly benefit our customers. Determining the value of improved reliability from a customer perspective is an inexact science. Different customers experiencing the same outage may incur disparate hardships due to socioeconomic or other personal circumstances. The variability of the impact of outages on our customers may be wide, nevertheless, there are some methods that are used across the industry to attempt to quantify the benefits of reliability improvements.

One concept used in reliability planning to compare the costs and benefits of system upgrades is “customer interruption costs” (CIC), which is the economic cost a customer occurs when they experience an interruption in electricity service. DTEE uses the Interruption Cost Estimation (ICE) Calculator developed by Nexant and Lawrence Berkeley National Lab, which is the most well-established tool in the industry, to estimate improvements in CIC. Over the next five years, the expected benefits the investments will deliver are calculated to be between \$9.8 billion and \$13.2 billion.

While the ICE calculator is useful to estimate of the benefits of improved reliability, it has some known limitations. For example, 1) much of the survey data collected to form the basis of the calculations is dated, 2) areas of the country, particularly those which experience extreme cold, are not well represented in the data, 3) the calculator has a bias toward undercounting residential customer costs, and 4) the impact of long duration (>24 hours) is not accounted for. In addition, it does not account for the disparate impact of long outages on our highly impacted communities, as discussed in Section 2.2 – Energy and environmental justice. DTEE will continue to evolve how it evaluates reliability impact to provide the most benefit to our customers. As discussed in Section 5.2 – GPM next steps, one initiative will be to incorporate energy and environmental justice into the GPM model, which enhance weight of reliability benefits to residential and highly impacted communities when prioritizing grid investments.

6 DGP investment summary



The grid modernization process discussed in Section 3 outlined both the current state of the grid and potential additional grid needs related to the three scenarios of increased electrification, increasing CAT storms, or higher customer adoption of DG/DS. The investment portfolio summarized in this section, which was developed and prioritized using the GPM model discussed in Section 5, is a no-regrets set of investments. In other words, the investments over the next five years not only address core reliability and safety needs of the current grid, they also will support the evolving customer needs if and when the various scenarios come to fruition.

Addressing aging assets is a key need of the grid in order to improve reliability and reduce reactive costs. Details of the age distributions and risk factors of our assets can be found in Section 8 – Asset Health Assessments, though in general, many of the assets are already near or past end of life and replacement programs are being accelerated to prevent further deterioration. For example, 43% of system cable and 41% of URD are candidates for replacement based on age and other factors specific to known failure mechanisms. Nearly 44% of poles are older than 50 years, which is considered the life expectancy of a wood pole.

Programs targeted at addressing the aging infrastructure, such as the 4.8kV Hardening program and Pole Top Maintenance and Modernization, drive significant reliability benefits that will enable DTEE to achieve second quartile SAIDI results in the near term and will help manage reactive costs. These programs also rebuild the overhead infrastructure to the latest standards, which will make them more resilient if the frequency or intensity of storms increases (e.g. the use of steel poles as a new standard for subtransmission).

Capacity constraints and overloading are another limitation of the current electrical system. Approximately 35% of substations have some loading limitation, either at the substation, equipment or circuit level. In addition, one-third of the subtransmission system has loading constraints. The lack of capacity decreases the resilience of the system by limiting flexibility (i.e., contingencies and redundancies) in the event of equipment failures. It is also a barrier to connecting new loads, whether through economic development, electrification, or other increasing demands on the system.

Sections 3.4.2 and 11.3.2 discuss the need to convert the 4.8kV system to a higher voltage, to add capacity and enable grid modernization. Investments to convert the 4.8kV system or upgrade the subtransmission system are targeted for areas that currently have loading constraints. However, these investments will also provide additional flexibility to expand if there is significant load growth due to electrification. For example, substations are often built with two transformers to support current load, but can be expanded to add a third transformer if additional load materializes in the area.

Overall, DTEE will invest \$7 billion of capital, approximately \$834 million in tree trimming, and \$50 million in preventative maintenance on the electric grid over the next five years. Of this spend, approximately \$3.8 billion is dedicated to strategic capital, which will deliver the benefits of improved reliability, reduced reactive cost, increased capacity to serve current and future load, and reduced system risk.

DTEE's strategic investments are categorized into four pillars: Infrastructure Resilience and Hardening, Tree Trimming, Infrastructure Redesign and Modernization, and Technology and Automation. Each pillar is described briefly below, and has a section of the DGP that will go into depth on the detail of each of the projects and programs, scope, investment levels and timing. Before the four pillars are discussed, a detailed assessment of the health of our assets is described in Section 8, including age distributions and risk factors. The asset health assessments are a key input to other investment programs, most specifically the capital replacement programs detailed in Section 9.1

The four pillars by which DTEE categorizes strategic investments are:

- **Infrastructure Resilience & Hardening (Section 9)** – These projects and programs are focused on replacing aging infrastructure, hardening the system, and addressing areas of the system with known poor reliability.
- **Tree Trimming (Section 10)** – The Tree Trimming investment includes both an enhanced specification to improve the impact of trimming trees, and a “surge” to bring the entire system ‘on-cycle’ to this enhanced specification by the end of 2025. The program brings significant reliability benefits, resilience to storms and ongoing reduction in reactive costs.
- **Infrastructure Redesign & Modernization (Section 11)** – These projects and programs make more fundamental changes to the electrical system, such as conversion of the 4.8kV and upgrades of the Subtransmission system that will increase system capacity. Overall redesign and modernization of the system also brings reliability and resilience improvements through the replacement of aging infrastructure.
- **Technology & Automation (Section 12)** – These programs are tightly linked to the grid needs identified as part of the grid modernization process, including investments to complete the ESOC and ADMS, expand the telecommunications infrastructure, increase the sensing and control on the system, and develop more advanced analytics capabilities. Advanced technology and automation on the grid not only improves reliability but is also a key factor in enabling greater customer accessibility to the grid.

The overall investment levels are shown in Exhibit 6.1, with more detail by pillar and program shown in Exhibit 6.2 These investments represent all of capital investments, and the two key maintenance programs (tree trimming and preventative maintenance).

Exhibit 6.1 Projected DTEE Five-Year Distribution Grid Plan Investments (Excluding PLD)

		\$ Millions					
Category		2021	2022	2023	2024	2025	5-Year Total
Capital Investments (\$ Millions)							
Base Capital	Emergent Replacements (Reactive Trouble and Storm Capital)	\$488	\$372	\$372	\$366	\$354	\$1,952
	Customer Connections, Relocations and Others	\$272	\$252	\$256	\$263	\$270	\$1,313
Strategic Capital Programs (details in Exhibit 6.2)		\$381	\$656	\$816	\$893	\$1,007	\$3,752
Total Capital Investments		\$1,141	\$1,280	\$1,443	\$1,522	\$1,631	\$7,017
Maintenance Programs (\$ Millions)							
Tree Trimming ¹		\$191	\$195	\$188	\$152	\$109	\$834
Preventive Maintenance		\$10	\$10	\$10	\$10	\$10	\$50
1. Tree trim spend profile subject to complete surge by 2024, subject to further refinement							

Exhibit 6.2 Projected Five-Year DTEE Distribution Grid Plan Investments
(Sorted by Reference Section Number)

Programs	\$ Millions						Reference Section #
	2021	2022	2023	2024	2025	5-Year Total	
Infrastructure Resilience & Hardening							
Pole and Pole Top Maintenance and Modernization	\$35	\$61	\$88	\$121	\$125	\$430	Section 9.1.1
4.8 kV Automatic Pole Top Switch (APTS)	\$2	\$5	\$6	\$6	\$6	\$26	Section 9.1.2
Subtransmission Disconnect Switches	\$2	\$3	\$3	\$3	\$3	\$12	Section 9.1.3
Circuit Switchers	\$1	\$1	\$1	\$1	\$1	\$5	Section 9.1.4
Circuit Breakers	\$11	\$15	\$18	\$20	\$20	\$84	Section 9.1.5
System Cable Replacement	\$13	\$35	\$55	\$65	\$75	\$243	Section 9.1.6
Underground Residential Distribution (URD) Cable	\$4	\$6	\$10	\$15	\$25	\$60	Section 9.1.7
SCADA Pole Top Device	\$1	\$1	\$2	\$2	\$3	\$9	Section 9.1.8
Substation Regulators	-	\$0	\$0	\$0	\$0	\$2	Section 9.1.9
Substation Outage Risk	\$6	\$11	\$12	\$11	\$12	\$52	Section 9.2.3
Station Upgrades	\$3	\$7	\$6	\$10	\$4	\$30	Section 9.2.3
4.8 kV Hardening	\$69	\$120	\$116	\$116	\$116	\$537	Section 9.3

Programs	\$ Millions						Reference Section #
	2021	2022	2023	2024	2025	5-Year Total	
Frequent Outage (CEMI) including Circuit Renewal	\$33	\$45	\$22	\$22	\$28	\$150	Section 9.4
4.8 kV Relay Improvement	\$2	\$3	\$3	\$3	\$3	\$15	Section 9.5
Mobile Fleet	\$5	\$3	\$3	\$3	\$1	\$14	Section 9.6
Pontiac Vaults	\$4	-	-	-	-	\$4	Section 11.5
Infrastructure Redesign							
System Loading	\$6	\$11	\$20	\$50	\$53	\$140	Section 11.1
Subtransmission Redesign & Rebuild	\$42	\$82	\$93	\$101	\$112	\$429	Section 11.2
4.8 kV Conversion and Consolidation	\$23	\$61	\$56	\$50	\$66	\$256	Section 11.3
City of Detroit Infrastructure (CODI)	\$44	\$54	\$95	\$104	\$107	\$404	Section 11.4
8.3 kV Conversion and Consolidation	\$1	\$4	\$12	\$34	\$38	\$90	Section 11.5
Strategic Undergrounding	\$2	\$11	\$30	\$60	\$100	\$203	Section 11.6
Technology & automation							
Advanced Distribution Management System (ADMS) ¹⁵	\$30	\$10	-	-	-	\$40	Section 12.1

¹⁵ In addition, the ADMS project requires \$17.1 million of regulatory asset spend from 2019 to 2023

Programs	\$ Millions						Reference Section #
	2021	2022	2023	2024	2025	5-Year Total	
System Operations Center (SOC) Modernization	\$18	\$34	\$55	-	-	\$107	Section 12.2
Grid Automation Telecommunications	\$5	\$15	\$19	\$17	\$15	\$71	Section 12.3
Distribution Sensing and Monitoring	\$2	\$6	\$12	\$13	\$16	\$49	Section 12.4
Substation Automation	<\$1	\$2	\$3	\$10	\$12	\$27	Section 12.5
Circuit Automation	-	\$2	\$5	\$10	\$12	\$30	Section 12.5
CVR / VVO	\$6	\$7	\$17	\$18	\$20	\$69	Section 12.6
NWA Pilots	\$7	\$10	\$15	\$3	\$1	\$35	Section 12.7
DER Control	-	\$3	\$3	\$3	\$3	\$11	Section 12.8
Mobile Technology	-	\$0	\$5	\$1	\$8	\$13	Section 12.9.1
Work Management and Integrated Scheduling	\$0	\$3	\$12	\$3	\$3	\$21	Section 12.9.2
Asset Management	-	\$1	\$2	\$2	\$1	\$6	Section 12.9.3
Distribution Planning Process Enablement	\$3	\$10	\$10	\$11	\$9	\$43	Section 12.9.4
Modernize Grid Management	\$3	\$11	\$6	\$6	\$9	\$35	Section 12.9.5
Substation Cybersecurity	-	\$2	-	-	-	\$2	Section 16.3

7 Distribution system overview



7.1 System overview

DTEE owns and operates approximately 31,000 miles of overhead distribution lines and 16,600 miles of underground distribution lines. Our service territory encompasses 7,600 square miles and includes approximately 2.2 million residential, commercial, and industrial customers. DTEE's distribution system consists of six voltage levels: 120kV, 40kV, 24kV, 13.2kV, 8.3kV and 4.8kV with the key statistics listed in Exhibits 7.1.1-7.1.6.

Exhibit 7.1.1 DTEE Substations

Substation Type	Total Number of Substations	Number of Substations by Low Side kV							
		4.8	8.3	13.2	4.8 13.2	24	40	24 40	Other
General Purpose	543	245	4	241	32	4	13	3	1
Single Customer	145	49	0	86	1	0	0	0	9
Customer Owned	96	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Total	784	294	4	327	33	4	13	3	10

N/A: Not Applicable

Exhibit 7.1.2 DTEE Transformers

Voltage Level	Number of Transformers	kVA Capacity
Substation - Subtransmission	176	12,870,000
Substation – Distribution	1,468	24,480,561
Distribution - Overhead and Padmount	445,848	32,438,078
Total	447,492	69,788,639

Exhibit 7.1.3 DTEE Subtransmission Circuits

Voltage	Number of Circuits	Overhead Miles	Underground Miles	Total Miles
120 kV	67	61	8	69
40 kV	320	2,302	382	2684
24 kV	246	181	682	863
Total	633	2,544	1,072	3,616

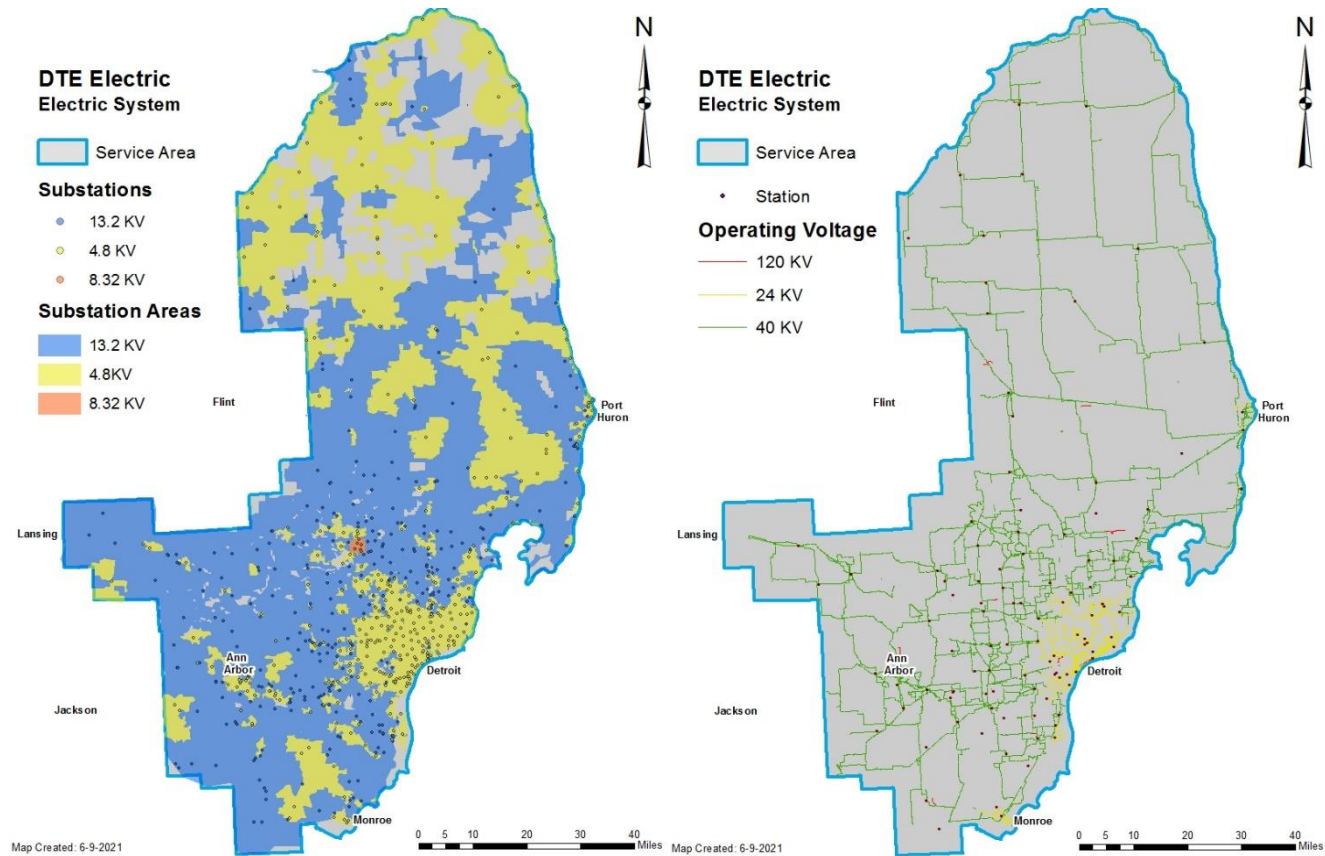
Exhibit 7.1.4 DTEE Distribution Circuits – Line Miles by Substation Bus Voltage

Voltage	Number of Circuits	Overhead Miles	Underground Miles	Total Miles
13.2 kV	1,244	17,320	13,001	30,321
8.3 kV	13	52	17	69
4.8 kV	1,982	11,176	2,522	13,698
Total	3,239	28,548	15,540	44,088

Exhibit 7.1.5 DTEE Distribution Circuits – Line Miles by Line Voltage

Bus Voltage	Number of Circuits	Overhead Miles			Underground			Total Miles
		13.2 kV	8.3 kV	4.8 kV	13.2 kV	8.3 kV	4.8 kV	
13.2 kV	1,244	11,742	19	5,559	12,602	1	398	30,321
8.3 kV	13	0	52	0	0	17	0	69
4.8 kV	1,982	32	0	11,144	269	0	2,253	13,698
Total	3,239	11,774	71	16,703	12,871	18	2,651	44,088

Exhibit 7.1.6 DTEE Distribution System



7.2 Reliability

7.2.1 Background

DTEE customers expect and deserve reliable service. In order to better serve the needs of customers, DTEE has made reliability one of the core planning objectives and recognizes that due to the aging infrastructure and capacity constraints in our current system, significant investments must be made to improve system reliability. Nearly all investments across DTEE's four strategic pillars that are part of the five-year DGP investment strategy will improve reliability. Measuring reliability performance helps drive operational performance improvement, prioritize strategic investments, and validate that those investments are delivering the desired benefits.

As illustrated in Section 5.3, DTEE has projected the impact on reliability from the Company's investment plan. The remainder of this section describes how DTEE measures its reliability performance and provides DTEE's historical reliability performance using the following industry standard measurements:

- SAIFI, SAIDI, and CAIDI – Sustained (greater than 5 minutes) outage frequency and duration-based metrics
- Causes of interruptions – Understanding what causes outages on DTEE's system
- MAIFI – Momentary (less than 5 minutes) interruption metrics
- CEMI_n and CELID_t – Customers experience multiple interruptions or long interruptions

While DTEE has long tracked outage data and the resulting reliability performance, wide deployment of AMI and the use of data analytics have enhanced its ability to measure customer reliability. For years 2017 and prior, the source for outage data was DTEE's outage management system. Beginning in 2018, AMI meter data became the source of the outage data. While the outage management system remains necessary to dispatch field personnel and track the restoration process, nearly full AMI saturation allows DTEE to more accurately know each individual customer's reliability experience without relying on customer or personnel input, or on a manual review of reliability data. However, it is impossible to know the cause of an interruption from AMI meter data alone (e.g., trees, equipment failure, public interference, etc.), as this can only be obtained by linking back to the outage management system and/or through further investigation. Currently, not all AMI outages can be linked with the outage management system; DTEE continues to investigate and resolve inconsistencies, which are attributable to several factors:

- One leg interruptions – one leg interruptions are when one of the two 120-volt service conductors is de-energized. AMI meter data may not indicate all instances of these interruptions due to the way in which the meter power supply is wired to the service conductors.
- Different interruption timestamps – for a given event, each meter may have a slightly different interruption time stamp, none of which may agree with the interruption time stamp in the outage management system (typically all are within seconds or minutes of each other). This may be a problem if there are multiple events occurring simultaneously.
- Outage management system interruption inaccuracies – even with manual reviews, the outage management data is not as accurate as data directly from the meters.
- Interruptions not captured by the outage management system – data from the AMI meters may indicate short duration, sustained interruptions (5 minutes or more in duration), while data from the outage management system may classify them as momentary interruptions (less than 5 minutes in duration) or low voltage events. This could be caused by lack of interruption verification by a customer from the outage management system (customer may be unaware of the interruption in real-time) or the failure of AMI meter real-time power outage notifications.

Data are obtained nightly from the AMI meter registry and are the most complete and accurate customer reliability data available. The use of AMI data for reliability statistics has resulted in higher SAIFI, SAIDI, CAIDI, CEMI and CELID values than the outage management system data used pre-2018. Prior to AMI, MAIFI was not able to be measured.

7.2.2 SAIFI, SAIDI and CAIDI

DTEE measures overall system reliability using the indices SAIFI, SAIDI, and CAIDI. These indices are defined in IEEE Standard 1366 and summarized in Exhibit 7.2.2.1 below.

Exhibit 7.2.2.1 Reliability Indices Definitions

Index	Full Name	Calculation
SAIFI	System Average Interruption Frequency Index	Total number of customer interruptions divided by the number of customers served
SAIDI	System Average Interruption Duration Index	Total minutes of interruption divided by the number of customers served
CAIDI	Customer Average Interruption Duration Index	Total minutes of interruption divided by the total number of customer interruptions

SAIFI, SAIDI, and CAIDI are reliability performance indices defined in IEEE Standard 1366. In addition to all weather conditions, these indices are also calculated excluding Major Event Days (MEDs), which is any 24-hour period in which there is a significant statistical difference in daily SAIDI; the details of the calculation are in IEEE Standard 1366. Excluding Major Event Days leads to a clearer picture of day-to-day system performance and the customers' experience absent significant weather events.

In addition to the IEEE Standard indices, DTEE also tracks reliability performance under various weather-related conditions: catastrophic storms, non-catastrophic storms, and excluding storms. Analysis using these stratifications allows better insight into reliability performance and root causes of customer outages, and therefore the remediations that are needed to improve performance under different conditions.

Catastrophic storms

In MPSC case U-12270, catastrophic conditions are defined as "severe weather conditions that result in service interruptions to 10% or more of a utility's customers." DTEE, however, has an internal catastrophic storm threshold of approximately 5% of its customers interrupted. That is the

level at which all internal resources, contract crews, local foreign crews, and mutual assistance from other utilities will be engaged, depending on the scale of the storm, to restore customers in a timely fashion. Typically, all restoration crews work 16-hour shifts per day with around-the-clock coverage until the restoration is complete.

Non-catastrophic storms

DTEE is in non-catastrophic storm mode whenever a catastrophic-level has not been reached as defined above, but weather conditions result in customer outage cases that cannot be restored in a timely manner by the on-shift crews. For non-catastrophic storms, in addition to the DTEE restoration personnel, contract crews and local foreign crews may also be mobilized. Typically, the restoration crews work 16-hour shifts until the restoration is complete.

Exhibits 7.2.2.2 – 7.2.2.6 show historical SAIFI, SAIDI, and CAIDI values for various conditions. Included in the system reliability charts are benchmarked quartiles from IEEE. Benchmark data is not published until the fall of the following year and thus 2020 benchmark data is currently unavailable. All the indices are also reported by service center in Appendix IV.

Exhibit 7.2.2.2 Reliability Statistics - All Weather

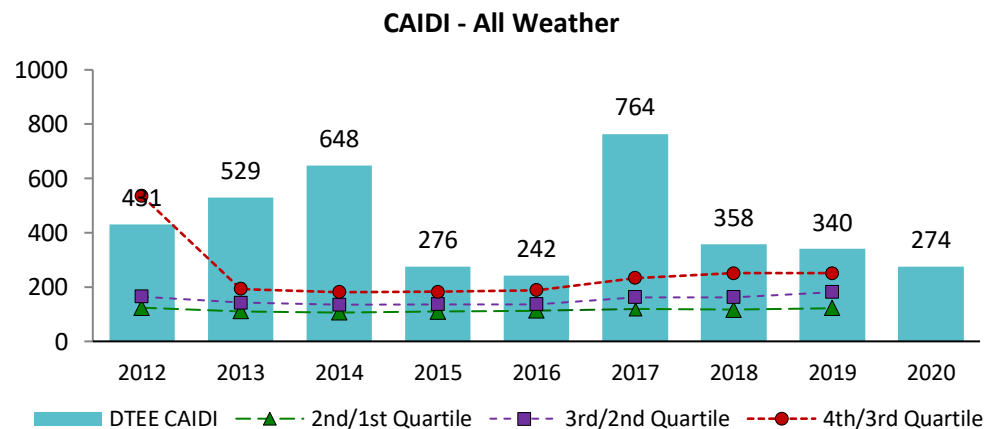
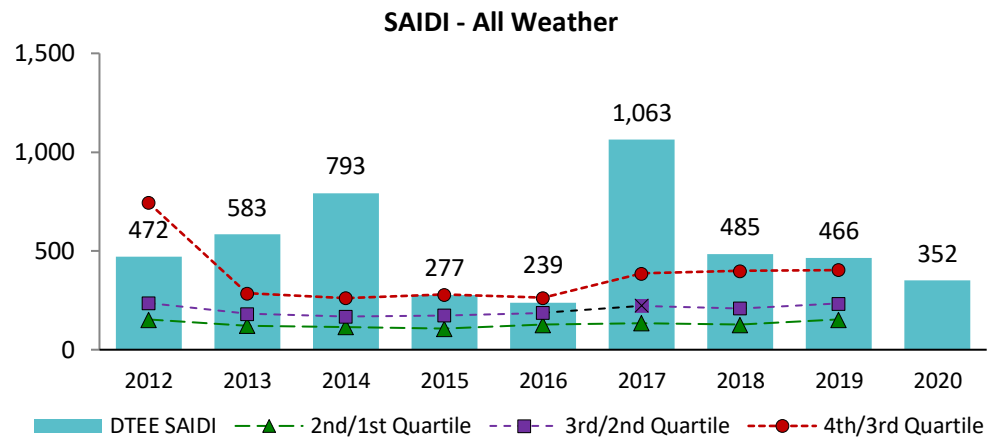
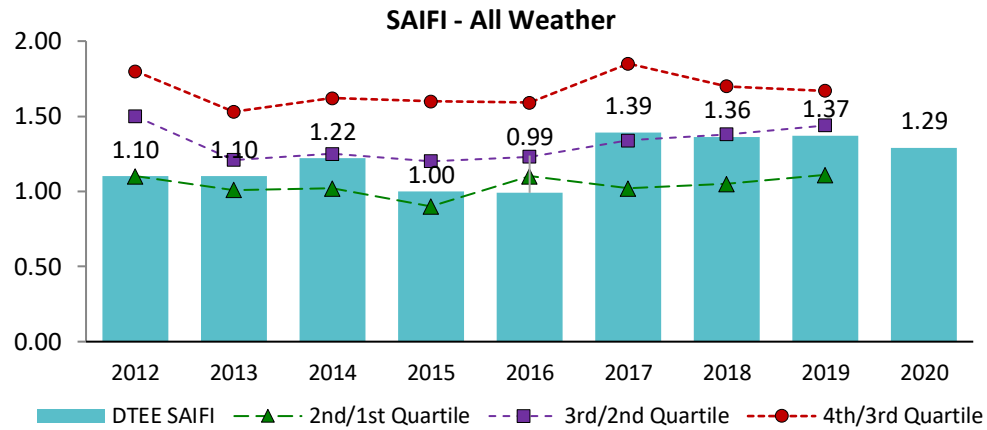
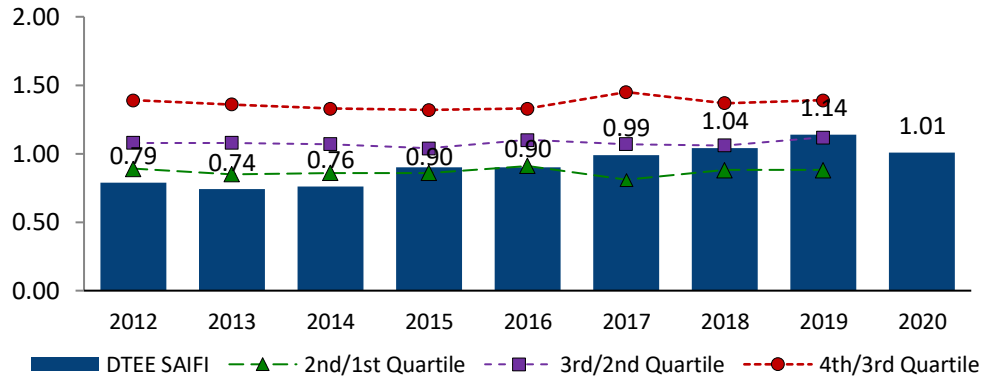
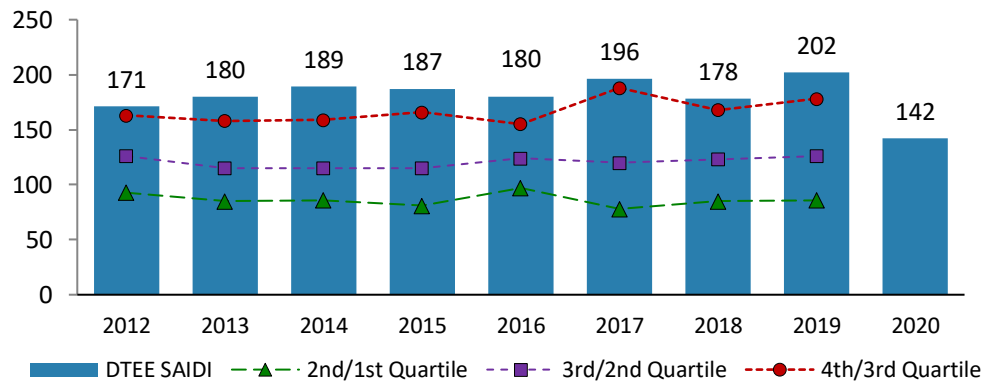


Exhibit 7.2.2.3 Reliability Statistics - Excluding MEDs

SAIFI - Excluding MEDs



SAIDI - Excluding MEDs



CAIDI - Excluding MEDs

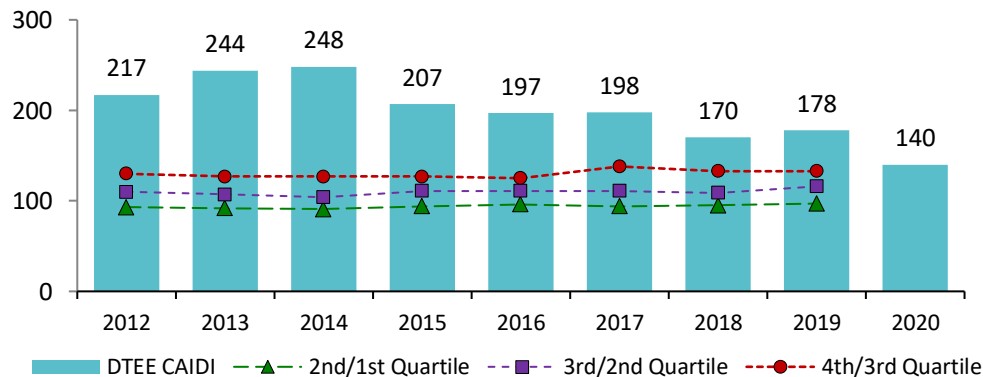


Exhibit 7.2.2.4 Reliability Statistics - Catastrophic Storms (DTEE Definition)

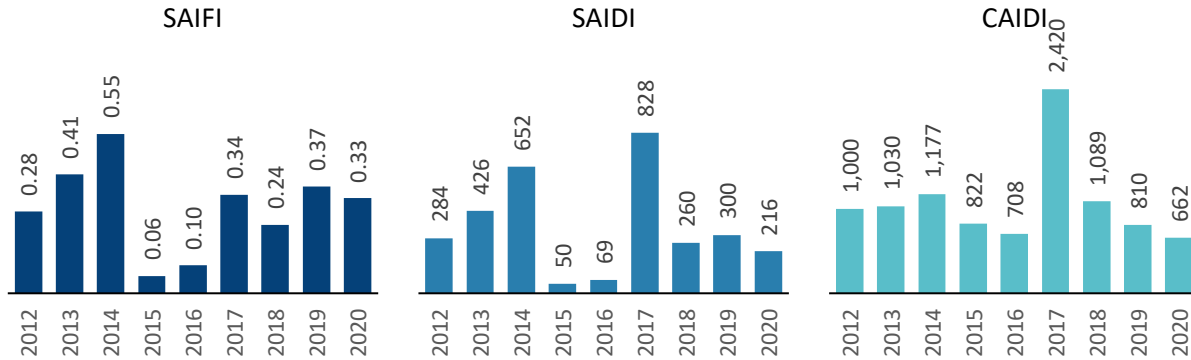


Exhibit 7.2.2.5 Reliability Statistics – Non-catastrophic Storms (DTEE Definition)

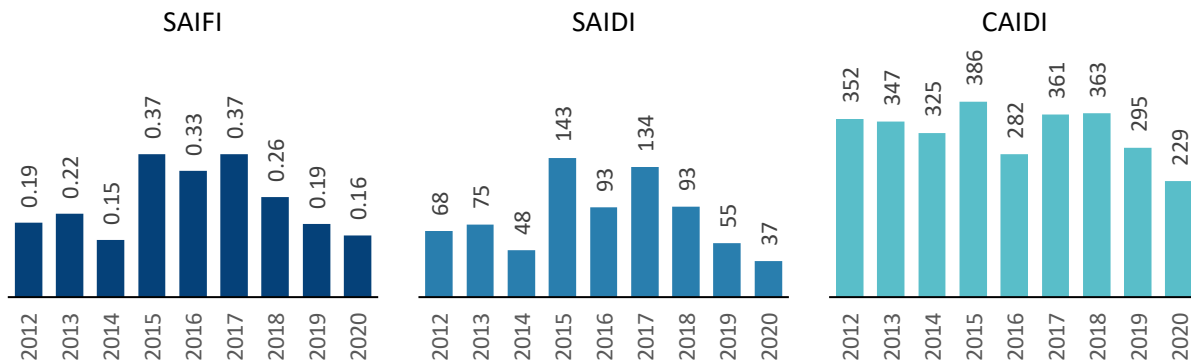
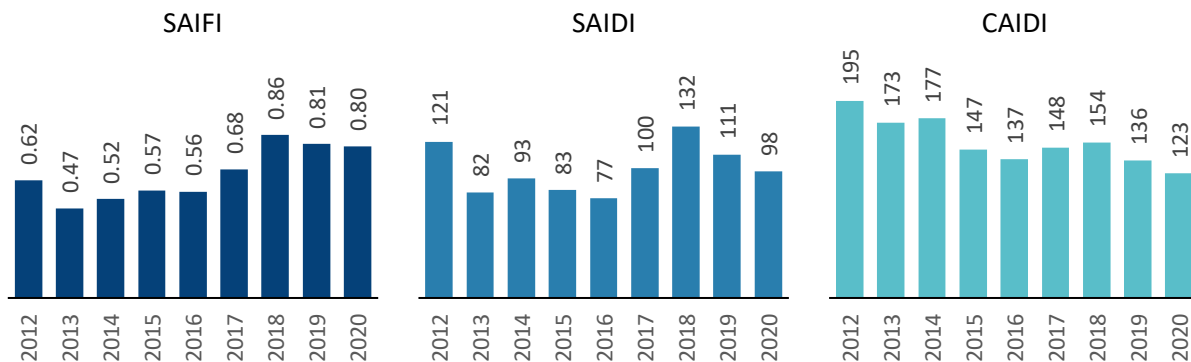


Exhibit 7.2.2.6 Reliability Statistics - Excluding All Storms



7.2.3 Causes of interruptions

Exhibits 7.2.3.1-7.2.3.3 illustrate the percent contribution to customer minutes of interruption, customer interruptions, and outage events by cause. Tree/wind interference is the leading cause of DTEE's customer minutes of interruption (SAIDI) and customer interruptions (SAIFI) as well as overall number of outage events (within DTEE's Outage Management System) on the DTEE system. Therefore, a robust tree trimming program is critical to addressing system reliability including customer minutes of interruption and customer interruptions. Equipment failures are the second leading cause of the SAIDI, SAIFI, and outage events on the DTEE system. In order to improve customer reliability and reduce outage events and durations, DTEE is planning to increase the investment in capital replacement programs for key equipment.

Exhibit 7.2.3.1 5-Year Average Customer Minutes of Interruption by Cause (SAIDI)

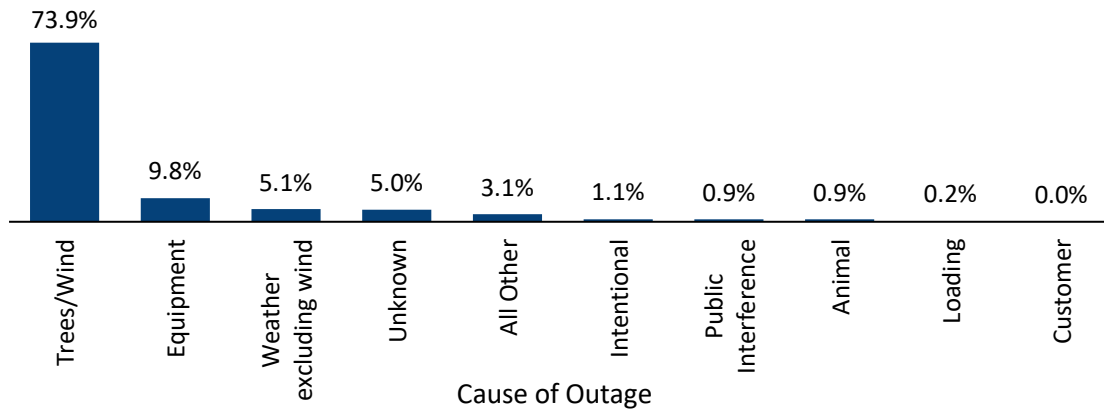


Exhibit 7.2.3.2 5-Year Average Customer Interruptions by Cause (SAIFI)

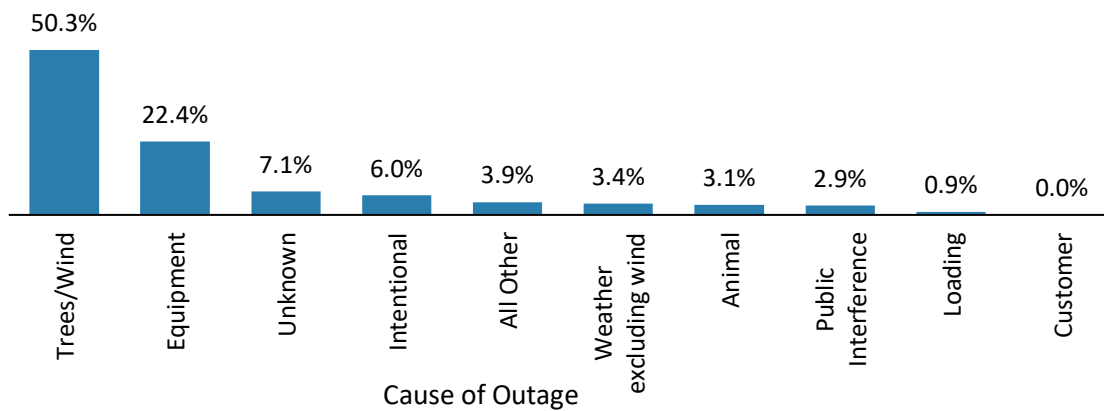
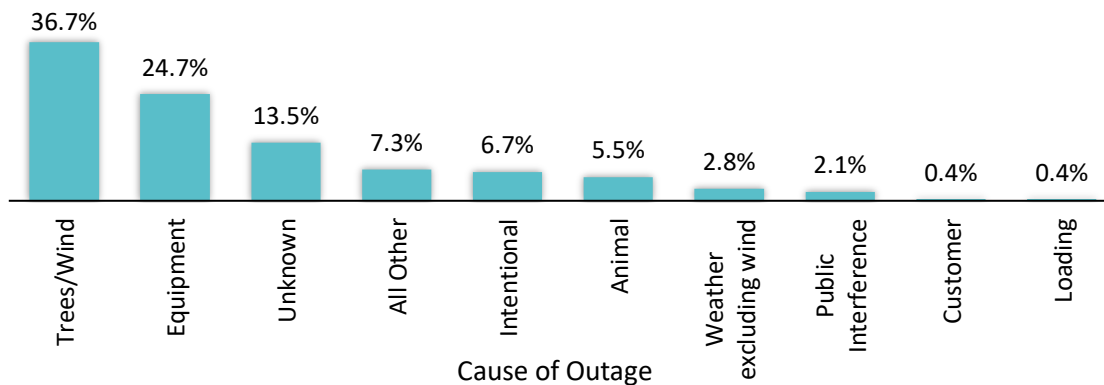


Exhibit 7.2.3.3 5-Year Average Outage Events by Cause



Note: Equipment includes substation, underground and overhead equipment and hardware

7.2.4 MAIFI

Sustained interruptions are not the only type of outage that impacts customers. Momentary interruptions, while brief in nature, are still a cause for concern and are tracked by DTEE. Customers need reliable power for the use of increasingly sensitive electronic devices that do not tolerate momentary interruptions. With the increase in remote work and schooling activities, customers are increasingly affected by momentary interruptions which often cause their internet connections to reboot.

IEEE Standard 1366 defines momentary interruptions and momentary interruption events.

Momentary interruption: The brief loss of power delivery to one or more customers caused by the opening and closing operation of an interrupting device.

NOTE—Two circuit breaker or recloser operations (each operation being an open followed by a close) that briefly interrupt service to one or more customers are defined as two momentary interruptions.”

Momentary interruption event: An interruption of duration limited to the period required to restore service by an interrupting device.

NOTE 1— Such switching operations must be completed within a specified time of 5 minutes or less. This definition includes all reclosing operations that occur within 5 minutes of the first interruption.

NOTE 2— If a recloser or circuit breaker operates two, three, or four times and then holds (within 5 minutes of the first operation), those momentary interruptions shall be considered one momentary interruption event.

The Momentary Average Interruption Event Frequency Index ($MAIFI_E$) indicates the average frequency of momentary interruption events. This index does not include the events immediately preceding a sustained interruption.”

The calculations of MAIFI and $MAIFI_E$ are shown in Exhibit 7.2.4.1 below. DTEE’s MAIFI calculation is for all weather conditions.

Exhibit 7.2.4.1 MAIFI Indices Calculations

Index	Full Name	Calculation
MAIFI	Momentary Average Interruption Frequency Index	Total number of customer momentary interruptions divided by the number of customers served
MAIFI _E	Momentary Average Interruption Event Frequency Index	Total number of customer momentary interruption events divided by the number of customers served; excluding events immediately preceding a sustained interruption

When DTEE fully implemented AMI meters in 2018, that technology enabled the ability to calculate MAIFI and MAIFI_E, however, due to data limitations currently DTEE only tracks MAIFI. While IEEE reporting is industry-accepted benchmarking for other reliability indices, MAIFI benchmarking is not yet included, and other widespread benchmark data regarding momentary outages are currently not published. DTEE's MAIFI for 2018 to 2020 is shown in Exhibit 7.2.4.2. As mentioned earlier in this section, MAIFI counts all momentary outages separately, which means an increased number of switching operations (which can occur in preventing sustained outages) will lead to an increase in MAIFI.

Exhibit 7.2.4.2 MAIFI by Year

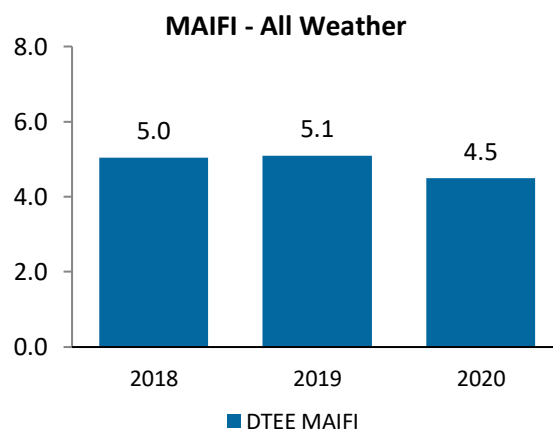


Exhibit 7.2.4.3 below contains MAIFI values by Service Center and for the System.

Exhibit 7.2.4.3 MAIFI by Year
(Average Number of Momentary Interruptions per Customer by Year)

Service Center	2018	2019	2020
Ann Arbor (ANN)	5.8	5.7	4.3
Caniff (CAN)	5.0	4.5	3.8
Howell (HWL)	4.7	6.9	6.2
Lapeer (LAP)	4.3	5.6	4.5
Marysville (MAR)	3.4	4.1	4.5
Mount Clemens (MTC)	4.2	4.5	3.6
North Area Energy Center (NAE)	7.4	6.6	5.3
Newport (NPT)	4.4	5.1	2.9
Pontiac (PON)	7.4	8.0	5.9
Redford (RFD)	5.5	4.6	4.4
Shelby (SBY)	4.1	4.1	4.4
Western Wayne (WWS)	4.3	4.4	4.7
System	5.0	5.1	4.5

7.2.5 CEMIn and CELIDt

CEMIn and CELIDt are measures that reflect customers' reliability experience with respect to multiple interruptions and long duration interruptions. These indices are defined in IEEE Standard 1366 and summarized in Exhibit 7.2.5.1 below. DTEE's CEMIn is based on all weather conditions; CELIDt is calculated for three conditions: all weather, normal, and catastrophic.

Exhibit 7.2.5.1 CEMIn and CELIDt Definitions

Index	Full Name	Calculation
CEMIn	Customers Experiencing Multiple Interruptions of n or More	Count of the number of Customers with n or more interruptions
CELIDt	Customers Experiencing Long Interruption Duration of t or More Hours	Count of the number of Customers with interruptions lasting t or more hours

The historical CEMIn and CELIDt values are listed in Exhibits 7.2.5.2 and 7.2.5.3, respectively.

Exhibit 7.2.5.2 CEMIn by Year
(Number of Customers Having “n” or More Outages by Year)

CEMIn	2012	2013	2014	2015	2016	2017	2017 ex Mar Storm	2018	2019	2020
CEMI1	1,482,953	1,443,749	1,421,322	1,273,402	1,245,866	1,528,465	1,302,141	1,399,468	1,415,643	1,377,066
CEMI2	788,531	786,657	779,176	622,180	596,814	859,466	648,574	765,711	779,763	719,205
CEMI3	380,037	399,349	388,076	282,709	270,676	423,990	297,999	413,715	399,023	353,989
CEMI4	175,709	192,366	178,812	125,496	121,110	193,470	129,299	208,042	201,835	181,175

CEMI5	79,285	97,726	76,180	58,121	51,279	86,695	54,959	99,455	105,862	90,837
CEMI6	32,892	44,130	30,307	25,957	24,692	37,508	24,485	45,757	54,591	47,317
CEMI7	13,662	18,916	11,152	11,485	12,334	16,524	10,613	19,844	28,106	26,685
CEMI8	5,420	7,092	5,143	5,432	4,634	7,711	4,958	9,647	15,235	15,710
CEMI9	2,371	2,741	1,943	3,236	1,712	2,813	2,228	4,867	8,021	9,251
CEMI10	1,562	1,395	1,204	2,737	605	1,483	1,134	2,623	4,814	4,798

Exhibit 7.2.5.3 CELIDt by Year

(Percent of Customers Experiencing t or More Hours of Interruption by Year)

Metric	2015	2016	2017	2017 Exclude March Storm	2018	2019	2020
CELID 8 Normal Conditions < = 10% Customers Interrupted	1%	0%	11%	2%	3%	4%	13%
CELID 36 All Conditions	14%	12%	10%	10%	9%	9%	2%
CELID 60 MPSC Catastrophic Conditions > 10% Customers Interrupted	N/A	N/A	23%	10%	2%	9%	N/A

7.3 Resiliency

Resiliency is a key concern for DTEE as well as the MPSC and other stakeholders. Emphasis was placed on this topic in the Commission's previous order in the U-20147 docket. Additionally, as described in Section 3.3.2 Increasing Catastrophic-Level Storms Scenario, there is evidence that DTEE is vulnerable to the increasing intensity and frequency of severe weather events. A strong focus on resiliency is necessary to ensure that impacts of severe weather events on DTEE's system are minimized and that systems are in place to address events when they do occur.

Resiliency for the electric grid is generally defined as the ability to withstand and recover quickly from an event that significantly impacts the grid. Typically, DTEE defines resilience as recovery from a weather-related event that causes power interruptions to at least 5% of customers. DTEE considers reliability metrics related to catastrophic storm, such as SAIFI, SAIDI and CAIDI, as the most relevant metrics to understand resiliency. These metrics are reported in Exhibit 7.2.2.4 and demonstrate that CAIDI during catastrophic storms, which measures the overall time for restoration, has improved by 40% from 2018 through 2020. The Company also considers the ability to respond to other more localized, high-impact outages, such as major equipment failures, as an important component of resiliency.

Most of the strategic investments in the DGP contribute to both reliability and resiliency. Efforts described in both Section 9 – Infrastructure resilience and hardening, such as 4.8kV Hardening, Pole Top Maintenance and Modernization (PTMM), as well as the Tree Trimming program (Section 10), provide significant benefit to withstand the impact of severe weather and catastrophic storms by strengthening the system and avoiding potential outages. The mobile fleet investments (Section 9.6) are key tools in responding to localized, high-impact outages such as major equipment failures, and aid in the restoration process. The investments in the Infrastructure redesign and modernization pillar both build the system to the latest standards, capable of withstanding severe weather, and provide additional capacity and flexibility to transfer load in the case of an equipment failure. In addition, the Strategic undergrounding initiative described in Section 11.6 will identify areas where access limitations such as rear-lot construction can be mitigated. Finally, many of the investments in Section 12 – Technology & automation, such as the Substation and Distribution Automation programs will help locate and isolate faults, speeding up restoration and minimizing the impact of outage events. These advanced technologies are enabled by the investments already made in ADMS and the Electric System Operations Center (ESOC).

In addition to specific investments, process improvements have improved DTEE's response to severe weather events and major equipment failures. Details of these process improvements are discussed in the next three sections (7.3.1 – 7.3.3.) and describe the prioritization process for restoration, which includes identifying and escalating the restoration of customer sites that impact public health, safety, and security.

7.3.1 Incident command system

One aspect of resiliency is having the ability to respond to outages and system events in a structured and efficient manner. While DTEE has always had a robust storm response structure in place, the Company has put in place the Incident Command System (ICS) in order to further improve storm and event response.

ICS is a best-practice method for responding to and managing emergencies that has been used by the US military, and more recently the private sector, for over 30 years. ICS is well suited for the utility sector due to its scalability, flexibility, and ability to manage a large influx of resources. ICS involves a response structure emphasizing chain of command and pushing decision-making down to the point of activity via a series of reporting sections.

DTEE has utilized an ICS structure since 2017 for significant events such as major equipment failures and implemented ICS for storm response in 2020. The ICS process has matured greatly over the course of 2020 and been documented into storm playbooks in early 2021. These playbooks ensure that best practices are used in every storm response regardless of the ICS leader, and the structure allows for improvements to be incorporated over time.

DTEE has seen significant improvement in storm response and storm metrics since implementing ICS and developing a best practices playbook. Specifically:

- Wire down response time during storm has improved by 58%, from 416 minutes in 2018 to 173 minutes in 2020
- Storm CAIDI has improved by 39% for CAT Storms and 37% for non-CAT storms between 2018 and 2020
- During the four CAT storms experienced in 2020, the accuracy of restoration estimates given to customers increased from 60% to 87% as process improvements were implemented. In 2021, estimate accuracy during the severe summer weather

was 72%. Gaps in processes and information systems were uncovered during the challenges of the largest CAT storm in August, and technology projects are being implemented to better align data and ensure that customers get accurate, error-free data during the restoration process.

7.3.2 Storm response

DTEE understands that losing power due to weather conditions is difficult for customers, particularly as outages during a storm tend to have longer restoration times than average. In order to provide an efficient response to potential weather events, DTEE has a multi-level classification of weather severity and initiates a response to a weather threat accordingly. Exhibit 7.3.2.1 shows DTEE’s five levels of severity related to potential weather, ranging from normal “Blue Sky” conditions to Level 5 – Crisis.

Exhibit 7.3.2.1 Weather Severity Levels

Level	Mode	Customer Interruptions	Impacts
5. Catastrophic	Crisis	> 420K	- An event that is extremely disruptive to a wide range of operational and business processes both within DTEE and the communities it serves - Available resources are insufficient to adequately address the response
4. Severe	Catastrophic Storm	110K – 420K	- A confirmed, active event resulting in significant damage to- or loss of company infrastructure or ability to perform vital business processes - Highly probable that additional internal and external resources are required
3. Serious	Serious Storm	25K – 110K	The event can impact multiple business operations or processes

			- Resources may need to move across regions or departments - Normal processes may not be able to address the response
2. Alert	High Impact Day	15K – 25K	- An event potentially limiting the ability to restore power using on-shift crews - Response strategies addressed with normally available local resources
1. Blue Sky	Routine	< 15K	- Common, day-to-day issues typically addressed through normal operating processes - Short in duration and result in little to no expectation of escalation

Storm response is managed through five phases: weather forecasting, ICS activation, ramp up, restoration, and ramp down.

1. Weather forecast

DTEE meteorologists prepare a two-day and five-day forecast by 8:00 am and 10:00 am each day, respectively. A potential weather threat noted in the five-day forecast that might lead to either a High-Impact Day or a Storm is escalated within the DO Emergency Preparedness and Response team (EP&R) to continue tracking it as it approaches. For threats with sufficient warning before impact, the forecast is refined each day leading into the two-day forecast as the Storm checkpoints and meetings begin.

The weather forecast is the first step in ensuring the organization is aware and ready for adverse conditions. The weather forecast is complemented by a customer outage prediction modeling tool.

- DTEE partnered with the University of Michigan-Dearborn graduate business analytics students to develop predictive outage models.
- Weather and outage data from the past 10 years for each service center was compiled and analyzed to identify impactful weather parameters. The result was the creation of a predictive model for each service center area.

- The output from the model shows how many weather-related jobs a service center can expect in a 24-hour period based on forecasted weather conditions. This is shared with DTEE Central Dispatch and Distribution Operations Leadership.

The number of predicted outage events initiates different readiness plans. One outage event could represent a single customer out of power or an entire circuit that is locked out, impacting thousands of customers. Each outage event in the Outage Management System (OMS) requires action by company personnel to resolve. This includes such tasks as dispatching of crews, updating crew status, adding notes to events, combining duplicate events in the system, adjusting event sizes based on information from the field, and creating associated events for other field resources such as tree trim or damage assessment. Thus, a higher number of events leads to a higher number of resources required to resolve the volume.

Predicted 0 – 150 outage events:

- Blue Sky day or normal operation day
- No internal escalations, each service center is self-sufficient in responding to trouble calls

Predicted 150 – 340 outage events:

- Gray Sky day, some level of escalation and central coordination exist
- Conference calls are initiated to ensure service centers are ready for higher trouble volume

Predicted 340 or more outage events:

- ICS activated in preparation for storm
- Emergency Headquarters is opened, and the formal storm processes and functions are initiated

2. ICS activation

If the weather forecast and modeling indicate that there is a potential for a storm, the on-call Incident Commander formally assumes command of the Storm event and officially activates all necessary Command and General Staff roles. Leading up to the Weather Update & Resource Ramp Up meeting, the activated Section Chiefs notify and/or activate necessary Functional Leads within their sections.

As previously mentioned, DTEE has documented its storm process and developed a series of playbooks for each of the Functional Leads to guide the storm restoration process. The Playbooks

outline the flow of an event from the initial weather and outage forecasts, through the weather briefing and storm leads ramp up, resource notification and activation, initial response and assessment, storm management, demobilization of all response resources, and the development of an After-Action Report.

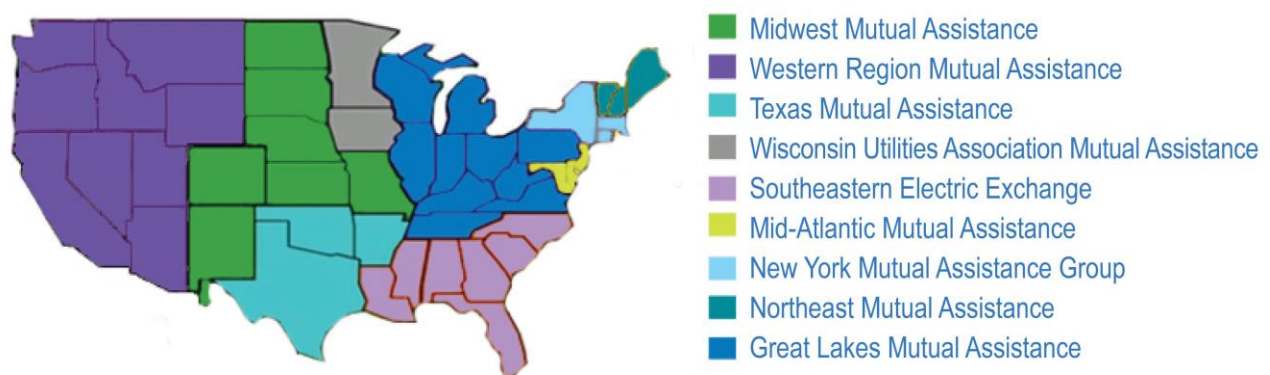
3. Ramp up

During the ramp up phase, the number of potential resources needed to address damage caused by the weather event and the strategy to secure those resources is determined. The Mutual Assistance (MA) Manager coordinates with the Operations Chief to gather initial reports from the field about the extent of damage to feed into the development of the Planning Scenario and Mutual Assistance Resource Plan. The Operations Chief, Logistics Chief, and MA Manager use this information to make a recommendation to the Incident Commander detailing the number and type of mutual assistance crews that should be requested through Regional Mutual Assistance Groups (RMAGs).

Depending on the severity and scope of the weather event, the MA Manager, Incident Commander, and Leadership may consider seeking resources from additional RMAGs if the initial request cannot be met through Great Lakes Mutual Assistance (GLMA).

DTEE participates in the GLMA group with neighboring utilities to assist with restoration after major events. A map of mutual assistance groups within the United States is shown in Exhibit 7.3.2.2

Exhibit 7.3.2.2 Mutual Assistance Groups within the United States



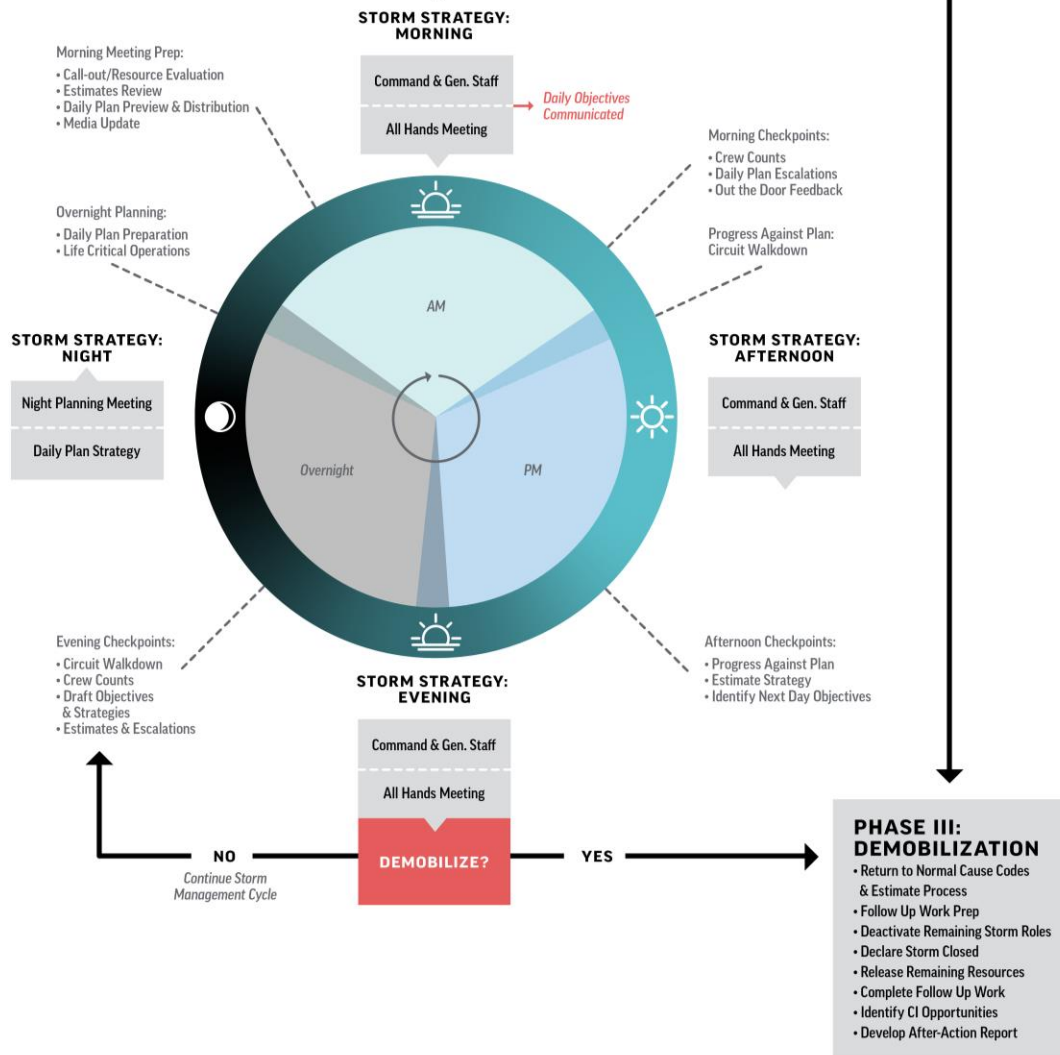
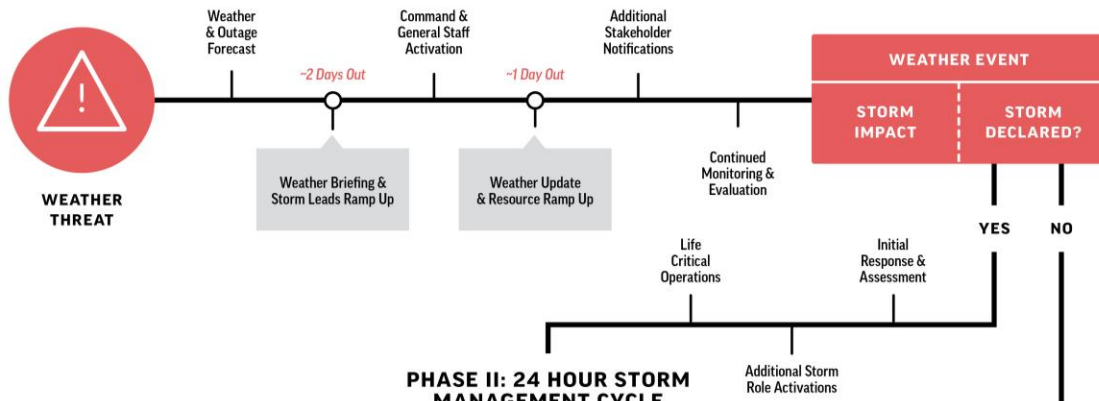
Map showing the seven mutual assistance group regions in the system of sharing resources to restore power in the U.S. Michigan is in the Great Lakes region.
Edison Energy Institute

4. Restoration

The restoration phase follows a structured 24-hour storm management cycle. This structured cycle allows the resources to be allocated to the restoration priorities in a methodical and predetermined manner. Exhibit 7.3.2.3 illustrates the 24-hour storm management cycle.

Exhibit 7.3.2.3 24-hour Storm Management Cycle

PHASE I: RAMP UP



Based on the forecast and outage prediction models, DTEE follows a pre-specified storm plan. Before the incoming weather hits the DTEE service area:

- The Systems Operations Center (SOC) and Central Dispatch work to complete critical open maintenance jobs on the electrical system
- Central Dispatch, SOC, Regional Customer Operations, and other key stakeholders organize and conduct several strategy planning sessions for the expected outages in order to coordinate and arrange resources as necessary
- Additional resources (overhead line contractors – including mutual assistance, tree trim crews, etc.) can be secured as needed depending on the expected size of the storm. Additional pull-out yards (areas that house non-DTEE overhead linemen trucks and supplies near the point of expected activity) may also be secured
- Public Protection crews are alerted, as addressing downed wires in the field is the number one priority to ensure the safety of the public and our employees

DTEE follows clear prioritization criteria in addressing potential hazards and restoring customers regardless of the severity of the event. Exhibit 7.3.2.4 illustrates DTEE's restoration plan.

Priority 1: Safety Events

- Public contact / public trapped
- Police and Fire events
- Pole to pole wire downs
- Other hazards and ground alarms

Priority 2: Critical Customers

- Hospitals, pumping stations, police and fire departments
- Depending on time of the year: schools, polling stations, etc.
- Any other critical infrastructure

Priority 3: Large Outages

- Substation level outages, including ties and trunks
- Multiple customer outages resulting from failed overhead or underground circuits
- Other large outages (fuse and recloser level)

Priority 4: All Other Outage and Trouble Events

- Transformer and single customer outages

- Non-outage events (low voltage, flickering lights, etc.)
- Others

Exhibit 7.3.2.4 DTEE's Restoration Plan



Mobile command center

DTEE's Mobile Command Center serves as an extension of the Emergency Headquarters in responding to complex forced outage events and severe storms. It provides for onsite oversight at localized events. It has the same tools and technology as the Emergency Headquarters.

Exhibit 7.3.2.5 DTEE's Mobile Command Center



Customer communication and education

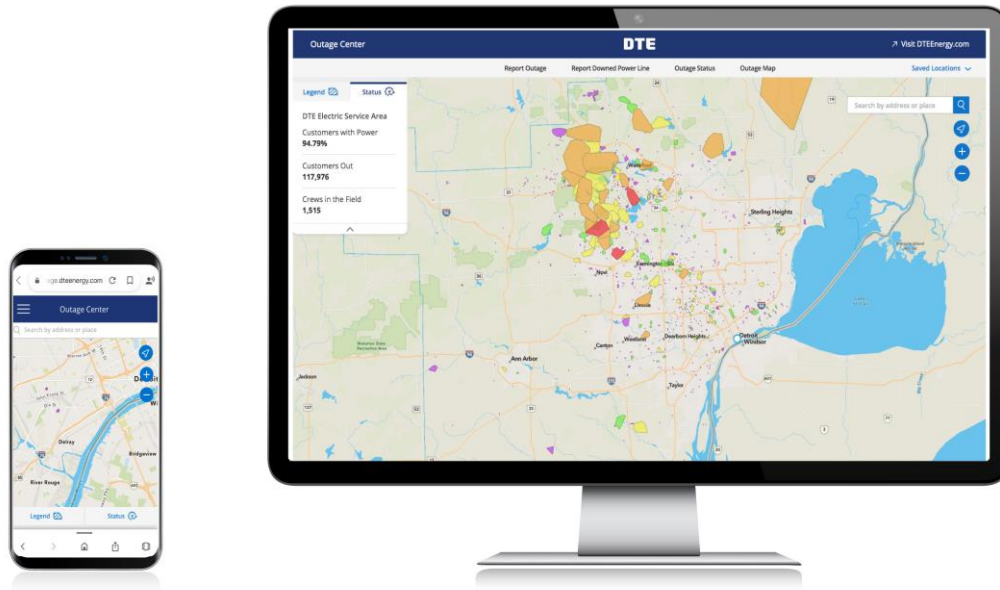
Losing power as a result of a weather event is a distressing situation for many customers. In addition to safety messages, knowledge about the cause and status of their outage as well as when to expect power to be restored is some of the information that DTEE communicates to customers during the storm. In addition, DTEE deploys resources such as our Community Vans and the Mobile Command center to provide relief in the form of water, blankets, and charging stations whenever possible.

Communication and education

To partner with and prepare communities for potential storm related outages, the emergency Managers in the 13 counties served by DTEE receive proactive communication whenever inclement weather is anticipated. Monthly problem-solving sessions are held with local firefighters and the Michigan State Fire Marshal to drive process improvements and improve communication of response times.

Customers are kept up to date about their outage through DTE's mobile app and website, as well as through notifications such as text messages and emails.

Exhibit 7.3.2.6 Outage Map on DTE's Mobile App and Website



Information available to customers includes:

- Estimated Restoration Time (ERT)
- Outage size and area affected
- Outage updates (on demand)
 - Crew en-route
 - Crew arrived
 - Cause of outage
 - Outage restored

DTE has partnered with National Energy Foundation (NEF) to create in-school educational program for K–5 students about energy safety, including important information such as staying away from downed power lines. To supplement verbal and written material, NEF also utilizes Hazard Hamlet simulator boards which simulate common hazards created by an energized down wire.

Exhibit 7.3.2.7 Hazard Hamlet Simulator



Community vans

DTEE maintains a fleet of Community Vans designed to offer support to customers in geographic areas impacted by planned and forced outage events, severe storms, customer excellence and community outreach events. The vans distribute outage kits to affected customers. The kits include helpful items such as: water, flashlights, battery chargers, blankets and literature¹⁶ like a brochure that explains DTEE's storm response procedures and safety information. The vans also provide a point of contact for customers to ask questions about their outage event.

¹⁶ "Guiding you through the storm" [23 Storm Info media brochure.pdf \(empoweringmichigan.com\)](https://empoweringmichigan.com/23%20Storm%20Info%20media%20brochure.pdf)

Exhibit 7.3.2.8 Community Van



7.3.3 Response to major (non-storm) events

The second type of event on the electrical system from which DTEE is prepared to recover quickly is a major event, such as a key equipment failure in a substation.

The first step to enhancing resiliency is proactively identifying vulnerabilities of the distribution system and preparing contingency plans. Asset health assessments, capital replacements, and preventative maintenance plans processes are some of the ways in which DTEE prevents equipment failures. These processes are described in Sections 8, 9.1, and 14 respectively. In addition, annual pre-peak assessments are performed to identify loading or voltage issues, and measures are taken to address them ahead of summer peak.

- Engineers identify voltage or overload violations through the Annual Area Load Analysis (ALA) process

- Relief plans are developed, and critical work is completed before the predicted summer peak period begins
- System Operations Center (SOC) and Engineers identify next contingency concerns and determine appropriate course of actions

A daily review of trouble work, such as temporary load transfers due to previous outages or equipment being offline, is performed to ensure the system is robust enough for the next contingency (system abnormality). This allows for awareness of equipment that is out of service, and for engineers to run N-1 contingency analysis daily to ensure system security. If issues are identified, corrective measures are developed to mitigate the next contingency with options including:

- Restrict timing for planned shutdowns
- Create must-do jobs for the day
- Use of throw-over circuits to transfer load
- Dispatch of mobile Distributed Generation (DG) units
- Dispatch of portable substations
- Dispatch of ISO-trailer (i.e., transformer trailer that connects systems with different voltages)
- Dispatch of mobile cable trailer
- Dispatch of mobile switchgear trailer
- Utilize manual and/or automatic under-frequency load shedding schemes for the subtransmission system
- Utilize local load shedding for issues on the distribution system

Despite the preparation and planning done to ensure resiliency in the system, equipment failures do occur that either create a large outage or put the system at risk of experiencing a large outage, were an additional equipment failure to occur. In this situation, the ICS process is implemented to manage the event. In addition, DTEE has a large mobile fleet of specific equipment that in many instances can provide back-up power to customers and support system needs during both trouble and planned work. Exhibit 7.3.3.1 provides a list of DTEE's mobile fleet of equipment while Exhibits 7.3.3.2 – 7.3.3.3 display two such pieces of equipment.

Exhibit 7.3.3.1 DTEE's Fleet of Mobile Restoration Equipment

Equipment	Units	MVA/MW
Mobile Generators	10	19.5
Portable Substations (+2 in 2020)	10	59.0
ISO-Trailers	9	35.0
Cable Trailer	1	N/A
Mobile Switchgear	1	N/A

Exhibit 7.3.3.2 Mobile Generator



Exhibit 7.3.3.3 Portable Substation



7.3.4 Low probability, high impact (LPHI) events

DTEE has robust planning, contingency procedures and protocols for LPHI events. As part of DTE's corporate emergency preparedness program, Distribution Operations completed simulation exercises on five LPHI events. Acting upon the lessons learned from these exercises and in conjunction with DTEE's standardized restoration process positions DTEE can efficiently address these situations in the unlikely event that they ever materialize.

Exhibit 7.3.4.1 Low Probably, High Impact Events

Event Exercised	Description	Lessons Learned
Electric Outage	420,000+ customer catastrophic storm	• Increase foreign crew visibility for safety • Maintain list of critical locations (hospitals and other critical infrastructure) per county • Design Storm Playbook including strategies around Operations, Planning, and Logistics
Mass Black Out	Loss of major portion of ITC transmission and DTEE systems caused by loss of generation capacity, transmission network, voltage collapse, or transmission system outside of territory	• Stand up satellite phone program in the event normal communication channels are lost • Improve integration of DTEE/ITC/State/County public information officers ensuring consistent dissemination of information during an incident
Geomagnetic Disturbance (GMD, EMP E3)	A geomagnetic disturbance (GMD), also known as a geomagnetic storm, is a major event in Earth's magnetosphere. It's caused by a very efficient transfer of energy from solar wind into the space environment surrounding Earth. Solar wind shockwaves result from a solar flare that is followed by	• MISO performs studies on the impacts of GMDs on Bulk Electric System transformers and DTEE supplies transformer data to them when requested, usually annually. • If the study indicates that a transformer may experience geomagnetically induced current (GIC) that exceeds NERC Standard TPL-007 criteria levels, then DTEE is required to perform a thermal impact assessment on that

	coronal mass ejections (CMEs) of charged and magnetized particles into space. GMDs are of particular concern to the reliability of the nation's power grid because they can cause geomagnetically induced currents (GICs)	transformer and determine appropriate mitigation. So far, none of DTEE's transformers have been identified from the MISO studies to date. Recently, NERC has initiated a process where they will make an annual request for GIC data from transformer monitors. Data will be requested only for strong GMD events (Kp=7 and greater).
Physical Attack	Attack on DTEE stations/substations or other operations facilities	Increase seating and technological capacity for Central Dispatch and System Operations Center at back-up location in the event the primary location is lost to a short-term disruption caused by a physical or cyber attack
Cyber Attack	Attack on critical computer systems	

7.3.5 Flood protection program

The typical storm impacts on the grid include the wind and tree damage to poles and wires. An additional weather impact, although rare, is significant rainfall that results in flooding events that can damage equipment and property. A recent example of this occurred due to a severe thunderstorm impacting DTEE's service territory with a total rainfall of over two inches on June 20, 2021 and then additional rainfall on June 25, 2021 that produced another six to eight inches of rain in a 24-hour period. Over ten inches of rain in less than a week caused flooding in some areas including the area near the intersection of I-94 and Wyoming Ave. This area includes McGraw and Scotten substations and resulted in the substations flooding, significant equipment damage and outages for customers. To prevent another occurrence of flooding resulting in equipment damage and customer outages at McGraw, Scotten or any other substation, the Company is initiating a Flood Protection Program. This work will progress over three phases.

In the first phase of this work, an assessment of the recent flooding events at McGraw and Scotten substations will be completed to fully understand the root causes and confirm that the countermeasures that the Company is planning will prevent a reoccurrence. With a complete understanding of the root causes of what occurred at McGraw and Scotten, a substation flood risk assessment will be completed for all the Company's substations and stations. The flood risk

assessment will identify the susceptibility and mechanism of potential substation flooding based upon regional United States Geological Survey (USGS) topographical data to determine storm runoff patterns. Federal Emergency Management Agency (FEMA) maps will be reviewed for each location to determine what properties have active floodplains. Substation layout and the surrounding substation area topography will also be considered during this study. With the risk of flooding well understood, the concerning cases will be prioritized based on the potential consequences of a flood event. Some of the factors that will be used in this prioritization are substation class, customers served, critical customers served, potential for stranded load, environmental concerns, and equipment impacted by flooding, among others. In parallel with the risk and impact assessment, benchmarking work will be completed with peer utilities to identify those that have taken on similar efforts, and the most effective countermeasures for addressing the concerning cases.

In the second phase, short- and long-term countermeasures will be developed for each substation with field visits to confirm executability and confirmation of assumptions. Some of the mitigations needed might include reconfiguration of the substation, storm water management systems, elevating equipment, additional flood barriers, and possible relocation/replacement of substations. Schedules and cost estimates will be developed considering the priorities developed in the first phase.

The execution phase will quickly follow phase two and will include designing and constructing the required capital projects.

7.4 Power quality

While reliability and the related metrics were discussed in Section 8.2 on reliability, power quality encompasses an additional set of disturbances or irregularities a customer may experience with their electrical service. Customer needs have shifted recently, including a potential permanent shift to working from home, schooling from home, and a reliance on a stable internet connection for many aspects of life, making power quality even more important to customers. A momentary interruption that resets an internet connection and disrupts a business meeting is much less acceptable to customers than needing to reset a blinking digital clock.

7.4.1 Defining power quality and power quality events

According to IEEE Standard 1159, power quality refers to “a wide variety of electromagnetic phenomena that characterize the voltage and current at a given time and at a given location on the power system.” IEEE breakdowns the phenomena into the following categories: Transients, Short-duration root-mean-square (rms) variations, Long duration rms variations, Imbalance, Waveform distortion, Voltage fluctuations, and Power frequency variations. DTEE has adapted these categories to assess power quality in terms of these characteristics of electric power:

1. Interruptions – momentary and sustained

Electric power interruptions may be sustained (lasting longer than 5 minutes) or momentary (lasting 5 minutes or less). While momentary interruptions can be bothersome to customers, in most instances the source of momentary interruptions is a purposeful equipment operation that actually is preventing a sustained interruption. Devices on the system (breakers, reclosers, and fuses) are designed to open when an electrical fault occurs to isolate and de-energize the faulted part of the system. Breakers and reclosers, however, are programmed and coordinated to open/close several times at defined intervals. If the fault is temporary (e.g. tree branch falls, then clears the wires) the device will ultimately close restoring power. However, the customer experiences these distinct open/close attempts as momentary outages.

2. Voltage – over/under, short and long duration

ANSI C84.1 defines voltage tolerance standards Range A and Range B for electrical systems with voltages of 600 or less. Acceptable sustained service voltage for Range A is between +5% to -5% and for Range B is between +6% to -8%. For nominal voltage of 120V these ranges are 114V to 126V for Range A and 110V to 127V for Range B. Although DTEE’s distribution system is designed to provide voltages within Range A, various conditions can cause the system to operate within Range B. Customer equipment should be designed to operate at voltages within either range. Under-voltage events occur when the voltage is below the acceptable range; if the duration is less than two minutes it is a voltage sag event. Over-voltage events occur when the voltage is above the acceptable range; if the duration is less than two minutes it is a voltage swell event.

3. Flicker – small and repetitive voltage fluctuations

Flicker is generally observed in changes in lighting intensity due to voltage variations within the acceptable range. The voltage variations are generally below 7 % of nominal voltage with frequencies of less than 25 Hz. Causes of flicker include but are not limited to arc furnaces, rolling mills, large motors, welders, capacitor switching, transformer load tap changers, and loose electrical connections.

4. Transients and Noise – large but very short duration voltage spikes

Transients or surges are events with significant changes in magnitudes of voltage, current, or both. These events may have durations of nanoseconds to milliseconds. Noise is a rapid series of transients, typically of less significant magnitude variation than an isolated event. Causes of transients and noise include but are not limited to lightning strikes, switching on the electrical system, recloser operations, transformer load tap changing, and loose electrical connections.

5. Harmonics and Distortion – non-smooth sinusoidal waveforms

Harmonics are sinusoidal voltages or currents having frequencies that are integer multiples of 60 Hz. The harmonics combine with the 60 Hz voltage or current and result in a distorted sinusoidal wave form. Total harmonic distortion levels are characterized by the complete spectrum of the individual harmonic components. Harmonics are mostly caused by non-linear loads and devices connected to the electrical grid. These non-linear loads include but are not limited to arc furnaces, welders, rectifiers, adjustable speed drives, switched mode power supplies, high efficiency lighting, and data processing equipment.

6. Frequency

The frequency of the interconnected electrical grid is usually very stable, and deviations are extremely rare – hence frequency is not an aspect of power quality that concerns customers on distribution systems.

Based on data collected from the Company's Customer Service organization, the majority of residential customers' power quality concerns in DTEE are related to the number and duration of power interruptions (approximately 87% of complaints), over/under voltages (approximately 13% of complaints), and to a lesser extent, flicker and transients. In addition to these concerns, commercial and industrial customers also have concerns related to distortion and harmonics. Transients and Noise, Harmonics and Distortion, and Frequency are critical issues relative to transmission and subtransmission planning and the overall electrical grid more so than the distribution system.

Power quality events or issues may originate from any part of DTEE's distribution system, the transmission system, or from the customers' equipment or operations. These events or issues may be isolated to a single customer or groups of customers in electrical proximity. Commercial and industrial customers may be more susceptible to power quality issues if their processes involve sophisticated machine controls.

7.4.2 Addressing distribution system power quality events

Distribution planning focuses on the two categories of power quality events that are both the most frequent and have the greatest customer impact: interruptions and high/low voltage.

Interruptions

Minimizing the number and duration of customer power interruptions is one of the highest priorities of DTEE. As discussed throughout this grid plan, DTEE has programs, processes, and plans for today and the future to improve reliability and resiliency and thereby minimize power interruptions to customers. The key programs that aim to reduce interruptions are Tree Trimming, Automation, 4.8 kV System Hardening, URD Cable Replacement, and the Pole/Pole Top Hardware program. Additionally, DTEE has established a Customer Excellence program described in Section 9.4 – Frequent Outage Program. This program identifies customers with excessive recent sustained and momentary outages and expedites a remediation plan to improve their reliability. The remediation plan typically includes repairs to overhead infrastructure and equipment and spot tree trim where necessary and is implemented on average within 60 days of the identification of the power quality issue.

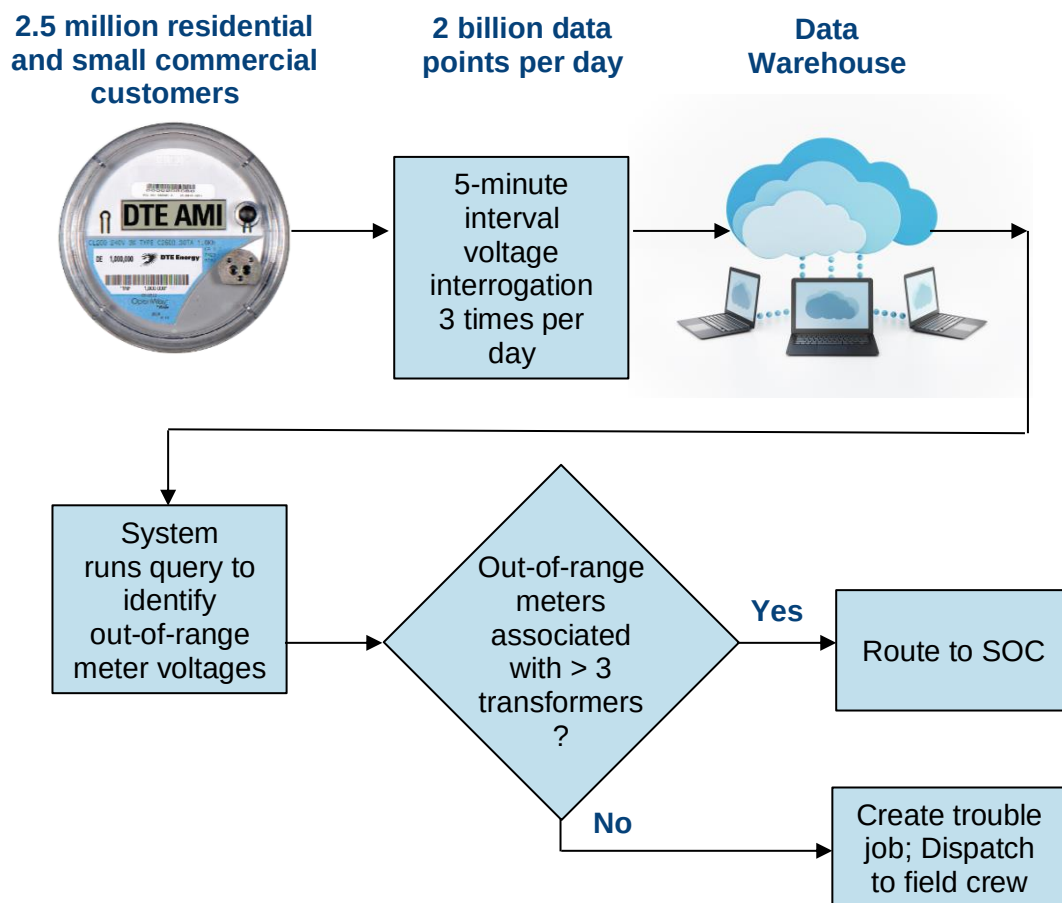
Voltage

DTEE has several initiatives to address voltage-related power quality issues. Historically, DTEE became aware of voltage issues when notified by the affected customer(s). Today, with AMI infrastructure and Power Quality meters, DTEE can pro-actively identify voltage events – even some that have not manifested to a level noticeable by customers.

AMI meters record 5-minute interval voltage. The AMI meters are interrogated three times per day. The data are then queried to identify meters with voltages outside the acceptable ANSI range. If the meters with abnormal voltages are associated with several distribution transformers, it may indicate a circuit problem; if only a few transformers are involved, it is likely a localized problem and

a trouble job is proactively created in InService, DTEE's Outage Management System. Problems affecting larger circuit areas are sent to the System Operations Center. In either case, an investigation is launched to identify the root cause and remediate the problem. Prior to the current process it would take several days to over a week to address high voltage cases through the planned work process; now the issues are typically identified and addressed within three days. Exhibit 7.4.2.1 illustrates the process.

Exhibit 7.4.2.1 Identifying Abnormal Voltage Conditions



Investigations have shown that high voltage cases are frequently the result of damaged windings inside distribution transformers, usually caused by lightning strikes. Investigations have shown that after two 100kA surges, lightning arrestors attached to the transformers disconnect and subsequent

lightning strikes can affect the actual transformer, leading to the damage that causes high voltage. Replacement of these transformers corrects the issue and are an example of work completed in the Emergent Capital category (Section 13 – Base Capital).

A case of high voltage due to ferroresonance, an electrical phenomenon caused by a non-linear interaction between capacitive and inductive elements on the circuit, was first raised in the Ann Arbor area by customer complaints and the cause was not immediately obvious when troubleshooting. This led to additional investigative methods being utilized, such as analyzing AMI data to detect patterns and affected areas including the creation of the process in Exhibit 8.5.1. Further study showed that AMI data of high voltage cases resulted in differing patterns for different causes; for example, the AMI data for the ferroresonance case did not follow the same patterns as in cases caused by damaged transformers. After extensive research the Engineering team suspected ferroresonance as all the conditions supported this conclusion. The team then sought support from industry experts to help confirm that this was the case.

In total, 94 circuits were identified as having potential ferroresonance issues utilizing the lessons learned from the original. A capacitor placement and control program has been developed to prevent any high voltage events on these circuits as a result of ferroresonance. Additional details and the investment profile can be found in Section 12.6 – CVR/VVO.

When the process to use AMI meters to seek out abnormal voltages was initiated in early 2020, nearly 3,000 customers meters were reporting a voltage over 106%. After investigating the initial cases and implementing an ongoing monitoring process, the number of customers meters reporting a high voltage is typically less than 10. During the same timeframe there were initially about 300 customer meters that were reporting a voltage over 110% and all cases have been addressed. DTEE now monitors and addresses cases on a regular basis.

Due to the success of the high voltage detection process, DTEE is in the process of developing a similar approach to handling Low Voltage and Momentary Interruption situations, utilizing AMI data to identify circuits or circuit sections that are experiencing low voltages or high numbers of momentary interruptions occurring above those expected due to circuit switching operations.

7.4.3 Addressing subtransmission system power quality events

In addition to the voltage analysis process described above, DTEE has an effective process in place that enables its large industrial and commercial customers typically served from the subtransmission or transmission system to report and resolve power quality issues. DTEE has various channels available to assist all Commercial and Industrial (C&I) customers which include account managers and a separate 24/7 customer support system. Account managers are assigned to industrial customers based on DTEE's Major Account Services six customer segments, which are defined using the industry standards, Standard Industrial Classification (SIC) / North American Industry Classification System (NAICS) codes. Each of the six customer segments are categorized from simple to complex and each customer is matched with an account manager depending on the complexity of service requirements, rates, business operations and electrical loads. Smaller sized C&I customers are eligible for an account manager when they meet two criteria: annual \$300,000 revenue and 5 million KWh threshold. C&I customers without an account manager have a dedicated team within the call center to handle only their calls. Using a special coding system for C&I customers allows for their calls to be held in a separate queue outside of residential customer calls. This ensures that all C&I customers' calls are processed in the timely manner. Community, municipal, or government channels are also available for C&I customers to submit requests and comments to DTEE. Often, C&I customers have established relationships with these agencies and are able to work through regional managers or account managers.

Although many power quality events warrant an investigation, it is not feasible to provide a root cause analysis for all reliability or power quality issues experienced by C&I customers. Many events clear themselves after a one-time occurrence, such as animals or temporary tree interferences because DTEE's subtransmission system is constructed with redundancies to protect against such instances. As such, an isolated event experienced by C&I customers may not necessarily trigger an investigation or patrol on a section of the electrical system. Instead, an investigation and patrol are typically conducted when multiple reliability or power quality events are experienced by a C&I customer over a short time period. Multiple events are a better indicator that a more persistent issue exists, for example a piece of failing equipment.

During a power quality or reliability investigation, replacements of cracked insulators or spot tree trimming are typically performed as potential solutions. However, it is not always clear if these

solutions resolve the root causes of the outage events. This is because DTEE’s experience suggests that a customers’ internal system is sometimes the root cause of the power quality and reliability problems and identifying and attributing any specific issue as the root cause of an interruption or voltage sag is occasionally challenging and uncertain.

To better identify and locate existing and potential power quality issues on the subtransmission system, it is necessary to have proper power quality (PQ) metering in place at critical substation and customer locations. Exhibit 7.4.3.1 shows the investment in PQ metering for 2021-2025.

Exhibit 7.4.3.1 Subtransmission Power Quality Meter Investments

	\$ Millions					
Category	2021	2022	2023	2024	2025	2021-2025 Total
PQ Metering	\$ <1	\$ <1	\$ 2	\$ 2	\$ 3	\$ 7

In addition to the investigations and troubleshooting in response to customer inquiries, DTEE conducts subtransmission modeling to identify power quality issues. Based on subtransmission modeling, there are 115 subtransmission circuits (18% of total) with potential thermal and/or voltage violations under certain loading and contingency conditions. A thermal overload indicates that equipment on the circuit or station exceeds its rating, and a voltage violation indicates that the voltage on at least part of the circuit is no longer within an acceptable range. This could lead to power interruptions or power quality issues for industrial, commercial, and/or residential customers. The investment that is planned to resolve these existing subtransmission planning criteria violations is discussed in more detail in Section 11.2 - Subtransmission Redesign and Rebuild Program.

7.4.4 Other power quality issues

Transients

Transients may result from lightning strikes or switching operations. Lightning arresters on the conductors are designed to shunt the lightning strike to ground eliminating the surge to customers. In some locations, sections of distribution circuits are attached on poles with subtransmission

overbuilt which shield the distribution wire from a direct lightning strike. And, subtransmission lines typically have an overhead shield wire to protect them from direct lightning strikes.

During pole top maintenance patrols, failed lightning arresters and missing sections of shield wire are identified for replacement.

Harmonics

Harmonics are rarely a problem with residential and most commercial customers. Commercial and industrial harmonic problems typically originate from customers' equipment or processes and may impact customers in electrical proximity. Harmonic problems are addressed by a power quality technician and/or engineer on a case by case basis once DTEE is aware of the problem.

8 Asset health assessment



The purpose of the asset health assessments is to gather information related to our assets and proactively address any identified equipment issues. In this section, we discuss the 20 assets that have the greatest potential impact on customer reliability. The assets described here represent approximately 80% of the distribution plant asset base.

The average age, range of ages and life expectancy for the 20 asset classes are summarized in Exhibit 8.1. The life expectancy is based on a combination of manufacturer recommendations, industry benchmarks, EPRI (Electric Power Research Institute) Industry Database, NEETRAC (National Electric Energy Testing, Research & Applications Center) Asset Survival Plots, and DTEE's own experience. Age can become a significant factor when replacement parts become unavailable or in the specific cases where asset health deteriorates sharply with age, however DTEE is also using health-based assessments to evaluate its assets. The remainder of this section provides a detailed description of individual asset health assessments.

In addition to the asset health assessment process, DTEE ensures reliable performance of individual asset classes through the combined measures of preventive maintenance and proactive replacements programs where justified. These measures are summarized in Exhibit 8.2 and detailed in individual asset subsections. Equipment that is a candidate for replacement is identified through the asset health assessments. The scope of work and capital funding for these replacements over the next five years are detailed in Section 9.1 – Capital Replacement Programs.

Exhibit 8.1 Asset Age Summary

Section	Asset	DTE Electric Average Age (Years)	DTE Electric Age Range (Years)	Life Expectancy (Years)
8.1	Substation Power Transformers	43	0 – 96	40 – 45
8.2	Network Banks	64 (structures) 46 (transformers)	0 – 85+	20 - 30 (transformers)
8.3	Circuit Breakers	42	0 – 85	30 – 40
8.4	Subtransmission Disconnect Switches	Not Available	0 - 85	Not available
8.5	Circuit Switchers	18	0 – 35	Not available
8.6	Relays	32	0 – 60+	15 – 50
8.7	Switchgear	37	0 – 67	40
8.8	Poles and Pole Top Hardware	46	0 – 90+	40 – 50
8.9	Small Wire (i.e., #4, #6, and #8)	70+	Not Available	Varies based on field conditions
8.10	Fuse Cutouts	19	0 – 50+	40
8.11	Three-phase Reclosers	9	0 – 31	20
8.12	SCADA (Supervisory Control and Data Acquisition) Pole Top Switches	14	0 – 38	20
8.13	40 kV Automatic Pole Top Switches	36	0 – 50+	40
8.14	Overhead Capacitors	Not Available	Oldest: 25+	20
8.15	Overhead Regulators	Not Available	Oldest: 25+	20
8.16	System Cable	45	0 – 100+	20 – 40
8.17	Underground Residential Distribution (URD) Cable	25	0 – 60+	40
8.18	Manholes	78	0 – 100+	Varies based on construction and field conditions
8.19	Vaults	Not Available	Not Available	Varies based on construction and field conditions
8.20	Advanced Metering Infrastructure (AMI meters)	6.5	0 – 10	20

Exhibit 8.2 Asset Program Summary

Section	Asset	Preventative Maintenance Programs (Yes/No)	----- Proactive Replacement Program -----		
			Yes/No	2021-2025 Investment (\$M)	2021-2025 Number of Units
8.1	Substation Power Transformers	Yes	No	Not Applicable	
8.2	Network Banks	Yes	Yes	Included in Downtown City of Detroit (CODI) Project	
8.3	Circuit Breakers	Yes	Yes	\$84 M	267 units
8.4	Subtransmission Disconnect Switches	Yes	Yes	\$12 M	194 units
8.5	Circuit Switchers	Yes	Yes	\$5 M	30 units
8.6	Relays	Yes	Yes	Not Applicable	
8.7	Switchgear	Yes	Yes	Included in Substation Risk Reduction Program	
8.8	Poles and Pole Top Hardware	Yes	Yes	\$430 M	Varies
8.9	Small Wire (i.e., #4, #6, and #8)	No	No	Not Applicable	
8.10	Fuse Cutouts	No	Yes	Included as part of Pole Top Hardware	
8.11	Three-phase Reclosers	Yes	Yes	\$9 M	95 units
8.12	SCADA Pole Top Switches	Yes	Yes	Combined with 8.11 in SCADA Pole Top Device Program	
8.13	40kV Automatic Pole Top Switches	Yes	Yes	\$26 M	70 units
8.14	Overhead Capacitors	Yes	No	Not Applicable	
8.15	Overhead Regulators	Yes	No	Not Applicable	
8.16	System Cable	No	Yes	\$243 M	98 miles
8.17	Underground Residential Distribution (URD) Cable	No	Yes	\$60 M	270 miles
8.18	Manholes	Yes	No	Not Applicable	
8.19	Vaults	Yes	Yes	\$4 M	2

Section	Asset	Preventative Maintenance Programs (Yes/No)	----- Proactive Replacement Program -----		
			Yes/No	2021-2025 Investment (\$M)	2021-2025 Number of Units
8.20	Advanced Metering Infrastructure	Yes	Yes	\$1 M	4,928

8.1 Substation power transformers

DTEE has approximately 1,600 substation power transformers, which connect the transmission system (120kV) to the subtransmission system (40kV and 24kV) and to the distribution system (4.8kV, 8.3kV, and 13.2kV) and also connect the subtransmission system (40kV and 24kV) to the distribution system (4.8kV, 8.3kV, and 13.2kV). Approximately 31% of the transformers have a high side voltage of 120kV. Exhibit 8.1.1 shows a substation power transformer. Exhibit 8.1.2 shows the age distribution of substation power transformers connected to the DTEE system. The average age of substation power transformers is approximately 42.5 years. Exhibit 8.1.3 shows the historical substation power transformer failures.

Exhibit 8.1.1 Typical Substation Power Transformer



Exhibit 8.1.2 Substation Power Transformer Age Distribution

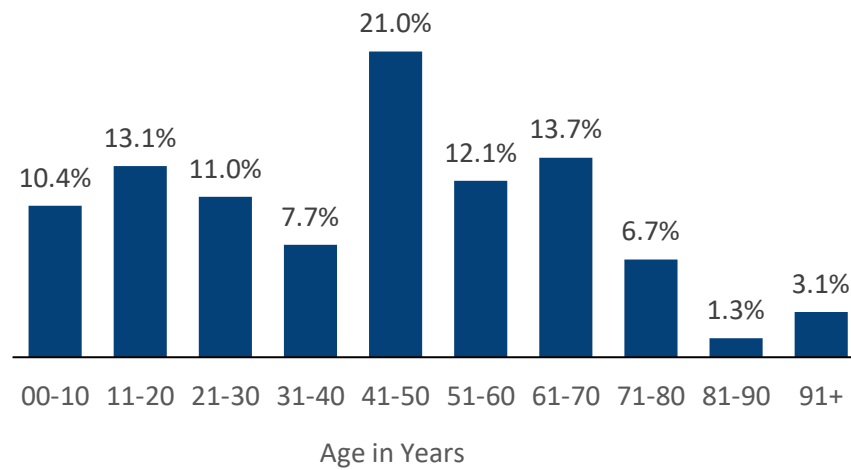
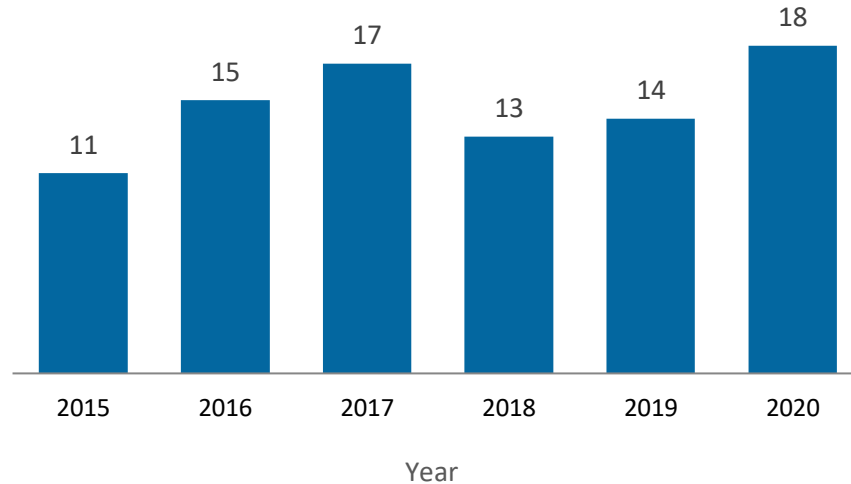


Exhibit 8.1.3 Substation Power Transformer Failures



The average number of substation power transformer failures over the past five years is over 15 per year – this includes units that were identified and proactively replaced prior to actual failure based on dissolved gas analysis (DGA) results. Bushings, load tap changers, and winding insulation breakdown account for approximately 65% of the failures. The transformer average age at time of failure is around 48 years.

Failures of substation power transformers can cause outages on multiple circuits simultaneously and can reduce system redundancy, effectively increasing outage risk to customers for extended periods of time, sometimes impacting thousands of customers. Based on the factors shown in Exhibit 8.1.4, approximately 15% of the substation power transformers are currently classified as candidates for replacement. These transformers are typically over 70 years old, have abnormal DGA ratios, and are water cooled with a potential active water leak into the unit.

Exhibit 8.1.4 Substation Power Transformers Replacement Factors

Factors	Impact
DGA results indicating gassing in the main tank	• Indication of early stages of failure
Water cooled transformers	• High costs to maintain water cooling

	• Greater chance of failures due to water leaking into the unit
High number of through faults	• High mechanical and electrical stress on the transformer core and winding insulation
70+ years old	• Deterioration of winding insulation • Uneconomical to repair
Chronic leaking of oil	• Environmental concern

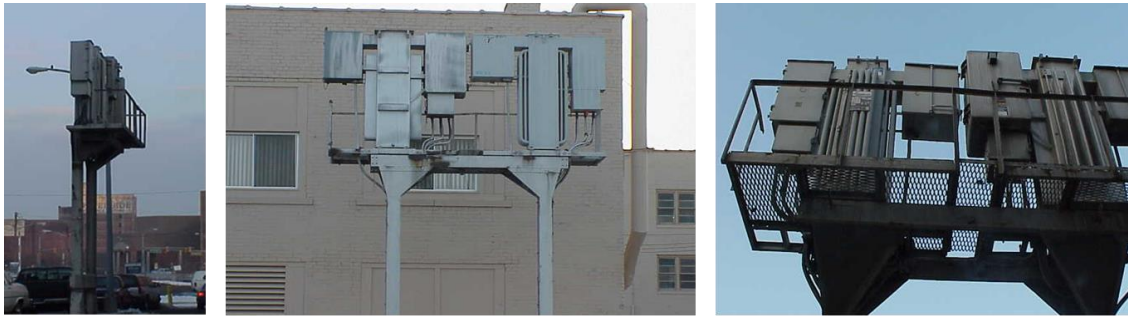
DTEE conducts regular inspections and DGA testing as part of our preventative maintenance (PM) plan on all substation power transformers once per year. DGA testing is an industry-proven Substation Predictive Maintenance Inspection (SPdM) methodology to identify internal transformer degradation and to predict incipient transformer failures. Based on results from routine inspection and DGA testing, DTEE determines when to either proactively replace or repair substation power transformers that do not pass inspection. Substation power transformers are sometimes also replaced reactively when they fail during service and are deemed uneconomical to repair. Preventative maintenance on ancillary components of power transformers such as load tap changers or bushings also takes place.

8.2 Network banks (Netbanks)

Netbanks are used to describe an installation consisting of specialized transformers, network protectors, and primary disconnect switches. A netbank system is recognized as the industry standard for the highest degree of continuity of service in heavy load-density city areas.

DTEE has netbank systems in the core downtown areas of Detroit and Ann Arbor as well as Mount Clemens and Port Huron. There is a total of 473 netbank transformers operating on the distribution system - the majority (90%) are located in Detroit. Netbank transformers are typically mounted on steel skids, steel columns, or wood poles, with a few located in underground vaults. The netbank wood pole supporting structures have an average age of 64 years, while the average age of steel supporting structures is 82 years. The netbank transformers have an average age of 46 years.

Exhibit 8.2.1 Netbank System Mounted on Steel Columns



DTEE conducts Electrical Integrity Inspections (EII) on a five-year cycle on netbank systems as part of the preventive maintenance program. It includes inspection of the primary disconnect and protector, visual (inside / outside) inspection of transformer tanks for corrosion, signs of water and oil sludge, PCB (Poly Chlorinated Biphenyl) and DGA oil analysis, and high voltage testing on the primary and secondary windings of the transformer.

The structural inspection of steel columns that support the netbanks is conducted to determine the replacement and repair needs based on the steel condition. In 2015 and 2016, a total of 115 structures were inspected. Of those, eight required replacements and 25 required repairs. Based on these results, DTEE expanded inspection efforts and completed inspections of the remaining 337 structures in 2017-2019, from which four required replacements and five required repairs. Netbanks found to require repair or replacement from 2015-2018, have been completed. Those found in 2019, will be completed by end of 2021.

As mentioned above, some netbanks are mounted on wood poles. Inspection of these poles is conducted as part of the pole inspection and replacement program, as discussed in Section 9.1.1 – Pole and Pole Top Maintenance and Modernization.

Ultimately, the entire netbank system will be replaced as part of the 4.8kV conversion and consolidation program with a 13.2kV netbank system, as discussed in Section 11.4 – CODI Program.

8.3 Circuit breakers

A circuit breaker is an electrical switch designed to isolate faults that occur on substation equipment, buses, or circuit positions. Its basic function is to interrupt current flow after a fault is detected to

minimize equipment damage due to high fault currents and to isolate the faulted asset from the electrical system. Exhibit 8.3.1 illustrates an oil breaker installed in the 1940s versus a recently installed vacuum breaker.

Exhibit 8.3.1 Circuit Breakers



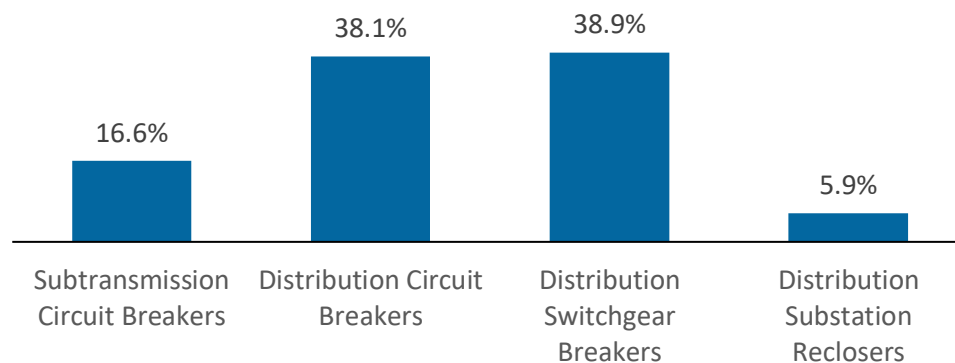
Traditional 120 kV Oil Breakers



Modern 120 kV Gas Breakers

DTEE has approximately 6,000 circuit breakers on its system, which include subtransmission circuit breakers, distribution circuit breakers, distribution switchgear breakers, and distribution substation reclosers (Exhibit 8.3.2).

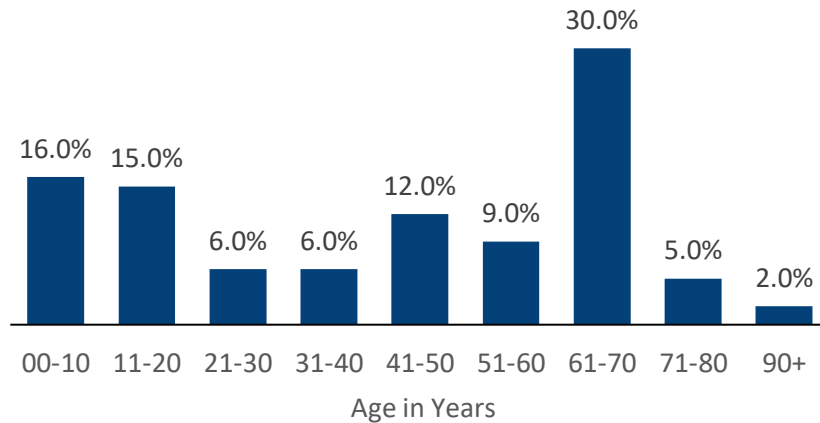
Exhibit 8.3.2 Circuit Breakers by Application



The life expectancy of a circuit breaker is approximately 40 years for early to middle 20th century equipment, while modern equipment has a life expectancy of 30 years and is designed for ease of

maintenance and replacement. Exhibit 8.3.3 shows that approximately 60% of DTEE's circuit breakers are beyond their life expectancy.

Exhibit 8.3.3 Circuit Breaker Age Distribution



Of the approximately 6,000 circuit breakers, approximately 58% are candidates for replacement based on the latest assessment shown in Exhibit 8.3.4. All oil filled breakers are considered for replacement due to a combination of multiple risk factors associated with that equipment.

Exhibit 8.3.4 Circuit Breaker Replacement Factors

Factors	Impact
Interrupting Medium	Specific interrupting mediums have known issues: - Oil breakers are an environmental concern due to possible leaks - Air magnetic breakers are prone to failure due issues extinguishing arcs within the breaker
High O&M Costs	High O&M costs due to more frequent repairs required, longer time to troubleshoot issues and/or shorter maintenance cycles
Unavailable Parts	Little or no vendor support and expensive to replace parts.
Known Performance Issues	Known higher failure rates or frequent failure to operate

Circuit breakers that are candidates for replacement could degrade the reliability and operability of the subtransmission and distribution systems. A failure of a circuit breaker can cause outages on multiple circuits and could reduce system redundancy for an extended period during repairs. Depending on the extent of the failure and possible adjacent collateral damage, thousands of customers could be impacted for an extended duration. Exhibit 8.3.5 shows failures in the last five years, as categorized by failure to operate, in-service failure, and failure during preventative maintenance activity. Based on the type of failure, repairs may be possible instead of replacement, but this could lead to customer outages with long outage durations. When replacement is required or there is a major in-service failure, there could be extended outages that would require the system to operate in an abnormal condition for an extended period.

Exhibit 8.3.5 Breaker Failures by Year

Failure Type	2016	2017	2018	2019	2020
Failures to Operate – Trip (repair)	6	17	16	18	11
Failures to Operate – Close (repair)	36	44	65	83	63
Other Failures in Service (repair)	22	12	29	36	25
PM Failures (replace)	30	19	20	7	8
Other Failures in Service (replace)	12	8	24	17	6
Total # of Failures	106	100	154	161	113

DTEE performs regular periodic inspections of circuit breakers based either on a time interval or the number of breaker fault operations. Circuit breakers that do not pass inspection are either repaired (if economical to do so) or replaced. Over the last three years, an average of approximately 19 breakers fail inspection annually.

As stated above, there is a large population (approximately 60%) of breakers beyond their life expectancy. Failures of these breakers not only increase reactive costs but pose system risks. DTEE has a proactive capital replacement program to replace breakers that is further discussed in Section 9.1.5 - Breakers.

8.4 Subtransmission disconnect switches

Subtransmission disconnect switches are used to manually sectionalize and provide isolation points on the electrical system for operational reasons or for service / maintenance. DTEE has approximately 3,600 disconnect switches that operate at 24kV, 40kV, and 120kV.

Failures of disconnect switches during service are infrequent and do not usually lead to customer outages. However, failures of disconnect switches during operation, when operators attempt to open or close a disconnect manually, can lead to safety hazards, reduce system operability, and force additional equipment to be taken out of service to allow critical work to continue. Approximately 132 Cap & Pin style (Exhibit 8.4.1), 58 PM-40 model (Exhibit 8.4.2) and 22 Westinghouse Type R (Exhibit 8.4.3) disconnect switches are candidates for replacement, with replacement factors detailed in Exhibit 8.4.4.

Exhibit 8.4.1 Cap & Pin Disconnect Switch



Exhibit 8.4.2 PM-40 Disconnect Switch



Exhibit 8.4.3 Westinghouse Type R Disconnect Switch

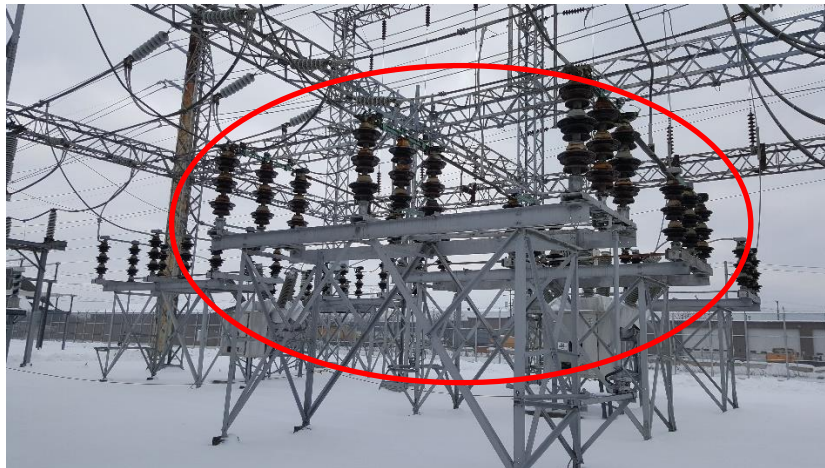


Exhibit 8.4.4 Subtransmission Disconnect Switch Replacement Factors

Factors	Impact
Cap & Pin Disconnects	- Insulators are problematic and could fail during operation - Employee safety concern upon failure during manual operation
Known Manufacturer Issues on PM-40 (120kV)	- Equipment could fail during operation due to bearing problems - Employee safety hazard
Westinghouse Type R	- PM inspections are revealing failed insulators, which could flash over during a fault condition. - Employee safety hazard - Disconnects are obsolete, up to 65 years old and in poor operating condition.

DTEE has a preventive maintenance program for subtransmission disconnect switches. In addition, DTEE has a proactive capital replacement program to replace the Cap & Pin, PM-40 and Westinghouse Type R disconnects that will be further discussed in Section 9.1.3 – Subtransmission Disconnect Switch.

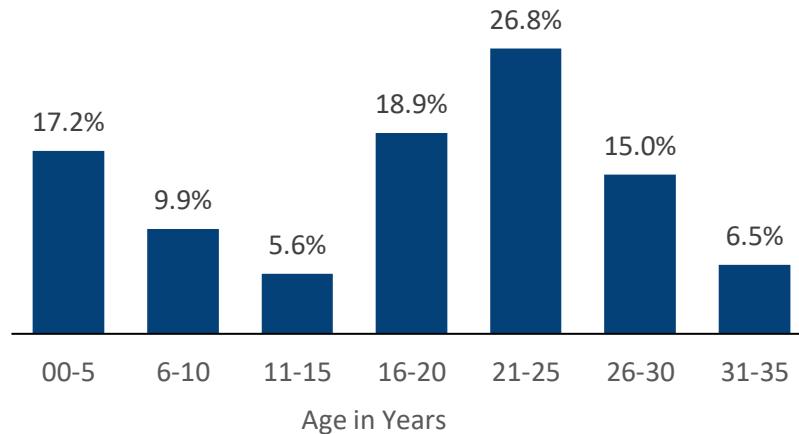
8.5 Circuit switchers

DTEE has approximately 354 circuit switchers, which connect the transmission system (120kV) and the subtransmission system (40kV) to the primary side of a substation power transformer. The purpose of the circuit switchers is to protect substation equipment from damage caused by excess fault current. It is used as a smaller and less expensive alternative to a circuit breaker when possible. The interrupting capability of a circuit switcher is not as high as a circuit breaker, so it cannot be used in all instances. Exhibit 8.5.1 shows two examples of circuit switchers. Exhibit 8.5.2 shows the age distribution of circuit switchers with the average age being 18 years old.

Exhibit 8.5.1 Circuit Switchers



Exhibit 8.5.2 Circuit Switcher Age Distribution



Based on the factors shown in Exhibit 8.5.3, approximately 22% of circuit switchers are classified as candidates for replacement. Failures of circuit switchers to trip when necessary can lead to system outages and damage to other critical equipment such as transformers. These circuit switchers will either undergo full replacement, component replacement, or inspection for further analysis.

Exhibit 8.5.3 Circuit Switcher Replacement Factors

Factors	Impact
Known failure modes	Insulating support columns manufactured prior to 1999 have a gasket that can fail, leading the columns to leak. This has the potential to cause the operating rods to seize resulting in failure to trip. All leaking units manufactured prior to 1999 will be replaced.
Potentially faulty components	The manufacturer has identified several components that may be faulty, based on manufactured year. Faulty parts could lead to failure to trip. Components will be tested and/or replaced as applicable.
Undersized units	These units do not have the interrupting capability to operate as designed. Delayed tripping is implemented until the proper rated switcher is installed.

Circuit switchers undergo periodic inspections after five automatic operations or when infrared survey programs produce unfavorable results. In addition, DTEE has a proactive capital replacement program to replace identified circuit switchers that is further discussed in Section 9.1.4 – Circuit Switcher.

8.6 Relays

The function of relays is to monitor system current and voltage, detect abnormal conditions (primarily fault current) and initiate breaker operations to isolate abnormalities.

DTEE relays operate on the transmission, subtransmission and distribution systems. There are three types of relays: electro-mechanical, solid state, and micro-processor. The majority (79%) of the approximately 27,000 relays are electro-mechanical relays as shown in Exhibit 8.6.1 and described in Exhibit 8.6.2.

Exhibit 8.6.1 Relays Design Types



Electro Mechanical



Solid State



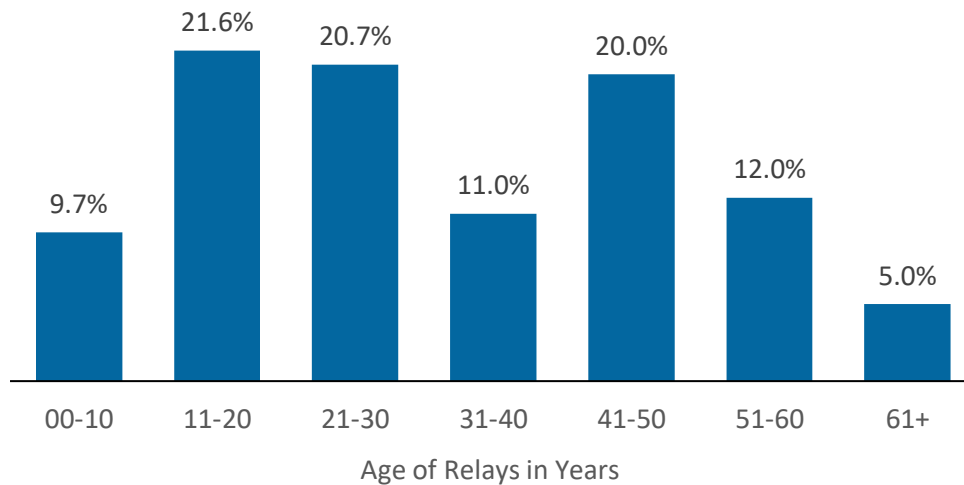
Micro-processor

Exhibit 8.6.2 Relay Design Type Benefits and Drawbacks

Relay Type	Life span	Benefits	Drawback
Electro-mechanical (79% of Total)	50 years	- Contain no electronics - Long life span - Low maintenance - Settings can be easily adjusted	- No SCADA or communication - No fault location detection - No metering or power flow monitoring - Unavailable parts
Solid state (2% of Total)	40 years	- Settings can be easily adjusted - No moving parts – mechanism does not wear out	- Obsolete technology - No SCADA or communication - No fault detection - No metering or power flow monitoring - Unavailable parts
Micro-processor (19% of Total)	15 – 20 years	- Provide SCADA, fault detection and metering functions - Replace 5+ electro-mechanical relay functions - Require limited preventive maintenance due to self-checking capability - Provide condition-based information for breakers and transformers - Relay settings can be applied remotely	- Relatively short life span - Early models have power supply issues and have limited capabilities as compared to newer models

The average age of DTEE relays is 33 years, with the age distribution illustrated in Exhibit 8.6.3. A relay failure can lead to loss of ability to return equipment to service. It may also result in a larger outage (e.g., a substation bus gets de-energized due to a distribution circuit relay failing to operate) or possibly result in damage to other control or power equipment.

Exhibit 8.6.3 Relay Age Distribution



Approximately 34% of the relays are classified as candidates for replacement based on factors listed in Exhibit 8.6.4.

Exhibit 8.6.4 Relay Replacement Factors

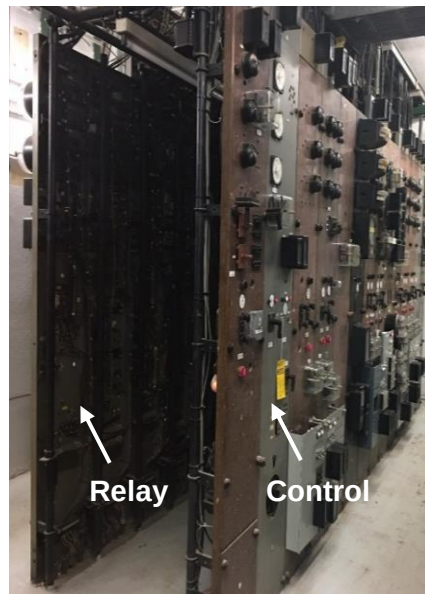
Factors	Impact
Known high impact failure modes	Poor operational performance that has led to customer outages Significant amount of time erroneously troubleshooting power equipment to find cause of fault indicated by the relay
Known high failure rate reclosing relays	Unnecessary sustained customer interruptions due to the inability of breakers to close automatically
Microprocessor relay 15+ Years (Distribution) and 20+ Years (Subtransmission)	Modern relays have a shorter life span and start failing after 15 years
Solid State 40+ Years	Solid state electronic components begin to fail with age Repetitive DC surges cause failures over time
Electromechanical Relays 50+ Years	Mechanical stress wears relay triggers (springs) out over time Electrolytic capacitors internal to relay dry out over time – causing non-correctable calibration issues Unavailable or expensive replacement parts

DTEE has a preventive maintenance program for relays. Relays found non-operational during maintenance or in service are replaced or repaired. Relay replacements are generally coordinated and included as part of the breaker or transformer replacements. In addition, DTEE plans to replace end-of-the-life marble relay panels (as shown in Exhibit 8.6.6). These marble panels exist at four stations and do not allow for replacing a single relay panel with a single breaker replacement in their current physical configuration. The schedule for these replacements is shown in Exhibit 8.6.5, and further details are included in Section 9.2.3. Additionally, as a part of the Substation Automation Program, electromechanical relays will be replaced with microprocessor relays. Today, roughly 32% of general-purpose substations in DTEE's territory have SCADA monitoring and control. Not all substations are good candidates for substation automation upgrades, but those that can support it will receive the upgrades which will include relay replacements. More details regarding how substations are prioritized for automation upgrades can be found in Section 12.5 – Distribution Automation.

Exhibit 8.6.5 Relay Panel Replacements as part of Station Upgrade Projects

Station	Years
Warren	Completion 2021
Northeast	2021-2023
Navarre	2023-2025
Lincoln	2022-2024

Exhibit 8.6.6 Relay Panels



**Old Marble Relay Panel
(Separate panels for relay
and control functions)**



**Modern Relay Panel
(Single panel for relay
and control functions)**

8.7 Switchgear

Switchgear is used at substations to house and protect from weather a combination of many equipment types including circuit breakers, power bus, relays, metering, SCADA control and communication support. The components within in a switchgear are housed in metal clad compartments / positions. Switchgear is used to de-energize equipment to allow work to be done and to isolate faults. Exhibits 8.7.1 and 8.7.2 provide exterior and interior views of switchgear.

Exhibit 8.7.1 Switchgear Design Types



Outdoor Single Row



Across-the-Aisle



Two-Tier Switchgear

For some switchgear models, a failure in a single switchgear position can cause damage to multiple adjacent switchgear positions. This type of failure has the potential to result in the loss of power for an entire bus or substation, and under some circumstances may affect thousands of customers. This could also result in the system being in an abnormal state for an extended period of time until all the positions are repaired or replaced. Abnormal states can introduce risk to a circuit.

Exhibit 8.7.2 Switchgear Interior View



DTEE has 261 sets of switchgear on its distribution system. As illustrated in Exhibit 8.7.3, the majority (76%) of the switchgear are operating at 13.2kV. Due to their smaller configuration, many 4.8kV substations do not require the use of metalclad switchgear. As illustrated in Exhibit 8.7.4, the average age of switchgear in DTEE's system is approximately 37 years old, with more than 55% of the switchgear more than 40 years old.

Exhibit 8.7.3 Switchgear by Voltage

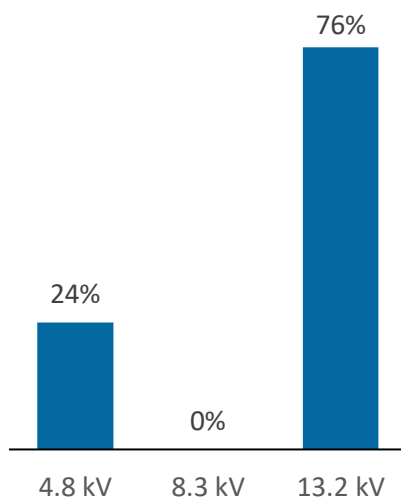
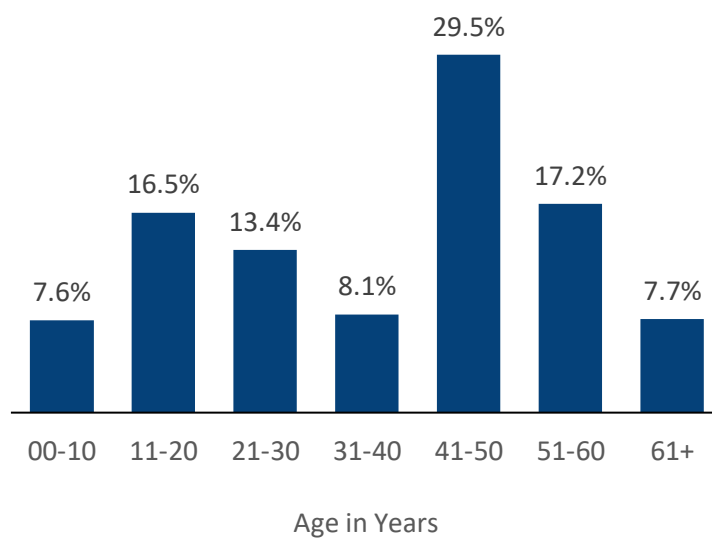


Exhibit 8.7.4 Switchgear Age Distribution



Based on the latest health assessment, approximately 27% of the switchgear sets are candidates for replacement. As shown in Exhibit 8.7.5, several risk factors are considered.

Exhibit 8.7.5 Switchgear Replacement Factors

Factors	Impact
Age	Bus insulation begins to degrade, which requires replacement to prevent bus failures. Replacing old insulation is a very expensive task and requires a lengthy shutdown of associated equipment
Calvert bus	Prone to failure due to exposure to the elements – see Exhibit 8.7.6
Design type	Some switchgear designs (outdoor single row, indoor across-the-aisle) are more likely to fail than others (two-tiered switchgear) – see Exhibit 8.7.1
Racking type	Some switchgear utilizes open door breaker racking which potentially increases arc flash hazards
Spare parts	Vendors/manufacturers no longer supply parts for some older vintage switchgear
Asbestos control wiring	Some control wiring inside the switchgear contain asbestos which is a health risk
Operating issues	Odd or unique situations like one-of-a-kind breakers or certain cell configurations
Spare land	If a major failure occurs on the site, no adjacent spare land is available to install mobile fleets and help restore customers
Previous failure	If the switchgear has experienced a major failure in its past, the likelihood of another major failure increases
Number of positions	The more positions connected in a single line-up without additional throw over capabilities, the larger the impact of the failure

Exhibit 8.7.6 Calvert Bus



DTEE does not have a stand-alone preventive maintenance program for switchgear; however, the different equipment types within the switchgear such as circuit breakers, power bus, or relays have their own preventive maintenance programs.

Replacing switchgear involves an extensive substation re-build. It includes pouring new concrete pads and executing a significant amount of underground work to install cable from the transformers, capacitors and circuit exits. Equipment control wiring and testing can take several weeks to complete. While the work is in progress, a significant amount of load must be transferred to adjacent substations or circuits, and/or fed by portable substations or distributed generation. The additional loading on adjacent circuits creates operational constraints on the system. Due to the complexity and cost requirements, switchgear replacement is typically executed as part of the broader Substation Risk Reduction Program discussed in Section 9.2 – Substation Risk Program.

8.8 Poles and pole top hardware

DTEE owns more than 1 million distribution and subtransmission poles and attaches to nearly 200,000 poles owned by other utilities (e.g., AT&T). The average pole age in the DTEE system is approximately 46 years. The age distribution for DTEE owned poles is illustrated in Exhibit 8.8.1, and the distribution of non-DTEE owned poles to which we attached equipment is shown in Exhibit

8.8.2. The life expectancy of a pole is 40 years for wood pine poles and 50 years for wood cedar poles, though the actual useful life expectancy can vary based on field conditions.

Exhibit 8.8.1 DTEE Owned Wood Pole Age Distribution

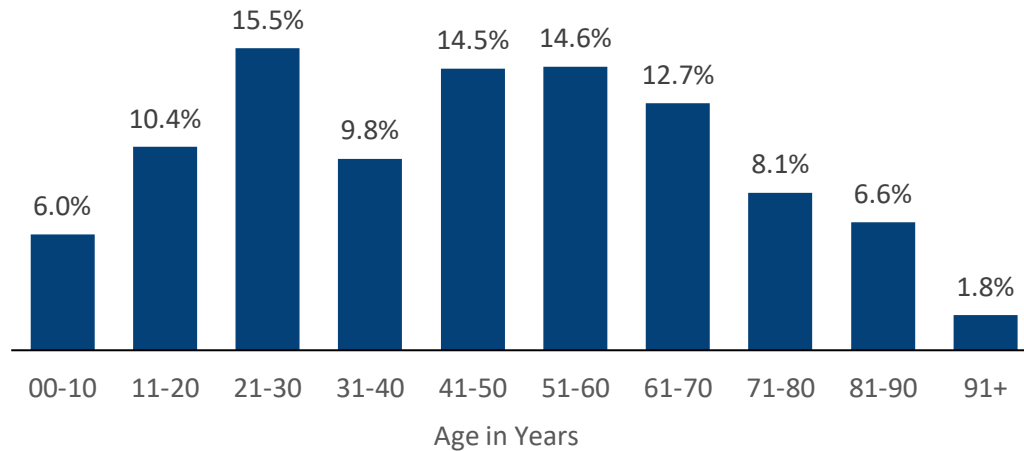


Exhibit 8.8.2 Non-DTEE Owned Wood Pole Age Distribution

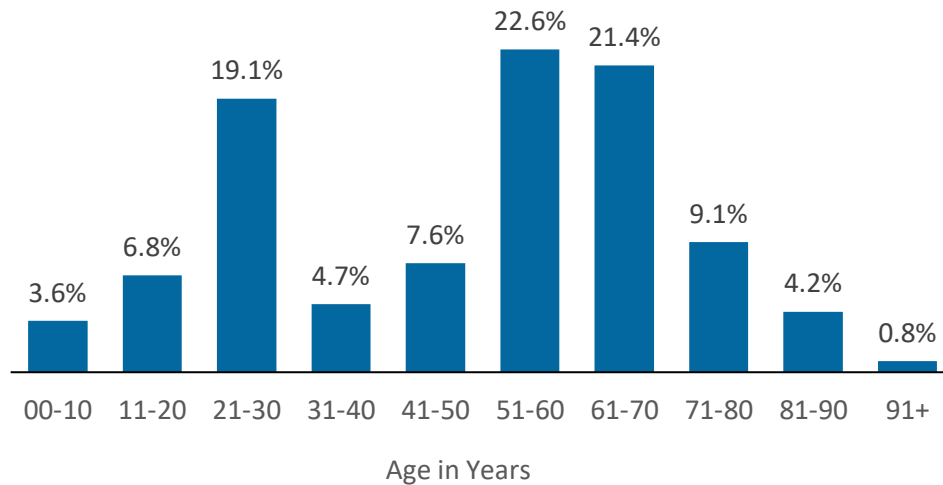
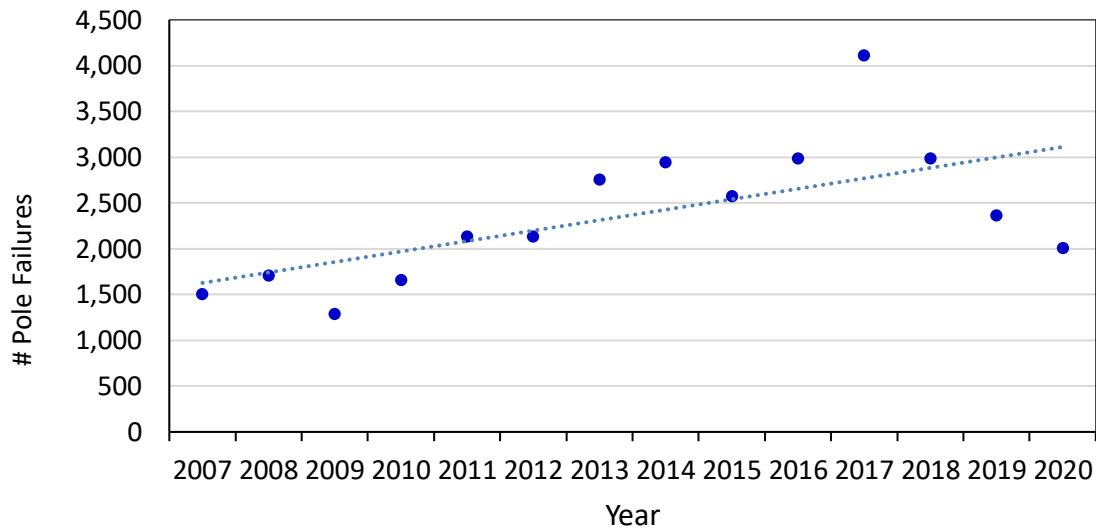


Exhibit 8.8.3 shows the increasing trend of broken poles. A broken pole does not always result in customer outages, in some cases the pole may be fractured or damaged, but the integrity of the conductors is maintained.

Exhibit 8.8.3 Annual Broken Pole Events (Failures during service)



Not all broken poles are the result of aging. A portion of them are due to vehicle strikes, trees, icing or wind loads above the design standard, etc. However, to identify and replace or remediate poles with deteriorating strength, DTEE conducts pole testing as part of the Pole and Pole Top Maintenance and Modernization Program (PTMM), as discussed in detail in Section 9.1.1. Non-DTEE owned poles that have DTEE assets attached are also visually inspected as part of the PTMM program.

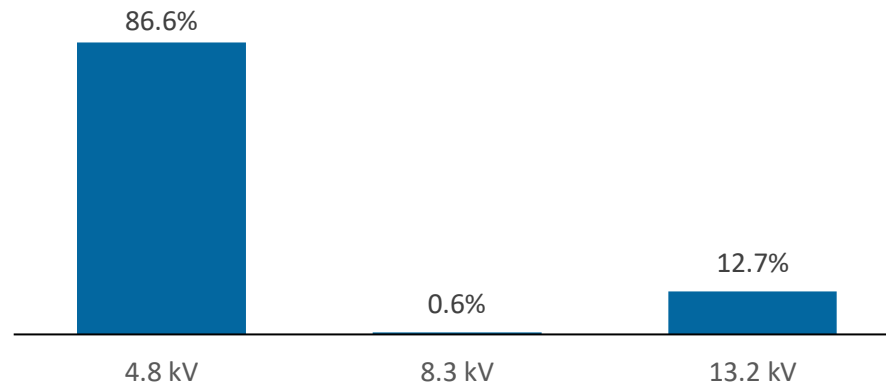
Pole top hardware is also inspected as part of the PTMM program. That hardware includes the following: cutouts, slack span sleeves, Blackburn hot taps, ground wires, arresters, cross-arm parts, transformer parts, insulators, primary line sag, guy wire components, spacer blocks, neutrals on secondary taps and secondary equipment.

DTEE is increasing investment in PTMM for multiple reasons. As shown earlier in this section, the average age of poles on the system is at or approaching life expectancy. Pole failures, as well as pole top equipment failures, lead to costly reactive work. The PTMM program, which is based on measurement of pole strength or observation of pole top equipment that is at risk for failure, allows for a much more efficient, planned replacement of assets that will be prone to failure in the near future. DTEE is also increasing investments in PTMM based on benchmarking and learnings from other key programs. The investments in this program will improve customer reliability and help manage emergent costs.

8.9 Small wire

Small wire is identified by the wire gauge, in this case #8, #6, and #4. The overhead distribution primary system has approximately 3,600 miles of small wire, representing 17% of the total overhead primary wire miles. Exhibit 8.9.1 shows the distribution of wire between the three operating voltages.

Exhibit 8.9.1 Small Wire by Voltage

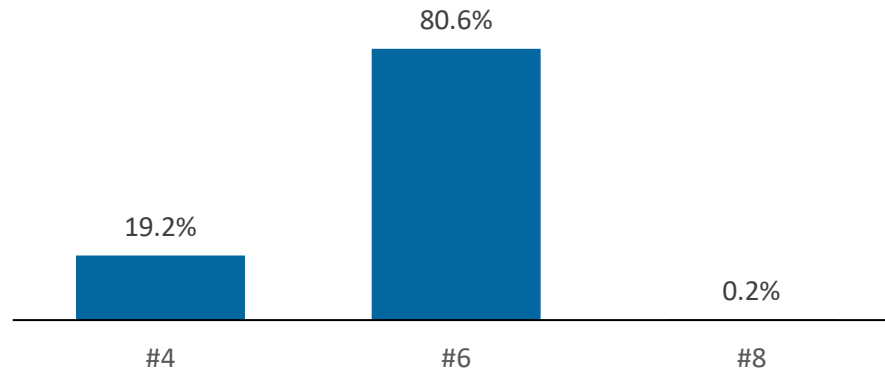


These wire sizes are no longer used for new installations, except some like-to-like replacements during emergent conditions (e.g., trouble or storm). Small wire is of concern because of its weaker mechanical strength.

To improve reliability, DTEE has upgraded standard sizes of conductors that are installed on all planned work. The new wire specifications provide strengths that are two to three times stronger than small wire sizes.

There is no stand-alone proactive replacement program for small wire currently. However, small wire replacement is performed during other capital work to support load growth, conversion and consolidation, or reliability improvements.

Exhibit 8.9.2 Small Wire by Size



Note: Small wire presents 17% of DTEE wire miles

8.10 Fuse cutouts

A fuse cutout is a combination of a fuse and a fuse carrier. This equipment provides overcurrent protection for the primary distribution system. The fuse opens (blows) when excessive current is produced by abnormal conditions such as line faults caused by trees, overloads, or equipment failures.

DTEE has approximately 668,000 fuse cutouts on its system. The fuse cutout replacement criteria are shown in Exhibit 8.10.1. Exhibit 8.10.2 illustrates the S&C R10/R11 Porcelain Cutouts. Over the past several years, DTEE has also made significant progress in replacing S&C Porcelain Cutouts. The population of this fuse cutout has reduced from 59,000 in 2018 to 11,000 as of today.

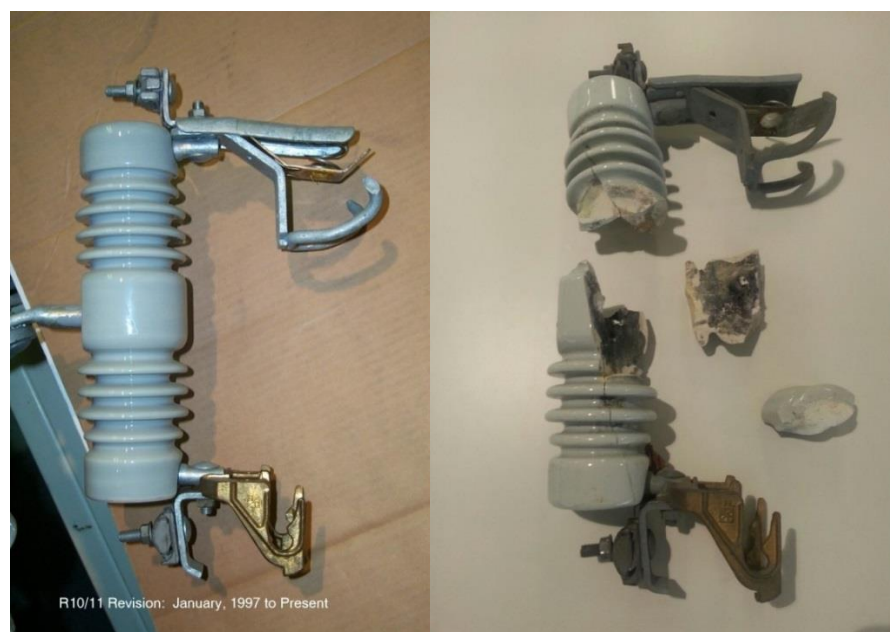
Other defective cutouts include certain vintages of AB Chance Porcelain and Durabute Polymer. Past programs replaced most of these devices.

The remaining defective cutouts will be identified and addressed through the Pole Top Maintenance and Modernization program as discussed in Section 9.1.1 – Pole and Pole Top Maintenance and Modernization.

Exhibit 8.10.1 Fuse Cutout Replacement Criteria

Type	Approximate population	Reason for replacement
S&C Porcelain R10/R11 (2005 – 2007 Vintage)	11,213	Premature failure due to latent defect
AB Chance Porcelain	21,000	Premature failure due to latent defect
Durabute Polymer	1,200	Improper operation
Total Defective Cutouts	33,413	

Exhibit 8.10.2 S&C R10/R11 Porcelain Cutout



8.11 Three-phase reclosers

An overhead three-phase recloser is a sectionalizing device that is located at key points on overhead circuits. It acts like a circuit breaker by opening under detection of high current due to a downstream fault, such as a tree branch across two phases. Reclosers localize the fault to the circuit section beyond the recloser (downstream of the recloser), leaving customers intact on the remainder of the circuit (upstream of the recloser). Unlike a fuse that will open and stay open, a recloser is designed to automatically attempt to reclose several times in order to restore the downstream customers if the fault has cleared. The open and reclose cycle allows a temporary fault, such as a small falling tree limb, to clear from the circuit and restores power to customers with only a momentary interruption. In the case of a sustained fault, the recloser will eventually remain open and isolate the fault from the rest of the circuit.

Reclosers with SCADA control capability can be used for automatic restoration in distribution system loop schemes. A typical loop scheme involves at least two separate but adjacent circuits: each of the circuits has a normally closed recloser installed near the midpoint; a normally open recloser is installed at the tie point of the two circuits (as shown in Exhibit 8.11.1). The scheme automatically operates to isolate a fault that occurs between a circuit breaker and the midpoint recloser. After the breaker locks open, the midpoint recloser opens and the tie recloser closes between the two circuits. This automatically restores half of the circuit by transferring it to the adjacent circuit, reducing the number of customers experiencing an extended outage.

Exhibit 8.11.1 Typical Loop Scheme Diagram



DTEE operates approximately 1,173 overhead three-phase reclosers on its distribution system. Of the total, 93.1% are installed on the 13.2kV system and 6.9% on the 4.8kV system. Approximately 87% of the three-phase reclosers have SCADA monitoring and control capability.

Exhibit 8.11.2 Three-phase Recloser Design Types



VWE Recloser

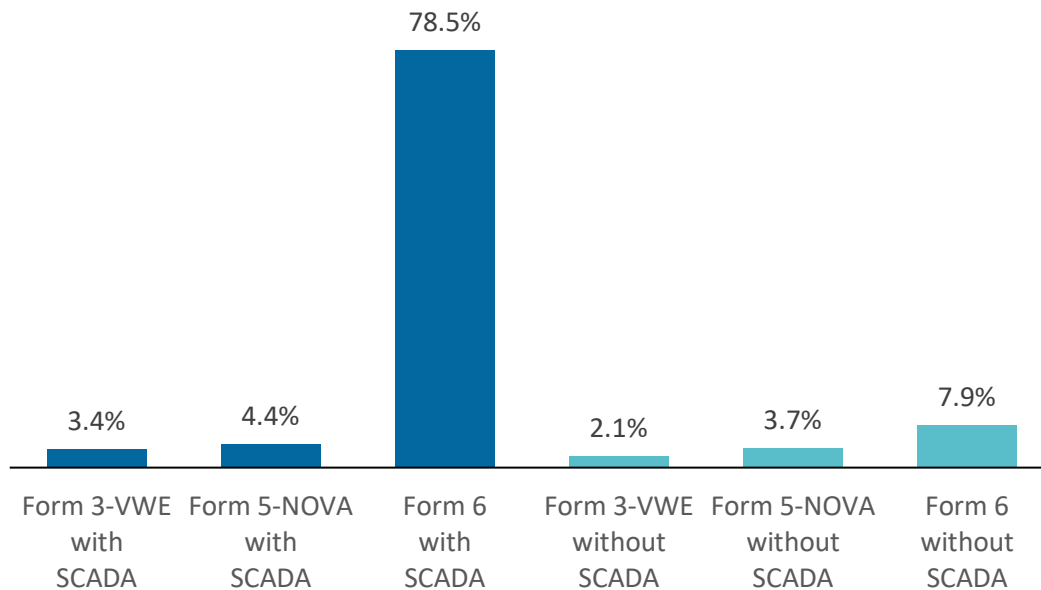


NOVA Recloser



**Triple Single Recloser &
Form 6 Control**

Exhibit 8.11.3 Three-phase Recloser Type Distribution

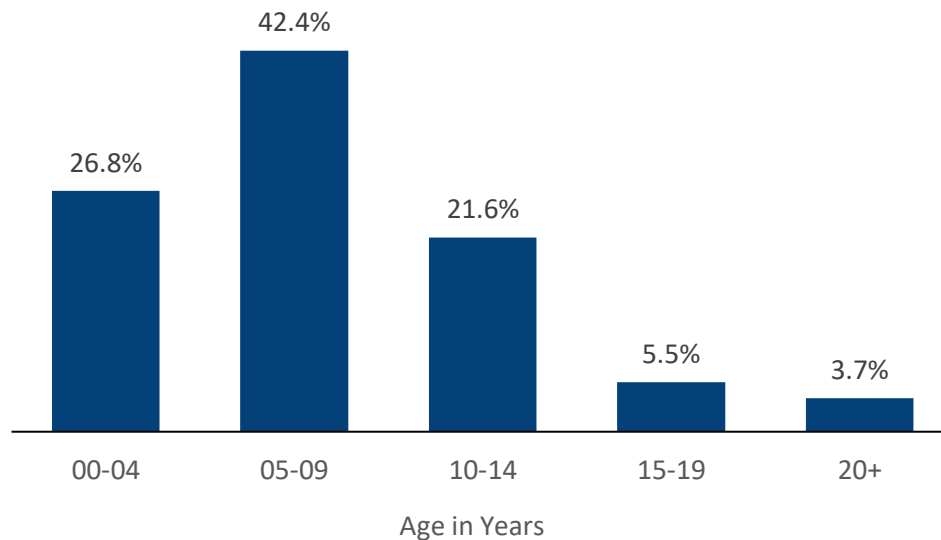


DTEE currently has Form 3-VWE, Form 5-Nova and Form 6-Triple Single reclosers on our system. The Form 3 and Form 5 reclosers operate all three phases simultaneously. DTEE now only installs Eaton Cooper Form 6 Triple Single reclosers for all new locations and all replacements. The Form

6 (Triple Single) reclosers can operate in single-phase mode as well as three-phase mode. The ability to operate single phase reduces the number of customers interrupted during an outage event. However, the single-phase mode is not used in areas that serve predominantly three-phase customers.

The average age of the three phase reclosers is approximately nine years. Exhibit 8.11.4 shows the age distribution.

Exhibit 8.11.4 Three-phase Recloser Age Distribution



Form 3-VWE and Form 5-NOVA reclosers (13.6% of the total population) are candidates for replacement, based on the factors shown in Exhibit 8.11.5. Depending on the failure mode (failed to close or open), the number of customers impacted can range between 300 and 1,500.

Exhibit 8.11.5 Three-phase Recloser Replacement Factors

Factors	Impacts
Unavailable parts	Many replacement parts for Form 3-VWE and Form 5-NOVA are unavailable; others are becoming increasingly unavailable
Known high failure rate	Form 3-VWE reclosers have seen high failures of batteries, controls and mechanical parts Form 5-NOVA reclosers have seen high failures of mechanical parts
Environmental issues	Form 3-VWE reclosers contain oil and can leak during service
SCADA capability	Form 3-VWE does not have diagnostic capability. It is expensive to retrofit SCADA on Form 3 controls

DTEE has a preventive maintenance (PM) program for overhead three-phase reclosers. Reclosers not passing PM inspection or failing in service are replaced or repaired.

8.12 4.8 kV & 13.2 kV SCADA distribution pole top switches

A complete SCADA pole top switch (PTS) includes a switch and a SCADA control box that provides an interface between the switch and the Energy Management System (EMS) master-station computer.

A distribution SCADA PTS allows the System Operations Center (SOC) to quickly isolate faults and restore customers remotely. Additional switches permit the circuit to be divided remotely, so load can be transferred to other circuits in the event of an outage.

DTEE operates 289 SCADA PTS's. There are two types of SCADA PTS's: S&C SCADA Mate switches and Bridges Auto Topper switches (Exhibit 8.12.1).

Exhibit 8.12.1 SCADA Pole Top Switch Types



S&C SCADA Mate switches



Bridges Auto Topper switches

Exhibit 8.12.2 SCADA Pole Top Switch Type Distribution

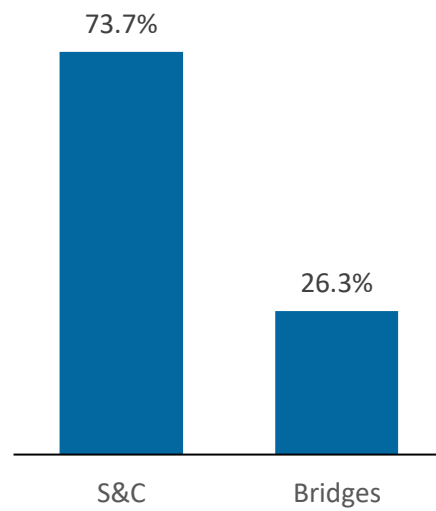
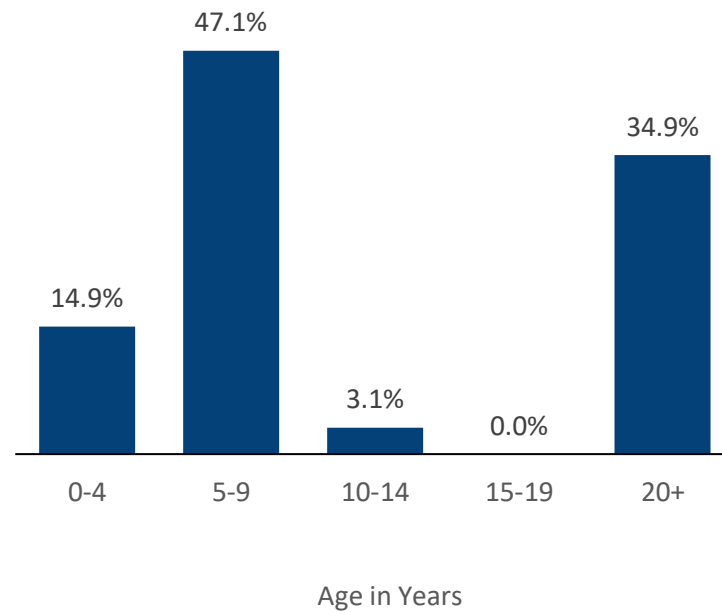


Exhibit 8.12.3 SCADA Pole Top Switch Age Distribution



The average age of all SCADA pole top switches is approximately 13 years. The Bridges switches average age is more than 25 years. Historically, Bridges switches have had a significantly higher failure rate. From 2017 – 2019, Bridges switches failed 21.9% of Preventative Maintenance inspections versus S&C failing 9% of inspections. Because of the factors in Exhibit 8.12.4, all Bridges SCADA PTS's (263% of the SCADA PTS population) are candidates for potential replacement.

Exhibit 8.12.4 SCADA Pole Top Switches Replacement Factors

Factors	Impacts
Switch design	Bridges switches have an open-air switch design vulnerable to adverse weather conditions, motor–timer failures, control board failures and Lindsey sensor failures
RTU control box design	Bridges switches have RTU control boxes with carbon steel cabinets, which are prone to rusting

When SCADA pole top switches fail to open or fail to close, remote restoration is unavailable and the switch must be opened manually by field personnel, resulting in longer restoration times. DTEE

has a preventive maintenance program on SCADA pole top devices (Section 9.1.8). S&C SCADA units that do not pass preventive maintenance inspections or fail in service are either repaired or replaced, while failed Bridges SCADA units are generally replaced.

8.13 40kV automatic pole top switches (40kV APTS)

DTEE has approximately 146 automatic pole top switches (APTS) with control boxes on the 40kV subtransmission system. Similar to the Pole Top Switches (PTS) described above in section 8.12 – 4.8kV & 13.2kV SCADA Distribution Pole Top Switches, the function of the APTS is to sectionalize, isolate, or connect portions of the system.

The failure rate of these switches is approximately 15% over a year. A failure of one of these switches has the potential to interrupt thousands of customers or result in significant operational constraints. The entire 40kV APTS population is considered for replacement as part of the APTS capital replacement program (Section 9.1.2) to address high failure rates and unavailable spare parts for early vintages.

The exact age of each 40kV APTS is difficult to determine as parts and controls have been replaced throughout their lives; however, Exhibit 8.13.1 shows a few of the APTS on our system and Exhibit 8.13.2 shows percentage of switches on the system by vintage.

Our current standard SEECO switch is manufactured as a single integrated component by only one provider which has resulted in a reliable cost-effective component with a shorter lead time.

40kV APTS recent failure modes are shown in Exhibit 8.13.3.

DTEE is also conducting a pilot study for the SEECO switch with updated technology. The current SEECO switch can only sense voltage on one side. A pilot study is being conducted with a switch with upgraded capabilities, namely the ability to sense voltage on both sides of the switch. This will allow for automatic fault isolation and prevent a larger outage from occurring if the fault were to cause the breaker to open, resulting in fewer customers losing power. Once this modification to the SEECO switch proves compatible with the subtransmission system and the SEECO switch, it will be installed on all future locations.

DTEE has a 40kV APTS preventive maintenance program which includes both verification of the mechanical operation of the device and testing of the relay and control settings.

In addition, DTEE has a proactive capital replacement program to replace APTS's which is further discussed in Section 9.1.2 – Automatic Pole Top Switch.

Exhibit 8.13.1 40kV Automatic Pole Top Switches



S&C R9



S&C R10



SEECO



S&C Horizontal

Exhibit 8.13.2 40kV Automatic Pole Top Switches by Vintage

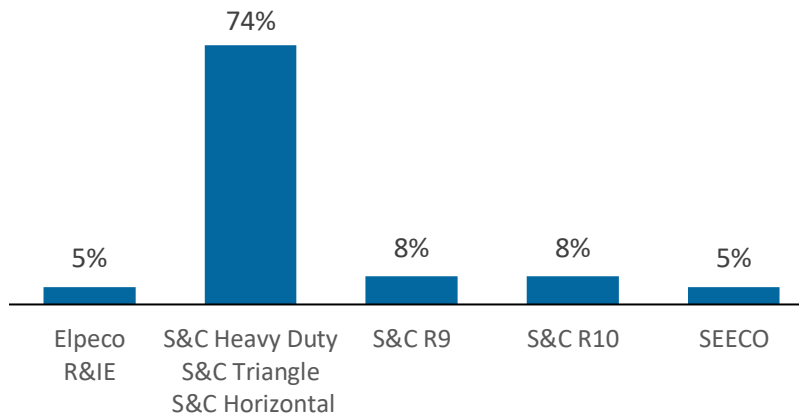
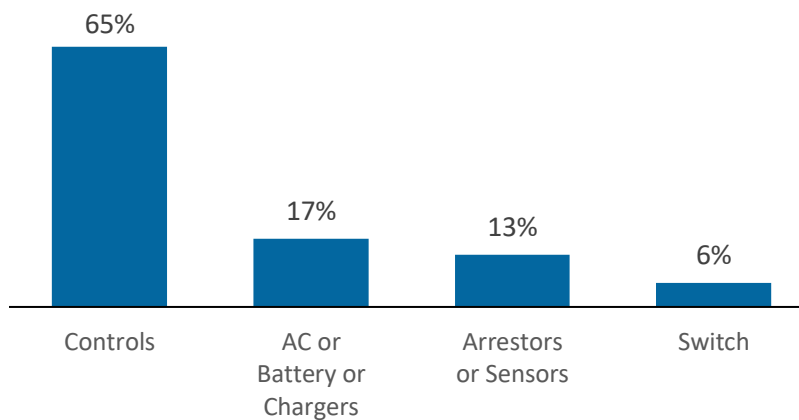


Exhibit 8.13.3 40kV Automatic Pole Top Switch Failure Modes



8.14 Overhead capacitors

Overhead capacitors are passive electronic components that provide a static source of reactive power to the distribution system. They are used to manage the reactive power and consequently voltage and losses on a distribution circuit.

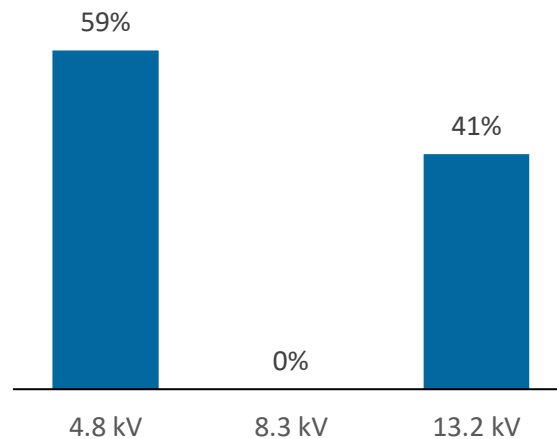
DTEE has approximately 3,160 overhead capacitor banks installed on 1,753 overhead circuits. They provide approximately 2,100 MVAR of reactive power. Not all circuits have or need capacitors. Engineering analyses are done to determine which circuits would benefit from the addition of capacitors and where those capacitors should be located on the circuit. Most capacitors were installed in the early 1990s.

Exhibit 8.14.1 Overhead Capacitor Bank



DTEE's System Operations Center (SOC) has control over capacitors through 11 different radio transmitters. These radio transmitters allow SOC to operate areas rather than individual units, however, this technology is becoming obsolete and is a candidate for replacement with updated technology. Smart capacitor control boxes, which function through the SCADA network, are in the pilot phase to take place of the radio transmitters. This would allow the SOC to control each capacitor individually. The capacitor banks will not be replaced but will be retrofitted with the new control box and sensors. There are currently 16 pilot control boxes on the system, we expect to install up to 300 per year for the next 10 years, primarily as part of the CVR/VVO program.

Exhibit 8.14.2 Overhead Capacitor Bank Voltage Distribution



Overhead capacitor failures typically impact localized power quality and line losses. If power quality is negatively affected, manual intervention may be taken in some circumstances to interrupt customers until the issues can be corrected.

DTEE has a preventive maintenance program (PM) for overhead capacitors. Reactive replacements are made for those units that fail inspection and for those that have failed in-service. The updated standard configuration using remote monitoring and control capability is used with any new installations of the capacitor banks installed due to failure, obsolescence, or capital programs such as Conservative Voltage Reduction / Volt Var Optimization (CVR/VVO). Details regarding the CVR/VVO program can be found in Section 12.6 – CVR/VVO.

8.15 Overhead regulators

Overhead regulators are used to maintain circuit voltage, particularly in situations where the load is distant from the substation. DTEE regulators adjust automatically based on the changing load to maintain circuit voltage level. Not all circuits have or need regulators. Engineering analyses are done to determine which circuits would benefit from the addition of regulators and where those regulators should be located on the circuit. DTEE has approximately 2,400 regulators on its overhead circuits.

Exhibit 8.15.1 Overhead Regulator

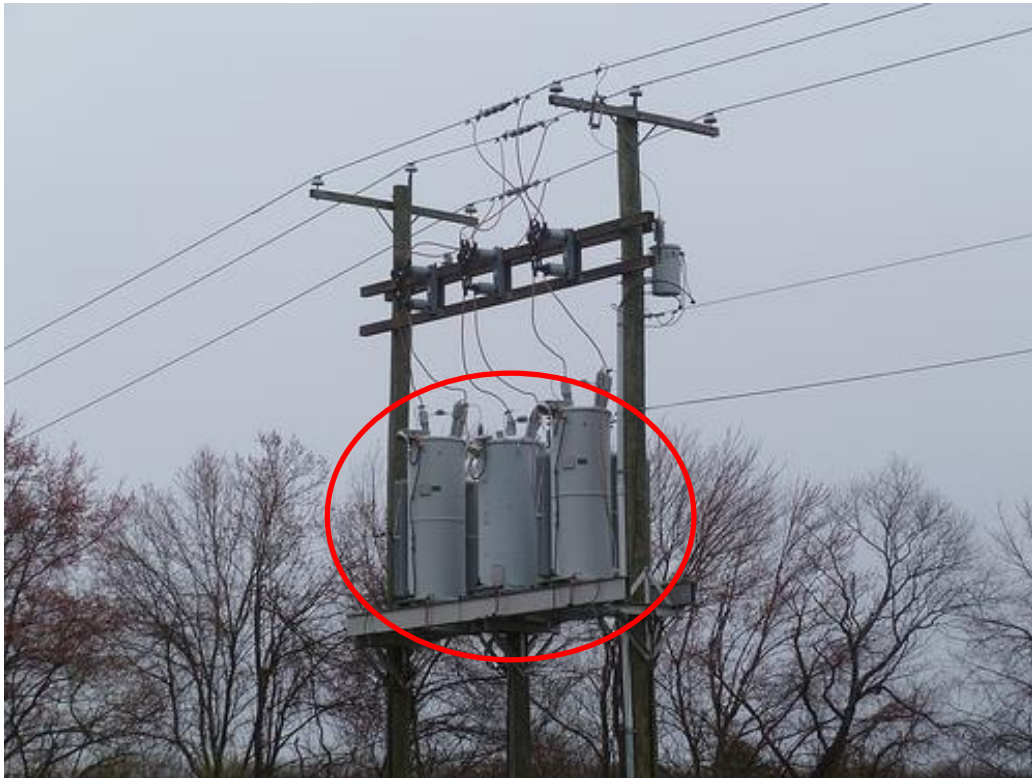
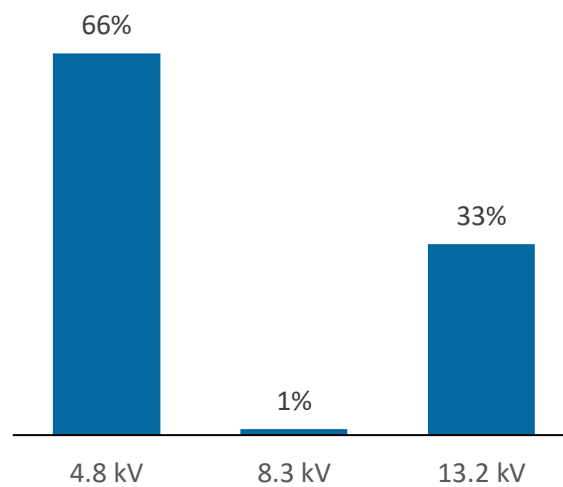


Exhibit 8.15.2 Overhead Regulator Voltage Distribution



Overhead voltage regulator failures typically do not lead to customer interruptions but may result in low voltage situations.

DTEE has a preventive maintenance program for overhead voltage regulators. Reactive replacements are made for units that do not pass PM inspection or fail in service. In addition to the PM program, remote monitoring, control, and condition-based monitoring for overhead regulators will be tested in late 2021 as part of the CVR/VVO pilot program. If the pilot results demonstrate the efficacy of the remote monitoring capabilities, new devices and devices replaced due to failure will be installed with these new control capabilities as standard. Details regarding the CVR/VVO program can be found in Section 12.6 – CVR/VVO.

8.16 System cable

DTEE's distribution and subtransmission system has over 3,200 miles of underground system cable. System cable is typically installed in conduit and requires manholes or switch-cabinets approximately every 100 to 800 feet depending on the cable type and path. Manholes and switch-cabinets provide locations where sections of cable can be pulled through the conduit and spliced together. System cable provides higher storm resiliency than overhead lines; however, the cost and time to install, repair or replace are much greater. System cable is especially useful to route multiple circuits through a small congested area (e.g., entrances and exits of a substation).

Exhibit 8.16.1 Underground System Cable Types



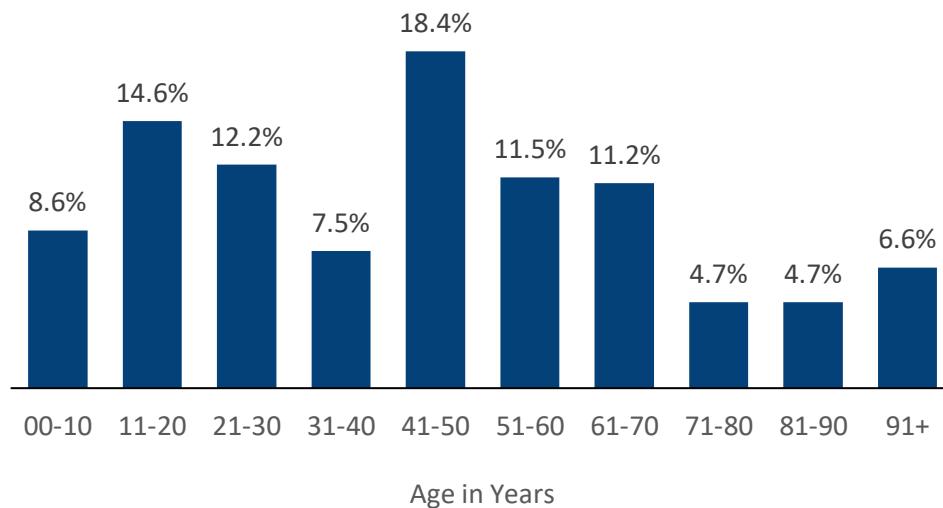
Exhibit 8.16.1 illustrates five of the six major types of underground system cable (BUYTL not pictured) installed in DTEE's distribution system. Exhibit 8.16.2 shows the types, quantities, average age, and life expectancy of system cable in the DTEE system. The average life expectancy of system cables is 40 years or less, although actual useful life varies depending on cable type and field conditions.

Exhibit 8.16.2 Underground System Cable Ages and Life Expectancy

Cable Type	PILC	EPR	VCL	Gas	XLPE Post 1990 (Tree retardant)	XLPE Pre 1990 (Non-tree retardant)	BUYTL
Miles	2,365	570	125	55	35	69	35
% of Total population	72.7%	17.5%	3.8%	1.7%	1.1%	2.1%	1.1%
Average age	50	16	59	54	20	36	52
Life Expectancy	40	35	40	40	40	25	20

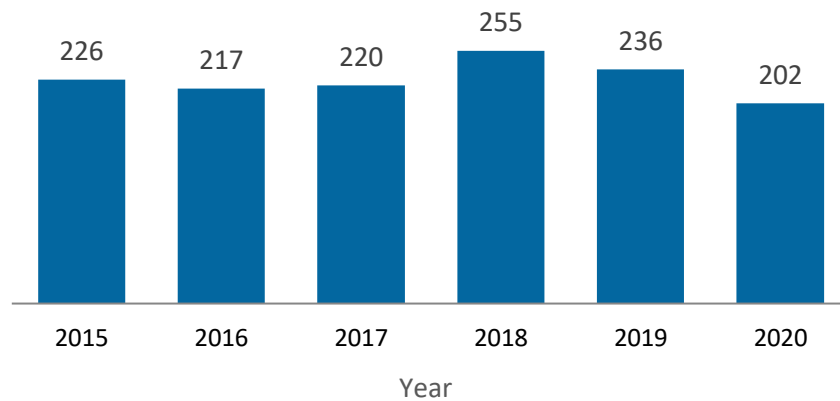
As shown in Exhibit 8.16.3, more than 57% of DTEE system cable is beyond its typical life expectancy.

Exhibit 8.16.3 Underground System Cable Age Distribution



System cable is a critical component for both the subtransmission and distribution systems. A cable failure reduces system redundancy and resiliency and can interrupt a large number of customers for an extended period of time. System cable failures over the past five years are shown in Exhibit 8.16.4.

Exhibit 8.16.4 System Cable Failures by Year



Based on the latest asset health assessment, 43% of system cable is a candidate for replacement. Replacement factors for system cable are detailed in Exhibit 8.16.5.

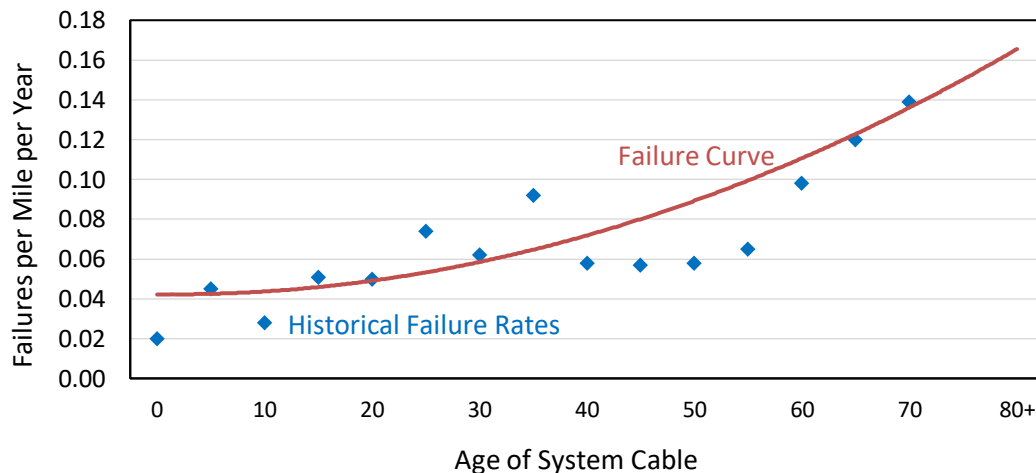
Exhibit 8.16.5 System Cable Replacement Factors

Factors	Impact
Insulation type	- Specific cable types have known issues that lead to failure: - XLPE (cross-linked polyethylene) cable manufactured before 1990 has a design defect that leads to premature insulation breakdown and dielectric failures (also called "Treeing"). The XLPE cable failure rate is approximately 0.06 failures per mile per year. - Gas (gas filled paper lead) cable has cavities within the insulating layer that are filled with gas (usually nitrogen gas) under pressure. This type of cable is prone to mechanical damage that leads to gas leaks and dielectric failures. Re-gassing this type of cable is very expensive. The Nitrogen Gas cable failure rate is approximately 0.15 failures per mile per year. - BUTYL (polyisobutylene) cable can be found mostly in the secondary networks. Under high temperatures, the cable releases gases which can cause arcing. - VCL (varnished cambric lead) cable is prone to failure under heavy loading and high temperature environment. The heavy loading or high temperature causes the insulation to degrade and lose its dielectric integrity. The VCL cable failure rate is approximately 0.08 failures per mile per year.

	PILC > 60 years old is prone to failure due to age. As age increases, failure rates increase, as shown in Exhibit 8.16.6.
Cable loading	Cables that are overloaded are prone to failure
Past failures	Cables that have experienced failures in the past are prone to another failure

Exhibit 8.16.6 shows the system cable failure rates based on age.

Exhibit 8.16.6 System Cable Failure Rates



Partial discharge testing of system cable can find defects in the cable before they result in failure and allow for proactive replacement. Partial discharges are small electrical sparks in the air-filled voids or cavities in the cable insulation. Defects in the system cable grow during over voltage conditions until it bridges the entire insulation. Partial discharge testing will catch these defects before it becomes an issue. Starting in 2021, this testing will be done on all newly installed EPR system cables to find any workmanship or cable defects before they result in premature failure. Testing will also be done on all system cable or network circuits consisting mostly of EPR or XLPE that has failed to better understand the cause of failure. Partial discharge testing does not give as accurate of results on gas, VCL, BUYTL, or PILC.

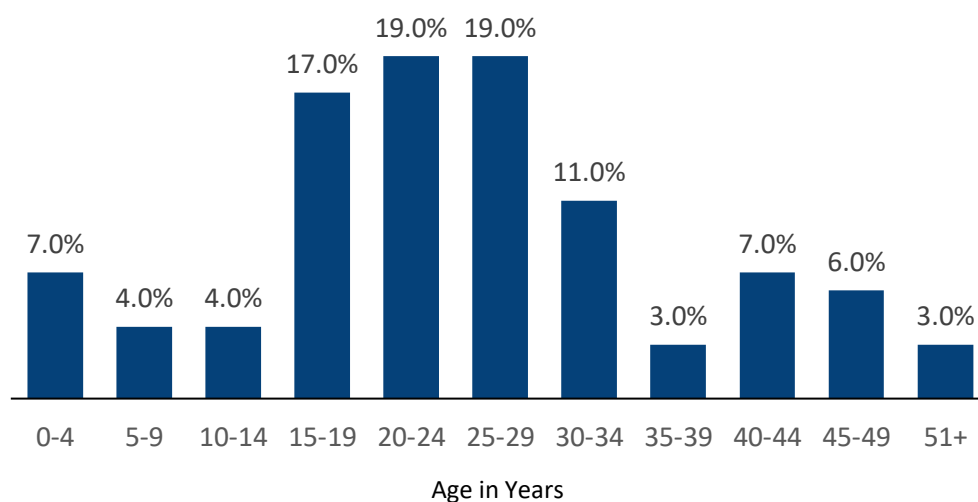
As described in this section, much of the system cable on the DTEE system has exceeded its life expectancy (approximately 57%). In additions to age, specific types of cables have known issues

that lead to failure. DTEE is increasing investments in cable replacement for two main reasons. First, replacing cable failures is more costly than doing planned cable replacements. To limit system risk during a cable failure, particularly a failure during the summer, additional measures must be taken to support loading, such as transferring load or deploying mobile generators. Many of these types of costs can be avoided if replacements are planned during non-peak times. Second, cable failures can interrupt a large number of customers for a lengthy period of time as well as reduce the level of redundancy on the system itself. In some cases, cable failures can damage adjacent equipment and lead to a larger outage, more costly repair, and additional system risk. More details about the proactive system cable program will be discussed in Section 9.1.6 – System Cable.

8.17 Underground residential distribution (URD) cable

DTEE has approximately 11,000 miles of underground residential distribution (URD) cable, with 89 % of the URD cable on the 13.2kV system. All subdivisions built since the early 1970s are served with URD cable. Buried underground, URD is not exposed to tree-related or other overhead related outages, however, it is more expensive to install and repair compared to overhead lines. Underground cable failures are less frequent but take a long time to repair or replace. DTEE is currently evaluating options to underground parts of circuits in order to improve the reliability for customers. More details on these projects can be found in Section 11.6 – Strategic Undergrounding. The average age of DTEE URD cable is 25 years. The age distribution is provided in Exhibit 8.17.1.

Exhibit 8.17.1 URD Cable Age Distribution



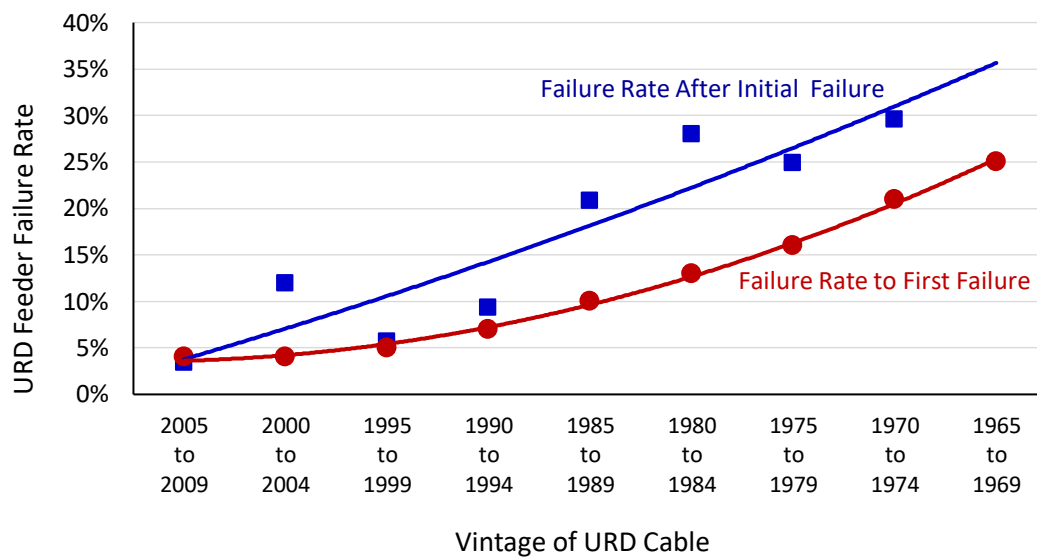
Based on the latest health assessment, 41% of URD is a candidate for replacement due to factors outlined in Exhibit 8.17.2.

Exhibit 8.17.2 URD Replacement Factors

Factors	Impact
Manufacturing year	Pre-1985, the insulation is XLPE (non-tree retardant) Post-1985, the insulation is TR-XLPE (tree retardant)
Number of outages	The rate of URD cable failures increases with the age of the cable; the rate further increases once a cable experiences its first failure as shown in Exhibit 8.17.3.

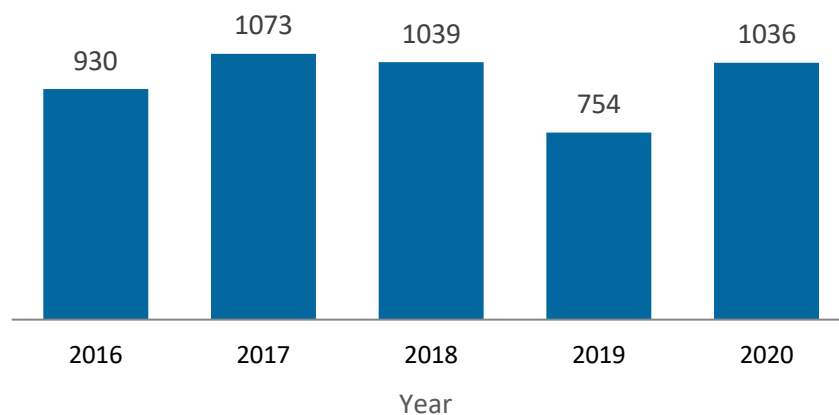
Year 1985 is a demarcation point in the manufacturing of URD cable. Pre-1985, the insulation is XLPE (non-tree retardant) and post-1985, the insulation is TR-XLPE (tree-retardant). In cable insulation, treeing refers to the tree-like pattern of insulation breakdown. The breakdown typically originates at an impurity or defect in the solid insulation and grows gradually over time to resemble the branches of a tree, ultimately leading to a cable failure.

Exhibit 8.17.3 URD Feeder Failure Rate (per mile)



As shown in Exhibit 8.17.4, for the five-year period from 2016 through 2019, there were on average nearly 1,000 URD feeder cable failures per year.

Exhibit 8.17.4 URD Cable Failures by Year



Not all URD failures lead to sustained outages; when a failure occurs on a URD in a looped configuration customers can still be fed by isolating the fault. Looped URD configurations with one failure are called open loops. When the failure is repaired the loop is returned to a normal operating configuration. Should a second failure occur prior to the open loop being repaired; customers will

experience outages and lengthy restoration times due to the nature of the underground loop. As URDs continue to fail, DTEE has seen a build-up of open loops on the system, increasing the risk of sustained customer interruptions due to a second failure. As shown in Exhibit 8.17.3, URD cables that have experienced a failure in the past are observed to have accelerated failure rates in the future.

Based on the asset health analysis described in this section, DTEE is increasing investment in proactively replacing URD to manage reactive costs and to improve reliability for customers. As shown earlier in this section, approximately 41% of cable is a candidate for replacement due to cable age or previous failures. URD failures require a lot of front-line resources to restore customers. Proactively replacing URD will help make the replacement process more efficient and cost effective. For example, a crew may be able to replace an entire loop of URD on a planned project, whereas following a failure, the crew may only replace the section that failed. A more comprehensive URD replacement program will reduce annual failures and additionally prevent a build-up of open loops on the system, which will reduce costly reactive work and reduce the risk of a lengthy outage for customers. This proactive URD cable replacement program is discussed more in Section 9.1.7 – Underground Residential Distribution (URD) Cable.

8.18 Manholes

Underground system cables are pulled and spliced inside manholes. The cables are mounted on support arms on the manhole walls to maintain separation and allow safe entry into the manholes for work, as illustrated in Exhibit 8.18.1. Failures of any part of the manhole infrastructure have the potential to impact a large number of customers because multiple cables typically run through the same manhole, and a fault on one cable can damage adjacent cables. DTEE has approximately 18,000 manholes. Their age distribution is shown in Exhibit 8.18.2.

Exhibit 8.18.1 Manhole and Cables

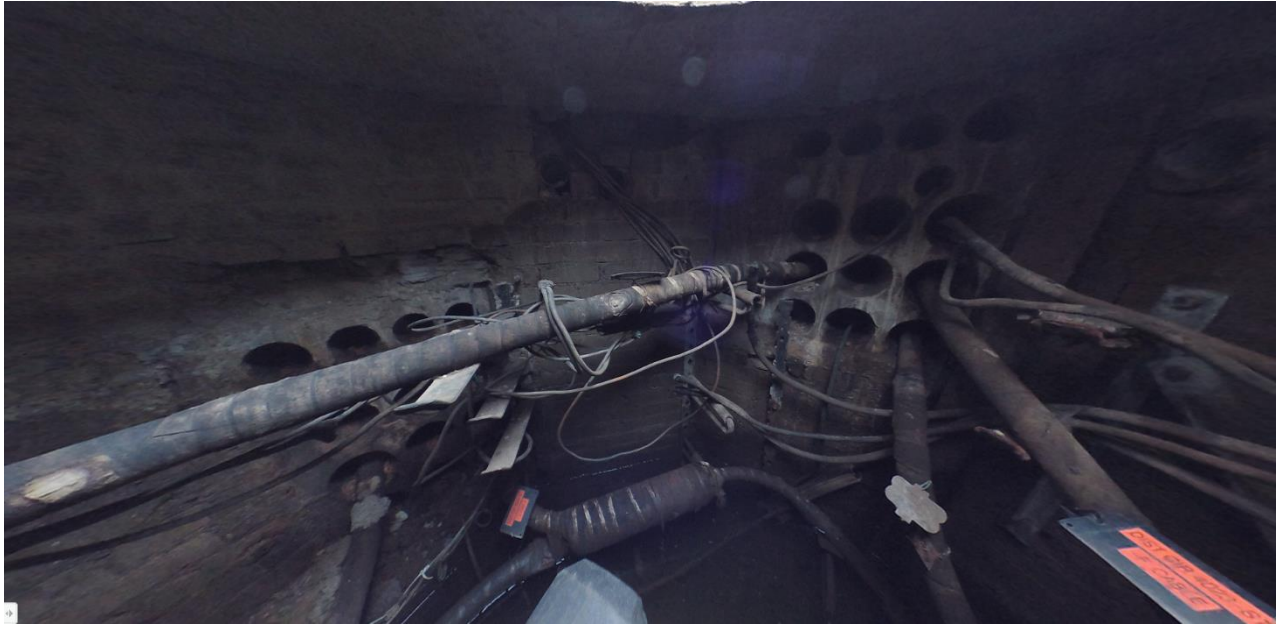
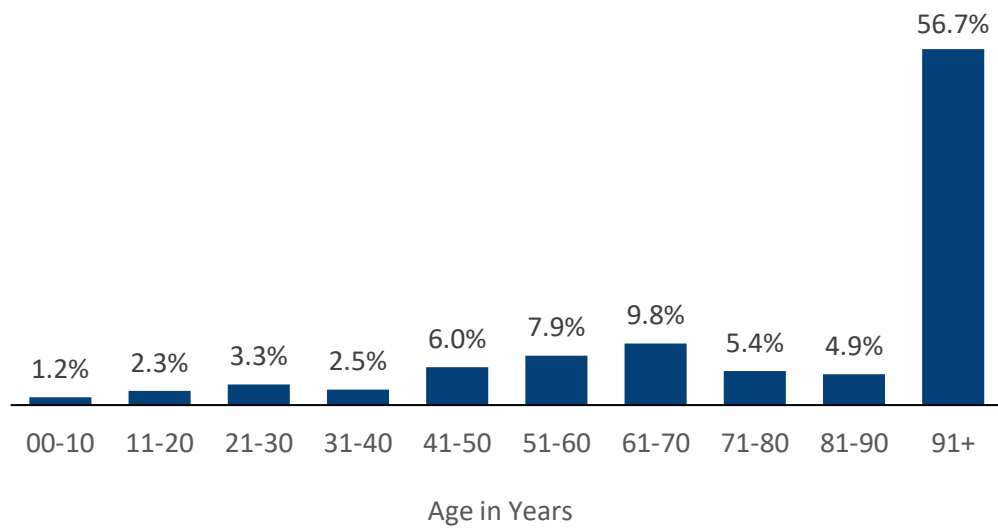


Exhibit 8.18.2 Manhole Age Distribution



Whenever a manhole is entered to perform cable work, it is inspected, and repairs are made as needed. In 2016, DTEE estimated that approximately 8,500 manholes have not been entered in over 20 years. As a preventative measure, DTEE started a manhole inspection program in 2016 for these 8,500 manholes. At the beginning of 2021, nearly 60% of the 8,500 have been inspected, with 3,605 remaining. DTEE plans to inspect the remaining manholes by 2026. Issues identified during manhole inspections include degradation of structural strength due to surface vibration, presence of ground water, etc. Any identified issues will be addressed either as part of the capital work or repair work.

8.19 Pontiac 8.3kV system vaults

DTEE has 33 vaults as part of the 8.3kV system in the city of Pontiac. DTEE acquired the Pontiac 8.3kV system from CMS Energy in the late 1980's. These vaults are typically located beneath city sidewalks with an overhead grating. Overhead style equipment (8.3kV) is used in the vaults. Much of the equipment used in the vaults is obsolete and spare parts are no longer available. Due to the confined space in the vaults, there is also a shock hazard to personnel entering the vaults. The minimum arc flash distance may not exist, making it difficult or impossible to operate within the vaults without causing customer interruptions during maintenance activities. This can add considerable restoration time to an outage event. Exhibit 8.19.1 provides an interior view of an underground vault.

Exhibit 8.19.1 Underground Vault and Equipment



To address the safety hazards and operating limitations associated with the vaults, DTEE has decommissioned nine vaults and replaced them as services directly to the customer between 2012 and 2020. Twelve switching vaults have also been replaced between 2019 and 2020, as shown in Exhibit 8.19.2 and 8.19.3. Switching vaults allow for isolation and replacement of cable or other failed equipment. In addition, 12 service vaults will be replaced with dual voltage (8.3kV and 13.2kV) submersible equipment over the next five years. The submersible equipment utilizes the latest technology. This not only provides safe, reliable operation of the equipment in the vaults, but also standardizes the vault equipment to reduce the cost of future repairs. The current live front equipment will be replaced with new dead front equipment reducing the risk of an arc flash. The operating and switching of this equipment can be done outside of the vault reducing the arc flash exposure to DTEE personnel. The dual voltage feature of the equipment will allow continued utilization of the equipment as the aging and islanded 8.3kV system is converted to 13.2kV. Additional details can be found in Section 11.5 – System Conversion and Consolidation.

Exhibit 8.19.2 Before Pictures of a Service Vault



Exhibit 8.19.3 After Picture of a Service Vault



8.20 Advanced metering infrastructure

DTEE replaced nearly all its electric meters with AMI (Advanced Metering Infrastructure) technology, also known as smart meters in 2010 – 2018. Based on set configurations, these meters are capable of measuring, recording, and uploading various parameters, as well as predefined events and event exceptions (i.e. alarms) that are transmitted upon occurrence. Ninety-nine percent of DTEE meters are solid-state meters with AMI capabilities and the remaining are either digital meters with no radios (0.3%) or digital AMR 2-way communication meters (0.2 %).

AMR 2-way communication meters require advanced registers for industrial customers and have their billing data collected wirelessly daily by a service provider.

To obtain data for digital meters with no radio, meter readers must walk up to the meter to record energy usage once per month. Disconnecting or connecting of a service must be done at the meter location. These types of meters serve customers using the DTEE Opt Out program.

AMI is considered fully deployed as of January 2018.

Exhibit 8.20.1 Electric Meter Types



Mechanical meter



AMI meter

The average age of the AMI meters is 6.5 years and expected life of the AMI meters is 20 years; however, meters may need to be replaced or upgraded earlier due to failures.

In order to ensure the health of our fleet of AMI meters, DTEE conducts a preventive maintenance program for meters includes meter testing as part of the MPSC mandated Sample Test Program, on-going meter replacement based on meter event conditions and refurbishment of equipment returned from field deployment. Routine review of meter conditions are also proactively administered and resultant actions taken (i.e. low battery, retired configurations). Meter communication functions are also independently reviewed and analyzed independent to ensure consistent operation. Similar to mechanical meters previously used, AMI meters are replaced if malfunctioning or failing during service.

The proactive programs mentioned above to maintain our AMI infrastructure is key to both current operations and future capabilities identified as part of the grid modernization process. AMI technology provides a number of current benefits to DTEE customers, such as:

1. Meter reading

AMI technology provides accurate daily and on-demand meter reads without the need for field visits or access to meters installed inside buildings. DTEE has seen a 90 % decrease in month-over-month estimated bills:

2018 per month average: 16,678

2019 per month average: 4,037

2020 per month average: 1,877

AMI provides customers with actual and accurate consumption data every month. Customers with multiple homes and or buildings can have all their sites combined in one bill with the readings on the same day. Starting and stopping billing services can also be done remotely with accurate meter reads.

2. Billing accuracy

AMI technology eliminates human errors in reading the meter and manual entry of the data. Customers also benefit in the elimination of estimated bills. Customers are billed on their actual usage, eliminating the need for estimated bills and related adjustments.

3. Outage restoration efficiency

a. Detection – Early outage notifications aid in damage assessment and initiation of service restoration operations.

b. Restoration efficiency – Timely initiation of restoration efforts coupled with an amplified ability to detect and assess nested trouble after restoration jobs are completed greatly reduces customer interruption durations. It also enables DTEE to assess the quality of restored electrical services to assure customers may return to their normal operations.

c. Daily storm and non-storm outage statistics – AMI data is currently used to create all daily outage statistics such as CAIDI, SAIFI, and SAIDI. This improves the accuracy of the outage data based on the outage experience at the customer site. The quantity and quality of the AMI data improves the overall storm modeling and restoration process.

- d. Enhanced automated storm job closures – AMI is used to automate single customer outages that are auto closed as electrical power is restored in an area. The auto close

algorithms are currently implemented for daily and storm day outages and are preventing numerous “ok on arrival” truck rolls. This feature also shortens the follow-up visits required after an outage and allows the crews to have a higher percentage work time on confirmed outages.

4. Power quality

a. DTEE has started developing analytics that leverage the minimum, maximum, and average voltage monitoring data from AMI meters. Currently, this data has been used to drive distribution transformer replacements. Although DTEE has begun to leverage this data, there remain additional uses that are in development. In process is a more comprehensive approach to proactively perform more robust analysis and respond to customer voltage inquiries and complaints.

b. Hazard detection – The Company is continuing to build on results from pilot studies modeling AMI alarms and voltage data along with phase connectivity, circuit design and protective device status to indicate possible human hazard cases from potentially broken conductors.

5. Safety risk reduction

The remote meter read capability eliminates the potential risk of injury to meter readers due to slips, trips, falls, vehicle accidents, and dog bites. Remote technologies also greatly assisted DTEE in maintaining physical distancing during the COVID-19 pandemic response.

6. Customer energy usage

Customers who use the free DTE Insight app with their AMI meters can see exactly how much energy they are using, which can help them find ways to be more energy efficient.

7. Turn on/turn off/restore

8. Remote capability to connect or disconnect electrical service enables more responsiveness to customer requests for Account Services as well as eliminating costs arising from the field visits.
- Theft and Tampering Notification

The technology continuously monitors for meter tampering. DTEE elected to further expand theft detection in conjunction with AMI data and launched a project in Spring 2020. The

Company is refreshing its theft analytics to also include usage patterns in addition to AMI tamper alarms which we anticipate will enable AMI data to better inform us when theft is occurring.

9. Meter testing

Annual meter testing takes place per MPSC Technical Standards rule 613 under DTEE's Sample Test program. Routine meter functional testing is also conducted as part of our Meter Refurbishment program. Per MPSC requirements, AMI meters must have an accuracy of 1 %.

Grid Security and Reliability Standards Workgroup – Technical Standards for Electric Service Rules U-20630 metering subgroup proposes two major changes that will impact DTEE. The first one is changing the meter accuracy requirement from 1 % to 0.8 % and the second one is a three-tier sample test plan, which based on current high accuracy meters would reduce the yearly volume of tested meters in half. A SAP IS-U enhancement is required to implement the new Sample Test program, along with new BOR reports.

Telecommunication providers (Verizon and AT&T) will retire 3G technology in the beginning of 2022. DTEE will replace approximately 6,000 large commercial and industrial customer 3G meters and 3,600 cellular 3G Cell Relays, serving 2.6 million commercial and residential customer meters. The Cell Relays serve as a communication “aggregator” or “gateway” for AMI meters. The Cell Relays communicate with DTEE data centers via either cellular telecommunications or Ethernet. Without the replacements, the meters serving 6,000 large commercial and industrial customers and 2.6 million commercial and residential customers would no longer be remotely accessible for billing, outage detection, voltage data or remote connect / disconnect capabilities, severely impacting customer service. This meter replacement project is summarized in Exhibit 8.20.2 with costs detailed in Exhibit 8.20.3. DTEE will also continue to expand capabilities to use data from AMI meters to enhance forecasting and provide additional information about grid operations. Investments in data analytics using AMI data are discussed in more depth in Section 12.9 – Technology Roadmap.

Exhibit 8.20.2 AMI Upgrade Projects Summary

Project	Scope of Work	Drivers
AMI 3G Cellular cell relay upgrade	Replace 3,600 3G Cell Relays with 4G devices due to obsolescence of 3G telecommunication	Telecommunication providers plan to phase out 3G in Michigan in 2022
AMI 3G Cellular large commercial and industrial meters upgrade	Replace 6,000 3G cellular meters serving large commercial and industrial customers with 4G devices Among the 6,000 sites, approximately 1,000 of the 6,000 will be replaced with advanced power quality meters	Telecommunication providers plan to phase out 3G in Michigan in 2022 SCADA and more advanced power quality data needed at selected sites, given the size of their load or the percentage of load of circuits they are on

Exhibit 8.20.3 Projected Costs and Timeline for AMI Upgrade Projects

Project	2021	2022	2023	2024	2025	2021-2025 Total
AMI 3G to 4G Upgrade	\$1.3 M					\$1.3M

9 Infrastructure resilience and hardening



As described in Section 6 – DGP Investment Summary, Infrastructure Resilience and Hardening is one of the four pillars in DTEE’s strategic investment framework. One of the fundamental grid needs identified through the grid modernization process discussed in Section 3 is addressing the aging physical infrastructure of DTEE’s system, which will be required to maintain reliable and affordable electric service. Investments in this pillar not only meet the current grid needs, they will also address the CAT storm scenario that projects a higher frequency and intensity of severe weather.

In working through the grid impacts of the three scenarios, DTEE identified that the existing grid infrastructure will need foundational investment in the near-term to support safety and reliability, even as some parts of the grid will start to see the enhancements including substation and circuit conversions and advanced sensing and control technologies that are needed to account for projected electrification and distributed resource adoption by customers over the coming decades. In the next phases of distribution planning, the near-term resilience and hardening strategy programs will be optimized along with the longer-term programs that support the 10-15 year grid vision. The no regrets investments that harden the grid and support the oldest infrastructure replacements will support reliability and resiliency until future investments are implemented that will allow for longer term capacity and other advanced grid needs. These investments also result in a reduction in emergent costs.

The 4.8kV Hardening and Pole Top Maintenance and Modernization (PTMM) programs replace current overhead equipment with equipment designed to our latest standard, which provides additional strength and resistance to failures. In addition, our Automatic Pole Top Switch and SCADA Pole Top Device programs (Sections 9.1.2 and 9.1.8) will help ensure we maintain the capability to isolate faults and minimize customer interruptions.

Projects and programs in this pillar include replacement of aging infrastructure, primarily due to asset health assessments or inspections, and upgrades to circuits in areas of poor reliability. The investments in this pillar will provide a reduction in reactive costs, as the 4.8kV Hardening and PTMM programs are two of the three highest ranked programs for emergent cost reduction (Tree Trimming, as described in Section 10, is also among the top programs for reducing emergent costs). The projects and programs are targeted specifically at equipment with known issues or in areas with known poor reliability and will help drive performance to second quartile within the next five years. Compared to the projects described in Infrastructure Redesign and Modernization, Infrastructure Resilience and Hardening programs can reach more customers in a shorter amount of time. For example, DTEE's goal for PTMM is to address pole top equipment on a five-year cycle, whereas rebuilding the 4.8kV system will likely take 15 years or longer. In addition, the Capital Replacement programs described in Section 9.1 will make the system more resilient and reduce the amount of risk by reducing the volume of failures of System Cable, URD, and breakers experienced by the system currently.

9.1 Capital replacement programs

Capital replacement programs for eight asset classes are summarized in Exhibit 9.1.1. The Pole and Pole Top Equipment program addresses equipment that is found defective as part of an inspection process. The remaining replacement programs were developed as a result of the Asset Health Assessments described in Section 8 – Asset Health Assessment.

Exhibit 9.1.1 Capital Replacement Program Summary

Section	Asset	2021-2025 investment (\$M)	2021-2025 number of units
9.1.1	Pole and Pole Top Equipment	\$430 M	Varies
9.1.2	Automatic Pole Top Switch	\$26 M	70 units
9.1.3	Subtransmission Disconnect Switch	\$12 M	194 units
9.1.4	Circuit Switcher	\$5 M	30 units
9.1.5	Breaker	\$84 M	267 units
9.1.6	System Cable	\$243 M	98 miles
9.1.7	URD (Underground Residential Distribution) Cable	\$60 M	270 miles
9.1.8	SCADA Pole Top Devices	\$9 M	95 Units
9.1.9	Substation Regulators	\$2 M	20 units

9.1.1 Pole and pole top maintenance and modernization

Poles and pole top hardware are one of the fundamental, most visible parts of the distribution system. Poles and their associated equipment are exposed to harsh conditions (e.g., ice, heat, sunlight and wind), causing them to degrade and weaken over time.

Long and costly customer outages result when this equipment fails unexpectedly. This program proactively identifies and replaces damaged or defective equipment before unexpected failures occur. The PTMM (Pole and Pole Top Maintenance and Modernization) program is designed to catch these issues prior to failures. This program was called the Pole Top Maintenance (PTM) program in the past, but with an enhanced specification that replaces old and outdated components with components of an enhanced specification, the term “Modernization” was added to the title. The enhanced specifications include higher grades of materials and updated more reliable design for individual components.

Annually, patrols are performed on a portion of the system to test and inspect poles and pole top hardware. DTEE strives to inspect poles on a 10- to 12-year cycle, as specified by the MPSC; however, in 2020 the 10- to 12-year cycle was not met. In 2020, DTEE’s planned onboarding of an

additional contractor to perform inspections was delayed due to COVID and contract negotiations. This contributed to the low number of poles inspected through the pole top maintenance program. DTEE is prioritizing this work and expects to get back on cycle in 2021. Results from these patrols have typically shown that approximately 8% of the total poles inspected have reduced strength and need to be remediated. These poles are either replaced or reinforced based on the remaining strength of the pole and shell thickness. During the patrols, pole top hardware is also inspected, and those that are broken or have failed are also addressed. Examples of replaced hardware include: cracked or broken insulators which can lead to pole fires, broken guy wires which can lead to excessive leaning and potentially to broken poles, and obsolete equipment that is prone to failures, such as cutouts and arrestors with known defects. The full list of hardware that is inspected for potential defects is detailed in Section 8.8 – Poles and Pole Top Hardware.

When DTEE replaces these items, it uses equipment that complies with its latest upgraded standards. For example, the minimum pole class for distribution poles with primary voltage wire (4.8kV and 13.2kV) is higher than previous standards (meaning that the pole strength is increased). Also, DTEE is now replacing wood crossarms with stronger fiberglass crossarms, porcelain cutouts with more resilient polymer cutouts, and porcelain insulators with more resilient polymer clamp-top insulators. Fiberglass crossarms have five times the mechanical strength of their wood counterparts, and polymer equipment has six times the mechanical strength of its porcelain counterparts.

DTEE has reviewed and enhanced its pole inspection specification and plans to increase its investments in the pole and pole top hardware program as a key lever to address overhead system related emergent failures. In addition to reliability and resiliency, there are two other main drivers for the increased investment plans in this area: benchmarking and the learnings from other key programs.

DTEE has benchmarked pole and pole top inspection practices with two utilities in the Midwest and two utilities in the Northeast. All four utilities inspected pole top equipment on a five or four-year cycle. Two utilities inspected poles on a 10-year cycle while the other two inspected poles on a five-year cycle. DTEE meets its 10- to 12-year pole inspection standards based on a mix of a dedicated PTMM program, the 4.8kV hardening program, and pole inspections that occur as a part of other processes.

Only the dedicated PTMM inspections and hardening program also address pole top equipment, so while the pole inspection is following a 10- to 12-year cycle, pole tops are currently inspected less frequently than the five-year cycle that is industry best practice. DTEE is reviewing the PTMM program and cycle times; and by 2025 investments will achieve a 10-year cycle on poles and PTMM. The ultimate vision for the PTMM program is to achieve a five-year cycle to be in line with industry best practices.

One of the improved specifications is for the pole itself in the inspection process, which is a part of the dedicated PTMM program. The age threshold for pole testing, as opposed to visual inspection alone, was modified from greater than 40 years old to 20 years and older, ensuring that the first signs of pole decay in older poles are identified and addressed. Also, the enhanced specification requires visible decay to be removed and the pole treated to prevent the spread of the decay on poles that are reinforceable. The enhanced specification allows for pole reinforcements in lieu of pole replacements whenever feasible, as pole reinforcement is widely adopted by the utility industry as a cost-effective measure to lengthen pole life. The enhanced pole inspection specification and process are aligned with industry best practices and expected to reduce the risk of pole failures, improve customer reliability and reduce reactive costs during trouble and storm events.

A second driver for an increased investment is the learnings from programs including the Customer Excellence (CE) and 4.8kV Hardening programs. As described in Section 9.4 – Frequent Outage below, the CE program has been successful in quickly addressing issues for customers experiencing poor reliability.

However, the issues that are commonly found during the evaluation process in CE would also commonly be remediated through a pole top maintenance program, if the program was on a shorter cycle. Common issues found through the CE program are broken cross arms, cracked insulators and other visibly defective pole top equipment. While the 4.8kV Hardening program has broader scope than PTMM (including arc wire removal and the removal of infrastructure associated with abandoned properties), much of the other scope around pole and pole top hardware is similar.

As a result of the planned improvements to the PTMM program, DTEE expects to see a significant reliability improvement. As proven in an examination of the reliability data and cause codes described in Section 7.2.3 – Causes of Interruptions, equipment-related issues are the second

leading cause for SAIDI, SAIFI, and outage events, trailing only tree-related outages. In fact, equipment related outages account for almost 25% of all events.

As shown in Exhibit 9.1.1.1 below, DTEE plans to invest \$430 million over the next five years in the Pole and Pole Top Maintenance and Modernization Program. The PTMM program ramp up will begin by increasing the volume of patrols in 2021. These patrols identify the specific pole top equipment and pole remediations needed and help gather information to plan and schedule the execution of the work in the field. In 2021, the increased volume and budget towards patrolling will support ramped up field construction in 2022. As the patrol cost per mile is less than the cost of construction, 2021 has a significantly lower cost per mile than future years. Due to aging infrastructure and the need to provide annual maintenance, this program will continue beyond five years.

Exhibit 9.1.1.1 Pole and Pole Top Program Future investment

	2021	2022	2023	2024	2025	2021-2025 Total
Investment Projection	\$35 M	\$61 M	\$88 M	\$121 M	\$125 M	\$430 M
Miles Inspected on PTMM	1,089	1326	1940	2,739	2,739	9,876
Miles Inspected on 4.8kV Hardening	195	375	340	352	334	1,596
Inspection Cycle	24	18	14	10	10	

9.1.2 Automatic pole top switch

As discussed in Section 8.13 – 40kV Automatic Pole Top Switches (40kV APTS), the entire 40kV APTS population, except for the SEECO switches, are considered for replacement to address high failure rates and unavailable spare parts for the early vintages based on the findings from the asset health assessment. There are 135 APTSs that remain to be replaced. The entire at-risk APTS population will be replaced by end of year 2029.

Replacing an APTS requires shutting down 40kV subtransmission and these shutdowns can typically only be performed from January through May or October through December to avoid peak loading of the system. And, depending on system constraints for a specific subtransmission line, such as weather or adjacent work at substations fed by the subtransmission line needing the shutdown, the shutdown may require extensive work and planning to complete.

In addition to the APTS replacements, shutdowns are also needed to perform routine maintenance and inspections. This can often require the need to deploy portable substations or generators to serve load during the shutdowns, which can limit the window for taking a shutdown, increase O&M costs, or delay maintenance. To avoid this, it is practical in some locations to build an overhead line extension to serve as a bypass for when maintenance or inspection work is done, or to perform repairs or replace equipment. The bypass eliminates the need to de-energize the entire line, so portable substation or generator deployments are not needed to continue serving the load. This results in lower O&M costs, scheduling flexibility, and efficiently performing replacements, maintenance, and inspections as required.

As shown in Exhibit 9.1.2.1, DTEE plans to invest \$21.2 million over the next five years in replacing automatic pole top switches and \$5 million to construct overhead bypasses.

Exhibit 9.1.2.1 APTS Replacement Investment

	2021	2022	2023	2024	2025	2021-2025
Investment projection – APTS replacement	\$2 M	\$4.8 M	\$4.8 M	\$4.8 M	\$4.8M	\$21.2
Quantity (units)	6	16	16	16	16	70
Investment projection - APTS overhead bypass	-	\$0.5 M	\$1.5 M	\$1.5 M	\$1.5 M	\$5 M

9.1.3 Subtransmission disconnect switch

As discussed in Section 8.4 – Subtransmission Disconnect Switches, 216 subtransmission disconnect switches remain to be replaced, as shown in the asset health assessment. Total cap & pin style subtransmission disconnects on the system totaled 424 in 2011 and there are 132

remaining. There are also 58 PM-40 and 22 Westinghouse Type R disconnects still left on the system to be replaced.

Replacing disconnect switches requires shutdowns the same as with the APTS. The shutdown allows crews to safely replace the disconnect switches. These shutdowns typically occur during specific months of the year and are limited by specific system constraints on the subtransmission line, weather and adjacent work at substations.

As shown in Exhibit 9.1.3.1, DTEE plans to invest \$12 million over the next five years to replace subtransmission disconnect switches. The program will continue past 2025 as additional issues with switches are identified during the annual health assessments.

Exhibit 9.1.3.1 Subtransmission Disconnect Switch Replacement Program Investment

	2021	2022	2023	2024	2025	Total investment 2021-2025
Investment Projection (\$ millions)	\$2.0	\$2.5	\$2.5	\$2.5	\$2.5	\$12
Quantity (units)	35	42	40	39	38	194

9.1.4 Circuit switcher

As discussed in Section 8.5 – Circuit switchers, 80 circuit switchers remain to be replaced as shown in the asset health assessment. The equipment identified for replacement is prioritized based on risk of failure and inadequate equipment sizing.

As shown in Exhibit 9.1.4.1, DTEE plans to invest \$5 million over the next five years to replace circuit switchers. This program will continue beyond 2025 to complete the replacement of the switchers identified through the asset health assessments.

Exhibit 9.1.4.1 Switcher Replacement Program Investment

	2021	2022	2023	2024	2025	Total investment 2021-2025
Investment projection (\$ millions)	\$1.0	\$1.0	\$1.0	\$1.0	\$1.0	\$5
Quantity (units)	6	6	6	6	6	30

9.1.5 Breakers

As discussed in Section 8.3 – Circuit breakers, 58% of breakers are candidates for replacement based on the findings in the asset health assessment. Due to the large at-risk breaker population and because these assets will continue to age, the replacement program will be continuous.

Exhibit 9.1.5.1 lists the types and quantities of the breaker replacements planned for the next five years. Subtransmission H breakers, subtransmission outdoor breakers and substation PR reclosers are prioritized for near-term replacements as allowed by field execution.

To the extent possible, breaker replacements are coordinated with other capital or maintenance work to reduce costs and minimize the overall time the equipment is out of service. In most cases, breaker replacements include relay replacements to make the breaker SCADA controllable and to increase the penetration rate of substation remote monitoring and control capability. This is expected to bring significant customer benefits from improved substation operability.

Exhibit 9.1.5.1 Breaker Replacements by Type 5-year Plan

Breaker type	Target quantity
Subtransmission H breaker	30
Subtransmission outdoor breakers	97
Substation recloser	24
Distribution breaker (oil and 1 st /2 nd generation air magnetic)	116
Total number of breakers	267

As shown in Exhibit 9.1.5.2 below, DTEE plans to invest \$84 million over the next five years on the Breaker Replacement program. Due to the number of breakers considered candidates for replacement, the program will continue beyond 2025.

Exhibit 9.1.5.2 Breaker Replacement Program Investment

	2021	2022	2023	2024	2025	Total investment 2021-2025
Investment projection (\$ millions)	\$11	\$15	\$18	\$20	\$20	\$84
Quantity (units)	54	55	54	53	51	267

9.1.6 System cable

As discussed in Section 8.16 – System cable, 43% of system cable, or approximately 1,300 miles, is a candidate for replacement based on the findings in the asset health assessment. Due to the large at-risk system cable population, and because these assets will continue to age, the replacement program will be continuous.

System cable is a critical component of the subtransmission and distribution systems. While underground cable provides a higher resiliency to storms than overhead lines, a failure can interrupt a large number of customers for an extended period of time. In-service failures are also more difficult to locate and repair than on overhead circuits.

Overall, the cost and time to install, replace and repair system cable issues are much greater. Cable failures can also cause failures in adjacent cables or other equipment, such as switchgear, which can then impact an even larger number of customers for a longer period of time. XLPE (cross-linked polyethylene) cable manufactured before 1990 has a design defect that leads to premature insulation breakdown and dielectric failures (also called “treeing”) and has contributed to switchgear failures. Gas cable (cable that has cavities within the insulating layer that are filled with nitrogen gas under pressure) is an obsolete design that is prone to mechanical damage that leads to leaks and dielectric failures and costly repairs.

DTEE is increasingly concerned about the level of failure risk posed by the age and condition of system cable. This concern is further escalated by the impact a system cable failure has on the overall system, because when a cable fails, it reduces the level of redundancy in the system itself. An acceleration in cable failures with simultaneous failures of multiple cables across the system could lead to very lengthy outages. In addition, planned replacements, compared to unplanned replacements, are more cost effective and reduce system risk.

In response to the need to increase investment in system cable replacements, DTEE created a dedicated team focused on underground work. This group has been building execution capacity and updating work processes to prepare for the needed levels of cable replacement. DTEE has developed partnerships with engineering, design, project management, and construction firms that have successfully helped other utilities build their cable replacement programs to significant levels.

DTEE has also conducted benchmarking with other utilities that have been involved in significant cable replacement efforts to learn best practices. An electric utility that is similar to DTEE averaged over 130 miles of system cable replacement per year between 2013 and 2017 and peaked at nearly 190 miles in a single year, demonstrating it is feasible to ramp up an efficient cable replacement program to address the risk on the system in an efficient manner.

As shown in Exhibit 9.1.6.1, DTEE plans to ramp up and invest \$243 million over the next five years in the Cable Replacement Program. While this represents a significant increase in cable replacement, it still does not fully address all at-risk cable on the system. Therefore, DTEE expects this program to continue to grow beyond 2025 to reduce the likelihood that customers experience lengthy outages, improve reliability, protect substation equipment from damage and reduce reactive costs.

Exhibit 9.1.6.1 System Cable Replacement Program Investment

	2021	2022	2023	2024	2025	Total Investment 2021-2025
Investment Projection (\$ millions)	\$13	\$35	\$55	\$65	\$75	\$243 M
Quantity (miles)	7	13	21	27	31	98

9.1.7 Underground residential distribution (URD) cable

Although underground cables are not subject to the storm impacts of trees and wind, they do have identified expected life and failure modes. As discussed in Section 8.17 – Underground residential distribution (URD) cable, 41% of URD cables (about 4,500 miles) are candidates for replacement based on findings in the asset health assessment. Currently, the system averages over 1,000 URD failures per year, which can create a build-up of open loops, and create the potential for extended outages if a concurrent failure were to occur. As planned replacements are more cost effective than unplanned replacements, and to help manage the build-up of open loops, DTEE is ramping up efforts to replace at risk URD cable. Even at the planned investment level by 2025, it still does not address all of the cable that is considered a candidate for replacement. Due to the large at-risk URD cable population and because these assets will continue to age, DTEE intends to continue to expand investment in the program.

As shown in Exhibit 9.1.7.1, DTEE plans to invest \$60 million over the next five years in the URD Cable Replacement Program.

Exhibit 9.1.7.1 URD Cable Replacement Investment

	2021	2022	2023	2024	2025	Total investment 2021-2025
Investment projection (\$ millions)	\$4	\$6	\$10	\$15	\$25	\$60
Quantity (miles)	24	28	45	66	107	270

9.1.8 SCADA pole top devices

The SCADA Pole Top Device Replacement program addresses the equipment that was discussed in Sections 8.11 - Three-phase reclosers and 8.12 – 4.8kV & 13.2kV SCADA Distribution pole top switches. This program targets two of the types of equipment that have experienced high rates of failure – the Form three reclosers and the Bridges pole top switches. There are 140 of these units on the system, and the program targets replacing 90 of these devices in the next five years.

In addition, the Circuit Automation program described in Section 12.5 – Distribution automation, will replace or upgrade reclosers as part of deploying increased sectionalize and control on the distribution system, a key component of grid modernization.

As shown in Exhibit 9.1.8.1, DTEE plans to invest \$9.4 million over the next five years in the SCADA Pole Top Device Replacement program.

Exhibit 9.1.8.1 SCADA Pole Top Device Replacement Investment

	2021	2022	2023	2024	2025	Total investment 2021-2025
Investment projection (\$ millions)	\$0.9	\$1.4	\$1.9	\$2.4	\$2.8	\$9.4
Quantity (units)	5	15	20	25	30	95

9.1.9 Substation regulators

DTEE uses substation regulators to regulate voltage at substations where the transformers are non-LTC (load tap changing).

The Substation Regulator Replacement program targets proactive replacement of regulator styles identified as being obsolete and technology that is no longer manufactured or supported. This includes induction, tri-delta, and tri-wye type regulators.

An increasing number of issues are occurring with these types of units with repair time continuing to increase due to a need to dry out the insulation in the units and repair or replace voltage controls.

DTEE plans to systematically replace obsolete units with modern step voltage regulators. This replacement will allow for proper voltage regulation at those substations and avoid out of range voltage damaging customer equipment or causing reliability problems.

As shown in Exhibit 9.1.9.1, DTEE plans to invest \$1.7 million over the next 5 years in the Substation Regulator Replacement program.

Exhibit 9.1.9.1 Substation Regulator Replacement Investment

	2021	2022	2023	2024	2025	Total investment 2021-2025
Investment projection (\$ millions)	-	\$0.4	\$0.4	\$0.4	\$0.4	\$1.7
Quantity (units)	-	5	5	5	5	20

9.2 Substation risk

The concept of substation risk is tied closely to resiliency, which was defined in Section 7.3 as the ability to withstand and recover quickly from an event that significantly impacts the grid. DTEE’s framework on substation risk considers the likelihood and impact of the complete loss of a substation, which is an event that can impact a significant number of customers (thousands) for an extended duration (a couple of days). A resilient system would have the flexibility to recover quickly from any type of failure or event that could potentially lead to a full substation outage.

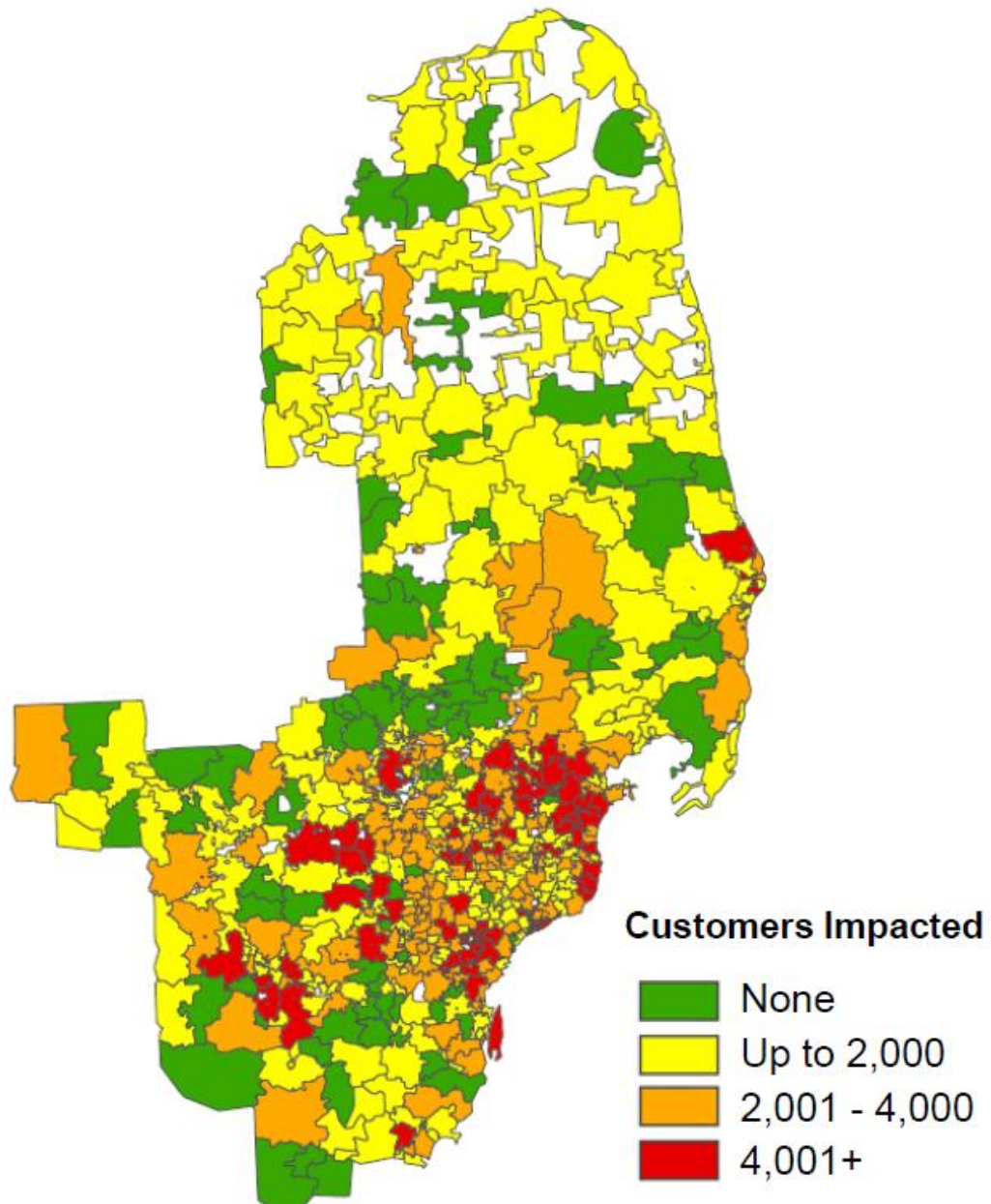
This section describes the overall substation risk framework and components, potential impact of future scenarios on substation risk, and the investments DTEE will make to manage risk and ensure the system can recover quickly from major events.

9.2.1 Substation outage risk model

To help DTEE identify and mitigate the risk of additional substation events, a substation outage risk model was developed. The model quantifies the relative substation outage risk scores and is used to help prioritize the capital investment to reduce this risk. Reduction in substation risk is one of the seven components quantified in the GPM methodology described in Section 5. The model calculates an outage risk score for each substation based on two factors: 1) impact, as measured by the stranded load at peak, and 2) risk, as measured by the asset conditions.

- 1) Substation stranded load at peak is the amount of load that cannot be restored by transferring load to adjacent circuits in the event of a substation outage during peak hours. There are multiple reasons why in a particular situation load cannot be transferred to adjacent circuits. First, in some cases the adjacent circuits operate at a different voltage, and therefore cannot be directly connected, which is called jumpering. While equipment in our portable fleet can allow for connections between different voltages, it adds complexity to the process. Second, the adjacent circuits and substations may not have capacity to take on additional load. As will be discussed more thoroughly in Section 11 – Infrastructure redesign and modernization, many of our substations and our subtransmission lines are over capacity now and would not be able to accept a load transfer if an adjacent substation had an outage. Exhibit 9.2.2.1 shows a stranded load map with red indicating substations that could result in 4,000 or more customers without power for more than eight hours (i.e., minimum time required to install mobile fleet to restore power) in a substation outage event during summer peak.

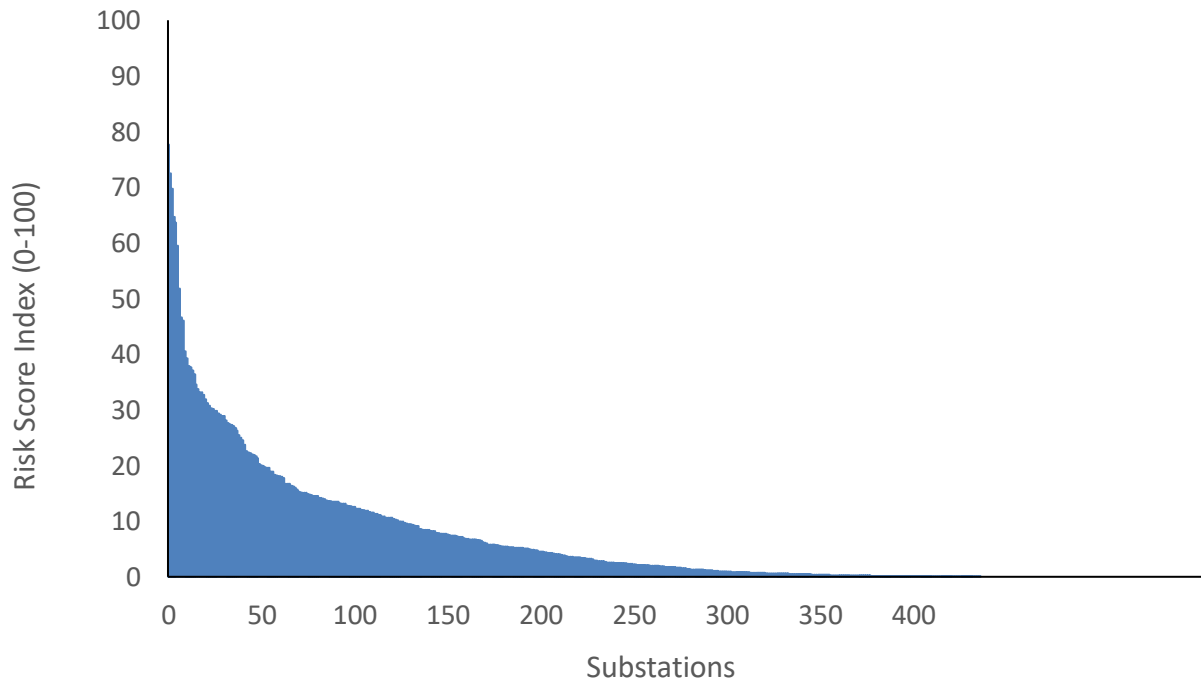
Exhibit 9.2.1.1 Potential Stranded Load Map During Complete Loss of a Substation



- 2) Substation asset condition risk represents the combined risk of four critical asset classes: system cable, transformers, breakers and switchgear. The condition of assets that feed substations, specifically subtransmission cable and subtransmission breakers, are also included when in the risk score. The basis for determining asset condition risk are the asset condition assessments (see Section 8 – Asset Health Assessments) with system considerations such as load impacted by the asset and the number of contingencies before the entire substation loses power also incorporated into the overall score.

Combining stranded load impact risk and asset condition risk, the risk profile of distribution substations is generated, as shown in Exhibit 9.2.2.2. Substation outage risk scores are indexed between 0 and 100, with 100 representing the score that Malta substation received when the model was first developed in 2018. The risk at Malta has been reduced through a switchgear replacement project, which was completed in 2019. The highest risk substation is currently Garfield with a score of 78. Load transfers at Garfield are currently taking place as part of the CODI project, which will improve Garfield's risk profile by reducing stranded load.

Exhibit 9.2.1.2 Substation Risk Score Distribution



The top five substations, listed in descending order by risk ranking, are shown in Exhibit 9.2.1.3 below, along with the project or program in the plan to address risk.

Exhibit 9.2.1.3 Top 5 Substations by Risk Ranking

Substation	Investment to address
Garfield	Garfield, Amsterdam and Madison are located within the city of Detroit (CODI) footprint, and will be addressed as part of the long-term CODI plan discussed in Section 11.4
Amsterdam	
Madison	
Chestnut	Chestnut switchgear replacement (Section 9.2.3)
Warren	Warren Station upgrade (Section 9.2.3)

9.2.2 Recent examples of substation events and future risks

Complete loss of a substation is a relatively rare event. While DTEE experienced approximately three major events per year from 2014-2018, there have been fewer events since then. Generally, due to the redundancy built into the system, multiple failures would be needed to cause the complete loss of a substation.

Despite the reduced frequency of total substation outages in recent years, the substation risk framework is still a useful tool to highlight examples of risk on the system and the programs and projects needed to reduce the risk of extended outages for our customers. Two examples of major events that occurred in the months before the filing of the DGP highlight the ongoing risk and costs of operating aging assets on a system with limited capacity.

Erin-Savoy-Baker

Savoy and Baker are two substations in St. Clair Shores. Both are fed via two subtransmission lines from Erin, which is a 120kV to 40kV subtransmission station. Combined, Savoy and Baker substations serve approximately 10,000 customers. In early June 2021 during very hot weather, a cable failure on one of the subtransmission lines left both Savoy and Baker substations with a single contingency. A concurrent failure on a second cable before the first cable was replaced (a process that can take a couple of days) would have led to a full substation outage at both Baker and Savoy. In addition, the cable that fed Baker and Savoy after the first failure was showing signs of failure as

it supported an increased load. Due to system limitations and capacity constraints in the area, Baker and Savoy had a combined 32 MVA of stranded load, indicating that jumpering load to adjacent substations based on the current configuration of the circuits was not a viable option. To mitigate the risk and potential of a substation outage, mobile generators were deployed to the site, and multiple jumpering points were built to allow for the load to be transferred in case of a second cable failure. While the second cable did not fail and therefore no customers experienced an outage, the overall cost of the contingency plans developed to avoid this outage was approximately \$800,000.

McGraw and Scotten flooding

A recent risk highlighted for DTEE's grid is flooding. Aligned with the scenario of increased catastrophic storms laid out in Section 3, climate change may drive increased precipitation and risk of flooding as a result of more frequent and intense storms. On June 26, 2021, the Detroit/Dearborn area near the I-94 corridor experienced a significant storm with rainfall in some areas in excess of 5 inches in a single day and 8 inches over two days. Two DTEE substations in Detroit, McGraw and Scotten, flooded as a result of this storm. The damage that resulted from multiple electrical faults caused by the flood left much of the equipment, including breakers, and buses unrepairable at McGraw. The damage at Scotten was not as extensive and the majority of the equipment was salvaged. McGraw and Scotten have a combined stranded load of 35 MVA, indicating there were limited options to transfer load to neighboring substations to quickly restore power. Instead, the restoration effort deployed mobile generators and portable substations and built temporary structures to enable customer restoration. In total, approximately 3,600 customers were without power for an average of 30 hours, with a cost of restoration for this event of approximately \$600k. As a result of the flooding, DTEE is conducting an assessment to evaluate flood risk for the key infrastructure at all substations and determine recommendations to mitigate risks that are found. A preliminary view of the system shows that other substations with below grade equipment are already planned for replacement as part of the CODI project.

Future risk and grid modernization scenarios

The substation risk model will be informed by our scenarios described in Section 3 in at least two key ways. First, the potential for increased CAT Storms can directly impact the likelihood of failure events that could lead a substation outage. In addition to increased intensity of wind, extreme heat will further strain the aging equipment, and flooding risk creates additional complications. The other

scenario that will impact the risk model is increased electrification. The loading that will come with higher adoption of EVs can further increase loading at substations. This will further increase the stranded load as both jumpering options will become more limited and the amount of load that needs to get transferred will increase. To continue to manage the risk of the system, asset replacements, hardening and capacity increases will be needed to avoid more frequent substation events, and avoid the costly contingency plans now associated with equipment failures.

9.2.3 Programs to address substation risk

The Substation Outage Risk Model provides a framework for DTEE to measure risk reduction as a result of investments. There are multiple needs identified by the model and recent examples: the need to replace assets identified by asset health assessments, the need for increased capacity, and the need for solutions to mitigate risk in the daily system operations. The following programs specifically help address substation risk:

- **Capital Replacement Programs (Section 9.1)** – Replacement of at-risk cable and breakers, at both substations and subtransmission system, will directly provide a risk reduction by reducing the probability of failure of key substation equipment.
- **Infrastructure Redesign and Modernization** – Increasing capacity of the system is a primary driver of the projects in the major categories of our Redesign and Modernization pillar. Subtransmission projects, conversion projects, and system loading projects all address the system's over-capacity areas and include load relief as one of the primary prioritization factors. Increasing system capacity will increase options to transfer load from substation to substation, thus reduce stranded load and improve the overall substation risk profile. In addition, the eventual conversion of the 4.8kV system (and 8.3kV) system, will eliminate one contributing factor to stranded load, which is having substations with different voltages adjacent to each other.
- **The Mobile Fleet program** – This program is expanding mobile generation, portable substations, and mobile switchgear in order to have more tools to decrease restoration time for stranded substation load or to support substations on a single contingency to avoid outages. The mobile fleet provides relatively quick restoration of customers and the ability to facilitate the repairs of the failed equipment inside the substation. Though expanding the mobile fleet capacity is relatively low cost, it does not reduce the substation outage risk, it does allow DTEE additional flexibility to restore customers. Moreover, the application of the

mobile fleet during restoration may be limited due to the feasibility of connecting it at substation sites and considerations on space, traffic, environmental and community impacts. Exhibit 9.2.3.1 shows a portable substation and mobile generator that are part of the fleet.

Exhibit 9.2.3.1 Mobile Fleet

Mobile generator Portable substation



- **The Substation Risk Reduction program** – This program involves replacing aging, at-risk equipment (mostly switchgear) to reduce the probability of a failure. This approach reduces substation outage risk; however, it can be costly and difficult to execute due to site-related construction constraints and the need to continue serving customers during the process. As such, the plan is to limit initial implementation to substations where deployment of mobile fleet assets is limited and cannot restore the entire substation load; in other words, load will be stranded for more than 24 hours.

Exhibit 9.2.3.2 shows the projected costs and timelines for the identified substation outage risk reduction projects. The timeline considers multiple factors including relative substation outage risk, resource availability, specific site conditions and other execution constraints. The cost and timeline estimates for the projects are based on the best knowledge and information known today by DTEE. Actual project costs and timeline could deviate from the projection due to various unforeseen factors or new information/learnings.

Exhibit 9.2.3.2 Projected Investments and Timeline for Substation Risk Reduction Projects

Project	2021	2022	2023	2024	2025	2021-2025 Total (\$ million)
Savage						\$0.4
Drexel						\$2
Belleville						\$2
Port Huron						\$2
Apache						\$19
Chestnut						\$14
Savage Switchgear						\$12
Total	\$6	\$11	\$12	\$11	\$12	\$52

- **The Station Upgrade program** – This program is an evolution of the former Relay program, which originally focused on end-of-life relaying. The first project executed as part of this

program intended to only replace relays at Warren Station. As the project progressed, the scope was reviewed and expanded to replace at-risk breakers within the substation as well. This helps reduce the long-term costs of having to revisit these substations later to replace equipment in need of replacement. With this expanded scope, the Station Upgrade program helps reduce substation risk on the system.

Exhibit 9.2.3.3 shows the projected investment and timelines for the identified Station Upgrade projects. The timeline considers multiple factors including relative substation risk, resource availability, specific site conditions and other execution constraints. The investment and timeline estimates for the projects are based on the best knowledge and information known today by DTEE. Actual project costs and timeline could deviate from the projection due to various unforeseen factors or new information/learnings.

Exhibit 9.2.3.3 Projected Investment and Timeline for Station Upgrade Projects

Project	2021	2022	2023	2024	2025	2021-2025 Cost Estimate (\$ million)
Warren						\$1
Northeast						\$9
Lincoln						\$10
Navarre						\$10

9.3 4.8kV Hardening program

9.3.1 Introduction to the 4.8kV Hardening program

The 4.8kV system in Detroit and its surrounding communities was the first part of DTEE's distribution network built. Due to its age, it has the highest volume of trouble events in our service territory. Those problems are exacerbated by the abandoned and overgrown alleys in the city of Detroit.

DTEE continues to work to maintain the electric grid across the entire service territory in a cost-effective manner. However, in many areas, general maintenance practices are simply no longer

sufficient and DTEE has begun investing more aggressively to harden and upgrade the infrastructure.

DTEE reviewed and evaluated four alternatives for improving the safety and reliability of the 4.8kV system (full conversion, Pre-conversion of overhead only, secondary program, and 4.8kV Hardening), as discussed in detail in MPSC Case No. U-20162. Because the full conversion of the 4.8kV system in the city of Detroit is estimated to cost over \$4 billion and take more than a decade to complete, DTEE has selected the 4.8kV Hardening program as this program provides improvements in safety and reliability at a much faster pace and more affordable cost.

Although the current scope of the 4.8kV Hardening program is focused on Detroit and its surrounding communities, DTEE will continue to evaluate the application of the program to other parts of the 4.8kV system in its service territory, as compared to the 4.8kV Conversion and Consolidation program. Such comparison and considerations are addressed in detail in Section 11.3 4.8kV Conversion.

The scope of work for the 4.8kV Hardening program includes:

- Testing all utility poles that have DTEE equipment attached and replace or reinforce those poles as needed
- Replace wooden crossarms with fiberglass crossarms
- Remove Detroit Public Lighting Department (Detroit PLD) arc wire from DTEE-owned equipment and ensure the remaining DTEE wires are left in a safe configuration
- Remove Detroit PLD distribution wire from DTEE-owned equipment when it can be confirmed that the wire is not serving customers
- Remove service lines to abandoned properties
- Trim trees as required to support construction activities
- Perform any additional necessary work as dictated by field conditions
- Conduct pilot projects to explore and assess alternatives to existing rear lot overhead configurations. The pilots include converting rear lot overhead construction to underground residential distribution construction, removing primary overhead construction, and adding secondary overhead construction to serve sparsely populated areas. The pilots aim at improving reliability and reducing trouble events in some areas of Detroit where inaccessible overhead construction in rear lots is encroached by fence lines and overgrown trees.

9.3.2 4.8kV Hardening program prioritization

DTEE prioritizes 4.8kV system hardening on four factors:

- Recorded wire downs per overhead line mile
- Estimated foot traffic within the substation service area
- Total substation SAIDI
- Total outage and non-outage events requiring the dispatch of a line crew

Data is collected and reviewed at the substation level as it is more efficient to plan and perform work for circuits fed by the same substation at the same time.

Ranking the substations follows DTEE's overall goal of reducing safety risk, improving reliability, and managing costs. Recorded wire downs and estimated foot traffic are measures of risk reduction metrics.

DTEE takes a three-year historic average of wire downs per overhead line mile multiplied by the estimated foot traffic for each substation. Foot traffic is estimated based on the number of each type of customers on a substation (residential, commercial, industrial) as well as schools, hospitals, and the occupancy rate for each type of customer, determined by census and other related business data. Substation SAIDI indicates the total minutes of interruptions experienced by customers served by a substation and represents a reliability metric. A three-year average of substation SAIDI is used for substation prioritization. Total outage and non-outage events requiring the dispatch of a line crew represents the cost management metric. The three-year historic average of the number of events per overhead line mile is used as a proxy for costs. The scores are normalized and then combined to provide an aggregate score, with greater weight given to safety risk reduction. The weighting is 45% for risk reduction and 27.5% each for reliability improvement and cost management.

The prioritized list of substations for the 4.8kV Hardening program is cross referenced with the schedule for planned 4.8kV Conversion and Consolidation program to eliminate overlap, as substations that are part of the conversion and consolidation in the next 5-10 years should not be considered for the 4.8kV System Hardening to achieve the maximum cost efficacy. It is important to note that prioritization factors and weightings may be adjusted over time based on the ongoing assessment of program effectiveness and customer needs.

9.3.3 4.8kV Hardening program progress and effectiveness

DTEE has aggressively ramped up the program since its beginning in 2018, as can be seen in Exhibit 9.3.3.1. In 2018, the first year of the program, DTEE hardened 105 miles. In 2021, DTEE will almost double that total, and will continue to ramp the program to achieve over 300 miles a year by 2022.

Results on three key metrics show that the 4.8kV Hardening program has been very effective thus far: (1) All-Weather SAIFI; (2) SAIDI excluding major event days (SAIDI ex-MEDs); and (3) unaudited wire down events.

All-weather SAIFI reflects the frequency of the outage events experienced by customers on the circuits regardless of weather conditions.

SAIDI ex-MEDs is an indicator of the amount of time customers are without power, excluding the most significant weather events, such as very large storms. SAIDI ex-MEDs contains multiple storm and weather impacts, with only the most extreme weather events removed.

The number of wire downs is a measure of the effectiveness of safety improvements for the circuits that were hardened. Circuits hardened in 2018 and 2019 show improvement over those not hardened during that time. (The control group consists of circuits in the city of Detroit not worked on in 2017 or 2020.)

Exhibit 9.3.3.1 shows the All-weather SAIFI improvement of hardened circuits vs. the control group, Exhibit 9.3.3.2 shows the SAIDI ex-MEDs improvement of hardened circuits vs. the control group, and Exhibit 9.3.3.3 shows the wire down event improvement of hardened circuits vs. the control group.

In addition to the circuit improvements, the 4.8kV Hardening program has successfully removed over 200 miles of arc wire from the system.

Exhibit 9.3.3.1 All Weather SAIFI Before and After Hardening

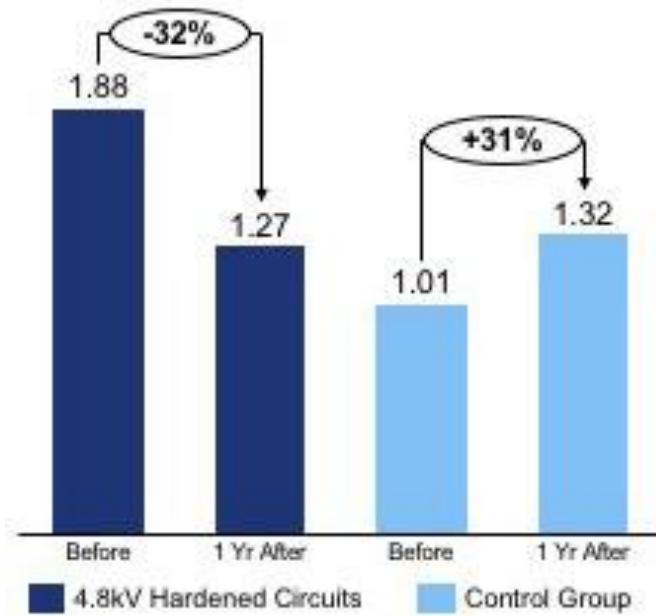


Exhibit 9.3.3.2 SAIDI ex-MEDs Before and After Hardening

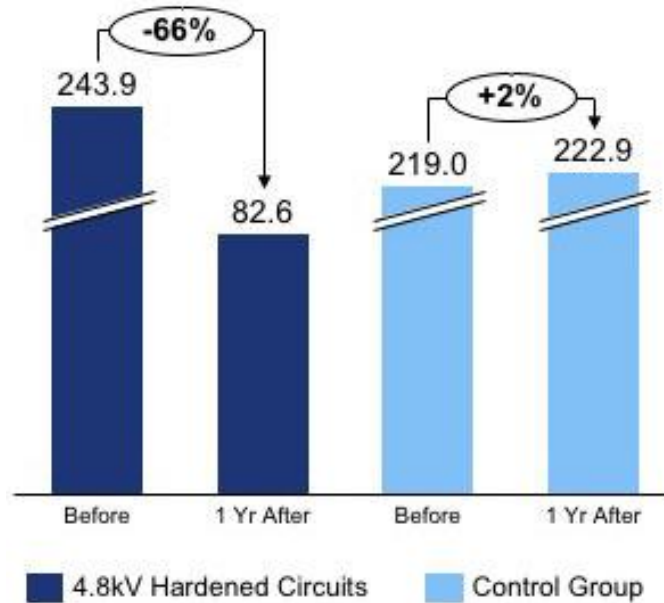
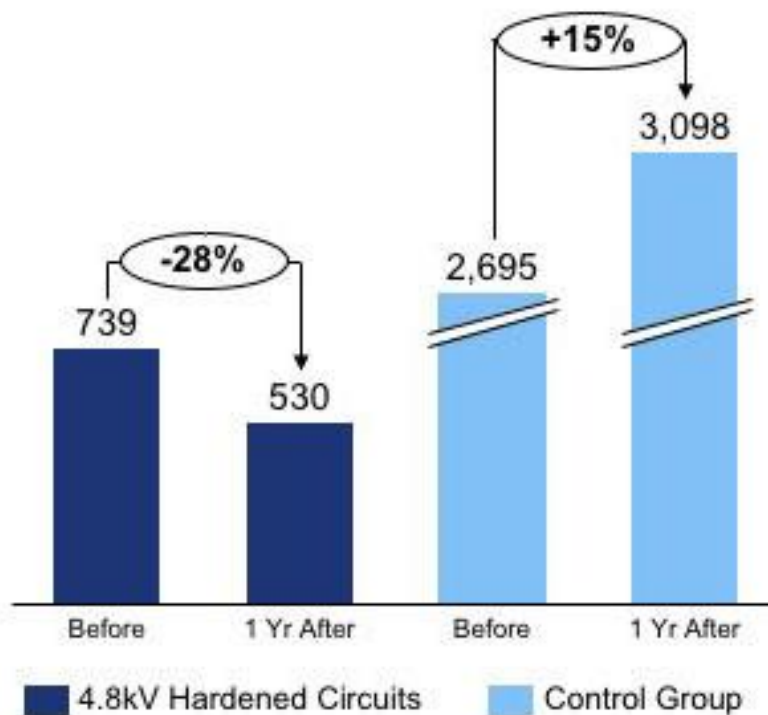


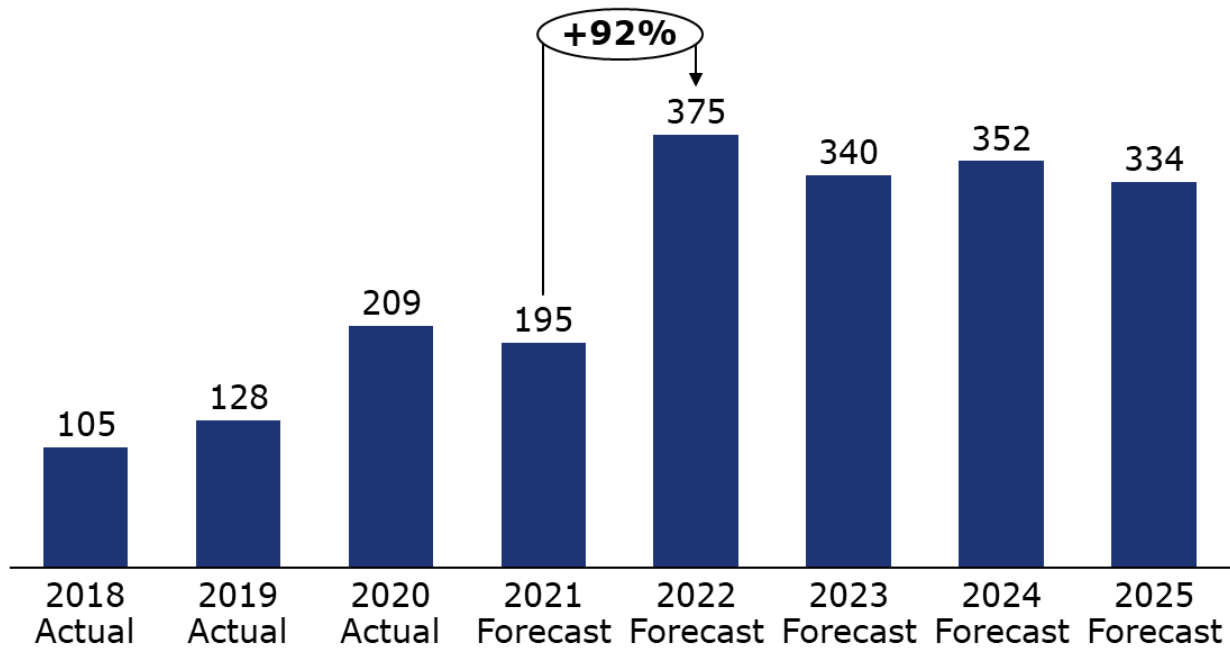
Exhibit 9.3.3.3 Wire Downs Before and After Hardening



9.3.4 4.8kV Hardening program plan and timeline

DTEE is expected to increase the number of miles completed annually through 2023 and is expected to complete over 2,000 miles by 2025.

Exhibit 9.3.4.1 4.8kV Miles Hardened



DTEE plans on continuing the 4.8kV Hardening program through 2029. DTEE expects to harden four to six substations and 200 to 300 overhead line miles per year. By the end of the program, DTEE will have hardened over 45 substations and over 2,220 overhead lines miles and removed 1,300 miles of arc wire within the city of Detroit and immediately adjacent suburbs.

Exhibit 9.3.4.2 4.8kV Hardening Program Investment

	2021	2022	2023	2024	2025	Total investment 2021-2025
Investment projection (\$ millions)	\$69	\$120	\$116	\$116	\$116	\$537

9.3.5 Alternatives methods to improve to 4.8kV reliability

The Company has considered five options to improve the reliability of 4.8kV circuits that were built with rear-lot overhead construction. These five options are shown below in Exhibit 9.3.5.1. Two of the pilots which are in progress evaluate the costs of converting rear-lot overhead to rear-lot

underground and front-lot underground. These pilots are discussed in more detail in Section 11.6 – Strategic Undergrounding. The third option being evaluated is selective deconductoring in sparsely populated areas (Option 5 in the table below).

Exhibit 9.3.5.1 Comparison of Rear-Lot OH Construction Alternatives

Rebuild Method	Description	Key Benefits	Key Considerations
1. Overhead Rear Lot	Install poles, overhead wires, and equipment in alleys, vacated alleys or rear lots	• Easy connection to existing meter boxes • Away from foot traffic	• Not truck accessible – climb required for tree trim, maintenance, and restorations • Maintenance tree trim required • Pole and Pole Top Maintenance (PTM) required
2. Underground Rear Lot	Install underground padmount transformers, pedestals, cable and conduit etc. in alleys, vacated alleys or rear lots	• No exposure to downed wires • Easy connection to existing meter boxes • No need for maintenance tree trim • No need for Pole and Pole Top Maintenance (PTM) • No need for 4.8 kV Hardening	• Encroached fence lines and obstacles in alleys make underground construction difficult • Difficulty in obtaining permits & customer approvals • Damage to landscape • Installation cost is significantly higher than overhead installation
3. Underground Front Lot	Install underground padmount transformers, pedestals, cable and conduit etc. in front lots adjacent to the road and reroute service wire to meter box in rear lot	• No exposure to downed wires • No need for maintenance tree trim • No need for Pole and PTM • No need for 4.8 kV Hardening	• Damage to customer property (driveways, landscape, etc.) • Interference with existing utilities (sewer, gas, water, etc.) • Difficulty in obtaining easements, permits and community approvals • Difficulty in connecting to existing meter boxes • Installation cost is significantly higher than overhead installation
4. Overhead Front Lot	Install poles, overhead wires, and equipment in front lots adjacent to the road and reroute service wire to meter box in rear lot	• Improved truck access • Reduced tree interference (typically front lots have less tree density) • Reduced cost in maintenance tree trim • Reduced cost in Pole and PTM • Reduced cost in 4.8kV Hardening	• Aesthetically not appealing • Damage to customer landscape • Interference with existing utilities (sewer, gas, water, etc.) • Difficulty in obtaining easements, permits and community approvals • Difficulty in connecting to existing meter boxes • Maintenance tree trim required • Higher foot traffic in vicinity of overhead wires

5. Primary De-conductoring	Remove primary wires in sparsely populated areas and reconductor secondary wires where necessary	• Reduced exposure to downed primary wires • Improved truck access • Reduced cost in maintenance tree trim • Reduced cost in Pole and PTM • Reduced cost in 4.8 kV Hardening • Radial secondary results in easier to locate and isolate trouble	• Potential low voltage issues for end customers • Rebuild required if the area repopulates
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Primary Deconductoring Pilot

DTEE is working on a pilot to deconductor specific circuits in areas that have seen significant population decline. For instance, the infrastructure in the city of Detroit was established to serve the peak population of approximately 1.8 million in the 1950s. With the population now at approximately 670,000, there are many unoccupied properties and underutilized and aging assets. The goal of this pilot is to remove underutilized infrastructure that is no longer needed. The primary wires, distribution transformers and other equipment only increase maintenance costs and the exposure for outages and other events for customers in these areas. When an area is redeveloped, the infrastructure can be rebuilt with newer standards that are consistent with new construction.

Exhibit 9.3.5.2 Unoccupied Areas in DTEE's Caniff Service Center Area

*(East side of the Detroit metropolitan area)
Unoccupied properties represented by dots*



In these sparsely populated areas, primary deconductoring may be an option to improve reliability and reduce risk. Primary deconductoring involves the removal of small-sized primary wire, arc circuit wire, overhead transformers and pole tops. Reconductoring secondary wires is also completed where necessary. The amount of infrastructure removal is dependent upon the sparsity and location of customers in the neighborhood. Maintaining adequate voltage on the secondary (in absence of the primary) and jumpering points for contingencies is the key engineering consideration. Two circuits were selected for the pilot: Wayburn DC 2136 and Grant DC 1598. Combined, the two circuits serve 1,300 customers and have 19 miles of overhead primary, or 68 customers per overhead mile of primary – sparse for an urban neighborhood.

One of the two deconductoring projects has been completed: Wayburn DC 1236. This circuit was completed ahead of schedule for several reasons, including the population density was such that there were limited issues. Grant DC 1598 is still under construction and has seen some additional challenges. The population density is higher on Grant, which caused some voltage issues when

deconductoring that were addressed by adding back some the removed infrastructure. Through these two projects, DTEE has learned several keys lessons, most importantly that removing some sections of primary has caused voltage issues on the circuits. As such, some primary has been left in place to maintain proper voltage. DTEE also learned that the less dense the population is in an area, the quicker that installation will proceed and the more effective the deconductoring project can be.

9.4 Frequent Outage program

DTEE has two programs specifically targeted toward a rapid solution for customers who are experiencing a high frequency of reliability issues. These reliability programs continually evolve to address emerging problems using the best available equipment and techniques. The main recurring reliability programs are listed in Exhibit 9.4.1.

Exhibit 9.4.1 Reliability Programs Summary

Program	Scope of work	Target areas
Customer Excellence	• Perform tree trim • Reconductoring • Replacement of broken pole top equipment • Pole replacement	• Circuit pockets with high SAIFI and/or MAIFI
Frequent Outage (CEMI)	• Perform tree trim • Underground portion of the circuits • Rebuild/reconductor/relocate overhead lines • Add or strengthen circuit ties • Provide circuit load relief • Install sectionalizing and switching devices, etc.	• Remediate small pockets of customers or single customers experiencing frequent outage events by focusing on removing root causes of reliability and power quality events • Address circuits with chronically poor reliability performance by focusing on removing root causes of reliability and power quality events

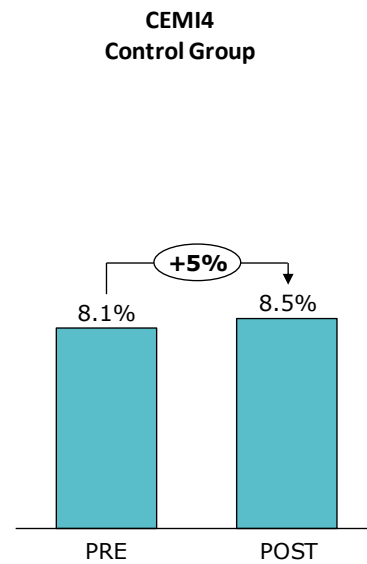
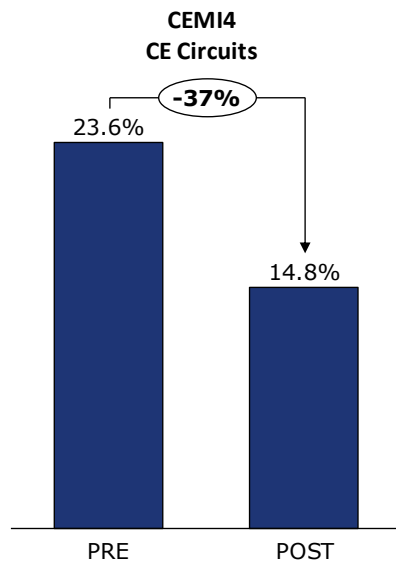
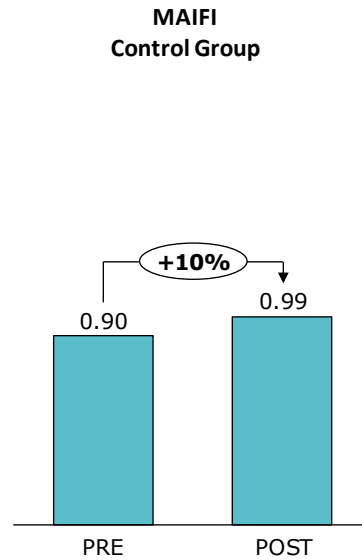
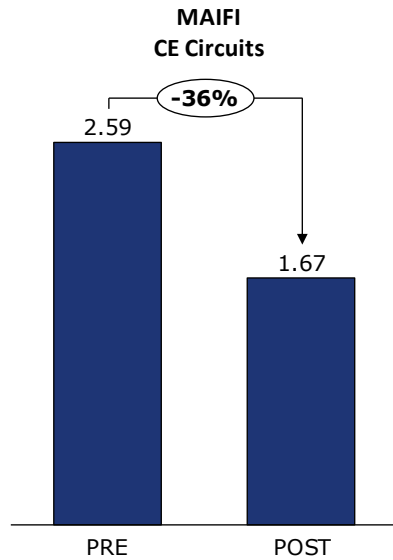
In 2019, the Customer Excellence (CE) program was established to provide rapid solutions to small pockets of customers experiencing poor reliability. These customers are identified as experiencing four sustained outages (SAIFI > 4.0) or nine momentary outages (MAIFI > 9.0).

The prioritization method for the Customer Excellence program relies on AMI data to identify these customers on a rolling 12-month basis to address issues more rapidly than other programs that rely annual analysis of reliability events. In addition to reliability event data, the prioritization also includes time since last tree trim, any other planned work on the circuits, and customer complaints.

Upon identification of a circuit that meets the CE prioritization criteria, DTEE conducts a field patrol to understand both equipment and tree conditions. After patrol, the scope of work is developed for both tree-related and equipment-related problems identified. In addition to the tree trimming and defective equipment replacements, the scope of work also includes checking operating equipment to ensure it is functioning properly, conducting fault studies to ensure fuses are properly sized, and installing additional equipment, such as reclosing devices and animal guards, to prevent future outages. On average, the solutions cost \$60,000 to \$80,000 per circuit to implement.

The cycle time from circuit selection to field completion averages less than 55 days, allowing for quick resolution of identified reliability problems. Throughout the CE process, customers on the selected circuits receive communication, starting with an initiating email. The CE team coordinates with the Corporate Communication team to collect photos of work being conducted in the field and utilizes social media to update the progress of the improvements. Once work is completed, customers receive a close-out email outlining the completed work. The CE program has had a positive impact on the customer experience and provided improvements in both CEMI and MAIFI, as shown in Exhibit 9.4.2 below. Examples of equipment issues found during patrols are shown in Exhibit 9.4.3.

Exhibit 9.4.2 Customer Excellence Reliability Impact



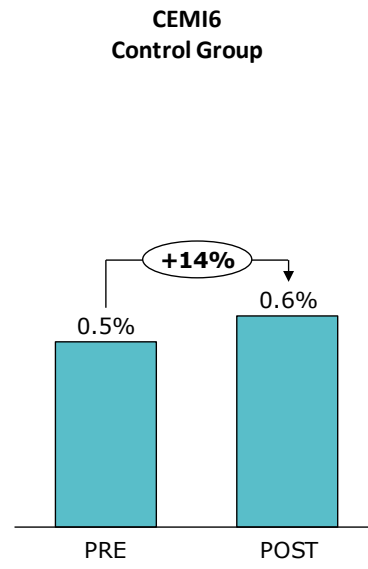
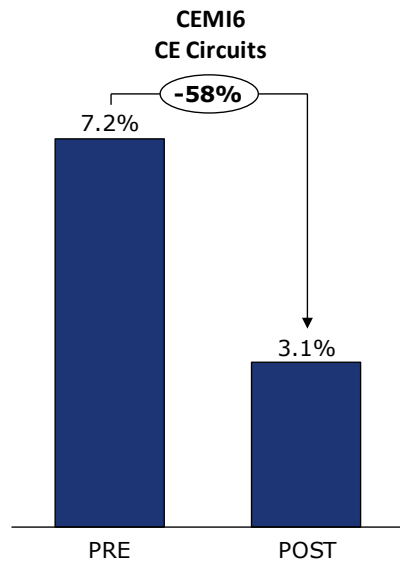


Exhibit 9.4.3 Example of Scope of Work for Customer Excellence Circuits

(Loose fuse cutout (left), broken cross arm (top right), broken fuse cutout (bottom right))



The other program that addresses customers who experience frequent reliability problems is the Frequent Outage program, also known as the CEMI (Customers Experiencing Multiple Interruptions) program. This program includes improvements to either circuit sections (customer pockets) or entire circuits. The primary distinctions between the Customer Excellence and the CEMI programs are

that circuits are selected for the CEMI program based on three-year average circuit SAIDI and SAIFI performance, MPSC complaints, and regional expertise on customer needs. In addition, the scope of work performed under the CEMI program is more comprehensive and typically requires investments between \$300,000 and \$250,000.

Exhibit 9.4.4 shows the projected annual capital investment for these reliability programs.

Exhibit 9.4.4 Projected Reliability Programs Capital Investment

	2021	2022	2023	2024	2025	Total investment 2021-2025
Frequent Outage (CEMI) including Customer Excellence (\$ millions)	\$33	\$45	\$22	\$22	\$28	\$150

9.5 4.8kV relay improvements

The Ground Detection program (4.8kV Relay Improvement program) allows the System Operations Center (SOC) to receive alarms for 4.8kV wire down events so actions can be taken quickly to remediate and make repairs.

The first step of the Ground Detection program is to install or upgrade telecommunications at 4.8kV substations, as most 4.8kV substations are of a vintage that precedes SCADA (a similar telecommunications program exists for 13.2kV substations, see Section 12.3). The second step is to upgrade to modern ground detection relaying utilizing microprocessor relaying. Ground detection relaying will identify real-time wire down events seen at the substation bus. Knowing that a ground wire exists in real-time allows for immediate crew dispatch to locate and repair the downed wire, reducing the amount of time needed for repair.

Exhibit 9.5.1 Projected 4.8kV Relay Improvements Capital Investment

4.8kV Ground Detection	2021	2022	2023	2024	2025	2021-2025
Investment (\$ millions)	\$2	\$3	\$3	\$3	\$3	\$15
Units	11	17	17	17	17	79
Remaining units	252	235	218	201	184	-

9.6 Mobile fleet

DTEE's mobile fleet program allows for system shutdowns to complete preventive maintenance at single transformer substations and safely executing planned strategic capital projects, while minimizing customer impacts. As referenced in Section 9.2.3 – Programs to address high risk substations, the program also provides options for restoring customers while making repairs of failed equipment at substations.

DTEE's current mobile fleet includes 11 portable substations, 10 portable distributed generation (DG) units, six manual portable ISO transformers, three portable ISO transformers including switchgear and pole, and two portable load banks.

DTEE plans to purchase five new portable substations between 2021 and 2024 to allow for the proper execution of preventive maintenance programs, including addressing the backlog of preventive maintenance for single tap substations.

In addition, DTEE will purchase a portable switchgear cable trailer and complete ongoing work to upgrade and integrate controls, support equipment purchases and refurbish and repackage DG units based on ongoing condition assessment. All of this work will support the ability to deploy distributed generation on the system when needed.

Exhibit 9.6.1 shows the projected annual capital investment for the mobile fleet program that enhances system reliability and resilience.

Exhibit 9.6.1 Projected Mobile Fleet Capital Investment

	2021	2022	2023	2024	2025	Total investment 2021-2025
Mobile fleet capital investment (\$ millions)	\$5	\$3	\$3	\$3	\$1	\$14

10 Tree Trimming program



10.1 Introduction

DTEE's Tree Trimming program addresses interference between vegetation and electric distribution facilities. The objectives of the program are to increase system reliability by reducing tree-related safety hazards and the volume of tree-related trouble cases. This program is particularly important to the grid modernization scenario envisioning "increasing CAT storms." If the volume and/or intensity of weather events increases, maintaining sufficient tree clearance from DTEE's electrical equipment and lines becomes even more critical to ensuring the safety and reliability of our customers. The Tree Trimming program addresses current system needs and is key to improving the overall resiliency of the DTEE grid.

Since 2016, DTEE has made two significant improvements to the Tree Trimming program. The first was moving to an enhanced trimming specification, based on industry best practices and the Company's experience, known as the Enhanced Tree Trimming program (ETTP). The ETTP was described in detail in testimony in the Company's last four rate cases: Case Nos. U-18014, U-18255, U-20162 and U-20561.

The second improvement was an increase in annual funding to help the Company achieve a five-year trimming cycle. In Case U-20162, DTEE proposed a "surge" in funding for 2019-2025 to reclaim areas that have yet to be trimmed under the ETTP, while maintaining the investment already made in the ETTP. The Commission was supportive of the surge, noting:

... the Commission agrees that falling behind in this area will cost more in the future and perpetuate reliability challenges. The record also shows direct, quantifiable benefits in terms of reliability improvements resulting from the ETP program. 3 Tr 200-206.

[MPSC Case No. U-20162, May 2, 2019 Order, p 79.]

The ETP has provided significant benefits to customers in improved reliability and safety. As outlined in DTEE's Annual Tree Trimming Report, customers that live on circuits that have been trimmed as part of this program experience significantly fewer outages, shorter outage durations when an outage does occur and significantly fewer wire downs. A circuit that has been trimmed as part of the ETP experiences 62.9% fewer customer interruptions and 57.3% fewer customer-outage minutes than circuits that have not been trimmed in the ETP within the first year of trimming. More of these results are discussed in detail later in this section.

In light of the severe storms DTEE's customers experienced during the summer of 2021, the Company has committed to accelerating the "surge" outlined in U-20162. An additional \$70 million of funding for the program will allow for an increased number of miles to be trimmed in 2021 and future years. The result of the additional funding is the completion of the surge program by the end of 2024, a year earlier than previously proposed. DTEE is not seeking cost recovery for this additional funding, providing an affordability benefit to our customers. The acceleration of the tree trim program will improve the safety, reliability and affordability of our electric system for our customers and make the system more resilient to severe weather sooner than previously planned.

10.2 The need for tree trimming

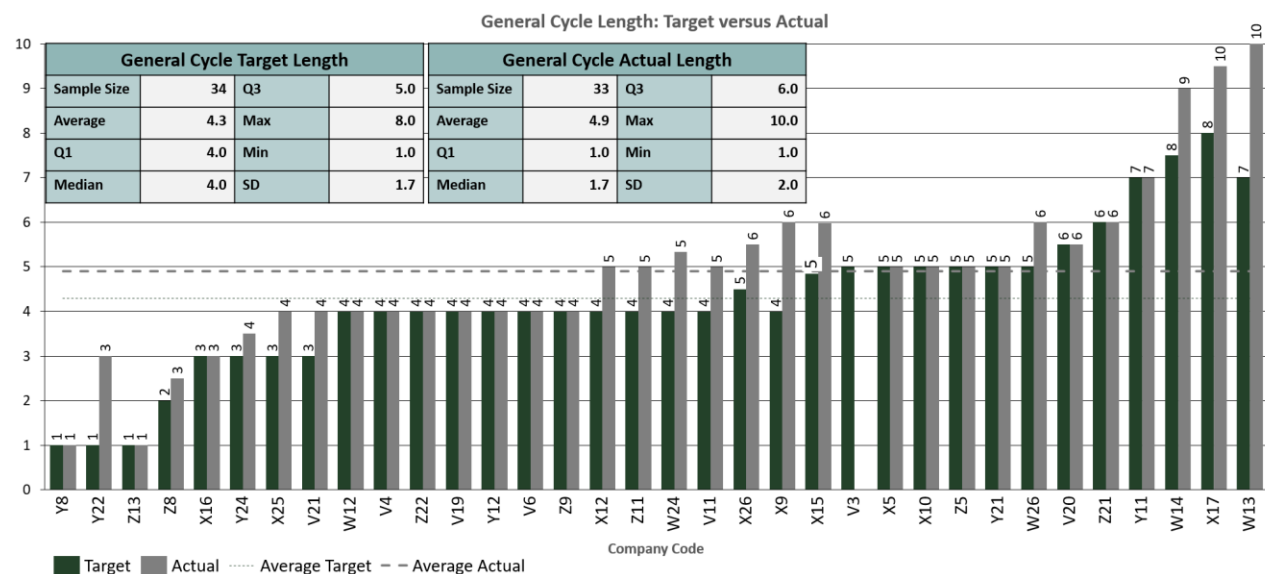
Tree interference is the leading cause of customer outages. Tree-caused outages account for two-thirds of the time that customers spend without power. A robust tree trimming program is needed to maintain and improve system reliability and resilience. It significantly reduces outage events, customer interruptions, customer minutes of interruption, wire downs and other non-outage trouble events. The goal is to achieve and maintain a five-year cycle for distribution circuits and a three-year cycle for subtransmission circuits. Tree trimming is one of the highest priority investments for Distribution Operations. No other program or project has a greater immediate impact on mitigating risks, improving system and customer reliability, and managing the electric distribution system operating costs.

The ETTP clears trees and branches to a distance of 15 feet on either side of the distribution pole centerline, or approximately 10 feet from the conductors (the actual clearance is species-specific). Because trees in our service area grow an average of 10 feet in five years, the five-year cycle provides a reasonable and acceptable level of tree-to-conductor contact comparable to the industry standard of 10%-15%. (Tree-to-conductor contact represents the likelihood of any portion of the tree touching the conductor. A tree-to-conductor contact level of 10%-15% denotes the estimated average percentage of trees in contact with the overhead electrical facilities across the entire distribution system when the recommended cycle length and clearance standards are reached.)

For subtransmission, a 3-year cycle is used because of the high customer impact associated with subtransmission line failures. An outage on a subtransmission line can potentially cause an entire substation to lose power, which would affect, on average, over 3,600 customers. By comparison, an outage on a distribution circuit would affect, on average, approximately 700 customers. The three-year cycle further reduces the likelihood of trees contacting overhead conductors.

The targeted goal of a five-year cycle on distribution circuits is comparable to the actual industry average of 4.9 years, per a report published by CN Utility Consulting, Inc. (CNUC), "Distribution Utility Vegetation Management Benchmark Survey Results 2016." All but six of the participating companies target a cycle of five years or less.

Exhibit 10.2.1 CNUC Benchmark Study (34 Respondents)



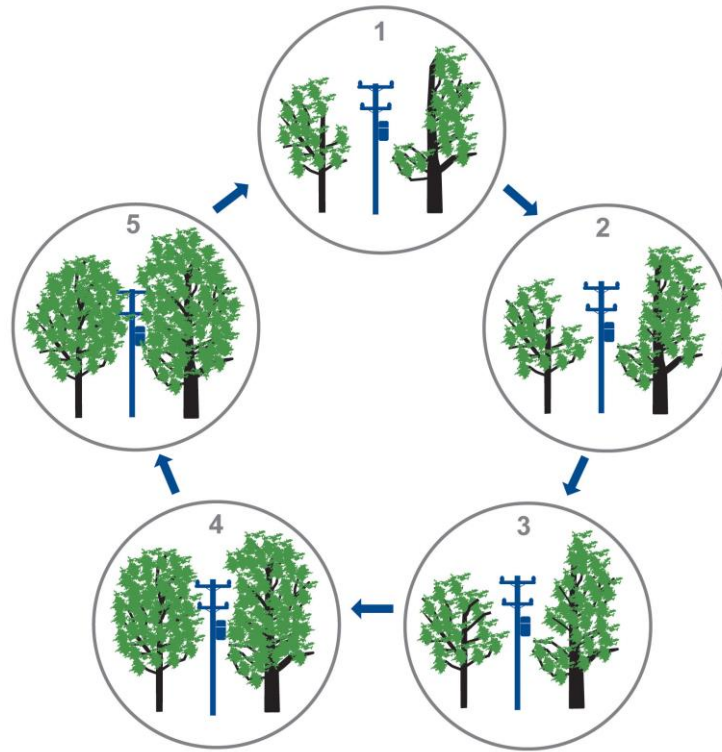
Target & actual cycle length (Years)

Deferring tree trimming maintenance beyond the five-year cycle results in cost escalation as described in a May 1997 study funded by International Society of Arboriculture (ISA) and conducted by consulting firm ECI, LLC, "The Economic Impacts of Deferring Electric Utility Tree Maintenance." The table below shows the relative cost, excluding inflation, of deferring tree trim maintenance beyond the optimum time – five years since last trim. As the time since last trim gets longer, the work becomes more complex as trees begin to interfere with the conductors.

Exhibit 10.2.2 Cost of Deferred Tree Trim

Timing of trimming	Year since last trim	Relative cost
Optimum	5	\$ 1.00
1 year past optimum	6	\$ 1.16 to \$ 1.23
2 years past optimum	7	\$ 1.30 to \$ 1.43
3 years past optimum	8	\$ 1.40 to \$ 1.59
4 years past optimum	9	\$ 1.47 to \$ 1.69

Exhibit 10.2.3 Illustration of Tree Growth Impact on Complexity Based on Years Since Last Tree Trim



The table below shows the number of miles per year necessary to be on-cycle. These numbers are averages and actual miles per year may vary based on density and complexity as discussed later in this document.

Exhibit 10.2.4 Tree-trimming Cycles

Circuit type	Overhead miles	Cycle length (years)	Trim miles per year
Distribution	28,459	5	5,692
Subtransmission	2,539	3	846
Total	30,998	4.75	6,538

10.3 Tree Trim Surge program

At year-end 2020, the Company had an average distribution cycle time of six years, with 16,152 circuit miles considered on-cycle. Our average subtransmission cycle time was three years, with all 2,539 circuit miles considered on-cycle. On-cycle means the circuit miles have been trimmed within the last three or five years (for subtransmission and distribution respectively) to either the ETPP specification or the Company's previous trim specification. The ETPP specification is the current and preferred specification. Circuit miles not yet trimmed to the ETPP specification are referred to as the "backlog." To expedite reaching the targeted goal of a five-year cycle, the Tree Trim Surge program was put in place. The Surge program is designed to ensure all miles are on-cycle and eliminate the backlog by the end of 2024.

Reducing the tree-trimming cycle length to three or five years respectively will provide several benefits and savings:

- Fewer wire-down events, resulting in improved safety.
- Fewer outage and non-outage events, leading to improved reliability and a positive impact on reactive O&M and capital costs. This will also allow reallocation of resources to other necessary work across the Company's distribution system.
- Lower future trimming costs as the number of trees growing within the right-of-way are trimmed or removed more frequently, resulting in less wood near the lines. In 2018, the Company commissioned ECI to perform a detailed study of the cost improvements during the second cycle of tree trimming. According to ECI the cost per line mile is expected to decrease by an average of 40% compared to the initial trimming conducted as part of the ETPP.
- Lower customer costs as tree-related outages are reduced. The improved reliability will reduce downtime for customers' manufacturing processes, allow commercial businesses to remain open and reduce inconveniences residential customers experience.

At the completion of the Surge program, tree-related O&M and capital costs for reactive maintenance and storm will be lower. With fewer tree-related events, the need for tree crews and overhead crews will be reduced. There will be less of a need to repair and replace assets on the system that have failed because of tree interference. The table below shows current O&M and capital costs compared to the projected costs upon completion of the Surge program, excluding inflation.

Exhibit 10.3.1 O&M and Capital Cost Comparisons

Estimated tree-related cost savings, excluding inflation				
Cost categories		Current cost	Post surge (2026)	Reduction
Tree-related O&M	Tree trim reactive	\$ 12.9 M	\$ 8.8 M	\$ 4.1 M
	Tree trim storm	\$ 11.5 M	\$ 7.9 M	\$ 3.6 M
	Storm Service operations storm and trouble	\$ 12.2 M	\$ 8.4 M	\$ 3.8 M
Tree-related capital	Tree trim reactive	\$ 4.5 M	\$ 3.1 M	\$ 1.4 M
	Tree trim storm	\$ 21.4 M	\$ 14.7 M	\$ 6.7 M
	Service operations storm and trouble	\$ 36.6 M	\$ 25.4 M	\$ 11.2 M

Exhibit 10.3.2 Average Annual Outage and Non-outage Events

	Pre-surge 2014-2018 average	Post surge 2026	Percent reduction
Tree-related outage and non-outage events	44,700	25,378	43%

In addition to a decrease in reactive trimming costs, at the completion of the Surge program the annual O&M for the Tree Trimming program is expected to be about \$100 million, not adjusted for inflation. This reduction will be realized because the entire system will be on a five-year cycle, and future trim cycles will require less wood removal (as discussed previously in this section). In 2025 and subsequent years, the Company expects to trim an average of 6,538 miles annually.

DTEE is committed to completing the Surge program by 2024. The program has had a very strong start and we are on track to meet this committed completion date.

10.4 Miles completed and spend

The charts below show ETP miles completed in recent years.

Exhibit 10.4.1 Annual ETP Miles Completed by System

Circuit Type	2015	2016	2017	2018	2019	2020	Total	% System
Distribution	257	2,270	2,733	3,128	3,181	4,583	16,152	56%
Subtransmission ¹	188	1,030	868	540	1,006	1,006	4,638	100%
Total	445	3,300	3,601	3,668	4,187	5,589	20,790	60% ²
% System	1%	12%	24%	35%	46%	60%	60%	

(1) 1,093 of these miles are subtransmission that are on their second cycle and show up in the numbers twice. Only unique miles are included in percent complete calculation.

(2) Numerator only includes unique miles – it does not double count subtransmission on second cycle.

The chart below shows the annual O&M spend on tree trimming.

Exhibit 10.4.2 Annual O&M Tree Trim Spend (millions)¹⁷

O&M spend	2016	2017	2018	2019	2020
Line clearance maintenance	\$59.2	\$69.3	\$71.8	\$116.7	\$151.0
Trouble	\$10.7	\$11.4	\$13.0	\$16.6	\$12.2
Tree Trim program and other	\$2.8	\$1.8	\$2.5	\$10.2	\$8.2
Program administration	\$1.5	\$1.7	\$1.9	\$2.7	\$3.7
Total	\$74.2	\$84.6	\$89.1	\$146.2	\$175.2

¹⁷ Table does not include all spot trimming expenses from other programs.

10.5 ETTP reliability results

The ETTP has provided significant benefits to customers in improved reliability and safety. Circuits that have been trimmed in the ETTP have fewer outage events, fewer customers interrupted, shorter outage duration and fewer wire downs compared to circuits that have not been trimmed in the ETTP.

The tables below compare the performance of ETTP circuits versus a control group (non-ETTP circuits) for four metrics: outage events, customers interrupted, customer-minutes interrupted and wire down events. Performance is measured as the percent difference between years one, two, three and four years after trimming compared to the average over the three-year period prior to trimming. Exhibits 10.5.1, 10.5.2, 10.5.3 and 10.5.4 illustrate the reduction/increase in performance for the ETTP and non-ETTP circuits compared to prior performance. Additionally, Exhibits 10.5.1, 10.5.2 and 10.5.3 state the difference between the ETTP and non-ETTP circuits compared to what was expected from the Surge Model. Comparison to a control group is done to account for natural weather variations or general deterioration/improvements to the system year-to-year. The ETTP circuits and the control group experience the same conditions each year, hence a comparison between the two negates the effect weather or system variations introduce into performance.

Exhibit 10.5.1 Outage Event Performance

Years after last trim	Number of distribution circuits with ETTP	Percent change in outage events			
		ETTP circuits	Non-ETTP circuits	Difference ETTP vs. non-ETTP	Surge model reduction
1	1651	-34.6%	38.7%	-73.3%	-57.0%
2	1108	-9.1%	62.1%	-71.2%	-57.0%
3	532	-11.2%	71.2%	-82.4%	-50.0%
4	180	16.6%	88.3%	-71.7%	-37.0%

Exhibit 10.5.2 Customer Interruption Performance

		Percent change in customers interrupted			
Years after last trim	Number of distribution circuits with ETPP	ETTP circuits	Non-ETTP circuits	Difference ETTP vs. non-ETTP	Surge model reduction
1	1651	-43.8%	29.3%	-73.1%	-57.0%
2	1108	-22.2%	45.8%	-68.0%	-57.0%
3	532	-40.1%	42.0%	-82.1%	-50.0%
4	180	-12.2%	45.0%	-57.2%	-37.0%

Exhibit 10.5.3 Customer-minute Interruption Performance

		Percent change in minutes of customer interruption			
Years after last trim	Number of distribution circuits with ETPP	ETTP circuits	Non-ETTP circuits	Difference ETTP vs. non-ETTP	Surge model reduction
1	1651	-53.8%	12.5%	-66.3%	-57.0%
2	1108	-19.3%	36.8%	-56.1%	-57.0%
3	532	-56.9%	-7.6%	-49.3%	-50.0%
4	180	-44.4%	-1.7%	-42.7%	-37.0%

Exhibit 10.5.4 Wire Down Event Performance

Years after last trim	Number of distribution circuits with ETPP	Percent change in actual wire down events		
		ETTP circuits	Non-ETTP circuits	Difference ETPP vs. non-ETTP
1	1651	-25.6%	9.6%	-35.2%
2	1108	-15.4%	21.7%	-37.1%
3	532	-13.0%	13.9%	-26.9%
4	180	-7.8%	4.9%	-12.7%

10.6 Program Improvements

DTEE has made several significant improvements since the beginning of the ETPP:

- The Company has dedicated a team to improve and advance tree trimming scheduling. Tree trimming supports several workstreams (e.g. maintenance trimming, trouble trimming, capital trimming). Effectively prioritizing and scheduling these workstreams has been disaggregated in years past, often done at the service center level. The scheduling team is working to develop an integrated scheduling tool that monitors all these workstreams in a centralized format. This will allow the Company to more efficiently schedule tree trimming required for customers and construction projects, while staying on-schedule for the annual maintenance trimming. The scheduling team has also developed dashboards to track maintenance miles trimmed on a weekly basis, and the dashboards are available to other DTEE stakeholders, such as our customer support teams, who use them to understand when tree trimming is taking place in different communities.

- In late 2018, the Company expanded the use of EPA-regulated herbicides to replace mechanical removal of vegetation from the right-of-way with a chemical treatment, which only controls tree species with the potential to grow into electrical wires. The Herbicide program uses industry best practices that were collected and developed through benchmarking and by working with an outside consultant, ECI. The Company has expanded its use of herbicides by implementing foliar herbicide treatment, basal herbicide treatment, and dormant stem treatment.
 - The herbicide treatment will reduce the cost of maintenance trimming in the right-of-way by reducing tree density. There are other advantages besides realizing cost savings. As tree density and brush height decreases, the electrical system becomes more reliable and the right-of-way becomes more accessible for overhead crews to repair a downed wire or broken equipment. Also, because grasses and shrubs are not affected by the herbicide treatment, the area will become a habitat for pollinators, birds and small mammals. The treatment will also target invasive plant species, limiting their spread.
- The Company is partnering with its tree-trim contractors on new ways to improve production using innovative tree-trim equipment.
 - An alley mowing pilot program in the city of Detroit has proven successful and has been expanded. Many Detroit alleys have become overgrown and impassable. Using mechanical mowers, our contractors have been able to clear brush and small trees allowing access for bucket equipment, which makes removals less expensive and easier. As part of this pilot, our contractors have obtained blanket permits with the city to remove alley trees. (Before and after pictured below.)
 - A pilot using a new insulated boom-mounted saw (one of two of its kind in the world) allows for the use of the saw within the “energized zone” near the wires. These saws can quickly and safely remove large tree branches without the need for climbing and rigging, saving significant amounts of time, improving safety and reducing the number of trimmers needed to accomplish the work. (Example pictured below)

- The Company is also exploring other ways to utilize specialty equipment in backyard areas where climbing crews are normally required. These include increased use of backyard buckets, backyard chippers, mowers and backyard Jarraffs.

Exhibit 10.6.1 Innovative Tree Trimming Equipment

Mowing Alleys



Before



After

Removing Unstable Trees



Removing a dead tree which has grown into the wires. Dead trees cannot be climbed, therefore specialty equipment is required

10.7 Resources

An average of approximately 1,200 tree trimmers are needed to execute the annual scope through the end of the Surge program, assuming average annual storm volume. Today, the Company primarily uses five tree-trimming contract companies.

Because there are not enough local tree trimmers to complete Surge program goals, DTEE must hire qualified tree-trimming crews from outside of our service territory to supplement the contract workforce. The long-term plan is to achieve an adequate level of qualified local workers to displace the non-local workers. A local trimmer lives within reasonable driving distance of the DTEE service

territory and makes union wages. Non-local trimmers normally work in other parts of the country but have come to work for DTEE contractors on a temporary basis. A non-local trimmer makes union wages but also requires a daily per diem rate to cover room and board. These requirements make non-local trimmers more expensive than local trimmers.

DTEE faces several risks that could erode its current non-local trimmer population:

- Due to the Company's location in a colder climate, many non-local trimmers do not want to work in the area during the winter months. To make up for the potential manpower shortfall in the winter months, a higher volume of workers is necessary during the warmer months.
- The market for non-local trimmers is tight and these workers are in high demand due to the utility industry's collective renewed focus on tree trimming. If DTEE does not stay on pace with market rates, non-local trimmers will leave for other utilities.
- DTEE requires specialized climbing tree-trim workers due to its high percentage of backlot construction. Safe and experienced climbing crews are difficult to acquire, and it could be difficult to grow the work force to the levels needed.
- Large weather events, such as hurricanes, often draw traveling non-local trimmers from across the country for the overtime hours and generous per diem rates. Not only could this delay implementation of our Surge program due to a temporary loss of workforce, but trimmers could choose not to return after the storm if other opportunities prove more attractive.

The Company has several initiatives underway to increase the local tree trimmer workforce and reduce dependency on non-local trimmers.

- The Company has partnered with the City of Detroit, IBEW Local 17 and its tree-trimming contractors to develop and implement a pre-woodsman training pilot program to satisfy the demand for qualified tree trimmers. The pilot tree trimming academy is located within the City of Detroit with the goal of preparing local resident candidates to work as woodsmen. Graduates of the six-week academy enter the nine-day boot camp that was designed in partnership with IBEW Local 17 and the Company's tree-trimming contractors. The boot camp gives participants intensive training and hands-on work experience on subjects such as safety, climbing systems, climbing techniques, arborist equipment, arborist tools, commercial vehicle operation, tree species identification, communication with line crews,

customer relations, and aerial rescue techniques. Boot camp graduates enter the Line Clearance Tree Trimming Apprenticeship Program. The 5,000-hour apprenticeship program, which includes 160 hours of classroom training, is recognized by the Department of Labor as an approved apprenticeship program and, as one of two programs in the United States, is benchmarked throughout the industry.

- The Company has implemented a pilot tree-trimming training program into the Vocational Village at Parnall Correctional Facility in Jackson. The training program was developed to allow returning citizens to directly enter the apprenticeship program upon leaving the correctional facility. In selecting applicants, the Vocational Village administration heavily weighs the applicant's county of residence to decrease the distance they would have to commute to work once they are released. Graduates are paired with jobs upon graduation.
- The Company is encouraging our local contractors to hire additional new trimmers through their normal processes, which in the past was used primarily to replace attrition.
- The Company is partnering with Local 17 and its contractors to reach out to local high schools, career fairs, and local nonprofit organizations to introduce the tree-trimming trade to interested candidates.

10.8 Conclusion

The tree-trimming program is one of the most impactful and important programs in the Company's commitment to reliability and safety. The program has shown that it significantly decreases system risk (specifically reduced wire downs), increases reliability (fewer and shorter outages), and decreases reactive trouble costs. Continuing to bring the system on-cycle and completing the Surge program will address the grid's current needs, while supporting the Company's long-term goals of resiliency and affordability.

11 Infrastructure Redesign and Modernization



Infrastructure Redesign and Modernization is one of the four pillars in DTEE's Distribution Grid Plan framework. It includes elimination of loading constraints, redesign and rebuild of the subtransmission system, and conversion and consolidation of the 4.8 kV and 8.3 kV systems to the more standard 13.2 kV system. The ultimate goal of these projects and investments is to enhance safety, reliability, and grid resilience for our customers.

As identified in the Grid Modernization section, the lack of capacity on significant portions of DTEE's distribution and subtransmission systems is a grid need that must be addressed to accommodate our customers' increased adoption of DERs and EVs. This section provides a detailed description for each of the infrastructure redesign projects identified for the next five years, based on current loading constraints. This area of redesign and modernization will likely grow in importance as DTEE continues to focus on the grid of the future and the role of increased electrification and DG/DS develop.

11.1 Distribution load relief projects

It is necessary to assess the load on the system and its impact on individual pieces of equipment in order to ensure that distribution system capacity exists to serve demand. Throughout this section, capacity refers to distribution assets' ability to deliver electrical power, not the ability to generate the power. There are areas within the system where the peak load is expected to increase. These increases may be the result of new load, or of customers relocating geographically from one area of the system to another. It is critical to identify expected capacity needs well in advance of the expected load increase in order to complete planning, siting, permitting, and construction of necessary infrastructure and to provide the expected service and reliability to customers.

Capacity needs are considered for two conditions: normal state and contingency states. The normal state exists when all equipment and components are in service and operating as designed. The contingency states exist when there is either a temporary planned equipment shutdown, the loss/failure of a component of the electric power system (e.g., subtransmission line), or the loss/failure of individual equipment (e.g., transformer, cable or breaker). Under contingency conditions, equipment in the rest of the system may see an increase in loading to compensate for the out-of-service equipment, therefore requiring additional capacity above normal state.

To meet the two capacity requirements, most components and equipment have two ratings: day-to-day and emergency. These ratings are calculated to maintain the viability of an asset throughout its expected useful life. Operating equipment above its designated ratings can cause immediate failure or accelerate end-of-life.

- The **day-to-day** rating (for normal state conditions) is the load level that the equipment can be operated at for its expected life span.
- The **emergency** rating (for contingency state conditions) is typically higher than the day-to-day rating and indicates the load level that the equipment can operate for short periods of time only. Operating at the emergency rating adds stress to the equipment and shortens its lifespan. If a piece of equipment exceeds its emergency rating, DTEE's System Operations Center takes immediate steps to transfer load or shed load if necessary.
- Substations also have a firm rating, which is the maximum load the substation can carry under a single contingency condition and is based on the lowest emergency rating of all the substation equipment that is required to serve the load.

To ensure that expected load growth can be served within the equipment ratings, DTEE planning engineers conduct annual Area Load Analyses (ALA). These analyses include verification of equipment ratings and substation firm ratings, past loading data, system conditions and configurations, known new loads, and input from large customers and municipal officials about potential development. Based on DTEE's 2020 ALA study, approximately 35% of distribution substations have loading constraints, including substations operating over its firm ratings, substation equipment working near or over its day-to-day rating during peak hours, or circuit equipment operating over its day-to-day rating during peak hours. Since the study is conducted for

peak load conditions, the over firm or overload assessments are usually of concern for just a small percent of hours annually, depending on loading characteristics of individual substations or subtransmission lines. As such, additional substation capacity is needed in certain areas to prevent customer interruptions during a single contingency event and to help maintain the useful life of the existing equipment.

For areas and cities that have experienced steady and/or strong load growth, capital investment is required to add or upgrade overhead or underground lines (subtransmission and/or distribution) and/or to expand or build new substation capacity.

A strategic load relief project is often the result of a combination of general load growth, specific customer connection requests, aging infrastructure replacement and reliability improvement needs. Some strategic load relief projects, such as building a new general purpose substation, may have the added benefits of mitigating substation risk at neighboring substations and improving overall reliability for the area.

11.1.1 Load relief prioritization

As mentioned above, approximately 35% of DTEE's distribution substations have loading constraints based on DTEE's 2020 ALA study. It is not feasible, nor cost effective, to address all the loading constraints at once due to other higher priority investments, as explained in DTEE's Global Prioritization Model (GPM) in Section 5. Therefore, DTEE has developed a prioritization methodology to assess substation loading constraints based on criticality. Results of the load relief prioritization are fed into DTEE's Global Prioritization Model to ultimately drive future capital planning.

This load relief prioritization methodology has evolved over time and is now based on four variables:

- **Substation equipment overload** (peak load nearing or exceeding substation equipment day-to-day ratings or nearing substation equipment emergency ratings)—The limiting element is often a substation transformer or regulator, but it can sometimes be other secondary equipment at the substation such as conductor, cable, bus bar, or breaker. A score is given to substations based on the ratio of load to rating with the highest score representing the most severe overload condition.

- **Substation Over Firm** (peak load exceeding substation firm rating under contingency conditions) – A score is given to substations that experience varying degrees of megavolt-amperes (MVA) load over their firm ratings with and without load transfer (or load jumpering). Under contingency conditions, any MVA load over substation firm rating represents customers that cannot be served by the substation itself. Some of this load can be subsequently transferred to adjacent substations for the duration of the contingency. This creates a difference in values and scoring in MVA over firm before load transfer, and MVA over firm after load transfer.
- **Circuit equipment overload** (peak load exceeding circuit equipment day-to-day ratings or exceeding circuit equipment emergency ratings – While the original load relief prioritization centered on substation equipment, the latest iteration has been expanded to include circuit equipment overloads, and thus provides visibility around the overall loading constraints of the distribution system. The limiting element for distribution circuits is often the underground cable exiting the substation, but it can also be other equipment such as reclosers or overhead conductor. A score is given to each circuit based on the ratio of load to rating with the highest score representing the most severe overload condition. The circuit scores are then summed for each substation.
- **Strong load growth prospect** – The overall score given to a substation is increased if the substation has known load growth in the area. Being able to create a project that not only serves the new customers but also resolves the substation overload or over firm situation is preferred from the economic perspective.

The final priority ranking of distribution load relief is a combination of the above four variables. Exhibit 11.1.1.1 lists the top 25 substations based on this methodology with applicable variables.

Exhibit 11.1.1.1 Top 25 Load Relief Substations Overview

Index	Substation	Voltage (kV)	Region	Community	Substation Equipment Overload	Substation Over Firm	Circuit Equipment Overload	Strong Load Growth Prospect
1	Alfred	13.2	SE	Detroit			✓	✓
2	Hawthorne	4.8	SE	Dearborn Heights	✓	✓	✓	
3	St. Antoine	13.2	SE	Detroit			✓	✓
4	Madison	4.8	SE	Detroit			✓	✓
5	Cato	13.2	SE	Detroit			✓	✓
6	Garfield	4.8	SE	Detroit			✓	✓
7	Temple	13.2	SE	Detroit			✓	✓
8	Walker	4.8	SE	Detroit			✓	✓
9	Cody	13.2	NW	South Lyon		✓	✓	✓
10	Prospect	4.8	SW	Ann Arbor	✓	✓	✓	
11	White Lake	4.8	NW	White Lake	✓	✓	✓	
12	Howard	4.8	SE	Detroit			✓	✓
13	Cato	4.8	SE	Detroit			✓	✓
14	Snover	4.8	NE	Snover	✓	✓	✓	
15	Shaw	4.8	NE	Imlay City	✓	✓	✓	
16	Roseville	4.8	NE	Roseville	✓		✓	
17	Golf	13.2	NE	Macomb			✓	

Index	Substation	Voltage (kV)	Region	Community	Substation Equipment Overload	Substation Over Firm	Circuit Equipment Overload	Strong Load Growth Prospect
18	Daly	4.8	SE	Dearborn Heights	✓	✓	✓	
19	Marlette	13.2	NE	Marlette	✓		✓	
20	North Branch	13.2	NE	North Branch	✓	✓	✓	
21	Brown City	4.8	NE	Brown City	✓	✓	✓	
22	Mayville	4.8	NE	Mayville	✓	✓	✓	
23	Carleton	4.8	SW	Carleton	✓	✓	✓	
24	Wolverine	13.2	SW	Ann Arbor	✓	✓	✓	
25	Goodison	13.2	NW	Oakland	✓		✓	✓

11.1.2 Consideration of non-wire alternatives

DTEE continues to investigate Non-wire alternatives (NWAs) to provide targeted load relief to the electrical system. If a substation or distribution circuit is identified as a potential candidate for NWA through the screening process, planning engineers would work with technology and EWR/DR experts to evaluate detailed designs. The process by which NWAs are evaluated by planning engineers as a potential load relief solution is discussed in detail in Section 4.3. Pilot projects as NWA solutions are discussed in Section 12.7.

DTEE will leverage learnings from in-house pilots and other utilities to refine its criteria, develop new tools and advance the overall process for pursuing NWAs as part of the distribution project review process.

11.1.3 Projects to address distribution loading

Exhibit 11.1.3.1 lists the scope of work planned to address the load relief for the top ranked substations. There are also other projects planned which address load relief along with additional project drivers. These projects are shown in Exhibit 11.1.3.2 including the additional drivers.

Exhibit 11.1.3.1 Top 25 Load Relief Substation - Project Summary

Index	Substation	Region	Community	Project and Scope of Work
1	Alfred	SE	Detroit	<u>CODI – Alfred Expansion</u> : Expand 13.2 kV Alfred substation by installing 3rd transformer and a 12-position switchgear
2	Hawthorne	SE	Dearborn Heights	<u>4.8 kV Conversion: Hawthorne Relief and Circuit Conversion</u> : - Extend overhead from Mallard distribution circuits and convert 4.8 kV areas out of Glendale and Hawthorne - Upgrade Biltmore 13.2 kV substation and establish 3 new distribution circuits - Replace switchgear at Villa substation - Expand Mallard substation and build two new general purpose substations to convert and remove load from Hawthorne, Glendale, Villa, and Daly substations
3	St. Antoine	SE	Detroit	Load relief will be provided by the <u>CODI – Alfred Expansion</u> project, see index #1, and <u>CODI – New Corktown Substation</u> : Construct new 13.2 kV Corktown substation which will support load relief of surrounding substations including St. Antoine
4	Madison	SE	Detroit	<u>CODI - Madison Conversion</u> : - Rebuild 31 miles of network feeder cable - Rebuild 30 miles of system cable

				• Replace or remove 92 netbank transformers • Convert 24 primary customers • Convert three miles of overhead • Convert and consolidate the circuits to 13.2 kV fed by Temple substation • Decommission Madison substation
5	Cato (13.2)	SE	Detroit	<u>CODI – Cato Substation Expansion:</u> Expand 13.2 kV Cato substation by installing 3 rd transformer and a 12-position switchgear
6	Garfield	SE	Detroit	<u>CODI – Garfield Network Conversion:</u> • Rebuild 36 miles of network feeder cable • Replace or remove 78 netbank transformers • Convert 24 miles of overhead • Convert and consolidate the circuits to 13.2 kV fed by Stone Pool substation • Remove 4.8 kV and 24 kV cable and decommission Garfield substation
7	Temple	SE	Detroit	<u>CODI – Midtown Expansion:</u> Expand 13.2 kV Midtown substation by installing 3 rd transformer and a 12-position switchgear
8	Walker	SE	Detroit	<u>CODI – Islandview Substation and Walker Conversion:</u> • Construct new 13.2 kV Islandview substation • Convert 32 existing 4.8 kV circuits from Walker and Pulford • Decommission Walker (built in 1923) substation

				Decommission aging 24kV cables and infrastructure
9	Cody	NW	South Lyon	<u>System Loading: Cody:</u> - Reconfigure circuits by performing phase balancing and load transfers - Rebuilding of circuit backbones to meet current standards - Establish new circuit backbones and build to current standards - Install sectionalizing and voltage correction equipment - Convert section of 4.8 kV to 13.2 kV
10	Prospect	SW	Ann Arbor	<u>System Loading: Prospect:</u> Replace existing transformer
11	White Lake	NW	White Lake	<u>4.8 kV Conversion: White Lake Decommissioning and Circuit Conversion:</u> - Build a new 40 kV to 13.2 kV substation - Convert distribution circuit 307 White Lake to a 13.2 kV circuit - Establish four load carrying circuits supporting White Lake Substation load and a portion of Clyde Substation - Establish new jumpering points and loop schemes between the new substation and Clyde, Wardlow, Teggerdine, Placid, and Osprey substations - Decommission White Lake substation
12	Howard	SE	Detroit	<u>CODI – Howard Conversion:</u> - Rebuild 6 miles of network feeder cable - Rebuild 12 miles of system cable

				• Replace or remove 89 netbank transformers • Convert 26 primary customers • Convert three miles of overhead • Convert and consolidate the circuits to 13.2 kV fed by Corktown, St. Antoine, Cato, and Temple substations • Decommission Howard substation
13	Cato (4.8)	SE	Detroit	<u>CODI – Cato/Orchard Conversion</u> (Cato portion): • Rebuild 17 miles of system cable • Convert 15 primary customers • Convert and consolidate the circuits to 13.2 kV fed by Temple substation • Decommission Cato 4.8 kV substation
14	Snover	NE	Snover	<u>4.8 kV Conversion: Snover Decommissioning and Circuit Conversion:</u> • Build a new 13.2 kV Class T substation • Reconductor approximately 13 miles on Snover circuits • Convert and transfer load from the Snover distribution circuits to the new substation • Decommission Snover Substation
15	Shaw	NE	Imlay City	Load relief will be provided by the <u>System Loading: Brown City</u> project, see index #21, and <u>System Loading: Otsego/Capac/Shaw:</u> • Install a new 13.2 kV skidmount substation

				Convert and transfer about 10 miles of Otsego, Capac, and Shaw circuits to the new skidmount substation
16	Roseville	NE	Roseville	A detailed area plan study will be performed for Roseville and the surrounding substations to determine a holistic solution that will address load relief for Roseville and is beneficial to the overall area
17	Golf	NE	Macomb	<u>System Loading: Macomb/Golf:</u> - Expand Macomb substation by adding a 3rd transformer and two 12-position switchgears - Install about 3 miles of underground conduit and 5 miles of underground cable for new distribution circuits - Reconductor about 5 miles of overhead line - Transfer load from Golf and Burbank substation areas to new Macomb distribution circuits
18	Daly	SE	Dearborn Heights	Load relief will be provided by the <u>4.8 kV Conversion: Hawthorne Relief and Circuit Conversion</u> project, see Index #2
19	Marlette	NE	Marlette	Load relief will be provided by the <u>System Loading: Brown City</u> project, see Index #21
20	North Branch	NE	North Branch	No additional load growth projected – will be re-evaluated during Fall 2021 Area Load Analysis
21	Brown City	NE	Brown City	<u>System Loading: Brown City:</u> - Build a new 13.2 kV Class T substation at Bennett station - Reconductor approximately 3.5 miles on Tacoma circuits - Reconductor approximately 4.5 miles on Marlette DC 8281

				- Rebuild and convert approximately 8 miles of Brown City distribution circuits from 4.8 kV to 13.2 kV - Transfer load from Brown City, Shaw, Tacoma, and Marlette substation areas to the new substation - Decommission Brown City Substation
22	Mayville	NE	Mayville	<u>System Loading: Mayville:</u> - Construct a new 13.2 kV Skidmount STDF and transfer 1200 kVA from Mayville and 1500 kVA from Clifford to new STDF - Reconductor 7 miles of overhead distribution - Install 5 miles of neutral - Convert about 12 miles of overhead distribution from 4.8 kV to 13.2 kV
23	Carleton	SW	Carleton	<u>System Loading: Carleton:</u> - Reroute 4.8kV and 40kV overhead on the substation property - Expand substation fence on the north side - Install new control center - Replace both transformers at Carleton and upgrade associated equipment - Re-establish jumpering point between the two circuits
24	Wolverine	SW	Ann Arbor	<u>System Loading: Grenada:</u> - Install new 13.2 kV Class A substation with two transformers and 12-position switchgear - Install one mile of underground conduit and 2 miles of underground cable

				• Rebuild two miles of overhead backbone • Convert iso-pockets on DC 9813 Grenada by replacing 10 miles of overhead conductors • Rebuild and convert 20 miles of overhead line from the Prospect substation area • Establish new distribution circuits to relieve load from Grenada, Wolverine, Prospect, and Price substations
25	Goodison	NW	Oakland	<u>System Loading: Goodison:</u> • Construct a 13.2 kV skidmount substation • Establish two new distribution circuits to relieve load from Goodison • Install underground conduit and cable to connect to existing circuits • Extend one primary main and create two new primary mains • Reconnector 1.3 miles of overhead line

Exhibit 11.1.3.2 Load Relief Projects with Additional Drivers

Substation(s) / Location	Region	Community	Drivers	Project and Scope of Work
Mack	SE	Detroit	- Inoperable tap changers on adjacent transformers - Long shutdowns causing increased risk of stranded load if energized transformer lost	<u>System Loading: Mack Transformer 102:</u> Install new 120 kV to 24 kV Transformer 102 at Mack to allow for repair of tap changers and future shutdowns
Mandalay	SE	Birmingham, Beverly Hills, Clawson	- Circuit overload - Poor reliability performance	<u>System Loading: Mandy 307 Load Transfer:</u> Convert 7,600' of 4.8kV distribution circuit 307 Mandalay and transfer to distribution circuit 9893 Ariel and install sectionalizing
Richmond / Armada	NE	Richmond, Armada	- Antiquated 4.8 kV switchgear at Richmond - Richmond and Armada 4.8 kV substations over firm - Circuit overload on DC 302 Armada - Expected load growth with lack of capacity - Limited jumpering	<u>System Loading: Richmond/Armada:</u> - Upgrade Richmond substation to a Class A substation, installing a new transformer, replacing the existing transformer, and installing new switchgear - Install 1,000 feet of underground conduit - Install 2,000 feet of underground cable - Decommission the existing Richmond 4.8kV substation and switchgear - Reconductor and convert 15 miles of overhead distribution line - Decommission Armada 4.8kV Substation

Port Sanilac	NE	Port Sanilac	- Port Sanilac and Foster substations over firm - Day-to-day circuit overload on DC 302 Port Sanilac and DC 301 Foster just under day-to-day with minimal differences between day-to-day and emergency ratings for both circuits - Limited jumpering and capacity	<u>System Loading: Port Sanilac:</u> - Install new Skidmount STDF between Foster & Port Sanilac - Establish new 13.2 kV distribution circuits - Rebuild and convert 3 miles of DC301 Foster - Rebuild and convert 3 miles of DC302 Port Sanilac - Decommission Foster substation
Jewell	NW	Washington	- Jewell substation just under firm, but has two circuits over day-to-day rating and three just under day-to-day rating - Load center doesn't allow for transferring load without causing issues on adjacent circuits - Expected load growth with lack of capacity - Closest adjacent substation is 4.8 kV Washington	<u>System Loading: Jewell:</u> - Build a new 13.2 kV substation with two transformers and 12-position switchgear - Build 6 miles of underground conduit and install 8 miles of underground cable - Establish 4 new distribution circuits to accommodate load transfers from Jewell circuits - Convert and transfer about 11 miles of overhead line from Washington and decommission Washington 4.8 kV substation
Wixom	NW	Wixom	- Wixom substation over firm - Circuit overload on DC 8239 with 4 other circuits	<u>System Loading: Wixom:</u> - Add a third transformer at Wixom substation - Replace existing switchgear with two 12-position switchgear

			approaching day to day rating - Expected load growth with lack of capacity - Existing switchgear nearing end of life	Build multiple miles of underground cable (22), conduit (5) and overhead lines (4)
New Baltimore / Chesterfield	NE	New Baltimore, Chesterfield	- New Baltimore and Chesterfield substations over firm with New Baltimore limited by subtransmission feed - Three circuits exceeding day-to-day rating with another at 99% - High load growth in Chesterfield substation area	<u>System Loading: New Baltimore/Chesterfield:</u> - Build a new 13.2 kV substation - Install 4,000 feet of underground conduit and 8,000 feet of underground cable - Reconductor 12 miles of overhead line - Establish 5 new distribution circuits
Diamond	SW	Dexter	- Diamond substation significantly over firm - Expected load growth with lack of capacity	<u>System Loading: Diamond:</u> - Upgrade one substation transformer with a larger unit - Rebuild primary and secondary tower to meet current standards - Install circuit switcher, transformer control panel in a new control center, and upgrade relays

Exhibit 11.1.3.3 lists the projected costs and timeline for load relief projects. This exhibit excludes the CODI and 4.8 kV conversion projects providing load relief as the projected investment for those projects is listed in Section 11.3 and Section 11.4 respectively. It is important to note that DTEE has based the cost and timeline estimates for the identified projects on the best knowledge and information known today. Actual project cost and timeline could deviate from the projection due to various unforeseen factors or new information/learnings.

Exhibit 11.1.3.3 Projected Costs and Timeline for Load Relief Projects

Project	2021	2022	2023	2024	2025	2021-2025 Cost Estimate (\$ million)	2026 to 2029 Cost Estimate (\$ million)
System Loading: Mack Transformer 102						\$0.2	\$-
System Loading: Carleton						\$2	\$-
System Loading: Mandy 307 Load Transfer						\$1	\$-
System Loading: Brown City						\$30	\$-
System Loading: Richmond/Armada						\$23	\$3
System Loading: Grenada						\$18	\$19
System Loading: Port Sanilac						\$18	\$-
System Loading: Prospect						\$1	\$-
System Loading: Cody						\$3	\$-
System Loading: Otsego/Capac/Shaw						\$12	\$2

Project	2021	2022	2023	2024	2025	2021-2025 Cost Estimate (\$ million)	2026 to 2029 Cost Estimate (\$ million)
System Loading: Macomb/Golf						\$19	\$21
System Loading: Jewell						\$5	\$51
System Loading: Mayville						\$3	\$28
System Loading: Goodison						\$1	\$13
System Loading: Wixom						\$1	\$83
System Loading: New Baltimore/Chesterfield						\$1	\$32
System Loading: Diamond						\$1	\$4

It should be noted that the load relief projects previously listed are proposed based on DTEE's assessments of area load growth and system loading conditions as of today and are subject to change in the future. Future area load growth is constantly evolving due to changes in general economic trends, utilization of demand response and energy efficiency measures, and changing customer trends such as adoption of EVs. In addition, as mentioned in Section 4.2, DTEE expects to enhance and refine substation and circuit level forecasting in the next three years. The loading information used in the current ALA process primarily relies on past years' peak load and known new load, based on customer or input from municipal officials. Planned future enhancements include developing forecasts at a more granular/hourly level and incorporating impacts of adoption of new technologies. Regardless of forecasting methodology, uncertainty remains in the nature of predicting future load growth, so the projects proposed to address loading concerns may change from year-to-year. Based on project proposals known today, DTEE estimates the annual capital investment on load relief projects as shown in Exhibit 11.1.3.4.

Exhibit 11.1.3.4 Projected Load Relief Capital Investment

	2021	2022	2023	2024	2025	Total Investment 2021-2025
Load Relief Capital Investment (\$ million)	\$6	\$11	\$20	\$50	\$53	\$140

11.2 Subtransmission Redesign and Rebuild Program

DTEE's subtransmission system is operated at a voltage of either 24kV, 4kV, or 120kV and is used to step down transmission voltage to serve distribution and industrial substations. The design of the system is intended to introduce redundancy to the feed points of the distribution and industrial substations, resulting in continued service to customers during a single contingency situation. DTEE's subtransmission system differs from that of most other utilities because it includes both radial and network designs. The radial configuration, called a trunk line, has one source station and can feed one or multiple substations. The network configuration, called a tie line, has multiple source stations and feeds multiple substations. DTEE utilizes a coordinated system of automatic pole top switches (APTS) as discussed in Section 8.13, and line section breakers on the networked tie lines to isolate faults and maintain service to customers in single contingency failure situations.

The subtransmission redesign and rebuild program is focused on installing station equipment, as well as rebuilding both the overhead and underground portions of the subtransmission system. The station work involves the installation of large transformers, capacitor banks and associated equipment. The overhead work includes the replacement of old wooden poles with new steel poles, porcelain insulators with polymer clamp top insulators, and small aging conductors – which are often damaged by multiple lightning strikes – with larger, stronger conductors. The underground work consists of replacing at-risk or overloaded cable with new sections and rebuilding cable poles to new specifications.

The need for a subtransmission redesign and rebuild program has become increasingly apparent during the attempted normal operations of the electric system, through the evaluation of new customer load requests, and during review of the increasing number of outages on DTEE's subtransmission system. The subtransmission system has seen increasing limitations to serve DTEE's customers during a single contingency situation.

Though the system was originally designed to efficiently serve customers with the necessary redundancy and in most cases to maintain service under a single contingency, system loading and the impact of outages on customer operations have increased over time. The increase in system loading has resulted in increasing limitations of the subtransmission system to serve customers during a single contingency situation. This condition has led to an increased number of outages and extended the length of time to restore large outages affecting one or multiple substations. Additionally, these system limitations, coupled with increasing customer sensitivity to outages, has led to the costly deployment of mobile generation or portable substations for normal planned work.

In addition, like the distribution system, the subtransmission system is aging, leading to an increased number of failures on both the overhead and underground system. These failures can result in either a large sustained outage or the loss of redundancy, depending on the system configuration and any existing violations of the planning criteria. Some areas of the overhead system are in deep-wooded areas and along railroads, increasing the difficulty for restoring service or maintaining equipment. The underground cables can be difficult to replace and restore within the conduit system, increasing the length of time that these lines are not available for use to multiple days.

Subtransmission Planning Engineers analyze the condition of the system on a yearly basis to determine existing and projected limitations to serving customers in a single contingency situation. This analysis is conducted by utilizing industry-standard modeling software (PSSE & TARA) with individual substation loads submitted by the Distribution Planning & Industrial Power Engineers and the models used by the Midwest Independent System Operator (MISO). The models provided by MISO include multiple electric system scenarios, including current and future peak loading conditions. Using these models, a study is run on each individual subtransmission line with all possible contingency

situations assessed to identify all violations of subtransmission planning criteria. The planning criteria focuses on both thermal overloads and voltage violations under normal system conditions and during a single contingency configuration.

A thermal overload indicates that equipment on the circuit or station exceeds its rating, and a voltage violation indicates that the voltage on at least part of the circuit is no longer within an acceptable range. The presence of planning criteria violations on a subtransmission circuits has two main implications. First, it means the system has additional operational risk, particularly during days with high load. Ideally, the subtransmission system would be able to serve all load at any time under a single contingency. If an equipment failure creates a thermal overload or voltage violation, load will be shed to protect equipment and maintain system integrity, leading to an outage. Second, the planning criteria violation means that the circuit has no capacity for additional load, either from a new large customer connection or general area load growth.

The results of the analysis on the subtransmission system determined that approximately one-third of the circuits on the subtransmission system violate a planning criteria under either normal or single contingency. A high-level estimate is that it would cost \$2 billion to address all of these violations. To begin addressing the constraints on the subtransmission system, the planning criteria violations have been ranked based on severity, and projects have been developed to address the limiting elements on the system, which will be discussed in the next two sub-sections.

The Subtransmission Planning Engineering group also closely monitors the reliability of the system, identifies circuits with multiple reliability issues, and works with communities that express reliability concerns to develop plans to address customer concerns. In developing a project plan, the Planning Engineers consider existing routes of the lines as well as sections where wire down events have occurred. Based on the analysis, the Planning Engineers identify which sections to focus on for redesign and rebuild. The project scope can include rebuilding the lines to the current more resilient construction standards and relocating the lines to road accessibility wherever possible.

In addition, DTEE performed an in-depth study of the subtransmission system in the Thumb area in 2019 by closely analyzing the current condition and performance of the

system, and evaluating what long-term options would best serve the needs of its customers with respect to upgrades to the subtransmission system in the area.

Through this in-depth 2019 study and close monitoring of reliability performance, DTEE has concluded that the condition of the subtransmission system has degraded to a point where a much more urgent and large-scale investment plan is required. A robust subtransmission system is required to serve large amount of dense load and provide the necessary reliability through well-designed redundant systems, where both large and small customers will benefit.

To achieve these improvements on the system, DTEE must rebuild the lines with larger conductors, and redesign the configuration of circuits to meet the load profiles of current customers, all while providing capacity for future expansion. The additional capacity provided by the subtransmission system redesign and rebuild would support the community by allowing for new business expansion and electrification integration in the area with reduced construction time and cost to connect or interconnect new customers.

The subtransmission system analysis has also shown the importance of taking a holistic view of an area when addressing loading issues on a line. This is especially evident when redesigning the networked tie system in rural areas. Improving capacity on a single circuit often requires work on multiple adjacent circuits, and the addition of transformers and capacitor banks at source stations due to the required redundancy of the system.

When designing system upgrades and executing rebuilds, DTEE will keep the future in mind by using strengthened standards (steel poles, larger conductors) and build in the appropriate system redundancy. For station upgrades, designs will ensure additional space, ground mats, oil containment, and the necessary below-grade infrastructure to allow for future expansion.

11.2.1 Subtransmission prioritization

DTEE considers multiple criteria when evaluating priority of the subtransmission redesign and rebuild projects. Consistent with the rest of the strategic investment portfolio, the load relief prioritization scores for subtransmission feed into the GPM model to help formulate the future capital plan. The Load Relief model was updated to capture considerations on

established planning criteria for subtransmission redesign & rebuild projects. Subtransmission load relief prioritization scores are assessed based on load loss for single contingency, load over allowable emergency rating for single contingency, load over day to day ratings, strong load growth prospect, and whether there is a voltage violation. Exhibit 11.2.2.1 below shows the details of the criteria. Exhibit 11.2.2.2 lists the top 25 subtransmission load relief projects based on this methodology with applicable dimensions.

Exhibit 11.2.2.1 Subtransmission Load Priority Criteria

Subtransmission Dimensions	Definition
Load Loss for Single Contingency	Total load that will be shed when a subtransmission line can no longer support the substation and does not have a back-up or the back-up cannot support the load
Load over allowable Emergency Rating for Single Contingency	Total load when a subtransmission line exceeds its emergency rating of its alternative route during an event (i.e., outage)
Load over Day to Day Rating, Normal Conditions	Total load that exceeds the rating of a subtransmission line during normal conditions
Strong Load Growth Prospect	Consideration given to subtransmission lines that are predicted to experience load growth
Voltage Violation	Consideration given to subtransmission lines that experience low voltage conditions when they are not in their normal configuration (i.e., due to an outage)

Exhibit 11.2.1.2 Top 25 Subtransmission Load Relief Project Dimensions Overview

Index	Project	Region	Community	Load loss for single contingency	Load over allowable Emergency Rating for Single Contingency	Load over Day to Day Rating	Strong Load Growth Prospect	Voltage Violation
1	Tie 3416 - Bad Axe	NE	Bad Axe	✓	✓	✓	✓	✓
2	Trunk 7333	NW	Madison Heights	✓	✓	✓		
3	Trunk 7106	SE	Southfield	✓	✓	✓	✓	
4	Trunk 4217	SE	Grosse Pointe, Harper Woods, Detroit	✓	✓	✓		
5	Boyne	NE	Macomb	✓	✓		✓	
6	Bad Axe - Transformer 102 Addition	NE	Bad Axe	✓	✓		✓	✓
7	Trunk 4266	SE	Eastpointe	✓	✓	✓		
8	Tie6907	NW	Rochester Hills	✓	✓		✓	✓
9	Tie 3416 - Pigeon	NE	Elkton, Pigeon	✓			✓	✓
10	Trunk 4911 Voltage	NE	Lenox, Chesterfield, New Baltimore, Ira	✓			✓	✓

11	Trunk 2255	SE	Detroit	✓	✓	✓		
12	Trunk 2237	SE	Redford	✓	✓	✓		
13	Tie 7504	NE	Novesta	✓			✓	✓
14	Derby	NE	Vassar				✓	✓
15	Sandusky Transformer 101 Breaker	NE	Sandusky			✓		
16	Sandusky - Transformer 102 addition	NE	Sandusky	✓	✓			✓
17	Pigeon Area	NE	Multiple	✓	✓			✓
18	Trunk 3509	NW	Royal Oak		✓	✓		
19	Trunk 3508	NW	Troy	✓	✓			
20	Tie 6602	SW	Lima Twp	✓	✓			
21	Trinity	SW	Monroe	✓				✓
22	Bernard	NE	Wales Twp	✓				✓
23	Trunk 4911 Load	NE	Lenox, Chesterfield, New Baltimore, Ira	✓	✓			
24	Tie 1568	SW	Ypsilanti	✓	✓			
25	Trunk 328	SE	Detroit	✓	✓			

11.2.2 Subtransmission projects and programs

As DTEE continues to increase investment to improve the subtransmission system, the portfolio of over 50 projects listed below includes the projects previously mentioned in Exhibit 11.2.1.2, additional subtransmission load relief projects, and projects developed to improve the reliability of the system and address planning criteria violations. Exhibit 11.2.2.1 lists the community and scope of work for the identified subtransmission redesign and rebuild projects. Exhibit 11.2.2.2 lists the projected costs and timeline for the projects. It is important to note that the cost and timeline estimates for the identified projects are based on the best knowledge and information known today by DTEE. Actual project cost and timeline could deviate from the projection due to various unforeseen factors or new information/learnings. The planned investments in the subtransmission system from 2021-2025 mark a significant increase from previous years. However, as shown in Exhibit 11.2.2.2, the total for the program over the next five years is less than \$500M. This means that it would require approximately two decades of spending at the existing spend rate to address current planning criteria violations.

DTEE intends to continue to grow the investment in the subtransmission system in future years, with the pace of that growth depending on the signposts that materialize for the scenarios discussed in the Grid Modernization Process (Section 3), specifically the Electrification and DG/DS scenarios which could cause additional circuits on the subtransmission system to violate planning criteria.

Exhibit 11.2.2.1 Subtransmission Projects Summary

Tie or Trunk	Community	Scope
Subtransmission Redesign & Rebuild: Maxwell Amherst	Detroit	- Decommission and remove Amherst Transformer 2 and associated station equipment and cable - Install new Transformer 2 and associated station equipment and cable
Ann Arbor System Improvements	Ann Arbor	Construct two new substations and five miles of 120 kV lines, and reconfigure subtransmission tie lines and trunk lines
Transformer High Side Protection Program	Various	- Install high side switching devices on 19 subtransmission and distribution transformers located at 15 different stations within DTEE territory 8 of these transformers will also require relocation and/or replacement due to space constraints at their current sites
Station Upgrade: Cortland Station Expansion	Highland Park	Purchase property adjacent to Cortland station and install two new transformers, 104 and 105 to replace existing transformers 101 and 102 to resolve physical replacement restrictions at existing Cortland site
40kV Capacitor Banks at Armada and Adair	Armada, Columbus	Install one 40 kV 9.6 MVAR capacitor at Armada substation and one 40 kV 9.6 MVAR capacitor at Adair substation
Subtransmission Breaker Short Circuit Violations	Multiple	Install reactors or replace breakers at multiple stations
Subtransmission Redesign and Rebuild: Trunk 7106	Southfield	Reconfigure Trunk 7106 into two trunk lines – Trunk 7106 and Trunk 7187

		Install 2.3 miles of underground cable for new Trunk 7187
Subtransmission Redesign and Rebuild: Trunk 7333	Madison Heights	• Reconfigure Trunk 7333T into two trunk lines - Trunk 7333 and Trunk 7370 • Install 3 miles of underground cable for new Trunk 7370 • Install a 40 kV breaker and two 40 kV disconnects (2000 Amp) at position SB Chestnut
Subtransmission Redesign and Rebuild: Trunk 2255	Detroit	Install 1.3 miles of new cable to split load of Trunk 2255 while removing old cable
Subtransmission Redesign and Rebuild: Boyne	Macomb	• Install new 120-40 kV transformer 102, 40 kV section breaker, and bus 2 at Boyne • Install one 40 kV line breaker, disconnect on the low side of transformer 102, install circuit switcher on the high side of transformer 102 • Install disconnects and breaker for the new position (new trunk) on Bus 2 and install relay panels • Install 1.5 miles of new conduit and 40 kV cable • Install 2.5 miles of new overhead 40 kV • Replace 2.1 miles of overhead conductor on Trunk 7909 • Replace 0.3 miles of overhead conductor and 0.4 miles of underground cable on Trunk 6759
Subtransmission Redesign and Rebuild: Badax Transformer 102 Addition	Bad Axe	• Expand Badax 40 kV Bus 2 and add two 40 kV positions • Install one 120-40 kV 45/60/75 MVA Transformer 102 from ITC's Bus 102 to new 40 kV position on Badax Bus two

		• Install one 120 kV S&C 2020 circuit switcher • Install one 40 kV 2000A breaker • Install two 40 kV 1200A disconnects
Subtransmission Redesign and Rebuild: Tie 4105	Lexington, Croswell	• Reconductor 35 miles of 40 kV overhead line • Rebuild 18.6 miles of distribution underbuild
Subtransmission Redesign and Rebuild: Trunk 2237-ST	Redford	Install 0.8 miles of underground cable to parallel overloaded Trunk 2237-ST cable
Subtransmission Redesign and Rebuild: Sandusky Transformer 101 Breaker	Sandusky	Replace existing Transformer 101 secondary disconnect switch, 40 kV breaker and associated control cables and current transformers
Subtransmission Redesign and Rebuild: Tie 810	Lenox	• Build a new station (Gramer Station) • Reconfigure Tie 810 into Tie 9505, Tie 9514, and Trunk 4938 • Complete the following work for Tie 9505, Tie 9514 and Trunk 4938: ○ Install 11 miles of overhead conductor ○ Reconductor 13 miles of overhead conductor ○ Decommission five miles of overhead conductor
Subtransmission Redesign & Rebuild: Slocum	Trenton	Remove onsite subtransmission lines and equipment and make modifications at Slocum station due to Trenton Channel Power Plant decommission
Subtransmission Redesign and Rebuild: Trunk 4217	Grosse Pointe, Harper Woods, and Detroit	Replace 2.1 miles of underground cable and associated station work

Subtransmission Redesign and Rebuild: Trunk 4266	Eastpointe	Replace 2.5 miles of underground cable
PQ Fault Meter Locating	Various	Install a total of 177 PQ meters throughout the system including 82 station transformer positions and 95 24 kV/ 40 kV tie lines
Subtransmission Redesign and Rebuild: Trunk 3509	Royal Oak	Install new Trunk 3562 by installing about one mile of new underground cable to relieve Trunk 3509 overloads and repurpose Trunk 3509 cable to improve reliability at Mandalay substation
Subtransmission Redesign and Rebuild: Tie 6602	Dexter	Rebuild 3.5 miles of overhead 40 kV line and replace all distribution underbuild conductor
Subtransmission Redesign and Rebuild: Tie 1568	Ypsilanti	Replace 1066' of underground cable
Subtransmission Redesign & Rebuild: Pigeon Area Improvement	Unionville, Sebewaing, Kilmanagh, Pigeon, Bayport, Caseville, Kinde, Port Austin	• Extend 120 kV line from Cosmo to Pigeon • Install 120 kV breaker at Bad Axe • Install 120 kV one way switch to new tap at Pigeon • Install new 120 kV to 40 kV transformer at Pigeon with associated station equipment
Subtransmission Redesign and Rebuild: Tie 7504	Novesta	• Tap 120 kV ITC line and extend the 120 kV line 12 miles to Wilmot • Expand Wilmot station by installing a 120-40 kV transformer, circuit switcher, and two 40 KV breakers, upgrading relays, • and reconfiguring busses • Reconductor 16 miles of Tie 7504

Subtransmission Redesign and Rebuild: Tie 6907	Rochester Hills	- Reconfigure Tie 6907 and Trunk 6941 Spokane ends to eliminate overload on Tie 6907 and abandon cable & conduit of Bridge Main 005 & Bridge Main 014 - Install 1,700' of overhead conductor - Replace existing breaker at SPKNE with 2000A gas breaker
Subtransmission Redesign & Rebuild: Trunk 4911 Voltage Correction	Lenox, Chesterfield, New Baltimore, Ira	- Install new 40 kV position on Bus 2 at Victor Station - Install 9 miles of overhead 40 kV line and upgrade the underbuild overhead distribution - Install 0.6 miles of underground conduit - Install 0.6 miles of underground cable
Subtransmission Redesign and Rebuild: Sandusky Transformer 102 Addition	Sandusky	- Install one 120-40kV 45/60/75 MVA Transformer 102 from a new 120kV source to a new 40kV position on Sandusky 40kV bus 2 - Install one 120kV S&C 2020 circuit switcher, one 40kV 2000A breaker, and two 40kV 1200A disconnects - Move Sandusky Trf 101 from bus 3 to bus 4 - Install 40kV section breaker between bus 2 and bus 4
Subtransmission Redesign & Rebuild: Trunk 2443	Southfield	Replace 0.9 miles of underground cable
Subtransmission Redesign and Rebuild: Trunk 4911 Load	Lenox	Replace about 0.1 miles of underground cable and 6.7 miles of overhead conductor

Subtransmission Redesign and Rebuild: Trunk 4245	Eastpointe	• Install 1,000' of new conduit under I-94 • Install 1,800' of new underground cable • Remove 1,700' of old underground cable
Subtransmission Redesign & Rebuild: Trunk 5838	Howell	Reconductor 2 miles of overhead 40 kV line
Subtransmission Redesign and Rebuild: Trunk 2308	Macomb	Rebuild 2.9 miles of overhead 40 kV line along with 2.4 miles of distribution underbuild
Subtransmission Redesign and Rebuild: Tie 4104 North	Sherman and Sand Beach	• Reconductor 7.3 miles of overhead 40kV line • Reconductor 5.1 miles of distribution underbuild • Install 13 miles of overhead 40 kV line • Decommission 17.8 miles of overhead 40 kV line • Decommission 5.5 miles of distribution underbuild
Subtransmission Redesign and Rebuild: Custer Republic	Monroe	• Decommission Republic transformer 3 • Reconfigure TRK1456 and TRK1470 to feed Republic transformer 1 • Decommission Custer positions RS and SN • Decommission TRK1459 • Replace relay panels at Custer position RM
Subtransmission Redesign & Rebuild: Trunk 2419	Detroit	Replace 0.3 miles of underground cable

Subtransmission Redesign & Rebuild: Tie 3416 – Bad Axe	Bad Axe	Replace breakers, trainers, disconnects, and relay panel for position RH at Bad Axe Reconductor 1.6 miles of overhead 40 kV line
Subtransmission Redesign & Rebuild: Trunk 3508	Troy	Replace 2.8 miles of underground cable
Subtransmission Redesign & Rebuild: Tie 3416 - Pigeon	Elkton, Pigeon	• Replace trainers at substation • Reconductor 6.3 miles of overhead 40 kV line
Subtransmission Redesign & Rebuild: Thumb Electric Fault Isolation	Kinde Watertown, Ubly, Owendale, Millington	Replace five 40 kV pole top switches with automatic pole top switches with 40 kV bypass to support shutdowns and future maintenance
Subtransmission Redesign & Rebuild: Trunk 362	Detroit	Replace 3.3 miles of underground cable
Subtransmission Redesign & Rebuild: Trunk 2659	Waterford	Reconductor 0.7 miles of overhead 40 kV line
Subtransmission Redesign and Rebuild: Derby	Vassar	• Install Transformers 101 and 102, 60/80/100 MVA 138x120- 40 kV series multiple switch transformers • Install two 120 kV S&C circuit switchers • Install two 40 kV 2000 A gas breakers • Install two 40 kV 1200 A disconnect switches
Subtransmission Redesign & Rebuild: Trunk 328	Detroit	Replace 3.4 miles of underground cable

Subtransmission Redesign & Rebuild: Trunk 3546	Troy, Birmingham, Royal Oak	• Install 0.9 miles of conduit • Replace 2.9 miles of underground cable
Subtransmission Redesign & Rebuild: Trunk 1444	Monroe	Install one 6.6 MVAR capacitor at Trinity Station
Subtransmission Redesign & Rebuild: Tie 3205	Pigeon, Caseville, Oak Beach, Port Austin	Reconductor 26 miles of overhead 40 kV line
Subtransmission Redesign & Rebuild: Mohican	Marysville	Complete decommission and equipment removal at Mohican Substation
Subtransmission Redesign & Rebuild: Bernard	Wales Twp	Install one 6.6 MVAR capacitor on Bus 2 at Bernard Station
Subtransmission Redesign & Rebuild: STPS6	Snover	Removed unused overhead subtransmission line feed and associated equipment from former STPS6 site
Subtransmission Redesign and Rebuild: Trunk 2455	Detroit	• Install 1,400 ft of 15-5" conduit • Install seven 2-way manholes • Install 1,860 ft 40 kV underground cable • Install 1,860 ft 15 kV underground cable • Remove 1.9 miles of underground cable
Subtransmission Redesign & Rebuild: Kennett	Pontiac	Complete decommission and equipment removal at Kennett Substation
Subtransmission Redesign & Rebuild - Small Projects	Various	Complete necessary work to address small overload issues

Exhibit 11.2.2.2 Projected Investment and Timeline for Subtransmission Projects

Project	\$ Millions					
	2021	2022	2023	2024	2025	Total Investment 2021-2025
Subtransmission Redesign & Rebuild: Maxwell Amherst						\$0
Ann Arbor System Improvements						\$35
Transformer High Side Protection Program						\$15
Cortland Station Expansion						\$9
40kV Capacitor Banks at Armada and Adair						\$3
Subtransmission Breaker Short Circuit Violations						\$9
Subtransmission Redesign & Rebuild: Trunk 7106						\$3
Subtransmission Redesign & Rebuild: Trunk 7333						\$2
Subtransmission Redesign & Rebuild: Trunk 2255						\$4
Subtransmission Redesign & Rebuild: Boyne						\$26
Subtransmission Redesign & Rebuild: Badax Transformer 102 Addition						\$4
Subtransmission Redesign & Rebuild: Tie 4105						\$43
Subtransmission Redesign & Rebuild: Trunk 2237-ST						\$2
Subtransmission Redesign & Rebuild: Sandusky Transformer 101 Breaker						\$2
Subtransmission Redesign & Rebuild: Tie 810						\$40
Subtransmission Redesign & Rebuild: Slocum						\$2

Project	\$ Millions					
	2021	2022	2023	2024	2025	Total Investment 2021-2025
Subtransmission Redesign & Rebuild: Trunk 4217						\$5
Subtransmission Redesign & Rebuild: Trunk 4266						\$4
PQ Meter Fault Locating						\$7
Subtransmission Redesign & Rebuild: Trunk 3509						\$2
Subtransmission Redesign & Rebuild: Tie 6602						\$4
Subtransmission Redesign & Rebuild: Tie 1568						\$1
Subtransmission Redesign & Rebuild: Pigeon Area Improvement						\$14
Subtransmission Redesign & Rebuild: Tie 7504						\$34
Subtransmission Redesign & Rebuild: Tie 6907						\$2
Subtransmission Redesign & Rebuild: Trunk 4911 Voltage Correction						\$15
Subtransmission Redesign & Rebuild: Sandusky Transformer 102 Addition						\$3
Subtransmission Redesign & Rebuild: Trunk 2443						\$2
Subtransmission Redesign & Rebuild: Trunk 4911						\$8
Subtransmission Redesign & Rebuild: Trunk 4245						\$5
Subtransmission Redesign & Rebuild: Trunk 5838						\$2
Subtransmission Redesign & Rebuild: Trunk 2308						\$10

Project	\$ Millions					
	2021	2022	2023	2024	2025	Total Investment 2021-2025
Subtransmission Redesign & Rebuild: Tie 4104 North						\$27
Subtransmission Redesign & Rebuild: Custer Republic						\$3
Subtransmission Redesign & Rebuild: Trunk 2419						\$1
Subtransmission Redesign & Rebuild: Tie 3416 Breaker Replacement and Reconductoring						\$2
Subtransmission Redesign & Rebuild: Trunk 3508						\$5
Subtransmission Redesign & Rebuild: Tie 3416 Reconductoring - Pigeon						\$7
Subtransmission Redesign & Rebuild: Thumb Electric Fault Isolation						\$1
Subtransmission Redesign & Rebuild: Trunk 362						\$5
Subtransmission Redesign & Rebuild: Trunk 2659						\$1
Subtransmission Redesign & Rebuild: Derby						\$5
Subtransmission Redesign & Rebuild: Trunk 328						\$5
Subtransmission Redesign & Rebuild: Trunk 3546						\$6
Subtransmission Redesign & Rebuild: Trunk 1444						\$1
Subtransmission Redesign & Rebuild: Tie 3205						\$21
Subtransmission Redesign & Rebuild: Mohican						\$1
Subtransmission Redesign & Rebuild: Bernard						\$1

	\$ Millions					
Project	2021	2022	2023	2024	2025	Total Investment 2021-2025
Subtransmission Redesign & Rebuild: STPS6						\$1
Subtransmission Redesign & Rebuild: Trunk 2455						\$5
Subtransmission Redesign & Rebuild: Kennett						\$1
Subtransmission Redesign & Rebuild - Small Projects						\$12
Total Investments	\$42	\$82	\$93	\$101	\$112	\$429

11.3 4.8kV Conversion

11.3.1 Introduction to the 4.8kV Conversion program

DTEE's original distribution system voltage was 4.8kV. The system was designed to an ungrounded delta configuration and banked secondary standard, which has both benefits and risks. Delta configuration is a design from the early 1900s, which provided very low occurrence of outage events for our customers. In most neighborhoods, the 4.8kV system was constructed as overhead rear-lot poles and wires, which is aesthetically preferable to front-lot construction. Initially, right-of-way truck access was readily available through municipally maintained alleys in many areas, including much of Detroit. Starting in the mid-1900s, many municipalities began to abandon alleys and allowed property owners to extend their fence lines, preventing DTEE trucks access to the poles and wires. Consequently, the limited access resulted in a significant increase in the time to locate and repair trouble on the 4.8kV system, as well as increases in time to perform tree trimming and maintenance work. The average restoration time for an outage on the 4.8kV system is 70% longer than on the 13.2kV system, which was typically constructed to provide better access to locate and repair trouble. In addition, other key issues impacting the reliability and operability of DTEE's 4.8kV system are summarized below:

- The 4.8kV system has most of the #6 and #4 conductor (the wires), which is weaker in strength compared to current higher standard wires.
- Inherent in the design, the 4.8kV substations and circuits have lower capacity than the 13.2kV system
- The 4.8kV system can experience more significant voltage drops than the 13.2kV system
- The 4.8kV system is an ungrounded delta configuration, making detection, location, and protection of single-phase downed wires challenging.
- Ringed circuit and banked secondary designs make maintenance, fault identification, troubleshooting, and restoration more difficult and longer in duration. Opening the rings on 4.8kV circuits may result in low voltage and/or more outage events for customers.
- The 4.8kV system has a limited amount of remote monitoring and control capability. Due to equipment age, the retrofits on 4.8kV substations and circuits to enhance

remote monitoring and control capability are challenging. The original 4.8kV substation design included individual relays for individual functions, usually on a 3-foot-by-7-foot panel. When a new breaker is installed, the entire relay panel and all the wiring must be replaced to accommodate the new technology.

In addition to the 4.8kV system limitations on reliability and operability in its current state, the constraints on the 4.8kV system make it incompatible with some of the requirements of grid modernization. As mentioned above, the 4.8kV system has less capacity to serve load, due to both the smaller conductor and the inherent limitations from the lower voltage. Increased loads at the grid edge, such as adoption of EVs and the likelihood of a significant portion of charging at home, can both thermally overload these conductors and create significant voltage drops. In addition, the challenges of enabling remote monitoring and control will present an even bigger challenge than it does today when more dynamic and unpredictable loading patterns from DERs become increasingly concentrated on a circuit. Overall, the constraints of the 4.8kV design negatively affect the operational flexibility of the system due to limited capacity needed to support contingency operations, which will only be exacerbated by moderate levels of EV and DER adoption. In addition, the system's uniqueness and complex design limit distribution automation opportunities. For further detail on 4.8kV conversion related to grid modernization, see Section 3.

The voltage map for DTEE's distribution system is shown in Exhibit 11.3.1.1. Comparisons among the three distribution system voltages are shown in the Exhibits 11.3.1.2-11.3.1.4.

Exhibit 11.3.1.1 DTEE Distribution System Voltage Map

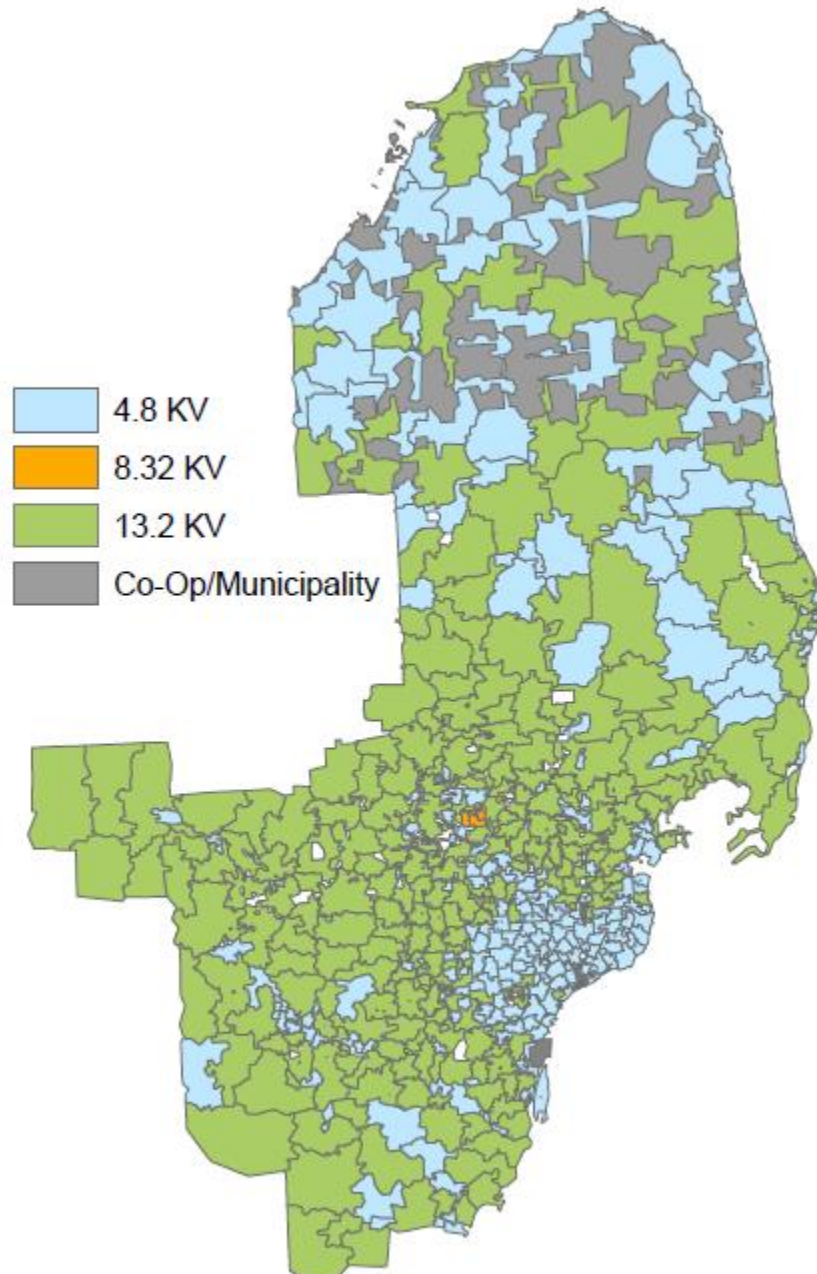


Exhibit 11.3.1.2 Percentage of Substations, Circuits, and Circuit Miles by Distribution Voltage

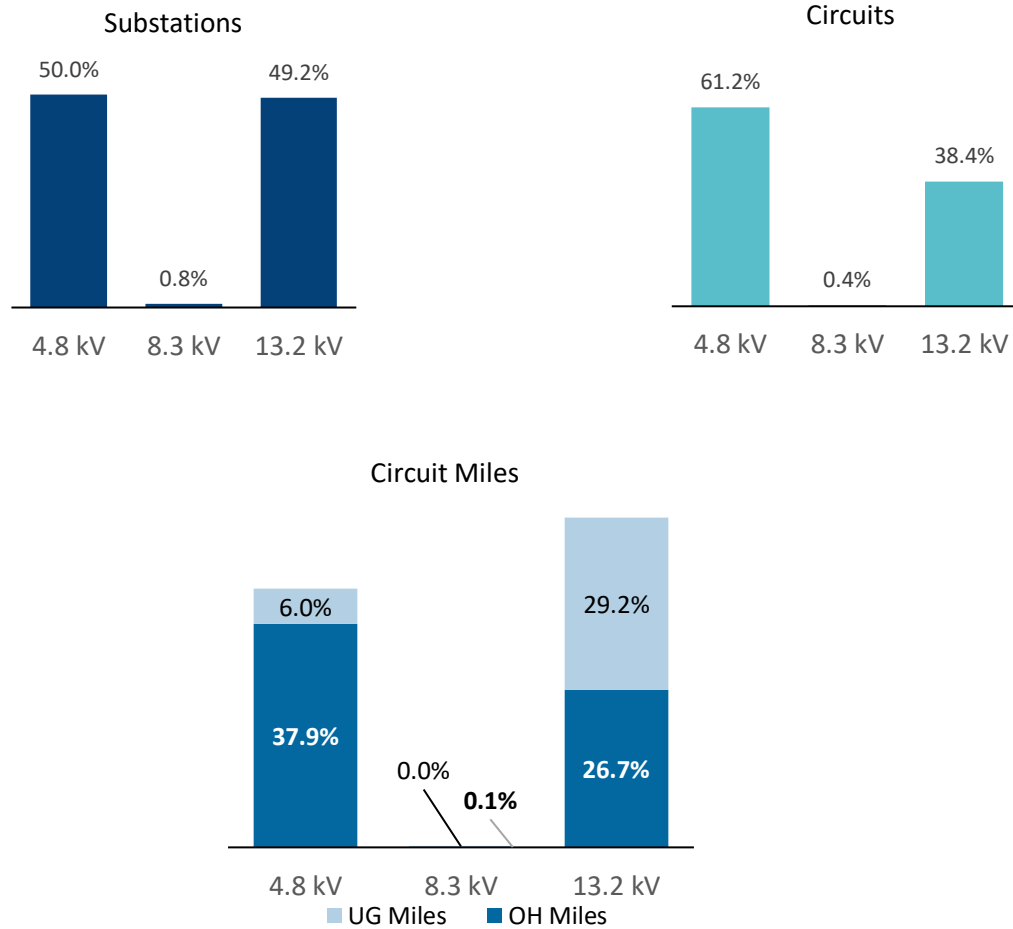
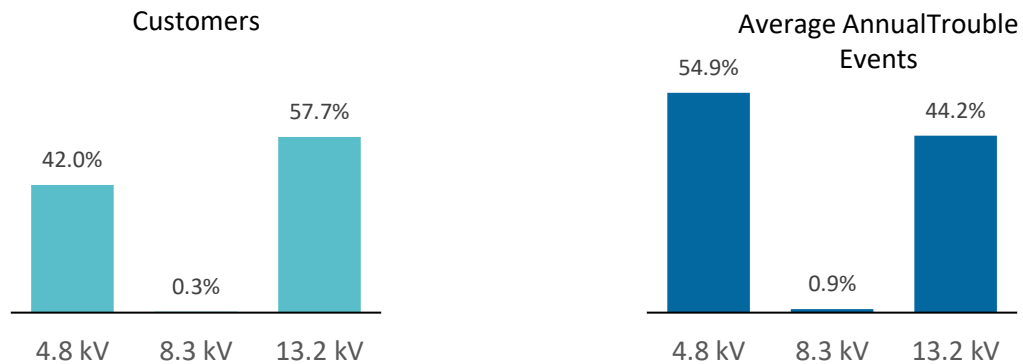
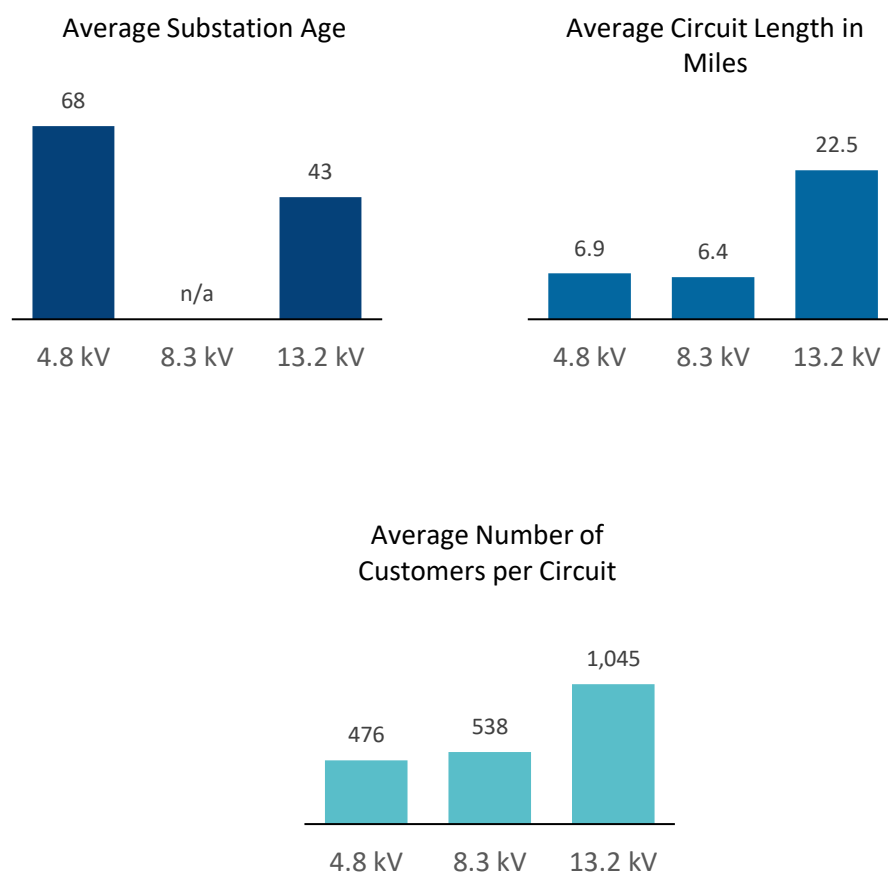


Exhibit 11.3.1.3 Percentage of Customers and Average Annual Trouble Events by Distribution Voltage



**Exhibit 11.3.1.4 Substation Average Age, Average Circuit Length in Miles, and Average
Number of Customers per Circuit by Distribution Voltage**



11.3.2 4.8kV Conversion prioritization

While DTEE addresses immediate challenges associated with the 4.8kV system through the hardening program discussed in Section 9.3, strategically phasing out the 4.8kV system has been an integral part of DTEE's long term strategy. Higher voltage systems are more efficient and allow for multiple 4.8kV substations to be taken out of service at the completion of a conversion project. In the previous section, the loading and operability limitations of the 4.8kV system were mentioned, including lower capacity for load and challenges in integrating remote monitoring control into the legacy equipment. The overall grid modernization assessment in Section 3 concluded that as customers adopt newer technology such as EVs, rooftop solar, and storage, a capacity constrained system will struggle to keep pace with customer needs.

As mentioned in Section 11.1, approximately 35% of our system has some form of loading constraint on substations, circuits, or equipment. On the 4.8kV system, approximately 25% of the substations are over the firm rating. Similar to some of the constraints seen on the subtransmission system, there are two major impacts of operating substations over their firm rating. The first is that during high loading periods, normally the summer season, an equipment failure in a substation that is over firm will likely lead to a customer outage until either the equipment is repaired, load is jumpered to a neighboring substation, or another mitigation plan is enacted such as deploying a mobile generator. In a system with limited capacity, the option of transferring load to an adjacent substation may also not be available. The second limitation is that substations over firm have limited capacity for new load. Under a scenario with increased electrification, capacity constraints are likely to get more severe over time.

It is through the primary lens of capacity that DTEE prioritizes the conversion of the 4.8kV system. In the current prioritization methodology, the following factors are considered when proposing a conversion project:

- Substation firm rating
- Circuit overloads
- Wire downs per overhead mile
- Substation risk ranking

While these criteria initially prioritize substations for conversion, many additional criteria are considered when developing and proposing a project. For example, adjacent substations are also evaluated for inclusion in the conversion, the feed for a new substation must be considered (either subtransmission or transmission), and property for expanding a current substation or building a new substation must be available and acquired.

Other considerations include avoiding projects that would create an island of 13.2kV substations surrounded by 4.8kV substations, which would add operational complexity and may be de-prioritized against a project that was already adjacent to a 13.2kV substation. In addition, constructing multiple projects in close proximity may not be feasible due to system shutdowns required to execute the work.

The conversion projects in the near-term consist of the completion of on-going projects that were driven by system capacity needs, and newer projects which meet the criteria listed above.

Prioritization and sequencing of a long-term conversion plan will be an iterative process. All of the near term (five-year) projects reflect no-regrets investments that address current loading constraints on the system. A summary of these projects and details can be found in sections 11.3.3 and 11.3.4, and in the discussion on the CODI project in Section 11.4. In the longer term, DTEE plans to convert the entire 4.8kV system to a higher voltage. While this may take decades, DTEE's aspiration is to convert it in the next 15 years, particularly as the grid needs from the electrification and increased DG/DS adoption scenarios materialize. Looking beyond the no-regrets investments of the next five years, DTEE will create longer term plans through the use of more advanced tools and analytics discussed in the DGP, such as the Integrated Forecasting Solution described in Section 4.2. These tools will enhance planning capabilities and incorporate additional data, such as propensity studies and hourly load shapes, into the current substation loading profiles, which typically only consider peak load during planning. In addition, the pace at which the electrification scenario develops, and the signposts of that scenario materialize, will impact prioritization results.

11.3.3 4.8kV Conversion plan and timeline

The conversion and consolidation projects are expected to bring multi-faceted benefits of safety improvements, load relief, risk reduction, reliability improvements, technology modernization and cost reduction.

The work performed as part of a 4.8kV Conversion includes:

- Building new 13.2kV substations or upgrading existing 13.2kV substations.
- Installing controls and automation in the substations and circuits to our latest design standards
- Completing overhead pre-conversion work including rebuilding pole tops, replacing poles and transformers as needed, and installing neutral wire.
- Rebuilding underground infrastructure.
- Reconductoring overhead lines as needed based on new circuit configurations.
- Establishing new distribution circuits from new, upgraded, or existing 13.2kV substations.
- Reconfiguring circuits and establishing new jumpering points.
- Converting and transferring the load off of the 4.8kV substations to the 13.2kV substations.
- Decommissioning of aging 4.8kV substations and associated subtransmission infrastructure

Due to the extensive scope of work involved in a conversion, converting the entirety of the 4.8kV system is estimated to cost in excess of \$30 billion, and may span multiple decades. DTEE evaluates conversion and consolidation projects on a substation-by-substation basis, using the prioritization criteria described in Section 11.3.2 – 4.8kV Conversion Prioritization. In the next five years, several conversion and consolidation projects are identified as listed in Exhibits 11.3.4.1 and 11.3.4.2. While the scope of work in Exhibit 11.3.4.1 refers to the entire project, which can extend beyond the next five years, the cost estimate in Exhibit 11.3.4.2 captures the projected investment in the next five years. Since many of these projects will extend beyond the next five years due to their size and complexity, the remaining cost to complete these projects beyond 2025 is also provided in Exhibit 11.3.4.2. As noted earlier, the cost estimates and timeline for these projects are

based on the best knowledge and information known today by DTEE. Actual project costs and timeline could deviate from the projections due to various unforeseen factors or new information/learnings.

11.3.4 Large current and near-term conversion projects

Exhibit 11.3.4.1 4.8 kV Conversion Projects

Project	Community	Drivers	Scope of Work
Hilton Substation and Circuit Conversion Phase One	Ferndale Hazel Park	- Provide load relief to Ferndale area - Replace aging infrastructure - Reduce trouble events and O&M expenses - Improve reliability in the Ferndale and Hazel Park areas	Started in 2013, the project constructed a new 13.2kV Hilton substation and is in the final stage of converting 10 existing 4.8kV circuits from Ferndale and Hazel Park to four new 13.2kV circuits
Ariel Substation and Circuit Conversion	Birmingham	- Provide load relief and capacity needs for downtown Birmingham - Improve reliability in the area	- Started in 2014 - Construct a new 13.2kV substation - Convert seven existing 4.8kV circuits from Dudley, Birmingham and Quarton Road substations to three new 13.2kV circuits at Ariel substation
Zenon Substation and Circuit Conversion	Detroit	- Provide load relief and capacity needs for the City of Detroit, west of downtown. - Improve reliability in the area	- Started in 2012 - Construct a new 13.2kV substation - Convert or transfer circuits from aging McKinstry, West End and Artillery substations to Zenon

Cortland / Oakman / Linwood Consolidation	Detroit	Reduce trouble events and O&M expenses by decommissioning two aging, underutilized 4.8kV substations	Started in 2016, this project consolidated 4.8kV Oakman and Linwood into 4.8kV Cortland substation to decommission aging substation equipment and system cable
Calla Circuit Conversion	Dexter	- Improve reliability in the area - Improve power quality	- Construct a new 13.2kV substation - Install underground cable and conduit for new circuits - Reconductor and transfer load from Lima substation to Calla
McKinstry Substation Decommission	Detroit	Decommission substation as all load has been transferred to Zenon or West End substations due to the Gordie Howe International Bridge project	McKinstry substation will be decommissioned and all equipment removed.
Quincy Conversion	Yale	Aging wood foundation supporting transformer at Quincy substation	- Construct new padmount substation - Convert 3 miles of overhead and install ISO-down transformers - Transfer load from Quincy to new padmount - Decommission Quincy substation
Pine Grove Substation Relocation and Conversion	Port Huron	Substation needs to be relocated due to bridge plaza expansion	- Construct a new 13.2kV substation - Re-route subtransmission lines to feed new substation - Convert and transfer all load from Pine Grove substation - Decommission Pine Grove substation

I-94 Substation and Circuit Conversion (Promenade)	Detroit	• Replace aging infrastructure • Reduce trouble events and O&M expenses • Provide capacity to emerging businesses such as I-94 industrial park	• Construct a new 13.2kV substation • Convert existing 4.8kV circuits from Lynch and Lambert • Decommission Lynch and Lambert substations
Buckler Circuit Conversion	Ann Arbor	• Provide load relief and capacity needs for downtown Ann Arbor • Increase jumpering capability	Transfer remaining circuits from Argo to Buckler, converting them to 13.2kV
Lapeer-Elba Expansion and Circuit Conversion (Apollo)	Lapeer / Elba Twp	• Provide load relief (Lapeer 13.2kV is 107% of firm rating and Elba is 103% of firm rating) • Replace aging infrastructure • Increase jumpering capability • Eliminate 40kV tap that has a history of poor reliability performance	• Build a new 13.2kV Apollo substation • Convert and consolidate 4.8kV circuits from Elba and Lapeer substations to 13.2kV • Decommission the 4.8kV portion of Lapeer substation • Decommission Elba and 40kV tap to substation
Almont Relief and Circuit Conversion (Midas)	Almont Twp	• Provide load relief (Almont is 160% of firm rating) • Replace aging infrastructure • Increase jumpering capability • Improve circuit voltage	• Build a new 120-13.2kV Midas substation • Transfer Almont load to new substation, converting it to 13.2kV • Reconnector 3 miles of overhead backbone • Establish new jumpering points
Novi Decommissioning and Circuit Conversion	Novi	• Address aging 4.8kV infrastructure • Islanded 4.8kV area with lack of jumpering options	• Convert and transfer Novi load to adjacent 13.2kV circuits • Decommission Novi substation
Hawthorne Relief and Circuit Conversion	Dearborn Heights	• Provide load relief (Hawthorne substation is 133% of firm rating and	• Extend overhead from Mallard distribution circuits and convert 4.8

		Daly substation is 115% of firm rating) • Address aging infrastructure • Provides capacity for new growth	kV areas out of Glendale and Hawthorne • Upgrade Biltmore 13.2 kV substation and establish 3 new distribution circuits • Replace switchgear at Villa substation • Expand Mallard substation and build two new general purpose substations to convert and remove load from Hawthorne, Glendale, Villa, and Daly substations
Birmingham Decommissioning and Circuit Conversion	Birmingham / Bloomfield Hills	• Substation outage risk for Birmingham • Provide capacity for new growth • Address aging infrastructure at Birmingham and Quarton Road substations	• Construct new 13.2kV substation in Birmingham area • Convert and transfer load from Birmingham and Quarton Road substations to new 13.2kV substation • Decommission Birmingham and Quarton Road substations
Belleville Substation and Circuit Conversion	Belleville	• Addresses aging infrastructure • Provides jumpering capability with surrounding 13.2kV substations • Provides capacity for new growth	• Install new 13.2kV transformer and replace transformer 3 with 13.2kV transformer • Rebuild and convert over 10 miles of overhead to 13.2kV • Create new jumpering points with adjacent substations
White Lake Decommission and Circuit Conversion	White Lake	• Provide load relief (White Lake 13.2kV substation is 108% of firm rating and White Lake 4.8kV substation is 110% of firm rating)	• Build a new 40kV to 13.2kV substation with four load carrying circuits • Convert distribution circuit 307 White Lake to a 13.2kV circuit

		• Replace aging infrastructure • Allows for jumpering (existing 4.8kV is islanded – surrounded by 13.2kV)	• Establish four load carrying circuits supporting White Lake Substation load and a portion of Clyde Substation • Establish new jumpering points and loop schemes between the new substation and Clyde, Wardlow, Teggerdine, Placid, and Osprey substations • Decommission White Lake substation
Calla Circuit Conversion Phase 2	Dexter	• Provide load relief and capacity needs for Dexter (Diamond substation is 144% of firm rating) • Improving reliability • Increase jumpering capability	Rebuild approximately 2 miles of 4.8kV Diamond circuits to 13.2kV and transfer this portion to Calla substation
Unionville Decommissioning and Circuit Conversion	Unionville	• Provide load relief (Unionville substation is 111% of firm rating and Fairgrove substation is 113% of firm rating) • Provide capacity for connecting new customers and accommodating load growth	• Install a new 13.2kV substation at existing Randolph Station • Reconductor 16 miles of overhead line • Convert and transfer load from Unionville and Fairgrove to the new 13.2kV substation • Decommission Unionville substation
Snover Decommissioning and Circuit Conversion	Snover / Sandusky / Marlette	• Provide load relief (Snover substation is 127% of firm rating and one Snover circuit and one Marlette circuit are over their ratings) • Lack of jumpering capability	• Construct a new 13.2kV substation • Reconductor over 12 miles of overhead line • Convert and transfer load from Snover to the new 13.2kV substation

			• Create new jumpering points with Marlette and Opal substations • Decommission Snover substation
Monroe Substation and Circuit Conversion	Monroe	• Address aging infrastructure • Provide load relief (Roosevelt substation is 107% of firm rating) • Provide capacity for new growth	• Construct a new 13.2kV substation • Convert and transfer load from Front and Roosevelt substations to the new 13.2kV substation • Decommission Front substation
Rochester Decommissioning and Tienken Relief	Rochester Hills	• Address aging infrastructure • Provide load relief (Tienken substation is 107% of firm rating)	• Construct a new 13.2kV substation • Convert and transfer load from Rochester and Tienken substations to the new 13.2kV substation • Decommission Rochester substation
Ann Arbor AC Network Conversion	Ann Arbor	• Address aging infrastructure • Provide capacity for new growth	• Install new underground conduit and cable including secondary cable • Replace existing netbank transformers with dual voltage netbank transformers

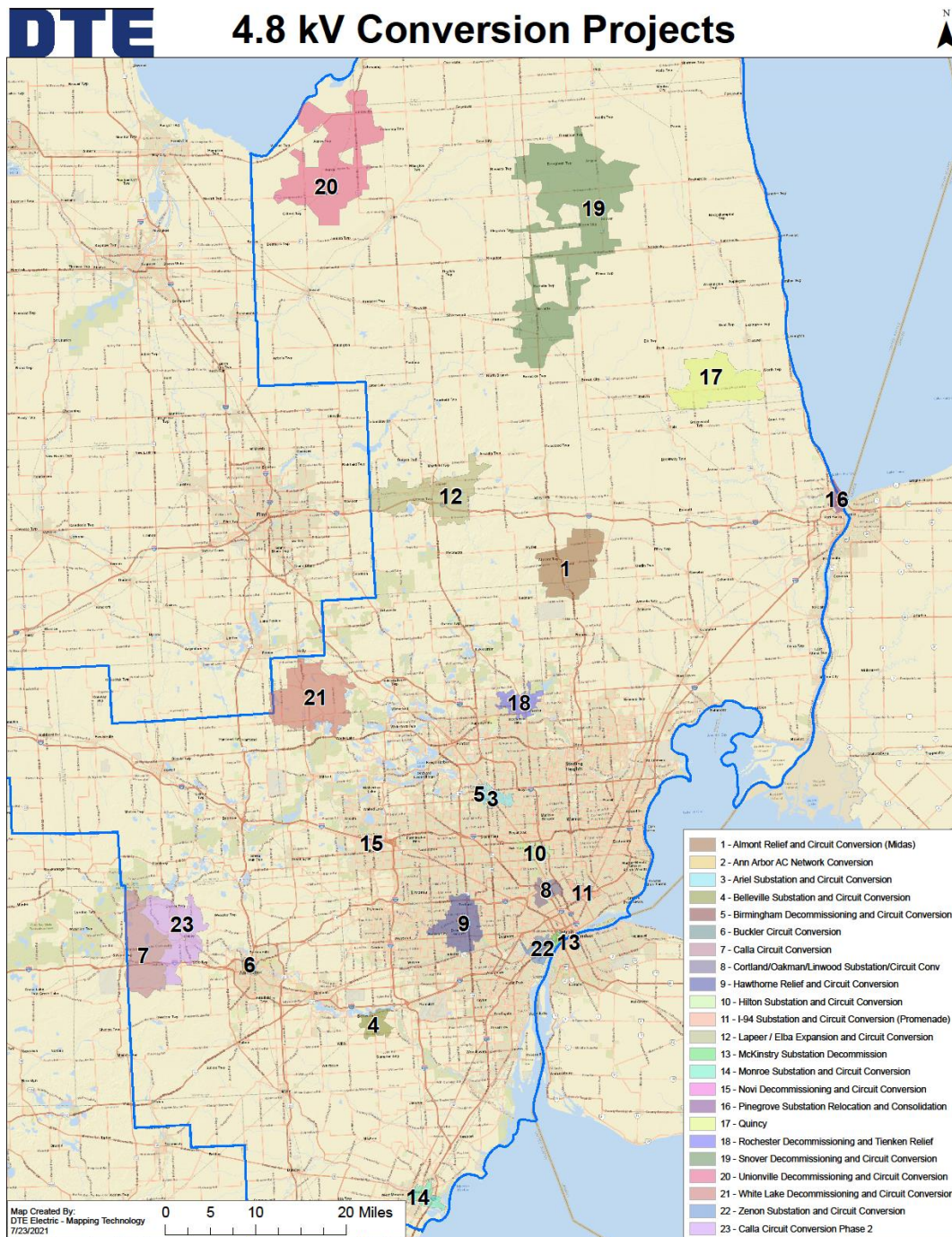
Exhibit 11.3.4.2 Projected Costs and Timeline for 4.8 kV Conversion Projects

Project	\$ Millions						2026 to 2030 Cost Estimate (\$ million)
	2021	2022	2023	2024	2025	Total Investment 2021-2025	
4.8 kV CC: Hilton Substation and Circuit Conversion						\$0	\$-
4.8 kV CC: Ariel Substation and Circuit Conversion						\$3	\$-
4.8 kV CC: Zenon Substation and Circuit Conversion						\$0	\$-
4.8 kV CC: Cortland / Oakman / Linwood Consolidation: Linwood						\$4	\$-
4.8 kV CC: Calla Circuit Conversion						\$0	\$-
4.8 kV CC: McKinstry Sub Decommission						\$4	\$-
4.8 kV CC: Quincy Conversion						\$4	\$-
4.8 kV CC: Pine Grove Substation Relocation and Conversion						\$22	\$-
4.8 kV CC: I-94 Substation and Circuit Conversion (Promenade)						\$54	\$72
4.8 kV CC: Buckler Circuit Conversion						\$18	\$-
4.8 kV CC: Lapeer - Elba Expansion and Circuit Conversion (Apollo)						\$36	\$-
4.8 kV CC: Almont Relief and Circuit Conversion (Midas Sub)						\$23	\$-

Project	\$ Millions						2026 to 2030 Cost Estimate (\$ million)
	2021	2022	2023	2024	2025	Total Investment 2021-2025	
4.8 kV CC: Novi Decommissioning and Circuit Conversion						\$5	\$-
4.8 kV CC: Hawthorne Relief and Circuit Conversion						\$37	\$85
4.8 kV CC: Birmingham Decommissioning and Circuit Conversion						\$18	\$35
4.8 kV CC: Belleville Substation and Circuit Conversion						\$1	\$12
4.8 kV CC: White Lake Decommissioning and Circuit Conversion						\$3	\$48
4.8 kV CC: Calla Circuit Conversion Phase 2						\$1	\$-
4.8 kV CC: Unionville Decommissioning and Circuit Conversion						\$3	\$15
4.8 kV CC: Snover Decommissioning and Circuit Conversion						\$2	\$15
4.8 kV CC: Monroe Substation and Circuit Conversion						\$2	\$34
4.8kV CC: Rochester Decommissioning and Tienken Relief						\$2	\$40
Ann Arbor AC Network Conversion						\$3	\$45
Total Investments	\$23	\$61	\$55	\$48	\$60	\$247	\$402

Exhibit 11.3.4.4 provides a map to illustrate the locations of the 4.8 kV Conversion and Consolidation projects.

Exhibit 11.3.4.4 4.8kV Conversion Map



11.3.5 4.8kV isolation down areas (ISO down) conversion

In some instance there is a need to address immediate overloading on a circuit or circuits fed from a substation at which other circuits are not overloaded. This issue can be addressed economically by building a higher voltage substation, typically 13.2kV, and converting only the overloaded circuits, or portions of the circuit, to the higher voltage and operating the remain circuits at the current voltage, typically 4.8kV. All the circuits on the new 13.2kV substation are fed at 13.2kV and circuits that were previously overloaded will be operated at 13.2kV relieving the overload; and circuits that were not overloaded will have the voltage dropped to 4.8kV by a transformer and will operate unconverted as 4.8kV circuits, thus isolating them from the 13.2kV circuits. This practice is known as “iso-downs.” There are iso-downs on over 400 circuits, Exhibit 11.3.5.1 provides the number of circuits with iso-downs and miles of 4.8kV overhead and underground lines.

Exhibit 11.3.5.1 Number of Circuits with ISO Downs and Miles of 4.8kV

	Number of Circuits	Number of Customers	Miles Overhead Wire	Miles Underground Wire
4.8kV ISO Downs	433	143,504	5,559	398

Circuits fed by a 13.2kV substation that have been iso-downed, operate at 4.8kV, have not been upgraded or modernized like the circuits that operate 13.2kV fed from the same 13.2kV substation. The 4.8kV circuits have the same characteristics of 4.8kV circuits fed from a 4.8kV substation, they have the same reliability issues, safety concerns, and operational concerns, and will face the same challenges to incorporate increasing EVs and DERs. As part of DTEE’s long term vision to convert the 4.8kV system, upgrading the ISO down areas to a higher voltage will also be required.

11.3.6 4.8kV ISO down conversion prioritization

The 4.8kV ISO down circuits are already fed from a 13.2kV substation which removes the substation firm rating and the substation risk ranking from prioritization consideration. Without need to consider substation factors, safety (wire down reduction), reliability (customer minute interruptions), and costs (avoided O&M and capital) become the driving factors for prioritization of 4.8kV ISO down circuits. These factors feed directly in to the GPM model and allow for the ranking of proposed 4.8kV ISO down conversion with other investments in the DGP.

11.3.7 4.8kV ISO down conversion plan and timeline

The 4.8kV ISO down conversion projects are expected to bring multi-faceted benefits of safety improvements, reliability improvements, technology modernization and cost reduction.

The work performed as part of a 4.8kV ISO down Conversion includes:

- Installing controls and automation on the circuits to our latest design standards
- Completing overhead pre-conversion work including rebuilding pole tops, replacing poles and transformers as needed, and installing neutral wire.
- Rebuilding underground infrastructure.
- Reconductoring overhead lines as needed.
- Reconfiguring circuits and establishing new jumpering points.
- Removing ISO down transformers.

Due to the extensive scope of work involved in converting a 4.8kV ISO down circuit, converting the entire population of the 4.8kV ISO downs on the system is estimated to cost in excess of \$2 billion and may span multiple decades. DTEE evaluates conversion and consolidation projects on a substation-by-substation basis, using the prioritization criteria described in Section 11.3.3 – 4.8kV ISO Down Conversion Prioritization. In the next five years, several conversion and consolidation projects are identified as listed in Exhibits 11.3.7.1. While the scope of work for these projects is still being identified (projects start with engineering and design as early as 2023), the cost estimate in Exhibit 11.3.7.1 captures the projected investment in the next five years. Since many of these projects will extend beyond the next five years due to their size and complexity, the total

cost to complete the project is also provided in Exhibit 11.3.7.1. The cost and timeline estimate for these projects is based on the best knowledge and information known today by DTEE. Actual project costs and timeline could deviate from the projections due to various unforeseen factors or new information/learnings.

Prioritization and sequencing of a long-term ISO down conversion plan will be an iterative process. All of the near term (five-year) projects reflect no-regrets investments that address current safety, reliability and emergent cost issues.

Exhibit 11.3.7.1 Projected Costs and Timeline of 4.8kV ISO Down Conversion Projects
(\$ Millions)

Project	2021	2022	2023	2024	2025	Total Investment 2021-2025
Camden	\$-	\$-	\$0.5	\$0.6	\$0.6	\$1.6 (total cost \$3.7)
Gilbert	\$-	\$-	\$0.5	\$1.3	\$1.3	\$3.1 (total cost \$7.2)
Kern	\$-	\$-	\$-	\$0.5	\$0.9	\$1.4 (total cost \$4.9)
Biddle	\$-	\$-	\$-	\$0.5	\$2.2	\$2.7 (total cost \$11.4)
Brazil	\$-	\$-	\$-	\$-	\$0.5	\$0.5 (total cost \$10.1)
Total Investment	\$-	\$-	\$1.0	\$2.9	\$5.4	\$9.3 (total cost \$36.9)

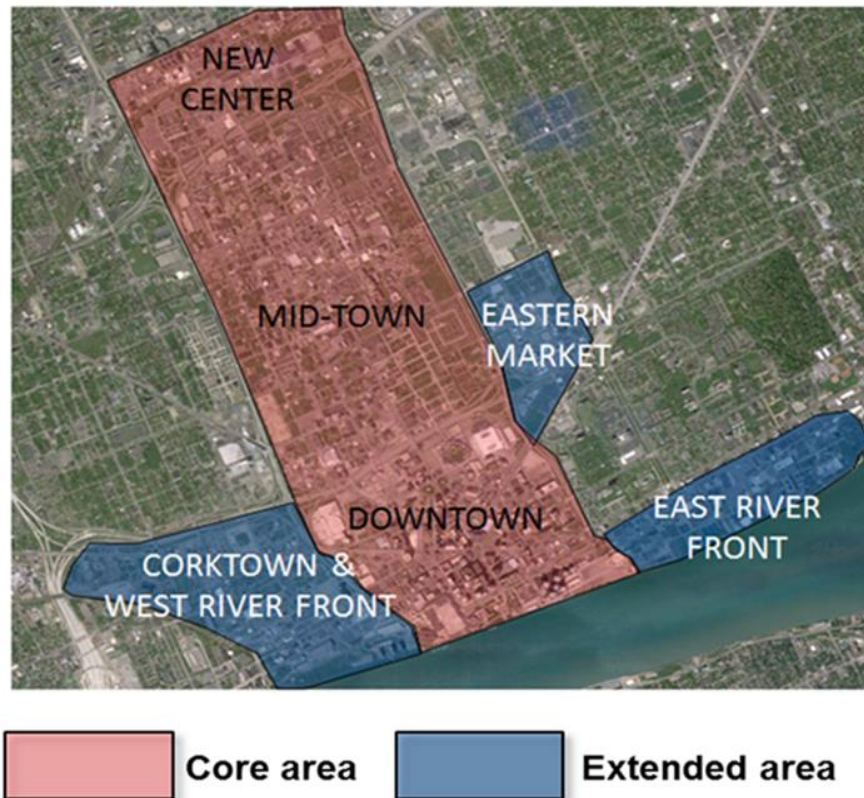
11.4 CODI Program

The City of Detroit Infrastructure (CODI) program is a subset of the overall 4.8kV conversion strategy targeted in the City of Detroit. Electrical system infrastructure in the city of Detroit dates to the early 20th century. A significant portion of the infrastructure is at end-of-life. The resurgence of development in the greater downtown Detroit area in recent years has resulted in increased loading on the end-of-life assets. The downtown Detroit Infrastructure (CODI) area has been experiencing load growth since 2012, with potential for up to 20% of additional load growth by 2023. Due to significant cable network configuration in the CODI area, substation and circuit upgrades must be sequenced and conducted in a robust, multi-year program as opposed to individual episodic projects to address the interdependency of the system. DTEE has developed the CODI program for that purpose. It is important to note that even without further load growth beyond what has already been added over the last 8 years, the substations, underground cables and manholes, and other assets which have served the area well over many decades, are experiencing higher failure rates, increasing the risk of long-duration outages that can lead to high reactive maintenance costs.

The downtown CODI program is also different from other conversion and consolidation projects due to presence of large amounts of system cable and secondary network cable, which add to the complexity of operating, maintaining, and upgrading this part of the system.

Exhibit 11.4.1 shows the areas of Detroit that will be addressed by the CODI program which includes a core area from Downtown to the Midtown and New Center areas and an extended area including Eastern Market, Corktown, and the West and East River Fronts. There are 31,800 customers served in this area including 27,486 residential, 4,299 commercial, and 15 industrial customers. The Detroit area upgrades identified in Exhibit 11.3.4.1 for 4.8 kV Conversion Projects, are separate from the CODI scope of work outlined in this section.

Exhibit 11.4.1 Downtown City of Detroit Infrastructure (CODI) Scope Area



The Downtown CODI program consists of 13 projects including the targeted network secondary cable replacement program. The high-level scope of work and estimated timeline is listed in Exhibit 11.4.2 with Exhibit 11.4.3 illustrating the substation areas for the CODI program. The yearly project estimates for the next five years are shown in Exhibit 11.4.4. The CODI program is expected to be complete by the end of 2035.

Exhibit 11.4.2 Downtown City of Detroit Infrastructure Modernization (Downtown Codi)

Project	Key Scope of Work	Estimated Timeline
CODI – Midtown Substation Expansion	Expand 13.2 kV Midtown substation by installing 3 rd transformer and a 12-position switchgear	2019-2021
CODI – Targeted Secondary Network Cable Replacement	Replace targeted sections of the secondary network cable system that have a higher probability of failure	2019-2025
CODI – New Corktown Substation	Build a new general purpose substation named Corktown	2019-2022
CODI – Alfred Substation Expansion	Expand 13.2 kV Alfred substation by installing 3 rd transformer and a 12-position switchgear	2021-2023
CODI – CATO Substation Expansion	Expand 13.2 kV Cato substation by installing 3 rd transformer and a 12-position switchgear	2022-2024
CODI – Charlotte Network Conversion	- Rebuild 30 miles of network feeder cable - Rebuild 7 miles of system cable - Replace or remove 83 netbank transformers - Convert 8 primary customers - Convert the circuits to 13.2 kV fed by Temple substation - Decommission Charlotte substation	2017-2024
CODI – Garfield Network Conversion	- Rebuild 36 miles of network feeder cable - Replace or remove 78 netbank transformers - Convert 24 miles of overhead - Convert and consolidate the circuits to 13.2 kV fed by Stone Pool substation - Remove 4.8 kV and 24 kV cable and decommission Garfield substation	2020-2030

CODI – New Islandview Substation and Walker Conversion	- Construct a new 13.2 kV Islandview substation - Convert 32 existing 4.8 kV circuits from Walker and Pulford - Decommission Walker (built in 1923) substation - Decommission aging 24kV cables and infrastructure	2020-2027
CODI – Kent/Gibson Conversion	<u>Kent Substation</u> - Rebuild six miles of system cable - Convert one primary customer - Convert seven miles of overhead - Convert and consolidate the circuits to 13.2 kV fed by Corktown substation - Decommission and remove 2 miles of 4.8 kV cable - Remove 24 kV cable and equipment - Remove 6 breakers and decommission Kent substation <u>Gibson Substation</u> - Rebuild 10 miles of system cable - Convert 22 miles of overhead - Convert and consolidate the circuits to 13.2 kV fed by Corktown substation - Decommission and remove 4 miles of 4.8 kV cable - Remove 24 kV cable and equipment - Remove 8 breakers and decommission Gibson substation	2021-2028
CODI – Howard Conversion	- Rebuild 6 miles of network feeder cable - Rebuild 12 miles of system cable - Replace or remove 89 netbank transformers - Convert 26 primary customers - Convert three miles of overhead - Convert and consolidate the circuits to 13.2 kV fed by Corktown, St. Antoine, Cato, and Temple substations - Decommission Howard substation	2023-2030

CODI – Amsterdam Conversion	• Rebuild 22 miles of network feeder cable • Rebuild 50 miles of system cable • Replace or remove 60 netbank transformers • Convert 28 primary customers • Convert seven miles of overhead • Convert and consolidate the circuits to 13.2 kV fed by Stone Pool substation • Decommission Amsterdam substation	2025-2034
CODI – Madison Conversion	• Rebuild 31 miles of network feeder cable • Rebuild 30 miles of system cable • Replace or remove 92 netbank transformers • Convert 24 primary customers • Convert three miles of overhead • Convert and consolidate the circuits to 13.2 kV fed by Temple substation • Decommission Madison substation	2028-2033
CODI – CATO/Orchard Conversion	<u>Cato 4.8 kV Substation</u> • Rebuild 17 miles of system cable • Convert 15 primary customers • Convert and consolidate the circuits to 13.2 kV fed by Temple substation • Decommission CATO 4.8 kV substation <u>Orchard Substation</u> • Rebuild 12 miles of network feeder cable • Rebuild 13 miles of system cable • Replace or remove 80 netbank transformers • Convert 12 primary customers • Convert and consolidate the circuits to 13.2 kV fed by Temple substation • Decommission Orchard substation	2029-2034

Exhibit 11.4.3 CODI Program Substation Areas

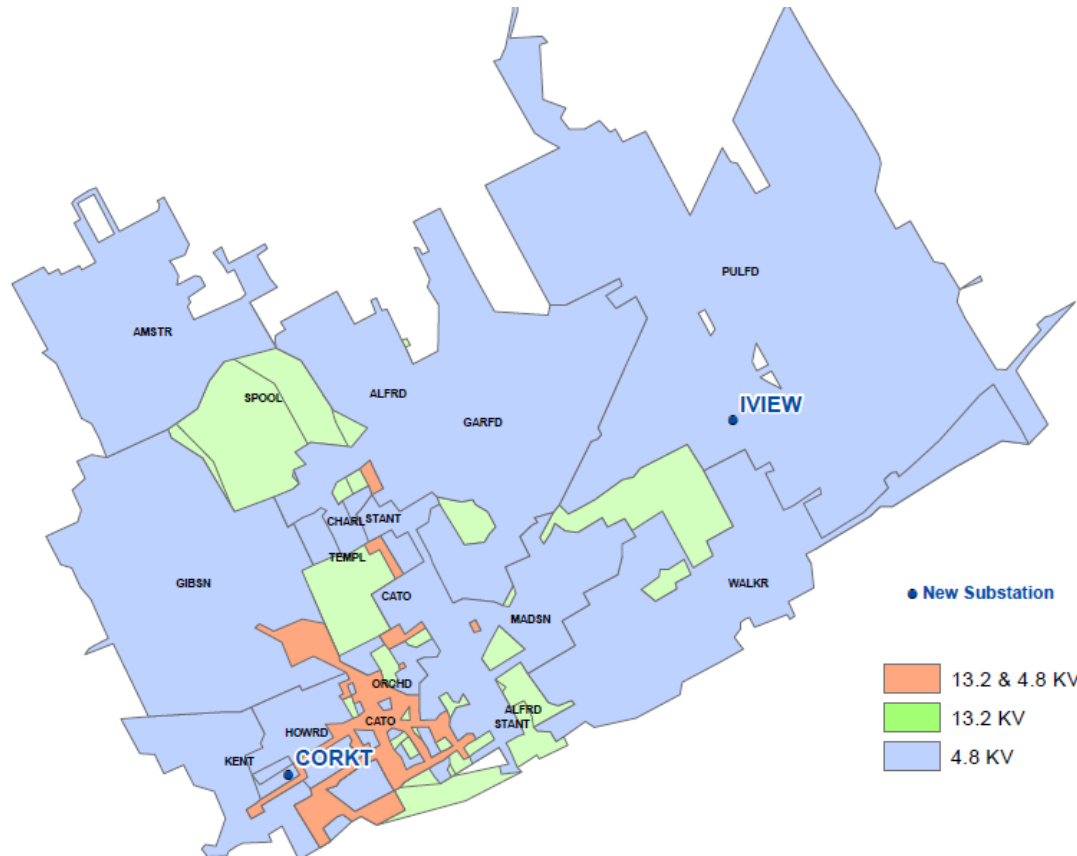


Exhibit 11.4.4 Projected Costs and Timeline for CODI Projects

Project	2021	2022	2023	2024	2025	2021-2025 Cost Estimate (\$ million)	2026 and Beyond Cost Estimate (\$ million)
CODI – Midtown Substation Expansion						\$7 (Total Cost ¹⁸ : \$11)	
CODI – Targeted Secondary Network Cable Replacement						\$15	
CODI – New Corktown Substation						\$16 (Total Cost: \$18)	
CODI – Alfred Substation Expansion						\$10	
CODI – Cato Substation Expansion						\$20	
CODI – Charlotte Network Conversion						\$49 (Total Cost: \$78)	
CODI – Garfield Network Conversion						\$108	\$120

¹⁸ Total cost represents the anticipated spend through 2025 including prior to 2021

Project	2021	2022	2023	2024	2025	2021-2025 Cost Estimate (\$ million)	2026 and Beyond Cost Estimate (\$ million)
CODI – New Islandview Substation and Walker Conversion						\$98	\$90
CODI – Kent/Gibson Network Conversion						\$44	\$51
CODI – Howard Conversion						\$36	\$108
CODI – Amsterdam Conversion						\$2	\$201
CODI – Madison Conversion							\$165
CODI – CATO/Orchard Conversion							\$122
Total Investments	\$44	\$54	\$95	\$104	\$107	\$404	\$858

11.5 8.3kV system conversion and consolidation

DTEE did not construct the 8.3kV system that serves the city of Pontiac. Located within DTEE's service territory, it was acquired from CMS Energy in the 1980s. The 8.3kV system is served by four substations: Bartlett, Paddock, Rapid Street, and Stockwell, and their 18 distribution circuits.

Unlike the 4.8kV and 13.2kV systems, contingency options are limited for the 8.3kV system. Because the 8.3kV system is an island surrounded by the 13.2kV system, it is impossible to transfer load from 8.3kV circuits to neighboring facilities. This results in a high risk for stranded load in the event of an 8.3kV substation outage event.

Adding to the operational challenges, replacement parts are no longer available for 8.3kV breakers and other substation equipment due to their obsolescence. Non-standard clearances require substation shutdowns for operations and maintenance. This leads to extended customer interruptions during outage events and leaves the system in an abnormal state for extended periods of time if any 8.3kV equipment fails. In addition, crews must be trained to operate and maintain the 8.3kV system, adding to training and operation and maintenance costs.

Meanwhile, the city of Pontiac has gradually increased in load in the past 5 years including seven new large service requests received in 2019 and 2020 from prospective new customers.

The plan to address the 8.3kV system has been developed, starting with the upgrading of the system vaults. As discussed in Section 8.19 – Pontiac 8.3kV System Vaults, the original underground vaults were a safety hazard, with live front equipment and often insufficient arc flash distances. Of the 15 vaults, three were eliminated and 10 were upgraded in 2020. The remaining two will be completed in 2021. To date, a small portion of the circuit conversion has been executed. In 2020, about 12.5 miles of overhead was converted and transferred from Stockwell and Paddock circuits to a newly extended distribution circuit out of the Catalina substation. The next phase will involve the extension of another Catalina distribution circuit with another 13.4 miles of overhead being converted and transferred from Stockwell, Paddock, and Bartlett substations. In addition, there will be a partial conversion and transfer of two underground Stockwell circuits.

Three main options were evaluated for building a new substation to complete the conversion of the 8.3kV system. These options included building a new 40kV fed 13.2kV substation in the Pontiac area at DTEE's Ninja property, building a new 120kV fed 13.2kV substation at the Ninja property, or upgrading the existing 13.2kV Wheeler substation. The Wheeler substation upgrade

was determined to be the most cost-effective solution at this time, as it will utilize existing 120kV feeds while providing the capacity needed in the area and highest number of potential new circuits. The Ninja property will be available for future use as demand requires. All remaining overhead and underground infrastructure from Bartlett, Paddock, Rapid Street and Stockwell substations will be converted and transferred to the upgraded Wheeler substation. In order for the conversions to take place, services in Pontiac fed through 8.3kV rated underground equipment will need to be upgraded. This effort will require replacement of customer-owned switchgear, fuses, transformers, and cables rated at less than 15kV class.

As a result of the project, all four 8.3kV substations, Bartlett, Rapid Street, Paddock, and Stockwell will be decommissioned by 2028. In addition, the Wheeler substation upgrade will enable load to be transferred off Bloomfield Substation. This will allow for Bloomfield Substation to be decommissioned, thus taking its at-risk switchgear out of service. Exhibit 11.5.1 illustrates the geographic locations of the substations under discussion. Exhibit 11.5.2 provides a summary of the project capital investment for the next 5 years with another \$67 million to be spent after 2025.

Exhibit 11.5.1 Pontiac Area Substations

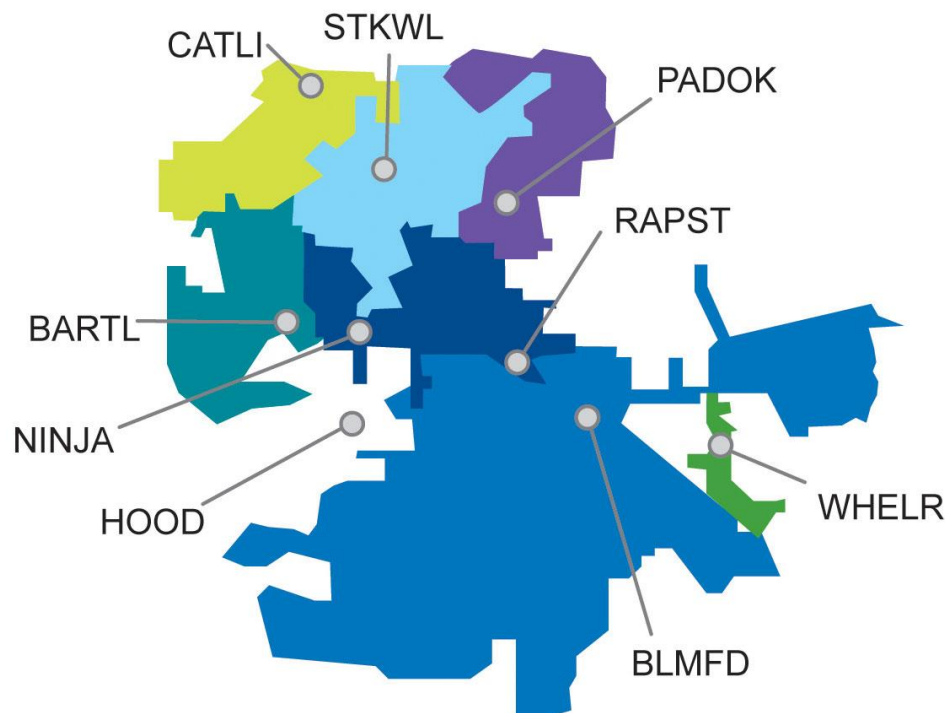


Exhibit 11.5.2 Pontiac 8.3 kV Overhead Conversion Capital Investment

	2021	2022	2023	2024	2025	5 Year Total	2026 and Beyond
Pontiac 8.3 kV Conversion Capital Investment (\$ millions)	\$1	\$4	\$12	\$34	\$38	\$90	\$67
Pontiac System Vaults	\$4	-	-	-	-	\$4	-
Total	\$5	\$4	\$12	\$34	\$38	\$93	\$67

11.6 Strategic undergrounding

Customers have been impacted by an unusually high number of severe weather events during the summer of 2021, as described in Section 1.2. To reduce the risk of future weather events impacting the Company’s customers as significantly as they did in 2021; DTEE is planning for a future that could include an increased number of weather-related storm events and for those events to have a higher level of severity, as discussed in Section 3 - Grid Modernization.

The weather impacts in the form of outages from the summer of 2021 have led many customers and other stakeholders to ask why DTEE does not bury the lines. The Company has received questions on this topic from a number of stakeholders, including direct meetings with state and local leaders, direct questions from customers in town hall meetings and comments left on social media channels. The answer is we do bury lines –well over 30% of the DTEE system is currently underground. Since the 1970s, underground lines have been installed during the construction phase of all new subdivisions and other construction projects where possible. What our customers and other stakeholders really want to understand is “why don’t you bury existing overhead lines.” DTEE employs a robust program to build resiliency on our system including tree trimming, infrastructure hardening, pole top maintenance and customer excellence. As we look to develop additional solutions to build resilience, Strategic Undergrounding is a program we are analyzing as an option. In addition to an undergrounding pilot that is underway, the Company is developing plans to partner with communities to convert overhead lines to underground lines where it makes

sense by implementing Strategic Undergrounding pilots. As part of program development, DTEE will undertake a study to look at the overall life cycle costs of underground construction.

Strategic Undergrounding is the practice of replacing overhead infrastructure with underground infrastructure. It must be implemented strategically to balance the reliability and other benefits of underground infrastructure to the higher installation cost when compared to overhead infrastructure. A pilot is currently underway, and additional pilots are being planned to study best practices that reduce costs and evaluate the overall total cost of ownership for underground infrastructure compared to overhead infrastructure. Depending on the results of the pilots, Strategic Undergrounding as an established program in the future might be applied to areas based on several factors including recent reliability and history of downed wires, storm resilience program status and effectiveness in a specific area, as well as coordination with city, DTE Gas and other infrastructure projects and community support.

Appoline Pilot

DTEE began an undergrounding pilot in 2018 on the Appoline DC 1346 circuit. The scope of this pilot is to move rear-lot overhead infrastructure to rear-lot Underground Residential Distribution (URD). This pilot was described by Company Witness Bruzzano in MPSC Case No. U-20561, and the goals of the pilot are to determine actual installation costs, understand customer acceptance, and determine opportunities to improve cost and construction efficiency in subsequent pilots. The Appoline DC 1346 Strategic Undergrounding pilot, which started in late 2018, includes approximately 60 residential customers on two city blocks in Detroit as shown in Exhibit 11.6.1. The scope of the pilot project includes the installation of a looped URD system with approximately 1,300 feet of primary, six transformers, and underground services to residences. When the underground equipment is completed and functional, the overhead infrastructure will be removed.

Exhibit 11.6.1 - Appoline 1346 Pilot:

Two City Blocks for Rear-Lot Overhead to Rear-Lot URD Conversion

Yellow Lines – Primary Conductors / Red Lines – Secondary Conductors / Green Lines – Edge of Right-of-Way



To install the pad mounted transformers and prepare the site for the installation of the URD loop, the area first needed to be cleared of debris. During this process DTEE had to address a higher level of tree and brush removal than expected, several fences and a few garages in the previous alley right of way, and significant garbage removal. These cost and project schedule impacts would have to be remediated or considered in any future rear-lot undergrounding projects. Another significant challenge impacting the schedule of this pilot was difficulty in contacting the customers to obtain approvals for modified attachments to their homes. After limited success using several mailings and door hangers, the Company is currently using door-to-door customer outreach, which has improved results, but reaching all property owners remains a challenge that will need to be addressed in this pilot and future work.

Customer Engagement in Detroit

As we mitigate the challenges identified in the Appoline pilot, Detroit continues to be a favorable location for Strategic Undergrounding deployments, as it meets many aspects of our current pilot criteria (wire downs, areas with little improvement after tree trim, community interest and

others). DTEE has engaged the Mayor's office on possible resilience projects, with favorable feedback on undergrounding existing overhead infrastructure. The City supports additional pilots to develop this program. For future pilots, the Company will work with existing city stakeholders including neighborhood leaders and customers.

Lessons Learned from Benchmarking

When developing and enhancing programs, the Company frequently engages in benchmarking efforts to identify best practices. Benchmarking has been done on Strategic Undergrounding among other infrastructure considerations with a dozen or so peer utilities. This work was to determine the optimum infrastructure configuration and construction methods that balance reliability, cost, constructability, and customer engagement when relocating infrastructure. The results overwhelmingly concluded that front-lot URD is the best option over other approaches to underground construction. Front-lot URD provides the best customer experience for both reliability and aesthetics.

Key lessons learned from benchmarking efforts on how to improve efficiency of execution include the importance of streamlined contracting for design, customer outreach, and construction, and utilizing public right-of-way. The greatest challenge identified by other utilities for other configuration options was community acceptance.

Circuit Construction Factors

The benchmarking results led DTEE to re-evaluate existing circuit construction as factors in a successful underground program. There are three main factors to evaluate when considering the construction of a circuit:

- Location: Rear-Lot versus Front-Lot
- Configuration: Overhead versus Underground
- Scope: Optimum Circuit Segments Convert

Location: Rear-Lot versus Front-Lot

Much of the Company's oldest infrastructure includes rear-lot overhead circuits, which have the poles and conductor located at the back of customer lots. This is compared to front-lot construction, where the electrical infrastructure is near the road. Poor accessibility of rear-lot circuit equipment in both alley and non-alley neighborhoods has resulted in challenges to crews

who are performing both planned and storm/emergent work. The Company's discussions with communities and benchmarking with other utilities have shown that customer acceptance is low for converting rear-lot overhead infrastructure to front-lot overhead infrastructure.

Configuration: Overhead versus Underground

As shown in Exhibit 11.6.2, overhead construction is typically completed at a lower cost when compared with underground construction. Overhead assets in the rear-lot offer challenges to locating trouble and maintaining the assets, especially in alleys that are not maintained, and front-lot overhead construction is often opposed by municipalities and customers. While overhead construction has initial lower cost of construction, the full longer-term life cycle cost of overhead circuits compared to underground circuits is something that will be further analyzed as part of developing an undergrounding program. As part of our Strategic Undergrounding pilots, DTEE electric will conduct studies of the costs as well as reliability and maintainability over the lifetime of the distribution assets.

Exhibit 11.6.2: Cost of Rebuilding Overhead Rear-Lot Overhead Assets

Rebuild Method	Cost per Mile (\$M)
Overhead Rear-Lot	1.1
Underground Rear-Lot	3.3*
Underground Front-Lot	TBD
Overhead Front-Lot	N/A**

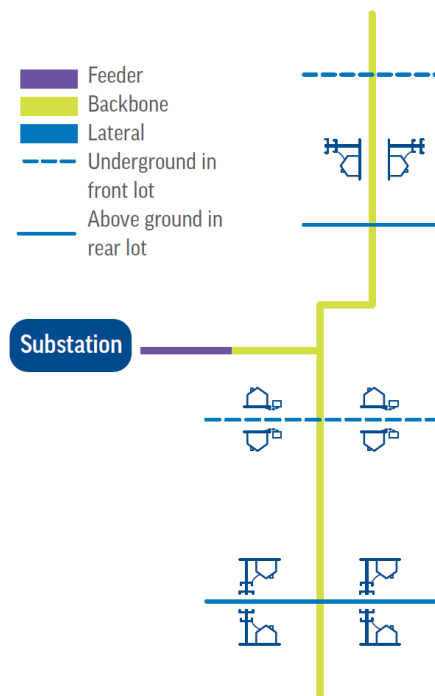
**Based on limited data from a single DTEE pilot project. As the Company enhances through benchmarking, cost efficiencies, and as the work grows using economies of scale, we can expect the cost to reduce between 20% to 30%*

***Discussions with community leadership and benchmarking results have shown that conversion to front-lot overhead will not be accepted by customers and will not provide the needed reliability benefits.*

Scope: Optimum Circuit Segments Convert

Circuits have different segments that need to be considered separately when considering construction. Circuit configuration is typically comprised of feeder, backbone and lateral segments as shown in Exhibit 11.6.3.

Exhibit 11.6.3 –Distribution Circuit Illustration



The backbone and feeder portions of circuits are most typically constructed as front-lot overhead, where they are accessible but are still vulnerable to storm and other impacts. The backbone and feeder portions of a circuit are the costliest to convert to underground because the construction requires trenching in active streets to place concrete incased conduit at approximately three or more times the cost of similar work on laterals. The standard since 1969 has been to provide service to all new subdivisions through URD construction, which would be on the lateral portion of the circuit.

Services

Services, sometimes called service drops, are the secondary voltage (typically 120V/240V) conductors connected to each customer's home. In areas with overhead laterals, the service wires run above ground from an overhead asset to customers' homes, and for URD areas, the service conductors run from equipment at ground level to customers' homes. As described by Witness Pogats in MPSC Case No. 17767, overhead residential services are approximately 16 times more likely to fail during storms than underground residential services. Witness Pogats also described how the Company prioritizes the largest outage cases, which leaves the restoration of single customer service outages to near the end of the storm restoration work. This led the Company to require any new, relocated or upgraded service to be placed underground. Given customer interest in underground electrical infrastructure, the Company is exploring extending this concept to proactively undergrounding existing service drops under some circumstances.

Circuit Construction Factors Conclusion

The Company has listened to feedback from its customers, analyzed the lessons learned from a current pilot and benchmarking efforts, and looked at the data on cost and reliability impacts to determine next steps with pilots in developing a Strategic Undergrounding program. In addition to other resilience programs, Strategic Undergrounding options will be explored and developed. Where it makes sense to address areas based on reliability, down wires, community interest and other factors, front-lot URD is the best construction alternative to replace overhead rear-lot construction on laterals. While a challenge is the significantly higher cost than overhead construction and the benefits and total cost of ownership must be further studied through the deployment of pilots, front-lot URD offers an opportunity to enhance resilience and reliability on those circuits that are difficult to maintain through traditional overhead circuit design. The construction would also be built to the higher 13.2 kV standards and with dual voltage transformers, which will allow for easier and less costly future conversion of the circuits to higher grid voltages. Pilots will also be explored for proactively undergrounding services in situations where it may make sense.

DTEE will continue benchmarking work with other utilities that have been successful with Strategic Undergrounding efforts. The purpose of these benchmarking discussions is to identify best practices that will reduce costs and better engage customers.

Ramp Up

For the next phase of Strategic Undergrounding pilots, DTEE is planning a project on Fairmount DC 1593. The circuit currently serves a well populated neighborhood with overhead rear-lot laterals. The customers on this circuit have experienced a higher than average number of down wires per mile, even after tree trim has been completed, so an additional solution is needed. The scope of the project will be to convert the current rear-lot overhead laterals to front-lot URD using the lessons learned from Appolline and benchmarking. For the portion of the circuit that has Strategic Undergrounding scope completed, we expect to achieve reduced emergent capital and O&M, improved reliability, and the elimination of wires down. This project will be used to gain a better understanding of the full life cycle cost of underground infrastructure compared to overhead, customer acceptance, cost effective methods and improved processes in order to develop a future effective, cost-efficient Strategic Undergrounding program.

Based on what is learned from the Fairmount project to improve processes and quantify the benefits expected for our customers, the Company will develop a program to implement Strategic Undergrounding where it makes sense, as one of our programs to improve reliability and resiliency. The Company plans to allocate the funding included in Exhibit 11.6.4 for moving rear-lot overhead assets to front-lot URD. The aspirational cost per mile view in Exhibit 11.6.4 is based on customer acceptance with easements, community engagement, and minimal unforeseen construction concerns. We will continue working to identify areas of cost reductions through benchmarking, continuous improvement, and realization of economies of scale. As the cost per mile improves, DTEE expects to achieve even more miles than are reflected in Exhibit 11.6.4.

Exhibit 11.6.4: Strategic Undergrounding Program Investment

Strategic Undergrounding	2021	2022	2023	2024	2025	2021-2025 Total
Cost	\$0.5 M	\$10 M	\$30 M	\$60 M	\$100 M	\$200.5 M
Miles	0	3	10	24	40	77

In addition to rear-lot overhead to front-lot undergrounding, the Company is investigating areas with a high number of service failures and is developing an additional pilot to underground the

services in one of these areas. The pilot will have the goals of determining the costs, the reliability and storm impact, customer interest, and the overall process for this work. Based on the results of the pilot the Company will consider future program development.

11.7 Impact of generation on the distribution system

As one part of the overall grid, the electrical distribution system is dependent on a robust transmission system and generation fleet to ensure consistent reliable power is available to all customers. The generation fleet as it exists today was strategically designed and located to provide the necessary capacity and redundancy to support the transmission and distribution system with power delivery to the customers.

Historically, when locating generation facilities, the goal was to balance locating the power plants near both load centers, and areas that improved the efficiency of production, such as cooling water sources and locations with ease of access to fuel delivery. Redundancy and consistency were other critical goals of the generation fleet design. It was deemed important to have multiple layers of redundancy both at a specific plant level and over the entire grid to prevent or minimize the impact to service to customers during one or multiple system failures that might take place concurrently.

Additionally, the overall system was designed to allow for shutdowns of certain equipment to perform regular maintenance and replacements while still being able to handle other equipment failures that may take place at the same time. A consistent power flow into the system supporting DTEE's customers has been achieved historically with traditional synchronous generators that helped maintain system strength due to their high inertia and ability to be dispatched to meet forecasted load demand. These characteristics, which today are often called ancillary services, are valuable and necessary for grid stability, and to ensure that the available short circuit level during disturbances is high enough for protective devices to operate and remove these disturbances from the system safely.

Another important task or ancillary service of the current generation fleet is to provide reactive power support to the system, which helps improve low voltages due to high demand or large industrial loads. The design of a robust generation fleet, coupled with a well-planned transmission system, has ensured that, during both normal operations and multiple equipment outages, the

distribution system has been able to provide service within expected voltage levels, system frequency, and without overloading any equipment.

11.7.1 Generation benefits for the distribution system

Generation resources have multiple benefits to the distribution system beyond simply supplying energy and powerflow to meet customer demands at any given time. The generation fleet plays a key role in multiple areas that have a direct impact on the reliability of the distribution system and the service provided to our customers.

Thermal overloads and low voltage

System overloads and low voltages can occur when power flows through certain parts of the system that are limited by the equipment's ability to carry the demanded powerflows. The dynamic nature of the location and magnitude of both the customers' load and power generation provided, along with the system impedance, will determine the direction of the power flow on the distribution system and loading of any specific piece of equipment. These changing paths and magnitude of the powerflow can have an adverse effect on the system resulting in overloads or low voltages, especially when powerflows have fewer available paths due to planned or unplanned system outages. The changing directions and large swings in magnitude of powerflows are mostly observed on the subtransmission tie lines when considering the distribution system. Additional generation sources and new lines or other equipment can be added to increase the number of powerflow paths and system upgrades to the limiting equipment can help strengthen the ability of existing paths.

Flicker

Some customers with large loads that require a significant amount of current inrush for their processes can cause a short-term voltage sag on the system resulting in a perceived flicker to their service and other customers served from the same line. The magnitude, duration and frequency of the voltage sag is determined by the customer's operation/current inrush needs and the robustness of the system in the area serving that customer. Distribution and transmission system strength, as well as proximity to generation sources, can help minimize the impact of flicker on the system. System upgrades may be required to provide the same level of voltage/flicker support as the generation portfolio shifts.

Stability

In addition to providing power during normal system conditions, the generation fleet is also responsible for ensuring that the grid remains stable during disturbances to the electrical system. This includes providing stable voltage and frequency support for many different types of issues, including equipment failures and malfunctions. Though this stability can be impacted locally by one or two local generators, it is the generation fleet at large that provides for the widespread stability of the electrical system.

In the past, the retirement or suspension of isolated generators did not typically have a large impact on system stability. However, with today's increasing pace of retiring older rotating generators, system stability has become more important than ever. Though current rotating generators provide voltage and frequency stability primarily through the inertia of their large rotating masses, newer technologies have sophisticated control systems, governed by IEEE 1547, that provide grid stability by constantly monitoring and adjusting their inputs to the grid. The electrical industry is still studying and improving the ability of newer generation technologies to effectively support the stability of the electrical system.

Short circuit

Traditional synchronous generators are critical for the reliability and safe operation of the power systems because they are sources of short circuit needed for protective relaying systems. Protective relaying systems are dependent on short circuit detection to clear disturbances safely from the system. Synchronous generators are excellent sources of short circuit due to their high inertia. A challenge faced by DTEE, other utilities, transmission companies, and ISO's in planning for a modernized grid is that with fewer of these resources and the addition of newer generation technologies that do not supply the equivalent short circuit, grid planning will need to adapt and evolve in order to continue to operate safely and reliably. These planning processes are described in section 11.7.2.

Reactive power

Though electrical loads always require a supply of “real” and much smaller amounts of “reactive” power, many large industrial loads that incorporate sizeable inductive machinery, such as motors and pumps, require large amounts of reactive or VAR support as well. Without this support, voltages can be impacted and begin violating local or NERC planning criteria. Additionally, large transmission lines require reactive power support to overcome their often high amounts of capacitive line charging. The generation fleet typically provides this by adjusting their reactive power controls to vary reactive input into the system. These reactive controls can both produce and absorb reactive power based on system needs, and the amount of reactive support is governed by the generator capability curves of individual generating units. Though large generation units can provide considerable amounts of reactive support, it is typically far less than the “real” output of these units. This is another area where the sophisticated control systems of newer generator technologies can prove beneficial to the system. Depending on the type of controls and power supplies installed, some newer generation sources can provide nearly their entire power output as reactive power. Depending upon system conditions, this support could prove crucial to avoid low voltage conditions or other potential power issues.

Peaker units and black start

In addition to supporting generation system load, the Peaking generator fleet serves many functions in support of the larger and local distribution electrical grid. Though the “base load” generation described above is typically dispatched in order to meet the expected load as forecasted in the MISO day ahead market, sometimes this generation is not enough to meet unexpected real-time increases in local load demand. In these situations, the local balancing authority dispatches these peaking generator units to help meet this unexpected demand growth. Additionally, these local peaking generators can be dispatched to provide some amount of reactive support if local loads are experiencing lower than normal voltages. Typically, peaking generators run for much shorter time periods than base load coal or nuclear generation, and use different fuel sources, often natural gas turbines or diesel in some cases.

In addition to the above load balancing function, often peaking generation can be used to support local operational concerns such as unplanned outages of local transmission or sub-transmission equipment. For unplanned outages, to avoid temporary load shed on the next contingency loss, area peakers can be operated prior to losing another piece of equipment to help reduce the

loading on specific lines / transformers. This helps avoid potential cascading outages and further equipment damages due to next contingency losses.

DTEE has developed and maintained several System Operating Practices (SOP) which documents the system load levels that would trigger the use of localized peaking generators. During additional system emergencies, DTEE uses Real-Time Contingency Analysis results displayed in the Energy Management System (EMS) software to further assist System Operations Center (SOC, and, in the near future, ESOC), and may deploy additional generation to alleviate the system concerns. DTEE is currently exploring the possibility of using alternative dispatchable sources, such as storage, to replace smaller and less efficient diesel generators that could provide all of the benefits outlined above. A peaker analysis that incorporates generation and distribution needs will be included in the next IRP.

Finally, local peaking generation is often used as a “black start” resource. A black start is the process of restoring an electric power station or a part of an electric grid to operation without relying on the external electric power transmission network to recover from a total or partial shutdown. This will occur during those periods when the grid has experienced a large-scale disturbance and large parts of it have gone offline. During these periods, large base load generators cannot start themselves due to the lack of local grid support to run pumps, conveyors, and other necessary equipment.

In these very rare circumstances , such as occurred with the August 2003 blackout in the Midwest and Northeast regions of the US, smaller peaking generators can often be started independently of the offline grid to provide the necessary power required by larger generation units. Typically, a special “cranking path” will be established between a peaking and base load generator pair that will be used to supply energy to start the larger unit, with voltages and frequency closely monitored by local operators.

Once the large base load unit comes online, it can be used to support other base load generator’s start up routines and the electrical grid, at large, can finally be reestablished. It should be noted that a robust and geographically diverse peaking generation fleet, with varying fuel sources, is critical to the safe and efficient operation of the electrical grid.

To summarize, there is an emerging challenge faced by DTEE, other utilities, transmission companies, and ISO’s in planning for a modernized grid, that bridges across all of the entities

responsible for generating, delivering, and managing the flow of power to customers. As baseload units retire and newer generation technologies with different grid support capabilities are added, grid planning will need to adapt and evolve in order to continue to operate safely and reliably.

11.7.2 Evaluating the impacts of generation retirement on the grid

As described in the Executive Summary, DTE and DTEE are on a path to cleaner energy. Consistent with announced decarbonization goals, DTEE developed and filed an integrated resource plan (IRP) in 2019 that included the retirement of several base-loaded coal plants. Periodic updates to the IRP plan will be filed in future years.

To identify potential impacts that retiring centralized generating units could have on the system and develop the necessary mitigations, multiple studies are required. This level of grid planning also requires close collaboration with MISO and the transmission provider in the DTEE service territory, ITC. When DTEE determines that it plans to retire a generation unit, the next step is for DTEE to initiate with MISO an Attachment-Y filing (notification of generator change of status, including notification of rescission). As part of the Attachment-Y process, MISO conducts studies to evaluate the potential reliability impact due to generator suspension. Sensitivities to generation and loading assumptions are evaluated through each individual study process. Multiple stakeholders are involved in the MISO study process to provide inputs and develop potential mitigations to the reliability issues identified in the study process. These studies can be complex, requiring a significant amount of time, especially when future operation of generating facilities is unclear and multiple scenarios of both continued and suspended generation need to be analyzed.

DTEE recognizes the importance of a collaborative approach to generation, transmission and distribution system planning and works proactively with MISO, ITC, and other stakeholders on generation retirement planning through multiple venues, for example:

- Individual generating unit Attachment-Y studies
- Annual MISO Transmission Expansion Plan (MTEP) reliability planning
- Integrated resource planning filings
- Long range transmission planning
- Within the established planning committee meetings cadence between the respective entities

Below are some examples of successful collaborations with both MISO and ITC to achieve the best project solutions for the area:

- DTEE initiated the Attachment-Y process at MISO several years before the intended retirement date for Trenton, St. Clair and River Rouge power plants. This allowed MISO and stakeholders to go through multiple studies, and more importantly provided sufficient time for robust mitigation development to resolve identified issues.
- Definitive resolution of the Trenton 9 System Support Resource. DTEE, in collaboration with ITC and MISO, developed a cost-effective, non-transmission solution that resolved the potential voltage concerns in the area of the Fermi nuclear plant. This solution entailed replacing two distribution transformers with LTC equipped transformers in the nuclear plant and allows for the safe retirement of Trenton 9 generation.
- Addressed grid issues associated with line ratings reductions, the concurrent retirement of multiple generating units (St. Clair, River Rouge, and Trenton), and the addition of the Blue Water Energy Center (BWEC).
- The BWEC reinforcement project to ensure system reliability when the plant becomes commercially available.
- City of Detroit Load interconnections and transmission overload mitigations.

Process for formally introducing a generation retirement and evaluating the impacts

The study process begins at the transmission level and then iteratively includes the distribution system. Issues on the higher voltage transmission system flow down to the lower voltage distribution system. As a starting point, a secure transmission system is needed before fully assessing distribution system concerns. If system issues are identified, then transmission projects can be proposed through the MISO MTEP planning process to address these issues.

After considering the impacts that the transmission projects will have on the system, a study on the distribution system is needed to identify any further impacts and mitigation projects. The type of projects that can be proposed range from reconductoring and adding new lines to adding new equipment like transformers, capacitors, reactors, or other voltage compensating devices.

The assessment of transmission system impacts begins with an Attachment-Y analysis. Attachment-Y studies are a specific study specified by MISO which seeks to identify serious grid issues that could cause cascading outages or wide-scale voltage collapse. The results of these

studies indicate if a generating unit is able to retire without system mitigations. Some of the Attachment-Y study limitations are that the studies are typically only steady-state and do not require system dynamics studies by default, they allow for load shed to mitigate system issues, and do not quantify the impact or cost of addressing all system issues to facilitate the generating unit retirement – only the most serious ones.

Additional studies required to identify system impacts

To properly identify all system impacts to the change in generation, additional studies that go beyond the typical planning review and scenarios must be completed. These additional studies are highlighted in the table below.

Exhibit 11.7.2.1 Types of Studies Performed to Assess Change in Generation

Studies	Reason
Transient stability analysis	Assess the stability of nearby generators and their ability to “ride through” various fault conditions
Voltage stability analysis	Ensure that system voltages can recover and remain stable after a wide range of system disturbances
Short circuit studies	Evaluate the grid to determine if there are locations that have low short circuit capacity. This low short circuit capacity can lead to relay mis-operation and flicker
Customer specific analysis (motor start/flicker)	Ensure end use large industrial customers are able to maintain operations after changes to the generation fleet
Transfer studies	Confirm no adverse impact on import / export limits. These types of studies will be increasingly important as more generation is retired
Black start studies	Performed if the retiring generation resource is currently included in the system restoration plan

11.7.3 Future of distribution grid with flexible generation resources

The electrical grid is going through a period of rapid transformation including large scale changes to the generation fleet, at DTEE, in Michigan, and across the US. new and different types of distributed generation resources, including solar, wind, and storage, are expected to interconnect to the electrical system, and, in many cases, replace the conventional rotating generators currently in-use. These sources support the goal of cleaner generation, but bring challenges because of their intermittent nature, which will require novel solutions for electric utilities that need to balance safety and reliability goals with green energy targets.

It is expected that adding storage to the generation fleet will be an important tool that will help to manage the variability inherently associated with the nature of some of the renewable resources. DTEE is already using storage in its generation portfolio through the Ludington Pumped Storage Plant, an 1100 MW hydroelectric facility. Other forms of storage are also showing promise and will be reviewed for applicability in future years. Additionally, locating these DER resources in geographically diverse areas will help to mitigate their intermittent nature and take advantage of the existing transmission infrastructure.

Though the changing nature of the generation fleet will undoubtedly require new technologies and new ways of supporting the electrical grid, these same technologies may offer new tools that can be used to ensure that DTEE continues to meet the energy needs of our customers in a way that is safe, clean, reliable and affordable.

12 Technology & automation



Technology and Automation is one of the four pillars in DTEE's Distribution Grid Plan framework. This pillar is tightly linked to the grid modernization process and includes investments that develop capabilities in observability, analytics and computing, controls, and communications. DTEE's Technology and Automation projects and programs over the next five years meet current grid needs and provide immediate benefits to our customers. For example, increased automation of our substations and circuits will reduce outage frequency and duration, and improve reliability, particularly when linked with the capabilities of the new ADMS. At the same time, these investments lay the foundation to support increased adoption of DERs and EVs that may materialize as part of planning scenarios. Increased sensing, monitoring and control of distributed assets is a necessary capability to manage the voltage and power quality across the system. In addition, many of the capabilities discussed in Section 4 – Distribution Planning Processes and Tools, such as improved forecasting and more streamlined interconnections, will be enabled by investments in this pillar.

DTEE is implementing multiple programs to enhance and modernize electric grid technology and in support of the DTEE's overall grid modernization strategy. Exhibit 12.1 provides the projected costs and timeline for the identified technology and automation programs.

Exhibit 12.1 Projected Investments and Timeline for Grid Technology Modernization Programs

Project	Section	2021	2022	2023	2024	2025	2021-2025 Cost Estimate (\$ million)
ADMS ¹⁹	12.1						\$40
SOC Modernization	12.2						\$107
Grid Automation Telecommunications	12.3						\$71
Distribution Sensing and Monitoring	12.4						\$49
Substation Automation	12.5						\$27
Circuit Automation	12.5						\$29
CVR / VVO	12.6						\$69
NWA Pilots	12.7						\$35
DER Control	12.8						\$11
Mobile Technology	12.9.1						\$13
Work Management and Integrated Scheduling	12.9.2						\$21
Asset Management	12.9.3						\$6
Distribution Planning Process Enablement	12.9.4						\$43

¹⁹ In addition, the ADMS project requires \$17.1 million of regulatory asset investment from 2019 to 2023

Project	Section	2021	2022	2023	2024	2025	2021-2025 Cost Estimate (\$ million)
Modernize Grid Management	12.9.5						\$35

12.1 ADMS

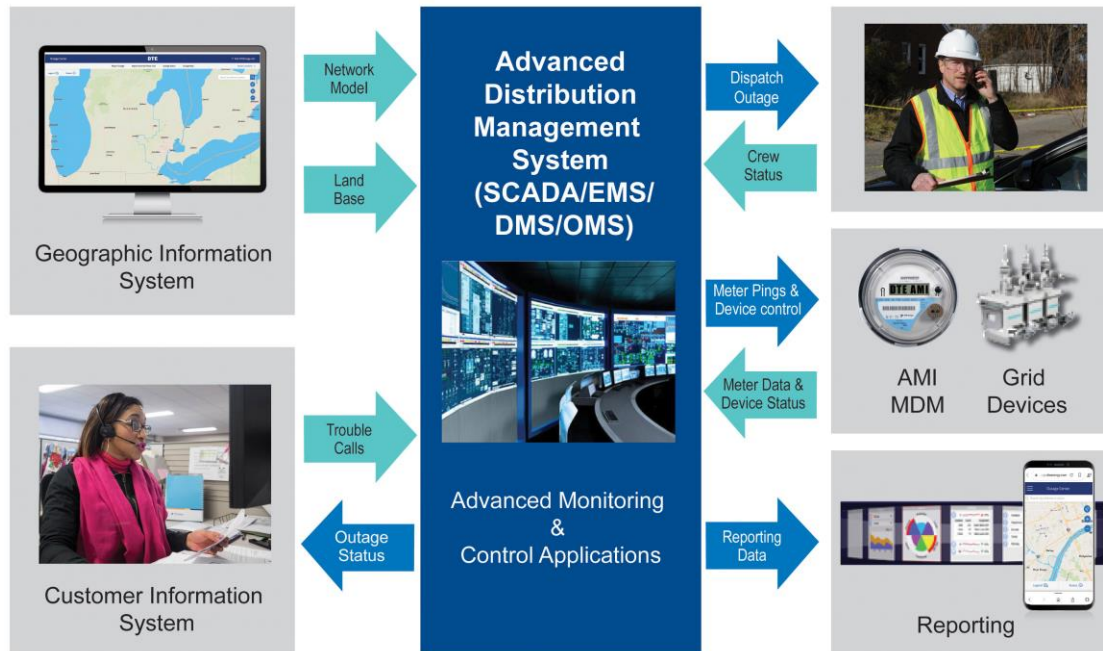
The Advanced Distribution Management System (ADMS) is an advanced operating technology platform that is essential to DTEE’s grid modernization efforts to improve system reliability and operational efficiency.

The ADMS leverages the enhanced technology deployed on the distribution system (such as line sensors, SCADA, etc.) and provides all users the same “as-operated” view of electrical system status and its performance. The ADMS will provide the foundation for integrating and managing distributed energy resources (DERs) in the future. The ADMS will also allow DTEE to fully integrate the Outage Management System (OMS), Distribution Management System (DMS), Energy Management System (EMS) and Supervisory Control and Data Acquisition System (SCADA) and provide seamless interface with the Geographical Information System (GIS), the Customer Relationship Manager (CRM), and AMI.

Exhibit 12.1.1, below, illustrates the ADMS as an integrated technology platform that will allow DTEE to perform advanced engineering and analytics and provide our system operators with industry-leading functionality to monitor and control the electric grid to meet customers’ needs.

DTEE began the ADMS implementation in 2018 and is expected to complete an EMS upgrade, OMS launch, and DMS application launches in 2022.

Exhibit 12.1.1 ADMS - An Integrated Technology Platform



The ADMS is comprised of the following functional components:

- **Energy Management System (EMS)** – Allows DTEE to manage the subtransmission system and the connections to the transmission system, provides the tools to perform state estimation and contingency analysis on the balanced electrical network, and allows DTEE to operate devices on the subtransmission system.
- **Outage Management System (OMS)** – Performs outage analysis based on the as-operated state of the electrical system, inputs from the AMI infrastructure, and customer channels. Outage and non-outage jobs are aggregated to ensure field crews are dispatched appropriately by the dispatchers and system supervisors working together to return the equipment back to normal operating state.

- **Distribution Management System (DMS)** – Starts with the as-operated Network Model (eMap) that serves as a building block for the use of advanced DMS applications such as Fault Locating, Switching Analysis, and Volt/Var Control functions that will provide for the safe, reliable and efficient management of the planned and unplanned events on the electrical system.
- **Network Management System (NMS)** – Allows DTEE to maintain high quality and accurate distribution and subtransmission locational and operational system data essential to the safe and effective monitoring and operations of the grid. See Section 12.9.4.3 – Network Model, for more information on the Network Model.

During the planning phase of this project, DTEE benchmarked multiple utilities that have implemented, or are in the process of implementing, an ADMS. These utilities shared the same goal of improving reliability by setting a foundation for technology deployed in the field and providing real-time views of the “as-operated” system.

The implementation of ADMS varied among the utilities, depending upon the components or modules needed or systems requiring replacement. One key “lesson learned” that was common among utilities is that data quality is crucial and requires process improvements to ensure sustainability once ADMS is fully functional.

Since successful ADMS implementation is dependent upon the accuracy of the data entered and stored in the systems, DTEE conducted a data gap analysis in 2017 to provide a clear understanding of potential issues in our current systems. The results, shown in Exhibit 12.1.2 below, indicate that DTEE’s data quality at the time was below the recommended level to achieve full benefits from an ADMS.

Exhibit 12.1.2 DTEE 2017 Data Quality Gap Analysis Results

Data Quality	DTEE Data Gaps	Third Party Assessment
Spatial integrity accuracy	Displacement up to 200 feet with further misalignment	Below Industry Standard
Single operating model	Many models. Including AC, PQ View, GIS	Below Industry Standard
Controlled operational data	Device and asset data lack location and system control	Below Industry Standard

Data Quality	DTEE Data Gaps	Third Party Assessment
SCADA integration	Data lacks operationally consistent nomenclature; lack of incremental registration	Below Industry Standard
AMI data in operations	Load, voltage and connectivity not in use for operations	On Par with Industry Standard
Power flow models	Lack of power flow model integration to GIS	Below Industry Standard
Underground network, cable and duct bank	AC networks are not modeled	Below Industry Standard
Visibility to quality	Lack of correlative validation and visualization	Below Industry Standard
GIS network connectivity	GIS distribution connectivity with transformer to meter connectivity supplying OMS	On Par with Industry Standard
Secondary system	18,000 miles of secondary is not connected in the system	Below Industry Standard

Based on this analysis, DTEE created a Network Management System for the as-built GIS Network Model to ensure all electrical network assets and their respective locations are accurately represented in one source. This project focused on creation of a network model, along with initial data cleansing and migration, hardware, and software upgrades to better track and maintain data going forward. DTEE has made significant strides in addressing spatial integrity accuracy, moving toward a single operating model, use of AMI data in operations, modeling of underground network cable and duct bank, visibility to quality, and connecting of the secondary system. The Network Model program is geared toward continuing to enhance the single operating model, expanding the use of work management and GIS integration for better asset data management, and visibility to quality by implementing solutions to enhance sustainability of data maintenance. See Section 12.9.4.3 – Network Model, for more information on the sustainability and continuous improvement of the Network Model.

The ADMS connects and aggregates several operational functions to provide real-time situational awareness and a controllable distribution grid. Implementation of the ADMS is expected to bring many benefits, including:

- **Enhanced visibility and operational safety** – The ADMS provides a tool for system supervisors to run “what-if” scenarios, perform pre-operation analysis checks for potential issues before issuing operating orders and validate switching orders in “study mode” prior to execution.
- **Improved system performance** – The ADMS supports Volt/VAR optimization and can automatically balance line voltage and system reactive power to reduce power line losses, reduce peak demand and improve the efficiency of the distribution grid.
- **Improved system reliability** – The ADMS reduces customer outage duration (SAIDI) by providing the following: automated outage verification; automated crew assignment optimization; automated power flow analysis and fault location; switching studies to isolate faults and restore the maximum number of customers; and eventually remote switching and automatic isolation and restoration operations.
- **Advanced data analytics** – The ADMS provides a platform to analyze significant volumes of data from various portals and provides system supervisors the information in one place.
- **Improved integration of distributed energy resources** – The ADMS increases grid visibility, allows flexible, real-time operation and provides appropriate balancing and control of power flow to lay the foundation for accommodating increased distributed energy resource penetration in the future.

The implementation of ADMS has been phased in over recent years. DTEE successfully launched the Generation Management System (GMS) in December 2018 and the EMS in May 2019. An EMS system upgrade is on track for completion in Q4 2022 to allow for decoupling of the EMS from the GMS.

Deployment of the new GMS and EMS systems has allowed DTEE to modernize IT infrastructure with virtualization technology and to increase cyber security design with physical and virtual networking, in addition to these other improvements:

- Data tables are stored in a consistent fashion, improving the user’s ability to search, find and correct entries when necessary.
- Application modules have improved visual and functional setting parameters.

- The display and database maintenance process has been improved, with new capabilities of the system to lock, edit and track data changes. Display archiving and database rollback options bring an additional level of security in case the data gets corrupted and needs to be re-populated from a backup repository.
- Hardware has been upgraded to a virtual environment (virtual servers) and is fully redundant from the data center perspective. The hardware has the latest processors, memory and graphics cards, greatly improving the user's experience when navigating the application on the client machines.
- Tools have additional features that help the users build situational awareness displays and dashboards.
- The system has a fully operational Operator Training Simulator domain that is ready to be used as a simulation tool for training and developing new system operators and engineers.
- The EpilogPro application module, planned to be implemented in 2021, will allow electronic logging, which is needed for electronic switching and tagging, and will also benefit compliance with improved record retention and archive searching capabilities.
- The system will be on the same platform as the Outage Management System (OMS) and Distribution Management System (DMS) once implemented, allowing transparency of the system to all appropriate personnel.
- The new system will allow for more straightforward system upgrades in the future, as it was configured with standard vendor-offered functionality and not custom built for DTEE.

The Outage Management System (OMS) and Compass Mobile field extension tool will be launched in Q4 2022. The new OMS will allow for improved outage identification time and visibility, which will enable DTEE to more safely and quickly work to restore power to customers. The Compass Mobile tool will be used to manage work and view the as-operated DMS Network Model and improvements planned for the future such as the Damage Assessment, Fault Location and Switch Order Management modules.

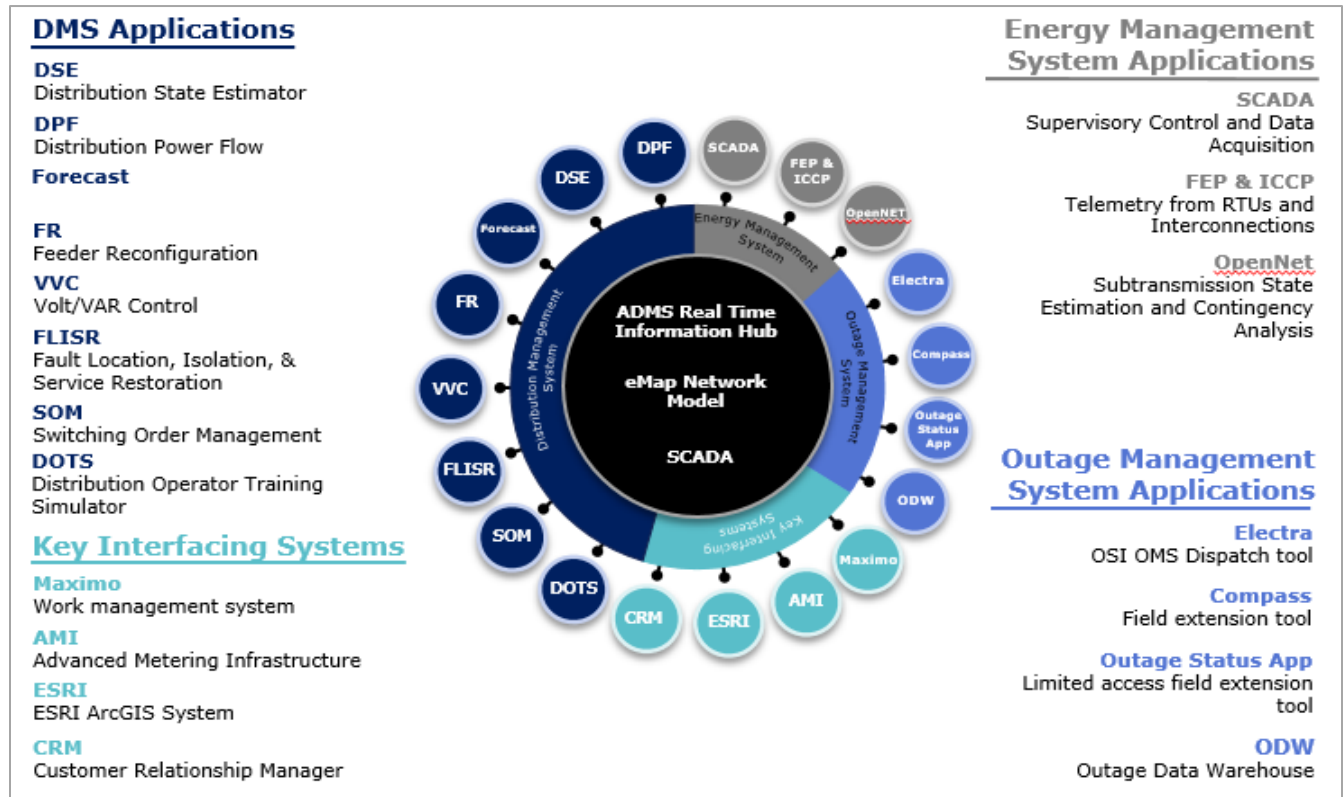
A key component of the Distribution Management System (DMS), eMap, will be launched in Q4 2022 and will provide a schematic of the as-operated network model representing the overall DTEE system.

Between Q3 and Q4 of 2022, DTEE has plans to roll out the remaining DMS applications listed in Exhibit 12.1.3, below. Exhibit 12.1.4 shows all the applications associated with the EMS, OMS, and DMS as part of the ADMS platform and the key interfacing systems.

Exhibit 12.1.3 Distribution Management System (DMS) Applications

Application	Description of Application Use
Distribution State Estimator (DSE)	Used to determine 3-phase voltage and load values on the system to feed into other advanced applications
Distribution Power Flow (DPF)	Used to determine 3-phase voltage and load values on the system and to run studies to analyze future conditions of the system
Forecast	Uses historical load and weather data to forecast electrical load for periods of one hour to 35 days to be used by System Operations and as inputs to other applications
Feeder Recognition (FR)	Used for response to system events to develop switching procedures to resolve overload and low voltage conditions, improve load balance, and improve system performance as it relates to losses and voltage profiles
Volt/Var Control (VVC)	Used to manage voltage and reactive supply levels and reduce system load to prevent load shedding
Fault Location, Isolation, & Service Restoration (FLISR)	Used to detect a fault or loss of voltage condition and recommend switching actions to isolate the fault and restore service
Switching Order Management (SOM)	Allows for quickly determining switching solutions for restore-before-repair
Distribution Operator Training Simulator (DOTS)	Used for scenario-based training on the Distribution Management System

Exhibit 12.1.4 ADMS Applications and Key Interfaces



As part of the implementation strategy, a focused change management approach is being deployed. A change management team was formed early in the project due to the significant changes in how work would be done with the implementation of the ADMS. The team identified the key internal stakeholders, has provided consistent messaging throughout the project, and has developed training plans for the affected roles.

The total capital investment for the ADMS project is \$72 million, including investments in 2021 and 2022 (shown in Exhibit 12.1.5), and investments made prior to 2021.

Exhibit 12.1.5 ADMS Capital Investment

	2021	2022	2023	2024	2025	Total Investment 2021-2025
ADMS (\$ millions)	\$30	\$10	-	-	-	\$40

12.2 SOC modernization

Another technology improvement that is essential to grid modernization is the System Operations Center (SOC) Modernization project. While the ADMS addresses the operating technology platform, the SOC Modernization project addresses the physical infrastructure and technology constraints within the current SOC. This project is aimed at replacing the outdated primary SOC and the smaller, outdated backup SOC by constructing two facilities designed using current industry security, resiliency and operability standards. The current SOC was built in the early 1980s and poses significant limitations including:

- **Outdated facility** – The facility lacks the redundancy in mechanical and electrical systems necessary to ensure continued operations in the event of a significant system event.
- **Outdated technology** – The current SOC, which is in service until the changeover later in 2021, uses a magnetic tile representation of the electric network, as opposed to an electronic display board of the transmission, subtransmission and distribution network, which is now very common in the industry. This severely limits situational awareness, which is always critical, but particularly during periods of significant system events, such as large storms. In addition to being outdated technology, the tile map board is running out of space to accommodate growth in the distribution system. The lack of modern technology also limits training opportunities.
- **Space limitations** – The SOC and dispatch personnel are physically separated, and their primary method of interaction is through repeated phone calls to share information and collaborate on dispatching field resources. The current SOC does not have sufficient space to collocate these resources that manage the system and dispatch field personnel to resolve operational issues. Colocation of SOC and dispatch personnel is a well-established industry best practice, as it provides significant customer benefits in terms of the speed at which issues can be addressed and electric service restored.
- **Limited visibility of telecommunication infrastructure performance** – The reliability of telecommunication paths from field devices to the SOC is critical for the effective monitoring of the grid and for remote operations. Developing the ability to separately monitor the condition of the telecommunication network through the construction of a Network Operations Center (NOC) is part of the SOC Modernization project.

Construction began on the new primary Electric SOC (ESOC) in June 2019 at DTEE's downtown Detroit headquarters complex. Building construction has been completed, the equipment and remaining infrastructure are being installed, and DTEE will begin using the ESOC in Q4 2021.

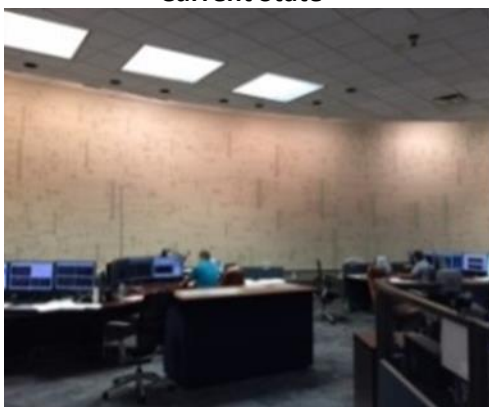
The ESOC facility will allow for co-location of dispatchers and operators and improve DTEE's ability to manage significant operational disruptions, thus improving the resiliency of the system.

The new Network Operations Center (NOC) built within the ESOC will improve visibility of the status and availability of field telecommunications, increasing DTEE's ability to respond to issues quickly. The facility includes electronic displays and improved console designs to meet dispatcher and operator needs, as illustrated in Exhibit 12.2.1. This improves situational awareness and improves safety and reliability.

The Alternate SOC (ASOC) is required to allow DTEE to move ESOC operations nearly seamlessly to a fully functional alternate site in the event the primary ESOC is unavailable. The design of both the ESOC and ASOC is based on extensive benchmarking discussions with many top utilities and industry experts. The ASOC is being designed in 2021 with plans to construct in 2022 and 2023 within DTEE's new Waterford Service Center.

Exhibit 12.2.1 Current and Future State Example of SOC Facility

Current State



Future State



The total investment for the ESOC is \$97 million with the new facility including:

12.3 Grid telecommunications

Grid modernization has increasing reliance on communication technology, making telecommunication foundational to a modern grid and a requirement for many of the advanced programs and features of the distribution system, especially automation. As an example, telecommunications enable the use of key functions of the ADMS, such as monitoring and device control. Such automation requires a resilient, secure, high-speed telecommunications network using interoperable protocols that provides very high-quality service.

To ensure a successful program, operability, security and resiliency must be built into the program from the start. This will allow future use cases and technologies beyond current grid technologies, such as DERs and coordination and control of customer and devices owned by third parties. To increase efficiency with other elements in DTEE's grid modernization efforts, the telecommunications plan is being coordinated with other related projects.

According to the Department of Energy's DSPx Grid Modernization Framework:

The operational communication network that enables the transformation of grid modernization efforts is an important aspect that cuts across the entire industry and is a key component in any modern grid. As such, there should be a consideration of the various components of operational networks based on needed capabilities and functions.

The DOE's DSPx Grid Modernization framework was discussed in the Section 3 – Grid Modernization Process and, as shown in Exhibit 3.4.2.1, Operational Communications is a Core Component that is foundational to all other aspects of grid modernization.

Starting in 2015, DTEE assessed the industry for best practices around telecommunications and developed Distribution Design Orders around network installations and grid equipment. For example:

- DDO 100-001: Requires remote Power quality metering at substations and industrial sites.
- DDO 100-003: Requires all new and retrofitted breakers and reclosers to be SCADA equipped.
- DDO 100-004: States all SCADA sites shall implement private and secure IP communications with fiber as the preferred medium to stations and substations and radio mesh for pole-top devices.

- DDO 100-005: States all interconnected generators must follow IEEE 1547-2018 and SunSpec communications standards.
- DDO 100-007: States all SCADA substations shall meet cyber security rules.

DTEE's Network Engineering has been implementing a tiered design to serve communication needs now and in the future in a highly resilient manner to provide a high quality of service:

- Tier 1 consists of microwave and fiber backbones for the Tier 1 and Tier 2 systems that traverse between critical facilities such as data centers, power plants, service centers and subtransmission stations.
- Tier 2 consists of wireless point-to-point and point-to-multi-point systems that allow data collection from the Tier 3 system, distribution substations and pole top devices.
- Tier 3 consists of wireless mesh networks that support end point SCADA and AMI meter Communications.

Many modern electrical components require communications that will be coordinated by the ADMS. Some of the ADMS functions requiring communications are:

- Remote operation/SCADA
- Electronic switching and tagging
- FLISR (fault location, isolation and service restoration)
- Volt/Var & CVR
- Distribution State Estimation
- Dynamic protection
- DER integration
- Field mobile and redlining

Among the thousands of existing device types that ADMS will connect over the network are:

- Inside the substation
 - Relays, meters, breaker controls, capacitor controls, transformer controls systems
 - RTUs (microprocessor-based devices that monitor and control field devices that then connects to plant control or SCADA systems) and communication processors
 - Alarms and Human-Machine Interfaces (HMI)
 - Asset health sensors such as voltage monitors

- Security Cameras and Access Control systems
- Outside the substation
 - Reclosers
 - Pole top switches
 - Capacitors
 - Regulators
 - Line sensors
 - Power quality meters
 - AMI meters

Telecommunications systems will be installed in accordance with the Distribution Design Orders to enable communications with devices being installed as part of the automation plan (see Section 12.5 – Distribution Automation) and other projects. All these devices require a secure, high speed data network to communicate with the ADMS and with each other.

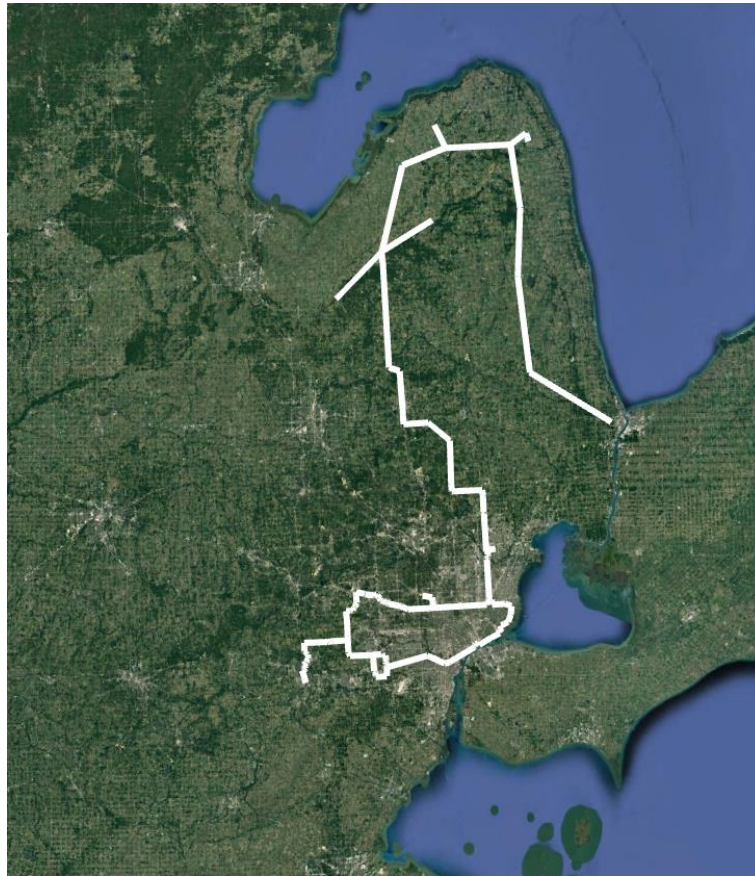
In addition to remote device management, configuration and operation, additional assets and functions that benefit from the telecommunications network are:

- Integration of DER control schemes and distributed generation (smart inverters)
- Future asset health monitoring devices such as voltage monitors and line sensors
- Collection of data from the AMI mesh of over two million AMI meters
- Future enhancements for field mobile for crews and planners
- High resolution security cameras and access control systems for infrastructure security at critical facilities

The first tier of DTEE's field network is made up of private fiber, leased fiber and point-to-point microwave systems, shown in Exhibit 12.3.1. This network was designed to interconnect data centers, power plants and critical distribution and generation systems across the service territory.

DTEE-owned fiber forms a loop around the center of the metro-Detroit area with taps to include the Ann Arbor and Lapeer Service Centers. The leased fiber is used to support the critical Thumb area substations and wind parks, the North Area Energy Center and the Lapeer Service Center. Microwave links extend the Tier 1 backbone to other service centers, power plants and critical subtransmission stations, however it is scheduled to be replaced in sections through 2025.

Exhibit 12.3.1 Map of existing Fiber systems – private and leased



DTEE's existing field communications network (FCN) does not currently have adequate capacity and does not have coverage over the entire service territory. In order to support the automation plan, it will be necessary to deploy fiber optic backbone and wireless systems to support the demand.

The majority of DTEE's service territory is without high bandwidth communications. Many of the outdated systems are radio-based systems utilizing serial protocols. This type of communication method is incompatible with modern systems, thereby limiting the system's usefulness when deploying new technologies.

Additionally, much of the equipment utilizing the radio-based systems are beyond end-of-life. Spare parts cannot be purchased or maintained. When the initial radio-based system was

designed, it was not intended for transmitting large volumes of data from the distribution system, and especially not the volume of data associated with modern network communications.

In addition to the stated gaps that will be filled with the telecommunications plan investment, there are also many associated potential O&M savings:

- For locations with cellular backhaul communications, the private network can be expanded there, saving between \$20-\$50 per month per cellular unit installed. Currently over 3,300 cellular units are supporting the existing field communications network.
- Leased lines can be eliminated, with costs ranging from hundreds to thousands of dollars a month. This includes phone lines for nearly 500 circuits, each costing between \$75 and \$200 a month, with specific data lines costing up to \$5,000 per month, plus 20-30% year over year increases due to tariff rates.
- Decreased latency and higher bandwidth will enable other uses in the substation, including high resolution security cameras and distributed energy management systems that can reduce local data storage and maintenance costs.
- Increased bandwidth availability in the mesh network may reduce cellular bills for mobile workers and line crews and will enable future applications, such as better mobile connections for new technology, including future cloud services, augmented reality and real-time system maps from the ADMS.

Projects to support new natural gas power plants and renewable energy generation sites will supplement a small portion of the fiber backbone system, however the DTEE telecommunications investment is essential in order to close the infrastructure gaps.

In evaluating alternatives for expansion, cellular technology was considered, however it has many limitations that make it unsuited for grid use:

- Due to the proximity between DTEE's service territory and Canada, cellular equipment attempts to connect to the wireless service available, sometimes between countries, degrading reliability.
- Coverage in rural areas is poor.
- Cellular network technology changes often, leading to frequent hardware replacements – for example the upcoming retirement of 3G.

- Cellular providers do not provide maintenance or unplanned outages or reliability statistics despite many requests. Tests of cellular services reliability over time have proved to be unacceptable use cases for critical control functions.
- Cellular Service Providers package utility data communications, which are ultimately sent over the public internet, leading to cyber security concerns.

Although fiber technology has many benefits, it is not practical to install fiber at every location, leading to the usage of other communication mediums to supplement connectivity. DTEE's Tier 2 telecommunication systems include wireless point-to-point and point-to-multi-point networks. To ensure these supplemental wireless systems can provide the necessary network connectivity, a fiber backbone system is needed to backhaul their communications. These fiber backbones, or trunks, are estimated to be needed within five miles of Tier 2 systems in urban areas and within 12 miles in rural areas.

To further develop the optimal solution, installation locations were weighed against the resiliency of the solution. For instance, underground fiber installations come at a very high cost unless there is existing conduit with space for additional cabling. Underground lines are still subject to dig-ins and animal interference in manholes. Optical Ground Wire (OPGW) is a specialized metal jacketed fiber that is embedded in the shield wire above the energized conductors at the top of the pole. This solution is most common for transmission system communications. However, it has limited bandwidth due to size constraints and failures typically require shutdowns and specialized equipment to fix. Modifications or new equipment installations in the transmission system space must be installed around energized conductors, leading to a much more hazardous working environment.

Locating the fiber in the internet services joint use space on the utility poles proves to be the more cost-effective area to install, as it is easy to repair and modify without shutdowns or specialized equipment, provides the easiest access to pole top equipment, and follows existing paths between circuits. Limitations include increased exposure to weather impacts, falling trees, animal interference and public interference. It also must follow existing pole leads and clearances may need to be adjusted to find room on crowded poles.

To facilitate connectivity between the overhead fiber network and substations, fiber is installed outside of the substation. Tier 2 and Tier 3 wireless systems are then used to provide last mile communications to the equipment within the substation property. This is the most cost-effective

means of data acquisition. In addition, it also has the added benefit of strengthening the Tier 3 network responsible for pole-top SCADA and AMI meter data acquisition. For many 4.8kV substations along fiber paths, this method allows those substations to get network connections for very low cost when compared to underground trenching and conduit and would not lead to stranded infrastructure should the substations be decommissioned. Substations with existing underground network communications entrances will utilize the existing access points.

Additionally, placing routers at junctions in the fiber at critical points will create independent rings in the network, reducing impacts of outages or damage and allowing communications to reroute along a different available path automatically. Newer routers have fault location technology embedded so that damage can be accurately located and addressed. An additional benefit to this is that it can supplement the electrical fault locating accuracy if damage brings down a pole or a major section of overhead wire.

Prioritization criteria for choice of new installations within grid telecommunication consider the following:

- Criticality of sites (peakers, service centers)
- RTU on SOC critical and cranking path (Cranking path is the portion of the electric system that can be isolated and then energized to deliver electric power from a generation source to enable the startup of one or more other generating units.)
- Number of devices at each location that have communications capabilities
- Number of customers served by the location
- Existing network infrastructure (towers, backhaul points)
- Planned upgrades or decommissioning at the location
- Number of pole top devices in the area
- Number of leased lines, Radio Communication Systems (DTEE's existing infrastructure and radio devices used for remote communications) radios, or sites with bandwidth issues
- Proximity to existing backbone
- Overlap with existing programs such as delta ground protection and substation physical security

DTEE has achieved cost savings by utilizing contractors already knowledgeable in the telecommunications industry. For example, there are many contractors who already install

communications equipment for joint users such as cable and telephone providers and are familiar with DTEE's specifications and requirements. In developing DTEE's standards, DTEE engaged fiber installers who provided specifications for telecom companies including Verizon and Comcast. Leveraging experienced installers alone has provided a 50% cost savings over other contractors.

Additionally, DTEE is planning for further cost reduction with additional improvements to standards and creating compatible units. Exhibit 12.3.2 shows the investment summary for Grid Telecommunications. The build out of telecommunications infrastructure will also continue beyond 2025.

Exhibit 12.3.2 Grid Telecommunications Investment Summary

	2021	2022	2023	2024	2025	2021-2025 Total
Investment Projection (\$ millions)	\$5	\$15	\$19	\$17	\$15	\$71

12.4 Distribution sensing and monitoring

DTEE's Distribution sensing and monitoring strategy has been developed as part of the Grid Needs Analysis (see Section 3.4) under the premise that increased grid visibility will improve the operation of the electrical grid and allow DTEE to maintain best-in-class safety and reliability given all the new technologies that are transforming the way the grid functions as well as customer expectations.

As discussed in Section 3 – Grid Modernization Process, grid changes, including the expected increase in the penetration of distributed energy resources (DERs), aggregations made possible under FERC Order 2222, and DTEE's pursuit of non-wire alternatives (NWAs) will present challenges to operation of the grid. Tools such as ADMS and its suite of advanced applications, as well as future Distributed Energy Resource Management Systems (DERMS), will be key to DTEE's ability to adapt. Real-time data from the distribution system will be vital to fully leverage the functionality and potential of these operational technologies and provide the necessary information to make decisions impacting safety, reliability and customer satisfaction every day.

Additional operational benefits will be achieved with enhanced sensing and monitoring data, such as ensuring more timely and accurate information regarding power quality is being delivered to customers, more granular data from an increasing number of monitoring points on the distribution system utilized to support a DERMS application, and the ability to manage distribution connected DERs as they participate in the energy markets and co-exist with traditional customer loads.

In addition to the operational benefits, there are other functions within DTEE that will benefit from robust sensing and monitoring capabilities. For example, the Distribution System Planning and Interconnections groups, as well as Corporate Energy Forecasting, will use more granular data for capacity planning (increased electric vehicle adoption, new DER loads) and impact studies, and Equipment Engineering can develop a more accurate assessment of asset health through enhanced monitoring of potentially harmful harmonics generated by DER devices on the system, which could lead to distortion in system voltage and currents.

The Sensing and Monitoring team has broken down DTEE's strategy into the following categories:

- Smart Fault Indicators/Line Sensors
- Low-cost Substation Monitoring
- SCADA and AMI Enhancement

Within these categories are the following key plans:

- Complete line sensor coverage of entire grid and pursue next generation sensor technologies
- Upgrade existing low-cost substation monitoring to provide SCADA-grade capabilities
- Enable SCADA on all overhead and substation voltage regulating equipment
- Enable SCADA on all sectionalizing equipment on distribution and sub-transmission grids
- Enable near-real-time load and power quality data from AMI meters

Overall, the investment in sensing and monitoring from 2021-2025 is \$53 million. Many of these efforts will likely extend through 2035 to provide comprehensive sensing coverage on the system. Several of these efforts will require specialized and skilled resources, which will be incorporated into the future workforce planning efforts.

To mitigate this, the work will steadily ramp up over the next five to 10 years to allow for necessary hiring and training of the resources required to support the installation and maintenance of new sensors, communication assets and smart grid devices. Exhibit 12.4.1 shows the projected investment summary for Sensing and Monitoring strategy. Sections 12.4.1-12.4.3 discuss the specific sensing technology that will be deployed in more detail.

Exhibit 12.4.1 Sensing and Monitoring Investment Summary

	2021	2022	2023	2024	2025	2021-2025 Total
Investment Projection (\$ millions)	\$2	\$6	\$12	\$13	\$16	\$49
Smart Fault Indicators/Line Sensors (\$ millions)	\$2	\$4	\$8.5	\$8.5	\$8.5	\$31.5
Low-cost Substation Monitoring (\$ millions)		\$1	\$2	\$3	\$3	\$8
ION Meter Data Integration (\$ millions)		\$1	\$1.5	\$2	\$5	\$9

12.4.1 Smart fault indicators/line sensors

Line sensors provide a low-cost and easily deployed means for DTEE to monitor the distribution system as well as provide valuable data such as periodic current readings, fault status and magnitude, and in some cases voltage and harmonic information. Line sensors can be installed at the head of circuits with no SCADA to provide near-real time load data to DTEE's Electric System Operations Center, as well as historical load-profile data that can be used by Planning and Engineering groups during their annual load analysis efforts. These sensors can also be deployed across overhead circuits for the purposes of fault locating. The real-time outage and fault data these line sensors provide has become an integral part of DTEE's outage restoration procedures, and allows operations teams to more quickly and accurately pinpoint the location of faults on the overhead system, thus reducing circuit patrol times and allowing for more timely isolation and repair.

In the future, the ADMS will rely heavily on smart fault indicator data to enable fault locating functionality such as that provided by the FLISR advanced application. To fully leverage those capabilities, DTEE will need to increase its penetration of overhead line sensors, begin to deploy more enhanced devices with voltage sensing and power quality data, and pursue an underground smart fault indicator solution.

12.4.1.1 Overhead line sensors/SFIs (current sensing)

Since 2013, DTEE has deployed over 11,000 current-sensing smart fault indicators on the distribution system, and the plan over the next five years includes completion of that deployment, as well as a transition to routine replacement, upgrades and repairs. The five-year Sensing and Monitoring plan calls for installing or replacing approximately 3,600 units per year between 2021-2025 to provide the highest level of device availability and geographical coverage of fault data for use by the ADMS fault locating functionality. Specific focus will be paid to circuits where fault status and magnitude data are not currently available and near key sectionalizing points such as A/B points.

12.4.1.2 New underground line sensor technology

DTEE has been actively pursuing an underground line sensor solution for fault locating and load monitoring on its underground distribution circuits. DTEE also plans to upgrade and expand upon the existing underground line sensor deployment on its A/C Network netbanks in downtown Detroit, an area where historically DTEE has not had any remote visibility. The benefits of this type of data are numerous, including understanding when netbanks have gone offline due to blown fuses or equipment issues. Because of the redundant build of the system, it has been difficult to understand loading on the various netbanks and allow for Planning and Engineering to accurately model the system.

Overall, DTEE expects to equip approximately 500 distribution circuits with underground smart fault indicators (line sensors) and another 500 A/C Network netbank transformers for a total of approximately 3,000 devices (three per location) over the next five years. Beyond that timeframe, around \$100,000 a year will be needed to cover device failure replacements and upgrades.

12.4.1.3 Voltage-sensing smart fault indicator technology

As part of grid modernization, there is a need for voltage data on the circuits and near the substations, in parts of the grid where there currently are no smart relays, SCADA-enabled overhead equipment or other means to collect that data and bring it back to EMS, ADMS and PI, the data archive. There are several line sensor products that have come to market in recent years that can provide this data at a relatively low cost for equipment and installation.

As part of the five-year and 10–15-year Sensing and Monitoring strategy, DTEE plans to deploy voltage-sensing line sensors at key locations on the overhead system to collect real-time current, voltage and harmonic data in of support programs including CVR/VVO (See Section 12.6 - CVR/VVO), DER monitoring/DERMS (See Section 12.8 – DER Control) and asset health.

For asset health, many of these voltage-sensing devices also provide harmonic data, which can be used to observe whether harmful harmonics are being introduced as result of DER devices or other causes on the distribution.

DTEE plans to deploy voltage-sensing devices out on the distribution system at approximately 500 locations over the next five years to provide ADMS and DERMS the necessary voltage information to make operational decisions to preserve the reliability of the grid. DTEE will also install voltage-sensing SFIs at another 150 locations to support CVR/VVO over the next five years, and then another 500 locations in the next 10-15 years.

12.4.2 Low-cost substation monitoring

Low-cost Substation Monitoring (LCM) has been a program run by the DR-SOC (Distributed Resources System Operations Center) with the goal of providing distribution substation data availability where there has traditionally been no SCADA telemetry, such as with many of the older 4.8kV substations. Instead of SCADA, DTEE will install low-cost meters that retrofit onto substation control circuitry to provide near real-time load and breaker status information at each feeder position, and voltage and current readings at each distribution transformer position. The data is transmitted over cellular back to the SOC at 5-minute intervals, as opposed to 2-4 second intervals with traditional SCADA data. Over 25% of all distribution feeders are now monitored using Low Cost Metering (LCM) technology.

The Sensing and Monitoring strategy calls for some necessary upgrades to the existing LCM substations as well as an expansion of the LCM deployment in order to provide more reliable

substation-level situational awareness for operational tools like ADMS as well as engineering resources like PI, which can make the data available for capacity planning and analysis.

12.4.2.1 Low-cost monitoring technology upgrades

In order to increase reliability of the monitoring platform and the rate at which data is transmitted back to ADMS via the DR-SOC, the LCM data loggers will be upgraded to SCADA grade Remote Terminal Units (RTUs) with improved telecom backhauls such as TropOS, Digi modems (another communications technology provider) or fiber. The metering and communicating wiring beyond the RTU will remain the same, however. The expectation is that with these upgrades, data from all LCM-equipped substations will be transmitted every 2-4 seconds, instead of every five minutes, resulting in much more real-time visibility into the state of the feeders and transformers at those substations. The cost to perform these upgrades includes a Real Time Automation Controller (RTAC) RTU, network switch and engineering and field labor to install and test. There are approximately 200 LCM substations that will be upgraded as part of the five-year plan.

12.4.2.2 Complete low-cost monitoring deployment

Low-cost Substation monitoring is a technology option that will defer or avoid SCADA retrofitting at older substations, but still provide SOC and Engineering with the real-time telemetry needed to operate and maintain the distribution grid until these substation breakers or substations themselves are replaced and fitted with traditional SCADA.

There are approximately 40 4.8kV substations and 75 13.2kV substations that do not have SCADA at which DTEE will explore deploying LCM technology. Many of those substations currently use line sensors at the head of circuits to provide the real-time load data to SOC but lack the breaker status and transformer data that is provided by LCM. Furthermore, line sensor data is transmitted every 5 minutes whereas LCM after the upgrades implemented in the Sensing and Monitoring plan will provide more data at the higher rate of every 2-4 seconds.

12.4.3 ION meter data integration

The DR-SOC has been deploying Schneider's ION meters at substation transformer positions as part of the low-cost monitoring installations to provide real-time current and voltage data to the SOC. There is potential for these meters to provide even more valuable data to DTEE, including more detailed power quality data, as well as fault information at the transformer/bus level. DTEE

plans to integrate these LCM substation transformer ION meters into ADMS as well as into the PQView power quality software system to collect data in the same way as Commercial & Industrial (C&I) and subtransmission ION meters collect data. There are also additional capabilities using Schneider's PME2020 software to achieve even further integration with DTEE's operational technology systems.

There are approximately 250 ION meter installations as part of the LCM program, but that number will likely double over time. Any new installations will be set up with the necessary integration steps during construction of the job.

12.4.3.1 SCADA and AMI enhancements

DTEE's deployment of an Advanced Distribution Management System (ADMS), along with an anticipated Distributed Energy Resource Management System (DERMS), will have a profound impact on the operation of the grid to provide safer, more reliable power to its customers. The suite of advanced applications these tools provide require timely, accurate and available data sources in order to inform correct operating decisions.

Two of the most vital sources of system data are SCADA and AMI. In addition to the remote-control capabilities SCADA provides, the variety of available data and alarms from all remote SCADA-enabled field devices provides crucial situational awareness and feedback to System Operations. AMI provides key data directly from the customer, but the current system is not real-time, other than power-off and power-restoration notification, limiting its use in operations.

The following sections describe some of the key enhancements to SCADA and AMI that are being targeted with Sensing and Monitoring strategy.

12.4.3.2 SCADA on all 3-phase operating devices

A key aspect of distribution sensing and monitoring includes providing 2-4 second interval SCADA updates from all distribution and subtransmission 3-phase operating devices including per phase voltage, current and power quantities (bidirectional and on both sides of the disconnect), as well as fault magnitude and waveform data.

On the distribution system, all older vintage Form 3 and Form 5 reclosers, which only provide basic information, need to be upgraded to a more modern device, such as Form 6 or Viper recloser, which includes the larger set of data required for ADMS and other application.

Additionally, some Form 6 controls will also need to be updated in order to be able to retrieve fault data. Additionally, installing automated, SCADA-capable devices at two jumpering points per circuit (approximately 6,000 total) is preferred.

It should be noted that the distribution overhead work is already included as part of the Distribution Automation program (Section 12.5 – Distribution Automation), therefore costs for this work are not captured as part of the Sensing and Monitoring strategy. However, enabling SCADA on all subtransmission 3-phase operating devices, such as pole top switches, is not included in the Distribution Automation program, so the costs to implement SCADA monitoring on the approximately 3,000 devices across 550 subtransmission lines are included in this program.

12.4.3.3 Substation transformer data enhancements

DTEE plans to equip all substation and station transformers with breaker duty monitoring for asset health, and sequence of events recording, which monitors collected data and records the time and sequences of changes, as well as power quality meters with harmonic monitoring to the 50th harmonic, including asset health monitoring. This will enable better situational awareness and predictive maintenance capabilities as well as understand the impact of increased DER penetration on these critical assets.

It should be noted that the 13.2kV substation work is already included in the Substation Automation program scope, therefore costs for this work are not captured as part of the Sensing and Monitoring strategy. However, enabling SCADA on other voltage class substation transformers is not included in the Distribution Automation program. This program will implement SCADA monitoring on the estimated 1,000 devices not covered by Substation Automation.

12.4.3.4 SCADA on all voltage regulating equipment

Having voltage data as well as state and condition data of voltage regulating equipment on DTEE's system is necessary to support programs like CVR/VVO, and to handle the increasing penetration of DERs on the distribution system. Tools like ADMS and DERMS will require accurate voltage data in order to facilitate critical decisions involving power quality and compliance with voltage standards.

Voltage regulating equipment, such as regulators and capacitors, can be found both in the substations and on the overhead circuits. As part of the Sensing and Monitoring strategy, DTEE expects to outfit all equipment with 2-4 second interval SCADA data. The cost of the devices themselves will be covered under other programs, but the cost of SCADA integration is expected to cost approximately \$10,000/device on the overhead. The Distribution Automation program will include equipment and SCADA integration for all capacitors and regulators within the substations, so that cost is not included here. That said, the combination of substation and overhead voltage regulating equipment data is necessary to leverage the full capabilities of the ADMS and DERMS.

12.4.3.5 C&I and large customer meter data enhancement

The goal with this aspect of the Sensing and Monitoring strategy is to gain real-time power quality and load data from most of DTEE's largest customers. These customers are 500kW and above in size, but not large enough to have industrial substations. This data will allow DTEE to make better operating decisions as well as to develop more accurate models of the commercial and industrial loads throughout the system for operating studies and capacity planning.

The five-year plan will be to retrieve five-minute data from all C&I AMI meters, using an updated Meter Data Management system (currently funded IT project) and an appropriate data retrieval tool based on meter type. There are two AMI vendors for C&I sites: Schneider and Honeywell. For the approximately 1,000 Schneider ION meters, DTEE will utilize the PME2020 software to access the required data. For the remaining Honeywell meters, a third-party system like Itron will be needed to access the same data.

Beyond 2025, DTEE will pursue two- to four-second interval data updates from the large customer meters which is more in line with SCADA-enabled metering. This will involve IT and potentially some meter upgrades.

12.4.3.6 Push Notifications for AMI near DER or CVR/VVO

The AMI enhancements DTEE is seeking to implement over the next 15 years will enable full situational awareness of customer load, power quality and outage status data in real-time, straight from the customer's meter.

The five-year plan calls for implementation of "push notification" functionality which is currently planned and under development. It enables near real-time updates of AMI data to a collection

engine where it can be routed to downstream systems like ADMS, PI, or a cloud-based data lake. The AMI group within DTEE has determined that, due to significant data and network requirements, this feature will be limited to approximately 10% of the meter population or around 260,000 AMI meters, so the limited subset will be prioritized for customers with bidirectional DER meters (DG/DER tariffs) and key customer locations on feeders that are part of the CVR/VVO program.

The operational benefits to ADMS are numerous, but specifically with DER customers, the data will allow DTEE to perform analysis of individual customer loads in near real-time. As DTEE develops plans to maintain grid stability under high DER penetration, it may be necessary to develop strategies that will permit/deny individual DER participation in market activities at a given time, such as those enabled by FERC Orders 841 and 2222 during abnormal system conditions or during circuit maintenance.

12.5 Distribution automation

As part of DTEE's grid modernization effort described in Section 3 – Grid Modernization Process, a large investment is required to upgrade the distribution system so that it enables automation capabilities. As explained in the DOE report, "Distribution Automation: Results from the Smart Grid Investment Grant Program²⁰," published in September 2016, "Distribution automation (DA) uses digital sensors and switches with advanced control and communication technologies to automate feeder switching, voltage and equipment health monitoring; and outage, voltage and reactive power management. Automation can improve the speed, cost, accuracy of these key distribution functions to deliver reliability improvements and cost savings to customers."

DTEE has identified five areas where automation can provide a significant positive impact to the grid:

- **Safety** – Increasing the amount of equipment that can be operated remotely to deenergize sections of lines that have potential hazards will ensure safety to employees and the public, including wire down detection and isolation on the 4.8kV system.
- **Reliability** – Remote identification and isolation of faults will be possible, reducing crew patrol time and enabling faster restoration for sections of circuits unaffected by the fault.

²⁰ "Distribution Automation: Results from the Smart Grid Investment Grant Program" (Sep 2016) [Distribution Automation: Results from the Smart Grid Investment Grant Program \(energy.gov\)](https://www.energy.gov/distribution-automation-results-smart-grid-investment-grant-program)

- **Operability** – Realize reconfigurations of the substations and circuits from the Electric System Operations Center to support daily operational needs, eliminating travel time to operating points and minimizing handoffs among trades.
- **Asset Health** – The health of advanced devices will be more visible, improving the availability and reliability of automatic restoration and switching.
- **Cost** – The benefits to reliability, operability, and asset health ultimately lead to cost saving opportunities for DTEE customers.

The Smart Grid Investment Grant Program as discussed by the DOE, demonstrated how distribution automation technologies can achieve substantial beneficial grid impacts, including improved Fault Location, Isolation and Service Restoration (FLISR) capabilities, improved distribution system resilience to extreme weather, more effective equipment monitoring and preventative maintenance, more efficient use of repair crews, reduced repair times and improved grid integration of DER.

As detailed in the DOE report on Distribution Automation, five utilities reported a 50-55% reduction in the number of customer interruptions and customer minutes of interruptions per outage events during FLISR operations and three utilities reported SAIFI improvements of 17-58% from pre-deployment baselines, thanks to the ability to more quickly identify the location of a fault and perform switching activities that allow the fault to be isolated so that fewer customers are impacted during the repair activities.

For DTEE, Distribution Automation is planned for substation automation and circuit automation, while Conservation Voltage Reduction / Volt-Var Optimization (CVR/VVO) has automation as part of the project but is identified as a separate program (Section 12.6 – CVR/VVO).

Substation automation involves control panel replacements with modern, standardized relays, installation of Remote Terminal Units (RTU), incorporation of the RTUs and automation controls in the same substation network and breaker replacements as needed.

Circuit automation involves installation of remotely controllable/automatic reclosers and sensing devices, replacements of end-of-life automation devices that cannot be retrofitted, and reconductoring as needed to enable circuit load transfer during outage conditions.

The automation technologies installed on substations and circuits will also be linked to the ADMS for enhanced applications. These include, but are not limited to, the FLISR application, which

combines fault location information with the circuit loading to optimize restoration and automatically restore service, and the Optimal Feeder Reconfiguration application, which analyzes load transfer options that can be executed across multiple substations to manage system loading conditions and improve system performance. While distribution automation applies to both circuits and substations, each area has specific requirements, functionalities and prioritization methods.

12.5.1 Circuit automation

DTEE's long-term goal for its Circuit Automation program is automating the entire overhead distribution. Because DTEE's distribution system has 4.8kV, 8.3kV, and 13.2kV infrastructure, which have vastly different configurations, short-term and long-term plans for each voltage type vary.

For the 13.2kV system, circuit automation involves installing SCADA reclosers and pole top switches, replacing outdated SCADA devices (such as Form 3 and Form 5 Reclosers), upgrading controls and protocols, or adding SCADA controllable devices to existing circuits.

All new circuits constructed today follow a standardized distribution design order and are installed with SCADA controllable reclosers and switches for every 4 MVA of load or circuit section containing 500 or more customers with a tie recloser between adjacent circuits. Any circuit improvement projects may also include SCADA reclosers, switches, regulators, and capacitors as part of the toolbox to improve circuit reliability. These devices will be used to isolate faults and provide service restoration automatically or under the direction of the ADMS and Electric System Operations Center. According to the DOE report referenced previously, other electric utilities have had significant reliability improvements after implementing circuit automation technology.

As discussed in Section 11.5 - 8.3kV Conversion and Consolidation, the 8.3kV system will be converted to 13.2kV by 2028, at which time automation will be added, as it is part of the distribution design orders.

Because of the complexity and uniqueness of the 4.8kV system, DTEE has done several equipment studies to determine how to address automation for this portion of the distribution system. For example, DTEE is currently testing a new recloser installed on the start of a 4.8kV circuit to be utilized to detect and isolate grounds on the circuit. As 4.8kV circuits are converted to 13.2kV, automation will be added as well (see Section 11.3 - 4.8kV Conversion).

Today, about 25% of the distribution circuits in DTEE's territory have SCADA monitoring and control. Approximately 5% of the distribution circuits in DTEE's territory have automatic loop schemes, which can automatically transfer sections of the circuits to adjacent circuits when an outage is detected. In order to achieve automation of the entire distribution system there will be a significant engineering effort to analyze and design the circuit equipment as well as Power Equipment and Relay Testing (PERT) SCADA resources for deployment of SCADA distribution devices.

Additionally, this program will require significant coordination with the long-term circuit planning process as the schemes increase operational flexibility but limit planning options to reconfigure circuits and increase distribution costs to adjust for new customers and changing loads. Significant portions of the system also require capacity upgrades to fully utilize automation once installed.

Exhibit 12.5.1.1 shows the anticipated investment and work to be done over the next five years.

Exhibit 12.5.1.1 Distribution Automation Circuit Investment Plan

	2021	2022	2023	2024	2025	2021-2025 Total
Circuit Automation (\$ millions)	-	\$2	\$5	\$10	\$12	\$29
Units (Circuit)		6	14	28	33	81
Units Remaining	2057	2051	2037	2009	1976	1976

Key considerations for prioritizing circuits for remote monitoring and control capabilities include historical circuit reliability performance, existing infrastructure considerations such as wire sizes, as well as the number of options for transferring load to adjacent circuits. For example, a circuit that has sufficiently large wire is more ready to automate than a circuit with smaller wire (see Section 8.9 – Small Wire). Establishing an objective prioritization method ensures that the program is implemented the most efficiently to ensure maximum benefits are achieved for DTEE's customers and the distribution system. While circuit automation will be executed strategically to maximize customer benefits, not all circuits are good candidates because of their circuit

configuration and cost effectiveness. For example, if a 4.8kV circuit is part of the 13.2kV conversion program in the short term, then it will be excluded from the automation program because automation will be included in the conversion. Additionally, circuits that are mainly underground, have low load, or small customer counts may not benefit from circuit automation upgrades.

12.5.2 Substation automation program

The Substation Automation program involves installing SCADA control at all DTEE substations to allow for full remote monitoring and control. The program is designed to retrofit existing substations with advanced monitoring and control technology. Over the past year, DTEE has worked on developing a standardized panel design for substation automation that involves replacing electromechanical relays with microprocessor relays, as well as upgrades to the SCADA and telecommunications infrastructure within the substation. This allows monitoring, control, and automation of each circuit position within the substation. Roughly 32% of general-purpose substations in DTEE's territory have SCADA monitoring and control.

This program is part of DTEE's long-term strategy to modernize the electric grid and will greatly improve operational efficiency during substation planned work and improve restoration efficiency during trouble and storm events. When advanced remote monitoring equipment is installed at substations, there is a potential to use breaker operation data to assess breaker and relay functions. If the equipment is assessed to be performing as designed, it may be possible to defer equipment maintenance activities.

Due to today's limited remote monitoring capability at substations, DTEE has not yet been able to widely apply this methodology to defer breaker or relay inspections. DTEE's telecommunications, breaker replacement and substation automation programs, despite targeting different areas of the substation assets, will need to be coordinated to increase the penetration of such advanced remote monitoring technology and enable the application of condition-based maintenance on breakers and relays. These programs are also highly coupled with the ADMS.

As with the Circuit Automation program, 13.2kV substations are better candidates for the Substation program as their configurations better facilitate the technology upgrades needed for automation. Additionally, the primary goal of 4.8kV automation is the detection and response to

ground alarms; therefore, the installation of reclosers as previously described also serves in part as the Substation automation for these substations.

Key considerations for prioritizing substations for remote monitoring and control capabilities primarily include historical substation reliability performance, gauged by number of circuit level lockouts, as well as the current level of automation at each substation. For example, if a substation already has some automation capabilities, then completing the automation for that substation may take precedence over a substation with no automation. However, reliability must also be taken into consideration when prioritizing which substations to automate; a substation with poor reliability may be chosen to be automated ahead of another substation with existing automation capabilities if the former's reliability performance could be improved with automation. As with Circuit Automation, establishing an objective prioritization method ensures that the program is implemented the most efficiently to ensure maximum benefits are achieved for DTEE's customers and the distribution system. While substation automation will be executed strategically to maximize customer benefits, not all substations are good candidates for automation given physical constraints or cost effectiveness. For instance, some 4.8kV substations do not have the physical infrastructure (breakers, relays, etc.) to support distribution automation. It will become possible to deploy automation to those sites as they are converted to 13.2kV. Additionally, coordination with other projects is an important consideration, such as conversion and consolidation and 4.8kV breaker replacements.

DTEE is currently upgrading its Jefferson substation in Trenton as part of the automation program. There have been many lessons learned along the way that will shape how DTEE executes the program, including development of standards for replacing existing relay panels, as well as working through resource constraints in design, panel fabrication and the field resources that complete the work in the substations.

There are 3,254 active distribution positions on DTEE's electrical system, with 1,050 already having SCADA control and monitoring (an additional 1,280 have monitoring without control). This leaves 352 13.2kV positions and 1901 4.8kV positions that require upgrades for this project.

Exhibit 12.5.2.1 shows the anticipated investment and work to be done over the next five years.

Exhibit 12.5.2.1 Distribution Automation Substation Investment Plan

	2021	2022	2023	2024	2025	2021-2025 Total
Substation Automation (\$millions)	<\$1 M	\$2	\$3	\$10	\$12	\$27
Units (13.2kV substation positions)	1	9	14	45	54	123
13.2kV positions remaining	351	342	328	283	229	229

12.6 CVR/VVO

To further upgrade DTEE's distribution system as part of its grid modernization efforts, DTEE has been evaluating Conservation Voltage Reduction/Volt-Var Optimization (CVR/VVO) as an option to reduce peak demand and energy consumption. The initial pilots were developed as part of DTEE's Integrated Resource Planning (IRP) filing in Case U-20471. DTEE is continuing to evaluate CVR/VVO through pilots both as an offset to peak generation and because of the potential benefits to the distribution grid. These benefits include improving grid efficiency as well as having sufficient capacity for demand.

Volt-Var Optimization (VVO) manages system-wide voltage levels and reactive power flow to achieve one or more specific operating objectives. The objectives can include reducing losses, managing voltage volatility due to intermittent renewable generation, optimizing operating parameters and/or optimizing power factors. Conservation Voltage Reduction (CVR), as one of the Volt-Var Control (VVO) options, is designed to maintain customer voltage levels in the lower portion of the allowable voltage ranges, thus reducing system losses, peak demand, or energy consumption.

CVR is achieved by utilizing equipment such as transformer load tap changers (LTC), overhead line regulators and capacitor banks. In addition, SCADA monitoring devices and line sensors are used to help manage voltage levels and advanced telecommunication and optimization tools can be used to achieve optimal demand and energy savings in the system.

DTEE has engaged industry experts (Burns & McDonnell and Dynamic Energy Group) and peer utilities (AEP, CMS, First Energy, and OG&E) that have implemented CVR/VVO methods to

achieve energy and demand savings for customers. DTEE is in the process of implementing two pilot studies across four substations to validate benefits realization for DTEE's system ahead of full-scale implementation.

12.6.1 First CVR/VVO pilot

The initial pilot began in 2020 as part of DTEE's IRP filing, at Beck (Roseville) and Medina (Clinton Township) substations, and follows a more static approach to CVR, as initially it will not be utilizing an advanced computer system, such as Advanced Distribution Management System (ADMS), for automatic device controls; however, construction will facilitate smooth transition to future ADMS control. These substations were chosen because they have a diverse base of residential, commercial and industrial customers, substation conditions (SCADA presence), voltage (mixed 4.8kV and 13.2kV) and minimal Distributed Generation.

The original scope for the project called for manually setting the substation transformer voltage, installing SCADA controlled capacitors with time-based settings, and then measuring customer voltage impacts through AMI on secondary lines and voltage sensors on primary lines. However, as the project went through its design phase, it was found that it was necessary to install SCADA controls at the substation transformers, allowing Engineering or the System Operations Center (SOC) to program and modify, if needed, the voltage schedule to better manage voltage conditions for the customers.

Additionally, using automatic voltage settings instead of time-based settings on the capacitors will lead to better risk mitigation with the CVR scheme if real-time conditions, such as low-voltage conditions, warrant a change in operations. Additionally, using smart capacitor controls may eliminate the need for annual capacitor inspections, increase capacitor availability and ultimately help improve power quality.

As additional substations are investigated during design phase of CVR, more improvements may be discovered as well. For example, there is investigation underway of potential benefits of adding additional capacitors at certain substations ahead of full CVR implementation that could lead to immediate system peak and loss reduction and prepare those substations for advanced CVR/VVO when in place.

As reported in DTEE's U-20471 Integrated Resource Plan Annual Report²¹, DTEE's CVR/VVO pilot is designed to use various technologies to balance line voltage and system reactive power to reduce system line losses, reduce peak demand and improve the efficiency of the distribution grid. This is accomplished by reducing the voltage on the distribution system and utilizing monitoring devices and line sensors to ensure the voltage remains within the allowable voltage range.

As part of this pilot, DTEE successfully enabled CVR/VVO technology on six pilot circuits on the distribution system in December 2020. The CVR/VVO pilot's scope was to choose up to 20 circuits for study and implementation, ultimately 6 circuits were chosen for the implementation phase. The identification criteria for these pilot circuits was based on a conceptual engineering study which identified that complete 13.2kV circuits (those without 4.8kV areas) in mixed urban/suburban areas were the most suitable candidates for CVR/VVO. Circuits with these types of characteristics are the least likely to require expensive circuit upgrades that were required to avoid experiencing voltages below the lower bounds.

CVR/VVO can achieve demand reduction and energy savings by lowering the voltage on all circuits fed by that transformer. This is achieved by changing the load tap setting at the substation transformer. To determine the magnitude of voltage reduction possible, detailed engineering studies and modeling must be performed to ensure that while the circuit voltage is lowered, no customers on the circuit experience voltage below the acceptable lower bounds.

In 2019 and early 2020, DTEE conducted field verification and performed detailed engineering studies on 18 circuits fed from six transformers. Field verification is an important step to achieve a high degree of circuit model accuracy and to ensure that low voltage conditions are avoided across the entire circuit. After the circuit models were updated, detailed engineering studies were performed to determine the voltage reductions possible at substation transformers and LTCs while at the same time avoiding any low voltage conditions for customers, particularly at peak load. The detailed screening analysis performed showed that out of the 18 circuits and six transformers studied, two substations, Beck in Roseville and Medina in Clinton Township, and six associated circuits were suitable for CVR.

²¹ See Exhibit in filing, "IRP REPORT OF DTE ELECTRIC COMPANY," [068t00000MNdSmAAL \(force.com\)](#)

After the screening analysis was completed, field implementation began for the six circuits. Deploying CVR in 2020 was subject to challenges which delayed implementation until December 1, 2020. The major causes of delays were equipment procurement and the additional time to complete engineering and field work. Both delays were due to COVID-19 related issues, which impacted the availability of the equipment and the ability to do field work as planned.

DTEE began the initial measurement and verification process in December 2020 after implementation on the six circuits was completed. DTEE will continue this process throughout 2021 to measure demand reduction and energy conservation for the full year and to determine the overall effectiveness of the CVR pilot.

Each of the four seasons has different customer load characteristics, which can cause markedly different CVR results. Analyzing results from spring, summer, fall and winter is needed to obtain a valid estimate of the pilot results. Demand reduction and energy conservation due to CVR are expected to be highest during the peak summer season.

DTEE plans to continue the investments on CVR/VVO beyond the pilots to cover additional circuits and substations with the goal of maximizing energy and demand savings for customers. The pilot application of CVR/VVO uses a static approach where the voltages are reduced permanently. Phase 2 of the CVR/VVO program will use a more advanced approach, which will be enabled by the VVC module of the ADMS to realize real-time dynamic control of the equipment involved. It is expected to produce higher demand and energy savings than the static approach and will also provide flexibility of adjusting voltages to better accommodate distributed energy resources.

12.6.2 Second CVR/VVO pilot

The second pilot, or advanced CVR/VVO, will use the new ADMS' built-in CVR/VVO modules to setup, monitor and control the CVR implementation dynamically, based on real-time conditions to achieve the maximum potential peak and energy savings at two substations. These substations, Augusta (Macomb) and Jefferson (Trenton), were chosen based on potential CVR savings and because additional grid modernization work is being done at Jefferson. Engineering activities were done in 2020 with the actual CVR implementation expected to occur in 2022 once the ADMS module is operational.

The ADMS is crucial to achieving full potential of CVR across any large-scale portion of the distribution system. There is close coordination ongoing with the ADMS technical and management teams on go-live timelines. Additionally, the SOC is actively engaged in identifying required procedures and documentation for the control room. Lastly, requirements for ADMS control of devices is being designed into all field devices that will be installed ahead of go-live. That way the hardware can be installed and brought online for seamless integration once ADMS is in operation. An example of such hardware would be capacitor controllers, which historically have been set to turn on/off on a schedule rather than on real-time data. Using sophisticated capacitor controllers in conjunction with ADMS will allow real-time system conditions to determine when to turn capacitors on/off rather than set times of day.

Highlighting the differences between the two pilots, with the first pilot approach of CVR/VVO, if a voltage reduction on substation transformer leads to low voltage conditions during any time period, the substation transformer would not be selected for CVR/VVO implementation, thus limiting its applicability. In contrast, the substation transformer could still be selected for the advanced approach of CVR/VVO because the substation transformer voltages would be adjusted up and down to maximize voltage reduction and still avoid low voltage conditions during certain time periods. The advanced approach can lead to energy consumption and peak demand savings ranging from 1.5-3.5 % based on industry benchmarks, depending upon the substation conditions. DTEE has hired an experienced third-party vendor (DEG) for the Evaluation, Measurement and Verification (EM&V) of the pilots.

It is important to note that it is neither feasible nor cost-effective to implement CVR/VVO across the entire system of DTEE. Based on recommendations from industry experts, due to low energy reduction potential, circuits on the 4.8kV system are not suitable for CVR projects. Additionally, some substations on the 13.2kV system would not yield a sufficient yearly energy reduction to justify the cost of CVR at those substations.

Based on DTEE's initial analysis, substations that feed less than 35 gigawatt-hours per year are not ideal candidates to be cost effective for an advanced CVR/VVO program; this results in approximately 740 circuits across approximately 140 substations as potentially viable candidates for the advanced CVR/VVO program.

Further prioritization and selection criteria for these circuits and substations include four steps:

- First, all the substations are analyzed and prioritized based on largest potential CVR savings.
- Second, a more in-depth feasibility and constructability assessment of the substation equipment at each substation would be completed to determine if the estimated savings matched modeling results.
- Third, an assessment of the respective circuits' AMI, SCADA and circuit models would be studied for customer voltage risks.
- Fourth, a preflight would occur to verify the study assumptions. As stated in the Power Quality section of this report (7.4 – Power Quality) additional attention regarding placement and operation of the capacitors needs to be taken into consideration to avoid potential ferroresonance on the electrical system.

Between 2021 and 2025, a total of \$69 million capital investment is projected, which would cover 100 substations, 590 circuits, and yield a max potential savings of between 200 and 270 GWh of energy savings and between 70 and 140 MW of peak demand reduction. Exhibit 12.6.1 shows the investment summary for CVR.

Exhibit 12.6.1 CVR Investment Summary

	2021	2022	2023	2024	2025	2021-2025 Total
Investment Projection	\$6 M	\$7 M	\$17 M	\$18 M	\$20 M	\$69 M

12.7 NWA pilots and other technology pilots

As discussed in Section 4.3 – Non-Wire Alternatives, a non-wires alternative (NWA) project is an investment or project that uses DER, energy waste reduction (EWR) or demand response (DR) to defer or offset the need for a traditional investment. The focus of DTEE's current suite of projects is to address a capacity need. There is currently a suite of six NWA projects planned in the near-term five-year plan which address the current grid needs. There are also two additional Technology Pilots (Small Solar and Storage Testing Facility and Electric Vehicle Extreme Fast

Charging and charge management) that will help DTEE understand the grid impacts of DERs and EVs and may help facilitate the development of future NWAs.

A table at the end of each pilot summarizes the key information on each project, including:

- Cost of traditional solution compared to the cost of the NWA solution
- Loading patterns with and without the NWA
- Implementation Timing
- Assumptions used in the analysis.

These pilots being pursued should be viewed as building blocks which will form a foundation for how to include NWAs into the overall distribution planning process. As the capabilities of the NWA technologies are confirmed, multiple NWAs can be combined together to further advance the Company's utilization of NWAs. This will depend to a large extent on the results of the different pilots and on the experience of other utilities that are pursuing similar pilots. Once the understanding of NWA applicability and economics has matured, the planning process will allow engineers to identify projects where NWAs could be effective options when compared to traditional wire solutions. If a project is identified as a potential candidate, planning engineers would work with storage and EWR/DR experts to evaluate detailed designs. As more is learned from the Company's pilots and other utilities, and as technology advances, the Company will refine its criteria, develop new tools, and advance the overall process for pursuing NWAs as part of the distribution project review process.

In addition, the design of these pilots will follow the guidelines developed as part of the MI Power Grid collaboration on pilots under docket U-20645, including providing details on the six key areas of 1) Need and goals, 2) Design and evaluation plan, 3) project costs, 4) timeline, 5) stakeholder engagement, and 6) public interest. Many of these pilots are currently under development and are expected to evolve and be refined as they are proposed for regulatory approval. Details on how the NWA pilots are being designed to follow the guidelines can be found in Appendix V.

Exhibit 12.7.1 summarizes the expected spend on these projects. Exhibit 12.7.2 summarizes the technologies included as part of the suite of NWA projects, as well as the high-level objective.

Exhibit 12.7.1 NWA Projects and Other Technology Pilots Investment Summary

Project	2021	2022	2023	2024	2025	2021-2025
NWA Pilot Projects						
Fisher	\$0.04 M	\$0.8 M	\$3.4 M	-	-	\$4.2 M
Port Austin	-	\$2.0 M	\$2.5 M	-	-	\$4.5 M
Omega	\$3.4 M	\$3.5 M	-	-	-	\$6.9 M
Veridian	-	\$1.8 M	\$5.6 M	\$0.3 M	\$0.3 M	\$8.1 M
O'Shea	\$1.3 M	-	-	-	-	\$1.3 M
Other Technology Pilots						
Mobile Energy Storage	\$1.5 M	\$0.5 M	\$1.5 M	\$1.5 M	-	\$5.0 M
Small Solar and Storage Test Bed	\$0.4 M	\$0.4 M	\$0.3 M	\$0.3 M	\$0.3 M	\$1.6 M
EV Fast Charging and Charge Management	\$0.7 M	\$0.8 M	\$1.3 M	\$0.4 M	\$0.4 M	\$3.6 M

Exhibit 12.7.2 Technology Projects Use Cases, Technologies and Objectives

Use Case	Project	Technology							Objective
		EWR	DR	Storage	Solar	BTM DER	EV Charging	Micro- grid	
Substation Loading	1. Hancock (Completed)	✓							Initial pilot to test geo-targeted EWR and develop DER Estimating tool
	2. Fisher	✓	✓						- Test geo-targeted EWR/DR costs - Measure implementation timing
	3. Port Austin			✓	✓			✓	- Test solar and storage to address substation capacity - Test redeployment of stationary battery from Omega
Subtransmission Loading	4. Omega			✓					- Deploy storage to address subtransmission loading - Install battery that can be re-located
Customer Load and DER Control	5. Veridian		✓	✓	✓	✓	✓	✓	Develop secure and effective methods to interface and control behind the meter (BTM) DER in conjunction with utility scale DER
Voltage Mitigation	6. O'Shea			✓					Test effectiveness of storage to address voltage instability due to intermittent solar
Portable Battery	7. Battery Trailer			✓					Test use of energy storage in place of traditional portable generators

12.7.1 Hancock substation NWA pilot

DTEE began its first NWA pilot at Hancock substation in 2018 as a collaborative team that consisted of DTEE and four external entities: ICF, Guidehouse, Energy Sciences, and the Natural Resources Defense Council (NRDC). The pilot had three main goals:

- To field test the effectiveness of energy efficiency (EWR) programs to address substation loading
- To understand the costs of deploying targeted energy efficiency
- To begin developing evaluation tools for future NWA projects

The design and implementation of the pilot ran approximately 18 months – from May 2018 to December 2019 – with field implementation (i.e., roll-out of EWR programs to the geo-targeted region) occurring during the last eight months of that period. Analysis of the program impact occurred in 2020. Initially, the Hancock substation was selected to address an overload of approximately 0.6 MVA. Over the course of pilot design, development in the area increased the overload projection to be 12 MVA by 2021. Due to this change, a traditional solution, including upgrading substation transformers and creating a new circuit to relieve load, was implemented.

The team selected a portfolio of EWR programs for residential, and commercial and industrial (C&I) customers. These programs included HVAC upgrades, appliance recycling and home energy survey programs for residential customers and lighting and HVAC upgrades for C&I customers.

Overall, the program set a target of 40kW of peak load reduction based on the limited implementation duration of the pilot (about eight months). An NWA project that targets achieving load reduction over a multiyear period should be able to achieve a proportionally higher reduction. After verification of peak savings, the pilot achieved 57kW of peak reduction, or approximately 141% of the goal.

There were multiple lessons learned from the pilot and processes developed that will be applied to future projects. The first is the development of the DER estimator tool, which allows for various technologies to be evaluated as a means to reduce peak demand. Secondly, the pilot allowed for some initial benefit-cost analysis that indicated that the C&I programs show more potential than residential programs. The takeaway is that having a representation of C&I customers on future

NWA circuits would be key to achieving peak reductions. Finally, there were multiple learnings from an implementation standpoint, such as the incentives necessary to drive the adoption of programs, the seasonality of customer interest in HVAC upgrades, and the type of marketing materials and approaches that were most successful.

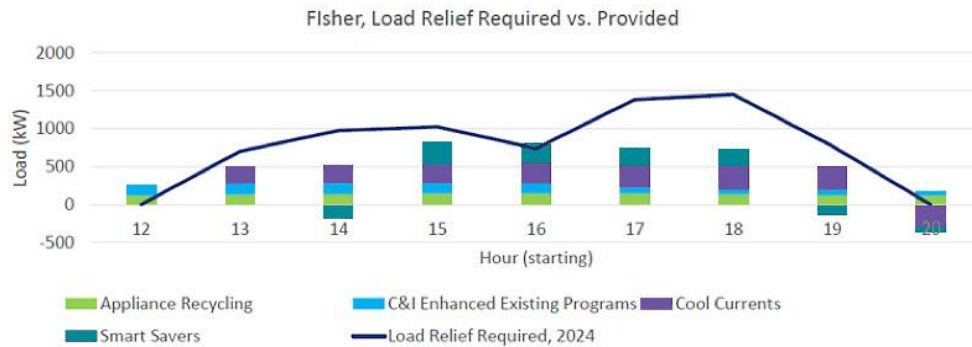
12.7.2 Fisher substation NWA pilot

Fisher substation is a 13.2kV substation that serves approximately 4,400 residential customers and 400 C&I customers in the Brownstown, Rockwood, Flat Rock and Gibraltar communities. This Fisher pilot is a direct follow-up to the Hancock substation pilot and will leverage several key learnings from Hancock, including the need to select an area with a representation of C&I customers to provide EWR benefit, and leveraging the DER Estimator tool and methodology. The portfolio of EWR and DR programs targeted for Fisher were developed and optimized using this DER Estimator tool.

Fisher substation is over its firm rating by approximately 2.5MVA. In addition, two circuits served by Fisher are over or expected to be over the 8MVA design standard for 13.2kV circuits. The traditional investment planned for Fisher is to expand the substation and do some reconfiguration at an estimated cost of \$8 million.

The Fisher project expands upon the scope of the Hancock project by incorporating geo-targeted demand response (DR) into the portfolio of programs. The two DR programs included are Cool Currents (or interruptible air conditioning) and Smart Savers (bring your own smart thermostats). The portfolio of geo-targeted EWR and DR programs is expected to provide approximately 56% of the load relief, as shown in Exhibit 12.7.2.1 below. The process for developing and optimizing the portfolio of programs was previously described in Section 4.3 – Non-Wire Alternatives.

Exhibit 12.7.2.1 – Fisher Overload and Load Relief Portfolio



This load relief portfolio assumes a participation rate in the EWR programs of 15% of residential customers and 18% of commercial and industrial customers. The demand response programs target residential customers at a rate of 5% for Smart Savers and 7% for Cool Currents.

The NWA work at Fisher will provide a portion of the load relief needed and allows for a lower cost traditional solution to provide the remaining needed load relief. A Subtransmission Distribution Facility (STDF), which is a facility that consists of a set of equipment which creates a single circuit from a subtransmission connection, will be installed to relieve load on one of the overall circuits.

In addition to the \$4.2 million capital cost of this project, an additional \$1.2 million of O&M related to the EWR/DR portfolio will be needed. This is less than the traditional substation expansion, which was estimated at \$8 million.

Overall, this pilot is expected to provide significant information on the effectiveness of geo-targeted EWR programs, including:

- Verifying the load relief achieved from demand response
- Determining if the targeted participation rates can be achieved
- Understanding the time to deployment and the level of incentives needed to achieve the desired participation rates
- Understanding customer acceptance of more frequent demand response interruptions
- Verifying deployment costs and progress toward optimizing targeted marketing and administrative costs

Exhibit 12.7.2.2 – Fisher NWA Pilot Summary

Fisher NWA Pilot information	
Cost and Scope of traditional solution	- Traditional Solution Scope - Upgrade Fisher Substation from a Class “F” to a Class “A” - Replace existing transformers with 15/20/25MVA transformers and install a 9 position PDC - Construct and commission two new circuits, split the load and throw-over circuit - Traditional Solution Cost \$8M – 2022 to 2024
Cost and Scope of proposed NWA solution	- DR Scope and Cost - Program implementation for (IAC & BYOD), technology platform (Energy Hub) and IAC equipment upgrades - EWR Scope and Cost - Residential Implementation, Incentives and installation labor - C&I Implementation and Incentives - STDF Scope and Cost - Establish STDF (Sub-transmission Distribution Facility) along same sub-transmission line that feeds Fisher Substation - Split FISHR DC 8188 with STDF circuit - Install new conduit and cable - The EWR and DR solution combined with traditional equipment solution defer the original traditional solution investment by four to eight years Total cost = \$4.2M in capital
Consumption / Loading patterns with and without NWA	Fisher substation overload, and EWR/DR contribution to address loading is shown in Exhibit 12.7.2.1
Implementation Timing	- STDF to be commissioned in 2022 - EWR/DR programs to run 2021 through 2024
Assumptions in analysis	Participation Rates: - EWR Residential (15% participation) - EWR Commercial (18% participation) - DR Cool currents (7% participation)

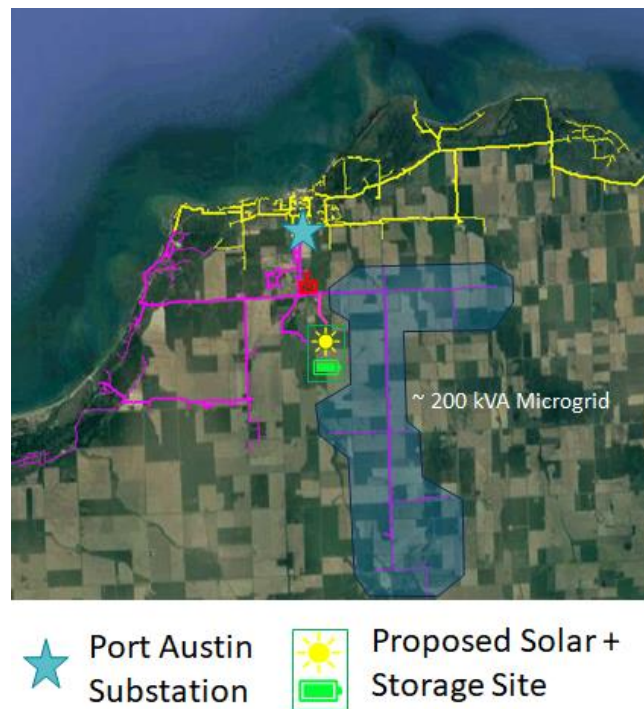
- DR Smart savers (5% participation)

Participation rates and demand impact based on models of customer demographic and loading data.

12.7.3 Port Austin substation NWA pilot

Port Austin substation serves approximately 2,000 customers in Port Austin in the Thumb region. The substation is at 120% of its firm rating. One circuit is at 108% of its day-to-day rating, indicating that during peak load some of the equipment is overloaded. A map of the general area and project is shown in Exhibit 12.7.3.1.

Exhibit 12.7.3.1 Port Austin Area Map



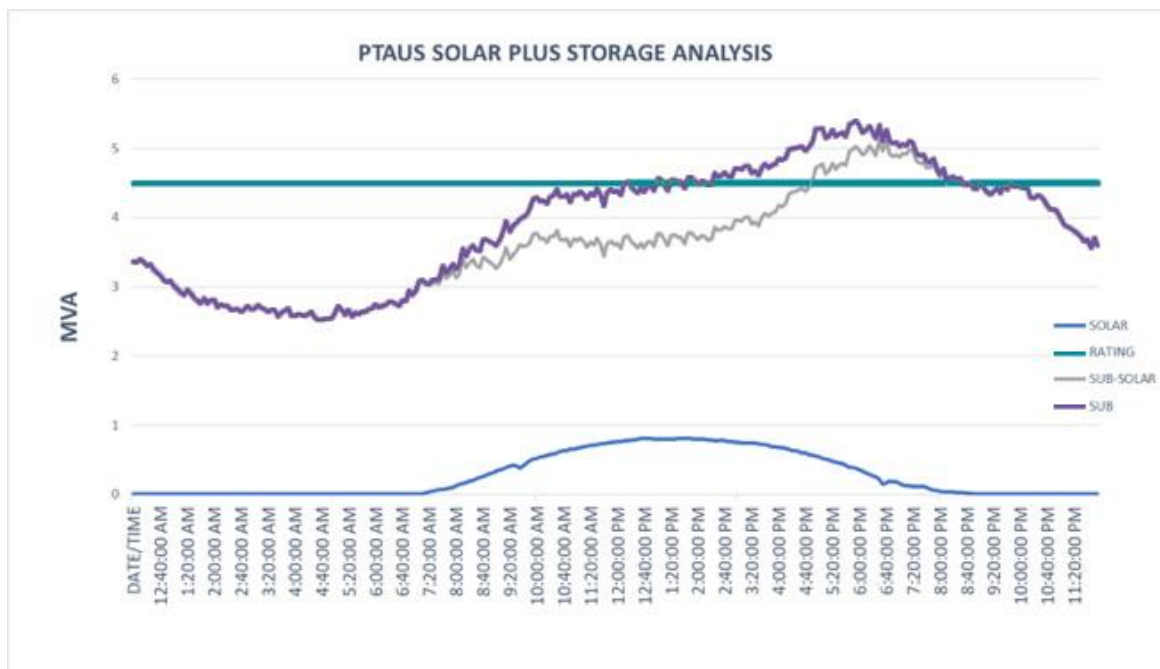
The traditional investment to alleviate load in this area would be to upgrade the substation and some of the circuits to 13.2kV, with an expected cost of approximately \$15 million.

The scope of the NWA pilot is to:

- Install a 500kW solar site
- Relocate a 1MW x 4MWh battery from Omega
- Explore feasibility of microgrid equipment and controls that could serve about 200kVA

Circuit modeling of the substation, as seen in Exhibit 12.7.3.2, shows the hourly load curve of the overloaded substation in yellow. A conceptual generation curve is shown in blue has been constructed using data from nearby arrays. The net load at the substation is shown in gray. Overall, the initial analysis indicates the solar could reduce the peak overload from 900kW to 600kW, and the total energy needed above the rating is reduced from 2.8MWh to 1.2MWh. Though more thorough study is needed, the analysis indicates that the 1MW x 4MWh battery could alleviate the remainder of the overload after accounting for operating margins of the battery.

Exhibit 12.7.3.2 Port Austin Loading with and without Solar



As described above, the overall goals for this pilot are to:

- Successfully re-deploy a battery from another site (Omega)
- Demonstrate the effectiveness of solar plus storage for addressing a loading concern
- Refine tools for evaluating solar and storage

Exhibit 12.7.3.3 – Port Austin NWA Pilot Summary

Port Austin NWA Pilot information	
Cost and Scope of traditional solution	Upgrade Port Austin substation to 13.2kV Cost is approximately \$14M to \$16M
Cost and Scope of proposed NWA solution	- Solar + Storage - Install approximately 500 kW solar site - Use one existing 1MW x 4MWh battery from OMEGA - Cost is approximately \$4.5M - Microgrid - Could be expanded for approximately 200 kVA Microgrid - Install Triple Single Recloser with SCADA - Total cost is forecast to be \$300k-\$400k
Consumption / Loading patterns with and without NWA	Overall Port Austin substation loading pattern is shown in Exhibit 12.7.3.2. The estimated generation from the solar array reduces peak overload from 0.9 MVA to 0.6 MVA, and the total energy needed from 2.8 MWh to 1.2 MWh. The remaining overload would be served by the installed battery.
Implementation Timing	2022-2024
Assumptions in analysis	The analysis is based on substation loading data, and the generation profiles of solar arrays in the same area

12.7.4 Omega substation NWA pilot

Omega substation serves approximately 4,000 customers in Harrison Township in Mount Clemens. This part of DTEE's service territory has experienced load growth due to indoor agriculture developments. Omega is served by two subtransmission lines which are currently overloaded, preventing the connection of new load.

A traditional distribution investment is in progress to address the loading in the area, as described in the Boyne project in Section 11.2.3 – Subtransmission Projects and Programs. However, the timing of these projects will not relieve load until after the 2022 peak.

The goal of this NWA project is to:

- Deploy batteries at Omega substation before 2022 peak to provide load relief and reduce risk of stranded load
- Test effectiveness of storage at reducing peak load, including developing and implementing operational procedures for load relief
- Install equipment that can be relocated and re-used in an efficient manner (equipment will be re-used at Port Austin or other sites)

Exhibit 12.7.4.1 – Omega NWA Pilot Summary

Omega NWA Pilot information	
Cost and Scope of traditional solution	There are multiple traditional solutions for the area, however the NWA solution will provide load relief before the traditional solutions (load transfer, subtransmission circuit rebuilt, and subtransmission station upgrades) can be completed
Cost and Scope of proposed NWA solution	Storage Purchase 2 mobile batteries for Omega (will be used for other projects) Install mobile batteries at Omega substation before 2022 peak Total cost of the batteries and installation is estimated at \$7M
Consumption / Loading patterns with and without NWA	Mobile batteries to provide temporary relief Provides about 1 MVA of load relief at peak times Reduces blocking hours and the risk of stranded load
Implementation Timing	2021/2022 Battery procurement 2022 Battery installation
Assumptions in analysis	The analysis is based on substation loading data

12.7.5 Veridian

Veridian is a planned development in Ann Arbor with net-zero community goals. A high penetration of solar, storage and EV charging is expected as part of this development. The area is served by Regent substation, which is a 4.8kV substation at 81% of firm rating. While substation

loading is not the primary concern, the overall load of the development is expected to approach 1.5MVA, and none of the existing circuits would have capacity to accommodate the full requested load. Traditional infrastructure upgrades to support the needs of the development could have included significant 4.8kV to 13.2kV conversion and construction of new distribution substation capacity.

Instead, by leveraging the onsite solar and storage capabilities as offsets to peak load, the proposed scope to support the load at Veridian is to reconductor existing circuit backbones and continue service from Regent substation.

The NWA portion of the project will involve the use of microgrid controls, which will manage the overall loading of the site through control of the solar, customer-owned storage, and potentially EV charging management.

The overall goals of the pilot are to understand:

- Effectiveness of solar and storage at reducing load
- Consumer behavior with microgrid and customer owned DER
- Control, protection and security requirements for storage
- Deployment costs, time and challenges
- Operating processes, procedures and staffing

Funding for the project will come from a mix of DTEE, developer and DOE grants. A final decision on DOE funding is expected in late 2021. If DOE funding is not approved, the timing, scope and cost of the project may undergo significant revision.

Exhibit 12.7.5.1 – Veridian NWA Pilot Summary

Veridian NWA Pilot information	
Cost and Scope of traditional solution	Short-Term: Upgrade circuit backbones to support load, back feed and voltage Perform load transfers to provide additional capacity Long-Term: Convert the entire substation area to 13.2kV for approximately \$50M+
Cost and Scope of proposed NWA solution	Create community microgrid utilizing demand response and grid controls Deploy additional large-scale generation and storage Total cost: \$8.3M (DTEE + Developer + DOE Grant) plus additional private and public funding
Consumption / Loading patterns with and without NWA	Loading patterns are still to be determined. Pilot will collect data from DG and DS, and test analytics and other technology for customer demand side management. Microgrid controller will also be implemented
Implementation Timing	Implementation is expected to take place from 2022 to 2026, and will be dependent on the progress of the development
Assumptions in analysis	In progress – details of the development and loading are still in early phases

12.7.6 O’Shea solar and storage

O’Shea Park in Detroit is the site of a 2MW solar array connected to the Chicago substation, which serves approximately 11,500 customers. The solar generation frequently exceeds circuit load, causing power to flow back through the substation. During periods where solar generation is intermittent or changing rapidly, voltage data as reported by AMI meters indicates there is some potential for small over-voltages or flicker.

The project will install a 1MW x 1MWh battery co-located with the solar array. There are multiple goals of this pilot, including:

- Improving power quality in the area
- Creating standardized designs, safety policies, and operating procedures for integrating storage
- Testing communication and control protocols
- Testing wholesale market participation for FERC Order 841 by responding to pricing signals and coordinating the priorities of multiple reliability use cases. Also developing the DTEE processes and framework for other energy storage resources to participate in FERC841
- Engaging communities on the permitting and approval process for siting batteries
- Receiving fire protection and UL certifications, including NFPA 855 compliance

The O'Shea battery project is expected to be complete by early 2022.

12.7.7 Other technology pilots

Mobile battery trailer

DTEE expects to receive and complete assembly of a portable 1MW x 4MWh hour battery trailer in 2021. Once the equipment is received, DTEE will test the unit and develop documentation and required operating procedures to deploy the mobile battery in the field. Once the new equipment has been fully vetted, it can be used for various use cases, including siting in place of traditional portable generators, supporting system needs during shutdowns or maintenance, or siting at substations or on circuits to reduce peak load as part of a broader NWA project. The Company plans to purchase four additional mobile trailers by 2025, pending results from the initial evaluation of the equipment.

Small solar and storage testing facility

Up to 60% of new projects for rooftop solar at residential and small commercial customer sites are proposed with some amount of battery storage. The MPSC has requested that DTEE provide a solution to properly meter and integrate these storage solutions into the grid while still maintaining compliance with distributed generation, net metering and other tariffs. Additionally, FERC and MISO are expected to require ride-through protection for primary frequency response as part of the FERC 841 order.

This requires that all inverters have a verified setting for specific protection parameters to support the bulk electric system stability. Smart inverters are also capable of providing voltage and reactive support in either a passive or actively controlled mode. As the testing standards in IEEE 1547.1-2020 on smart inverters are finalized and interoperability standards such as IEEE2030.5 evolve and reach the market. DTEE must develop the capability to validate the settings and communicate with the DERs to ensure any new installations with smart inverters follow the standards to prevent any potential power quality, voltage and safety issues in the electric system and to determine the best method to utilize these new capabilities.

This project is to install solar arrays, commonly used inverters, storage units, switching and metering systems, and utility gateway communication systems at a DTEE site. The project will allow validation of behavior, interaction and compliance of the new features for smart inverters and act as a location to test new features and capabilities while also serving as a training platform for DTEE engineers, technicians and field employees. Standards can be developed around these technologies to ensure safety and reliable operation.

The testing facility costs \$2.1 million and most of the infrastructure work is scheduled to be completed in 2022, however incremental improvements will continue to support new projects and technology integration efforts. The use of this facility will assist customers seeking to integrate solar with battery storage into DTEE's distribution system by providing clear standards and thereby streamlining the interconnection process for this type of technology application.

The lab facility will support several evaluations and will be continually updated to demonstrate technology interoperability and the processes and technologies to integrate customer resources into the grid. These evaluations will have project specific targets and be measured on the ability of the products and solutions to meet DTEE operational requirements, work practices and industry standards and interoperability.

EV Extreme fast charging demonstration at the American Center for Mobility (ACM)

The ACM is a unique, purpose-built facility focused on testing, validation and self-certification of connected and automated vehicles and other mobility technologies at the 500-plus-acre historic Willow Run site in Ypsilanti Township in Southeast Michigan. DTEE will leverage this charging network to get a better understanding of the capabilities to control charging and predict the need for charging from autonomous vehicles. DTEE is supporting the implementation of the Delta

Power Electronics DC Xtreme 400KW fast charger when it becomes available in 2021 as part of a DOE project. This DOE pilot program, EVS-at-RISC, will help DTEE develop cyber secure monitoring and control capabilities with smart charge controls and inductive road charging to ensure reliable and secure interfaces and will further enhance knowledge of the grid impact of these new technologies. This pilot project is expected to:

- Install a utility gateway and communications portal to enable charge management
- Develop control algorithm and conduct testing on a Delta Extreme fast charger and its interfaces
- Develop cyber security interfaces and control, install additional sensing capabilities, and monitor performance on charging network
- Develop monitoring and control algorithm for in-road inductive charging at the ACM site

The initial Delta fast charger will be available by the end of 2021 with an anticipated second phase to be awarded in late 2021 with implementation in 2023 of a commercially available charger. Smart charge management will be implementing in three phases through 2024 and will be available during the later phases of the project for wider testing.

These projects are predominately research and development to establish the capabilities of utilizing charge management on charge control in extreme fast charging environments and the scope of evaluation with the Department of Energy is to demonstrate technical feasibility and meet demonstration milestones.

12.8 DER control

As discussed in Section 3.3.3 – Distributed generation/distributed storage growth scenario, there is a likelihood that many more distributed energy resources (DERs) will be introduced to the grid. Furthermore, with the implementation of FERC Order 2222/2222A, DERs that were previously only able to participate in retail programs are now able to participate in the wholesale market via aggregations. DTEE's DER Control strategy has been developed to help manage the grid stability reliability concerns that stem from the interconnection of large numbers of these DERs across its grid.

As noted in Section 12.4 – Distribution Sensing and Monitoring, proper operation of any distribution grid starts with the timely monitoring of distribution network assets. As more energy

sources are connected to the grid, voltage variability is introduced that, if unmanaged, affects the quality of power being delivered to loads connected to the affected circuits.

The DER Control workgroup has identified several areas of focus that will yield the visibility and control required to ensure that DTEE is delivering safe, reliable, and affordable electric service to all its customers.

The projected cost of implementation is \$44.1 million through 2036, of which \$12 million will fall in the 2021-2025 timeframe. The resources required to support the proposed changes span an array of disciplines from SCADA, (Network Data & Application Support (NDAS), Power Equipment & Relay Test (PERT), Engineering), IT (SCADA Realtime Support (SRS)) and Regional Planning. Exhibit 12.8.1 shows the investment summary for DER Control.

Exhibit 12.8.1 DER Control Investment Summary

	2021	2022	2023	2024	2025	2021-2025 Total
Investment Projection (\$ millions)	-	\$3	\$3	\$3	\$3	\$12

12.8.1 Large and medium-sized DER monitoring & control

Interconnection Control System Enclosure Integration

The telecommunications and SCADA package developed by DTEE's Interconnections team is intended to be a drop-in solution for monitoring larger DER installations. This package contains all the equipment required for SCADA and performs the following functions:

- Metering using ION power quality meters
- Communications (network switches)
- Automation (Schweitzer Engineering Labs (SEL) RTAC Remote Terminal Unit (RTU))
- Security (firewalls)

DTEE expects to equip key DER installations on its distribution system with this package. These sites include Demille Solar Park (Lapeer), Turrill Solar Park (Lapeer), and IKEA (Canton). The intent is to demonstrate that DTEE utilizes the same package for its sites as it is requiring its larger customers to use.

12.8.2 Upgrade of DR-SOC monitored solar sites to RTAC RTU

DTEE-owned solar sites installed as part of the SolarCurrents program were outfitted with DR-SOC low-cost monitoring (LCM) very similar to the low-cost substation monitoring outlined in Section 12.4 – Distribution Sensing and Monitoring. LCM has provided data at 5-minute intervals since its inception, but as the needs of the System Operations Center (SOC) and Merchant Operations Center (MOC) have evolved over the years, the systems put in place have not yielded the 2-4 second interval data that is typically available in SCADA installations.

Under this initiative, the data logger installed at 31 Renewable Energy solar sites would be replaced with an SEL RTAC RTU capable of yielding the required scan rates, thus increasing visibility into the operations and impact these systems have on their hosting circuits.

12.8.3 Viper recloser sensing/data

The ability of larger DERs to affect their hosting circuit continues to be a source of concern for the SOC and Regional Planning. Having a SCADA-operable device such as a Viper recloser available for the SOC's use at these larger sites provides a necessary isolation capability if the DER is adversely impacting the reliable operation of the circuit to which it is connected or if the circuit is in an abnormal condition, such as during outage repairs or maintenance.

Additional benefits include SCADA telemetry from the device monitoring the recloser. This data represents the interconnection point on the system and would be very useful for use with the ADMS state estimator, power flow and contingency analysis algorithms.

It is expected that the installation costs of such operable devices be borne by the customers meeting Engineering's size criteria, with a small trailing O&M component required for the telecommunications and maintenance of the equipment.

12.8.4 Small DER monitoring

While current MPSC rules primarily focus on metering, they do not offer clear guidelines for when telemetry is required. DTEE's Interconnection Standards need to clearly delineate what assets have a mandate to provide telemetry. The SOC requires EMS/ADMS to have accurate power injections for all DERs systems connected to the Distribution Network to properly model power flow, as well as keep its State Estimator (SE) and Contingency Analysis (CA) systems converged.

This translates into a need for near-real time visibility into DER units as small as 100kW and SCADA for individual DER units 500kW or larger in size.

Aggregations made possible by FERC Order 2222 will pose a whole new set of challenges and opportunities for DTE's Distribution System. With DER Aggregations (DERAs) being dispatched on a purely economic basis, on-peak charging of energy storage resources (ESRs) with the resultant higher peak demand is theoretically made unattractive via unfavorably high locational marginal prices (LMPs). This does not, however, guarantee that on-peak charging will not happen, and this is an area of concern which requires careful consideration and planning for future circuit loads based on forecasted per-circuit DER penetration levels.

12.8.5 DERA monitoring integration

Proper integration of DER Aggregations includes per-circuit sums of kW and kvar for each aggregator. Data points provided by an aggregator via Inter-Control Center Protocol (ICCP) should be formatted as follows for each circuit on which they are participating:

- *<DERA Name>_<Substation>_<Circuit>_KW* (eg. DERA1_LINCEN_2800_KW)
- *<DERA Name>_<Substation>_<Circuit>_KVAR* (eg. DERA1_LINCEN_2800_KVAR)

The costs associated with this implementation are primarily in the NDAS area and cover the creation and maintenance of points and their mappings via ICCP.

12.8.6 ICCP connectivity – DERs in a DERA

Though it may seem appropriate to simplify integration by solely summarizing data from DER aggregators at the electric pricing node, doing so comes with a significant challenge for the grid operator of reduced visibility, and is therefore not advisable. Accounting for all DER injections/withdrawals on the system is key to ensuring that state estimator and contingency analysis run properly. An example of how this could be achieved would be via ICCP exchange with DER aggregators for each of their connected DER units:

- *<DERA Name>_DER<DER Number>_KW* (eg. DERA1_DER127_KW)
- *<DERA Name>_DER<DER Number>_KVAR* (eg. DERA1_DER127_KVAR)
- *<DERA Name>_DER<DER Number>_VOLT* (eg. DERA1_DER127_VOLT)

As with the previous area of focus, NDAS resources constitute the bulk of the expenditure for this endeavor.

12.8.7 ICCP connectivity – load shed in a DERA

In the event of a load shed situation, the MOC and SOC control rooms both have a need for a tally of traditional Demand-Response Resources (DRR) being dispatched. The MOC's need centers on NSO load reconciliation, while the SOC needs to know how much load may appear on a per-circuit basis once a load shedding event is complete. Mapping this data via ICCP could be achieved as follows:

- *<DERA Name>_<Substation>_<Circuit>_DRR_COUNT* (eg. DERA1_LINCN_2800_DRR_COUNT)
- *<DERA Name>_<Substation>_<Circuit>_DRR_KW* (eg. DERA1_LINCN_2800_DRR_KW)

12.8.8 AMI and de-aggregated load data

For proper and required differentiation of retail vs. wholesale market transactions, submetering of generation assets considered Behind The Meter (BTM) is necessary. De-aggregating this data also permits SOC's Operations Engineering to better identify overload conditions on circuits with high DER penetration and an anticipated large drop in generation (such as inclement weather setting in on circuits with high solar penetration).

12.8.9 DERMS module

While in some circumstances it may be possible to operate without a DERMS system, as outlined in the ICF "DTEE Grid Modernization" report, there is a need for DTEE to acquire DERMS functionality to assist in the management of DERs in the long run. Distribution Operations (DO) is working with Corporate IT and Generation Optimizations to identify a suitable DERMS for implementation.

The primary drivers for this need are:

- DTEE currently has 5,000 DERs on its distribution system, and this number has been growing at an increasing rate as measured by the interconnection volume. DO expects 2,000 additional DERs to seek interconnection this year.
- Extrapolating the numbers from the Interconnections Group, the DERMS will need to support at minimum of 25,000 DERs to cover the 10-year horizon using the most conservative estimates.

The possibility of developing tools in-house has been evaluated, but the concerns raised about the scalability of a home-grown solution remain an unknown. This risk was deemed unacceptable and DTEE approached multiple vendors regarding their offerings.

DTEE will be working to determine the specific needs for a DERMS platform and plans to develop a project within the next year. Essential investment areas to enable this capability include software licensing, compute capacity planning, engineering and implementation services.

12.8.10 DERMS module modelling

One of the key areas of focus for the DER Control strategy involves the accurate modeling of DERs on the distribution system. This cross-functional effort will utilize resources from NDAS, Electrical Engineering's Interconnections Group, Mapping and Regional Planning to ensure that units are placed properly on the network model for ADMS and DERMS' use. As this work depends in part on the operational status of the DERMS detailed in Section 12.8.9, it is anticipated that work performed under this effort will begin Q1 2023.

12.8.11 OSI enterprise customer agreement

The current EMS licensing makes a fixed number of analog and digital points available for SOC's use. Though neither of these limits have been hit to date, the limits could easily be reached in the future if there are a large number of DER installations that require aggregation under FERC Order 2222. As outlined in the ICCP Connectivity - DERs in a DERA Section (12.8.6), DTEE will require a minimum of three data points per DER for such participation. It is unlikely that the licensed limits will be hit before 2024 based on present projects, however, these limits will need to be reviewed yearly.

12.9 Technology Strategy Roadmap

Strategic Goals

DTEE operates across a large service territory, serving over 2.2 million customers with more than 600 substations and thousands of miles of distribution lines managed from a central System Operations Center (SOC). DTEE, with the help of supporting software, must continue to integrate new streams of data and information in near real-time to better respond to customer requests for

information about their energy usage and to leverage that same data to increase reliability and manage distribution operations.

As detailed in Grid Modernization, Section 3 – Grid Modernization Process, DTEE’s long-term planning efforts include integrating new technologies including EVs, DERs and storage.

Meanwhile, planning engineers leverage all available information to identify grid needs, develop forecasts, analyze trends, inform investment strategies and ensure that the grid and its operators are prepared to meet evolving customer needs and expectations.

Finally, field crews can currently leverage mobile technology to more efficiently interface with grid operators and distribution engineers to assess situations and determine the best path forward as field crews are more likely to use this information to interface with others. Mobile technology improves operational efficiencies and keeps customers and employees safe. To ensure the technology approach remains aligned with business objectives, the technology roadmap for Electric Distribution focuses on the key business objectives:

- **Operate the grid safely** – Ensure employees consistently display the proper behaviors to safely operate and maintain the grid and minimize operational risks and challenges, with a target of top-decile industry performance
- **Support customers** – Build a culture of customer service using the communication, collaboration, leadership and recognition enablers to ensure all internal and external transactions deliver on DTE’s Service Keys – Safe, Caring, Dependable and Efficient
- **Work management** – Create and implement a well-defined process for all strategic and emergent work among all asset families to ensure the safe and efficient work execution
- **Distribution system reliability** – Achieve the Company’s “Best-Operated” and aspirational reliability targets (SAIDI ex MED, CEMI6)
- **Capital efficiency and work productivity** – Continue to focus on capital cost efficiency for both planned and reactive work
- **Grid of the future and new technology** – Develop a flexible grid to serve the ever-evolving needs of customers, through upgrades, system monitoring and automation

To deliver these business objectives, the technology investments for Electric Distribution focus on five goals, described below:

1. Improve productivity of employees through mobile technology

- The first goal is to provide seamless transitions between online and offline operations using mobile applications and devices and to provide integrated workflows between the control room, back office, field leaders and field crews.

2. Deliver best-in-class work management and scheduling

- The second goal is to achieve best-in-class work management in the “Initiate, Plan, Schedule, Execute, Close” (IPSEC) model of workflow, ensuring the highest level of safe, efficient and reliable grid management and operations.

3. Drive top-quartile DO asset management

- The third goal is to enable rapid decision making and improve service delivery with an improved understanding of asset performance by aggregating and analyzing current and historical operational and geospatial data to assess and visualize asset health.

4. Transform capabilities for distribution planning processes

- The fourth goal is to safely and efficiently interconnect DERs, while incorporating scenario analysis and performing planning to achieve network and economic optimization across the grid, improve grid reliability, ensure power quality and enable ancillary services.

5. Modernize grid management

- The fifth goal is for operators, dispatchers, planning engineers and supervisors to act in a coordinated manner, accessing real-time, as-built and as-operated representation of network information that underpins sensing and control for efficient, safe and reliable management of grid operations. ADMS is a key component of this goal.

12.9.1 Goal 1: Improve employee productivity through mobile technology

Mobile technology allows collecting and distributing data at the point of activity through mobile devices, across all personas and use cases. Mobile capabilities include both devices and applications that are engineered for mobile access and usage. Applications include dispatch, work

execution and forms digitization, location tracking, route navigation, analytics and secure file sharing.

To benefit from the opportunities and business value that mobility will offer DTEE, it is important to understand the current state of those assets and to clearly highlight the future investments in this area in a planful way. Today, the mobile technology deployed for DTEE is sufficient to handle the needs of the Company's existing corporate systems and pre-Covid working model.

DTEE is actively investing to improve and upgrade the level of mobile technology available to the field force as new systems come online and in coordination with the Company's ADMS roll out and its associated capabilities.

Additionally, with the onset of the Covid-19 pandemic, Distribution Operations has accelerated plans to deploy more capable technology into the field with an emphasis on system offerings that provide the frontline workers and leaders the flexibility of remaining in the field for extended periods without having system constraints that require them to return to the service centers or to congregate as frequently as was needed in the past. Even after the pandemic, these new skills and processes will continue to drive efficiency.

- The implementation and upgrades to DTEE's grid operating systems, such as ADMS, ESRI Mapping and AMI investments, all demand expanded device capabilities as the new investments take the Company beyond the capabilities of the currently deployed technologies. While the existing mobility computing fleet has served the enterprise well up until now, it is no longer sufficient, without additional investment, to meet the future needs of the Company's workforce.
- Mobility is ultimately most useful when it is used to maximize the amount of time that workers can remain mobile and more often be on site where the actual work is to occur. This means that deploying technology securely and effectively to allow increased time onsite as a key priority in this area. Deployment of up-to-date systems and equipment improves the efficiency and value of the Company's service delivery. To realize this value, DTEE will need ongoing investments to operate and replace field equipment as new capabilities are delivered by system and hardware partners.

- As DTEE deploys more system tools into the hands of the field force and remote digital workers, the Company remains responsible for all the regulations and compliance mandates that govern the Company's usage of both operating data and customer data, which is held in trust. These include payment card industry, grid operating information and any interconnection with NERC or SOX systems. When developing the Company's mobile enterprise strategy, all compliance requirements must be evaluated and considered as the Company moves toward a more off-site employee presence.
- The move to more mobility-based systems presents an ever-changing threat landscape often requiring additional security measures. This move also needs to account for a wider variety of device types and requires more use cases than ever before. Concurrently, mobility exposes the Company's systems to different forms of possible exploitation even while it offers better ways of serving customers. As a result, heightened security vigilance is a part of our mobility expansion.
- DTEE will ensure that mobility enhances daily work by providing timely and accurate information, as well as allowing quick and easy capture of critical data in the field, including photos, markups and videos for planned work, outages and equipment failures.
- DTEE plans to deploy optimized task-based mobile-first applications and general-purpose apps, and mobile access to all enterprise systems and documents.
- DTEE will better track the time that field workers spend completing work orders and other manhours tracking scenarios.
- In 2023, DTEE will select and implement a new and improved vegetation management platform that will provide a more reliable and scalable mobile end-to-end tree trimming operations solution to support initiation, planning, execution and audit close-out. This solution will include computer hardware, licensing, configuration and integrations to work management systems.
- In 2025, DTEE will implement an interface between the AMI system and the mobile dispatch and work management solution for the Electric Field Operations organization to perform meter reads and other operations remotely, allowing for safer operations and the ability to verify work completed while in the field.

Exhibit 12.9.1 Mobility Capital Investment

Mobility (\$ millions)	2021	2022	2023	2024	2025	2021-2025 Total
Vegetation Management Platform	-	-	\$4.3	-	-	\$4.3
AMI Integration with Mobile	-	-	-	-	\$4.7	\$4.7
Other Mobility	-	\$0.5	\$0.3	\$0.7	\$2.8	\$4.7
Total	-	\$0.5	\$4.6	\$0.7	\$7.5	\$13.3

12.9.2 Goal 2: Deliver best-in-class work management and integrated scheduling

Work management includes scheduled and unscheduled work, work order creation, resource scheduling, and work execution for employees, vendors and third-party crews. This includes job planning, labor, material, tools and services for both large and complex work efforts and smaller, short-term jobs. Upon completion, work must be audited and verified to reconcile time capture and payment.

An effort is underway in DTEE to improve work management and scheduling processes to ensure field resources have the processes and tools to support the right jobs, with the right people, at the right time. This includes ensuring unscheduled work is reflected in the work management system, Maximo, and field resources can receive and status jobs in real-time to create a feedback loop on asset condition and performance in alignment with the asset management capabilities. Process improvements, reduction of jobs held due to resource constraints, higher schedule compliance rate, and CWIP backlog reduction have resulted in savings.

Very similar to the Company's enterprise mobility systems, DTEE employs a platform of tools and systems centered around Maximo to enable effective work management. The core Maximo system is out of date and requires an upgrade to the current version before vendor support of the existing system is discontinued in late 2021. The upgrade is required before portions of the platform can be moved to the cloud, a more efficient means of storing information in the longer term. Investments are already underway to return Maximo to health, improve stability and prepare it for modernization over the next three years, including working to provide each major business

area that uses the platform its own asset. This will decrease operational complexity and align each asset more directly with the business outcomes of that area.

- The Company's currently deployed enterprise system is several version upgrades behind the industry standard and will go out of support in October 2021. This will result in increased operational fragility until upgraded.
- A project is active to return this to a supported, on-premise version, by the Q4 2021.
- The current system version limits the ability to utilize the mobile aspects, making it more difficult to take advantage of operational efficiencies in the field.
- The existing production system implementation has all business units across the Company enterprise using a single system, which imposes operating requirements on all business units when only one unit has that requirement. For example, a decision in Generation impacts Distribution and vice versa. As the system moves to the cloud platform, each business unit will have its own implementation and will only be affected by its specific operational needs, functions, security requirements and schedules, which will improve efficiency and the ability to incorporate change.
- To facilitate a future move to a cloud version of the product, the current version must first be updated.
- Future enhanced functionality will include better internal resource planning and Alassisted automatic work scheduling.

Exhibit 12.9.2 Work Management and Integrated Scheduling Capital Investment

Work Management and Integrated Scheduling (\$ millions)	2021	2022	2023	2024	2025	5 Year Total
Stabilize Distribution Operations' Maximo Instance	-	-	\$5.2	-	-	\$5.2
Other Work Management and Integrated Scheduling	\$0.1	\$2.9	\$6.8	\$3.0	\$3.2	\$16.1
Total	\$0.1	\$2.9	\$12.0	\$3.0	\$3.2	\$21.3

12.9.3 Goal 3: Drive top-quartile DO asset management

The overall goal of effectively managing assets is to minimize the total cost of ownership and operations while delivering the desired service levels. This includes maintaining access to accurate digital data about the asset, algorithms to process and analyze data, and a visual context of an asset's location and condition.

Asset management includes managing the full lifecycle of an asset, engineering and financial management, as well as linking spatial information in the ESRI Mapping system to an asset.

Managing the electric grid is an asset-intensive operation, and as such, relies heavily on the capabilities of its Asset Management systems. Several of these systems are common to both Work Management and Asset Management with Maximo and ESRI Mapping systems sitting at the core of this capability. Critical issues include:

- Both currently deployed systems are several version upgrades behind the industry standard and will no longer be supported by the vendor after October 2021. This will result in increased operational fragility until upgraded.
- Projects are active to return them to a supported on-premise version by Q4 2021.
- Once the ESRI Mapping system is upgraded to a vendor supported version in 2021, additional investment will be made to in the ESRI platform to improve digital asset management processes. This includes investment in ESRI-based design tools that provide a streamlined asset management process from design to field completion through integration with existing mobile mapping technology, ArcFM Mobile.
- Investment in DTEE's centralized asset data repository is required to obtain the value of a 360-degree view of assets including master, transactional and performance data. This would include the capability to fully link and deliver supporting documents such as asset manuals and engineering drawings to off-site field workers digitally.
- The current system has differing levels of capability to manage asset processes online, offline and via mobile activities. This requires investment to digitally deliver the highest service levels to the mobile workforce.
- Investment is required to improve the timeliness of asset updates from the field and support for map-centric asset management.

- Investment is needed to improve the system's ability to track labor, materials, and tool charges by all asset types and linear measures in a fully automated fashion that reduces/eliminates manual errors and delays.

As part of DTEE's ongoing technology plan, the Maximo and ESRI Mapping systems are in the process of being upgraded to both return them to health and to expand their capability in conjunction with the ADMS investment.

With the implementation of ADMS, the Maximo version upgrade, and the ESRI Mapping upgrade over the next three years, systems will be both up-to-date and able to support the anticipated business outcomes.

Exhibit 12.9.3 Asset Management Capital Investment

Asset Management (\$ millions)	2021	2022	2023	2024	2025	5 Year Total
ESRI System Upgrades	-	\$0.8	\$1.1	-	-	\$1.8
Other Asset Management	-	\$0.5	\$1.0	\$1.5	\$0.8	\$3.8
Total	-	\$1.3	\$2.1	\$1.5	\$0.8	\$5.6

12.9.4 Goal 4: Transform capabilities for distribution planning processes

As discussed in Section 4 – Distribution Planning Processes and Tools, one of the critical aspects for a smart and integrated grid is the capability to connect customer equipment, run alternative scenario analysis at scale, and provide situational awareness for targeted system upgrades. These capabilities rely on systems to continuously manage and process large volumes of data from many sources as both the grid and customers connect devices and sensors.

An emerging issue for most utilities, including DTEE, is that the planning tools that are currently being used to analyze the loading data are not sufficient to meet this quickly growing challenge.

To enable the maturing distribution planning business processes discussed in detail in Section 4 – Distribution Planning Processes and Tools, the existing technology platforms supporting DER

interconnection analysis and distribution planning will need to be upgraded and new analytical tools, models and platforms will need to be implemented in five key areas:

- **Interconnection process enablement** – This investment includes upgrading the new interconnection application software to provide seamless customer logon, payment and transparent application experience. Additional investment includes tools and integrations supporting the application review process. The Interconnection process is discussed in further detail in Section 4.4 of the DGP.
- **Hosting capacity enablement** – This requires deploying tools to support stakeholder hosting capacity visualization and enhanced analysis tools, and integration with existing planning tools. More on the Hosting Capacity Analysis (HCA) process is discussed in detail in Section 4.5 of the DGP.
- **Network model** – The foundation for the evolving Distribution Planning processes is a high-quality network model including grid topology and electrical characteristics. Network model investments include technology to better align field conditions and maps to the digital representation of the grid, integrations between asset systems, new data models to support planning and operations topology and characteristics, and advanced analytics to leverage sensor data to continuously improve the network model.
- **Distribution load forecasting** – Investment is required to develop DER and load forecasting capability at a granular level to support studies and scenario planning that enables DTEE circuit design to be responsive to meet ever-changing customer needs. For further details on the Forecasting process, see Section 4.2 of the DGP.
- **Situational awareness analytics** – Key investments are planned for analytics platforms to provide situational awareness and proactive recommendations of grid upgrades to improve customer reliability, power quality and DER connectivity.

12.9.4.1 Interconnection process enablement

Technology investment is needed to achieve the maturing processes discussed in Section 4.4 – Interconnection Process. The interconnection application software, Power Clerk, currently requires customers to maintain separate login information and provide payments via check. Investment is required to upgrade the portal to provide a seamless customer logon, electronic payment and a transparent application experience. Funding will include investments to upgrade the Power Clerk software to accommodate new processes and updated forms. This also includes

investments to implement single sign-on and customer information system integrations, as well as integration with Maximo to help streamline the work management process of interconnection applications.

As DTEE sees increasing volumes of interconnection applications, investment must support internal processes for interconnection application review by integrating the application system with the planning study tool, CYME planning tool, as well as the Company enterprise data lake for circuit analysis, application trending reporting, and project optimization.

Investments are also required to support the interconnection process in the screening steps for qualifying projects. This includes specific integrations to the CYME planning tool to support automating screens for loading, transformer size compatibility, and voltage constraints, as well as to DTEE's spatial solution, ESRI Mapping, to support screening within the service territory.

The Interconnection Process Enablement program also includes technology enhancements that will be needed to ensure DTEE is equipped to meet emergent regulatory requirements including FERC orders 841 and 2222. Study tools will need to be enhanced to support the integration of aggregated resource across locations on the grid, while evaluating interrelated impacts. DTEE will also integrate the planning tools with scenario-based forecasts to determine opportunity and assessment.

Exhibit 12.9.4.1.1 Interconnection Process Enablement Capital Investment

Interconnection Process (\$ millions)	2021	2022	2023	2024	2025	5 Year Total
New Interconnection Application	-	\$1.1	\$0.5	\$0.6	\$1.1	\$3.4
Internal Process Integrations	-	\$1.8	\$1.4	\$1.3	\$1.9	\$6.3
Other Tools and Integrations	-	\$0.9	\$1.7	\$3.5	\$1.5	\$7.6
Total	-	\$3.8	\$3.6	\$5.5	\$4.5	\$17.4

12.9.4.2 Hosting capacity enablement

Technology investment is needed to achieve the maturing processes discussed in Section 4.5 – Hosting Capacity Analysis. To improve the speed in which customers can evaluate the ability to interconnect DERs with the grid, investment is required in both customer-facing and internal evaluation tools. The technology investment will include a system to store information on likelihood of system upgrades needed to support DER interconnection, as well as a visual interface that is externally facing for customers and developers to access.

To support DTEE’s processes to evaluate hosting capacity at a granular level, investments in tools and integrations are needed. This will begin with tools such as EPRI DRIVE and CYME integration capacity analysis to evaluate areas with high penetration of DER. Further years will see investments to link the hosting capacity into the interconnection application portal.

Exhibit 12.9.4.2.1 Hosting Capacity Enablement Capital Spend

Hosting Capacity (\$ millions)	2021	2022	2023	2024	2025	5 Year Total
Total	-	\$0.25	\$0.25	\$0.15	\$0.15	\$0.8

12.9.4.3 Network model

The network model is a series of IT and data architecture that will provide DTEE with the foundation needed to deploy advanced grid simulations and analytics called for by the evolving distribution network. The Electric Power Research Institute (EPRI), which refers to this as “Grid Model Data Management (GMDM),” has provided a reference framework for utilities and vendors to leverage in addressing the network model challenges.

As discussed in Section 3 – Grid Modernization Process, grid modernization has brought significantly higher requirements for the tools and analytics used to plan, operate and manage the electric grid. Due to the increases in volume, velocity and the granularity of data needed to support the planning tools, ADMS, and distribution system real-time controls, DTEE needs to continuously improve the network model systems to enhance detailed modeling and better support data flows for systems of records to provide historical, as-built, and future scenario-based network models. In addition to the tools, models and interfaces required to meet network models

needs for operations and planning, advanced analytics will also be leveraged to ensure data integrity.

Although the grid model data management relies heavily on ESRI Mapping functionality, this program does not include basic ESRI Mapping software upgrades, which is supported by the corporate IT investment portfolio. This program also excludes work specifically needed to migrate data to the ESRI Mapping utility network.

Common platform operating maps

Currently, most operating diagrams are maintained in computer-aided design (CAD) software in addition to ESRI Mapping, DTEE's Geographic Information System (GIS). This can cause discrepancies between sources, which leads to extra field verification or rework during the planning, construction and operation of the electrical system. There are over 14,000 map documents for the subtransmission, Underground Residential Distribution (URD), AC Network, and cable conduit systems that reside and are maintained in both CAD files and the ESRI Mapping network model.

To ensure data consistency, all the data or information related to subtransmission, URD, AC Network and cable conduit need to be transferred, reconciled and validated within the ESRI Mapping platform. This program track will create a unified network model with ESRI Mapping becoming the single source of data for the DTEE as-built network model and operational maps.

Asset data integrations

The data lifecycle of an asset is the digital representation of an electrical asset from installation through decommissioning. The current process to update asset and network model information includes many manual and duplicated steps between systems, such as creation of the digital representation in Maximo and ESRI. To support the increasing velocity of network model updates for real-time grid operations and planning, DTEE plans to expand integrations among its enterprise asset management systems (Maximo, ESRI, SAP). These integrations help efficiently manage asset data lifecycles for asset management and grid model requirements.

These integrations will focus initially on data flows needed for current planning and operations functionality, such as power flow and area load analysis (CYME, ADMS, PSSE), with focus shifting to advanced integrations with detailed control settings for equipment (Relay and device

settings) to support bidirectional power flow, dynamic protection, and even advanced Distributed Energy Management System (DERMS) in the future as ADMS and Distribution Automation functionality mature.

Network model enhancements

With the switch to electronic operational orders and advanced functionality of ADMS required to manage bidirectional power flow, current network model granularity will not suffice. Additional details needed range from schematics to support switching and tagging in the short term, to engineering and control setting information to support advanced functionality, such as smart inverters functionality in ADMS in the long term.

This program track will enhance the data model in the ESRI Mapping system to meet the needs of planning and operations functions in DTEE's Grid modernization plan. DTEE will also perform first-time data capture and consolidation for digital representations that do not currently exist.

Grid model analytics

With the increasing volume of data needed to support grid models for planning and operations, there will be a need for increased staffing in order to maintain high quality data. For example, the impedance per mile of conductor and the substation transformer phase shift information is needed to support real-time load flow in ADMS; millions of these data points will need to be provided with sufficient quality in near-real time. To increase efficiency and accuracy of maintaining the grid models for planning and operations, DTEE will need technology solutions to continuously improve data integrity.

DTEE will leverage the big data cloud computing infrastructure to explore and develop machine learning algorithms to address data quality gaps based on existing and derived information. Once the algorithms are tested and proven, DTEE will integrate the machine learning workflow with source systems to raise overall data effectiveness.

Network model orchestrator

In order to study the impacts of proposed system upgrade projects, there is a significant amount of manual work required in the current distribution planning processes and tools to create historical analysis and network models for studies performed on an individual basis. Although

there is no commercialized industry product to manage historical, current and future network models, including planning scenario analysis, DTEE sees this functionality will be necessary in the five-to-fifteen-year outlook. The technology architecture recommended by EPRI, which includes management and orchestration services for various planning and operational scenarios, will enable stage two and stage three planning tools, such as time-series power flow, scenario-based project evaluation, and asset replacement optimization.

The Network Model program will deliver both the grid network model and model management technologies needed as a foundation for many functionalities of integrated distribution planning and a modern grid.

Exhibit 12.9.4.3.1 Network Model Capital Investment

Network Model (\$ millions)	2021	2022	2023	2024	2025	5 Year Total
Common Platform	\$0.4	\$0.8	\$1.2	\$1.3	\$1.3	\$5.1
Asset Data Integrations	\$0.7	\$0.8	\$0.6	\$0.6	\$0.4	\$3.2
Network Model Enhancements	\$0.3	\$0.4	\$0.3	\$0.2	\$0.2	\$1.5
Grid Model Analytics	\$0.6	\$0.8	\$0.8	\$0.8	\$0.8	\$3.9
Total	\$2.0	\$2.8	\$2.9	\$2.9	\$2.7	\$13.3

12.9.4.4 Distribution load forecasting

As discussed in Section 4.2 – Forecasting, new developments in distributed generation resources, electric vehicle adoption, and energy waste reduction/demand response are requiring a change in process for distribution load and generation forecasting. The process, and thus data infrastructure, need to evolve into a more integrated and granular approach to distribution load and generation forecasting.

To address the needs to forecast changing customer-side equipment adoption, such as DER, energy waste reduction, and microgrids, which will inform the scenario-based distribution planning, DTEE will need to collect data on customer adoption and propensity of future adoption for various customer-side equipment.

DTEE will need to implement machine learning models to forecast and verify expected system impacts, including customer energy usage and system efficiency. Additional models will need to be developed to provide decision support to system planners by optimizing system upgrade options based on known and forecasted scenarios. These models must support the ability to optimize multiple likely scenarios to efficiently plan for changing customer needs within DTEE's Distribution Planning processes.

To enable these loading and forecasting functionalities, DTEE will implement big data tools in the Azure and Databricks data platform in conjunction with the PowerRunner Energy Platform to continuously validate, process and prepare the loading and voltage data for automatic analysis and system modeling, forecast DER adoption, and provide decision support for system planning. Features of the big data forecasting solution include:

- **Load curve generation** – Functionality to prepare representative disaggregated usage curves for loads and generation at customer levels at hourly values for an entire year is necessary to ensure that forecasting system and customer conditions are accurately incorporating the separate impacts of DER on the grid, such as with solar panels varying output based on the amount of light. Once load curves are generated at individual customer level, the toolset will aggregate to various levels of system equipment, such as service transformer and fuses.
- **Development of DER and EV propensity forecasting tools** – Creation of machine learning models will incorporate external factors and demographics to predict customer adoption of DER, such as energy efficiency equipment, electric vehicles, solar inverters, and micro-grids.
- **Verification of expected impacts and benefits** – Analysis tools will allow system planners to validate the impacts of DER on energy usage, system efficiency and reliability. These verification tools and models are needed as DER becomes more widespread, making current methods difficult to disaggregate from aggregated loads across time.
- **System planning decision support:** Optimization analytics will be developed to consume the forecasted scenarios and run numerous plausible scenarios for customer adoption across the system to determine system investments that will result in a reliable grid for all connected customers while also considering costs and project timing. These decision support models based on load forecasting scenarios will interface with the

situational awareness analytics tools (see Section 12.9.4.5 – Situational Awareness Analytics, below) for detailed planning at the individual component level.

Exhibit 12.9.4.4.1 Distribution Load Forecasting Capital Investment

Distribution Load Forecasting (\$ millions)	2021	2022	2023	2024	2025	5 Year Total
Total	\$0.5	\$2.2	\$1.6	\$1.8	\$1.5	\$7.5

12.9.4.5 Situational awareness analytics

Advanced situational awareness analytics are needed for planning system upgrade decision support. This investment will build on the existing Company enterprise data lake to develop a data analytics architecture based on use case capability, while integrating systems to bring additional power quality data and other sensor information into the data architecture in the data lake. Feature engineering will be performed to summarize data aligned with integrated Distribution Planning use cases. Investments in this space include developing artificial intelligence and machine learning technologies that describe, predict and prescribe suggested actions to take during the integrated distribution planning process. This will improve reliability, power quality and customer flexibility. Anticipated scope and timing of investments is as follows:

- 2022: Investments include development and deployment of data engineering pipelines and visualization applications to create toolsets for integrated grid power flow modeling with load projections, constraints based on aging infrastructure, and planning for multiyear projects, as well as spatial and time-based contextualization of grid overloads, capacity, planned work and asset conditions to aggregate and visualize data to support distribution investment prioritization.
- 2023: Investments include development and deployment of data engineering pipelines to integrate detailed financial data, electric vehicle grid services and charging data, as well as commercial and industrial power quality data into the enterprise data lake to support analysis of reliability and power quality performance. Investments also include licenses and configurations of advanced artificial and machine learning capabilities in the enterprise data platform to support future investments in predictive and prescriptive

analytics, such as evaluation of system upgrades in Section 12.9.4.1 - Interconnection Process Enablement and in Section 12.9.4.4 - Distribution Load Forecasting .

- 2024: Investments include licenses, configuration and development of analytics to process meter and edge monitoring devices within context of the connected grid to trigger action for targeted system upgrades.
- 2025: Investment includes development and deployment of analytic tools that will process power quality information such as flicker, harmonics and advanced fault waveforms to integrate analysis of the impacts of modern grid devices into the integrated distribution planning analysis process.

Exhibit 12.9.4.5.1 Situational Awareness Analytics Capital Investment

Situational Awareness Analytics (\$ millions)	2021	2022	2023	2024	2025	5 Year Total
Total	-	\$0.9	\$1.5	\$0.8	\$0.4	\$3.6

12.9.5 Goal 5: Modernize grid management

Grid management includes the full suite of distribution management applications, including advanced metering infrastructure, distribution automation, outage response, and advanced control toolsets. Mature grid management enables distribution operators to model and manage the distribution network, monitor the power system, manage planned and unplanned outages, and analyze and optimize the quality and reliability of the network.

In this area, DTEE also relies upon an existing platform of systems to deliver superior operating results and contribute to the Company's commitments surrounding power quality, reliability and safety. These systems require the following investment:

- The existing communications infrastructure that DTEE relies upon to monitor and manage the grid in real time will require additional investment as new or expanded use cases and data requirements are identified which surpass the systems' current implementation.

- The Advanced Metering System and its associated infrastructure are operationally stable, however with new use cases, they are being relied upon to provide operational data beyond their original design specifications. Initially deployed as a daily usage collection system and given its success, AMI is now relied upon provide additional information at a much higher intra-day frequency. DTEE is investing in this system to bring this deployment to a more real-time data network to meet both current and future demands. This will be a significant long-term effort.
- As the first meters installed during DTEE's Advanced Metering Infrastructure deployment approach end of life, DTEE will evaluate the next generation of meter and meter communication technology that will meet the needs for advanced planning and operations functionality. In 2025, DTEE will invest in meters to deploy limited scale tests of viable next generation mesh meters to evaluate functionality and performance.
- The microwave network is used as a primary and secondary WAN communications for corporate and SCADA networks. The modernization of telecom microwave tower equipment will optimize performance and align DTEE's frequency usage with the FCC. Due to the FCC deregulation of privately owned frequencies, a number of DTEE's microwave systems must be replaced to maintain a reliable network and avoid collisions with other entities using the same frequency. DTEE will design and implement new microwave backhaul communications, install modern dishes for tower weight reduction, install hardware for network redundancy and resiliency, and replace end-of-life hardware.
- Investment is needed to support operational use of sensor technology on the grid, including devices being deployed at the grid edge to provide information to both DTEE and its customers about their real-time and historical energy information for the submetering customer programs.
- Utilizing sensor information for operational analytics will require investment in real-time streaming and analytics in DTEE's data platform to proactively respond to customer outage and power quality issues.
- Investment is needed to implement an industry standard configuration and test records platform to manage the increasing amount of grid control device configurations. This investment will include procuring an industry standard database

tool with integration into Maximo for work and asset management functionalities, as well as purchasing advanced test sets to be used by field technicians to automate relay testing and recording of test records to the database.

Exhibit 12.9.5.1 Modernize Grid Management Capital Investment

Modernize Grid Management (\$ millions)	2021	2022	2023	2024	2025	5 Year Total
Configuration and Test Records Platform	\$0.3	\$2.3	\$1.7	\$0.7	-	\$5.0
Grid Edge Insights	-	\$0.8	\$0.9	\$0.9	\$0.6	\$3.1
AMI Mesh Network Enhancements	-	\$0.5	\$0.5	\$0.5	\$0.5	\$2.0
Microwave End of Life	-	\$0.5	\$0.5	\$0.5	\$0.5	\$2.0
Evaluate Next Generation Metering and Communication Network	-	-	-	-	\$4.8	\$4.8
Other Modernize Grid Management	\$2.8	\$7.3	\$2.9	\$3.4	\$2.2	\$18.6
Total	\$3.1	\$11.3	\$6.5	\$5.9	\$8.5	\$35.3

12.9.6 Technology strategy roadmap capital investment summary

Exhibit 12.9.6.1 Technology Strategy Roadmap Investments (\$ millions)

Program Track	2021	2022	2023	2024	2025	2021-2025 Total
Mobility	\$-	< \$1	\$5	\$1	\$8	\$13
Work Management and Integrated Scheduling	<\$1	\$3	\$12	\$3	\$3	\$21

Asset Management	\$-	\$1	\$2	\$2	\$1	\$6
Distribution Planning Processes Enablement	\$2	\$10	\$10	\$11	\$9	\$43
Modernize Grid Management	\$3	\$11	\$6	\$6	\$9	\$35
Total	\$6	\$26	\$35	\$22	\$29	\$118

13 Base capital



Base capital refers to work activities that DTEE performs as part of the normal business of serving our customers. Activities include, but are not limited to, restoration of customers after a storm or equipment failure, connection of new customers, and relocation of existing assets.

Base capital is divided into two major categories: 1) emergent replacements and 2) connections, relocations and other.

13.1 Emergent replacement

Emergent replacements are expenditures that are used to address immediate customer needs (outages, broken equipment, etc.) or immediate system needs (loss of a contingency, unsafe operating conditions, etc.). Exhibit 13.1.1 provides the forecasted emergent spend for 2021-2025.

If the Distribution Grid Plan investments, summarized in Section 6, are implemented as described through the DGP, then DTEE forecasts that capital required for emergent replacements will be reduced compared to what it otherwise would have been, which includes offsetting inflation; and is projected to be lower than current emergent spend. The savings were calculated during the development of the DGP by modeling the reduction in emergent events due to strategic capital investments and a base level of tree trimming and other maintenance. These savings were then applied to the forecasted emergent capital spend projected. As was shown in Section 5.3, the projected emergent costs are \$67 million lower per year than the baseline scenario, which projected \$420 million in emergent costs by 2025.

Emergent replacements fall into three categories of work: storm, non-Storm, and substation reactive. Each category is described in detail in the following sections.

Exhibit 13.1.1 Emergent Replacement Expenditures (\$ millions)

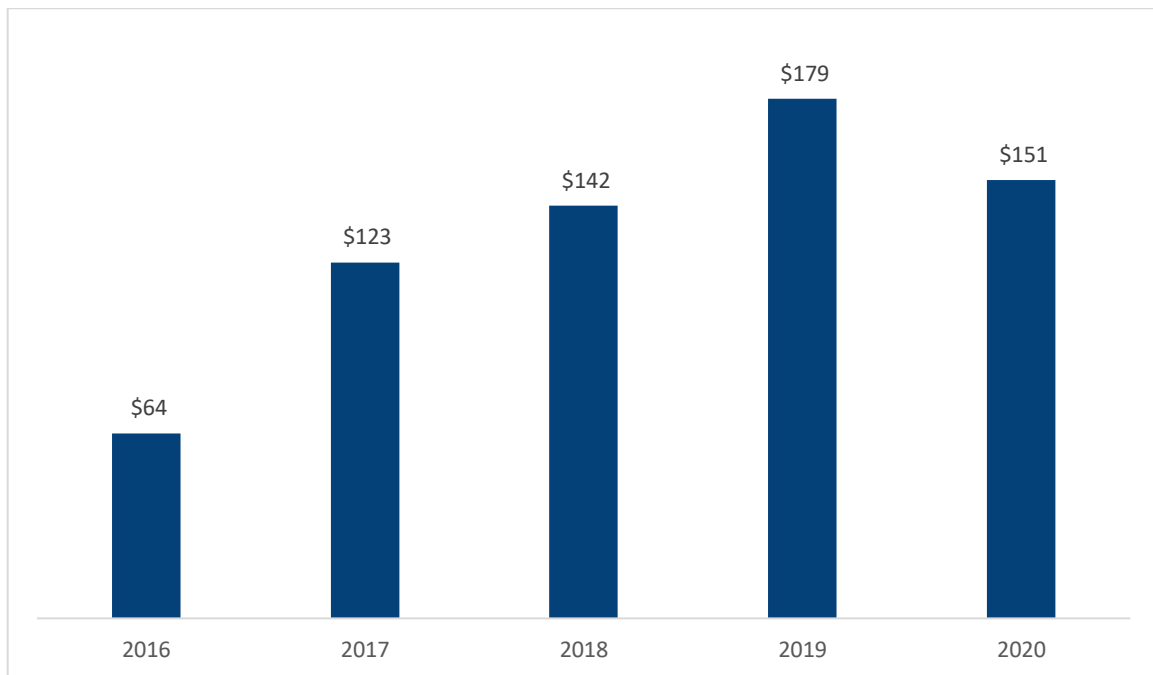
Category	2021	2022	2023	2024	2025	5-Year Total
Emergent Replacements	\$488	\$372	\$372	\$366	\$354	\$1,952

13.1.1 Emergent replacement during storm conditions

This category includes expenditures required to replace damaged assets on DTEE's overhead and underground distribution and subtransmission systems during storm conditions, which is defined as having more than 340 outage events impacting at least 125 circuits across the entire system. This threshold is typically equivalent to 25,000 customers affected over a 24-hour window. Several events can cause storm conditions distribution failures, such as wind causing trees or tree branches to come in contact with poles and power lines causing outages. It should be noted that not all weather related incidents create storm conditions.

Capital investments in emergent replacements during storm conditions has been increasing over the past five years. Exhibit 13.1.1.1 shows the amount invested from 2016 through 2020.

**Exhibit 13.1.1.1 Capital Spend on Emergent Replacement during Storm Conditions
(\$ millions)**



The primary driver of the increased investment is the increasing number of catastrophic (CAT) storms DTEE has experienced over the last five years.

There are two CAT storm classifications: a CAT-1 storm is defined as more than approximately 5% of DTEE customers without power and a CAT-2 storm is defined as more than approximately 10% of DTEE customers without power.

Exhibit 13.1.1.2 shows the number of CAT storms DTEE has experienced since 2015 and Exhibit 13.1.1.3 shows the number of customers impacted by the CAT storms.

Exhibit 13.1.1.2 Number of CAT Storms 2016 – 2020

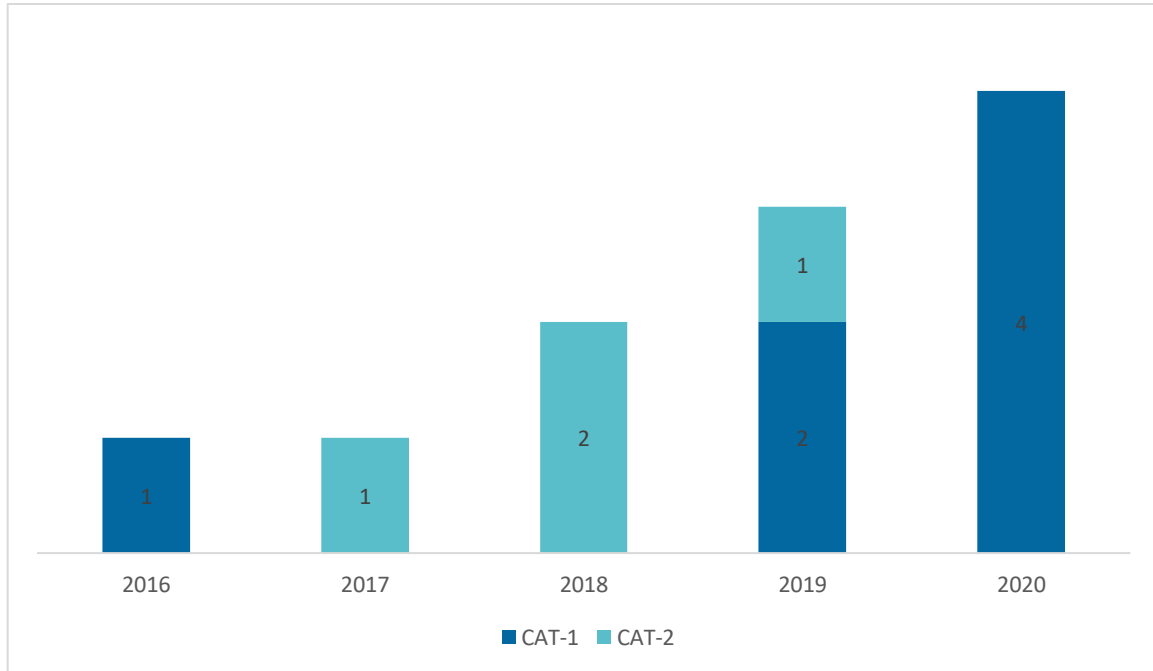
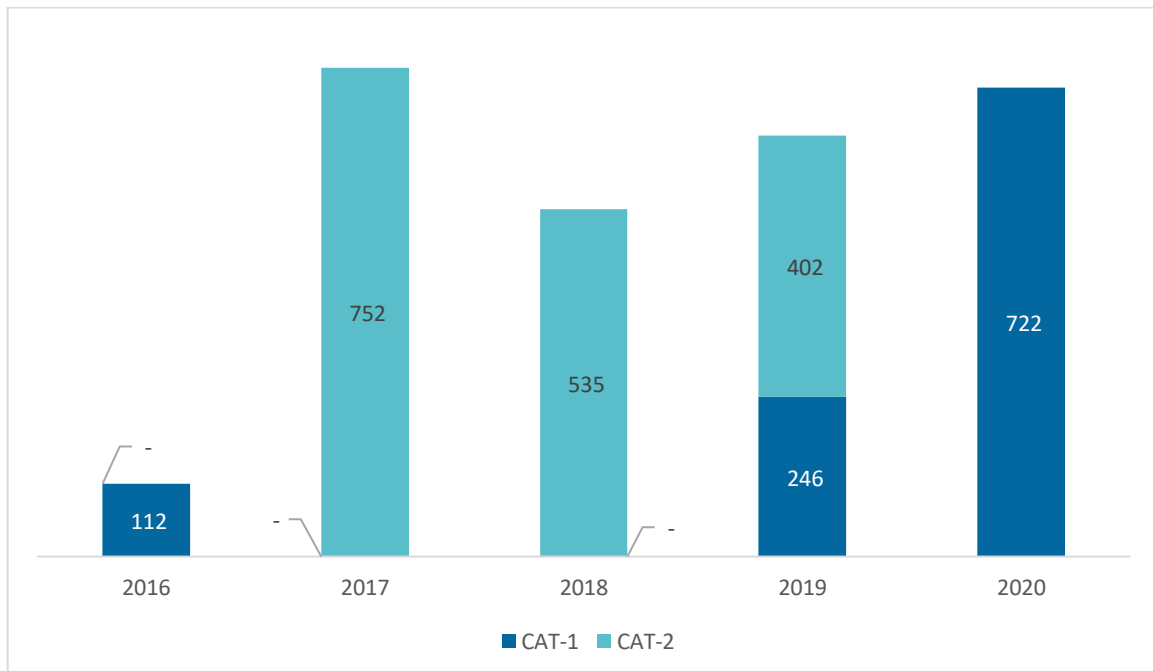


Exhibit 13.1.1.3 Number of Customers Impacted 2015 – 2019 (thousands)



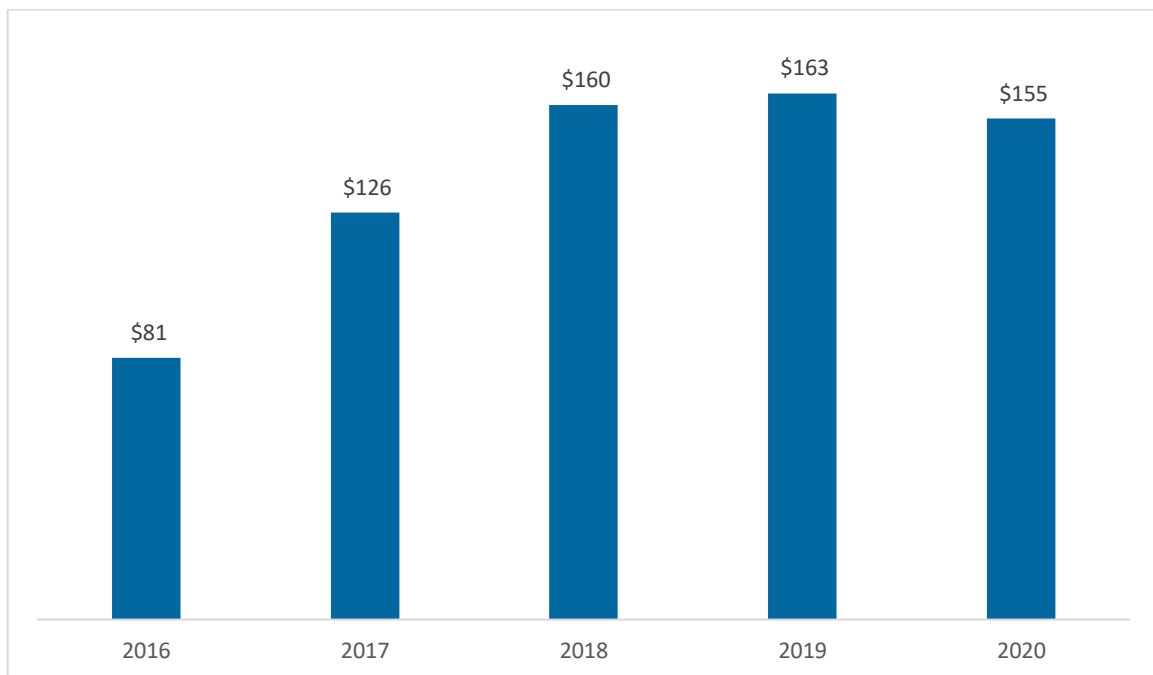
13.1.2 Emergent replacement during non-storm conditions

This category includes expenditures required to replace damaged assets on DTEE's overhead and underground distribution and subtransmission systems during non-storm conditions, defined as less than 340 outage events (approximately 20,000 customers) for the entire system.

The "non-storm" description does not imply that these events are not caused by thunderstorms and high winds. It simply means that there were not enough system-wide outages to be considered a declared storm for the system; yet localized thunderstorms and weather events could occur. Assets can also fail due to age, overloading, other electrical or mechanical issues, vehicles striking poles, or animal interference.

Capital spend on emergent replacements during non-storm conditions has been increasing over the past five years. Exhibit 13.1.2.1 shows the spend from 2016 through 2020.

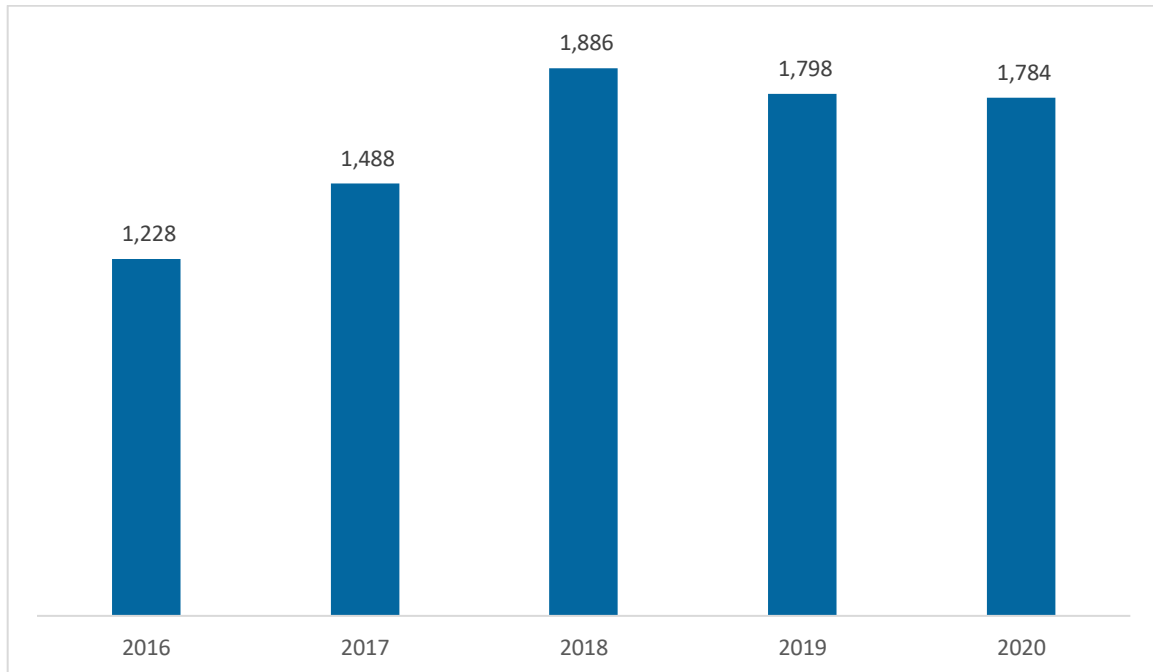
Exhibit 13.1.2.1 Capital Spend on Emergent Replacement during Non-storm Conditions
(\$ millions)



There are several factors driving the increase in non-storm capital spend. A key contributing factor has been the increase in the number of customers experiencing interruptions in non-storm conditions. This increase has been caused by aging infrastructure as well as an increase in

weather events that do not rise to the level of a storm threshold as defined in Section 13.1.1. As shown in Exhibit 13.1.2.2, there has been over a 40 % increase in the number of customers that lost power during non-storm conditions since 2016.

Exhibit 13.1.2.2 Number of Customer Interruptions during Non-storm Conditions (thousands)



Another driver for increased emergent capital spend has been the increase in asset failures and replacements due to system aging, deterioration, and other factors like tree damage. As the system ages and damage accumulates, there is a need to replace equipment instead of repairing. For example, a span of wire that has experienced repeated failures and repaired with multiple sleeves should be replaced; as opposed to adding an additional sleeve and increasing the probability of future failure. At-risk underground residential distribution (URD) cable should be replaced upon failure to avoid increased probability of experiencing another failure and the extensive restoration time needed to repair. And a service line that is past its useful life or not up to current standard should be replaced once it has failed in order to avoid additional failures that are driven by that same service line.

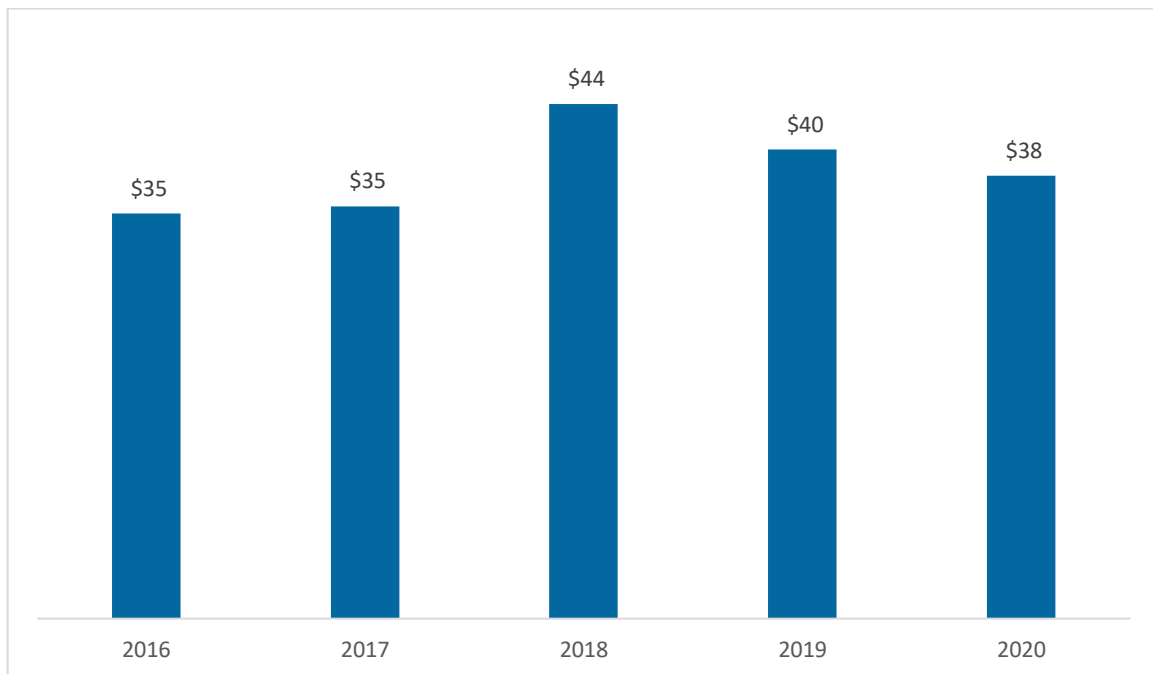
Using materials designed to higher engineering standards to improve system reliability and resiliency has also contributed to increased emergent capital spend. Examples include the use of fiberglass crossarms, which cost more than wood crossarms, but provide much greater longevity

and resistance to damage; higher wood pole class and armless construction, which increases the strength of poles by a factor of more than 2; and use of the more rugged K-line clamp-top polymer insulators, which have six times the mechanical strength of the older design F-neck porcelain insulator and can better secure the wires to mitigate downed wire risk.

13.1.3 Emergent replacement for substation reactive

This category includes expenditures required to replace damaged assets or assets failed during inspections in stations and substations. These expenditures were made to replace broken assets that either led to customer outages, elevated the risk of wide-spread outage events or impacted the redundancy or operability of the electric system. Substation reactive expenditures have increased slightly over the last five years as shown in Exhibit 13.1.3.1.

Exhibit 13.1.3.1 Capital Spend on Emergent Replacements for Substation Reactive
(\$ millions)



Factors that have contributed to the increased substation reactive capital include:

- Inspections and consequent replacements of defective netbank structures and associated system cable since 2016
- A more integrated approach to replace failed substation assets as well as adjacent at-risk assets to achieve higher cost efficiency since 2016
- Major substation failures and reactive replacement work, such as Plymouth substation event in 2018, Gay substation event in 2019, and Stephens substation event in 2020
- Increasing utilization of mobile fleets in support of restoration work, to preserve grid reliability and resiliency when a substation is in an abnormal condition due to equipment failures; for example should a substation experience a switchgear failure, portable switchgear can be mobilized to the site and connected to the circuits, restoring power to the customer and providing time for DTEE to replace or repair the failed switchgear

13.1.4 Measures to reduce emergent capital spend

As described in Section 5 – Benefit Cost Analysis; DTEE investments are prioritized based on their ability to meet the planning objectives of Safety, Reliability and Resiliency and Affordability. Affordability includes the impact investments have on avoided emergent capital costs. The investments detailed in this DGP are expected to reduce emergent capital spend by \$67 million in 2025.

Among the entire investment portfolio, tree trimming, 4.8kV hardening, 4.8kV conversion and consolidation, and pole and pole top hardware are among the top contributors to improve storm resiliency, reduce day-to-day trouble and event volumes, and ultimately reduce emergent capital spend. A robust pole top maintenance and modernization program, discussed in Section 9.1.1 - Pole and Pole Top Maintenance and Modernization, is critical to maintain the health of the overhead system and keep the emergent spend under control on a long-term basis.

Technology programs can also have a positive impact on avoided costs, improved reliability, and improved operational efficiency by incorporating enhanced sensing, communication and control of the electric grid that will allow for faster identification of faults and outages, quicker customer restorations, reduced patrol times, and increased remote operation of the grid. These technologies include: Grid technology such as Advanced Distribution Management System

(ADMS) discussed in Section 12.1, Telecommunications in Section 12.3, and Distribution Automation in Section 12.5.

13.1.5 Emergent spend forecast

DTEE uses a historic average spend for emergent capital forecast to normalize for weather and other factors that change year to year. In MPSC cases U-20162 and U-20561, DTEE forecasted emergent capital spend by calculating a historic average in each category, adding inflation through the “test year,” and subtracting a calculated avoided cost based on the level of proposed strategic capital spend.

Based on the recent historic trends, such as increase in catastrophic storms, number of customers out during non-storm conditions, etc., as noted in the proceeding subsections; DTEE has decided to use a three-year historic average for emergent capital spend as opposed to the previous method of using a five-year average. A three-year historic average better represents the current conditions of the system and weather-related patterns that DTEE has experienced and allows for more accurate projections of future emergent expenditures.

13.2 Connections, relocations and other

The connections, relocations and other investment category is broken down in to five major sub-categories: Connections and New Load; Relocations; Electric System Equipment; NRUC and Improvement Blankets; and General Plant, Tools & Equipment and Miscellaneous. These sub-categories are discussed in more detail below. Exhibit 13.2.1 provides the connections, relocations and other spend for 2021-2025. Except for the specific projects described in Section 13.2.6, the forecast in Exhibit 13.2.1 for 2022-2025 is based on the last full year of spend (2020) plus inflation. The 2021 budget is based on known projects and requests.

Exhibit 13.2.1 Connections, Relocations and Other Spend (\$ Millions)

Category	2021	2022	2023	2024	2025	5-Year Total
Connections, Relocations and Other	\$272	\$252	\$256	\$263	\$270	\$1,313

13.2.1 Connections and new load

The Connections and New Load sub-category expenditures are divided into two categories: Small Load Growth projects & Customer Connections, and New Business projects.

Small Load Growth & Customer Connections projects are projects that typically cost less than \$500,000 to complete and are developed and executed by distribution regional service centers. These are projects required to serve new load, customer connections, or an upgrade needed to address a limited level of loading issues on the electric system. Activities to support these customer needs may include reconductoring lines, expanding a substation, or transferring load to provide load relief. DTEE could also install additional overhead and underground lines to provide service to small commercial businesses or housing developments. Small Load Growth & Customer Connections projects are often requested by customers and can carry a customer contribution in aid of construction (CIAC) cost wherein the customer pays an allocated cost of new service depending on the load requested and service requirements.

New Business investment projects are larger customer-driven projects, in excess of \$500,000, and require a central engineering approach to design and implementation. Like Small Load Growth and Customer Connections, these are projects required to serve new load, connections or to address loading issues, just on a larger scale. Activities to support the customer new business needs often include installing additional overhead and underground lines, upgrading or building new substations to provide services to large commercial or industrial customers. As with Customer Connections, New Business projects are often requested by customers and can carry a CIAC cost wherein the customer may be responsible for an allocated cost of system upgrades, depending on the existing system conditions, the load requested and the impact to the electrical system.

These investments connect new customers, support area load growth for both existing and new customers, enable economic growth, and improve customer satisfaction.

13.2.2 Relocation

Relocation projects are requests from customers or municipalities and other governmental entities, including the Michigan Department of Transportation, to relocate existing DTEE facilities. These requests are typically related to construction activities in specific geographic areas, including the

modification of roadways, bridges, water mains, public sanitation, alleys, or other customer activities.

DTEE facilities being relocated could include overhead lines and poles, or underground lines and padmount transformers, or possibly substations and system cable. With expected federal investment in infrastructure, it's expected that these types of projects will become more frequent in the coming decade.

These investments can enable economic growth, meet customer needs, are required by regulation, or at the request of municipalities. Similar to Customer Connections and New Business, some projects requested by customers or municipalities can carry a CIAC cost wherein the requester pays for part of the costs of relocating DTEE assets.

13.2.3 Electric System equipment

DTEE maintains an inventory of critical spare equipment in order to both support emergent replacements, and planned projects. The Electric System Equipment sub-category of investment includes investments in equipment, particularly long lead time items, to provide proper inventory levels of major electric system equipment. Equipment purchased under this sub-category includes substation transformers, distribution transformers, regulators, and meters, among other items.

These investments provide inventory to help serve customer needs in a timely fashion, so that urgent work is not slowed down due to a need for critical equipment.

13.2.4 NRUC and improvement blankets

Normal Retirement Unit Change-out (NRUC) consists of projects to perform scheduled work for replacement of assets determined to be end-of-life. Improvement Blankets include several types of small projects aimed at addressing improvements to the electrical system, software applications for field use, and animal mitigation.

The scope of work for this sub-category includes small improvements to address localized reliability issues, projects to perform scheduled work for change out of items on the subtransmission and distribution systems (e.g. replacement of reclosers and switches) determined to be at end-of-life, batteries and charger replacements at substations that are at end-of-life, development of software applications for field use, and animal mitigation (e.g. installing

squirrel guards on bus insulators). Capital replacement programs detailed in Section 9.1 are designed to handle larger replacement projects, in addition multiple strategic projects that fall under Infrastructure Redesign and Modernization (Section 11) include replacements of larger assets such as substation breakers, etc.

These projects are focused on improving operating conditions to reduce the frequency and duration of outage events. Example of projects performed including installing/upgrading fusing on a circuit and installing fault indicators.

13.2.5 General plant, tools & equipment, and miscellaneous

General Plant, Tools & Equipment and Miscellaneous includes the tools and replacement tools required for linemen and splicers to perform their work and test equipment for engineers. In addition, this sub-category includes substation physical security.

These expenditures ensure that DTEE work crews are equipped for their day to day field work. Examples of items included in General Plant, Tools & Equipment and Miscellaneous include: very low frequency (VLF) trucks used to test cables at frequencies between 0.02 and 0.1 Hz, DC hi-potential testers (Hippoters) used to test insulation on cable, breakers, transformers, etc., oil processing trailers, insulated screwdrivers, pencil tip soldering iron, and other tools and equipment.

13.2.6 Other known major projects

DTEE supports some grid work that is outside of the more typical categories already described in this plan. The Gordie Howe bridge project is a large relocation project example. The United States, Canada, and the State of Michigan agreed upon a new U.S. customs plaza and international bridge, now known as the Gordie Howe International Bridge (GHIB), linking Detroit, Michigan and Windsor, Ontario.

In order to build the GHIB and U.S. customs plaza, the Windsor Detroit Bridge Authority (WDBA) and Michigan Department of Transportation (MDOT) requested that DTEE relocate all its facilities outside the plaza footprint. DTEE agreed to a joint venture with MDOT wherein MDOT would pay half the cost to relocate DTEE facilities. The project is expected to be completed by the end of 2021.

Another example of a major one-time project in the New Business project is Warren Service Center (WSC) Transformer Yard Reorganization. The transformer yard located at WSC has reached capacity and needs to be upgraded to provide better handling and storage of these assets (transformers and regulators). In order to support more efficient transformer replacements in the future, the WSC transformer yard will be reorganized and upgraded to accommodate a total of 193 transformers/regulators and will include an assembly area.

Stone will be added as a base to aide in stabilization of the stored transformers and electric power will added so that the transformer control panel will be continually operational, and the equipment health information can be monitored.

14 Preventative maintenance



Preventive maintenance (PM) is performed to achieve three primary objectives:

- 1) Control or reduce unplanned equipment-related outages.
- 2) Maximize the total life expectancy of the equipment at minimum lifecycle cost while maintaining an expected level of reliability and safety.
- 3) Acquire timely asset performance and condition data to support trend data analysis and identify potential issues.

An effective PM program is critical in managing asset conditions and capital and O&M costs. Risks of not performing preventive maintenance include compromised equipment performance during service and/or increased trouble costs due to equipment failing prematurely.

The PM program was developed by DTEE Engineering based on DTEE's maintenance policies which set the inspection scope and intervals based on:

- Manufacturer's recommendations
- Industry standards and best practices
- Historical inspection results and equipment condition data
- Engineering studies
- Benchmarking with utility peers

DTEE continues to refine its maintenance policies in order to find efficiencies and reduce failures. Some examples of the continuous improvement integrated into the PM program are:

- Changing inspection intervals based on engineering analyses of inspection results and equipment failures (condition-based maintenance).

- Using substation predictive maintenance (SPdM) data to drive inspection timelines instead of time-based inspections, to be aligned with condition-based maintenance principles and to achieve cost efficiency.
- Reducing maintenance activities by using new technologies (e.g., smart relays).
- Coordinating inspection intervals on different equipment to reduce the need for multiple shutdowns.

Furthermore, DTEE has developed condition-based maintenance and substation predictive maintenance (SPdM) plans to maximize maintenance efficiency. By evaluating certain monitored data available on some assets, engineers can prioritize equipment on the maintenance schedule and where appropriate, may defer some equipment to the next inspection cycle. One successful example is the prioritization of substation power transformer maintenance using dissolved gas analyses, which takes a sample of the oil in the transformer or regulator and looks for gases that are early indicators of insulation failure. Predictive maintenance includes taking pictures of substation equipment taken with infrared cameras to detect thermal hot spots, operational checks such as stepping through a regulator's settings, verifying oil circulation, and ensuring operational counters are functional. Information collected from these types of activities help drive the priority of maintenance activities on substation equipment.

The previous activities require personnel to interact with the equipment. Alternatively, when advanced remote monitoring equipment is installed at substations, breaker operation data can be used to assess breaker and relay functions. If the equipment is assessed to be performing as designed, it may be possible to defer equipment maintenance activities. Currently, remote monitoring capability at substations is limited, and DTEE has not yet been able to widely apply this methodology to defer breaker or relay inspections. However, DTEE's investments in telecommunications, breaker replacement, and substation automation programs described throughout Section 12 will increase the penetration of such advanced remote monitoring technology and will enable the application of condition-based maintenance on breakers and relays.

DTEE's preventive maintenance programs are categorized in three groups, substations, overhead, and underground, as shown in Exhibits 14.2.1– 14.2.3. The projected O&M spend associated with each preventive maintenance Exhibit 14.2.4.

Exhibit 14.2.1 Preventative Maintenance Activities - Substations

#	Asset	Population (approx)	General inspection cycle (years)
1	120kV disconnects	352	10
2	120kV breakers	32	10/12
3	24/40kV breakers	1,021	3/10/12
4	Distribution breakers	5,334	3/10/12
5	Buses	250	10
6	Regulators	269	10
7	Portable substations	10	1
8	Single tap substations	118	10
9	13.2kV capacitor banks	224	1
10	AC netbank transformers	505	5
11	AC netbank structures	384	10
12	DC systems	589	1
13	Dissolved gas analysis (DGA)	3,323	1
14	Substation predictive maintenance (SPdM)	1,640	3
15	Voltage Control periodics	1,435	1
16	Relay periodics	9,703	5/7/10
	Total	25,198	

Exhibit 14.2.2 Preventative Maintenance Activities - Overhead

#	Asset	Population (approx)	General inspection cycle (years)
1	40kV pole top switches	862	8
2	Distribution remote devices	1,455	4/8
3	OH capacitor & regulator controls	4,505	1
4	OH distribution device battery replacements	1,455	4
5	Towers	332	10
6	DTEE equipment in high rise structures	45	20
	Total	8,654	

Exhibit 14.2.3 Preventative Maintenance Activities - Underground

#	Asset	Population (approx)	General inspection cycle (years)
1	Primary switch cabinets	1,731	5/10/15
2	Manholes	19,356	10
3	Vaults	30	5
	Total	21,087	

Exhibit 14.2.4 Preventative Maintenance O&M Spend Projections

	2021	2022	2023	2024	2025	Total
Preventative maintenance O&M spend	\$10M	\$10M	\$10M	\$10M	\$10M	\$50M

15 Performance-based rate making



Performance-based rate-making (PBR) refers to regulatory frameworks that aim to create a stronger connection between a utility's achieved returns and its performance. This section will describe the recent history of PBR in Michigan, the Commission's Order in Case No. U-20561, and DTEE's approach to benchmarking, developing and proposing a holistic PBR plan for the Commission's consideration. The section is organized into three primary areas and reflects the direction set forth in Case Nos. U-20561 and U-20147:

- **Background and context** – The Michigan context and discussion has shaped DTEE's proposal, reflecting perspectives articulated by the Commission, Commission Staff, other stakeholders, and DTEE over the last several years.
- **Benchmarking** – DTEE undertook an extensive benchmarking effort following the Commission's Order in Case No. U-20561 with the goal of understanding how jurisdictions around the country have approached PBR both from a process perspective and a substantive perspective.
- **DTEE's proposed PBR plan** – The proposed plan focuses on additional transparency of key reliability metrics and offers a flexible pathway to manage continued reliability and grid accessibility investments.

15.1 Background and context

In 2018, the Commission completed a "Report on the Study of Performance-Based Regulation." The report discussed several pertinent topics related to PBR, with particular focus on performance incentive metrics and their associated goals, multiyear rate plans (MRP) and examples of broad-based PBR in the United Kingdom and New York state. Michigan utilities and Commission staff

supported the creation of this report through additional survey reporting of PBR around the country and the key considerations identified. One of the key recommendations from the report was, “The Commission’s review of PBR mechanisms indicates that they can be used to augment the existing cost-of-service approach provided that they are tailored to the specific requirements associated with utility regulation in Michigan,” and further observed that, “[a] variety of approaches are available.”

The Commission’s Order in Case No. U-20561 said “...the Company should consider the following:

1. The utility’s financial PBR system should include both incentives and disincentives based on performance; incentive structures should be holistically considered in terms of impacts on potential earnings;
2. The utility should consider the pros and cons of a comprehensive PBR system, which would avoid concurrent regular annual rate cases and separate PBR reconciliations;
3. Performance metrics should include outcome measures (e.g., customer average interruption duration index (CAIDI)) and not be limited to output metrics such as number of poles replaced;
4. Performance metrics should be linked to regional, national, and/or peer utility benchmarks, where possible;
5. Data and calculation methodologies should be well defined, transparent, and open for auditing/verification purposes;
6. Targets should be utility specific; and
7. Potential areas of performance focus are safety, customer service (end-use customers, builders, interconnecting generators, etc.), timeliness and quality, reliability and resiliency, long-term costs, and innovation.”

The proposal contained in this section addresses considerations specified by the Commission.

As the Commission observed in the 2018 report, “PBR is complex and has both advantages and disadvantages.” DTEE agrees with this perspective and, as described in more detail below, is supportive of a deliberate approach to PBR that allows for a thoughtful and measured implementation as a first step. Benchmarking indicates that measured deployment over time can create broad stakeholder buy-in on the elements, while ensuring that each is well understood and the sum of the parts can be reasonably estimated. Over time, PBR is something that may grow

toward an expanded implementation in future years as DTEE learns and adopts best practices. As such, and with the venue for the initial PBR plan in the Distribution Grid Plan, the focus on distribution reliability and customer outcomes and engagement reflects what DTEE believes is the appropriate entry point for PBR.

DTEE views the included proposal as foundational groundwork for future PBR and an opportunity to explore and apply many of the PBR concepts discussed in the Commission's report and e observed elsewhere. Continued stakeholder engagement as the Company learns more about the applied impacts and effects of the proposal will serve to enhance the plan going forward. This may be informal engagement or under the purview of the Commission.

15.2 Benchmarking

Benchmarking is a key input to the development of new approaches to problem solving and growth. A robust understanding of how different states and regulatory environments have considered PBR supports a proposal that leverages those insights while simultaneously reflecting the Michigan-specific context of the concepts. Beginning in mid-2020, DTEE undertook an extensive benchmarking effort to understand the diverse applications of PBR around the country, focusing on 1) the process of how states engaged the conversation and 2) the substance generated by those processes. The goal of these inquiries was two-fold. The first was to baseline how the various states thought about and approached process and substance, and what the key considerations were. The second was to highlight the learnings generated from those states as a means to leverage best practices as a starting point for our own work here in Michigan, without starting from scratch.

The benchmarking included two key modes of understanding:

- **Research** – In conjunction with stakeholders, DTEE synthesized existing dockets, publications and other available materials from numerous states. The initial list of states to review was generated through internal research and input from local and national stakeholders. This step was critical in narrowing down the focus to those states which provided useful comparability to Michigan and also represented diverse approaches, goals and objectives for PBR.

- **Engagement** – Primary perspectives on PBR from involved parties provide context and insight that is not necessarily available in published documents. DTEE coordinated live discussions with organizations in multiple states to better understand both how they viewed the role of PBR, but also to learn about their experiences and advice. This live engagement included regulators, utilities, the financial community and non-governmental organizations representing customer and other stakeholder perspectives.

Key state focus

- **Minnesota** – Minnesota’s PBR journey has unfolded over a decade of focused collaboration between the utilities, regulators, stakeholders and facilitators. Minnesota is an important benchmarking focus for three reasons: 1) the long record of development offers many insights into successes and pitfalls, 2) the state’s process provides a strong example of broad stakeholder engagement, and 3) Minnesota’s approach to performance incentive mechanism (PIM) development and the iterative perspective on PIMs provides an instructive framework.
- **Colorado** – Colorado has had elements of performance regulation for more than a decade, including for demand side management, revenue decoupling, earnings sharing, and quality of service plans (QSPs). The recent move to a more structured and comprehensive investigation of performance regulation was spurred by state legislation in 2019, which directed the Colorado Commission to investigate financial performance incentive mechanisms and metrics and return a report to the state senate. The legislation specified the forum for gathering information and certain specific pieces of content, including existing and potential metrics, future test years and new performance incentives.
- **Illinois** – Illinois is an instructive state for benchmarking, with a highly prescribed approach to PBR. Legislation in Illinois was first enacted in 2011 with the Energy Infrastructure Modernization Act (“EIMA”) and expanded on in 2016 with the Future Energy Jobs Act (“FEJA”). Illinois utilizes formula ratemaking and has multiyear performance incentives, each which alter allowed ROE. There are no defined dollar penalties in the Illinois PBR approach.
- **Massachusetts** – Massachusetts has legislatively enabled performance-based regulation premised on several pieces of legislation. Its inclusion in recent Eversource and National Grid rate cases is closely tied with Massachusetts’ grid modernization docket opened in late 2013. In 2015, the large utilities filed company specific plans to address the requirements, which

included individual and statewide performance metrics related to grid modernization. The combined 2018 order in the grid modernization docket directed the inclusion of additional PBR elements in future rate cases, requiring utilities to include “performance metrics that measure progress toward the objectives of grid modernization.”

- **Pennsylvania** – Pennsylvania’s PBR process is legislative enabled and non-prescriptive. The utilities have the discretion to introduce PBR, which method(s) to request, and deployment approaches. The Pennsylvania Public Utility Commission (PPUC) efforts, which began in 2016, have since developed into guidelines on how PBR will be considered by the Commission. Separately but in parallel action, PBR legislation was adopted and the PPUC approved a Final Implementation Order of Act 58 in April 2019. Act 58 gives the PPUC authority to approve PBR plans and while it does outline specific PBR methodologies, it does not limit the Commission’s authority to approve any other methodologies proposed.

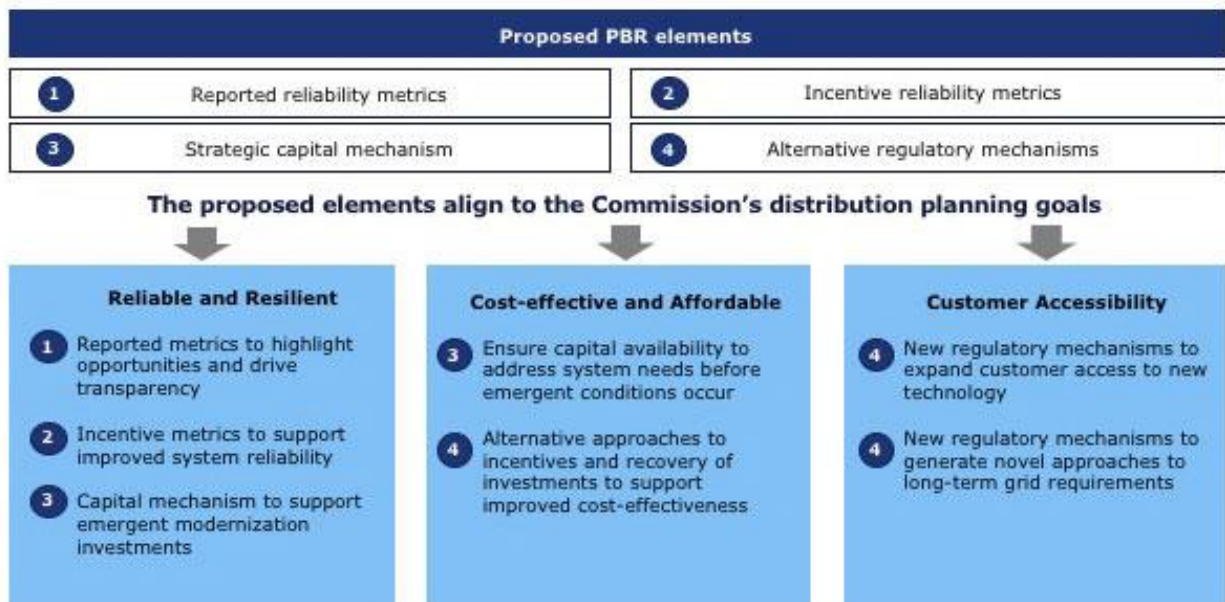
Key findings

- An impactful but measured pace serves all stakeholders and drives win-win outcomes for customers, utilities, regulators and other stakeholders. Minnesota did not move toward a “comprehensive” PBR plan at the outset but stress-tested and evaluated each step before evolving to the next iteration, for example. Other states each had a form of iteration or reflection between the initial requirement or proposal and the formal implementation of a PBR element in their ratemaking process. This is instructive for a Michigan approach and is consistent with how DTEE has considered this proposal, as a focused and initial step into what may evolve into a broader PBR framework.
- The processes undertaken by each state are unique and reflect their past engagement with performance ratemaking, stakeholder perspectives and the intended outcomes of the efforts. This diversity is evident in the types of PBR under discussion (multiyear rates vs. performance metrics vs. other), the financial mechanisms (dollar basis vs. allowed ROE), and in who has been the leading facilitator and enabler of these efforts. The Michigan background is punctuated by certain PBR elements, such as a future test year, certain shared savings mechanisms, and penalties associated with certain reliability performance. Each of these, in addition to the many discussions, workgroups and past explorations provides the foundation for the present effort.

15.3 DTEE's proposed PBR plan

DTEE's proposed PBR plan provides a balanced and forward-looking pathway to expanded linkages between utility performance and returns, as well as increased transparency of reporting and communication of performance. The proposal is designed to align with the overarching goals of DTEE's Distribution Grid Plan as well as the Commission's distribution planning goals as outlined in Case No. U-20147. These key goals are achieved through two primary and two support elements within the PBR plan, as shown below. The Plan will further highlight the cross-cutting themes of customer experience and equity throughout the four elements.

Exhibit 15.3.1 DTEE PBR Goals and Plan²²



- **Reported reliability metrics** – Submit a new, standalone annual report that will include new metrics and expanded context and discussion of metric performance. The result of increased transparency and learnings will support more effective metrics and a better understanding of how the Company can continue to improve the impact of investments. Implementation of reporting would commence in the year following an Order in this docket.
- **Incentive reliability metrics** — Associating incentives with key metrics continue to drive alignment of operational performance and financial performance for the Company while providing additional consideration to customers if performance is significantly below target.

²² Safety is a key DTEE Planning Objective as shown in Exhibit 4.2.2, and should remain outside of this PBR framework

Implementation of reliability metrics with financial incentives would follow an Order in a contested case subsequent to this case.

- **Strategic capital mechanism** – A capital tracking mechanism is one approach to investment flexibility in response to quickly evolving grid usage. DTEE is not proposing the near-term implementation of a mechanism, but the Company does highlight the role it could have going forward in a comprehensive PBR framework.
- **Alternative regulatory mechanism** – Alternative regulatory mechanisms are a key component of a broad-based and holistic approach to performance regulation. Beyond traditional COSR and a reliability focused PBR plan are alternative regulatory mechanisms which, in tandem with COSR and other metrics, can drive additional focus on 1) policy goals, 2) customer engagement and accessibility and 3) cost management.

Reported reliability metrics

Performance metrics may be published for additional transparency and communication, or they may be tied to incentives; DTEE's PBR Plan includes both. The reporting metrics in DTEE's PBR Plan have been structured to support two goals:

- **Learning** – Reporting metrics before incentivizing them encourages forward thinking approaches to measure and reflect performance before attaching the risk of financial incentives. This approach creates learning periods as specific metrics are defined and the supporting methodologies are refined. This allows the Company to better consider metrics to accurately reflect system performance and customer experience, and also to understand the conditions that drive those metrics. A period of learning will also allow the Company to calibrate forecasted improvement with actual improvements over the same historical period, which, over time, will support improved understanding of how specific investments impact reliability.
- **Transparency and communication** – Reported metrics support enhanced transparency of the Company's performance on a range of key reliability indicators. Many of the metrics proposed for annual reporting are currently included in compliance filings made by the Company each year. The filings are highly technical documents and were a step forward in communicating system performance; the additional reporting will provide further opportunity for context and understanding of system reliability metrics. As discussed below, an enhanced approach to reliability reporting will provide an opportunity to better communicate with customers and share how the Company is addressing reliability considerations on a more regular cadence.

Existing metrics

DTEE proposes the following, currently reported metrics and information for inclusion in a new PBR Report:

- System Average Interruption Duration Index (SAIDI)
- System Average Interruption Frequency Index (SAIFI)
- Customers Experiencing More than Six Interruptions (CEMI-6)
- Customers Experiencing Long Duration Interruptions (CELID)
- Worst Performing Circuits (SAIDI and SAIFI)

New metrics

DTEE additionally proposes to include metrics that are not reflected in existing reporting. Legacy electric reliability metrics are often focused on system considerations, which is an important indicator for overall performance. The Company is also focused on the customer experience, and leveraging available technology allows reporting to achieve a better focus on that experience. These new metrics each serve to further highlight the customer reliability experience as a critical element of service excellence:

- **Momentary Average Interruption Frequency Index (MAIFI)** – Traditional reliability indicators, such as SAIFI, include only those outages above 5 minutes and do not account for very short duration, or momentary, outages. However, customers experience brief service interruptions during momentary outages and those may have noticeable impact. Thus, this proposal includes MAIFI as one opportunity to further highlight the customer reliability experience.
- **Enhanced granularity of SAIDI and SAIFI reporting** – DTEE's broad service territory is subject to diverse weather impacts, system conditions, voltages and other characteristics. While system-level reliability metrics remain a critical indicator for the Company, it is not always intuitive for customers to connect those figures to their own experience. SAIDI and SAIFI reported at the zip code level begins to facilitate those connections, while also ensuring that the Company continues its equity focus in maintaining and improving the reliability of the system for all customers. This approach is consistent with the principles outlined in Section 2.2 – Energy and Environmental Justice, which in part describes the

Michigan Environmental Justice Screen and Company's efforts to expand the availability and framing of reliability data.

Further defined in Section 7 – Distribution System Overview, of the Distribution Grid Plan, DTEE calculates the metrics as follows:

SAIDI	$(\sum \text{Customer interruption durations}) / \text{Total number of customers served}$
SAIFI	Total number of customer interruptions / Total number of customers served
CEMI - "n"	$(\sum \text{Customer experiencing "n" or more interruptions}) / \text{Total number of customers served}$
CELID - "x"	$(\sum \text{Customer experiencing interruptions longer than "x" hours}) / \text{Total number of customers served}$
MAIFI	Total number of momentary outages / Total number of customers served

Annual reporting will include the intent and methodology for each of the included metrics, and forecasted and actual performance of the metrics, and discussion of what is driving the performance outcomes of the metrics. It may additionally highlight investment themes that could address some portion of the reported output metrics DTEE intends to provide in the report to Commission staff and share it on the DTEE website to make it accessible. The metrics within the report will be refreshed concurrent with subsequent Distribution planning cycles – while targets and forecasts for performance may update annually, the metrics themselves will not.

Incentive reliability metrics

Incentivized reliability metrics are often the first thing included in a discussion of PBR. In other contexts, they may be referred to as performance incentive metrics (PIMs) or targeted performance incentives (TPIs). The purpose of these mechanisms is to create a direct linkage between a Company's performance on a given metric and a corresponding financial reward or penalty. The underlying theory asserts that by associating financial impacts with performance, as opposed to the non-financial impacts of reduced customer satisfaction, utilities will be more proactive in reducing incidents of underperformance. Based on this definition, the Company is

currently subject to incentive metrics through the outage credits provided to customers for long duration outages.

DTEE proposes to design new incentive structures for two reliability metrics: SAIDI excluding major event days (MEDs), and CEMI6.

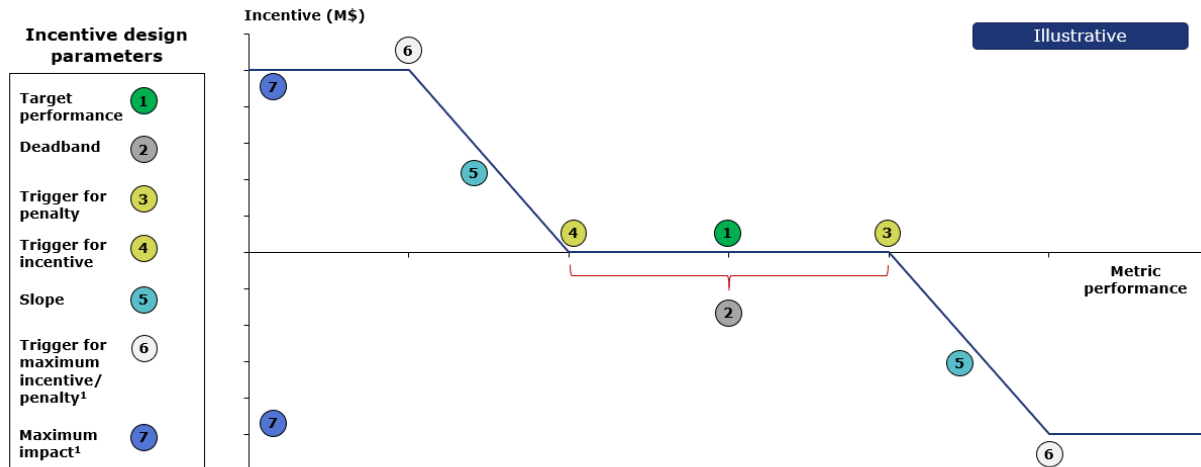
- **SAIDI excluding MEDs** – SAIDI is representative of overall system performance and is a clear entry point for new reliability incentive metrics. There are two primary versions of SAIDI – all weather and excluding MEDs. While investments in distribution infrastructure should, over time, improve the performance of both, DTEE has significantly less control over all weather metrics on a year-to-year basis. The unpredictability of weather, and the associated volatility of all-weather SAIDI, suggests that removing the most severe weather impacts on the metric is prudent and focusing on SAIDI excluding MEDs. This construct best aligns the incentives with performance that is most directly under the Company's control. Major event days are defined in Section 7 – Distribution System Overview.
- **CEMI-6** – CEMI is complimentary to SAIDI and takes a customer-focused view of reliability. The proposal reflects a focus on CEMI-6 because of both the importance of enhancing reliability for those customers and because six outages is a threshold that can be reasonably addressed and managed by the Company. Over time, as the Company is able to reduce the tail of CEMI customers (i.e., the proportion of customers experiencing repetitive outages), it may make sense to tension either the performance of CEMI-6 or the observed metric (e.g., an evolution to CEMI-5).

Incentive design

The design of the incentive structure should accomplish two objectives. First, it should appropriately account for volatility in the metric, understanding that the Company does not have full control over metric outcomes in a given year. And second, both the target and the point at which a penalty is applied should reflect enough tension to drive improved performance while still being achievable. The key elements of the incentive design are described in Exhibit 15.3.2 below, which represents an example of symmetric design. However, as discussed in the descriptions of specific incentive design elements below, the incentive need not be symmetrical. Depending on the specific metric, existing incentive structures, or other factors, the incentive may be designed in an asymmetric fashion. Incentive design proposals specific to CEMI-6 will be designed in

conjunction with a resolution to the Commission's ongoing discussions of outage credit valuation and methodology.

Exhibit 15.3.2 Illustrative Incentive Design



1. The triggers and maximum impacts do not need to be symmetrical. This example is illustrative only

- **Target performance** – This is the intended outcome in a given year for a given metric. The targets for the proposed incentive metrics are defined by the projected system improvements as shown in Section 6.3.
- **Deadband** – Deadbands are used in incentive design as an acknowledgement that neither forecasts nor Company control of metric outcomes are perfect. The deadband allows for performance within a pre-defined range above or below the target and which generates neither a penalty nor a reward. Given their role managing volatility, DTEE proposes that the deadband for SAIDI excluding MED be based on one standard deviation of historical performance, as represented in minutes. This band is initially set at +/- 21 minutes. The historical information used to determine the deadband can be found in Section 7 – Distribution System Overview.
- **Trigger for penalty or incentive** – These are the points at which penalties or incentives begin to accrue, and which are defined by the edges of the deadband. Thus, the trigger for a penalty or incentive to begin accruing is +/- 21 minutes of target performance.

- **Slope** – The slope describes the rate at which a penalty or incentive accrue, up to the maximum impact. DTEE proposes a linear slope, beginning at one standard deviation and concluding at two standard deviations. With this slope, penalties or incentives would accrue between one standard deviation and two standard deviations at a constant rate. Reflected in minutes, the slope is linear between 21 minutes and 42 minutes.
- **Maximum incentive** – This is the total dollars available for an incentive for a given metric.
- **Maximum penalty** – This is the total dollars at risk for a given metric.

Over time, as incentive metrics such as these become more familiar and understood, tension can be added either through changes to the targets, reducing the deadbands or modifying the slopes to achieve the desired structure. The proposed deadbands and maximums are intended to remain in place through the next Distribution Grid Plan, therefore the +/- 21-minute deadband will not be recalculated annually and will be updated in the next Plan filing.

Incentive timing and accounting

The metrics proposed for incentives would be measured on a calendar year basis. It is a defined, well understood boundary for measuring performance, however, it may be misaligned with general rate case timing. Therefore, DTEE proposes to describe any incentives or penalties applied from the year prior in its annual PBR report and reflect them as regulatory assets (for incentives) or regulatory liabilities (for penalties) as of March 1, and carry them until the next opportunity to reflect the impacts in rates. Actual implementation of reliability metrics with financial incentives would follow an Order in a contested case subsequent to this case. Thus, it is expected that the metrics, targets, triggers and maximum amounts, as well as the accounting treatment, would be addressed through a specific proposal by DTEE in a contested case prior to implementation.

Capital tracking mechanism

Performance-based regulation, as noted in the introduction, is intended to more closely link operational performance with financial performance. There are diverse drivers of operational performance discussed in the preceding sections of the distribution grid plan. The investments reflected in this plan support continued achievement of reliability targets and system

improvements. However, it also reflects a point-in-time view of system conditions, load growth and locational impacts that might be forecasted over the planning period.

In the coming years, the ability for DTEE to deploy capital quickly and flexibly may become necessary as the adoption of distributed energy resources (DERs), such as electric vehicles and rooftop solar, become more widespread. The potentially substantial changes in how and when customers use energy could drive large changes in usage patterns and peak demands. This may require flexible and responsive investments in locations with high activity. For example, if electric vehicle uptake occurs more quickly than expected and is not uniformly distributed, it could create circuit-level load in excess of equipment capacities. The available interventions would be to improve the infrastructure to support and promote customer accessibility or manage load and potentially risk curtailing customer activity.

A capital tracking mechanism is one option for investment flexibility to be responsive to quickly evolving grid usage. DTEE is not proposing the near-term implementation of this mechanism, but the Company does highlight the role it could have going forward in a comprehensive PBR framework. A tracker would provide specific benefits to customers:

- Improved reliability and grid performance through targeted investments on circuits
- Ensure no reduction in overall grid performance while supporting customer access to new technologies
- Support investments that can be measured and reporting in a systematic and transparent way

A future capital tracking mechanism could be designed as a two-way tracker with an upper limit of spend. It would have a target spending level for urgent and targeted circuit improvements, a reconciliation for underspend (a refund to customers) or incremental spend (additional recovery) and a limit on allowable incremental spend. There would be no minimum spending requirement. DTEE may propose a similar mechanism to the one described here at such time it becomes prudent for the operational integrity of the distribution system and the ability to seamlessly integrate customer adoption of DERs.

Alternative regulatory mechanisms

Beyond traditional COSR and a reliability focused PBR plan are alternative regulatory mechanisms which, in tandem with COSR and other metrics, can drive additional focus on: 1) policy goals, 2) customer engagement and accessibility, 3) cost management or other yet to be defined goals. They reflect a key component of a broad-based and holistic approach to performance regulation. The term “alternative regulatory mechanism” does not refer to a single tool, but a broad range of options available to achieve preferred outcomes. This section highlights three such mechanisms that could be utilized to generate specific outcomes in specific contexts, but highlights that the list here is not exhaustive. DTEE is not proposing a specific mechanism or approach in this document and is including the overview here to generate discussion and engagement with the many stakeholders in Michigan.

Exhibit 15.3.3 Alternative Regulatory Mechanisms

Mechanism	Description
Incentives for exceeding policy driven targets	Policy-related incentives provide additional earnings opportunities for achieving or exceeding specific adoption targets, thresholds, or other measures of performance. These incentives are structured as “one-way,” without penalty risk
Rate-basing of assets or payments to customer	Provides an opportunity to capitalize contracts with customers for certain grid services and which may cost effectively displace capital investment
Earnings mechanism for targeted O&M spend	Provides an opportunity to earn on O&M spend that may displace capital investments. Any resulting cost savings could reduce upward rate pressure

Future areas of opportunity

This proposal is oriented toward reliability metrics and the distribution system based on the venue of the filing and the robust discussions in Michigan around reliability and distribution. Improvements in the electric distribution system are also essential to supporting a more dynamic, and clean electric system with increased customer engagement. Going forward, there may be additional areas that support an exploration of PBR structures and opportunities. These could include customer engagement, long-term planning, environmental considerations, or expanded areas of distribution reliability. The Company looks forward to these discussions at the appropriate time and in the appropriate venue.

16 Enablers



16.1 Distribution design standards

DTEE develops and maintains a comprehensive set of distribution design standards (DDS) that promote uniform construction practices and provide the foundation to design, build and maintain a safe, resilient electric infrastructure. These standards comply with National Electric Safety Code (NESC) and have evolved to incorporate industry best practices gained from DTEE's involvement in the Edison Electric Institute (EEI), Electric Power Research Institute (EPRI), Institute of Electrical and Electronics Engineers (IEEE), National Electric Energy Testing, Research & Applications Center (NEETRAC) and benchmarking with peer utilities.

DDS is the "single source of knowledge," that consist of detailed specifications which contribute to the resiliency of the electric infrastructure.

Additionally, the DDS consist of specific distribution design orders (DDOs), which contain detail design orders to guide DTEE engineers when planning and designing the electric system. These orders allow for maintaining system design consistency throughout DTEE's service territory. Furthermore, the DDOs specify the equipment to be used and contain methods and procedures which determine how the distribution system is designed and modified. There are also Job Aids to supplement the DDOs. The job aids contain step-by-step instructions and any additional information useful for the planners or engineers.

At times, the planning engineer may encounter extenuating circumstances where implementation of the DDS or DDOs may be impossible, impractical or result in excessive cost. In these cases, the planning engineer may develop an alternate approach. This exception to the DDS or DDOs must be presented to a technical peer group for discussion and review. The peer group then

makes the decision to accept or deny the exception prior implementation. Exceptions are tracked as a basis for potential future modifications to the DDS and DDOs.

To ensure DDS and DDOs are compliant and up to date with code requirements, DTEE established the Five-year Cycle Review process in 2018. The Five-year Cycle Review process calls for the DDS and DDOs to be reviewed and updated every five years as NESC (National Electric Safety Code) requirements are updated. Additionally, DDS and DDOs are updated based on new equipment or technologies, as dictated by developments in the industry, or based on asset and system performance/field experience. Updating DDS and DDOs is a cross-functional effort and collaboration with Engineering and other DO groups is essential in the development and successful implementation of these standards. Over the years, the Company has enhanced the minimum standards for construction to further harden the system and better sustain impacts from inclement weather and tree interference. These improved design standards, grade B construction, are based on NESC (National Electric Safety Code) factors and industry best practices.

The structural upgrades include using armless construction or fiberglass crossarms, larger conductors, polymer insulators that are four times stronger than pin insulators, and standard poles that are one and a half times stronger, as illustrated in Exhibit 16.1.1. These structural upgrades are intended to standardize overhead construction materials and specifications to significantly streamline the design and construction of overhead facilities, and to increase the structural strength to improve system reliability and storm resiliency.

In addition to hardening, DTEE continues to develop standards that will support our vision of grid modernization. DTEE is developing 69kV standards for the subtransmission system in replacement of 40kV standards. Although the system will remain energized at 40kV in the near term, 69kV standards development allows for the rebuilding of the existing 40kV system to 69kV standards. This new standard construction will enable the grid to have the capability to be energized at 69kV, if that level of capacity is needed to accommodate growth in the future.

The implementation of wood pole alternatives is another example of DTEE's effort to further harden the electric infrastructure and commit to system resiliency. Wood pole alternatives are a good option when evaluating ways to strengthen parts of the distribution and subtransmission system and make it more tolerant to catastrophic events. Wood poles can lose strength overtime due to deterioration of organic material. Wood pole alternatives, such as steel, concrete and

composite poles, are engineered to retain their strength overtime if installed and manufactured properly. DTEE continues to evaluate these poles and evaluate their use case where it makes sense. As one example, steel poles have been standardized for all truck accessible locations for subtransmission rebuilds.

Furthermore, updated circuit designs improve circuit configuration and allow crews to more quickly restore customers during outage events. Examples of updated design standards include ensuring valid jumpering points on circuits with properly sized operating equipment, installing fault indicators to reduce outage patrol time and installing loop schemes for automatic restoration of sections of circuits during an outage, as illustrated in Exhibit 16.1.2.

The topics covered by the DDS and DDO are summarized in Appendix VI.

Exhibit 16.1.1 DDO Structural Upgrades to Distribution Circuits

(stronger poles, stronger conductors, and polymer insulators)

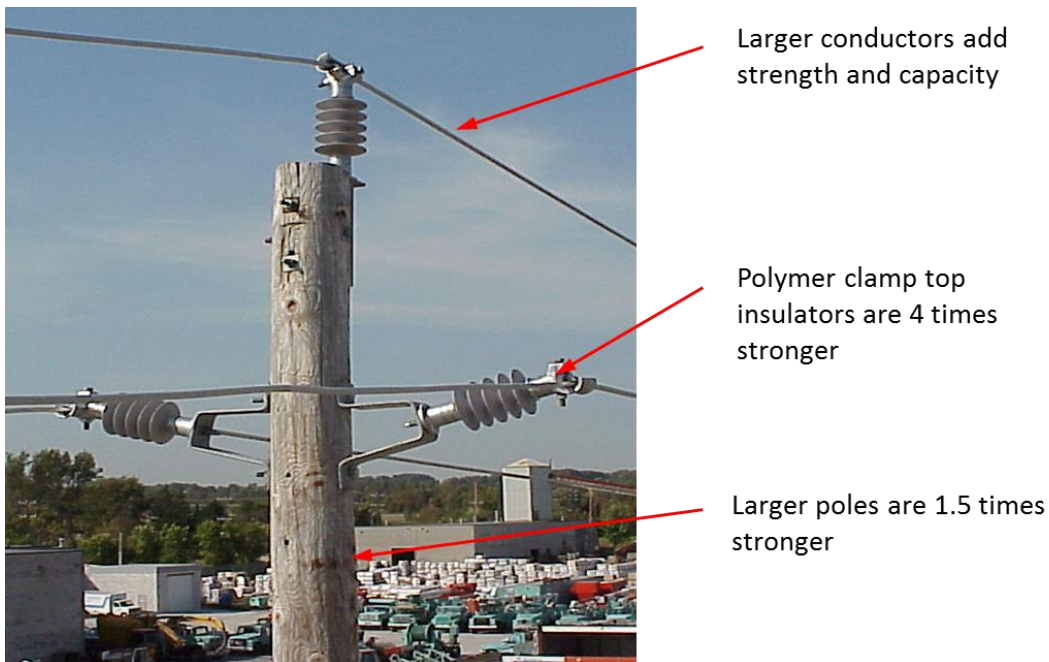
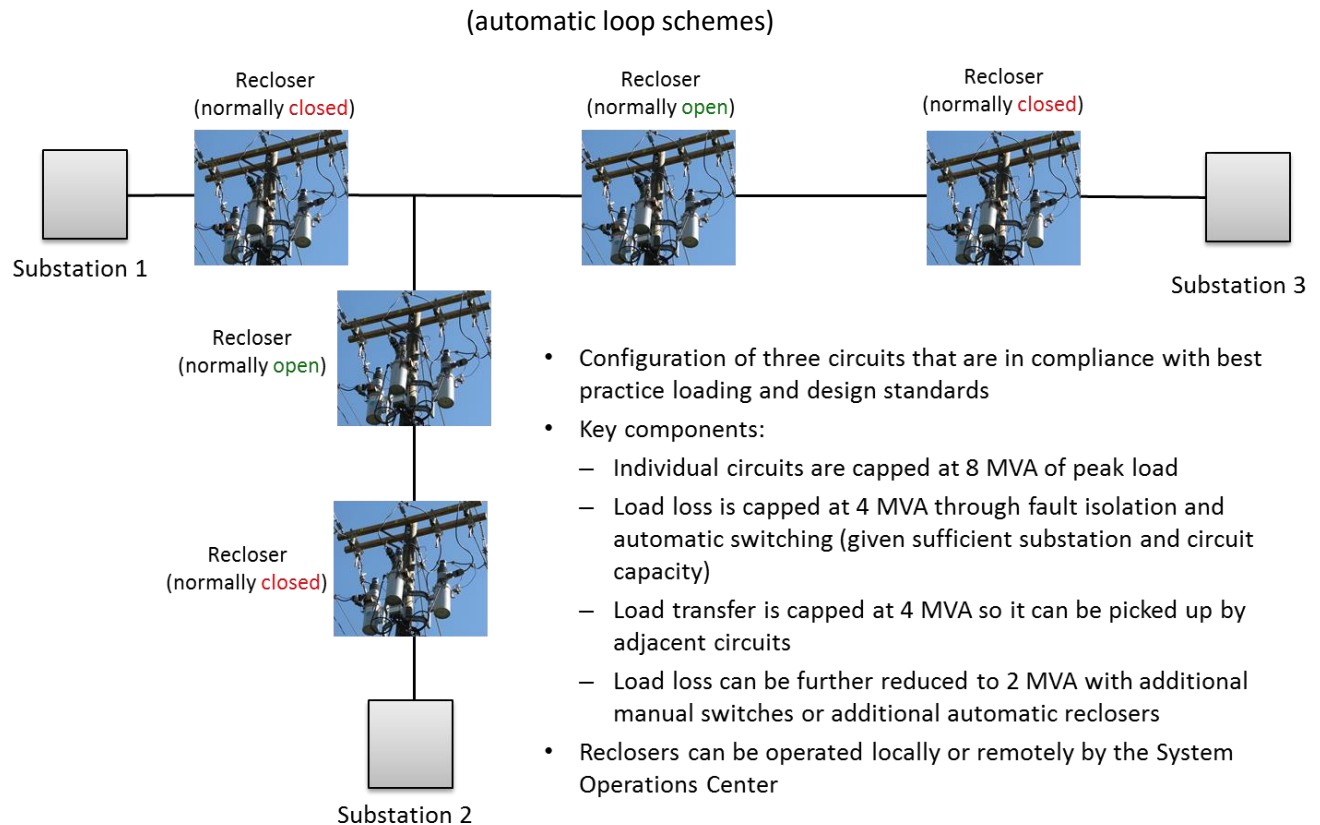


Exhibit 16.1.2 13.2 kV DDO Restore-before-repair Design



The use of the DDS/DDO framework is vital to ensuring that as work is designed and complete in the field it is consistent with the long-term plans for the distribution system. Examples of this include the use of standard conductor sizes for primary and secondary conductors, circuit load limits, and automation standards to new investments are consistent with grid moderation efforts to allow for increasing electrification, adoption of DERs, automatic restoration, and better system visibility. Standards enforcing the use of 15kV equipment and grade B construction standards also support long term effort to reduce the cost of future conversion from 4.8kV to 15kV and other traditional infrastructure improvements. The DDS/DDO framework is useful in both helping to understand how to integrate future technologies such as distributed resource interconnection, electrification, and automation within the existing system as well being able to understand what standards changes might help better facilitate future developments.

16.2 Workforce and resource planning

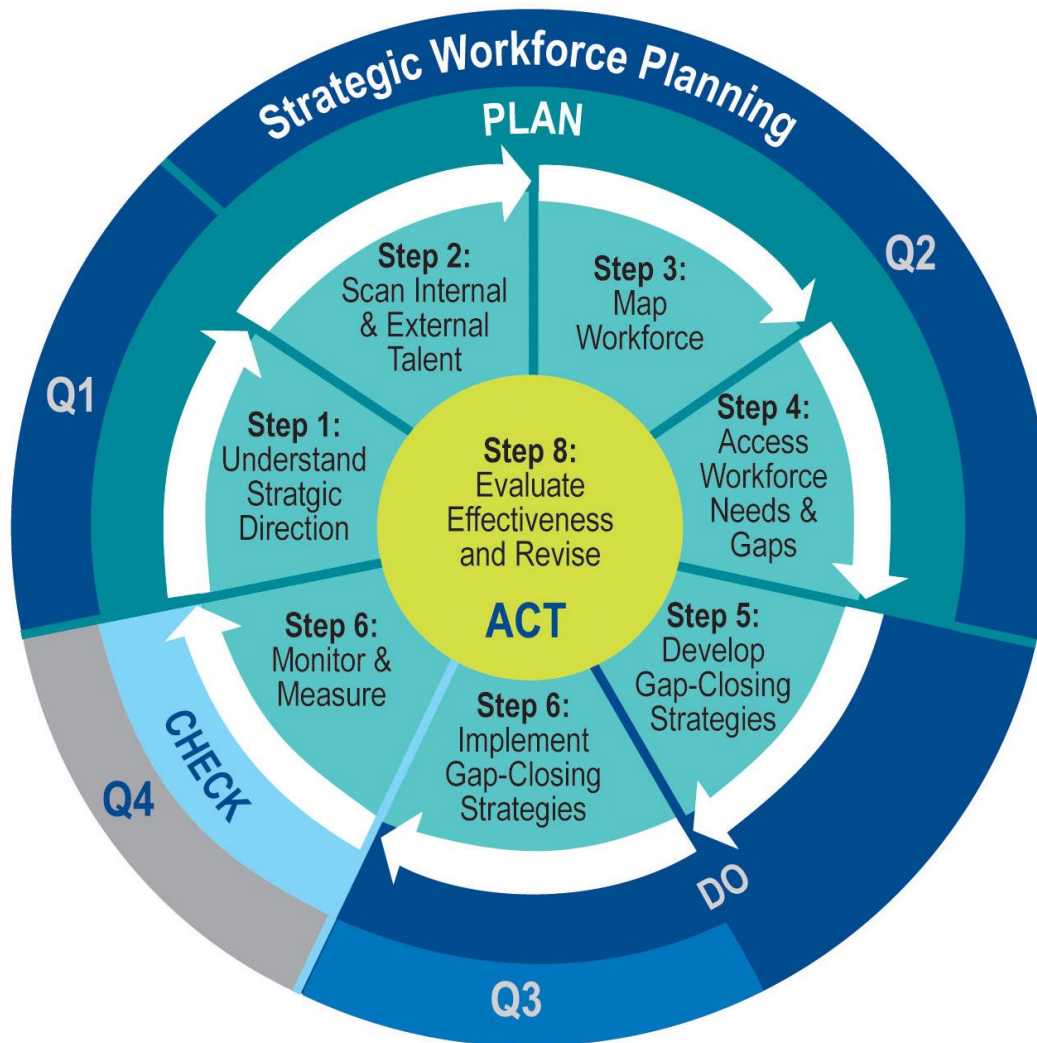
Workforce and resource planning is a key factor for the successful implementation of the investment plan and grid modernization strategy outlined in this filing.

Workforce planning process

As DTEE is planning for the increase in investment that will rebuild and modernize the distribution grid, an essential element of success will be to plan for the workforce of the future. The Company is at the beginning of a process to determine the needed workforce, the specific skill sets, and the timing for successful implementation of the five-year investment plan as well as the long-term vision.

The overall objective for DTEE is to build a detailed workforce model that will support operational decisions to successfully identify workforce needs based on planned work, criticality of role, and skillset needed to effectively execute the plan. The workforce needed for planned work must be coordinated with expected emergent work that results for storm restoration, day-to-day outage and non-outage events, customer requests for connections, relocation requests and others. Exhibit 16.2.1 illustrates the Company's overall workforce planning process, which was also described in the Distribution Operations Five-year (2018 to 2022) Investment and Maintenance Plan Final Report.

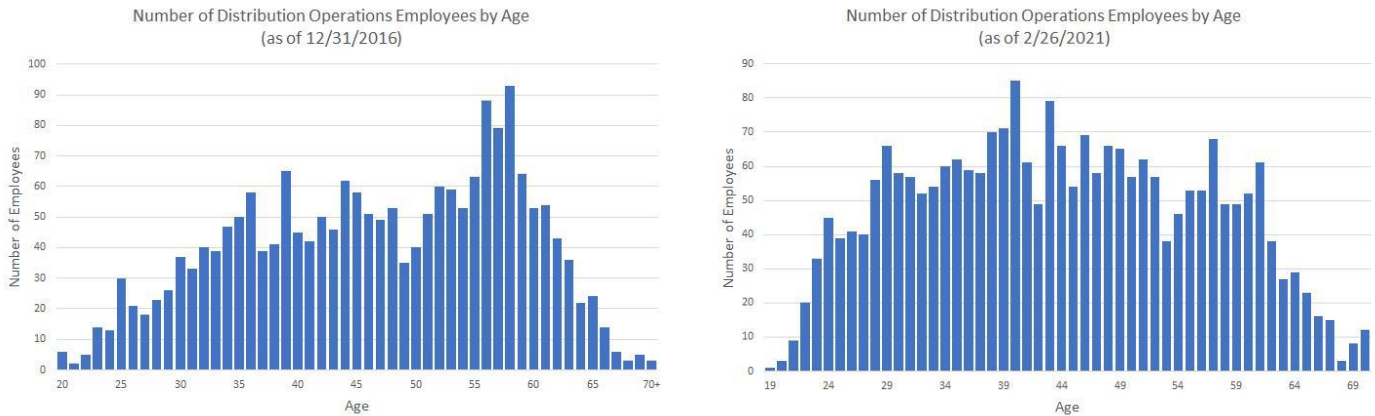
Exhibit 16.2.1: Workforce Planning Process



Realized improvements

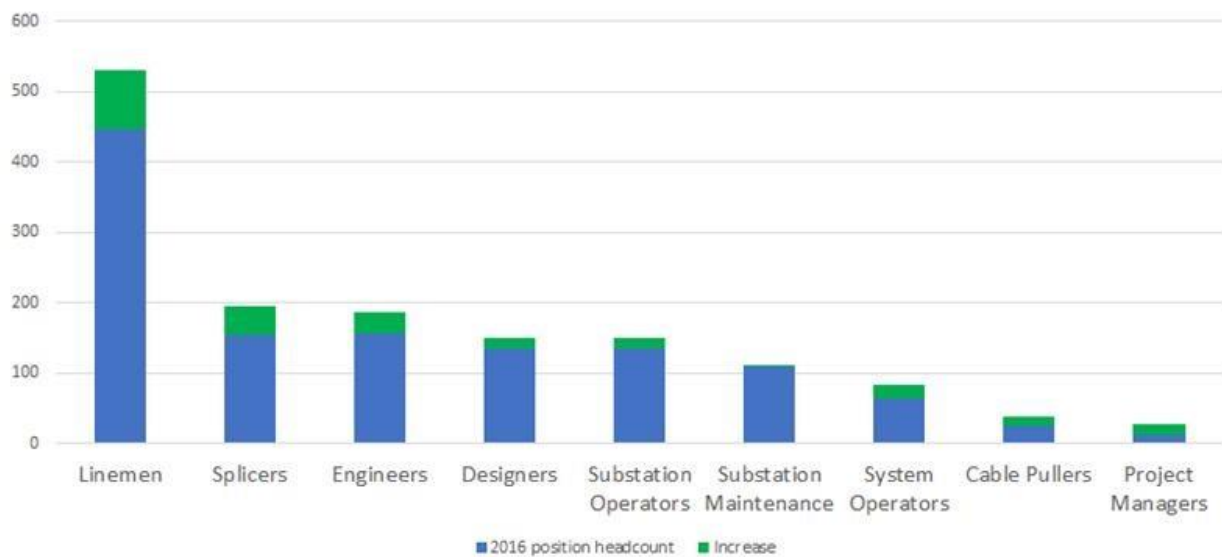
The Company has leveraged this process over the last few years to address several workforce related risks to the execution of its plans. One of the most concerning aspects in 2018 was expected retirements. Exhibit 16.2.2 shows that the established workforce process has reduced this risk by shifting the workforce makeup to be less skewed toward near-term retirements.

Exhibit 16.2.2: Reduced Retirement Risk



Another area of improvement has been adding workforce to key job classifications that are critical to execute the plans described. Exhibit 16.2.3 shows the increases to key positions.

Exhibit 16.2.3: Additions to Key Positions to Implement the DGP



Planned improvement

Building upon these improvements, the Company is continuing to enhance its processes and tools for workforce planning. Further integration of processes that consider different positions is one aspect of continuous improvement. Since the plan involves several different work groups, there will be benefits from a more defined, coordinated process and model. DTEE is leveraging benchmarking data and best practices to develop a capital organization structure that will integrate with DTEE's internal systems to provide full costs, labor and unit information inputs to advance financial modeling capabilities. The results of the integrated planning workstreams will increase collaboration within the decision-making process to recommend the best approach for meeting resource needs. Another component of workforce planning will include strategies to retire our power plants with pride, which provides opportunities for employees to transition to other roles at the Company. A transition team has conducted initial focus groups to report findings and recommendations to identify needs to support our future workforce. DTEE will focus on providing tools, communication and aligning skillset with career or training opportunities to plan for this aspect of workforce transformation.

An additional aspect of workforce planning improvement is the development of a skilled resource pool. Section 10.7 describes what the Company has done to develop a local tree trimming workforce pool. In addition to that, the Company has increased the number of linemen apprentices and cable splicer apprentices. DTEE has developed relationship with local universities to expand the number of engineers and data scientists with an interest in power systems work. DTEE is also developing role-based training to provide engineers and data scientists with the skills necessary to be successful.

Increasing workforce demands

Despite the robust workforce process and the improvements it is bringing to the Company's personnel makeup, the level of work described in this document will present a challenge for the Company to address. Ramping up investments in overhead line work and system cable replacement, for example, will increase the demand for highly skilled positions. The training timeframe to develop these resources tests the ability to ramp-up quickly. DTEE maintains partnerships with the unions to develop this workforce, which can be seen by the increase in apprentices in these areas over the last few years (65% increase in linemen apprentices and 48% in splicer apprentices since 2016). Other areas, such as engineering, design, construction and

project management, will also present a challenge for the Company, and the Company is increasing hiring in these areas as well.

In addition to the hiring and training of direct employees, the Company has developed partnerships with engineering, design, project management and construction firms that have successfully helped other utilities build their capacity. For example, the cable replacement programs described in this document will require significant increases available staff. The Company has conducted benchmarking with other utilities that have been involved in significant cable replacement efforts to learn best practices. An electric utility in a very large Midwest city averaged over 700,000 feet of system cable replacement per year between 2013 and 2017 and peaked at nearly 1,000,000 feet in a single year. DTEE is partnering with the same service providers that helped that utility build its capacity.

Material & equipment resource planning

The increased investment to rebuild and modernize the distribution grid will drive the need to consider and plan procurement of additional levels of materials and equipment and understand and mitigate associated risks.

Procurement processes

DTEE Supply Chain proactively anticipates and mitigates both cost and material availability issues through the following processes:

- Forecasting material and equipment demand and securing supply or shop time early;
- Holding weekly meetings with our top suppliers (including material integrators) to stay ahead of any market fluctuations;
- Monitoring the market and increasing stock inventories when market tightening is observed;
- Building long-term relationships with our key suppliers to get preferred customer status;
- Putting our eyes and ears on key suppliers and sub-suppliers via site visits to validate availability of critical materials; and
- Evaluating current contract structures to quantify market inflation.

DTEE Supply Chain monitors material requirements for critical inventory continuously and adjusts for changes in market conditions.

Procurement risks

DTEE Supply Chain works with its integrator to continuously monitor and minimize risk of both key materials as well as long lead equipment. Our close collaboration with our integrator helps us minimize our risk for key materials. We monitor risk throughout the supply chain including raw materials, manufacturer capacity, and distribution supplies. Mitigation strategies can include some of the following examples:

- Buying additional inventory where possible
- Sourcing additional capacity at current/alternative suppliers
- Securing production space in advance
- Securing contracts on future raw material commitments

DTEE Supply Chain also works with our integrator to survey the industry and identify trends in pricing and lead times for key items such as transformers, switchgear, poles, underground cable and meters. We leverage our supplier relationships to mitigate price and lead time uncertainty. Below is an example of some of the risk exposure and mitigation plans put in place.

Workforce & resource planning conclusion

With the capital investments outlined in this distribution grid plan, the workforce and resource planning processes will optimize constrained resources to successfully execute the plan. The process that the Company has established is already showing a positive impact on the workforce and the Company is continuing to improve its workforce plans. DTEE is confident that the process can provide the needed workforce to implement that plan, and the Company is already reviewing current headcount, determining resource gaps, assessing the criticality of positions and preparing estimates of future resource needs for the plan described in this document. DTEE Supply Chain has developed processes to forecast and monitor the increased material and equipment inventory. The Company is already working with integrators and key suppliers to identify risks of inflation and lead times and develop risk mitigation plans where necessary to ensure availability.

16.3 Cybersecurity

The importance of maintaining cybersecurity in electric utilities has long been recognized as reflected in the industry's NERC-CIP (National Electric Reliability Corporation – Critical Infrastructure Protection) standards. The increasing intelligence and interconnected nature of the electric distribution system will continue to add to the importance of cybersecurity. Cyber-attacks targeting the electric grid are becoming more numerous and aggressive. The increasing threat, paired with the increasing deployment of interconnected intelligent devices, is leading to an increased focus on securing distribution assets. At DTEE, the focus of the next five years will be to ensure that as distribution assets increasingly become cyber assets the necessary policies, standards, and protections are in place to ensure the safe, reliable operation of the electrical distribution grid.

Securing the distribution system from cyber-attacks requires a defense in depth approach to protect against internal and external and as well as amateur and sophisticated attackers. This requires enhancing the cyber security and physical security capabilities. Adding these defenses requires updating existing design standards to ensure that newly constructed sites have adequate protection as well as identifying and addressing latent vulnerabilities in the existing system. This defense will continue to evolve as the threats change and the industry progresses and learns from new events.

Increasing physical defense of cyber assets includes improving fencing, layered protection of network connected devices with multiple locked barriers, and increased access restrictions on exposed devices. Limiting electronic access includes allowing only the necessary network traffic, limiting group or individual access through password protection on intelligent devices, restricting electronic and physical access from unapproved computing devices (such as laptops) and portable media, standardized device hardening processes to ensure consistent protection across the system increasing the ability to detect and respond to cyber-attacks, performing anti-virus scanning, sound patching and firmware management procedures, scanning of network connected devices for vulnerabilities and detecting unauthorized accesses or changes to configuration and settings files. The steps are being included in the updates of engineering and design standards, and will be piloted in 2021 and 2022 to facilitate potential future retrofitting options for 2023 and beyond. The initial focus will be on developing the process and infrastructure to support these

programs in the future and the preliminary estimate will require \$1-2M to initiate the program and will expand based on pilot results.

The initial efforts will focus on securing sites that DTEE considers high impact beyond those that meet the more restrictive NERC high impact definitions. These will include sub transmission stations and other high capacity, or critical load facilities. In addition, priority will be given to distribution substations that have significant connectivity and intelligent electronic devices already deployed. Future efforts will expand to distribution system devices including overhead and underground components with SCADA access.

Additional concerns will be reviewed and assessed over the next five years to address the growing importance of distributed energy resources and methods will be developed to ensure the cyber protection between new devices, the electric grid and other necessary parties.

16.4 Physical security

The DTEE Distribution Site Security program is a multi-year effort designed to reduce the risk associated with intentional and unintentional security events that could affect DTEE Distribution infrastructure, affecting the Company's ability to distribute energy to its customers. Substations have been targeted internationally and within the US, requiring improved security measures. The Site Security program has two components that are tightly integrated 1) Implementation of baseline security and 2) Security Risk Assessment (SRA) process which focuses on site specific vulnerabilities and threats.

Baseline security standards

An enterprise-wide Security Assessment model is used to prioritize sites. It uses multiple attributes to rank distribution sites as either high, moderate or low criticality to our operations. The assessment also determines the application of baseline security standards (Standards) commensurate with the ranking.

DTE security teams and distribution personnel conducted initial site walk-downs in 2018/19 to identify security gaps. Security measures are currently being implemented to address the gaps identified and to bring the security levels at each site in line with the standards. Additionally, a sustainability program is being developed to maintain these security measures.

SRA

Critical/high risk facilities are further assessed for site-specific risk through the SRA process which involves an in-depth analysis of specific attack paths, likely targets, potential threat actors (design basis threats), and impacts of successful attacks. Security countermeasures are recommended to reduce the likelihood of a successful attack taking place and implementation is prioritized by risk. SRAs will be conducted on a periodic basis, not to exceed five years, within 12 months after completion of a significant enhancement or modification to the facility, within 12 months of a newly identified critical facility, or newly constructed facility identified as critical.

Scope:

The scope of security work that is in progress includes: installation of new fence, upgrade of existing fences, automation and/or other upgrades of main gates, brushing/removal of trees along fence lines, lighting upgrades, installation of cameras for intrusion detection and alarming, critical area access control through badging system installation and new doors with automatic locking systems. Additional target hardening measures are applied that are site specific as identified during the SRA. DTE has also begun assessing cyber security gaps at these sites and the measures required to address these gaps. This can be reviewed in Section 16.3 – Cybersecurity, of the Distribution Grid Plan.

Future state solutions under consideration include the implementation of a preventative maintenance program for security equipment to ensure longevity and reliability, and the application of controls as needed to further protect distribution infrastructure in accordance with the Company's Insider Risk program. The Insider Risk program is designed with the intent of deterring employees from becoming insider threats, detecting those who pose a risk, and mitigating the risks through response protocols.

17 Key next steps



Throughout this Distribution Grid Plan, DTEE has outlined a comprehensive five-year plan and 15-year vision which: 1) assesses the current state of the electric system, 2) defines the desired future state of the electric system, and 3) describes key initiatives to move from the current to future state. Executed successfully, this strategy will deliver on key objectives of a safe, reliable, resilient, affordable, accessible, and clean power system for the customers and communities that DTEE serves. Listed below are some initiatives and steps that we will take in the next couple of years to continue to prepare for the grid of the future:

Forecasting

DTEE's integrated forecasting solution is a project made up of experts from Corporate Forecasting and Distribution Operations, Corporate Strategy and other support teams from across the organization. As a first step, the project team produced an integrated forecast modeling framework that can be leveraged for both generation resource and distribution planning.

This forecast development process has three distinct elements, with the first step being completed in Q2 2021. The initial project step involved finalizing the organization-wide baseline forecast. While many of the data inputs are the same, modeling outputs for distribution planning requires more granularized forecasts relative to generation resource planning. For instance, when considering Electric Vehicles (EVs) in the respective planning processes, resource planners can make effective use of a model detailing the total number of EVs and corresponding load on the system. However, for distribution planners it is crucial to also estimate the charging patterns and location of EVs on substations and circuits across the system.

Next, the team is currently focused on developing the work process for creating substation and circuit-level 8760 forecasts. Modeling of the substation and circuits will require the use of AMI

data, modeling weather response functions at the circuit and substation level and incorporating changes due to the adoption of new technologies (EV, PV, Battery, EWR, etc.) A key element of this work will involve creating forecasted adoption rates by technology at the zip code/circuit level and defining a formal process for the development and prioritization of sensitivity analysis. This work is currently ongoing with a distribution forecast Proof of Concept model for Walnut and Sunset substations to be completed by Q4 2021.

In the final phase of the project, the Proof-of-Concept work will inform next steps in which the output of the circuit-level forecasts will be reconciled to the overall system-level forecast. DTEE expects to refine the modeling output over multiple iterations before publishing the forecasts. One of the primary reasons for this is the availability of AMI/SCADA data across all circuits and substations. While modeling is an effective tool in estimation, validation with actual data will take some time.

Upon completion of the model validation work, the project will integrate 8760 hours per year modeling into distribution planning processes. Currently, the distribution planning process is largely based on peak load planning. The availability of 8760 forecast will allow DTEE to plan the distribution system by optimizing the distributed generation, storage and loads. Moving to hourly forecast instead of annual peak will be a fundamental shift in distribution planning and will require changes to most of the planning criteria and processes.

Develop accelerated conversion plan

The scenarios developed through the grid modernization process, specifically the electrification and distributed generation scenarios, identified the potential need to convert the 4.8kV system to a higher voltage at an accelerated pace. One of the next steps is to develop the tools and processes necessary to create and execute a plan for conversion beyond the projects identified in the DGP. The development of these tools and processes began in Q2 2021 and is expected to be fully integrated into distribution planning by Q4 2023. The project will be completed in two phases: the first phase will be complete by the end of 2021 and will define the parameters for developing a long-term conversion plan. During this phase, the project team will design and run several cross-functional workshops with key internal subject matter experts to outline key design elements that will impact the total cost of the conversions, including assumptions on customers per transformer, sectionalizing, looping circuits and voltages. Additionally, this first phase will include the development of the conversion tool. The tool is expected to outline the total estimated

cost, resources and timeline of undertaking a conversion (circuit and substation) of the entire 4.8kV system.

The second phase of the project is still being refined in terms of key activities and timing. However, it is expected that this phase will result in the prioritization of circuits/substations for conversion in the five- to 15-year timeframe, more detailed estimates of costs for the specific projects that will be pursued, and benchmarking peer utilities' most recent conversion efforts. It should also be noted that developing a full conversion plan for the 4.8kV system will be an iterative process that will need to be updated periodically, especially as the signposts identified during the scenario planning processes materialize. For example, as the pace and locational impacts of the adoption of electrification and distributed generation resources become more certain, the areas of the grid where the 4.8kV system can no longer meet customer needs may become more apparent and will need to be prioritized accordingly.

Higher distribution voltage

In the near future, the Company will continue to evaluate the introduction of a higher distribution voltage onto its system. Internal discussions have begun on the potential to develop a new 34.5kV voltage class along with continued research and benchmarking. By Q2 2022, an implementation plan will be created by a cross functional team of stakeholders within the Company. Next, through data analysis and stakeholder input an ideal candidate profile for a pilot will be defined. Pilot candidates will be narrowed down by Q4 2022 through internal stakeholder interactions and by cross-referencing investment plan to maintain capital efficiency. With a pilot candidate identified, engineering will begin for all components required for implementation. Engineering will include but is not limited to geographic location selection and site due diligence, operating procedures, circuit layouts, construction standard development, protection scheme development, and equipment selection and testing. This engineering phase of the project will last until Q1 2025. After the engineering phase completes, design and construction will begin sometime in 2025.

Secondary Loading and Electric Vehicle Monitoring

While electric vehicles will increase loading on all levels of the system, the effects of their adoption will first materialize on DTEE's customer facing assets such as URD (Underground Residential Distribution) loops and service transformers. To confidently integrate the grid impacts of EV (Electric Vehicles) growth, DTEE will require advanced analytics tools to increase its situational

awareness of customer facing, “grid edge” distribution assets. To help achieve this DTEE is currently in the beginning phases of developing an AMI (Advanced Metering Infrastructure) based tool to assess loading on its service transformers. By the end of 2021, DTEE plans to have refined the project plan, scope, data needed, and business unit collaboration required for developing the AMI tool. From the beginning of 2022 until the end of Q1 2023, DTEE will develop the planning tool and ensure that it can determine the magnitude and location of existing transformer overloads and current EV locations. In the beginning of 2023, DTEE will verify that the tool works as intended and integrate the tool within the Distribution Operations (DO) organization. While this process is wrapping up in Q3 2023, the Company will also begin expanding the tool to assess the impacts of EV loading on more grid edge assets such as secondary and URD loops, as well as future overloads from increasing EV adoption.

In tandem with development of the secondary loading tool, the Company also plans to thoroughly assess what standards changes are required to accommodate the increasing penetration of electric vehicles within DTEE’s service territory. From the beginning of Q3 2022 until the end of Q3 2023, DTEE will determine current and probable DDO (Distribution Design Orders)/standards violations resulting from EV growth, and in turn develop and distribute the upgraded and new standards within the DO organization.

Additional efforts

In addition to developing new planning tools and long-term investment plans, DTEE will expand the current investment prioritization framework (GPM), to ensure that all planning objectives and customer needs are included in the evaluation and prioritization of investments. As discussed in Section 5 on benefit cost analysis, two of DTEE’s planning objectives, Clean and Customer Accessibility and Community Focus, do not have specific dimensions aligned to them in the current Global Prioritization Model. By the end of 2022, DTEE anticipates updating GPM to fully align with the five planning objectives. Part of this expansion of GPM dimensions will also incorporate a focus on equity and improving reliability for all customers. Changes to the GPM will include robust stakeholder discussion, starting with the planned MPSC workgroups on BCA in Q4 2021, to ensure transparency in how these additional factors are incorporated into the prioritization of the investment portfolio.

Advanced analytic capabilities derived from grid modernization efforts will support the Company’s PBR plan and goals which are aligned with the Distribution Grid Plan and the Commission’s

Distribution planning goals. DTEE will continue to work with stakeholders and the Commission to further enhance the PBR plan through formal and informal engagement. The Company will continue to enhance transparency by reporting and communicating performance annually through a new PBR report. DTEE's first PBR report will be filed by July 1, 2022 and will include the intent and methodology for each identified metric including the drivers behind the performance outcome. DTEE's proposed report will expand available reliability information and ensure that the Company continues its equity focus in maintaining and improving the reliability of the system for all customers. DTEE will continue to explore the role of capital mechanisms and alternative regulatory approaches to drive policy outcomes, cost-beneficial grid solutions, and expanded customer engagement and accessibility.

Successful development and execution of the grid modernization roadmap is reliant upon continued efforts around staffing of current and future positions within DTEE. While much progress has been made in this space, enhanced modeling of workforce needs will assist in identifying resource gaps to allow DTEE to proactively attract and retain a diverse and highly skilled workforce.

In summary

With changes in climate, customer needs and emerging technology, the electric grid of the future is a constantly evolving target. While this presents challenges, the strategies outlined in the DGP are designed to navigate the uncertainty with a combination of flexible and no-regrets investments in modernizing the electric grid. Just as prior efforts and current initiatives have laid the groundwork to redesign DTEE's electric system, the Company is confident that the grid modernization roadmap will establish the path to deliver an efficient electrical system to serve the evolving needs of customers well into the future.

18 Conclusion

DTE Electric is committed to providing the Michigan families and businesses we serve with energy that is safe, reliable, clean, affordable and accessible. This report has outlined the process we used to plan for grid modernization, the investments our Company will make over the next five years to improve and modernize our electric system to ensure we deliver the service and value they expect and deserve, as well as our vision for the future of the grid.

We would like to thank the Michigan Public Service commissioners and staff for their leadership and guidance in this effort. We also would like to thank the residential and business customers who participated in the focus groups that were an important part of our planning process (Section 2 – Stakeholder Engagement and input), our technical stakeholders who joined us in webinars and individual discussions, and our state, local and community partners who shared feedback through one-on-one conversations. Their input helped shape this plan, as well as our vision for the future, which is referenced throughout this document.

The electric industry is in the midst of fundamental change – in the way we generate and deliver energy, and the way our 2.2 million customers use that energy. Our customers are more dependent than ever on our service, not just to keep the lights on, but also to power an ever-growing array of technologies and services that are now essential to their lives and businesses. Demand for our services will continue to grow, driven by emerging innovations like electric vehicles, and customer interest in advances like private generation, storage and residential electrification.

This, and other factors are driving the need for continued, significant investment in DTEE's electric system. While demands for the electricity we provide are surging, portions of our grid – some more than 90 years old – are beyond life expectancy and are in dire need of replacement. Climate change and increasingly severe weather are presenting serious challenges to grid reliability, and the summer storms of 2021 are a good example of that. In addition, state of Michigan sustainability goals, DTEE's own sustainability goals and commitment to net-zero carbon emissions by 2050, and customer demand for cleaner energy have led our Company to increase investments in renewable energy and energy efficiency.

These are challenges – and opportunities. And in response, DTEE has accelerated investment in grid hardening and system upgrades. Since submitting our Distribution Operations Five-Year

Investment and Maintenance Plan to the MPSC in 2018, we've invested nearly \$1 billion per year to upgrade aging infrastructure, enhance our tree-trimming program, and install leading-edge equipment to improve safety, and better detect, prevent and manage outages. We've built new substations and updated others to expand capacity and improve power quality and reliability. We also broke ground on a new state-of-the-art Electric System Operations Center that will significantly improve our ability to manage system disruptions and response time.

We know these improvements are working. In areas where our grid hardening work has been completed, reliability has improved by 50-70%. Projects that have made our infrastructure more resilient have reduced time and money spent on outages caused by equipment failure and storms. With continued investment and infrastructure upgrades, we'll be able to improve reliability for our customers by 60%, helping to bring DTEE's electric reliability to best-in-class standards.

While we're proud of the progress we have made, we know there is more we can and must do. We want all of our customers to have this level of service, and we believe our new Distribution Grid Plan will bring us closer to that goal.

Guided by the MPSC's goals and our own planning objectives, DTEE's plan calls for increasing spending on grid improvements from \$1 billion to \$1.5 billion a year by 2025, focused on four key areas: infrastructure resilience and hardening, infrastructure redesign, technology and automation, and tree trimming. The new plan builds on the strength of our 2018 strategy, but also takes into account plausible, future scenarios that DTEE developed with an outside consultant.

Clearly, there are challenges we have to overcome, but continued investment will help us meet our goals while meeting the demands of our customers. It has to be done thoughtfully, efficiently and over time to keep the energy we deliver affordable for our customers, communities, and especially those who are most vulnerable. It needs to address environmental concerns and climate change in a meaningful way. And, we agree with our customers who told us our investments must ensure fairness for all customers.

There's a great deal of work ahead, but we believe that offers us a tremendous opportunity to rebuild and modernize our distribution system to provide safe, reliable, clean energy that is affordable and accessible to all our customers.

Appendix I Distribution Plan Requirements

Category	Distribution Plan Requirement (MPSC Order text)	Order /Date Page	Incorporation into the DGP
General Requirements	The Commission reiterates its support for the four objectives (safety, reliability/resiliency, cost-effectiveness/affordability, and accessibility) outlined above to guide this next iteration of distribution planning. Although some stakeholders suggested replacing these objectives with the MI Power Grid focus on customer engagement, integrating emerging technologies, and optimizing grid performance and investment, the Commission finds that this is unnecessary and prefers to retain the existing distribution planning objectives.	U-20147 2020-08-20 (p38)	DTEE's planning objectives, described in the Executive Summary and Section 3 - Grid Modernization, are well aligned with the Commissions' objectives
	Commission agrees with the Staff and stakeholders that it is important to articulate the Commission's vision for the future of the grid, and specifically adopts Mr. De Martini's concept that the elements of distribution planning should all relate to a core objective of "customer needs" that binds everything together.	U-20147 2020-08-20 (p39)	DTEE's four planning objectives are well aligned with customer needs
	Staff's final report, p. 34. In its executive summary, the Staff also recommends that utilities include measurable goals and objectives as part of their long-term strategic vision and plan. Id., Executive Summary, p. vii.	U-20147 2020-08-20 (p22-23)	Measurable goals and system performance projections for SAIFI, SAIDI, and emergent cost reductions are including in Section 5
	Therefore, the Commission directs DTE Electric, Consumers, and I&M to continue to develop detailed distribution plans over a five-year period, but also include in the plan their vision and high-level investment strategies 10 and 15 years out. This approach is consistent with the planning horizons used in IRPs.	U-20147 2019-09-11 (p4)	The scenario-based planning approach looked at grid impacts with a 15-year planning horizon, as well as a summary of grid needs in the next 5 and 5-15 year timeframe.

	The Commission concurs with the ALJ that Soulardarity's concerns were substantially addressed in Case No. U-20162. However, the Commission also agrees with the ALJ that, in its next general rate case, DTE Electric shall provide a more detailed explanation of the factors and scoring process the company uses to prioritize the circuits to be hardened. In addition, DTE Page 111 U-20561 Electric shall provide a plan and timeline for system hardening and conversion in its next distribution investment and maintenance plan to be filed by June 30, 2021.	U-20561 2020-05-08 (p110)	Prioritization criteria and plans for Hardening and Conversion are discussed in detail in Sections 9.3 and 11.3-5 respectively
Dynamic System Load Forecasting	The Commission seeks to avoid prescribing specific methods or approaches in the next round of distribution plans but acknowledges that the Staff's recommended dynamic approach to load forecasting with scenario analysis could help better understand and accommodate uncertainty associated with DERs, PEV charging, and other factors. The Commission encourages continued discussion of forecasting methods to inform the next iteration of distribution plans	U-20147 2018-11-21 (p32)	Scenario analysis, including forecasts for DER and EV adoption, was fundamental to the Grid Modernization Process (Section 3)
	The Commission agrees with the Staff's recommendation for the utilities to consider EWR in their upcoming distribution plans due next year. The Commission finds it important to run sensitivities in load forecasts for distribution planning and to start modeling locational impacts from customer behavior (whether through plug-in electric vehicles, EWR, storage, solar DG, DR, etc.). The Commission recognizes that the purpose of distribution planning is not to design EWR programs or to conduct localized EWR/DR potential studies, but finds that a stronger linkage between EWR and DR efforts and distribution planning would facilitate the identification of potentially cost-effective NWA's that could defer to displace an expensive distribution upgrade. Indeed, in the 2019 SEA, Recommendation E-5 states that a framework should be developed "to evaluate non-wires alternatives such as targeted energy waste reduction and demand response in IRPs and distribution plans." 2019 SEA, p. 196. As such, the	U-20147 2020-08-20 (p49-50)	Section 4.2 provides a clear explanation of the updates to forecasting methodologies that will take place in the near term, and coordinate with other resource planning efforts, including EWR and DR

	Commission is already on record recommending exactly the type of targeted demand-side programs supported by the joint commenters, ACEEE, and the Staff. Moreover, the Commission agrees with joint commenters, ACEEE, and the Staff that geotargeting demand-side programs to impact distribution-level system needs is consistent with the statutory framework for delivering EWR and DR. The system-wide EWR requirements contained in statute, as well as the EWR and DR commitments approved as part of utility IRPs, represent a baseline for demand-side resources. However, as the joint commenters and ACEEE point out, demand-side investments may provide additional and situation-specific cost savings and other benefits for customers in specific distribution system contexts. While these situation specific applications can certainly be included as part of utility compliance strategies with statutory mandates and IRP commitments, the fulfillment of those other obligations does not relieve utilities from considering—and ultimately implementing—demand-side options where doing so results in cost savings or other quantifiable benefits.		
	The Commission should require a dynamic approach to load forecasting for the purpose of distribution planning which considers multiple scenarios and probabilistic planning to properly accommodate uncertainty around distributed energy resource penetrations.; The Commission emphasizes the importance of accurate forecasting in planning and investment decisions and the need to ensure best practices in forecasting methods as technologies and customer behavior evolve with the adoption of DERs and PEV charging, which may include scenario-based forecasting to account for uncertainties and identify least regret solutions. Whether it is at the bulk transmission system or the individual distribution circuit level, the Commission believes prudent planning and investments will require more sophisticated forecasting approaches to develop best practices and mitigate risks. The Commission seeks to	U-20147 2018-11-21 (p4,32)	

	avoid prescribing specific methods or approaches in the next round of distribution plans but acknowledges that the Staff's recommended dynamic approach to load forecasting with scenario analysis could help better understand and accommodate uncertainty associated with DERs, PEV charging, and other factors. The Commission encourages continued discussion of forecasting methods to inform the next iteration of distribution plans		
Hosting Capacity Analysis	The Staff specifically recommends the following be adopted by utilities for HCA pilots under this framework: Adopt streamlined interconnection of DER and improved utility distribution mapping capabilities as the use-case for HCA	U-20147 2020-08-20 (p15)	Section 4.5 provides a status update on the Hosting Capacity analysis, including both the zonal go/no-go analysis and more specific analysis. Noting that the HCA pilots have not yet been completed. Section 4.4 covers the interconnection process, with additional capabilities to streamline interconnection of DER provided in Section 12.9
	Adopt a phased implementation approach for HCA pilots to allow utilities to focus on providing cost-effectively obtained, basic system-level information and at the same time highlighting areas of their system that cannot safely accommodate an increase in DER penetration by doing the following: - Perform base-level approach with a zonal go/no-go map. - Conduct specific, detailed analyses on areas of the distribution system with high DER penetration and incorporate this information into a more detailed map with feeder voltage level information as DER penetration continues to increase - Examine HCA best practices and methods for cost reduction, as demonstrated by other jurisdictions nationally. - Benchmark projected and actual HCA pilot costs against HCA costs nationally - HCA information should be publicly available with a downloadable map and spreadsheet.	U-20147 2020-08-20 (p15)	
	The Staff specifically recommends the following be adopted by utilities for HCA pilots under this framework. The recommended HCA activities should be accomplished within the next two years, while resulting information is made available publicly throughout the two year period. A detailed	U-20147 2020-08-20 (p14-15, 41)	

	status update should be provided in the electric distribution plans filed in 2021		
Non wires alternatives	The Commission agrees with the Staff that recommendations stemming from the MI Power Grid Energy Programs and Technology Pilots workgroup should inform the pilots proposed in the utility distribution plans. The Commission further notes that the Staff's draft report on Energy Programs and Technology Pilots was circulated to stakeholders on July 31, 2020, and the final report is scheduled to be filed on September 30, 2020. The Commission will provide additional guidance on structuring of pilots included in utility distribution plans following the filing of the Staff's final report on this issue, and encourages utilities to begin considering how to apply the recommendations from that stakeholder process as they begin to develop and refine the pilots to be included in their distribution plans next year	U-20147 2020-08-20 (p48)	The NWA pilots developed in Section 12.7 are being designed with the MI Power Grid pilots requirements in mind. Initial responses to the questions are found in Appendix V, with further refinement needed as the projects are developed
	Indeed, in the 2019 SEA, Recommendation E-5 states that a framework should be developed "to evaluate non-wires alternatives such as targeted energy waste reduction and demand response in IRPs and distribution plans." 2019 SEA, p. 196. As such, the Commission is already on record recommending exactly the type of targeted demand-side programs supported by the joint commenters, ACEEE, and the Staff.	U-20147 2020-08-20 (p49-50)	The NWA screening process is discussed in Section 4.3, including development of an analytical tool to evaluate the ability of targeted EWR and DR to solve distribution needs
	In the distribution plans to be filed in 2021, the Commission expects further progress on articulation of decision criteria used by utilities to screen projects for NWA analysis, as well as additional pilots that could be considered.	U-20147 2020-08-20 (p42)	
	Regarding the distribution plans to be filed in 2021, the Commission agrees that, in the short-term, NWA projects should focus on capacity and substation projects. As the Staff and interested parties work to integrate NWAs into the utility planning process, limiting the number of capacity and	U-20147 2020-08-20 (p42-43)	A suite a pilots is included in Section 12.7 that incorporate a wide range of technologies beyond EWR/DR

	substation projects considered for NWAs, as well as the scope of the eventual NWA pilots, in the next round of distribution plans is prudent. With this in mind, as noted in the May 8 order, p. 112, the Commission expects to be presented with “a robust suite of NWAs that may be evaluated for prudence as possible programs.” Longer term, the Commission looks forward to continuing to refine the criteria under which NWAs would be considered, drawing on the experience of other jurisdictions from around the country, and directs the Staff to continue to work with utilities and other stakeholders in developing this criteria.		
	- As a threshold matter, the Commission notes the Staff’s endorsement of a series of questions that should be addressed by utilities “prior to refining and implementing additional NWA pilots.” Staff’s final report, p. 28 (footnote omitted). These questions, taken from Mr. De Martini’s recommendations as well as the Staff’s addition, focus on a “customer-centric approach to NWA” and include the following: - Why are non-wires alternatives being pursued? - What are the pressing issues? - What are the desired outcomes? - Optimize utility distribution expenditures? - Enable greater value for customer/developer DER investments? - Enable greater adoption of DER to meet renewable/customer choice goals? - What are the range of potential solutions? - Pricing, programs and procurements (3P’s)? - What is the role of customers, DER developers, utilities, aggregators[,] and others? - Are the benefits and costs of NWAs accruing to all customers on an equitable basis?	U-20147 2020-08-20 (p44)	Staff and Mr. De Martini’s questions are addressed in Section 4.3
	The Commission specifically directs the utilities mentioned above to include in their draft plan responses to the questions	U-20147	

	posed by Mr. De Martini and the Staff, and invites comment from the Staff and stakeholders on these issues as well. While the Commission will not be approving the distribution plans, it continues to emphasize the role a robust consideration of alternatives will play in the consideration of alternatives—including NWAs—in specific proposed investments included in future rate cases.	2020-08-20 (p45)	
	Further, as part of the NWA analysis, the Commission would like greater transparency into the need for, costs, and timing of traditional solutions. For proposed NWAs, the Commission expects information regarding costs and savings, impact of the NWA in offsetting the need for traditional investment, customer consumption patterns with and without the NWA, implementation timing, and assumptions used in the analysis, including minimum customer participation levels. The Commission expects this to be an iterative process, with additional guidance on specifics relating to the affected project areas, substation characteristics, and details on both proposed wired and NWAs on a going-forward basis.	U-20147 2020-08-20 (p44)	The information requested here is listed in table form at the end of each NWA pilot detailed in Section 12.7
Integration of IRPs and distribution plans	The Commission sees value in aligning distribution plans and IRP filings and is interested in such an alignment to the best extent possible. The Commission, however, recognizes that a perfect line up between the two may not be achievable given the statutory framework of MCL 460.6t and the utility-specific nature of each IRP case. See, e.g., Case Nos. U-20165 and U-20471. Moreover, given the pressing need to respond to the coronavirus pandemic during 2020 and the timing of this order, the Commission finds additional time is warranted to prepare the distribution plans and extends the previous deadline of June 30, 2021 to September 30, 2021.	U-20147 2020-08-20 (p51)	Collaboration between the IRP and Distribution planning processes are described in Section 4.6. The impact of generation planning on distribution planning is also discussed in Section 11.7
	The Commission expects it will take several iterations of plans to more fully align integrated resource planning and distribution planning and is not envisioning such alignment to	U-20147 2019-09-11	

	be completed with this next round of distribution plans. It is nonetheless important to raise this longer-term objective as utilities, the Staff, and stakeholders are working on both IRPs and distribution plans. These ongoing planning activities may shed light on opportunities and barriers that affect alignment, particularly as it relates to forecasting, modeling tools, and analysis of alternatives. This, in turn, could help inform updates to the Michigan Integrated Resource Planning Parameters for the next round of IRPs pursuant to 2016 PA 341. To be clear, however, the Commission is not initiating a process to update the IRP parameters in this docket as such updates would be handled in a separate proceeding at a later time.	(p3)	
Alternative Regulatory Approaches	Directed DTE Electric to include in its next distribution plan “proposed PBR [performance-based regulation] elements with reasonable metrics tied to utility financial performance, improvement targets, and timelines for achievement,” and specified the following elements for consideration: - The utility should consider the pros and cons of a comprehensive PBR system, which would avoid concurrent regular annual rate cases and separate PBR reconciliations; - The utility’s financial PBR system should include both incentives and disincentives based on performance; incentive structures should be holistically considered in terms of impacts on potential earnings - Performance metrics should include outcome measures (e.g., customer average interruption duration index (CAIDI)) and not be limited to output metrics such as number of poles replaced; - Performance metrics should be linked to regional, national, and/or peer utility benchmarks, where possible; - Data and calculation methodologies should be well defined, transparent, and open for auditing/verification purposes; - Targets should be utility specific; and - Potential areas of performance focus are safety, customer service (end use customers, builders, interconnecting	U-20147 2020-08-20 (p47)	Section 15 discusses DTEE's PBR plan in detail

	generators, etc.), timeliness and quality, reliability and resiliency, long-term costs, and innovation.		
Cost/Benefit Analysis	While the Commission finds this an important topic to tackle in the near term, this is not a topic that can be resolved before the next round of distribution plans are due next year. The Commission appreciates the Staff's recommendation to convene stakeholders on the BCA framework after the plans are filed next year, but the Commission refrains from making this commitment so that it can evaluate priorities in 2021 in light of other MI Power Grid activities; the Commission directs the Staff to continue to work with utilities and other stakeholders in continuing to explore the appropriate framework for evaluating BCA, including consideration of experiences from other jurisdictions and recommendations related to the issues highlighted in the Staff's final report. The Commission expects these additional details to inform and be integrated into future utility distribution plans.	U-20147 2020-08-20 (p45)	Current cost benefit approach (GPM) as well as proposed next steps involving stakeholder discussion included in Section 6
Resilience	The Commission agrees with DTE Electric on the description of resilience, in terms of the ability to restore power following a major catastrophic event. The Commission also thinks about this term more broadly such as planning to mitigate more localized, high-impact outages caused by equipment issues, access limitations, or system configurations that inhibit timely restoration or backup capabilities, e.g., specialized equipment necessary to replace substation failure cannot be replaced in timely manner and the area cannot receive backup power through an alternative means or obstructions limit access to distribution equipment (e.g., closed in alleyways in Detroit). Resilience in this context should also consider the vulnerability of loads that would affect public health, safety, or security under an extended outage, and related mitigation strategies to ensure continuity of service. With the potential for supply chain disruptions due the pandemic or other factors, as well as cyber or physical security threats, the Commission underscores the	U-20147 2020-08-20 (p48)	Resiliency is covered in many sections, including Section 7.3. High-impact outages can be mitigated through the use of the mobile fleet, or longer-term investments driven by the substation risk framework (9.2). The storm restoration process incorporates vulnerable loads in restoration priority (Section 7.3). Additionally, pilots and further efforts to underground or move circuits in areas with access issues is discussed in Section 11.6

	importance of robust, risk-based resilience evaluations and mitigation strategies as part of distribution planning efforts.		
Workforce adequacy plans	The Commission acknowledges that an adequate utility workforce is key to the distribution system. However, the Commission believes that a focus more on implementation considerations generally, with workforce being a component, would be sufficient for the next round of distribution plans. The Commission is also sensitive to publishing detailed workforce information that could affect bidding and contracting by utilities.	U-20147 2018-11-21 (p36-37)	DTEE's process and general approach to workforce and resource planning is discussed in Section 16.2

Appendix II Strategic Capital Programs excluded from Global Prioritization Model prioritization

Investment Name	DGP Section	2021-2025 Investment (\$M)	Notes
Strategic Pillar: Infrastructure Hardening and Resilience			
Substation Risk: Drexel	9.2	\$ 1.9	Completing construction on switchgear replacement which was damaged due to failure
Belleville Switchgear Decommissioning	9.2	\$1.7	Switchgear is end of life with no replacement breakers available and safety concern with control wiring
Disconnect and Switcher: Disconnect	9.1	\$12.0	Replacement of equipment that is obsolete or prone to failure during operation
Disconnect and Switcher: Switcher	9.1	\$5.3	Replacement of equipment with known failure modes, potentially faulty components, or undersized for the need
Mobile Fleet Program: Portable Equip / DG	9.6	\$7.0	Allows for timely restoration of customers during major events and improves system operability while providing for completion of planned work and preventive maintenance.
Mobile Fleet Program: Portable Substation	9.6	\$7.4	Allows for timely restoration of customers during major events and improves system operability while providing for completion of planned work and preventive maintenance.
SCADA Pole Top Device Replacement	9.1	\$9.4	Replacement of equipment with unavailable parts and known high failure rate
Substation Regulator Replacement	9.1	\$1.7	Replacement of equipment that is obsolete and employs technology that is no longer manufactured or supported
Strategic Pillar: Infrastructure Redesign and Modernization			
Transformer High Side Protection Program	11.2	\$15.0	Installation of high side protection as required by NERC standards
Station Upgrade: Cortland Station Expansion	11.2	\$9.3	Existing transformers 101 and 102 cannot be replaced in existing location due to physical restrictions leaving high amount of load at risk in the event of a failure
40kV Capacitor Banks at Armada and Adair	11.2	\$3.3	Work is necessary to maintain adequate voltage for multiple contingencies on Tie 810 and facilitate shutdowns for the Tie 810 (Gramer) project
Subtransmission Redesign & Rebuild: Slocum	11.2	\$2.2	Needed due to decommissioning of Trenton Channel Power Plant
Subtransmission Redesign & Rebuild - Small Projects & Reserve	11.2	\$11.5	Needed to implement small projects aimed at addressing overloads and voltage violations on the subtransmission system.
4.8 kV Conversion: Ariel Substation and Circuit Conversion	11.3	\$2.8	Project is under construction

4.8 kV Conversion: Cortland / Oakman / Linwood Consolidation: Linwood	11.3	\$4.3	Project is under construction
4.8 kV Conversion: Pinegrove Substation Relocation and Conversion	11.3	\$22.4	Project is driven by MDOT work requiring relocation of Pinegrove substation
Ann Arbor AC Network Conversion	11.3	\$3.1	Upgraded capacity of downtown Ann Arbor network to accommodate increased load
Pontiac Vaults	11.5	\$3.5	Necessary to meet operational and safety needs
Strategic Pillar: Technology and Automation			
ESOC	12.2	\$15.0	Upgrades of physical infrastructure and technology constraints in the current SOC needed to fully utilize the new ADMS platforms
ASOC	12.2	\$92.6	Needed to provide for full redundancy with the new ESOC
CVR/VVO	12.6	\$65.5	Currently a pilot program being evaluated as an offset to peak generation and for grid efficiency and having sufficient capacity for demand
DER Control	12.8	\$11.3	Comply with upcoming regulatory requirements such as FERC2222
Mobile Technology	12.9	\$13.3	Improve data availability to field resources to improve operational efficiency
Work Management & Scheduling	12.9	\$21.3	Upgrade work management systems to streamline project execution
Asset Management	12.9	\$5.6	Upgrade mapping system
Distribution Planning Process Capabilities	12.9	\$42.6	Support processes such as Interconnections, HCA, load forecasting
Modernize Grid Management	12.9	\$35.6	New technology to enable and support grid modernization efforts
2021-2025 Investment Total		\$425	11% of the \$3.8B in Strategic capital is not prioritized by GPM

Appendix III Historical Storm Events

Exhibit A-2 Storm Event and Restoration Duration History

Storm #	<u>Conditions</u>	Storm Start	Restoration Duration in Hours	Customers Affected	Cost \$ million
	W – Wind L – Lightning R – Rain I – Ice S – Snow H – Heat C – Cold				
2016001	W - - - S -	01/10/16 05:30	39.0	42,871	\$ 5.746
2016002	W - - - -	02/19/16 12:00	77.5	112,419	\$ 10.416
2016003	W - R I S C	02/24/16 07:30	53.5	26,590	\$ 4.509
2016004	W - R I S C	02/28/16 19:00	36.5	20,052	\$ 2.058
2016005	- - R - - -	03/13/16 10:00	35.0	20,519	\$ 1.757
2016006	W - R - - -	03/16/16 07:30	75.5	54,601	\$ 10.970
2016007	W - R - - -	06/04/16 12:00	71.0	41,376	\$ 6.288
2016008	W - - - - H	06/11/16 12:00	35.0	33,444	no w/o
2016009	W - - - - H	06/19/16 14:00	56.0	46,495	\$ 6.477
2016010	W L R - - H	07/08/16 00:00	91.0	100,228	\$ 11.194
2016011	W L R - - H	07/12/16 16:00	76.0	54,799	\$ 7.357
2016012	W - - - - H	07/17/16 15:00	50.0	26,933	\$ 7.363
2016013	W L R - - H	07/21/16 12:00	59.5	36,313	no w/o
2016014	W L R - - -	07/29/16 14:00	57.5	23,158	no w/o
2016015	- L R - - -	08/12/16 15:00	59.5	43,902	\$ 4.725
2016016	W - R - - -	08/16/16 00:00	37.0	28,692	\$ 5.330
2016017	W L R - - H	09/07/16 13:00	32.0	24,553	\$ 4.613
2016018	W L R - - -	09/10/16 04:00	44.5	17,312	no w/o
2016019	W L R - - -	09/29/16 05:00	58.0	32,104	no w/o

Storm #	<u>Conditions</u>	Storm Start	Restoration Duration in Hours	Customers Affected	Cost \$ million
	W – Wind L – Lightning R – Rain I – Ice S – Snow H – Heat C – Cold				
2016020	W L R - - -	11/18/16 19:00	80.0	57,216	\$ 6.682
2016021	W - - - - -	11/28/16 20:00	33.5	52,256	\$ 4.339
2016022	W - - - - -	12/26/16 00:00	24.0	33,644	\$ 1.758
2017001	W - - - - -	01/10/17 15:00	65.9	74,371	\$ 8.337
2017002	- L R - S -	01/16/17 15:00	36.1	19,301	\$ 0.870
2017003	W - R - s -	02/12/17 10:00	34.3	17,934	\$ 1.180
2017004	- L R - S -	02/24/17 16:00	26.5	8,223	\$ 0.896
2017005	W - - - - -	02/28/17 22:00	46.5	39,339	\$ 3.831
2017006	W - - - - -	03/08/17 08:00	202.0	749,511	\$ 91.426
2017007	W - - - s -	03/18/17 02:00	40.2	26,915	\$ 6.373
2017008	W - R - s -	03/30/17 09:00	51.8	26,179	\$ 0.732
2017009	W - - - - -	04/05/17 21:00	142.4	100,839	\$ 10.732
2017010	W - - - - -	04/26/17 15:00	53.0	24,702	\$ 1.900
2017011	W L R - - -	04/30/17 22:00	48.0	15,748	\$ 0.801
2017012	W - - - - -	05/17/17 08:00	133.3	92,221	\$ 11.422
2017013	W - - - - H	06/10/17 07:00	229.0	128,406	\$ 11.939
2017014	W - R - - -	06/22/17 09:00	92.0	93,330	\$ 8.419
2017015	- - R - - H	06/29/17 21:00	66.8	28,651	\$ 3.647
2017016	W - R - - -	07/06/17 23:00	52.0	56,145	\$ 4.887
2017017	W L R - - -	07/10/17 10:00	83.1	54,192	\$ 7.017
2017018	- L R - - H	07/23/17 03:00	40.1	26,099	\$ 0.921
2017019	W - - - - H	07/28/17 15:00	32.5	10,688	\$ 0.678

Storm #	<u>Conditions</u>	Storm Start	Restoration Duration in Hours	Customers Affected	Cost \$ million
	W – Wind L – Lightning R – Rain I – Ice S – Snow H – Heat C – Cold				
2017020	W L R - - -	08/02/17 16:00	85.4	67,461	\$ 9.310
2017021	W L R - - H	08/11/17 05:00	58.5	14,189	\$ 1.051
2017022	W L R - - H	08/16/17 20:00	46.0	25,497	\$ 1.600
2017023	- L R - - -	08/28/17 18:00	44.5	26,299	\$ 1.526
2017024	W L R - - H	09/04/17 16:00	55.3	19,118	\$ 2.228
2017025	W - R - - -	10/07/17 13:00	61.9	30,103	\$ 4.904
2017026	W - R - - -	10/11/17 05:00	38.2	34,854	\$ 4.252
2017027	W L R - - -	10/14/17 18:00	80.1	79,164	\$ 7.237
2017028	W - R - - -	10/23/17 18:00	73.3	51,957	\$ 9.955
2017029	W L R - - -	11/15/17 15:00	27.6	13,636	\$ 1.890
2017030	W - - - - -	12/05/17 04:00	20.0	27,597	\$ 2.165
2018001	W - R - S - -	03/01/18 13:00	79.5	87,922	\$ 10.9
2018002	W - R - S - -	04/04/18 07:30	35.8	5,894	\$ 3.2
2018003	W - R I S - -	04/15/18 05:30	126.2	284,535	\$ 39.1
2018004	W - R - - - -	05/04/18 12:00	107.8	250,063	\$ 35.7
2018005	W - R - - H -	05/30/18 17:00	55	53,095	\$ 9.7
2018006	W - R - - H -	06/13/18 14:00	32	23,201	\$ 4.6
2018007	- - R - - H -	06/18/18 9:00	51.3	47,425	\$ 7.9
2018008	- L R - - H -	07/01/18 15:00	37	20,200	\$ 7.5
2018009	W L R - - H -	07/16/18 09:00	38.3	29,667	\$ 7.6
2018010	W L R - - H -	08/01/18 0:00	45	35,461	\$ 5.3
2018011	W L R - - H -	08/06/18 13:00	80	54,956	\$ 13.1

Storm #	<u>Conditions</u>	Storm Start	Restoration Duration in Hours	Customers Affected	Cost \$ million
	W – Wind L – Lightning R – Rain I – Ice S – Snow H – Heat C – Cold				
2018012	W L R -- H -	09/01/18 16:00	78	52,560	\$ 6.8
2018013	W L R -- H -	09/05/18 20:00	27.5	21,984	\$ 7.2
2018014	W - R ----	09/21/18 10:00	64.5	66,447	\$ 16.7
2018015	W L R ----	10/20/18 13:00	58	69,431	\$ 15.8
2019001	W - R ----	02/07/19 21:00	42.7	34,238	\$ 5.8
2019002	W - R I ---	02/12/19 8:00	36	43,075	\$ 8.5
2019003	W - R ----	02/24/19 8:00	59.2	128,515	\$ 21.6
2019004	W - R ----	05/23/19 6:00	39.5	50,070	\$ 7.3
2019005	W - R ----	06/1/19 13:00	32	40,432	\$ 8.9
2019006	W L R -- H -	07/02/19 9:00	139.2	117,854	\$ 17.2
2019007	W L R -- H -	07/19/19 16:00	151.8	402,442	\$ 51.8
2019008	W L R -- H -	07/28/19 15:00	71	97,173	\$ 16.5
2019009	W L R -- H -	08/18/19 12:00	41.5	41,059	\$ 8.5
2019010	W L R ----	08/26/19 18:30	42	40,847	\$ 10.0
2019011	W L R ----	09/11/19 14:00	49.5	59,054	\$ 7.5
2019012	W L R ----	09/13/19 15:31	48	28,877	\$ 13.5
2019013	W - R ----	10/27/19 01:00	47.8	44,480	\$ 10.3
2019014	W - R ----	11/27/19 10:00	64	74,376	\$ 19.3
2019015	W - R ----	12/30/19 06:00	37.5	30,883	\$ 12.7
2020001	W -- I ---	01/11/20 05:00	56	35,717	\$ 10.5
2020002	W L R ----	03/29/20 13:00	34	31,287	\$ 7.6
2020003	W L R -- H -	06/03/20 03:00	32.6	30,384	\$ 4.5

Storm #	<u>Conditions</u>	Storm Start	Restoration Duration in Hours	Customers Affected	Cost \$ million
	W – Wind L – Lightning R – Rain I – Ice S – Snow H – Heat C – Cold				
2020004	W L R - - H -	06/10/20 13:00	102.3	187,824	\$ 31.1
2020005	W L R - - H -	06/23/20 08:00	37	34,978	\$ 5.0
2020006	W L R - - H -	06/26/20 23:00	42	51,598	\$ 8.1
2020007	W L R - - H -	07/07/20 19:00	131	143,875	\$ 24.6
2020008	W L R - - H -	07/19/20 09:00	80	184,726	\$ 29.7
2020009	W L R - - H -	08/02/20 5:30	39.8	41,510	\$ 6.2
2020010	W L R - - - -	08/27/20 17:00	69.7	48,785	\$ 16.2
2020011	W - R - - - -	10/23/20 17:30	48.3	38,762	\$ 4.9
2020012	W - R - - - -	11/01/20 08:30	55	47,311	\$ 8.0
2020013	W - R - - - -	11/15/20 02:30	112	203,919	\$ 46.3

Appendix IV Reliability Indices for Service Centers

Exhibit A.3.1 SAIFI, SAIDI, CAIDI – All Weather, All Causes

SAIFI - All Weather, All Causes

SC	2015	2016	2017	2018	2019	2020
ANN	1.07	1.19	1.37	1.52	1.81	1.40
CAN	0.68	0.62	1.45	1.22	1.19	0.96
HWL	1.31	1.57	1.63	1.87	2.52	2.10
LAP	1.58	1.50	1.45	1.15	1.89	1.90
MAR	1.88	1.19	1.22	1.22	1.37	1.33
MTC	0.76	0.89	1.13	1.03	1.36	1.47
NAE	1.69	1.29	2.08	2.09	2.52	2.38
NPT	0.82	0.66	1.27	1.25	1.41	0.97
PON	1.00	0.98	1.55	1.68	1.82	1.60
RFD	1.13	0.90	1.46	1.34	0.97	0.90
SBY	0.72	1.17	1.19	1.06	1.00	1.10
WWS	0.84	0.95	1.46	1.33	1.29	1.28

SAIDI - All Weather, All Causes

SC	2015	2016	2017	2018	2019	2020
ANN	255	290	1,023	449	467	323
CAN	397	169	1,829	543	684	498
HWL	219	377	1,241	304	1,220	421
LAP	428	431	883	551	897	440
MAR	486	313	983	437	416	375
MTC	151	181	553	195	318	258
NAE	1,147	483	1,041	1,109	769	1,193
NPT	158	185	876	317	273	181
PON	248	209	983	366	484	334
RFD	322	190	1,586	788	473	288

SBY	100	177	496	238	192	203
WWS	228	304	914	466	403	422

CAIDI – All Weather, All Causes

SC	2015	2016	2017	2018	2019	2020
ANN	238	244	747	295	258	230
CAN	584	273	1,261	445	575	519
HWL	167	240	761	163	484	200
LAP	271	287	609	479	475	232
MAR	259	263	806	358	304	282
MTC	199	203	489	189	234	175
NAE	679	374	500	531	305	501
NPT	193	280	690	254	194	186
PON	248	213	634	218	266	209
RFD	285	211	1,086	588	488	319
SBY	139	151	417	225	192	185
WWS	271	320	626	350	312	329

Exhibit A.3.2 SAIFI, SAIDI, CAIDI – Excluding MEDs, All Causes

SAIFI - Excluding MEDs, All Causes

SC	2015	2016	2017	2018	2019	2020
ANN	1.01	1.15	1.03	1.17	1.49	1.07
CAN	0.53	0.59	0.90	0.84	0.91	0.67
HWL	1.23	1.42	1.19	1.62	1.90	1.67
LAP	1.45	1.35	1.03	0.82	1.43	1.38
MAR	1.72	1.06	0.76	0.89	1.09	1.00
MTC	0.69	0.81	0.87	0.84	1.19	1.30
NAE	1.32	1.25	1.53	1.54	2.09	1.61
NPT	0.74	0.60	0.92	0.97	1.33	0.88

PON	0.94	0.92	1.13	1.39	1.59	1.28
RFD	1.03	0.84	0.91	0.93	0.74	0.70
SBY	0.69	1.13	0.98	0.89	0.90	0.91
WWS	0.73	0.82	1.08	0.96	1.07	0.99

SAIDI – Excluding MEDs, All Causes

SC	2015	2016	2017	2018	2019	2020
ANN	215	255	175	192	220	129
CAN	195	147	323	231	199	140
HWL	160	284	196	204	393	216
LAP	359	319	172	140	266	210
MAR	381	270	126	103	187	122
MTC	103	115	155	114	190	144
NAE	453	441	392	351	426	380
NPT	125	142	115	159	226	104
PON	208	175	194	202	289	158
RFD	219	154	217	184	151	116
SBY	81	158	125	123	124	112
WWS	145	165	212	146	180	137

CAIDI – Excluding MEDs, All Causes

SC	2015	2016	2017	2018	2019	2020
ANN	213	222	170	164	148	120
CAN	368	249	359	275	219	209
HWL	130	200	165	126	207	130
LAP	248	236	167	171	186	152
MAR	222	255	166	116	172	122
MTC	149	142	178	136	160	111
NAE	343	353	256	228	204	237
NPT	169	237	125	164	170	119

PON	221	190	172	145	182	124
RFD	213	183	238	198	204	167
SBY	117	140	128	138	138	123
WWS	199	201	196	152	168	138

Exhibit A.3.3 SAIFI, SAIDI, CAIDI – Catastrophic Storms, All Causes

SAIFI – Catastrophic Storms, All Causes

SC	2015	2016	2017	2018	2019	2020
ANN	0.04	0.07	0.29	0.13	0.43	0.35
CAN	0.12	0.05	0.49	0.32	0.40	0.29
HWL	0.02	0.23	0.38	0.06	0.89	0.35
LAP	0.04	0.24	0.33	0.32	0.65	0.30
MAR	0.02	0.17	0.37	0.32	0.45	0.16
MTC	0.04	0.10	0.22	0.15	0.25	0.14
NAE	0.00	0.04	0.19	0.52	0.53	0.49
NPT	0.08	0.07	0.26	0.19	0.19	0.08
PON	0.06	0.07	0.34	0.11	0.48	0.27
RFD	0.07	0.07	0.50	0.36	0.36	0.18
SBY	0.02	0.03	0.19	0.11	0.20	0.15
WWS	0.09	0.16	0.33	0.27	0.32	0.30

SAIDI – Catastrophic Storms, All Causes

SC	2015	2016	2017	2018	2019	2020
ANN	26	41	837	81	269	160
CAN	152	36	1,457	268	525	309
HWL	3	111	1,021	14	938	105
LAP	23	140	644	407	692	83

MAR	18	54	742	330	264	54
MTC	29	72	373	63	137	66
NAE	1	43	311	711	384	410
NPT	37	47	719	121	64	24
PON	38	39	764	100	275	99
RFD	70	44	1,345	554	369	108
SBY	17	19	363	87	91	49
WWS	62	164	661	265	235	229

CAIDI – Catastrophic Storms, All Causes

SC		2015	2016	2017	2018	2019	2020
ANN	650	586	2,886	623	626	450	
CAN	1,267	720	2,973	838	1,313	1,079	
HWL	150	483	2,687	233	1,054	301	
LAP	575	583	1,952	1,272	1,065	272	
MAR	900	318	2,005	1,031	587	340	
MTC	725	720	1,695	420	548	462	
NAE	0	1,075	1,637	1,367	725	831	
NPT	463	671	2,765	637	337	291	
PON	633	557	2,247	909	573	365	
RFD	1,000	629	2,690	1,539	1,025	588	
SBY	850	633	1,911	791	455	330	
WWS	689	1,025	2,003	981	734	759	

Exhibit A.3.4 SAIFI, SAIDI, CAIDI – Non-Catastrophic Storms, All Causes

SAIFI – Non-Catastrophic Storms, All Causes

SC	2015	2016	2017	2018	2019	2020
ANN	0.38	0.45	0.40	0.33	0.21	0.24
CAN	0.26	0.24	0.44	0.30	0.19	0.21
HWL	0.60	0.54	0.43	0.51	0.43	0.49
LAP	0.65	0.55	0.29	0.11	0.27	0.47
MAR	0.70	0.35	0.26	0.17	0.19	0.47
MTC	0.27	0.21	0.30	0.18	0.22	0.19
NAE	1.03	0.67	0.79	0.23	0.36	0.74
NPT	0.22	0.21	0.37	0.28	0.32	0.15
PON	0.36	0.36	0.46	0.38	0.17	0.30
RFD	0.36	0.30	0.33	0.23	0.11	0.20
SBY	0.33	0.34	0.27	0.17	0.12	0.17
WWS	0.28	0.30	0.39	0.22	0.17	0.25

SAIDI – Non-Catastrophic Storms, All Causes

SC	2015	2016	2017	2018	2019	2020
ANN	127	135	93	131	71	79
CAN	185	75	259	162	54	113
HWL	147	190	113	144	127	165
LAP	256	199	129	40	52	203
MAR	300	145	167	30	72	224
MTC	66	37	111	46	56	67
NAE	1,054	326	515	56	96	589
NPT	63	83	89	72	87	82
PON	99	95	110	103	57	117
RFD	151	76	132	108	34	109

SBY	46	60	58	55	27	61
WWS	97	71	131	86	58	102

CAIDI – Non-Catastrophic Storms, All Causes

SC	2015	2016	2017	2018	2019	2020
ANN	334	300	233	397	338	332
CAN	712	313	589	540	284	541
HWL	245	352	263	282	295	340
LAP	394	362	445	364	193	433
MAR	429	414	642	176	379	479
MTC	244	176	370	256	255	357
NAE	1,023	487	652	243	267	801
NPT	286	395	241	257	272	555
PON	275	264	239	271	335	386
RFD	419	253	400	470	309	546
SBY	139	176	215	324	225	351
WWS	346	237	336	391	341	408

Exhibit A.3.5 SAIFI, SAIDI, CAIDI – Blue Sky, All Causes

SAIFI – Blue Sky, All Causes

SC	2015	2016	2017	2018	2019	2020
ANN	0.64	0.68	0.67	1.05	1.16	0.81
CAN	0.29	0.33	0.52	0.60	0.60	0.47
HWL	0.70	0.80	0.81	1.30	1.19	1.26
LAP	0.88	0.71	0.84	0.72	0.98	1.12
MAR	1.16	0.67	0.60	0.73	0.73	0.70
MTC	0.45	0.59	0.60	0.70	0.89	1.14
NAE	0.66	0.58	1.10	1.34	1.63	1.15
NPT	0.52	0.37	0.64	0.78	0.91	0.74
PON	0.57	0.55	0.75	1.19	1.17	1.02
RFD	0.70	0.53	0.64	0.75	0.49	0.52
SBY	0.37	0.80	0.73	0.78	0.68	0.77
WWS	0.47	0.49	0.74	0.84	0.80	0.73

SAIDI – Blue Sky, All Causes

SC	2015	2016	2017	2018	2019	2020
ANN	102	114	93	237	126	84
CAN	60	57	113	113	104	76
HWL	68	76	108	146	155	151
LAP	148	92	110	104	153	154
MAR	169	113	74	76	80	97
MTC	55	72	68	85	125	125
NAE	92	115	216	342	289	194
NPT	58	55	67	124	122	75

PON	112	74	108	164	153	118
RFD	101	70	110	126	69	70
SBY	37	97	75	96	73	93
WWS	69	69	121	116	110	91

CAIDI – Blue Sky, All Causes

SC	2015	2016	2017	2018	2019	2020
ANN	159	168	139	226	109	104
CAN	207	173	217	188	173	164
HWL	97	95	133	112	130	119
LAP	168	130	131	144	156	137
MAR	146	169	123	104	110	138
MTC	122	122	113	121	140	109
NAE	139	198	196	255	177	169
NPT	112	149	105	159	134	100
PON	196	135	144	138	131	115
RFD	144	132	172	168	141	136
SBY	100	121	103	123	107	120
WWS	147	141	164	138	138	124

Exhibit A.3.6 SAIFI, SAIDI, CAIDI – All Weather, Cause = Tree/Wind

SAIFI - All Weather, Cause = Tree/Wind

SC	2015	2016	2017	2018	2019	2020
ANN	0.43	0.65	0.78	0.70	0.83	0.68
CAN	0.42	0.27	0.98	0.58	0.63	0.50
HWL	0.62	0.71	0.98	0.81	1.38	0.93

LAP	0.64	0.76	0.83	0.44	0.92	1.16
MAR	0.62	0.50	0.73	0.48	0.60	0.59
MTC	0.25	0.32	0.53	0.49	0.59	0.47
NAE	0.81	0.71	1.31	1.02	1.06	1.34
NPT	0.35	0.27	0.66	0.62	0.57	0.47
PON	0.38	0.47	0.92	0.76	0.98	0.66
RFD	0.44	0.44	0.95	0.61	0.56	0.48
SBY	0.24	0.30	0.45	0.39	0.34	0.30
WWS	0.38	0.46	0.82	0.60	0.63	0.66

SAIDI - All Weather, Cause = Tree/Wind

SC	2015	2016	2017	2018	2019	2020
ANN	160	179	862	256	328	207
CAN	322	109	1,469	309	512	405
HWL	132	219	1,142	150	1,000	263
LAP	225	277	738	331	695	323
MAR	275	171	850	235	273	235
MTC	82	110	441	97	209	149
NAE	718	345	901	546	492	941
NPT	99	104	758	173	137	132
PON	123	128	812	204	356	195
RFD	198	124	1,276	454	392	212
SBY	49	60	394	113	103	99
WWS	152	226	766	259	299	298

CAIDI – All Weather, Cause = Tree/Wind

SC	2015	2016	2017	2018	2019	2020
ANN	372	275	1,105	366	395	304
CAN	767	404	1,499	533	813	810
HWL	213	308	1,165	185	725	283
LAP	352	364	889	752	755	279
MAR	444	342	1,164	490	455	396
MTC	328	344	832	198	354	320
NAE	886	486	688	535	464	704
NPT	283	385	1,148	279	240	280
PON	324	272	883	268	363	293
RFD	450	282	1,343	744	700	441
SBY	204	200	876	290	303	331
WWS	400	491	934	432	475	448

Exhibit A.3.7 SAIFI, SAIDI, CAIDI – Excluding MEDs, Cause = Tree/Wind

SAIFI - Excluding MEDs, Cause = Tree/Wind

SC	2015	2016	2017	2018	2019	2020
ANN	0.38	0.62	0.48	0.51	0.58	0.45
CAN	0.28	0.25	0.53	0.37	0.41	0.24
HWL	0.55	0.59	0.58	0.63	0.92	0.60
LAP	0.58	0.65	0.47	0.22	0.60	0.71
MAR	0.56	0.45	0.30	0.28	0.40	0.38
MTC	0.19	0.26	0.29	0.40	0.47	0.31
NAE	0.56	0.68	0.84	0.73	0.80	0.66
NPT	0.27	0.22	0.34	0.45	0.51	0.40
PON	0.33	0.42	0.55	0.58	0.79	0.44

RFD	0.35	0.40	0.51	0.36	0.39	0.31
SBY	0.21	0.28	0.27	0.28	0.26	0.17
WWS	0.30	0.34	0.47	0.41	0.45	0.45

SAIDI – Excluding MEDs, Cause = Tree/Wind

SC	2015	2016	2017	2018	2019	2020
ANN	125	147	97	101	111	73
CAN	123	89	234	131	99	72
HWL	86	131	129	89	258	93
LAP	180	199	89	45	142	114
MAR	201	152	76	41	81	53
MTC	41	54	81	64	102	39
NAE	280	306	294	216	266	184
NPT	67	61	62	72	105	62
PON	87	99	111	101	172	69
RFD	100	93	143	103	102	66
SBY	31	47	43	43	49	28
WWS	79	93	105	78	111	74

CAIDI – Excluding MEDs, Cause = Tree/Wind

SC	2015	2016	2017	2018	2019	2020
ANN	329	237	202	198	191	162
CAN	439	356	442	354	241	293
HWL	156	222	222	141	280	155
LAP	310	306	189	205	237	161
MAR	359	338	253	146	203	139

MTC	216	208	279	160	217	127
NAE	500	450	350	296	333	280
NPT	248	277	182	160	206	156
PON	264	236	202	174	218	157
RFD	286	233	280	286	262	210
SBY	148	168	159	154	188	165
WWS	263	274	223	190	247	162

Exhibit A.3.8 SAIFI, SAIDI, CAIDI – Catastrophic Storms, Cause = Tree/Wind

SAIFI – Catastrophic Storms, Cause = Tree/Wind

SC	2015	2016	2017	2018	2019	2020
ANN	0.04	0.05	0.27	0.04	0.31	0.23
CAN	0.12	0.04	0.40	0.17	0.29	0.23
HWL	0.01	0.13	0.37	0.04	0.67	0.19
LAP	0.04	0.18	0.31	0.22	0.42	0.25
MAR	0.02	0.06	0.34	0.20	0.28	0.09
MTC	0.04	0.06	0.18	0.05	0.15	0.10
NAE	0.00	0.03	0.17	0.22	0.30	0.40
NPT	0.08	0.06	0.24	0.12	0.08	0.06
PON	0.04	0.05	0.30	0.07	0.37	0.15
RFD	0.06	0.05	0.40	0.19	0.27	0.15
SBY	0.02	0.02	0.17	0.06	0.11	0.09
WWS	0.07	0.15	0.30	0.12	0.23	0.20

SAIDI – Catastrophic Storms, Cause = Tree/Wind

SC	2015	2016	2017	2018	2019	2020
ANN	23	37	753	35	235	103
CAN	147	34	1,193	149	441	288
HWL	1	93	992	9	833	82
LAP	22	101	597	285	588	66
MAR	18	22	666	192	206	39
MTC	25	60	335	17	115	59
NAE	0	40	292	281	253	377
NPT	36	45	669	80	42	21

PON	33	32	677	62	242	61
RFD	65	36	1,108	309	329	93
SBY	16	13	346	47	65	39
WWS	58	153	623	142	207	175

CAIDI – Catastrophic Storms, Cause = Tree/Wind

SC		2015	2016	2017	2018	2019	2020
ANN	575	740	2,789	875	758	452	
CAN	1,225	850	2,983	876	1,521	1,238	
HWL	100	715	2,681	225	1,243	439	
LAP	550	561	1,926	1,295	1,400	271	
MAR	900	367	1,959	960	736	419	
MTC	625	1,000	1,861	340	767	592	
NAE	0	1,333	1,718	1,277	843	945	
NPT	450	750	2,788	667	525	356	
PON	825	640	2,257	886	654	412	
RFD	1,083	720	2,770	1,626	1,219	638	
SBY	800	650	2,035	783	591	434	
WWS	829	1,020	2,077	1,183	900	860	

Exhibit A.3.9 SAIFI, SAIDI, CAIDI – Non-Catastrophic Storms, Cause = Tree/Wind

SAIFI – Non-Catastrophic Storms, Cause = Tree/Wind

SC	2015	2016	2017	2018	2019	2020
ANN	0.21	0.26	0.22	0.15	0.09	0.16
CAN	0.18	0.12	0.35	0.18	0.11	0.14

HWL	0.40	0.36	0.30	0.28	0.22	0.38
LAP	0.31	0.33	0.18	0.03	0.09	0.30
MAR	0.37	0.18	0.20	0.10	0.11	0.25
MTC	0.11	0.12	0.19	0.11	0.12	0.11
NAE	0.63	0.49	0.56	0.16	0.15	0.53
NPT	0.14	0.09	0.26	0.12	0.13	0.08
PON	0.19	0.20	0.29	0.23	0.09	0.20
RFD	0.18	0.16	0.22	0.14	0.08	0.16
SBY	0.14	0.15	0.10	0.11	0.07	0.07
WWS	0.17	0.15	0.22	0.13	0.10	0.16

SAIDI – Non-Catastrophic Storms, Cause = Tree/Wind

SC	2015	2016	2017	2018	2019	2020
ANN	90	79	67	66	40	61
CAN	150	51	219	105	33	92
HWL	109	94	92	86	82	134
LAP	135	142	90	7	32	169
MAR	213	90	150	17	33	154
MTC	43	26	82	31	29	60
NAE	686	261	463	39	53	460
NPT	43	34	64	37	40	71
PON	72	65	83	70	44	89
RFD	94	55	101	81	28	89
SBY	25	31	25	41	19	40
WWS	69	46	96	53	35	78

CAIDI – Non-Catastrophic Storms, Cause = Tree/Wind

SC	2015	2016	2017	2018	2019	2020
ANN	429	304	305	440	444	395
CAN	833	425	626	583	300	644
HWL	273	261	307	307	373	348
LAP	435	430	500	233	356	562
MAR	576	500	750	170	300	608
MTC	391	217	432	282	242	563
NAE	1,089	533	827	244	353	876
NPT	307	378	246	308	308	880
PON	379	325	286	304	489	457
RFD	522	344	459	579	350	565
SBY	179	207	250	373	271	606
WWS	406	307	436	408	350	487

Exhibit A.3.10 SAIFI, SAIDI, CAIDI – Blue Sky, Cause = Tree/Wind

SAIFI – Blue Sky, Cause = Tree/Wind

SC	2015	2016	2017	2018	2019	2020
ANN	0.19	0.34	0.30	0.50	0.43	0.30
CAN	0.13	0.12	0.23	0.22	0.23	0.12
HWL	0.20	0.22	0.31	0.48	0.49	0.36
LAP	0.29	0.24	0.35	0.19	0.41	0.61
MAR	0.24	0.26	0.19	0.19	0.21	0.25
MTC	0.10	0.14	0.15	0.33	0.32	0.26
NAE	0.17	0.19	0.58	0.65	0.61	0.41
NPT	0.13	0.12	0.15	0.38	0.35	0.33
PON	0.14	0.22	0.33	0.45	0.52	0.32

RFD	0.19	0.23	0.33	0.27	0.21	0.18
SBY	0.08	0.13	0.18	0.23	0.16	0.14
WWS	0.14	0.16	0.29	0.35	0.29	0.30

SAIDI – Blue Sky, Cause = Tree/Wind

SC	2015	2016	2017	2018	2019	2020
ANN	48	63	42	155	53	43
CAN	25	24	57	56	39	25
HWL	22	32	59	55	85	48
LAP	68	34	51	40	75	87
MAR	45	59	34	26	34	42
MTC	14	24	24	49	65	30
NAE	32	45	146	226	185	104
NPT	20	25	25	56	55	40
PON	19	31	52	73	70	45
RFD	38	33	66	65	36	30
SBY	8	15	23	25	19	20
WWS	24	28	47	63	57	44

CAIDI – Blue Sky, Cause = Tree/Wind

SC	2015	2016	2017	2018	2019	2020
ANN	253	185	140	310	123	143
CAN	192	200	248	255	170	200
HWL	110	145	190	115	173	133
LAP	234	142	146	211	183	143
MAR	188	227	179	137	162	171

MTC	140	171	160	148	203	115
NAE	188	237	252	348	303	252
NPT	154	208	167	147	157	121
PON	136	141	158	162	135	140
RFD	200	143	200	241	171	170
SBY	100	115	128	109	119	137
WWS	171	175	162	180	197	148

Exhibit A.3.11 SAIFI, SAIDI, CAIDI – All Weather, Cause = Equipment

SAIFI - All Weather, Cause = Equipment

SC	2015	2016	2017	2018	2019	2020
ANN	0.15	0.23	0.34	0.35	0.27	0.26
CAN	0.12	0.19	0.23	0.18	0.07	0.16
HWL	0.31	0.38	0.34	0.59	0.40	0.36
LAP	0.34	0.23	0.21	0.28	0.19	0.36
MAR	0.21	0.25	0.20	0.31	0.18	0.16
MTC	0.24	0.27	0.22	0.21	0.32	0.50
NAE	0.22	0.22	0.44	0.27	0.37	0.32
NPT	0.13	0.19	0.29	0.29	0.22	0.22
PON	0.25	0.26	0.34	0.37	0.34	0.42
RFD	0.22	0.30	0.31	0.26	0.10	0.19
SBY	0.21	0.40	0.46	0.30	0.34	0.33
WWS	0.18	0.25	0.39	0.26	0.23	0.28

SAIDI - All Weather, Cause = Equipment

SC	2015	2016	2017	2018	2019	2020
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ANN	34	48	76	71	41	55
CAN	31	33	111	54	27	28
HWL	51	57	42	79	63	62
LAP	62	53	45	42	40	57
MAR	49	60	43	44	35	34
MTC	34	35	48	29	40	67
NAE	45	44	58	75	45	45
NPT	19	55	80	59	32	23
PON	60	46	78	67	50	68
RFD	53	46	103	54	14	28
SBY	28	62	68	49	40	34
WWS	37	44	89	41	37	53

CAIDI – All Weather, Cause = Equipment

SC	2015	2016	2017	2018	2019	2020
ANN	227	209	224	203	152	210
CAN	258	174	483	300	386	177
HWL	165	150	124	134	158	173
LAP	182	230	214	150	211	161
MAR	233	240	215	142	194	217
MTC	142	130	218	138	125	133
NAE	205	200	132	278	122	141
NPT	146	289	276	203	145	103
PON	240	177	229	181	147	161
RFD	241	153	332	208	140	151
SBY	133	155	148	163	118	103

WWS	206	176	228	158	161	192
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Exhibit A.3.12 SAIFI, SAIDI, CAIDI – Excluding MEDs, Cause = Equipment

SAIFI - Excluding MEDs, Cause = Equipment

SC	2015	2016	2017	2018	2019	2020
ANN	0.15	0.23	0.33	0.33	0.27	0.22
CAN	0.11	0.19	0.21	0.17	0.07	0.16
HWL	0.29	0.36	0.31	0.58	0.32	0.35
LAP	0.31	0.22	0.20	0.26	0.18	0.32
MAR	0.21	0.25	0.20	0.30	0.15	0.14
MTC	0.24	0.27	0.21	0.20	0.31	0.50
NAE	0.21	0.22	0.42	0.26	0.35	0.30
NPT	0.13	0.19	0.26	0.25	0.22	0.22
PON	0.24	0.25	0.33	0.32	0.33	0.39
RFD	0.22	0.29	0.28	0.25	0.09	0.18
SBY	0.21	0.39	0.44	0.29	0.34	0.32
WWS	0.17	0.25	0.37	0.24	0.23	0.25

SAIDI – Excluding MEDs, Cause = Equipment

SC	2015	2016	2017	2018	2019	2020
ANN	32	48	54	50	35	26
CAN	28	33	64	44	17	26
HWL	40	55	35	74	45	47
LAP	52	50	34	32	36	50
MAR	47	59	22	36	30	16
MTC	30	34	32	26	37	65

NAE	39	41	46	61	43	37
NPT	19	55	30	47	30	20
PON	58	41	49	47	46	52
RFD	52	43	54	44	13	22
SBY	26	61	53	42	38	31
WWS	35	42	77	31	34	32

CAIDI – Excluding MEDs, Cause = Equipment

SC	2015	2016	2017	2018	2019	2020
ANN	213	209	164	152	130	116
CAN	255	174	305	259	243	165
HWL	138	153	113	128	141	138
LAP	168	227	170	123	200	159
MAR	224	236	110	120	200	114
MTC	125	126	152	130	119	131
NAE	186	186	110	235	123	125
NPT	146	289	115	188	136	92
PON	242	164	148	147	139	135
RFD	236	148	193	176	144	125
SBY	124	156	120	145	112	97
WWS	206	168	208	129	148	129

Exhibit A.3.13 SAIFI, SAIDI, CAIDI – Catastrophic Storms, Cause = Equipment

SAIFI – Catastrophic Storms, Cause = Equipment

SC	2015	2016	2017	2018	2019	2020
ANN	0.00	0.00	0.01	0.00	0.01	0.07

CAN	0.01	0.00	0.01	0.01	0.01	0.00
HWL	0.00	0.02	0.00	0.00	0.09	0.01
LAP	0.00	0.01	0.01	0.01	0.02	0.04
MAR	0.00	0.02	0.00	0.01	0.02	0.00
MTC	0.00	0.01	0.00	0.01	0.03	0.01
NAE	0.00	0.00	0.00	0.03	0.07	0.02
NPT	0.00	0.00	0.01	0.02	0.02	0.01
PON	0.01	0.01	0.01	0.01	0.04	0.04
RFD	0.00	0.01	0.02	0.01	0.01	0.01
SBY	0.00	0.00	0.01	0.01	0.05	0.03
WWS	0.01	0.01	0.01	0.02	0.01	0.03

SAIDI – Catastrophic Storms, Cause = Equipment

SC	2015	2016	2017	2018	2019	2020
ANN	2	1	22	3	6	29
CAN	3	1	41	6	13	1
HWL	1	9	4	1	24	3
LAP	1	7	10	10	8	12
MAR	0	6	14	7	6	2
MTC	4	2	15	2	7	2
NAE	0	3	9	17	11	8
NPT	1	1	38	9	5	1
PON	2	7	29	9	10	7
RFD	2	4	49	13	2	3
SBY	1	1	11	8	9	4
WWS	2	4	11	6	3	14

CAIDI – Catastrophic Storms, Cause = Equipment

SC	2015	2016	2017	2018	2019	2020
ANN	0	0	2,200	0	600	404
CAN	300	0	4,100	600	1,300	360
HWL	0	450	0	0	267	349
LAP	0	700	1,000	1,000	400	298
MAR	0	300	0	700	300	592
MTC	0	200	0	200	233	185
NAE	0	0	0	567	157	322
NPT	0	0	3,800	450	250	119
PON	200	700	2,900	900	250	188
RFD	0	400	2,450	1,300	200	509
SBY	0	0	1,100	800	180	136
WWS	200	400	1,100	300	300	563

Exhibit A.3.14 SAIFI, SAIDI, CAIDI – Non-Catastrophic Storms, Cause = Equipment

SAIFI – Non-Catastrophic Storms, Cause = Equipment

SC	2015	2016	2017	2018	2019	2020
ANN	0.07	0.08	0.12	0.07	0.04	0.01
CAN	0.04	0.09	0.05	0.03	0.01	0.04
HWL	0.09	0.10	0.09	0.12	0.04	0.02
LAP	0.11	0.06	0.05	0.01	0.02	0.06
MAR	0.07	0.08	0.03	0.03	0.02	0.04
MTC	0.07	0.05	0.06	0.03	0.02	0.01
NAE	0.10	0.05	0.11	0.03	0.03	0.03

NPT	0.02	0.08	0.07	0.05	0.04	0.02
PON	0.06	0.05	0.07	0.06	0.05	0.07
RFD	0.07	0.10	0.08	0.02	0.01	0.02
SBY	0.07	0.08	0.09	0.02	0.03	0.01
WWS	0.03	0.10	0.10	0.04	0.04	0.03

SAIDI – Non-Catastrophic Storms, Cause = Equipment

SC	2015	2016	2017	2018	2019	2020
ANN	13	23	17	16	10	5
CAN	13	18	30	21	4	7
HWL	25	26	12	22	7	15
LAP	24	24	14	4	6	12
MAR	19	27	11	8	15	18
MTC	8	6	14	7	4	1
NAE	32	12	12	9	2	5
NPT	5	40	16	11	5	7
PON	10	11	15	15	6	17
RFD	27	14	22	7	1	6
SBY	11	11	20	4	4	1
WWS	7	17	24	10	8	14

CAIDI – Non-Catastrophic Storms, Cause = Equipment

SC	2015	2016	2017	2018	2019	2020
ANN	186	288	142	229	250	414
CAN	325	200	600	700	400	169
HWL	278	260	133	183	175	778

LAP	218	400	280	400	300	177
MAR	271	338	367	267	750	511
MTC	114	120	233	233	200	179
NAE	320	240	109	300	67	185
NPT	250	500	229	220	125	314
PON	167	220	214	250	120	255
RFD	386	140	275	350	100	381
SBY	157	138	222	200	133	243
WWS	233	170	240	250	200	415

Exhibit A.3.15 SAIFI, SAIDI, CAIDI – Blue Sky, Cause = Equipment

SAIFI – Blue Sky, Cause = Equipment

SC	2015	2016	2017	2018	2019	2020
ANN	0.08	0.14	0.22	0.28	0.22	0.17
CAN	0.07	0.10	0.17	0.15	0.06	0.11
HWL	0.22	0.25	0.25	0.46	0.27	0.33
LAP	0.23	0.16	0.15	0.26	0.14	0.25
MAR	0.14	0.15	0.18	0.27	0.14	0.12
MTC	0.17	0.22	0.15	0.18	0.27	0.49
NAE	0.12	0.17	0.33	0.21	0.27	0.27
NPT	0.11	0.11	0.21	0.22	0.16	0.19
PON	0.18	0.19	0.26	0.30	0.25	0.32
RFD	0.16	0.19	0.21	0.23	0.08	0.17
SBY	0.14	0.31	0.36	0.27	0.27	0.29
WWS	0.15	0.15	0.28	0.20	0.18	0.22

SAIDI – Blue Sky, Cause = Equipment

SC	2015	2016	2017	2018	2019	2020
ANN	19	25	37	52	25	20
CAN	14	14	40	26	10	20
HWL	25	23	26	56	32	43
LAP	37	22	21	29	26	34
MAR	30	27	18	29	13	14
MTC	22	27	18	20	29	64
NAE	13	29	37	50	32	32
NPT	13	15	26	39	22	15
PON	49	29	35	43	34	44
RFD	24	28	32	35	10	20
SBY	16	50	37	38	27	28
WWS	28	23	55	25	25	25

CAIDI – Blue Sky, Cause = Equipment

SC	2015	2016	2017	2018	2019	2020
ANN	238	179	168	186	114	113
CAN	200	140	235	173	167	174
HWL	114	92	104	122	119	132
LAP	161	138	140	112	186	134
MAR	214	180	100	107	93	118
MTC	129	123	120	111	107	131
NAE	108	171	112	238	119	120
NPT	118	136	124	177	138	78
PON	272	153	135	143	136	138

RFD	150	147	152	152	125	118
SBY	114	161	103	141	100	97
WWS	187	153	196	125	139	115

Appendix V Pilot guidelines and information

U-20645 Pilot Plan Criteria	
1. Pilot need and goals detailed	a. Need for the pilot is expressed. Results of past similar pilots and findings are shared to justify the need for the proposed pilot. b. Pilot goals and desired learnings detailed. c. Reference any pending applicable regulatory dockets, legislation, or other consideration relevant to the pilot project.
2. Pilot design and evaluation plan designed and presented together	a. Pilot program design and evaluation plans are designed together so examined metrics and collected data support evaluation of the pilot in meeting goals and desired learnings b. If applicable, define target customer population, selection rationale (including those for location-driven programs), recruitment plans, and evaluation plans for customer adoption and satisfaction. c. If statistical analysis will be conducted on pilot results, a statistically significant sample size must be selected, supported, and detailed. If a statistically significant sample size is not selected, justification must be provided. d. If statistical analysis will not be conducted, justification must be provided as well as an approach for evaluating pilot goals. e. If changes are required during implementation, pilot design, and evaluation impacts are shared.
3. Pilot project costs detailed	a. Project costs are detailed by source and amount for applicable periods. b. Availability of non-utility funding and whether any was pursued (such as state or federal funding opportunities) described. c. Anticipated cost-effectiveness and net benefits when deployed at scale described. i. Quantification of expected benefits of the pilot and the evaluation criteria/methods used. d. Proposed rate recovery approach detailed.
4. Pilot timeline detailed	a. Proposed timeline for the pilot project and any related reports or evaluations delineated
5. Stakeholder engagement plan detailed	a. Stakeholder engagement plan before, during, and after pilot takes place detailed. b. Interim and final stakeholder reporting described. c. Expected publicly available data from pilot shared under proper protections and privacy.
6. Public interest detailed	

- a. Public interest justification, including supporting the transition to clean, distributed energy resources; enhancing reliability, safety, affordability, or equity; or other related goals, and the pilot's expected impacts described.
- b. Any added benefits to ratepayers or the energy delivery system, either due to proposed site selection or through other pilot variables, especially if any system weaknesses or forecasted needs are addressed, shared.
- c. Expected impacts of the piloted measure on reliability, resilience, safety, and ratepayer bills detailed.
 - i. Pilot reduction goals for metrics like customer bill, outage minutes/frequency, and OSHA reportable, as well as the translation to full deployment expectations.
- d. Expected local or Michigan-based employment and business opportunities created by pilot described.
- e. Any potential impacts or added benefits of the pilot on low-income customers, seniors or other vulnerable populations described.
- f. Any other public benefits detailed.

Fisher NWA Pilot Design	
1. Pilot need and goals detailed	- Fisher substation pilot project will leverage several learnings of acquired from the Phase I, Hancock substation cost benefit model. These learnings include leveraging a newly developed DER Estimator tool to plan and target high impact, cost-effective programs with tested geographically targeted marketing to maximize participation results and minimize administrative costs in driving and tracking toward a set of highly analyzed peak load goals. - Determine the effectiveness of EWR and DR at addressing a substation over firm rating. Combining NWA options with traditional solutions will also be tested. The lessons learned will help develop future pilots and lasting process changes that reduce costs and carbon emissions.
2. Pilot design and evaluation plan designed and presented together	- Measures will include the following: - Demand impact to expectation - Participation rate - Deployment time and incentives relationship to participation rates - Customer acceptance of more frequent demand response interruptions - Deployment costs
3. Pilot project costs detailed	- DR Scope and Cost - Marketing (IAC & BYOD) - Platform (EnergyHub) - IAC equipment upgrades \$208K – 2021 to 2024 - EWR Scope and Cost - Residential Implementation, incentive, and labor - C&I Implementation and Incentive \$1.8M – 2021 to 2024 - STDF Scope and Cost - Establish STDF (Sub-transmission Distribution Facility) along same sub-transmission line that feeds Fisher Substation - Split FISHR DC 8188 with STDF circuit - Install new conduit and cable \$4.1M – 2021 to 2022

4. Pilot timeline detailed	• STDF Schedule • 2021 – Conceptual design • 2022 – Detail design and begin construction • 2023 – Complete construction and commission • EWR Programs Schedule • 2021 ○ Residential programs, September launch ○ C&I programs, September ramp up and launch • 2022-2024 ○ EWR programs continue through September 2024 • DR Schedule • 2021 ○ Establish program baseline ○ Marketing ▪ Introduce programs ▪ Equipment upgrades • 2022 – 2023 ○ Marketing ▪ New customer acquisition ▪ Existing customer education ○ Equipment upgrades and maintenance ○ Energy hub portal and subscription • 2024 ○ Marketing ▪ Customer retention and attrition management ○ Equipment upgrades and maintenance ○ Energy hub portal and subscription
5. Stakeholder engagement plan detailed	• Partner with NRDC for pilot review • Partner with Staff for pilot review
6. Public interest detailed	• Corporate and Government Affairs team will work with local businesses and community organizations to discuss the scope of the pilot and benefits from participating

Port Austin NWA Pilot Design	
1. Pilot need and goals detailed	Needs • Port Austin substation is 120% of firm rating • Over firm for approximately 60 hours per year • One circuit is 108% of day-to-day Goals • Effectiveness of solar + storage at reducing blocking hours • Refine evaluation tool for storage • Control, protection and security requirements for storage • Deployment costs, time and challenges • Operating processes, procedures and staffing • Microgrid development, operations and benefits
2. Pilot design and evaluation plan designed and presented together	• Measures will include the following: • Demand impact compared to expectation • Deployment time • Deployment costs • Reliability impact compared to expectation
3. Pilot project costs detailed	• Solar plus storage Installation: \$4.5M • Potential Microgrid addition \$0.4M
4. Pilot timeline detailed	• 2021 – 2022 ◦ Property search ◦ Conceptual engineering • 2022 ◦ Detail design ◦ Purchase property ◦ Site prep ◦ Start construction • 2023 ◦ Complete battery installation ◦ Complete solar installation ◦ Evaluation of solar/storage to relief loading conditions • 2024 ◦ Microgrid implementation and testing
5. Stakeholder engagement plan detailed	• Partner with MPSC Staff for pilot review
6. Public interest detailed	• Corporate and Government Affairs team will engage with local community leaders and officials

Omega NWA Pilot Design	
1. Pilot need and goals detailed	Needs • Two subtransmission lines feeding Omega had more than 1000 blocking hours in 2020 • High load growth in the area due to indoor agriculture Goals • Effectiveness of storage at reducing blocking hours • Refine evaluation tool for storage • Control, protection and security requirements for storage • Deployment costs, time and challenges • Operating processes, procedures and staffing
2. Pilot design and evaluation plan designed and presented together	• Measures will include the following: • Demand impact compared to expectation • Deployment time • Deployment costs • Reliability impact compared to expectation
3. Pilot project costs detailed	• Batteries – 5.7M • Install Batteries – 1.0M • Engineering – 0.5M • Total – 7.2M
4. Pilot timeline detailed	• Install batteries – Summer 2022 • Operate batteries – Summer 2022 to December 2023
5. Stakeholder engagement plan detailed	• Partner with MPSC Staff for pilot review
6. Public interest detailed	• Corporate and Government Affairs team will engage with local community leaders and officials

Veridian NWA Pilot Design	
1. Pilot need and goals detailed	Pilot Need • Net-zero community with solar, storage and EV charging • Base load + EV load could approach 1.5MVA • Fed by two circuits from Regent substation at 4.8kV, 81% firm rating Pilot Goals • Effectiveness of solar and storage at reducing load • Consumer behavior with microgrid and customer owned DER • Control, protection and security requirements for storage • Deployment costs, time and challenges • Operating processes, procedures and staffing • Microgrid development, operations and benefits
2. Pilot design and evaluation plan designed and presented together	• To be developed following customer application for service
3. Pilot project costs detailed	• Estimated cost is approximately \$8.3M (DTEE + Developer + DOE Grant) plus additional private and public funding • Estimate to be revised during design
4. Pilot timeline detailed	• 2021 • Customer to submit formal request (Q3) • URD detail design (Q4) • 2022 • OH circuit upgrades • URD construction • Start of Microgrid implementation • 2022 -2026 • Expand Microgrid as customer units are constructed
5. Stakeholder engagement plan detailed	• Refine scope and expectations from the developer • Partner with MPSC Staff for pilot review
6. Public interest detailed	• Partner with the Ann Arbor community for pilot engagement

Front Lot URD - FRMNT1593 Pilot Design

1. Pilot need and goals detailed	Pilot Need - The weather impacts from the summer of 2021 have generated customers interest in undergrounding lines. The Company is developing plans to partner with communities to convert overhead lines to underground lines where it makes sense by implementing Strategic Undergrounding pilots - In order to improve customer reliability and resilience to severe weather, alternatives to rear lot overhead construction are being piloted to determine the overall cost and performance of modifying the current infrastructure - In this pilot, overhead lateral infrastructure from rear-lot alleys will be removed and replaced with a looped URD system in the front lot. - Primary, secondary and services will be undergrounded - Poles will be removed from rear lot alleyway Pilot Goals - Determine down wire improvements of the strategic undergrounding for the customers on the FRMNT circuit - Develop a customer engagement plan and determine best practices for customer engagement - Refine estimates for URD costs, project duration, and uncover unforeseen challenges before ramping the program - Document operating processes, procedures, and staffing - Develop turnkey contractor for program level strategic undergrounding - Evaluate the overall total cost of ownership for underground infrastructure compared to overhead infrastructure
2. Pilot design and evaluation plan designed and presented together	Evaluation of the pilot will include comparing down wires, reliability and emergent costs on the circuit before and after construction is complete, as well as completing an overall assessment of the total lifecycle cost of an underground lateral against an overhead lateral
3. Pilot project costs detailed	- Estimated cost is approximately \$30M (URD, services, restoration, tree trimming, customer engagement) - Estimate to be revised during design
4. Pilot timeline detailed	2021 - Complete Engineering (Q4) - Onboard Turnkey Contractor (Q4) 2022 - URD detail design (Q1) - Start FRMNT construction (Q23) - Deploy Customer Engagement (Q1- Q4) 2023 - Continue Customer Engagement - Complete FRMNT Construction

5. Stakeholder engagement plan detailed	• Develop Customer Engagement Plan • Partner with MPSC Staff for pilot review
6. Public interest detailed	• Partner with the City of Detroit and Community Organizations for pilot engagement

Appendix VI Distribution Design Standards and Distribution Design Orders

Exhibit A-5 Distribution Design Standards and Distribution Design Orders

DDS Section 10 DDO Number	General Information Topic
DDO-0010-001	General Information
DDO-0010-002	DDO Writer's Guide
DDO-0010-003	DDO Procedures and Attachments
DDO-0010-004	DDO Index

DDS Section 20 DDO Number	Overhead Topic
DDO-0020-001	OH Construction Standards
DDO-0020-002	Subtransmission Systems Construction Voltage
DDO-0020-003	Distribution System Construction Voltage
DDO-0020-004	Overhead Construction Type
DDO-0020-005	Distribution Circuit Construction Grade
DDO-0020-006	Circuit Accessibility Construction
DDO-0020-007	Distribution Construction Pole Size
DDO-0020-008	Subtransmission Circuit Standard Conductor Sizes
DDO-0020-009	Distribution Circuit Backbone Standard Conductor Sizes
DDO-0020-010	Distribution Circuit Lateral Standard Conductor Sizes
DDO-0020-011	Distribution Circuit Crossings Standard Conductor Sizes
DDO-0020-012	Distribution Circuit Tree Exposure Standard Conductor Sizes
DDO-0020-013	Distribution Circuit Lightning Protection
DDO-0020-014	Overhead Construction Shield Wire
DDO-0020-015	Overhead Circuit Sectionalizing- Reclosers
DDO-0020-016	Overhead Circuit Sectionalizing- Switches

DDS Section 20 DDO Number	Overhead Topic
DDO-0020-017	Overhead Loop Schemes
DDO-0020-018	Overhead Jumpering Points
DDO-0020-019	Overhead Circuit- DG Interconnection Point
DDO-0020-020	Sectionalizing and Jumpering Points Requirements
DDO-0020-021	Circuit Laterals Sectionalizing
DDO-0020-022	Overhead Fault Indicators
DDO-0020-023	Secondary Construction
DDO-0020-024	Secondary Standard Conductor Type
DDO-0020-025	Distribution Transformers
DDO-0020-026	Services
DDO-0020-027	Circuit Tree Trimming

DDS Section 30 DDO Number	Underground Topic
DDO-0030-001	Underground Construction Standards
DDO-0030-002	Underground Conduit Design
DDO-0030-003	Underground System Cable
DDO-0030-004	URD Front Lot Design
DDO-0030-005	Cable Size and Type for Three Phase URD Loops
DDO-0030-006	Cable Size and Type for Single Phase URD Loops
DDO-0030-007	Residential Padmount Transformer Size
DDO-0030-008	UG Fault Indicators

DDS Section 40 DDO Number	Substation Topic
DDO-0040-001	Substation Construction Standards

DDO-0040-002	Substation Property
DDO-0040-003	Substation Blocking
DDO-0040-004	Station Class Standard
DDO-0040-005	Substation Class Standard
DDO-0040-006	Industrial Substation Standard
DDO-0040-007	Class A and R Substation Transformer Size
DDO-0040-008	Class T Substation Transformer Size
DDO-0040-009	Portable Substation
DDO-0040-010	Subtransmission Distributed Resource

DDS Section 50 DDO Number	Circuit Protection Topic
DDO-0050-001	Substation Fault Current
DDO-0050-002	Minimum Fault Current
DDO-0050-003	Protection Device Coordination
DDO-0050-004	Fuses and Oil Reclosers Loading
DDO-0050-005	Fault Current Interrupting
DDO-0050-006	Minimum Fault Current
DDO-0050-007	Primary Fuse Saving
DDO-0050-008	Maximum Number of Devices

DDS Section 70 DDO Number	Voltage Topic
DDO-0070-001	Primary Voltage Limits
DDO-0070-002	Secondary Voltage Limits
DDO-0070-003	Voltage Unbalance
DDO-0070-004	Voltage Flicker
DDO-0070-005	Power Factor

DDS Section 70 DDO Number	Voltage Topic
DDO-0070-006	Harmonics
DDO-0070-007	Subtransmission Power Service

DDS Section 80 DDO Number	Project Benefit Evaluation Topic
DDO-0080-001	Project Benefit Evaluation

DDS Section 90 DDO Number	Loading Topic
DDO-0090-001	Maximum Outage Duration
DDO-0090-002	Peak Demand Load
DDO-0090-003	Substation Loading
DDO-0090-004	Load Imbalance
DDO-0090-005	13.2 kV Normal Circuit Loading
DDO-0090-006	13.2 kV Emergency Circuit Loading
DDO-0090-007	4.8 kV Normal Circuit Loading
DDO-0090-008	4.8 kV Emergency Circuit Loading
DDO-0090-009	13.2 kV Load Loss
DDO-0090-010	13.2 kV Single Phase URD Loading
DDO-0090-011	13.2 kV Three Phase URD Loading
DDO-0090-012	Redundancy

DDS Section 100 DDO Number	System Automation Topic
DDO-0100-001	Device Monitoring, Power Quality, AMI and Industrial Metering
DDO-0100-002	System Network Model and Asset Data Model

DDS Section 100 DDO Number	System Automation Topic
DDO-0100-003	System Automation
DDO-0100-004	Telecommunications
DDO-0100-005	Interconnection
DDO-0100-006	Technology Maturity

DDS Section 110 DDO Number	Urban Networks Topic
DDO-0110-001	Network Feeder Configuration
DDO-0110-002	Urban AC Network Substation Configuration
DDO-0110-003	Network Feeder Routing
DDO-0110-004	Contingency Design Criteria
DDO-0110-005	Feeder Group Configuration
DDO-0110-006	Feeder Cable Size
DDO-0110-007	Feeder Sectionalizing
DDO-0110-008	Network Platforms
DDO-0110-009	In-Building Installations
DDO-0110-010	Netbank Cable Size
DDO-0110-011	Netbank Voltage Configuration
DDO-0110-012	Netbank Size
DDO-0110-013	Netbank Protector SCADA
DDO-0110-014	Secondary Areas
DDO-0110-015	Secondary Main Size
DDO-0110-016	Secondary Main Protection
DDO-0110-017	Secondary Moles
DDO-0110-018	Secondary Services

The following appendices are attached separately to this document

Appendix VII Customer focus groups

Appendix VIII Community leader engagement

Appendix IX DTEE/ICF Grid Modernization Study 2021-2035

Table of Exhibits

EXHIBIT 2.1.3.1 FEEDBACK FROM INDIVIDUAL DGP STAKEHOLDER DISCUSSIONS.....	24
EXHIBIT 2.2.1.1 – ILLUSTRATIVE SAIFI BY CENSUS TRACT WITH HIGHLY IMPACTED COMMUNITIES IDENTIFIED	27
EXHIBIT 3.1.1 FIVE STEP APPROACH TO DTEE’S GRID MODERNIZATION PLANNING PROCESS.....	31
EXHIBIT 3.2.1 MPSC DISTRIBUTION PLANNING OBJECTIVES	33
EXHIBIT 3.2.2 DTEE PLANNING OBJECTIVES.....	34
EXHIBIT 3.3.1 DTEE 2035 GRID MODERNIZATION SCENARIOS.....	36
EXHIBIT 3.3.1.1 ELECTRIFICATION SCENARIO FORECASTS (2025-2035)	37
EXHIBIT 3.3.1.2 SUBSTATION LOADING	38
EXHIBIT 3.3.2.1 UPPER MIDWEST EXTREME WEATHER IS INCREASING IN FREQUENCY AND INTENSITY.....	39
EXHIBIT 3.3.3.1 DG/DS SCENARIO FORECAST (2025-2035)	41
EXHIBIT 3.4.2.1 DTEE GRID MODERNIZATION NEAR-TERM STATUS	45
EXHIBIT 3.5.1 MAPPING OF GRID NEEDS TO DGP SECTIONS.....	54
EXHIBIT 4.2.1 CURRENT STATE OF ANNUAL DISTRIBUTION PLANNING CYCLE.....	57
EXHIBIT 4.2.1.1 INTEGRATED FORECASTING SOLUTION (IFS) STEPS	59
EXHIBIT 4.3.1 NWA SCREENING PROCESS	66
EXHIBIT 4.5.1 ILLUSTRATIVE IMAGE OF HOSTING CAPACITY ON A DISTRIBUTION CIRCUIT IN THE SECOND PHASE	74
EXHIBIT 5.1.1 DTEE PLANNING OBJECTIVES.....	83
EXHIBIT 5.1.2 PROGRAM IMPACT DIMENSIONS.....	85
EXHIBIT 5.1.3 SELECTED PROGRAMS PROJECT AND ALIGNMENT TO OBJECTIVES	86
EXHIBIT 5.1.4 IMPACT DIMENSION WEIGHTS	87
EXHIBIT 5.1.5 OVERALL BENEFIT-COST SCORES FOR STRATEGIC CAPITAL PROGRAMS AND PROJECTS.....	88
EXHIBIT 5.1.6 TOP 50 STRATEGIC CAPITAL PROGRAMS AND PROJECTS BASED ON BENEFIT-COST PRIORITIZATION RANKING	89
EXHIBIT 5.3.1 SYSTEM ALL WEATHER SAIDI.....	93
EXHIBIT 5.3.2 ALL WEATHER SAIFI.....	93
EXHIBIT 5.3.3 SAIDI EXCLUDING MAJOR EVENT DAYS (EX-MEDS).....	94
EXHIBIT 5.3.4 SAIFI EXCLUDING MAJOR EVENT DAYS (EX-MEDS).....	95
EXHIBIT 5.3.5 CEMI6	96
EXHIBIT 5.3.6 EMERGENT CAPITAL SPEND PROJECTION	97
EXHIBIT 6.1 PROJECTED DTEE FIVE-YEAR DISTRIBUTION GRID PLAN INVESTMENTS (EXCLUDING PLD)	102
EXHIBIT 6.2 PROJECTED FIVE-YEAR DTEE DISTRIBUTION GRID PLAN INVESTMENTS (SORTED BY REFERENCE SECTION NUMBER)	103
EXHIBIT 7.1.1 DTEE SUBSTATIONS	106

EXHIBIT 7.1.2 DTEE TRANSFORMERS	107
EXHIBIT 7.1.3 DTEE SUBTRANSMISSION CIRCUITS	107
EXHIBIT 7.1.4 DTEE DISTRIBUTION CIRCUITS – LINE MILES BY SUBSTATION BUS VOLTAGE	107
EXHIBIT 7.1.5 DTEE DISTRIBUTION CIRCUITS – LINE MILES BY LINE VOLTAGE.....	108
EXHIBIT 7.1.6 DTEE DISTRIBUTION SYSTEM	109
EXHIBIT 7.2.2.1 RELIABILITY INDICES DEFINITIONS	112
EXHIBIT 7.2.2.2 RELIABILITY STATISTICS - ALL WEATHER	114
EXHIBIT 7.2.2.3 RELIABILITY STATISTICS - EXCLUDING MEDS	115
EXHIBIT 7.2.2.4 RELIABILITY STATISTICS - CATASTROPHIC STORMS (DTEE DEFINITION)	116
EXHIBIT 7.2.2.5 RELIABILITY STATISTICS – NON-CATASTROPHIC STORMS (DTEE DEFINITION)	116
EXHIBIT 7.2.2.6 RELIABILITY STATISTICS - EXCLUDING ALL STORMS	116
EXHIBIT 7.2.3.1 5-YEAR AVERAGE CUSTOMER MINUTES OF INTERRUPTION BY CAUSE (SAIDI)	118
EXHIBIT 7.2.3.2 5-YEAR AVERAGE CUSTOMER INTERRUPTIONS BY CAUSE (SAIFI)	118
EXHIBIT 7.2.3.3 5-YEAR AVERAGE OUTAGE EVENTS BY CAUSE.....	118
EXHIBIT 7.2.4.1 MAIFI INDICES CALCULATIONS	120
EXHIBIT 7.2.4.2 MAIFI BY YEAR.....	120
EXHIBIT 7.2.4.3 MAIFI BY YEAR (AVERAGE NUMBER OF MOMENTARY INTERRUPTIONS PER CUSTOMER BY YEAR)	121
EXHIBIT 7.2.5.1 CEMIN AND CELIDT DEFINITIONS	122
EXHIBIT 7.2.5.2 CEMIN BY YEAR (NUMBER OF CUSTOMERS HAVING “N” OR MORE OUTAGES BY YEAR)	122
EXHIBIT 7.2.5.3 CELIDT BY YEAR (PERCENT OF CUSTOMERS EXPERIENCING T OR MORE HOURS OF INTERRUPTION BY YEAR).....	123
EXHIBIT 7.3.2.1 WEATHER SEVERITY LEVELS	126
EXHIBIT 7.3.2.2 MUTUAL ASSISTANCE GROUPS WITHIN THE UNITED STATES	129
EXHIBIT 7.3.2.3 24-HOUR STORM MANAGEMENT CYCLE	131
EXHIBIT 7.3.2.4 DTEE’S RESTORATION PLAN	133
EXHIBIT 7.3.2.5 DTEE’S MOBILE COMMAND CENTER	134
EXHIBIT 7.3.2.6 OUTAGE MAP ON DTE’S MOBILE APP AND WEBSITE	135
EXHIBIT 7.3.2.7 HAZARD HAMLET SIMULATOR	136
EXHIBIT 7.3.2.8 COMMUNITY VAN.....	137
EXHIBIT 7.3.3.1 DTEE’S FLEET OF MOBILE RESTORATION EQUIPMENT	139
EXHIBIT 7.3.3.2 MOBILE GENERATOR	139
EXHIBIT 7.3.3.3 PORTABLE SUBSTATION	139
EXHIBIT 7.3.4.1 LOW PROBABLY, HIGH IMPACT EVENTS	140
EXHIBIT 7.4.2.1 IDENTIFYING ABNORMAL VOLTAGE CONDITIONS	146
EXHIBIT 7.4.3.1 SUBTRANSMISSION POWER QUALITY METER INVESTMENTS	149

EXHIBIT 8.1 ASSET AGE SUMMARY	152
EXHIBIT 8.2 ASSET PROGRAM SUMMARY	153
EXHIBIT 8.1.1 TYPICAL SUBSTATION POWER TRANSFORMER.....	155
EXHIBIT 8.1.2 SUBSTATION POWER TRANSFORMER AGE DISTRIBUTION	155
EXHIBIT 8.1.3 SUBSTATION POWER TRANSFORMER FAILURES.....	156
EXHIBIT 8.1.4 SUBSTATION POWER TRANSFORMERS REPLACEMENT FACTORS	156
EXHIBIT 8.2.1 NETBANK SYSTEM MOUNTED ON STEEL COLUMNS	158
EXHIBIT 8.3.1 CIRCUIT BREAKERS	159
EXHIBIT 8.3.2 CIRCUIT BREAKERS BY APPLICATION	159
EXHIBIT 8.3.3 CIRCUIT BREAKER AGE DISTRIBUTION	160
EXHIBIT 8.3.4 CIRCUIT BREAKER REPLACEMENT FACTORS	160
EXHIBIT 8.3.5 BREAKER FAILURES BY YEAR	161
EXHIBIT 8.4.1 CAP & PIN DISCONNECT SWITCH	163
EXHIBIT 8.4.2 PM-40 DISCONNECT SWITCH	163
EXHIBIT 8.4.3 WESTINGHOUSE TYPE R DISCONNECT SWITCH	164
EXHIBIT 8.4.4 SUBTRANSMISSION DISCONNECT SWITCH REPLACEMENT FACTORS	164
EXHIBIT 8.5.1 CIRCUIT SWITCHERS	165
EXHIBIT 8.5.2 CIRCUIT SWITCHER AGE DISTRIBUTION	166
EXHIBIT 8.5.3 CIRCUIT SWITCHER REPLACEMENT FACTORS.....	166
EXHIBIT 8.6.1 RELAYS DESIGN TYPES	167
EXHIBIT 8.6.2 RELAY DESIGN TYPE BENEFITS AND DRAWBACKS	168
EXHIBIT 8.6.3 RELAY AGE DISTRIBUTION	169
EXHIBIT 8.6.4 RELAY REPLACEMENT FACTORS.....	169
EXHIBIT 8.6.5 RELAY PANEL REPLACEMENTS AS PART OF STATION UPGRADE PROJECTS	170
EXHIBIT 8.6.6 RELAY PANELS	171
EXHIBIT 8.7.1 SWITCHGEAR DESIGN TYPES	171
EXHIBIT 8.7.2 SWITCHGEAR INTERIOR VIEW	172
EXHIBIT 8.7.3 SWITCHGEAR BY VOLTAGE	173
EXHIBIT 8.7.4 SWITCHGEAR AGE DISTRIBUTION.....	173
EXHIBIT 8.7.5 SWITCHGEAR REPLACEMENT FACTORS	174
EXHIBIT 8.7.6 CALVERT BUS.....	175
EXHIBIT 8.8.1 DTEE OWNED WOOD POLE AGE DISTRIBUTION.....	176
EXHIBIT 8.8.2 NON-DTEE OWNED WOOD POLE AGE DISTRIBUTION	176
EXHIBIT 8.8.3 ANNUAL BROKEN POLE EVENTS (FAILURES DURING SERVICE).....	177
EXHIBIT 8.9.1 SMALL WIRE BY VOLTAGE	178
EXHIBIT 8.9.2 SMALL WIRE BY SIZE	179

EXHIBIT 8.10.1 FUSE CUTOUT REPLACEMENT CRITERIA	180
EXHIBIT 8.10.2 S&C R10/R11 PORCELAIN CUTOUT	180
EXHIBIT 8.11.1 TYPICAL LOOP SCHEME DIAGRAM	181
EXHIBIT 8.11.2 THREE-PHASE RECLOSER DESIGN TYPES	182
EXHIBIT 8.11.3 THREE-PHASE RECLOSER TYPE DISTRIBUTION	182
EXHIBIT 8.11.4 THREE-PHASE RECLOSER AGE DISTRIBUTION	183
EXHIBIT 8.11.5 THREE-PHASE RECLOSER REPLACEMENT FACTORS	184
EXHIBIT 8.12.1 SCADA POLE TOP SWITCH TYPES	185
EXHIBIT 8.12.2 SCADA POLE TOP SWITCH TYPE DISTRIBUTION	185
EXHIBIT 8.12.3 SCADA POLE TOP SWITCH AGE DISTRIBUTION	186
EXHIBIT 8.12.4 SCADA POLE TOP SWITCHES REPLACEMENT FACTORS	186
EXHIBIT 8.13.1 40kV AUTOMATIC POLE TOP SWITCHES	188
EXHIBIT 8.13.2 40kV AUTOMATIC POLE TOP SWITCHES BY VINTAGE	189
EXHIBIT 8.13.3 40kV AUTOMATIC POLE TOP SWITCH FAILURE MODES	189
EXHIBIT 8.14.1 OVERHEAD CAPACITOR BANK	190
EXHIBIT 8.14.2 OVERHEAD CAPACITOR BANK VOLTAGE DISTRIBUTION	191
EXHIBIT 8.15.1 OVERHEAD REGULATOR	192
EXHIBIT 8.15.2 OVERHEAD REGULATOR VOLTAGE DISTRIBUTION	192
EXHIBIT 8.16.1 UNDERGROUND SYSTEM CABLE TYPES	193
EXHIBIT 8.16.2 UNDERGROUND SYSTEM CABLE AGES AND LIFE EXPECTANCY	194
EXHIBIT 8.16.3 UNDERGROUND SYSTEM CABLE AGE DISTRIBUTION	194
EXHIBIT 8.16.4 SYSTEM CABLE FAILURES BY YEAR	195
EXHIBIT 8.16.5 SYSTEM CABLE REPLACEMENT FACTORS	195
EXHIBIT 8.16.6 SYSTEM CABLE FAILURE RATES	196
EXHIBIT 8.17.1 URD CABLE AGE DISTRIBUTION	197
EXHIBIT 8.17.2 URD REPLACEMENT FACTORS	198
EXHIBIT 8.17.3 URD FEEDER FAILURE RATE (PER MILE)	199
EXHIBIT 8.17.4 URD CABLE FAILURES BY YEAR	199
EXHIBIT 8.18.1 MANHOLE AND CABLES	201
EXHIBIT 8.18.2 MANHOLE AGE DISTRIBUTION	201
EXHIBIT 8.19.1 UNDERGROUND VAULT AND EQUIPMENT	203
EXHIBIT 8.19.2 BEFORE PICTURES OF A SERVICE VAULT	204
EXHIBIT 8.19.3 AFTER PICTURE OF A SERVICE VAULT	205
EXHIBIT 8.20.1 ELECTRIC METER TYPES	206
EXHIBIT 8.20.2 AMI UPGRADE PROJECTS SUMMARY	210
EXHIBIT 8.20.3 PROJECTED COSTS AND TIMELINE FOR AMI UPGRADE PROJECTS	210

EXHIBIT 9.1.1 CAPITAL REPLACEMENT PROGRAM SUMMARY	213
EXHIBIT 9.1.1.1 POLE AND POLE TOP PROGRAM FUTURE INVESTMENT	216
EXHIBIT 9.1.2.1 APTS REPLACEMENT INVESTMENT	217
EXHIBIT 9.1.3.1 SUBTRANSMISSION DISCONNECT SWITCH REPLACEMENT PROGRAM INVESTMENT	218
EXHIBIT 9.1.4.1 SWITCHER REPLACEMENT PROGRAM INVESTMENT	219
EXHIBIT 9.1.5.1 BREAKER REPLACEMENTS BY TYPE 5-YEAR PLAN.....	219
EXHIBIT 9.1.5.2 BREAKER REPLACEMENT PROGRAM INVESTMENT	220
EXHIBIT 9.1.6.1 SYSTEM CABLE REPLACEMENT PROGRAM INVESTMENT	221
EXHIBIT 9.1.7.1 URD CABLE REPLACEMENT INVESTMENT	222
EXHIBIT 9.1.8.1 SCADA POLE TOP DEVICE REPLACEMENT INVESTMENT	223
EXHIBIT 9.1.9.1 SUBSTATION REGULATOR REPLACEMENT INVESTMENT	224
EXHIBIT 9.2.1.1 POTENTIAL STRANDED LOAD MAP DURING COMPLETE LOSS OF A SUBSTATION	226
EXHIBIT 9.2.1.2 SUBSTATION RISK SCORE DISTRIBUTION	227
EXHIBIT 9.2.1.3 TOP 5 SUBSTATIONS BY RISK RANKING	228
EXHIBIT 9.2.3.1 MOBILE FLEET	231
EXHIBIT 9.2.3.2 PROJECTED INVESTMENTS AND TIMELINE FOR SUBSTATION RISK REDUCTION PROJECTS	232
EXHIBIT 9.2.3.3 PROJECTED INVESTMENT AND TIMELINE FOR STATION UPGRADE PROJECTS	233
EXHIBIT 9.3.3.1 ALL WEATHER SAIFI BEFORE AND AFTER HARDENING	237
EXHIBIT 9.3.3.2 SAIDI EX-MEDS BEFORE AND AFTER HARDENING	237
EXHIBIT 9.3.3.3 WIRE DOWNS BEFORE AND AFTER HARDENING.....	238
EXHIBIT 9.3.4.1 4.8kV MILES HARDENED.....	239
EXHIBIT 9.3.4.2 4.8kV HARDENING PROGRAM INVESTMENT.....	239
EXHIBIT 9.3.5.1 COMPARISON OF REAR-LOT OH CONSTRUCTION ALTERNATIVES	240
EXHIBIT 9.3.5.2 UNOCCUPIED AREAS IN DTEE’S CANIFF SERVICE CENTER AREA.....	242
EXHIBIT 9.4.1 RELIABILITY PROGRAMS SUMMARY	243
EXHIBIT 9.4.2 CUSTOMER EXCELLENCE RELIABILITY IMPACT.....	245
EXHIBIT 9.4.3 EXAMPLE OF SCOPE OF WORK FOR CUSTOMER EXCELLENCE CIRCUITS	247
EXHIBIT 9.4.4 PROJECTED RELIABILITY PROGRAMS CAPITAL INVESTMENT	248
EXHIBIT 9.5.1 PROJECTED 4.8kV RELAY IMPROVEMENTS CAPITAL INVESTMENT	249
EXHIBIT 9.6.1 PROJECTED MOBILE FLEET CAPITAL INVESTMENT	250
EXHIBIT 10.2.1 CNUC BENCHMARK STUDY (34 RESPONDENTS)	253
EXHIBIT 10.2.2 COST OF DEFERRED TREE TRIM	254
EXHIBIT 10.2.3 ILLUSTRATION OF TREE GROWTH IMPACT ON COMPLEXITY BASED ON YEARS SINCE LAST TREE TRIM.....	255
EXHIBIT 10.2.4 TREE-TRIMMING CYCLES.....	255
EXHIBIT 10.3.1 O&M AND CAPITAL COST COMPARISONS.....	257

EXHIBIT 10.3.2 AVERAGE ANNUAL OUTAGE AND NON-OUTAGE EVENTS	257
EXHIBIT 10.4.1 ANNUAL ETPP MILES COMPLETED BY SYSTEM	258
EXHIBIT 10.4.2 ANNUAL O&M TREE TRIM SPEND (MILLIONS)	258
EXHIBIT 10.5.1 OUTAGE EVENT PERFORMANCE	259
EXHIBIT 10.5.2 CUSTOMER INTERRUPTION PERFORMANCE	260
EXHIBIT 10.5.3 CUSTOMER-MINUTE INTERRUPTION PERFORMANCE	260
EXHIBIT 10.5.4 WIRE DOWN EVENT PERFORMANCE	261
EXHIBIT 10.6.1 INNOVATIVE TREE TRIMMING EQUIPMENT	263
EXHIBIT 11.1.1.1 TOP 25 LOAD RELIEF SUBSTATIONS OVERVIEW	270
EXHIBIT 11.1.3.1 TOP 25 LOAD RELIEF SUBSTATION - PROJECT SUMMARY	273
EXHIBIT 11.1.3.2 LOAD RELIEF PROJECTS WITH ADDITIONAL DRIVERS	280
EXHIBIT 11.1.3.3 PROJECTED COSTS AND TIMELINE FOR LOAD RELIEF PROJECTS	283
EXHIBIT 11.1.3.4 PROJECTED LOAD RELIEF CAPITAL INVESTMENT	285
EXHIBIT 11.2.2.1 SUBTRANSMISSION LOAD PRIORITY CRITERIA	289
EXHIBIT 11.2.1.2 TOP 25 SUBTRANSMISSION LOAD RELIEF PROJECT DIMENSIONS OVERVIEW	290
EXHIBIT 11.2.2.1 SUBTRANSMISSION PROJECTS SUMMARY	293
EXHIBIT 11.2.2.2 PROJECTED INVESTMENT AND TIMELINE FOR SUBTRANSMISSION PROJECTS	301
EXHIBIT 11.3.1.1 DTEE DISTRIBUTION SYSTEM VOLTAGE MAP	307
EXHIBIT 11.3.1.2 PERCENTAGE OF SUBSTATIONS, CIRCUITS, AND CIRCUIT MILES BY DISTRIBUTION VOLTAGE	308
EXHIBIT 11.3.1.3 PERCENTAGE OF CUSTOMERS AND AVERAGE ANNUAL TROUBLE EVENTS BY DISTRIBUTION VOLTAGE	308
EXHIBIT 11.3.1.4 SUBSTATION AVERAGE AGE, AVERAGE CIRCUIT LENGTH IN MILES, AND AVERAGE NUMBER OF CUSTOMERS PER CIRCUIT BY DISTRIBUTION VOLTAGE	309
EXHIBIT 11.3.4.1 4.8 kV CONVERSION PROJECTS	313
EXHIBIT 11.3.4.2 PROJECTED COSTS AND TIMELINE FOR 4.8 kV CONVERSION PROJECTS	319
EXHIBIT 11.3.4.4 4.8kV CONVERSION MAP	321
EXHIBIT 11.3.5.1 NUMBER OF CIRCUITS WITH ISO DOWNS AND MILES OF 4.8kV	322
EXHIBIT 11.3.7.1 PROJECTED COSTS AND TIMELINE OF 4.8kV ISO DOWN CONVERSION PROJECTS (\$ MILLIONS)	324
EXHIBIT 11.4.1 DOWNTOWN CITY OF DETROIT INFRASTRUCTURE (CODI) SCOPE AREA	326
EXHIBIT 11.4.2 DOWNTOWN CITY OF DETROIT INFRASTRUCTURE MODERNIZATION (DOWNTOWN CODI)	327
EXHIBIT 11.4.3 CODI PROGRAM SUBSTATION AREAS	330
EXHIBIT 11.4.4 PROJECTED COSTS AND TIMELINE FOR CODI PROJECTS	331
EXHIBIT 11.5.1 PONTIAC AREA SUBSTATIONS	334
EXHIBIT 11.5.2 PONTIAC 8.3 kV OVERHEAD CONVERSION CAPITAL INVESTMENT	335
EXHIBIT 11.6.1 - APPOLINE 1346 PILOT:	337

EXHIBIT 11.6.2: COST OF REBUILDING OVERHEAD REAR-LOT OVERHEAD ASSETS	339
EXHIBIT 11.6.3 –DISTRIBUTION CIRCUIT ILLUSTRATION.....	340
EXHIBIT 11.6.4: STRATEGIC UNDERGROUNDING PROGRAM INVESTMENT	342
EXHIBIT 11.7.2.1 TYPES OF STUDIES PERFORMED TO ASSESS CHANGE IN GENERATION	350
EXHIBIT 12.1 PROJECTED INVESTMENTS AND TIMELINE FOR GRID TECHNOLOGY MODERNIZATION PROGRAMS	353
EXHIBIT 12.1.1 ADMS - AN INTEGRATED TECHNOLOGY PLATFORM	355
EXHIBIT 12.1.2 DTEE 2017 DATA QUALITY GAP ANALYSIS RESULTS	356
EXHIBIT 12.1.3 DISTRIBUTION MANAGEMENT SYSTEM (DMS) APPLICATIONS.....	360
EXHIBIT 12.1.4 ADMS APPLICATIONS AND KEY INTERFACES.....	361
EXHIBIT 12.1.5 ADMS CAPITAL INVESTMENT	361
EXHIBIT 12.2.1 CURRENT AND FUTURE STATE EXAMPLE OF SOC FACILITY	363
EXHIBIT 12.2.2 SOC MODERNIZATION CAPITAL INVESTMENT	364
EXHIBIT 12.3.1 MAP OF EXISTING FIBER SYSTEMS – PRIVATE AND LEASED	368
EXHIBIT 12.3.2 GRID TELECOMMUNICATIONS INVESTMENT SUMMARY	372
EXHIBIT 12.4.1 SENSING AND MONITORING INVESTMENT SUMMARY.....	374
EXHIBIT 12.5.1.1 DISTRIBUTION AUTOMATION CIRCUIT INVESTMENT PLAN	384
EXHIBIT 12.5.2.1 DISTRIBUTION AUTOMATION SUBSTATION INVESTMENT PLAN	387
EXHIBIT 12.6.1 CVR INVESTMENT SUMMARY	392
EXHIBIT 12.7.1 NWA PROJECTS AND OTHER TECHNOLOGY PILOTS INVESTMENT SUMMARY	394
EXHIBIT 12.7.2 TECHNOLOGY PROJECTS USE CASES, TECHNOLOGIES AND OBJECTIVES	395
EXHIBIT 12.7.2.1 – FISHER OVERLOAD AND LOAD RELIEF PORTFOLIO.....	398
EXHIBIT 12.7.2.2 – FISHER NWA PILOT SUMMARY	399
EXHIBIT 12.7.3.1 PORT AUSTIN AREA MAP	400
EXHIBIT 12.7.3.2 PORT AUSTIN LOADING WITH AND WITHOUT SOLAR	401
EXHIBIT 12.7.3.3 – PORT AUSTIN NWA PILOT SUMMARY	402
EXHIBIT 12.7.4.1 – OMEGA NWA PILOT SUMMARY	403
EXHIBIT 12.7.5.1 – VERIDIAN NWA PILOT SUMMARY	405
EXHIBIT 12.8.1 DER CONTROL INVESTMENT SUMMARY.....	409
EXHIBIT 12.9.1 MOBILITY CAPITAL INVESTMENT.....	418
EXHIBIT 12.9.2 WORK MANAGEMENT AND INTEGRATED SCHEDULING CAPITAL INVESTMENT	419
EXHIBIT 12.9.3 ASSET MANAGEMENT CAPITAL INVESTMENT.....	421
EXHIBIT 12.9.4.1.1 INTERCONNECTION PROCESS ENABLEMENT CAPITAL INVESTMENT.....	423
EXHIBIT 12.9.4.2.1 HOSTING CAPACITY ENABLEMENT CAPITAL SPEND	424
EXHIBIT 12.9.4.3.1 NETWORK MODEL CAPITAL INVESTMENT.....	427
EXHIBIT 12.9.4.4.1 DISTRIBUTION LOAD FORECASTING CAPITAL INVESTMENT.....	429
EXHIBIT 12.9.4.5.1 SITUATIONAL AWARENESS ANALYTICS CAPITAL INVESTMENT	430

EXHIBIT 12.9.5.1 MODERNIZE GRID MANAGEMENT CAPITAL INVESTMENT	432
EXHIBIT 12.9.6.1 TECHNOLOGY STRATEGY ROADMAP INVESTMENTS (\$ MILLIONS)	432
EXHIBIT 13.1.1 EMERGENT REPLACEMENT EXPENDITURES (\$ MILLIONS)	435
EXHIBIT 13.1.1.1 CAPITAL SPEND ON EMERGENT REPLACEMENT DURING STORM CONDITIONS (\$ MILLIONS)...	436
EXHIBIT 13.1.1.2 NUMBER OF CAT STORMS 2016 – 2020	437
EXHIBIT 13.1.1.3 NUMBER OF CUSTOMERS IMPACTED 2015 – 2019 (THOUSANDS)	437
EXHIBIT 13.1.2.1 CAPITAL SPEND ON EMERGENT REPLACEMENT DURING NON-STORM CONDITIONS (\$ MILLIONS)	438
EXHIBIT 13.1.2.2 NUMBER OF CUSTOMER INTERRUPTIONS DURING NON-STORM CONDITIONS (THOUSANDS)...	439
EXHIBIT 13.1.3.1 CAPITAL SPEND ON EMERGENT REPLACEMENTS FOR SUBSTATION REACTIVE (\$ MILLIONS) ..	440
EXHIBIT 13.2.1 CONNECTIONS, RELOCATIONS AND OTHER SPEND (\$ MILLIONS)	442
EXHIBIT 14.2.1 PREVENTATIVE MAINTENANCE ACTIVITIES - SUBSTATIONS	449
EXHIBIT 14.2.2 PREVENTATIVE MAINTENANCE ACTIVITIES - OVERHEAD	450
EXHIBIT 14.2.3 PREVENTATIVE MAINTENANCE ACTIVITIES - UNDERGROUND	450
EXHIBIT 14.2.4 PREVENTATIVE MAINTENANCE O&M SPEND PROJECTIONS	450
EXHIBIT 15.3.1 DTEE PBR GOALS AND PLAN	456
EXHIBIT 15.3.2 ILLUSTRATIVE INCENTIVE DESIGN	461
EXHIBIT 15.3.3 ALTERNATIVE REGULATORY MECHANISMS	464
EXHIBIT 16.1.1 DDO STRUCTURAL UPGRADES TO DISTRIBUTION CIRCUITS	468
EXHIBIT 16.1.2 13.2 kV DDO RESTORE-BEFORE-REPAIR DESIGN	469
EXHIBIT A-2 STORM EVENT AND RESTORATION DURATION HISTORY	498
EXHIBIT A.3.1 SAIFI, SAIDI, CAIDI – ALL WEATHER, ALL CAUSES	503
EXHIBIT A.3.2 SAIFI, SAIDI, CAIDI – EXCLUDING MEDs, ALL CAUSES	504
EXHIBIT A.3.3 SAIFI, SAIDI, CAIDI – CATASTROPHIC STORMS, ALL CAUSES	506
EXHIBIT A.3.4 SAIFI, SAIDI, CAIDI – NON-CATASTROPHIC STORMS, ALL CAUSES	507
EXHIBIT A.3.5 SAIFI, SAIDI, CAIDI – BLUE SKY, ALL CAUSES	510
EXHIBIT A.3.6 SAIFI, SAIDI, CAIDI – ALL WEATHER, CAUSE = TREE/WIND	511
EXHIBIT A.3.7 SAIFI, SAIDI, CAIDI – EXCLUDING MEDs, CAUSE = TREE/WIND	513
EXHIBIT A.3.8 SAIFI, SAIDI, CAIDI – CATASTROPHIC STORMS, CAUSE = TREE/WIND	516
EXHIBIT A.3.9 SAIFI, SAIDI, CAIDI – NON-CATASTROPHIC STORMS, CAUSE = TREE/WIND	517
EXHIBIT A.3.10 SAIFI, SAIDI, CAIDI – BLUE SKY, CAUSE = TREE/WIND	519
EXHIBIT A.3.11 SAIFI, SAIDI, CAIDI – ALL WEATHER, CAUSE = EQUIPMENT	521
EXHIBIT A.3.12 SAIFI, SAIDI, CAIDI – EXCLUDING MEDs, CAUSE = EQUIPMENT	523
EXHIBIT A.3.13 SAIFI, SAIDI, CAIDI – CATASTROPHIC STORMS, CAUSE = EQUIPMENT	524
EXHIBIT A.3.14 SAIFI, SAIDI, CAIDI – NON-CATASTROPHIC STORMS, CAUSE = EQUIPMENT	526
EXHIBIT A.3.15 SAIFI, SAIDI, CAIDI – BLUE SKY, CAUSE = EQUIPMENT	528

EXHIBIT A-5 DISTRIBUTION DESIGN STANDARDS AND DISTRIBUTION DESIGN ORDERS	540
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Glossary

ADMS	Advanced Distribution Management System
ALA	Area Load Analysis
AMI	Advanced Metering Infrastructure
APTS	Automatic Pole Top Switch
C&I	Commercial and Industrial
CAIDI	Customer Average Interruption Duration Index
Capacity	Amount of electrical demand that a single piece or group of electrical equipment can deliver based on safety and preservation of the asset
CELIDt	Customers Experiencing Long Interruption Durations of t hours or more
CEMIn	Customers Experiencing Multiple Interruptions of n or more
CHP	Combined Heat and Power
CODI	City of Detroit Infrastructure (Downtown)
Customer 360	DTEE's customer information and billing system
DDO	Distribution Design Orders
DDS	Distribution Design Standards
DER	Distributed Energy Resource
DERMS	Distributed Energy Resource Management Systems
DG	Distributed Generation
DGA	Dissolved Gas Analysis
DO	Distribution Operations Organization of DTE Electric Company
DOE	Department of Energy
DS	Distributed Storage

DTEE	DTE Electric Company
EEI	Edison Electric Institute
E-ISAC	Electricity Information Sharing and Analysis Center
EMS	Emergency Management System
EPR	Ethylene Propylene Rubber
EPRI	Electric Power Research Institute
ETTP	Enhanced Tree Trimming Program
FLISR	Fault Location, Isolation and Service Restoration
Gas Breaker	Circuit breaker where the interrupting arc quenching is done with compressed gas
Gas Cable	Underground cable which requires pressurized gas to maintain the insulation integrity
GIS	Geographical Information System
ICS	Incident Command System
IEEE	Institute of Electrical and Electronics Engineers
ISO	Isolation transformer
Industrial Control System	A general term that encompasses several types of control systems and associated instrumentation used in industrial production technology, including supervisory control and data acquisition (SCADA) systems, distribution management systems (DMS), and other smaller control system configurations often found in the industrial sectors and critical infrastructure
IRP	Integrated Resource Planning
Line losses	Electrical power loss resulting from an electric current passing through a resistive element (e.g., conductor)

Jumpering Point	A location on a distribution circuit in proximity to a second distribution circuit where the two can be electrically tied together
Line Sensors	Devices installed on distribution circuits that provide load and fault data
Manhole	An underground structure for cable pulling and splicing
MED	Major Event Day - defined in IEEE Standard 1366 as any day in which the daily SAIDI exceeds a threshold value
MPSC	Michigan Public Service Commission
NDAS	Network Database and Applications Support
NEETRAC	National Electric Energy Testing, Research and Applications Center
NERC	North American Electric Reliability Corporation
NERC CIP	North American Electric Reliability Corporation Critical Infrastructure Protection
NESC	National Electric Safety Code
Netbank	Distribution network design used in heavy-load-density city areas which provides high reliability
Oil Breaker	Circuit breaker where the interrupting arc quenching is done in oil
O&M	Operation and Maintenance
OMS	Outage Management System
Overload	Electrical demand that exceeds the electrical capacity to serve
PCB	Polychlorinated biphenyl - now considered an environmental contaminant
PERT	Power Equipment and Relay Testing
PI	Data collection software
PILC	Paper in Lead Cable - refers to the type of insulation/jacket on an underground cable
PM	Preventative maintenance - routine scheduled maintenance based on time or number of operations

PON	Power Outage Notification
Primary	Any part of the electrical system energized at 4.8 kV, 8.3 kV, or 13.2 kV
PSSE	Data modeling software
PTMM	Pole Top Maintenance and Modernization
PTS	Pole Top Switch
RBR	Restore Before Repair. It is the practice that customers (load) are transferred to adjacent circuits or substations to restore power before repair can be completed on the failed section of the circuit
Recloser	Sectionalizing device which opens upon detection of fault current
Redundancy	Ability to continue to serve in the event of a contingency condition
Relay	Electrical switch used to initiate operations of other electrical equipment
ReliabilityFirst	One of the eight regional entities that are responsible for ensuring the reliability of the North American bulk power system under Federal Energy Regulatory Commission approved delegation agreements with the North American Electric Reliability Corporation, pursuant to the Energy Policy Act of 2005. DTEE's service territory is in ReliabilityFirst region
RM	Reactive maintenance resulting from a misoperation or malfunction
ROW	Right-of-Way
RTU	Remote Terminal Unit that sends or receives telemetry data to or from a master control
SAIDI	System Average Interruption Duration Index
SAIFI	System Average Interruption Frequency Index
SCADA	Supervisory Control and Data Acquisition
Secondary	Any part of the electrical system energized at 120/240 volts
Service	The conductor / cable and equipment that connects a customer to the electrical system

SFI	Smart Fault Indicators
SOC	System Operations Center
STDF	Subtransmission Distribution Facility
Stranded load	Under contingency conditions, electrical demand that cannot be readily served through available jumpering or mobile generation
Subtransmission	Any part of the electrical system energized at 24 kV, 40 kV, 120 kV, or higher
Substation	A facility of the electrical power grid that allows for the connection and/or switching of circuits and/or the transformation of voltage from one level to another
TARA	Data modeling software
Through Fault	A fault occurring on the secondary side of a power transformer which may damage the insulation of the transformer
TIE (Tie line)	A subtransmission circuit that interconnects two or more substations with power flow normally from any of the substations
TRK (Trunk line)	A radial subtransmission circuit with power flow normally in one direction to serve substation or individual customer loads at 24 kV or 40 kV
TR-XLPE cable	Tree retardant cross-linked polyethylene - refers to the type of insulation on an underground cable
URD	Underground Residential Distribution
Vacuum Breaker	Circuit breaker where the interrupting arc quenching is done in a vacuum
Vault	An underground structure for cable pulling and splicing that also contains power equipment such as transformers and switches
VCL Cable	Varnished Cambric Lead - refers to the type of insulation/jacket on an underground cable
XLPE cable	Cross-linked polyethylene - refers to the type of insulation on an underground cable

1.0 Capital Category Name Changes

As projects progress sometimes there is a need to change the name to better reflect updated scope, and sometimes the Company expands on existing program or project and changes the name or category of spend to better align with these changes. In order to avoid confusion and be transparent with such changes, the Company has chosen to identify major changes in the capital categories between MPSC Case No. U-20561 and the instant case.

1.1 High Level Categorization (Exhibit A-12 Schedule B5.4 Pgs. 1 and 2)

The Company continues to classify investments as either Base Capital programs or Strategic Capital programs which are further broken down in to subcategories. Within the subcategories there have been two changes in Exhibit A-12 Schedule B5.4: one, the final phase Public Lighting Department Project, which was not part of U-20561, is now included in Distribution Operations investments and has been added in Base Capital; two, the name of the Strategic Capital category Infrastructure Redesign has been changed to Infrastructure Redesign and Moderization to reflect the Company's priority to upgrade and update the grid to meet the needs of the future. Figure 1 shows Exhibit A-12, Schedule B5.4 Pages 1 and 2 Description with changes highlighted.

Figure 1: Exhibit A-12, Schedule B5.4 Pgs. 1 and 2 Description

Line No.	Description
1	<u>Base Capital Programs</u>
2	Emergent Replacements
3	Storm
4	Non - Storm
5	Substation Reactive
6	Emergent Replacement Reduction Based on Strategic Spend
7	Subtotal Emergent Replacements
8	Customer Connections, Relocations & Other
9	Connections and New Load
10	Relocations
11	Electric System Equipment
12	NRUC and Improvement Blankets
13	General Plant, Tools & Equipment and Miscellaneous
14	Public Lighting Department Project
15	Subtotal Customer Connections, Relocations & Other
16	Customer Advances for Construction
17	Total Base Capital Programs
18	<u>Strategic Capital Programs</u>
19	Infrastructure Resilience and Hardening
20	Infrastructure Redesign and Modernization
21	Technology and Automation
22	Subtotal Strategic Capital Programs
23	Total Capital
24	<u>Regulatory Asset</u>
25	Advanced Distribution Management System (ADMS)

1.2 Emergent Replacements (Exhibit A-12 Schedule B5.4 Pg. 3)

As noted in direct testimony, Emergent Replacements (storm, non-storm, and substation reactive) are now being forecasted using a three-year historic average instead of the five-year historic average method used in MPSC Case No. U-20561.

This page supports investments in emergent replacements and are in line with those proposed in MPSC Case no. U-20561, Exhibit A-12, Schedule B5.4, Page 3.

1.3 Connections, Relocations and Other (Exhibit A-12 Schedule B5.4 Pgs. 4 and 5)

Public Lighting Department project (PLD) was added to Base Capital as noted above and was added to Exhibit A-12 Schedule B5.4 Page 5 Line 39. This was done as described in direct testimony and as supported in Exhibit A-23 Schedule M3.

These pages support investments in new business, relocations, and basic tools and equipment and are in line with those proposed in MPSC Case No. U-20561, Exhibit A-12, Schedule B5.4, Page 4.

1.4 Prior Year's Projects (Exhibit A-12 Schedule B5.4 Pgs. 6 and 7)

As in MPSC Case No. U-20561, these pages support the historic year's New Business and Relocations.

1.5 Infrastructure Resilience and Hardening (Exhibit A-12 Schedule B5.4 Pg. 8)

As in MPSC Case No. U-20561 this page supports investments in programs and projects that meet some of the fundamental grid needs as described in direct testimony and the DGP. Investments on this page are in line with those proposed in MPSC Case No. U-20561, Exhibit A-12, Schedule B5.4, Page 7.

A change to this category is the addition of Station Upgrade projects (Line No. 25 to 28). In MPSC Case No. U-20561 these projects were considered Infrastructure Redesign Modernization projects and listed on MPSC Case No. U-20561, Exhibit A-12, Schedule B5.4, Page 8.

1.6 Infrastructure Redesign and Modernization (Exhibit A-12 Schedule B5.4 Pgs. 9 and 10)

As in MPSC Case No. U-20561 these pages support investments in programs and projects that are often driven by need capacity additions in growth areas or areas with overloaded equipment and grid modernization needs, as discussed in direct testimony and the DGP. Investments on these pages are in line with those proposed in MPSC Case No. U-20561, Exhibit A-12, Schedule B5.4, Page 8.

Notable change to this category is the heading, in MPSC Case No. U-20561 the category was labeled Infrastructure Redesign. The Company now considers modernization to be a key factor in these investments and has renamed the category to reflect this: Infrastructure Redesign and Modernization.

Additionally, Subtransmission Redesign projects were renamed to Subtransmission Redesign & Rebuild to reflect the need to reconstruct the subtransmission system.

1.7 Technology and Automation (Exhibit A-12 Schedule B5.4 Pg. 11)

As in MPSC Case No. U-20561, technology and automatin programs and projects are tighly linked to grid modernization, as discussed in direct testimony and the DGP. Investments on this page are in line with those proposed in MPSC Case No. U-20561, Exhibit A-12, Schedule B5.4, Page 9.

Notable change to this category is the SOC Modernization project, as labeled in MPSC Case No. U-20561, Exhibit A-12, Schedule B5.4, Page 9, Line No. 2, has now been separated in to two projects: SOC: ESOC and SOC: ASOC. Support for these projects is provided by the Company
Witness: Elliott Andahazy.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 1 of 19

(a)

(b)

Category	Distribution Plant Capital Project Detail - Base Capital
	Capital Exhibit:
pages: 2-3 of 19	Exhibit A-12, Schedule B5.4, Page 3 of 12, Lines 2-24 Emergent Replacements
pages: 4-5 of 19	Exhibit A-12, Schedule B5.4, Page 4 of 12, Lines 2-5, Small Load Growth Blankets & Customer Connections
pages: 6-7 of 19	Exhibit A-12, Schedule B5.4, Page 4 of 12, Lines 6-8, New Business Projects
pages: 8-9 of 19	Exhibit A-12, Schedule B5.4, Page 4 of 12, Lines 13-21, Relocations
pages: 12-13 of 19	Exhibit A-12, Schedule B5.4, Page 4 of 12, Lines 23-26, Electric System Equipment
pages: 12-13 of 19	Exhibit A-12, Schedule B5.4, Page 5 of 12, Lines 28-33, NRUC and Improvement Blankets
pages: 14-15 of 19	Exhibit A-12, Schedule B5.4, Page 5 of 12, Line 34, General Plant, Tools & Equipment
pages: 16-17 of 19	Exhibit A-12, Schedule B5.4, Page 5 of 12, Lines 2-42, Public Lighting of Detroit
page: 18 of 19	Exhibit A-12, Schedule B5.4, Page 6 of 12, Lines 2-42, Prior Year's New Business Projects
page: 19 of 19	Exhibit A-12, Schedule B5.4, Page 7 of 12, Lines 2-9, Prior Year's Relocation Projects

**Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital**

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 2 of 19

(a)

(b)

Program:	Emergent Replacements (Pg. 1 of 2)						
Purpose and Necessity:	Spend required to perform emergency replacement work for retirement units on the overhead and underground distribution and subtransmission systems, and substation equipment. Drivers of this work include, but are not limited to, damage from storms, wind, trees, ice, and equipment failures.						
Category:	Emergent Replacements						
Line Number:	Exhibit A-12, Schedule B5.4, Page 3, Lines 2-24						
DGP Section:	13.1						
Scope:	• Reactive replacement of broken or damaged equipment						
Customer benefit / Effect on cost of operation and reliability:	• Restore power to customers in a timely fashion during outage events • Remove hazards from the field after outage events and restore the electrical system to normal operating conditions						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Substation Outage Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	X				X		
Current Projects:	• Work performed to address storm events • Work performed to address non-storm events • Work performed to address emergent substation events						

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 3 of 19

(a) (b)

Program:	Emergent Replacements (Pg. 2 of 2)		
Cost:	Estimated program spend in 2021-2023: \$1,356 million The program will continue beyond the Test Year		
Test Year (\$000's):			
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23
Total Labor (DTE Electric & Contractors)	54,080	238,938	293,018
Material	14,421	63,717	78,138
Other	3,605	15,929	19,534
Subtotal	72,106	318,584	390,690
Reduction based on Strategic Spend	(10,148)	(8,831)	(18,979)
Total	61,958	309,753	371,711

**Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital**

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 4 of 19

(a)

(b)

Program:	Small Load Growth Projects & Customer Connections (Pg. 1 of 2)						
Purpose and Necessity:	Small projects (<\$350k) required to serve new load, connections, or to address loading issues on the electric system.						
Category:	Connections, Relocations & Other						
Line Number:	Exhibit A-12, Schedule B5.4, Page 4, Lines 2-5						
DGP Section:	13.2.1						
Scope:	• Reconductor lines, expand substations or transfer load to provide small load relief • Install additional overhead and underground lines to provide electric services to small commercial business or housing developments						
Customer benefit / Effect on cost of operation and reliability:	• Connect new customers • Support area load growth for existing and new customers • Enable economic growth in the service territory • Improve customer satisfaction						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Substation Outage Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:		X					
Current Projects:	Estimated test year volumes: • 3,300 line extensions • 16,700 new services						

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 5 of 19

(a) (b)

Program:	Small Load Growth Projects & Customer Connections (Pg. 2 of 2)		
Cost:	Estimated program spend in 2021-2023: \$294 million (net of CIAC) The program will continue beyond the Test Year		
Test Year (\$000's):			
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23
Total Labor (DTE Electric & Contractors)	24,340	80,293	104,634
Material	4,868	16,059	20,927
Other	3,245	10,706	13,951
Subtotal	32,453	107,058	139,511
CIAC	(12,940)	(6,657)	(19,597)
Total	19,513	100,400	119,914

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 6 of 19

(a)

(b)

Program:	New Business Projects (Pg. 1 of 2)						
Purpose and Necessity:	Large projects (\$>350k) required to serve new load.						
Category:	Connections, Relocations & Other						
Line Number:	Exhibit A-12, Schedule B5.4, Page 4, Lines 6-8						
DGP Section:	13.2.1						
Scope:	• Install additional overhead and underground lines, upgrade or build new substations to provide service to large commercial and industrial customers						
Customer benefit / Effect on cost of operation and reliability:	• Connect new customers • Support area load growth for existing and new customers • Enable economic growth in the service territory • Improve customer satisfaction						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Substation Outage Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:		X					

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 7 of 19

(a) (b)

Program:	New Business Projects (Pg. 2 of 2)																														
Current Projects:	• Quaker Transformer Upgrade • New business fed from Hood Substation • New business hospital expansion • New business szephyr sub (allen park) • New business data center • New business health system campus • New business hospital Class I substation • New business dairy MOS • New business Lombard subs • New business kern substation upgrades • New business CJC CUP • New business 120kV feed • New residential load • New business health grocery store & hotel • New business compersion station • Brush park development																														
Cost:	Estimated program spend in 2021-2023: \$79 million (net of CIAC) The program will continue beyond the Test Year																														
Test Year (\$000's):	<table> <tr> <th></th><th>2 mos. 11/01/22 12/31/22</th><th>10 mos. 01/01/23 10/31/23</th><th>Test Year 12 mos. ending 10/31/23</th></tr> <tr> <td>Total Labor (DTE Electric & Contractors)</td><td>3,973</td><td>20,439</td><td>24,412</td></tr> <tr> <td>Material</td><td>795</td><td>4,088</td><td>4,882</td></tr> <tr> <td>Other</td><td>530</td><td>2,725</td><td>3,255</td></tr> <tr> <td>Subtotal</td><td>5,297</td><td>27,252</td><td>32,549</td></tr> <tr> <td>CIAC</td><td>(833)</td><td>(4,284)</td><td>(5,117)</td></tr> <tr> <td>Total</td><td>4,464</td><td>22,968</td><td>27,432</td></tr> </table>				2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	3,973	20,439	24,412	Material	795	4,088	4,882	Other	530	2,725	3,255	Subtotal	5,297	27,252	32,549	CIAC	(833)	(4,284)	(5,117)	Total	4,464	22,968	27,432
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Total Labor (DTE Electric & Contractors)	3,973	20,439	24,412																												
Material	795	4,088	4,882																												
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Total	4,464	22,968	27,432																												

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 8 of 19

(a)

(b)

Program:	Relocations (Pg. 1 of 2)						
Purpose and Necessity:	Requests from customers and municipalities to relocate existing facilities.						
Category:	Connections, Relocations & Other						
Line Number:	Exhibit A-12, Schedule B5.4, Page 4, Lines 13-21						
DGP Section:	13.2.2						
Scope:	Relocation of DTE Electric facilities due to construction or modification of roadways, water mains, public sanitation, alleys, or at the request of municipalities or customers						
Customer benefit / Effect on cost of operation and reliability:	- Enable economic growth in the service territory - Meet customer requests as required by regulation						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Substation Outage Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:			X				

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 9 of 19

(a) (b)

Program:	Relocations (Pg. 2 of 2)																														
Current Projects:	• Small Relocation Projects (Blanket) • Gordie Howe International Bridge • New Boston Relocation • Relocations for rail right of way		• Warren Road Relocation • Cass I-94 Bridge Relocation • Mound Road Relocation																												
Cost:	Estimated program spend in 2021-2023: \$78 million (net of CIAC) The program will continue beyond the Test Year																														
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>3,440</td><td>17,059</td><td>20,499</td></tr><tr><td>Material</td><td>688</td><td>3,412</td><td>4,100</td></tr><tr><td>Other</td><td>459</td><td>2,275</td><td>2,733</td></tr><tr><td>Subtotal</td><td>4,587</td><td>22,746</td><td>27,332</td></tr><tr><td>CIAC</td><td>(341)</td><td>(1,755)</td><td>(2,096)</td></tr><tr><td>Total</td><td>4,246</td><td>20,991</td><td>25,236</td></tr></table>				2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	3,440	17,059	20,499	Material	688	3,412	4,100	Other	459	2,275	2,733	Subtotal	4,587	22,746	27,332	CIAC	(341)	(1,755)	(2,096)	Total	4,246	20,991	25,236
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																												
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Subtotal	4,587	22,746	27,332																												
CIAC	(341)	(1,755)	(2,096)																												
Total	4,246	20,991	25,236																												

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 10 of 19

(a)

(b)

Program:	Electric System Equipment (Pg. 1 of 2)						
Purpose and Necessity:	Spend required for large transformers and other equipment needed for emergent replacements, customer connections and relocations and to provide proper inventory level for major electric system equipment, particularly for long lead time items.						
Category:	Connections, Relocations & Other						
Line Number:	Exhibit A-12, Schedule B5.4, Page 4, Lines 23-26						
DGP Section:	13.2.3						
Scope:	• Purchase major equipment, i.e. substation transformers • Purchase and install meters						
Customer benefit / Effect on cost of operation and reliability:	• Make equipment readily available to serve customers						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Substation Outage Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:		X			X		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 11 of 19

(a) (b)

Program:	Electric System Equipment (Pg. 2 of 2)																						
Current Projects:	Estimated test year volumes: • ~30 large transformers • ~34,000 meters																						
Cost:	Estimated program spend in 2019-2021: \$165 million The program will continue beyond the Test Year																						
Test Year (\$000's):	<table> <tr> <th></th><th>2 mos. 11/01/22 12/31/22</th><th>10 mos. 01/01/23 10/31/23</th><th>Test Year 12 mos. ending 10/31/23</th></tr> <tr> <td>Total Labor (DTE Electric & Contractors)</td><td>1,694</td><td>8,716</td><td>10,411</td></tr> <tr> <td>Material</td><td>4,540</td><td>23,360</td><td>27,900</td></tr> <tr> <td>Other</td><td>542</td><td>2,789</td><td>3,331</td></tr> <tr> <td>Total</td><td>6,777</td><td>34,865</td><td>41,642</td></tr> </table>				2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	1,694	8,716	10,411	Material	4,540	23,360	27,900	Other	542	2,789	3,331	Total	6,777	34,865	41,642
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																				
Total Labor (DTE Electric & Contractors)	1,694	8,716	10,411																				
Material	4,540	23,360	27,900																				
Other	542	2,789	3,331																				
Total	6,777	34,865	41,642																				

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 12 of 19

(a)

(b)

Program:	NRUC and Improvement Blankets (Pg. 1 of 2)						
Purpose and Necessity:	This category includes several types of small projects aimed at addressing planned replacements, improvements to the electrical system, software applications for field use and animal mitigation.						
Category:	Connections, Relocations & Other						
Line Number:	Exhibit A-12, Schedule B5.4, Page 4, Lines 28-33						
DGP Section:	13.2.4						
Scope:	• System Improvements: small system improvement blankets to address small and localized reliability issues • Normal Retirement Unit Changeout (NRUC): projects to perform scheduled work for normal retirement unit change out items on the subtransmission and distribution systems, for example the replacement of reclosers and switches determined to be at end-of-life during routine preventive maintenance inspections • Batteries and Chargers: replace substation batteries and chargers which are at end-of-life • Operational Technology: development of software applications for field use • Animal Mitigation: install bus insulator squirrel guards and other equipment on transformers to mitigate animals interfering with the operations of the transformers						
Customer benefit / Effect on cost of operation and reliability:	• Projects are focused on improving operating conditions to reduce the frequency and duration of outage events						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Substation Outage Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	X				X		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 13 of 19

(a) (b)

Program:	NRUC and Improvement Blankets (Pg. 2 of 2)																						
Current Projects:	• System Improvements • Normal Retirement Unit Changeouts (NRUC) • Batteries and Chargers • Operational Technologies • Animal Mitigation																						
Cost:	Estimated program spend in 2021-2023: \$84 million The program will continue beyond the Test Year																						
Test Year (\$000's):	<table> <tr> <th></th><th>2 mos. 11/01/22 12/31/22</th><th>10 mos. 01/01/23 10/31/23</th><th>Test Year 12 mos. ending 10/31/23</th></tr> <tr> <td>Total Labor (DTE Electric & Contractors)</td><td>4,066</td><td>20,920</td><td>24,986</td></tr> <tr> <td>Material</td><td>508</td><td>2,615</td><td>3,123</td></tr> <tr> <td>Other</td><td>508</td><td>2,615</td><td>3,123</td></tr> <tr> <td>Total</td><td>5,082</td><td>26,150</td><td>31,232</td></tr> </table>				2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	4,066	20,920	24,986	Material	508	2,615	3,123	Other	508	2,615	3,123	Total	5,082	26,150	31,232
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																				
Total Labor (DTE Electric & Contractors)	4,066	20,920	24,986																				
Material	508	2,615	3,123																				
Other	508	2,615	3,123																				
Total	5,082	26,150	31,232																				

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 14 of 19

(a) (b)

Program:	General Plant, Tools & Equipment and Miscellaneous (Pg. 1 of 2)						
Purpose and Necessity:	Capital expenditures for tools and test equipment required to support field resources.						
Category:	Connections, Relocations & Other						
Line Number:	Exhibit A-12, Schedule B5.4, Page 4, Line 34						
DGP Section:	13.2.5						
Scope:	• Purchase tools and test equipment						
Customer benefit / Effect on cost of operation and reliability:	• Provide employees required tools and test equipment to support field work						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Substation Outage Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	X						

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 15 of 19

(a) (b)

Program:	General Plant, Tools & Equipment and Miscellaneous (Pg. 2 of 2)																						
Current Projects:	• Tools and equipment for linemen and splicer apprentices • Replacement tools for linemen, splicers, electric maintenance journeyman • Test equipment for engineers and technicians																						
Cost:	Estimated program spend in 2021-2023: \$31 million The program will continue beyond the Test Year																						
Test Year (\$000's):	<table> <tr> <th></th><th>2 mos. 11/01/22 12/31/22</th><th>10 mos. 01/01/23 10/31/23</th><th>Test Year 12 mos. ending 10/31/23</th></tr> <tr> <td>Total Labor (DTE Electric & Contractors)</td><td>309</td><td>1,141</td><td>1,450</td></tr> <tr> <td>Material</td><td>1,546</td><td>5,703</td><td>7,249</td></tr> <tr> <td>Other</td><td>206</td><td>760</td><td>966</td></tr> <tr> <td>Total</td><td>2,061</td><td>7,604</td><td>9,665</td></tr> </table>				2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	309	1,141	1,450	Material	1,546	5,703	7,249	Other	206	760	966	Total	2,061	7,604	9,665
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																				
Total Labor (DTE Electric & Contractors)	309	1,141	1,450																				
Material	1,546	5,703	7,249																				
Other	206	760	966																				
Total	2,061	7,604	9,665																				

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 15 of 19

(a)

(b)

Program:	Public Lighting Department Program (Pg. 1 of 2)						
Purpose and Necessity:	These capital investments support the transition of Public Lighting Department (PLD) customers to the Company's distribution system.						
Category:	Connections, Relocations & Other						
Line Number:	Exhibit A-12, Schedule B5.4, Page 5, Line 39						
DGP Section:	Project coming to an end, not part of the long term plan						
Scope:	Extending the Company's overhead and underground distribution system to transition customers from the PLD system to the Company's distribution system						
Customer benefit / Effect on cost of operation and reliability:	Improve safety and reliability in the area of the former PLD distribution system						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Substation Outage Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	X		X		X		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 16 of 19

(a) (b)

Program:	Public Lighting Department Program (Pg. 2 of 2)																						
Current Projects:	Conversion of former PLD customers to the Company's distribution system will be complete before the start of the projected test period.																						
Cost:	Estimated program spend in 2021-2023: \$29 million The program will not continue beyond the Test Year																						
Test Year (\$000's):	<table> <tr> <th></th><th>2 mos. 11/01/22 12/31/22</th><th>10 mos. 01/01/23 10/31/23</th><th>Test Year 12 mos. ending 10/31/23</th></tr> <tr> <td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr> <tr> <td>Material</td><td>-</td><td>-</td><td>-</td></tr> <tr> <td>Other</td><td>-</td><td>-</td><td>-</td></tr> <tr> <td>Total</td><td>-</td><td>-</td><td>-</td></tr> </table>				2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total	-	-	-
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																				
Total Labor (DTE Electric & Contractors)	-	-	-																				
Material	-	-	-																				
Other	-	-	-																				
Total	-	-	-																				

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 18 of 19

(a)

(b)

Program:	Prior Year's New Business Projects
Purpose and Necessity:	Large projects (\$>350k) required to serve new load.
Category:	Connections, Relocations & Other
Line Number:	Exhibit A-12, Schedule B5.4, Page 6, Lines 2-59
Scope:	Completed capital projects to connect new customer load
Customer benefit / Effect on cost of operation and reliability:	- Connect new customers - Support area load growth for existing and new customers - Meet economic growth in the service territory - Improve customer satisfaction
Current Projects:	Listed in Exhibit A-12, Schedule B5.4, page 6
Cost:	2020 spend \$30 million (net of CIAC)
Test Year (\$000's):	Not Applicable

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Base Capital

Case No.: U-20836
Exhibit: A-23
Schedule: M3
Witness: S. G. Pfeuffer
Page: 19 of 19

(a)

(b)

Program:	Prior Year's Relocation Projects
Purpose and Necessity:	Large projects (\$>350k) for requests from customers and municipalities to relocate existing facilities.
Category:	Connections, Relocations & Other
Line Number:	Exhibit A-12, Schedule B5.4, Page 7, Lines 2-30
Scope:	Relocation of DTE Electric facilities due to construction or modification of roadways, water mains, public sanitation, alleys, or at the request of municipalities or customers
Customer benefit / Effect on cost of operation and reliability:	- Enable economic growth in the service territory - Meet customer requests as required by regulation
Current Projects:	Listed in Exhibit A-12, Schedule B5.4, page 7
Cost:	2020 spend \$10 million (net of CIAC)
Test Year (\$000's):	Not Applicable

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 1 of 101

(a) Category	(b) Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening
	Capital Exhibit:
pages: 2-5 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 2 Mobile Fleet Program
pages: 6-9 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 3 Substation Risk: Drexel
pages: 10-13 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 4 Substation Risk: Chestnut
pages: 14-17 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 5 Substation Risk: Savage
pages: 18-21 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 6 Substation Risk: Apache
pages: 22-25 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 7 Substation Risk: Port Huron
pages: 26-29 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 8 Substation Risk: Belleville Switchgear Decommission
pages: 30-33 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 9 4.8 kV Hardening
pages: 34-37 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 10 Pole and Pole Top Hardware
pages: 38-41 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 11 & 12 Cable Replacement Program
pages: 42-45 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 13 Frequent Outage Program (CEMI) including Circuit Renewal
pages: 46-49 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 14 System Resiliency - Efficient Frontier
pages: 50-53 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 15 Breaker Replacement Program
pages: 54-57 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 16 Pontiac Vaults
pages: 58-61 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 17 & 18 URD Replacement Program
pages: 62-65 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 19 4.8 kV Relay Improvements (Delta Ground Detection Program)
pages: 62-65 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 20 40 kV: Automatic Pole Top Switch
pages: 66-69 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 21 Disconnect and Switcher Replacement
pages: 70-73 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 22 SCADA Pole Top Device Replacement
pages: 74-77 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 23 Substation Regulator Replacement
pages: 78-81 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 24 Steel Pole Highway Crossings
pages: 86-89 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 25 Station Upgrade: Warren (Relay Replacement)
pages: 90-93 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 26 Station Upgrade: Northeast (Relay Replacement)
pages: 94-97 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 27 Station Upgrade: Lincoln
pages: 98-101 of 101	Exhibit A-12, Schedule B5.4, Page 8 of 12, Line 28 Station Upgrade: Navarre

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 2 of 101

(a)	(b)
Program:	Mobile Fleet Program (Pg. 1 of 4)
Purpose and Necessity:	Mobile Fleet program (aka portable equipment) provides timely restoration to customers stranded during major outage events. The Mobile Fleet program also enables system shut downs during planned capital and O&M work to minimize customers affected. It will increase existing fleet and would allow for the proper execution of the PM program - including addressing the 46% backlog of single tap substations. This does not include unplanned failures or obsolescence of the existing fleet of portable substations.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 3 of 101

(a)	(b)
Program:	Mobile Fleet Program (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 2 : 9.6
DGP Section:	9.6
Scope:	Portable equipment/DG • Acquire a portable battery trailer 1MW x 4MWH for use as a non-wire alternative and test validate storage use cases. Purchase (4) additional units to support NWA projects at Malta, Fisher and Omega substations • Commissioning of switchgear trailer Portable Substation: • To purchase enough portable substations to supplement existing fleet and replace some of the existing units to allow for maintenance of DTE single tap substations (per our policy). Currently, there are 118 single tap substations and 46% of them are due or overdue for inspection.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 4 of 101

(a)	(b)						
Program:	Mobile Fleet Program (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Mitigate major substation outage risk by reducing customer restoration time during major substation events • Improve system operability under contingency conditions or planned work • Allow for the proper execution of the preventative maintenance program						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
				x	x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 5 of 101

(a)	(b)																				
Program:	Mobile Fleet Program (Pg. 4 of 4)																				
Current Projects:	Mobile Fleet Program: Portable Equip / DG Mobile Fleet Program: Portable Substation																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$11.5 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>560</td><td>2,066</td><td>2,626</td></tr><tr><td>Material</td><td>112</td><td>413</td><td>525</td></tr><tr><td>Other</td><td>75</td><td>275</td><td>350</td></tr><tr><td>Total</td><td>747</td><td>2,754</td><td>3,501</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	560	2,066	2,626	Material	112	413	525	Other	75	275	350	Total	747	2,754	3,501
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	560	2,066	2,626																		
Material	112	413	525																		
Other	75	275	350																		
Total	747	2,754	3,501																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 6 of 101

(a)

(b)

Program:	Substation Risk: Drexel (Pg. 1 of 4)
Purpose and Necessity:	Drexel experienced a failure in position 'I' which caused major damage to surrounding positions. As a result, Bus 11 and Bus 12 were decommissioned and load was jumpered to surrounding substations.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 7 of 101

(a)

(b)

Program:	Substation Risk: Drexel (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 3
DGP Section:	9.2
Scope:	• Decommission and remove the damaged switchgear lineup • Install nine position switchgear • Install one 15/20/25 MVA transformer to feed the new switchgear • Install 6900 feet of EPR cable from new switchgear lineup to existing cable poles • Restore circuits to pre-fire configurations

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 8 of 101

(a)	(b)						
Program:	Substation Risk: Drexel (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce substation outage risk by replacing aging infrastructure and failed equipment						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x			x			

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 9 of 101

(a)	(b)																				
Program:	Substation Risk: Drexel (Pg. 4 of 4)																				
Current Projects:	Substation Risk: Drexel																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$1.9 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>106</td><td>-</td><td>106</td></tr><tr><td>Material</td><td>21</td><td>-</td><td>21</td></tr><tr><td>Other</td><td>14</td><td>-</td><td>14</td></tr><tr><td>Total</td><td>141</td><td>-</td><td>141</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	106	-	106	Material	21	-	21	Other	14	-	14	Total	141	-	141
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	106	-	106																		
Material	21	-	21																		
Other	14	-	14																		
Total	141	-	141																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 10 of 101

(a)	(b)
Program:	Substation Risk: Chestnut (Pg. 1 of 4)
Purpose and Necessity:	Chestnut substation is located in Madison Heights and is at risk for failure. It is a two switchgear lineup that is connected in the same building, creating more of a risk in the event of a major failure. A switchgear failure could result in loss of all switchgear positions and a potential of stranded load of up to 32 MVA.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 11 of 101

(a)	(b)
Program:	Substation Risk: Chestnut (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 4
DGP Section:	9.2
Scope:	• Replace existing 19-position PDC with (2) 12-position PDCs • Install two capacitor banks • Install 500 feet of 15-5" concrete encased duct bank • Install four new manholes • Install 10,000 feet 1000 AL EPR cable & associated branch and straight joints

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 12 of 101

(a)	(b)						
Program:	Substation Risk: Chestnut (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce substation outage risk by replacing aging infrastructure and failed equipment • Reduction in O&M cost due to removal of aging infrastructure • Provide sufficient capacity needed to serve fast growing Detroit core downtown areas						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x			x			

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 13 of 101

(a)	(b)		
Program:	Substation Risk: Chestnut (Pg. 4 of 4)		
Current Projects:	Substation Risk: Chestnut		
Budget Basis:	• Engineering Estimate		
Cost:	Estimated project spend in 2021-2023: \$4.5 M This project is expected to continue beyond 2023		
Test Year (\$000's):	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23
	Total Labor (DTE Electric & Contractors)	63	1,833
	Material	13	367
	Other	8	244
	Total	83	2,444
			2,527

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 14 of 101

(a)	(b)
Program:	Substation Risk: Savage (Pg. 1 of 4)
Purpose and Necessity:	Savage substation has high substation outage risk. The stranded load is estimated at 30 MVA after possible load transfer and mobile fleet deployments. This project is to address the substation outage risk at Savage substation. In addition, it will establish means for jumper point connections to circuits from neighboring circuits in similar conditions.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 15 of 101

(a)	(b)
Program:	Substation Risk: Savage (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 5
DGP Section:	9.2
Scope:	Stranded Load Scope • Extend 770 feet of conduit and 3,489 feet of cable to feed seven new primary switch cabinets and overhead connections • Extend overhead 770 feet to establish two portable ready locations • Install two sets of 3-333KVA regulators

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 16 of 101

(a)	(b)						
Program:	Substation Risk: Savage (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce substation outage risk by replacing aging infrastructure and failed equipment • Enhance safety and operability						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x			x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 17 of 101

(a)	(b)																				
Program:	Substation Risk: Savage (Pg. 4 of 4)																				
Current Projects:	Substation Risk: Savage Stranded Load																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.2 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>5</td><td>-</td><td>5</td></tr><tr><td>Material</td><td>1</td><td>-</td><td>1</td></tr><tr><td>Other</td><td>1</td><td>-</td><td>1</td></tr><tr><td>Total</td><td>7</td><td>-</td><td>7</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	5	-	5	Material	1	-	1	Other	1	-	1	Total	7	-	7
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	5	-	5																		
Material	1	-	1																		
Other	1	-	1																		
Total	7	-	7																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 18 of 101

(a)	(b)
Program:	Substation Risk: Apache (Pg. 1 of 4)
Purpose and Necessity:	In 2015, a breaker position failure at Apache Substation caused a significant area-wide outage for the entire substation area. The downtown business district of Troy and the surrounding commercial/residential area were without electrical service for several days. This switchgear was classified as at risk and needs to be replaced to prevent a second switchgear failure. A switchgear failure could result in loss of all switchgear positions and a potential of stranded load of up to 16 MVA.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 19 of 101

(a)	(b)
Program:	Substation Risk: Apache (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 6
DGP Section:	9.2
Scope:	• Replace existing 19 position switchgear with two 12 position switchgear • Replace three circuit switchers with three new S&C 2020 circuit switchers • Build system cable termination in new switchgear for all 13 distribution circuits • Decommission/remove existing 19 position switchgear once all load is cut over to new switchgear • Install 325 feet of 12-5" conduit on the north side of the substation • Build one two-way manhole • Build two three-way manholes • Remove 9,100 feet of 1000 EPR 1CX3 cable (including straight and branch joints) • Install 9,950 feet of 1000 EPR 1CX3 cable (including straight and branch joints) • Cable testing, manhole cleaning/pumping, and, if necessary, asbestos abatement

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 20 of 101

(a)	(b)						
Program:	Substation Risk: Apache (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce substation outage risk due to switchgear failure and stranded load upwards to 16.0 MVA by replacing aging infrastructure and failed equipment • Future opportunity to relieve Apache Substation circuits that exceed day to day circuit limits						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x			x	x	x	

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 21 of 101

(a)	(b)																				
Program:	Substation Risk: Apache (Pg. 4 of 4)																				
Current Projects:	Substation Risk: Apache																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$18.10 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>1,062</td><td>5,766</td><td>6,828</td></tr><tr><td>Material</td><td>212</td><td>1,153</td><td>1,366</td></tr><tr><td>Other</td><td>142</td><td>769</td><td>910</td></tr><tr><td>Total</td><td>1,416</td><td>7,688</td><td>9,104</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	1,062	5,766	6,828	Material	212	1,153	1,366	Other	142	769	910	Total	1,416	7,688	9,104
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	1,062	5,766	6,828																		
Material	212	1,153	1,366																		
Other	142	769	910																		
Total	1,416	7,688	9,104																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 22 of 101

(a)	(b)
Program:	Substation Risk: Port Huron (Pg. 1 of 4)
Purpose and Necessity:	Port Huron substation has exposed energized equipment causing safety and operating concerns. Work is not possible unless the equipment is deenergized. Jumpering options are very limited, making it difficult to deenergize the equipment and still serve customers. Energized equipment extends beyond the rope barrier. Along with the safety concerns at the substation, the breakers are at risk.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 23 of 101

(a)	(b)
Program:	Substation Risk: Port Huron (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 7
DGP Section:	9.2
Scope:	• Install customized nine position indoor, non-walk in, 15kV metal-clad switchgear line-up on main floor of Port Huron Substation building • Decommission and remove existing breakers, disconnects and bus work • Install distribution primary cable from new switchgear to first manhole • Maintain current distribution circuit configurations

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 24 of 101

(a)	(b)						
Program:	Substation Risk: Port Huron (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Eliminate safety issues and operating constraints • Reduce substation outage risk by replacing aging infrastructure and failed equipment						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x			x			x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 25 of 101

(a)	(b)																				
Program:	Substation Risk: Port Huron (Pg. 4 of 4)																				
Current Projects:	Substation Risk: Port Huron																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$2.9 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>1</td><td>-</td><td>1</td></tr><tr><td>Material</td><td>0</td><td>-</td><td>0</td></tr><tr><td>Other</td><td>0</td><td>-</td><td>0</td></tr><tr><td>Total</td><td>1</td><td>-</td><td>1</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	1	-	1	Material	0	-	0	Other	0	-	0	Total	1	-	1
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	1	-	1																		
Material	0	-	0																		
Other	0	-	0																		
Total	1	-	1																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 26 of 101

(a)

(b)

Program:	Substation Risk: Belleville Switchgear Decommission (Pg. 1 of 4)
Purpose and Necessity:	Belleville Substation 4.8kV switchgear is at end-of-life and there are no replacements for breakers if they were to fail. Belleville is currently without the automatic throw over. The control wiring in every position is falling apart and requires weeks of work to re-wire.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 27 of 101

(a)	(b)
Program:	Substation Risk: Belleville Switchgear Decommission (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 8
DGP Section:	9.2
Scope:	• This work will bypass BELVL's 4.8kV switchgear by installing overhead NOVA Triple-Single reclosers just outside of the substation. The 4.8kV portion of BELVL substation will be converted from a Class 'A' to a Class 'T' operating at 4.8kV. • Reconfigure the 3 load carrying circuits and throw over circuit into 2 load carrying circuits

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 28 of 101

(a)	(b)						
Program:	Substation Risk: Belleville Switchgear Decommission (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce substation outage risk by eliminating aging infrastructure						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:		X		X	X		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 29 of 101

(a)	(b)																				
Program:	Substation Risk: Belleville Switchgear Decommission (Pg. 4 of 4)																				
Current Projects:	Substation Risk: Belleville Switchgear Decommission																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$1.6 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>142</td><td>-</td><td>142</td></tr><tr><td>Material</td><td>28</td><td>-</td><td>28</td></tr><tr><td>Other</td><td>19</td><td>-</td><td>19</td></tr><tr><td>Total</td><td>189</td><td>-</td><td>189</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	142	-	142	Material	28	-	28	Other	19	-	19	Total	189	-	189
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	142	-	142																		
Material	28	-	28																		
Other	19	-	19																		
Total	189	-	189																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 30 of 101

(a)	(b)
Program:	4.8 kV Hardening (Pg. 1 of 4)
Purpose and Necessity:	The 4.8 kV Hardening Program is a program designed to harden and stabilize the 4.8 kV distribution system to enhance safety, reliability and storm resiliency, and extend the life of the 4.8 kV until DTE Electric completes conversion to 13.2 kV over the next several decades as supported primarily by load growth or cost reductions. Additionally, there has been a lack of proper maintenance within the City of Detroit's alleys, which over time have become overgrown and filled with debris, impeding access and making maintenance activities difficult and costly, which the 4.8 kV Hardening Program will help address.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 31 of 101

(a)	(b)
Program:	4.8 kV Hardening (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 9
DGP Section:	9.3
Scope:	• Test all utility poles that have Company equipment attached and replace or reinforce those poles as needed • Replace wooden cross arms with fiberglass cross arms • Remove Detroit Public Lighting Department (DPLD) arc wire from Company-owned equipment and ensure the remaining Company wires are left in a safe configuration • Remove DPLD distribution wire from Company-owned equipment when it can be confirmed that the wire is not serving customers • Remove service lines to abandoned properties • Trim trees as required to support construction activities • Perform any additional necessary work as dictated by field conditions • Conduct pilot projects to explore and assess alternatives to existing rear lot overhead configurations. The pilots include converting rear lot overhead construction to underground residential distribution construction; and removing primary overhead construction and adding secondary overhead construction to serve sparsely populated areas. The pilots aim at improving reliability and reducing trouble events in some areas of the City of Detroit where inaccessible overhead construction in rear lots is encroached by fence lines and overgrown trees.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 32 of 101

(a)	(b)						
Program:	4.8 kV Hardening (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Avoid 61 M customer minute interruptions annually by 2026 • Avoid \$4 M costs annually for both circuit tree related and non-tree related reactive capital by 2026 • Extend the life of the 4.8 kV system, allowing the deferral of more expensive conversions to 13.2 kV						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x		x		x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 33 of 101

(a)	(b)			
Program:	4.8 kV Hardening (Pg. 4 of 4)			
Current Projects:	4.8 kV Hardening			
Budget Basis:	• Ranked 3rd based on the Global Prioritization Model • Spend based on engineering estimates to harden approximately 50% of the circuits in the City of Detroit, within 10 years			
Cost:	Estimated project spend in 2021-2023: \$298.1 M This project is expected to continue beyond 2023			
Test Year (\$000's):	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	
	Total Labor (DTE Electric & Contractors)	14,482	71,250	85,732
	Material	2,896	14,250	17,146
	Other	1,931	9,500	11,431
	Total	19,310	95,000	114,310

(a) (b)

Program:	Pole and Pole Top Hardware (Pg. 1 of 4)																						
Purpose and Necessity:	DTE Electric owns more than 1 millions distribution and subtransmission poles and attaches to nearly 200,000 poles owned by other utilities. The average pole age in the DTEE system is approximately 46 years. The life expectancy of a pole is 40 years for wood pine poles and 50 years for wood cedar poles. The Pole and Pole Top Hardware programs are designed to identify and remediate poles with deteriorating strength and identify and replace defective pole top hardware prior to causing outages. DTEE Owned Wood Pole Age Distribution <table border="1"> <thead> <tr> <th>Age in Years</th> <th>Percentage</th> </tr> </thead> <tbody> <tr> <td>00-10</td> <td>6.0%</td> </tr> <tr> <td>11-20</td> <td>10.4%</td> </tr> <tr> <td>21-30</td> <td>15.5%</td> </tr> <tr> <td>31-40</td> <td>9.8%</td> </tr> <tr> <td>41-50</td> <td>14.5%</td> </tr> <tr> <td>51-60</td> <td>14.6%</td> </tr> <tr> <td>61-70</td> <td>12.7%</td> </tr> <tr> <td>71-80</td> <td>8.1%</td> </tr> <tr> <td>81-90</td> <td>6.6%</td> </tr> <tr> <td>91+</td> <td>1.8%</td> </tr> </tbody> </table>	Age in Years	Percentage	00-10	6.0%	11-20	10.4%	21-30	15.5%	31-40	9.8%	41-50	14.5%	51-60	14.6%	61-70	12.7%	71-80	8.1%	81-90	6.6%	91+	1.8%
Age in Years	Percentage																						
00-10	6.0%																						
11-20	10.4%																						
21-30	15.5%																						
31-40	9.8%																						
41-50	14.5%																						
51-60	14.6%																						
61-70	12.7%																						
71-80	8.1%																						
81-90	6.6%																						
91+	1.8%																						
Category:	Infrastructure Resilience & Hardening																						

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 35 of 101

(a)	(b)
Program:	Pole and Pole Top Hardware (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 10
DGP Section:	9.1
Scope:	• Perform foot patrols to visually and physically inspect poles for signs of damage or rot, and to test mechanical strength • Reinforce or replace poles based on testing results • Perform foot patrols to visually inspect pole top hardware for signs of defective or damaged equipment • Replace defective or damaged pole top hardware

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 36 of 101

(a)	(b)						
Program:	Pole and Pole Top Hardware (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Comply with MPSC recommendation of inspecting poles on a 10-12 year cycle • Enhance safety • Avoid 40 M customer minute interruptions annually by 2026 • Avoid \$4.8 M costs annually for both circuit tree related and non-tree related reactive capital by 2026						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x		x		x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 37 of 101

(a)	(b)																				
Program:	Pole and Pole Top Hardware (Pg. 4 of 4)																				
Current Projects:	Pole and Pole Top Equipment (including Fuse Cutouts)																				
Budget Basis:	• Pole Replacement program and Pole Top Hardware program ranked 4th based on the Global Prioritization Model • Adherence to the industry's best practice of 10-12 year inspection cycle																				
Cost:	Estimated project spend in 2021-2023: \$185.9 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>7,364</td><td>58,438</td><td>65,802</td></tr><tr><td>Material</td><td>1,473</td><td>11,688</td><td>13,160</td></tr><tr><td>Other</td><td>982</td><td>7,792</td><td>8,774</td></tr><tr><td>Total</td><td>9,819</td><td>77,917</td><td>87,735</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	7,364	58,438	65,802	Material	1,473	11,688	13,160	Other	982	7,792	8,774	Total	9,819	77,917	87,735
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	7,364	58,438	65,802																		
Material	1,473	11,688	13,160																		
Other	982	7,792	8,774																		
Total	9,819	77,917	87,735																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 38 of 101

(a)

(b)

Program:	Cable Replacement Program (Pg. 1 of 4)																																								
Purpose and Necessity:	DTE Electric distribution and subtransmission systems have around 3,200 miles of underground system cable. System cable is a critical component for both the subtransmission and distribution systems. A failure can interrupt a large number of customers for an extended period of time and remove important operational contingency. Approximately 43% of the system cable is identified as candidate for replacement. Underground System Cable Ages and Life Expectancy <table> <tr> <th>Cable Type</th> <th>PILC</th> <th>EPR</th> <th>VCL</th> <th>Gas</th> <th>XLPE Post 1990 (Tree retardant)</th> <th>XLPE Pre 1990 (Non-tree retardant)</th> <th>BUYTL</th> </tr> <tr> <td>Miles</td> <td>2,365</td> <td>570</td> <td>125</td> <td>55</td> <td>35</td> <td>69</td> <td>35</td> </tr> <tr> <td>% of Total population</td> <td>72.7%</td> <td>17.5%</td> <td>3.8%</td> <td>1.7%</td> <td>1.1%</td> <td>2.1%</td> <td>1.1%</td> </tr> <tr> <td>Average age</td> <td>50</td> <td>16</td> <td>59</td> <td>54</td> <td>20</td> <td>36</td> <td>52</td> </tr> <tr> <td>Life Expectancy</td> <td>40</td> <td>35</td> <td>40</td> <td>40</td> <td>40</td> <td>25</td> <td>20</td> </tr> </table>	Cable Type	PILC	EPR	VCL	Gas	XLPE Post 1990 (Tree retardant)	XLPE Pre 1990 (Non-tree retardant)	BUYTL	Miles	2,365	570	125	55	35	69	35	% of Total population	72.7%	17.5%	3.8%	1.7%	1.1%	2.1%	1.1%	Average age	50	16	59	54	20	36	52	Life Expectancy	40	35	40	40	40	25	20
Cable Type	PILC	EPR	VCL	Gas	XLPE Post 1990 (Tree retardant)	XLPE Pre 1990 (Non-tree retardant)	BUYTL																																		
Miles	2,365	570	125	55	35	69	35																																		
% of Total population	72.7%	17.5%	3.8%	1.7%	1.1%	2.1%	1.1%																																		
Average age	50	16	59	54	20	36	52																																		
Life Expectancy	40	35	40	40	40	25	20																																		
Category:	Infrastructure Resilience & Hardening																																								

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 39 of 101

(a)	(b)
Program:	Cable Replacement Program (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 11/ Exhibit A-12, Schedule B5.4, Page 8, Line 12
DGP Section:	9.1
Scope:	• Prioritize and replace system cable based on multiple factors including cable type, vintage, previous failure history, impacts to the system, cable loading, and critical infrastructure customers • Remove underground hazards and replace defective conduits if needed

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 40 of 101

(a)	(b)						
Program:	Cable Replacement Program (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Enhance safety • Reduce major substation outage risk • Avoid 0.6 M customer minute interruptions annually by 2026 • Avoid \$1.8 M costs annually for both circuit tree related and non-tree related reactive capital by 2026						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x			x	x		x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 41 of 101

(a)	(b)																				
Program:	Cable Replacement Program (Pg. 4 of 4)																				
Current Projects:	Cable Replacement Program Cable Replacement: I-696/Dequindre Overpass Cable Replacement Program: Evergreen VCL TRK 7104 Cable Replacement Cable Replacement: Harsen's Island																				
Budget Basis:	• Ranked 29th based on the Global Prioritization Model • Avoid further acceleration of system cable failures																				
Cost:	Estimated project spend in 2021-2023: \$92.8 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>4,391</td><td>27,625</td><td>32,016</td></tr><tr><td>Material</td><td>878</td><td>5,525</td><td>6,403</td></tr><tr><td>Other</td><td>585</td><td>3,683</td><td>4,269</td></tr><tr><td>Total</td><td>5,854</td><td>36,834</td><td>42,688</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	4,391	27,625	32,016	Material	878	5,525	6,403	Other	585	3,683	4,269	Total	5,854	36,834	42,688
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	4,391	27,625	32,016																		
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Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 42 of 101

(a)	(b)
Program:	Frequent Outage Program (CEMI) including Circuit Renewal (Pg. 1 of 4)
Purpose and Necessity:	The Frequent Outage program, also known as the CEMI (Customers Experiencing Multiple Interruptions) program, includes improvements to either circuit sections (customer pockets) or entire circuits. The number of customers experiencing either sustained or momentary outages, historical reliability performance, power quality complaints, and MPSC complaints are considered in determining inclusion in the Frequent Outage program.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 43 of 101

(a)	(b)
Program:	Frequent Outage Program (CEMI) including Circuit Renewal (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 13
DGP Section:	9.4
Scope:	Work scope may include: • Rebuild / reconductor / relocate overhead lines • Replace condemned poles and other pole top hardware • Install sectionalizing and switching devices • Add or strengthen circuit ties • Underground portion of the circuits

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 44 of 101

(a)	(b)						
Program:	Frequent Outage Program (CEMI) including Circuit Renewal (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Enhance safety • Avoid 75 M customer minute interruptions annually by 2026 • Avoid \$0.2 M costs annually for both circuit tree related and non-tree related reactive capital by 2026						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:			x		x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 45 of 101

(a)	(b)																												
Program:	Frequent Outage Program (CEMI) including Circuit Renewal (Pg. 4 of 4)																												
Current Projects:	Frequent Outage Program: Plymouth 4.8 kV CC: Globe 8034 Power Quality Frequent Outage Program: Customer Excellence Frequent Outage Program: CEMI Overloaded Transformers Pilot 2020 LOC DL Delphi Reliability Frequent Outage Program: Lark - Spruce 2024																												
Budget Basis:	• Target number of circuits and engineering estimates of cost per circuit																												
Cost:	Estimated project spend in 2021-2023: \$98.2 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>5,755</td><td>13,796</td><td>19,551</td></tr><tr><td>Material</td><td>1,151</td><td>2,759</td><td>3,910</td></tr><tr><td>Other</td><td>767</td><td>1,840</td><td>2,607</td></tr><tr><td>Total</td><td>7,673</td><td>18,395</td><td>26,068</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	5,755	13,796	19,551	Material	1,151	2,759	3,910	Other	767	1,840	2,607	Total	7,673	18,395	26,068
	2 mos.	10 mos.	Test Year																										
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Total	7,673	18,395	26,068																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 46 of 101

(a)	(b)
Program:	System Resiliency - Efficient Frontier (Pg. 1 of 4)
Purpose and Necessity:	Circuits selected for the System Resiliency Program are circuits with low penetration of switching and sectionalizing devices and little opportunity to localize outage events to perform Restore Before Repair (RBR). The target of the system resiliency program is to reduce long-outage durations for circuits with few sectionalizing points.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 47 of 101

(a)

(b)

Program:	System Resiliency - Efficient Frontier (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 14
DGP Section:	9.4
Scope:	• Install sectionalizing and switching devices to localize outage events and enable restore before repair

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 48 of 101

(a)	(b)						
Program:	System Resiliency - Efficient Frontier (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Improved customer reliability • Remediate long-outage durations for circuits with few sectionalizing points						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
					x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 49 of 101

(a)	(b)																				
Program:	System Resiliency - Efficient Frontier (Pg. 4 of 4)																				
Current Projects:	System Resiliency - Efficient Frontier																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.0 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td>-</td><td>-</td><td>-</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total	-	-	-
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	-	-	-																		
Material	-	-	-																		
Other	-	-	-																		
Total	-	-	-																		

(a)	(b)																				
Program:	Breaker Replacement Program (Pg. 1 of 4)																				
Purpose and Necessity:	DTE Electric has approximately 6,000 circuit breakers. Out of the total population, 58% are candidates for replacement due to factors such as equipment condition, age, lack of available parts (no support from vendors), and environmental concerns (oil spills). A failure of a circuit breaker can cause outages on multiple circuits and could reduce system redundancy for an extended period of time during repairs. Approximately 60% of DTE Electric's circuit breakers are beyond industry life expectancy and should also be replaced over time. DTE works to extend asset life as much as possible through its preventative maintenance programs and uses equipment condition, not just age, to determine which breakers to replace and when. The industry life expectancy of a circuit breaker is approximately 40 years for early to mid-20th century equipment and 30 years for modern equipment. Graph below illustrates Circuit Breaker Age Distribution. Circuit Breaker Age Distribution <table border="1"> <thead> <tr> <th>Age in Years</th> <th>Percentage</th> </tr> </thead> <tbody> <tr> <td>00-10</td> <td>16.0%</td> </tr> <tr> <td>11-20</td> <td>15.0%</td> </tr> <tr> <td>21-30</td> <td>6.0%</td> </tr> <tr> <td>31-40</td> <td>6.0%</td> </tr> <tr> <td>41-50</td> <td>12.0%</td> </tr> <tr> <td>51-60</td> <td>9.0%</td> </tr> <tr> <td>61-70</td> <td>30.0%</td> </tr> <tr> <td>71-80</td> <td>5.0%</td> </tr> <tr> <td>90+</td> <td>2.0%</td> </tr> </tbody> </table>	Age in Years	Percentage	00-10	16.0%	11-20	15.0%	21-30	6.0%	31-40	6.0%	41-50	12.0%	51-60	9.0%	61-70	30.0%	71-80	5.0%	90+	2.0%
Age in Years	Percentage																				
00-10	16.0%																				
11-20	15.0%																				
21-30	6.0%																				
31-40	6.0%																				
41-50	12.0%																				
51-60	9.0%																				
61-70	30.0%																				
71-80	5.0%																				
90+	2.0%																				
Category:	Infrastructure Resilience & Hardening																				

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 51 of 101

(a)	(b)
Program:	Breaker Replacement Program (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 15
DGP Section:	9.1
Scope:	• Remove and replace at-risk breakers including • H breakers - 30 • Subtransmission breakers - 97 • Substation reclosers - 24 • Distribution breakers - 116

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 52 of 101

(a)	(b)						
Program:	Breaker Replacement Program (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Enhance safety • Reduce major substation outage risk • Improve customer reliability • Avoid 2.9 M customer minute interruptions annually by 2026 • Avoid \$2.8 M costs annually for both circuit tree related and non-tree related reactive capital by 2026 • Avoid further acceleration of breaker failures • Address most at risk breakers through aggressive prioritization						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x			x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 53 of 101

(a)	(b)																				
Program:	Breaker Replacement Program (Pg. 4 of 4)																				
Current Projects:	Essex 24kV H-Breaker Decom & Bus Consolidation Breaker Replacement Program																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$45.1 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>1,930</td><td>11,250</td><td>13,180</td></tr><tr><td>Material</td><td>386</td><td>2,250</td><td>2,636</td></tr><tr><td>Other</td><td>257</td><td>1,500</td><td>1,757</td></tr><tr><td>Total</td><td>2,573</td><td>15,000</td><td>17,573</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	1,930	11,250	13,180	Material	386	2,250	2,636	Other	257	1,500	1,757	Total	2,573	15,000	17,573
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	1,930	11,250	13,180																		
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Other	257	1,500	1,757																		
Total	2,573	15,000	17,573																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 54 of 101

(a)	(b)
Program:	Pontiac Vaults (Pg. 1 of 4)
Purpose and Necessity:	DTE Electric acquired from CMS Energy the 8.3KV Distribution System serving the City of Pontiac (late 1970's). The underground system contained fifteen system switching vaults. These (confined space) vaults, containing obsolete overhead equipment, are typically located beneath city sidewalks with surface mounted grates. Shock hazard to personnel entering the vaults and inadequate arc flash distance make it difficult or impossible to operate within the vaults. Restoration time to an outage event was considerably extended due to system complexity and hazardous work environment conditions. DTE Electric eliminated three vaults and upgraded ten vaults by 2020. The remaining two vaults will be completed by March 2021. The project objective is complete redesign of system configuration, vault equipment replacement and structural reconditioning, and replacement of prevalent undersized, end of life cables.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 55 of 101

(a)	(b)
Program:	Pontiac Vaults (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 16
DGP Section:	11.5
Scope:	• Redesign underground system configuration • Remove live front overhead equipment from the below grade vaults and replace with dead front (arc flash contained) submersible equipment • Upgrade / repair structural condition of these vaults • Replace 39,167 feet of paper and lead cable with 500 EPR cable • Install SCADA communication / monitoring equipment

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 56 of 101

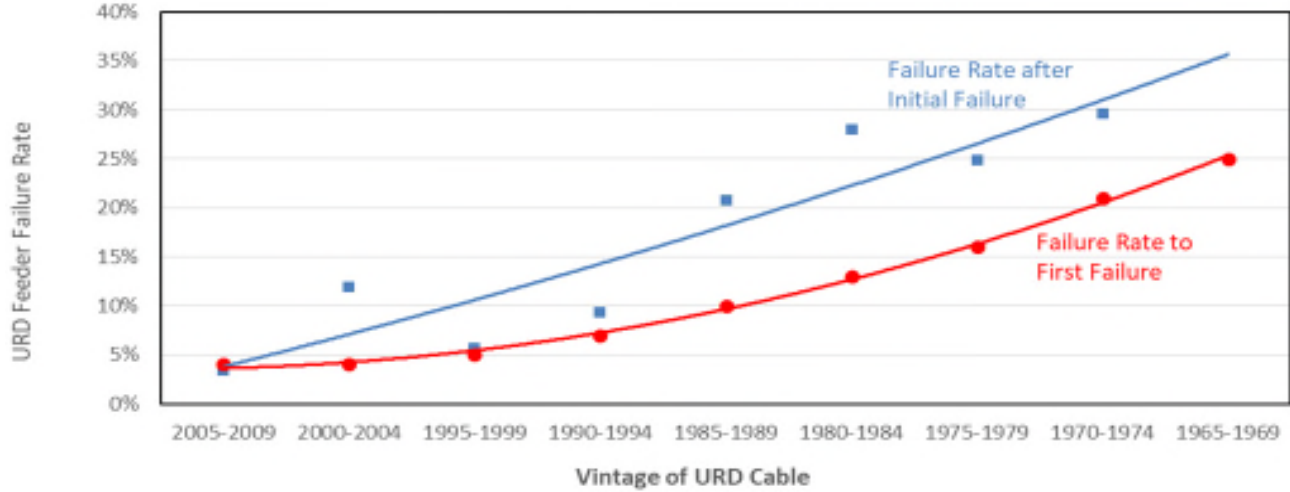
(a)	(b)						
Program:	Pontiac Vaults (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Enhance safety • Improve operability • Replace aging infrastructure and improve reliability • Reduce reactive cost by replacing obsolete, special order equipment						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x				x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 57 of 101

(a)	(b)		
Program:	Pontiac Vaults (Pg. 4 of 4)		
Current Projects:	Pontiac Vaults		
Budget Basis:	• Engineering Estimate		
Cost:	Estimated project spend in 2021-2023: \$4.3 M		
Test Year (\$000's):	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23
	Total Labor (DTE Electric & Contractors)	77	-
	Material	15	-
	Other	10	-
	Total	103	-
			103

(a) (b)

Program:	URD Replacement Program (Pg. 1 of 4)																														
Purpose and Necessity:	DTE Electric has approximately 11,000 miles of underground residential distribution (URD) cable, of which approximately 41% (4,500 miles) is cross-linked polyethylene (XLPE) cable installed prior to 1985. This cable is prone to high failure rates due to a manufacturing flaw known as "water treeing"; which is a breakdown of the insulation that allows water to enter the cable and causes faults. All cable insulation will breakdown eventually as URD cable is typically direct buried and more susceptible to environmental interference. As URD cable ages, the number of failures goes up as shown in the graph below. In addition, after the URD loop has experienced one failure, the risk for a second failure also goes up. The average age of DTE Electric's URD cable is 28 years. URD Feeder Failure Rate (per mile) <table><caption>URD Feeder Failure Rate Data (Estimated from Graph)</caption><thead><tr><th>Vintage of URD Cable</th><th>Failure Rate to First Failure (%)</th><th>Failure Rate after Initial Failure (%)</th></tr></thead><tbody><tr><td>2005-2009</td><td>~4%</td><td>~4%</td></tr><tr><td>2000-2004</td><td>~4%</td><td>~12%</td></tr><tr><td>1995-1999</td><td>~5%</td><td>~6%</td></tr><tr><td>1990-1994</td><td>~7%</td><td>~9%</td></tr><tr><td>1985-1989</td><td>~10%</td><td>~21%</td></tr><tr><td>1980-1984</td><td>~13%</td><td>~28%</td></tr><tr><td>1975-1979</td><td>~16%</td><td>~25%</td></tr><tr><td>1970-1974</td><td>~21%</td><td>~30%</td></tr><tr><td>1965-1969</td><td>~25%</td><td>~35%</td></tr></tbody></table>	Vintage of URD Cable	Failure Rate to First Failure (%)	Failure Rate after Initial Failure (%)	2005-2009	~4%	~4%	2000-2004	~4%	~12%	1995-1999	~5%	~6%	1990-1994	~7%	~9%	1985-1989	~10%	~21%	1980-1984	~13%	~28%	1975-1979	~16%	~25%	1970-1974	~21%	~30%	1965-1969	~25%	~35%
Vintage of URD Cable	Failure Rate to First Failure (%)	Failure Rate after Initial Failure (%)																													
2005-2009	~4%	~4%																													
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1980-1984	~13%	~28%																													
1975-1979	~16%	~25%																													
1970-1974	~21%	~30%																													
1965-1969	~25%	~35%																													
Category:	Infrastructure Resilience & Hardening																														

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 59 of 101

(a)	(b)
Program:	URD Replacement Program (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 17/ Exhibit A-12, Schedule B5.4, Page 8, Line 18
DGP Section:	9.1
Scope:	• Prioritize and replace pre-1985 non-tree retardant URD cable based on multiple factors including previous failure history, impacts to the system and customers, loading, and partial discharge testing results • Replace live front transformers with dead front (arc flash contained)

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 60 of 101

(a)	(b)						
Program:	URD Replacement Program (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Enhance safety by replacing live front transformers with dead front transformers • Avoid 5 M customer minute interruptions annually by 2026 • Avoid \$4.4 M costs annually for both circuit tree related and non-tree related reactive capital by 2026						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x				x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 61 of 101

(a)	(b)																												
Program:	URD Replacement Program (Pg. 4 of 4)																												
Current Projects:	URD Replacement Program URD Replacement Program: PLY 86 URD Replacement Program: Outer Drive 1299 URD Replacement Program: Detroit URD 1-1																												
Budget Basis:	• Ranked 41st based on the Global Prioritization Model • Manage URD cable failures and avoid an acceleration of asset failures																												
Cost:	Estimated project spend in 2021-2023: \$21.2 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>754</td><td>6,250</td><td>7,004</td></tr><tr><td>Material</td><td>151</td><td>1,250</td><td>1,401</td></tr><tr><td>Other</td><td>100</td><td>833</td><td>934</td></tr><tr><td>Total</td><td>1,005</td><td>8,333</td><td>9,338</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	754	6,250	7,004	Material	151	1,250	1,401	Other	100	833	934	Total	1,005	8,333	9,338
	2 mos.	10 mos.	Test Year																										
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Total Labor (DTE Electric & Contractors)	754	6,250	7,004																										
Material	151	1,250	1,401																										
Other	100	833	934																										
Total	1,005	8,333	9,338																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 62 of 101

(a)	(b)
Program:	4.8 kV Relay Improvements (Delta Ground Detection Program) (Pg. 1 of 4)
Purpose and Necessity:	Reconfigure alarm panel to allow System Operations Center (SOC) to automatically detect and receive alerts of wire down events. This will allow quick response to these events.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 63 of 101

(a)	(b)
Program:	4.8 kV Relay Improvements (Delta Ground Detection Program) (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 19
DGP Section:	9.5
Scope:	• Install and/or upgrade telecommunication and Remote Terminal Unit (RTU) for substation remote monitoring • Reconfigure alarm panels at 4.8 kV substations • Connect system to FCN network to provide data to the Systems Operation Center • Upgrade cable trays as needed

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 64 of 101

(a)	(b)						
Program:	4.8 kV Relay Improvements (Delta Ground Detection Program) (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Enhance safety • Allow SOC to automatically detect and receive alerts of wire down events and isolate the ground alarms from other station alarms						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x						

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 65 of 101

(a)	(b)																				
Program:	4.8 kV Relay Improvements (Delta Ground Detection Program) (Pg. 4 of 4)																				
Current Projects:	4.8 kV Relay Improvements (Delta Ground Detection Program)																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$14.3 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>384</td><td>5,313</td><td>5,697</td></tr><tr><td>Material</td><td>77</td><td>1,063</td><td>1,139</td></tr><tr><td>Other</td><td>51</td><td>708</td><td>760</td></tr><tr><td>Total</td><td>513</td><td>7,083</td><td>7,596</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	384	5,313	5,697	Material	77	1,063	1,139	Other	51	708	760	Total	513	7,083	7,596
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	384	5,313	5,697																		
Material	77	1,063	1,139																		
Other	51	708	760																		
Total	513	7,083	7,596																		

(a)	(b)												
Program:	40 kV: Automatic Pole Top Switch (Pg. 1 of 4)												
Purpose and Necessity:	DTE Electric has approximately 144 automatic pole top switches (APTS) with control boxes on the 40 kV subtransmission system. In recent years, the failure rate of these switches has been increasing, ranging from 15 percent to 20 percent annually. A failure of one of these switches has the potential to interrupt thousands of customers or result in significant operational constraints. The entire 40 kV APTS population is considered for replacement to address high failure rates and unavailable spare parts for early vintages. The following graph shows the vintages of the APTS in the Company's system. 40 kV Automatic Pole Top Switches by Vintage <table border="1"> <thead> <tr> <th>Vintage</th> <th>Percentage</th> </tr> </thead> <tbody> <tr> <td>Elpeco R&IE</td> <td>5%</td> </tr> <tr> <td>S&C Heavy Duty S&C Triangle S&C Horizontal</td> <td>74%</td> </tr> <tr> <td>S&C R9</td> <td>8%</td> </tr> <tr> <td>S&C R10</td> <td>8%</td> </tr> <tr> <td>SEECO</td> <td>5%</td> </tr> </tbody> </table>	Vintage	Percentage	Elpeco R&IE	5%	S&C Heavy Duty S&C Triangle S&C Horizontal	74%	S&C R9	8%	S&C R10	8%	SEECO	5%
Vintage	Percentage												
Elpeco R&IE	5%												
S&C Heavy Duty S&C Triangle S&C Horizontal	74%												
S&C R9	8%												
S&C R10	8%												
SEECO	5%												
Category:	Infrastructure Resilience & Hardening												

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 67 of 101

(a)

(b)

Program:	40 kV: Automatic Pole Top Switch (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 20
DGP Section:	9.1
Scope:	• Replace the entire population of the obsolete 40 kV APTSs with modern SCADA controlled equipment

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 68 of 101

(a)	(b)						
Program:	40 kV: Automatic Pole Top Switch (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Improve reliability by effectively sectionalizing subtransmission lines and localizing faults • Provide SCADA capability • Improve system shutdown capability and allow for safe operation and maintenance • Avoid 20 M customer minute interruptions annually by 2026 • Avoid \$1 M costs annually for both circuit tree related and non-tree related reactive capital by 2026						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:					X		X

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 69 of 101

(a)	(b)																				
Program:	40 kV: Automatic Pole Top Switch (Pg. 4 of 4)																				
Current Projects:	40 kV: Automatic Pole Top Switch Sub transmission OH Bypass for PTS Maintenance																				
Budget Basis:	• Engineering estimate																				
Cost:	Estimated project spend in 2021-2023: \$13.1 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>663</td><td>3,938</td><td>4,600</td></tr><tr><td>Material</td><td>133</td><td>788</td><td>920</td></tr><tr><td>Other</td><td>88</td><td>525</td><td>613</td></tr><tr><td>Total</td><td>883</td><td>5,250</td><td>6,133</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	663	3,938	4,600	Material	133	788	920	Other	88	525	613	Total	883	5,250	6,133
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
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Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 70 of 101

(a)	(b)
Program:	Disconnect and Switcher Replacement (Pg. 1 of 4)
Purpose and Necessity:	Disconnects: Subtransmission disconnect switches are used to sectionalize and provide isolation points on the electrical system for operational reasons or for service / maintenance. Failures of disconnect switches during operation, when operators attempt to open or close a disconnect manually, can lead to safety concerns, reduce system operability, and force additional equipment to be taken out of service to allow critical work to continue. DTE Electric is replacing at risk subtransmission disconnect switches. Switchers: Series 2000 circuit switchers are installed for primary side transformer protection on 120kV systems. Failures of circuit switchers to trip when necessary can lead to system outages and damage to other critical equipment such as transformers. DTE has a strategy to replace certain at risk circuit switchers. Also, 35 installations have been identified where fault currents at the station now exceed the interrupting rating of the circuit switcher. The circuit switchers need to be replaced with ones with a higher interrupting rating in order to operate as designed to provide protection to the transformer.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 71 of 101

(a)

(b)

Program:	Disconnect and Switcher Replacement (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 21
DGP Section:	9.1
Scope:	• Replace at risk Cap & Pin and PM-40 disconnects • Replace at risk and undersized Circuit Switchers

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 72 of 101

(a)	(b)						
Program:	Disconnect and Switcher Replacement (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Enhance safety • Improving system operability • Improve system reliability						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x				x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 73 of 101

(a)	(b)																												
Program:	Disconnect and Switcher Replacement (Pg. 4 of 4)																												
Current Projects:	Disconnect and Switcher: Disconnect Disconnect and Switcher: Switcher																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$10.4 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>441</td><td>2,219</td><td>2,660</td></tr><tr><td>Material</td><td>88</td><td>444</td><td>532</td></tr><tr><td>Other</td><td>59</td><td>296</td><td>355</td></tr><tr><td>Total</td><td>588</td><td>2,959</td><td>3,547</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	441	2,219	2,660	Material	88	444	532	Other	59	296	355	Total	588	2,959	3,547
	2 mos.	10 mos.	Test Year																										
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	12/31/22	10/31/23	10/31/23																										
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Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 74 of 101

(a)	(b)
Program:	SCADA Pole Top Device Replacement (Pg. 1 of 4)
Purpose and Necessity:	Through Asset Health Assessments, DTE has identified two types of SCADA pole-top equipment that have experienced high rates of failure - the Form 3 reclosers and the Bridges pole top switches.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 75 of 101

(a)	(b)
Program:	SCADA Pole Top Device Replacement (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 22
DGP Section:	9.1
Scope:	• Replace end-of-life VWE Form 3 Reclosers with Form 6 Triple Single reclosers • Replace end-of-life PTSs with Second Generation S&C PTS

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 76 of 101

(a)	(b)						
Program:	SCADA Pole Top Device Replacement (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Eliminate aging infrastructure/equipment						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:					x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 77 of 101

(a)	(b)																												
Program:	SCADA Pole Top Device Replacement (Pg. 4 of 4)																												
Current Projects:	SCADA Pole Top Device Replacement																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$4.2 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>178</td><td>1,188</td><td>1,366</td></tr><tr><td>Material</td><td>36</td><td>238</td><td>273</td></tr><tr><td>Other</td><td>24</td><td>158</td><td>182</td></tr><tr><td>Total</td><td>238</td><td>1,583</td><td>1,821</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	178	1,188	1,366	Material	36	238	273	Other	24	158	182	Total	238	1,583	1,821
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
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Other	24	158	182																										
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Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 78 of 101

(a)	(b)
Program:	Substation Regulator Replacement (Pg. 1 of 4)
Purpose and Necessity:	Through Asset Health Assessments, DTE has identified induction, tri-delta and tri-wye type regulators as being obsolete with technology that is no longer manufactured or supported. An increasing number of issues are occurring with these types of units with repair time continuing to increase due to a need to dry out the insulation in the units and repair or replace voltage controls.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 79 of 101

(a)	(b)
Program:	Substation Regulator Replacement (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 23
DGP Section:	9.1
Scope:	• Replace 5 obsolete three-phase substation regulators per year for several years (tri-delta, tri-wye, or induction) with new three-phase step regulators.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 80 of 101

(a)	(b)						
Program:	Substation Regulator Replacement (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Allow for proper voltage regulation at those substations and avoid out-of-range voltage that could damage customer equipment or cause reliability problems. • Eliminate aging infrastructure						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
					X	X	X

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 81 of 101

(a)	(b)																				
Program:	Substation Regulator Replacement (Pg. 4 of 4)																				
Current Projects:	Substation Regulator Replacement																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.9 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>53</td><td>266</td><td>319</td></tr><tr><td>Material</td><td>11</td><td>53</td><td>64</td></tr><tr><td>Other</td><td>7</td><td>35</td><td>43</td></tr><tr><td>Total</td><td>71</td><td>354</td><td>425</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	53	266	319	Material	11	53	64	Other	7	35	43	Total	71	354	425
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	53	266	319																		
Material	11	53	64																		
Other	7	35	43																		
Total	71	354	425																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 82 of 101

(a)	(b)
Program:	Steel Pole Highway Crossings (Pg. 1 of 4)
Purpose and Necessity:	DTE has been subject to a few incidents that have resulted in collapse of freeway/highway crossings this year. As a result all wood poles at freeway and highway crossings should be replaced with steel poles as a multi-year phased program as opposed to a project. 5-7 locations will be targeted in 2022.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 83 of 101

(a)	(b)
Program:	Steel Pole Highway Crossings (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 24
DGP Section:	Not Available
Scope:	For the future installations of steel poles at Freeway/Highway crossings, the following work will be needed. • 2-4 poles to be replaced with new steel poles at each crossing. • Approximately 400-600 feet of OH Conductor to be replaced at each crossing • 636 AL or 945 ACSR Conductor will be used for Subtransmission lines • Distribution lines and Communication lines will be replaced, as necessary

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 84 of 101

(a)	(b)						
Program:	Steel Pole Highway Crossings (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Switching from Wood to Steel Poles increases reliability of the structure. • Design stress limited to 75% for increased strength at crossings • Damage limiter structures at or near the crossings will be required to prevent cascading failures • Re-Designing the crossings will allow us the ability to ensure the crossings have adequate strengths due to the addition of joint use post initial design • Future rebuilds of adjacent circuits that the crossings are associated with will go smoother as the crossing will be brought up to current standards and DDOs • Pending: Installation of an overhead conduit to receive a future DTE communication fiber						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	X			X	X		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 85 of 101

(a)	(b)																				
Program:	Steel Pole Highway Crossings (Pg. 4 of 4)																				
Current Projects:	Steel Pole Highway Crossings																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$11.1 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>375</td><td>5,000</td><td>5,375</td></tr><tr><td>Material</td><td>75</td><td>1,000</td><td>1,075</td></tr><tr><td>Other</td><td>50</td><td>667</td><td>717</td></tr><tr><td>Total</td><td>500</td><td>6,667</td><td>7,167</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	375	5,000	5,375	Material	75	1,000	1,075	Other	50	667	717	Total	500	6,667	7,167
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	375	5,000	5,375																		
Material	75	1,000	1,075																		
Other	50	667	717																		
Total	500	6,667	7,167																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 86 of 101

(a)	(b)
Program:	Station Upgrade: Warren (Relay Replacement) (Pg. 1 of 4)
Purpose and Necessity:	Existing 24kV relay panels are marbled and cannot be maintained or modified when relay upgrades are needed. There is not enough space to install new panels and new configuration is needed to create space. Existing relays are nearly 60 years old and obsolete. There are at risk breakers, disconnects, and other equipment. Replacements of the breakers and disconnects are challenging due to space constraints. A comprehensive station upgrade is needed to address these issues. In addition, ITC is upgrading their 120 kV bus relaying and removing high speed protection for the transformers. DTE needs to add that functionality by installing new relay panels.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 87 of 101

(a)	(b)
Program:	Station Upgrade: Warren (Relay Replacement) (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 25
DGP Section:	9.2
Scope:	• Replace (21) 24 kV oil circuit breakers with gas circuit breakers • Replace existing relay panels with 43 new microprocessor based relay panels • New relay panels include three bus panels and one under frequency panel

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 88 of 101

(a)	(b)						
Program:	Station Upgrade: Warren (Relay Replacement) (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Remove at risk relays • Improve grid-wide situational awareness • Enable SCADA, fault detection, and metering capabilities in substations • Enable remote modification on relay settings can be applied remotely, resulting in fewer field visits • Reduce reactive costs during non-storm and storm events						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:				X	X	X	X

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 89 of 101

(a)	(b)																				
Program:	Station Upgrade: Warren (Relay Replacement) (Pg. 4 of 4)																				
Current Projects:	Station Upgrade: Warren (Relay Replacement)																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$1.0 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td>-</td><td>-</td><td>-</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total	-	-	-
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	-	-	-																		
Material	-	-	-																		
Other	-	-	-																		
Total	-	-	-																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 90 of 101

(a)	(b)
Program:	Station Upgrade: Northeast (Relay Replacement) (Pg. 1 of 4)
Purpose and Necessity:	Existing 24kV relay panels are marbled and cannot be maintained or modified when relay upgrades are needed. There is not enough space to install new panels and new configuration is needed to create space. Existing relays are nearly 60 years old and obsolete. There are at risk breakers, disconnects, and other equipment. Replacements of the breakers and disconnects are challenging due to space constraints. A comprehensive station upgrade is needed to address these issues. In addition, ITC is upgrading their 120 kV bus relaying and removing high speed protection for the transformers. DTE needs to add that functionality by installing new relay panels.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 91 of 101

(a)	(b)
Program:	Station Upgrade: Northeast (Relay Replacement) (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 26
DGP Section:	9.2
Scope:	• Replace (21) 24 kV oil circuit breakers with gas circuit breakers • Replace existing relay panels with 37 new microprocessor based relay panels • New relay panels include three bus panels and one under frequency panel

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 92 of 101

(a)	(b)						
Program:	Station Upgrade: Northeast (Relay Replacement) (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Remove at risk relays • Improve grid-wide situational awareness • Enable SCADA, fault detection, and metering capabilities in subtransmission stations • Enable remote modification on relay settings can be applied remotely, resulting in fewer field visits • Reduce reactive costs during trouble						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:				x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 93 of 101

(a)	(b)																				
Program:	Station Upgrade: Northeast (Relay Replacement) (Pg. 4 of 4)																				
Current Projects:	Station Upgrade: Northeast (Relay Replacement)																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$10.5 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>1,029</td><td>-</td><td>1,029</td></tr><tr><td>Material</td><td>206</td><td>-</td><td>206</td></tr><tr><td>Other</td><td>137</td><td>-</td><td>137</td></tr><tr><td>Total</td><td>1,372</td><td>-</td><td>1,372</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	1,029	-	1,029	Material	206	-	206	Other	137	-	137	Total	1,372	-	1,372
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	1,029	-	1,029																		
Material	206	-	206																		
Other	137	-	137																		
Total	1,372	-	1,372																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 94 of 101

(a)	(b)
Program:	Station Upgrade: Lincoln (Pg. 1 of 4)
Purpose and Necessity:	Existing 24kV relay panels are marbled and cannot be maintained or modified when relay upgrades are needed. There is not enough space to install new panels and new configuration is needed to create space. Existing relays are nearly 60 years old and obsolete. There are at risk breakers, disconnects, and other equipment. Replacements of the breakers and disconnects are challenging due to space constraints. A comprehensive station upgrade is needed to address these issues.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 95 of 101

(a)	(b)
Program:	Station Upgrade: Lincoln (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 27
DGP Section:	9.2
Scope:	• Replace (14) 24 kV oil circuit breakers with gas circuit breakers • Replace existing relay panels with 20 new microprocessor based relay panels • New relay panels include three bus panels and one under frequency panels

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 96 of 101

(a)	(b)						
Program:	Station Upgrade: Lincoln (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Remove at risk relays • Improve grid-wide situational awareness • Enable SCADA, fault detection, and metering capabilities in substations • Enable remote modification on relay settings can be applied remotely, resulting in fewer field visits • Reduce reactive costs during non-storm and storm events						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:				X	X	X	X

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 97 of 101

(a)	(b)																				
Program:	Station Upgrade: Lincoln (Pg. 4 of 4)																				
Current Projects:	Station Upgrade: Lincoln																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$5.6 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>41</td><td>3,313</td><td>3,354</td></tr><tr><td>Material</td><td>8</td><td>663</td><td>671</td></tr><tr><td>Other</td><td>6</td><td>442</td><td>447</td></tr><tr><td>Total</td><td>55</td><td>4,417</td><td>4,472</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	41	3,313	3,354	Material	8	663	671	Other	6	442	447	Total	55	4,417	4,472
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	41	3,313	3,354																		
Material	8	663	671																		
Other	6	442	447																		
Total	55	4,417	4,472																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 98 of 101

(a)	(b)
Program:	Station Upgrade: Navarre (Pg. 1 of 4)
Purpose and Necessity:	Existing 24kV relay panels are marbled and cannot be maintained or modified when relay upgrades are needed. There is not enough space to install new panels and new configuration is needed to create space. Existing relays are nearly 60 years old and obsolete. There are at risk breakers, disconnects, and other equipment. Replacements of the breakers and disconnects are challenging due to space constraints. A comprehensive station upgrade is needed to address these issues.
Category:	Infrastructure Resilience & Hardening

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 99 of 101

(a)	(b)
Program:	Station Upgrade: Navarre (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 8, Line 28
DGP Section:	9.2
Scope:	• Replace (20) 24 kV oil circuit breakers with gas circuit breakers • Replace existing relay panels with 35 new microprocessor based relay panels. New relay panels include three bus panels and one under frequency panel

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 100 of 101

(a)	(b)						
Program:	Station Upgrade: Navarre (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Remove at risk relays • Improve grid-wide situational awareness • Enable SCADA, fault detection, and metering capabilities in substations • Enable remote modification on relay settings can be applied remotely, resulting in fewer field visits • Reduce reactive costs during non-storm and storm events						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:				x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Resilience & Hardening

Case No.: U-20836
Exhibit: A-23
Schedule: M4
Witness: S. G. Pfeuffer
Page: 101 of 101

(a)	(b)																				
Program:	Station Upgrade: Navarre (Pg. 4 of 4)																				
Current Projects:	Station Upgrade: Navarre																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.3 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>206</td><td>206</td></tr><tr><td>Material</td><td>-</td><td>41</td><td>41</td></tr><tr><td>Other</td><td>-</td><td>28</td><td>28</td></tr><tr><td>Total</td><td>-</td><td>275</td><td>275</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	206	206	Material	-	41	41	Other	-	28	28	Total	-	275	275
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	-	206	206																		
Material	-	41	41																		
Other	-	28	28																		
Total	-	275	275																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 1 of 347

(a)	(b)
Category	Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization
	Capital Exhibit:
pages: 4-7 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 2 Ann Arbor System Improvements: State Substation
pages: 8-11 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 3 Ann Arbor System Improvements: Apex (Blue) Substation
pages: 12-15 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 4 Ann Arbor System Improvements: Argo 40kV Reconfiguration
pages: 16-19 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 5 Transformer High Side Protection Program
pages: 20-23 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 6 PQ Meter Fault Locating
pages: 24-27 of 355	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 7 Subtransmission Redesign & Rebuild: Prior Year's
pages: 28-31 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 8 Subtransmission Redesign & Rebuild: Tie 4104
pages: 32-35 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 9 Subtransmission Redesign & Rebuild: Tie 3416
pages: 36-39 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 10 Subtransmission Redesign & Rebuild: Maxwell Amherst
pages: 40-43 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 11 Subtransmission Redesign & Rebuild: Tie 810
pages: 44-47 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 12 Subtransmission Redesign & Rebuild: Trunk 7106
pages: 48-51 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 13 Subtransmission Redesign & Rebuild: Trunk 2255
pages: 52-55 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 14 Subtransmission Redesign & Rebuild: Trunk 2237-ST
pages: 56-59 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 15 Subtransmission Redesign & Rebuild: Trunk 7333
pages: 60-63 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 16 Subtransmission Redesign & Rebuild: Sandusky Transformer 101 Breaker
pages: 64-67 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 17 Subtransmission Redesign & Rebuild: Sandusky Transformer 102 Addition
pages: 68-71 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 18 Subtransmission Redesign & Rebuild: Boyne
pages: 72-75 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 19 Subtransmission Redesign & Rebuild: Tie 6907
pages: 76-79 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 20 Subtransmission Redesign & Rebuild: Tie 7504
pages: 80-83 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 21 Subtransmission Redesign & Rebuild: Tie 4105
pages: 84-87 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 22 Subtransmission Redesign & Rebuild: Trunk 2308
pages: 88-91 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 23 Subtransmission Redesign & Rebuild: Tie 1568
pages: 92-95 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 24 Subtransmission Redesign & Rebuild: Trunk 4217
pages: 96-99 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 25 Subtransmission Redesign & Rebuild: Trunk 3509
pages: 100-103 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 26 Subtransmission Redesign & Rebuild: Trunk 4266
pages: 104-107 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 27 Subtransmission Redesign & Rebuild: Badax Transformer 102 Addition
pages: 108-111 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 28 Subtransmission Redesign & Rebuild: Trunk 4245
pages: 112-115 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 29 Subtransmission Redesign & Rebuild: Tie 6602
pages: 116-119 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 30 Subtransmission Redesign & Rebuild: Trunk 4911
pages: 120-123 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 31 Subtransmission Redesign & Rebuild: Custer Republic

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 2 of 347

(a)	(b)
Category	Distribution Plant Capital Project Detail - Infrastructure Redesign
	Capital Exhibit:
pages: 124-127 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 32 Subtransmission Redesign & Rebuild: Pigeon Area Improvement
pages: 128-131 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 33 Subtransmission Redesign & Rebuild: Trunk 2443
pages: 132-135 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 34 Subtransmission Redesign & Rebuild: Trunk 5838
pages: 136-139 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 35 Subtransmission Redesign & Rebuild: Trunk 2419
pages: 140-143 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 36 Subtransmission Redesign & Rebuild : Small Projects & Reserve
pages: 144-147 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 37 Subtransmission Breaker Short Circuit Violations
pages: 148-151 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 38 Subtransmission Redesign & Rebuild: Slocum
pages: 152-155 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 39 Station Upgrade: Cortland Station Expansion
pages: 156-159 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 40 Subtransmission Redesign & Rebuild: 40kV Capacitor Banks at Armada and Adair
pages: 160-163 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 41 CODI: Charlotte Network Upgrade
pages: 164-167 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 42 CODI: Targeted Network Secondary Cable Replacement
pages: 168-171 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 43 CODI: Corktown Substation
pages: 172-175 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 44 CODI: Islandview Substation
pages: 176-179 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 45 CODI: CATO Substation Expansion
pages: 180-183 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 46 CODI: Howard Conversion
pages: 184-187 of 347	Exhibit A-12, Schedule B5.4, Page 9 of 12, Line 47 CODI: Midtown Substation Expansion
pages: 188-191 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 48 CODI: Alfred Substation Expansion
pages: 192-195 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 49 CODI: Garfield Network Upgrade
pages: 196-199 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 50 CODI: Kent/Gibson Conversion
pages: 200-203 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 51 4.8 kV CC: Hilton Substation and Circuit Conversion
pages: 204-207 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 52 4.8 kV CC: Zenon Substation and Circuit Conversion
pages: 208-211 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 53 4.8 kV CC: Cortland / Oakman / Linwood Consolidation
pages: 212-215 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 54 4.8 kV CC: Ariel Substation and Circuit Conversion
pages: 216-219 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 55 4.8 kV CC: I-94 Substation and Circuit Conversion
pages: 220-223 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 56 4.8 kV CC: Almont Relief and Circuit Conversion (Midas)
pages: 224-227 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 57 4.8 kV CC: Buckler Circuit Conversion
pages: 228-231 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 58 4.8 kV CC: Calla Circuit Conversion
pages: 232-235 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 59 4.8 kV CC: Northville Decommissioning and Circuit Conversion
pages: 236-239 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 60 4.8 kV CC: Lapeer - Elba Expansion and Circuit Conversion (Apollo)
pages: 240-243 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 61 4.8 kV CC: McKinstry Sub Decommission
pages: 244-247 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 62 4.8 kV CC: White Lake Decommissioning and Circuit Conversion

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 3 of 347

(a)	(b)
Category	Distribution Plant Capital Project Detail - Infrastructure Redesign
	Capital Exhibit:
pages: 248-251 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 63 4.8 kV CC: Quincy Conversion
pages: 252-255 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 64 & 65 4.8 kV CC: Pinegrove Substation Relocation and Conversion
pages: 256-259 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 66 4.8 kV CC: Promenade Site Prep and Below Grade
pages: 260-263 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 67 4.8 kV CC: Hawthorne Relief and Circuit Conversion
pages: 264-267 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 68 4.8 kV CC: Birmingham Decommissioning and Circuit Conversion
pages: 268-271 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 69 4.8kV CC: ISO Camden
pages: 272-275 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 70 4.8kV CC: ISO Gilbert
pages: 276-279 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 71 8.3 kV CC: Pontiac Overhead Conversion
pages: 280-283 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 72 8.3 kV CC: Pontiac Underground Conversion
pages: 284-287 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 73 System Loading: Prior Year
pages: 288-291 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 74 System Loading: Brest Substation
pages: 292-295 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 75 System Loading: Carleton
pages: 296-299 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 76 System Loading: Grayling
pages: 300-303 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 77 System Loading: Prospect
pages: 304-307 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 78 System Loading: Mandy 307 Load Transfer
pages: 308-311 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 79 System Loading: Brown City
pages: 312-315 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 80 System Loading: Mack Transformer 102
pages: 316-319 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 81 System Loading: Port Sanilac
pages: 320-323 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 82 System Loading: Richmond/Armada
pages: 324-327 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 83 System Loading: Cody
pages: 328-331 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 84 System Loading: Otsego/Capac/Shaw
pages: 332-335 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 85 System Loading: Macomb/Golf
pages: 336-339 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 86 System Loading: Grenada
pages: 340-343 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 87 Pilot: Strategic and Service Undergrounding
pages: 344-347 of 347	Exhibit A-12, Schedule B5.4, Page 10 of 12, Line 88 Pilot: Primary Deconductoring

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 4 of 347

(a) (b)

Program:	Ann Arbor System Improvements: State Substation (Pg. 1 of 4)
Purpose and Necessity:	The existing system in the Greater Ann Arbor area experiences frequent reliability and power quality issues and is not adequately sized to accommodate existing load and accelerated load growth in the area. Voltage sags and outages are frequent in the area, negatively affecting customers. The overloaded system is an impediment to economic growth. There is currently limited ability to jumper/transfer load to adjacent substations during emergency situations or for routine maintenance. This project, coupled with the other Ann Arbor System Improvements Projects Apex, Argo, and Buckler Circuit Conversions, will address these issues.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 5 of 347

(a)	(b)
Program:	Ann Arbor System Improvements: State Substation (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 2
DGP Section:	11.2
Scope:	• Construct a new 120/40 kV class 'D' station (including two 45/60/75 MVA, 120-40 kV transformers and seven Subtransmission breakers) • Establish four Subtransmission circuits out of State Station. Load from three existing stations will be transferred to State. Substations impacted include: Burton, Hoover, University, Ingalls, and Academy. • Construct a new 120/13.2 kV class 'A' substation (including two 24/32/40 MVA, 120-13.8 kV transformers and ten circuit breakers) • Decommission existing State substation and cutover eight distribution circuits from existing substation to the new substation.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 6 of 347

(a)	(b)						
Program:	Ann Arbor System Improvements: State Substation (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide sufficient capacity to serve existing and new customer load. • Reduce equipment risk by providing load relief to the Subtransmission system in the Greater Ann Arbor area to be compliant with the Subtransmission Planning Criteria and Distribution Design Orders. • Improve system shutdown capability and operability. • Improve customer reliability and power quality.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x	x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 7 of 347

(a)	(b)																				
Program:	Ann Arbor System Improvements: State Substation (Pg. 4 of 4)																				
Current Projects:	Ann Arbor System Improvements: State Substation																				
Budget Basis:	• MEP Estimate																				
Cost:	Estimated project spend in 2021-2023: \$18.1 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>1,683</td><td>-</td><td>1,683</td></tr><tr><td>Material</td><td>337</td><td>-</td><td>337</td></tr><tr><td>Other</td><td>224</td><td>-</td><td>224</td></tr><tr><td>Total</td><td>2,244</td><td>-</td><td>2,244</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	1,683	-	1,683	Material	337	-	337	Other	224	-	224	Total	2,244	-	2,244
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	1,683	-	1,683																		
Material	337	-	337																		
Other	224	-	224																		
Total	2,244	-	2,244																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 8 of 347

(a)	(b)
Program:	Ann Arbor System Improvements: Apex (Blue) Substation (Pg. 1 of 4)
Purpose and Necessity:	The existing system in the Greater Ann Arbor area experiences frequent reliability and power quality issues and is not adequately sized to accommodate existing load and accelerated load growth in the area. Voltage sags and outages are frequent in the area, negatively affecting customers. The overloaded system is an impediment to economic growth. There is currently limited ability to jumper/transfer load to adjacent substations during emergency situations or for routine maintenance. This project, coupled with the other Ann Arbor System Improvements Projects State, Argo, and Buckler Circuit Conversions, will address these issues.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 9 of 347

(a)	(b)
Program:	Ann Arbor System Improvements: Apex (Blue) Substation (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 3
DGP Section:	11.2
Scope:	• ITC to design and construct five miles of 120 kV line to Apex Station. • Construct a new 120/40 kV class 'D' station (including two 60/80/100 MVA, 120-40kV transformers and 11 Subtransmission breakers). • Reconfigure the existing 40 kV system in the area. Establish eight Subtransmission circuits out of Apex station. Load from three existing stations will be transferred to Apex Station. Substations impacted include: University, Ingalls, Academy, Campus, Leland, and Bates.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 10 of 347

(a)

(b)

Program:	Ann Arbor System Improvements: Apex (Blue) Substation (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide sufficient capacity to serve existing and new customer load. • Reduce equipment risk by providing load relief to Subtransmission system in the Greater Ann Arbor area to be compliant with the Distribution Design Orders. • Improve system shutdown capability and operability. • Improve customer reliability and power quality.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x	x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 11 of 347

(a)	(b)																				
Program:	Ann Arbor System Improvements: Apex (Blue) Substation (Pg. 4 of 4)																				
Current Projects:	Ann Arbor System Improvements: Apex (Blue) Substation																				
Budget Basis:	• MEP Estimate																				
Cost:	Estimated project spend in 2021-2023: \$6.6 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>506</td><td>-</td><td>506</td></tr><tr><td>Material</td><td>101</td><td>-</td><td>101</td></tr><tr><td>Other</td><td>68</td><td>-</td><td>68</td></tr><tr><td>Total</td><td>675</td><td>-</td><td>675</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	506	-	506	Material	101	-	101	Other	68	-	68	Total	675	-	675
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	506	-	506																		
Material	101	-	101																		
Other	68	-	68																		
Total	675	-	675																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 12 of 347

(a)	(b)
Program:	Ann Arbor System Improvements: Argo 40kV Reconfiguration (Pg. 1 of 4)
Purpose and Necessity:	Due to high growth in Ann Arbor, ARGO does not have capacity to serve downtown Ann Arbor. This is part of the ARGO decommissioning project. Both Power Delivery Planning and South West Central Distribution Engineering will work together to decommission and remove Argo from service (see related project Buckler Circuit Conversions). This project also avoids \$5M of work that would be required at the ARGO yard should the reconfiguration not take place.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 13 of 347

(a)	(b)
Program:	Ann Arbor System Improvements: Argo 40kV Reconfiguration (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 4
DGP Section:	11.2
Scope:	• Decommission and remove all 40kV busses at Argo: Bus 1, Bus 2, and Bus 3. • Decommission all 40kV Breakers and Disconnects at Argo Station. • Decommission TIE 4501 Argo source, re-tag circuit as TRK 4567 out of Lark Station and TRK 5595 out of Phoenix Station. • Decommission TIE 5517 Argo source, and Re-tag circuit as TRK 5595A-B from Phoenix Station. • Reconfigure TIE 5513, remove Argo source. • Reconfigure TIE 1540, remove Argo source and redesignate as TRK 1572 out of Superior. • Reconfigure TIE 1531, remove Argo source and redesignate as TRK 1510 out of Superior. • Decommission TRK 3150 and TRK 3160A-3161A.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 14 of 347

(a)	(b)						
Program:	Ann Arbor System Improvements: Argo 40kV Reconfiguration (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide sufficient capacity to serve existing and new customer load • Reduce equipment risk by providing load relief to subtransmission system in the Greater Ann Arbor area to be compliant with DTE Electric's Distribution Design Standards • Improve system shutdown capability and operability • Improve customer reliability and power quality						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:		X		X	X		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 15 of 347

(a)	(b)																				
Program:	Ann Arbor System Improvements: Argo 40kV Reconfiguration (Pg. 4 of 4)																				
Current Projects:	Ann Arbor System Improvements: Argo 40kV Reconfiguration																				
Budget Basis:	• MEP Estimate																				
Cost:	Estimated project spend in 2021-2023: \$23.3 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>831</td><td>10,423</td><td>11,254</td></tr><tr><td>Material</td><td>166</td><td>2,085</td><td>2,251</td></tr><tr><td>Other</td><td>111</td><td>1,390</td><td>1,501</td></tr><tr><td>Total</td><td>1,108</td><td>13,897</td><td>15,006</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	831	10,423	11,254	Material	166	2,085	2,251	Other	111	1,390	1,501	Total	1,108	13,897	15,006
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	831	10,423	11,254																		
Material	166	2,085	2,251																		
Other	111	1,390	1,501																		
Total	1,108	13,897	15,006																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 16 of 347

(a)

(b)

Program:	Transformer High Side Protection Program (Pg. 1 of 4)
Purpose and Necessity:	Several subtransmission and distribution transformers require the installation of high side protection, either a circuit switcher or breaker, to mitigate observed NERC reportable system issues during simulated contingency conditions that may result in the loss of service to 100+ MVA of customer load. These issues were discovered during annual studies performed by ITC and are required to be mitigated by NERC TPL-001-4 standards.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 17 of 347

(a)	(b)
Program:	Transformer High Side Protection Program (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 5
DGP Section:	11.2
Scope:	• Install high side switching devices, circuit switchers or breakers, on 22 subtransmission and distribution transformers located at 16 different stations • 10 of these transformers will also require relocation and/or replacement due to space constraints at their current sites with the possibility of additional relaying and monitoring equipment for operation of the high side switching devices

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 18 of 347

(a)

(b)

Program:	Transformer High Side Protection Program (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• DTE Transformers will be in compliance with NERC TPL-001-4 criteria • Mitigate the impact of potential DTE equipment failures impacting the ITC Bulk Electric System • Improve Bulk Electric System reliability and resiliency to widespread system outage of 100+ MVA of customer load • Increase DTE operational flexibility by allowing remote operation of protective devices at the transformers						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
			x	x	x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 19 of 347

(a)	(b)																				
Program:	Transformer High Side Protection Program (Pg. 4 of 4)																				
Current Projects:	Transformer High Side Protection Program																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$7.9 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>375</td><td>1,875</td><td>2,250</td></tr><tr><td>Material</td><td>75</td><td>375</td><td>450</td></tr><tr><td>Other</td><td>50</td><td>250</td><td>300</td></tr><tr><td>Total</td><td>500</td><td>2,500</td><td>3,000</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	375	1,875	2,250	Material	75	375	450	Other	50	250	300	Total	500	2,500	3,000
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	375	1,875	2,250																		
Material	75	375	450																		
Other	50	250	300																		
Total	500	2,500	3,000																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 20 of 347

(a)	(b)
Program:	PQ Meter Fault Locating (Pg. 1 of 4)
Purpose and Necessity:	PQ Meters provide power quality (voltage, current, fault current, fault location, and duration of event) information for the subtransmission system in order to improve the analysis of system performance. This information will help provide a more accurate record of what circuits on the system are impacted by events/outages and further target specific locations on those circuits where the events originate. This detail will help with scoping of the redesign of the circuit or with short-term mitigation plans to address reoccurring issues. Furthermore, the System Operations Center will be able to fault locate using PQ meters to optimize patrol efforts and improve operational efficiency.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 21 of 347

(a)	(b)
Program:	PQ Meter Fault Locating (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 6
DGP Section:	11.2
Scope:	A total of 177 meters will be installed throughout the system: • 82 station transformer positions • 95 24 kV/ 40 kV tie lines

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 22 of 347

(a)	(b)						
Program:	PQ Meter Fault Locating (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Obtain accurate power quality data to diagnose issues on the subtransmission system • Improve capability to address trouble root causes and customer complaints.						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:					x	x	

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 23 of 347

(a)	(b)																				
Program:	PQ Meter Fault Locating (Pg. 4 of 4)																				
Current Projects:	PQ Meter Fault Locating																				
Budget Basis:	• Consultant estimate																				
Cost:	Estimated project spend in 2021-2023: \$1.8 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>6</td><td>1,087</td><td>1,093</td></tr><tr><td>Material</td><td>1</td><td>217</td><td>219</td></tr><tr><td>Other</td><td>1</td><td>145</td><td>146</td></tr><tr><td>Total</td><td>8</td><td>1,449</td><td>1,457</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	6	1,087	1,093	Material	1	217	219	Other	1	145	146	Total	8	1,449	1,457
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	6	1,087	1,093																		
Material	1	217	219																		
Other	1	145	146																		
Total	8	1,449	1,457																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 24 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Prior Year's (Pg. 1 of 4)
Purpose and Necessity:	DTE Electric's subtransmission system is experiencing increasing failures due to corrosion of steel cores in the conductors, arc strike (lightning) damage, and other storm and trouble events that cause multiple splices along sections of the trunk and tie lines. Additionally, multiple trunk and tie lines on the subtransmission system are over their emergency ratings during contingency events. Any trunk or tie line failure could impact multiple substations and a significant number of customers. The program was established to address both reliability and loading issues associated with subtransmission tie lines and trunk lines.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 25 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Prior Year's (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 7
DGP Section:	Not Available
Scope:	• RP Defective Transformers (HV): Replace defective OH distribution transformers that caused voltage in excess of 110% of nominal voltage • TIE 4512 / 120-40 kV station in Lima Twp. (Chelsea): Build a 120-40 kV station and split TIE 4512 into two 40 kV lines

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 26 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Prior Year's (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce equipment risk by providing load relief to substations and circuits • Provide capacity to serve existing and new customer load • Improve system shutdown capability and operability • Address voltage and power quality issues • Reduce major substation outage risk by reducing stranded load • Improve reliability by adding contingencies to the system						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x			x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 27 of 347

(a)	(b)																																				
Program:	Subtransmission Redesign & Rebuild: Prior Year's (Pg. 4 of 4)																																				
Current Projects:	RP Defective Transformers (HV) Tie 4512/120-40kV Station Lima Twp. (Chelsea)																																				
Budget Basis:	• Project Management Estimate																																				
Cost:	Estimated project spend in 2021-2023: \$0.0 M																																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td></td><td colspan="3"> </td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td> </td><td> </td><td> </td></tr><tr><td></td><td>-</td><td>-</td><td>-</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23					Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total					-	-	-
	2 mos.	10 mos.	Test Year																																		
	11/01/22	01/01/23	12 mos. ending																																		
	12/31/22	10/31/23	10/31/23																																		
Total Labor (DTE Electric & Contractors)	-	-	-																																		
Material	-	-	-																																		
Other	-	-	-																																		
Total																																					
	-	-	-																																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 28 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Tie 4104 (Pg. 1 of 4)
Purpose and Necessity:	The north end of TIE 4104 has experienced four sustained outages impacting customer service due to wire down events during the first quarter of 2019. Testing of the wire confirmed lightning arc strike damage, corrosion on the steel core, and cracked zinc coating (galvanization) on the conductor. Distribution Design Order-0070-008 states that the Standard conductor size for subtransmission system shall be 636 AL SW-ACSR. There are 17.8 miles of 3/0 ACSR SW-AL wire that need to be replaced.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 29 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Tie 4104 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 8
DGP Section:	11.2
Scope:	• Rebuild Delaware tap - remove 3.1 miles of 3/0 ACSR 40 KV and install 3.1 Miles of 636 AL -SW-AL, replace all poles with steel poles, replace Automatic Pole Top Switch 1694 with new SEEEO 600 AMPS Automatic Pole Top Switch, rebuild underbuilt 3.1 miles of TLBOT 8051 3-phase 13.2KV and replaced distribution circuit equipment as needed. • Rebuild Talbot tap - remove 0.2 miles of 3/0 ACSR and install 636 ALL SW-AL. Substation work - Install 636 AL trainer to the TLBOT Transformer 1. Portable will be needed for shutdown or generation. Reconductor 0.2 miles of underbuilt TLBOT 8051 3-phase 13.2KV. • Rebuild 3.6 Miles of 40 KV with 636 AL SW-AL from Automatic Pole Top Switch 1681 to Ruth Rd and under build 1.8 Miles DC TLBOT 8051. Install 13 miles of new 40 KV using 636 AL-SW going North on Ruth RD and East on Section RD. Rebuild 0.4 Miles of 40 KV LW Shock RD and Underbuild NEFF 0314 4.8 KV 3 phase. Decommission & Remove 17.8 miles of 40kV 3/0 ACSR, which 5.5 miles has distribution under build.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 30 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Tie 4104 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce safety risk, to customers and crews, of wire down events with new conductor • Improve customer reliability and power quality with less outage events and better voltage support with new conductor • Reduce reactive trouble cost by approximately 20% for TIE 4104 and associated underbuilt distribution circuits						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x			x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 31 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: Tie 4104 (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: Tie 4104																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$5.6 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>38</td><td>3,329</td><td>3,367</td></tr><tr><td>Material</td><td>8</td><td>666</td><td>673</td></tr><tr><td>Other</td><td>5</td><td>444</td><td>449</td></tr><tr><td>Total</td><td>50</td><td>4,439</td><td>4,489</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	38	3,329	3,367	Material	8	666	673	Other	5	444	449	Total	50	4,439	4,489
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	38	3,329	3,367																		
Material	8	666	673																		
Other	5	444	449																		
Total	50	4,439	4,489																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 32 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Tie 3416 (Pg. 1 of 4)
Purpose and Necessity:	TIE 3416 Breaker Replacement and Reconductoring Distribution Design Order-0090-013 (Subtransmission Loading Criteria): Allowable Summer Normal configuration (day to day) loading on subtransmission equipment is 100% and Summer Emergency loading on subtransmission equipment is 90%. • TIE 3416 I-K section is at 100.24% of the Summer Normal rating (0.1 MVA over the allowable loading). Limiting elements are CTs with a Summer Normal rating of 23.8 MVA currently loaded at 100.24 %, and a trainer/conductor 1/0 AWG with a Summer Normal rating of 25.2 MVA currently loaded at 95% • TIE 3416 I-K section is loaded at 109% of the Summer Emergency rating (2.6 MVA over the allowable load) upon the contingency event of a breaker operation on the Randolph end and Automatic Pole Top Switch 1781. Three other contingencies result in lower overload. The limiting element is CTs with a Summer Emergency rating of 28.9 MVA, currently loaded at 109%. Current potential load shed, to protect from overloading equipment and causing damage, for a single contingency event is 5.5 MVA TIE 3416 Reconductoring - Pigeon Distribution Design Order-0070-008 Subtransmission Voltage Criteria : Under emergency conditions, for single contingency events (N-1), Subtransmission voltage limits are 92%-105% of the nominal voltage • SULVN, SILO, STPS29, STPS4, STPS10, STPS15, ELKTN2, ELKTN1 (all substations fed by TIE 3416) the voltage is at 83% of nominal for a single contingency event of a breaker operation on the Bad Axe end and APTS 1662. Six more single contingency events on the system cause low voltages for every substation on TIE 3416 in the range of 86-91% of nominal. Current potential load shed, to protect customer equipment from damage caused by low voltage, for a single contingency event is 9.5 MVA. TIE 3416 Reconductoring - Pigeon DDO-0070-008 Subtransmission Voltage Criteria : Under emergency conditions, for single contingency events (N-1), Subtransmission voltage limits are 92%-105% of the nominal voltage • SULVN, SILO, STPS29, STPS4, STPS10, STPS15, ELKTN2, ELKTN1 (all substations fed by TIE 3416) voltage is at 83% for single contingency loss of TIE 3416 Bad Axe POS RH and APTS 1662. Six more single contingencies on the system causes low voltages for every substation on TIE 3416 in the range of 86-91%. Load shed for single contingency is 9.5 MVA.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 33 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Tie 3416 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 9
DGP Section:	11.2
Scope:	TIE 3416 Breaker Replacement and Reconductoring • Bad Axe Substation: Replace oil breaker 0221 and CTs; replace trainers: 250KCM, 1/0 AWG; replace disconnects; replace relay panel at position RH BADAX. Overhead Line: replace 1.57 miles of 3/0 ACSR with 636 AL SW-ACSR (enhanced 40 KV, armless, steel pole construction) and replace underbuilt distribution voltage lines from Bad Axe substation to Van Dyke and Pigeon. Replace distribution equipment as needed. TIE 3416 Reconductoring - Pigeon • Substation: replace 3/0 trainers coming to substation for Transformer 1 and Transformer 2 Elkton. Overhead Line: replace 6.33 Miles of 3/0 ACSR with 636 AL SW-ACSR (enhanced 40 KV, armless, steel pole construction) from Pigeon substation to the buck pole at the Subtransmission Power Service (STPS)s tap including tap to STPSs, Elkton Transformer 1 and Elkton Transformer 2 taps and replace underbuilt distribution voltage lines. Replace distribution equipment as needed. Replace Pole Top Switch 314, Pole Top Switch 85 & Pole Top Switch 206.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 34 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Tie 3416 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	TIE 3416 Breaker Replacement and Reconductoring: • Alleviate thermal loading violations, and the possibility of shedding customer load during a single contingency event on TIE 3416 I-K section. New equipment ratings lower the percent of capacity utilized by the current system loading from the Summer Emergency rating of 109% to 84% and the Summer Normal rating of 100.2 % to 81%. Replace 3/0 ACSR wire, that is prone to failures that result in wire down events, with standard 636AL. The new conductor creates 26.5 MVA of additional capacity on TIE 3416 I-K section for the customers in the area. TIE 3416 Reconductoring - Pigeon : • Alleviate voltage violations, and the possibility of shedding customer load during a single contingency event on TIE 3416 from 83% to 94% of nominal voltage. 40 KV wire up to latest standard 636 AL per Distribution Design Order-0070-008, which will also improve the reliability of service to the customers for those sections.						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 35 of 347

(a)	(b)																																				
Program:	Subtransmission Redesign & Rebuild: Tie 3416 (Pg. 4 of 4)																																				
Current Projects:	Tie 3416 Breaker Replacement and Reconductoring Tie 3416 Reconductoring - Pigeon																																				
Budget Basis:	• Project Management Estimate																																				
Cost:	Estimated project spend in 2021-2023: \$0.0 M																																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td></td><td colspan="3"> </td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td> </td><td> </td><td> </td></tr><tr><td></td><td>-</td><td>-</td><td>-</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23					Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total					-	-	-
	2 mos.	10 mos.	Test Year																																		
	11/01/22	01/01/23	12 mos. ending																																		
	12/31/22	10/31/23	10/31/23																																		
Total Labor (DTE Electric & Contractors)	-	-	-																																		
Material	-	-	-																																		
Other	-	-	-																																		
Total																																					
	-	-	-																																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 36 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Maxwell Amherst (Pg. 1 of 4)
Purpose and Necessity:	Maxwell is a switching station which includes buses and circuit switchers. Amherst is an industrial substation dedicated to serving a specific customer. There are two transformers serving this customer and one of them failed. The customer requires both transformers to be in service due to the necessary current inrush of the motor startups and redundancy. To get this customer back up and running, the easiest place to install a transformer was the Maxwell switching station on trouble. Now, transformer two at Amherst must be moved to Maxwell and follow up work must be completed to sustain the customer being served from Maxwell substation. Moving the transformers to this location and installing a similar sister unit ensures the correct impedance for the customer.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 37 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Maxwell Amherst (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 10
DGP Section:	11.2
Scope:	Work at Amherst: • Decommission and remove Amherst Transformer 2 • Decommission and remove capacitor 11A and 11B • Decommission and remove 120 kV cable between Maxwell - Amherst • Decommission old audio-tone transfer trip scheme between DTEE and the customer sites Work at Maxwell: • Install 120-13.8 kV, 24/32/40 MVA Transformer 2 at Maxwell position HR • Decommission and remove Maxwell position HP circuit switcher • Install/Replace three circuit switchers at Maxwell position HQ, HR, and HD • Install 2,000 feet 13.2 kV 1500 CU EPR 1C X 3 from Transformer 2 secondary to customer's manhole and splice with customer's cable • Install ION 7650 SCADA/PQ meter with ethernet connection at Maxwell • Install new transfer trip scheme between the customer and Maxwell using radio communication and SEL 2506/751A relays

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 38 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Maxwell Amherst (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Correct impedance and improved equipment serviceability for the customer due to both transformers being the same type and distance from the customer, as well as located at the same site • Reduce equipment risk of failure by replacing Transformer 2 serving the customer						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x			x			

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 39 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: Maxwell Amherst (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: Maxwell Amherst																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.3 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td>-</td><td>-</td><td>-</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total	-	-	-
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	-	-	-																		
Material	-	-	-																		
Other	-	-	-																		
Total	-	-	-																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 40 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Tie 810 (Pg. 1 of 4)
Purpose and Necessity:	Distribution Design Order-0090-013 (Subtransmission Loading Criteria): Allowable emergency loading on subtransmission equipment is 90%. • 21 separate violations have been identified. 6 greater than 100% of the Summer Emergency rating. Distribution Design Order-0090-013 (Subtransmission Loading Criteria): A single contingency event should not result in load loss. • 31.4MVA of potential load loss through different single contingency events. Distribution Design Order-0070-008 Subtransmission Voltage Criteria: Under emergency conditions, for single contingency events (N-1), Subtransmission voltage limits are 92%-105% of the nominal voltage. • 30 single contingency events (N-1) result in voltage violations of the above criteria.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 41 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Tie 810 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 11
DGP Section:	11.2
Scope:	• Build a new subtransmission voltage station (Gramer Station) • Reconfigure TIE 810 into 3 separate circuits: TIE 9505, TIE 9514, and TRK 4938 • Complete the following work for TIE 9505, TIE 9514 and TRK 4938: • Install 11 miles of new overhead conductor with 636 AL SW-ACSR (enhanced 40 KV, armless, steel pole construction) • Rebuild 16 miles of overhead lines with 636 AL SW-ACSR (enhanced 40 KV, armless, steel pole construction) and replace underbuilt distribution voltage lines and equipment • Decommission and remove 5 miles of overhead lines

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 42 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Tie 810 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide additional capacity to serve existing and new customer load • Reduce risk of thermal overload to equipment and the potential to shed customer load • Improve system shutdown capability and operability to perform regular maintenance work • Improve customer reliability with new overhead lines built along the road and improve power quality with voltage changes remaining within the Distribution Design Order limits						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x	x			x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 43 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: Tie 810 (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: Tie 810																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$35.8 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>2,271</td><td>10,111</td><td>12,382</td></tr><tr><td>Material</td><td>454</td><td>2,022</td><td>2,476</td></tr><tr><td>Other</td><td>303</td><td>1,348</td><td>1,651</td></tr><tr><td>Total</td><td>3,028</td><td>13,481</td><td>16,509</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	2,271	10,111	12,382	Material	454	2,022	2,476	Other	303	1,348	1,651	Total	3,028	13,481	16,509
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	2,271	10,111	12,382																		
Material	454	2,022	2,476																		
Other	303	1,348	1,651																		
Total	3,028	13,481	16,509																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 44 of 347

(a) (b)

Program:	Subtransmission Redesign & Rebuild: Trunk 7106 (Pg. 1 of 4)
Purpose and Necessity:	Southfield 40kV TRK 7106T is 134% of the Summer Normal rating (N-0), 126% of the Summer Emergency rating (N-1) upon a contingency event loss of TRK 2237-ST, and 112% of the Summer Emergency rating (N-1) upon contingency outage event of TRK 7151A. The limiting element is the underground cable of TRK 7106. Distribution Design Order Violations: Distribution Design Order-0040-003: Automatic throwover Blocking schemes are not acceptable • Blocking schemes are in place to prevent substation automatic throwovers, due to overloads on the 40 kV feeds to FRMTN & INDIN substations. FRMTN substation automatic throwover has been placed on manual blocking to prevent equipment failure on TRK 7106 due to overloads. 12.7 MVA of customer load at FRMTN substation will be dropped in the event of an outage of TRK 7151 for momentary or sustained outage events. Distribution Design Order-0090-013 (Subtransmission Loading Criteria): Allowable day to day (summer Normal) loading on subtransmission equipment is 100%. • TRK 7106 is at 134% (6.6 MVA over the allowable load) of the Summer Normal rating. Distribution Design Order-0090-013 (Subtransmission Loading Criteria): Allowable emergency loading on subtransmission equipment is 90%. • TRK 7106 is 126% of its Summer Emergency rating. The current potential load shed, to protect from overloading equipment and causing damage, for a single contingency event is 10.3 MVA. Distribution Design Order-0090-013 (Subtransmission Loading Criteria): Single contingency events on the Subtransmission System should not result in load loss. • Approximately 12.7 MVA of customer load would be shed due to the manual blocking of automatic throwover at Farmington substation for an outage event of TRK 7151.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 45 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Trunk 7106 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 12
DGP Section:	11.2
Scope:	Reconfigure 40kV TRK 7106 into two circuits TRK 7106 and TRK 7187: • TRK 7106 will serve Farmington substation Transformer 4 utilizing the existing cable Create new TRK 7187 at Southfield station position RC that will serve Subtransmission Distribution Facility 38, West Service Center Transformer 1 and Indian Transformer 2: • Install 40kV breaker and two 40kV disconnects at Southfield station position RC • Install 2.3 miles of 40kV 650-1Cx3 EPR cable for TRK 7187 from Southfield station position RC to the overhead cable pole • Install a manual Pole Top Switch on TRK 7187

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 46 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Trunk 7106 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Alleviate thermal loading violations, and the possibility of shedding customer load during a single contingency event • Eliminate automatic throwover blocking schemes that can result in customer outages during a single contingency event, improve operability and enable load transfers during planned and trouble events						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:		X		X	X	X	X

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 47 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: Trunk 7106 (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: Trunk 7106																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$3.2 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>1</td><td>-</td><td>1</td></tr><tr><td>Material</td><td>0</td><td>-</td><td>0</td></tr><tr><td>Other</td><td>0</td><td>-</td><td>0</td></tr><tr><td>Total</td><td>2</td><td>-</td><td>2</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	1	-	1	Material	0	-	0	Other	0	-	0	Total	2	-	2
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	1	-	1																		
Material	0	-	0																		
Other	0	-	0																		
Total	2	-	2																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 48 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Trunk 2255 (Pg. 1 of 4)
Purpose and Necessity:	TRK 2255 ST section B-C is 98% of the Summer Normal ratings (N-0) and 110% of the Summer Emergency rating (N-1) upon a contingency event outage of TRK 2276. The limiting element is the C cable of TRK 2255-ST. Distribution Design Order Violations: Distribution Design Order-0040-003 (Automatic throwover Blocking schemes are not acceptable). • Blocking schemes are in place to prevent overloads on the 40 kV feeds to Glendale substation & Villa substation. Distribution Design Order-0090-013 Violations (Subtransmission Loading Criteria): Allowable Summer Normal on subtransmission equipment is 100%. • TRK 2255 is 98% of its Summer Normal rating. Distribution Design Order-0090-013 Violations (Subtransmission Loading Criteria): Allowable Summer Emergency loading on subtransmission equipment is 90%. • TRK 2255 is at 110% of its Summer Emergency rating (~2.2 MVA over the allowable loading). Distribution Design Order-0090-013 Violations (Subtransmission Loading Criteria): Single contingency events on the Subtransmission System should not result in load loss: • Approximately 11 MVA of customer load at Villa substation would be shed with the current automatic throwover blocking scheme in place, and the contingency event of an outage on TRK 2276.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 49 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Trunk 2255 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 13
DGP Section:	11.2
Scope:	A new underground cable will be installed to split the load of the "C" cable & replace the "E" Cable of TRK 2255-ST: • Break down branch joint "C", "D" & "E" cable in manhole 15434 and install one straight joint "C" to "D" cable • Remove 4,765 feet of 350 CU PLSJ 40kV cable on TRK 2255 from manhole 15434 to Villa Transformer 2 • Install 40kV cable pole and 350 EPR termination • Install 15 feet of 1x2 5" conduit for cable pole sweep • Install 6,687 feet of 40kV 350 -1Cx3- EPR cable from cable pole to Villa Transformer 2 utilizing the existing conduit

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 50 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Trunk 2255 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Alleviate the Sumer Normal and Summer Emergency overloads • Remove the automatic throwover blocking schemes and the potential for 11 MVA of customer load shed during a contingency outage event on TRK 2276 • Improve operability and enable load transfers during planned work and trouble events • Replace aging cable with new cable reducing the risk of outages						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		X		X			X

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 51 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: Trunk 2255 (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: Trunk 2255																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$4.3 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>494</td><td>22</td><td>516</td></tr><tr><td>Material</td><td>99</td><td>4</td><td>103</td></tr><tr><td>Other</td><td>66</td><td>3</td><td>69</td></tr><tr><td>Total</td><td>659</td><td>29</td><td>689</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	494	22	516	Material	99	4	103	Other	66	3	69	Total	659	29	689
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	494	22	516																		
Material	99	4	103																		
Other	66	3	69																		
Total	659	29	689																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 52 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Trunk 2237-ST (Pg. 1 of 4)
Purpose and Necessity:	Evergreen station 40kV TRK 2237-ST is at 100% of its Summer Normal rating (N-0) and 110% of its Summer Emergency rating (N-1) upon the contingency event of an outage on TRK 2250. It is also at 96% of its Summer Emergency rating (N-1) upon the contingency event of an outage on TRK 7106T. Distribution Design Order Violations: Distribution Design Order-0040-003 (Automatic throwover Blocking schemes are not acceptable): • A Blocking scheme is in place to prevent overloads on the 40 kV feed to SIXMI substation. Distribution Design Order-0090-013 (Subtransmission Loading Criteria): Allowable Summer Normal loading on subtransmission equipment is 100% • TRK 2237-ST is at 100% of its Summer Normal rating. Distribution Design Order-0090-013 (Subtransmission Loading Criteria): Allowable Summer Emergency loading on subtransmission equipment is 90%. • TRK 2237-ST is at 110% of its Summer Emergency rating (5.7 MVA over the allowable loading). Distribution Design Order-0090-013 (Subtransmission Loading Criteria): Single contingency events on the Subtransmission System should not result in load loss. • Approximately 10.3 MVA of customer load would be shed at SIXMI substation for the contingency event outage of TRK 2250AC-BC.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 53 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Trunk 2237-ST (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 14
DGP Section:	11.2
Scope:	• Install 4,400 feet 350-1Cx3 EPR cable to parallel overloaded TRK 2237-ST "E" cable in the vicinity of Redford substation • Install a new cable pole • Remove branch joint and install (2) straight joints in manhole 7157 • Install 15 feet of 1x2 5" conduit sweep to the new cable pole

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 54 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Trunk 2237-ST (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Alleviate the Summer Normal and Summer Emergency overloads • Remove the Automatic throwover blocking schemes and the potential for 10.3 MVA of customer load shed during a contingency outage event on TRK 2250AC-BC • Improve operability and enable load transfers during planned work and trouble events						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:		X		X	X	X	X

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 55 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: Trunk 2237-ST (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: Trunk 2237-ST																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$2.4 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>280</td><td>-</td><td>280</td></tr><tr><td>Material</td><td>56</td><td>-</td><td>56</td></tr><tr><td>Other</td><td>37</td><td>-</td><td>37</td></tr><tr><td>Total</td><td>373</td><td>-</td><td>373</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	280	-	280	Material	56	-	56	Other	37	-	37	Total	373	-	373
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	280	-	280																		
Material	56	-	56																		
Other	37	-	37																		
Total	373	-	373																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 56 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Trunk 7333 (Pg. 1 of 4)
Purpose and Necessity:	TRK 7333T A-B section is at 129% of the Summer Normal rating (N-0) and is at 126% Summer Emergency rating (N-1) upon the contingency event of an outage on TRK 7315 (limited by cable). Distribution Design Order Violations: Distribution Design Order -0040-003 (Automatic throwover Blocking schemes are not acceptable): • A Blocking scheme is in place to prevent overloads on the 40 kV feeds to Brazil & Kenney substations. Distribution Design Order -0090-013 (Subtransmission Loading Criteria): Allowable Summer Normal loading on subtransmission equipment is 100% • TRK 7333 is at 129% of its Summer Normal rating (6.5 MVA over the allowable loading). Distribution Design Order -0090-013 (Subtransmission Loading Criteria): Allowable Summer Emergency loading on subtransmission equipment is 90%. • TRK 7333 is at 126% of its Summer Emergency rating (12 MVA over the allowable loading). Distribution Design Order -0090-013 (Subtransmission Loading Criteria): Single contingency events on the Subtransmission System should not result in load loss • Approximately 12 MVA of customer load would be shed at Brazil substation for a contingency outage event of TRK 7315
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 57 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Trunk 7333 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 15
DGP Section:	11.2
Scope:	• Reconfigure TRK 7333T into two separate trunk lines - TRK 7333 and TRK 7370 • Rebuild TRK 7333 underground cable joint with a straight joint in manhole 17429 • For TRK 7370, install 16,391 feet of 650 -1C x 3- EPR 40 kV cable from Chestnut station position SB to manhole 17429 via existing conduit • Install a 40 kV breaker and two 40 kV disconnects (2000 Amp) at position SB in Chestnut station

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 58 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Trunk 7333 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Alleviate Summer Normal and Summer Emergency overloads • Remove the Automatic throwover blocking schemes and the potential for 12 MVA of customer load shed during a contingency outage event on TRK 7315 • Improve operability and enable load transfers during planned work and trouble events						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:		X		X	X	X	X

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 59 of 347

(a)	(b)																																
Program:	Subtransmission Redesign & Rebuild: Trunk 7333 (Pg. 4 of 4)																																
Current Projects:	Subtransmission Redesign & Rebuild: Trunk 7333																																
Budget Basis:	• Engineering Estimate																																
Cost:	Estimated project spend in 2021-2023: \$3.1 M																																
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td></td><td colspan="3"> </td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>2</td><td>-</td><td>2</td></tr><tr><td>Material</td><td>0</td><td>-</td><td>0</td></tr><tr><td>Other</td><td>0</td><td>-</td><td>0</td></tr><tr><td>Total</td><td>3</td><td>-</td><td>3</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23					Total Labor (DTE Electric & Contractors)	2	-	2	Material	0	-	0	Other	0	-	0	Total	3	-	3
	2 mos.	10 mos.	Test Year																														
	11/01/22	01/01/23	12 mos. ending																														
	12/31/22	10/31/23	10/31/23																														
Total Labor (DTE Electric & Contractors)	2	-	2																														
Material	0	-	0																														
Other	0	-	0																														
Total	3	-	3																														

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 60 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Sandusky Transformer 101 Breaker (Pg. 1 of 4)
Purpose and Necessity:	The station equipment for Sandusky Transformer 101 exceeds its thermal rating by 126% under normal system conditions. This thermal overload is a violation of the Subtransmission Planning Criteria and the Distribution Design Order-0090-013: Day to day loading (normal, N-0) on the Subtransmission system shall not exceed 100% of the day to day rating of equipment). In the event an equipment failure occurs at Sandusky Transformer 101 due to overload, customers would experience low voltage and be at risk of load shed for an extended period until permanent repairs are made. Wind park generation in the area would also be restricted or taken offline. The existing station configuration at Sandusky does not include a section breaker between the two 40kV buses. Without a section breaker in place, bus shutdowns needed for maintenance or reactive work require the entire station to be outaged.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 61 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Sandusky Transformer 101 Breaker (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 16
DGP Section:	11.2
Scope:	• Replace Sandusky Transformer 101 secondary disconnect switch. • Replace Sandusky Transformer 101 secondary breaker and associated control cables and current transformers. • Replace trainers at Sandusky Transformer 101 and Sandusky Bus 4 with 750 CU MCM trainers. • Install new relay panel for Sandusky Transformer 101. • Install a new section breaker between Sandusky Bus 2 and Bus 4.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 62 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Sandusky Transformer 101 Breaker (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Eliminate thermal loading violation of the Subtransmission Planning Criteria and Distribution Design Order-0090-013. • Prevent equipment failure at Sandusky Transformer 101 due to thermal overload which could result in customer low voltage events, curtailed wind park generation, and extensive contingency plans and portable generation to restore customers. • Improved operability to maintain equipment at Sandusky station by the installation of a section breaker.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x	x	x	

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 63 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: Sandusky Transformer 101 Breaker (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: Sandusky Transformer 101 Breaker																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$1.6 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>188</td><td>-</td><td>188</td></tr><tr><td>Material</td><td>38</td><td>-</td><td>38</td></tr><tr><td>Other</td><td>25</td><td>-</td><td>25</td></tr><tr><td>Total</td><td>250</td><td>-</td><td>250</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	188	-	188	Material	38	-	38	Other	25	-	25	Total	250	-	250
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	188	-	188																		
Material	38	-	38																		
Other	25	-	25																		
Total	250	-	250																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 64 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Sandusky Transformer 102 Addition (Pg. 1 of 4)
Purpose and Necessity:	For the single contingency event where Sandusky Transformer 101 is taken out of service, several substations result in customer low voltage events. This contingency of Sandusky Transformer 101 violates the Subtransmission Planning Criteria as well as Distribution Design Order-0070-008: Under emergency conditions, for single contingency events (N-1), Subtransmission voltage limits are 92%-105% of the nominal voltage. Customers would experience low voltage and be at risk of load shed. Approximately 8.2 MVA of load would be shed to operate within required voltage limits. Wind park generation in the area would also be restricted or taken offline.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 65 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Sandusky Transformer 102 Addition (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 17
DGP Section:	11.2
Scope:	• Install one 120-40kV 45/60/75 MVA TRANSFORMER 102 from a new 120kV source to a new 40kV position on Sandusky 40kV Bus 2. • Install one 120kV S&C 2020 circuit switcher, one 40kV 2000A breaker, two 40kV 1200A disconnects, and a new relay panel.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 66 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Sandusky Transformer 102 Addition (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Elimination the violation of the Subtransmission Planning Criteria & Distribution Design Order-0070-008. • Prevent customer low voltage events that occur when Sandusky Transformer 101 is taken out of service. • Prevent load shed, curtailed wind park generation, and extensive contingency plans and portable generation to restore customers for the single contingency event when Sandusky Transformer 101 is taken out of service. • Improved operability to maintain equipment at Sandusky station by the installation of a 2nd 120kV source for the area.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x	x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 67 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: Sandusky Transformer 102 Addition (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: Sandusky Transformer 102 Addition																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.1 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>73</td><td>73</td></tr><tr><td>Material</td><td>-</td><td>15</td><td>15</td></tr><tr><td>Other</td><td>-</td><td>10</td><td>10</td></tr><tr><td>Total</td><td>-</td><td>97</td><td>97</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	73	73	Material	-	15	15	Other	-	10	10	Total	-	97	97
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	-	73	73																		
Material	-	15	15																		
Other	-	10	10																		
Total	-	97	97																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 68 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Boyne (Pg. 1 of 4)
Purpose and Necessity:	Boyne Station is a single transformer station. Boyne Transformer 101 has system load limitations for shutdowns to do maintenance/equipment upgrades. In the event Boyne Transformer 101 breaker opens due to a fault, up to 24.6 MVA of load will be shed for this single contingency. This violates the Subtransmission Planning Criteria and Distribution Design Order-0090-013, stating the forced outage of any single element should not result in load loss. There have been several requests to serve new load in the Harrison Township area currently served by TRK 7909 (out of Boyne Station) and TRK 6759 (out of Medina Station). These two trunk lines have no remaining capacity. TRK 7909 is loaded at 120% of its summer emergency thermal rating when TRK 6759 is taken out of service (N-1). TRK 6759 is loaded at 107% of its summer emergency thermal rating when TRK 7909 is taken out of service (N-1). This violates the Subtransmission Planning Criteria and Distribution Design Order-0090-013: Emergency loading on the Subtransmission system (resulting from the forced outage of any single element, N-1) shall not exceed 90% of the emergency rating of equipment. When these outage events occur (scheduled shutdown or breaker operation due to fault), customer load shed as high as 13.5 MVA will occur to prevent permanent damage to the Subtransmission equipment.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 69 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Boyne (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 18
DGP Section:	11.2
Scope:	• Install new 40kV Bus 2 at Boyne Station. • Install new 120-40 kV Transformer 102, disconnect on the low side of Transformer 102, circuit switcher on the high side of Transformer 102, and install 40kV section breaker. • Install new 18 Megavar Capacitor on new Boyne Bus 2 for voltage support. • Install one 40 kV line breaker, install disconnects and breaker for the new position (new 40kV trunk line, TRK 7921) on Bus 2, install relay panel for new TRK 7921 position • Install 1.5 miles of new underground conduit 3x5 5" ducts • Reconductor 2.2 miles of TRK 7909. Remove 2.2 miles of 3/0 ACSR conductor and install 2.2 miles of 636 ACSR conductor with 1/0 ACSR shield wire lightning protection. • Replace approximately 0.2 miles of 750 Copper EPR cable with 0.2 miles of parallel 650 Copper EPR cable on TRK 6759 and reconductor approximately 0.03 miles of 3/0 ACSR conductor with 636 ACSR conductor with shield wire lightning protection. • Establish new TRK 7921 out of Boyne Station to serve the load of Selfridge Transformers 2 and 3, Omega Transformer 1, and Beach Transformer 3. Install 1.5 miles of 40kV 650 Copper EPR cable. Install 2.5 miles of new 40kV overhead conductor 636 ACSR with 1/0 ACSR shield wire lightning protection.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 70 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Boyne (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Eliminate equipment thermal overloads for TRK 7909, TRK 6759, and Boyne Transformer 101. Eliminate violations of the Subtransmission Planning Criteria and Distribution Design Order-0090-013. • Eliminate load shed for a single contingency outage or shutdown events. • Provide capacity for future load growth.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x	x	x	

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 71 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: Boyne (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: Boyne																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$25.7 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>2,255</td><td>1,816</td><td>4,072</td></tr><tr><td>Material</td><td>451</td><td>363</td><td>814</td></tr><tr><td>Other</td><td>301</td><td>242</td><td>543</td></tr><tr><td>Total</td><td>3,007</td><td>2,421</td><td>5,429</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	2,255	1,816	4,072	Material	451	363	814	Other	301	242	543	Total	3,007	2,421	5,429
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	2,255	1,816	4,072																		
Material	451	363	814																		
Other	301	242	543																		
Total	3,007	2,421	5,429																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 72 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Tie 6907 (Pg. 1 of 4)
Purpose and Necessity:	TIE 6907 is loaded at 110% of its summer emergency thermal rating. Approximately 5.8 MVA of load to be shed to operate within thermal loading limits upon the contingency event of the TIE 6907 breaker at Adams Station opening (N-1). This violates the Subtransmission Planning Criteria as well as Distribution Design Order-0090-013: Emergency loading on the Subtransmission system (resulting from the forced outage of any single element, N-1) shall not exceed 90% of the emergency rating of equipment. This contingency event (TIE 6907 breaker operation at Adams Station) also causes low voltage for customers served out of Washington Substation (91% of nominal voltage) and Disco Substation (89% of nominal voltage). This violates the Subtransmission Planning Criteria as well as Distribution Design Order-0070-008: Under emergency conditions, for single contingency events (N-1), Subtransmission voltage limits are 92%-105% of the nominal voltage. Buildings have been constructed over top of the cable section tagged as Bridge Main 05 on TIE 6907 and the cable section tagged as Bridge Main 014 on TRK 6941. In the event a cable failure occurs on either of these cables, an extended outage would occur, and customers would be at risk for load shed.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 73 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Tie 6907 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 19
DGP Section:	11.2
Scope:	• Reconfigure TIE 6907 and TRK 6941 Spokane ends to eliminate the thermal overload on TIE 6907 and abandon cable & conduit of Bridge Main 005 & Bridge Main 014. • Install 1,700 feet of overhead 636 ACSR conductor with 1/0 ACSR lightning protection for the line extension to Parkdale substation. • Re-tag TIE 6907 cables and Spokane positions. • Replace existing breaker at Spokane position RP with 2000A gas breaker.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 74 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Tie 6907 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Eliminate equipment thermal overloads for TIE 6907. • Eliminate customer low voltage events. • Prevent extended outage/load shed in the event of a cable failure on TRK 6941 or TIE 6907.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x			

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 75 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: Tie 6907 (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: Tie 6907																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.1 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><thead><tr><th></th><th>2 mos. 11/01/22 12/31/22</th><th>10 mos. 01/01/23 10/31/23</th><th>Test Year 12 mos. ending 10/31/23</th></tr></thead><tbody><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>44</td><td>44</td></tr><tr><td>Material</td><td>-</td><td>9</td><td>9</td></tr><tr><td>Other</td><td>-</td><td>6</td><td>6</td></tr><tr><td>Total</td><td>-</td><td>59</td><td>59</td></tr></tbody></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	44	44	Material	-	9	9	Other	-	6	6	Total	-	59	59
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	-	44	44																		
Material	-	9	9																		
Other	-	6	6																		
Total	-	59	59																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 76 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Tie 7504 (Pg. 1 of 4)
Purpose and Necessity:	Customers served from Cass City Substation and Proctor Substation have low voltage upon the contingency event of the TIE 7504 breaker operation at Arrowhead Station (Cass City will be at 86% of nominal voltage and Proctor will be at 91% of nominal voltage). This violates the Subtransmission Planning Criteria and Distribution Design Order-0070-008: Under emergency conditions, for single contingency events (N-1), Subtransmission voltage limits are 92%-105% of the nominal voltage. Currently up to 7 MVA of load is shed to prevent this low voltage and keep the remaining customers served from this Tie line within the voltage criteria. Recent new load requests have been denied due to existing voltage violations on this Tie line. Customer complaints have also been received due to frequent outages on TIE 7504 in areas where circuit is in deep right-of-way and has had repeated outages due to tree interference and failure of 3/0 ACSR conductor. For the contingency event when Arrowhead Transformer 102 is taken out of service, customers in the area will experience low voltage. Arrowhead Transformer 102 has system load restrictions for maintenance/equipment upgrades due to these low voltage violations. This violates the Subtransmission Planning Criteria and Distribution Design Order-0070-008: Under emergency conditions, for single contingency events (N-1), Subtransmission voltage limits are 92%-105% of the nominal voltage.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 77 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Tie 7504 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 20
DGP Section:	11.2
Scope:	• Reconductor 16 miles of overhead conductor on TIE 7504. Remove 3/0 ACSR conductor and install 636 ACSR conductor with 1/0 ACSR lightning protection. Where TIE 7504 is built in deep right-of-way, relocate the Tie line to be truck accessible for improved reliability and maintenance. • ITC to tap the 120 kV line Tuscola-Klam and extend the 120 kV line 12 miles to Wilmot Station. • Expand Wilmot station by installing a 120-40 kV transformer, circuit switcher, and two 40 KV breakers, upgrading relays, and reconfiguring busses.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 78 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Tie 7504 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Eliminate customer low voltage events. • Prevent customer load shed for a single contingency event. • Improve system shutdown capability and operability. • Improve customer reliability and power quality. • Provide capacity for new load growth.						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:		X	X	X	X		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 79 of 347

(a)	(b)																												
Program:	Subtransmission Redesign & Rebuild: Tie 7504 (Pg. 4 of 4)																												
Current Projects:	Subtransmission Redesign & Rebuild: Tie 7504																												
Budget Basis:	• Project Management Estimate																												
Cost:	Estimated project spend in 2021-2023: \$1.1 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>671</td><td>671</td></tr><tr><td>Material</td><td>-</td><td>134</td><td>134</td></tr><tr><td>Other</td><td>-</td><td>90</td><td>90</td></tr><tr><td>Total</td><td>-</td><td>895</td><td>895</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	-	671	671	Material	-	134	134	Other	-	90	90	Total	-	895	895
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	-	671	671																										
Material	-	134	134																										
Other	-	90	90																										
Total	-	895	895																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 80 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Tie 4105 (Pg. 1 of 4)
Purpose and Necessity:	TIE 4105 is a 61-mile Tie line between Lee and Sandusky stations that has received escalated customer complaints due to repeat power outages. Frequent lightning strikes have been correlated with breaker operations on TIE 4105 to demonstrate that the lack of lightning protection is causing lightning to strike the phase conductors resulting in phase to phase faults on the Subtransmission conductor. Review of the TIE 4105 pole inspection data has shown that a high number of wood poles are deteriorated and require replacement. Review of the sustained and momentary outages in recent years has identified this circuit as a poor performer, and this data was used to identify which segments of TIE 4105 have experienced the most outages. Sections of the 3/0 ACSR conductor have been taken from this Tie line for evaluation of remaining tensile strength in the conductor. It has been determined that the steel core is deteriorated, and the remaining tensile strength greatly reduced. This conductor was determined to be at the end of its life based on this evaluation. The existing TIE 4105 is built in deep Right-of-way, which contribute to frequent outages from tree interference as well as outage duration due to the line being not truck accessible. There are 3,543 customers directly impacted by the reliability of TIE 4105.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 81 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Tie 4105 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 21
DGP Section:	11.2
Scope:	• Reconductor 37.6 miles of 3/0 ACSR conductor on TIE 4105 with 636 Al conductor with 1/0 ACSR lightning protection. Relocate the circuit from deep right-of-way to become truck accessible to allow for improved reliability and maintenance. • Rebuild approximately 18.6 miles of distribution under build on the new poles for TIE 4105.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 82 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Tie 4105 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce wire down events. • Reduce total outage events and durations. • Improve customer reliability and power quality • Provide additional capacity for customer load growth						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x			x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 83 of 347

(a)	(b)																												
Program:	Subtransmission Redesign & Rebuild: Tie 4105 (Pg. 4 of 4)																												
Current Projects:	Subtransmission Redesign & Rebuild: Tie 4105																												
Budget Basis:	• Project Management Estimate																												
Cost:	Estimated project spend in 2021-2023: \$35.7 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>1,250</td><td>12,468</td><td>13,718</td></tr><tr><td>Material</td><td>250</td><td>2,494</td><td>2,744</td></tr><tr><td>Other</td><td>167</td><td>1,662</td><td>1,829</td></tr><tr><td>Total</td><td>1,667</td><td>16,624</td><td>18,290</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	1,250	12,468	13,718	Material	250	2,494	2,744	Other	167	1,662	1,829	Total	1,667	16,624	18,290
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	1,250	12,468	13,718																										
Material	250	2,494	2,744																										
Other	167	1,662	1,829																										
Total	1,667	16,624	18,290																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 84 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Trunk 2308 (Pg. 1 of 4)
Purpose and Necessity:	TRK2308 A-B section is at 99% (5.8 MVA over the allowable load) of the Summer Emergency rating upon the contingency event of breaker operation on the Macomb-Stephens 120kV transmission line. RK 2308 C-D section is at 97% (2.2 MVA over the allowable load) of its Summer Emergency rating upon the contingency event of breaker operation on the Macomb-Stephens 120kV transmission line. There is an Emergency Load Control scheme at BENSN substation that will automatically shed load (5.4 MVA) to protect from overloading equipment and causing damage for a single contingency event. Distribution Design Order-0090-013 (Subtransmission Loading Criteria Violations): Allowable summer normal configuration (day to day) loading on subtransmission equipment is 100% and summer emergency loading on subtransmission equipment is 90%. Distribution Design Order-0040-003 (Substation automatic throwover Blocking): Substation Load blocking schemes are not acceptable
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 85 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Trunk 2308 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 22
DGP Section:	11.2
Scope:	• Reconductor 1.8 miles of 350AL overhead lines with 636AL SW-ST (enhanced 40kV standard, armless, steel pole construction) and 1.8 miles of distribution under build. • Reconductor 0.6 miles of OCU overhead lines with 636AL SW-ST (enhanced 40kV standard, armless, steel pole construction) and 0.6 miles of distribution under build. • Reconductor 0.5 miles of 3/0ACSR overhead lines with 636AL SW-ST (enhanced 40kV standard, armless, steel pole construction) and 0.5 miles of distribution under build.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 86 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Trunk 2308 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Alleviate thermal loading violations, and the possibility of shedding customer load during a single contingency event. • New equipment ratings lower the percent of capacity utilized by the current system loading. • Replace 3/0 ACSR wire that is prone to failures that result in wire down events. • Improve system shutdown capability and operability						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x	x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 87 of 347

(a)	(b)																												
Program:	Subtransmission Redesign & Rebuild: Trunk 2308 (Pg. 4 of 4)																												
Current Projects:	Subtransmission Redesign & Rebuild: Trunk 2308																												
Budget Basis:	• Project Management Estimate																												
Cost:	Estimated project spend in 2021-2023: \$1.9 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>1,186</td><td>1,186</td></tr><tr><td>Material</td><td>-</td><td>237</td><td>237</td></tr><tr><td>Other</td><td>-</td><td>158</td><td>158</td></tr><tr><td>Total</td><td>-</td><td>1,582</td><td>1,582</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	-	1,186	1,186	Material	-	237	237	Other	-	158	158	Total	-	1,582	1,582
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	-	1,186	1,186																										
Material	-	237	237																										
Other	-	158	158																										
Total	-	1,582	1,582																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 88 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Tie 1568 (Pg. 1 of 4)
Purpose and Necessity:	TIE 1568 B-C section is at 99% (8.7 MVA over the allowable load) of the Summer Emergency rating upon the contingency event of breaker operation on the Noble-Superior 120kV transmission line. There is an Emergency Load Control scheme at CROWN substation that will automatically shed load (10.1 MVA) to protect from overloading equipment and causing damage for a single contingency event. Distribution Design Order-0090-013 (Subtransmission Loading Criteria Violations): Allowable summer normal configuration (day to day) loading on subtransmission equipment is 100% and summer emergency loading on subtransmission equipment is 90%. Distribution Design Order-0040-003 (Substation Blocking): Substation Load blocking schemes are not acceptable
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 89 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Tie 1568 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 23
DGP Section:	11.2
Scope:	• Install 0.2 miles of 650EPR cable on CROWN tap. • Remove 0.2 miles of existing TIE 1568 500CU cable. • Replace existing Cable Pole 1 with steel pole.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 90 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Tie 1568 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Alleviate thermal loading violations, and the possibility of shedding customer load during a single contingency event. • New equipment ratings lower the percent of capacity utilized by the current system loading. • Improve system shutdown capability and operability.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x	x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 91 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: Tie 1568 (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: Tie 1568																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.9 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>6</td><td>539</td><td>545</td></tr><tr><td>Material</td><td>1</td><td>108</td><td>109</td></tr><tr><td>Other</td><td>1</td><td>72</td><td>73</td></tr><tr><td>Total</td><td>8</td><td>718</td><td>726</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	6	539	545	Material	1	108	109	Other	1	72	73	Total	8	718	726
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	6	539	545																		
Material	1	108	109																		
Other	1	72	73																		
Total	8	718	726																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 92 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Trunk 4217 (Pg. 1 of 4)
Purpose and Necessity:	TRK 4217 is at 123% (4.3 MVA over the allowable load) of the Summer Normal rating. TRK 4217 is at 116% (3.5 MVA over the allowable load) of the Summer Emergency rating upon the contingency event of breaker operation on TRK 4210. TRK 4217 is at 103% (2.3 MVA over the allowable load) of the Summer Emergency rating upon the contingency event of breaker operation on TRK 4218. TRK 4217 is at 100% (2.3 MVA over the allowable load) of the Summer Emergency rating upon the contingency event of breaker operation on TRK 1633. There is an Emergency Load Control scheme at GRPTE and VERNR substations that will automatically shed load (3.5 MVA) to protect from overloading equipment and causing damage for a single contingency event. Distribution Design Order-0090-013 (Subtransmission Loading Criteria Violations): Allowable summer normal configuration (day to day) loading on subtransmission equipment is 100% and summer emergency loading on subtransmission equipment is 90%. Distribution Design Order-0040-003 (Substation Blocking): Substation Load blocking schemes are not acceptable
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 93 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Trunk 4217 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 24
DGP Section:	11.2
Scope:	• At ERIN station install a new 40kV, 2000A breaker and two 40kV, 1200A disconnects for new station position. • Install 2.1 miles of 650EPR cable from new station position to manhole 14768. • Remove 2.1 miles of existing TRK4217 500CU cable. • Replace VERNR transformer 3 secondary breaker and cable terminations.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 94 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Trunk 4217 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Alleviate thermal loading violations, and the possibility of shedding customer load during a single contingency event. • New equipment ratings lower the percent of capacity utilized by the current system loading. • Improve system shutdown capability and operability. • Reduced substation outage risk by replacing aging infrastructure.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 95 of 347

(a)	(b)																												
Program:	Subtransmission Redesign & Rebuild: Trunk 4217 (Pg. 4 of 4)																												
Current Projects:	Subtransmission Redesign & Rebuild: Trunk 4217																												
Budget Basis:	• Project Management Estimate																												
Cost:	Estimated project spend in 2021-2023: \$5.3 M																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>7</td><td>3,125</td><td>3,132</td></tr><tr><td>Material</td><td>1</td><td>625</td><td>626</td></tr><tr><td>Other</td><td>1</td><td>417</td><td>418</td></tr><tr><td>Total</td><td>9</td><td>4,167</td><td>4,176</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	7	3,125	3,132	Material	1	625	626	Other	1	417	418	Total	9	4,167	4,176
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	7	3,125	3,132																										
Material	1	625	626																										
Other	1	417	418																										
Total	9	4,167	4,176																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 96 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Trunk 3509 (Pg. 1 of 4)
Purpose and Necessity:	TRK 3509 is at 100% (0 MVA over the allowable load) of the Summer Normal rating. TRK 3509 is at 109% (6.2 MVA over the allowable load) of the Summer Emergency rating upon the contingency event of breaker operation on TRK 3536. There is an Emergency Load Control scheme at PATON substation that will automatically shed load (4.1 MVA) to protect from overloading equipment and causing damage for a single contingency event. Distribution Design Order-0090-013 (Subtransmission Loading Criteria Violations): Allowable summer normal configuration (day to day) loading on subtransmission equipment is 100% and summer emergency loading on subtransmission equipment is 90%. Distribution Design Order-0040-003 (Substation Blocking): Substation Load blocking schemes are not acceptable
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 97 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Trunk 3509 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 25
DGP Section:	11.2
Scope:	• At TROY station install a new 40kV, 2000A breaker and two 40kV, 1200A disconnects for new station position. • Install 0.1 miles of 5-inch conduit. • Install 1.1 miles of 650 EPR from new station position to manhole 16357. • Install 0.1 miles of 650 EPR from manhole 16357 to cable pole 5. • Install 0.1 miles of 650 EPR from manhole 16357 to manhole 16358. • Remove 1.3 miles of existing TRK3509 750CU cable.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 98 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Trunk 3509 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Alleviate thermal loading violations, and the possibility of shedding customer load during a single contingency event. • New equipment ratings lower the percent of capacity utilized by the current system loading. • Improve system shutdown capability and operability. • Improved reliability by reducing total circuit length.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x			

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 99 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: Trunk 3509 (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: Trunk 3509																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$1.6 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>8</td><td>940</td><td>947</td></tr><tr><td>Material</td><td>2</td><td>188</td><td>189</td></tr><tr><td>Other</td><td>1</td><td>125</td><td>126</td></tr><tr><td>Total</td><td>10</td><td>1,253</td><td>1,263</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	8	940	947	Material	2	188	189	Other	1	125	126	Total	10	1,253	1,263
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	8	940	947																		
Material	2	188	189																		
Other	1	125	126																		
Total	10	1,253	1,263																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 100 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Trunk 4266 (Pg. 1 of 4)
Purpose and Necessity:	TRK 4266 is at 102% (4 MVA over the allowable load) of the Summer Normal rating. TRK 4266 is at 112% (5.8 MVA over the allowable load) of the Summer Emergency rating upon the contingency event of breaker operation on TRK 4227. There is an Emergency Load Control scheme at SAVOY substation that will automatically shed load (6.2 MVA) to protect from overloading equipment and causing damage for a single contingency event. Distribution Design Order-0090-013 (Subtransmission Loading Criteria Violations): Allowable summer normal configuration (day to day) loading on subtransmission equipment is 100% and summer emergency loading on subtransmission equipment is 90%. Distribution Design Order-0040-003 (Substation Blocking): Substation Load blocking schemes are not acceptable
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 101 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Trunk 4266 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 26
DGP Section:	11.2
Scope:	• Install 2.5 miles of 650 EPR cable to replace TRK4227 "A" cable, Bridge Main 066, SAVOY transformer 1 tap, and BAKER transformer 1 tap. • Remove 1.3 miles of 500 CU cable. • Remove 1.2 miles of 750 CU cable.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 102 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Trunk 4266 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Alleviate thermal loading violations, and the possibility of shedding customer load during a single contingency event. • New equipment ratings lower the percent of capacity utilized by the current system loading. • Improve system shutdown capability and operability.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x	x		x			

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 103 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: Trunk 4266 (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: Trunk 4266																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$3.8 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>38</td><td>2,218</td><td>2,255</td></tr><tr><td>Material</td><td>8</td><td>444</td><td>451</td></tr><tr><td>Other</td><td>5</td><td>296</td><td>301</td></tr><tr><td>Total</td><td>50</td><td>2,957</td><td>3,007</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	38	2,218	2,255	Material	8	444	451	Other	5	296	301	Total	50	2,957	3,007
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	38	2,218	2,255																		
Material	8	444	451																		
Other	5	296	301																		
Total	50	2,957	3,007																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 104 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Badax Transformer 102 Addition (Pg. 1 of 4)
Purpose and Necessity:	BADAX station serves approximately 16,700 customers and four tie lines. Upon the contingency event of breaker operation on BADAX Transformer 101, 60 voltage violations and three thermal violations. The system is at risk for these violations when total system load exceeds 80% of summer peak load.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 105 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Badax Transformer 102 Addition (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 27
DGP Section:	11.2
Scope:	• Extend BADAX 40 kV Bus 2 • Install one 120-40 kV 45/60/75 MVA Transformer 102 from ITC's Bus 102 to new 40 kV position on BADAX Bus 2 • Install one 120 kV S&C 2020 circuit switcher • Install new 40kV BADAX Bus 3 • Install (4) 40kV, 2000A breakers and two 1200A • Install (8) 40kV, 1200A disconnects on new 40kV • Install new 6.6 MVAR Cap 3 on Bus 3

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 106 of 347

(a) Program:	(b) Subtransmission Redesign & Rebuild: Badax Transformer 102 Addition (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Alleviate thermal loading violations, and the possibility of shedding customer load during a single contingency event. • Alleviate voltage violations, and the possibility of shedding customer load during a single contingency event. • New equipment ratings lower the percent of capacity utilized by the current system loading. • Improve system shutdown capability and operability.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x	x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 107 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: Badax Transformer 102 Addition (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: Badax Transformer 102 Addition																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$3.6 M																				
Test Year (\$000's):	<table><thead><tr><th></th><th>2 mos. 11/01/22 12/31/22</th><th>10 mos. 01/01/23 10/31/23</th><th>Test Year 12 mos. ending 10/31/23</th></tr></thead><tbody><tr><td>Total Labor (DTE Electric & Contractors)</td><td>278</td><td>856</td><td>1,134</td></tr><tr><td>Material</td><td>56</td><td>171</td><td>227</td></tr><tr><td>Other</td><td>37</td><td>114</td><td>151</td></tr><tr><td>Total</td><td>370</td><td>1,141</td><td>1,512</td></tr></tbody></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	278	856	1,134	Material	56	171	227	Other	37	114	151	Total	370	1,141	1,512
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	278	856	1,134																		
Material	56	171	227																		
Other	37	114	151																		
Total	370	1,141	1,512																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 108 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Trunk 4245 (Pg. 1 of 4)
Purpose and Necessity:	TRK 4245 cable failed and is stuck in conduit under the bridge crossing over I-94. All three subtransmission feeds to SHORS substation utilize this conduit which no longer has any spare ducts. Any future cable failure in this conduit will result in a long outage and load shed (6.2 MVA) at SHORS. Distribution Design Order-0040-003 (Substation Blocking): Substation Load blocking schemes are not acceptable
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 109 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Trunk 4245 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 28
DGP Section:	11.2
Scope:	• Install 1,000 feet of new conduit (15-5" ducts) using directional drilling under I-94 • Install 1,800 feet of new 650 EPR cable for TRK 4245 • Remove 1,700 feet of 500 CU and 750 CU PLSJ cable

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 110 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Trunk 4245 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Eliminate risk of customers experiencing a long outage due to cable failure in a duct bank with no spare conduit. • Improve system shutdown capability and operability.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
				x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 111 of 347

(a)	(b)																												
Program:	Subtransmission Redesign & Rebuild: Trunk 4245 (Pg. 4 of 4)																												
Current Projects:	Subtransmission Redesign & Rebuild: Trunk 4245																												
Budget Basis:	• Project Management Estimate																												
Cost:	Estimated project spend in 2021-2023: \$4.6 M																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>38</td><td>2,680</td><td>2,717</td></tr><tr><td>Material</td><td>8</td><td>536</td><td>543</td></tr><tr><td>Other</td><td>5</td><td>357</td><td>362</td></tr><tr><td>Total</td><td>50</td><td>3,573</td><td>3,623</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	38	2,680	2,717	Material	8	536	543	Other	5	357	362	Total	50	3,573	3,623
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	38	2,680	2,717																										
Material	8	536	543																										
Other	5	357	362																										
Total	50	3,573	3,623																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 112 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Tie 6602 (Pg. 1 of 4)
Purpose and Necessity:	TIE 6602 is at 110% (3.5 MVA over the allowable load) of the Summer Emergency rating upon the contingency event of breaker operation on CVTRY TRF 301. TIE 6602 is at 107% (2.5 MVA over the allowable load) of the Summer Emergency rating upon the contingency event of breaker operation on LARK TRF 101. TIE 6602 is at 101% (0.4 MVA over the allowable load) of the Summer Emergency rating upon the contingency event of breaker operation at CHLSE and APTS 1713. There is an Emergency Load Control scheme at DIMND substation that will automatically shed load to protect from overloading equipment and causing damage for a single contingency event. Distribution Design Order-0090-013 (Subtransmission Loading Criteria Violations): Allowable summer normal configuration (day to day) loading on subtransmission equipment is 100% and summer emergency loading on subtransmission equipment is 90%. Distribution Design Order-0040-003 (Substation Blocking): Substation Load blocking schemes are not acceptable
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 113 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Tie 6602 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 29
DGP Section:	11.2
Scope:	• Rebuild approximately 3.5 miles of 3/0 ACSR with 636 AL • Rebuild all under build distribution with new conductor

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 114 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Tie 6602 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Alleviate thermal loading violations, and the possibility of shedding customer load during a single contingency event. • New equipment ratings lower the percent of capacity utilized by the current system loading. • Improve system shutdown capability and operability						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x	x		x	x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 115 of 347

(a)	(b)		
Program:	Subtransmission Redesign & Rebuild: Tie 6602 (Pg. 4 of 4)		
Current Projects:	Subtransmission Redesign & Rebuild: Tie 6602		
Budget Basis:	• Engineering Estimate		
Cost:	Estimated project spend in 2021-2023: \$0.7 M This project is expected to continue beyond 2023		
Test Year (\$000's):	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23
	Total Labor (DTE Electric & Contractors)	- 413	413
	Material	- 83	83
	Other	- 55	55
	Total	- 550	550

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 116 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Trunk 4911 (Pg. 1 of 4)
Purpose and Necessity:	TRK 4911 is at 108% of the Summer Emergency rating upon the contingency event of breaker operation on TRK 4962. TRK 4911 is at 108% of the Summer Emergency rating upon the contingency event of breaker operation on TRK 7913. There is an Emergency Load Control scheme at NBALT and CHEST substations that will automatically shed load (11.5 and 11.6 MVA respectively) to protect from overloading equipment and causing damage for a single contingency event. Low voltages at IRA - 87.8%, NBALT - 88.1%, COTTN - 89.4% upon single contingency Loss of TRK 4971. Low voltages at IRA - 89.69%, NBALT - 91.76%, COTTN - 89.96% upon single contingency loss of TRK 7913. Four more single contingency events result in low voltages for the same substations in the range of 90% - 91.7%. Distribution Design Order-0090-013 (Subtransmission Loading Criteria Violations): Allowable summer normal configuration (day to day) loading on subtransmission equipment is 100% and summer emergency loading on subtransmission equipment is 90%. Distribution Design Order-0040-003 (Substation Blocking): Substation Load blocking schemes are not acceptable
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 117 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Trunk 4911 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 30
DGP Section:	11.2
Scope:	• Reconductor 4.2 miles of 350AL with 636AL SW-ST (enhanced 40kV standard, armless, steel pole construction) and 4.2 miles of distribution under build. • Reconductor 2.5 miles of 3/0ACSR with 636AL SW-ST (enhanced 40kV standard, armless, steel pole construction) and 4.2 miles of distribution under build. • Install 9.0 miles of new 636AL SW-ST (enhanced 40kV standard, armless, steel pole construction) and replace distribution under build along route. • Install (1) 40kV, 2000A breakers and two 1200A • Install (2) 40kV, 1200A disconnects on new 40kV

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 118 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Trunk 4911 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Alleviate thermal loading violations, and the possibility of shedding customer load during a single contingency event. • Alleviate voltage violations, and the possibility of shedding customer load during a single contingency event. • New equipment ratings lower the percent of capacity utilized by the current system loading. • Improve system shutdown capability and operability. • Replace 3/0 ACSR wire known for bad condition, that is prone to failures that result in wire down events.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x			x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 119 of 347

(a)	(b)																												
Program:	Subtransmission Redesign & Rebuild: Trunk 4911 (Pg. 4 of 4)																												
Current Projects:	Sub transmission Redesign & Rebuild: Trunk 4911 Sub transmission Redesign & Rebuild: Trunk 4911 Voltage Correction																												
Budget Basis:	• Project Management Estimate																												
Cost:	Estimated project spend in 2021-2023: \$0.4 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>279</td><td>279</td></tr><tr><td>Material</td><td>-</td><td>56</td><td>56</td></tr><tr><td>Other</td><td>-</td><td>37</td><td>37</td></tr><tr><td>Total</td><td>-</td><td>373</td><td>373</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	-	279	279	Material	-	56	56	Other	-	37	37	Total	-	373	373
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	-	279	279																										
Material	-	56	56																										
Other	-	37	37																										
Total	-	373	373																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 120 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Custer Republic (Pg. 1 of 4)
Purpose and Necessity:	Republic transformer 3 is 71 years old and carrying no load. The breaker equipment at Custer positions RS and SN is 68 years old. Reconfiguring the trunk lines between Custer and Republic will allow these under-utilized assets to be decommissioned.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 121 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Custer Republic (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 31
DGP Section:	11.2
Scope:	• Decommission Republic transformer 3 • Reconfigure TRK1456 and TRK1470 to feed Republic transformer 1 and 2. • Decommission Custer position RS and SN • Decommission TRK 1459 • Replace relay panels at Custer position RM

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 122 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Custer Republic (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce substation outage risk by replacing aging infrastructure • Reduce O&M costs by removing under-utilized equipment						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
			x	x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 123 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: Custer Republic (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: Custer Republic																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.1 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>56</td><td>56</td></tr><tr><td>Material</td><td>-</td><td>11</td><td>11</td></tr><tr><td>Other</td><td>-</td><td>7</td><td>7</td></tr><tr><td>Total</td><td>-</td><td>74</td><td>74</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	56	56	Material	-	11	11	Other	-	7	7	Total	-	74	74
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	-	56	56																		
Material	-	11	11																		
Other	-	7	7																		
Total	-	74	74																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 124 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Pigeon Area Improvement (Pg. 1 of 4)
Purpose and Necessity:	The Pigeon Area is in the North Western part of the Thumb. This area consists of multiple class B stations in series as well as undersized conductor which has resulted in multiple thermal and voltage violations. Upon contingency events load shed is required to restore voltage to usable levels for the customer and avoid overloads. • 15 unique thermal violations for single contingency. • 66 unique voltage violations for single contingency. • 14MVA load shed required to mitigate voltage violations. The majority of the overloaded conductor in this area is 3/0ASCR.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 125 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Pigeon Area Improvement (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 32
DGP Section:	11.2
Scope:	• Extend 120kV approximately 1 mile from COSMO to PIGON utilizing 954ACSR. • Install approximately 0.5 miles 120kV 954ACSR between COSMO and BADAX. • Install 120kV breaker at BADAX, along with associated disconnects, current transformers and relay panel. • Install 120kV one way switch at new tap to PIGON. • Install 60/80/100MVA 120-40kV transformer. • Install 120kV high side circuit switcher. • Install 40k transformer secondary breaker along with associated current transformers, disconnects and trainers. • Install new transformer relay panel. • Reconductor 17 miles of various conductor sizes with 636AL SW-ST (enhanced 40kV standard, armless, steel pole construction) and 17 miles of distribution under build.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 126 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Pigeon Area Improvement (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Alleviate thermal loading violations, and the possibility of shedding customer load during a single contingency event. • 78% reduction in total # of thermal violations for the area. • Alleviate voltage violations, and the possibility of shedding customer load during a single contingency event. • 33% reduction in total # of voltage violations for the area. • New equipment ratings lower the percent of capacity utilized by the current system loading. • Improve system shutdown capability and operability. • Replace 3/0 ACSR wire known for bad condition, that is prone to failures that result in wire down events.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x			x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 127 of 347

(a)	(b)																												
Program:	Subtransmission Redesign & Rebuild: Pigeon Area Improvement (Pg. 4 of 4)																												
Current Projects:	Subtransmission Redesign & Rebuild: Pigeon Area Improvement																												
Budget Basis:	• Project Management Estimate																												
Cost:	Estimated project spend in 2021-2023: \$3.6 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>58</td><td>1,929</td><td>1,988</td></tr><tr><td>Material</td><td>12</td><td>386</td><td>398</td></tr><tr><td>Other</td><td>8</td><td>257</td><td>265</td></tr><tr><td>Total</td><td>78</td><td>2,573</td><td>2,650</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	58	1,929	1,988	Material	12	386	398	Other	8	257	265	Total	78	2,573	2,650
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	58	1,929	1,988																										
Material	12	386	398																										
Other	8	257	265																										
Total	78	2,573	2,650																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 128 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Trunk 2443 (Pg. 1 of 4)
Purpose and Necessity:	The underground cable 'E' located on section D-F on TRK 2443 was observed to be loaded to 111% of its summer emergency thermal rating when the load of Angola substation thrown over to this circuit in an emergency scenario. The limiting element causing this overload scenario is 0.9 miles of 500 cooper underground cable. This scenario violates our Distribution Design Order-0090-013 (Subtransmission Loading Criteria) which states that allowable summer normal configuration (day to day) loading on subtransmission equipment is 100% and summer emergency loading on subtransmission equipment is 90%. The current potential load shed, to protect from overloading equipment and causing damage, for a single contingency event is 7 MVA affecting 1,879 customers.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 129 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Trunk 2443 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 33
DGP Section:	11.2
Scope:	• Replace existing 0.9 miles of 500 cooper underground cable (cable 'E') using new 650 copper underground cable on section D-F on TRK 2443. • Use existing 5" conduit from manhole 18055 to manhole 16856 (12 MHs).

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 130 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Trunk 2443 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	The overload on the underground cable 'E' on TRK 2443 section D-F will be eliminated. The newly rebuilt section will have summer emergency loading reduced from 111% to 54% with total emergency capacity increased from 30 MVA to 61 MVA.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x			

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 131 of 347

(a)	(b)																												
Program:	Subtransmission Redesign & Rebuild: Trunk 2443 (Pg. 4 of 4)																												
Current Projects:	Subtransmission Redesign & Rebuild: Trunk 2443																												
Budget Basis:	• Project Management Estimate																												
Cost:	Estimated project spend in 2021-2023: \$0.1 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>78</td><td>78</td></tr><tr><td>Material</td><td>-</td><td>16</td><td>16</td></tr><tr><td>Other</td><td>-</td><td>10</td><td>10</td></tr><tr><td>Total</td><td>-</td><td>103</td><td>103</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	-	78	78	Material	-	16	16	Other	-	10	10	Total	-	103	103
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	-	78	78																										
Material	-	16	16																										
Other	-	10	10																										
Total	-	103	103																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 132 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Trunk 5838 (Pg. 1 of 4)
Purpose and Necessity:	An overhead wire located on section A-B on TRK 5838 is loaded to 97% of its summer emergency thermal rating when an outage event occur on TRK 5817. The limiting element causing this overload scenario is 0.43 miles of 0 aluminum conductor overhead wire. This scenario violates our Distribution Design Order-0090-013 (Subtransmission Loading Criteria) which states that allowable summer normal configuration (day to day) loading on subtransmission equipment is 100% and summer emergency loading on subtransmission equipment is 90%.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 133 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Trunk 5838 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 34
DGP Section:	11.2
Scope:	Replace existing 0.43 miles (approximately) of 0 aluminum conductor and 1.57 miles (approximately) of 0 copper overhead wire with new 636 aluminum conductor, 3/0 aluminum conductor shield wire.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 134 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Trunk 5838 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	The overload on the overhead wire on TRK5838 A-B section will be eliminated. The section loading will be reduced from 97% to 68%.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x					

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 135 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: Trunk 5838 (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: Trunk 5838																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.2 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>111</td><td>111</td></tr><tr><td>Material</td><td>-</td><td>22</td><td>22</td></tr><tr><td>Other</td><td>-</td><td>15</td><td>15</td></tr><tr><td>Total</td><td>-</td><td>148</td><td>148</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	111	111	Material	-	22	22	Other	-	15	15	Total	-	148	148
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	-	111	111																		
Material	-	22	22																		
Other	-	15	15																		
Total	-	148	148																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 136 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Trunk 2419 (Pg. 1 of 4)
Purpose and Necessity:	An underground cable located on section B-G on TRK 2419 is loaded to 97% of its summer emergency thermal rating when an outage event occur on TRK 2404. The limiting element causing this overload scenario is 0.32 miles of 350 cooper underground cable. This scenario violates our Distribution Design Order-0090-013 (Subtransmission Loading Criteria) which states that allowable summer normal configuration (day to day) loading on subtransmission equipment is 100% and summer emergency loading on subtransmission equipment is 90%.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 137 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: Trunk 2419 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 35
DGP Section:	11.2
Scope:	• Replace existing 0.32 miles of 350 copper underground cable using new 600 copper underground cable on section B-G on TRK 2419. • Use existing 5" conduit from MH 14246 to MH 13938 (7 Manholes).

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 138 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Trunk 2419 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	The overload on the underground cable on TRK 2419 will be eliminated. The loading on section B-G will decrease from 97% to 58% of its summer emergency thermal rating. The circuit capacity on this section will increase from 14.5 MVA to 24.3 MVA.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x			

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 139 of 347

(a)	(b)																												
Program:	Subtransmission Redesign & Rebuild: Trunk 2419 (Pg. 4 of 4)																												
Current Projects:	Subtransmission Redesign & Rebuild: Trunk 2419																												
Budget Basis:	• Project Management Estimate																												
Cost:	Estimated project spend in 2021-2023: \$0.2 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>106</td><td>106</td></tr><tr><td>Material</td><td>-</td><td>21</td><td>21</td></tr><tr><td>Other</td><td>-</td><td>14</td><td>14</td></tr><tr><td>Total</td><td>-</td><td>141</td><td>141</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	-	106	106	Material	-	21	21	Other	-	14	14	Total	-	141	141
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	-	106	106																										
Material	-	21	21																										
Other	-	14	14																										
Total	-	141	141																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 140 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild : Small Projects & Reserve (Pg. 1 of 4)
Purpose and Necessity:	This category includes small projects aimed at addressing thermal overloads and voltage violations on the Subtransmission system. Circuits Identified violate Subtransmission Planning Criteria and Distribution Design Order-0090-013 or Distribution Design Order-0070-00.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 141 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild : Small Projects & Reserve (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 36
DGP Section:	11.2
Scope:	This category represents a blanket to address small and localized overload conditions and voltage violations on the Subtransmission system Individual project scope will be small, such as upgrade a relay panel, reconductor approximately 0.2 miles of overhead conductor, replace approximately 0.1 miles of cable, replace trainers, etc. Some examples identified for this program include: • TRK 2559 – Replace 0.003 miles of 350 Copper PLJ cable to eliminate thermal overload. • TIE 5611 – Replace Relay Panel at Neff • TRK 2508 – Replace 0.1 miles of 500 Copper PLJ cable to eliminate thermal overload • TRK 5127 – Replace 0.1 miles of 350 Copper PLJ cable to eliminate thermal overload.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 142 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild : Small Projects & Reserve (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Eliminate thermal loading violation of the Subtransmission Planning Criteria and Distribution Design Order-0090-013. • Eliminate customer low voltage events and violations of the Subtransmission Planning Criteria and Distribution Design Order-0070-008. • Prevent customer load shed for a single contingency event.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x	x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 143 of 347

(a)	(b)																												
Program:	Subtransmission Redesign & Rebuild : Small Projects & Reserve (Pg. 4 of 4)																												
Current Projects:	Subtransmission Redesign & Rebuild : Small Projects & Reserve																												
Budget Basis:	• Blanket funding																												
Cost:	Estimated project spend in 2021-2023: \$3.5 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>2,188</td><td>2,188</td></tr><tr><td>Material</td><td>-</td><td>438</td><td>438</td></tr><tr><td>Other</td><td>-</td><td>292</td><td>292</td></tr><tr><td>Total</td><td>-</td><td>2,917</td><td>2,917</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	-	2,188	2,188	Material	-	438	438	Other	-	292	292	Total	-	2,917	2,917
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	-	2,188	2,188																										
Material	-	438	438																										
Other	-	292	292																										
Total	-	2,917	2,917																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 144 of 347

(a)	(b)
Program:	Subtransmission Breaker Short Circuit Violations (Pg. 1 of 4)
Purpose and Necessity:	In 2020 Fault duties were evaluated at approximately 989 subtransmission breaker locations on the DTE Electric System. Momentary and Interrupting fault duties were determined for comparison to each breaker's capabilities. Twenty breakers could be exposed to fault currents higher than their interrupting ratings. Stations include: Badax, Elm, Erin, Evergreen, Lincoln, Milford, Northwest, and Red Run. Of these 20 breakers, 9 work orders were issued in 2020 under the breaker replacement program to upgrade these breakers.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 145 of 347

(a)	(b)
Program:	Subtransmission Breaker Short Circuit Violations (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 37
DGP Section:	11.2
Scope:	• Erin: Replace breaker at position SA with 48PM31-20. • Evergreen: Install 1.5 ohm grounding device on EVRGN TRF 101, TRF 102, & TRF 103. • Lincoln: Replace breakers at positions RG & RP with 72PM44-20. • Northwest: Replace breaker at position RZ with ABB 72PM44-20. • Red Run: Replace breakers at position RI, RN, & RO with ABB 48PM31-20. • Include relay panel replacements with breaker replacements. • Temporary operating mitigations will be necessary to reduce the fault current these breakers are exposed to, to prevent breaker failure. The proposed temporary mitigations include: • Badax: Disable automatic reclosing on TIE 3416, TIE 3402, TIE 3417, & TIE 3410 to prevent testing of the line during permanent faults. • Elm: Open SB1-2, ERIN: Open SB2-3, EVRGN: Open SB3-4, need to evaluate the Evergreen Capacitors 1, 2, 4, & 6 ON / OFF switching times to ensure adequate voltage / Var flows on the 40kV & 120kV system depending on the system load. • Lincoln: TBD, MILFD: Disable automatic reclosing on TIE 6117 & TIE 6147 to prevent testing of the line during permanent faults. • Northwest: Open SB2-3, REDRN: Open SB1-2

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 146 of 347

(a)	(b)						
Program:	Subtransmission Breaker Short Circuit Violations (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Subtransmission breakers will not be exposed to faults they are not capable to interrupt based on the existing breaker's ratings. • Eliminate the possibility for a breaker failure at these locations. • Upgrade breakers and relay panels at these locations to meet current design standards.						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x			x	x		x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 147 of 347

(a)	(b)																												
Program:	Subtransmission Breaker Short Circuit Violations (Pg. 4 of 4)																												
Current Projects:	Subtransmission Breaker Short Circuit Violations																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$6.1 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>364</td><td>1,500</td><td>1,864</td></tr><tr><td>Material</td><td>73</td><td>300</td><td>373</td></tr><tr><td>Other</td><td>49</td><td>200</td><td>249</td></tr><tr><td>Total</td><td>486</td><td>2,000</td><td>2,486</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	364	1,500	1,864	Material	73	300	373	Other	49	200	249	Total	486	2,000	2,486
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	364	1,500	1,864																										
Material	73	300	373																										
Other	49	200	249																										
Total	486	2,000	2,486																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 148 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Slocum (Pg. 1 of 4)
Purpose and Necessity:	The Trenton Channel Power Plant, including all outlying structures and buildings are planned for retirement in 2022 and subsequent demolition. TRK 1110, TRK 1116, TRK 1140 will be redesigned to remove all unused equipment.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 149 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: Slocum (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 38
DGP Section:	11.2
Scope:	TRK 1116 Scope: • Decommission and remove approximately 0.8 miles of 40kV overhead conductor from Slocum tap to system service 71 transformer. • Decommission and remove approximately 0.41 miles of 24kV underground cable from cable pole 1 to system service 60 transformer. • Decommission and remove pole top switch (PTS) 542 and pole top switch (PTS) 556. • Decommission and remove cable pole 1 and fuse on cable pole 1. TRK 1140 Scope (will be decommissioned): • Decommission and remove TRK 1140. • Decommission and remove approximately 2,530 feet of 24kV 350 paper and lead cable from Slocum position RF to system service 62 pothead at Trenton Channel Power Plant. SLOCM Station Scope: • Remove section breaker 1-2 and tie bus 1 and bus 2. • Jumper both busses together after removed of section breaker. • Remove disconnect switch at position RH. • Install 150 feet of 3x4-5" duct.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 150 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: Slocum (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	The redesign of the subtransmission circuits will help improve the reliability by eliminating unnecessary exposure to potential outages in the electric system and by removing unused aging infrastructure. The removal of unused underground cables will also free up underground ducts that can be used for future projects in the area.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x	x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 151 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: Slocum (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: Slocum																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$2.2 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>63</td><td>1,088</td><td>1,150</td></tr><tr><td>Material</td><td>13</td><td>218</td><td>230</td></tr><tr><td>Other</td><td>8</td><td>145</td><td>153</td></tr><tr><td>Total</td><td>83</td><td>1,450</td><td>1,533</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	63	1,088	1,150	Material	13	218	230	Other	8	145	153	Total	83	1,450	1,533
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	63	1,088	1,150																		
Material	13	218	230																		
Other	8	145	153																		
Total	83	1,450	1,533																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 152 of 347

(a)	(b)
Program:	Station Upgrade: Cortland Station Expansion (Pg. 1 of 4)
Purpose and Necessity:	In the event of a transformer failure at Cortland, removing and replacing a failed transformer would take an excessive amount of time due to the station configuration which puts the customers served from the station at risk (approximately 115 MVA, 49,255 customers & two industrial customers). The station currently possesses physical restrictions that will not allow the installation of new equipment following current design standards.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 153 of 347

(a)	(b)
Program:	Station Upgrade: Cortland Station Expansion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 39
DGP Section:	11.2
Scope:	• Install two new 120-24 kV TRF 104 and TRF 105 to replace existing Cortland transformers 101 and 102 on property adjacent to Cortland Station with provisions to relocate Cortland TRF 103 on new property at a future date. • Install two S&C 2020 120kV circuit switchers • Install approximately 300 feet of 3x4 5" conduit from transformer 104 to 24kV position RJ • Install approximately 300 feet of 24kV secondary cable from transformer 104 to 24kV position RJ • Install approximately 300 feet of 3x4 5" conduit from transformer 105 to Cortland position SJ • Install approximately 300 feet of 24kV secondary cable from transformer 105 to Cortland position SJ

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 154 of 347

(a)	(b)						
Program:	Station Upgrade: Cortland Station Expansion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	Significantly reduce the risk of losing approx. 49,255 customers & two industrial substations in Highland Park and Detroit served from Cortland transformer 101, 102, and 103.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x	x		x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 155 of 347

(a)	(b)																				
Program:	Station Upgrade: Cortland Station Expansion (Pg. 4 of 4)																				
Current Projects:	Station Upgrade: Cortland Station Expansion																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$9.4 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>236</td><td>3,972</td><td>4,208</td></tr><tr><td>Material</td><td>47</td><td>794</td><td>842</td></tr><tr><td>Other</td><td>31</td><td>530</td><td>561</td></tr><tr><td>Total</td><td>314</td><td>5,296</td><td>5,610</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	236	3,972	4,208	Material	47	794	842	Other	31	530	561	Total	314	5,296	5,610
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	236	3,972	4,208																		
Material	47	794	842																		
Other	31	530	561																		
Total	314	5,296	5,610																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 156 of 347

(a)

(b)

Program:	Subtransmission Redesign & Rebuild: 40kV Capacitor Banks at Armada and Adair (Pg. 1 of 4)
Purpose and Necessity:	TIE 810 is a three ended tie line consisting of 50 miles of overhead lines serving 10 substations. Low voltages are currently seen on this circuit due to significant circuit mileage and load. This work is necessary to maintain adequate voltage along TIE 810 for multiple emergency contingencies and to help facilitate shutdowns for the TIE 810 GRAMER project.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 157 of 347

(a)	(b)
Program:	Subtransmission Redesign & Rebuild: 40kV Capacitor Banks at Armada and Adair (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 40
DGP Section:	11.2
Scope:	Install a new 40kV 9.6MVAR capacitor bank at Adair substation: Install new control house, extend substation fence and a new 40kV 9.6MVAR capacitor bank at Armada substation.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 158 of 347

(a)	(b)						
Program:	Subtransmission Redesign & Rebuild: 40kV Capacitor Banks at Armada and Adair (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	The voltage on the circuit will be improved increasing operational flexibility on TIE 810. This project will also help facilitate the circuit shutdowns needed to construct TIE 810 reconductoring project.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x			

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 159 of 347

(a)	(b)																				
Program:	Subtransmission Redesign & Rebuild: 40kV Capacitor Banks at Armada and Adair (Pg. 4 of 4)																				
Current Projects:	Subtransmission Redesign & Rebuild: 40kV Capacitor Banks at Armada and Adair																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$3.3 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>284</td><td>-</td><td>284</td></tr><tr><td>Material</td><td>57</td><td>-</td><td>57</td></tr><tr><td>Other</td><td>38</td><td>-</td><td>38</td></tr><tr><td>Total</td><td>378</td><td>-</td><td>378</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	284	-	284	Material	57	-	57	Other	38	-	38	Total	378	-	378
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	284	-	284																		
Material	57	-	57																		
Other	38	-	38																		
Total	378	-	378																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 160 of 347

(a)	(b)
Program:	CODI: Charlotte Network Upgrade (Pg. 1 of 4)
Purpose and Necessity:	Charlotte substation is over 90 years old. A significant portion of the associated infrastructure (system cable, netbank transformers) are at end-of-life and unable to support potential load increases in the downtown area of Detroit.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 161 of 347

(a)	(b)
Program:	CODI: Charlotte Network Upgrade (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 41
DGP Section:	11.4
Scope:	• Convert and consolidate existing Charlotte 4.8kV Network to 13.2kV from Temple Substation • Transfer radial power line customers to adjacent substations • Eliminate network service in areas where it makes sense to provide a different method of service (i.e. URD, OH, etc.) • Decommission Charlotte substation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 162 of 347

(a)	(b)						
Program:	CODI: Charlotte Network Upgrade (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide sufficient capacity to serve growing Detroit core downtown areas • Reduce network cable failures and improve manhole safety • Decommission aging infrastructure and reduce major event risk • Increase protection for the distribution system • Enhance grid technology and automation • Reduce new customer connection timeframe and cost						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 163 of 347

(a)	(b)																												
Program:	CODI: Charlotte Network Upgrade (Pg. 4 of 4)																												
Current Projects:	CODI: Charlotte Network Upgrade																												
Budget Basis:	• Project Management Estimate																												
Cost:	Estimated project spend in 2021-2023: \$32.8 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>1,454</td><td>8,045</td><td>9,500</td></tr><tr><td>Material</td><td>291</td><td>1,609</td><td>1,900</td></tr><tr><td>Other</td><td>194</td><td>1,073</td><td>1,267</td></tr><tr><td>Total</td><td>1,939</td><td>10,727</td><td>12,666</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	1,454	8,045	9,500	Material	291	1,609	1,900	Other	194	1,073	1,267	Total	1,939	10,727	12,666
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	1,454	8,045	9,500																										
Material	291	1,609	1,900																										
Other	194	1,073	1,267																										
Total	1,939	10,727	12,666																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 164 of 347

(a)

(b)

Program:	CODI: Targeted Network Secondary Cable Replacement (Pg. 1 of 4)
Purpose and Necessity:	This cable program is designed to replace targeted 4.8 kV secondary network cable system in Detroit that have a higher probability of failure. This will reduce the secondary network cable failures and manhole events in downtown City of Detroit.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 165 of 347

(a)	(b)
Program:	CODI: Targeted Network Secondary Cable Replacement (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 42
DGP Section:	11.4
Scope:	• Replace targeted sections of the secondary network cable system that have a higher probability of failure

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 166 of 347

(a)	(b)						
Program:	CODI: Targeted Network Secondary Cable Replacement (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce secondary network cable failures and enhance manhole safety • Replace aging infrastructure and reduce major event risk • Enhance safety, reliability and power quality • Reduce reactive costs associated with secondary cable failures						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x				x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 167 of 347

(a)	(b)																												
Program:	CODI: Targeted Network Secondary Cable Replacement (Pg. 4 of 4)																												
Current Projects:	CODI: Targeted Network Secondary Cable Replacement																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$6.9 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>231</td><td>2,375</td><td>2,606</td></tr><tr><td>Material</td><td>46</td><td>475</td><td>521</td></tr><tr><td>Other</td><td>31</td><td>317</td><td>348</td></tr><tr><td>Total</td><td>308</td><td>3,167</td><td>3,475</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	231	2,375	2,606	Material	46	475	521	Other	31	317	348	Total	308	3,167	3,475
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	231	2,375	2,606																										
Material	46	475	521																										
Other	31	317	348																										
Total	308	3,167	3,475																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 168 of 347

(a)	(b)
Program:	CODI: Corktown Substation (Pg. 1 of 4)
Purpose and Necessity:	Corktown is a general purpose substation that was proposed as part of the CODI upgrade to provide capacity needed to serve fast growing Detroit core downtown areas. This substation will also help consolidate, convert and eliminate the 4.8 kV distribution and 24 kV subtransmission system in the Corktown/West River Front area. A significant portion of the associated infrastructure (system cable/conductor, circuit breakers,) are at end-of-life.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 169 of 347

(a)	(b)
Program:	CODI: Corktown Substation (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 43
DGP Section:	11.4
Scope:	• Install a 120/13.2 kV Class 'A' general purpose substation with two 24/32/40MVA transformers • Install one twelve position switchgear • Install two capacitor banks • Install two circuit breakers • Install two house service transformers • Install 400 feet of 9-5" concrete encased duct bank • Install four three-way manholes • Two (2) duct break-ins (1-labrosse & 1-Porter) • Misc. costs (manhole/duct break-in, permit fees, traffic control, etc.) • Install fiber optic cable between Corktown and downtown complex

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 170 of 347

(a)	(b)						
Program:	CODI: Corktown Substation (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide sufficient capacity needed to serve fast growing Detroit core downtown areas • Provide load relief for existing and new customers • Provide sufficient capacity to serve fast growing Detroit core downtown areas • Decommission aging infrastructure and reduce major event risk • Enhance safety, reliability and power quality for upgraded circuits - approximately 90% reduction in circuit wire down events - approximately 85% reduction in circuit customer interruptions - approximately 85% reduction in circuit customer minute interruptions • Reduce trouble events and reactive costs for upgraded circuits - approximately 80% reduction in circuit tree related reactive capital - approximately 90% reduction in circuit non-tree related reactive capital • Increase protection & sectionalizing capabilities for the distribution system • Enhance grid technology and automaton, including DER integration, SCADA capabilities , and fault detection/location						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x		x	x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 171 of 347

(a)	(b)																				
Program:	CODI: Corktown Substation (Pg. 4 of 4)																				
Current Projects:	CODI: Corktown Substation																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$14.1 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>162</td><td>-</td><td>162</td></tr><tr><td>Material</td><td>32</td><td>-</td><td>32</td></tr><tr><td>Other</td><td>22</td><td>-</td><td>22</td></tr><tr><td>Total</td><td>216</td><td>-</td><td>216</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	162	-	162	Material	32	-	32	Other	22	-	22	Total	216	-	216
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	162	-	162																		
Material	32	-	32																		
Other	22	-	22																		
Total	216	-	216																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 172 of 347

(a)	(b)
Program:	CODI: Islandview Substation (Pg. 1 of 4)
Purpose and Necessity:	New customer load has been added along the Detroit riverfront and additional development is anticipated in the future. The new substation is to be sited east of downtown Detroit near the riverfront and allow for the decommissioning of Walker substation and half of Pulford substation. Both Pulford and Walker substations have been in service since the 1920's and cannot reliably serve new load. The new substation will also provide capacity to serve customers being transitioned from the Detroit Public Lighting Department system to the DTE Electric system.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 173 of 347

(a)

(b)

Program:	CODI: Islandview Substation (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 44
DGP Section:	11.4
Scope:	• Construct a new Class 'A' 13.2 kV substation with two transformers and a setup for a future third transformer position • Pre-Convert existing 4.8 kV circuits from Walker Substation - 22 circuits • Pre-Convert existing 4.8 kV circuits from Pulford Substation - 10 circuits • Convert and transfer WALKR and PULFD circuits to new substation • Decommission cable for four trunk lines (24 kV) • Decommission WALKR Substation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 174 of 347

(a)	(b)						
Program:	CODI: Islandview Substation (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide sufficient capacity to serve growing areas • Decommission aging infrastructure and reduce major event risk • Enhance safety, reliability and power quality for upgraded circuits by reducing 85% of circuit wire down events, customer interruptions, and customer minutes • Reduce trouble events and reactive costs for upgraded circuits by 85% for both circuit tree related and non-tree related reactive capital • Increase protection for the distribution system • Enhance grid technology and automation • Reduce new customer connection timeframe and cost						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 175 of 347

(a)	(b)																				
Program:	CODI: Islandview Substation (Pg. 4 of 4)																				
Current Projects:	CODI: Islandview Substation																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$42.4 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>2,000</td><td>15,715</td><td>17,715</td></tr><tr><td>Material</td><td>400</td><td>3,143</td><td>3,543</td></tr><tr><td>Other</td><td>267</td><td>2,095</td><td>2,362</td></tr><tr><td>Total</td><td>2,667</td><td>20,954</td><td>23,621</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	2,000	15,715	17,715	Material	400	3,143	3,543	Other	267	2,095	2,362	Total	2,667	20,954	23,621
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	2,000	15,715	17,715																		
Material	400	3,143	3,543																		
Other	267	2,095	2,362																		
Total	2,667	20,954	23,621																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 176 of 347

(a)	(b)
Program:	CODI: CATO Substation Expansion (Pg. 1 of 4)
Purpose and Necessity:	CATO is a 2 transformer Class 'A' general purpose substation that serves downtown Detroit New Center and Midtown areas and is currently 74% of its firm rating. Future load will cause the substation to exceed its firm rating. Reliability issues have been identified due to only two underground 120kV lines feeding the substation causing planned switching for customers during ITC work and extensive contingency plans.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 177 of 347

(a)	(b)
Program:	CODI: CATO Substation Expansion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 45
DGP Section:	11.4
Scope:	• Expand CATO substation building to accept a third 120kV line and space on the second floor for third 13.2kV transformer and second 13.2kV PDC • Install a third 120/13.2 kV 24/32/40 MVA transformer • Install a second 12 position 13.2kV PDC and 13.2kV cap bank • Transfer existing 13.2kV circuit to new PDC • Replace the existing PDC and transfer four 13.2kV circuits to the replaced PDC • Install SCADA on 4.8kV transformers and PDC

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 178 of 347

(a)	(b)						
Program:	CODI: CATO Substation Expansion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Eliminate need for contingency planning during ITC work on 120kV lines feeding CATO sub and reduced substation outage risk • Sufficient capacity needed to serve fast growing Detroit core downtown areas • Provide load relief for existing and new customers • Decommission aging infrastructure and reduce major event risk • Enhance safety, reliability and power quality for upgraded circuits						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	X	X		X	X		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 179 of 347

(a)	(b)																				
Program:	CODI: CATO Substation Expansion (Pg. 4 of 4)																				
Current Projects:	CODI: CATO Substation Expansion																				
Budget Basis:	• Engineering estimate																				
Cost:	Estimated project spend in 2021-2023: \$11.1 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>81</td><td>6,505</td><td>6,586</td></tr><tr><td>Material</td><td>16</td><td>1,301</td><td>1,317</td></tr><tr><td>Other</td><td>11</td><td>867</td><td>878</td></tr><tr><td>Total</td><td>108</td><td>8,673</td><td>8,782</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	81	6,505	6,586	Material	16	1,301	1,317	Other	11	867	878	Total	108	8,673	8,782
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	81	6,505	6,586																		
Material	16	1,301	1,317																		
Other	11	867	878																		
Total	108	8,673	8,782																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 180 of 347

(a)	(b)
Program:	CODI: Howard Conversion (Pg. 1 of 4)
Purpose and Necessity:	Howard is a 4.8 kV substation commissioned in 1930 and has aging equipment that is at end of life. Aging equipment such as oil breakers hold higher risk of failure, potentially causing extensive outages. In order to provide sufficient capacity to serve the growing Detroit areas and improve reliability and power quality, this substation will be converted and consolidated to the new 13.2 kV Corktown substation.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 181 of 347

(a)	(b)
Program:	CODI: Howard Conversion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 46
DGP Section:	11.4
Scope:	• Rebuild 6 miles of network feeder cable • Rebuild 12 miles of radial powerline system cable • Replace or remove 89 netbank transformers • Convert 26 primary customers from 4.8kV to 13.2kV • Convert 3 miles of overhead from 4.8kV to 13.2kV • Convert and consolidate the circuits to 13.2 kV fed by Corktown, Temple, St. Antoine, and Cato substations • Decommission Howard substation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 182 of 347

(a)	(b)						
Program:	CODI: Howard Conversion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce network cable failures and improve manhole safety • Decommission aging infrastructure and reduce major event risk • Enhance safety, reliability and power quality for upgraded circuits • Reduce trouble events and reactive costs for upgraded circuits • Approximately 90% reduction in circuit non-tree related reactive capital • Increase protection for the distribution system • Reduce new customer connection timeframe and cost of completing this project. • Increase protection & sectionalizing capabilities for the distribution system • Enhance grid technology and automaton, including DER integration, SCADA capabilities , and fault detection/location						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 183 of 347

(a)	(b)		
Program:	CODI: Howard Conversion (Pg. 4 of 4)		
Current Projects:	CODI: Howard Conversion		
Budget Basis:	• Project Management Estimate		
Cost:	Estimated project spend in 2021-2023: \$0.7 M This project is expected to continue beyond 2023		
Test Year (\$000's):	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23
	Total Labor (DTE Electric & Contractors)	-	438
	Material	-	88
	Other	-	58
	Total	-	583

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 184 of 347

(a)	(b)
Program:	CODI: Midtown Substation Expansion (Pg. 1 of 4)
Purpose and Necessity:	Midtown (MIDTN) is a 2 transformer Class 'A' general purpose substation constructed in 1973 that serves downtown Detroit New Center and Midtown areas and is currently 81% of firm. MIDTN's current configuration cannot support approximately 20MVA of future known load growth. Without any substation work, this will take the substation to 146% of firm, excluding CODI conversions.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 185 of 347

(a)	(b)
Program:	CODI: Midtown Substation Expansion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 9, Line 47
DGP Section:	11.4
Scope:	• Install a third 120/13.2kV 24/32/40MVA transformer • Install 12 position switchgear • Install (2) capacitors • Install ~1,300 feet of 15-5" concrete encased duct bank • Install (7) new three-way manholes • Misc. costs (manhole/duct break-in, permit fees, traffic control, etc.)

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 186 of 347

(a)	(b)						
Program:	CODI: Midtown Substation Expansion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide sufficient capacity needed to serve fast growing Detroit core downtown areas • Provide load relief for existing and new customers • Increase SCADA capabilities at Midtown substation						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x			x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 187 of 347

(a)	(b)																												
Program:	CODI: Midtown Substation Expansion (Pg. 4 of 4)																												
Current Projects:	CODI: Midtown Substation Expansion																												
Budget Basis:	• Project Management Estimate																												
Cost:	Estimated project spend in 2021-2023: \$9.5 M																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>376</td><td>-</td><td>376</td></tr><tr><td>Material</td><td>75</td><td>-</td><td>75</td></tr><tr><td>Other</td><td>50</td><td>-</td><td>50</td></tr><tr><td>Total</td><td>502</td><td>-</td><td>502</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	376	-	376	Material	75	-	75	Other	50	-	50	Total	502	-	502
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	376	-	376																										
Material	75	-	75																										
Other	50	-	50																										
Total	502	-	502																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 188 of 347

(a)	(b)
Program:	CODI: Alfred Substation Expansion (Pg. 1 of 4)
Purpose and Necessity:	Alfred (ALFRD) is a 2 transformer Class 'A' general purpose substation constructed in 1968 that serves Detroit's Central Business District and East Riverfront Areas. ALFRD's current configuration cannot support the 9MVA of known future load and the anticipated 9MVA load in Eastern Market's Core Development. Doing so would take the substation to 153% of firm, excluding CODI conversions.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 189 of 347

(a)	(b)
Program:	CODI: Alfred Substation Expansion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 48
DGP Section:	11.4
Scope:	• Install a third 120/13.2kV 24/32/40MVA transformer • Install 12 position switchgear • Install (2) capacitors • Install ~1,300 feet of 15-5" concrete encased duct bank • Install (7) New three-way manholes • Misc. costs (manhole/duct break-in, permit fees, traffic control, etc.)

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 190 of 347

(a)	(b)						
Program:	CODI: Alfred Substation Expansion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide sufficient capacity needed to serve fast growing Detroit core downtown areas • Provide load relief for existing and new customers • Increase SCADA capabilities at Alfred substation						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x			x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 191 of 347

(a)	(b)			
Program:	CODI: Alfred Substation Expansion (Pg. 4 of 4)			
Current Projects:	CODI: Alfred Substation Expansion			
Budget Basis:	• Project Management Estimate			
Cost:	Estimated project spend in 2021-2023: \$9.5 M			
Test Year (\$000's):	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	
	Total Labor (DTE Electric & Contractors)	951	883	1,834
	Material	190	177	367
	Other	127	118	245
	Total	1,268	1,178	2,446

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 192 of 347

(a)	(b)
Program:	CODI: Garfield Network Upgrade (Pg. 1 of 4)
Purpose and Necessity:	Garfield substation is a 4.8 kV substation that is over 90 years old. It serves the Eastern Market, Forest Park and parts of Art Center and Midtown. Due to its aging infrastructure and loading limitations, it is unable to provide sufficient capacity to serve potential developments in the City of Detroit. Garfield substation is to be decommissioned and its circuits converted to the newly constructed Stone Pool 13.2 kV substation.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 193 of 347

(a)	(b)
Program:	CODI: Garfield Network Upgrade (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 49
DGP Section:	11.4
Scope:	• Replace 36 miles of network feeder cable • Replace or remove 78 netbank transformers • Convert 26 primary customers from 4.8 kV to 13.2 kV • Convert 24 miles of overhead from 4.8 kV to 13.2 kV • Convert and consolidate the circuits to 13.2 kV fed by Stone Pool substation • Decommission Garfield substation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 194 of 347

(a)	(b)						
Program:	CODI: Garfield Network Upgrade (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide sufficient capacity to serve growing Detroit core downtown areas • Reduce network cable failures and improve manhole safety • Decommission aging infrastructure and reduce major event risk • Enhance safety, reliability and power quality for upgraded circuits by reducing 85% of circuit wire down events, customer interruptions, and customer minutes • Reduce trouble events and reactive costs for upgraded circuits by 85% for both circuit tree related and non-tree related reactive capital • Increase protection for the distribution system • Enhance grid technology and automation • Reduce new customer connection timeframe and cost						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 195 of 347

(a)	(b)																				
Program:	CODI: Garfield Network Upgrade (Pg. 4 of 4)																				
Current Projects:	CODI: Garfield Network Upgrade																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$54.10 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>1,875</td><td>18,750</td><td>20,625</td></tr><tr><td>Material</td><td>375</td><td>3,750</td><td>4,125</td></tr><tr><td>Other</td><td>250</td><td>2,500</td><td>2,750</td></tr><tr><td>Total</td><td>2,500</td><td>25,000</td><td>27,500</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	1,875	18,750	20,625	Material	375	3,750	4,125	Other	250	2,500	2,750	Total	2,500	25,000	27,500
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	1,875	18,750	20,625																		
Material	375	3,750	4,125																		
Other	250	2,500	2,750																		
Total	2,500	25,000	27,500																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 196 of 347

(a)	(b)
Program:	CODI: Kent/Gibson Conversion (Pg. 1 of 4)
Purpose and Necessity:	Kent substation was built in 1958 and Gibson substation was built in 1927. Both have reached end of life. Kent and Gibson substations have limited capacity and are unable to support strong load growth in downtown Detroit.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 197 of 347

(a)	(b)
Program:	CODI: Kent/Gibson Conversion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 50
DGP Section:	11.4
Scope:	Kent Substation • Rebuild 6 miles of radial powerline system cable • Convert one primary customer from 4.8 kV to 13.2 kV • Convert 7 miles of overhead from 4.8 kV to 13.2 kV • Convert and consolidate the circuits to 13.2 kV fed by Corktown substation • Decommission and remove 11,000 feet of 4.8 kV cable • Remove six 4.8 kV breakers and decommission Kent substation Gibson Substation • Rebuild 10 miles of radial powerline system cable • Convert 22 miles of overhead from 4.8 kV to 13.2 kV • Convert and consolidate the circuits to 13.2 kV fed by Corktown substation • Decommission and remove 21,000 feet of 4.8 kV cable • Remove eight 4.8 kV breakers and decommission Gibson substation • Removal of 24kV cables and equipment

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 198 of 347

(a)	(b)						
Program:	CODI: Kent/Gibson Conversion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce substation outage risk by replacing aging infrastructure • Enhance safety, reliability, and power quality • Provide sufficient capacity to serve existing and new customer load						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 199 of 347

(a)	(b)																												
Program:	CODI: Kent/Gibson Conversion (Pg. 4 of 4)																												
Current Projects:	CODI: Kent/Gibson Conversion																												
Budget Basis:	• Project Management Estimate																												
Cost:	Estimated project spend in 2021-2023: \$16.1 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>875</td><td>5,547</td><td>6,422</td></tr><tr><td>Material</td><td>175</td><td>1,109</td><td>1,284</td></tr><tr><td>Other</td><td>117</td><td>740</td><td>856</td></tr><tr><td>Total</td><td>1,167</td><td>7,396</td><td>8,563</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	875	5,547	6,422	Material	175	1,109	1,284	Other	117	740	856	Total	1,167	7,396	8,563
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
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Total	1,167	7,396	8,563																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 200 of 347

(a)	(b)
Program:	4.8 kV CC: Hilton Substation and Circuit Conversion (Pg. 1 of 4)
Purpose and Necessity:	Both Ferndale and Hazel Park substations are 60-90 years old and represent a significant reliability and reputational risk. Both substations are significantly over their firm ratings. A large portion of the underlying assets are considered "at risk" due to both age and persistent overloads. There are limited options to restore customers in the event of a major equipment failure.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 201 of 347

(a)	(b)
Program:	4.8 kV CC: Hilton Substation and Circuit Conversion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 51
DGP Section:	11.3
Scope:	• Construct a two transformer 24/32/40 MVA 13.2 kV Class 'A' substation • Project completion entails construction of a modern 13.2kV infrastructure and removal of the old 5kV class infrastructure • Rebuild and convert ten existing 4.8 kV circuits (37 miles of overhead) into four 13.2 kV circuits • This is phase one of a two phase project to fully decommission Ferndale and Hazel Park substations - Phase two will convert the remaining circuits out of Ferndale and Hazel Park to 13.2kV and decommission the substations

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 202 of 347

(a)	(b)						
Program:	4.8 kV CC: Hilton Substation and Circuit Conversion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Alleviates the loading issue on Ferndale and Hazel Park substations, leading to: - Elimination of loading related safety risks - Improved reliability - Reduction of long duration outage risks - Reduced O&M and Capital expenses • Elimination of aging OH infrastructure, leading to: - Improved reliability - Fewer truck roles - Reduced Capital expenses - Reduced O&M expenses • Reduce new customer connection timeframe and cost						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 203 of 347

(a)	(b)																				
Program:	4.8 kV CC: Hilton Substation and Circuit Conversion (Pg. 4 of 4)																				
Current Projects:	4.8 kV CC: Hilton Substation and Circuit Conversion																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.1 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td>-</td><td>-</td><td>-</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total	-	-	-
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	-	-	-																		
Material	-	-	-																		
Other	-	-	-																		
Total	-	-	-																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 204 of 347

(a)

(b)

Program:	4.8 kV CC: Zenon Substation and Circuit Conversion (Pg. 1 of 4)
Purpose and Necessity:	McKinstry is an antiquated substation with much of the equipment beyond normal service life and with unavailable replacement or spare parts. A large portion of the underlying assets are considered "at risk" due to age. The infrastructure is experiencing more frequent planned and trouble related outages, both of which reduces system availability, reliability, and customer satisfaction.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 205 of 347

(a)	(b)
Program:	4.8 kV CC: Zenon Substation and Circuit Conversion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 52
DGP Section:	11.3
Scope:	• Construct a two 24/32/40 MVA transformer 4.8 kV & 13.2 kV Class 'A' substation • Project completion entails construction of a modern 13.2kV infrastructure and removal of the old 5kV class infrastructure • Convert nine 4.8kV circuits to four 13.2kV circuits and transfer the remaining four 4.8kV circuits to Zenon • This is phase one of a three phase project - Phase two will decommission McKinstry Substation - Phase three will convert the remaining four 4.8kV Zenon ckts to 13.2kV

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 206 of 347

(a)

(b)

Program:	4.8 kV CC: Zenon Substation and Circuit Conversion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Elimination of aging OH infrastructure, leading to: - Improved reliability - Fewer truck roles - Reduced Capital expenses - Reduced O&M expenses • Removal of antiquated substation equipment at McKinstry leading to: - Elimination of safety risks - Improved reliability - Reduction of long duration outage risks - Reduced equipment to be maintained - Reduced O&M and Capital expenses						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x			x			

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 207 of 347

(a)	(b)																				
Program:	4.8 kV CC: Zenon Substation and Circuit Conversion (Pg. 4 of 4)																				
Current Projects:	4.8 kV CC: Zenon Substation and Circuit Conversion																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.0 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td>-</td><td>-</td><td>-</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total	-	-	-
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	-	-	-																		
Material	-	-	-																		
Other	-	-	-																		
Total	-	-	-																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 208 of 347

(a)	(b)
Program:	4.8 kV CC: Cortland / Oakman / Linwood Consolidation (Pg. 1 of 4)
Purpose and Necessity:	Linwood and Oakman substations, and their associated circuits, are significantly under loaded. To eliminate this underutilization, Linwood and Oakman substations will be decommissioned and their circuits will be consolidated into Cortland Substation. Cortland is a much newer 120kV fed substation and will provide a higher level of reliability. Consolidation of these substations; and subsequent removal of the associated breakers, system cable, regulators, and transformers; will eliminate the need for preventive maintenance on this equipment.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 209 of 347

(a)	(b)
Program:	4.8 kV CC: Cortland / Oakman / Linwood Consolidation (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 53
DGP Section:	11.3
Scope:	• Consolidate Linwood & Oakman circuits into Cortland Substation • Remove twenty-seven breakers, cables and OH wire from the associated circuits, six induction regulators, and six substation transformers • Decommission Linwood and Oakman Substations

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 210 of 347

(a)	(b)						
Program:	4.8 kV CC: Cortland / Oakman / Linwood Consolidation (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Eliminating old cable, breakers, and transformers will: - Eliminate safety risks associated with the obsolete equipment - Reduce the overall substation outage risk - Avoid maintenance related O&M expenses - Reduce failure related O&M & Capital expenses • Transferring the load to a newer substation will: - Improve reliability • Eliminating overhead primary and secondary on circuits will: - Reduce truck rolls - Reduce O&M and Capital expenses						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	X	X		X	X	X	X

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 211 of 347

(a)	(b)																												
Program:	4.8 kV CC: Cortland / Oakman / Linwood Consolidation (Pg. 4 of 4)																												
Current Projects:	4.8 kV CC: Cortland / Oakman / Linwood Consolidation																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$4.3 M																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>356</td><td>-</td><td>356</td></tr><tr><td>Material</td><td>71</td><td>-</td><td>71</td></tr><tr><td>Other</td><td>48</td><td>-</td><td>48</td></tr><tr><td>Total</td><td>475</td><td>-</td><td>475</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	356	-	356	Material	71	-	71	Other	48	-	48	Total	475	-	475
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	356	-	356																										
Material	71	-	71																										
Other	48	-	48																										
Total	475	-	475																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 212 of 347

(a)	(b)
Program:	4.8 kV CC: Ariel Substation and Circuit Conversion (Pg. 1 of 4)
Purpose and Necessity:	BIHAM is a one line-up switchgear substation (exposed to 26.8 MVA of stranded load) neighboring a smaller heavily loaded DUDLY substation. This project is in progress to complete conversion and transfer load of BIHAM and DUDLY circuits to new ARIEL overhead circuits (DC 9893, DC 9815) and underground (DC 9879) ARIEL circuits.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 213 of 347

(a)	(b)
Program:	4.8 kV CC: Ariel Substation and Circuit Conversion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 54
DGP Section:	11.3
Scope:	• Completed overhead conversion and transfer of 3.1 MVA of load (from PL 0487 and 0141 Dudley to distribution DC 9893 Ariel) • Complete underground conversion and transfer of 4.3 MVA of load (from DC 2734, 2720, and 1867 Birmingham to DC 9879 Ariel) • Removal of 4.8 kV cables (DC 2665 and 2734 Birmingham, DC 1669, 1650, 0487, and 0141 Dudley) from converted 13.2KV system area - 2.4 miles of 4.8 kV system cables • Project completion entails construction of a modern 13.2kV infrastructure and removal of the old 5kV class infrastructure

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 214 of 347

(a)	(b)						
Program:	4.8 kV CC: Ariel Substation and Circuit Conversion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Commercial load (east of Old Woodward) continuous expansion to be reliably served by new Ariel substation • BIHAM load to be reduced from 98% to 78% firm • DUDLY 4.8KV load to be reduced from 89% to 43% firm - This firm reduction will decrease the potential stranded load and risk of a major event at both substations • Fewer truck roles due to elimination of aging OH infrastructure, leading to: - Improved reliability - Reduced Capital expenses - Reduced O&M expenses • Reduce new customer connection timeframe and cost						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x			x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 215 of 347

(a)	(b)																												
Program:	4.8 kV CC: Ariel Substation and Circuit Conversion (Pg. 4 of 4)																												
Current Projects:	4.8 kV CC: Ariel Substation and Circuit Conversion																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$0.9 M																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td>-</td><td>-</td><td>-</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total	-	-	-
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	-	-	-																										
Material	-	-	-																										
Other	-	-	-																										
Total	-	-	-																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 216 of 347

(a)	(b)
Program:	4.8 kV CC: I-94 Substation and Circuit Conversion (Pg. 1 of 4)
Purpose and Necessity:	The Detroit Economic Growth Corporation (DEGC) has assembled the I-94 Industrial Park site – a 186-acre site north of I-94 between Mt. Elliott and Van Dyke on Detroit's east side. It is part of the 3,203-acre Mt. Elliot Development Zone, which is the single largest industrial district in Detroit and encompasses automotive, metal, transportation and logistics clusters. It offers access to major transportation assets. The site is a federally designated Historically Underutilized Business (HUB) Zone and a state designated Renaissance Zone. Recent new loads were fed from the aging Lynch and Lambert 4.8 kV substations. These substations have little capacity for additional load, have limited throw-over and jumpering options and cannot reliably serve the new load coming to the I-94 industrial park. The new substation is required to serve the load growth in and around I-94 industrial park and allows for decommissioning of the aging Lynch, Lambert, and Pulford 4.8 kV substations. Recent new loads were fed from the aging Lynch and Lambert 4.8 kV substations. These substations have little capacity for additional load, have limited throw-over and jumpering options and cannot reliably serve the new load coming to the I-94 industrial park. The new substation is required to serve the load growth in and around I-94 industrial park and allows for decommissioning of the aging Lynch and Lambert 4.8 kV substations.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 217 of 347

(a)	(b)
Program:	4.8 kV CC: I-94 Substation and Circuit Conversion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 55
DGP Section:	11.3
Scope:	• Project completion entails construction of a modern 13.2kV infrastructure and removal of the old 5kV class infrastructure • Construct a new Class 'A', two transformer, 12 position switchgear 13.2 kV substation with 2- capacitor banks with option to add third transformer, second set of switchgear, and one capacitor bank. • Convert 24 existing 4.8 kV circuits and three throw over circuits from Lynch, Lambert, and Pulford substations to 4-13.2 kV circuits with one throw over circuit. • Decommission Lynch (78 years old), Lambert (88 years old), and Pulford (88 years old) • Convert 112 miles of 4.8kV OH • Decommission 14.5 miles of 4.8kV cable • Decommission 31 miles of 24kV cable (trunk lines from Essex and NEAST) • Convert 40 primary service customers • Deconductor blighted areas The substation site prep and below grade is performed under a separate Promenade Site project

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 218 of 347

(a)	(b)						
Program:	4.8 kV CC: I-94 Substation and Circuit Conversion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide sufficient capacity to serve new load growth and Economic Development • Elimination of aging OH infrastructure, leading to: - Improved reliability - Fewer truck roles - Reduced Capital expenses - Reduced O&M expenses • Removal of antiquated substation equipment leading to: - Elimination of safety risks - Improved reliability - Reduction of long duration outage risks - Reduced equipment to be maintained - Reduced O&M and Capital expenses • Reduce new customer connection timeframe and cost						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 219 of 347

(a)	(b)																				
Program:	4.8 kV CC: I-94 Substation and Circuit Conversion (Pg. 4 of 4)																				
Current Projects:	4.8 kV CC: I-94 Substation and Circuit Conversion																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$4.10 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>3,107</td><td>3,107</td></tr><tr><td>Material</td><td>-</td><td>621</td><td>621</td></tr><tr><td>Other</td><td>-</td><td>414</td><td>414</td></tr><tr><td>Total</td><td>-</td><td>4,142</td><td>4,142</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	3,107	3,107	Material	-	621	621	Other	-	414	414	Total	-	4,142	4,142
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	-	3,107	3,107																		
Material	-	621	621																		
Other	-	414	414																		
Total	-	4,142	4,142																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 220 of 347

(a)	(b)
Program:	4.8 kV CC: Almont Relief and Circuit Conversion (Midas) (Pg. 1 of 4)
Purpose and Necessity:	Almont substation is 3.4 MVA over its substation firm rating. Both of its circuits are at or above their day-to-day ratings during summer peak hours. Limited/No jumpering options are available because Almont is a 4.8 kV substation surrounded by 13.2 kV substations. Five new Method of Service requests (MOS) have been received for new sub divisions and grow houses, with no capacity to feed them. Commitments to the community have been made to improve reliability. Breakers are 1956 and spare parts are not available. Each circuit is over 55 overhead miles. A cable failure occurred in 2019 caused breaker to fail and extensive outage for the Almont area due to no jumpering available and, due to high loads, the substation was being blocked from automatic throwover
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 221 of 347

(a)	(b)
Program:	4.8 kV CC: Almont Relief and Circuit Conversion (Midas) (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 56
DGP Section:	11.3
Scope:	• Build a new Class 'R' 13.2 kV substation with 4 circuits • Reconductor 2.7 miles of overhead primary • Convert 21 miles of overhead primary • Establish new jumpering points • Reconfigure Almont circuits • Transfer Almont load to the new substation • Convert and transfer remaining Almont load to the new substation • Decommission Almont substation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 222 of 347

(a)	(b)						
Program:	4.8 kV CC: Almont Relief and Circuit Conversion (Midas) (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Alleviates the loading issue on Almont substation, leading to: - Elimination of loading related safety risks - Improved reliability - Reduction of long duration outage risks - Reduced O&M and Capital expenses • Elimination of aging OH infrastructure, leading to: - Improved reliability - Fewer truck roles - Reduced Capital expenses - Reduced O&M expenses • Reduce new customer connection timeframe and cost						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 223 of 347

(a)	(b)																												
Program:	4.8 kV CC: Almont Relief and Circuit Conversion (Midas) (Pg. 4 of 4)																												
Current Projects:	4.8 kV CC: Almont Relief and Circuit Conversion (Midas Sub) - MEP 4.8 kV CC: Almont Relief and Circuit Conversion (Midas OH) - DO																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$21.4 M																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>1,701</td><td>664</td><td>2,365</td></tr><tr><td>Material</td><td>340</td><td>133</td><td>473</td></tr><tr><td>Other</td><td>227</td><td>88</td><td>315</td></tr><tr><td>Total</td><td>2,268</td><td>885</td><td>3,153</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	1,701	664	2,365	Material	340	133	473	Other	227	88	315	Total	2,268	885	3,153
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	1,701	664	2,365																										
Material	340	133	473																										
Other	227	88	315																										
Total	2,268	885	3,153																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 224 of 347

(a)	(b)
Program:	4.8 kV CC: Buckler Circuit Conversion (Pg. 1 of 4)
Purpose and Necessity:	Ann Arbor continues to see economic growth resulting in increased loads. Argo substation is not capable of serving downtown Ann Arbor due to the loading limitations of the 4.8kV system. Argo circuits are 4.8 kV circuits limited to 3.0 MVA. ARGO is currently at 92% of firm. Argo substation load will continue to rise, circuits will become overloaded and the substation will be above firm. Buckler substation was built to support the current and future loads of Argo and the surrounding substations.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 225 of 347

(a)	(b)
Program:	4.8 kV CC: Buckler Circuit Conversion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 57
DGP Section:	11.3
Scope:	• Establish distribution circuit 9936 Buckler transferring a portion of distribution circuit 2942 Argo • Establish distribution circuit 9927 Buckler by converting and transferring distribution circuit 326 Argo • Convert and transfer distribution circuits 321 & 323 Argo to distribution circuit 9927 Buckler • Establish distribution circuit 9984 Buckler by converting and transferring distribution circuits 319 & 333 Argo • Establish distribution circuit 9978 Buckler • Establish Burton PL 0070

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 226 of 347

(a)	(b)						
Program:	4.8 kV CC: Buckler Circuit Conversion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide sufficient capacity to serve new load growth and Economic Development • Elimination of aging OH infrastructure, leading to: - Improved reliability - Fewer truck roles - Reduced Capital expenses - Reduced O&M expenses • Removal of aged substation equipment leading to: - Elimination of safety risks - Improved reliability - Reduction of long duration outage risks - Reduced equipment to be maintained - Reduced O&M and Capital expenses • Reduce new customer connection timeframe and cost						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 227 of 347

(a)	(b)																				
Program:	4.8 kV CC: Buckler Circuit Conversion (Pg. 4 of 4)																				
Current Projects:	4.8 kV CC: Buckler Circuit Conversion																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$19.10 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>1,444</td><td>3,832</td><td>5,276</td></tr><tr><td>Material</td><td>289</td><td>766</td><td>1,055</td></tr><tr><td>Other</td><td>192</td><td>511</td><td>703</td></tr><tr><td>Total</td><td>1,925</td><td>5,109</td><td>7,034</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	1,444	3,832	5,276	Material	289	766	1,055	Other	192	511	703	Total	1,925	5,109	7,034
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	1,444	3,832	5,276																		
Material	289	766	1,055																		
Other	192	511	703																		
Total	1,925	5,109	7,034																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 228 of 347

(a)	(b)
Program:	4.8 kV CC: Calla Circuit Conversion (Pg. 1 of 4)
Purpose and Necessity:	Diamond substation is well over its firm rating. DIMND9875 is significantly impacted by poor reliability. Customers at the end of this circuit are about ten miles from the substation and regularly experience low voltage/power quality issues. DIMND9874 also has a pocket of poor reliability in an ISO down area.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 229 of 347

(a)	(b)
Program:	4.8 kV CC: Calla Circuit Conversion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 58
DGP Section:	11.3
Scope:	• DIMND9875 (future CALLA9330): Install approximately 2400 feet of new infrastructure to serve as future CALLA9330 backbone • DIMND9875 (future CALLA9330): Install regulators and reclosers • DIMND9874: Replace 10,000 feet overhead to convert the 'M4' iso down - Project completion entails construction of a modern 13.2kV infrastructure and removal of the old 5kV class infrastructure

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 230 of 347

(a)	(b)						
Program:	4.8 kV CC: Calla Circuit Conversion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Relieves load off DIMND, thus reducing percent over firm • Splits DIMND9875 which spans to just under 10 miles, thus: - Improving customer power quality - Allowing for more jumpering in the area - Improving reliability • Fewer truck roles due to elimination of aging 4.8kV OH infrastructure, leading to: - Reduced Capital expenses - Reduced O&M expenses - Improving reliability						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:		X			X		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 231 of 347

(a)	(b)			
Program:	4.8 kV CC: Calla Circuit Conversion (Pg. 4 of 4)			
Current Projects:	4.8 kV CC: Calla Circuit Conversion			
Budget Basis:	• Engineering Estimate			
Cost:	Estimated project spend in 2021-2023: \$0.3 M			
Test Year (\$000's):		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23
	Total Labor (DTE Electric & Contractors)	-	-	-
	Material	-	-	-
	Other	-	-	-
	Total	-	-	-

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 232 of 347

(a)

(b)

Program:	4.8 kV CC: Northville Decommissioning and Circuit Conversion (Pg. 1 of 4)
Purpose and Necessity:	The 4.8 kV switchgear at Northville substation is sixty years old and is at the end of its life. Westinghouse no longer supports this equipment. Three positions have failed and spares are no longer available. The foundation has settled and run off drains towards the switchgear, causing water intrusion issues.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 233 of 347

(a)	(b)
Program:	4.8 kV CC: Northville Decommissioning and Circuit Conversion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 59
DGP Section:	11.3
Scope:	• Convert and transfer three circuits to the Northville 13.2kV side • Project completion entails construction of a modern 13.2kV infrastructure and removal of the old 5kV class infrastructure • Convert and transfer remaining load to surrounding 13.2kV substations • Remove the 4.8 kV substation equipment, including: transformers, switchgear, and cable

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 234 of 347

(a)	(b)						
Program:	4.8 kV CC: Northville Decommissioning and Circuit Conversion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Elimination of aging OH infrastructure, leading to: - Improved reliability - Fewer truck roles - Reduced Capital expenses - Reduced O&M expenses • Removal of antiquated and failing substation equipment leading to: - Elimination of safety risks - Improved reliability - Reduction of long duration outage risks - Reduced equipment to be maintained - Reduced O&M and Capital expenses • Increased jumpering between circuits due to all circuits being the same operating voltage						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x			x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 235 of 347

(a)	(b)																												
Program:	4.8 kV CC: Northville Decommissioning and Circuit Conversion (Pg. 4 of 4)																												
Current Projects:	4.8 kV CC: Northville Decommissioning and Circuit Conversion																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$0.0 M																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td>-</td><td>-</td><td>-</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total	-	-	-
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	-	-	-																										
Material	-	-	-																										
Other	-	-	-																										
Total	-	-	-																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 236 of 347

(a)	(b)
Program:	4.8 kV CC: Lapeer - Elba Expansion and Circuit Conversion (Apollo) (Pg. 1 of 4)
Purpose and Necessity:	This project is to provide capacity to serve existing and new customer load and address various issues associated with aging infrastructure and the islanded 4.8 kV system at Lapeer and Elba substations. In addition, this project will transfer part of Hunter's Creek load onto the new substation to assist with reliability issues. • Lapeer substation is at 107% of the firm rating • One of its circuits is over day-to-day rating with the other approaching its day-to-day rating. • Elba substation is at or slightly above its transformer day to day rating during summer peak hours. • The general load growth has been strong in the area (6-8% in 2017 and 2018). • Elba has problematic subtransmission infrastructure that has a history of poor reliability performance and limits shutdown capability for operation and maintenance • The Lapeer-Elba area has experienced a number of low voltage and power quality issues • Limited jumpering options are available at the Elba substation since it is a 4.8 kV substation surrounded by 13.2 kV substations
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 237 of 347

(a)	(b)
Program:	4.8 kV CC: Lapeer - Elba Expansion and Circuit Conversion (Apollo) (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 60
DGP Section:	11.3
Scope:	• Build a new Class 'A' 13.2 kV substation named Apollo • Convert and consolidate 4.8 kV circuits from Elba and Lapeer substations to 13.2 kV • Decommission 4.8 kV portion of Lapeer substation • Decommission Elba and 6.2 miles of 40 kV tap to the substation • Transfer part of Hunter's Creek to the new substation to improve reliability

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 238 of 347

(a)	(b)						
Program:	4.8 kV CC: Lapeer - Elba Expansion and Circuit Conversion (Apollo) (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Alleviates the loading issue on Lapeer and Elba substations, leading to: - Elimination of loading related safety risks - Improved reliability - Reduction of long duration outage risks - Reduced O&M and Capital expenses • Elimination of aging OH infrastructure, leading to: - Improved reliability - Fewer truck roles - Reduced Capital expenses - Reduced O&M expenses • Reduce new customer connection timeframe and cost • Increased jumpering between circuits due to all circuits being the same operating voltage						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 239 of 347

(a)	(b)																				
Program:	4.8 kV CC: Lapeer - Elba Expansion and Circuit Conversion (Apollo) (Pg. 4 of 4)																				
Current Projects:	4.8 kV CC: Lapeer - Elba Expansion and Circuit Conversion (Apollo)																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$27.9 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>817</td><td>13,051</td><td>13,868</td></tr><tr><td>Material</td><td>163</td><td>2,610</td><td>2,774</td></tr><tr><td>Other</td><td>109</td><td>1,740</td><td>1,849</td></tr><tr><td>Total</td><td>1,089</td><td>17,401</td><td>18,490</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	817	13,051	13,868	Material	163	2,610	2,774	Other	109	1,740	1,849	Total	1,089	17,401	18,490
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	817	13,051	13,868																		
Material	163	2,610	2,774																		
Other	109	1,740	1,849																		
Total	1,089	17,401	18,490																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 240 of 347

(a)	(b)
Program:	4.8 kV CC: McKinstry Sub Decommission (Pg. 1 of 4)
Purpose and Necessity:	• Decommission McKinstry substation • This completes phase two of three of the Zenon Substation construction project • In support of the Gordie Howe International Bridge Project, all load has been transferred to Zenon Substation or Westend Substation
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 241 of 347

(a)

(b)

Program:	4.8 kV CC: McKinstry Sub Decommission (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 61
DGP Section:	11.3
Scope:	• McKinstry Substation will be decommissioned and equipment will be removed.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 242 of 347

(a)	(b)						
Program:	4.8 kV CC: McKinstry Sub Decommission (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Eliminate aging infrastructure substation • This completes phase two of three of the Zenon Substation construction project						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x					x	

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 243 of 347

(a)	(b)																				
Program:	4.8 kV CC: McKinstry Sub Decommission (Pg. 4 of 4)																				
Current Projects:	4.8 kV CC: McKinstry Sub Decommission																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$5.2 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>574</td><td>294</td><td>868</td></tr><tr><td>Material</td><td>115</td><td>59</td><td>174</td></tr><tr><td>Other</td><td>77</td><td>39</td><td>116</td></tr><tr><td>Total</td><td>765</td><td>392</td><td>1,158</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	574	294	868	Material	115	59	174	Other	77	39	116	Total	765	392	1,158
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	574	294	868																		
Material	115	59	174																		
Other	77	39	116																		
Total	765	392	1,158																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 244 of 347

(a)	(b)
Program:	4.8 kV CC: White Lake Decommissioning and Circuit Conversion (Pg. 1 of 4)
Purpose and Necessity:	White Lake Substation is an older class T-T substation with multiple at-risk equipment that needs to be upgraded or replaced. There are two transformers each with a single circuit, but operate at two different operating voltages (13.2kV and 4.8kV). The two circuits cannot be jumpered to one another. 13.2kV White Lake Class 'T' substation is 101% of the firm rating. DC 307 White Lake is 102% of its day to day rating. 4.8 kV White Lake Class 'T' substation peak load is 194% of its firm rating. There is also a 40kV capacitor (CAP1-1) that also has multiple at-risk equipment which needs to be upgraded or replaced. The existing substation site is too small to support replacing major equipment at the White Lake Substation.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 245 of 347

(a)	(b)
Program:	4.8 kV CC: White Lake Decommissioning and Circuit Conversion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 62
DGP Section:	11.3
Scope:	• Purchased property in 2019 • Build a new 40kv fed Class 'R-R' substation with four load carrying circuits • Convert DC 307 White Lake to a 13.2 kV circuit • Establish four load carrying circuits supporting White Lake Substation load and a portion of Clyde Substation • Establish new jumpering points and loop schemes between the new substation and Clyde, Widlow, Teggerdine, Placid, and Osprey substations • Convert ISO down locations on DC 8255 Wardlow and DC 8265 Osprey • Decommission White Lake substation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 246 of 347

(a)	(b)						
Program:	4.8 kV CC: White Lake Decommissioning and Circuit Conversion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce equipment risk by providing load relief • Provide sufficient capacity to serve existing and new customers • Create circuit jumpering point to improve operability • Elimination of aging OH infrastructure, leading to: - Improved reliability - Fewer truck roles - Reduced Capital expenses - Reduced O&M expenses • Removal of antiquated and failing substation equipment leading to: - Elimination of safety risks - Improved reliability - Reduction of long duration outage risks - Reduced equipment to be maintained - Reduced O&M and Capital expenses						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:		x					

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 247 of 347

(a)	(b)																												
Program:	4.8 kV CC: White Lake Decommissioning and Circuit Conversion (Pg. 4 of 4)																												
Current Projects:	4.8 kV CC: White Lake Decommissioning and Circuit Conversion																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$0.0 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td>-</td><td>-</td><td>-</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total	-	-	-
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	-	-	-																										
Material	-	-	-																										
Other	-	-	-																										
Total	-	-	-																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 248 of 347

(a)	(b)
Program:	4.8 kV CC: Quincy Conversion (Pg. 1 of 4)
Purpose and Necessity:	Transformer has been leaking for a few years and has a very small amount of oil in it. There are numerous leaks located at various spots. There is no oil containment on site which exposes us to environmental issues. Transformer was installed on top of railroad ties. These railroad ties have rotted and are breaking down (some part crumble as you touch) causing the transformer to list to one side. Because of the list/tilt, it's stretching the conductor causing strain on the cross arms and insulators that guide the conductors to the transformer bushings.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 249 of 347

(a)	(b)
Program:	4.8 kV CC: Quincy Conversion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 63
DGP Section:	11.3
Scope:	• Install a pad mount STDF (Subtransmission Distribution Facility) at existing Quincy property • Install ~ 3 miles of neutral conductor • Reconductor ~2 miles • Install 8 ISO down transformers • Replace ~ 20 transformers • Decommission QUINCY Substation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 250 of 347

(a)	(b)						
Program:	4.8 kV CC: Quincy Conversion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Replace damage/high risk equipment • Setup to relieve Lexington substations which are 120% (13.2KV) and 97% (4.8kV) of firm, or Worth that is at 88% and possibly Lake Port substation at 112% • Replace leaking transformer with new transformer • Eliminate environmental risk						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 251 of 347

(a)	(b)																				
Program:	4.8 kV CC: Quincy Conversion (Pg. 4 of 4)																				
Current Projects:	4.8 kV CC: Quincy Conversion																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$4.5 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>444</td><td>-</td><td>444</td></tr><tr><td>Material</td><td>89</td><td>-</td><td>89</td></tr><tr><td>Other</td><td>59</td><td>-</td><td>59</td></tr><tr><td>Total</td><td>592</td><td>-</td><td>592</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	444	-	444	Material	89	-	89	Other	59	-	59	Total	592	-	592
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	444	-	444																		
Material	89	-	89																		
Other	59	-	59																		
Total	592	-	592																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 252 of 347

(a)

(b)

Program:	4.8 kV CC: Pinegrove Substation Relocation and Conversion (Pg. 1 of 4)
Purpose and Necessity:	To accommodate MDOT request to relocate Pinegrove Substation to make way for the expansion of the Blue water bridge Plaza.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 253 of 347

(a)	(b)
Program:	4.8 kV CC: Pinegrove Substation Relocation and Conversion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 64/ Exhibit A-12, Schedule B5.4, Page 10, Line 65
DGP Section:	11.3
Scope:	• The current substation will need to be decommissioned and a new substation will be built. Pinegrove is currently a three transformer 4.8kV Class 'A' substation. A two transformer 13.kV Class 'A' substation will replace it.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 254 of 347

(a)	(b)						
Program:	4.8 kV CC: Pinegrove Substation Relocation and Conversion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Improved and additional capacity • Creates Capacity to improve load growth • Replaced aging infrastructure to improve substation operability and reliability • Creates circuit jumpering options to utilize for planned construction or outage restoration						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:		X			X		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 255 of 347

(a)	(b)																				
Program:	4.8 kV CC: Pinegrove Substation Relocation and Conversion (Pg. 4 of 4)																				
Current Projects:	4.8 kV CC: Pinegrove Substation Relocation and Conversion 4.8 kV CC: Pinegrove Substation Relocation and Conversion CIAC																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$56.3 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>1,188</td><td>8,456</td><td>9,644</td></tr><tr><td>Material</td><td>238</td><td>1,691</td><td>1,929</td></tr><tr><td>Other</td><td>158</td><td>1,128</td><td>1,286</td></tr><tr><td>Total</td><td>1,583</td><td>11,275</td><td>12,858</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	1,188	8,456	9,644	Material	238	1,691	1,929	Other	158	1,128	1,286	Total	1,583	11,275	12,858
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	1,188	8,456	9,644																		
Material	238	1,691	1,929																		
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Total	1,583	11,275	12,858																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 256 of 347

(a)	(b)
Program:	4.8 kV CC: Promenade Site Prep and Below Grade (Pg. 1 of 4)
Purpose and Necessity:	The Detroit Economic Growth Corporation (DEGC) has assembled the I-94 Industrial Park site – a 186-acre site north of I-94 between Mt. Elliott and Van Dyke on Detroit's east side. It is part of the 3,203-acre Mt. Elliot Development Zone, which is the single largest industrial district in Detroit and encompasses automotive, metal, transportation and logistics clusters. It offers access to major transportation assets. The site is a federally designated Historically Underutilized Business (HUB) Zone and a state designated Renaissance Zone. Recent new loads were fed from the aging Lynch and Lambert 4.8 kV substations. These substations have little capacity for additional load, have limited throw-over and jumpering options and cannot reliably serve the new load coming to the I-94 industrial park. The new substation is required to serve the load growth in and around I-94 industrial park and allows for decommissioning of the aging Lynch, Lambert, and Pulford 4.8 kV substations.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 257 of 347

(a)	(b)
Program:	4.8 kV CC: Promenade Site Prep and Below Grade (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 66
DGP Section:	11.3
Scope:	• Construct a new Class 'A', two transformer, 12 position switchgear 13.2 kV substation with 2- capacitor banks with option to add third transformer, second set of switchgear, and one capacitor bank. • Convert 24 existing 4.8 kV circuits and three throw over circuits from Lynch, Lambert, and Pulford substations to 4-13.2 kV circuits with one throw over circuit. • Decommission Lynch (78 years old), Lambert (88 years old), and Pulford (88 years old) • Convert 112 miles of 4.8kV OH • Decommission 14.5 miles of 4.8kV cable • Decommission 31 miles of 24kV cable (trunk lines from Essex and NEAST) • Convert 40 primary service customers • Deconductor blighted areas This project scope is limited to prepare the substation site and below grade work for I-94 Substation project. Electrical infrastructure work is included in I-94 substation project.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 258 of 347

(a)	(b)						
Program:	4.8 kV CC: Promenade Site Prep and Below Grade (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provides additional capacity for load growth in the area • Convert thirteen 4.8kV circuits to 13.2kV and decommission the associated high risk system cables with known failures • Relieve load on the Subtransmission by 22.1MVA and decommission 7 trunk lines • Retire Lynch, Lambert, and Pulford Substations • Increase protection & sectionalizing capabilities for the distribution system • Enhance grid technology and automaton, including DER integration, SCADA capabilities , and fault detection/location						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 259 of 347

(a)	(b)																												
Program:	4.8 kV CC: Promenade Site Prep and Below Grade (Pg. 4 of 4)																												
Current Projects:	4.8 kV CC: Promenade Site Prep and Below Grade																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$13.1 M																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>1,504</td><td>18</td><td>1,522</td></tr><tr><td>Material</td><td>301</td><td>4</td><td>304</td></tr><tr><td>Other</td><td>201</td><td>2</td><td>203</td></tr><tr><td>Total</td><td>2,005</td><td>24</td><td>2,030</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	1,504	18	1,522	Material	301	4	304	Other	201	2	203	Total	2,005	24	2,030
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	1,504	18	1,522																										
Material	301	4	304																										
Other	201	2	203																										
Total	2,005	24	2,030																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 260 of 347

(a)	(b)
Program:	4.8 kV CC: Hawthorne Relief and Circuit Conversion (Pg. 1 of 4)
Purpose and Necessity:	Hawthorne substation area is a dense, 4.8 kV service territory with many commercial and residential customers. The substation's load exceeds 130% of its firm rating and must be addressed. Adjacent substations are Daly, Biltmore, Villa, and Glendale. Daly, Biltmore, and Villa all have excessive loads. Glendale has some capacity, but it is a significant distance away from most Hawthorne substation circuits.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 261 of 347

(a)	(b)
Program:	4.8 kV CC: Hawthorne Relief and Circuit Conversion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 67
DGP Section:	11.3
Scope:	Near-Term 1: Mallard OH Work • Build approximately 4,200 feet overhead extension from MALRD9375 along Ann Arbor Trail • Reconductor approximately 1,060 feet backbone on MALRD9375 and 2,380 feet of backbone MALRD9384 • Convert approximately 78,000 feet of overhead lines on HAWTH1392 on HAWTH2112 from 4.8 kV to 13.2 kV • Convert approximately 71,000 feet of overhead lines on GLEND1974 and GLEND 2213 from 4.8 kV to 13.2 kV Near-Term 2: Biltmore Substation • Convert 13.2 kV Class 'F' substation to temporary overhead Class 'R', then convert Class 'F' to Class 'A' substation. • Convert approximately 99,000 feet of overhead lines on BLTMR1442, BLTMR2050, and HAWTH1176 from 4.8 kV to 13.2 kV • Install approximately 7,500 feet of new cable for three (3) new distribution circuits Near-Term 3: Villa PDC Replacement • Install new 12-position PDC. Cut over circuits from old PDC and install onto new PDC. • Cutover around 1,000 feet of cable from old PDC to new PC Near-Term 4: Property Procurement • Purchase two (2) properties for future substations Near-Term 5: Non-Wires Alternatives (NWA) for Villa • Develop NWA solution for Villa substation to address loads during peak hours. Long-Term 1: Mallard Substation Expansion • Expand substation to accommodate 2-24/32/40 MVA transformers and 12-position PDC • Complete below grade for future third transformer, future second PDC, and two (2) replacement PDC locations Long-Term 2: New General-Purpose Class 'A' substation along Warren Ave. • Build new Class 'A' with 3-24/32/40 MVA transformers, two (2) 12-positions PDC locations and two (2) PDC replacement locations Long-Term 3: New General-Purpose substation near Glendale and Villa substations • Develop new substation and integrate new substation and circuits near Glendale and Villa substations.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 262 of 347

(a)	(b)						
Program:	4.8 kV CC: Hawthorne Relief and Circuit Conversion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Convert dense 4.8 kV service territory to 13.2 kV. • Remove aging infrastructure and replace with new equipment in line with DDO standards. • Install poles tall enough for clearances that will allow for conversion. • Prepare area for load transfer off of Hawthorne substation. • Expanded Mallard substation provides additional circuit capacity for area covered by the Hawthorne Initiative. • Purchasing property allows DTE to plan new substations and integrate them into the existing power grid. • Integrate NWA solutions into the power grid to assist during peak load hours. • Eliminate stranded load east of Telegraph Rd. and north of Hines Park. • New substations provide for capacity coming from the east to support the impacted area.						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	X	X			X	X	

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 263 of 347

(a)	(b)																				
Program:	4.8 kV CC: Hawthorne Relief and Circuit Conversion (Pg. 4 of 4)																				
Current Projects:	4.8 kV CC: Hawthorne Relief and Circuit Conversion																				
Budget Basis:	• High Level Conceptual Estimate																				
Cost:	Estimated project spend in 2021-2023: \$6.4 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>45</td><td>3,750</td><td>3,795</td></tr><tr><td>Material</td><td>9</td><td>750</td><td>759</td></tr><tr><td>Other</td><td>6</td><td>500</td><td>506</td></tr><tr><td>Total</td><td>60</td><td>5,000</td><td>5,060</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	45	3,750	3,795	Material	9	750	759	Other	6	500	506	Total	60	5,000	5,060
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	45	3,750	3,795																		
Material	9	750	759																		
Other	6	500	506																		
Total	60	5,000	5,060																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 264 of 347

(a)	(b)
Program:	4.8 kV CC: Birmingham Decommissioning and Circuit Conversion (Pg. 1 of 4)
Purpose and Necessity:	Birmingham substation, mainly serving downtown Birmingham, has a high substation outage risk. The substation's outage risk (probability of experiencing a loss of the entire substation) is estimated at 2.3%; the stranded load is estimated at 21 MVA after possible load transfers and 19 MVA after mobile fleet deployments. The substation is operating at its substation firm rating. Any new customer load or hot summer days will place the substation over its firm rating and in violation of DTE Electric's Distribution Design Standards. This project is to address the loading, aging infrastructure and substation outage risk at Birmingham substation. In addition Quarton Rd. substation is a 1948 vintage switchgear. QTNRD circuits will be left as an island with no jumper capability after BIHAM conversion. QTNRD has significant number of customer complaints due to aging infrastructure.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 265 of 347

(a)	(b)
Program:	4.8 kV CC: Birmingham Decommissioning and Circuit Conversion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 68
DGP Section:	11.3
Scope:	• Purchase property • Build new 40/13.2 kV substation in the Birmingham area • Convert existing 4.8 kV circuits to 13.2 kV • Transfer customer load from Birmingham and Quarton Road to the new substation • Decommission Birmingham and Quarton Road substations

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 266 of 347

(a)	(b)						
Program:	4.8 kV CC: Birmingham Decommissioning and Circuit Conversion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide sufficient capacity to serve 42.1MVA load presently supported by aging infrastructure • Provide load relief to Birmingham and Quarton Rd substations to be compliant with DTE Electric's Distribution Design Standards • Create jumper points with new ARIEL substation and existing, LNGLK, SAVGE substations • Reduce substation outage risk by replacing aging infrastructure • Enhance safety, reliability and power quality for upgraded circuits • Reduce non-storm events and reactive costs for upgraded circuits • Enhance grid technology and automaton, including DER integration						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x		x	x	x	

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 267 of 347

(a)	(b)																				
Program:	4.8 kV CC: Birmingham Decommissioning and Circuit Conversion (Pg. 4 of 4)																				
Current Projects:	4.8 kV CC: Birmingham Decommissioning and Circuit Conversion																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$2.1 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>28</td><td>1,158</td><td>1,187</td></tr><tr><td>Material</td><td>6</td><td>232</td><td>237</td></tr><tr><td>Other</td><td>4</td><td>154</td><td>158</td></tr><tr><td>Total</td><td>38</td><td>1,544</td><td>1,582</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	28	1,158	1,187	Material	6	232	237	Other	4	154	158	Total	38	1,544	1,582
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	28	1,158	1,187																		
Material	6	232	237																		
Other	4	154	158																		
Total	38	1,544	1,582																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 268 of 347

(a)	(b)
Program:	4.8kV CC: ISO Camden (Pg. 1 of 4)
Purpose and Necessity:	The 4.8kV ISO down circuits are already fed from a 13.2kV substation which removes the substation firm rating and the substation risk ranking from prioritization consideration. Without need to consider substation factors, safety (wire down reduction), reliability (customer minute interruptions), and costs (avoided O&M and capital) become the driving factors for prioritization of 4.8kV ISO down circuits.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 269 of 347

(a)	(b)
Program:	4.8kV CC: ISO Camden (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 69
DGP Section:	11.3
Scope:	• Installing controls and automation on the circuits to our latest design standards • Completing overhead conversion work including rebuilding pole tops, replacing poles and transformers as needed, and installing neutral wire. • Rebuilding underground infrastructure. • Reconductoring overhead lines as needed. • Reconfiguring circuits and establishing new jumpering points. • Removing ISO down transformers. • Rebuilding underground infrastructure as needed

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 270 of 347

(a)	(b)						
Program:	4.8kV CC: ISO Camden (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• The 4.8kV ISO down conversion projects are expected to bring multi-faceted benefits of safety improvements, reliability improvements, technology modernization, potential circuit load relief and cost reduction. • Elimination of aging OH infrastructure, leading to: - Improved reliability - Fewer truck roles - Reduced Capital expenses - Reduced O&M expenses • Removal of antiquated and failing substation equipment leading to: - Elimination of safety risks - Improved reliability - Reduction of long duration outage risks - Reduced equipment to be maintained - Reduced O&M and Capital expenses						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x	x			x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 271 of 347

(a)	(b)																				
Program:	4.8kV CC: ISO Camden (Pg. 4 of 4)																				
Current Projects:	4.8kV CC: ISO Camden																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.5 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>313</td><td>313</td></tr><tr><td>Material</td><td>-</td><td>63</td><td>63</td></tr><tr><td>Other</td><td>-</td><td>42</td><td>42</td></tr><tr><td>Total</td><td>-</td><td>417</td><td>417</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	313	313	Material	-	63	63	Other	-	42	42	Total	-	417	417
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	-	313	313																		
Material	-	63	63																		
Other	-	42	42																		
Total	-	417	417																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 272 of 347

(a)	(b)
Program:	4.8kV CC: ISO Gilbert (Pg. 1 of 4)
Purpose and Necessity:	The 4.8kV ISO down circuits are already fed from a 13.2kV substation which removes the substation firm rating and the substation risk ranking from prioritization consideration. Without need to consider substation factors, safety (wire down reduction), reliability (customer minute interruptions), and costs (avoided O&M and capital) become the driving factors for prioritization of 4.8kV ISO down circuits.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 273 of 347

(a)	(b)
Program:	4.8kV CC: ISO Gilbert (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 70
DGP Section:	11.3
Scope:	• Installing controls and automation on the circuits to our latest design standards • Completing overhead conversion work including rebuilding pole tops, replacing poles and transformers as needed, and installing neutral wire. • Rebuilding underground infrastructure. • Reconductoring overhead lines as needed. • Reconfiguring circuits and establishing new jumpering points. • Removing ISO down transformers. • Rebuilding underground infrastructure as needed.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 274 of 347

(a)	(b)						
Program:	4.8kV CC: ISO Gilbert (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• The 4.8kV ISO down conversion projects are expected to bring multi-faceted benefits of safety improvements, reliability improvements, technology modernization, potential circuit load relief and cost reduction. • Elimination of aging OH infrastructure, leading to: - Improved reliability - Fewer truck roles - Reduced Capital expenses - Reduced O&M expenses • Removal of antiquated and failing substation equipment leading to: - Elimination of safety risks - Improved reliability - Reduction of long duration outage risks - Reduced equipment to be maintained - Reduced O&M and Capital expenses						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x			x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 275 of 347

(a)	(b)		
Program:	4.8kV CC: ISO Gilbert (Pg. 4 of 4)		
Current Projects:	4.8kV CC: ISO Gilbert		
Budget Basis:	• Engineering Estimate		
Cost:	Estimated project spend in 2021-2023: \$0.5 M This project is expected to continue beyond 2023		
Test Year (\$000's):	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23
	Total Labor (DTE Electric & Contractors)	- 313	313
	Material	- 63	63
	Other	- 42	42
	Total	- 417	417

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 276 of 347

(a)	(b)
Program:	8.3 kV CC: Pontiac Overhead Conversion (Pg. 1 of 4)
Purpose and Necessity:	DTE Electric did not construct the 8.3 kV system but acquired it from CMS Energy in the 1980s. Because the 8.3 kV system is an island surrounded by the 13.2 kV system, it is impossible to transfer load from 8.3 kV circuits to neighboring facilities. This results in a high risk for stranded load in the event of an 8.3kV substation outage event. In addition, replacement parts are no longer available for 8.3 kV breakers, other substation equipment and equipment in the underground vaults due to their obsolescence. Non-standard clearances require substation shutdowns for operations and maintenance. This leads to extended customer interruptions during outage events and leaves the system in an abnormal state for extended periods of time if any 8.3 kV equipment fails. Crews must be trained to operate and maintain the 8.3 kV system, adding to training, operation and maintenance costs. Meanwhile, the City of Pontiac is experiencing an economic rebound, with an estimated 40 MVA (37 percent) load growth in the next five to ten years. Catalina Phase 2 & 3: Aging Infrastructure - Substation at Risk - 8.3KV circuit conversion. Steady diminishing capacity. Incompatible, obsolete, at risk, failing 8.32KV distribution system infrastructure supports the City of Pontiac (29.5MVA of load / 7,001 customers). Current substantial stranded load exposure (13.9MVA) Added exposure: BLMFD (2nd highest at-risk switchgear) supports 3MVA of transferred 8.32KV load. Unable to perform needed breaker replacements. Potential exposure: six breakers operating to failure Disconnect switches (at multiple substations) exhibit visibly damaged insulators Spare substation transformers (40-8.3kV) are not available Massive footprint to implement portable substation support (ISO DN trailer required) CAIDI - Combined unfavorable factors (unavailable replacement parts, multiple abnormal configurations, unfamiliar configuration) significantly impact duration of shutdowns and restoration efforts Overhead extension will enhance reliability (loop schemes) for deployed CATLI circuits Significant UG Automation investment (\$26.1M) supported by failing substations
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 277 of 347

(a)	(b)
Program:	8.3 kV CC: Pontiac Overhead Conversion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 71
DGP Section:	11.5
Scope:	Catalina Phase 2: • Extend / replace 5.2 miles conductors with 3-636, B to transfer (converted) Paddock circuits to Catalina substation • Establish two new loop schemes and sectionalizing points Catalina Phase 3 • Extend (~6,550 feet) of conduit (eight 2 way MHs, four 3 way MHs, two PMH-11 PSCs) • Extend (~14,720 feet) of 1000, EPR cables to extend DC 9131 CATLI UG to support conversion and partial transfer of PL 4030 STKWL and DC 9147 CATLI to establish throw over capabilities and supports conversion and partial transfer of PL 4014 STKWL • Extend / replace (50,250 feet) of undersized (smaller than #2CU) conductors; replace 268 OH service transformers. DC 9131 CATLI OH extension enables conversion and transfer of DC 4010 STKWL, and spatial conversion and transfer of DC 4002 STKWL, DC 4011 BARTL, DC 8276 CAMDN (8.3KV) load. DC 9147 CATLI establishes three <J1> points for extended CATLI circuits. Replace (~6340 feet) of URD cable and 27 pad-mounted transformers. Establish eight loop-scheme reclosers Establish infrastructure to support decommission of (8.3KV) substations: • Expand 120-13.2KV Wheeler Substation • Extend conduit (~7,250 feet) and cables to transfer all Bloomfield circuits (~32MVA of load) to expanded Wheeler substation • Extend cables and conductors to transfer (converted) Rapid Street circuits to expanded Wheeler substation • Extend cables and conductors to transfer (converted) Bartlett circuits to expanded Wheeler substation • Decommission end-of-life PADOK, RAPST, BARTL, STKWL, BLMFD substations

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 278 of 347

(a)	(b)						
Program:	8.3 kV CC: Pontiac Overhead Conversion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide sufficient capacity to serve growing areas • Decommission aging infrastructure and reduce major event risk • Enhance safety, reliability and power quality for upgraded circuits by reducing 85% of circuit wire down events, customer interruptions, and customer minutes • Reduce trouble events and reactive costs for upgraded circuits by 85% for both circuit tree related and non-tree related reactive capital • Increase protection for the distribution system • Enhance grid technology and automaton, including DER integration • Reduce new customer connection timeframe and cost • Corrects front line employees' exposure to unsafe work environments						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 279 of 347

(a)	(b)																												
Program:	8.3 kV CC: Pontiac Overhead Conversion (Pg. 4 of 4)																												
Current Projects:	8.3 kV CC: Catalina Phase 2 8.3 kV CC: Catalina Phase 3 8.3 kV CC: Infrastructure to support decommission of (8.3kV) substations																												
Budget Basis:	• Project Management Estimate																												
Cost:	Estimated project spend in 2021-2023: \$15.2 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>426</td><td>6,844</td><td>7,270</td></tr><tr><td>Material</td><td>85</td><td>1,369</td><td>1,454</td></tr><tr><td>Other</td><td>57</td><td>913</td><td>969</td></tr><tr><td>Total</td><td>568</td><td>9,125</td><td>9,693</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	426	6,844	7,270	Material	85	1,369	1,454	Other	57	913	969	Total	568	9,125	9,693
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
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Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 280 of 347

(a)	(b)
Program:	8.3 kV CC: Pontiac Underground Conversion (Pg. 1 of 4)
Purpose and Necessity:	City of Pontiac supporting substations (failing, end-of-life 8.3kV, supporting 29.5MVA of load / 7,001 customers) will be decommissioned and replaced with 13.2kV facilities. Largest services are delivered through 8.3 kV rated UG equipment. These services contain customer owned switchgear, fuses and transformers. Replacement of 8.3kV substations with 13.2KV infrastructure requires replacement of these services with 15kV class equipment. Extension of four 13.2kV distribution circuits will secure vital redundant sources for UG Automation (\$26.1M investment upgrade). Safety related working conditions at remaining service vaults: Below grade confined space which is subject to frequent water and debris intrusion; contains energized, exposed, primary voltage overhead equipment; large amounts of combustible fuel are present.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 281 of 347

(a)	(b)
Program:	8.3 kV CC: Pontiac Underground Conversion (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 72
DGP Section:	11.5
Scope:	• Pre-convert current 8.3KV system to enable deployment of scheduled 13.2kV substation expansion • In preparation for 8.3KV substations upgrade to 13.2KV substations, create ability to support 15KV class services (i.e. McLaren Hospital). • Establish strong working relationships with affected businesses and City officials to secure optimal services & infrastructure replacement cadence Equipment will be removed and replaced based on field inspections and may include some or all of the following: • Replace customer equipment: 4 switchgears, 2 metal clad switchgear, 4 fuse positions • Replace customer transformers: twenty-eight single-phase, ten three-phase, fourteen special order (>750KVA) transformers • Replace eighteen customer cables (estimated 800 feet per site) rated less than 15KV class • Coordinate efforts to deploy DGs (2 weeks minimum) to support conversion and transfer at Redundant Services - Critical Customer facilities

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 282 of 347

(a)	(b)						
Program:	8.3 kV CC: Pontiac Underground Conversion (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Replacement of non-compatible, outdated, non-repairable, non-standard voltage class infrastructure, leading to: - Elimination of safety risks - Improved reliability - Reduction of long duration outage risks - Reduced equipment to be maintained & stocked - Reduced O&M and Capital expenses • Increased jumpering between circuits due to all circuits being the same operating voltage • Reduce new customer connection timeframe and cost						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x			x	x	

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 283 of 347

(a)	(b)																				
Program:	8.3 kV CC: Pontiac Underground Conversion (Pg. 4 of 4)																				
Current Projects:	8.3 kV CC: Pontiac Underground Conversion																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$2.1 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>100</td><td>755</td><td>855</td></tr><tr><td>Material</td><td>20</td><td>151</td><td>171</td></tr><tr><td>Other</td><td>13</td><td>101</td><td>114</td></tr><tr><td>Total</td><td>133</td><td>1,007</td><td>1,140</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	100	755	855	Material	20	151	171	Other	13	101	114	Total	133	1,007	1,140
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	100	755	855																		
Material	20	151	171																		
Other	13	101	114																		
Total	133	1,007	1,140																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 284 of 347

(a)	(b)
Program:	System Loading: Prior Year (Pg. 1 of 4)
Purpose and Necessity:	DTE Electric evaluates system loading on a regular basis as part of its annual area load analysis (ALA). Based on DTE Electric's recent ALA studies, approximately 30 percent of distribution substations have various loading constraints, either substations being over their firm ratings or equipment being over its day-to-day rating during peak hours. As such, additional capacity is needed to prevent customer interruptions during a single contingency event and help maintain the useful life of the existing equipment.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 285 of 347

(a)

(b)

Program:	System Loading: Prior Year (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 73
DGP Section:	Not Available
Scope:	• BRAZL 8402 Overload: Establish new distribution circuit 8495 BRAZL and split the load of distribution circuit 8402 • Milford Substation Project: Convert Milford to a Class A substation with 9 position switchgear

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 286 of 347

(a)	(b)						
Program:	System Loading: Prior Year (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Enhance safety • Improve customer reliability and power quality • Reduce major customer outages by eliminating failing subtransmission line • Reduce reactive costs by replacing aging or defective infrastructure						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x				x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 287 of 347

(a)	(b)																																				
Program:	System Loading: Prior Year (Pg. 4 of 4)																																				
Current Projects:	BRAZL 8402 Overload Milford Substation Project-C																																				
Budget Basis:	• Project Management Estimate																																				
Cost:	Estimated project spend in 2021-2023: \$0.0 M																																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td></td><td colspan="3"> </td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td> </td><td> </td><td> </td></tr><tr><td></td><td>-</td><td>-</td><td>-</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23					Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total					-	-	-
	2 mos.	10 mos.	Test Year																																		
	11/01/22	01/01/23	12 mos. ending																																		
	12/31/22	10/31/23	10/31/23																																		
Total Labor (DTE Electric & Contractors)	-	-	-																																		
Material	-	-	-																																		
Other	-	-	-																																		
Total																																					
	-	-	-																																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 288 of 347

(a) (b)

Program:	System Loading: Brest Substation (Pg. 1 of 4)
Purpose and Necessity:	Operators are concerned about safety as the switchgear pad deteriorates further and causes the switchgear to shift (on the 4.8kV side of the substation). Several concrete equipment mats require replacement or significant repair. Switchgear breakers have reached end-of-life and the manufacturer no longer supports this breaker product.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 289 of 347

(a)	(b)
Program:	System Loading: Brest Substation (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 74
DGP Section:	Not Available
Scope:	• Convert the substation from a Class 'E' 4.8 kV substation and a Class 'T' 13.2kV substation to a Class 'A' 13.2kV substation.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 290 of 347

(a)	(b)						
Program:	System Loading: Brest Substation (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce safety concerns at the substation • Reduce stranded load • Provide capacity for future load growth						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x				x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 291 of 347

(a)	(b)																				
Program:	System Loading: Brest Substation (Pg. 4 of 4)																				
Current Projects:	System Loading: Brest Substation																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.1 M																				
Test Year (\$000's):	<table><thead><tr><th></th><th>2 mos. 11/01/22 12/31/22</th><th>10 mos. 01/01/23 10/31/23</th><th>Test Year 12 mos. ending 10/31/23</th></tr></thead><tbody><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td>-</td><td>-</td><td>-</td></tr></tbody></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total	-	-	-
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	-	-	-																		
Material	-	-	-																		
Other	-	-	-																		
Total	-	-	-																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 292 of 347

(a)

(b)

Program:	System Loading: Carleton (Pg. 1 of 4)
Purpose and Necessity:	Carleton is approximately 190% of the maximum loadability. TRF 2 is at 106% of its day to day rating. DC 312 CARLT is at 117% of its day to day rating.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 293 of 347

(a)	(b)
Program:	System Loading: Carleton (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 75
DGP Section:	11.1
Scope:	• Reroute 4.8kV and 40kV overhead on the substation property • Expand substation fence on the north side • Install new control center • Replace both transformers with 5/6.3MVA transformers and upgrade associated equipment • Re-establish J1 point between the two circuits

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 294 of 347

(a)	(b)						
Program:	System Loading: Carleton (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce equipment risk by alleviating the day to day overloads on the circuit, bus, and transformer • Re-establish the J1 point between the two circuits						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x	x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 295 of 347

(a)	(b)																												
Program:	System Loading: Carleton (Pg. 4 of 4)																												
Current Projects:	System Loading: Carleton																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$2.1 M																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>0</td><td>-</td><td>0</td></tr><tr><td>Material</td><td>0</td><td>-</td><td>0</td></tr><tr><td>Other</td><td>0</td><td>-</td><td>0</td></tr><tr><td>Total</td><td>0</td><td>-</td><td>0</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	0	-	0	Material	0	-	0	Other	0	-	0	Total	0	-	0
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	0	-	0																										
Material	0	-	0																										
Other	0	-	0																										
Total	0	-	0																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 296 of 347

(a)	(b)
Program:	System Loading: Grayling (Pg. 1 of 4)
Purpose and Necessity:	Loading at Grayling Substation was approximately 68.1 MVA in 2020, which is 119% of FIRM and is expected to rise over the next five years to as much as 138% of FIRM. DC 8517 GRAYL and DC 8539 GRAYL circuits are over the 12.0 MVA DDO Emergency Limit with the largest circuit at 13.8 MVA. The average circuit loading across all seven load carrying circuits is approximately 10.5 MVA. Additionally, there is new business that is still being developed across the Grayling Substation service area. The project includes constructing a new 120-13.2kV Class 'A' Substation, which has been named Sigma (SIGMA) Substation to relieve Grayling Substation, constructing six load carrying distribution circuits and one throw-over circuit.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 297 of 347

(a)

(b)

Program:	System Loading: Grayling (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 76
DGP Section:	11.1
Scope:	• Build a new 120/13.2 kV Class 'A' substation. • Build approximately six miles of cable • Build approximately two and a half miles of conduit with manholes • Build approximately four miles of overhead lines • Install one primary switch cabinet

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 298 of 347

(a)

(b)

Program:	System Loading: Grayling (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide load relief to Grayling substation and its circuits to be compliant with DTE Electric Distribution Design Standards • Provide capacity to serve existing and new customer load • Reduce major substation outage risk • Create circuit jumpering point to improve operability						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x	x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 299 of 347

(a)	(b)																												
Program:	System Loading: Grayling (Pg. 4 of 4)																												
Current Projects:	System Loading: Grayling																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$0.0 M																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td>-</td><td>-</td><td>-</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total	-	-	-
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	-	-	-																										
Material	-	-	-																										
Other	-	-	-																										
Total	-	-	-																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 300 of 347

(a)	(b)
Program:	System Loading: Prospect (Pg. 1 of 4)
Purpose and Necessity:	Prospect substation was at 131% of substation loading before 0.7 MVA was temporarily transferred to DC 9789 MOTT. The transformer is currently at 112% of its D-D rating and experienced higher loads in summer of 2020 due to many customers working from home. The transformer is expected to rise to 144% overloaded once the 0.7 MVA is transferred back. DC 0306 PROSP is currently at 130% of its D-D Rating. The transformer replacement is to relieve existing overload conditions at the substation and provide additional capacity for existing and new customer load. In the summer of 2018, the load at PROSP exceeded emergency rating of the transformer. Load shedding was required until a DG unit was installed from June - October. DO spend approximately \$107K on the DG installation.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 301 of 347

(a)

(b)

Program:	System Loading: Prospect (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 77
DGP Section:	11.1
Scope:	• Upgrade existing transformer • Extend the width of the substation fence and relocate 40kV line entrance to allow space for the new transformer • Install reclosers on pole outside of substation fence to serve as protection for the substation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 302 of 347

(a)	(b)						
Program:	System Loading: Prospect (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce equipment risk by providing load relief to Prospect substation • Create circuit jumpering points to improve operability • Reduce customer complaints from DC 9789 MOTT once able to transfer load back to PROSP after transformer is replaced						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x			

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 303 of 347

(a)	(b)																												
Program:	System Loading: Prospect (Pg. 4 of 4)																												
Current Projects:	System Loading: Prospect																												
Budget Basis:	• Project Management Estimate																												
Cost:	Estimated project spend in 2021-2023: \$0.5 M																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>67</td><td>-</td><td>67</td></tr><tr><td>Material</td><td>13</td><td>-</td><td>13</td></tr><tr><td>Other</td><td>9</td><td>-</td><td>9</td></tr><tr><td>Total</td><td>90</td><td>-</td><td>90</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	67	-	67	Material	13	-	13	Other	9	-	9	Total	90	-	90
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	67	-	67																										
Material	13	-	13																										
Other	9	-	9																										
Total	90	-	90																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 304 of 347

(a)

(b)

Program:	System Loading: Mandy 307 Load Transfer (Pg. 1 of 4)
Purpose and Necessity:	Mandalay substation is over firm due to overloads on circuits fed from TRF 1. (MANDY0307 and MANDY1390) Project driver is to permanently transfer 0.9MVA of load from MANDY0307 to ARIEL9893.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 305 of 347

(a)

(b)

Program:	System Loading: Mandy 307 Load Transfer (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 78
DGP Section:	11.1
Scope:	• Convert 7,600 feet of 4.8kV distribution circuit 307 Mandalay and transfer to distribution circuit 9893 Ariel • Transfer approximately 1.5 MVA from Mandalay substation to Ariel substation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 306 of 347

(a)	(b)						
Program:	System Loading: Mandy 307 Load Transfer (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Bring Substation under firm rating • Bring TRF1, MANDY0307, and MANDY1390 under D-D Ratings • Provide capacity to emerging businesses and subdivisions • Reduce substation outage risk by replacing aging infrastructure • Enhance safety, power quality and reliability for upgraded circuit						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x			x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 307 of 347

(a)	(b)																				
Program:	System Loading: Mandy 307 Load Transfer (Pg. 4 of 4)																				
Current Projects:	System Loading: Mandy 307 Load Transfer																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.9 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td>-</td><td>-</td><td>-</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total	-	-	-
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	-	-	-																		
Material	-	-	-																		
Other	-	-	-																		
Total	-	-	-																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 308 of 347

(a)	(b)
Program:	System Loading: Brown City (Pg. 1 of 4)
Purpose and Necessity:	Marlette 13.2kV is 113% of Firm, Brown City is 103% of firm, Tacoma is 79% of Firm and Shaw is 122% of Firm. Marlette 13.2kV is 95% of Emergency, Brown City is 92% of Emergency, Tacoma is at 65% of Emergency and Shaw is 100% of Emergency. MARL8281 is 101% of is DD rating and 95% of Emergency Rating. BRCTY0305 is 101% of is DD rating and 73% of Emergency Rating. SHAW0312 is at 120% if its DD rating and 92% of Emergency Rating. Brown City is an islanded 4.8kV Class 'T' substation surrounded by Tacoma 13.2kV circuit/substation and has no jumpering options. Tacoma substation is a 13.2kV Class 'T' substation with one load carrying circuit. With a recent method of service request, Tacoma substation is expected to be at 100% of its firm rating in 2021. Customers are not moving forward with requests to add load due to high C.I.A.C . Tacoma8311 has had many PQ complaints due to being 176 OH miles and conductor size is primarily #2 or 1/0 ACSR. Expected to have more load growth in Tacoma/Shaw area. TIE 9205 is not reliable and would require extensive reconductoring and a 40kV cap to allow for a load transfer of 1.2MVA from SHAW to TCOMA. TIE 3241 also has low voltage issues at SHAW and STDF4. With additional projected load growth, we will need to install 2 of 40kV Caps, Relay Upgrades, and some 40kV OH work.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 309 of 347

(a)	(b)
Program:	System Loading: Brown City (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 79
DGP Section:	11.1
Scope:	• Install 120-13.2kV 15/20/25MVA Class 'T' substation at Peck & Gosline Rd. Property purchase will be required • Reconductor approximately 4 miles on TCOMA8311, install iso-dn (2) at Peck and Maple Valley • Reconductor approximately 3.75 miles on MARLT8281 • Replace existing 41.6-13.2, 5MVA transformer at Tacoma Substation with 41.6-13.2kV, 10/12.5MVA transformer • Rebuild and convert approximately 8 miles of Brown City distribution circuits from 4.8 kV to 13.2 kV • Transfer the load from Brown City Substation to Tacoma Substation • Decommission Brown City Substation • Reconductor and convert approximately 5 miles TCOMA to transfer 1.2MVA from SHAW

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 310 of 347

(a)	(b)						
Program:	System Loading: Brown City (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide capacity to allow load growth • Decommission aging infrastructure and reduce major event risk • Enhance safety, reliability and power quality for upgraded circuits • Provide capacity for potential new developments in areas surrounding BRCTY, TCOMA and SHAW Substations • Relief SHAW from being 122% of Firm and 100% of EM to 78% of firm and 63% of EM • Relief SHAW DC312 from 123% of Firm and 92% of EM to 70% of DD and 52% of EM • Relief MARLT DC8281 from being 101% of DD and 95% of EM to 64% of DD and 60% of EM • Relief MARLT Substation being 117% of firm and 95% of EM to 74% of firm and 60% EM • Convert existing 4.8kV distribution circuits to 13.2kV • Provide jumpering options when currently there is limited/none jumpering options in the area						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:		X			X	X	X

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 311 of 347

(a)	(b)			
Program:	System Loading: Brown City (Pg. 4 of 4)			
Current Projects:	System Loading: Brown City			
Budget Basis:	• Project Management Estimate			
Cost:	Estimated project spend in 2021-2023: \$10.3 M This project is expected to continue beyond 2023			
Test Year (\$000's):	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	
	Total Labor (DTE Electric & Contractors)	342	4,676	5,018
	Material	68	935	1,004
	Other	46	623	669
	Total	456	6,234	6,690

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 312 of 347

(a)

(b)

Program:	System Loading: Mack Transformer 102 (Pg. 1 of 4)
Purpose and Necessity:	The load tap changers on TRF 101 and TRF 104 at Mack are currently inoperable. Maintenance to make necessary repairs requires lengthy shutdowns. Loss of an energized transformer while the other is shutdown for maintenance will cause significant stranded load to the system.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 313 of 347

(a)

(b)

Program:	System Loading: Mack Transformer 102 (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 80
DGP Section:	11.1
Scope:	• Install a new 60/80/120 MVA, 120kV-24KV TRF 102 to allow for the repair of the inoperable load tap changers and to allow for shutdowns in the future • Install 24kV secondary breaker

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 314 of 347

(a)	(b)						
Program:	System Loading: Mack Transformer 102 (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduction in substation outage risk • Improve system operability under contingency conditions or planned work						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x			x			

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 315 of 347

(a)	(b)																				
Program:	System Loading: Mack Transformer 102 (Pg. 4 of 4)																				
Current Projects:	System Loading: Mack Transformer 102																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.2 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td>-</td><td>-</td><td>-</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total	-	-	-
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	-	-	-																		
Material	-	-	-																		
Other	-	-	-																		
Total	-	-	-																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 316 of 347

(a)	(b)
Program:	System Loading: Port Sanilac (Pg. 1 of 4)
Purpose and Necessity:	Port Sanilac substation is 106% of its firm and Foster substation is at 117% of firm. Both Port Sanilac substation and Foster substation are Class 'T'. DC 301 Foster is at 97% of its day to day rating and 100kVA from its emergency rating and DC 302 Port Sanilac is at 129% of its day to day rating and is 200kVA from its emergency rating. The area surrounding Port Sanilac hasn't seen significant load growth in recent years, but the loading on it and surrounding substations has increased to a point at which new loading cannot be adequately served. Also, there is limited/no jumpering options during peak since Port Sanilac is surrounded by Forester and Applegate substations - both of which are approaching/exceeding their substation transformer limits.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 317 of 347

(a)	(b)
Program:	System Loading: Port Sanilac (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 81
DGP Section:	11.1
Scope:	• Install new Skidmount STDF between Foster & Port Sanilac Substations • Install one 40kV PTS and one 40kV Cable pole • Install one distribution cable pole • Install two triple single reclosers • Install 150 feet of 3-5" conduit and 1000AL cable for 40kV and 13.2kV • Rebuild/convert 3 miles of DC301 FOSTR • Rebuild/convert 3 miles of DC302 PTSAN • Relocate existing 40kV LSB at FOSTR • Decommission FOSTR

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 318 of 347

(a)

(b)

Program:	System Loading: Port Sanilac (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide sufficient capacity to serve existing and new customer load • Provide load relief to the substation to be compliant with DTE Electric Distribution Design Standards • Reduce substation outage risk by replacing aging infrastructure • Increase jumpering capability and eliminate stranded load associated with the islanded 4.8 kV system • Relocate aging 40kV from deep ROW to more truck accessible location • Enhance safety, reliability and power quality for upgraded circuits • Reduce non-storm events and reactive costs for upgraded circuits • Enhance grid technology and automaton, including DER integration						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x			x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 319 of 347

(a)	(b)																				
Program:	System Loading: Port Sanilac (Pg. 4 of 4)																				
Current Projects:	System Loading: Port Sanilac																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$3.7 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>56</td><td>2,024</td><td>2,080</td></tr><tr><td>Material</td><td>11</td><td>405</td><td>416</td></tr><tr><td>Other</td><td>7</td><td>270</td><td>277</td></tr><tr><td>Total</td><td>75</td><td>2,698</td><td>2,773</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	56	2,024	2,080	Material	11	405	416	Other	7	270	277	Total	75	2,698	2,773
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	56	2,024	2,080																		
Material	11	405	416																		
Other	7	270	277																		
Total	75	2,698	2,773																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 320 of 347

(a)	(b)
Program:	System Loading: Richmond/Armada (Pg. 1 of 4)
Purpose and Necessity:	Richmond 4.8 kV antiquated switchgear is the only oil filled truck type outdoor switchgear that we have left on our system. The switchgear is 1956 vintage equipped with oil filled circuit breakers along with CT's and other pieces that there are no spare parts for. The nine position switchgear at Spokane was installed about 10 years ago, but has no load connected. Richmond 4.8 kV substation is at 108% of its firm, rising to 117% by end of summer. Distribution circuit 2496 Richmond is at 94% of its day to day rating. Armada 4.8 kV substation is at 113% of firm with distribution circuit 302 at 112% of its day to day rating. Armada 13.2 kV substation is at 96% of firm and its transformer is at 95% of its day to day rating. Due to limited/no jumpering options, when the substation needs a shutdown for maintenance, a portable substation is required.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 321 of 347

(a)	(b)
Program:	System Loading: Richmond/Armada (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 82
DGP Section:	11.1
Scope:	Upgrade Richmond substation to a 40kV-13.2kV Class 'A' substation with 9 position switchgear (utilizing unused switchgear at Spokane substation): • Relocate Spokane switchgear to Richmond substation • Install new 15/20/25 transformer • Replace existing 7.5MVA transformer with 15/20/25 XFMR • Install 2 new circuit switchers • Install 1,000 feet of 3x4-5" duct • Install 2,000 feet of 1000AL cables to cut the 4.8kV and 13.2kV circuits to the new switchgear • Reroute about 500 feet of 40 kV overhead (include 2 new poles) • Decommission the existing Richmond 4.8kV substation and switchgear • Reconductor and convert 15 miles of overhead circuit • Decommission Armada 4.8kV Substation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 322 of 347

(a)	(b)						
Program:	System Loading: Richmond/Armada (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Eliminates insertion and removal of breakers using a cart set on an outdoor elevated platform. Breakers rack out directly outdoors. • Replaces old oil filled breakers with modern vacuum style breakers. • Reduces time and cost to perform a breaker inspection • Substation will be more reliable with the ability to pickup load from a dedicated T.O. circuit • Decommission 4.8kV substations and bring ARMAD substation and RICHM substation • All circuits will be below 90% of DD Rating • Provide reliable jumpering option to ARMAD 13.2kV instead of using the portable substation for any work require shutdown of the substation • Provides more capacity for growth and relieve adjacent substations • Cost reduction for pre conversion scope due to incorporating TIE810 subtransmission project (Distribution project will include 9 miles of pre-conversion instead of 16.1 miles) • Provide more capacity for growth and relieve adjacent substations						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:		X		X	X	X	X

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 323 of 347

(a)	(b)			
Program:	System Loading: Richmond/Armada (Pg. 4 of 4)			
Current Projects:	System Loading: Richmond/Armada			
Budget Basis:	• Project Management Estimate			
Cost:	Estimated project spend in 2021-2023: \$9.4 M This project is expected to continue beyond 2023			
Test Year (\$000's):	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	
	Total Labor (DTE Electric & Contractors)	626	2,406	3,033
	Material	125	481	607
	Other	84	321	404
	Total	835	3,208	4,043

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 324 of 347

(a)

(b)

Program:	System Loading: Cody (Pg. 1 of 4)
Purpose and Necessity:	Two of the distribution circuits at Cody substation are loaded above 14.0 MVA. According to the DDO standard, circuits should not be loaded above 8.0 MVA. One of the circuits is loaded so high that it could cause false trip on breakers.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 325 of 347

(a)	(b)
Program:	System Loading: Cody (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 83
DGP Section:	11.1
Scope:	• Circuit phase balancing • Circuit load transfers/reconfigurations • Rebuild 2.6 miles 13.2kV, 3-phase, overhead conductor to 636,B to support current circuit backbone • Rebuild 1.7 miles 13.2kV, single-phase, OH conductor to 3-phase, 636,B to create new circuit backbone • Convert a half mile of single-phase, 4.8kV to 3-phase, 13.2kV, 636 OH conductor to create new circuit backbone • Install sectionalizing and voltage correction equipment

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 326 of 347

(a)

(b)

Program:	System Loading: Cody (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Balance phasing of circuits to avoid breaker operation caused by high-phase load • Relieve heavy loaded circuits to avoid circuit ratings						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x	x					

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 327 of 347

(a)	(b)																				
Program:	System Loading: Cody (Pg. 4 of 4)																				
Current Projects:	System Loading: Cody																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.2 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>6</td><td>103</td><td>109</td></tr><tr><td>Material</td><td>1</td><td>21</td><td>22</td></tr><tr><td>Other</td><td>1</td><td>14</td><td>15</td></tr><tr><td>Total</td><td>8</td><td>137</td><td>145</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	6	103	109	Material	1	21	22	Other	1	14	15	Total	8	137	145
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	6	103	109																		
Material	1	21	22																		
Other	1	14	15																		
Total	8	137	145																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 328 of 347

(a)	(b)
Program:	System Loading: Otsego/Capac/Shaw (Pg. 1 of 4)
Purpose and Necessity:	OTSGO existing class 'R' 13.2kV substation is loaded at 20.9MVA at 138% of Firm and 77% of EM rating. 3.7MVA additional load is expected by this year which will cause the substation to be at 163% of firm and 90% of EM rating. There is limited to no jumpering options to surrounding circuits. OTSGO has a blocking scheme and is set at 15.1MVA Summer, with load substation will be 4-9MVA stranded CAPAC existing class 'R' 13.2kV substation is loaded at 15.7MVA at 116% of firm and 73% of EM rating. There has been 7 Method of Service requests (approximately total 2MVA) in the CAPAC area in 2020. The majority of them are small so we are expecting the load at CAPAC to increase. CAPAC has a blocking scheme and is set at 13.5MVA summer, with load, substation will be 2-5MVA stranded. There is no jumpering option outside of the CAPAC circuits SHAW is an existing class 'T' 4.8kV substation equipped with a 2.5MVA transformer. Substation is 123% of Firm and 100% of EM Rating. The Brown City/TCOMA project will help relieve the northern portion of SHAW and this will help relieve the Southern portion of Shaw.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 329 of 347

(a)	(b)
Program:	System Loading: Otsego/Capac/Shaw (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 84
DGP Section:	11.1
Scope:	• Install a new 120kV fed Skidmount substation. Property Purchase will be required. • Pre-convert and convert about 10 miles (includes part of CAPAC, OTSGO, and SHAW). New substation load initial load will be ~5MVA • Transfer ~0.76MVA of load from SHAW312 • Transfer ~2.4MVA of load from CAPAC8756 (open at R2) • Transfer ~1.7MVA of load from OTSGO8803 (open at L) • Transfer ~2MVA of load from CAPAC 8013 to new MIDAS circuit and pre-convert/Convert and about 2 miles • Transfer ~1.5MVA of load from OTSGO to ALMOT. Pre-convert/convert ~1 mile

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 330 of 347

(a)

(b)

Program:	System Loading: Otsego/Capac/Shaw (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	With the new Class 'R' substation, we will be able to relieve a lot more load from SHAW, CAPAC, and OTSGO. OTSGO 13.2kV substation is projected to be at 163% of Firm this year, after this project, OTSGO will be at 114% of firm. Another project will be required on the west in maybe 10 years to further relief OTSGO, ROBIN, and SDF4 and perhaps decommission SHAW and IMLAY4.8kV. CAPAC 13.2kV substation is at 116% of firm and after load transfer it will be at 82% of firm. Shaw will be at about 70% of firm after the Brown City project and this project						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 331 of 347

(a)	(b)																												
Program:	System Loading: Otsego/Capac/Shaw (Pg. 4 of 4)																												
Current Projects:	System Loading: Otsego/Capac/Shaw																												
Budget Basis:	• Project Management Estimate																												
Cost:	Estimated project spend in 2021-2023: \$1.5 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>41</td><td>714</td><td>755</td></tr><tr><td>Material</td><td>8</td><td>143</td><td>151</td></tr><tr><td>Other</td><td>5</td><td>95</td><td>101</td></tr><tr><td>Total</td><td>54</td><td>953</td><td>1,007</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	41	714	755	Material	8	143	151	Other	5	95	101	Total	54	953	1,007
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	41	714	755																										
Material	8	143	151																										
Other	5	95	101																										
Total	54	953	1,007																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 332 of 347

(a)	(b)
Program:	System Loading: Macomb/Golf (Pg. 1 of 4)
Purpose and Necessity:	GOLF is a 3 transformer class 'A' substation with 85MVA of load. It's at 89% of firm. Six out of 10 circuits are over 90% of DD rating. MACMB Sub is class 'TT' at 112% of Firm High Load Density Area - MACMB and GOLF carry 143 MVA combined. No capacity for new load on distribution system. Energized 1.1 MVA of new customers in 2020 on MACMB (MACMB8277). 3.1 MVA MOS requested but unable to serve due to capacity limitations on MACMB
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 333 of 347

(a)	(b)
Program:	System Loading: Macomb/Golf (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 85
DGP Section:	11.1
Scope:	• Upgrade the existing Macomb substation to three transformers with 2 PDC Class 'A' sub. (Keep the existing 2 transformers). Add 1 more transformer, two 12 position PDC's, 4 substation 13.2kV distribution caps at sub. • Install 25,000 feet underground distribution cable for circuit exits and throw over cable • Install 15,000 feet of conduit • Circuit 1: Rebuild 2.5 miles of overhead infrastructure with 636AL double build and install 3 reclosers to operate as a loop scheme. • Circuit 2: Install 1.5 miles of neutral to convert 13.2 kV • Circuit 3: Rebuild 1.5 miles of overhead with 636 AL • Circuit 4: Rebuild and convert 1 mile with 636 AL double build • Circuit 5: Install a recloser

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 334 of 347

(a)	(b)						
Program:	System Loading: Macomb/Golf (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• MACMB SUB 112% of firm to 79% of firm • GOLF SUB 89% of firm to 75% • GOLF 8552 from 112% D-D to 32% D-D • GOLF8531 from 103% D-D to 85% D-D • MACMB 8510 from 109% D-D to 54% D-D • BURBK2438 from 130% D-D to 91% D-D • Relieve overloads in the area • Support load growth • Eliminate 4.8kV area • Provide Jumpering options in the area						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x	x	x	

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 335 of 347

(a)	(b)																				
Program:	System Loading: Macomb/Golf (Pg. 4 of 4)																				
Current Projects:	System Loading: Macomb/Golf																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$2.8 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>90</td><td>1,314</td><td>1,403</td></tr><tr><td>Material</td><td>18</td><td>263</td><td>281</td></tr><tr><td>Other</td><td>12</td><td>175</td><td>187</td></tr><tr><td>Total</td><td>120</td><td>1,752</td><td>1,871</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	90	1,314	1,403	Material	18	263	281	Other	12	175	187	Total	120	1,752	1,871
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	90	1,314	1,403																		
Material	18	263	281																		
Other	12	175	187																		
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Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 336 of 347

(a)

(b)

Program:	System Loading: Grenada (Pg. 1 of 4)
Purpose and Necessity:	This project aligns with the proactive area plan. There is increased load within the existing GRENA substation area.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 337 of 347

(a)	(b)
Program:	System Loading: Grenada (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 86
DGP Section:	11.1
Scope:	This is preliminary scope and will be refined during detailed engineering: • Install a new 120-13.2kV Class 'A' substation by installing (2) 24/32/40MVA transformers and a 12-position switchgear • Install 5,000 feet of new 5" conduit • Install 10,000 feet of new cable • Rebuild 10,000 feet of backbone • Convert iso pockets on GRENA9813 by replacing 50,000 feet of overhead conductors • Rebuild and convert up to 1000,000 feet of overhead from PROSP • Install sectionalizing and operating devices

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 338 of 347

(a) Program:	(b) System Loading: Grenada (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide capacity for existing and new customers • Bring firm below 100%, eliminating the risk of stranded load • Help eliminate 4.8kV circuits						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x		x			

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 339 of 347

(a)	(b)																				
Program:	System Loading: Grenada (Pg. 4 of 4)																				
Current Projects:	System Loading: Grenada																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$2.8 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>93</td><td>1,250</td><td>1,343</td></tr><tr><td>Material</td><td>19</td><td>250</td><td>269</td></tr><tr><td>Other</td><td>12</td><td>167</td><td>179</td></tr><tr><td>Total</td><td>125</td><td>1,667</td><td>1,791</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	93	1,250	1,343	Material	19	250	269	Other	12	167	179	Total	125	1,667	1,791
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	93	1,250	1,343																		
Material	19	250	269																		
Other	12	167	179																		
Total	125	1,667	1,791																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 340 of 347

(a)	(b)
Program:	Pilot: Strategic and Service Undergrounding (Pg. 1 of 4)
Purpose and Necessity:	One of the planning scenarios described in the 2021 DGP considers changing weather patterns that lead to an increasing number of catastrophic storms. Although storm resiliency programs like tree trimming, hardening, PTMM, and customer excellence are established to provide customers the needed reliability for this scenario, in some areas they may not completely resolve the issue. In these situations, Strategic Underground may be a good option. Furthermore, the Company's customers have been impacted by an unusually high number of severe weather events during the summer of 2021. This has driven significant customer interest in undergrounding, and the Company is developing plans to partner with communities to implement based on their interest.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 341 of 347

(a)	(b)
Program:	Pilot: Strategic and Service Undergrounding (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 87
DGP Section:	11.6
Scope:	• Apply lessons learned from Appoline pilot undergrounding to developing pilots. • Replacing overhead services with underground services • Replacing overhead laterals with URD • Working with communities that have specific needs, and those who may be willing to partner in funding projects

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 342 of 347

(a)	(b)						
Program:	Pilot: Strategic and Service Undergrounding (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce tree trim costs • Improve safety by reducing wire down events • Improve reliability						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x				x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 343 of 347

(a)	(b)																				
Program:	Pilot: Strategic and Service Undergrounding (Pg. 4 of 4)																				
Current Projects:	Pilot: Strategic and Service Undergrounding																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$61.0 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>2,587</td><td>25,000</td><td>27,587</td></tr><tr><td>Material</td><td>517</td><td>5,000</td><td>5,517</td></tr><tr><td>Other</td><td>345</td><td>3,333</td><td>3,678</td></tr><tr><td>Total</td><td>3,450</td><td>33,333</td><td>36,783</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	2,587	25,000	27,587	Material	517	5,000	5,517	Other	345	3,333	3,678	Total	3,450	33,333	36,783
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	2,587	25,000	27,587																		
Material	517	5,000	5,517																		
Other	345	3,333	3,678																		
Total	3,450	33,333	36,783																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 344 of 347

(a)	(b)
Program:	Pilot: Primary Deconductoring (Pg. 1 of 4)
Purpose and Necessity:	This is a pilot to deconductor specific circuits, in areas that have seen significant population decline with a goal of removing underutilized infrastructure that is no longer needed. Work is being done to reduce the number of wire downs and assets in unoccupied areas. This work will decrease the wire down exposure and outage events. This program will reduce the number of assets which will result in lower tree trimming and PTM costs. This program will also reduce the scope of 4.8kV hardening by reducing the number of pole tops. Circuits will be evaluated based on their peak demand and vacant parcels to determine if they fit this program. After selecting circuits, voltage drop will be calculated per lateral to see effects of removing primary conductor and transformers. After determining voltage drop, primary conductor and transformers on laterals will be removed, secondary will be reconducted where necessary, and transformers will be installed in truck accessible circuit backbone.
Category:	Infrastructure Redesign & Modernization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 345 of 347

(a)

(b)

Program:	Pilot: Primary Deconductoring (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 10, Line 88
DGP Section:	11.6
Scope:	• Deconductor pockets of sparsely occupied areas within the City of Detroit by removing primary wire and hardware. • Remove approximately 19,000 feet of small primary wire and arc circuit in rear lots along with pole tops • Remove rear lot distribution transformers • Install truck accessible transformers • Reconductor approximately 10,500 feet of secondary conductor • Perform necessary spot line clearance

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 346 of 347

(a)	(b)						
Program:	Pilot: Primary Deconductoring (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce the number of wire downs • Reduce the number of outage events • Lower tree trimming and PTM costs • Reduce the need for 4.8kV hardening • To reduce restoration timing						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x				x	x	
Targeted:							

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Infrastructure Redesign & Modernization

Case No.: U-20836
Exhibit: A-23
Schedule: M5
Witness: S. G. Pfeuffer
Page: 347 of 347

(a)	(b)																				
Program:	Pilot: Primary Deconductoring (Pg. 4 of 4)																				
Current Projects:	Pilot: Primary Deconductoring Pilot: WAYBN 1163 Deconductor																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$2.0 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>171</td><td>-</td><td>171</td></tr><tr><td>Material</td><td>34</td><td>-</td><td>34</td></tr><tr><td>Other</td><td>23</td><td>-</td><td>23</td></tr><tr><td>Total</td><td>228</td><td>-</td><td>228</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	171	-	171	Material	34	-	34	Other	23	-	23	Total	228	-	228
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	171	-	171																		
Material	34	-	34																		
Other	23	-	23																		
Total	228	-	228																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
M. Elliott Andahazy 1/
P.Smith 2/
Page: 1 of 158

(a)	(b)
Category	Distribution Plant Capital Project Detail - Technology and Automation
	Capital Exhibit:
pages: 3-6 of 158	1/ Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 2 ADMS: DMS/OMS
pages: 7-10 of 158	1/ Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 3 ADMS: Network Management System
pages: 11-14 of 158	1/ Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 4 SOC Modernization:ESOC
pages: 15-18 of 158	1/ Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 5 SOC Modernization:ASOC
pages: 19-22 of 158	Exhibit A-13, Schedule B5.5, Page 11 of 12, Line 6 Grid Automation Telecommunications (13.2kV Telecommunications)
pages: 23-26 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 7 Distribution Sensing and Monitoring (inclgd Line Sensors)
pages: 27-30 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 8 SCADA / AMI Enhancements
pages: 31-34 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 9 Automation: Substation
pages: 35-38 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 10 Automation: Distribution
pages: 39-42 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 11 CVR/VVO
pages: 43-46 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 12 NWA: O'Shea Energy Storage
pages: 47-50 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 13 NWA: Battery Trailer
pages: 51-54 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 14 NWA: Omega Load Relief
pages: 55-58 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 15 NWA: Fisher Load Relief
pages: 59-62 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 16 NWA: Port Austin Load Relief
pages: 63-66 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 17 NWA: Veridian
pages: 67-70 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 18 NWA: Small Solar and Storage Testbed
pages: 71-74 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 19 NWA: EV Charging Demonstration at ACM
pages: 75-78 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 20 Technology Programs & NWA
pages: 79-82 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 21 DERMS
pages: 83-86 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 22 DER Control

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
M. Elliott Andahazy 1/
P. Smith 2/
Page: 2 of 158

(a)

(b)

Category	Distribution Plant Capital Project Detail - Technology and Automation
	Capital Exhibit:
pages: 87-90 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 23 Mobile Technology
pages: 91-94 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 24 Work Management & Scheduling Upgrades
pages: 95-98 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 25 Substation Design Tool Upgrades
pages: 99-102 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 26 Asset Management Upgrades
pages: 103-106 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 27 Load Forecasting & Analytics
pages: 107-110 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 28 Interconnection Process Enablement
pages: 111-114 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 29 Hosting Capacity Enablement
pages: 115-118 of 158	2/ Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 30 AMI: Installations
pages: 119-122 of 158	2/ Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 31 AMI: Meter Communications Upgrades
pages: 123-126 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 32 Microwave End of Life
pages: 127-130 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 33 SCADA Remote Access and Configuration Platform
pages: 131-134 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 34 Automation Configuration and Test Record Database
pages: 135-138 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 35 Grid Edge Insights & New Technology
pages: 139-142 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 36 Sensor, Network and Algorithm Development (Solar Deployment)
pages: 143-146 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 37 Other Modernize Grid Management
pages: 147-150 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 38 Substation Cybersecurity
pages: 151-154 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 39 Operational Technology and Error Free Communication
pages: 155-158 of 158	Exhibit A-12, Schedule B5.4, Page 11 of 12, Line 40 Analog Lines Elimination

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: M. Elliott Andahazy
Page: 3 of 158

(a)

(b)

Program:	ADMS: DMS/OMS (Pg. 1 of 4)
Purpose and Necessity:	The current DMS and OMS systems are outdated, non-integrated and with partial functionality. The current DMS system is at end of life. The existing infrastructure supporting the software is no longer commercially available, increasing the risk of recoverability from system failures and putting the safe operation of the electric system at risk. The current DMS and OMS were developed by separate companies and not built on a common data platform. The OMS is not connected to the DMS, which makes real time operation and analysis inefficient and difficult. The Company cannot upgrade further without replacing the hardware (which is ~15 years old) and performing a significant upgrade of the application software. The hardware and software support is currently being phased out by vendors, which increases risks due to the lack of replacement hardware, software patches, and support.
Category:	Technology and Automation

(a) (b)

Program:	ADMS: DMS/OMS (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 2
DGP Section:	12.1
Scope:	• Purchase software and hardware for DMS/OMS systems • Install DMS/OMS portion of the ADMS to integrate various operational technologies and analytical tools • Integrate new systems with other key systems, such as mapping and customer information systems like AMI to provide potential switching and dispatching solutions to these personnel to reduce the cost and time of outages • Install functions to perform all electrical system analysis from a single view of the system • Install functions to perform digital switch orders to enhance employee safety in the field

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: M. Elliott Andahazy
Page: 5 of 158

(a) (b)

Program:	ADMS: DMS/OMS (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Enhance safety and restoration times by providing real-time situational awareness to all resources • Enhance safety, reliability, and efficiency by eliminating paper maps and switching orders • Provide platform for Fault Location, Isolation and Restoration • Enable remote switching operations • Support integration of Distributed Energy Resources						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: M. Elliott Andahazy
Page: 6 of 158

(a)	(b)																				
Program:	ADMS: DMS/OMS (Pg. 4 of 4)																				
Current Projects:	ADMS: DMS/OMS																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$64.2 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>4,267</td><td>5,055</td><td>9,323</td></tr><tr><td>Material</td><td>853</td><td>1,011</td><td>1,865</td></tr><tr><td>Other</td><td>569</td><td>674</td><td>1,243</td></tr><tr><td>Total</td><td>5,690</td><td>6,741</td><td>12,430</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	4,267	5,055	9,323	Material	853	1,011	1,865	Other	569	674	1,243	Total	5,690	6,741	12,430
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	4,267	5,055	9,323																		
Material	853	1,011	1,865																		
Other	569	674	1,243																		
Total	5,690	6,741	12,430																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: M. Elliott Andahazy
Page: 7 of 158

(a)

(b)

Program:	ADMS: Network Management System (Pg. 1 of 4)
Purpose and Necessity:	The foundation for the evolving Distribution Planning processes is a high-quality network model including grid topology and electrical characteristics. Network model investments include technology to better align field conditions and maps to the digital representation of the grid, integrations between asset systems, new data models to support planning and operations topology and characteristics, and advanced analytics to leverage sensor data to continuously improve the network model.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: M. Elliott Andahazy
Page: 8 of 158

(a)

(b)

Program:	ADMS: Network Management System (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 3
DGP Section:	12.9
Scope:	• Common platform operating maps - Configuration and software scripts developed to support cartographic representations of subtransmission, AC Network, and URD maps. Transfer, reconciliation, and field validation of the new configuration within the ESRI platform. • Asset data integrations - software integration among enterprise asset systems (Maximo, ESRI, SAP) for electric asset data. • Network model enhancements - additional detailed data model configuration and initial attribution to support electronic switching and tagging, as well as future distribution planning processes such as hosting capacity and scenario analysis. • Grid model analytics - leverage the enterprise data platform to explore and develop machine learning algorithms to address data quality gaps based on existing and derived information. Once the algorithms are tested and proven, DTEE will integrate the machine learning workflow with source systems to raise overall data effectiveness. • Network Model orchestrator (beyond 2025) - select and implement software solution to manage network model versions for planned work, including scenario configurations

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: M. Elliott Andahazy
Page: 9 of 158

(a)

(b)

Program:	ADMS: Network Management System (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provide foundation needed to deploy advanced grid simulations and analytics called for by the evolving distribution network. • Improve situational awareness by generating operating maps used in the field from the same source that is powering planning and operations electronic tools • Perform data input and maintenance efficiently through asset data integrations and algorithms to address data quality gaps						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x				x	x	

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: M. Elliott Andahazy
Page: 10 of 158

(a)	(b)																				
Program:	ADMS: Network Management System (Pg. 4 of 4)																				
Current Projects:	Network Model																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$6.3 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>350</td><td>1,813</td><td>2,163</td></tr><tr><td>Material</td><td>70</td><td>363</td><td>433</td></tr><tr><td>Other</td><td>47</td><td>242</td><td>288</td></tr><tr><td>Total</td><td>467</td><td>2,417</td><td>2,883</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	350	1,813	2,163	Material	70	363	433	Other	47	242	288	Total	467	2,417	2,883
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	350	1,813	2,163																		
Material	70	363	433																		
Other	47	242	288																		
Total	467	2,417	2,883																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: M. Elliott Andahazy
Page: 11 of 158

(a)	(b)
Program:	SOC Modernization:ESOC (Pg. 1 of 4)
Purpose and Necessity:	SOC is the most critical facility in Distribution Operations. Personnel in the SOC operate the generation, subtransmission, and distribution systems in southeast Michigan, monitor alarms and system conditions, and direct field personnel to operate electrical elements for routine switching, risk mitigation, and outage restoration. SOC also works with Central Dispatch personnel to ensure appropriate crews are assigned to address system issues. The current SOC poses several limitations which the utilities DTE Electric has benchmarked have already addressed. Outdated facility: The facility lacks the redundancy in mechanical and electrical systems that are necessary to ensure continued operations in the event of a crisis. Outdated technology: SOC utilizes a magnetic tile representation of the electric network, as opposed to an electronic display board of the transmission, subtransmission and distribution network as is now very common in the industry. This severely limits situational awareness, which is critical at all times, but particularly during periods of crisis (for example during large storms). The current tile map board is also running out of space to accommodate growth of the system. The lack of modern technology also limits training opportunities. Space limitations: DTE Electric's SOC and dispatch personnel are currently physically separated, and their primary method of interaction is through repeated phone calls to share information and collaborate on dispatching field resources. The current SOC does not have sufficient space to achieve the co-location of these resources that manage the system and dispatch field personnel to resolve operational issues. The co-location of SOC and dispatch personnel is a well-established best practice, as it provides significant customer benefits in terms of the speed at which issues can be addressed and electric service can be restored. Limited visibility of telecommunication infrastructure performance: The reliability of telecommunication paths from field devices to the SOC is critical for the effective monitoring of the grid and for remote operations. Developing the ability to separately monitor the condition of the telecommunication network through the construction of a Smart Grid Network Operation Center (SGNOC) is part of the SOC Modernization project.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: M. Elliott Andahazy
Page: 12 of 158

(a)

(b)

Program:	SOC Modernization:ESOC (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 4
DGP Section:	12.2
Scope:	• Complete design and construct primary • Complete design and construct the Smart Grid Network Operation Center

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: M. Elliott Andahazy
Page: 13 of 158

(a) (b)

Program:	SOC Modernization:ESOC (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Upgrade System Operations Center (SOC) to meet industry standards • Support faster restoration times during outages and improve system resiliency during major storms • Focused monitoring capabilities for hardware, software, network, security, endpoints, and field devices specific to the operation of the electric system through the SGNOC						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: M. Elliott Andahazy
Page: 14 of 158

(a)	(b)																												
Program:	SOC Modernization:ESOC (Pg. 4 of 4)																												
Current Projects:	SOC Modernization:ESOC																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$14.4 M																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>46</td><td>-</td><td>46</td></tr><tr><td>Material</td><td>9</td><td>-</td><td>9</td></tr><tr><td>Other</td><td>6</td><td>-</td><td>6</td></tr><tr><td>Total</td><td>62</td><td>-</td><td>62</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	46	-	46	Material	9	-	9	Other	6	-	6	Total	62	-	62
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	46	-	46																										
Material	9	-	9																										
Other	6	-	6																										
Total	62	-	62																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: M. Elliott Andahazy
Page: 15 of 158

(a) (b)

Program:	SOC Modernization:ASOC (Pg. 1 of 4)
Purpose and Necessity:	Though it does meet minimum regulatory requirements for NERC regulated Balancing Authority and Generator Operator tasks, managing the distribution system and recovering from a storm or other disaster from the existing backup SOC would be extraordinarily challenging and lead to very slow restoration of the distribution system. The new backup SOC, also known as the Alternate System Operations Center (ASOC) will have the appropriate square footage required to collocate personnel and will have the appropriate mechanical and electrical system redundancies. In addition, the new ASOC will also be outfitted with the same ADMS technology (including a video wall) as the new ESOC for seamless operations during the transition between facilities. This will allow the Company to continue the use of the electronic records of the Network Model rather than reverting to a paper version as is used in the current backup facility
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: M. Elliott Andahazy
Page: 16 of 158

(a)

(b)

Program:	SOC Modernization:ASOC (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 5
DGP Section:	12.2
Scope:	• Complete design and construct an Alternative SOC site

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: M. Elliott Andahazy
Page: 17 of 158

(a) (b)

Program:	SOC Modernization:ASOC (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Upgraded Alternative System Operations Center (ASOC) to meet industry standards • Supports faster restoration times during outages and improve system resiliency during major storms						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: M. Elliott Andahazy
Page: 18 of 158

(a)	(b)																				
Program:	SOC Modernization:ASOC (Pg. 4 of 4)																				
Current Projects:	SOC Modernization:ASOC																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$34.5 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>1,248</td><td>14,979</td><td>16,227</td></tr><tr><td>Material</td><td>250</td><td>2,996</td><td>3,245</td></tr><tr><td>Other</td><td>166</td><td>1,997</td><td>2,164</td></tr><tr><td>Total</td><td>1,664</td><td>19,972</td><td>21,636</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	1,248	14,979	16,227	Material	250	2,996	3,245	Other	166	1,997	2,164	Total	1,664	19,972	21,636
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	1,248	14,979	16,227																		
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Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 19 of 158

(a)	(b)
Program:	Grid Automation Telecommunications (13.2kV Telecommunications) (Pg. 1 of 4)
Purpose and Necessity:	Many devices, such as pole-top and substation equipment, on DTE Electric's system are either not connected for remote monitoring and control or are connected through a communication network that has been developed in patchwork fashion over many decades. Where the communication network is available, it is frequently made up of diverse equipment and technology, such as fiber, microwave, leased phone lines, radio and other. Furthermore, much of the communication available is completed through a point-to-point network that is less reliable than current mesh technologies. Also, because the equipment that enables the current network has been installed over an extended timeframe, some of the equipment has become obsolete and spare parts are no longer available. The Grid Automation Telecommunication program is designed to address these communication gaps and to deploy a consistent channel with sufficient and reliable bandwidth to meet the current and growing requirements of a modern electrical system and to allow the deployment of the appropriate cybersecurity protocols.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 20 of 158

(a)	(b)
Program:	Grid Automation Telecommunications (13.2kV Telecommunications) (Pg. 2 of 4)
Line Number:	Exhibit A-13, Schedule B5.5, Page 11, Line 6
DGP Section:	12.3
Scope:	• Extend the telecommunication network through a combination of supplementing the existing Wi-Fi Mesh, building fiber to existing substations sites, and establishing a telecommunications network inside existing substations to connect relays and metering • Install standardized telecommunications and SCADA cabinets in each substation and build the telecommunications path to the substations • Replace microwave tower transmitters and WiMAX pole top transmitters with transmitters operating on different frequencies (5G & Wi-Fi upgrade) (FCC mandated reassignment of Microwave and WiMAX frequencies to 5G cellular and Wi-Fi devices)

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 21 of 158

(a)

(b)

Program:	Grid Automation Telecommunications (13.2kV Telecommunications) (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Allows for SCADA upgrades and automation • Allows for ADMS to monitor and control devices • Enhances cyber security and resiliency • Forms the basis of a smart grid • Allows for greater resiliency during weather events and in cases of equipment failure • Eliminates dial-up modems for substation remote access • Enables many future use cases and technologies beyond grid devices						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x				x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 22 of 158

(a)	(b)																				
Program:	Grid Automation Telecommunications (13.2kV Telecommunications) (Pg. 4 of 4)																				
Current Projects:	Grid Automation Telecommunications Communication Upgrade - TCHPP Demo																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$43.0 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>1,850</td><td>11,938</td><td>13,788</td></tr><tr><td>Material</td><td>370</td><td>2,388</td><td>2,758</td></tr><tr><td>Other</td><td>247</td><td>1,592</td><td>1,838</td></tr><tr><td>Total</td><td>2,467</td><td>15,917</td><td>18,383</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	1,850	11,938	13,788	Material	370	2,388	2,758	Other	247	1,592	1,838	Total	2,467	15,917	18,383
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
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Total	2,467	15,917	18,383																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 23 of 158

(a)	(b)
Program:	Distribution Sensing and Monitoring (inclgd Line Sensors) (Pg. 1 of 4)
Purpose and Necessity:	Line sensors (smart fault indicators) and low-cost substation monitoring equipment provides key situational awareness data to Distribution Operations, such as real time fault status and magnitude as well as near-real-time (5 minute) circuit loading information. These tools allow DTE to gain important visibility into parts of the grid where there is no existing SCADA solution and can defer costly equipment upgrades needed to implement a long-term SCADA solution. Furthermore, in an operational context, these devices enable shorter restoration times through a better understanding of the causes and locations of outages throughout our distribution system.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 24 of 158

(a)	(b)
Program:	Distribution Sensing and Monitoring (inclg Line Sensors) (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 7
DGP Section:	12.4
Scope:	• Install overhead line sensors on distribution circuits that do not have any source of fault locating data • Replace defective line sensors and currently-installed sensors that have obsolete telecommunication technology (cellular 3G) • Install low-cost substation monitoring equipment at 4.8kV substations that don't have SCADA, including both feeder and transformer positions • Deploy voltage-sensing smart fault indicators at key distribution locations, particularly those with high DER penetration • Deploy underground smart fault indicators on underground circuits or portions of circuits with no other source of fault locating data • Deploy underground smart fault indicators on A/C Network netbanks that don't have SCADA monitoring capability • Integrate PQ data from existing and new ION meters installed at low-cost monitoring substation transformer positions

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 25 of 158

(a)	(b)						
Program:	Distribution Sensing and Monitoring (inclgd Line Sensors) (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Help identify and locate faults and dispatch crews to the right location • Reduce patrol time and time to perform switching analysis and load jumpering • Provide more accurate load data for system planning and design • Enhance grid-wide situational awareness as a low-cost alternative to SCADA • Understand DER impacts to customer voltage and power quality • Improve customer reliability by avoiding approximately 20 M customer minute interruptions annually by 2026						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
					x	x	

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 26 of 158

(a)	(b)																				
Program:	Distribution Sensing and Monitoring (inclgd Line Sensors) (Pg. 4 of 4)																				
Current Projects:	Line Sensors Underground Line Sensor Technology Voltage-Sensing Smart Fault Indicator Technology Low-Cost Substation Monitoring Ion Meter Data Integration																				
Budget Basis:	• Engineering estimate																				
Cost:	Estimated project spend in 2021-2023: \$17.4 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>608</td><td>6,562</td><td>7,171</td></tr><tr><td>Material</td><td>122</td><td>1,312</td><td>1,434</td></tr><tr><td>Other</td><td>81</td><td>875</td><td>956</td></tr><tr><td>Total</td><td>811</td><td>8,750</td><td>9,561</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	608	6,562	7,171	Material	122	1,312	1,434	Other	81	875	956	Total	811	8,750	9,561
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	608	6,562	7,171																		
Material	122	1,312	1,434																		
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Total	811	8,750	9,561																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 27 of 158

(a)

(b)

Program:	SCADA / AMI Enhancements (Pg. 1 of 4)
Purpose and Necessity:	DTEE's deployment of an Advanced Distribution Management System (ADMS), along with an anticipated Distributed Energy Resource Management System (DERMS), will have a profound impact on the operation of the grid to provide safer, more reliable power to its customers. These tools provide advanced applications that require timely, accurate and available data sources in order to inform correct operating decisions. Two of the most vital sources of system data are SCADA and AMI.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 28 of 158

(a)	(b)
Program:	SCADA / AMI Enhancements (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 8
DGP Section:	12.4
Scope:	• SCADA on all 3-phase operating devices: • Substation transformer data enhancements: DTEE plans to equip all substation and station transformers with breaker duty monitoring for asset health, and sequence of events recording. • SCADA on all voltage regulating equipment: Voltage regulating equipment, such as regulators and capacitors, can be found both in the substations and on the overhead circuits. As part of the Sensing and Monitoring strategy, DTEE expects to outfit all equipment with 2-4 second interval SCADA data. • C&I and large customer meter data enhancement • Push notifications for AMI near DER or CVR/VVO

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 29 of 158

(a)	(b)						
Program:	SCADA / AMI Enhancements (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• In addition to the remote-control capabilities SCADA provides, the variety of available data and alarms from all remote SCADA-enabled field devices provides crucial situational awareness and feedback to System Operations. • AMI provides key data directly from the customer • Provides safer, more reliable power to its customers						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x				x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 30 of 158

(a)	(b)																												
Program:	SCADA / AMI Enhancements (Pg. 4 of 4)																												
Current Projects:	SCADA / AMI Enhancements																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$2.3 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>94</td><td>938</td><td>1,031</td></tr><tr><td>Material</td><td>19</td><td>188</td><td>206</td></tr><tr><td>Other</td><td>13</td><td>125</td><td>138</td></tr><tr><td>Total</td><td>125</td><td>1,250</td><td>1,375</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	94	938	1,031	Material	19	188	206	Other	13	125	138	Total	125	1,250	1,375
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	94	938	1,031																										
Material	19	188	206																										
Other	13	125	138																										
Total	125	1,250	1,375																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 31 of 158

(a)	(b)
Program:	Automation: Substation (Pg. 1 of 4)
Purpose and Necessity:	Distribution automation technologies can achieve substantial grid impacts including improved Fault Location, Isolation and Service Restoration (FLISR) capabilities, improved distribution system resilience to extreme weather, more effective equipment monitoring and preventative maintenance, more efficient use of repair crews and truck rolls and improved grid integration of selected distributed energy resources (DER). Distribution Automation, spanning from automated field devices, advanced protection, to SCADA, is identified as one of the core components for grid modernization as seen in the DOE's DSPx framework, referenced in Part III of my direct testimony. Distribution Automation provides immediate system benefits, while also supporting other functional capabilities in the DSPx framework that may be added in the future, as discussed above.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 32 of 158

(a)	(b)
Program:	Automation: Substation (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 9
DGP Section:	12.5
Scope:	Substation Automation • Control panel replacements with modern, standardized relays • Installation of Remote Terminal Units (RTU) • Incorporation of the RTUs and automation controls in the same substation network • Breaker replacements as needed

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 33 of 158

(a)	(b)						
Program:	Automation: Substation (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduction in the number customer interruptions and outage duration • Improved distribution system resilience to extreme weather • More effective equipment monitoring and preventative maintenance • More efficient use of repair crews and truck rolls and improved grid integration of selected distributed energy resources (DER)						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 34 of 158

(a)	(b)			
Program:	Automation: Substation (Pg. 4 of 4)			
Current Projects:	Automation: Substation			
Budget Basis:	• Engineering Estimate			
Cost:	Estimated project spend in 2021-2023: \$5.2 M This project is expected to continue beyond 2023			
Test Year (\$000's):	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	
	Total Labor (DTE Electric & Contractors)	255	1,875	2,130
	Material	51	375	426
	Other	34	250	284
	Total	340	2,500	2,840

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 35 of 158

(a)	(b)
Program:	Automation: Distribution (Pg. 1 of 4)
Purpose and Necessity:	The Smart Grid Investment Grant Projects, as discussed in the DOE report “Distribution Automation: Results from the Smart Grid Investment Grant Program”, demonstrated distribution automation technologies can achieve substantial grid impacts including improved Fault Location, Isolation and Service Restoration (FLISR) capabilities, improved distribution system resilience to extreme weather, more effective equipment monitoring and preventative maintenance, more efficient use of repair crews and truck rolls and improved grid integration of selected distributed energy resources (DER). Distribution Automation, spanning from automated field devices, advanced protection, to SCADA, is identified as one of the core components for grid modernization as seen in the DOE's DSPx framework, referenced in Figure 3, Part III of my direct testimony and discussed in detail in the EPRI report included in this testimony as Exhibit A-23, Schedule M9. Distribution Automation provides immediate system benefits, while also supporting other functional capabilities in the DSPx framework that may be added in the future.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 36 of 158

(a)

(b)

Program:	Automation: Distribution (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 10
DGP Section:	12.5
Scope:	• Installation of remotely controllable / automatic reclosers and sensing devices • Reconductoring as needed to enable circuit load transfer during outage conditions

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 37 of 158

(a)	(b)						
Program:	Automation: Distribution (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduction in the number customer interruptions and outage duration • Improved distribution system resilience to extreme weather • More effective equipment monitoring and preventative maintenance • More efficient use of repair crews and truck rolls and improved grid integration of selected distributed energy resources (DER)						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 38 of 158

(a)	(b)																				
Program:	Automation: Distribution (Pg. 4 of 4)																				
Current Projects:	Circuit Automation																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$7.0 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>253</td><td>3,125</td><td>3,378</td></tr><tr><td>Material</td><td>51</td><td>625</td><td>676</td></tr><tr><td>Other</td><td>34</td><td>417</td><td>450</td></tr><tr><td>Total</td><td>337</td><td>4,167</td><td>4,504</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	253	3,125	3,378	Material	51	625	676	Other	34	417	450	Total	337	4,167	4,504
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	253	3,125	3,378																		
Material	51	625	676																		
Other	34	417	450																		
Total	337	4,167	4,504																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 39 of 158

(a)	(b)
Program:	CVR/VVO (Pg. 1 of 4)
Purpose and Necessity:	Volt Var Optimization (VVO) manages system-wide voltage levels and reactive power flow to achieve one or more specific operating objectives. The objectives can include reducing losses, managing voltage volatility due to intermittent renewable generation, optimizing operating parameters and/or optimizing power factors, etc. Conservation Voltage Reduction (CVR), as one of the VVO options, is designed to maintain customer voltage levels in the lower portion of the allowable voltage ranges, thus reducing system losses, peak demand or energy consumption.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 40 of 158

(a)	(b)
Program:	CVR/VVO (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 11
DGP Section:	12.6
Scope:	Pilot CVR/VVO on six distribution circuits by end of 2020: • Installing Remote Terminal Units and SCADA at substations in order to lower bus voltage • Installing advanced line sensors on circuits • Installing remote controllable capacitor banks to improve circuit voltage profile during peak hours • Upgrading existing capacitor banks to improve circuit voltage during peak hours Scope for 2021/2022 program that leverages ADMS to achieve advanced CVR/VVO: • Upgrade substation transformer load tap changers (LTC) with remote control capability in order to control bus voltage • Install remote controls for existing overhead capacitor banks or install new smart capacitor banks at targeted locations on the circuits • Install remote controls for line regulators • Install voltage sensors at strategic locations to monitor primary voltage levels • Implement the Volt-Var Control (VVC) module from the ADMS to allow real-time voltage control and optimization

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 41 of 158

(a)	(b)						
Program:	CVR/VVO (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	Pilot CVR VVO: • 0.9%-1.4% for peak demand and 0.7%-1.0% on energy consumptions on substations installed with pilot CVR/VVO Advanced CVR VVO: • 1.5-3.5% savings in energy consumption and peak demand on substations installed with advanced CVR/VVO						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x				x	

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 42 of 158

(a)	(b)																				
Program:	CVR/VVO (Pg. 4 of 4)																				
Current Projects:	CVR/VVO Capacitor Placement and Control Program																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$28.9 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>875</td><td>10,876</td><td>11,751</td></tr><tr><td>Material</td><td>175</td><td>2,175</td><td>2,350</td></tr><tr><td>Other</td><td>117</td><td>1,450</td><td>1,567</td></tr><tr><td>Total</td><td>1,167</td><td>14,502</td><td>15,668</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	875	10,876	11,751	Material	175	2,175	2,350	Other	117	1,450	1,567	Total	1,167	14,502	15,668
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	875	10,876	11,751																		
Material	175	2,175	2,350																		
Other	117	1,450	1,567																		
Total	1,167	14,502	15,668																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 43 of 158

(a)	(b)
Program:	NWA: O'Shea Energy Storage (Pg. 1 of 4)
Purpose and Necessity:	O'Shea Park in Detroit is the site of a 2MW solar array connected to Chicago Substation. The solar generation frequently exceeds circuit load, causing power to flow back through the substation. During periods where solar generation is intermittent or changing rapidly, voltage data as reported by the AMI meters indicates there is some potential for small over-voltages or flickers.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 44 of 158

(a)	(b)
Program:	NWA: O'Shea Energy Storage (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 12
DGP Section:	12.7
Scope:	• Install a 1MW x 1MWh battery co-located with the solar array.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 45 of 158

(a)	(b)						
Program:	NWA: O'Shea Energy Storage (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Improve power quality in the area by addressing voltage instability						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
					x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 46 of 158

(a)	(b)																				
Program:	NWA: O'Shea Energy Storage (Pg. 4 of 4)																				
Current Projects:	NWA: O'Shea Energy Storage																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$1.4 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>16</td><td>-</td><td>16</td></tr><tr><td>Material</td><td>3</td><td>-</td><td>3</td></tr><tr><td>Other</td><td>2</td><td>-</td><td>2</td></tr><tr><td>Total</td><td>21</td><td>-</td><td>21</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	16	-	16	Material	3	-	3	Other	2	-	2	Total	21	-	21
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	16	-	16																		
Material	3	-	3																		
Other	2	-	2																		
Total	21	-	21																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 47 of 158

(a)

(b)

Program:	NWA: Battery Trailer (Pg. 1 of 4)
Purpose and Necessity:	Explore and understand non-wire alternatives to provide relief to the electrical system.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 48 of 158

(a)	(b)
Program:	NWA: Battery Trailer (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 13
DGP Section:	12.7
Scope:	• Build a 1MW x 4MWh trailer-mounted mobile battery storage unit • Test the unit and develop documentation and required operating procedures to deploy the mobile battery in the field • Purchase 4 additional trailers by 2025 pending results of initial evaluation of the equipment

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 49 of 158

(a)	(b)						
Program:	NWA: Battery Trailer (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Use in place of traditional generators • Support system needs during shutdowns or maintenance • Use at substations or on circuits to reduce peak load as part of a broader NWA project						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x					

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 50 of 158

(a)	(b)																												
Program:	NWA: Battery Trailer (Pg. 4 of 4)																												
Current Projects:	NWA: Battery Trailer																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$2.0 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>63</td><td>938</td><td>1,000</td></tr><tr><td>Material</td><td>13</td><td>188</td><td>200</td></tr><tr><td>Other</td><td>8</td><td>125</td><td>133</td></tr><tr><td>Total</td><td>83</td><td>1,250</td><td>1,333</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	63	938	1,000	Material	13	188	200	Other	8	125	133	Total	83	1,250	1,333
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	63	938	1,000																										
Material	13	188	200																										
Other	8	125	133																										
Total	83	1,250	1,333																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 51 of 158

(a)	(b)
Program:	NWA: Omega Load Relief (Pg. 1 of 4)
Purpose and Necessity:	Recently in the Harrison Twp./Mount Clemens area there has seen a large influx of indoor agriculture operations. Omega substation currently feeds this area. The 40kv truck lines feeding Omega substation are currently exceeding their emergency ratings increasing the risk of stranded load in the area. the Blocking at this sub is currently set at 10.9 MVA. The 2020 peak at OMEGA was 16.3 MVA with over 980 hours of blocking. There has been a total of 19 load requests in this area since 2019. There is an additional 1.5 MVA of committed new customer load expected to energize in 2021. There is an additional 3.6 MVA expected by Q2 2022. The current system in the area cannot handle the expected load growth without increasing the risk of stranded load and equipment overloads. This project will utilize NWA's to reduce loading on OMEGA substation and mitigate overloads on the subtransmissions feeds.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 52 of 158

(a)	(b)
Program:	NWA: Omega Load Relief (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 14
DGP Section:	12.7
Scope:	• Purchase 2 mobile batteries for Omega • Install mobile batteries at Omega Substation • 2MW utility scale solar site at Gibraltar trade center. If site not available rooftop solar options will be pursued

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 53 of 158

(a)	(b)						
Program:	NWA: Omega Load Relief (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Provides approximately 1 MVA of load relief at peak times • Reduces blocking hours and the risk of stranded load						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x					

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 54 of 158

(a)	(b)																				
Program:	NWA: Omega Load Relief (Pg. 4 of 4)																				
Current Projects:	NWA: Omega Load Relief																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$7.8 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>503</td><td>-</td><td>503</td></tr><tr><td>Material</td><td>101</td><td>-</td><td>101</td></tr><tr><td>Other</td><td>67</td><td>-</td><td>67</td></tr><tr><td>Total</td><td>670</td><td>-</td><td>670</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	503	-	503	Material	101	-	101	Other	67	-	67	Total	670	-	670
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	503	-	503																		
Material	101	-	101																		
Other	67	-	67																		
Total	670	-	670																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 55 of 158

(a)

(b)

Program:	NWA: Fisher Load Relief (Pg. 1 of 4)
Purpose and Necessity:	Fisher Substation is over firm by approximately 2.5 MVA. In addition, two circuits served by Fisher are over or expected to be over the 8MVA design standard for 13.2kV circuits.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 56 of 158

(a)	(b)
Program:	NWA: Fisher Load Relief (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 15
DGP Section:	12.7
Scope:	• EWR Scope - Residential implementation, incentives and installation labor, C&I (commercial and industrial) implementation and incentives • DR Scope - Program implementation for Smart Savers and Cool Currents, technology platform and IAC equipment upgrades • STDF Scope - Establish STDF along same subtransmission line that feeds Fisher Substation, split DC 8188 FISHR with STDF circuit, install new cable and conduit

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 57 of 158

(a)	(b)						
Program:	NWA: Fisher Load Relief (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce equipment risk by providing load relief • Provide sufficient capacity to serve existing and new customers • Defer traditional load relief projects						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:		x			x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 58 of 158

(a)	(b)		
Program:	NWA: Fisher Load Relief (Pg. 4 of 4)		
Current Projects:	NWA: Fisher Load Relief		
Budget Basis:	• Engineering Estimate		
Cost:	Estimated project spend in 2021-2023: \$4.3 M		
Test Year (\$000's):	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23
	Total Labor (DTE Electric & Contractors)	105	2,114
	Material	21	423
	Other	14	282
	Total	140	2,819
			2,958

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 59 of 158

(a)

(b)

Program:	NWA: Port Austin Load Relief (Pg. 1 of 4)
Purpose and Necessity:	Port Austin substation is over its firm rating and one circuit is over its day-to-day rating. The traditional investment to alleviate load in this area would be to upgrade the substation and some of the circuits to 13.2kV. This program is to explore and understand non-wire alternatives to provide load relief to the electrical system.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 60 of 158

(a)	(b)
Program:	NWA: Port Austin Load Relief (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 16
DGP Section:	12.7
Scope:	• Install 500kW solar site • Use one existing 1MW x 4MWh battery from Omega • Explore feasibility of microgrid equipment and controls that could serve about 200 kVA

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 61 of 158

(a)	(b)						
Program:	NWA: Port Austin Load Relief (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce equipment risk by providing load relief						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x			x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 62 of 158

(a)	(b)																				
Program:	NWA: Port Austin Load Relief (Pg. 4 of 4)																				
Current Projects:	NWA: Port Austin Load Relief																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$4.5 M																				
Test Year (\$000's):	<table><thead><tr><th></th><th>2 mos. 11/01/22 12/31/22</th><th>10 mos. 01/01/23 10/31/23</th><th>Test Year 12 mos. ending 10/31/23</th></tr></thead><tbody><tr><td>Total Labor (DTE Electric & Contractors)</td><td>237</td><td>1,563</td><td>1,801</td></tr><tr><td>Material</td><td>47</td><td>313</td><td>360</td></tr><tr><td>Other</td><td>32</td><td>208</td><td>240</td></tr><tr><td>Total</td><td>317</td><td>2,084</td><td>2,401</td></tr></tbody></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	237	1,563	1,801	Material	47	313	360	Other	32	208	240	Total	317	2,084	2,401
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	237	1,563	1,801																		
Material	47	313	360																		
Other	32	208	240																		
Total	317	2,084	2,401																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 63 of 158

(a)

(b)

Program:	NWA: Veridian (Pg. 1 of 4)
Purpose and Necessity:	Veridian is a planned development in Ann Arbor. The area is served by Regent substation, which is a 4.8kV substation. While the substation loading is not the primary concern, the overall load of the development is expected to approach 1.5MVA, and none of the existing circuits would have capacity to accommodate that load.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 64 of 158

(a)	(b)
Program:	NWA: Veridian (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 17
DGP Section:	12.7
Scope:	• Leverage customer owned solar and storage • Create community microgrid utilizing demand response and grid controls

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 65 of 158

(a)	(b)						
Program:	NWA: Veridian (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduce equipment risk by providing load relief • Provide capacity to serve new development • Defers conversion to 13.2kV						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x			x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 66 of 158

(a)	(b)																				
Program:	NWA: Veridian (Pg. 4 of 4)																				
Current Projects:	NWA: Veridian																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$7.4 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>230</td><td>3,484</td><td>3,714</td></tr><tr><td>Material</td><td>46</td><td>697</td><td>743</td></tr><tr><td>Other</td><td>31</td><td>465</td><td>495</td></tr><tr><td>Total</td><td>307</td><td>4,645</td><td>4,952</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	230	3,484	3,714	Material	46	697	743	Other	31	465	495	Total	307	4,645	4,952
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	230	3,484	3,714																		
Material	46	697	743																		
Other	31	465	495																		
Total	307	4,645	4,952																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 67 of 158

(a)

(b)

Program:	NWA: Small Solar and Storage Testbed (Pg. 1 of 4)
Purpose and Necessity:	Many residential and small commercial rooftop solar interconnections are being proposed with some level of battery storage. The MPSC has requested DTEE provide a solution to properly meter and integrate these storage solutions into the grid and allow battery systems to export onto the grid while still maintaining compliance with distributed generation, net metering and other tariffs.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 68 of 158

(a)	(b)
Program:	NWA: Small Solar and Storage Testbed (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 18
DGP Section:	12.7
Scope:	• Install solar arrays, commonly used inverters, storage units, switching and metering systems, and utility gateway communication systems at a DTEE site and allow for understanding the impact it will have on the electric grid.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 69 of 158

(a)

(b)

Program:	NWA: Small Solar and Storage Testbed (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Allows for validation of behavior, interaction and compliance of the new features for smart inverters and acts as a location to test future new features and capabilities while also serving as a training platform for DTEE field services, engineers and technicians • Standards can be developed around these technologies to ensure safety and reliable operation						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x			x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 70 of 158

(a)	(b)																				
Program:	NWA: Small Solar and Storage Testbed (Pg. 4 of 4)																				
Current Projects:	NWA: Small Solar and Storage Testbed																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$1.0 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><thead><tr><th></th><th>2 mos. 11/01/22 12/31/22</th><th>10 mos. 01/01/23 10/31/23</th><th>Test Year 12 mos. ending 10/31/23</th></tr></thead><tbody><tr><td>Total Labor (DTE Electric & Contractors)</td><td>47</td><td>172</td><td>219</td></tr><tr><td>Material</td><td>9</td><td>34</td><td>44</td></tr><tr><td>Other</td><td>6</td><td>23</td><td>29</td></tr><tr><td>Total</td><td>63</td><td>229</td><td>292</td></tr></tbody></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	47	172	219	Material	9	34	44	Other	6	23	29	Total	63	229	292
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	47	172	219																		
Material	9	34	44																		
Other	6	23	29																		
Total	63	229	292																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 71 of 158

(a)	(b)
Program:	NWA: EV Charging Demonstration at ACM (Pg. 1 of 4)
Purpose and Necessity:	The purpose of this project is to explore and understand the impacts of EV charging on the electrical grid.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 72 of 158

(a)	(b)
Program:	NWA: EV Charging Demonstration at ACM (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 19
DGP Section:	12.7
Scope:	• ACM (American Center for Mobility) has the capability to test autonomous electric vehicles. DTEE is supporting the implementation of the Delta Power Electronics DC Xtreme fast charger when it becomes available in 2021 as part of a DOE project. • DTEE will use this partnership to understand the impacts on the electrical grid.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 73 of 158

(a)	(b)						
Program:	NWA: EV Charging Demonstration at ACM (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Ability to develop monitoring and control capabilities with smart charge controls and inductive road charging to ensure reliable and secure interfaces • Enhance knowledge of these new technologies on grid impacts						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x			x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 74 of 158

(a)	(b)																												
Program:	NWA: EV Charging Demonstration at ACM (Pg. 4 of 4)																												
Current Projects:	NWA: EV Charging Demonstration at ACM																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$2.6 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>100</td><td>813</td><td>913</td></tr><tr><td>Material</td><td>20</td><td>163</td><td>183</td></tr><tr><td>Other</td><td>13</td><td>108</td><td>122</td></tr><tr><td>Total</td><td>133</td><td>1,083</td><td>1,217</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	100	813	913	Material	20	163	183	Other	13	108	122	Total	133	1,083	1,217
	2 mos.	10 mos.	Test Year																										
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Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 75 of 158

(a)	(b)
Program:	Technology Programs & NWA (Pg. 1 of 4)
Purpose and Necessity:	Test new technologies and concepts on a small scale to ensure system compatibility and economic justification before executing them system-wide.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 76 of 158

(a)	(b)
Program:	Technology Programs & NWA (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 20
DGP Section:	12.9
Scope:	• Automatic pole top devices for the 4.8 kV system • SCADA-enabled overhead capacitors • SCADA-enabled overhead regulators

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 77 of 158

(a)	(b)						
Program:	Technology Programs & NWA (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Test new technology for compatibility and economic justification prior to system-wide implementation • Develop experience and knowledge base on new technologies • Provide learning prior to system wide deployment to improve project execution						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
					x		x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 78 of 158

(a)	(b)																																				
Program:	Technology Programs & NWA (Pg. 4 of 4)																																				
Current Projects:	Pilot: Secondary and Service Undergrounding IST Solar Energy Storage New Technology Pilots Trip Saver Program Capacitor Placement & Control Program																																				
Budget Basis:	• Engineering Estimate																																				
Cost:	Estimated project spend in 2021-2023: \$0.0 M																																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td></td><td colspan="3"> </td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td>-</td><td>-</td><td>-</td></tr><tr><td></td><td colspan="3"> </td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23					Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total	-	-	-				
	2 mos.	10 mos.	Test Year																																		
	11/01/22	01/01/23	12 mos. ending																																		
	12/31/22	10/31/23	10/31/23																																		
Total Labor (DTE Electric & Contractors)	-	-	-																																		
Material	-	-	-																																		
Other	-	-	-																																		
Total	-	-	-																																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 79 of 158

(a)	(b)
Program:	DERMS (Pg. 1 of 4)
Purpose and Necessity:	This program aims to modernize the company's existing telemetry and control capabilities for medium and large Distributed Energy Resources using industry-standard equipment to lower the cost, maximize visibility and ultimately support the safe and efficient integration of these resources.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 80 of 158

(a)	(b)
Program:	DERMS (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 21
DGP Section:	12.8
Scope:	• DERA integration and ICCP connectivity

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 81 of 158

(a)	(b)						
Program:	DERMS (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Improved visibility into DER output translates into more efficient operations of the distribution circuits to which these units are connected. More timely data acquisition supports the situational awareness of the control room, which translates into safer and more reliable operation. • Refer to the testimony of Pankaj Sharma for further explanation of customer benefits						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x	x			x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 82 of 158

(a)	(b)																												
Program:	DERMS (Pg. 4 of 4)																												
Current Projects:	Large / Medium Sized DER Monitoring & Control Upgrade DR-SOC Monitored Solar Sites to RTAC RTU Viper Reclosing Sensing / Data																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$5.1 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>318</td><td>1,588</td><td>1,905</td></tr><tr><td>Material</td><td>64</td><td>318</td><td>381</td></tr><tr><td>Other</td><td>42</td><td>212</td><td>254</td></tr><tr><td>Total</td><td>423</td><td>2,117</td><td>2,540</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	318	1,588	1,905	Material	64	318	381	Other	42	212	254	Total	423	2,117	2,540
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
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Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 83 of 158

(a)	(b)
Program:	DER Control (Pg. 1 of 4)
Purpose and Necessity:	This program is intended to provide the SCADA and back-office side of the changes required to support dispatchable DER integration at scale, for support of FERC Orders 2222 and 841.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 84 of 158

(a)

(b)

Program:	DER Control (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 22
DGP Section:	12.8
Scope:	• DER Control installed in over thirty solar sites

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 85 of 158

(a)	(b)						
Program:	DER Control (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	DERMS are designed to streamline the integration of large amounts of DER into utility systems. This integration improves the data fed to Contingency Analysis and State Estimator systems, permitting them to converge more readily and provide solutions that more accurately reflect the distribution network's power flow. In turn, this improves the Control Room's situational awareness, which translates into safer and more reliable operation of the distribution system.						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x	x	x		x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 86 of 158

(a)	(b)																												
Program:	DER Control (Pg. 4 of 4)																												
Current Projects:	DERA Monitoring Integration ICCP Connectivity DERs in a DERA ICCP Connectivity - Load shed in a DERA AMI, Secondary, De-Aggregated Load Data OSI DERMS Module OSI DERMS Module Modeling OSE Enterprise Customer Agreement																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$0.6 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>36</td><td>181</td><td>218</td></tr><tr><td>Material</td><td>7</td><td>36</td><td>44</td></tr><tr><td>Other</td><td>5</td><td>24</td><td>29</td></tr><tr><td>Total</td><td>48</td><td>242</td><td>290</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	36	181	218	Material	7	36	44	Other	5	24	29	Total	48	242	290
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
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Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 87 of 158

(a)	(b)
Program:	Mobile Technology (Pg. 1 of 4)
Purpose and Necessity:	Mobile technology allows collecting and distributing data at the point of activity through mobile devices, across all personas and use cases. Mobile capabilities include both devices and applications that are engineered for mobile access and usage. Applications include dispatch, work execution and forms digitization, location tracking, route navigation, analytics and secure file sharing. Additionally, with the onset of the Covid-19 pandemic, Distribution Operations has accelerated plans to deploy more capable technology into the field with an emphasis on system offerings that provide the frontline workers and leaders the flexibility of remaining in the field for extended periods without having system constraints that require them to return to the service centers or to congregate as frequently as was needed in the past. Even after the pandemic, these new skills and processes will continue to drive efficiency.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 88 of 158

(a)	(b)
Program:	Mobile Technology (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 23
DGP Section:	12.9
Scope:	• Select and implement a new and improved vegetation management platform that will provide a more reliable and scalable mobile end-to-end tree trimming operations solution to support initiation, planning, execution and audit close-out. This solution will include computer hardware, licensing, configuration and integrations to work management systems. • Implement an interface between the AMI system and the mobile dispatch and work management solution for the Electric Field Operations organization to perform meter reads and other operations remotely, allowing for safer operations and the ability to verify work completed while in the field. • Implement various application configurations, interfaces, hardware, and software to improve mobile work in the field.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 89 of 158

(a)	(b)						
Program:	Mobile Technology (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Enable field workforce to operate safely from both Covid-19 distancing requirements, as well as enabling safe work by providing information to frontline workers at the point of activity • Deploy tools to enable efficient work at the point of activity, reducing drive-time to service centers						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x					x	

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 90 of 158

(a)	(b)																												
Program:	Mobile Technology (Pg. 4 of 4)																												
Current Projects:	Vegetation Management Platform AMI Integration with Mobile Other Mobile Technology																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$5.0 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>56</td><td>2,859</td><td>2,916</td></tr><tr><td>Material</td><td>11</td><td>572</td><td>583</td></tr><tr><td>Other</td><td>8</td><td>381</td><td>389</td></tr><tr><td>Total</td><td>75</td><td>3,813</td><td>3,888</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	56	2,859	2,916	Material	11	572	583	Other	8	381	389	Total	75	3,813	3,888
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	56	2,859	2,916																										
Material	11	572	583																										
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Total	75	3,813	3,888																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 91 of 158

(a)	(b)
Program:	Work Management & Scheduling Upgrades (Pg. 1 of 4)
Purpose and Necessity:	An effort is underway in DTEE to improve work management and scheduling processes to ensure field resources have the processes and tools to support the right jobs, with the right people, at the right time. This includes ensuring unscheduled work is reflected in the work management system, Maximo, and the field resources can receive and status jobs in real-time to create a feedback loop on asset condition and performance in alignment with the asset management capabilities.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 92 of 158

(a)	(b)
Program:	Work Management & Scheduling Upgrades (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 24
DGP Section:	12.9
Scope:	• Acquire software licenses for additional platform functionality • Server installations and replacements • Software configuration and IT software development

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 93 of 158

(a)

(b)

Program:	Work Management & Scheduling Upgrades (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Process improvements, reduction of jobs held due to resource constraints, higher schedule compliance rate and CWIP backlog reduction resulting in capital efficiency and O&M reduction • Additionally, the "Stabilizing DO's Maximo Instance" project in 2023 will improve overall experience with Maximo as follows: 1) improve experience for users to make modifications in Maximo, 2) transfer ownership to business units to lessen the time it takes to make system modifications, 3) reduce administrative burden in utilizing IT resources to make the system modifications, and 4) provide end-users ability to access Maximo on their mobile devices						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
					x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 94 of 158

(a)	(b)																				
Program:	Work Management & Scheduling Upgrades (Pg. 4 of 4)																				
Current Projects:	Stabilize DO Maximo Instance Other Work Management & Scheduling Maximo Return to Support																				
Budget Basis:	• IT Estimate																				
Cost:	Estimated project spend in 2021-2023: \$12.4 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>188</td><td>6,813</td><td>7,000</td></tr><tr><td>Material</td><td>38</td><td>1,363</td><td>1,400</td></tr><tr><td>Other</td><td>25</td><td>908</td><td>933</td></tr><tr><td>Total</td><td>250</td><td>9,083</td><td>9,333</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	188	6,813	7,000	Material	38	1,363	1,400	Other	25	908	933	Total	250	9,083	9,333
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	188	6,813	7,000																		
Material	38	1,363	1,400																		
Other	25	908	933																		
Total	250	9,083	9,333																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 95 of 158

(a)	(b)
Program:	Substation Design Tool Upgrades (Pg. 1 of 4)
Purpose and Necessity:	The current process for designing any substation work or updating schematics utilizes a 2D CAD system. These drawings have no links to any additional data and all notes and comments are essentially text on the prints with no ability to dynamically search and update the drawing. There is no automatic way to generate different views or cross sections of a design, requiring significant rework for each required design and construction drawing. There is also no automatic way to generate a bill of materials or design cost estimate from the drawings. Furthermore, capabilities are needed to automatically detect clearance and set back conflicts, to take into account high resolution site plan information and to reduce the effort needed to design and construct substations and upgrades to substation equipment. Wire and conduit routing must be done manually and validated through each drawing and every reference. Schematics must be manually compared to all design drawings and other schematics to ensure that reference numbers, phasing and continuity is correct. The current process has no automated checking or data validation and each print must be reviewed manually and against each other to ensure that dimensions, wiring terminations and equipment position during every change. Quality control is done completely through a manual review process. While standard drawing elements are available in a library, any modification must be checked manually and any dimensional or elevation drawing must be carefully constructed each time it is needed. When working with outside design firms, multiple file conversions are needed to coordinate drawings and bring them into DTE formats. Industry tools have improved and capabilities now exist in the software to have 3D models of equipment that have relevant attributes and configurable parameters. These sub assemblies can be placed in spatially accurate configurations and views and drawings from any angle can be generated quickly and consistently. The sub assembly parts and bill of materials can be aggregated into project level lists and interfaced with estimating systems like Maximo to order and plan the work in conjunction with the design. Vendor designs can be quickly integrated and validated through quality tools and wire and conduit routing can be checked for conflicts and clearance concerns from the same model with minimal effort. Schematic drawings are tied directly to structural and physical layouts so that the consistency checks are continuous throughout the process saving significant review time. Changes to standards can be propagated into new designs from the sub assembly libraries and validation of clearances and set backs can be performed quickly for all components effected.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 96 of 158

(a)	(b)
Program:	Substation Design Tool Upgrades (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 25
DGP Section:	12.9
Scope:	• Identify scope of upgraded software platform • Procure software and hardware for project • Work on cell libraries and metadata • Work with IT to configure platform

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 97 of 158

(a)	(b)						
Program:	Substation Design Tool Upgrades (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Reduction in rework and improvement to consistency • Automatic generation of bill of materials and cost estimate framework reduces manual work and errors • Automated quality control checks and automatic cross referencing of drawings reduces review time and errors						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x				x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 98 of 158

(a)	(b)																				
Program:	Substation Design Tool Upgrades (Pg. 4 of 4)																				
Current Projects:	Substation Design Tool Upgrades																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$2.7 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>178</td><td>680</td><td>858</td></tr><tr><td>Material</td><td>36</td><td>136</td><td>172</td></tr><tr><td>Other</td><td>24</td><td>91</td><td>114</td></tr><tr><td>Total</td><td>237</td><td>906</td><td>1,144</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	178	680	858	Material	36	136	172	Other	24	91	114	Total	237	906	1,144
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	178	680	858																		
Material	36	136	172																		
Other	24	91	114																		
Total	237	906	1,144																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 99 of 158

(a)	(b)
Program:	Asset Management Upgrades (Pg. 1 of 4)
Purpose and Necessity:	The overall goal of effectively managing assets is to minimize the total cost of ownership and operations while delivering the desired service levels. This includes maintaining access to accurate digital data about the asset, algorithms to process and analyze data, and a visual context of an asset's location and condition. Asset management includes managing the full lifecycle of an asset, engineering and financial management, as well as linking spatial information in the ESRI Mapping system to an asset. Managing the electric grid is an asset-intensive operation, and as such, relies heavily on the capabilities of its Asset Management systems..
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 100 of 158

(a)	(b)
Program:	Asset Management Upgrades (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 26
DGP Section:	12.9
Scope:	• Acquire software licenses for additional platform functionality • Server installation and replacements • Software configuration and IT software development • Development of software API's and integration between platforms • Development of machine learning models and deployment to real-time data pipeline

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 101 of 158

(a)	(b)						
Program:	Asset Management Upgrades (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Have single view into asset health on the distribution system, across equipment and system cross-sections • Bring equipment databases to supported enterprise platforms						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x			x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 102 of 158

(a)	(b)																				
Program:	Asset Management Upgrades (Pg. 4 of 4)																				
Current Projects:	ESRI System Upgrades Other Asset Management																				
Budget Basis:	• High Level IT Estimates																				
Cost:	Estimated project spend in 2021-2023: \$3.4 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>163</td><td>1,297</td><td>1,459</td></tr><tr><td>Material</td><td>33</td><td>259</td><td>292</td></tr><tr><td>Other</td><td>22</td><td>173</td><td>195</td></tr><tr><td>Total</td><td>217</td><td>1,729</td><td>1,946</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	163	1,297	1,459	Material	33	259	292	Other	22	173	195	Total	217	1,729	1,946
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	163	1,297	1,459																		
Material	33	259	292																		
Other	22	173	195																		
Total	217	1,729	1,946																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 103 of 158

(a)	(b)
Program:	Load Forecasting & Analytics (Pg. 1 of 4)
Purpose and Necessity:	Advanced situational awareness analytics are needed for planning system upgrade decision support. New developments in distributed generation resources, electric vehicle adoption, and energy waste reduction/demand response are requiring a change in process for distribution load and generation forecasting. The process, and thus data infrastructure, need to evolve into a more integrated and granular approach to distribution load and generation forecasting. To address the needs to forecast changing customer-side equipment adoption, such as DER, energy waste reduction, and microgrids, which will inform the scenario-based distribution planning, DTEE will need to collect data on customer adoption and propensity of future adoption for various customer-side equipment.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 104 of 158

(a)	(b)
Program:	Load Forecasting & Analytics (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 27
DGP Section:	12.9
Scope:	• Selecting of Loading Analytics tool for implementation to provide load curve generation. Configuration of tool and integration with sources of data (AMI, GIS, PI, weather, DER adoption). Integration of load curve data with planning tool (CYME) with automatic analysis. Aggregation algorithms to evaluate loading of equipment at all levels of the grid. • Configuration of CYME's data gateway to align with network model updates, specifically substations and secondary models. Upgrade to CYME 8.2. Addition of the Advanced Project Management module. Integrate AMI data and dynamic Data Pull plugin to better analysis grid conditions during planning. • Development of DER and EV propensity forecasting tools within the DTE Enterprise Data platform, Databricks. These machine learning models will incorporate external factors and demographics to predict customer adoption of DER, such as energy efficiency equipment, electric vehicles, solar inverters, and micro-grids. • Creation of analysis tools and models to allow system planners to validate impacts of DER on energy usage, system efficiency, and reliability. • Software development for optimization analytics to consume forecasted scenarios and run numerous plausible scenarios for customers adoption across the system • Feature engineering to summarize data aligned with integrated Distribution Planning use cases. Developing artificial intelligence and machine learning technologies that describe, predict and prescribe suggested actions to take during the integrated distribution planning process, including spatial and temporal contextualization of conditions and scenarios to provide situational awareness in the planning process.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 105 of 158

(a)	(b)						
Program:	Load Forecasting & Analytics (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Improve reliability, power quality and customer flexibility through enabling distribution planning process capabilities in alignment with grid modernization. • Load curve generation - 8760 hour load curves representing disaggregated usage and customer generation curves • Planning processes to include multiple scenarios, including forecasting of DER and EV propensity across the system, with automated analysis support for planning engineers • Ability to forecast and sequence multi-year projects within the distribution planning toolset • Improved accuracy of system planning results with integration of AMI data						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x	x		x	x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 106 of 158

(a)	(b)																												
Program:	Load Forecasting & Analytics (Pg. 4 of 4)																												
Current Projects:	CYME Planning Tool Upgrades Loading Allocation Analytics Distribution Load Forecasting Situational Awareness Analytics																												
Budget Basis:	• IT evaluation of platform and data interfaces																												
Cost:	Estimated project spend in 2021-2023: \$6.9 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>394</td><td>1,953</td><td>2,347</td></tr><tr><td>Material</td><td>79</td><td>391</td><td>469</td></tr><tr><td>Other</td><td>53</td><td>260</td><td>313</td></tr><tr><td>Total</td><td>525</td><td>2,604</td><td>3,129</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	394	1,953	2,347	Material	79	391	469	Other	53	260	313	Total	525	2,604	3,129
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	394	1,953	2,347																										
Material	79	391	469																										
Other	53	260	313																										
Total	525	2,604	3,129																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 107 of 158

(a)	(b)
Program:	Interconnection Process Enablement (Pg. 1 of 4)
Purpose and Necessity:	DERs are expected to be an increasingly common technology deployed on the electric grid. In particular, the DG/DS scenario from the Distribution Grid Plan considers directionally plausible forecasts where customer-owned generation, predominantly expected to be rooftop solar PVs and storage in the form of batteries, is expected to increase in adoption throughout DTEE's service territory. As the number of applications for DER increases, a critical component in providing accessibility to the grid will be to ensure that the process to connect to the distribution grid is streamlined and efficient. Enhancements to the interconnection process and technology to effectively capture and utilize information like the location, size and type of DER will help distribution planning efforts.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 108 of 158

(a)	(b)
Program:	Interconnection Process Enablement (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 28
DGP Section:	12.9
Scope:	• Improve customer experience with application process - upgrade Power Clerk and interconnection application portal to provide seamless customer logon between sites (single sign on), implement electronic payment for applications, provide interface to transparently display application status to customer including integration with Maximo internal work tracking system for status updates. • Improve internal system integrations for processing interconnection applications - system integration from planning application to planning study tool, CYME, and enterprise data lake. Automation scripts will support automated circuit analysis, application trending reporting, and project optimization opportunities. Automated screening tools will include loading, transformer size compatibility, and voltage constraints on the system, as well as verifying customer is within DTEE's service territory. • Study tools to support aggregated resources - enhance study tools, CYME and the enterprise data lake, to support emergent regulatory requirements including FERC 841 and FERC 2222. Study tools will need to evaluate aggregated resources across locations on the grid while evaluating interrelated impacts.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 109 of 158

(a)	(b)						
Program:	Interconnection Process Enablement (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Improve customer experience during the interconnection application process • Maintain reliable electric service by providing technology to study interrelated grid impacts of DER • Improve process efficiency evaluating interconnection applications as requests scale up						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:			X		X	X	

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 110 of 158

(a)	(b)																				
Program:	Interconnection Process Enablement (Pg. 4 of 4)																				
Current Projects:	Interconnection Process Enablement																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$7.4 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>471</td><td>2,260</td><td>2,731</td></tr><tr><td>Material</td><td>94</td><td>452</td><td>546</td></tr><tr><td>Other</td><td>63</td><td>301</td><td>364</td></tr><tr><td>Total</td><td>628</td><td>3,013</td><td>3,641</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	471	2,260	2,731	Material	94	452	546	Other	63	301	364	Total	628	3,013	3,641
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
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Material	94	452	546																		
Other	63	301	364																		
Total	628	3,013	3,641																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 111 of 158

(a)	(b)
Program:	Hosting Capacity Enablement (Pg. 1 of 4)
Purpose and Necessity:	Hosting capacity analysis (HCA) is the amount of DER that can be accommodated without adversely impacting operational criteria such as power quality, reliability, and safety under existing grid control and operations without requiring infrastructure upgrades. In addition, in Case No. U-20147, the MPSC has requested that DTEE adopt a phased implementation approach for HCA pilots. As the adoption rate of DERs on the system increases, HCA will be a critical tool for distribution planning to identify initial grid constraints. Furthermore, a hosting capacity map can give DER developers the ability to better select sites and possibly reduce interconnection applications for non-viable areas with grid constraints. However, as DTEE has noted in prior discussions and filings, HCA is not a substitute for the developer submitting an application to study interconnecting to the grid. An application to interconnect is still necessary to identify impacts caused by the interconnect and propose solutions to mitigate those impacts.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 112 of 158

(a)	(b)
Program:	Hosting Capacity Enablement (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 29
DGP Section:	12.9
Scope:	• Technology investment is needed to achieve the maturing Hosting Capacity Analysis processes. To improve the speed in which customers can evaluate the ability to interconnect DERs with the grid, investment is required in both customer-facing and internal evaluation tools. The technology investment will include a system to store information on likelihood of system upgrades needed to support DER interconnection, as well as a visual interface that is externally facing for customers and developers to access. • To support DTEE's processes to evaluate hosting capacity at a granular level, investments in tools and integrations are needed. This will begin with tools such as EPRI DRIVE and CYME integration capacity analysis to evaluate areas with high penetration of DER. Further years will see investments to link the hosting capacity into the interconnection application portal. • Various server installation • Software configuration and IT software development • IT interface development and deployment

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 113 of 158

(a)	(b)						
Program:	Hosting Capacity Enablement (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Customers able to see proactive information for their interconnection applications • Screening and application tools improved for efficiency • Helps DTE meet regulatory requirement						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
			x		x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 114 of 158

(a)	(b)																				
Program:	Hosting Capacity Enablement (Pg. 4 of 4)																				
Current Projects:	Hosting Capacity Enablement																				
Budget Basis:	• High Level IT Estimates																				
Cost:	Estimated project spend in 2021-2023: \$0.5 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><thead><tr><th></th><th>2 mos. 11/01/22 12/31/22</th><th>10 mos. 01/01/23 10/31/23</th><th>Test Year 12 mos. ending 10/31/23</th></tr></thead><tbody><tr><td>Total Labor (DTE Electric & Contractors)</td><td>31</td><td>156</td><td>188</td></tr><tr><td>Material</td><td>6</td><td>31</td><td>38</td></tr><tr><td>Other</td><td>4</td><td>21</td><td>25</td></tr><tr><td>Total</td><td>42</td><td>208</td><td>250</td></tr></tbody></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	31	156	188	Material	6	31	38	Other	4	21	25	Total	42	208	250
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	31	156	188																		
Material	6	31	38																		
Other	4	21	25																		
Total	42	208	250																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: P. Smith
Page: 115 of 158

(a)

(b)

Program:	AMI: Installations (Pg. 1 of 4)
Purpose and Necessity:	This project will complete the installation of the remaining hard-to-reach AMI meters as completion of the company's AMI program, and replace the remaining electromechanical meters with AMI meters, automating meter reading.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: P. Smith
Page: 116 of 158

(a)	(b)
Program:	AMI: Installations (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 30
DGP Section:	Not Available
Scope:	• Updated AMI technology in back office systems • Replaced the remaining 630 meters with AMI metering • Provided call center support for customer appointments, scheduling and communication

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: P. Smith
Page: 117 of 158

(a)	(b)						
Program:	AMI: Installations (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Increase bill accuracy • Enable remote alarm on energy theft and tampering • Increase customer outage notification and restoration efficiency • Enable remote power quality monitoring						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:						x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: P. Smith
Page: 118 of 158

(a)	(b)																				
Program:	AMI: Installations (Pg. 4 of 4)																				
Current Projects:	AMI: Installations																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.0 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td>-</td><td>-</td><td>-</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total	-	-	-
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	-	-	-																		
Material	-	-	-																		
Other	-	-	-																		
Total	-	-	-																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: P. Smith
Page: 119 of 158

(a)	(b)
Program:	AMI: Meter Communications Upgrades (Pg. 1 of 4)
Purpose and Necessity:	Public Wireless Telecommunication providers are phasing out 3G cellular in Michigan by the end of 2021. After this date, 3G technology will no longer be supported, and existing 3G devices will not be operable with the cellular 4G platforms. The existing AMI metering deployment is dependent on 3G cellular public carrier service for approximately 1 million of the 2.6 million customer electric meters and all data for customer billing, outage analysis, electric grid analytics, etc. Without this technology transition from 3G cellular to 4G cellular, all AMI benefits related to 1.1 million residential and small commercial meters will cease and these meters will no longer be remotely accessible for billing, outage detection, voltage data or remote connect / disconnect capabilities, severely impacting customer service.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: P. Smith
Page: 120 of 158

(a)	(b)
Program:	AMI: Meter Communications Upgrades (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 31
DGP Section:	12.9
Scope:	• Update technology to operate on 4G public cellular networks: - Replace 2,700 3G cell relays with 4G devices - Optimize deployments to reduce seasonal vegetation impact - Sustain network performance & remediate coverage gaps

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: P. Smith
Page: 121 of 158

(a)

(b)

Program:	AMI: Meter Communications Upgrades (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Mitigate the impact of cellular 3G phase out by public wireless carriers on the existing AMI network, and attendant customer focused services. • Increase reliability of AMI reporting, by providing faster data collection during outages, increase the frequency of normal day-to-day data receipt and reduce network disruptions caused by vegetation. • Enable use of the AMI infrastructure as a shared infrastructure for future expansion of circuit automation.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
					x		x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: P. Smith
Page: 122 of 158

(a)	(b)																				
Program:	AMI: Meter Communications Upgrades (Pg. 4 of 4)																				
Current Projects:	AMI Enhanced Support AMI: 3G to 4G Communication Upgrade AMI: 3G to 4G Industrial Upgrade																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$2.3 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>63</td><td>313</td><td>375</td></tr><tr><td>Material</td><td>13</td><td>63</td><td>75</td></tr><tr><td>Other</td><td>8</td><td>42</td><td>50</td></tr><tr><td>Total</td><td>83</td><td>417</td><td>500</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	63	313	375	Material	13	63	75	Other	8	42	50	Total	83	417	500
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	63	313	375																		
Material	13	63	75																		
Other	8	42	50																		
Total	83	417	500																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 123 of 158

(a)	(b)
Program:	Microwave End of Life (Pg. 1 of 4)
Purpose and Necessity:	The microwave network is used as a primary and secondary WAN communications for corporate and SCADA (FCN) networks. The modernization of telecom microwave tower equipment will optimize performance and align our frequency usage with the FCC. Due to the FCC deregulation of privately owned frequencies, a number of DTE's microwave systems must be replaced to maintain a reliable network and avoid collisions with other entities using the same frequency.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 124 of 158

(a)	(b)
Program:	Microwave End of Life (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 32
DGP Section:	12.9
Scope:	• Design and implement new microwave backhaul (WAN) communications • Install modern dishes for tower weight reduction • Install hardware for network reliability and redundancy • Replace end of life hardware

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 125 of 158

(a)	(b)						
Program:	Microwave End of Life (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Improved reliability and control, alleviating dependency on third party systems • Network modernization and resiliency						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x			x	x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 126 of 158

(a)	(b)																				
Program:	Microwave End of Life (Pg. 4 of 4)																				
Current Projects:	Microwave End of Life																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$1.0 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>63</td><td>313</td><td>375</td></tr><tr><td>Material</td><td>13</td><td>63</td><td>75</td></tr><tr><td>Other</td><td>8</td><td>42</td><td>50</td></tr><tr><td>Total</td><td>83</td><td>417</td><td>500</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	63	313	375	Material	13	63	75	Other	8	42	50	Total	83	417	500
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	63	313	375																		
Material	13	63	75																		
Other	8	42	50																		
Total	83	417	500																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 127 of 158

(a)	(b)
Program:	SCADA Remote Access and Configuration Platform (Pg. 1 of 4)
Purpose and Necessity:	The existing SCADA tool server is the only mechanism to remotely access and configure relays, RTU's, and other field equipment. Without this server all changes or validation of settings require a field visit. The SCADA tool server allows engineers and technicians to remotely perform many day to day operations to check on the status and health of devices on a detailed level far beyond the monitoring capabilities in SCADA. The existing tool server is not supported by the vendor.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 128 of 158

(a)	(b)
Program:	SCADA Remote Access and Configuration Platform (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 33
DGP Section:	12.9
Scope:	• Provisioning of upgraded and supported hardware at downtown and alternate data center in resilient configuration • Integration of SUBNET solutions Power SYSTEM Center • Integration of ION Enterprise • Configuration of PI interfaces for PQ • Configuration of PI interfaces to support increased data load • Configuration of SCADA and System Protection & Automation tools

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 129 of 158

(a)	(b)						
Program:	SCADA Remote Access and Configuration Platform (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Remote configuration management of SCADA devices resulting in less field visits • Immediate verification and health monitoring of assets • Centralization of power quality data from SCADA into single management platform allowing for quick and easy access						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x	x			x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 130 of 158

(a)	(b)																				
Program:	SCADA Remote Access and Configuration Platform (Pg. 4 of 4)																				
Current Projects:	SCADA Remote Access and Configuration Platform																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$1.9 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>159</td><td>-</td><td>159</td></tr><tr><td>Material</td><td>32</td><td>-</td><td>32</td></tr><tr><td>Other</td><td>21</td><td>-</td><td>21</td></tr><tr><td>Total</td><td>212</td><td>-</td><td>212</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	159	-	159	Material	32	-	32	Other	21	-	21	Total	212	-	212
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	159	-	159																		
Material	32	-	32																		
Other	21	-	21																		
Total	212	-	212																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 131 of 158

(a)	(b)
Program:	Automation Configuration and Test Record Database (Pg. 1 of 4)
Purpose and Necessity:	Relay and SCADA Configuration Data was migrated from an obsolete FOCUS database when WINDOWS XP was retired. The current database tool is functioning, but is at risk of not being improved or supported due to its custom design. When migrating from FOCUS to the new tool, SCADA data was not added to the database and is currently stored on network drives. Due to process issues and lack of software support, the status of certain protection settings is not accurate and requires manual data entry and conversion. This project would procure a tool that would perform those functions and augment the asset and work management processes already developed. Relay and SCADA Configuration data is important to accurately record. NERC CIP dictates how protection settings need to be calculated for assets and these settings would be stored in the database. The integration with this software will streamline work in System Protection and Automation. Streamlined tools also have built in workflow tools that allow data to be automatically update status. This will help to ensure the accuracy of the status of the data in the database and the coordination with the work management system, Maximo.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 132 of 158

(a)	(b)
Program:	Automation Configuration and Test Record Database (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 34
DGP Section:	12.9
Scope:	• Data Preparation and Conversion - Review test records and update relay settings status' in current database - Identify missing settings and add to the existing database - Flag NERC relay configurations • Evaluate Software Options - Reach out to vendors for quotes, specifications, and capabilities, conduct RFP - Compare and select software • Purchase software, order hardware, install and configure • Develop standards and job unit materials for engineers and technicians • User implementation • Go live

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 133 of 158

(a)	(b)						
Program:	Automation Configuration and Test Record Database (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Automate relay testing and test record process • Improved accuracy of data which will ensure NERC compliance • Rapid discovery of equipment configuration via search tools • Audit and version history for data and configuration settings • Automatic and preconfigured test routines • Streamline coordination studies with data link to short circuit modeling tools						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x				x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 134 of 158

(a)	(b)																				
Program:	Automation Configuration and Test Record Database (Pg. 4 of 4)																				
Current Projects:	Automation Configuration and Test Record Database																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$4.2 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>281</td><td>1,089</td><td>1,370</td></tr><tr><td>Material</td><td>56</td><td>218</td><td>274</td></tr><tr><td>Other</td><td>38</td><td>145</td><td>183</td></tr><tr><td>Total</td><td>375</td><td>1,452</td><td>1,827</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	281	1,089	1,370	Material	56	218	274	Other	38	145	183	Total	375	1,452	1,827
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
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Total	375	1,452	1,827																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 135 of 158

(a)	(b)
Program:	Grid Edge Insights & New Technology (Pg. 1 of 4)
Purpose and Necessity:	As more grid interactive customer equipment is connected to the electrical system in the form of smart inverters, energy storage, electric vehicles (including Vehicle to Grid), demand response and load control resources, non-wire alternatives and microgrids, the complexity of real time grid management increases with each new interaction of Distributed Energy Resources. DER Resources will need to work together with interoperable communications that may be local, but will still need to be interfaced to a central system like the ADMS. Not all of the capability of DER resources can be coordinated in real time by a central system such as the ADMS. There are elements of DER operations, optimization and protection that have timing requirements that cannot be addressed without local intelligence. The safe, reliable and efficient operation of these DER resources requires a distributed platform that can take control from and report information to utility backend systems such as the ADMS and future DERMs system while locally managing DER endpoints to achieve a system management goal. This mix of local and remote control, utility and customer owned resources also requires a significant emphasis on cyber security.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 136 of 158

(a)	(b)
Program:	Grid Edge Insights & New Technology (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 35
DGP Section:	12.9
Scope:	• This project addresses the real time interoperability, cyber security communications and control of distributed resources and their interaction with local and remote control systems such as the ADMS and DERMS. It is focused on the 'Grid Edge' where real time optimization of local DER resources is required across operational and ownership boundaries. • The project will design, test, standardize and implement cyber secure control and communications schemes for DER resources and microgrids using an interoperable platform based on industry standards such as Open Field Message Bus. • The platform will be modular and project scope will build on functionality developed in conjunction with the Department of Energy, National Labs and other utilities that are facing similar DER integration goals. • The platform will integrate current and future standards for customer owned DER to enable the broadest possible enablement of the DER capabilities while maintaining a cyber security first design. • The platform will be iterated upon over multiple years to optimize cost, reduce maintenance and integrate new developments in the industry, DER capabilities and standards. • The platform will be used in a standardized and repeatable manner to allow for the greatest possible efficiency in deployment to areas that require the grid edge capability.

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 137 of 158

(a)	(b)						
Program:	Grid Edge Insights & New Technology (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	The Grid Edge platform will allow for safe and reliable interaction with DER resources in complex implementations where the coordinated interaction of Smart Inverters, energy storage, load control and microgrids is required. Standardization of the platform will reduce per project costs and the variety of different systems that need to be implemented and maintained. The platform will also allow for simulation of DER interaction during project design phases to reduce on site testing and provide more reliable operation of the DER solutions.						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x				x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 138 of 158

(a)	(b)																												
Program:	Grid Edge Insights & New Technology (Pg. 4 of 4)																												
Current Projects:	Grid Edge Insights & New Technology																												
Budget Basis:	• Engineering Estimate																												
Cost:	Estimated project spend in 2021-2023: \$4.1 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>213</td><td>1,125</td><td>1,338</td></tr><tr><td>Material</td><td>43</td><td>225</td><td>268</td></tr><tr><td>Other</td><td>28</td><td>150</td><td>178</td></tr><tr><td>Total</td><td>283</td><td>1,500</td><td>1,783</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	213	1,125	1,338	Material	43	225	268	Other	28	150	178	Total	283	1,500	1,783
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
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Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 139 of 158

(a)

(b)

Program:	Sensor, Network and Algorithm Development (Solar Deployment) (Pg. 1 of 4)
Purpose and Necessity:	This project is in support of the MPSC's "MI Power Grid" initiative and to prepare for the greater penetration of customer owned solar and FERC Order 2222.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 140 of 158

(a)	(b)
Program:	Sensor, Network and Algorithm Development (Solar Deployment) (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 36
DGP Section:	12.9
Scope:	• Develop low-cost sensors • Deploy distributed computing elements • Grow a centralized machine learning platform • Install a low-cost supplemental network

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 141 of 158

(a)	(b)						
Program:	Sensor, Network and Algorithm Development (Solar Deployment) (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	The project will develop functionality in three target areas to benefit DTE customers including: • Net load forecasting • Power quality and reliability • Transformer health						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
		x	x		x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 142 of 158

(a)	(b)																				
Program:	Sensor, Network and Algorithm Development (Solar Deployment) (Pg. 4 of 4)																				
Current Projects:	Sensor, Network and Algorithm Development (Solar Deployment)																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$1.6 M This project is expected to continue beyond 2023																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>55</td><td>736</td><td>791</td></tr><tr><td>Material</td><td>11</td><td>147</td><td>158</td></tr><tr><td>Other</td><td>7</td><td>98</td><td>105</td></tr><tr><td>Total</td><td>73</td><td>982</td><td>1,055</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	55	736	791	Material	11	147	158	Other	7	98	105	Total	73	982	1,055
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	55	736	791																		
Material	11	147	158																		
Other	7	98	105																		
Total	73	982	1,055																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 143 of 158

(a)	(b)
Program:	Other Modernize Grid Management (Pg. 1 of 4)
Purpose and Necessity:	Grid management includes the full suite of distribution management applications, including advanced metering infrastructure, distribution automation, outage response, and advanced control toolsets. Mature grid management enables distribution operators to model and manage the distribution network, monitor the power system, manage planned and unplanned outages, and analyze and optimize the quality and reliability of the network. This program includes components to enable real-time situational awareness and analytics as well as IT systems to manage the increasing amount of smart devices on the grid.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 144 of 158

(a)	(b)
Program:	Other Modernize Grid Management (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 37
DGP Section:	12.9
Scope:	• Acquire software licenses for additional platform functionality • Server installation and replacements • Software configuration and IT software development • Development of software API's and integration between platforms • Development of machine learning models and deployment to real-time data pipeline

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 145 of 158

(a)	(b)						
Program:	Other Modernize Grid Management (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Improved real-time situational awareness from grid devices • Ability to manage IED configurations in central location • Real-time detection and prediction of power quality issues impacting customers						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:					x	x	x

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 146 of 158

(a)	(b)																												
Program:	Other Modernize Grid Management (Pg. 4 of 4)																												
Current Projects:	Other Modernize Grid Management																												
Budget Basis:	• High Level IT Estimates																												
Cost:	Estimated project spend in 2021-2023: \$5.4 M This project is expected to continue beyond 2023																												
Test Year (\$000's):	<table><tr><td></td><td>2 mos.</td><td>10 mos.</td><td>Test Year</td></tr><tr><td></td><td>11/01/22</td><td>01/01/23</td><td>12 mos. ending</td></tr><tr><td></td><td>12/31/22</td><td>10/31/23</td><td>10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>583</td><td>469</td><td>1,052</td></tr><tr><td>Material</td><td>117</td><td>94</td><td>210</td></tr><tr><td>Other</td><td>78</td><td>63</td><td>140</td></tr><tr><td>Total</td><td>777</td><td>625</td><td>1,402</td></tr></table>		2 mos.	10 mos.	Test Year		11/01/22	01/01/23	12 mos. ending		12/31/22	10/31/23	10/31/23	Total Labor (DTE Electric & Contractors)	583	469	1,052	Material	117	94	210	Other	78	63	140	Total	777	625	1,402
	2 mos.	10 mos.	Test Year																										
	11/01/22	01/01/23	12 mos. ending																										
	12/31/22	10/31/23	10/31/23																										
Total Labor (DTE Electric & Contractors)	583	469	1,052																										
Material	117	94	210																										
Other	78	63	140																										
Total	777	625	1,402																										

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 147 of 158

(a)

(b)

Program:	Substation Cybersecurity (Pg. 1 of 4)
Purpose and Necessity:	This work is necessary to avoid and reduce system vulnerability to cyber attacks on the distribution system.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 148 of 158

(a)	(b)
Program:	Substation Cybersecurity (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 38
DGP Section:	16.3
Scope:	This is a pilot program to implement Substation Cyber security solutions at Stone Pool Substation. The scope includes the following solutions to correct security gaps: • Patching • HMI Anti-Virus • Special purpose devices • Alerting • Vulnerability scanning • HMI hardening • Asset and configuration management

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 149 of 158

(a)	(b)						
Program:	Substation Cybersecurity (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Increase security posture and reduce cybersecurity risk						
Impact Dimension: Targeted:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
	x						

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 150 of 158

(a)	(b)																				
Program:	Substation Cybersecurity (Pg. 4 of 4)																				
Current Projects:	Substation Cybersecurity																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$2.3 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>288</td><td>-</td><td>288</td></tr><tr><td>Material</td><td>58</td><td>-</td><td>58</td></tr><tr><td>Other</td><td>38</td><td>-</td><td>38</td></tr><tr><td>Total</td><td>383</td><td>-</td><td>383</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	288	-	288	Material	58	-	58	Other	38	-	38	Total	383	-	383
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	288	-	288																		
Material	58	-	58																		
Other	38	-	38																		
Total	383	-	383																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 151 of 158

(a)	(b)
Program:	Operational Technology and Error Free Communication (Pg. 1 of 4)
Purpose and Necessity:	Customers are frustrated because based on the data we have, we cannot quickly and definitively answer this question - do we know if a customer has power. Therefore, we cannot feed that information into our workflows and this creates errors in communication with our customers about the current state of their restoration. Additionally, we have prioritized operational efficiency over restoration data accuracy - we must balance operational efficiency with customer experience. - No single view of customer outage status based on all sources of information available to DTE (Customer, AMI, Equipment, Jobs, Restoration Estimate) - Currently, closing job in OMS directly translates into the incorrect assumption that customers are restored which is driving our channels to have errors (website, mobile app, outage map) - We are using AMI numbers publicly during a storm which is inconsistent with OMS numbers on the outage map - media, poweroutage.us, and EEI are unclear of the correct number which is reducing our credibility
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 152 of 158

(a)	(b)
Program:	Operational Technology and Error Free Communication (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 39
DGP Section:	Not Available
Scope:	The following key scope items will be delivered as part of the Error Free Communications project: • For the first time, we will combine meter data (AMI), work order data, customer data and equipment hierarchy that makes up our entire grid to create single source of truth for outage data • Using the single source of truth, we will be able to further validate outage data with Smart Meter Data and more accurately close jobs • We will move and develop quickly to make meaningful impacts. We will expose insights to existing applications, with the ability to reliably scale 24/7, 365

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 153 of 158

(a)	(b)						
Program:	Operational Technology and Error Free Communication (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Single data source for each customer outage status (Yes/No/Don't Know) to be available all-weather (not just storm) • Customer status and outage map API service layer created, deployed, and can be used by operations and customer systems • Outage Status app, SEL-UOL updated with information to be viewable to all DO employees (office and field)						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:	x				x		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 154 of 158

(a)	(b)																				
Program:	Operational Technology and Error Free Communication (Pg. 4 of 4)																				
Current Projects:	Operational Technology and Error Free Communication																				
Budget Basis:	• Project Management Estimate																				
Cost:	Estimated project spend in 2021-2023: \$12.9 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>250</td><td>-</td><td>250</td></tr><tr><td>Material</td><td>50</td><td>-</td><td>50</td></tr><tr><td>Other</td><td>33</td><td>-</td><td>33</td></tr><tr><td>Total</td><td>333</td><td>-</td><td>333</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	250	-	250	Material	50	-	50	Other	33	-	33	Total	333	-	333
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	250	-	250																		
Material	50	-	50																		
Other	33	-	33																		
Total	333	-	333																		

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 155 of 158

(a)	(b)
Program:	Analog Lines Elimination (Pg. 1 of 4)
Purpose and Necessity:	Analog lines are obsolete telecommunication technology that results in O&M expense. This project is to eliminate the usage of analog lines in our system.
Category:	Technology and Automation

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 156 of 158

(a)	(b)
Program:	Analog Lines Elimination (Pg. 2 of 4)
Line Number:	Exhibit A-12, Schedule B5.4, Page 11, Line 40
DGP Section:	Not Available
Scope:	• Removal of analog lines

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 157 of 158

(a)	(b)						
Program:	Analog Lines Elimination (Pg. 3 of 4)						
Customer benefit / Effect on cost of operation and reliability:	• Remove aging and obsolete equipment and reduce O&M						
Impact Dimension:	Safety	Load Relief	Regulatory Compliance	Major Event Risk	Reliability	O&M Avoidance	Reactive Capital Avoidance
Targeted:						x	

Michigan Public Service Commission
DTE Electric Company
Distribution Plant Capital Project Detail - Technology and Automation

Case No.: U-20836
Exhibit: A-23
Schedule: M6
Witness: S. G. Pfeuffer
Page: 158 of 158

(a)	(b)																				
Program:	Analog Lines Elimination (Pg. 4 of 4)																				
Current Projects:	Analog Lines Elimination																				
Budget Basis:	• Engineering Estimate																				
Cost:	Estimated project spend in 2021-2023: \$0.0 M																				
Test Year (\$000's):	<table><tr><td></td><td>2 mos. 11/01/22 12/31/22</td><td>10 mos. 01/01/23 10/31/23</td><td>Test Year 12 mos. ending 10/31/23</td></tr><tr><td>Total Labor (DTE Electric & Contractors)</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Material</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Other</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Total</td><td>-</td><td>-</td><td>-</td></tr></table>		2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23	Total Labor (DTE Electric & Contractors)	-	-	-	Material	-	-	-	Other	-	-	-	Total	-	-	-
	2 mos. 11/01/22 12/31/22	10 mos. 01/01/23 10/31/23	Test Year 12 mos. ending 10/31/23																		
Total Labor (DTE Electric & Contractors)	-	-	-																		
Material	-	-	-																		
Other	-	-	-																		
Total	-	-	-																		

1.0 Benefit-Cost Analysis and Program / Project Prioritization

The Company assesses the impacts of strategic investment programs and projects on each of its three objectives: risk mitigation, reliability improvement and cost management; which align with the Company's Planning Objectives of safe, reliable and resilient, and affordable, as described as the first step of the Grid Modernization processes. The expected benefits of each program and project are used to develop an initial ranking so that capital investments can be evaluated against each other. The cost of each program and project is used to refine the prioritization. Finally, execution factors such as resource availability and system operational constraints are considered in the development of detailed execution plans.

1.1.1 Investment Programs Evaluation Methodology

Strategic investment programs are evaluated against seven impact dimensions, as described in Table 1.1, for the Distribution Operations Global Prioritization Model (GPM). Quantitative assessments are developed for all the impact dimensions to score and rank programs.

Table 1.1 Program Impact Dimensions

Index	Impact Dimension	Major Drivers
1	Safety	- Reduction in wire down events - Reduction in secondary network cable manhole events - Reduction in major substation events
2	Load Relief	- System capability to meet area load growth and system operability needs - Elimination of system overload or over firm
3	Regulatory Compliance	- MPSC staff's recommendation (March 30, 2010 report) on utilities' pole inspection program - Docket U-12270 – Service restoration under normal conditions within 8 hours - Docket U-12270 – Service restoration under catastrophic conditions within 60 hours - Docket U-12270 – Service restoration under all conditions within 36 hours - Docket U-12270 – Same circuit repetitive interruption of less than 5 within a 12-month period
4	Major Event Risk	Reduction in extensive substation outage events that lead to a large amount of stranded load for more than 24 hours
5	Reliability	Reduction in number of outage events experienced by customers

Index	Impact Dimension	Major Drivers
		Reduction in restoration duration for outage events
6	O&M Cost Avoidance	- Trouble event reduction and truck roll reduction - Preventive maintenance spend reduction
7	Reactive Capital Avoidance	- Trouble event reduction and truck roll reduction - Reduction in capital replacement during equipment failures

Strategic programs are assessed, scored and ranked in a quantitative manner on each impact dimension. Table 1.2 shows the benefit mapping of the programs to each of the impact dimensions.

Table 1.2 Selected Programs and Projects' Benefit Mapping

Program	Safety	Load Relief	Regulatory Compliance	Substation Outage Risk	Reliability	O&M Cost	Reactive Capital
Tree Trimming to the Enhanced Specification	X		X		X	X	X
4.8/8.3 kV Conversion and Consolidation	X	X		X	X	X	X
Substation Risk	X	X		X	X		X
Load Relief		X		X			
System Cable Replacement	X			X	X		X
Breaker Replacement	X			X	X	X	X
ADMS	X	X		X	X	X	X
System Automation	X			X	X	X	X
Subtransmission Hardening	X	X			X	X	X
4.8 kV System Hardening	X		X		X	X	X
Frequent Outage (CEMI)	X		X		X	X	X
URD Cable Replacement					X	X	X
Pole/PTMM	X		X		X	X	X

Some strategic projects are excluded from the prioritization model due to their unique benefits and situations addressed. For instance, subtransmission disconnect switches and the Pontiac vault projects are necessary to meet operational safety needs. The Ann Arbor System Improvement project is necessary to address subtransmission system integrity and power quality concerns in the City of Ann Arbor. Projects that are excluded from the GPM are subject to close examination of their criticality before being deemed as “must fund” in the capital plan.

Additionally, pilots are excluding from the ranking process as they are used to determine what the true benefits and cost will be to complete an activity that the Company believes will be beneficial to its customers. An example is strategic undergrounding. The Company understands that overhead residential services are approximately 16 times more likely to fail during storms, as described in direct testimony. The goal of the pilot will be to determine whether it will be cost-effective to underground existing overhead services, and if so what are the most cost effective methods for construction, and what will be the benefit in reduction of storm related service events.

Detailed analyses based on historical data, engineering assessments and field feedback were performed to quantify programs’ benefits within each impact dimension. The quantified benefits are then compared to the programs’ costs to derive benefit-cost ratios.

Unit measurements used for benefit-cost analysis are different for each impact dimension. Reliability benefits are captured in customer minutes of interruption reduction. O&M and reactive capital benefits are captured in dollars avoided. Safety, load relief, regulatory compliance and major substation outage benefits are rated in indexed scores.

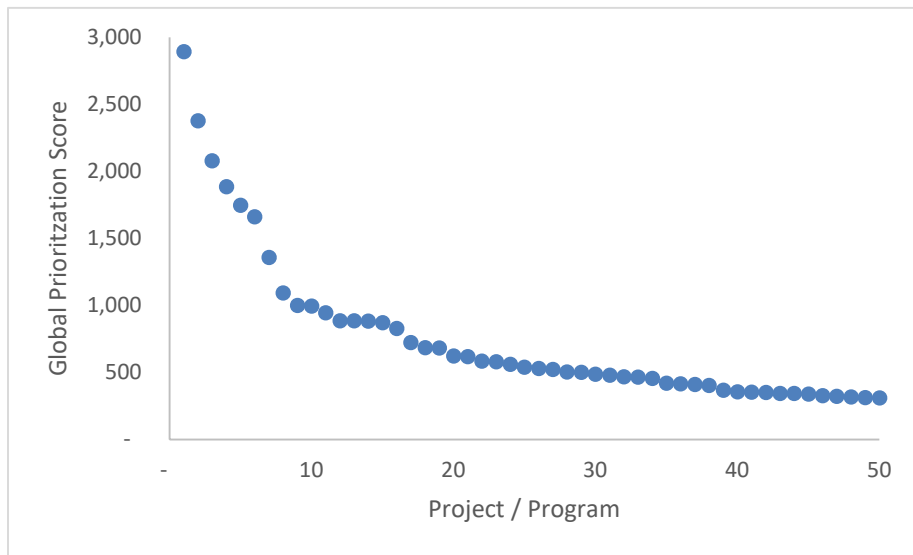
To aggregate a program’s benefit-cost ratios across all the impact dimensions, reliability, O&M and reactive capital, benefit-cost ratios are indexed to benefit-cost scores of 0-100. However, a score of 100 in one dimension is not considered equal to a score of 100 in another dimension. A program’s overall benefit-cost score is calculated as the weighted summation of the program’s benefit-cost scores across all the impact dimensions. Table 1.3 lists the different weights given to different impact dimensions that reflect our customer priorities.

Table 1.3 Impact Dimension Weights

Impact Dimension	Safety	Load Relief	Regulatory Compliance	Substation Outage Risk	Reliability	O&M Cost	Reactive Capital
Weight	10	4	4	4	3	3	3

The benefit-cost scores for strategic capital programs, ranked from highest to lowest, are illustrated in Figure 1.1 and Table 1.4. Tree trimming to the enhanced specification, although excluded from the Tables, continues to provide the highest customer benefits of any program in the five-year investment portfolio.¹

Figure 1.1 Overall Benefit-Cost Scores for Strategic Capital Programs and Projects



¹ Tree trimming, as an O&M program, is excluded from the Exhibits. Nonetheless, tree trimming to the enhanced specification has the highest benefit-cost among all the programs.

**Table 1.4 Top 50 Strategic Capital Programs and Projects Based on Benefit-Cost
Prioritization Ranking**

Rank	Capital Program/ Project	Rank	Capital Program / Project
1	ADMS	26	System Loading: Marlette/Tacoma/Brown City/Shaw
2	CODI: Charlotte Network Conversion	27	Line Sensors
3	Substation Risk: Port Huron	28	APTS
4	System Loading: Mandy 307 Load Transfer	29	4.8kV CC: I-94 (Promenade)
5	4.8kV CC: ISO Kern	30	4.8kV CC: ISO Waterford
6	4.8kV Hardening	31	URD Replacement
7	4.8kV CC: ISO Camden	32	PQ Meter Fault Locating
8	Poles / PTMM	33	4.8kV CC: Belle Isle (Island View)
9	Substation Risk: Novi Decommission	34	4.8kV CC - Birmingham
10	CODI: Garfield Upgrade	35	Relay Replacement: Northeast
11	4.8kV CC: ISO Gilbert	36	Subtransmission Redesign & Rebuild: Trunk 7106
12	4.8kV CC: ISO Brazil	37	Subtransmission Redesign & Rebuild: Trunk 7333
13	Subtransmission Redesign & Rebuild: Trunk 4217	38	Subtransmission Redesign & Rebuild: TRK 3508
14	4.8kV CC: ISO Venoy	39	Breaker Replacement
15	4.8kV CC: ISO Bidle	40	Subtransmission Redesign & Rebuild: Trunk 2255
16	Automation	41	4.8kV Conversion: Belleville Circuit Conversion
17	4.8kV Conversion: Hawthorne Conversion	42	System Loading: Cody Circuits
18	CODI: Amsterdam Upgrade	43	Relay Replacement: Navarre
19	Customer Excellence/ CEMI	44	System Loading: OTSGO/CAPAC/SHAW
20	8.3 kV CC: Pontiac Projects	45	Subtransmission Redesign & Rebuild: Boyne Station
21	CODI: Howard Conversion	46	Subtransmission Redesign & Rebuild: Tie 3416 (OH PIGON)
22	Subtransmission Redesign & Rebuild: Trunk 4266	47	Subtransmission Redesign & Rebuild: Thumb Electric Fault Isolation
23	Cable Replacement	48	System Loading: Macomb/Golf
24	Substation Risk: Apache	49	Subtransmission Redesign & Rebuild: Trunk 3509
25	Subtransmission Redesign & Rebuild: Tie 3416 (Badax Breaker & OH Reconductoring)	50	System Loading: Richmond/Armada

1.1.2 Other Considerations

The benefit-cost scores of programs and projects and their prioritization ranking provide the foundation for the Company's strategic investment decisions. However, there are other key considerations that impact capital allocation and project timing, as listed below.

- Capital spend profiles for new projects are subject to development milestones, especially in the conceptual design and early development stages, including land availability and property purchases, municipal approvals and construction permits, right-of-way acquisition and easements, and lead times when procuring major equipment. While the Company takes proactive measures to mitigate some of these execution risks, many of these early stage milestones are out of the Company's control and can introduce schedule delays or cost increases. Therefore, the Company's investment and maintenance plan is designed to be flexible to accommodate these unpredictable variations in timing and cost.
- Funding decisions for programs and projects need to consider the implication on workforce needs. Resources required for engineering, design, project management, and scheduling and construction need to be evaluated not only by project type (substation, overhead or underground) but also by geographic location. Resource gaps and balancing need to be evaluated before funding decisions are made. Funding decisions must also consider the Company's capacity to manage the work, as the Company must partner with third parties to plan and execute projects. The Company bears the responsibility for overseeing project scope, schedule, cost, and to ensure adherence to its standards.
- As the Company continues to make investments in distribution infrastructure, the effectiveness of the capital spend is examined on a continuous basis. The benefit-cost scores of the programs and projects may change over time as new data become available. The prioritization ranking of the programs and projects may change accordingly.
- Some capital replacement programs are funded annually despite having lower benefit-cost scores. This is done to avoid an acceleration of asset failures and the risk that a large quantity of assets would reach end-of-life concurrently, thus exceeding available resources to replace them (e.g., underground residential distribution cable program).

It is important to note that some programs and projects may not receive immediate funding due to their lower benefit-cost scores, but this does not indicate that these programs or projects are not beneficial to customers. Rather, all the programs and projects identified by the Company provide system improvements and should be funded over the next several years. While the strategic capital investment plan is primarily driven by the GPM, the Company may adjust the annual plans based on changing circumstances.

1.0 Lawrence Berkeley National Laboratory Study

Utilities incur direct costs due to power outages, such as the cost to restore service, which are easily quantifiable. However, there are external costs borne by customers that are often ignored by investment benefit-cost analysis because they are difficult to quantify. By recognizing these costs, the Company is better able to take a holistic view of the full benefit of reliability investment decisions and account for both the internal and external financial benefits associated with improvements in electric reliability. The 2015 Lawrence Berkeley National Laboratory (LBNL) study titled “Updated Value of Service Reliability Estimates for Electric Utility Customers in the United States” provides an aggregate model incorporating nearly all large power interruption cost studies that have been conducted in the US. This study, along with the associated Department of Energy “Interruption Cost Estimate Calculator” (ICECalculator), allow for estimating these customer-borne costs.

The LBNL study and the successive updates, incorporating new service reliability studies, is the most comprehensive effort to estimate the cost, by customer segment, associated with loss of power. Other studies do not represent all segments of customers when determining the customer outage cost. The LBNL study is the industry standard in this field and has been referenced in multiple reports, including the White House sanctioned report entitled “Economic Benefits of Increasing Electric Grid Resilience to Weather Outages.” The LBNL report has also been used by other utilities in their reliability improvement business cases, including Pacific Gas & Electric, Southern Company, National Grid, EPB of Chattanooga, and Central Maine Power. San Diego Gas & Electric has referenced the study in their annual regulatory status report for smart grid deployments and investments dating back to 2012.

1.1 Estimating Customer Costs

The LBNL report’s analysis is based on surveys sent to consumers asking them to estimate the costs, damages, and savings from each outage. Commercial and industrial customers were asked for direct costs, while residential customers were asked about their willingness to pay to avoid outages. The commercial and industrial surveys inquired about major drivers of cost, such as

damage to equipment, damage to materials, wages paid without production, lost sales, percentage of sales to be recouped, wages saved, and energy cost saved.

Survey responses were divided into three different customer groups: Medium and Large Commercial and Industrial (Medium and Large C&I), Small Commercial and Industrial (Small C&I), and Residential. Customers in the Medium and Large C&I group have an annual usage of 50,000 kWh or more.

Through regression analysis, LBNL was able to identify the key variables driving outage costs. The results enabled LBNL to estimate the cost associated with outage duration for each customer group. Table 1 summarizes the results.

Table 1: Customer Savings due to Avoided Outages

Customer Type	Momentary	30 Minutes	1 Hour	4 Hours	8 Hours	16 Hours
Residential	3.9	4.5	5.1	9.5	17.2	32.4
Small Commercial & Industrial	412	520	647	1,880	4,690	9,055
Medium & Large Commercial & Industrial	12,952	15,241	17,804	39,458	84,083	165,482

The DOE commissioned LBNL to develop the ICECalculator based on the study findings. The ICECalculator is an online tool that can be used to estimate the cost of reliability at any level of SAIDI and the ICECalculator can determine cost savings based on two different SAIDI levels (before and after investment, for example). In addition to the LBNL study, the ICECalculator utilizes census and other government geographic data to estimate the economic impacts of outages in each state. This calculator was co-created with LBNL and Nexant, a utility software, consulting and services company.

1.2 Methodology and Customer Savings Projections

The Company projected reliability improvements by modeling the reduction in emergent events due to strategic capital investments and the tree trimming surge (i.e., the spend above what is included in base O&M). Reliability improvements in SAIDI and SAIFI were then used in the ICE Calculator to determine the annual savings for the 2021 to 2025, adjusting projected benefits for inflation. The present value was determined for both the “constrained scenario”, in which only Base Capital investments are made, and the current strategic capital investment plan. The economic benefit for customers is the present value of the difference between the two scenarios.

Customer benefits are expected to be between \$9.8 billion and \$13.2 billion, as illustrated in Figure 1. The low end of the range conservatively assumes that industrial customers are able to recover losses incurred from service interruptions.

Figure 1: Customer Outage Costs (\$billion)

